



CH32H417/H416/H415 Reference Manual

V1.8

<https://wch-ic.com>

Overview

The CH32H417 is an interconnected general-purpose microcontroller based on a dual-core design featuring QingKe RISC-V5F and RISC-V3F architectures. It integrates a USB3.2 Gen1 controller and transceiver, 100/1000M Ethernet MAC and 10/100M PHY, a SerDes high-speed isolated transceiver, and a Type-C/PD controller and PHY. It Provides SD/eMMC controllers, 500MBytes Universal High-Speed Interface (UHSIF), DVP digital image interface, Single Wire Protocol master interface (SWPMI), Programmable Interface Controller (PIOC), Flexible Memory Controller (FMC), DFSDM, LTDC, GPHA, DMA controller, multiple timer groups, 8 serial ports, I3C, 4 groups of I2C, 2 groups of QSPI, 4 groups of SPI, 2 groups of I2S, 3 groups of CAN, and other peripheral resources. It incorporates a 5MHz sampling rate dual 12-bit ADC unit, a 20MHz sampling rate 10-bit high-speed HSADC unit, 16 Touchkey channels, dual DAC units, 3 groups of OPAs, and CMP for analog resources. Supports 10M/100M Ethernet communication, USB2.0 and USB3.0, USB Host and USB Device functionality, Type-C and PD USB fast charging, SerDes high-speed isolation and long-distance transmission, and dual-core task division to enhance network protocol processing efficiency and communication response speed.

This manual provides detailed usage information for the CH32H417, H416, and H415 chips to support user application development.

For device specifications of the CH32H417 series chips, please refer to the CH32H417DS0 manual.

RISC-V core version comparison overview

Features Core versions	Instruction set	Hardware stack levels	Interrupt nesting levels	Fast interrupt channels	Pipeline	Vector table model	Extended instruction	Instruction cache	Memory protection	Interface debug
QingKeV3F	IMAFCB-X	2	2	4	3	Address or command	Supported	-	Standard	1/2-wire
QingKeV5F	IMAFCB-X	8	8	4	7-9	Address or command	Supported	Supported	Standard	1/2-wire

For information regarding the RISC-V core, refer to Chapter 4: Central Processing Unit (CPU).

Abbreviated description of the bit attribute in the register:

Register bit attributes	Property description
RF	Read-only attribute, read a fixed value.
RO	Read-only attribute, changed by hardware.
RZ	Read-only attribute, auto bit clear 0 after read operation.
WO	Write only attribute (not readable, read value uncertain)
WA	Write-only attribute, writable in Safe mode.
WZ	Write only attribute, auto bit clear 0 after write operation.
RW	Readable and writable.
RWA	Readable, writable in Safe mode.
RW1	Readable, write 1 valid, write 0 invalid.
RW0	Readable, write 0 valid, write 1 invalid.
RW1T	Readable, write 0 invalid, write 1 flipped.
RW1Z	Readable, write 1 to clear, write 0 invalid.
SC	Auto cleared.

Chapter 1 Memory and Bus Architecture

1.1 Bus Architecture

The CH32H417 series chips feature an internally integrated dual-core architecture based on the QingKe V3F and V5F designs. Within this architecture, components such as the cores, arbitration unit, DMA module, and SRAM storage interact via multiple bus groups. In the design, a universal DMA controller is integrated to alleviate the CPU load and enhance access efficiency, while also incorporating data protection mechanisms and clock automatic switching safeguards, thereby increasing system stability. The system block diagram is shown in Figure below.

Figure 1-1-1 CH32H417 system block diagram

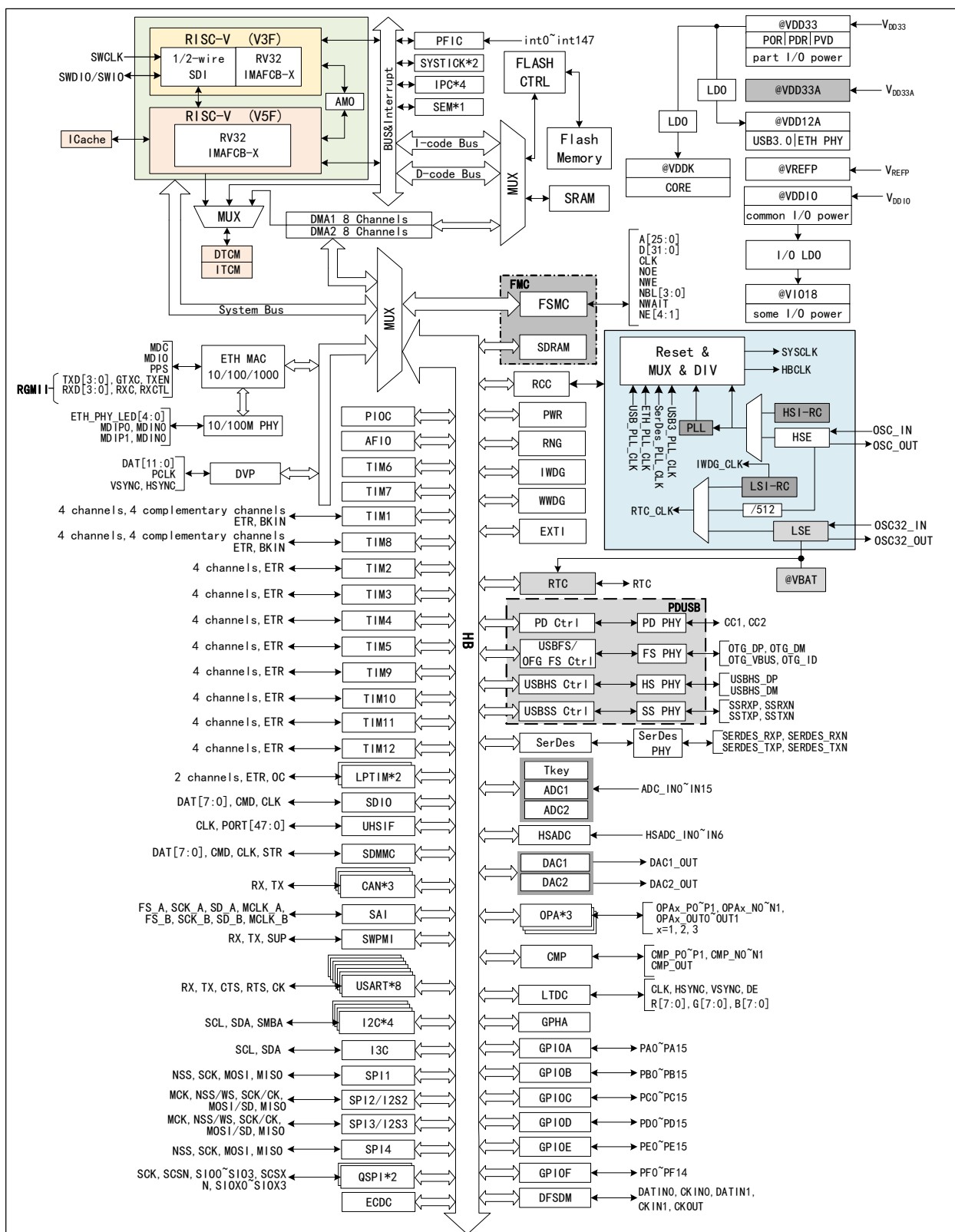


Figure 1-1-2 CH32H416 System block diagram

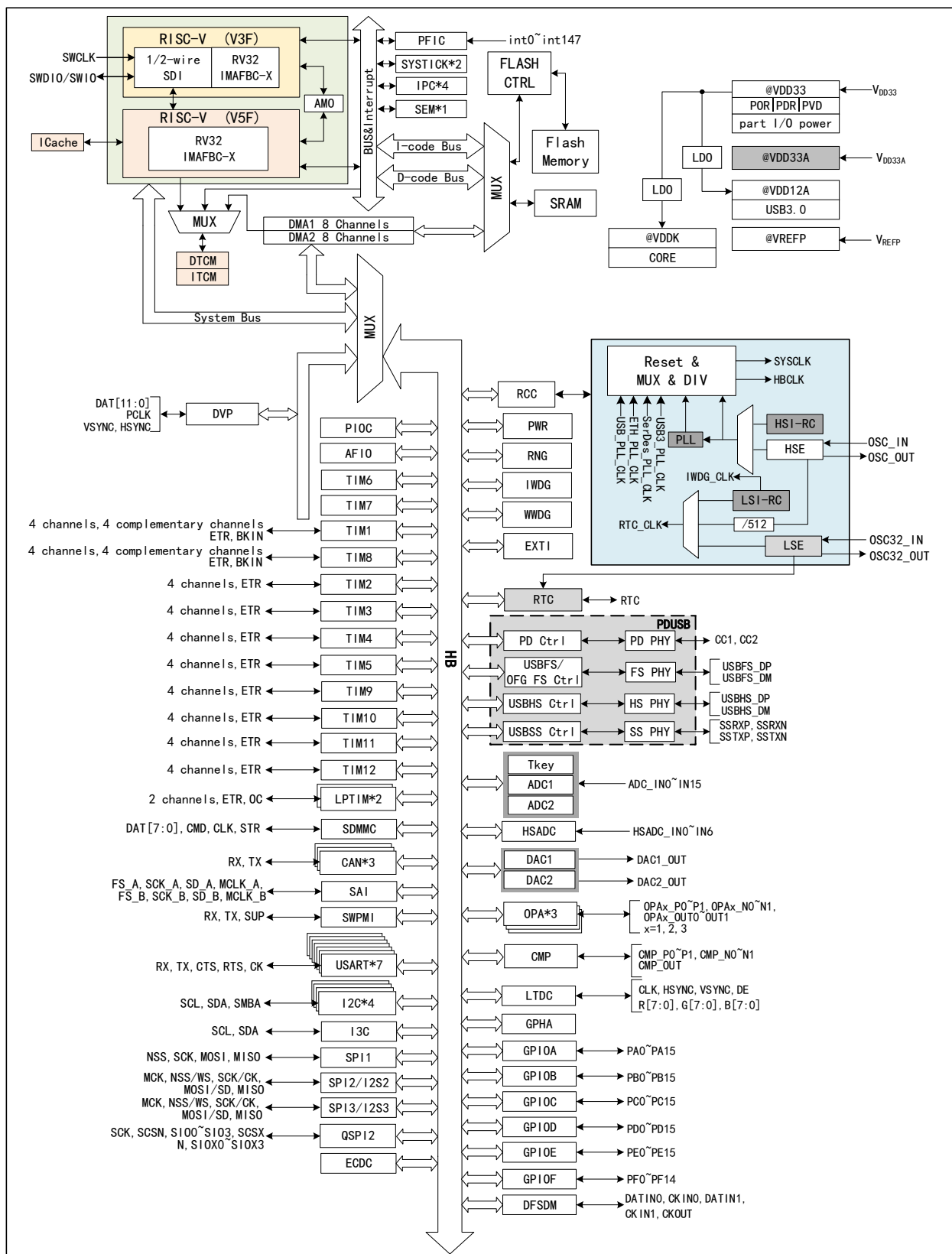
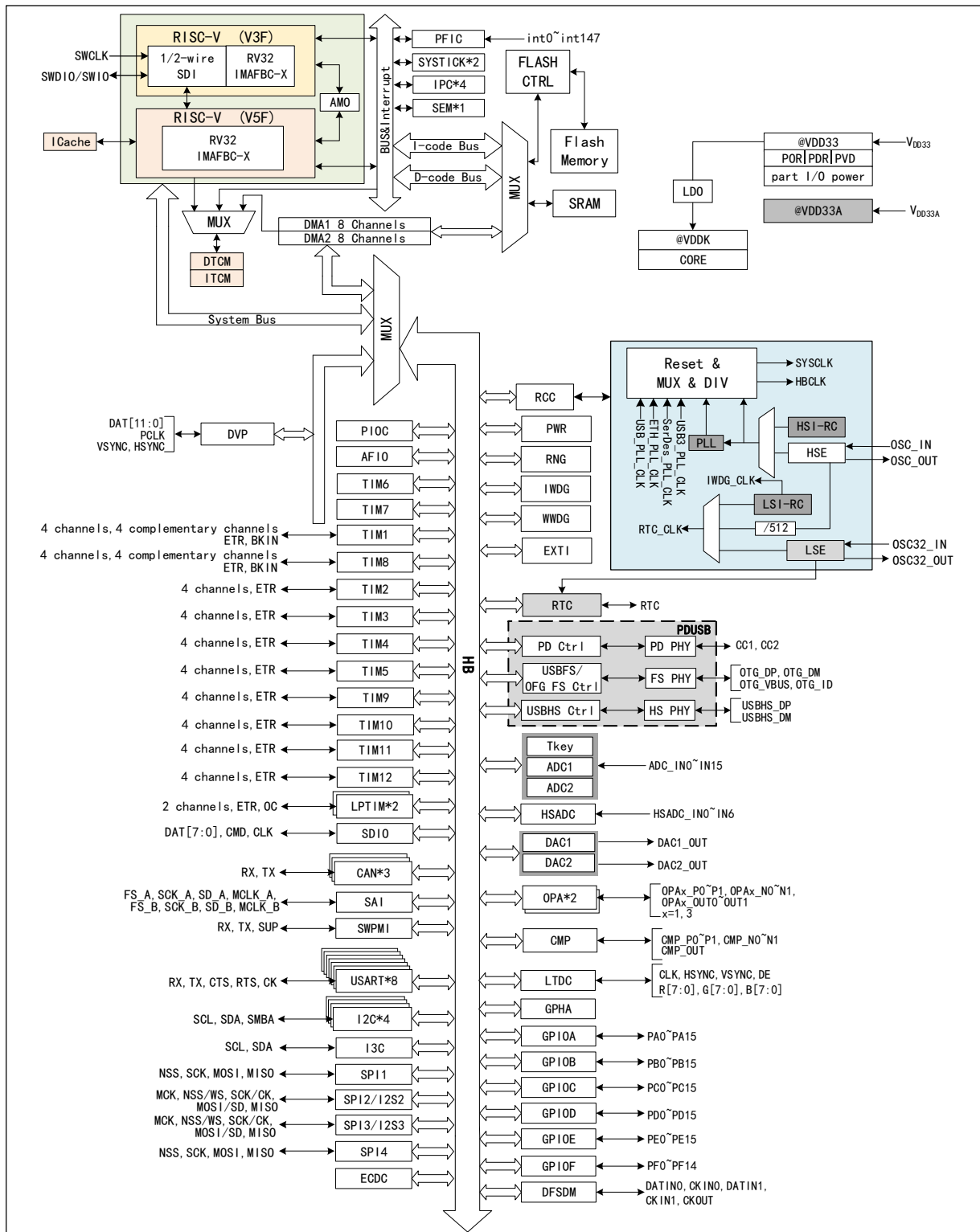


Figure 1-1-3 CH32H415 System block diagram



The system is equipped with: Flash access pre-fetching mechanism is used to accelerate code execution speed; General-purpose DMA controller to reduce the CPU burden and improve efficiency; clock tree hierarchy management to reduce the total power consumption of peripherals, as well as data protection mechanisms, clock security system protection mechanisms and other measures to increase system stability.

- The instruction bus (I-Code) connects the core to the FLASH instruction interface, where instruction prefetching occurs.
- The data bus (D-Code) connects the core to the FLASH data interface, used for constant loading and debugging.

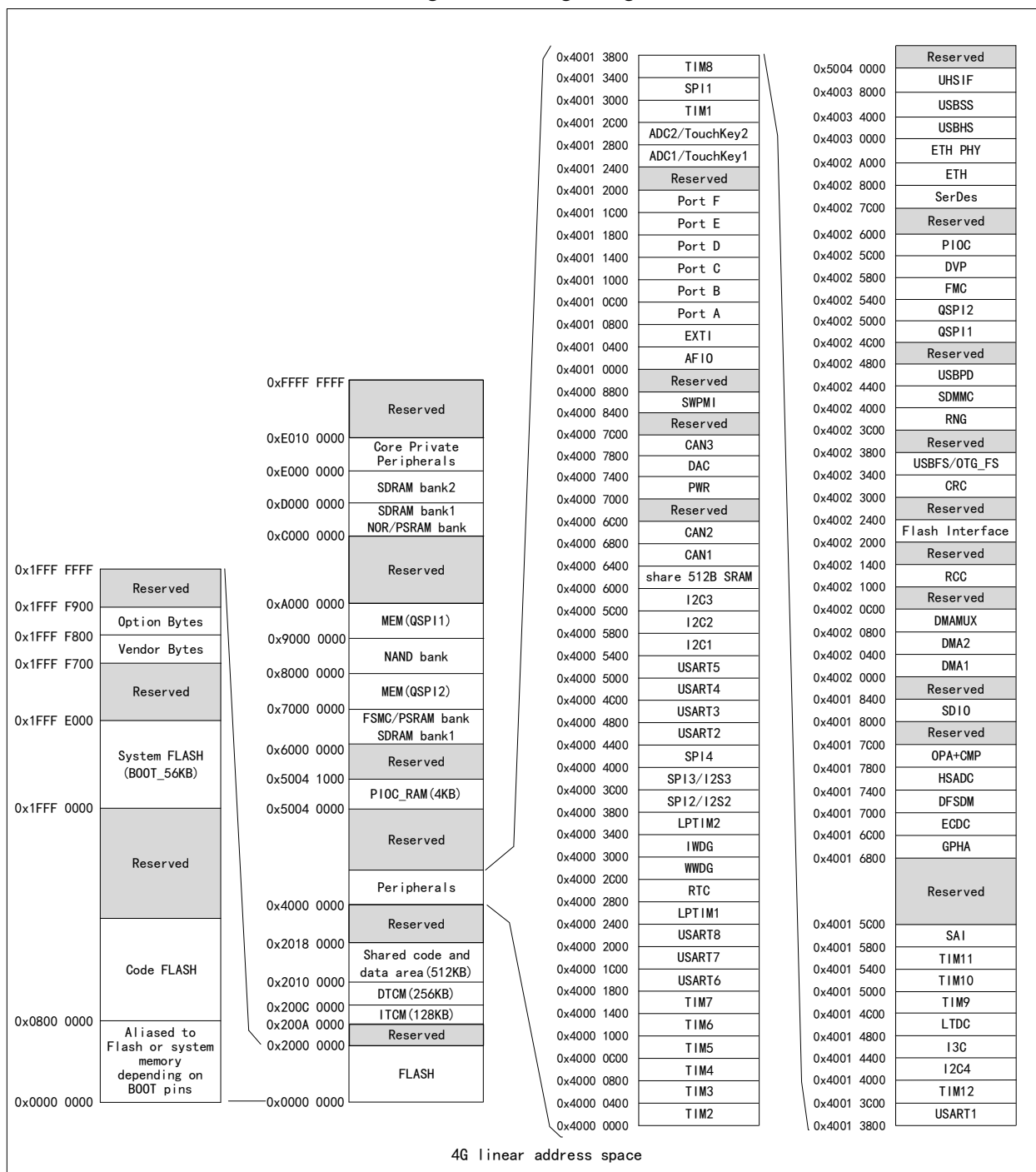
- The system bus connects the core to the bus matrix, coordinating access between the core, DMA, SRAM, and peripherals.
- The DMA bus connects the HB controller interface for DMA operations to the bus matrix. This bus accesses FLASH data, SRAM, and peripherals.
- The bus matrix coordinates access between the system bus, data bus, DMA bus, and SRAM.

1.2 Memory Map

The CH32H417 family contains program memory, data memory, core registers, peripheral registers, and more, all addressed in a 4GB linear space.

System storage stores data in small-end format, i.e., low bytes are stored at the low address and high bytes are stored at the high address.

Figure 1-2 Storage image



1.2.1 Memory Allocation

Built-in SRAM with a total capacity of 896K bytes, partially allocated for the data area and partially for the zero-

wait code area. Data is lost upon power loss. This SRAM is divided into 3 blocks: a 128KB ITCM core 1 tightly coupled zero-wait code area, a 256KB DTCM core 1 tightly coupled zero-wait data area, and the remaining 512KB shared code and data area. Additionally, core 1 features a 32KB high-speed instruction cache.

The 512KB shared area can be configured as the zero-wait code and data regions for RISC-V3F. It is recommended to allocate this space flexibly in 128KB increments as needed.

As an option, both the 128KB ITCM and 256KB DTCM can be configured as the RISC-V5F code segment, totaling 384KB. When the DTCM is used as the code segment, it introduces an additional clock wait during jumps.

RISC-V3F can access ITCM or DTCM with 2 HCLK clock cycles.

RISC-V5F and RISC-V3F can access the 512KB shared region with zero HCLK clock cycles.

Built-in 256-byte system non-volatile configuration information storage area for vendor configuration word storage, fixed at factory and non-modifiable by users.

Built-in 256-byte user-defined information storage area for user selection word storage.

The sizes of the user area and BOOT area are read via the DBMODE bit in the R32_FLASH_CFGR0 register:

When DBMODE=1, the chip incorporates a 960K-byte non-zero-wait program flash memory area (Code FLASH), i.e., the user area, for storing user applications and constant data, with an equivalent frequency of approximately 25MHz. It also incorporates a maximum of 56K bytes of system memory area (System FLASH), i.e., the BOOT area, for storing the system bootloader and the built-in bootstrap loader.

When DBMODE=0, the chip incorporates a 480K-byte non-zero-wait program flash memory area (Code FLASH), serving as the user area for storing user applications and constant data, with an equivalent frequency of approximately 12.5MHz. It also includes a maximum 28K-byte system memory area (System FLASH), functioning as the BOOT area for storing the system bootloader, with an integrated bootstrap loader.

For information on using the chip's BOOT function and related precautions, please refer to the *CH32H417 Evaluation Board Reference*.

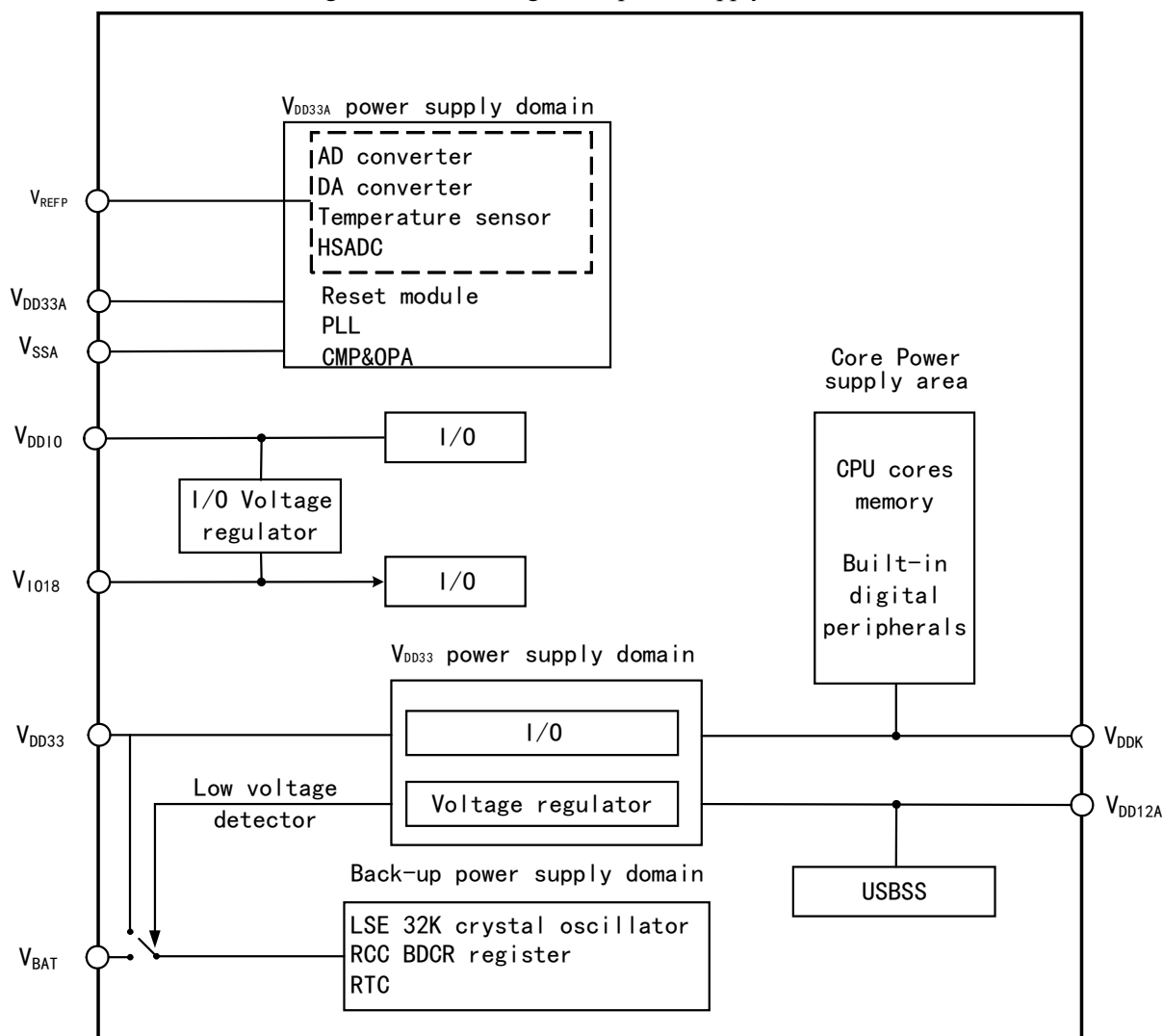
Chapter 2 Power Control (PWR)

2.1 Overview

The CH32H417 system operating voltage V_{DD} ranges from 2.4 to 3.6 V. The built-in voltage regulator provides the low-voltage power supply required by the core. When the main power supply, V_{DD33} , is powered down, a backup power source such as a battery can provide power for the real-time clock (RTC) through the V_{BAT} pin. If no backup power is required, it is recommended that V_{DD33} be connected directly to the V_{BAT} pin.

The V_{DD33A} and V_{SSA} pins are specifically designed to power analog-related circuits within the system, including ADC, DAC, temperature sensors, and similar components. V_{REFP} serves as a reference point for the upper voltage limit of certain analog circuits, such as ADCs. In some packages, it is provided as a separate pin; in others, it remains unpackaged and internally equals V_{DD33A} within the chip.

Figure 2-1 Block diagram of power supply structure

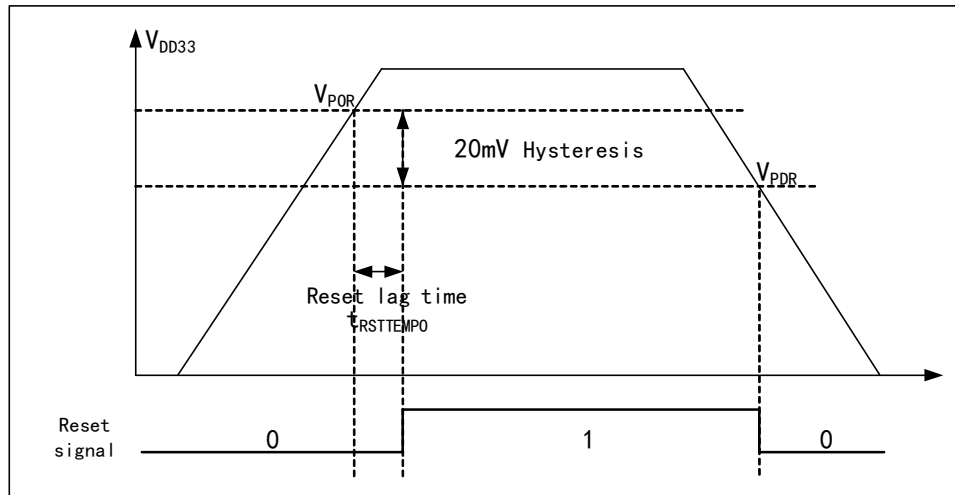


2.2 Power Management

2.2.1 Power-on Reset and Power-down Reset

The system integrated a power-on reset POR and a power-down reset PDR circuit. When the chip supply voltage V_{DD33} falls below the corresponding threshold voltage, the system is reset by the relevant circuit, and no additional external reset circuit is required. Please refer to the corresponding datasheet for the parameters of the power-on threshold voltage V_{POR} and the power-down threshold voltage V_{PDR} .

Figure 2-2 Schematic diagram of the operation of POR and PDR



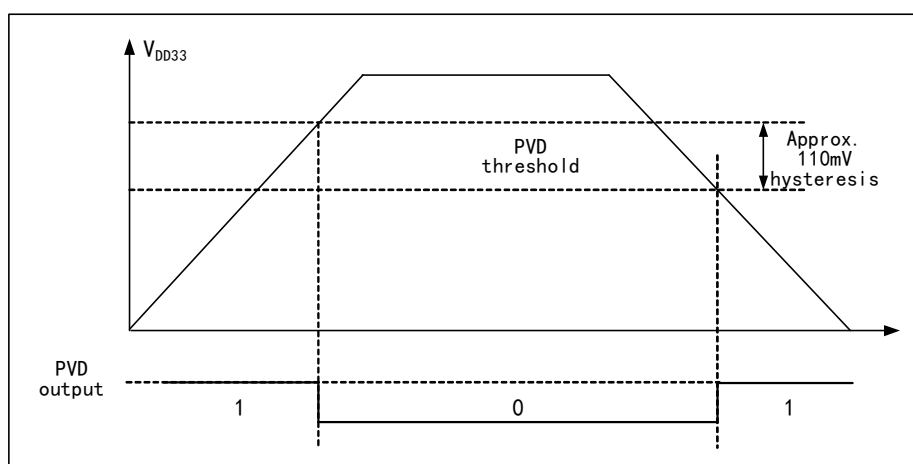
2.2.2 Programmable Voltage Detector

The programmable voltage detector, PVD, is mainly used to monitor the change of the main power supply of the system, compared with the threshold voltage set by PLS[2:0] of the power control register PWR_CTLR, and together with the setting of the external interrupt register (EXTI), the relevant interrupt can be generated in order to notify the system to carry out the pre-dropout operation, such as the data saving, in time.

The specific configuration is as follows:

- 1) Set the PLS[2:0] fields of the PWR_CTLR register to select the voltage threshold to be monitored.
- 2) Set the PVD bit of the PWR_CTLR register to enable the PVD function.
- 3) Read the PVD0 bit of PWR_CSR status register to obtain the relationship between the current system main power supply and the threshold value set by PLS[2:0], and perform the corresponding soft processing. When the V_{DD33} voltage is higher than the PLS[2:0] setting threshold, the PVD0 position is 0; when the V_{DD33} voltage is lower than the PLS[2:0] setting threshold, the PVD0 position is 1.

Figure 2-3 Schematic diagram of PVD operation



2.3 Low-power Modes

After a system reset, the microcontroller is in a normal operating state (run mode), where system power can be saved by reducing the system main frequency or turning off the unused peripheral clock or reducing the operating peripheral clock. If the system does not need to work, you can set the system to enter low-power mode and let the system jump out of this state by specific events.

Microcontrollers currently offer 2 low-power modes, divided in terms of operating differences between processors, peripherals, voltage regulators, etc.

- Sleep mode: The core stops running and all peripherals (including core private peripherals) are still running.
- Stop mode: The stop mode only takes effect when both V3F and V5F enter stop mode. Stops all clocks and the system continues to run after waking up.

Table 2-1 Low-power mode list

Mode	Entry	Wake-up source	Effect on clock	Voltage regulator
SLEEP	WFI	Any interrupt	Core clock OFF, no effect on other clocks	Normal
	WFE	Wake-up event		
STOP	Set SLEEPDEEP to 1 WFI or WFE	Any external interrupt/event, external reset signal on NRST pin, IWDG reset	Both V3F and V5F enter stop mode: HSE, HSI, PLL, and peripheral clocks are disabled.	Normal: LPDS=0 or Low power: LPDS=1

2.3.1 Low-power Configuration Options

- WFI and WFE

WFI: The microcontroller is woken up by an interrupt source with interrupt controller response, and the interrupt

service function will be executed first after the system wakes up (except for microcontroller reset).

WFE: The wakeup event triggers the microcontroller to exit low-power mode. Wake-up events include:

- 1) Configure an external or internal EXTI line to event mode, when no interrupt controller needs to be configured.
- 2) Interrupt response and interrupt exit markers are maintained as events within the interrupt controller.
- 3) Or configure the SEVONPEND bit to turn on peripheral interrupt enable, but not interrupt enable in the interrupt controller, and the interrupt pending bit needs to be cleared after the system wakes up.
- 4) Debug request of debugger

- **SLEEPONEXIT**

Enable: After executing the WFI or WFE instruction, the microcontroller ensures that all pending interrupt services are exited and then enters low-power mode.

Disable: The microcontroller enters low-power mode immediately after executing the WFI or WFE command.

- **SEVONPEND**

Enable: All interrupts or wake-up events can wake up the low-power consumption entered by executing WFE.

Disable: Only interrupts or wake-up events enabled in the interrupt controller can wake up the low-power consumption entered by executing WFE.

2.3.2 Sleep Mode

In this mode, all IO pins keep their state in Run mode and all peripheral clocks are normal, so try to turn off useless peripheral clocks before entering Sleep mode to reduce low-power consumption. This mode takes the shortest time to wake up.

Enter: Configure core register control bit SLEEPDEEP=0, LPDS determines internal regulator status, execute WFI or WFE, optionally SEVONPEND and SLEEPONEXIT.

Exit: Any interrupt or wakeup event.

2.3.3 Stop Mode

Sleep mode combines the clock control mechanisms of the peripherals with the fact that both the V3F and V5F cores enter deep sleep (SLEEPDEEP).

When both V3F and V5F enter this mode: the high-frequency clock (HSE/HSI/PLL) domains are disabled, SRAM and register contents are preserved, and I/O pin states are maintained. Upon waking from this mode, the system can resume operation, with HSI serving as the default system clock.

You can set LPDS=1 to switch the LDO regulator from normal mode to low-power mode, thereby achieving a state of lower power consumption.

If flash programming is in progress, the system does not enter stop mode until access to memory is complete; if access to the HB is in progress, the system does not enter stop mode until access to the HB is completed.

Modules that can operate in Stop mode: Independent Watchdog Dog (IWDG), Real Time Clock (RTC), Low Frequency Clock (LSI/LSE).

Entry: Configuration of the core register control bit SLEEPDEEP=1, optional LPDS bit, execute WFI or WFE, optional SEVONPEND and SLEEPONEXIT.

Exit: Any external interrupt/event, external reset on the RST pin, IWDG reset.

In stop mode, the LPDS bit is selectable, LPDS=0 indicates the voltage regulator operates in normal mode; LPDS=1 indicates the voltage regulator operates in low-power mode.

2.3.4 RTC Auto-wakeup

RTC can automatically wake up without external interruption. By programming the time base, it can be awakened periodically from Stop or Standby mode.

The accurate external low frequency 32.768KHz crystal oscillator LSE can be selected as the RTC clock source, or the internal LSI oscillator can be selected as the RTC clock source. The accuracy and power consumption index of LSI is worse than that of LSE.

The RTC alarm clock event can wake the MCU from stop mode. In order to achieve this function, the external break line 17 needs to be configured, and the RTC needs to be set to generate alarm clock events.

2.4 Register Description

Table 2-2 PWR-related registers list

Name	Access address	Description	Reset value
R32_PWR_CTLR	0x40007000	Power control register	0x00000000
R32_PWR_CSR	0x40007004	Power control/status register	0x00000X00
R32_SYS_CFGR0	0x5003C000	System configuration register 0	0xFFFFFFFF

2.4.1 Power Control Register (PWR_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				VSEL_VIO18[2:0]		VIO_SW_CR	DBP	PLS[2:0]		PVDE	Reserved			LPDS	

Bit	Name	Access	Description	Reset value
[31:13]	Reserved	RO	Reserved	0
[12:10]	VSEL_VIO18[2:0]	RW	V_{IO18} power regulation bit: 000: 1.2V; 001: 1.8V; 010: 2.5V; 011: 3.3V; 100: V_{IO18} power down, 10mA discharging; 101: V_{IO18} power down, output floating; 110: V_{IO18} power down, output floating; 111: V_{IO18} power down, output floating. <i>Note: When V_{IO18} switches from low voltage to high</i>	0

			voltage, the capacitance of the V_{DDIO} pin must be significantly greater than the total capacitance of the V_{IO18} pin to mitigate the voltage drop on V_{DDIO} during the V_{IO18} voltage rise, and V_{DD33} must be at least 2.7 V.	
9	VIO_SW_CR	RW	V_{IO18} power regulation mode selection bit: 1: Configure by VSEL_VIO18 software; 0: XO external pull-down resistor hardware configuration. Note: The initial value upon power-up for VIO18 is determined by hardware configuration, specifically via the external pull-down resistor on XO: Float: 1.8V; Pull-down 330K: 2.5V; Pull-down 100K: 1.2V. Note: 1. For the chip with external V_{IO18}, users need to short-circuit V_{DDIO} and V_{IO18} externally when using 3.3V. Additionally, after the power-up procedure is completed, the LDO should be disabled by setting VSEL_VIO18 to 111b in the software to reduce power consumption. 2. For chips without an external V_{IO18} pin, V_{DDIO} and V_{IO18} are internally shorted. After power-up, the LDO must be disabled via software by setting VSEL_VIO18=111b to reduce power consumption.	0
8	DBP	RW	Write enable for backup domain: In the reset state, all registers in the backup domain are protected against write access. This bit must be set to 1 to enable write access to these registers. 1: Enable access to registers in the backup domain; 0: Disable access to registers in the backup domain.	0
[7:5]	PLS[2:0]	RW	PVD voltage monitoring threshold settings. See the Electrical Characteristics section of the data sheet for details. 000: Rising edge 2.54V/Falling edge 2.44V; 001: Rising edge 2.6V/Falling edge 2.49V; 010: Rising edge 2.7V/Falling edge 2.59V; 011: Rising edge 2.8V/Falling edge 2.69V; 100: Rising edge 2.9V/Falling edge 2.79V; 101: Rising edge 3.0V/Falling edge 2.89V; 110: Rising edge 3.1V/Falling edge 2.99V; 111: Rising edge 3.2V/Falling edge 3.09V.	0
4	PVDE	RW	Power supply voltage monitoring enable bit: 1: Enable power supply voltage monitoring function; 0: Disable power supply voltage monitoring function.	0
[3:1]	Reserved	RO	Reserved	0
0	LPDS	RW	Stop mode voltage regulator operating mode selection bit.	0

			1: Voltage regulator operates in low-power mode; 0: Voltage regulator operates in normal mode.	
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2.4.2 Power Control/Status Register (PWR_CSR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						VIO18_SR [1:0]		Reserved						PVD O	

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
[9:8]	VIO18_SR[1:0]	RO	V _{IO18} power initial state indicator bit: 00: 1.2V; 01: 2.5V; 1x: 1.8V. <i>Note: The initial power-on value of V_{IO18} is determined by VIO_SW_CR and VSEL_VIO18[2:0].</i>	x
[7:1]	Reserved	RO	Reserved	0
0	PVDO	RO	PVD output status flag. This bit is active when PVDE=1 in the PWR_CTLR register. 1: V _{DD33} is below the PVD falling edge threshold set by PLS[2:0]; 0: V _{DD33} is above the PVD rising edge threshold set by PLS[2:0].	0

2.4.3 System Configuration Register 0 (SYS_CFGR0)

Offset address: 0x5003C000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									LDO_VDDK[2:0]			Reserved			

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RW	Reserved. The original value must be maintained when writing. When external power supply is used in the application, bit[16:14] should be set to 001b and bit[19:17] should be set to 001b.	x
[6:4]	LDO_VDDK[2:0]	RW	Adjust the digital core voltage value, LDO_VDDK voltage	011b

			value adjust bit: 011: Rated voltage; 101: Improved voltage-optimized performance, only suitable for commercial-grade applications with $T_A \leq 70^\circ\text{C}$ and good heat dissipation; Other: Reserved. When external power is used in the application, it is recommended to set 001b or 000b.	
[3:0]	Reserved	RW	Reserved. The original value must be maintained when writing.	x

Chapter 3 Reset and Clock Control (RCC)

The controller provides different forms of resets and configurable clock tree structures based on the division of power areas and peripheral power management considerations in the application. This section describes the scope of each clock in the system.

3.1 Main Features

- Multiple reset forms
- Multiple clock sources, bus clock management
- Built-in external crystal oscillation monitoring and clock security system
- Independent management of each peripheral clock: Reset, On, Off
- Supports internal clock output

3.2 Reset

The controller provides 3 forms of reset: Power Reset, System Reset, and Backup Domain Reset.

3.2.1 Power Reset

When a power reset occurs, all registers except the backup domain (which is powered by V_{BAT}) will be reset.

The conditions for its production include:

- Power-on/power-off reset (POR/PDR reset)

3.2.2 System Reset

When a system reset occurs, all registers except the reset flag and the backup domain in the control / status register `RCC_RSTSCKR` are reset. The source of the reset event is identified by looking at the reset status flag bit in the `RCC_RSTSCKR` register.

The conditions for its production include:

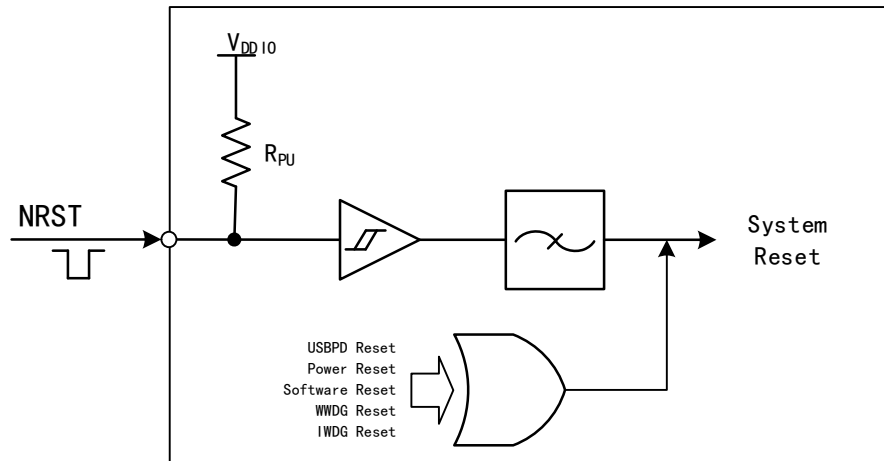
- Low level signal on NRST pin (External reset)
- Window watchdog count terminated (WWDG reset)
- Independent watchdog count terminated (IWDG reset)
- Software reset (SW reset)
- USBPD reset

Window/independent watchdog reset: Triggered by window/independent watchdog peripheral timer count cycle overflow, see the corresponding chapter for a detailed description.

Software reset: This product resets the system through the `SYSRST` position 1 of the interrupt configuration register `PFIC_CFGR` in the programmable interrupt controller `PFIC` or the `RSTSYS` position 1 of the configuration register `PFIC_SCTLR`, refer to the corresponding chapter.

USBPD Reset: When `PD_RST_EN` is 1, the CH32H417 supports the reset generated by the Hard Reset of the USB PD signal frame. If `IE_RX_RESET` is also 1, the Reset generated by Cable Reset of signal frames is also supported. USB PD does not have a reset flag, but the resulting reset effect is the same as a software reset.

Figure 3-1 System reset structure



3.2.3 Backup Domain Reset

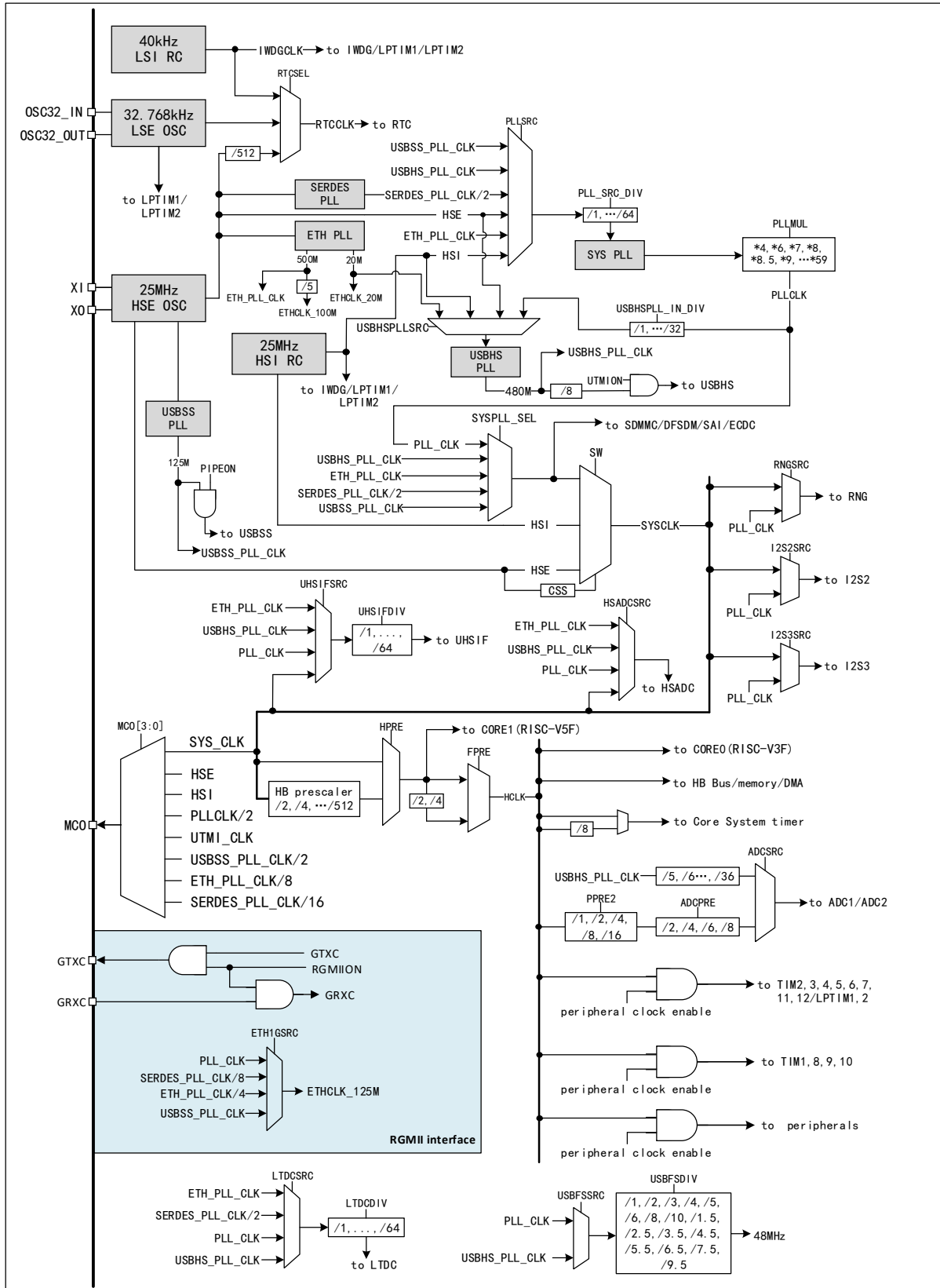
When a backup domain reset occurs, only the backup domain registers are reset, including the backup registers, the RCC_BDCTL registers (RTC enable and LSE oscillator). The conditions for its generation include:

- Caused by V_{DD33} or V_{BAT} power-up with both V_{DD33} and V_{BAT} powered down
- Set BDRST to 1 of the RCC_BDCTL register

3.3 Clock

3.3.1 System Clock Structure

Figure 3-2 Clock tree block diagram



3.3.2 High-speed Clock (HSI/HSE)

HSI is a high-speed clock signal generated by the RC oscillator of 25MHz in the system. The HSI RC oscillator can provide the system clock without any external devices. Its start-up time is very short, but the clock frequency accuracy is poor. The HSI is turned on and off by setting the HSION bit in the RCC_CTLR register, and the HSIRDY bit indicates whether the HSI RC oscillator is stable. The system defaults to HSION and HSIRDY setting 1 (it is recommended that you do not turn it off). If the HSIRDYIE bit of the RCC_INTR register is set, the corresponding interrupt will be generated.

- **Factory calibration:** differences in manufacturing processes will lead to different RC oscillation frequencies of each chip, so HSI calibration is performed for each chip before it leaves the factory. After the system is reset, the factory calibration value is loaded into the HSICAL[7:0] in the RCC_CTLR register.
- **User adjustment:** depending on the voltage or ambient temperature, the application can adjust the HSI frequency through the HSITRIM[4:0] bit in the RCC_CTLR register.

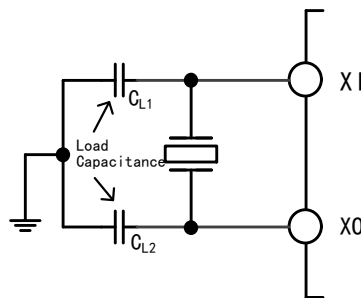
Note: If the HSE crystal oscillator fails, the HSI clock will be used as a backup clock source (clock security system).

HSE is an external high-speed clock signal, including external crystal/ceramic resonator generation or external high-speed clock input.

HSE crystal oscillator can enter HSELP position 1 into low-power mode through the RCC_CTRL register.

- **External Crystal/Ceramic Resonator (HSE Crystal):** An external oscillator provides a more accurate clock source for the system. Further information can be found in the Electrical Characteristics section of the datasheet. The HSE crystal can be turned on and off by setting the HSEON bit in the RCC_CTLR register. The HSERDY bit indicates whether the HSE crystal oscillation is stable or not, and the hardware sends the clock into the system only after the HSERDY bit is set to one. If the HSERDYIE bit in the RCC_INTR register is set, the appropriate interrupt will be generated.

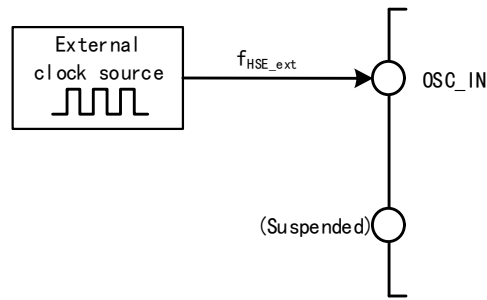
Figure 3-3 High-speed external crystal circuit



Note: The load capacitance needs to be as close as possible to the oscillator pins and the capacitance value selected according to the crystal vendor's parameters.

- **External High Speed Clock Source (HSE Bypass):** this mode feeds the clock source directly from the external source to the OSC_IN pin, with the OSC_OUT pin is left floating. A maximum frequency of 25MHz is supported. The application program needs to set the HSEBYP bit to turn on the HSE bypass function with the HSEON bit at 0, and then set the HSEON bit again.

Figure 3-4 High-speed clock source circuit



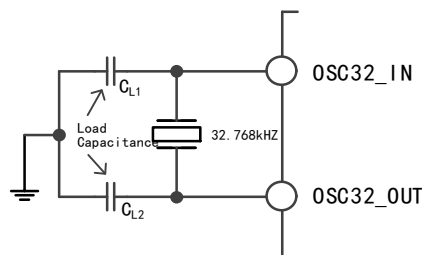
3.3.3 Low-speed Clock (LSI/LSE)

LSI is a low-speed clock signal generated by a RC oscillator in the system. It can be kept running in stop mode, providing clock references for RTC clocks, independent watchdogs, and wake-up units. For further information, please refer to the electrical characteristics section of the data manual. LSI can be turned on and off by setting the LSION bit in the RCC_RSTSCKR register, and then check whether the LSIRC oscillation is stable by querying the LSIRDY bit, and the hardware sends the clock in after LSIRDY position 1. If the LSIRDYIE bit of the RCC_INTR register is set, the corresponding interrupt will be generated.

LSE is an external low-speed clock signal, including external crystal/ceramic resonator generation or external low-speed clock input. It provides a low-power and accurate clock source for RTC clocks or other timing functions.

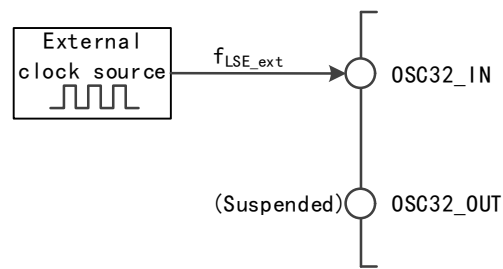
- External crystal/ceramic resonator (LSE crystal): External low-speed oscillator with external 32.768kHz. LSE is turned on and off by setting the LSEON bit in the RCC_BDCTL register, and the LSERDY bit indicates whether the LSE crystal oscillation is stable or not, and the hardware sends the clock into the system after LSERDY position 1. If the LSERDYIE bit of the RCC_INTR register is set, the corresponding interrupt will be generated.

Figure 3-5 Low-speed external crystal circuit



- External low-speed clock source (LSE bypass): this mode feeds the clock source directly from the outside to the OSC32_IN pin, and the OSC32_OUT pin is suspended. The application needs to set the LSEBYP bit when the LSEON bit is 0, turn on the LSE bypass function, and then set the LSEON bit.

Figure 3-6 Low-speed clock source circuit



3.3.4 PLL Clock

By configuring the RCC_CFGR0 register and RCC_PLLCFGR register, the internal PLL clock can select 6 clock sources and multiplication factors. These settings must be completed before each PLL is turned on. Once the PLL is turned on, these parameters cannot be changed.

Set the PLLON bit in RCC_CTLR register to be turned on and off, and the PLLRDY bit indicates whether the PLL clock is stable, and the hardware sends the clock to the system after the PLLRDY bit is set.

Set the USBHS_PLLON bit in the RCC_CTLR register to be turned on and off. The USBHS_PLLRDY bit indicates whether the USBHS_PLLRDY clock is stable, and the hardware sends the clock to the system after the USBHS_PLLRDY bit is set.

Set the USBSS_PLLON bit in RCC_CTLR register to be turned on and off, and the USBSS_PLLRDY bit indicates whether the USBSS_PLLRDY clock is stable, and the hardware sends the clock to the system after the USBSS_PLLRDY bit is set.

Set the ETH_PLLON bit in RCC_CTLR register to be turned on and off, and the ETH_PLLRDY bit indicates whether the ETH_PLLRDY clock is stable, and the hardware sends the clock to the system after the ETH_PLLRDY bit is set.

Set the SERDES_PLLON bit in RCC_CTLR register to be turned on and off. The SERDES_PLLRDY bit indicates whether the SERDES_PLLRDY clock is stable, and the hardware sends the clock to the system after the SERDES_PLLRDY bit is set.

If the PLLRDYIE bit, ETHPLLRDYIE bit or SERDESPLLRDYIE bit of RCC_INTR register is set, a corresponding interrupt will be generated.

PLL clock source:

- HSI clock input
- HSE clock input
- USBHS_PLL (480MHz) clock input
- ETH_PLL (500MHz) clock input
- USBSS_PLL (125MHz) clock input
- SERDES_PLL clock output through 2 frequency division

3.3.5 Bus/Peripheral Clock

3.3.5.1 System Clock (SYSCLK)

By configuring the RCC_CFGR0 register SW[1:0] bit to configure the system clock source, SWS[1:0] indicates the current system clock source.

- HSI as the system clock.
- HSE as the system clock.
- PLL clock as the system clock.

After the controller is reset, the default HSI clock is selected as the system clock source. The switching between clock sources does not occur until the target clock source is ready.

3.3.5.2 HB Bus Peripheral Clock (HCLK)

By configuring the HPRE[3:0] and PPRE2[2:0] bits of the RCC_CFGR0 register, the clocks of the HB bus can be configured respectively. These bus clocks determine the peripheral interface mounted below them to access the clock reference. The application can adjust different values to reduce the power consumption of some peripherals.

Different peripheral modules can be reset and restored to the initial state through each bit in the RCC_HBRSTR, RCC_HB1PRSTR and RCC_HB2PRSTR registers.

The communication clock interface of different peripheral modules can be turned on or off separately through each bit in the RCC_HBPCENR, RCC_HB1PCENR and RCC_HB2PCENR registers. When using a peripheral, you first need to turn on its clock enable bit before you can access its register.

3.3.5.3 RTC Clock (RTCCLK)

By setting the RTCSEL[1:0] bit of the RCC_BDCTLR register, the RTCCLK clock source can be provided by the HSE, LSE, or LSI clock. Before modifying this bit, make sure that the DBP position 1 in the power control register (PWR_CTLR) can be reset only if the backup domain is reset.

- LSE acts as the RTC clock: Since LSE is powered by V_{BAT} in the backup domain, RTC continues to work as long as V_{BAT} maintains power, even though V_{DD33} power is cut off.
- LSI as the RTC clock: If the V_{DD33} power is cut off, RTC automatic wake up is not guaranteed.
- HSE is used as RTC clock after frequency division of 512: If the V_{DD33} power supply is cut off or the internal voltage regulator is turned off (the power supply in the 1.8V domain is cut off), the RTC state is uncertain.

3.3.5.4 Independent Watchdog Clock

If the independent watchdog has been started by the hardware configuration or software, the LSI oscillator will be forced to open and cannot be turned off. After the LSI oscillator is stabilized, the clock is supplied to the IWDG.

3.3.5.5 Microcontroller Clock Output (MCO)

The microcontroller allows clock signals to be output to MCO pins. The alternate push-pull output mode is configured in the corresponding GPIO port register. by configuring the RCC_CFGR0 register MCO[3:0] bit, the following 8 clock signals can be selected as MCO clock output:

- System clock (SYSCLK) output.
- HSI clock output.
- HSE clock output.
- PLL clock output through 2 frequency division
- UTMI clock output.
- USBSS_PLL clock output through 2 frequency division
- ETH_PLL clock output through 8 frequency division
- SERDES_PLL clock output through 16 frequency division

3.3.5.6 USB Clock

The USBHS clock source originates from the USBHSPLL clock. The USBHS_PLL clock source is selected by configuring the USBHSPLLSRC bit in the RCC_PLLCFGR2 register. If the input clock is selected as the divided SYS_PLL_CLK, the division factor can be chosen via the USBHSPLL_IN_DIV[4:0] bits. The clock frequency is selected by configuring the USBHSPLL_REFSEL[1:0] bits. The clock frequency must match the USBHS_PLL input clock. The USBHS_PLL clock is enabled by configuring the USBHS_PLLON bit in the RCC_CTLR register.

The USBFS/OTG_FS 48MHz clock source originates from either the PLL clock through a configurable divider or the USBFS_PLL clock, selectable by configuring the USBFSSRC bit in the RCC_CFGR2 register. The USBOTG_FS clock is enabled by configuring the OTGFSEN bit in the RCC_HBPCENR register.

The USBSS clock source originates from a USBSS_PLL clock, which outputs a 125MHz clock to USBSS. The USBSS_PLL clock is enabled by configuring the USBSS_PLLON bit in the RCC_CTLR register.

3.3.5.7 ETH Clock

Refer to Section 31.4.3 for the ETH clock configuration.

3.3.5.8 I2S and RNG Clock

The clock source for I2S and RNG originates from PLL_CLK or the system clock (SYSCLK). The clock sources for I2S2, I2S3, and RNG can be selected via the I2S2SRC, I2S3SRC, and RNGSRC bits in the configuration register RCC_CFGR2, respectively.

3.3.6 Clock Security System

The clock security system is a running protection mechanism of the controller, which can switch to the HSI clock in the case of HSE clock transmission failure, and generate interrupt notification, allowing application software to complete the rescue operation.

Activate the clock security system by setting CSSON position 1 of the RCC_CTLR register. At this point, the clock monitor will be enabled after the HSE oscillator startup (HSERDY=1) delay and turned off after the HSE clock is turned off. Once the HSE clock fails during the operation of the system, the HSE oscillator will be turned off, the clock failure event will be sent to the break input of the advanced timer (TIM1 and TIM8), and the clock security interrupt will be generated, CSSF position 1, and the application will enter the non-maskable interrupt (NMI). By setting the CSSC bit, the CSSF bit flag can be cleared and the NMI interrupt suspension bit can be revoked.

When CSSON is enabled, an HSE failure event does not affect HSE or PLL enable states, and software may handle it within the NMI interrupt. However, if the CSS_HSE_DIS bit in the RCC_CTLR register is set to 1 before activating the clock security system (CSSON=1), an HSE failure event occurring while CSSON is enabled will cause the hardware to disable HSE and all PLLs utilizing HSE.

3.3.7 RTC Calibration

This function must be configured with PC13 pin as a normal IO port.

- Pulse output

Configure the ASOE bit in the RCC_BDCTLR register to turn on the RTC pulse output, set the ASOS bit, and select whether the second pulse output or the alarm pulse output.

- RTC Calibration

When the CCO bit of the RCC_BDCTLR register is configured, the internal RTC clock is divided by 64 and output to the PC13 pin. Through actual testing, software adjusts the clock to calibrate the RTC in conjunction with modifying the RTCCAL[6:0] bits.

3.3.8 BKP Interface Reset

The BKP region can be powered independently by V_{BAT} during a V_{DD33} main power supply failure. During application code-controlled reset of the BKP region registers, the ASOS and ASOE bits are reset under the BDRST bit of the software-configured RCC_BDCTLR register and are unaffected by the BKPRST bit of the RCC peripheral interface control.

3.4 Register Description

Table 3-1 RCC-related registers list

Name	Access address	Description	Reset value
R32_RCC_CTLR	0x40021000	Clock control register	0x0000xx83
R32_RCC_CFGR0	0x40021004	Clock configuration register 0	0x00000000
R32_RCC_PLLCFGR	0x40021008	PLL configuration register	0x00000004
R32_RCC_INTR	0x4002100C	Clock interrupt register	0x00000000
R32_RCC_HB2PRSTR	0x40021010	HB2 peripheral reset register	0x00000000
R32_RCC_HB1PRSTR	0x40021014	HB1 peripheral reset register	0x00000000
R32_RCC_HBPCENR	0x40021018	HB peripheral clock enable register	0x00000000
R32_RCC_HB2PCENR	0x4002101C	HB2 peripheral clock enable register	0x00000000
R32_RCC_HB1PCENR	0x40021020	HB1 peripheral clock enable register	0x00000000
R32_RCC_BDCTLR	0x40021024	Backup domain control register	0x00000000
R32_RCC_RSTSCKR	0x40021028	Control/status register	0x08000000
R32_RCC_HBRSTR	0x4002102C	HB peripheral reset register	0x00000000
R32_RCC_CFGR2	0x40021030	Clock configuration register 2	0x00000000
R32_RCC_PLLCFGR2	0x40021034	PLL configuration register 2	0x00080020

3.4.1 Clock Control Register (RCC_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CSS_HSE_DIS	Reserved	SERDES_PL_LRDY	SERDES_PL_LON	ETH_PLLR_DY	ETH_PLLO_N	PLL_RDY	PLLO_N	USBS_PL_LRDY	USBS_PL_LON	USB_HS_PL_LRDY	USB_HS_PL_LLO_N	CSSO_N	HSE_BYP	HSE_RDY	HSE_ON
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]				Reserved	HSI_RDY	HSI_ON	

Bit	Name	Access	Description	Reset value
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31	CSS_HSE_DIS	RW	Configuration for HSE failure events occurring when the clock security system is enabled: 1: When CSSON is enabled and an HSE failure event occurs, the hardware disables HSE and all PLLs using HSE; 0: When CSSON is enabled and an HSE failure event occurs, HSE and PLL enable states remain unaffected; software may handle the event within the NMI interrupt. <i>Note: This bit is writable only when CSSON is 0.</i>	0
30	Reserved	RO	Reserved	0
29	SERDES_PLLRDY	RO	SERDES_PLL lock status bit: 1: SERDES_PLL clock locked; 0: SERDES_PLL clock not locked.	0
28	SERDES_PLLOEN	RW	SERDES_PLL clock enable control bit: 1: Enable the SERDES_PLL clock; 0: Disable the SERDES_PLL clock. The hardware automatically resets upon entering deep sleep.	0
27	ETH_PLLRDY	RO	ETH_PLL lock status bit: 1: ETH_PLL clock locked; 0: ETH_PLL clock not locked.	0
26	ETH_PLLOEN	RW	ETH_PLL enable control bit: 1: Enable ETH_PLL clock; 0: Disable ETH_PLL clock. Hardware automatically resets upon entering deep sleep.	0
25	PLLRDY	RO	PLL clock ready flag (set by hardware): 1: PLL locked; 0: PLL unlocked	0
24	PLLON	RW	PLL enable: 1: Enable PLL clock; 0: Disable PLL clock. <i>Note: Cleared by hardware when entering Stop mode.</i>	0
23	USBSS_PLLRDY	RO	USBSS_PLL clock ready locked flag: 1: USBSS_PLL clock locked; 0: USBSS_PLL clock not locked.	0
22	USBSS_PLLOEN	RW	USBSS_PLL clock enable control bit: 1: Enable USBSS_PLL clock; 0: Disable USBSS_PLL clock. Hardware automatically resets upon entering deep sleep.	0
21	USBHS_PLLRDY	RO	USBHS_PLL clock ready locked flag: 1: USBHS_PLL clock locked; 0: USBHS_PLL clock not locked.	0
20	USBHS_PLLOEN	RW	USBHS_PLL clock enable control bit: 1: Enable USBHS_PLL clock;	0

			0: Disable USBHS_PLL clock. Hardware automatically resets upon entering deep sleep.	
19	CSSON	RW	Clock security system enable: 1: Enable clock security system. When HSE is ready (HSERDY is set to 1), the hardware turns on the clock monitoring function for HSE and finds that HSE abnormal triggers the CSSF flag and NMI interrupt; when HSE is not ready, the hardware turns off the clock monitoring function for HSE. 0: Disable clock security system.	0
18	HSEBYP	RW	External high-speed clock bypass: 1: Bypass the external high-speed crystal/ceramic resonator (if the HSEON bit is 0, set the HSEBYP bit to turn on the HSE bypass function, and then set the HSEON bit again to take effect); 0: No bypass high-speed external crystal/ceramic resonator.	0
17	HSERDY	RO	External high-speed crystal oscillation stability ready flag (set by hardware): 1: External high-speed crystal oscillation is stable. 0: External high-speed crystal oscillation is not stable. <i>Note: After the HSEON bit is cleared, it needs 6 HSE cycles to clear 0.</i>	0
16	HSEON	RW	HSE clock enable: 1: Enable HSE oscillator; 0: Disable HSE oscillator. <i>Note: Clear by hardware to stop HSE oscillator when entering Stop low-power mode.</i>	0
[15:8]	HSICAL[7:0]	RO	Internal high-speed clock calibration value These bits are automatically initialized at system startup.	x
[7:3]	HSITRIM[4:0]	RW	Internal high-speed clock adjustment: The user can enter an adjustment value that is superimposed on the HSICAL [7:0] value to adjust the frequency of the internal HIS RC oscillator according to changes in voltage and temperature. The default value is 16, which can adjust the HSI to 25MHz±0.25%; the change of HSICAL is adjusted about 50kHz per step.	10000b
2	Reserved	RO	Reserved	0
1	HSIRDY	RO	Internal high-speed HSI (25MHz) stable ready flag (set by hardware) 1: Internal high-speed HSI (25MHz) is stable; 0 Internal high-speed HSI (25MHz) is not stable. <i>Note: After the HSION bit is cleared, it needs 6 HSI</i>	1

			<i>cycles to clear 0.</i>	
0	HSION	RW	Internal high-speed clock (25MHz) enable control bit: 1: Enable the HSI oscillator; 0: Disable the HSI oscillator. <i>Note: This bit is set by hardware to 1 to start the internal RC oscillator when returning from Stop mode or when the external oscillator HSE used as the system clock fails.</i>	1

3.4.2 Clock Configuration Register 0 (RCC_CFGR0)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADCSRC	ADC_DUTY_SEL	Reserved			MCO[3:0]			UTMION	PIPEON	RGMIION		Reserved			FPRE[1:0]
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADCPRE[1:0]		PPRE2[2:0]			Reserved				HPRE[3:0]			SWS[1:0]			SW[1:0]

Bit	Name	Access	Description	Reset value
31	ADCSRC	RW	ADC input clock source selection: 1: USBHS_PLL (480MHZ); 0: HCLK.	0
30	ADC_DUTY_SEL	RW	ADC Clock Duty Cycle Selection: 1: ADC clock duty cycle is 75%; 0: ADC clock duty cycle is 50%. <i>Note: This bit is only effective when the ADC input clock source is selected as HCLK.</i>	0
[29:28]	Reserved	RO	Reserved	0
[27:24]	MCO[3:0]	RW	Microcontroller MCO pin clock output control: 00xx: No clock output; 0100: System clock (SYSCLK) output; 0101: Internal 25MHz oscillator clock (HSI) output; 0110: External oscillator clock (HSE) output; 0111: PLL clock output after 2 frequency division; 1000: UTMI clock output; 1001: USBSS_PLL clock output after 2 frequency division; 1010: ETH_PLL clock output after 8 frequency division; 1011: SERDES_PLL clock output after 16 frequency division;	0

			Others: Reserved.	
23	UTMION	RW	UTMI clock gating enable: 1: Enable; 0: Disable.	0
22	PIPEON	RW	PIPE clock gating enable: 1: Enable; 0: Disable.	0
21	RGMIION	RW	1000M Ethernet RGMII Interface Enable and Clock Enable: 1: Enable the 1000M Ethernet RGMII interface and enable the clock; 0: Disable the interface and disable the clock.	0
[20:18]	Reserved	RO	Reserved	0
[17:16]	FPRE[1:0]	RW	HCLK input clock source prescaler control: 00: No prescaling; 01: Divided by 2; 1x: Divided by 4.	0
[15:14]	ADCPRE[1:0]	RW	ADC clock source selection USBHS_PLL prescaler control high 2 bits (used in conjunction with RCC_CFGR0 register bits [13:11] PPRE2[2:0]): 00000: Divided by 5; 00001: Divided by 6; 00010: Divided by 7; 00011: Divided by 8; 11110: Divided by 35; 11111: Divided by 36. <i>Note: When using TIM1/8/9/10, do not configure PPRE2[2:0] to 101b/110b/111b.</i> ADC clock source selection HCLK prescaler control: 00: Divided by 2; 01: Divided by 4; 10: Divided by 6; 11: Divided by 8; <i>Note: The ADC clock should not exceed 80MHz.</i>	0
[13:11]	PPRE2[2:0]	RW	ADC clock source selection USBHS_PLL prescaler control low 3 bits (Used in conjunction with RCC_CFGR0 register bits [15:14] ADCPRE[1:0]). When applied to the ADC, the clock source selection enables HCLK prescaler control:	0

			0xx: No frequency division; 100: Divided by 2; 101: Divided by 4; 110: Divided by 8; 111: Divided by 16. <i>Note: When using TIM1/8/9/10, do not configure PPRE2[2:0] to 101b/110b/111b.</i>	
[10:8]	Reserved	RO	Reserved. Must be 0.	0
[7:4]	HPRE[3:0]	RW	HB clock source prescaler control: 0xxx: No frequency division; 1000: SYSCLK divided by 2; 1001: SYSCLK divided by 4; 1010: SYSCLK divided by 8; 1011: SYSCLK divided by 16; 1100: SYSCLK divided by 64; 1101: SYSCLK divided by 128; 1110: SYSCLK divided by 256; 1111: SYSCLK divided by 512.	0
[3:2]	SWS[1:0]	RO	System clock (SYSCLK) status (set by hardware): 00: System clock source is HSI; 01: System clock source is HSE; 10: System clock source is PLL; 11: Not available.	0
[1:0]	SW[1:0]	RW	Select the system clock source: 00: HSI as system clock; 01: HSE as system clock; 10: PLL output as system clock; 11: not available. <i>Note: With Clock Safe System enabled (CSSON=1), the selection of HSI as the system clock is forced by hardware when returning from Standby and Stop mode or when the external oscillator HSE used as the system clock fails.</i>	0

3.4.3 PLL Configuration Register (RCC_PLLCFGR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SYSP LL_G ATE	SYSPLL_SEL[2:0]		Reserved												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		PLL_SRC_DIV[5:0]					PLLSRC[2:0]			PLLMUL[4:0]					

Bit	Name	Access	Description	Reset value
31	SYSPLL_GATE	RW	This bit needs to be set before the system clock switches to PLL.	0
[30:28]	SYSPLL_SEL[2:0]	RW	The system clock is switched to PLL selection, and writing is allowed only when SYSPLL_GATE is 0: 0xx: PLL_CLK; 100: USBHS_PLL (480MHz); 101: ETH_PLL (500MHz); 110: SERDES_PLL/2; 111: USBSS_PLL (125MHz).	0
[27:14]	Reserved	RO	Reserved	0
[13:8]	PLL_SRC_DIV[5:0]	RW	The PLL input clock source is divided by frequency, and can only be written when PLLON is 0: 000000: divided by 1; 000001: divided by 2; 000010: divided by 3; ... 111111: divided by 64.	0
[7:5]	PLLSRC[2:0]	RW	PLL input clock source, which can only be written when PLLON is 0: 000: HIS; 001: HSE; 100: USBHS_PLL (480MHz); 101: ETH_PLL (500MHz); 110: USBSS_PLL (125MHz); 111: SERDES_PLL/2. <i>Note: The input reference clock for the main PLL is derived from the aforementioned clock source through PLL_SRC_DIV frequency division and PLLMUL frequency multiplication.</i>	0
[4:0]	PLLMUL[4:0]	RW	PLL clock frequency multiplication coefficient, which can only be written when PLLON is 0: 00000: PLL 4x frequency output; 00001: PLL 6x frequency output; 00010: PLL 7x frequency output; 00011: PLL 8x frequency output; 00100: PLL 8.5x frequency output; 00101: PLL 9x frequency output; 00110: PLL 9.5x frequency output; 00111: PLL 10x frequency output; 01000: PLL 10.5x frequency output; 01001: PLL 11x frequency output; 01010: PLL 11.5x frequency output; 01011: PLL 12x frequency output;	00100b

		01100: PLL 12.5x frequency output; 01101: PLL 13x frequency output; 01110: PLL 14x frequency output; 01111: PLL 15x frequency output; 10000: PLL 16x frequency output; 10001: PLL 17x frequency output; 10010: PLL 18x frequency output; 10011: PLL 19x frequency output; 10100: PLL 20x frequency output; 10101: PLL 22x frequency output; 10110: PLL 24x frequency output; 10111: PLL 26x frequency output; 11000: PLL 28x frequency output; 11001: PLL 30x frequency output; 11010: PLL 32x frequency output; 11011: PLL 34x frequency output; 11100: PLL 36x frequency output; 11101: PLL 38x frequency output; 11110: PLL 40x frequency output; 11111: PLL 59x frequency output;	
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3.4.4 Clock Interrupt Register (RCC_INTR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CSSC	SER DESP LLR DYC	ETHP LLR DYC	PLL RDY C	HSE RDY C	HSI RDY C	LSE RDY C	LSI RDY C
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	SER DESP LLR DYIE	ETHP LLR DYIE	PLL RDYI E	HSE RDYI E	HSI RDYI E	LSE RDYI E	LSI RDYI E	CSSF	SER DESP LLR DYF	ETHP LLR DYF	PLL RDY F	HSE RDY F	HSI RDY F	LSE RDY F	LSI RDY F

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23	CSSC	WO	Clear clock security system interrupt flag (CSSF): 1: Clear CSSF interrupt flag; 0: No effect.	0
22	SERDESPLLRDYC	WO	Clear SERDES_PLL ready interrupt flag: 1: Clear SERDESRDYF interrupt flag; 0: No action.	0
21	ETHPLLRDYC	WO	Clear ETH_PLL ready interrupt flag: 1: Clear ETHRDYF interrupt flag;	0

			0: No action.	
20	PLLRDYC	WO	Clear PLL ready interrupt flag: 1: Clear PLLRDYF interrupt flag; 0: No action.	0
19	HSERDYC	WO	Clear HSE oscillator ready interrupt flag: 1: Clear HSERDYF interrupt flag; 0: No action.	0
18	HSIRDYC	WO	Clear HSI oscillator ready interrupt flag: 1: Clear HSIRDYF interrupt flag; 0: No action.	0
17	LSERDYC	WO	Clear LSE oscillator ready interrupt flag: 1: Clear LSERDYF interrupt flag; 0: No action.	0
16	LSIRDYC	WO	Clear LSI oscillator ready interrupt flag: 1: Clear LSIRDYF interrupt flag; 0: No action.	0
15	Reserved	RO	Reserved	0
14	SERDESPLLRDYIE	RW	SERDES_PLL ready interrupt enable: 1: Enable SERDES_PLL ready interrupt; 0: Disable SERDES_PLL ready interrupt.	0
13	ETHPLLRDYIE	RW	ETH_PLL ready interrupt enable: 1: Enable ETH_PLL ready interrupt; 0: Disable ETH_PLL ready interrupt.	0
12	PLLRDYIE	RW	PLL ready interrupt enable: 1: Enable PLL ready interrupt; 0: Disable PLL ready interrupt.	0
11	HSERDYIE	RW	HSE ready interrupt enable: 1: Enable HSE ready interrupt; 0: Disable HSE ready interrupt.	0
10	HSIRDYIE	RW	HSI ready interrupt enable: 1: Enable HSI ready interrupt; 0: Disable HSI ready interrupt.	0
9	LSERDYIE	RW	LSE ready interrupt enable: 1: Enable LSE ready interrupt; 0: Disable LSE ready interrupt.	0
8	LSIRDYIE	RW	LSI ready interrupt enable: 1: Enable LSI ready interrupt; 0: Disable LSI ready interrupt.	0
7	CSSF	RO	Clock security system interrupt flag bit: 1: HSE clock failure, resulting in clock security interrupt CSSI. 0: No clock security system interrupt. Set by hardware, clear when software write CSSC bit 1.	0
6	SERDESPLLRD	RO	SERDES_PLL clock ready lock interrupt flag:	0

	YF		1: SERDES_PLL clock lock generates an interrupt; 0: No SERDES_PLL clock lock interrupt. Set by hardware, clear when software write SERDES_RDYC bit 1.	
5	ETHPLL RDYF	RO	ETH_PLL clock ready lock interrupt flag: 1: ETH_PLL clock lock generates an interrupt; 0: No ETH_PLL clock lock interrupt. Set by hardware, clear when software write ETHPLL_RDYC bit 1.	0
4	PLL RDYF	RO	PLL clock ready lock interrupt flag: 1: PLL clock lock generates an interrupt; 0: No PLL clock lock interrupt. Set by hardware, clear when software write PLL_RDYC bit 1.	0
3	HSE RDYF	RO	HSE clock ready interrupt flag: 1: HSE clock ready generates an interrupt; 0: No HSE clock ready interrupt. Set by hardware, clear when software write HSE_RDYC bit 1.	0
2	HSI RDYF	RO	HSI clock ready interrupt flag: 1: HSI clock ready generates an interrupt; 0: No HSI clock ready interrupt. Set by hardware, clear when software write HSI_RDYC bit 1.	0
1	LSE RDYF	RO	LSE clock ready interrupt flag: 1: LSE clock ready generates an interrupt; 0: No LSE clock ready interrupt. Set by hardware, clear when software write LSE_RDYC bit 1.	0
0	LSI RDYF	RO	LSI clock ready interrupt flag: 1: LSI clock ready generates an interrupt; 0: No LSI clock ready interrupt. Set by hardware, clear when software write LSI_RDYC bit 1.	0

3.4.5 HB2 Peripheral Reset Register (RCC_HB2PRSTR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
I3C RST	LTDC RST	Reserved		GPH A RST	ECD CRST	DFS DM RST	Reser ved	OPC MRS T	TIM1 2 RST	TIM1 1 RST	TIM1 0 RST	TIM9 RST	SDIO RST	Reser ved	SAI RST
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
I2C4 RST	USAR T1	TIM 8	SPI1 RST	TIM1 RST	ADC 2	ADC 1	Reser ved	IOPF RST	IOPE RST	IOPD RST	IOPC RST	IOPB RST	IOPA RST	HSA DC	AFIO RST

	RST	RST			RST	RST								RST	
--	-----	-----	--	--	-----	-----	--	--	--	--	--	--	--	-----	--

Bit	Name	Access	Description	Reset value
31	I3CRST	RW	I3C module reset control: 1: Reset module; 0: No effect.	0
30	LTDCRST	RW	LTDC module reset control: 1: Reset module; 0: No effect.	0
[29:28]	Reserved	R0	Reserved	0
27	GPHARST	RW	GPHA module reset control: 1: Reset module; 0: No effect.	0
26	ECDCRST	RW	ECDC module reset control: 1: Reset module; 0: No effect.	0
25	DFSDMRST	RW	DFSDM module reset control: 1: Reset module; 0: No effect.	0
24	Reserved	R0	Reserved	0
23	OPCMRST	RW	OPCM module reset control: 1: Reset module; 0: No effect.	0
22	TIM12RST	RW	TIM12 module reset control: 1: Reset module; 0: No effect.	0
21	TIM11RST	RW	TIM11 module reset control: 1: Reset module; 0: No effect.	0
20	TIM10RST	RW	TIM10 module reset control: 1: Reset module; 0: No effect.	0
19	TIM9RST	RW	TIM9 module reset control: 1: Reset module; 0: No effect.	0
18	SDIORST	RW	SDIO module reset control: 1: Reset module; 0: No effect.	0
17	Reserved	R0	Reserved	0
16	SAIRST	RW	SAI module reset control: 1: Reset module; 0: No effect.	0
15	I2C4RST	RW	I2C4 module reset control: 1: Reset module; 0: No effect.	0
14	USART1RST	RW	USART1 module reset control: 1: Reset module; 0: No effect.	0
13	TIM8RST	RW	TIM8 module reset control: 1: Reset module; 0: No effect.	0
12	SPI1RST	RW	SPI1 module reset control: 1: Reset module; 0: No effect.	0
11	TIM1RST	RW	TIM1 module reset control: 1: Reset module; 0: No effect.	0
10	ADC2RST	RW	ADC2 module reset control: 1: Reset module; 0: No effect.	0
9	ADC1RST	RW	ADC1 module reset control: 1: Reset module; 0: No effect.	0

8	Reserved	R0	Reserved	0
7	IOPFRST	RW	IO's PF port module reset control: 1: Reset module; 0: No effect.	0
6	IOPERST	RW	IO's PE port module reset control: 1: Reset module; 0: No effect.	0
5	IOPDRST	RW	IO's PD port module reset control: 1: Reset module; 0: No effect.	0
4	IOPCRST	RW	IO's PC port module reset control: 1: Reset module; 0: No effect.	0
3	IOPBRST	RW	IO's PB port module reset control: 1: Reset module; 0: No effect.	0
2	IOPARST	RW	IO's PA port module reset control: 1: Reset module; 0: No effect.	0
1	HSADCRST	RW	HSADC module reset control: 1: Reset module; 0: No effect.	0
0	AFIORST	RW	IO auxiliary function module reset control: 1: Reset module; 0: No effect.	0

3.4.6 HB1 Peripheral Reset Register (RCC_HB1PRSTR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWP MI RST	I2C3 RST	DAC RST	PWR RST	BKP RST	CAN 2 RST	CAN 1 RST	CAN 3 RST	Reser ved	I2C2 RST	I2C1 RST	USART RST	USART RST	USART RST	USART RST	SPI4 RST
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 RST	SPI2 RST	QSPI 2 RST	QSPI 1 RST	WW DG RST	LPTI M2 RST	LPTI M1 RST	USA RT8R ST	USA RT7R ST	USA RT6R ST	TIM7 RST	TIM6 RST	TIM5 RST	TIM4 RST	TIM3 RST	TIM2 RST

Bit	Name	Access	Description	Reset value
31	LPTIMRST	RW	LPTIM module reset control: 1: Reset module; 0: No effect.	0
30	I2C3RST	RW	I2C3 module reset control: 1: Reset module; 0: No effect.	0
29	DACRST	RW	DAC module reset control: 1: Reset module; 0: No effect.	0
28	PWRRST	RW	Power interface module reset control: 1: Reset module; 0: No effect.	0
27	BKPRST	RW	Backup unit reset control: 1: Reset module; 0: No effect.	0
26	CAN2RST	RW	CAN2 module reset control: 1: Reset module; 0: No effect.	0
25	CAN1RST	RW	CAN1 module reset control: 1: Reset module; 0: No effect.	0

24	CAN3RST	RW	CAN3 module reset control: 1: Reset module; 0: No effect.	0
23	Reserved	RO	Reserved	0
22	I2C2RST	RW	I2C 2 module reset control: 1: Reset module; 0: No effect.	0
21	I2C1RST	RW	I2C 1 interface reset control: 1: Reset module; 0: No effect.	0
20	USART5RST	RW	USART5 module reset control: 1: Reset module; 0: No effect.	0
19	USART4RST	RW	USART4 module reset control: 1: Reset module; 0: No effect.	0
18	USART3RST	RW	USART3 module reset control: 1: Reset module; 0: No effect.	0
17	USART2RST	RW	USART2 module reset control: 1: Reset module; 0: No effect.	0
16	SPI4RST	RW	SPI4 module reset control: 1: Reset module; 0: No effect.	0
15	SPI3RST	RW	SPI3 module reset control: 1: Reset module; 0: No effect.	0
14	SPI2RST	RW	SPI2 module reset control: 1: Reset module; 0: No effect.	0
13	QSPI2RST	RW	QSPI2 module reset control: 1: Reset module; 0: No effect.	0
12	QSPI1RST	RW	QSPI1 module reset control: 1: Reset module; 0: No effect.	0
11	WWDGRST	RW	WWDG reset control: 1: Reset module; 0: No effect.	0
10	LPTIM2RST	RW	LPTIM2 module reset control: 1: Reset module; 0: No effect.	0
9	LPTIM1RST	RW	LPTIM1 module reset control: 1: Reset module; 0: No effect.	0
8	USART8RST	RW	USART8 module reset control: 1: Reset module; 0: No effect.	0
7	USART7RST	RW	USART7 module reset control: 1: Reset module; 0: No effect.	0
6	USART6RST	RW	USART6 module reset control: 1: Reset module; 0: No effect.	0
5	TIM7RST	RW	Timer 7 module reset control: 1: Reset module; 0: No effect.	0
4	TIM6RST	RW	Timer 6 module reset control: 1: Reset module; 0: No effect.	0
3	TIM5RST	RW	Timer 5 module reset control: 1: Reset module; 0: No effect.	0
2	TIM4RST	RW	Timer 4 module reset control:	0

			1: Reset module; 0: No effect.	
1	TIM3RST	RW	Timer 3 module reset control: 1: Reset module; 0: No effect.	0
0	TIM2RST	RW	Timer 2 module reset control: 1: Reset module; 0: No effect.	0

3.4.7 HB Peripheral Clock Enable Register (RCC_HBPCENR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									PIOC EN	Reser ved	SER DESE N	USBP D EN	UHSI F EN	USB OTG EN	Reserv ed
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	ETH MAC EN	DVP EN	USBS S EN	USB HS EN	SDM MC EN	RNG EN	FMC EN	Reser ved	CRC EN	Reserved				DMA 2 EN	DMA 1 EN

Bit	Name	Access	Description	Reset value
[31:23]	Reserved	RO	Reserved	0
22	PIOCEN	RW	PIOC module clock enable: 1: Module clock on; 0: Module clock off.	0
21	Reserved	RO	Reserved	0
20	SERDESEN	RW	SERDES module clock enable: 1: Module clock on; 0: Module clock off.	0
19	USBPDEN	RW	USBPD module clock enable: 1: Module clock on; 0: Module clock off.	0
18	UHSIFEN	RW	HSPP module clock enable: 1: Module clock on; 0: Module clock off.	0
17	OTGFSEN	RW	USBOTG_FS module clock enable: 1: Module clock on; 0: Module clock off.	0
[16:15]	Reserved	RO	Reserved	0
14	ETHMACEN	RW	Ethernet MAC Clock Enable: 1: Ethernet MAC clock enabled; 0: Ethernet MAC clock disabled.	0
13	DVPEN	RW	DVP module clock enable: 1: Module clock on; 0: Module clock off.	0
12	USBSEN	RW	USBSS module clock enable: 1: Module clock on; 0: Module clock off.	0
11	USBHSEN	RW	USBHS module clock enable: 1: Module clock on; 0: Module clock off.	0
10	SDMMCEN	RW	SDMMC module clock enable: 1: Module clock on; 0: Module clock off.	0
9	RNGEN	RW	RNG module clock enable: 1: Module clock on; 0: Module clock off.	0

8	FMCEN	RW	FMC module clock enable: 1: Module clock on; 0: Module clock off.	0
7	Reserved	RO	Reserved	0
6	CRCEN	RW	CRC module clock enable: 1: Module clock on; 0: Module clock off.	0
[5:2]	Reserved	RO	Reserved	0
1	DMA2EN	RW	DMA2 module clock enable: 1: Module clock on; 0: Module clock off.	0
0	DMA1EN	RW	DMA1 module clock enable: 1: Module clock on; 0: Module clock off.	0

3.4.8 HB2 Peripheral Clock Enable Register (RCC_HB2PCENR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
I3C EN	LTDC EN	Reserved		GPH A EN	ECD C EN	DFS DM EN	Reser ved	OPC M EN	TIM1 2 EN	TIM1 1 EN	TIM1 0 EN	TIM9 EN	SDIO EN	Reser ved	SAI EN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
I2C4 EN	USAR T1 EN	TIM8 EN	SPI1 EN	TIM1 EN	ADC 2 EN	ADC 1 EN	Reser ved	IOPF EN	IOPE EN	IOPD EN	IOPC EN	IOPB EN	IOPA EN	HSA DCE N	AFIO EN

Bit	Name	Access	Description	Reset value
31	I3CEN	RW	I3C module clock enable: 1: Module clock on; 0: Module clock off.	0
30	LTDCEN	RW	LTDC module clock enable: 1: Module clock on; 0: Module clock off.	0
[29:28]	Reserved	RO	Reserved	0
27	GPHAEN	RW	GPHA module clock enable: 1: Module clock on; 0: Module clock off.	0
26	ECDCEN	RW	ECDC module clock enable: 1: Module clock on; 0: Module clock off.	0
25	DFSDMEN	RW	DFSDM module clock enable: 1: Module clock on; 0: Module clock off.	0
24	Reserved	RO	Reserved	0
23	OPCMEN	RW	OPCM module clock enable: 1: Module clock on; 0: Module clock off.	0
22	TIM12EN	RW	TIM12 module clock enable: 1: Module clock on; 0: Module clock off.	0
21	TIM11EN	RW	TIM11 module clock enable: 1: Module clock on; 0: Module clock off.	0
20	TIM10EN	RW	TIM10 module clock enable: 1: Module clock on; 0: Module clock off.	0
19	TIM9EN	RW	TIM9 module clock enable:	0

			1: Module clock on; 0: Module clock off.	
18	SDIOEN	RW	SDIO module clock enable: 1: Module clock on; 0: Module clock off.	0
17	Reserved	RO	Reserved	0
16	SAIEN	RW	SAI module clock enable: 1: Module clock on; 0: Module clock off.	0
15	I2C4EN	RW	I2C4 module clock enable: 1: Module clock on; 0: Module clock off.	0
14	USART1EN	RW	USART1 module clock enable: 1: Module clock on; 0: Module clock off.	0
13	TIM8EN	RW	TIM8 module clock enable: 1: Module clock on; 0: Module clock off.	0
12	SPI1EN	RW	SPI1 module clock enable: 1: Module clock on; 0: Module clock off.	0
11	TIM1EN	RW	TIM1 module clock enable: 1: Module clock on; 0: Module clock off.	0
10	ADC2EN	RW	ADC2 module clock enable: 1: Module clock on; 0: Module clock off.	0
9	ADC1EN	RW	ADC1 module clock enable: 1: Module clock on; 0: Module clock off.	0
8	Reserved	RO	Reserved	0
7	IOPFEN	RW	IO'PF port module clock enable: 1: Module clock on; 0: Module clock off.	0
6	IOPEEN	RW	IO'PE port module clock enable: 1: Module clock on; 0: Module clock off.	0
5	IOPDEN	RW	IO'PD port module clock enable: 1: Module clock on; 0: Module clock off.	0
4	IOPCEN	RW	IO'PC port module clock enable: 1: Module clock on; 0: Module clock off.	0
3	IOPBEN	RW	IO'PB port module clock enable: 1: Module clock on; 0: Module clock off.	0
2	IOPAEN	RW	IO'PA port module clock enable: 1: Module clock on; 0: Module clock off.	0
1	HSADCEN	RW	HSADC module clock enable: 1: Module clock on; 0: Module clock off.	0
0	AFIOEN	RW	IO auxiliary function module clock enable: 1: Module clock on; 0: Module clock off.	0

3.4.9 HB1 Peripheral Clock Enable Register (RCC_HB1PCENR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWP MI EN	I2C3 EN	DAC EN	PWR EN	BKP EN	CAN 2 EN	CAN 1 EN	CAN 3 EN	Reser ved	I2C2 EN	I2C1 EN	USAR T5 EN	USAR T4 EN	USAR T3 EN	USAR T2 EN	SPI4 EN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 EN	SPI2 EN	QSPI 2 EN	QSPI 1 EN	WW DG EN	LPTI M2E N	LPTI M1E N	USA RT8E N	USA RT7E N	USA RT6E N	TIM7 EN	TIM6 EN	TIM5 EN	TIM4 EN	TIM3 EN	TIM2 EN

Bit	Name	Access	Description	Reset value
31	SWPMIEN	RW	SWPMI module clock enable: 1: Module clock on; 0: Module clock off.	0
30	I2C3EN	RW	I2C3 module clock enable: 1: Module clock on; 0: Module clock off.	0
29	DACEN	RW	DAC module clock enable: 1: Module clock on; 0: Module clock off.	0
28	PWREN	RW	Power interface module clock enable: 1: Module clock on; 0: Module clock off.	0
27	BKPEN	RW	Backup unit clock enable: 1: Module clock on; 0: Module clock off.	0
26	CAN2EN	RW	CAN2 module clock enable: 1: Module clock on; 0: Module clock off.	0
25	CAN1EN	RW	CAN1 module clock enable: 1: Module clock on; 0: Module clock off.	0
24	CAN3EN	RW	CAN3 module clock enable: 1: Module clock on; 0: Module clock off.	0
23	Reserved	RO	Reserved	0
22	I2C2EN	RW	I2C2 module clock enable: 1: Module clock on; 0: Module clock off.	0
21	I2C1EN	RW	I2C1 module clock enable: 1: Module clock on; 0: Module clock off.	0
20	USART5EN	RW	USART5 module clock enable: 1: Module clock on; 0: Module clock off.	0
19	USART4EN	RW	USART4 module clock enable: 1: Module clock on; 0: Module clock off.	0
18	USART3EN	RW	USART3 module clock enable: 1: Module clock on; 0: Module clock off.	0
17	USART2EN	RW	USART2 module clock enable: 1: Module clock on; 0: Module clock off.	0
16	SPI4EN	RW	SPI4 module clock enable: 1: Module clock on; 0: Module clock off.	0
15	SPI3EN	RW	SPI3 module clock enable: 1: Module clock on; 0: Module clock off.	0
14	SPI2EN	RW	SPI2 module clock enable: 1: Module clock on; 0: Module clock off.	0
13	QSPI2EN	RW	QSPI2 module clock enable: 1: Module clock on; 0: Module clock off.	0
12	QSPI1EN	RW	QSPI1 module clock enable:	0

			1: Module clock on; 0: Module clock off.	
11	WWDGEN	RW	Window watchdog clock enable: 1: Module clock on; 0: Module clock off.	0
10	LPTIM2EN	RW	LPTIM2 module clock enable: 1: Module clock on; 0: Module clock off.	0
9	LPTIM1EN	RW	LPTIM1 module clock enable: 1: Module clock on; 0: Module clock off.	0
8	USART8EN	RW	USART8 module clock enable: 1: Module clock on; 0: Module clock off.	0
7	USART7EN	RW	USART7 module clock enable: 1: Module clock on; 0: Module clock off.	0
6	USART6EN	RW	USART6 module clock enable: 1: Module clock on; 0: Module clock off.	0
5	TIM7EN	RW	Timer 7 module clock enable: 1: Module clock on; 0: Module clock off.	0
4	TIM6EN	RW	Timer 6 module clock enable: 1: Module clock on; 0: Module clock off.	0
3	TIM5EN	RW	Timer 5 module clock enable: 1: Module clock on; 0: Module clock off.	0
2	TIM4EN	RW	Timer 4 module clock enable: 1: Module clock on; 0: Module clock off.	0
1	TIM3EN	RW	Timer 3 module clock enable: 1: Module clock on; 0: Module clock off.	0
0	TIM2EN	RW	Timer 2 module clock enable: 1: Module clock on; 0: Module clock off.	0

3.4.10 Backup Domain Control Register (RCC_BDCTLR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															BDR ST
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTCCAL[6:0]							RTCE N	RTCSEL [1:0]	ASO S	ASO E	CCO	LSEB YP	LSEB EDY	LSEB N	

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	BDRST	RW	Backup domain software reset control: 1: Reset the entire backup domain. 0: Undo the reset.	0
[15:9]	RTCCAL[6:0]	RW	RTC Calibration Value Register The value in this register indicates how many clock pulses are skipped per 2^{20} clock pulses. This function is used to calibrate the RTC	0

			clock. The RTC clock can be slowed by 0 to 121 ppm.	
8	RTCEN	RW	RTC clock enable control: 1: Enable the RTC clock; 0: Disable RTC clock.	0
[7:6]	RTCSEL[1:0]	RW	RTC clock source selection: 00: No clock; 01: LSE oscillator as RTC clock; 10: LSI oscillator as RTC clock; 11: HSE oscillator divided by 512 as RTC clock.	0
5	ASOS	RW	PC13 Pin alarm/second pulse output selection: 1: Outputs second pulses; 0: Outputs alarm pulses. <i>Note: This bit is reset only by the Backup Domain Reset (BDRST).</i>	0
4	ASOE	RW	PC13 Pin Enable Pulse Output Bit: 1: Enables output of alarm pulses or second pulses; 0: Disables output of alarm pulses or second pulses. <i>Note: This bit is reset only by the Backup Domain Reset (BDRST).</i>	0
3	CCO	RW	Calibration clock output selection bit: 1: Outputs RTC clock divided by 64 via PC13 pin; 0: Does not output calibration clock. <i>Note: This bit is cleared when V_{DD33} power supply is disconnected.</i>	0
2	LSEBYP	RO	External Low-speed crystal (LSE) bypass control bit: 1: Bypass external low-speed crystal/ceramic resonator (use external clock source); 0: Do not bypass the low-speed external crystal/ceramic resonator.	0
1	LSERDY	RO	External low-speed crystal oscillation stabilization ready flag bit (set by hardware): 1: External low-speed crystal oscillation is stabilized; 0: External low-speed crystal oscillation is not stabilized. <i>Note: After the LSEON bit is cleared to 0, this bit takes 6 LSE cycles to clear to 0.</i>	0
0	LSEON	RO	External low-speed crystal oscillation enable control bit: 1: Enable the LSE oscillator; 0: Disable the LSE oscillator.	0

Note: The LSEON, LSEBYP, CCO, ASOE, ASOS, RTCSEL, RTCEN and RTCCAL bits within the Backup Domain Control Register (RCC_BDCTL) reside in the backup domain. Consequently, these bits remain in write-protected state after reset and can only be modified after setting the DBP bit to 1 in the Power Control Register (PWR_CR). These bits can be cleared by a backup domain reset. Neither internal nor external resets affect these bits.

3.4.11 Control/Status Register (RCC_RSTSCKR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LKUP RSTF	WWDG RSTF	IWDG RSTF	SFT RSTF	POR RSTF	PIN RSTF	Reserved	RMVF	Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													LSI RDY	LSIO N	

Bit	Name	Access	Description	Reset value
31	LKUPRSTF	RO	LOCKUP Reset Flag: 1: System reset occurred due to LOCKUP; 0: Normal operation. Set to 1 by hardware during LOCKUP reset; cleared by software writing the RMVF bit.	0
30	WWDGRSTF	RO	Window watchdog reset flag: 1: Occurrence of a window watchdog reset. 0: No window watchdog reset occurs. Set to 1 by hardware when a window watchdog reset occurs; cleared by software writing of the RMVF bit.	0
29	IWDGRSTF	RO	Independent watchdog reset flag: 1: Occurrence of an independent watchdog reset. 0: No independent watchdog reset occurs. Set to 1 by hardware when an independent watchdog reset occurs; cleared by software writing of the RMVF bit.	0
28	SFTRSTF	RO	Software reset flag: 1: Software reset occurs. 0: No software reset occurs. Set to 1 by hardware when a software reset occurs; software write RMVF bit cleared.	0
27	PORRSTF	RO	Power-up/power-down reset flag: 1: Power-up/power-down reset occurs. 0: No power-up/power-down reset occurs. Set to 1 by hardware when power-up/power-down reset occurs; cleared by software writing of RMVF bit.	1
26	PINRSTF	RO	External manual reset (NST pin) flag: 1: Occurrence of NST pin reset. 0: No NST pin reset occurs. Set to 1 by hardware when NST pin reset occurs; cleared by software writing of RMVF bit.	0
25	Reserved	RO	Reserved	0
24	RMVF	WO	Clear reset flag control: 1: Clear the reset flag.	0

			0: No effect.	
[23:2]	Reserved	RO	Reserved	0
1	LSIRDY	RO	Internal low-speed clock (LSI) stabilization ready flag bit (set by hardware): 1: The internal low-speed clock (40KHz) is stabilized; 0: The internal low-speed clock (40KHz) is not stabilized. <i>Note: After the LSION bit is cleared, it takes 3 LSI cycles to clear it.</i>	0
0	LSION	RW	Internal low-speed clock (LSI) enable control bits: 1: Enables the LSI (40KHz) oscillator; 0: Disables the LSI (40KHz) oscillator.	0

Note: Except for BIT1, which is cleared by power-on reset, other write-clear reset flags can be cleared.

3.4.12 HB Peripheral Reset Register (RCC_HBRSTR)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									PIOC RST	Reser ved	SER DES RST	USBP D RST	UHSI FRST	OTG FSRS T	Reser ved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	ETH MAC RST	DVP RST	USBS SRST	USB HS RST	SDM MC RST	RNG RST	FMC RST	Reserved						DMA 2 RST	DMA 1 RST

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
22	PIOCRST	RW	PIOC module reset control: 1: Reset module; 0: No effect.	0
21	Reserved	RO	Reserved	0
20	SERDES RST	RW	SERDES module reset control: 1: Reset module; 0: No effect.	0
19	USBPDRST	RW	USBPD module reset control: 1: Reset module; 0: No effect.	0
18	UHSIFRST	RW	UHSIF module reset control: 1: Reset module; 0: No effect.	0
17	OTGFSRST	RW	USBOTG_FS module reset control: 1: Reset module; 0: No effect.	0
[16:15]	Reserved	RO	Reserved	0
14	ETHMACRST	RW	Ethernet MAC module reset control: 1: Reset module; 0: No effect.	0
13	DVPRST	RO	DVP module reset control: 1: Reset module; 0: No effect.	0
12	USBSSRST	RW	USBSS module reset control: 1: Reset module; 0: No effect.	0

11	USBHSRST	RW	USBHS module reset control: 1: Reset module; 0: No effect.	0
10	SDMMCRST	RW	SDMMC module reset control: 1: Reset module; 0: No effect.	0
9	RNGRST	RW	RNG module reset control: 1: Reset module; 0: No effect.	0
8	FMCIRST	RW	FMC module reset control: 1: Reset module; 0: No effect.	0
[7:2]	Reserved	RO	Reserved	0
1	DMA2RST	RW	DMA2 module reset control: 1: Reset module; 0: No effect.	0
0	DMA1RST	RW	DMA1 module reset control: 1: Reset module; 0: No effect.	0

3.4.13 Clock Configuration Register 2 (RCC_CFGR2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ETH1GSRC [1:0]		HSADCSRC [1:0]		Reserved		I2S3S RC	I2S2S RC	RNG SRC	Reserved		USBF SSRC	USBFSDIV[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LTDCSRC [1:0]		LTDCDIV[5:0]						UHSIFSRC [1:0]		UHSIFDIV[5:0]					

Bit	Name	Access	Description	Reset value
[31:30]	ETH1GSRC[1:0]	RW	Gigabit Ethernet 125M clock source selection: 00: PLL Clock (PLL_CLK); 01: USBSS_PLL Clock; 10: ETH_PLL Clock output after 4-division; 11: SERDES_PLL Clock output after 8-division.	0
[29:28]	HSADCSRC[1:0]	RW	HSADC clock source selection: 00: System clock (SYSCLK); 01: PLL clock (PLL_CLK); 10: USBHS_PLL clock (480MHz); 11: ETH_PLL clock (500MHz).	0
[27:26]	Reserved	RO	Reserved	0
25	I2S3SRC	RW	I2S3 clock source: 1: PLL clock (PLL_CLK); 0: System clock (SYSCLK).	0
24	I2S2SRC	RW	I2S2 clock source: 1: PLL clock (PLL_CLK); 0: System clock (SYSCLK).	0
23	RNGSRC	RW	RNG clock source: 1: PLL clock (PLL_CLK);	0

			0: System clock (SYSCLK).	
[22:21]	Reserved	RO	Reserved	
20	USBFSRC	RW	USBFS 48M clock source selection: 1: USBHS_PLL clock; 0: PLL clock.	0
[19:16]	USBFSDIV[3:0]	RW	USBFS 48M clock source prescaler: 0000: No frequency division; 0001: Divided by 2; 0010: Divided by 3; 0011: Divided by 4; 0100: Divided by 5; 0101: Divided by 6; 0110: Divided by 8; 0111: Divided by 10; 1000: Divided by 1.5; 1001: Divided by 2.5; 1010: Divided by 3.5; 1011: Divided by 4.5; 1100: Divided by 5.5; 1101: Divided by 6.5; 1110: Divided by 7.5; 1111: Divided by 9.5.	0
[15:14]	LTDCSRC[1:0]	RW	LTDC clock source selection: 00: PLL clock (PLL_CLK); 01: SERDES_PLL clock divided by 2; 10: ETH_PLL clock; 11: USBHS_PLL clock.	0
[13:8]	LTDCDIV[5:0]	RW	LTDC clock source prescaler control: 000000: No frequency division; 000001: Divided by 2; 000010: Divided by 3; 000011: Divided by 4; 000100: Divided by 5; ... 111110: Divided by 63; 111111: Divided by 64.	0
[7:6]	UHSIFSRC[1:0]	RW	UHSIF clock source selection: 00: System clock (SYSCLK); 01: PLL clock (PLL_CLK); 10: USBHS_PLL clock (480MHz); 11: ETH_PLL clock (500MHz).	0
[5:0]	UHSIFDIV[5:0]	RW	UHSIF clock source prescaler control: 000000: No frequency division; 000001: Divided by 2; 000010: Divided by 3; 000011: Divided by 4; 000100: Divided by 5; ... 111110: Divided by 63; 111111: Divided by 64.	0

3.4.14 PLL Configuration Register 2 (RCC_PLLCFGR2)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												SERDESPLL_MUL[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			USBHSPLL_IN_DIV[4:0]				Reserved	USBSSPLL_REFSEL[2:0]			USBHSPLL_REFSEL[1:0]		USBHSPLL_SRC[1:0]		

Bit	Name	Access	Description	Reset value
[31:20]	Reserved	RO	Reserved	0
[19:16]	SERDESPLL_MUL[3:0]	RW	SERDES_PLL clock multiplication factor, write permitted only when SERDES_PLLON is 0: 0000: 25x frequency output; 0001: 28x frequency output; 0010: 30x frequency output; 0011: 32x frequency output; 0100: 35x frequency output; 0101: 38x frequency output; 0110: 40x frequency output; 0111: 45x frequency output; 1000: 50x frequency output; 1001: 56x frequency output; 1010: 60x frequency output; 1011: 64x frequency output; 1100: 70x frequency output; 1101: 76x frequency output; 1110: 80x frequency output; 1111: 90x frequency output.	1000b
[15:13]	Reserved	RO	Reserved	0
[12:8]	USBHSPLL_IN_DIV[4:0]	RW	The frequency division settings for the USBHS_PLL input clock source SYS_PLL are only writable when USBHS_PLLON is 0: 00000: No frequency division; 00001: Divided by 2; ... 11111: Divided by 32.	0
7	Reserved	RO	Reserved	0
[6:4]	USBSSPLL_REFSEL[1:0]	RW	USBSS_PLL reference clock frequency selection is writable only when USBSS_PLLON is 0: 000: 20MHz; 001: 24MHz; 010: 25MHz; 011: 30MHz;	010b

			100: 32MHz; 101: 40MHz; 110: 60MHz; 111: 80MHz.	
[3:2]	USBHSPLL_REFSEL[1:0]	RW	USBHS_PLL reference clock frequency selection is writable only when USBHS_PLLON is 0: 00: 25MHz; 01: 20MHz; 10: 24MHz; 11: 32MHz.	0
[1:0]	USBHSPLLSRC[1:0]	RW	USBHS_PLL clock source selection: 00: HSE; 01: HIS; 10: ETHCLK_20M; 11: The SYS_PLL_CLK after frequency division uses the division factor referenced from RCC_PLLCFGR2[12:8].	0

Chapter 4 Central Processing Unit (CPU)

The CH32H417 series integrates a dual-core architecture, with both RISC-V5F and RISC-V3F cores supporting 32-bit I, C, M, A, B, F, and extended instruction sets.

The RISC-V3F core (C0/Core 0) employs a sequential single-issue architecture internally. Its fetch mode is configurable, supporting both pipelined and non-pipelined modes. Operating in non-pipelined mode reduces core power consumption.

The RISC-V5F core (C1/Core 1) employs an out-of-order dual-issue architecture internally and integrates multiple branch predictors with instruction cache support, enabling superior performance for the V5. Additionally, RISC-V5 implements the WCH-X extended instruction set, delivering improvements in both code density and computational performance.

Private peripherals include system timers, inter-core communication, and hardware semaphore modules. External debugger support is provided, with the debugger enabling single-wire/dual-wire communication modes. Single-wire communication speed is adjustable across 4 levels.

4.1 Main Features

RISC-V3F

- Sequential single-issue architecture
- 3-stage pipeline
- Low clock frequency, low power consumption, control-focused
- Provides one non-maskable interrupt (NMI)
- 4-channel physical memory protection (PMP)
- 4-channel hardware breakpoints
- 2-level interrupt nesting
- Programmable Fast Interrupt Controller (PFIC)
- Custom extendable instructions
- Hardware stack

RISC-V5F

- Out-of-order multiple-issue architecture
- Built-in 32KB instruction cache
- High clock frequency, high performance, focused on computation
- Provides one non-maskable interrupt (NMI)
- 4-channel physical memory protection (PMP)
- 4-channel hardware breakpoints
- 8-level interrupt nesting
- Programmable Fast Interrupt Controller (PFIC)
- Customizable extended instructions
- Hardware stack

4.1.1 V5F Core Memory Operation Model

V5F core is a kind of out-of-order superscalar processing, and its access to memory has reordering function.

The memory types of V5F are divided into 2 categories:

- **Normal**

CPU's access to this kind of memory will be reordered to improve performance.

- **Device and Strongly-ordered**

CPU access to this type of memory will be strictly executed in program order.

Table 4-1 lists the default attributes and types of memory regions. The Memory Protection Unit (MPU) can disable the Read (R), Write (W), and Execute (X) attributes.

Table 4-1 Default attributes and types of memory areas

	CODE	ITCM	DTCM	SRAM	PERI	MEM0	MEM1	DEV0	DEV1	PPB	RSV
Attribute	RWX	RWX	RWX	RWX	RW	RWX	RWX	RW	RW	RW	RW
Bus	HB	TCM_BUS	TCM_BUS	HB	HB	HB	HB	HB	HB	HB	HB
Type	Normal	Normal	Normal	Normal	Device	Device	Device	Normal	Device	Device	Device

Note: (1) TCM_BUS refers to the bus customized within this chip to interface with the TCM memory.

(2) Access to the PPB region operates at a fixed priority, with the V5 core taking precedence over the V3 core.

Note that bandwidth limitations may impact the performance of the V3 core.

The address range of each memory area is shown in Table 4-2:

Table 4-2 Address range of memory area

Memory area	Address range
CODE	0x00000000 - 0x1FFFFFFF
ITCM	0x200A0000 - 0x200BFFFF
DTCM	0x200C0000 - 0x200FFFFF
SRAM	0x20100000 - 0x3FFFFFFF
PERI	0x40000000 - 0x5FFFFFFF
MEM0	0x60000000 - 0x7FFFFFFF
MEM1	0x80000000 - 0x9FFFFFFF
DEV0	0xA0000000 - 0xBFFFFFFF
DEV1	0xC0000000 - 0xDFFFFFFF
PPB	0xE0000000 - 0xE00FFFFF
RSV	Other address

In the out-of-order core, the actual memory access sequence is not always the same as that in the program. In order to improve the access efficiency, the core will adopt reordering, buffer and multi-bus interface to speed up the access speed without violating the data correlation.

For 2 unrelated memory access instructions A1 and A2 in the program, assuming that the A1 instruction sequence is earlier than A2 instruction and there is no instruction/data synchronization instruction between them, the actual request sequence of A1/A2 depends on the memory type (Normal, Device) accessed by A1/A2 and its instruction type (Load, Store). Possible situations as shown in Table 4-3.

Table 4-3 A1/A2 Memory access types and instruction types

A1 \ A2	store device	store normal	load device	load normal
store device	order	re-order	order	re-order
store normal	re-order	re-order	re-order	re-order
load device	order	order	order	re-order
load normal	order	order	re-order	re-order

Note: The order and reordering described above are based on the execution sequence of the CPU's top-level

interface bus operations. For all out-of-order accesses, sequential execution can be enforced by inserting fence instructions.

4.2 Privileged Register (CSR)

Both the RISC-V3F and RISC-V5F cores comply with the RISC-V standard privilege structure, supporting M and U modes, debug mode, and more. The cores define a subset of standard registers and custom registers.

4.2.1 General Privilege Register

The RISC-V architecture defines a set of Control and Status Registers (CSRs) used to configure, identify, or record operational states. CSR registers are internal to the core and utilize a dedicated 12-bit address space.

Note: Registers marked with the "MRW, MRO, MRWI" attributes require the system to be in machine mode for access.

Table 4-4 General privilege register list

Name	CSR address	Description	Reset
MVENDORID	0xF11	Chip vendor ID register	0x00000000
MARCHID	0xF12	Chip structure ID register	V3F: 0xDC68D86D V5F: 0xDC68D8AE
MIMPID	0xF13	Machine implementation ID register	V3F: 0xDC688002 V5F: 0xDC688001
MHARTID	0xF14	Core ID register	V3F: 0x00000000 V5F: 0x00000001
MSTATUS	0x300	Machine status register	0x00000000
MISA	0x301	Machine instruction set register	0x40901127
MTVEC	0x305	Machine mode exception base address register	0x00000000
MSCRATCH	0x340	Machine mode temporary register	0xFFFFFFFF
MEPC	0x341	Machine abnormal program counter register	0xFFFFFFFF
MCAUSE	0x342	Machine exception register	0x000000XX
MTVAL	0x343	Machine self-trapping value register	0xFFFFFFFF
TINFO	0x7A4	Trigger information register	0x00000004
DPC	0x7B1	Debugging PC address register	0xFFFFFFFF
DSCRATCH0	0x7B2	Debug mode temporary register 0	0xFFFFFFFF
DSCRATCH1	0x7B3	Debug mode temporary register 1	0xFFFFFFFF
DBGMCU_0	0x7C0	Debug register 0	0x00000000
FFLAGS	0x001	Floating-point cumulative exception flag register	0x00000000
FRM	0x002	Floating-point rounding mode register	0x00000000
FCSR	0x003	Floating point control and status register	0x00000000
UACCES_MSTATUS	0x800	User access to machine status register	0x00000000
HW_POPDM_CTLR	0x804	Hardware stack control register	0x00000004
TSELECT	0x7A0	Trigger channel selection register	0x0000000X
TDATA1	0x7A1	Trace trigger data register	0x23E00048

TDATA2	0x7A2	Save trigger data register	0xFFFFFFFF
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4.2.1.1 Chip Vendor ID Register (MVENDORID)

CSR address: 0xF11

Bit	Name	Access	Description	Reset value
[31:0]	mvendorid[31:0]	MRO	Chip vendor ID, fixed value is 0.	0

4.2.1.2 Chip Structure ID Register (MARCHID)

CSR address: 0xF12

Bit	Name	Access	Description	Reset value
[31:0]	marchid[31:0]	MRO	Chip structure ID, fixed value cannot be modified.	V3F: 0xDC68D86D V5F: 0xDC68D8AE

4.2.1.3 Machine Implementation ID Register (MIMPID)

CSR address: 0xF13

Bit	Name	Access	Description	Reset value
[31:0]	mimpid[31:0]	MRO	Machine implementation ID, fixed value cannot be modified.	V3F: 0xDC688002 V5F: 0xDC688001

4.2.1.4 Core ID Register (MHARTID)

CSR address: 0xF14

Bit	Name	Access	Description	Reset value
[31:0]	mhartid[31:0]	MRO	Core ID, indicating the core ID number of this core in the chip structure.	V3F: 0 V5F: 1

4.2.1.5 Machine Status Register (MSTATUS)

CSR address: 0x300

Bit	Name	Access	Description	Reset value
31	SD	MRO	Used to mark whether a floating-point instruction is in a dirty state.	0
[30:15]	Reserved	MRO	Reserved	0
[14:13]	FS[1:0]	MRW	Floating-point instruction set status, when the value is not 0, single-precision floating-point instruction (f) is allowed to execute: 00: OFF; 01: Initial; 10: Clean; 11: Dirty.	0
[12:11]	MPP[1:0]	MRW	Save/restore the machine state before entering abnormality, and update it to the machine state before entering abnormality:	0

			00: Set machine status to U mode upon exit exception; 01: Reserved; 10: Reserved; 11: Set machine status to M mode upon exit exception.	
[10:8]	Reserved	MRO	Reserved	0
7	MPIE	MRW	Preserve the interrupt enable status upon entering an interrupt: 1: Enable global interrupts after exiting the interrupt; 0: Disable global interrupts after exiting the interrupt.	0
[6:4]	Reserved	MRO	Reserved	0
3	MIE	MRW	Global interrupts enabled; updates to MPIE value upon interrupt exit: 1: Global interrupts enabled, allowing the processor to respond to interrupt requests; 0: Global interrupts disabled, preventing the processor from responding to interrupt requests.	0
[2:0]	Reserved	MRO	Reserved	0

4.2.1.6 Machine Instruction Set Register (MISA)

CSR address: 0x301

Bit	Name	Access	Description	Reset value
[31:30]	XLEN[1:0]	MRO	Machine basic integer bit width and basic instruction bit width information, 32-bit wide core, fixed value is 1.	0x1
[29:24]	Reserved	MRO	Reserved	0
23	X	MRO	This core supports the execution of custom instruction sets.	1
[22:21]	Reserved	MRO	Reserved	0
20	U	MRO	Support user mode operation.	1
[19:13]	Reserved	MRO	Reserved	0
12	M	MRO	This core supports the execution of multiplication instruction sets.	1
[11:9]	Reserved	MRO	Reserved	0
8	I	MRO	This core supports the execution of basic integer instruction sets.	1
[7:6]	Reserved	MRO	Reserved	0
5	F	MRO	This core supports the execution of single-precision floating-point instruction sets.	1
[4:3]	Reserved	MRO	Reserved	0

2	C	MRO	This core supports the execution of compressed instruction sets.	1
1	B	MRO	This core supports the execution of bit operation instruction sets.	1
0	A	MRO	This core supports the execution of atomic instruction sets.	1

4.2.1.7 Machine Mode Exception Base Address Register (MTVEC)

CSR address: 0x305

Bit	Name	Access	Description	Reset value
[31:10]	mtvec_base[21:0]	MRW	The high-order bits of the base address for jump instructions when processor exceptions occur, aligned to 1kB. The actual 32-bit base address $BASE = \{mtvec_base, 10'h0\}$.	0
[9:2]	Reserved	MRO	Reserved	0
1	mode1	RW	Vector Table Mode 1: 1: The interrupt vector table stores jump addresses; 0: The interrupt vector table stores jump instructions.	0
0	mode0	RW	Vector Table Mode 0: 1: All interrupts/exceptions have independent entry points, with jump addresses set to $BASE + 4 * IRQ/EXC ID$; 0: All interrupts/exceptions share a unified entry point, all jumping to the exception jump base address.	0

4.2.1.8 Machine Mode Temporary Register (MSCRATCH)

CSR address: 0x340

Bit	Name	Access	Description	Reset value
[31:0]	mscratch[31:0]	MRW	Machine mode temporary register.	x

4.2.1.9 Machine Abnormal Program Counter Register (MEPC)

CSR address: 0x341

Bit	Name	Access	Description	Reset value
[31:0]	mepc[31:0]	MRW	Save the PC address recovered when exiting the interrupt/exception, and mepc[0] is fixed to 0.	x

4.2.1.10 Machine Exception Register (MCAUSE)

CSR address: 0x342

Bit	Name	Access	Description	Reset value
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31	Interrupt	RW	Interrupt/Exception Flag: 1: Exception/interrupt generated externally; 0: Exception/interrupt generated internally by the core.	x
[30:8]	Reserved	RO	Reserved	0
[7:0]	Exception Code[7:0]	RW	Save the source information that caused the exception or interrupt.	x

Interrupt exception source code:

Interrupt/Exception	Exception Code	Interrupt/exception source
1	2	Keep NMI
1	3	Machine mode software interrupt
1	4	User mode timer interrupt
1	5	Monitor mode timer interrupt
1	6	Reserved
1	7	Machine mode timer interrupt
1	8	User mode external interrupt
1	11	Machine mode external interrupt
1	12	System timer interrupt
1	≥16	Reserved platform interrupt id
0	0	Instruction address misalignment
0	1	Instruction access error
0	2	Illegal instruction
0	3	Breakpoint
0	4	Read address misalignment
0	5	Read access error
0	6	Write/atomic operation address misalignment
0	7	Write/atomic operation access error
0	8	User mode ecall
0	11	Machine mode ecall
0	≥16	Reserved

4.2.1.11 Machine Self-trapping Value Register (MTVAL)

CSR address: 0x343

Bit	Name	Access	Description	Reset value
[31:0]	mtval[31:0]	RW	Save additional information about self-trapping.	x

Additional information saved by mtval register in trap:

Abnormal type	Mcause	Save information
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Instruction address misalignment	0	Instruction PC
Instruction access failed	1	Instruction PC
Illegal instruction	2	Instruction machine code
Read address misalignment	4	Read data address
Read access failed	5	Read data address
Write/atomic access failed	6	Write data address
Write/atomic address misalignment	7	Write data address

4.2.1.12 Trigger Information Register (TINFO)

CSR address: 0x7A4

Bit	Name	Access	Description	Reset value
[31:0]	tinfo[31:0]	MRO	Trigger information register, corresponding to the type value in the TDATA1 register. The type value is 2, and the second bit of tinfo is set to 1.	0x4

4.2.1.13 Debug PC Address Register (DPC)

CSR address: 0x7B1

Bit	Name	Access	Description	Reset value
[31:0]	dpc[31:0]	RW	Save the PC address recovered when exiting debug mode.	x

The addresses saved by dpc for jumping to debugging mode caused by different debugging request sources are as follows:

Cause	Save address
ebreak	The instruction address of Ebreak.
One step	The address of the next instruction to be executed when saved in non-debug mode (V5F only).
Trigger module	tdata1.timing=0: Instruction address that triggers the breakpoint (V5F only).
Pause request	The address of the next instruction to be executed when entering debug mode.

4.2.1.14 Debug Mode Temporary Register 0 (DSCRATCH0)

CSR address: 0x7B2

Bit	Name	Access	Description	Reset value
[31:0]	Dscratch0[31:0]	RW	Debug mode temporary register 0.	x

4.2.1.15 Debug Mode Temporary Register 1 (DSCRATCH1)

CSR address: 0x7B3

Bit	Name	Access	Description	Reset value
[31:0]	Dscratch0[31:0]	RW	Debug mode temporary register 1.	x

4.2.1.16 Debug Register 0 (DBGMCU_0)

CSR address: 0x7C0

Bit	Name	Access	Description	Reset value
[31:8]	dbg_mode_stop	MRW	Registered modules cease operation when entering debug mode: 1: Registered peripherals cease operation when the core enters debug mode; 0: Registered peripherals continue normal operation when the core enters debug mode.	0
[7:0]	Reserved	MRO	Reserved	0

4.2.1.17 Floating-point Cumulative Exception Flag Register (FFLAGS)

CSR address: 0x001

Bit	Name	Access	Description	Reset value																		
[31:5]	Reserved	RO	Reserved	0																		
[4:0]	FFlags[4:0]	RW	Cumulative exception flag, used to indicate exceptions occurring on floating-point instructions since the last reset (hard reset or software clear). <table><tr><th>name</th><th>bit</th><th>description</th></tr><tr><td>NV</td><td>4</td><td>Illegal operation</td></tr><tr><td>DZ</td><td>3</td><td>Divider is zero</td></tr><tr><td>OF</td><td>2</td><td>Overflow upward</td></tr><tr><td>UF</td><td>1</td><td>Overflow downward</td></tr><tr><td>NX</td><td>0</td><td>Inaccuracy</td></tr></table>	name	bit	description	NV	4	Illegal operation	DZ	3	Divider is zero	OF	2	Overflow upward	UF	1	Overflow downward	NX	0	Inaccuracy	0
name	bit	description																				
NV	4	Illegal operation																				
DZ	3	Divider is zero																				
OF	2	Overflow upward																				
UF	1	Overflow downward																				
NX	0	Inaccuracy																				

4.2.1.18 Floating-point Rounding Mode Register (FRM)

CSR address: 0x002

Bit	Name	Access	Description	Reset value
[31:3]	Reserved	RO	Reserved	0
[2:0]	Frm[2:0]	RW	Floating-point operation rounding mode register: 000: Round to nearest value (round toward even numbers when intermediate); 001: Round toward zero; 010: Round down; 011: Round up; 100: Round to nearest value (round toward the largest magnitude when intermediate); 101, 110: Preserve; 111: Dynamic rounding, selects rounding mode based on the FRM bit configuration	0

			in the instruction.	
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4.2.1.19 Floating-point Control and Status Register (FCSR)

CSR address: 0x003

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:5]	Frm[2:0]	RW	Floating-point operation rounding mode register.	0
[4:0]	FFlags[4:0]	RW	Cumulative anomaly flag.	0

4.2.1.20 User Access Machine Status Register (UACCES_MSTATUS)

CSR address: 0x800

Bit	Name	Access	Description	Reset value
31	SD	RO	Used to mark whether a floating-point instruction is in a dirty state.	0
[30:15]	Reserved	RO	Reserved	0
[14:13]	FS[1:0]	RO	Floating-point instruction set status, when the value is not 0, single-precision floating-point instruction (f) is allowed to execute: 00: OFF; 01: Initial; 10: Clean; 11: Dirty.	0
[12:11]	MPP[1:0]	RO	Preserve/restore the machine state prior to entering an abnormal condition; upon exiting the abnormal state, update to the pre-abnormal machine state: 00: Set machine state to U mode upon exiting abnormal state; 01: Reserved 10: Reserved 11: Set machine state to M mode upon exiting abnormal state.	0
[10:8]	Reserved	RO	Reserved	0
7	MPIE	RW	Preserve the interrupt enable status upon entering an interrupt: 1: Enable global interrupts after exiting the interrupt; 0: Disable global interrupts after exiting the interrupt.	0
[6:4]	Reserved	RO	Reserved	0
3	MIE	RW	Global interrupts enabled; updates to MPIE value upon interrupt exit:	0

			1: Global interrupts enabled, allowing the processor to respond to interrupt requests; 0: Global interrupts disabled, preventing the processor from responding to interrupt requests.	
[2:0]	Reserved	RO	Reserved	0

Note: This register is used by the core to configure the MSTATUS register when running in user mode. This address becomes accessible when the non_usta bit in the CPU_RUN_CTLR register is enabled. The register settings are identical to those of the MSTATUS (0x300) register.

4.2.1.21 Hardware Stack Control Register (HW_POPDM_CTLR)

CSR address: 0x804

Bit	Name	Access	Description	Reset value
31	lock	RW	User Mode Lock Flag When locked, user mode cannot rewrite this register group configuration: 1: Locked; configurable only in machine mode. 0: Unlocked; configurable in machine/user mode.	0
[30:6]	Reserved	RO	Reserved	0
5	hw_pop_off	RW	Disable hardware push upon exit from interrupt, automatically reset: 1: Disable hardware push upon next interrupt exit; 0: Do not disable hardware push.	0
4	Reserved	RO	Reserved	0
[3:2]	preempt[1:0]	RW	The preemption priority bit width configuration register is used for configuring the preemption priority bit width in the interrupt priority: 00: The preemption bit width is 0, and interrupts with any priority cannot be nested; 01: The preemption bit width is 1, that is, bit [7] of the interrupt priority register is the preemption priority; 10: The preemption bit width is 2, that is, [7:6] bits of the interrupt priority register are preemption priority; 11: The preemption bit width is 3, that is, [7:5] bits of the interrupt priority register are preemption priority.	0x1
1	nest_en	RW	Interrupt Nesting Enable Register:	0

			1: Enable interrupt nesting 0: Disable interrupt nesting	
0	hw_stk_en	RW	Hardware stack protection enable: 1: Use hardware stack protection when entering and exiting interrupts; 0: Do not use hardware stack protection when entering and exiting interrupts.	0

4.2.1.22 Trigger Channel Selection Register (TSELECT)

CSR address: 0x7A0

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	MRO	Reserved	0
[1:0]	mscratch[1:0]	MRW	Trigger channel selection register.	x

Note: (1) Both Core 1 and Core 2 implement four-channel triggers. The trigger channel is configured via the TSELECT register, where the lower two bits of TSELECT are active, while the remaining bits are permanently set to 0.

4.2.1.23 Trace Trigger Data Register (TDATA1)

CSR address: 0x7A1

Bit	Name	Access	Description	Reset value
[31:28]	type[3:0]	MRO	Trigger type definition.	0x2
27	dmode	MRO	Both machine mode and debug mode can modify the trigger-related registers.	0
[26:21]	maskmax	MRO	When machth=1, the maximum exponential power range of matching is allowed.	0x1F
[20:19]	Reserved	MRO	Reserved	0
18	timing	MRO	Triggered before the instruction/operation is executed.	0
[17:13]	Reserved	MRO	Reserved	0
12	action	MRW	Set the processing mode when triggering: 1: When triggered, enter the debugging mode, and dpc saves the instruction address that needs to be restored when exiting debugging; 0: When triggered, an exception is generated that leads the thread to M mode.	0
[11:8]	Reserved	MRO	Reserved	0
7	match	MRO	Matching policy configuration: Allow address trigger with comparison value equal to tdata2 value.	0
6	m	MRO	Trigger enable in M mode: 1: Enable; 0: Disable.	1

[5:4]	Reserved	MRO	Reserved	0
3	u	MRO	Trigger enable in U mode: 1: Enable; 0: Disable.	1
2	execute	MRW	Triggered by the virtual address of the executed instruction 1: The trigger compares the instruction address; 0: The trigger does not compare the instruction address.	0
1	store	MRW	Triggered based on the virtual address of stored data 1: Trigger compares the write data address; 0: Trigger does not compare the write data address.	0
0	load	MRW	Triggered based on the virtual address of the read data 1: Trigger compares the read data address; 0: Trigger does not compare the read data address	0

Note: The processor trigger configuration complies with the RISC-V debugging standard. The trigger configuration type is 2, matching based on the least significant bit of the accessed virtual address. The trigger operation executes before the matched instruction or data read/write.

4.2.1.24 Save Trigger Data Register (TDATA2)

CSR address: 0x7A2

Bit	Name	Access	Description	Reset value
[31:0]	tdata2[31:0]	MRW	Used to save trigger data (instruction PC or data address).	x

4.2.2 RISC-V3F Core Custom Register

Table 4-5 RISC-V3F core custom registers list

Name	CSR address	Description	Reset value
CPU_RUN_CTLR	0xBC0	Processor run control register	0x12370000
INEST_CTLR	0xBC1	Interrupt nested control register	0x00000000
MIE	0xBC8	Machine interrupt register	0x00000000
DCSR	0x7B0	Debug control and status register	0x40000000

4.2.2.1 Processor Run Control Register (CPU_RUN_CTLR)

CSR address: 0xBC0

Bit	Name	Access	Description	Reset value
[31:28]	fadd_clkdiv[3:0]	MRW	Floating-point plus instruction frequency division coefficient, the default value is 1, that is, the	0x1

			floating-point plus instruction frequency is divided by 2 of the main frequency.	
[27:24]	fmul_clkdiv[3:0]	MRW	The frequency division coefficient of floating-point multiplication instruction is 2 by default, that is, the frequency of floating-point multiplication instruction is 3 of the main frequency.	0x2
[23:20]	fmac_clkdiv[3:0]	MRW	The frequency division coefficient of floating-point multiply-add instruction is 3 by default, that is, the frequency of floating-point multiply-add instruction is 4 of the main frequency.	0x3
[19:16]	fdiv_clkdiv[3:0]	MRW	The frequency division coefficient of floating-point divide instruction is 7 by default, that is, the frequency of floating-point divide instruction is 8 of the main frequency.	0x7
7	nmi_ie	MRW	Non-maskable interrupt enable: 1: Triggers NMI interrupt when an exception causes nested overflow; 0: Does not trigger NMI interrupt when an exception causes nested overflow.	0
6	int_fence	MRW	fence interrupt enable: 1: Clear interrupt requests when executing fence instructions; 0: Do not clear interrupt requests when executing fence instructions.	0
5	non_usta	MRW	UACCES_MSTATUS register enable: 1: Bits 3 and 7 of CSR address 0x800 map to MIE and MPIE bits of the MSTATUS register, respectively; 0: CSR address 0x800 is a read-only register, returning the value of MSTATUS.	0
[4:2]	Reserved	MRO	Reserved	0
[1:0]	pipe_acc[1:0]	MRW	Fetching mode: 00: Prefetch disabled. Disables instruction prefetching to avoid invalid fetches, allowing a maximum of one valid instruction in the CPU pipeline. This mode consumes the least power but reduces performance by approximately 2 to 3 times. 01: Prefetch enabled. Instruction prefetching is enabled. The CPU continuously accesses the instruction memory until the number of pending instructions in the internal instruction buffer exceeds a certain threshold, or until the buffer is full, at which point instruction fetching pauses. This mode consumes high power but delivers strong	0

			performance. (CPU prediction failures may introduce redundant fetch operations; in some cases, the execution unit may incur an additional 0–2 cycles of bubble, though performance degradation is generally negligible for most programs.) Other: Reserved.	
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Core	fadd_freq (max)	fmul_freq (max)	fmac_freq (max)	fdiv_freq (max)
V3F	80MHz	53MHz	40MHz	20MHz

Floating-point arithmetic instructions are multi-cycle instructions. For different types of floating-point instructions, the highest running frequency is shown in the table above. According to the operating frequency of the processor, the user needs to configure the floating-point operation frequency division coefficient so that the frequency after frequency division is not higher than the maximum operating frequency shown in the above table.

$$f_{xxx_freq(max)} \geq \text{Core main frequency} / f_{xxx_clkdiv}$$

4.2.2.2 Interrupt Nesting Control Register (INEST_CTLR)

CSR address: 0xBC1

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
30	nest_ovr	MRW	Interrupt/Exception nesting overflow flag: Set to 1 when nesting depth exceeds maximum depth; write 1 to clear. <i>Note: Interrupt overflow occurs only during instruction exceptions generated by level 2 interrupt service functions and NMI interrupts. In these cases, the exception and NMI interrupts proceed normally, but the CPU stack overflows, preventing exit from these exceptions and NMI interrupts.</i>	0
[29:12]	Reserved	MRO	Reserved	0
[11:8]	nest_sta[3:0]	MRO	Nested status flag: 0000: No interrupt; 0001: Level 1 interrupt; 0011: Level 2 interrupt nested; 0111: Level 3 interrupt nested, overflow; Others: Reserved.	0
[7:2]	Reserved	MRO	Reserved	0
[1:0]	nest_max[1:0]	MRW	Maximum nesting level: 00: Nesting disabled (disable nesting function); 01: 2-level nesting (enable nesting function); 10: Unwritable; 11: Unwritable.	0

			<i>Note: (1) Writing 10b or 11b to this field sets the register to 01b;</i> <i>(2) When 11b is written, reading this register returns the chip's maximum nesting level.</i>	
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4.2.2.3 Machine Interrupt Register (MIE)

CSR address: 0xBC8

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
[4:0]	nest_mie[4:0]	RW	Machine interrupt register, storing the MIE information for each level of nested interrupts: status.mie=nest_mie[0]; status.mpie=nest_mie[1].	0

4.2.2.4 Debug Control and Status Register (DCSR)

CSR address: 0x7B0

Bit	Name	Access	Description	Reset value
[31:28]	exdebug[3:0]	MRO	There is an external debugger that conforms to the debugger protocol.	0x4
[27:16]	Reserved	MRO	Reserved	0
15	ebreakm	MRW	Enable debug mode via ebreak instruction in machine mode: 1: Execute ebreak instruction in machine mode to enter debug mode; 0: Execute ebreak instruction in machine mode to trigger an exception.	0
[14:13]	Reserved	MRO	Reserved	0
12	ebreaku	MRW	Enable debug mode via ebreak instruction in user mode: 1: Execute ebreak instruction in user mode to enter debug mode; 0: Execute ebreak instruction in user mode to trigger an exception.	0
[11:10]	Reserved	MRO	Reserved	0
9	stoptime	MRW	Timer pause enable: 1: Counter stops counting in debug mode; 0: Counter operates normally in debug mode.	0
[8:6]	dcause[2:0]	MRW	Debug source register, used to mark the specific event triggering entry into debug mode 000: Reset value; 001: ebreak instruction in program triggers debugging (priority 3);	0

			010: Trigger module causes a breakpoint exception (priority 4) (V5F only); 011: Debugger generates a pause request (priority 1); 100: Pause in single-step mode (priority 0, lowest) (V5F only); 101: Not implemented; 110: Not implemented; 111: Reserved.	
[5:4]	Reserved	MRO	Reserved	0
3	nmip	MRO	Core NMI interrupt pending flag.	0
2	Reserved	MRO	Reserved	0
[1:0]	prv[1:0]	MRW	Record the machine state before entering debug mode: 00: Conversion from user mode to debug mode; upon recovery, the processor returns to user mode. 01, 10: Reserved. 11: Conversion from machine mode to debug mode; upon recovery, the processor returns to machine mode.	0

4.2.3 RISC-V5F Core Custom Register

Table 4-6 RISC-V5F core custom registers list

Name	CSR address	Description	Reset value
MCOUNT_INHIBIT	0x320	Counter mask register	0x00000000
DCSR	0x7B0	Debug control and status register	0x40000000
MCYCLE	0xB00	Machine cycle counter register	0x00000000
MINSTRET	0xB02	Machine instruction counter register	0x00000000
UCYCLE	0xC00	User mode cycle counter register	0x00000000
UINSTRET	0xC02	User mode instruction counter register	0x00000000
CPU_RUN_CTLR	0xBC0	Processor run control register	0x12370300
INEST_CTLR	0xBC1	Interrupt nested control register	0x00000000
CACHE_STRTG_CTLR	0xBC2	Cache policy control register	0x0F000003
CACHE_PMP_OVR	0xBC3	Cache policy PMP override register	0x00000000
HW_POPDM_ADDR	0xBC4	Hardware stack address register	0x20060000
MEMORY_CFGR	0xBC5	Memory configuration register	0x600F0FFF
TCM_RRDUTY_CFGR	0xBC6	TCM priority access duration configuration register	0x00000000
MIE	0xBC8	Machine interrupt register	0x00000000
OPCACHE_CTLR	0xBD0	Cache operation register	0x00000000
MEMINFO	0xFC0	Memory information register	0x12220030

4.2.3.1 Machine Mask Register (MCOUNT_INHIBIT)

CSR address: 0x320

Bit	Name	Access	Description	Reset value
[31:3]	Reserved	MRO	Reserved	0
2	IR	MRW	Instruction counter mask register: 1: Masks the instruction counter; minstret and unstret cease counting. 0: Does not mask the instruction counter; minstret and unstret count normally.	0
1	Reserved	MRO	Reserved	0
0	CY	MRW	Period counter mask register: 1: Mask instruction counter, mcycle and ucycle stop counting; 0: Without masking the period counter, mcycle and ucycle count normally.	0

4.2.3.2 Debug Control and Status Register (DCSR)

CSR address: 0x7B0

Bit	Name	Access	Description	Reset value
[31:28]	exdebug[3:0]	MRO	There is an external debugger that conforms to the debugger protocol.	0x4
[27:16]	Reserved	MRO	Reserved	0
15	ebreakm	MRW	Enable debug mode via ebreak instruction in machine mode: 1: Execute ebreak instruction in machine mode to enter debug mode; 0: Execute ebreak instruction in machine mode to trigger an exception.	0
[14:13]	Reserved	MRO	Reserved	0
12	ebreaku	MRW	Enable debug mode via ebreak instruction in user mode: 1: Execute ebreak instruction in user mode to enter debug mode; 0: Execute ebreak instruction in user mode to trigger an exception.	0
11	stepie	MRW	Single-step mode interrupt enable: 1: Enable interrupt response in single-step mode; 0: Disable interrupt response in single-step mode.	0
10	Reserved	MRO	Reserved	0
9	stoptime	MRW	Timer pause enable: 1: Counter stops counting in debug mode; 0: Counter operates normally in debug mode.	0

[8:6]	dcause[2:0]	MRW	Debug source register, used to mark the specific event triggering entry into debug mode 000: Reset value; 001: ebreak instruction in program triggers debugging (priority 3); 010: Trigger module causes a breakpoint exception (priority 4) (V5F only); 011: Debugger generates a pause request (priority 1); 100: Pause in single-step mode (priority 0, lowest) (V5F only); 101: Not implemented; 110: Not implemented; 111: Reserved.	0
[5:4]	Reserved	MRO	Reserved	0
3	nmip	MRO	Core NMI interrupt pending flag.	0
2	step	MRW	Single-step mode: 1: Single-step mode; 0: Non-single-step mode.	0
[1:0]	prv[1:0]	MRW	Record the machine state prior to entering debug mode: 00: Conversion from user mode to debug mode; upon recovery, the processor returns to user mode. 01, 10: Reserved. 11: Conversion from machine mode to debug mode; upon recovery, the processor returns to machine mode.	0

4.2.3.3 Machine Cycle Counter Register (MCYCLE)

CSR address: 0xB00

Bit	Name	Access	Description	Reset value
[31:0]	mcyle[31:0]	MRW	Machine cycle counter, used to count the clock cycles of machine operation. Counting occurs when the CY bit in the MCOUNT_INHIBIT register is set to 0. Counting is disabled when the counter is stopped in debug mode.	0

4.2.3.4 Machine Instruction Counter Register (MINSTRET)

CSR address: 0xB02

Bit	Name	Access	Description	Reset value
[31:0]	minstret[31:0]	MRW	Instruction count register, used to count the number of instructions correctly executed and	0

			completed by the machine. Counting occurs when the IR bit in the MCOUNT_INHIBIT register is set to 0. Counting is disabled when the counter is stopped in debug mode.	
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4.2.3.5 User Mode Cycle Counter Register (UCYCLE)

CSR address: 0xC00

Bit	Name	Access	Description	Reset value
[31:0]	ucyle[31:0]	RO	User mode cycle counter, used to count the clock cycles during machine operation. Counting occurs when the CY bit in the MCOUNT_INHIBIT register is set to 0. Counting is disabled when the counter is stopped in debug mode.	0

Note: the user mode cycle counter has the same function as the mcycle register, and is used to read the number of running cycles of the machine in user mode.

4.2.3.6 User Mode Instruction Counter Register (UINSTRET)

CSR address: 0xC02

Bit	Name	Access	Description	Reset value
[31:0]	uinstret[31:0]	RO	Instruction count register, used to count the number of instructions correctly executed and completed by the machine. Counting occurs when the IR bit in the MCOUNT_INHIBIT register is set to 0. Counting is disabled when the counter is stopped in debug mode.	0

Note: The user mode instruction counter, which has the same function as the minstret register, is used to read the number of instructions correctly executed by the machine in user mode.

4.2.3.7 Processor Run Control Register (CPU_RUN_CTLR)

CSR address: 0xBC0

Bit	Name	Access	Description	Reset value
[31:28]	fadd_clkdiv[3:0]	MRW	Floating-point plus instruction frequency division coefficient, the default value is 1, that is, the floating-point plus instruction frequency is divided by 2 of the main frequency.	0x1
[27:24]	fmul_clkdiv[3:0]	MRW	Floating-point plus instruction frequency division coefficient, the default value is 2, that is, the floating-point plus instruction frequency is divided by 3 of the main frequency.	0x2
[23:20]	fmac_clkdiv[3:0]	MRW	Floating-point plus instruction frequency division coefficient, the default value is 3,	0x3

			that is, the floating-point plus instruction frequency is divided by 4 of the main frequency.	
[19:16]	fdiv_clkdiv[3:0]	MRW	Floating-point plus instruction frequency division coefficient, the default value is 7, that is, the floating-point plus instruction frequency is divided by 8 of the main frequency.	0x7
15	nlp_en	MRW	Next branch prediction function enabled: 1: Enabled; 0: Disabled.	0
14	ghr_en	MRW	Global history register function enable: 1: Branch global history register enabled; 0: Branch global history register disabled.	0
[13:11]	Reserved	MRO	Reserved	0
10	lsu_dual	MRW	Parallel transmission of memory access instructions is enabled. When set to 1, it allows the transmitting unit to transmit memory access instructions in parallel, which improves the transmission speed of memory access instructions.	0
[9:8]	Reserved	MRO	Reserved	0
7	nmi_ie	MRW	Non-maskable interrupt enable: 1: Triggers NMI interrupt when an exception causes nested overflow; 0: Does not trigger NMI interrupt when an exception causes nested overflow.	0
6	int_fence	MRW	fence interrupt enable: 1: Clear interrupt requests when executing fence instructions; 0: Do not clear interrupt requests when executing fence instructions.	0
5	non_usta	MRW	UACCES_MSTATUS register enable: 1: Bits 3 and 7 of CSR address 0x800 map to MIE and MPIE bits of the MSTATUS register, respectively; 0: CSR address 0x800 is a read-only register, returning the value of MSTATUS.	0
[4:0]	Reserved	MRO	Reserved	0

Core	fadd_freq (max)	fmul_freq (max)	fmac_freq (max)	fdiv_freq (max)
V5F	128MHz	96MHz	76MHz	38MHz

Floating-point arithmetic instructions are multi-cycle instructions. For different types of floating-point instructions,

the highest running frequency is shown in the table above. According to the operating frequency of the processor, the user needs to configure the floating-point operation frequency division coefficient so that the frequency after frequency division is not higher than the maximum operating frequency shown in the above table.

$$fxxx_freq(max) \geq \text{Core main frequency} / fxxx_clkdiv$$

4.2.3.8 Interrupt Nested Control Register (INEST_CTLR)

CSR address: 0xBC1

Bit	Name	Access	Description	Reset value
31	lsu_nmi_sta	MRW	The memory access unit generates an unmasked interrupt flag (usually a write bus return error), and writes a 1 to reset.	0
30	nest_ovr_sta	MRW	The interrupt/abnormal nesting overflow flag bit is set to 1 when the nesting depth exceeds the maximum depth, and cleared by writing 1. Note: Interrupt overflow will only occur when executing the maximum interrupt level service function allowed by the core to generate instruction exception and NMI interrupt. At this time, the exception and NMI interrupt enter normally, but the CPU stack overflows, so you cannot exit from this exception and NMI interrupt.	0
[29:16]	Reserved	MRO	Reserved	0
[15:8]	nest_sta	MRO	Nesting status flag bit: [0x00]: Non-interrupt. [0x01]: 1-level interrupt; [0x03]: 2-level interrupt nesting; [0x07]: 3-level interrupt nesting; [0x0F]: 4-level interrupt nesting; [0x1F]: 5-level interrupt nesting; [0x3F]: 6-level interrupt nesting; [0x7F]: 7-level interrupt nesting; [0xFF]: 8-level interrupt nesting; Others: Reserved.	0
[7:3]	Reserved	MRO	Reserved	0
[2:0]	nest_max	MRW	Maximum nesting level: 000: Nesting is prohibited (nesting function is turned off); 001: Allow 2-level nesting; 010: Allow 3-level nesting; 011: Allow 4-level nesting; 100: Allow 5-level nesting; 101: Allow 6-level nesting; 110: Allow 7-level nesting; 111: Allow 8-level nesting;	0

4.2.3.9 Cache Policy Control Register (CACHE_STRTG_CTLR)

CSR address: 0xBC2

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
27	ic_mem1_strtg	MRW	Instruction Cache Enable for the 0x80000000–0x9fffffff Range: 1: Enables caching of instructions within this range; 0: Disables caching of instructions within this range.	1
26	ic_mem0_strtg	MRW	Enable the instruction cache for the 0x60000000–0x7fffffff address range.	1
25	ic_sram_strtg	MRW	Enable the instruction cache for the 0x20000000–0x3fffffff address range.	1
24	ic_code_strtg	MRW	Enable the instruction cache for the 0x00000000–0x1fffffff address range.	1
[23:2]	Reserved	RO	Reserved	0
1	ic_disable	MRW	Instruction cache disable flag bit, which is 0 to turn on the instruction cache function.	1
0	Reserved	RO	Reserved	1

4.2.3.10 Cache Policy PMP Coverage Registration (CACHE_PMP_OVR)

CSR address: 0xBC3

Bit	Name	Access	Description	Reset value
[31:13]	Reserved	MRO	Reserved	0
12	ic_pmp3cache_strtg	MRW	Instruction cache policy register in PMP channel 3 address area.	0
[11:9]	Reserved	MRO	Reserved	0
8	ic_pmp2cache_strtg	MRW	Instruction cache policy register in PMP channel 2 address area.	0
[7:5]	Reserved	MRO	Reserved	0
4	ic_pmp1cache_strtg	MRW	Instruction cache policy register in PMP channel 1 address area.	0
[3:1]	Reserved	MRO	Reserved	0
0	ic_pmp0cache_strtg	MRW	Instruction cache policy register for PMP channel 0 address range: 1: When the instruction address is protected by PMP Channel 0, the current instruction may be cached. 0: When the instruction address is protected by PMP Channel 0, the current instruction may not be cached.	0

Note: The cache policy PMP covers the register. When the instruction or data address matches the address

controlled by the PMP channel, the policy in the cache policy control register (0xBC2) is not executed, but the policy matched with the PMP channel in this register is executed.

4.2.3.11 Hardware Stack Address Register (HW_POPDM_ADDR)

CSR address: 0xBC4

Bit	Name	Access	Description	Reset value
[31:20]	hw_stk_vector	MRO	Fixed header address of DTCM area.	0x200
[19:9]		MRW	DTCM area block address, 512bit alignment.	0x600
[8:0]		MRO	Fixed to 0.	0

Note: The hardware stack pressing address register points to the DTCM area. When the hardware stack pressing function is enabled and an interrupt/exception is entered, the stack value is quickly saved to the designated DTCM area, usually pointing to the bottom of the DTCM area.

4.2.3.12 Memory Configuration Register (MEMORY_CFGR)

CSR address: 0xBC5

Bit	Name	Access	Description	Reset value
[31:20]	MEM0_addr_hi[11:0]	MRW	The branch predictor address matching register is configured with the upper 12 bits of external memory address information. When the upper bits of the external memory address do not match, some branch prediction functions are restricted. <i>Note: Do not configure with upper bits of internal memory addresses (0x000, 0x200, 0x201).</i>	0x600
[19:18]	dtcm_rr_mode[1:0]	MRW	DTCM Memory Access Priority Mode: 00: Fixed priority, other requests (DMA, C0_lsu, C0_ifu) take precedence over C1 requests; 01: Fixed priority, C1 requests take precedence over other requests (DMA, C0_lsu, C0_ifu); 10: Polling priority: one channel continuously obtains the operation priority for a period of time at most, and then switches to another channel after timeout. When the prioritized channel has no operation request, it switches to another channel and re-counts. See TCM_RRDUTY_CFGR register for the configuration of priority duration. 11: Forced polling priority: one channel is forced to obtain the operation priority for a period of time, and then it is switched to another channel after timeout. See TCM_RRDUTY_CFGR register for the configuration of priority duration.	0x3
[17:16]	itcm_rr_mode[1:0]	MRW	ITCM memory access priority mode: 00: Fixed priority, other requests (DMA, C0_lsu, C0_ifu) are higher than C1 requests. 01: Fixed priority, C1 request is higher than other	0x3

			requests (DMA, C0_lsu, C0_ifu). 10: Polling priority: one channel continuously obtains the operation priority for a period of time at most, and then switches to another channel after timeout. When the prioritized channel has no operation request, it switches to another channel and re-counts. See TCM_RRDUTY_CFGR register for the configuration of priority duration. 11: Forced polling priority: one channel is forced to obtain the operation priority for a period of time, and then it is switched to another channel after timeout. See TCM_RRDUTY_CFGR register for the configuration of priority duration.	
[15:12]	Reserved	RO	Reserved	0
[11:10]	C0_ifu_dtcn_mode	MRW	C0_IFU DTCM memory access permissions configuration: C0_ifu_dtcn_mode[1]: 1: Read enabled; 0: Read disabled. C0_ifu_dtcn_mode[0]: 1: Write enabled; 0: Write disabled.	0x3
[9:8]	C0_ifu_itcm_mode	MRW	C0_IFU ITCM memory access permissions configuration: C0_ifu_itcm_mode[1]: 1: Read enabled; 0: Read disabled. C0_ifu_itcm_mode[0]: 1: Write enabled; 0: Write disabled.	0x3
[7:6]	C0_lsu_dtcn_mode	MRW	C0_LSU DTCM memory access permissions configuration: C0_lsu_dtcn_mode[1]: 1: Read enabled; 0: Read disabled. C0_lsu_dtcn_mode[0]: 1: Write enabled; 0: Write disabled.	0x3
[5:4]	C0_lsu_itcm_mode	MRW	C0_LSU ITCM memory access permissions configuration: C0_lsu_itcm_mode[1]: 1: Read enabled; 0: Read disabled. C0_lsu_itcm_mode[0]: 1: Write enabled;	0x3

			0: Write disabled.	
[3:2]	DMA_dtcn_mode	MRW	DMA DTCM memory access permissions configuration: DMA_dtcn_mode[1]: 1: Read enabled; 0: Read disabled. DMA_dtcn_mode[0]: 1: Write enabled; 0: Write disabled.	0x3
[1:0]	DMA_itcm_mode	MRW	DMA ITCM memory access permissions configuration: DMA_itcm_mode[1]: 1: Read enabled; 0: Read disabled. DMA_itcm_mode[0]: 1: Write enabled; 0: Write disabled.	0x3

4.2.3.13 TCM Priority Access Duration Configuration Register (TCM_RRDUTY_CFGR)

CSR address: 0xBC6

Bit	Name	Access	Description	Reset value
[31:24]	C1_dtcn_rr_cycle [7:0]	MRW	High-priority continuous clock cycle configuration from C1 for DTCM requests.	0
[23:16]	Cx_dtcn_rr_cycle [7:0]	MRW	Persistent clock cycle configuration for high-priority DTCM requests originating from non-C1 (C0, DMA) pairs.	0
[15:8]	C1_itcm_rr_cycle [7:0]	MRW	High-priority continuous clock cycle configuration request from C1 to ITCM.	0
[7:0]	Cx_itcm_rr_cycle [7:0]	MRW	Persistent clock cycle configuration for high-priority ITCM requests originating from non-C1 (C0, DMA) sources.	0

Note: The TCM region access priority duration configuration takes effect when itcm_rr_mode and dtcm_rr_mode in TCM_CFGR are set to 10b or 11b, controlling the duration of the corresponding channel's high priority.

4.2.3.14 Machine Interrupt Register (MIE)

CSR address: 0xBC8

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
[8:0]	nest_mie	RW	Machine interrupt register, storing the mie information for each level of nested interrupts: MSTATUS register mie bit = nest_mie[0]; MSTATUS register mpie bit = nest_mie[1].	0

4.2.3.15 Cache Operation Register (OPCACHE_CTLR)

CSR address: 0xBD0

Bit	Name	Access	Description	Reset value
[31:5]	Vaddr[27:0]	WO	Operation address or index information.	0
[4:3]	Reserved	RO	Reserved	0
2	Idx_mode	WO	Indexing Mode 1: Execute operations using the vaddr value as the address information; 0: Execute operations using the index information within the vaddr value as the starting address.	0
[1:0]	Opcode[1:0]	WO	00: Icache invalidate; Other: invalid.	0

4.2.3.16 Memory Information register (MEMINFO)

CSR address: 0xFC0

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:26]	DTCM datasize[3:0]	MRO	Data memory capacity information: $64KB * 2^{DTCM\ datasize} = 256KB$	0100b
[25:24]	DTCM LineSize[1:0]	MRO	Data memory line length information: 00: 8-byte; 01: 16-byte; 10: 32-byte; 11: 64-byte.	10b
[23:22]	Reserved	RO	Reserved	0
[21:18]	ITCM datasize[3:0]	MRO	Instruction memory capacity information: $64KB * 2^{ITCM\ datasize} = 128KB$	0010b
[17:16]	ITCM LineSize[1:0]	MRO	Instruction memory line length information: 00: 8-byte; 01: 16-byte; 10: 32-byte; 11: 64-byte.	10b
[15:7]	Reserved	RO	Reserved	0
[6:5]	Icache_way[1:0]	MRO	Number of instruction cache ways: 00: 1-way; 01: 2-way; 10: 4-way; 11: 8-way.	01b
[4:2]	Icache_datasize [2:0]	MRO	Instruction cache capacity information: 000: NONE; 001: 4KB; 010: 8KB; 011: 16KB; 100: 32KB;	100b

			101: 64KB; 110: 128KB; 111: 256KB.	
[1:0]	Icache linesize [1:0]	MRO	Instruction cache line length information: 00: 8-byte; 01: 16-byte; 10: 32-byte; 11: 64-byte.	0

4.3 Physical Memory Protection (PMP)

In order to improve the security of the system, a set of physical address access restrictions is defined in the architecture of RISC-V, which can set its read, write and execute attributes for the physical memory in the area, and the minimum length of the area is 4 bytes. PMP unit always takes effect in user mode, and it can take effect optionally in machine mode. If it violates the current memory limit, it will generate system abnormal interrupt (EXC).

The PMP unit of Core 1 (RISC-V5F) contains four sets of 8-bit configuration registers (CSR address 0x3A0) and four sets of address registers (CSR addresses 0x3B0–0x3B3). These require access via the "csr" instruction and must be performed in machine mode.

The PMP unit of Core 0 (RISC-V3F) contains four sets of 8-bit configuration registers (CSR address 0x3A0) and four sets of address registers (CSR addresses 0x3B0–0x3B3). Access requires the "csr" instruction and must be performed in machine mode.

4.3.1 PMP Register

Table 4-7 PMP registers list

Name	CSR address	Description	Reset value
PMPCFG0	0x3A0	PMP configuration register	0x00000000
PMPADDR0	0x3B0	PMP address 0 register	0x00000000
PMPADDR1	0x3B1	PMP address 1 register	0x00000000
PMPADDR2	0x3B2	PMP address 2 register	0x00000000
PMPADDR3	0x3B3	PMP address 3 register	0x00000000

4.3.1.1 PMP Configuration Register (PMPCFG0)

CSR address: 0x3A0

Bit	Name	Access	Description	Reset value
[31:24]	pmp3cfg[7:0]	MRW	See pmp0cfg.	0
[23:16]	pmp2cfg[7:0]	MRW	See pmp0cfg.	0
[15:8]	pmp1cfg[7:0]	MRW	See pmp0cfg.	0

[7:0]	pmp0cfg[7:0]	MRW	Bit	Name	Description	0
			7	L	Lock Enable, Unlockable in Machine Mode 0: Unlocked; 1: Locks the corresponding register.	
			6	-	Reserved	
			5	IC_Str	The instruction cache policy is executed during address matching, with the policy value corresponding to the CACHE_PMP_OVR register.	
			[4:3]	A	Address alignment and protection area range selection.	
			2	X	Executable attribute.	
			1	W	Writeable attribute.	
			0	R	Readable attribute.	

Specifically, for address alignment and protected region selection, memory protection is applied to the range where $A_ADDR \leq \text{region} < B_ADDR$ (requiring both A_ADDR and B_ADDR to be aligned to 4-byte boundaries):

1. If $B_ADDR - A_ADDR == 2^2$, it is based on NA4;
2. If $B_ADDR - A_ADDR == 2^{(G+2)}$, $G \geq 1$, and A_ADDR is aligned to $2^{(G+2)}$, then use the NAPOT method;
3. Otherwise it is based on TOR.

A value	Name	Description
00b	OFF	No region to protect.
01b	TOR	Top aligned region protection: region = $BUS_ADDR \gg 2$; For pmp0cfg, $0 \leq \text{region} < \text{pmpaddr0}$; For pmp1cfg, $\text{pmpaddr0} \leq \text{region} < \text{pmpaddr1}$; For pmp2cfg, $\text{pmpaddr1} \leq \text{region} < \text{pmpaddr2}$; For pmp3cfg, $\text{pmpaddr2} \leq \text{region} < \text{pmpaddr3}$. $\text{pmpaddr}_{i-1} = A_ADDR \gg 2$; $\text{pmpaddr}_i = B_ADDR \gg 2$.
10b	NA4	Fixed 4-byte region protection. $\text{pmp0cfg} \sim \text{pmp3cfg}$ to $\text{pmpaddr0} \sim \text{pmpaddr3}$ as start address. $\text{pmpaddr}_i = A_ADDR \gg 2$.
11b	NAPOT	$2^{(G+2)}$ region protection, $G \geq 1$, in this case, A_ADDR is aligned on $2^{(G+2)}$. $\text{pmpaddr}_i = ((A_ADDR \gg 2) (2^{(G-1)} - 1))$.

4.3.1.2 PMP Address 0 Register (PMPADDR0)

CSR address: 0x3B0

Bit	Name	Access	Description	Reset value
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[31:0]	ADDR0[31:0]	MRW	The [33:2] bits of the PMP set address0, actually the higher 2 bits are not used.	0
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4.3.1.3 PMP Address 1 Register (PMPADDR1)

CSR address: 0x3B1

Bit	Name	Access	Description	Reset value
[31:0]	ADDR1[31:0]	MRW	The [33:2] bits of the PMP set address1, actually the higher 2 bits are not used.	0

4.3.1.4 PMP Address 2 Register (PMPADDR2)

CSR address: 0x3B2

Bit	Name	Access	Description	Reset value
[31:0]	ADDR2[31:0]	MRW	The [33:2] bits of the PMP set address2, actually the higher 2 bits are not used.	0

4.3.1.5 PMP Address 3 Register (PMPADDR3)

CSR address: 0x3B3

Bit	Name	Access	Description	Reset value
[31:0]	ADDR1[31:0]	MRW	The [33:2] bits of the PMP set address3, actually the higher 2 bits are not used.	0

4.4 Inter-process Communication (IPC)

Inter-core communication functionality provides four communication channels, each with 8 status bits. Status bits are independently enabled, supporting 2 interrupt modes, and featuring 32bit*4 communication data registers.

- 2 types of channel interrupt modes: receive interrupt and transmit interrupt.
- Each channel features 8 status bits, with independent enable capability and automatic status bit updates.
- Receive interrupt and transmit interrupt are independently enabled, with configurable interrupt cores.
- 4 groups of inter-core information registers.

4.4.1 IPC Function Description

This module provides 2 types of interrupts:

- Receiving end interrupt
- Transmitting end interrupt

IPC channel interrupts are jointly controlled by the interrupt enable bits (RX_IERx/TX_IERx), status enable bits (IPC_ENAx[7:0]), and status bits (IPC_STSx[7:0]).

Receive interrupt

When the status bit value is 1, the corresponding status enable bit value is 1, and the corresponding bit value in the IPC_ISR register is 1. If the channel receive interrupt enable is active (RX_IERx = 1), an interrupt request can be generated to the RX_CID core configured for the channel, and the corresponding bit value in the IPC_ISM register is 1.

Transmit interrupt

When the status bit value is 0, the corresponding status enable bit value is 1, and the corresponding bit value in the IPC_ISR register is 1. If the channel transmit interrupt is enabled (TX_IERx = 1), an interrupt request can be generated to the TX_CID core configured for the channel, and the corresponding bit value in the IPC_ISM register is 1.

When the AUTOENx bit is configured for automatic updating, writing/setting the IPC_STS register at the transmit end simultaneously updates the IPC_ENA register to the same value. If the channel receive interrupt is enabled, a receive interrupt can be immediately generated at the receive end. Clearing the IPC_STS flag after completing the reception interrupt response does not affect the IPC_ENA register. If the channel transmission interrupt is active, a transmission interrupt can be immediately generated for the transmitter. The transmitter disables the IPC_ENA enable within the interrupt and clears the transmitter interrupt flag.

When the AUTOENx bit is configured for non-auto-update, the sequence of first writing the status enable bit followed by writing the IPC_STS may introduce additional transmitter interrupts.

4.4.2 IPC Register

Table 4-8 IPC registers list

Name	Access	Description	Reset value
R32_IPC_CTLR	0xE000D000	IPC control register	0x00000000
R32_IPC_ISR	0xE000D004	IPC interrupt status register	0x00000000
R32_IPC_ISM	0xE000D008	IPC ISM channel interrupt mask register	0x00000000
R32_IPC_ENA	0xE000D010	IPC status bit interrupt enable register	0x00000000
R32_IPC_STS	0xE000D014	IPC channel status register	0xFFFFFFFF
R32_IPC_SET	0xE000D018	IPC status flag bit setting register	0x00000000
R32_IPC_CLR	0xE000D01C	IPC status flag bit reset register	0x00000000
R32_IPC_MSG0	0xE000D020	IPC information register 0	0xFFFFFFFF
R32_IPC_MSG1	0xE000D024	IPC information register 1	0xFFFFFFFF
R32_IPC_MSG2	0xE000D028	IPC information register 2	0xFFFFFFFF
R32_IPC_MSG3	0xE000D02C	IPC information register 3	0xFFFFFFFF

4.4.2.1 IPC Control Register (IPC_CTLR)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
31	LOCK3	RW	Channel 3 configuration is locked. After writing a 1, only a reset can clear it. Once locked, other control registers for Channel 3 cannot be rewritten. 1: Channel 3 control is locked; 0: Channel 3 control is unlocked.	0
30	AUTOEN3	RW	Channel 3 Status Bit Enables Automatic Update. When set to 1, upon the sender writing IPC_STS, IPC_ENA automatically updates to the same value. When the sender writes IPC_SET to set	0

			IPC_STS, IPC_ENA sets the corresponding bit position. 1: IPC_ENA register auto-update enabled; 0: IPC_ENA register auto-update disabled.	
29	RX_IER3	RW	Channel 3 receive interrupt enable: When set to 1, Channel 3 can generate a receive interrupt: 1: Channel 3 receive interrupt enabled; 0: Channel 3 receive interrupt disabled.	0
28	TX_IER3	RW	Channel 3 transmit interrupt enable: When set to 1, Channel 3 can generate transmit interrupts. 1: Channel 3 transmit interrupt enabled; 0: Channel 3 transmit interrupt disabled.	0
27	Reserved	RO	Reserved	0
26	RX_CID3	RW	Receive-side core ID, configure the core to respond to receive interrupts: 1: Generate a receive interrupt request to core 1; 0: Generate a receive interrupt request to core 0.	0
25	Reserved	RO	Reserved	0
24	TX_CID3	RW	Transmit core ID, configure the core to respond to transmit interrupts: 1: Generate a transmit interrupt request to core 1; 0: Generate a transmit interrupt request to core 0.	0
23	LOCK2	RW	Channel 2 configuration lock.	0
22	AUTOEN2	RW	Channel 2 status bit enables automatic update.	0
21	RX_IER2	RW	Channel 2 receive interrupt is enabled. When set to 1, channel 2 can generate receive interrupt.	0
20	TX_IER2	RW	Channel 2 transmit interrupt is enabled. When set to 1, channel 2 can generate a transmit interrupt.	0
19	Reserved	RO	Reserved	0
18	RX_CID2	RW	The receiver core ID, which configures the core that responds to the receiving interrupt.	0
17	Reserved	RO	Reserved	0
16	TX_CID2	RW	The transmitter core ID, which configures the core that responds to the sending interrupt.	0
15	LOCK1	RW	Channel 1 configuration lock.	0
14	AUTOEN1	RW	Channel 1 status bit enables automatic update.	0
13	RX_IER1	RW	Channel 1 receive interrupt is enabled. When set to 1, channel 1 can generate receive interrupt.	0
12	TX_IER1	RW	Channel 1 transmit interrupt is enabled. When set to 1, channel 1 can generate a sending interrupt.	0
11	Reserved	RO	Reserved	0
10	RX_CID1	RW	The receiver core ID, which configures the core that responds to the receiving interrupt.	0
9	Reserved	RO	Reserved	0

8	TX_CID1	RW	The transmitter's core ID, which configures the core that responds to the sending interrupt.	0
7	LOCK0	RW	Channel 0 configuration lock.	0
6	AUTOEN0	RW	Channel 0 status bit enables automatic update.	0
5	RX_IER0	RW	Channel 1 receive interrupt is enabled. When set to 1, channel 1 can generate receive interrupt.	0
4	TX_IER0	RW	Channel 1 transmit interrupt is enabled. When set to 1, channel 1 can generate a sending interrupt.	0
3	Reserved	RO	Reserved	0
2	RX_CID0	RW	The receiver core ID, which configures the core that responds to the receiving interrupt.	0
1	Reserved	RO	Reserved	0
0	TX_CID0	RW	The transmitter core ID, which configures the core that responds to the transmitting interrupt.	0

4.4.2.2 IPC Interrupt Status Register (IPC_ISR)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[31:24]	IPC_ISR3[7:0]	RO	IPC channel 3 interrupt status register: A non-zero value indicates an active interrupt on IPC Channel 3. When IPC_ENA is enabled, a transmit interrupt occurs when the IPC_STS register value is 0; a receive interrupt occurs when the IPC_STS register value is 1.	0
[23:16]	IPC_ISR2[7:0]	RO	IPC channel 2 interrupt status register.	0
[15:8]	IPC_ISR1[7:0]	RO	IPC channel 1 interrupt status register.	0
[7:0]	IPC_ISR0[7:0]	RO	IPC channel 0 interrupt status register.	0

4.4.2.3 IPC ISM Channel Interrupt Mask Register (IPC_ISM)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[31:24]	IPC_ISM3[7:0]	RO	IPC channel 3 interrupt mask register: When the value is non-zero, IPC Channel 3 generates an interrupt request. When RX_IER is enabled, the bit with IPC_STS set to 1 and IPC_ENA set to 1 generates a receive interrupt to RX_CID. When TX_IER is enabled, the bit where IPC_STS is 0 and IPC_ENA is 1 generates a transmit interrupt to TX_CID.	0
[23:16]	IPC_ISM2[7:0]	RO	IPC channel 2 interrupt mask register.	0
[15:8]	IPC_ISM1[7:0]	RO	IPC channel 1 interrupt mask register.	0

[7:0]	IPC_ISM0[7:0]	RO	IPC channel 0 interrupt mask register.	0
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4.4.2.4 IPC Status Bit Interrupt Enable Register (IPC_ENA)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[31:24]	IPC_ENA3[7:0]	RW	IPC channel 3 flag bit interrupt enable register enables interrupt functionality for the corresponding bit when set to 1. When IPC_STS is 0 and IPC_ENA is 1, the TX_CID bit generates a transmit interrupt. When IPC_STS is 1 and IPC_ENA is 1, the RX_CID bit generates a receive interrupt. The TX_CID bit generates a transmit interrupt when IPC_STS is 0 and IPC_ENA is 1.	0
[23:16]	IPC_ENA2[7:0]	RW	IPC channel 2 flag bit interrupt enable register.	0
[15:8]	IPC_ENA1[7:0]	RW	IPC channel 1 flag bit interrupt enable register.	0
[7:0]	IPC_ENA0[7:0]	RW	IPC channel 0 flag bit interrupt enable register.	0

4.4.2.5 IPC Channel Status Register (IPC_STS)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[31:24]	IPC_STS3[7:0]	RW	IPC channel 3 status register: Bits with an IPC_STS value of 1 can generate receive interrupts; Bits with an IPC_STS value of 0 can generate transmit interrupts.	X
[23:16]	IPC_STS2[7:0]	RW	IPC channel 2 status register.	X
[15:8]	IPC_STS1[7:0]	RW	IPC channel 1 status register.	X
[7:0]	IPC_STS0[7:0]	RW	IPC channel 0 status register.	X

4.4.2.6 IPC Status Flag Bit Setting Register (IPC_SET)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[31:24]	IPC_SET3[7:0]	WO	IPC channel 3 status flag bit setting register, sets the corresponding bit in the IPC_STS register when written with a value of 1. If the channel's AUTOEN bit is enabled, the corresponding bit in the IPC_ENA register is also set simultaneously.	0
[23:16]	IPC_SET2[7:0]	WO	IPC channel 2 status flag bit setting register.	0
[15:8]	IPC_SET1[7:0]	WO	IPC channel 1 status flag bit setting register.	0
[7:0]	IPC_SET0[7:0]	WO	IPC channel 0 status flag bit setting register.	0

4.4.2.7 IPC Status Flag Bit Reset Register (IPC_CLR)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
[31:24]	IPC_CLR3[7:0]	WO	IPC channel 3 status flag bit clear register, write 1 to reset the corresponding bit in the IPC_STS register.	0
[23:16]	IPC_CLR2[7:0]	WO	IPC channel 2 status flag bit clear register, write 1 to reset the corresponding bit in the IPC_STS register.	0
[15:8]	IPC_CLR1[7:0]	WO	IPC channel 1 status flag bit clear register, write 1 to reset the corresponding bit in the IPC_STS register.	0
[7:0]	IPC_CLR0[7:0]	WO	IPC channel 0 status flag bit clear register, write 1 to reset the corresponding bit in the IPC_STS register.	0

4.4.2.8 IPC Information Register (IPC_MSGx) (x=0/1/2/3)

Offset address: 0x20+0x4*n

Bit	Name	Access	Description	Reset value
[31:0]	IPC_MSGx[31:0]	RW	IPC information register, used to store and transmit information required for inter-process communication.	X

4.5 Hardware Semaphore (HSEM)

The hardware semaphore function provides 32 hardware semaphore channels. The semaphore function can be used for synchronous operation between different processes running in multi-core and provides non-blocking single instruction locking mode. Functions include the following:

- An interrupt was generated when the semaphore channel was released.
- Semaphore single channel unlocking: to unlock the channel, the operating core should match the CID, and the write data should match the PID value.
- Lot unlocking of semaphore channels: lot unlocking by matching key values, CID and PID.
- 2 ways to lock semaphore channels:
 - 2-step locking: Acquires a lock by writing the PID to the HSEM channel, automatically updating the CID to the core ID of the acquiring process, followed by reading data for verification.
 - 1-step locking: Acquires a lock by reading the HSEM channel, automatically updating the CID to the core ID of the acquiring process.

4.5.1 HSEM Function Description

4.5.1.1 Locking Mode

This module provides 2 locking modes:

- 2-step (write) lock
- 1-step (read) lock

A value of 0 in the HSEM channel LOCK register indicates that the channel is idle, and the CID and PID values of this channel are also 0. When the channel LOCK register value is 1, it means that the channel is locked, the CID value is updated to the core ID of the locked channel, and the PID register value is the bus write value.

When writing a locked channel, the CID is updated to the core ID of the locked channel, and the PID value is

updated to write data. When reading the locked channel, the CID is updated to the core ID of the locked channel, and the PID value remains 0. Any PID value can be written when writing lock.

2-step (write) lock process

The 2-step lock process locks the channel by writing the lock information to the HSEM_RX register, and then checks whether the lock is successful by reading the same data as the write data.

- Write the PID to the HSEM_RX register with LOCK = 1. If the channel is idle, locking succeeds, and the PID, CID, and LOCK values are updated.
- Perform a read data check: the read PID matches the written PID, the CID matches the core ID, and a LOCK value of 1 indicates a successful operation.
- Otherwise, repeat the above steps.

The HSEM channel can only be locked when idle.

Attempting to lock an already locked channel will have no effect.

1-step (read) lock process

The 1-step lock process checks for successful locking by reading the HSEM_RLRX register and comparing the returned value.

- Read the HSEM_RLRX register to obtain the return value.
- If the read data is 0, the lock was successful; if non-zero, the lock failed, indicating the channel is already locked by another process. For successfully locked channels, the HSEM_RLRX register updates to LOCK = 1, CID = the core ID performing the operation, and PID = 0.
- Otherwise, repeat the above operation.
- Channel unlocking process:
- Write unlock:

Write the PID to the HSEM_RX register corresponding to the locked HSEM channel with LOCK=0. If this channel is locked by the current core, writing a PID equal to the channel's PID unlocks the channel, clearing PID, CID, and LOCK. When the channel interrupt is enabled, an interrupt request may be generated.

- Match unlock:

Write a key value to the HSEM_CLR register to configure matching regulars. Channels matching the regulars can be unlocked in lot. When the channel interrupt is enabled, an interrupt request can be generated.

4.5.1.2 HSEM Interrupt

When the HSEM channel is unlocked, an interrupt can be generated. The interrupt of HSEM has the following characteristics:

- Each HSEM channel can be enabled independently.
- Each HSEM channel has an independent interrupt flag.
- The interrupt flag of each HSEM channel can be cleared separately.
- Each core has independent interrupt enable, interrupt flag and clear bit.

When the channel interrupt is enabled (HSEM_IER), an interrupt request can be generated when the channel is unlocked by other cores, and the corresponding channel interrupt flag register (HSEM_ISR) is set to 1.

Write a 1 to the corresponding bit of the channel interrupt flag (HSEM_ISR) to clear the interrupt flag.

The channel interrupt mask register (HSEM_ISM) can be used to obtain the source channel that generates the

interrupt, and the channel with the channel interrupt enabled and the interrupt flag set to 1 can generate the interrupt request, and the corresponding HSEM_ISM bit value is 1.

4.5.2 HSEM Register

Table 4-9 HSEM registers list

Name	Access address	Description	Reset value
R32_HSEM_RX0	0xE000C000	HSEM lock channel register 0	0x00000000
R32_HSEM_RX1	0xE000C004	HSEM lock channel register 1	0x00000000
R32_HSEM_RX2	0xE000C008	HSEM lock channel register 2	0x00000000
R32_HSEM_RX3	0xE000C00C	HSEM lock channel register 3	0x00000000
R32_HSEM_RX4	0xE000C010	HSEM lock channel register 4	0x00000000
R32_HSEM_RX5	0xE000C014	HSEM lock channel register 5	0x00000000
R32_HSEM_RX6	0xE000C018	HSEM lock channel register 6	0x00000000
R32_HSEM_RX7	0xE000C01C	HSEM lock channel register 7	0x00000000
R32_HSEM_RX8	0xE000C020	HSEM lock channel register 8	0x00000000
R32_HSEM_RX9	0xE000C024	HSEM lock channel register 9	0x00000000
R32_HSEM_RX10	0xE000C028	HSEM lock channel register 10	0x00000000
R32_HSEM_RX11	0xE000C02C	HSEM lock channel register 11	0x00000000
R32_HSEM_RX12	0xE000C030	HSEM lock channel register 12	0x00000000
R32_HSEM_RX13	0xE000C034	HSEM lock channel register 13	0x00000000
R32_HSEM_RX14	0xE000C038	HSEM lock channel register 14	0x00000000
R32_HSEM_RX15	0xE000C03C	HSEM lock channel register 15	0x00000000
R32_HSEM_RX16	0xE000C040	HSEM lock channel register 16	0x00000000
R32_HSEM_RX17	0xE000C044	HSEM lock channel register 17	0x00000000
R32_HSEM_RX18	0xE000C048	HSEM lock channel register 18	0x00000000
R32_HSEM_RX19	0xE000C04C	HSEM lock channel register 19	0x00000000
R32_HSEM_RX20	0xE000C050	HSEM lock channel register 20	0x00000000
R32_HSEM_RX21	0xE000C054	HSEM lock channel register 21	0x00000000
R32_HSEM_RX22	0xE000C058	HSEM lock channel register 22	0x00000000
R32_HSEM_RX23	0xE000C05C	HSEM lock channel register 23	0x00000000
R32_HSEM_RX24	0xE000C060	HSEM lock channel register 24	0x00000000
R32_HSEM_RX25	0xE000C064	HSEM lock channel register 25	0x00000000
R32_HSEM_RX26	0xE000C068	HSEM lock channel register 26	0x00000000
R32_HSEM_RX27	0xE000C06C	HSEM lock channel register 27	0x00000000
R32_HSEM_RX28	0xE000C070	HSEM lock channel register 28	0x00000000
R32_HSEM_RX29	0xE000C074	HSEM lock channel register 29	0x00000000
R32_HSEM_RX30	0xE000C078	HSEM lock channel register 30	0x00000000
R32_HSEM_RX31	0xE000C07C	HSEM lock channel register 31	0x00000000
R32_HSEM_RLRX0	0xE000C100	HSEM read lock register 0	0x00000000
R32_HSEM_RLRX1	0xE000C104	HSEM read lock register 1	0x00000000
R32_HSEM_RLRX2	0xE000C108	HSEM read lock register 2	0x00000000
R32_HSEM_RLRX3	0xE000C10C	HSEM read lock register 3	0x00000000
R32_HSEM_RLRX4	0xE000C110	HSEM read lock register 4	0x00000000

R32_HSEM_RLRX5	0xE000C114	HSEM read lock register 5	0x00000000
R32_HSEM_RLRX6	0xE000C118	HSEM read lock register 6	0x00000000
R32_HSEM_RLRX7	0xE000C11C	HSEM read lock register 7	0x00000000
R32_HSEM_RLRX8	0xE000C120	HSEM read lock register 8	0x00000000
R32_HSEM_RLRX9	0xE000C124	HSEM read lock register 9	0x00000000
R32_HSEM_RLRX10	0xE000C128	HSEM read lock register 10	0x00000000
R32_HSEM_RLRX11	0xE000C12C	HSEM read lock register 11	0x00000000
R32_HSEM_RLRX12	0xE000C130	HSEM read lock register 12	0x00000000
R32_HSEM_RLRX13	0xE000C134	HSEM read lock register 13	0x00000000
R32_HSEM_RLRX14	0xE000C138	HSEM read lock register 14	0x00000000
R32_HSEM_RLRX15	0xE000C13C	HSEM read lock register 15	0x00000000
R32_HSEM_RLRX16	0xE000C140	HSEM read lock register 16	0x00000000
R32_HSEM_RLRX17	0xE000C144	HSEM read lock register 17	0x00000000
R32_HSEM_RLRX18	0xE000C148	HSEM read lock register 18	0x00000000
R32_HSEM_RLRX19	0xE000C14C	HSEM read lock register 19	0x00000000
R32_HSEM_RLRX20	0xE000C150	HSEM read lock register 20	0x00000000
R32_HSEM_RLRX21	0xE000C154	HSEM read lock register 21	0x00000000
R32_HSEM_RLRX22	0xE000C158	HSEM read lock register 22	0x00000000
R32_HSEM_RLRX23	0xE000C15C	HSEM read lock register 23	0x00000000
R32_HSEM_RLRX24	0xE000C160	HSEM read lock register 24	0x00000000
R32_HSEM_RLRX25	0xE000C164	HSEM read lock register 25	0x00000000
R32_HSEM_RLRX26	0xE000C168	HSEM read lock register 26	0x00000000
R32_HSEM_RLRX27	0xE000C16C	HSEM read lock register 27	0x00000000
R32_HSEM_RLRX28	0xE000C170	HSEM read lock register 28	0x00000000
R32_HSEM_RLRX29	0xE000C174	HSEM read lock register 29	0x00000000
R32_HSEM_RLRX30	0xE000C178	HSEM read lock register 30	0x00000000
R32_HSEM_RLRX31	0xE000C17C	HSEM read lock register 31	0x00000000
R32_HSEM_LSE	0xE000C200	HSEM lock status register	0x00000000
R32_HSEM_CLR	0xE000C208	HSEM unlock register	0x00000000
R32_HSEM_KEY	0xE000C20C	HSEM unlock key register	0x5AA50003
R32_HSEM_IER	0xE000C300	HSEM interrupt enable register	0x00000000
R32_HSEM_ISR	0xE000C308	HSEM interrupt status register	0x00000000
R32_HSEM_ISM	0xE000C310	HSEM interrupt mask register	0x00000000
R32_HSEM_LSM	0xE000C318	HSEM lock status mask register	0x00000000

Note: Registers marked as "private registers for the core" are multiple groups of registers with the same function and independent data, and the data cannot be accessed or rewritten by other cores, and only the cores have the same access address.

4.5.2.1 HSEM Lock Channel Register y (HSEM_RXy) (y=0-31)

Offset address: 0x000 + 4y

Bit	Name	Access	Description	Reset value
31	LOCK	RW	Channel lock status indicator: 1: Channel locked;	0

			0: Channel idle.	
[30:10]	Reserved	RO	Reserved	0
[9:8]	CID	RW	Core ID of locked channel: 1: Channel locked by core1; 0: Channel idle or locked by core 0.	0
[7:0]	PID	RW	Process ID of the locked channel: Reset value or loaded as 0 after channel release; updated to the written bus data when write lock is acquired.	0

4.5.2.2 HSEM Read Lock Register y (HSEM_RLRXy) (y=0-31)

Offset address: $0x100 + 4y$

Bit	Name	Access	Description	Reset value
31	LOCK	RO	Channel lock status indicator: 1: Channel locked; 0: Channel idle.	0
[30:10]	Reserved	RO	Reserved	0
[9:8]	CID	RO	Core ID of locked channel: 1: Channel locked by core 1; 0: Channel idle or locked by core 0.	0
[7:0]	PID	RO	The process ID of the locked channel, which is 0 when reading the lock.	0

4.5.2.3 HSEM Lock Status Register (HSEM_LSE)

Offset address: $0x200$

Bit	Name	Access	Description	Reset value
[31:0]	HSEM_LSE	RO	HSEM channel lock status register, used to obtain the lock status of 32 channels: 1: Channel locked; 0: Channel idle.	0

4.5.2.4 HSEM Unlock Register (HSEM_CLR)

Offset address: $0x208$

Bit	Name	Access	Description	Reset value
[31:16]	CLR_KEY	WO	Unlock the keyword. The operation is valid when the written data is equal to the matching keyword, otherwise it is invalid. The key value is 0x5AA5.	0
[15:14]	Reserved	RO	Reserved	0
13	CID_MASK	WO	CID matching mask: 1: Disable CID matching; all CID values are unlocked.	0

			0: Enable CID matching; channels with locked CID values matching the match_CID value will unlock.	
12	PID_MASK	WO	PID matching mask: 1: Disable PID matching; all PID values are unlocked. 0: Enable PID matching; channels with PID values matching match_PID will unlock.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	match_CID	WO	CID value used for matching unlocking is compared with the CID value of each locked channel.	0
[7:0]	match_PID	WO	Used to match the unlocked PID value and compare it with the PID value of each locked channel.	0

4.5.2.5 HSEM Unlock Key Register (HSEM_KEY)

Offset address: 0x20C

Bit	Name	Access	Description	Reset value
[31:16]	KEY_VALUE	RO	Unlock keyword, which is used to match the keyword comparison of unlocking mode.	0x5AA5
[15:2]	Reserved	RO	Reserved	0
1	auto_2step_clr	RW	Automatically clear the corresponding channel interrupt status bit when locking the channel in 2step mode fails. Enable: 1: Automatically clears the corresponding channel interrupt status bit upon lock failure; 0: Lock failure does not affect the corresponding channel interrupt status bit.	1
0	auto_1step_clr	RW	Enable automatic clearing of the corresponding channel interrupt status bit when locking fails using the 1-step method: 1: Automatically clear the corresponding channel interrupt status bit upon locking failure; 0: Locking failure does not affect the corresponding channel interrupt status bit.	1

4.5.2.6 HSEM Interrupt Enable Register (HSEM_IER)

Offset address: 0x300

Bit	Name	Access	Description	Reset value
[31:0]	HSEM_IER	RW	HSEM channel interrupt enable register, a core-private register. Setting this bit to 1 permits interrupt to occur when the corresponding channel is unlocked by another core. It works in	0

			conjunction with the HSEM_ISR register to generate interrupts.	
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4.5.2.7 HSEM Interrupt Status Register (HSEM_ISR)

Offset address: 0x308

Bit	Name	Access	Description	Reset value
[31:0]	HSEM_ISR	RW1Z	HSEM channel interrupt status register, a core-private register. When a locked channel is unlocked by another core, the corresponding bit is set. When used in conjunction with the HSEM_IER register, it can generate an interrupt.	0

4.5.2.8 HSEM Interrupt Mask Register (HSEM_ISM)

Offset address: 0x310

Bit	Name	Access	Description	Reset value
[31:0]	HSEM_ISM	RO	HSEM channel interrupt mask register, a core-private register. Its value equals HSEM_ISR & HSEM_IER. A non-zero value indicates a SEM interrupt has occurred, with the corresponding channel bit set to 1.	0

4.5.2.9 HSEM Lock Mask Register (HSEM_LSM)

Offset address: 0x318

Bit	Name	Access	Description	Reset value
[31:0]	HSEM_LSM	RO	HSEM channel lock status register, used to determine whether the HSEM channel is locked by this core: 1: Channel is locked by this core; 0: Channel is idle or locked by another core.	0

4.6 System Counter (SysTick)

The core incorporates two 32-bit up/down counters (SysTick) that can independently generate interrupt signals and are configurable to send interrupt requests to specified cores. The system counter operates at the same frequency as HCLK (V3F core frequency), while the RTC counter operates at HCLK/8.

- Dual 32-bit timers
- Up/down counting capability
- •Configurable main clock and RTC clock counting
- Configurable interrupt request core
- • Configurable automatic reload

4.6.1 RISC-V-SysTick Register

Table 4-10 STK registers list

Name	Access	Description	Reset value
R32_STK_CTLR_0	0xE000F000	System count control register 0	0x00000000
R32_STK_ISR	0xE000F004	System counter interrupt status register	0x00000000
R32_STK_CNT_0	0xE000F008	System counter 0 count register	0x00000000
R32_STK_CMP_0	0xE000F010	System counter 0 compare register	0x00000000
R32_STK_CTLR_1	0xE000F080	System count control register 1	0x00000000
R32_STK_CNT_1	0xE000F088	System counter 1 count register	0x00000000
R32_STK_CMP_1	0xE000F090	System counter 1 compare register	0x00000000

4.6.1.1 System Count Control Register 0 (STK_CTLR_0)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
6	CID_0	RW	Counter 0 allocates a core register to configure its interrupt request to the specified core. When the enable bit for Counter 0 is set, it automatically updates to the operating core ID.	0
5	Reserved	RO	Reserved	0
4	DOWN_MODE_0	RW	Counting mode: 1: Count down; 0: Count up.	0
3	AUTO_RELOAD_0	RW	Auto reload enable bit: 1: Auto reload enabled; 0: Auto reload disabled.	0
2	NO_RTC_0	RW	Counter clock source selection bit: 1: Use HCLK as the clock source; 0: Use HCLK/8 as the clock source.	0
1	IE_0	RW	Counter 0 interrupt enable control bit: 1: Enable counter interrupt; 0: Disable counter interrupt.	0
0	EN_0	RW	System counter 0 enable control bit: 1: Enables system counter STK. After setting to 1, configuration of TIM_CTL_0 can only be modified when CID matches the operating core. 0: Disables system counter STK, stopping counter operation.	0

4.6.1.2 System Counter Interrupt Status Register (STK_ISR)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved	0

1	STK_ISR1	RW	Systick1 interrupt flag: 1: Timer 1 has an unprocessed interrupt request; 0: Timer 1 has no interrupt request.	0
0	STK_ISR0	RW	Systick0 interrupt flag: 1: Timer 0 has an unprocessed interrupt request; 0: Timer 0 has no interrupt request.	0

4.6.1.3 System Counter 0 Count Register (STK_CNT_0)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[31:0]	CNT_0[31:0]	RW	Systick0 counter register, which accumulates/decrements and automatically reloads based on configuration after counting is enabled.	0

4.6.1.4 System Counter 0 C0mpare Register (STK_CMP_0)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[31:0]	CMP_0[31:0]	RW	Systick0 Compare Register, used for comparison during upward counting and automatic reloading during downward counting.	0

4.6.1.5 System Count Control Register 1 (STK_CTLR_1)

Offset address: 0x80

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
6	CID_1	RW	Counter 1 allocates a core register to configure interrupt requests from Counter 1 to the designated core. When the Counter 1 enable is configured as active, it automatically updates to the operating core ID.	0
5	Reserved	RO	Reserved	0
4	DOWN_MODE_1	RW	Counting mode: 1: Count down; 0: Count up.	0
3	AUTO_RELOAD_1	RW	Auto reload enable bit: 1: Auto reload enabled; 0: Auto reload disabled.	0
2	NO_RTC_1	RW	Counter clock source selection bit: 1: Use HCLK as the clock source; 0: Use HCLK/8 as the clock source.	0
1	IE_1	RW	Counter 1 interrupt enable control bit: 1: Enable counter interrupt;	0

			0: Disable counter interrupt.	
0	EN_1	RW	System counter 1 enable control bit: 1: Enables system counter STK. After setting to 1, configuration of TIM_CTL_0 can only be modified when CID matches the operating core. 0: Disables system counter STK, stopping counter operation.	0

4.6.1.6 System Counter 1 Count Register (STK_CNT_1)

Offset address: 0x88

Bit	Name	Access	Description	Reset value
[31:0]	CNT_1[31:0]	RW	Systick1 counter register, which accumulates/decrements and automatically reloads based on configuration after counting is enabled.	0

4.6.1.7 System Counter 1 Compare Register (STK_CMP_1)

Offset address: 0x90

Bit	Name	Access	Description	Reset value
[31:0]	CMP_1[31:0]	RW	Systick1 Compare Register, used for comparison during upward counting and automatic reloading during downward counting.	0

4.7 Interrupt and Event (PFIC)

Built-in programmable fast interrupt controller (PFIC—programmable fast interrupt controller) supports up to 256 interrupt vectors, in which RISC-V3F and RISC-V5F manage 32 private interrupts and share other interrupts. It can be allocated to one of the cores by configuring the interrupt allocation register. Some peripherals such as IPC and SysTick in the low-32 interrupt need to be allocated to the designated cores by configuring the registers in the peripherals, and an interrupt can only generate an interrupt request to one core.

The interrupt control platform can control the core's sleep state and wake-up. When the core executes the WFI instruction, the core enters the low power consumption state. The interrupt control platform will set the core sleep flag and monitor the interrupt or event. When there is an interrupt or event, the interrupt control platform will wake the core to execute the interrupt and other programs.

4.7.1 PFIC Controller Main characteristics

- Up to 256 interrupt arbitrations are supported, and the lower 32 bits are private interrupts.
- Shared interrupt allocation
- Support Hardware Prologue/Epilogue (HPE) without instruction overhead.
- Provides 4-channel of Vector Table Free (VTF)
- Support low-power sleep and wake-up management
- The core RISC-V3F supports configurable level 2 interrupt nesting at most, and the core RISC-V5F supports configurable level 8 interrupt nesting at most.

4.7.2 Vector Table of Interrupt and Exception

Table 4-11 CH32H417 series chip vector table

No.	Priority	Type	Name	Description	Offset address
0	-6	Fixed	RESET	Reset interrupt	0x000
1	-	-	-	Reserved	0x004
2	-5	Fixed	NMI	Non-maskable interrupt	0x008
3	-4	Fixed	HardFault	Abnormal interrupt	0x00C
4	-	-	-	Reserved	0x010
5	-3	Fixed	Ecall-M	Machine mode callback interrupt	0x014
6-7	-	-	-	Reserved	0x018-0x01C
8	-2	Fixed	Ecall-U	User mode callback interrupt	0x020
9	-1	Fixed	BreakPoint	Breakpoint callback interrupt	0x024
10-11	-	-	-	Reserved	0x028-0x02C
12	0	Programmable	SysTick0	System timer interrupt 0	0x030
13	1	Programmable	SysTick1	System timer interrupt 1	0x034
14	2	Programmable	SW	Software interrupt	0x038
15	-	-	-	Reserved	0x03C
16	3	Programmable	IPC_CH0	Inter-process communication interrupt 0	0x040
17	4	Programmable	IPC_CH1	Inter-process communication interrupt 1	0x044
18	5	Programmable	IPC_CH2	Inter-process communication interrupt 2	0x048
19	6	Programmable	IPC_CH3	Inter-process communication interrupt 3	0x04C
20-27	-	-	-	Reserved	0x050-0x06C
28	7	Programmable	HSEM	Hardware semaphore interrupt	0x070
29-31	-	-	-	Reserved	0x074-0x07C
32	8	Programmable	WWDG	Window timer interrupt	0x080
33	9	Programmable	EXTI15_8	EXTI line [15:8] interrupt	0x084
34	10	Programmable	FLASH	Flash global interrupt	0x088
35	11	Programmable	RCC	Reset and clock control interrupt	0x08C
36	12	Programmable	EXTI7_0	EXTI line [7:0] interrupt	0x090
37	13	Programmable	SPI1	SPI1 global interrupt	0x094
38	14	Programmable	DMA1_CH2	DMA1 channel 2 global interrupt	0x098
39	15	Programmable	DMA1_CH3	DMA1 channel 3 global interrupt	0x09C
40	16	Programmable	DMA1_CH4	DMA1 channel 4 global interrupt	0x0A0
41	17	Programmable	DMA1_CH5	DMA1 channel 5 global interrupt	0x0A4
42	18	Programmable	DMA1_CH6	DMA1 channel 6 global interrupt	0x0A8
43	19	Programmable	DMA1_CH7	DMA1 channel 7 global interrupt	0x0AC
44	20	Programmable	DMA1_CH8	DMA1 channel 8 global interrupt	0x0B0
45	21	Programmable	USART2	USART2 global interrupt	0x0B4

46	22	Programmable	I2C1_EV	I ² C1 event interrupt	0x0B8
47	23	Programmable	I2C1_ER	I ² C1 error interrupt	0x0BC
48	24	Programmable	USART1	USART1 global interrupt	0x0C0
49	25	Programmable	SPI2	SPI2 global interrupt	0x0C4
50	26	Programmable	SPI3	SPI3 global interrupt	0x0C8
51	27	Programmable	SPI4	SPI4 global interrupt	0x0CC
52	28	Programmable	I2C2_EV	I ² C2 event interrupt	0x0D0
53	29	Programmable	I2C2_ER	I ² C2 error interrupt	0x0D4
54	30	Programmable	USBPD	USBPD global interrupt	0x0D8
55	31	Programmable	USBPDWake UP	USBPD wakeup interrupt	0x0DC
56	32	Programmable	USBHS	USBHS global interrupt	0x0E0
57	33	Programmable	DMA1_CH1	DMA1 channel 1 global interrupt	0x0E4
58	34	Programmable	CAN1_SCE	CAN1_SCE global interrupt	0x0E8
59	35	Programmable	CAN1_TX	CAN1_TX global interrupt	0x0EC
60	36	Programmable	CAN1_RX0	CAN1_RX0 global interrupt	0x0F0
61	37	Programmable	CAN1_RX1	CAN1_RX1 global interrupt	0x0F4
62	38	Programmable	USBSS	USBSS global interrupt	0x0F8
63	39	Programmable	USBSS_LIN K	USBSS LINK interrupt	0x0FC
64	40	Programmable	USBHSWake UP	USBHS wakeup interrupt	0x100
65	41	Programmable	USBSSWake UP	USBSS wakeup interrupt	0x104
66	42	Programmable	RTCArm	RTC alarm clock interrupt (EXTI)	0x108
67	43	Programmable	USBFS	USBFS global interrupt	0x10C
68	44	Programmable	USBFSWake UP	USBFS wakeup interrupt	0x110
69	45	Programmable	ADC1_2	ADC1_2 global interrupt	0x114
70	46	Programmable	TIM1_BRK	TIM1 break interrupt	0x118
71	47	Programmable	TIM1_UP	TIM1 update interrupt	0x11C
72	48	Programmable	TIM1_TRG_ COM	TIM1 triggering and communication interruption	0x120
73	49	Programmable	TIM1_CC	TIM1 capture compare interrupt	0x124
74	50	Programmable	TIM2	TIM2 global interrupt	0x128
75	51	Programmable	TIM3	TIM3 global interrupt	0x12C
76	52	Programmable	TIM4	TIM4 global interrupt	0x130
77	53	Programmable	TIM5	TIM5 global interrupt	0x134
78	54	Programmable	I2C3_EV	I ² C3 event interrupt	0x138
79	55	Programmable	I2C3_ER	I ² C3 error interrupt	0x13C
80	56	Programmable	I2C4_EV	I ² C4 event interrupt	0x140
81	57	Programmable	I2C4_ER	I ² C4 error interrupt	0x144
82	58	Programmable	QSPI1	QSPI1 global interrupt	0x148

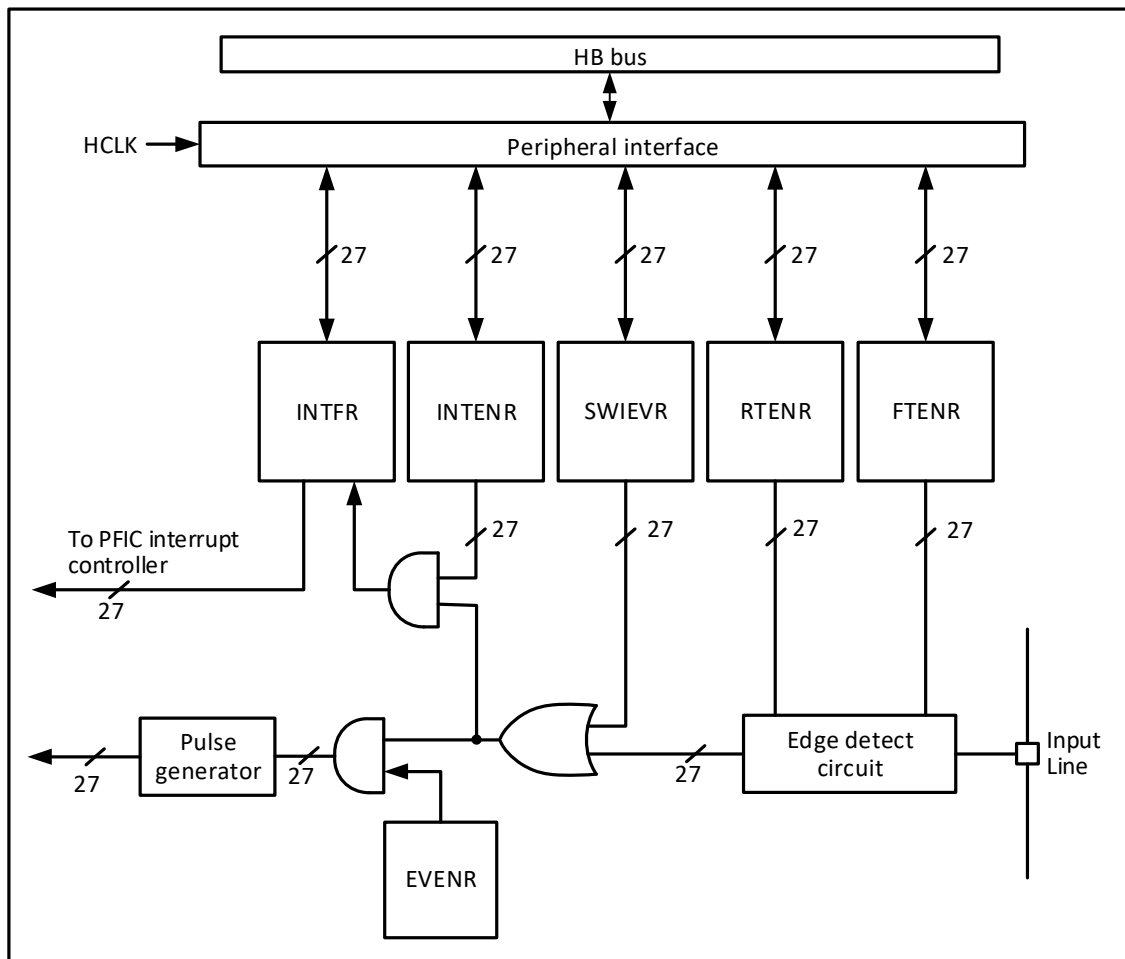
83	59	Programmable	SERDES	SERDES global interrupt	0x14C
84	60	Programmable	USART3	USART3 global interrupt	0x150
85	61	Programmable	USART4	USART4 global interrupt	0x154
86	62	Programmable	TIM8_BRK	TIM8 break interrupt	0x158
87	63	Programmable	TIM8_UP	TIM8 update interrupt	0x15C
88	64	Programmable	TIM8_TRG_COM	TIM8 triggering and communication interruption	0x160
89	65	Programmable	TIM8_CC	TIM8 capture compare interrupt	0x164
90	66	Programmable	TIM9	TIM9 global interrupt	0x168
91	67	Programmable	TIM10	TIM10 global interrupt	0x16C
92	68	Programmable	TIM11	TIM11 global interrupt	0x170
93	69	Programmable	TIM12	TIM12 global interrupt	0x174
94	70	Programmable	FMC	FMC global interrupt	0x178
95	71	Programmable	SDMMC	SDMMC global interrupt	0x17C
96	72	Programmable	LPTIM1	LPTIM1 global interrupt	0x180
97	73	Programmable	LPTIM2	LPTIM2 global interrupt	0x184
98	74	Programmable	USART5	USART5 global interrupt	0x188
99	75	Programmable	USART6	USART6 global interrupt	0x18C
100	76	Programmable	TIM6	TIM6 global interrupt	0x190
101	77	Programmable	TIM7	TIM7 global interrupt	0x194
102	78	Programmable	DMA2_CH1	DMA2 channel 1 global interrupt	0x198
103	79	Programmable	DMA2_CH2	DMA2 channel 2 global interrupt	0x19C
104	80	Programmable	DMA2_CH3	DMA2 channel 3 global interrupt	0x1A0
105	81	Programmable	DMA2_CH4	DMA2 channel 4 global interrupt	0x1A4
106	82	Programmable	DMA2_CH5	DMA2 channel 5 global interrupt	0x1A8
107	83	Programmable	DMA2_CH6	DMA2 channel 6 global interrupt	0x1AC
108	84	Programmable	DMA2_CH7	DMA2 channel 7 global interrupt	0x1B0
109	85	Programmable	DMA2_CH8	DMA2 channel 8 global interrupt	0x1B4
110	86	Programmable	ETH	ETH global interrupt	0x1B8
111	87	Programmable	ETHWakeUP	ETH wakeup interrupt	0x1BC
112	88	Programmable	CAN2_SCE	CAN2_SCE global interrupt	0x1C0
113	89	Programmable	CAN2_TX	CAN2_TX global interrupt	0x1C4
114	90	Programmable	CAN2_RX0	CAN2_RX0 global interrupt	0x1C8
115	91	Programmable	CAN2_RX1	CAN2_RX1 global interrupt	0x1CC
116	92	Programmable	USART7	USART7 global interrupt	0x1D0
117	93	Programmable	USART8	USART8 global interrupt	0x1D4
118	94	Programmable	I3C_EV	I ³ C event interrupt	0x1D8
119	95	Programmable	I3C_ER	I ³ C error interrupt	0x1DC
120	96	Programmable	DVP	DVP global interrupt	0x1E0
121	97	Programmable	ECDC	ECDC global interrupt	0x1E4
122	98	Programmable	PIOC	PIOC global interrupt	0x1E8
123	99	Programmable	SAI	SAI global interrupt	0x1EC
124	100	Programmable	LTDC	LTDC global interrupt	0x1F0

125	101	Programmable	GPHA	GPHA global interrupt	0x1F4
126	-	-	-	Reserved	0x1F8
127	102	Programmable	DFSDM0	DFSDM0 global interrupt	0x1FC
128	103	Programmable	DFSDM1	DFSDM1 global interrupt	0x200
129	-	-	-	Reserved	0x204
130	-	-	-	Reserved	0x208
131	104	Programmable	SWPMI	SWPMI global interrupt	0x20C
132	-	-	-	Reserved	0x210
133	-	-	-	Reserved	0x214
134	105	Programmable	QSPI2	QSPI2 global interrupt	0x218
135	106	Programmable	SWPMIWake UP	SWPMI wakeup interrupt	0x21C
136	107	Programmable	CAN3_SCE	CAN3_SCE global interrupt	0x220
137	108	Programmable	CAN3_TX	CAN3_TX global interrupt	0x224
138	109	Programmable	CAN3_RX0	CAN3_RX0 global interrupt	0x228
139	110	Programmable	CAN3_RX1	CAN3_RX1 global interrupt	0x22C
140	111	Programmable	LPTIM2Wake UP	LPTIM2 wakeup interrupt	0x230
141	112	Programmable	LPTIM1Wake UP	LPTIM1 wakeup interrupt	0x234
142	113	Programmable	I3CWakeUP	I ³ C wakeup interrupt	0x238
143	114	Programmable	RTC	RTC global interrupt	0x23C
144	115	Programmable	HSADC	HSADC global interrupt	0x240
145	116	Programmable	UHSIF	UHSIF global interrupt	0x244
146	117	Programmable	RNG	RNG global interrupt	0x248
147	118	Programmable	SDIO	SDIO global interrupt	0x24C
148	119	Programmable	USARTWake UP	USART wakeup interrupt	0x250

4.7.3 External Interrupt and Event Controller (EXTI)

4.7.3.1 Overview

Figure 4-1 External interrupt (EXTI) interface block diagram



As can be seen from figure 4-1, the trigger source of the external interrupt can be either a software interrupt (SWIEVR) or an actual external interrupt channel, and the signal of the external interrupt channel is first screened by the edge detection circuit. As long as one of the software interrupts or external interrupt signals is generated, it will be output to both event enabling and interrupt enabling circuits through the OR gate circuit in the diagram. As long as an interrupt is enabled or an event is enabled, an interrupt or event will occur. The six registers of EXTI are accessed by the processor through the HB interface.

4.7.3.2 Wakeup Event

The system can wake up sleep patterns caused by WFE instructions through wake-up events. Wake-up events are generated through the following 3 configurations:

- Enable an interrupt in the register of the peripheral, but not in the PFIC of the core, as well as the SEVONPEND bit in the core. Reflected in EXTI, it enables EXTI interrupts, but does not enable EXTI interrupts in PFIC, while enabling SEVONPEND bits. After the CPU wakes up from WFE, it must clear the EXTI interrupt flag and the PFIC suspend bit.
- Enable an EXTI channel to be an event channel, and after CPU wakes up from WFE, there is no need to clear the operation of the interrupt flag bit and PFIC hang bit.

- A write event is generated by another running core writing to the SENDEVENT register.

4.7.3.3 Description

The use of external interrupt needs to configure the corresponding external interrupt channel, that is, select the appropriate trigger edge to enable the corresponding interrupt. When a set trigger edge appears on the external interrupt channel, an interrupt request will be generated and the corresponding interrupt flag bit will be set. Write 1 to the flag bit to clear it.

Use external hardware interrupt steps:

- 1) Configure GPIO operation.
- 2) Configure the interrupt enable bit (EXTI_INTENR) of the corresponding external interrupt channel.
- 3) Configure trigger edge (EXTI_RTENR or EXTI_FTENR), select rising edge trigger, falling edge trigger or double edge trigger.
- 4) Configure EXTI interrupts in the PFIC of the core to ensure that they respond correctly.

To use external hardware events:

- 1) Configure GPIO operation.
- 2) Configure the event enable bit (EXTI_EVENTR) of the corresponding external interrupt channel.
- 3) Configure trigger edge (EXTI_RTENR or EXTI_FTENR), select rising edge trigger, falling edge trigger or double edge trigger.

Use software interrupt/event steps:

- 1) Enable external interrupt (EXTI_INTENR) or external event (EXTI_EVENTR).
- 2) If you use the interrupt service function, you need to set the EXTI interrupt in the PFIC of the core.
- 3) Set the software interrupt trigger (EXTI_SWIEVR), that is, an interrupt will occur.

4.7.3.4 External Event Map

Table 4-12 EXTI interrupt map

External interrupt/event line	Mapping event description
EXTI0~EXTI15	Px0~Px15 (x=A/B/C/D/E), any of the IO ports can be enabled for external interrupt/event functionality, configured by the AFIO_EXTICRx register.
EXTI16	USBHS wakeup event
EXTI17	RTC alarm event
EXTI18	SWPMI wakeup event
EXTI19	USBFSOTG wakeup event
EXTI20	USBPD wakeup event
EXTI21	ETH wakeup event
EXTI22	USBSS wakeup event
EXTI23	LPTIM1 wakeup event
EXTI24	LPTIM2 wakeup event
EXTI25	I3C wakeup event
EXTI26	USART wakeup event

4.7.4 EXTI Register

Table 4-13 EXTI registers list

Name	Access	Description	Reset value
R32_EXTI_INTENR	0x40010400	Interrupt enable register	0x00000000
R32_EXTI_EVENTR	0x40010404	Event enable register	0x00000000
R32_EXTI_RTENR	0x40010408	Rising edge trigger enable register	0x00000000
R32_EXTI_FTENR	0x4001040C	Falling edge trigger enable register	0x00000000
R32_EXTI_SWIEVR	0x40010410	Software interrupt event register	0x00000000
R32_EXTI_INTFR	0x40010414	Interrupt flag register	0x0000XXXX

4.7.4.1 Interrupt Enable Register (EXTI_INTENR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					MR2 6	MR2 5	MR2 4	MR2 3	MR2 2	MR2 1	MR2 0	MR1 9	MR1 8	MR1 7	MR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR1 5	MR1 4	MR1 3	MR12	MR1 1	MR1 0	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:0]	MRx	RW	Enable the interrupt request signal for external interrupt channel x: 1: Enable interrupt for this channel; 0: Mask interrupt for this channel.	0

4.7.4.2 Event Enable Register (EXTI_EVENTR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					MR2 6	MR2 5	MR2 4	MR2 3	MR2 2	MR2 1	MR2 0	MR1 9	MR1 8	MR1 7	MR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR1 5	MR1 4	MR1 3	MR12	MR1 1	MR1 0	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:0]	MRx	RW	Enable the event request signal for external interrupt channel x: 1: Enable event for this channel; 0: Mask event for this channel.	0

4.7.4.3 Rising Edge Trigger Enable Register (EXTI_RTENR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					TR26	TR25	TR24	TR23	TR22	TR21	TR20	TR19	TR18	TR17	TR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:0]	TRx	RW	Enable rising edge triggering of external interrupt channel x: 1: Enable rising edge triggering for this channel; 0: Disable rising edge triggering for this channel.	0

4.7.4.4 Falling Edge Trigger Enable Register (EXTI_FTEENR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					TR26	TR25	TR24	TR23	TR22	TR21	TR20	TR19	TR18	TR17	TR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:0]	TRx	RW	Enable falling edge triggering of external interrupt channel x: 1: Enable falling edge triggering for this channel; 0: Disable falling edge triggering for this channel.	0

4.7.4.5 Software Interrupt Event Register (EXTI_SWIEVR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SWIER26	SWIER25	SWIER24	SWIER23	SWIER22	SWIER21	SWIER20	SWIER19	SWIER18	SWIER17	SWIER16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWIER15	SWIER14	SWIER13	SWIER12	SWIER11	SWIER10	SWIER9	SWIER8	SWIER7	SWIER6	SWIER5	SWIER4	SWIER3	SWIER2	SWIER1	SWIER0

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:0]	SWIERx	RW	A software interrupt is set on the corresponding	0

			externally triggered interrupt channel. Setting it here causes the interrupt flag bit (EXTI_INTFR) to be set to the corresponding position bit, and an interrupt or event is generated if the interrupt enable (EXTI_INTENR) or event enable (EXTI_EVENTR) is turned on.	
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4.7.4.6 Interrupt Flag Register (EXTI_INTFR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					IF26	IF25	IF24	IF23	IF22	IF21	IF20	IF19	IF18	IF17	IF16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IF15	IF14	IF13	IF12	IF11	IF10	IF9	IF8	IF7	IF6	IF5	IF4	IF3	IF2	IF1	IF0

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:0]	IFx	RW1	Interrupt Flag Bit, this location bit flag indicates that the corresponding external interrupt has occurred. Writing 1 clears this bit.	0

4.7.5 PFIC Registers

Table 4-14 PFIC-related registers

Name	Access address	Description	Reset value
R32_PFIC_ISR1	0xE000E000	PFIC interrupt enable status register 1	0x0000032C
R32_PFIC_ISR2	0xE000E004	PFIC interrupt enable status register 2	0x00000000
R32_PFIC_ISR3	0xE000E008	PFIC interrupt enable status register 3	0x00000000
R32_PFIC_ISR4	0xE000E00C	PFIC interrupt enable status register 4	0x00000000
R32_PFIC_ISR5	0xE000E010	PFIC interrupt enable status register 5	0x00000000
R32_PFIC_IPR1	0xE000E020	PFIC interrupt pending status register 1	0x00000000
R32_PFIC_IPR2	0xE000E024	PFIC interrupt pending status register 2	0x00000000
R32_PFIC_IPR3	0xE000E028	PFIC interrupt pending status register 3	0x00000000
R32_PFIC_IPR4	0xE000E02C	PFIC interrupt pending status register 4	0x00000000
R32_PFIC_IPR5	0xE000E030	PFIC interrupt pending status register 5	0x00000000
R32_PFIC_ITHRESDR	0xE000E040	PFIC interrupt priority threshold configuration register	0x00000000
R32_PFIC_CFGR	0xE000E048	PFIC interrupt configuration register	0x00000000
R32_PFIC_GISR	0xE000E04C	PFIC interrupt global status register	0x00000000
R32_PFIC_VTFIDR	0xE000E050	PFIC VTF interrupt ID configuration register	0xFFFFFFFF
R32_PFIC_VTFADDRR0	0xE000E060	PFIC VTF interrupt 0 offset address register	0xFFFFFFFF
R32_PFIC_VTFADDRR1	0xE000E064	PFIC VTF interrupt 1 offset address register	0xFFFFFFFF

		register	
R32_PFIC_VTFADDR2	0xE000E068	PFIC VTF interrupt 2 offset address register	0xFFFFFFFF
R32_PFIC_VTFADDR3	0xE000E06C	PFIC VTF interrupt 3 offset address register	0xFFFFFFFF
R32_PFIC_IENR1	0xE000E100	PFIC interrupt enable set register 1	0x00000000
R32_PFIC_IENR2	0xE000E104	PFIC interrupt enable set register 2	0x00000000
R32_PFIC_IENR3	0xE000E108	PFIC interrupt enable set register 3	0x00000000
R32_PFIC_IENR4	0xE000E10C	PFIC interrupt enable set register 4	0x00000000
R32_PFIC_IENR5	0xE000E110	PFIC interrupt enable set register 5	0x00000000
R32_PFIC_IRER1	0xE000E180	PFIC interrupt enable clear register 1	0x00000000
R32_PFIC_IRER2	0xE000E184	PFIC interrupt enable clear register 2	0x00000000
R32_PFIC_IRER3	0xE000E188	PFIC interrupt enable clear register 3	0x00000000
R32_PFIC_IRER4	0xE000E18C	PFIC interrupt enable clear register 4	0x00000000
R32_PFIC_IRER5	0xE000E190	PFIC interrupt enable clear register 5	0x00000000
R32_PFIC_IPSR1	0xE000E200	PFIC interrupt pending set register 1	0x00000000
R32_PFIC_IPSR2	0xE000E204	PFIC interrupt pending set register 2	0x00000000
R32_PFIC_IPSR3	0xE000E208	PFIC interrupt pending set register 3	0x00000000
R32_PFIC_IPSR4	0xE000E20C	PFIC interrupt pending set register 4	0x00000000
R32_PFIC_IPSR5	0xE000E210	PFIC interrupt pending set register 5	0x00000000
R32_PFIC_IPRR1	0xE000E280	PFIC interrupt pending clear register 1	0x00000000
R32_PFIC_IPRR2	0xE000E284	PFIC interrupt pending clear register 2	0x00000000
R32_PFIC_IPRR3	0xE000E288	PFIC interrupt pending clear register 3	0x00000000
R32_PFIC_IPRR4	0xE000E28C	PFIC interrupt pending clear register 4	0x00000000
R32_PFIC_IPRR5	0xE000E290	PFIC interrupt pending clear register 5	0x00000000
R32_PFIC_IACTR1	0xE000E300	PFIC interrupt activation register 1	0x00000000
R32_PFIC_IACTR2	0xE000E304	PFIC interrupt activation register 2	0x00000000
R32_PFIC_IACTR3	0xE000E308	PFIC interrupt activation register 3	0x00000000
R32_PFIC_IACTR4	0xE000E30C	PFIC interrupt activation register 4	0x00000000
R32_PFIC_IACTR5	0xE000E310	PFIC interrupt activation register 5	0x00000000
R32_PFIC_IPRIORx	0xE000E400	PFIC interrupt priority configuration register	0x00000000
R32_PFIC_IALLOCx	0xE000E600	PFIC interrupt allocation register	0xFFFFFFFF
R32_PFIC_IAUTR1	0xE000E700	PFIC interrupt authority register 1	0xFFFFFFFF
R32_PFIC_IAUTR2	0xE000E704	PFIC interrupt authority register 2	0xFFFFFFFF
R32_PFIC_IAUTR3	0xE000E708	PFIC interrupt authority register 3	0xFFFFFFFF
R32_PFIC_IAUTR4	0xE000E70C	PFIC interrupt authority register 4	0xFFFFFFFF
R32_PFIC_IAUTR5	0xE000E710	PFIC interrupt authority register 5	0xFFFFFFFF
R32_PFIC_WAKEIP0	0xE000E720	PFIC wakeup instruction pointer register 0	0x00000001
R32_PFIC_WAKEIP1	0xE000E724	PFIC wakeup instruction pointer register 1	0x00000000
R32_PFIC_CSTAR0	0xE000E780	PFIC core status register 0	0x00000000
R32_PFIC_CSTAR1	0xE000E784	PFIC core status register 1	0x00000000
R32_PFIC_EENR	0xE000EC80	PFIC event enable register	0xFFFFFFFF

R32_PFIC_EPR	0xE000EC84	PFIC event pending register	0x00000000
R32_PFIC_EWUPR	0xE000EC88	PFIC event wakeup register	0x00000000
R32_PFIC_SCTLR	0xE000ED10	PFIC system control register	0x00000000

Note: 1. NMI, EXC, ECALL-M, ECALL-U, BREAKPOINT interrupts are always enabled by default.

2. ECALL-M, ECALL-U, and BREAKPOINT are all EXC cases, and the status is indicated by EXC status bit

3.

3. NMI and EXC support interrupt suspend clear and set operations, but not interrupt enable clear and set operations.

4. ECALL-M, ECALL-U, and BREAKPOINT do not support interrupt hang clear and set, interrupt enable clear and set operations.

4.7.5.1 PFIC Interrupt Enable Status Register 1 (PFIC_ISR1)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTE NST A15	INTE NST A14	INTE NST A13	INTE NST A12	Reserved		INTE NST A9	INTE NST A8	Reserved		INTE NST A5	Reser ved	INTE NST A3	INTE NST A2	Reserved	

Bit	Name	Access	Description	Reset value
[31:12]	INTENSTA	RO	12#-31#Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	INTENSTA	RO	8#-9# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0x3
[7:6]	Reserved	RO	Reserved	0
5	INTENSTA	RO	5# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	1
4	Reserved	RO	Reserved	0
[3:2]	INTENSTA	RO	2#-3# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0x3
[1:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.2 PFIC Interrupt Enable Status Register 2 (PFIC_ISR2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	INTENSTA	RO	32#-63# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0

4.7.5.3 PFIC Interrupt Enable Status Register 3 (PFIC_ISR3)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	INTENSTA	RO	64#-95# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0

4.7.5.4 PFIC Interrupt Enable Status Register 4 (PFIC_ISR4)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[111:96]															

Bit	Name	Access	Description	Reset value
[31:0]	INTENSTA	RO	96#-127# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0

4.7.5.5 PFIC Interrupt Enable Status Register 5 (PFIC_ISR5)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											INTENSTA[148:144]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[143:128]															

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	INTENSTA	RO	128#-148# Interrupt current enable status. 1: Current numbered interrupt is enabled; 0: Current numbered interrupt not enabled.	0

4.7.5.6 PFIC Interrupt Pending Status Register 1 (PFIC_IPR1)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PEN DST A15	PEN DST A14	PEN DST A13	PEN DST A12	Reserved		PEN DST A9	PEN DST A8	Reserved		PEN DST A5	Reser ved	PEN DST A3	PEN DST A2	Reserved	

Bit	Name	Access	Description	Reset value
[31:12]	PENDSTA	RO	16#-31#interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	PENDSTA	RO	8#-9#interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
[7:6]	Reserved	RO	Reserved	0
5	PENDSTA	RO	5# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
4	Reserved	RO	Reserved	0
[3:2]	PENDSTA	RO	2#-3# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
[1:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.7 PFIC Interrupt Pending Status Register 2 (PFIC_IPR2)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

PENDSTA[47:32]

Bit	Name	Access	Description	Reset value
[4:0]	PENDSTA	RO	32#-63# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0

4.7.5.8 PFIC Interrupt Pending Status Register 3 (PFIC_IPR3)

Offset address: 0x28

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

PENDSTA[95:80]

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

PENDSTA[79:64]

Bit	Name	Access	Description	Reset value
[31:0]	PENDSTA	RO	64#-95# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0

4.7.5.9 PFIC Interrupt Pending Status Register 4 (PFIC_IPR4)

Offset address: 0x2C

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

PENDSTA[127:112]

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

PENDSTA[111:96]

Bit	Name	Access	Description	Reset value
[31:0]	PENDSTA	RO	96#-127# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0

4.7.5.10 PFIC Interrupt Pending Status Register 5 (PFIC_IPR5)

Offset address: 0x30

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Reserved

INTENSTA[148:144]

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

INTENSTA[143:128]

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0

[20:0]	PENDSTA	RO	128#-148# interrupts currently pending status. 1: Current numbered interrupt is pending; 0: Current numbered interrupt is not pending.	0
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4.7.5.11 PFIC Interrupt Priority Threshold Configuration Register (PFIC_ITHRESDR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																								THRESHOLD[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	THRESHOLD	RW	Interrupt priority threshold setting value. Interrupt priority values lower than the current set value do not perform interrupt service when hung; a 0 in this register indicates that the threshold register function is invalid. [7:4]: Priority threshold value. [3:0]: Reserved, fixed to 0, write invalid.	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.12 PFIC Interrupt Configuration Register (PFIC_CFGR)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEYCODE[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								RSTS YS	Reserved						

Bit	Name	Access	Description	Reset value
[31:16]	KEYCODE[15:0]	WO	The 'key' control keyword is valid when writing to 'beef'; otherwise, the operation is invalid.	0
[15:8]	Reserved	RO	Reserved	0
7	RSTSYS	WO	Reset register: System reset occurs when 0xBEEF is written to register 0080. Other values are invalid. <i>Note: Same function as PFIC_SCTLR register SYSRST bit.</i>	0
[6:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.13 PFIC Interrupt Global Status Register (PFIC_GISR)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EX_STATE [1:0]	LOC K_UP	DBG _MODE	GLO BL_IE	Reser ved	GPE NDS TA	GAC T STA	NESTSTA[7:0]								

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	EX_STATE[1:0]	RO	Core status register, used to obtain the core's operational state 00: Running; 01: Light sleep; 10: Deep sleep; 11: Deep sleep (Locked).	0
13	LOCK_UP	RO	Core lock status register, used to determine whether the core is in a locked state. 1: Locked; 0: Unlocked.	0
12	DBG_MODE	RO	Core debug mode register, used to determine whether the core is in debug mode 1: Core is in debug mode; 0: Core is in non-debug mode.	0
11	GLOBL_IE	RO	Core global interrupt enable register, used to query whether core global interrupts are enabled 1: Core global interrupts enabled; 0: Core global interrupts disabled.	0
10	Reserved	RO	Reserved	0
9	GPENDSTA	RO	Whether an interrupt is currently pending: 1: Yes; 0: No.	0
8	GACTSTA	RO	Whether the interrupt is executed currently: 1: Yes; 0: No.	0
[7:0]	NESTSTA	RO	Current interrupt nesting state, used to query the core interrupt nesting state. Corresponds to the nest_sta bit in the INEST_CTRLR register of the CSR register.	0x00

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.14 PFIC VTF Interrupt ID Configuration Register (PFIC_VTFIDR)

Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VTFID3								VTFID2							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VTFID1								VTFID0							

Bit	Name	Access	Description	Reset value
[31:24]	VTFID3	RW	Configure VTF interrupt 3 ID	X
[23:16]	VTFID2	RW	Configure VTF interrupt 2 ID	X
[15:8]	VTFID1	RW	Configure VTF interrupt 1 ID	X
[7:0]	VTFID0	RW	Configure VTF interrupt 0 ID	X

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.15 PFIC VTF Interrupt 0 Address Register (PFIC_VTFADDR0)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR0[15:1]														VTF0EN	

Bit	Name	Access	Description	Reset value
[31:1]	ADDR0	RW	VTF interrupt 0 service program address bit[31:1], bit0 is 0.	X
0	VTF0EN	RW	VTF interrupt 0 enable bit: 1: Enabled VTF interrupt 0 channel; 0: Disabled.	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.16 PFIC VTF Interrupt 1 Address Register (PFIC_VTFADDR1)

Offset address: 0x64

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR1[15:1]														VTF1EN	

Bit	Name	Access	Description	Reset value
[31:1]	ADDR1	RW	VTF interrupt 1 service program address bit[31:1], bit0 is 0.	X
0	VTF1EN	RW	VTF interrupt 1 enable bit: 1: Enabled VTF interrupt 1 channel; 0: Disabled.	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.17 PFIC VTF Interrupt 2 Address Register (PFIC_VTFADDRR2)

Offset address: 0x68

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR2[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR2[15:1]															VTF2EN

Bit	Name	Access	Description	Reset value
[31:1]	ADDR2	RW	VTF interrupt 2 service program address bit[31:1], bit0 is 0.	X
0	VTF2EN	RW	VTF interrupt 2 enable bit: 1: Enabled VTF interrupt 2 channel; 0: Disabled.	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.18 PFIC VTF Interrupt 3 Address Register (PFIC_VTFADDRR3)

Offset address: 0x6C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR3[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR3[15:1]															VTF3EN

Bit	Name	Access	Description	Reset value
[31:1]	ADDR3	RW	VTF interrupt 3 service program address bit[31:1], bit0 is 0.	X
0	VTF3EN	RW	VTF interrupt 3 enable bit: 1: Enabled VTF interrupt 3 channel;	0

			0: Disabled.	
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Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.19 PFIC Interrupt Enable Set Register 1 (PFIC_IENR1)

Offset address: 0x100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	INTEN1 4	Reser ved	INTEN1 2	Reserved											

Bit	Name	Access	Description	Reset value
[31:12]	INTEN	WO	16#-31# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0
[11:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.20 PFIC Interrupt Enable Set Register 2 (PFIC_IENR2)

Offset address: 0x104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTEN[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	INTEN	WO	32#-63# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0

4.7.5.21 PFIC Interrupt Enable Set Register 3 (PFIC_IENR3)

Offset address: 0x108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTEN[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	INTEN	WO	64#-95# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0

4.7.5.22 PFIC Interrupt Enable Set Register 4 (PFIC_IENR4)

Offset address: 0x10C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTEN[111:96]															

Bit	Name	Access	Description	Reset value
[31:0]	INTEN	WO	96#-127# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0

4.7.5.23 PFIC Interrupt Enable Set Register 5 (PFIC_IENR5)

Offset address: 0x10C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											INTEN[148:144]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTEN[143:128]															

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	INTEN	WO	128#-148# interrupt enable control. 1: Current numbered interrupt enable; 0: No effect.	0

4.7.5.24 PFIC Interrupt Enable Clear Register 1 (PFIC_IRER1)

Offset address: 0x180

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTR SET1 5	INTR SET1 4	INTR SET1 3	INTR SET1 2	Reserved											

Bit	Name	Access	Description	Reset value
[31:12]	INTRSET	WO	12#-31# interrupt shutdown control.	0

			1: Current numbered interrupt shutdown; 0: No effect.	
[11:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.25 PFIC Interrupt Enable Clear Register 2 (PFIC_IRER2)

Offset address: 0x184

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	INTRSET	WO	32#-63# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0

4.7.5.26 PFIC Interrupt Enable Clear Register 3 (PFIC_IRER3)

Offset address: 0x188

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	INTRSET	WO	64#-95# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0

4.7.5.27 PFIC Interrupt Enable Clear Register 4 (PFIC_IRER4)

Offset address: 0x18C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[111:96]															

Bit	Name	Access	Description	Reset value
[31:0]	INTRSET	WO	96#-127# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0

4.7.5.28 PFIC Interrupt Enable Clear Register 5 (PFIC_IRER5)

Offset address: 0x190

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											INTRSET[148:144]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[143:128]															

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	INTRSET	WO	128#-148# interrupt shutdown control. 1: Current numbered interrupt shutdown; 0: No effect.	0

4.7.5.29 PFIC Interrupt Pending Set Register 1 (PFIC_IPSR1)

Offset address: 0x200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PEND SET1 5	PEND SET1 4	PEND SET1 3	PEND SET1 2	Reserved		PEN D SET9	PEN D SET8	Reserved		PEN D SET5	Reser ved	PEN D SET3	PEN D SET2	Reserved	

Bit	Name	Access	Description	Reset value
[31:12]	PENDSET	WO	12#-31# interrupt pending settings. 13# and 15# reserved. 1: Current numbered interrupt pending; 0: No effect.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	PENDSET	RO	8#-9# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0
[7:6]	Reserved	RO	Reserved	0
5	PENDSET	RO	5# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

4	Reserved	RO	Reserved	0
[3:2]	PENDSET	WO	2#-3# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0
[1:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.30 PFIC Interrupt Pending Set Register 2 (PFIC_IPSR2)

Offset address: 0x204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[63:48]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDSET	WO	32#-63# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

4.7.5.31 PFIC Interrupt Pending Set Register 3 (PFIC_IPSR3)

Offset address: 0x208

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[95:80]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDSET	WO	64#-95# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

4.7.5.32 PFIC Interrupt Pending Set Register 4 (PFIC_IPSR4)

Offset address: 0x20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[127:112]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

PENDSET[111:96]

Bit	Name	Access	Description	Reset value
[31:0]	PENDSET	WO	96#-127# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

4.7.5.33 PFIC Interrupt Pending Set Register 5 (PFIC_IPSR5)

Offset address: 0x210

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											PENDSET[148:144]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET[143:128]															

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	PENDSET	WO	128#-148# interrupt pending settings. 1: Current numbered interrupt pending; 0: No effect.	0

4.7.5.34 PFIC Interrupt Pending Clear Register 1 (PFIC_IPRR1)

Offset address: 0x280

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PEND RST1 5	PEND RST1 4	PEND RST1 3	PEND RST1 2	Reserved		PEN D RST9	PEN D RST8	Reserved		PEN D RST5	Reser ved	PEN D RST3	PEN D RST2	Reserved	

Bit	Name	Access	Description	Reset value
[31:12]	PENDRST	WO	12#-31# interrupt pending clear. 13# and 15# reserved. 1: Current numbered interrupt pending; 0: No effect.	0
[11:10]	Reserved	RO	Reserved	0
[9:8]	PENDRST	RO	8#-9# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

[7:6]	Reserved	RO	Reserved	0
5	PENDRST	RO	5# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0
4	Reserved	RO	Reserved	0
[3:2]	PENDRST	WO	2#-3# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0
[1:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.35 PFIC Interrupt Pending Clear Register 2 (PFIC_IPRR2)

Offset address: 0x284

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDRST	WO	32#-63# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

4.7.5.36 PFIC Interrupt Pending Clear Register 3 (PFIC_IPRR3)

Offset address: 0x288

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDRST	WO	64#-95# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

4.7.5.37 PFIC Interrupt Pending Clear Register 4 (PFIC_IPRR4)

Offset address: 0x28C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[111:96]															

Bit	Name	Access	Description	Reset value
[31:0]	PENDRST	WO	96#-127# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

4.7.5.38 PFIC Interrupt Pending Clear Register 5 (PFIC_IPRR5)

Offset address: 0x290

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											PENDRST[148:144]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[143:128]															

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	PENDRST	WO	128#-148# interrupt pending clear. 1: Current numbered interrupt clears the pending state; 0: No effect.	0

4.7.5.39 PFIC Interrupt Activation Register 1 (PFIC_IACR1)

Offset address: 0x300

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACT S15	IACT S14	IACT S13	IACT S12	Reserved								IACT S3	IACT S2	Reserved	

Bit	Name	Access	Description	Reset value
[31:12]	IACTS	RO	12#-31# interrupt execution status.	0

			1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	
[11:4]	Reserved	RO	Reserved	0
[3:2]	IACTS	RO	2#-3# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0
[1:0]	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.40 PFIC Interrupt Activation Register 2 (PFIC_IACR2)

Offset address: 0x304

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	IACTS	RO	32#-63# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0

4.7.5.41 PFIC Interrupt Activation Register 3 (PFIC_IACR3)

Offset address: 0x308

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	IACTS	RO	64#-95# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0

4.7.5.42 PFIC Interrupt Activation Register 4 (PFIC_IACR4)

Offset address: 0x30C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[127:112]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[111:96]															

Bit	Name	Access	Description	Reset value
[31:0]	IACTS	RO	96#-127# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0

4.7.5.43 PFIC Interrupt Activation Register 5 (PFIC_IACR5)

Offset address: 0x310

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											IACTS[148:144]				

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[143:128]															

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	IACTS	WO	128#-148# interrupt execution status. 1: Current numbered interrupt is executing; 0: Current numbered interrupt not executed.	0

4.7.5.44 PFIC Interrupt Priority Configuration Register (PFIC_IPRIORx) (x=0-63)

Offset address: 0x400 - 0x4FF

The controller supports 256 interrupts (0-255), and the priority of each interrupt is controlled by 8bits.

	31	24	23	16	15	8	7	0
IPRIOR63	PRIO_255		PRIO_254		PRIO_253		PRIO_252	
...	...		...		...		...	
IPRIORx	PRIO_ (4x+3)		PRIO_ (4x+2)		PRIO_ (4x+1)		PRIO_ (4x)	
...	...		...		...		...	
IPRIOR0	PRIO_3		PRIO_2		PRIO_1		PRIO_0	

Bit	Name	Access	Description	Reset value
[2047:2040]	IP_255	RW	Same as IP_0 description.	0
...	...	...	...	...
[31:24]	IP_3	RW	Same as IP_0 description.	0
[23:16]	IP_2	RW	Same as IP_0 description.	0

[15:8]	IP_1	RW	Same as IP_0 description.	0
[7:0]	IP_0	RW	No. 0 Interrupt Priority Configuration: [7:4]: Priority control bits: the lower the value, the higher the priority. The preemption bit is controlled by the CSR register's preempt flag. When the preempt register value is 0, there is no preemption bit. When the preempt register value is 1, bit 7 is the preemption bit. When the preempt register value is 2, bits 7 through 6 are the preemption bits. When the preempt register value is 3, bits 7 through 5 are the preemption bits. The V3F core supports up to 2 levels of nesting. Preemption cannot occur when nesting reaches its maximum level. The V5F core supports up to 8 levels of nesting. The nesting level is configurable. Preemption cannot occur when nesting reaches its maximum level. [3:0]: fixed to 0.	0

4.7.5.45 PFIC Interrupt Allocation Register (PFIC_IALLOCx) (x=0-63)

Offset address: 0x600 - 0x6FF

	31	24	23	16	15	8	7	0
IALLOC63	IALLOC_255		IALLOC_254		IALLOC_253		IALLOC_252	
...	...		...		...		...	
IALLOCx	IALLOC_(4x+3)		IALLOC_(4x+2)		IALLOC_(4x+1)		IALLOC_(4x)	
...	...		...		...		...	
IALLOC0	IALLOC_3		IALLOC_2		IALLOC_1		IALLOC_0	

Bit	Name	Access	Description	Reset value
[2047:2040]	IALLOC_255	RW	Same as IALLOC_32 description.	X
...	...	...	...	...
[263:256]	IALLOC_33	RW	Same as IALLOC_32 description.	X
[255:248]	IALLOC_32	RW	Interrupt allocation register #32 This system contains 2 cores, with only bit 0 configurable. When the ENA bit for this interrupt is enabled by either core, this register automatically updates to reflect the	X

			core ID that set the ENA register. 1: This interrupt is assigned to C1 for processing. 0: This interrupt is assigned to C0 for processing.	
[247:240]	IALLOC_31	RO	Same as IALLOC_0 description.	X
...	...	...	...	...
[31:24]	IALLOC_3	RO	Same as IALLOC_0 description.	X
[23:16]	IALLOC_2	RO	Same as IALLOC_0 description.	X
[15:8]	IALLOC_1	RO	Same as IALLOC_0 description.	X
[7:0]	IALLOC_0	RO	Interrupt allocation for Pin#0: This system has 2 cores. Only pin 0 is configurable. The C0 core reads a value of 0, while the C1 core reads a value of 1. 1: This interrupt is assigned to C1 for handling. 0: This interrupt is assigned to C0 for handling.	X

4.7.5.46 PFIC Interrupt Authority Register 1 (PFIC_IATR1)

Offset address: 0x700

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IAUT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IAUT[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	IAUT	RO	31#-0# Interrupt allocation register Used to determine whether an interrupt is assigned to this core for response. 1: Interrupt assigned to this core; 0: Interrupt not assigned to this core.	0xFFFFFFFF

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.47 PFIC Interrupt Authority Register 2 (PFIC_IATR2)

Offset address: 0x704

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IAUT[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IAUT[47:32]															

Bit	Name	Access	Description	Reset value
[31:0]	IAUT	RO	32#-63# Interrupt allocation register Used to determine whether an interrupt is assigned to this core for response. 1: Interrupt assigned to this core; 0: Interrupt not assigned to this core.	X

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.48 PFIC Interrupt Authority Register 3 (PFIC_IAUTR3)

Offset address: 0x708

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IAUT[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IAUT[79:64]															

Bit	Name	Access	Description	Reset value
[31:0]	IAUT	RO	64#-95# Interrupt allocation register Used to determine whether an interrupt is assigned to this core for response. 1: Interrupt assigned to this core; 0: Interrupt not assigned to this core.	X

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.49 PFIC Interrupt Authority Register 4 (PFIC_IAUTR4)

Offset address: 0x70C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IAUT[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IAUT[111:96]															

Bit	Name	Access	Description	Reset value
[31:0]	IAUT	RO	96#-127# Interrupt allocation register Used to determine whether an interrupt is assigned to this core for response. 1: Interrupt assigned to this core; 0: Interrupt not assigned to this core.	X

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.50 PFIC Interrupt Authority Register 5 (PFIC_IAUTR5)

Offset address: 0x710

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											IAUT[148:144]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IAUT[143:128]															

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[31:0]	IAUT	RO	128#-148# Interrupt allocation register Used to determine whether an interrupt is assigned to this core for response. 1: Interrupt assigned to this core; 0: Interrupt not assigned to this core.	X

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.51 PFIC Wakeup Instruction Pointer Register 0 (PFIC_WAKEIP0)

Offset address: 0x720

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IP_RELOAD0[30:15]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IP_RELOAD0[14:0]														SHU TDO WN0	

Bit	Name	Access	Description	Reset value
[31:1]	IP_RELOAD0	RW	PC address register during core C0 wake-up, used for reloading the PC value after power-on wake-up and reset wake-up.	0
0	SHUTDOWN0	RW	Core C0 Deep Sleep (Lock) Unlock Register: Used to unlock a core that entered deep sleep (lock) state after power-on, allowing it to be woken by the timer: 1: Locked; 0: Unlocked.	1

4.7.5.52 PFIC Wakeup Instruction Pointer Register 1 (PFIC_WAKEIP1)

Offset address: 0x724

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IP_RELOAD1[30:15]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IP_RELOAD1[14:0]														SHU TDO WN0	

Bit	Name	Access	Description	Reset value
[31:1]	IP_RELOAD1	RW	PC address register during core C1 wake-up, used for reloading the PC value after power-on wake-up and reset wake-up.	0
0	SHUTDOWN1	RW	Core C1 Deep Sleep (Lock) Unlock Register: Used to unlock a core that entered deep sleep (lock) state after power-on, allowing it to be woken by the timer: 1: Locked; 0: Unlocked.	1

4.7.5.53 PFIC Core Status Register 0 (PFIC_CSTAR0)

Offset address: 0x780

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CPU_ex_state_0	CPU_lock_up_0	CPU_dbg_mode_0	CPU_globl_ie_0	Reserved	CPU_irq_pending_0	CPU_irq_active_0	CPU_nest_sta_0								

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	CPU_ex_state_0	RO	Core C0 Status Register, used to obtain the operational status of core C0: 00: Running; 01: Light Sleep; 10: Deep Sleep; 11: Deep Sleep (Locked).	0
13	CPU_lock_up_0	RO	Core C0 Lock Status Register, used to determine whether core C0 is in a locked state: 1: Locked; 0: Unlocked.	0

12	CPU_dbg_mode_0	RO	Core C0 Debug Mode Register, used to determine whether the core C0 is in debug mode: 1: Core is in debug mode; 0: Core is in non-debug mode.	0
11	CPU_globl_ie_0	RO	Core C0 Global Interrupt Enable Register, used to query whether the core C0 global interrupt is enabled: 1: Core global interrupt enable is active; 0: Core global interrupt enable is inactive.	0
10	Reserved	RO	Reserved	0
9	CPU_irq_pend_0	RO	Core C0 Interrupt Pending Flag Register, used to check whether the core C0 has any unprocessed interrupts: 1: The core has unprocessed interrupts; 0: The core has no unprocessed interrupts.	0
8	CPU_irq_active_0	RO	Core C0 Interrupt Active Flag Register, used to determine whether the core is currently handling the C0 interrupt 1: Core is handling the interrupt; 0: Core is not handling the interrupt.	0
[7:0]	CPU_nest_sta_0	RO	The core C0 Interrupt Nesting Status Register, used to query the nesting status of core C0 interrupts. It corresponds to the nest_sta bit in the INEST_CTLR register of the CSR register.	0

4.7.5.54 PFIC Core Status Register 1 (PFIC_CSTAR1)

Offset address: 0x784

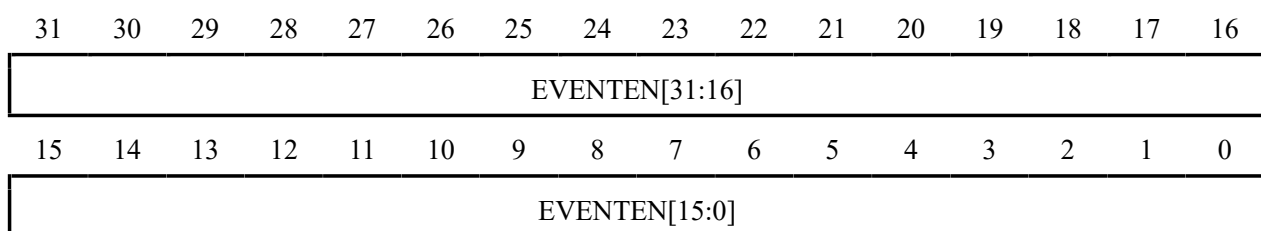
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CPU_ex_state_1	CPU_lock_up_1	CPU_dbg_mode_1	CPU_globl_ie_1	Reserved	CPU_irq_pend_1	CPU_irq_active_1	CPU_nest_sta_1								

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	CPU_ex_state_1	RO	Core C1 Status Register, used to obtain the operational status of core C1: 00: Running; 01: Light Sleep; 10: Deep Sleep;	0

			11: Deep Sleep (Locked).	
13	CPU_lock_up_1	RO	Core C1 Lock Status Register, used to determine whether core C1 is in a locked state: 1: Locked; 0: Unlocked.	0
12	CPU_dbg_mode_1	RO	Core C1 Debug Mode Register, used to determine whether the core C1 is in debug mode: 1: Core is in debug mode; 0: Core is in non-debug mode.	0
11	CPU_globl_ie_1	RO	Core C1 Global Interrupt Enable Register, used to query whether the core C1 global interrupt is enabled: 1: Core global interrupt enable is active; 0: Core global interrupt enable is inactive.	0
10	Reserved	RO	Reserved	0
9	CPU_irq_pend_1	RO	Core C1 Interrupt Pending Flag Register, used to check whether the core C0 has any unprocessed interrupts: 1: The core has unprocessed interrupts; 0: The core has no unprocessed interrupts.	0
8	CPU_irq_active_1	RO	Core C1 Interrupt Active Flag Register, used to determine whether the core is currently handling the C1 interrupt 1: Core is handling the interrupt; 0: Core is not handling the interrupt.	0
[7:0]	CPU_nest_sta_1	RO	The core C1 Interrupt Nesting Status Register, used to query the nesting status of core C1 interrupts. It corresponds to the nest_sta bit in the INEST_CTLR register of the CSR register.	0

4.7.5.55 PFIC Event Enable Register (PFIC_EENR)

Offset address: 0xC80



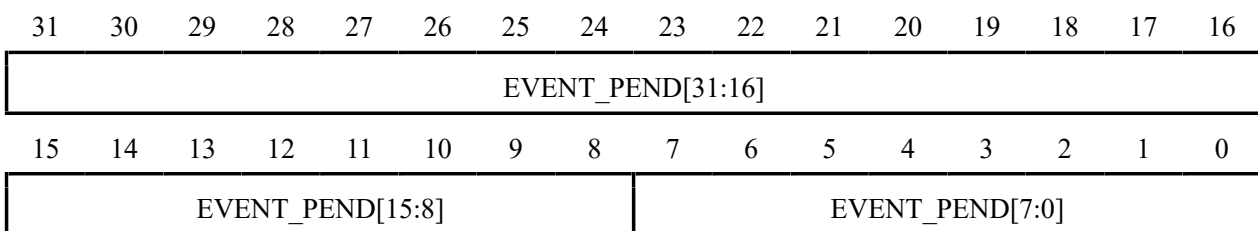
Bit	Name	Access	Description	Reset value
[31:0]	EVENTEN	RW	31#-0# event wake-up enable:	0xFFFFFFFF

			1: Allow the corresponding event to wake up the core in sleep; 0: Mask the corresponding event, and it is not allowed to wake up the core when this event occurs. Up to 32 event sources are supported, of which 0-7 is level triggered (high active) and 8-31 is pulse triggered: 0#: Core pause request; 1#: Core interrupt request; 7#-2#: Reserved; 8#: Core enters interrupt; 9#: Core exit interrupt; 10#: A new interrupt is pending; (For shared interrupts, it is not distinguished whether the interrupt is routed to the local core.) 11#: The core sends an event; (Send event of local core) 12#: The core receives an event; (Send events from other cores) 13#: The system receives the event; (System-configured events, such as GPIO flip events ...) 31#-14#: Reserved.	F
--	--	--	---	---

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.56 PFIC Event Pending Register (PFIC_EPR)

Offset address: 0xC84



Bit	Name	Access	Description	Reset value
[31:8]	EVENT_PEND	RW1Z	31#-8# event pending status, write 1 to clear it: 1: The occurrence of the corresponding event was not handled; 0: The corresponding event did not occur.	0

[7:0]	EVENT_PEND	RO	7#-0# event pending status, cannot be cleared: 1: The occurrence of the corresponding event was not handled; 0: The corresponding event did not occur.	0
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Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.57 PFIC Event Wakeup Register (PFIC_EWUPR)

Offset address: 0xC88

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVENT_WUP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENT_WUP[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	EVENT_WUP	RO	31#-0# event wake-up register: 1: Corresponding events wake the core from sleep; 0: Corresponding event is not the event that wakes up the core this time.	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

4.7.5.58 PFIC System Control Register (PFIC_SCTLR)

Offset address: 0xD10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SYS RST	Reserved							HART_ID							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										SET EVE NT	SEV ONP END	WFIT O WFE	SLEE P DEEP	SLEE P ONE XIT	Reser ved

Bit	Name	Access	Description	Reset value
31	SYSRST	WO	System reset, auto clear 0. Write 1 is valid, write 0 is invalid, same effect as PFIC_CFGR	0

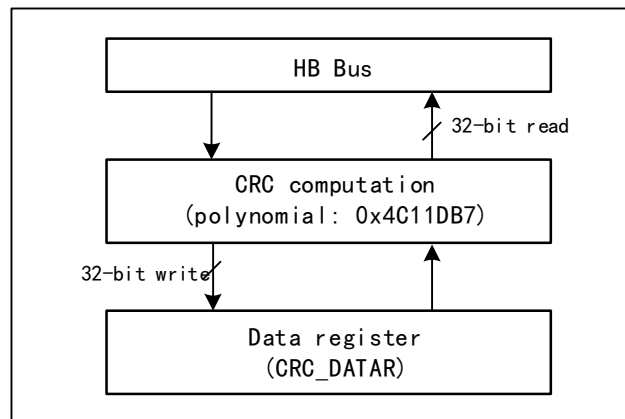
			register.	
[30:24]	Reserved	RO	Reserved	0
[23:16]	HART_ID	RO	Core ID Register, used to obtain the core ID of the register being read: 1: This core is C1; 0: This core is C0.	0
[15:6]	Reserved	RO	Reserved	0
5	SETEVENT	WO	Set up events that can wake up the WFE situation.	0
4	SEVONPEND	RW	When an event occurs or interrupts the pending state, the system can be woken up from the WFE instruction, and if the WFE instruction is not executed, the system will be awakened immediately after the next execution of the instruction. 1: Enabled events and all interrupts (including unopened interrupts) can wake up the system. 0: Only enabled events and enabled interrupts can wake up the system.	0
3	WFIOWFE	RW	The WFI instruction executes as WFE, which is a custom instruction. When this register is set to 1, the core enters a low-power state after executing the WFI instruction and waits for an external event to wake up the core. 1: After the WFI instruction, wait for an event to wake up the core; 0: After the WFI instruction, wait for an interrupt to wake up the core.	0
2	SLEEPDEEP	RW	Control the low-power mode of the system: 1: deepsleep 0: sleep	0
1	SLEEPONEXIT	RW	Controls the state of the system after leaving the interrupt service program: 1: The system enters low-power mode; 0: The system enters the main program.	0
0	Reserved	RO	Reserved	0

Note: This register is a core-private register. That is, within the peripheral, multiple sets of registers with identical functions exist, each containing mutually independent data. These data cannot be accessed or modified by other cores; only the core shares the same access address for them.

Chapter 5 Cyclic Redundancy Check (CRC)

The cyclic redundancy check (CRC) computation unit is used to obtain the result of the CRC computation for any 32-bit data based on a fixed generating polynomial. It is generally used in the field of data storage and data communication to verify the correctness of data. The system provides hardware CRC calculation unit which can greatly save CPU and RAM resources to improve efficiency.

Figure 5-1 CRC structure diagram



5.1 Main Features

- CRC32 polynomial (0x4C11DB7): $X^{32}+X^{26}+X^{23}+X^{22}+X^{16}+X^{12}+X^{11}+X^{10}+X^8+X^7+X^5+X^4+X^2+X+1$;
- Same 32-bit register as input for data and output for CRC32 calculation
- Single conversion time: 4 HB clock cycles (HCLK)

5.2 Function Description

- CRC unit reset

To start a CRC calculation for a new data set, the CRC calculation unit needs to be reset. Writing a 1 to the RST bit of the control register CRC_CTLR will reset the data register by the hardware, restoring the initial value 0xFFFFFFFF.

- CRC calculation

The CRC unit calculates the CRC result of the previous CRC calculation and the CRC result of the newly involved data. The CRC_DATAR data register, for which a write operation will feed new data to the hardware calculation unit; and a read operation will get the value of the latest round of CRC calculation. The hardware calculation interrupts the system write operation, so new values can be written continuously.

Note: The CRC unit calculates the entire 32-bit data, not byte-by-byte.

- Independent data buffer

The CRC unit provides an 8-bit independent data register, CRC_IDATAR, which is used for the application code to temporarily store 1 byte of data independent of the CRC unit reset.

5.3 Register Description

Table 5-1 CRC-related registers list

Name	Access address	Description	Reset value
R32_CRC_DATAR	0x40023000	Data register	0xFFFFFFFF
R8_CRC_IDATAR	0x40023004	Independent data buffer	0x00
R32_CRC_CTLR	0x40023008	Control register	0x00000000

5.3.1 Data Register (CRC_DATAR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	DR[31:0]	RW	Write raw data; read calculations.	0xFFFFFFFF

5.3.2 Independent Data Buffer (CRC_IDATAR)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[7:0]	IDR[7:0]	RW	An 8-bit general-purpose register that can be used as a data cache, this register is not affected by the RST field of the control register.	0

5.3.3 Control Register (CRC_CTLR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														RST	

Bit	Name	Access	Description	Reset value
[31:1]	Reserved	RO	Reserved	0
0	RST	WO	CRC calculation unit reset control, write 1 execution, hardware auto clear, after execution, data register is 0xFFFFFFFF.	0

Chapter 6 Real Time Clock (RTC)

The Real Time Clock (RTC) is a standalone timer module with a programmable counter up to 32 bits, which can be used with software to realize the real time clock function, and the counter value can be modified to reconfigure the current time and date of the system. The RTC module is in the backup power supply area, and the system reset and wake-up from Standby mode do not have any effect on it.

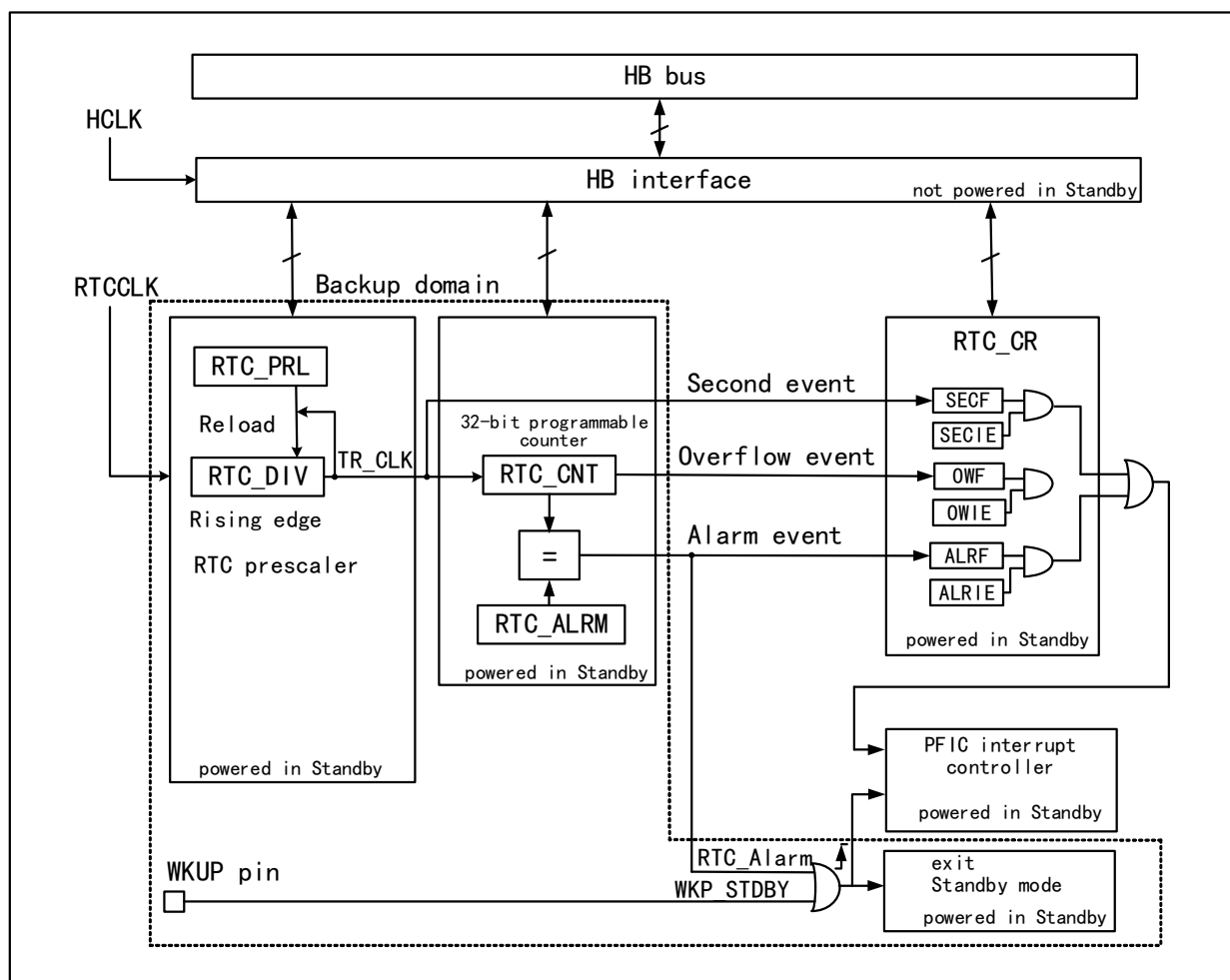
6.1 Main Features

- Prescaler coefficients up to 2^{20}
- 32-bit programmable counter
- Multiple clock sources, interrupts
- Independent reset

6.2 Function Description

6.2.1 Overview

Figure 6-1 RTC structure diagram



As shown in figure 6-1, the RTC module is mainly composed of 3 parts: HB bus interface, frequency divider and counter, control and status register, in which the frequency divider and counter are in the backup area and can be

powered by V_{BAT} . After inputting the frequency divider (RTC_DIV), the RTCK is divided into TR_CLK. It is worth noting that the inside of the frequency divider (RTC_DIV) is a self-subtractive counter, which will output a TR_CLK from the overrun, then take the default value from the reload value register (RTC_PSCR) and reinstall it into the frequency divider, the read divider actually reads its real-time value (read only), and the write division factor should be written to the reload value register (RTC_PSCR). Generally, when the period of TR_CLK is set to 1 second, TR_CLK will trigger the second event and increase the main counter (RTC_CNT) by 1; when the main counter is increased to the same value as the alarm clock register, the alarm clock event will be triggered; when the master counter increases to overflow, the overflow event will be triggered. All of the above three events can trigger interrupts and correspond to the corresponding interrupts to enable level control.

6.2.2 Reset

Due to the special purpose of the real-time clock, the four sets of registers that are in the backup domain: the prescaler, prescaler reload value, master counter, and alarm clock, can only be reset by the reset signal in the backup domain, refer to the RCC's Backup Domain Reset chapter. The control registers of the real-time clock are controlled by a system reset or a power reset.

6.2.3 Special Read-Write Register Operation

Due to the special use of the real-time clock, RTC and HB buses are independent, and the reading of RTC by HB is not necessarily real-time. Reading the register of RTC through HB must be after HB startup and passing through a RTC rising edge. This situation may occur after system reset and power reset, after waking up from Standby or Stop mode. It is convenient to wait for the RSF bit of the control register (CTLR) to be set high. The write operator for RTC must wait for the end of the last write operation and must enter configuration mode. The specific steps are as follows:

- 1) Query the RTOFF bit until it becomes 1.
- 2) Set the CNF bit and enter the configuration mode.
- 3) Write to one or more RTC registers.
- 4) Clear the CNF bit, exit the configuration mode, and the HB interface starts to write the RTC register.
- 5) Query the RTOFF bit until it becomes 1.

6.3 Register Description

Table 6-1 BKP-related registers list

Name	Access address	Description	Reset value
R16_RTC_CTLRH	0x40002800	RTC control register high	0x0000
R16_RTC_CTLRL	0x40002804	RTC control register low	0x0020
R16_RTC_PSCRH	0x40002808	Prescaler reload register high	0x0000
R16_RTC_PSCRL	0x4000280C	Prescaler reload register low	0xFFFF
R16_RTC_DIVH	0x40002810	Divider register high	0xFFFF
R16_RTC_DIVL	0x40002814	Divider register low	0xFFFF
R16_RTC_CNTH	0x40002818	RTC counter high	0xFFFF
R16_RTC_CNTL	0x4000281C	RTC counter low	0xFFFF
R16_RTC_ALRMH	0x40002820	Alarm clock register high	0xFFFF
R16_RTC_ALRML	0x40002824	Alarm clock register low	0xFFFF

6.3.1 RTC Control Register High (RTC_CTLRH)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	RO	Reserved	0
2	OWIE	RW	Overflow interrupt enable: 1: Enable; 0: Disable.	0
1	ALRIE	RW	Alarm interrupt enable: 1: Enable; 0: Disable.	0
0	SECIE	RW	Second interrupt enable: 1: Enable; 0: Disable.	0

6.3.2 RTC Control Register Low (RTC_CTLRL)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:6]	Reserved	RO	Reserved	0
5	RTOFF	RO	RTC operation status indication bit, indicates the execution status of the last operation to the RTC, the operation to the RTC must wait for this bit to be 1. 1: The last operation on the RTC has been completed; 0: The last operation on the RTC is still in progress.	1
4	CNF	RW	Configuration flag bit, writing this bit by 1 enters the configuration mode, thereby allowing values to be written to the Counter (R16_RTC_CNTx), the Alarm Clock Register (R16_RTC_ALRMx) and the Prescaler Reload Value Register (R16_RTC_PSCRx). The write operation is performed only after the bit is written 1 and re-cleared 0 by the software: 1: Enter configuration mode; 0: Exit configuration mode and start updating the RTC registers.	0
3	RSF	RW0	Register synchronization flag bit, before reading and writing registers such as pre-division frequency (PSCRx), alarm clock (ALRMx) and counter (CNTx) of RTC module, make sure that this bit has been set by hardware to make sure that these registers have been synchronized; when reading and writing these registers, or after HB reset or HB clock stops, the first step should reset this bit. 1: The register is synchronized; 0: The register is not synchronized.	0

2	OWF	RW0	Counter overflow flag, this bit is set by hardware when the 32-bit counter overflows. An overflow interrupt is also generated if the OWIE bit is set. This bit can only be cleared by software and cannot be set by software.	0
1	ALRF	RW0	Alarm Clock Flag, this bit is set by hardware when the counter value reaches the value of the Alarm Clock Register (ALRMx), and an alarm clock interrupt is also generated if the Alarm Clock Interrupt Enable Bit (ALRIE) is set. This bit can only be cleared by software and cannot be set by software.	0
0	SECF	RW0	The second event flag, when the clock is divided by the prescaler after each falling edge, will cause the counter to increment by one, and at the same time generate a second event, this bit will be set, if the second interrupt is enabled (SECIE is set), and at the same time will also generate a second interrupt. This bit can only be cleared by software and cannot be set by software.	0

6.3.3 Prescaler Reload Register High (RTC_PSCRH)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[15:4]	Reserved	RO	Reserved	0
[3:0]	PRL[19:16]	WO	Reload value high	0

6.3.4 Prescaler Reload Register Low (RTC_PSCRL)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:0]	PRL[15:0]	WO	The reloading value is low. The actual frequency division factor is (PRL [19:0] + 1). If the RTC input frequency is 32768Hz, then this value is set to 0x7fff, you can divide the signal with a period of 1 second.	X

6.3.5 Divider Register High (RTC_DIVH)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[15:4]	Reserved	RO	Reserved	0
[3:0]	DIV[19:16]	RO	Divider register high.	x

6.3.6 Divider Register Low (RTC_DIVL)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:0]	DIV[15:0]	RO	Divider register low. DIV is actually a self-subtractive counter. Every time a clock comes to RTC_CLK, the DIV counter is subtracted by 1. After the overflow, a TR_CLK is output and the value is reloaded from the PSCR. DIV can only read and read the remaining value of the counter of the current frequency divider.	X

6.3.7 RTC Counter High (RTC_CNTH)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[15:0]	CNT[31:16]	RW	Counter High	X

6.3.8 RTC Counter Low (RTC_CNTL)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	Counter low, the core device of the RTC timer, the clock is provided by the TRCLK (the period is generally set to 1 second). Calculate the current time by reading CNT [31:0]. To write this value, you need to enter configuration mode.	X

6.3.9 Alarm Register High (RTC_ALRMH)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[15:0]	ALR[31:16]	WO	Alarm register high	X

6.3.10 Alarm Register Low (RTC_ALRML)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[15:0]	ALR[15:0]	WO	Alarm clock register is low. An alarm clock event occurs when the value of the alarm clock register ALRM [31:0] is the same as that of the counter CNT [31:0]. Changing this value requires you to enter configuration mode.	X

Chapter 7 Independent Watchdog (IWDG)

The system is equipped with an independent watchdog (IWDG) to detect software failures caused by logic errors and external environment interference. The IWDG clock source comes from LSI and can be run independently of the main program, so it is suitable for situations with low precision requirements.

7.1 Main Features

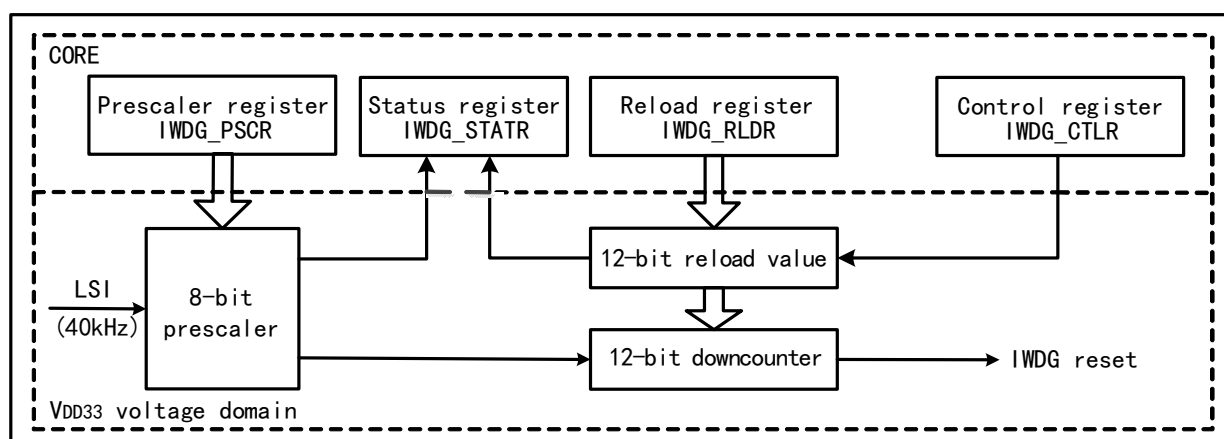
- 12-bit self-subtractive counter.
- Clock source LSI frequency division, can run in low-power mode.
- Reset condition: counter value reduced to 0

7.2 Functional Description

7.2.1 Principle and Application

The independent watchdog is clocked from the LSI clock and functions in shutdown and standby modes. When the watchdog counter decrements itself to 0, a system reset will be generated, so the timeout is (reload value + 1) clock.

Figure 7-1 Structure block diagram of Independent Watchdog



- Enable independent watchdog

After the system reset, the watchdog is OFF. Write 0xCCCC to the IWDG_CTLR register to enable the watchdog, and then it can no longer be disabled unless a reset occurs.

If the hardware independent watchdog enable bit (IWDG_SW) is enabled in User Option Bytes, the IWDG is permanently enabled after the microcontroller reset.

- Watchdog configuration

Inside the watchdog is a 12-bit counter running progressively. When the value of the counter is reduced to 0, a system reset will occur. To enable the IWDG function, you need to perform the following actions:

1) Count time base: IWDG clock source LSI, through the IWDG_PSCR register to set the LSI frequency division value clock as the IWDG count time base. The operation method first writes 0x5555 to the IWDG_CTLR register,

and then modifies the frequency division value in the IWDG_PSCR register. The PVU bit in the IWDG_STATR register indicates the update status of the frequency division value, and the frequency division value can only be modified and read out when the update is completed.

2) Reload value: used to update the current value of the counter in the independent watchdog, and the counter is decremented by this value. The operation method first writes 0x5555 to the IWDG_CTLR register, and then modifies the IWDG_RLDR register to set the target reload value. The RUV bit in the IWDG_STATR register indicates the update status of the reload value, and the IWDG_RLDR register can be modified and read out only after the update is completed.

3) Watchdog enable: write 0xCCCC to the IWDG_CTLR register to turn on the watchdog function.

4) Feeding the dog: that is, before the watchdog counter decreases to 0, refresh the current counter value to prevent system reset. Write 0xAAAA to the IWDG_CTLR register and have the hardware update the IWDG_RLDR register value to the watchdog counter. This action needs to be performed regularly after the watchdog function is turned on, otherwise the watchdog reset action will occur.

7.2.2 Debug Mode

When the system enters the debug mode, the IWDG counter can be configured by the debug module register to continue working or stop.

7.3 Register Description

Table 7-1 IWDG-related registers

Name	Access address	Description	Reset value
R16_IWDG_CTLR	0x40003000	IWDG Control register	0x0000
R16_IWDG_PSCR	0x40003004	IWDG Prescaler register	0x0000
R16_IWDG_RLDR	0x40003008	IWDG Reload Value register	0x0FFF
R16_IWDG_STATR	0x4000300C	IWDG Status register	0x0000

7.3.1 IWDG Control Register (IWDG_CTLR)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
[15:0]	KEY[15:0]	WO	Operate the key value lock. 0xAAAA: Feed the dog. Load the IWDG_RLDR register value into the independent watchdog counter. 0x5555: Allow modification of R16_IWDG_PSCR and R16_IWDG_RLDR registers. 0xCCCC: Start the watchdog, which is not subject to this restriction if the hardware watchdog is enabled (user chooses word configuration).	0

7.3.2 IWDG Prescaler Register (IWDG_PSCR)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	RO	Reserved	0
[2:0]	PR[2:0]	RW	IWDG clock division factor, write 0x5555 to KEY before modifying this field. 000: Divided by 4; 001: Divided by 8; 010: Divided by 16; 011: Divided by 32; 100: Divided by 64; 101: Divided by 128; 110: Divided by 256; 111: Divided by 256. IWDG count time base = LSI/division factor. <i>Note: Before reading the value of this field, make sure that the PVU bit in the IWDG_STATR register is 0, otherwise the read value is invalid.</i>	000b

7.3.3 IWDG Reload Value Register (IWDG_RLDR)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[15:12]	Reserved	RO	Reserved	0
[11:0]	RL[11:0]	RW	Counter reload value. Write 0x5555 to KEY before modifying this field. When 0xAAAA is written to KEY, the value of this field is loaded into the counter by the hardware, and the counter is then decremented from that value. <i>Note: Before reading and writing the field value, make sure that the RUV bit in the IWDG_STATR register is 0, otherwise it is invalid to read and write this field.</i>	FFFh

7.3.4 IWDG Status Register (IWDG_STATR)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:2]	Reserved	RO	Reserved	0
1	RVU	RO	The reinstall value updates the flag bit. Hardware setting or clearing 0. 1: Reload value update is in progress. 0: Reload update ends (up to 5 LSI cycles). <i>Note: The reload value register IWDG_RLDR can be read and written only after the RVU bit has been cleared.</i>	0
0	PVU	RO	Clock frequency division coefficient updates flag bits. Hardware setting or clearing 0. 1: Clock division value update is in progress.	0

			0: Clock division value update ends (up to 5 LSI cycles). <i>Note: The frequency division factor register IWDG_PSCR can be read and written only after the PVU bit has been cleared.</i>	
--	--	--	---	--

Note: After the pre-division or reinstallation value is updated, you do not have to wait for RVU or PVU to reset, you can continue to execute the following code. (This write operation continues to be completed even in low power mode.)

Chapter 8 Window Watchdog (WWDG)

The window watchdog is generally used to monitor the software failures of the system, such as external interference, unforeseen logic errors and so on. It needs to refresh the counter (feed the dog) within a specific window time (with upper and lower limits), otherwise the watchdog circuit will produce a system reset earlier or later than this window time.

8.1 Main Features

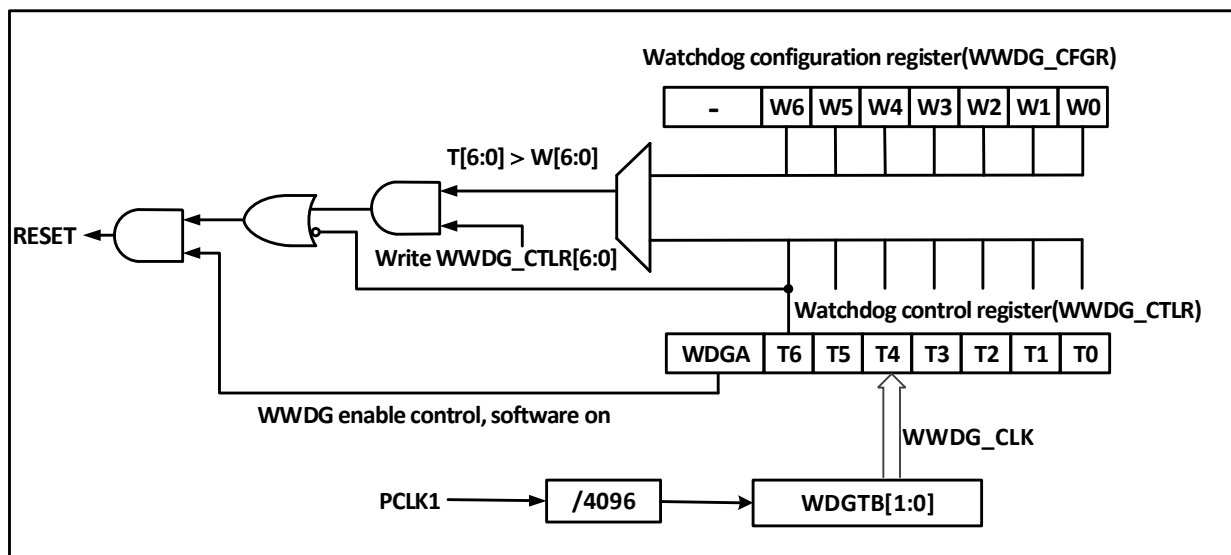
- Programmable 7-bit self-subtractive counter.
- Double conditional reset: the current counter value is less than 0x40, or the counter value is reloaded outside the window time.
- Wake up advance notice function (EWI), which is used to feed the dog in time to prevent system reset

8.2 Function Description

8.2.1 Principle and Application

The window watchdog runs based on a 7-bit decrement counter, which is mounted on the HB bus to count the frequency division of the time-based WWDG _ CLK source (PCLK1/4096) clock, and the frequency division factor is set in the WDG TB [1:0] field in the configuration register WWDG_CFGR. The decrement counter is in a state of free operation, regardless of whether the watchdog function is turned on or not, the counter has been cyclically decreasing counting. As shown in figure 8-1, the internal structure block diagram of the window watchdog.

Figure 8-1 Window Watchdog structure diagram



- Enable window watchdog

After the system reset, the watchdog is disabled. Set the WDGA bit in the WWDG_CTLR register to switch on the watchdog, and then it can no longer be disabled unless a reset occurs.

Note: WWDG clock source can be disabled by setting the RCC_HB1PCENR register to suspend WWDG_CLK counting and indirectly stop the watchdog function. Or reset the WWDG module by setting the RCC_HB1PRSTR register, which is equivalent to the function of reset.

- Watchdog configuration

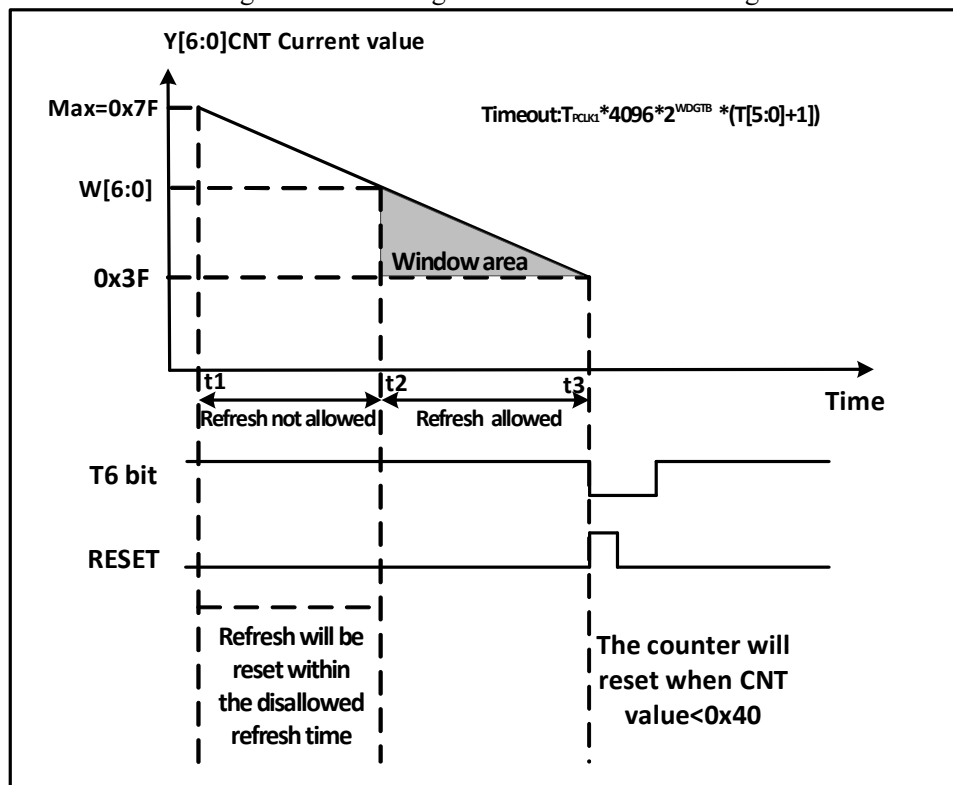
Inside the watchdog is a 7-bit counter running in a continuous cycle, which supports read and write access. To use the watchdog reset function, you need to perform the following actions:

- 1) Counter time base: The WDGTB[1:0] bits in the WWDG_CFGR register. Note to switch on the WWDG module clock of the RCC unit.
- 2) Window counter: Set the W[6:0] bits in the WWDG_CFGR register. This counter is used to be compared with the current counter by hardware, the value is configured by the user software and will not change. It serves as the maximum value of window time.
- 3) Watchdog enable: The WDGA bit in the WWDG_CTLR register is set to 1 by software, and the watchdog function is enabled to reset the system.
- 4) Feed dog: Refresh the current counter value and configure the T[6:0] bits in the WWDG_CTLR register. This action needs to be executed in the periodic window time after the watchdog function is enabled. Otherwise, the watchdog reset action occurs.

- Feed dog window time

As shown in Figure 8-2, the gray area is the detector window area of the window watchdog. Its maximum timeout (t_2) corresponds to the time point when the current counter value reaches the window value W[6:0]. Its minimum timeout (t_3) corresponds to the time point when the current counter value reaches 0x3F. Within this area time ($t_2 < t < t_3$), the feed dog operation can be performed (write T[6:0]) to refresh the current counter value.

Figure 8-2 Counting mode of window watchdog



- Watchdog reset:

- 1) When the feed dog operation is not performed in time, the value of the T[6:0] counter changes from 0x40 to 0x3F, a "Window Watchdog Reset" occurs, and a system reset occurs. I.e., when T6-bit is detected as 0 by hardware, the system reset occurs.

Note: The application program can write 0 to the T6-bit by software to implement system reset, which is equivalent to software reset function.

2) When the counter refresh action is executed when the feed dog operation is disabled, i.e., when write operation is performed on the T[6:0] bits when $t_1 \leq t \leq t_2$, a "window watchdog reset" occurs, and a system reset occurs.

- Early wake-up

To prevent the system from resetting due to failure to refresh the counter in time, the watchdog module provides early wake-up interrupt (EWI) notification. When the counter is reduced to 0x40, an early wake-up signal is generated, and the WEIF flag is set to 1. If the EWI bit is set, the window watchdog will be triggered to interrupt at the same time. At this point, there is a counter clock cycle (self-reduced to 0x3F) from the hardware reset, during which the application can immediately feed the dog.

8.2.2 Debug Mode

When the system enters debug mode, the WWDG counter can either continues to work normally or stops, depending on the debugging module register.

8.3 Register Description

Table 8-1 WWDG-related registers

Name	Access address	Description	Reset value
R16_WWDG_CTLR	0x40002C00	WWDG Control Register	0x007F
R16_WWDG_CFGR	0x40002C04	WWDG Configuration Register	0x007F
R16_WWDG_STATR	0x40002C08	WWDG Status Register	0x0000

8.3.1 WWDG Control Register (WWDG_CTLR)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	WDGA	RW1	The window watchdog resets the enable position. 1: Enable the watchdog function (can generate reset signal). 0: Disable the watchdog function. Software write 1 is on, but only hardware is allowed to clear 0 after reset.	0
[6:0]	T[6:0]	RW	7-bit self-subtractive counter, minus 1 per $4096 * 2^{WDGTB}$ PCLK1 cycle. When the counter is reduced from 0x40 to 0x3F, that is, when T6 jumps to 0, a watchdog reset is generated.	0x7F

8.3.2 WWDG Configuration Register (WWDG_CFGR)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0

9	EWI	RW1	Early wakeup interrupt: If it set to 1, interrupt is generated when the counter reaches 0x40. It can only be cleared by hardware after reset.	0
[8:7]	WDGTB[1:0]	RW	Window watchdog time base: 00: Divided by 1, counter time base =PCLK1/4096; 01: Divided by 2, counter time base =PCLK1/4096/2; 10: Divided by 4, counter time base = PCLK1/4096/4; 11: Divided by 8, counter time base = PCLK1/4096/8.	0
[6:0]	W[6:0]	RW	Window watchdog 7-bit window value. It is used to be compared with the counter value. The feed dog operation can be performed only when the counter value is less than the window value and is greater than 0x3F.	0x7F

8.3.3 WWDG Status Register (WWDG_STATR)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[15:1]	Reserved	WO	Reserved	0
0	EWIF	RW0	Early wakeup interrupt flag. When the counter reaches 0x40, this bit is set by hardware, and it must be cleared by software. User setting is invalid. Even if the EWI is not set, this bit is still set as usual when the event occurs.	0

Chapter 9 GPIO and Alternate Functions (GPIO/AFIO)

The chip incorporates six sets of GPIO ports (PA0~PA15, PB0~PB15, PC0~PC15, PD0~PD15, PE0~PE15, PF0~PF14), totaling 95 GPIO pins. Most pins can be software-configured as output (push-pull or open-drain), input (with or without pull-up/pull-down resistors), or as multiplexed peripheral function ports.

Pins PA9 to PA12, PC13 to PC15, and PE3 to PE6 are powered by V_{DD33} , rated at 3.3V. When V_{DD33} is de-energized, pins PC13 to PC15 automatically switch to being powered by V_{BAT} .

Pins PA0~PA8, PB2~PB7, PB15, PC4~PC5, PD8, PE2, and PF6~PF10 are powered by V_{DDIO} , rated at 3.3V, and support 1.8V, 2.5V, and 3.3V power supplies.

High-speed pins PA13~PA15, PB0~PB1, PB10~PB14, PC0~PC3, PC6~PC12, PD0~PD7, PD9~PD15, PE0~PE1, PE7 to PE15, PF0 to PF5, PF11 to PF14 are powered by V_{IO18} . The built-in I/O pin voltage regulator supports 1.2V, 1.8V, 2.5V, and 3.3V voltages with dynamic power supply voltage switching. It supports configuring the default voltage on the XO pin after power-up. For specific configuration details, refer to the [12:9] field of the PWR_CTLR register.

9.1 Main Features

Each pin of the port can be configured into one of the following modes:

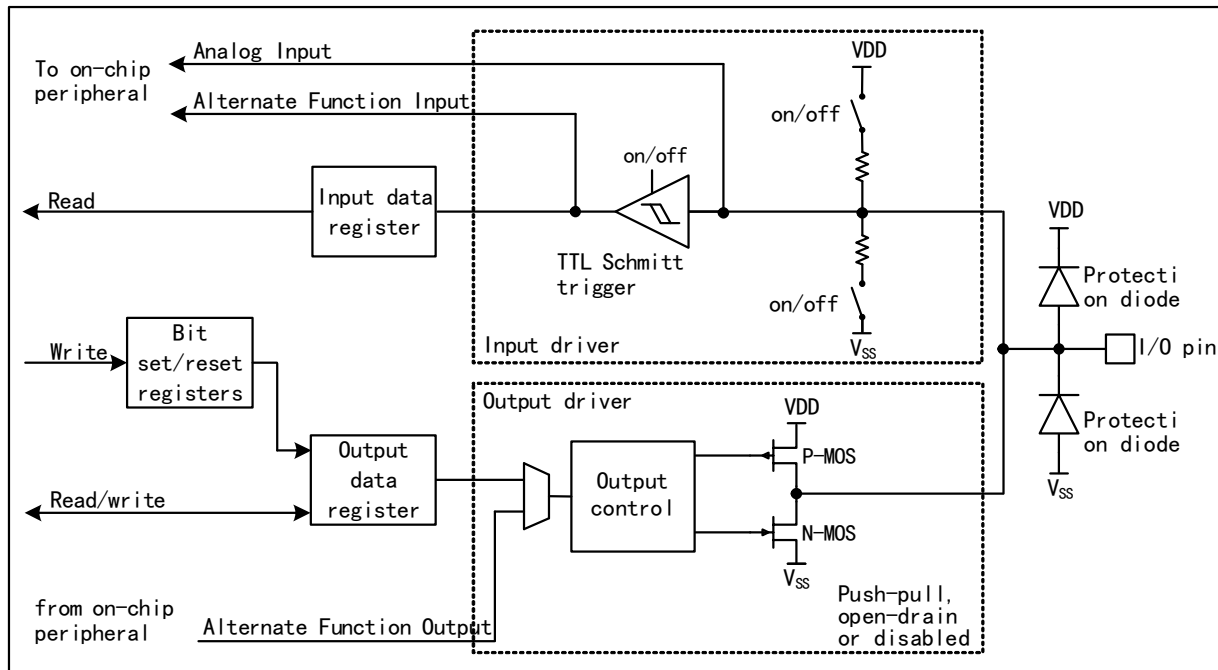
- Floating input
- Pull-up input
- Pull-down input
- Analog input
- Open-drain output
- Push-pull output
- Input and output of alternate function

Many pins have alternate functions, and many other peripherals map their own output and input channels to these pins. The specific application of these alternate pins needs to be with reference to each peripheral, and this chapter shall specify whether these pins are alternate and remapped.

9.2 Function Description

9.2.1 Overview

Figure 9-1 Basic structure of GPIO module



Note: V_{DD} is the supply voltage. The voltage type depends on the specific pin; refer to Chapter 9 for details.

As shown in figure 9-1, the IO port structure, each pin has 2 protection diodes inside the chip, and the IO port can be divided into input and output drive modules. The input driver has a weak pull-up resistor, which can be connected to AD and other analog input peripherals; if you input to a digital peripheral, you need to go through a TTL Schmitt trigger, and then connect to the GPIO input register or other multiplexed peripherals. The output driver has a pair of MOS tubes, and the IO port can be configured to open leakage or push-pull output by configuring whether the upper and lower MOS tubes are enabled or not; the output driver can also be configured to be controlled by GPIO or by other peripherals that are reused.

9.2.2 GPIO Initialization

Just after the reset, the GPIO port is running in the initial state, at this time, most IO ports are running in the floating input state, but there are also HSE and other peripheral-related pins that run on the peripheral reuse function. For specific initialization functions, please refer to the relevant sections of the pin description.

9.2.3 External Interrupt

All GPIO ports can be configured as external interrupt input channels, but each external interrupt input channel can only be mapped to one GPIO pin. Furthermore, the external interrupt channel number must correspond to the bit number of the GPIO port. For example, PA1 (or PB1, PC1, PD1, PE1, etc.) can only be mapped to EXTI1, and EXTI1 can only accept the mapping of either PA1, PB1, PC1, PD1, or PE1. Both sides maintain a one-to-one relationship.

9.2.4 multiplexing function multiplexer and mapping

Alternate function

The I/O pins are connected to the onboard peripherals/modules via a multiplexer that allows only one peripheral's multiplexing function (AF) to be connected to the I/O pins at a time. This ensures that there is no conflict between peripherals sharing the same I/O pin.

Each I/O pin has a multiplexer with up to 16 multiplexed functional inputs (AF0 to AF15) that can be configured via the GPIOx_AFLR and GPIOx_AFHR registers:

After reset, the multiplexer is selected as multiplexing function 0(AF0). I/O is configured by GPIOx_CFGLR/GPIOx_CFGHR register in multiplexing mode.

The specific multiplexing function assignment of each pin is detailed in the CH32H417DS0 manual.

Alternate function configuration

- To use the alternate function of input direction, the port must be configured in input mode, and the floating, pull-up or pull-down mode can be configured according to the actual situation.
- To use the alternate function in the output direction, the port must be configured in the multiplexing output mode, and the multiplexing push-pull or multiplexing open-drain mode can be configured according to the actual situation.
- For the bidirectional multiplexing function, the port must be configured to multiplex the output mode, and then the driver is configured to float the input mode.

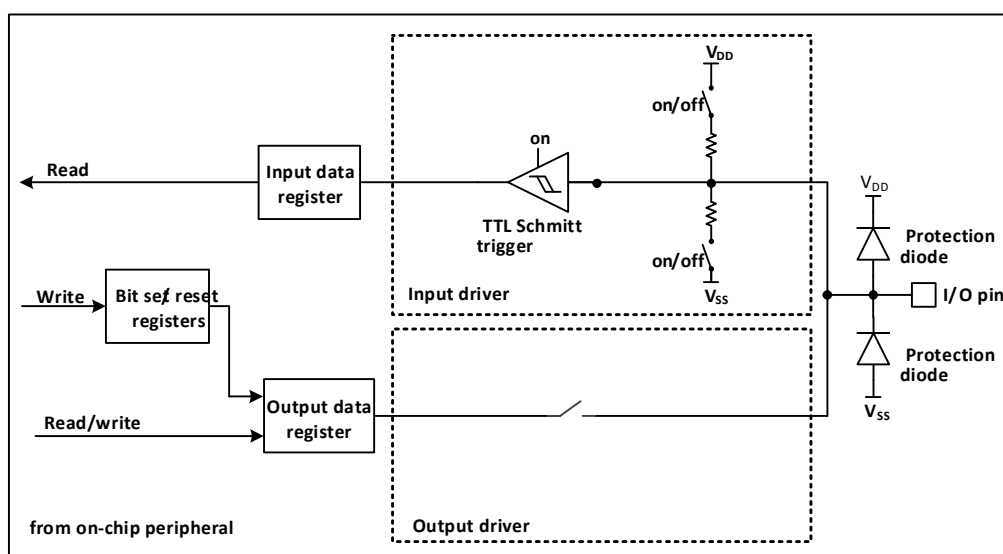
There may be multiple peripherals multiplexed to this pin in the same IO port, so in order to maximize the space for each peripheral, in addition to the default multiplexed pin, the multiplexed pin of the peripheral can be remapped to other pins to avoid the occupied pin.

9.2.5 Locking Mechanism

The locking mechanism can lock the configuration of the IO port. After a specific write sequence, the selected IO pin configuration is locked and cannot be changed until the next reset.

9.2.6 Input Configuration

Figure 9-2 Input configuration structure block diagram of GPIO module

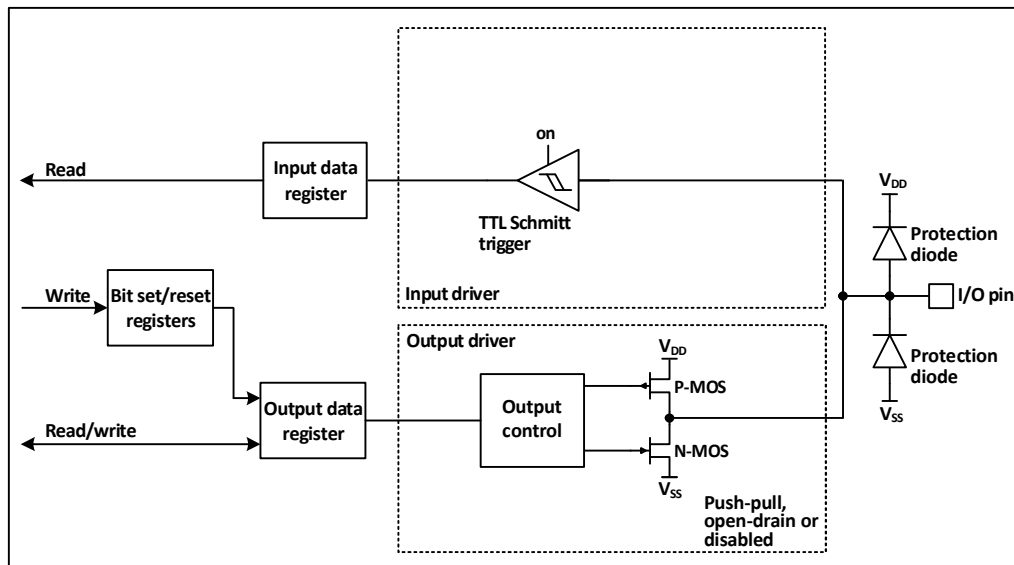


Note: V_{DD} is the supply voltage, and the voltage type depends on the specific pin. Please refer to Chapter 9 for details.

When the IO port is configured in input mode, the output driver is disconnected, the input up and down is optional, and the alternate function and analog input are not connected. The data on each IO port is sampled to the input data register at each HB clock, and the level state of the corresponding pin is obtained by reading the corresponding bit of the input data register.

9.2.7 Output Configuration

Figure 9-3 Output configuration structure block diagram of GPIO module

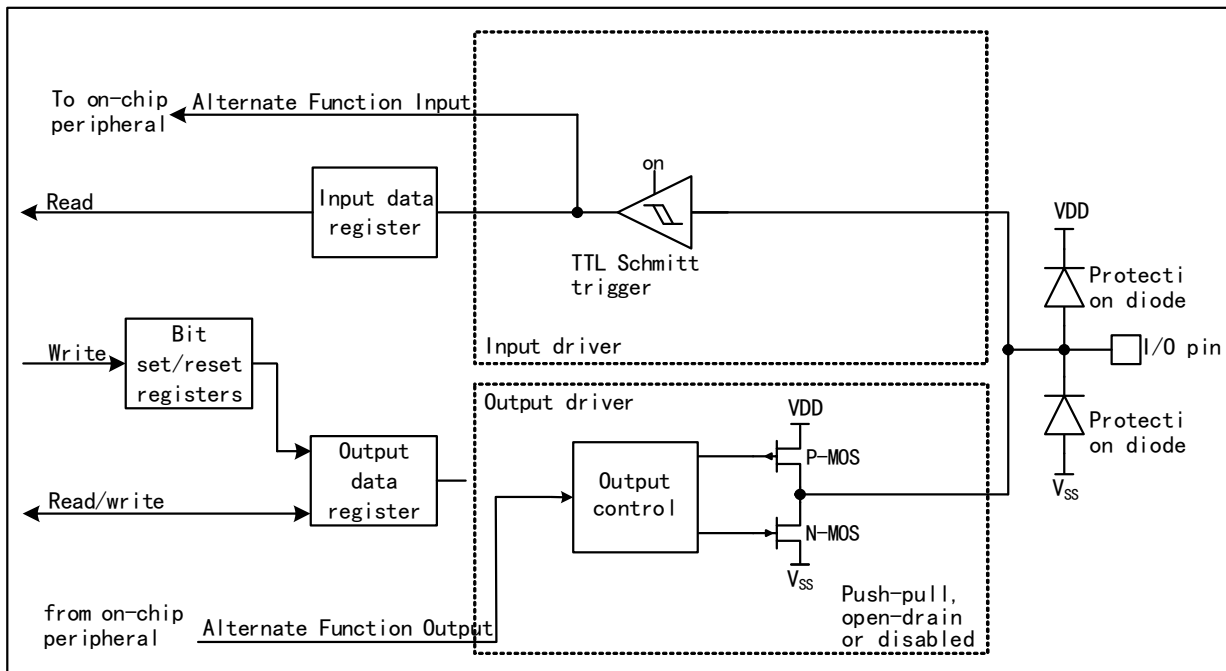


Note: V_{DD} is the supply voltage, and the voltage type depends on the specific pin. Please refer to Chapter 9 for details.

When the IO port is configured in output mode, a pair of MOS in the output driver can be configured in push-pull or open-drain mode as needed, without using the alternate function. The input-driven pull-up resistor is disabled, the TTL Schmitt trigger is activated, and the level that appears on the IO pin will be sampled to the input data register at each HB clock, so reading the input data register will get the IO state, and in push-pull output mode, access to the output data register will get the last written value.

9.2.8 Alternate Function Configuration

Figure 9-4 Structure block diagram when GPIO module is multiplexed by other peripherals

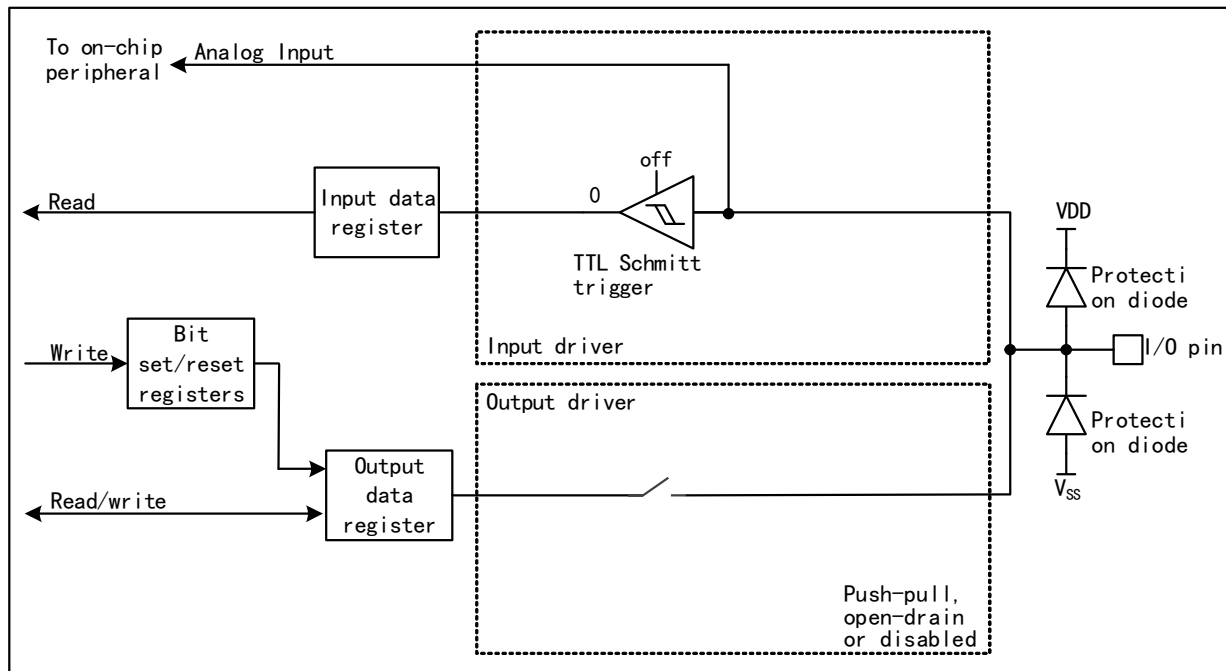


Note: V_{DD} is the supply voltage, and the voltage type depends on the specific pin. Please refer to Chapter 9 for details.

When the alternate function is enabled, the output driver is enabled and can be configured in open-drain or push-pull mode as needed, the Schmitt trigger is also turned on, the input and output lines of the alternate function are connected, but the output data register is disconnected, and the level that appears on the IO pin will be sampled to the input data register at each HB clock. In open-drain mode, reading the input data register will get the current state of the IO port. In push-pull mode, reading the output data register will get the last written value.

9.2.9 Analog Input Configuration

Figure 9-5 Configuration structure block diagram of GPIO module as analog input



Note: V_{DD} is the supply voltage, and the voltage type depends on the specific pin. Please refer to Chapter 9 for details.

When the analog input is enabled, the output buffer is disconnected, the input of the Schmitt trigger in the input driver is disabled to prevent consumption on the IO port, the pull-up resistor is prohibited, and the read input data register will always be 0.

9.2.10 Peripheral GPIO Setting

The following table recommends the corresponding GPIO port configuration of each peripheral pin.

Table 9-1 Advanced-control timer (TIM1/8)

TIM1	Configuration	GPIO configuration
TIMx_CHx	Input capture channel x	Floating input
	Output compare channel x	Push-pull alternate output
TIMx_CHxN	Complementary output channel x	Push-pull alternate output
TIMx_BKIN	Break input	Floating input
TIMx_BKIN2	Break input	Floating input
TIMx_ETR	External trigger clock input	Floating input

Table 9-2 General-purpose timer (TIM2/3/4/5/9/10/11/12)

TIMx pin	Configuration	GPIO configuration
TIMx_CHx	Input capture channel x	Floating input
	Output compare channel x	Push-pull alternate output
TIMx_ETR	External trigger clock input	Floating input

Table 9-3 Low-power timer (LPTIM1/2)

LPTIMx pin	Configuration	GPIO configuration
LPTIMx_CHx	Input capture channel x	Floating input
	Output capture channel x	Push-pull alternate output
LPTIMx_ETR	External trigger clock input	Floating input
LPTIMx_OC	PWM output	Push-pull alternate output

Table 9-4 Universal Synchronous/Asynchronous Receiver Transmitter (USART1/2/3/4/5/6/7/8)

USART pin	Configuration	GPIO configuration
USARTx_TX	Full-duplex mode	Push-pull alternate output
	Half-duplex synchronous mode	Open-drain alternate output
USARTx_RX	Full-duplex mode	Floating input or pull-up input
	Half-duplex synchronous mode	Not used
USARTx_CK	Synchronous mode	Push-pull alternate output
USARTx_RTS	Hardware flow control	Push-pull alternate output
USARTx_CTS	Hardware flow control	Floating input or pull-up input

Table 9-5 Serial Peripheral Interface (SPI1/2/3/4) module

SPI pin	Configuration	GPIO configuration
SPIx_SCK	Master mode	Push-pull alternate output
	Slave mode	Floating input
SPIx_MOSI	Full-duplex master mode	Push-pull alternate output
	Full-duplex slave mode	Floating input or pull-up input
	Simplex bidirectional data line/master mode	Push-pull alternate output
	Simplex bidirectional data line/slave mode	Not used
SPIx_MISO	Full duplex master mode	Floating input or pull-up input
	Full duplex slave mode	Push-pull alternate output
	Simplex bidirectional data line/master mode	Not used
	Simplex bidirectional data line/slave mode	Push-pull alternate output
SPIx_NSS	Hardware master or slave mode	Floating input or pull-up or pull-down input
	Hardware master mode/NSS output enable	Push-pull alternate output
	Software mode	Not used

Table 9-6 Built-in audio bus (I2S2/3) module

I2Sx pin	Configuration	GPIO configuration
I2Sx_WS	Master mode	Push-pull alternate output
	Slave mode	Floating input

I2Sx_CK	Master mode	Push-pull alternate output
	Slave mode	Floating input
I2Sx_SD	Transmitter	Push-pull alternate output
	Receiver	Floating input or pull-up or pull-down input
I2Sx_MCK	Master mode	Push-pull alternate output
	Slave mode	Not used

Table 9-7 Inter IC Sound (I2C1/2/3/4) module

I2C pin	Configuration	GPIO configuration
I2C_SCL	I2C clock	Open-drain alternate output
I2C_SDA	I2C data	Open-drain alternate output

Table 9-8 Inter IC Sound (I2C1/2/3/4) module

I ³ C pin	Configuration	GPIO configuration
I ³ C_SCL	I ³ C clock	Push-pull alternate output
I ³ C_SDA	I ³ C data	Push-pull alternate output

Table 9-9 Controller LAN (CAN1/2/3) module

CAN pin	GPIO configuration
CANx_TX	Push-pull alternate output
CANx_RX	Floating input or pull-up input

Table 9-10 USB PD Controller

USBPD pin	GPIO configuration
USBPD_CC1/USBPD_CC2	Push-pull alternate output

Table 9-11 USB 2.0 High-speed Host Device (USBHS) Controller

USBHS pin	GPIO configuration
USBHS_DM/USBHS_DP	After the USB module is enabled, alternate IO port connects to the internal USBHS transceiver automatically

Table 9-12 USB Host Device (USBFS) controller

USBFS pin	GPIO configuration
USBFS_DM/USBFS_DP	After the USB is enabled, alternate IO port connects to the internal USBFS transceiver automatically

Table 9-13 USB OTG_FS controller

USB OTG_FS pin	GPIO configuration
OTG_FS_VBUS	Analog input
OTG_FS_ID	Pull-up input
OTG_FS_DM	Automatically controlled by USB power failure

OTG_FS_DP	Automatically controlled by USB power failure
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Table 9-14 SDIO controller module

SDIO pin	Configuration	GPIO configuration
CK	Clock	Master mode: Push-pull alternate output Slave mode: Floating input
CMD	Command	Push-pull alternate output
D[7:0]	Data	Push-pull alternate output

Table 9-15 SD/EMMC controller module

SD/EMMC pin	Configuration	GPIO configuration
CK	Clock	Master mode: Push-pull alternate output Slave mode: Push-pull alternate output
CMD	Command	Push-pull alternate output
D[7:0]	Data	Push-pull alternate output
STR	Return clock signal for HS400 mode.	Push-pull alternate output

Table 9-16 Digital filter for $\Sigma\Delta$ modulator (DFSDM) module

DFSDM pin	Configuration	GPIO configuration
DFSDM_DATINx	Serial data input	Floating input
DFSDM_CKINx	Serial clock input	Floating input
DFSDM_CKOUT	Internal clock output	Push-pull alternate output

Table 9-17 Serial-Parallel interchange controller and transceiver (SerDes) module

SerDes pin	Configuration	GPIO configuration
SerDes_TXP	Differential transmission	Floating input
SerDes_TXN		Floating input
SerDes_RXP	Differential transmission	Floating input
SerDes_RXN		Floating input

Table 9-18 Flexible Memory Controller (FMC/SDRAM)

FMC pin	GPIO configuration
FMC_A[25:0] FMC_D[31:0]	Push-pull alternate output
FMC_CK	Push-pull alternate output
FMC_NOE FMC_NWE	Push-pull alternate output
FMC_NE1 FMC_NCE2	Push-pull alternate output

FMC_NWAIT	Floating input or pull-up input
FMC_NBL[1:0]	Push-pull alternate output
FMC_SDCKE[1:0]	Push-pull alternate output
FMC_SDNE[1:0]	Push-pull alternate output
FMC_BA[1:0]	Push-pull alternate output
FMC_NRAS	Push-pull alternate output
FMC_NCAS	Push-pull alternate output
FMC_SDNWE	Push-pull alternate output
FMC_DQM[3:0]	Push-pull alternate output

Table 9-19 Analog-to-Digital Converter (ADC) and Digital-to-Analog Converter (DAC)

ADC/DAC pin	GPIO configuration
ADC/DAC	Analog input

Table 9-20 High speed analog-to-digital converter (HSADC)

HSADC pin	GPIO configuration
HSADC	Analog input

Table 9-21 Digital image interface (DVP) module

DVP pin	Configuration	GPIO configuration
PCLK	Pixel clock input	Floating input or pull-up input
DATA[11:0]	Pixel data input	Floating input or pull-up input
VSYN	Frame synchronization signal input	Floating input or pull-up input
HSYN	Line synchronization signal input	Floating input or pull-up input

Table 9-22 Serial audio interface (SAI) module

SAI pin	Configuration	GPIO configuration
SAI_SD	SD data line input and output	Push-pull alternate output
SAI_SCK	Bit clock input and output	Push-pull alternate output
SAI_FS	FS signal input and output	Push-pull alternate output
SAI_MCLK	Master clock output of external decoder	Push-pull alternate output

Table 9-23 QSPI interface (QSPI) module

QSPI pin	GPIO configuration
QSPI_SCK	Push-pull alternate output
QSPI_SCSN	Push-pull alternate output
QSPI_SIO[3:0]	Push-pull alternate output
QSPI_SCSXN	Push-pull alternate output
QSPI_SIOX[3:0]	Push-pull alternate output

Table 9-24 QSPI interface (QSPI) module

SWPMI pin	Configuration	GPIO configuration
SWP_RX	(Non-1-wire mode) Receive signal input	Floating input
SWP_TX	(Non-1-wire mode) Transmit signal output	Push-pull alternate output
SWP_SUP	(Non-1-wire mode) Suspend signal output	Push-pull alternate output
SWPMI_IO	(1-wire mode) input and output	Push-pull alternate output

Table 9-25 QSPI interface (QSPI) module

LTDC pin	Configuration	GPIO configuration
LCD_CLK	Clock output	Push-pull alternate output
LCD_VSYNC	Vertical synchronous output	Push-pull alternate output
LCD_HSYNC	Horizontal synchronous output	Push-pull alternate output
LCD_DE	Non-data enabled output	Push-pull alternate output
LCD_R[7:0]	Data: 8-bit red data output	Push-pull alternate output
LCD_G[7:0]	Data: 8-bit green data output	Push-pull alternate output
LCD_B[7:0]	Data: 8-bit blue data output	Push-pull alternate output

Note: The LCD-TFT controller pin must be configured through the user program. Unused pins can be used for other functions.

For LTDC outputs up to 24 bits (RGB888), when outputting RGB565 or RGB666 to 16-bit or 18-bit displays using a pixel depth lower than 8bpp, the RGB display data lines must be connected to the MSB of the LCD-TFT controller's RGB data lines. For example, when the LCD-TFT controller is connected to an RGB565 16-bit display, the LCD display's R[4:0], G[4:0], and B[4:0] data line pins must connect to the LCD-TFT controller's LCD_R[7:3], LCD_G[7:3], and LCD_B[7:3] pins.

Table 9-26 Other IO function settings

Pin	Configuration function	GPIO configuration
(PC13) RTC	RTC output	Hardware automatic setting
MCO	Clock output	Push-pull alternate output
EXTI	External interrupt input	Floating input or pull-up or pull-down input

9.2.11 Alternate Function Remapping GPIO Settings

9.2.11.1 OSC32_IN/OSC32_OUT as GPIO Port PC14/PC15

When LSEON=0, the LSE oscillator pin OSC32_IN/OSC32_OUT can be used as PC14/PC15 of GPIO respectively.

When LSEON=1, it is used as LSE pin.

9.2.11.2 OSC_IN/OSC_OUT as GPIO port PD0/PD1

OSC_IN/OSC_OUT can be used as GPIO PD0/PD1, which is realized by setting Remap Register 1 (AFIO_PCFR1).

9.2.11.3 ADC Alternate Function Remapping

Table 9-27 ADC1 external trigger injection conversion alternate function remapping

Alternate function	ADC1_ETRGINJ_RM=0	ADC1_ETRGINJ_RM=1
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	Default mapping	Remapping
ADC1 external trigger injection conversion	ADC1 external trigger injection conversion connect to EXTI15	ADC1 external trigger injection conversion connect to TIM8_CH4

Table 9-28 ADC1 external trigger regular conversion alternate function remapping

Alternate function	ADC1_ETRGREG_RM=0 Default mapping	ADC1_ETRGREG_RM=1 Remapping
ADC1 external trigger regular conversion	ADC1 external trigger regular conversion connect to EXTI11	ADC1 external trigger regular conversion connect to TIM8_TRGO

Table 9-29 ADC2 external trigger injection conversion alternate function remapping

Alternate function	ADC1_ETRGINJ_RM=0 Default mapping	ADC1_ETRGINJ_RM=1 Remapping
ADC1 external trigger injection conversion	ADC1 external trigger injection conversion connect to EXTI15	ADC1 external trigger injection conversion connect to TIM8_CH4

Table 9-30 ADC2 external trigger regular conversion alternate function remapping

Alternate function	ADC2_ETRGREG_RM=0 Default mapping	ADC2_ETRGREG_RM=1 Remapping
ADC2 external trigger regular conversion	ADC1 external trigger regular conversion connect to EXTI11	ADC1 external trigger regular conversion connect to TIM8_TRGO

9.2.11.4 TIM2 Alternate Function Remapping

Table 9-31 TIM1 alternate function remapping

Alternate function	TIM2ITR1_RM=0	TIM2ITR1_RM=1
TIM2 internal trigger 1	Connect RIM2_ITR1 internally to the Ethernet PTP output.	Connect RIM2_ITR1 internally to the SOF output of the full-speed USB OTG.

9.2.11.5 SDMMC Alternate Function Remapping

Table 9-32 SDMMC alternate function remapping

Alternate function	SDMMC_RM=00 Default mapping	SDMMC_RM=01 Remapping	SDMMC_RM=1x Remapping
STS/CMD	PD2	PD12	PC10
SDCK/SLVCK	PC12	PD11	PC12

STR	PD3	PD10	PC11
D0	PC8	PB13	PD0
D1	PC9	PC9	PD1
D2	PC10	PB10	PD2
D3	PC11	PB11	PD3
D4	PA14	PA14	PD4
D5	PA15	PA15	PD5
D6	PC6	PC6	PD6
D7	PC7	PC7	PD7

9.2.11.6 UHSIF Alternate Function Remapping

Table 9-33 UHSIF PORT alternate function remapping

Alternate function	UHSIF_PORT_RM=00 Default mapping	UHSIF_PORT_RM=01 Remapping	UHSIF_PORT_RM=1x Remapping
PORT0	PF12	PF12	PC1
PORT1	PF13	PF13	PC2
PORT2	PE7	PE7	PC3
PORT3	PE8	PC1	PB0
PORT4	PE9	PC2	PB1
PORT5	PE10	PC3	PE10
PORT6	PE11	PB0	PE11
PORT7	PE12	PB1	PE12
PORT8	PE13		
PORT9	PE14		
PORT10	PE15		
PORT11	PB10		
PORT12	PB11		
PORT13	PB12		
PORT14	PB13		
PORT15	PB14		
PORT16	PD10		
PORT17	PD11		
PORT18	PD12		
PORT19	PD13		
PORT20	PD14		
PORT21	PD15		
PORT22	PF0		

PORT23	PF1
PORT24	PF2
PORT25	PC6
PORT26	PC7
PORT27	PC8
PORT28	PC9
PORT29	PA13
PORT30	PA14
PORT31	PA15
PORT32	PC10
PORT33	PC11
PORT34	PC12
PORT35	PD0
PORT36	PD1
PORT37	PD2
PORT38	PD3
PORT39	PD4
PORT40	PD5
PORT41	PD6
PORT42	PD7
PORT43	PF3
PORT44	PF4
PORT45	PF5
PORT46	PE0
PORT47	PE1

Table 9-34 UHSIF CLK alternate function remapping

Alternate function	UHSIF_CLK_RM=00 Default mapping	UHSIF_CLK_RM=01 Remapping	UHSIF_CLK_RM=10 Remapping	UHSIF_CLK_RM=11 Remapping
UHSIF_CLK	PF11	PC0	PF14	PD9

9.3 Register Description

9.3.1 GPIO Registers

Unless otherwise specified, the GPIO registers must be operated in words (operate these registers in 32 bits).

Table 9-35 GPIO-related registers

Name	Access address	Description	Reset value
R32_GPIOA_CFGLR	0x40010800	PA port configuration register low	0x44444444
R32_GPIOB_CFGLR	0x40010C00	PB port configuration register low	0x44444444
R32_GPIOC_CFGLR	0x40011000	PC port configuration register low	0x44444444
R32_GPIOD_CFGLR	0x40011400	PD port configuration register low	0x44444444

R32_GPIOE_CFGLR	0x40011800	PE port configuration register low	0x44444444
R32_GPIOF_CFGLR	0x40011C00	PF port configuration register low	0x44444444
R32_GPIOA_CFGHR	0x40010804	PA port configuration register high	0x44444444
R32_GPIOB_CFGHR	0x40010C04	PB port configuration register high	0x44444444
R32_GPIOC_CFGHR	0x40011004	PC port configuration register high	0x44444444
R32_GPIOD_CFGHR	0x40011404	PD port configuration register high	0x44444444
R32_GPIOE_CFGHR	0x40011804	PE port configuration register high	0x44444444
R32_GPIOF_CFGHR	0x40011C04	PF port configuration register high	0x44444444
R32_GPIOA_INDR	0x40010808	PA port input data register	0x0000XXXX
R32_GPIOB_INDR	0x40010C08	PB port input data register	0x0000XXXX
R32_GPIOC_INDR	0x40011008	PC port input data register	0x0000XXXX
R32_GPIOD_INDR	0x40011408	PD port input data register	0x0000XXXX
R32_GPIOA_OUTDR	0x4001080C	PA port output data register	0x00000000
R32_GPIOB_OUTDR	0x40010C0C	PB port output data register	0x00000000
R32_GPIOC_OUTDR	0x4001100C	PC port output data register	0x00000000
R32_GPIOD_OUTDR	0x4001140C	PD port output data register	0x00000000
R32_GPIOE_OUTDR	0x4001180C	PE port output data register	0x00000000
R32_GPIOF_OUTDR	0x40011C0C	PF port output data register	0x00000000
R32_GPIOA_BSHR	0x40010810	PA port set/reset register	0x00000000
R32_GPIOB_BSHR	0x40010C10	PB port set/reset register	0x00000000
R32_GPIOC_BSHR	0x40011010	PC port set/reset register	0x00000000
R32_GPIOD_BSHR	0x40011410	PD port set/reset register	0x00000000
R32_GPIOE_BSHR	0x40011810	PE port set/reset register	0x00000000
R32_GPIOF_BSHR	0x40011C10	PF port set/reset register	0x00000000
R32_GPIOA_BCR	0x40010814	PA port reset register	0x00000000
R32_GPIOB_BCR	0x40010C14	PB port reset register	0x00000000
R32_GPIOC_BCR	0x40011014	PC port reset register	0x00000000
R32_GPIOD_BCR	0x40011414	PD port reset register	0x00000000
R32_GPIOE_BCR	0x40011814	PE port reset register	0x00000000
R32_GPIOF_BCR	0x40011C14	PF port reset register	0x00000000
R32_GPIOA_LCKR	0x40010818	PA port lock configuration register	0x00000000
R32_GPIOB_LCKR	0x40010C18	PB port lock configuration register	0x00000000
R32_GPIOC_LCKR	0x40011018	PC port lock configuration register	0x00000000
R32_GPIOD_LCKR	0x40011418	PD port lock configuration register	0x00000000
R32_GPIOE_LCKR	0x40011818	PE port lock configuration register	0x00000000
R32_GPIOF_LCKR	0x40011C18	PF port lock configuration register	0x00000000
R32_GPIOA_SPEED	0x4001081C	PA port speed register	0x00000000
R32_GPIOB_SPEED	0x40010C1C	PB port speed register	0x00000000
R32_GPIOC_SPEED	0x4001101C	PC port speed register	0x00000000
R32_GPIOD_SPEED	0x4001141C	PD port speed register	0x00000000
R32_GPIOE_SPEED	0x4001181C	PE port speed register	0x00000000
R32_GPIOF_SPEED	0x40011C1C	PF port speed register	0x00000000

9.3.1.1 GPIO Configuration Register Low (GPIOx_CFGLR) (x=A/B/C/D/E/F)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF7[1:0]	MODE7[1:0]	CNF6[1:0]	MODE6[1:0]	CNF5[1:0]	MODE5[1:0]	CNF4[1:0]	MODE4[1:0]	CNF3[1:0]	MODE3[1:0]	CNF2[1:0]	MODE2[1:0]	CNF1[1:0]	MODE1[1:0]	CNF0[1:0]	MODE0[1:0]
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF3[1:0]	MODE3[1:0]	CNF2[1:0]	MODE2[1:0]	CNF1[1:0]	MODE1[1:0]	CNF0[1:0]	MODE0[1:0]								

Bit	Name	Access	Description	Reset value
[31:30] [27:26] [23:22] [19:18] [15:14] [11:10] [7:6] [3:2]	CNFy[1:0]	RW	(y=0-7), the configuration bits of port x, through which the corresponding ports are configured. When entering a mode (MODE=00b): 00: Analog input mode. 01: Floating input mode. 10: With pull-up mode. 11: Reserved. In output mode (MODE > 00b): 00: Universal push-pull output mode. 01: Universal open-leak output mode. 10: Alternate function push-pull output mode. 11: Alternate function open-leak output mode.	01b
[29:28] [25:24] [21:20] [17:16] [13:12] [9:8] [5:4] [1:0]	MODEy[1:0]	RW	(y=0-7), port x mode selection, configure the corresponding port through these bits. 00: Input mode. 01: Output mode, maximum speed 10MHz. 10: Output mode, maximum speed 2MHz. 11: Output mode, maximum speed 50MHz.	00b

9.3.1.2 GPIO Configuration Register High (GPIOx_CFGHR) (x=A/B/C/D/E/F)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF15[1:0]	MODE15[1:0]	CNF14[1:0]	MODE14[1:0]	CNF13[1:0]	MODE13[1:0]	CNF12[1:0]	MODE12[1:0]	CNF11[1:0]	MODE11[1:0]	CNF10[1:0]	MODE10[1:0]	CNF9[1:0]	MODE9[1:0]	CNF8[1:0]	MODE8[1:0]
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF11[1:0]	MODE11[1:0]	CNF10[1:0]	MODE10[1:0]	CNF9[1:0]	MODE9[1:0]	CNF8[1:0]	MODE8[1:0]								

Bit	Name	Access	Description	Reset value
[31:30] [27:26] [23:22] [19:18]	CNFy[1:0]	RW	(y=8-15), the configuration bits for port x, by which the corresponding port is configured. When in input mode (MODE=00b): 00: Analog input mode;	01b

[15:14] [11:10] [7:6] [3:2]			01: Float input mode; 10: With pull-up and pull-down modes. 11: Reserved. In output mode (MODE>00b): 00: General-purpose push-pull output mode; 01: General-purpose open-drain output mode; 10: Alternate function push-pull output mode; 11: Alternate function open-drain output mode.	
[29:28] [25:24] [21:20] [17:16] [13:12] [9:8] [5:4] [1:0]	MODEy[1:0]	RW	(y=8-15), the mode bits of port x, by which the corresponding port is configured. 00: Input mode; 01: Output mode, max. speed 10MHz. 10: Output mode, maximum speed 2MHz; 11: Output mode, maximum speed 50MHz.	00b

9.3.1.3 Port Input Data Register (GPIOx_INDR) (x=A/B/C/D/E/F)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDR1 5	IDR1 4	IDR1 3	IDR1 2	IDR1 1	IDR1 0	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	IDRy	RO	(y=0-15), port input data. These bits are read-only and can only be read out in 16-bit form. The value read out is the high and low state of the corresponding bit.	X

9.3.1.4 Port Output Data Register (GPIOx_OUTDR) (x=A/B/C/D/E/F)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR1 5	ODR1 4	ODR1 3	ODR1 2	ODR1 1	ODR1 0	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	ODRy	RW	For output mode:	0

			(y=0-15). the port outputs data. This data can only be manipulated in 16-bit form. the IO port externally outputs the values of these registers. For input modes with pull-down: 0: Pull-down input; 1: Pull-up input.	
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9.3.1.5 Port Set/Reset Register (GPIOx_BSHR) (x=A/B/C/D/E/F)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0

Bit	Name	Access	Description	Reset value
[31:16]	BRy	WO	(y=0-15), the corresponding OUTDR bits are cleared for these location bits; writing 0 has no effect. These bits can only be accessed in 16-bit format. If both BR and BS bits are set, the BS bit acts.	0
[15:0]	BSy	WO	(y=0-15), for which positional bits cause the corresponding OUTDR positional bits, and writing 0 has no effect. These bits can only be accessed in 16-bit format. If both BR and BS bits are set, the BS bit acts.	0

9.3.1.6 Port Reset Register (GPIOx_BCR) (x=A/B/C/D/E/F)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	BRy	WO	(y=0-15), the corresponding OUTDR bits are cleared for these location bits; writing 0 has no effect. These bits can only be accessed in 16-bit form.	0

9.3.1.7 Lock Configuration Register (GPIOx_LCKR) (x=A/B/C/D/E/F)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved															LCKK
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCK1 5	LCK1 4	LCK1 3	LCK1 2	LCK1 1	LCK1 0	LCK9	LCK8	LCK7	LCK6	LCK5	LCK4	LCK3	LCK2	LCK1	LCK0

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LCKK	RW	Lock key, which can be locked by writing a specific sequence, but it can be read out at any time. When it reads 0, it means that the lock is not in effect, and read 1 means that the lock is in effect. The write sequence of the lock key is: write 1-write 0-write 1-read 0-read 1, the last step is not necessary, but can be used to confirm that the lock key has been activated. Any error when writing a sequence does not activate a lock, and the value of LCK [15:0] cannot be changed when writing a sequence. After the lock is in effect, the configuration of the port can be changed only after the next reset.	0
[15:0]	LCKy	RW	(y=0-15), where 1 indicates that the configuration of the corresponding port is locked. These bits can only be changed before the LCKK is locked. Locked configurations refer to the configuration registers GPIOx_CFGLR and GPIOx_CFGHR.	0

Note: After the LOCK sequence is performed on the corresponding port bit, the configuration of the port bit cannot be changed until the next system reset.

9.3.1.8 Configuration Lock Register (GPIOx_SPEED) (x=A/B/C/D/E/F)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SPEED15 [1:0]	SPEED14 [1:0]	SPEED13 [1:0]	SPEED12 [1:0]	SPEED11 [1:0]	SPEED10 [1:0]	SPEED9 [1:0]	SPEED8 [1:0]								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPEED7 [1:0]	SPEED6 [1:0]	SPEED5 [1:0]	SPEED4 [1:0]	SPEED3 [1:0]	SPEED2 [1:0]	SPEED1 [1:0]	SPEED0 [1:0]								

Bit	Name	Access	Description	Reset value
[31:0]	SPEEDy[1:0]	RW	(y=0-15), port x speed selection, configure the corresponding port through these bits. 00: 10MHz; 01: 50MHz;	00b

			10: 100MHz; 11: 180MHz.	
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9.3.2 AFIO Registers

Unless otherwise specified, the AFIO registers must be operated in words (operate these registers in 32 bits).

Table 9-36 AFIO-related registers

Name	Access address	Description	Reset value
R32_AFIO_PCFR1	0x40010004	Remap register1	0x00000000
R32_AFIO_GPIOA_AFLR	0x40010004	PA port alternate function register low	0x00000000
R32_AFIO_GPIOA_AFHR	0x40010008	PA port alternate function register high	0x00000000
R32_AFIO_GPIOB_AFLR	0x4001000C	PB port alternate function register low	0x00000000
R32_AFIO_GPIOB_AFHR	0x40010010	PB port alternate function register high	0x00000000
R32_AFIO_GPIOC_AFLR	0x40010014	PC port alternate function register low	0x00000000
R32_AFIO_GPIOC_AFHR	0x40010018	PC port alternate function register high	0x00000000
R32_AFIO_GPIOD_AFLR	0x4001001C	PD port alternate function register low	0x00000000
R32_AFIO_GPIOD_AFHR	0x40010020	PD port alternate function register high	0x00000000
R32_AFIO_GPIOE_AFLR	0x40010024	PE port alternate function register low	0x00000000
R32_AFIO_GPIOE_AFHR	0x40010028	PE port alternate function register high	0x00000000
R32_AFIO_GPIOF_AFLR	0x4001002C	PF port alternate function register low	0x00000000
R32_AFIO_GPIOF_AFHR	0x40010030	PF port alternate function register high	0x00000000
R32_AFIO_EXTICR1	0x40010008	External interrupt configuration register 1	0x00000000
R32_AFIO_EXTICR2	0x4001000C	External interrupt configuration register 2	0x00000000

9.3.2.1 Remap Register 1 (AFIO_PCFR1)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SW_CFG[2:0]			Reserved			USBP D_C C_H VT	Reser ved	VDD 33_I O_HS LV	VDDI O_IO _HSL V	VIO1 8_IO _HSL V
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			TIM2I T R1_R M	SDMMC_R M [1:0]		UHSIF_PO RT_RM[1:0]		UHSIF_CL K_RM[1:0]		Reser ved	ADC 2_ET RGIN J_RM	ADC 2_ET RGR EG_R M	ADC 1_ET RGIN J_RM	ADC 1_ET RGR EG_R M	PD0P D1_R M

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:24]	SW_CFG[2:0]	WO	These bits are used to configure the IO ports for the SW function and the trace function. SDI is the	000b

			debugging interface for accessing the core. It is always used as the SDI port after a system reset. 0xx: Enable SDI; 100: Disable SDI as a GPIO function; Other: Invalid.	
[23:21]	Reserved	RO	Reserved	0
20	USBPD_CC_HVT	RW	CC pin input channel threshold adjustment. USB PD CC pin high threshold input mode enable: 1: High threshold input, reduces power consumption during PD communication; 0: Normal GPIO threshold input.	0
19	Reserved	RO	Reserved	0
18	VDD33_IO_HSLV	RW	VDD33 IO speed configuration at low voltage: 1: When IO voltage is below 2.7V, IO speed can be enhanced; 0: When IO operates at low voltage, speed will not be enhanced.	0
17	VDDIO_IO_HSLV	RW	VDDIO IO speed configuration at low voltage: 1: When IO voltage is below 2.7V, IO speed can be enhanced; 0: When IO operates at low voltage, speed will not be enhanced.	0
16	VIO18_IO_HSLV	RW	VIO18 IO speed configuration at low voltage: 1: When IO voltage is below 2.7V, IO speed can be enhanced; 0: When IO operates at low voltage, speed will not be enhanced.	0
[15:13]	Reserved	RO	Reserved	0
[12]	TIM2ITR1_RM	RW	TIM2 internal triggered 1 remapping. 1: Internally connect TIM2_ITR1 to the SOF output for full-speed USB OTG; 0: Internally connect TIM2_ITR1 to the PTP output for Ethernet.	0
[11:10]	SDMMC_RM[1:0]	RW	SDMMC remapping: 00: Default mapping (STS/CMD/PD2, SDCK/SLVCK/PC12, STR/PD3, D0/PC8 D1/PC9, D2/PC10, D3/PC11, D4/PA14, D5/PA15, D6/PC6, D7/PC7);; 01: Remapping (STS/CMD/PD12, SDCK/SLVCK/PD11, STR/PD10, D0/PB13, D1/PC9, D2/PB10 D3/PB11, D4/PA14, D5/PA15, D6/PC6, D7/PC7); 1x: Remapping (STS/CMD/PC10,	00b

			SDCK/SLVCK/PC12, STR/PC11, D0/PD0 D1/PD1, D2/PD2, D3/PD3, D4/PD4, D5/PPD5, D6/PD6, D7/PD7).	
[9:8]	UHSIF_PORT_RM[1:0]	RW	UHSIF PORT remapping: 00: Default mapping (PORT0/PF12, PORT1/PF13, PORT2/PE7, PORT3/PE8, PORT4/PE9, PORT5/PE10, PORT6/PE11, PORT7/PE12, PORT8/PE13, PORT9/PE14, PORT10/PE15, PORT11/PB10, PORT12/PB11, PORT13/PB12, PORT14/PB13, PORT15/PB14, PORT16/PD10, PORT17/PD11, PORT18/PD12, PORT19/PD13, PORT20/PD14, PORT21/PPD15, PORT22/PF0, PORT23/PF1, PORT24/PF2, PORT25/PC6, PORT26/PC7, PORT27/PC8, PORT28/PC9, PORT29/PA13, PORT30/PA14, PORT31/PA15, PORT32/PC10, PORT33/PC11, PORT34/PC12, PORT35/PD0, PORT36/PD1, PORT37/PD2, PORT38/PD3, PORT39/PD4, PORT40/PD5, PORT41/PD6, PORT42/PD7, PORT43/PF3, PORT44/PF4, PORT45/PF5, PORT46/PE0, PORT47/PE1); 01: Remapping (PORT0/PF12, PORT1/PF13, PORT2/PE7, PORT3/PC1, PORT4/PC2, PORT5/PC3, PORT6/PB0, PORT7/PB1, PORT8/PE13, PORT9/PE14, PORT10/PE15, PORT11/PB10, PORT12/PB11, PORT13/PB12, PORT14/PB13, PORT15/PB14, PORT16/PD10, PORT17/PD11, PORT18/PD12, PORT19/PD13, PORT20/PD14, PORT21/PPD15, PORT22/PF0, PORT23/PF1, PORT24/PF2, PORT25/PC6, PORT26/PC7, PORT27/PC8, PORT28/PC9, PORT29/PA13, PORT30/PA14, PORT31/PA15, PORT32/PC10, PORT33/PC11, PORT34/PC12, PORT35/PD0, PORT36/PD1, PORT37/PD2, PORT38/PD3, PORT39/PD4, PORT40/PD5, PORT41/PD6, PORT42/PD7, PORT43/PF3, PORT44/PF4, PORT45/PF5, PORT46/PE0, PORT47/PE1); 1x: Remapping (PORT0/PC1, PORT1/PC2, PORT2/PC3, PORT3/PB0, PORT4/PB1, PORT5/PE10, PORT6/PE11, PORT7/PE12, PORT8/PE13, PORT9/PE14, PORT10/PE15, PORT11/PB10, PORT12/PB11, PORT13/PB12, PORT14/PB13, PORT15/PB14, PORT16/PD10, PORT17/PD11, PORT18/PD12, PORT19/PD13,	00b

			PORT20/PD14, PORT21/PPD15, PORT22/PF0, PORT23/PF1, PORT24/PF2, PORT25/PC6, PORT26/PC7, PORT27/PC8, PORT28/PC9, PORT29/PA13, PORT30/PA14, PORT31/PA15, PORT32/PC10, PORT33/PC11, PORT34/PC12, PORT35/PD0, PORT36/PD1, PORT37/PD2, PORT38/PD3, PORT39/PD4, PORT40/PD5, PORT41/PD6, PORT42/PD7, PORT43/PF3, PORT44/PF4, PORT45/PF5, PORT46/PE0, PORT47/PE1). <i>Note: The chip packaged with 56/68 pins is recommended to use the mapping configuration of 01b; It is recommended to use 1xb mapping configuration for chips packaged as 88 pins.</i>	
[7:6]	UHSIF_CLK_RM[1:0]	RW	UHSIF CLK remapping: 00: Default mapping (CLK/PF11); 01: Remapping (CLK/PC0); 10: Remapping (CLK/PF14); 11: Remapping (CLK/PD9).	00b
5	Reserved	RO	Reserved	0
4	ADC2_ETRGINJ_RM	RW	Remapping bits for ADC2 external trigger injection conversion. 1: ADC2 external trigger injection conversion connected to TIM8_CH4; 0: ADC2 external trigger injection conversion connected to EXTI15.	0
3	ADC2_ETRGREG_RM	RW	Remapping bits for ADC2 external trigger regular conversion. 1: ADC2 external trigger regular conversion connected to TIM8_TRGO; 0: ADC2 external trigger regular conversion connected to EXTI11.	0
2	ADC1_ETRGINJ_RM	RW	Remapping bits for ADC1 external trigger injection conversion. 1: ADC1 external trigger injection conversion connected to TIM8_CH4; 0: ADC1 external trigger injection conversion connected to EXTI15.	0
1	ADC1_ETRGREG_RM	RW	Remapping bits for ADC1 external trigger regular conversion. 1: ADC1 external trigger regular conversion connected to TIM8_TRGO; 0: ADC1 external trigger regular conversion connected to EXTI11.	0

0	PD0PD1_RM	RW	Pin PD0&PD1 remapping bit This bit is user-readable and writable. It controls whether the GPIO functions of PD0 and PD1 are remapped, i.e., whether PD0&PD1 are mapped to XI&XO. 1: Pins function as GPIO ports; 0: Pins function as crystal oscillator pins.	0
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9.3.2.2 Port Alternate Function Register Low Bit (x=A/B/C/D/E/F)

Offset address: 0x04, 0x0C, 0x14, 0x1C, 0x24, 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFR7[3:0]				AFR6[3:0]				AFR5[3:0]				AFR4[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFR3[3:0]				AFR2[3:0]				AFR1[3:0]				AFR0[3:0]			

Bit	Name	Access	Description	Reset value
[31:28]	AFRy[3:0]	RW	(y=0-7), Port X pin Y alternate function selection:	0
[27:24]			0000: AF0; 0001: AF1;	
[23:20]			0010: AF2; 0011: AF3;	
[19:16]			0100: AF4; 0101: AF5;	
[15:12]			0110: AF6; 0111: AF7;	
[11:8]			1000: AF8; 1001: AF9;	
[7:4]			1010: AF10; 1011: AF11;	
[3:0]			1100: AF12; 1101: AF13;	
			1110: AF14; 1111: AF15.	

9.3.2.3 Port Alternate Function Register High Bit (x=A/B/C/D/E/F)

Offset address: 0x08, 0x10, 0x18, 0x20, 0x28, 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFR15[3:0]				AFR14[3:0]				AFR13[3:0]				AFR12[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFR11[3:0]				AFR10[3:0]				AFR9[3:0]				AFR8[3:0]			

Bit	Name	Access	Description	Reset value
[31:28]	AFRy[3:0]	RW	(y=8-15), Port X pin Y alternate function selection:	0
[27:24]				
[23:20]			0000: AF0; 0001: AF1;	
[19:16]			0010: AF2; 0011: AF3;	
[15:12]			0100: AF4; 0101: AF5;	
[11:8]			0110: AF6; 0111: AF7;	
[7:4]			1000: AF8; 1001: AF9;	

[3:0]			1010: AF10; 1100: AF12; 1110: AF14;	1011: AF11; 1101: AF13; 1111: AF15.	
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9.3.2.4 External Interrupt Configuration Register 1 (AFIO_EXTICR1)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			

Bit	Name	Access	Description	Reset value
[31:28] [27:24] [23:20] [19:16] [15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=0-7), External interrupt input pin configuration bit. Used to determine which port the external interrupt pin is mapped to: 0000: X pin of PA pin; 0001: X pin of PB pin; 0010: X pin of PC pin; 0011: X pin of PD pin; 0100: X pin of PE pin; 0101: X pin of PF pin; 0110: CMP output; Other: Reserved.	0000b

9.3.2.5 External Interrupt Configuration Register 2 (AFIO_EXTICR2)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			

Bit	Name	Access	Description	Reset value
[31:28] [27:24] [23:20] [19:16] [15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=8-15), External interrupt input pin configuration bit. Used to determine which port the external interrupt pin is mapped to: 0000: X pin of PA pin; 0001: X pin of PB pin; 0010: X pin of PC pin; 0011: X pin of PD pin; 0100: X pin of PE pin;; 0101: X pin of PF pin;	0000b

			0110: CMP output; Other: Reserved.	
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Chapter 10 Direct Memory Access Control (DMA)

Direct Memory Access (DMA) controllers provide a high-speed means of converted data between a peripheral and memory or between memory and memory without CPU intervention, and data can be moved quickly through the DMA to conserve CPU resources for other operations.

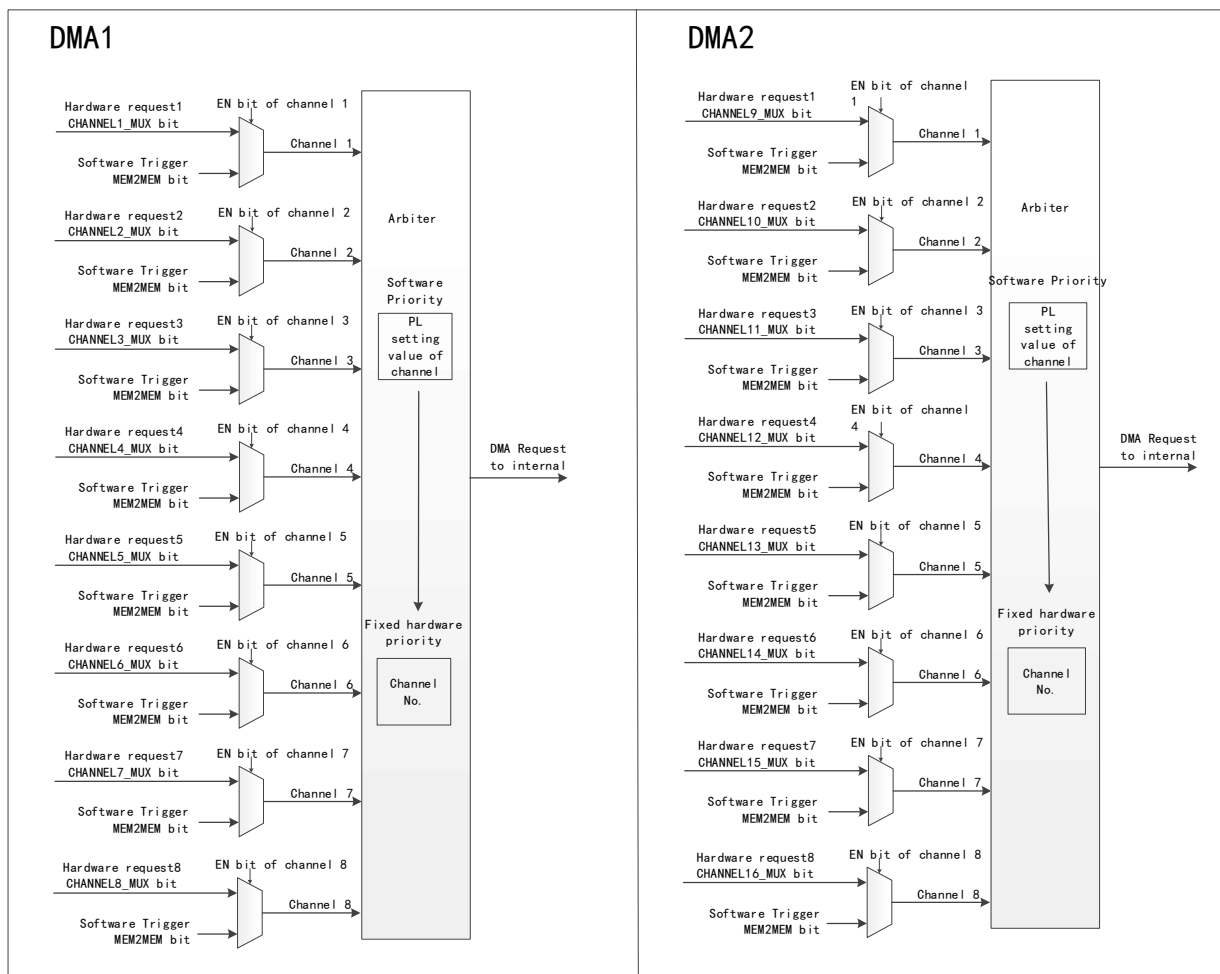
The DMA controller supports the reconfiguration of DMA request lines between peripherals and DMA. There is also an arbiter to coordinate the priorities between channels.

10.1 Main Features

- Multiple independent configurable channels
- Support the reconfiguration of DMA request lines between peripherals and DMA, and support software triggering.
- Support cyclic buffer management
- The priority of requests between multiple channels can be set by software programming (very high, high, medium and low level). When the priority settings are equal, it is determined by the channel number (the lower the channel number, the higher the priority)
- Support transmission from peripheral to memory, memory to peripheral, and memory to memory
- Flash memory, SRAM, peripheral SRAM, HB peripherals can all be used as the sources and targets of access
- Number of programmable data transfer: 65535 at most

10.2 Functional Description

Figure 10-1 Schematic diagram of DMA channel operation



10.2.1 DMA Channel Processing

1) Arbitration priority

DMA requests generated by multiple independent channels are logically or structurally input to the DMA controller, and currently only one channel request is answered. The arbitrator within the module selects the access of the peripheral / memory to be initiated according to the priority of the channel request.

In software management, applications can independently configure priority levels for each channel, including the highest, high, medium and low levels, by setting the PL [1:0] bit of the DMA_CFGRx register. When the software settings between the channels are the same, the modules will be selected according to the fixed hardware priority, and the lower channel number will have higher priority than the higher one.

2) DMA configuration

When the DMA controller receives a request signal, it accesses the requesting peripheral or memory to establish the peripheral or data transfer between the memory and the memory. It mainly includes the following 3 steps:

- 1) Data is taken from the memory address indicated by the peripheral data register or the current peripheral / memory address register, and the starting address of the first transmission is the peripheral base address or memory address specified by the DMA_PADDRx or DMA_MADDRx register.
- 2) Data is stored in the peripheral data register or the memory address indicated by the current peripheral / memory

address register, and the starting address of the first transmission is the peripheral base address or memory address specified in the DMA_PADDRx or DMA_MADDRx register.

3) Perform a decrement operation of the values in the DMA_CNTRx register, which indicates the current number of outstanding operations.

Each channel has 3 DMA data transfer modes:

- Peripheral to memory (MEM2MEM=0, DIR=0)
- Memory to peripheral (MEM2MEM=0, DIR=1)
- Memory to memory (MEM2MEM=1)

Note: Memory-to-memory mode does not require a peripheral request signal, after configuring this mode (MEM2MEM=1), the channel is turned on (EN=1) to start data transfer. This method does not support cyclic mode.

The configuration process is as follows:

- 1) Set the first address of the peripheral register or the memory data address in the memory-to-memory mode (MEM2MEM=1) in the DMA_PADDRx register. When an DMA request occurs, this address will be the source or destination address of the data transfer.
- 2) The memory address is selected via the FLAG_CUR_MEM bit, and the memory data address is set in the DMA_MADDRx register. When a DMA request occurs, data will be read from or written to this address.
- 3) Set the amount of data to be transferred in the DMA_CNTRx register. After each data transmission, this value decreases.
- 4) Set the channel priority in the PL [1:0] bit of the DMA_CFGRx register.
- 5) Set the data transfer direction, loop mode, incremental mode of peripherals and memory, data width of peripherals and memory, transmission halfway, transmission completion, transmission error interrupt enable level in DMA_CFGRx register.
- 6) Set the EN bit of the DMA_CFGRx register and start the channel x.

Note: The control bits in the DMA_PADDRx/DMA_MADDRx/DMA_CNTRx register and the data transfer direction (DIR), loop mode (location), peripheral and memory incremental mode (MINC/PINC) in the DMA_CFGRx register can be configured to write only when the DMA channel is turned off.

3) Cycle mode

Set the CIRC location 1 of the DMA_CFGRx register to enable the circular mode function of channel data transfer. In circular mode, when the number of data transfers becomes 0, the contents of the DMA_CNTRx register are automatically reloaded to their initial values, the internal peripherals and memory address registers are also reloaded to the initial address values set by the DMA_PADDRx and DMA_MADDRx registers, and the DMA operation continues until the channel is closed or DMA mode is turned off.

4) DMA processing status

- Transfer more than half: corresponding to the HTIFx bit hardware setting in the DMA_INTFR register. When the number of DMA transmissions is reduced to less than half of the initial setting value, more than half of the DMA transfer flag will be generated, and if HTIE is set in the DMA_CFGRx register, an interrupt will occur. The hardware uses this flag to remind the application that it can prepare for a new round of data transfer.
- Transfer completed: corresponding to the TCIFx bit hardware setting in the DMA_INTFR register. When the number of DMA transmissions is reduced to 0, the DMA transfer completion flag will be generated, and if TCIE is set in the DMA_CFGRx register, an interrupt will occur.
- Transfer error: corresponding to the TEIFx bit hardware setting in the DMA_INTFR register. Reading and

writing a reserved address area will result in a DMA transmission error. At the same time, the module hardware will automatically clear the EN bits of the DMA_CFGRx register corresponding to the channel where the error occurred, and the channel is closed. If TEIE is set in the DMA_CFGRx register, an interrupt will occur.

When querying the status of the DMA channel, the application can first access the GIFx bit of the DMA_INTFR register, determine which channel has the DMA event, and then deal with the specific DAM event content of the channel.

10.2.2 Programmable Data Transmission Total Size/Data Bit Width/Alignment

The total amount of data transmitted by DMA in one round of each channel is programmable, up to 65535 times. The DMA_CNTRx register indicates the number of transfers to be transmitted. In EN=0, the setting value is written, and after the EN=1 opens the DMA transmission channel, the register becomes read-only, and the value decreases after each transfer.

The transmission data values of peripherals and memory support the function of automatic increment of address pointer, and the pointer increment is programmable. The first transmitted data address they access is stored in the DMA_PADDRx and DMA_MADDRx registers. By setting the PINC bit or MINC location 1 of the DMA_CFGRx register, you can turn on the peripheral address self-increment mode or the memory address self-increment mode, respectively. PSIZE [1:0] sets the peripheral address to take data size and address self-increase, and MSIZE [1:0] sets the memory address to take data size and address self-increase. There are 4 options: 8-bit, 16-bit, 32-bit, 256-bit. The specific data transfer methods are as follows:

Table 10-1 DMA transfer under different data bit width (PINC=MINC=1)

Source bit width	Target bit width	Transfer data Number	Source: address/data	Target: address/data	Transfer operation
8	8	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/B0 0x01/B1 0x02/B2 0x03/B3	- The source address increment is aligned with the bit width of the data set by the source, and the size of the value is equal to the bit width of the source data. - The target address increment is aligned with the bit width of the target set data, and the size of the value is equal to the bit width of the target data. - DMA transfer to the target side of the data based on the principle: data size is not enough to make up the high bit 0, data size overflow high bit removed
8	16	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/00B0 0x02/00B1 0x04/00B2 0x06/00B3	
8	32	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/000000B0 0x04/000000B1 0x08/000000B2 0x0C/000000B3	
8	256	4	0x00/B0 0x01/B1 0x02/B2 0x03/B3	0x00/00...0000B0 0x20/00...0000B1 0x40/00...0000B2 0x60/00...0000B3	
16	8	4	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	0x00/B0 0x01/B2 0x02/B4 0x03/B6	
16	16	4	0x00/B1B0	0x00/B1B0	

			0x02/B3B2 0x04/B5B4 0x06/B7B6	0x02/B3B2 0x04/B5B4 0x06/B7B6	Data storage mode: small end mode, low address stores low bytes, high address stores high bytes.
16	32	4	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	0x00/0000B1B0 0x04/0000B3B2 0x08/0000B5B4 0x0C/0000B7B6	
16	256	4	0x00/B1B0 0x02/B3B2 0x04/B5B4 0x06/B7B6	0x00/00...00B1B0 0x20/00...00B3B2 0x40/00...00B5B4 0x60/00...00B7B6	
32	8	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/B0 0x01/B4 0x02/B8 0x03/BC	
32	16	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/B1B0 0x02/B5B4 0x04/B9B8 0x06/BDBC	
32	32	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	
32	256	4	0x00/B3B2B1B0 0x04/B7B6B5B4 0x08/BBBAB9B8 0x0C/BFBEBDBC	0x00/00...B3B2B1B0 0x20/00...B7B6B5B4 0x40/00...BBBAB9B8 0x60/00...BFBEBDBC	
256	8	4	0x00/2F...13121110 0x20/4F...33323130 0x40/6F...53525150 0x60/8F...73727170	0x00/10 0x01/30 0x02/50 0x03/70	
256	16	4	0x00/2F...13121110 0x20/4F...33323130 0x40/6F...53525150 0x60/8F...73727170	0x00/1110 0x02/3130 0x04/5150 0x06/7170	
256	32	4	0x00/2F...13121110 0x20/4F...33323130 0x40/6F...53525150 0x60/8F...73727170	0x00/13121110 0x04/33323130 0x08/53525150 0x0C/73727170	
256	256	4	0x00/2F...13121110 0x20/4F...33323130 0x40/6F...53525150 0x60/8F...73727170	0x00/2F...13121110 0x20/4F...33323130 0x40/6F...53525150 0x60/8F...73727170	

10.2.3 DMA Request Multiplexer

The DMAMUX multiplexer reconfigures (routes) DMA request lines between the chip's peripherals and the DMA controller. This routing capability is achieved through a programmable multi-channel DMA request line multiplexer. Each channel can either select a DMA request line without restriction or synchronize its selection with events from its DMAMUX synchronous input.

The DMA controller provides 16 channels, where DMAMUX channels 1 through 8 are connected to DMA1 channels 1 through 8, and DMAMUX channels 9 through 16 are connected to DMA2 channels 1 through 8.

Table 10-2 DMA multiplexer input to resource allocation table

DMA request input	Peripheral	DMA request input	Peripheral	DMA request input	Peripheral
1	TIM1_CH1	42	TIM9_CH4	83	I3C_TX
2	TIM1_CH2	43	TIM9_UP	84	I3X_RX
3	TIM1_CH3	44	TIM9_TRIG	85	USART1_TX
4	TIM1_CH4	45	TIM10_CH1	86	USART1_RX
5	TIM1_UP	46	TIM10_CH2	87	USART2_TX
6	TIM1_COM	47	TIM10_CH3	88	USART2_RX
7	TIM1_TRIG	48	TIM10_CH4	89	USART3_TX
8	TIM2_CH1	49	TIM10_UP	90	USART3_RX
9	TIM2_CH2	50	TIM10_TRIG	91	USART4_TX
10	TIM2_CH3	51	TIM11_CH1	92	USART4_RX
11	TIM2_CH4	52	TIM11_CH2	93	USART5_TX
12	TIM2_UP	53	TIM11_CH3	94	USART5_RX
13	TIM2_TRIG	54	TIM11_CH4	95	USART6_TX
14	TIM3_CH1	55	TIM11_UP	96	USART6_RX
15	TIM3_CH2	56	TIM11_TRIG	97	USART7_TX
16	TIM3_CH3	57	TIM12_CH1	98	USART7_RX
17	TIM3_CH4	58	TIM12_CH2	99	USART8_TX
18	TIM3_UP	59	TIM12_CH3	100	USART8_RX
19	TIM3_TRIG	60	TIM12_CH4	101	SWPMI_TX
20	TIM4_CH1	61	TIM12_UP	102	SWPMI_RX
21	TIM4_CH2	62	TIM12_TRIG	103	DAC1
22	TIM4_CH3	63	SPI1_TX	104	DAC2
23	TIM4_CH4	64	SPI1_RX	105	Reserved
24	TIM4_UP	65	SPI2_TX	106	Reserved
25	TIM4_TRIG	66	SPI2_RX	107	DFSDM_DMA0
26	TIM5_CH1	67	SPI3_TX	108	DFSDM_DMA1
27	TIM5_CH2	68	SPI3_RX	109	Reserved
28	TIM5_CH3	69	SPI4_TX	110	Reserved
29	TIM5_CH4	70	SPI4_RX	111	SDIO
30	TIM5_UP	71	QSPI1_DMA	112	SAI_A_TX
31	TIM5_TRIG	72	QSPI2_DMA	113	SAI_A_RX
32	TIM8_CH1	73	I2C1_TX	114	SAI_B_TX

33	TIM8_CH2	74	I2C1_RX	115	SAI_B_RX
34	TIM8_CH3	75	I2C2_TX	116	Reserved
35	TIM8_CH4	76	I2C2_RX	117	Reserved
36	TIM8_UP	77	I2C3_TX	118	Reserved
37	TIM8_COM	78	I2C3_RX	119	Reserved
38	TIM8_TRIG	79	I2C4_TX	120	ADC1
39	TIM9_CH1	80	I2C4_RX	121	ADC2
40	TIM9_CH2	81	I3C_RS	122	TIM6_UP
41	TIM9_CH3	82	I3C_TC	123	TIM7_UP

10.2.4 Double Buffer Mode

Dual-buffered mode is used for DMA1 and DMA2 channels. While dual-buffered mode features two memory pointers, the channel operation remains identical to single-buffered mode.

Enable double-buffered mode by setting the DOUBLE_MODE bit to 1 in the DMA_CFGRx register. This automatically enters the cyclic mode and switches memory addresses (alternating between DMA_MADDRx and DMA_M1ADDRx) after completing one cycle.

At the end of each cycle, the DMA controller switches from one memory target to another. This allows software to process one memory region while the DMA transfer simultaneously fills or utilizes the second memory region. The dual-buffer channel can operate bidirectionally, as shown in the table below.

Table 10-3 Source and destination address registers in double buffer mode

DMA_CFGRx.DIR	Direction	Source address	Target address
0	Read from peripheral	DMA_PADDRx	DMA_MADDRx/DMA_M1ADDRx
1	Read from memory	DMA_MADDRx/DMA_M1ADDRx	DMA_PADDRx

Note: Configuring memory-to-memory mode is prohibited when double buffering mode is enabled.

10.3 Register Description

Table 10-4 DMA1 registers

Name	Access address	Description	Reset value
R32_DMA1_INTFR	0x40020000	DMA1 interrupt status register	0x00000000
R32_DMA1_INTFCR	0x40020004	DMA1 interrupt flag clear register	0x00000000
R32_DMA1_CFGR1	0x40020008	DMA1 channel 1 configuration register	0x00000000
R32_DMA1_CNTR1	0x4002000C	DMA1 channel 1 transfer data number register	0x00000000
R32_DMA1_PADDR1	0x40020010	DMA1 channel 1 peripheral address register	0x00000000
R32_DMA1_MADDR1	0x40020014	DMA1 channel 1 memory address register	0x00000000
R32_DMA1_M1ADDR1	0x40020018	DMA1 channel 1 memory address register 1	0x00000000
R32_DMA1_CFGR2	0x4002001C	DMA1 channel 2 configuration register	0x00000000
R32_DMA1_CNTR2	0x40020020	DMA1 channel 2 transfer data number register	0x00000000
R32_DMA1_PADDR2	0x40020024	DMA1 channel 2 peripheral address register	0x00000000

R32_DMA1_MADDR2	0x40020028	DMA1 channel 2 memory address register	0x00000000
R32_DMA1_M1ADDR2	0x4002002C	DMA1 channel 2 memory address register 1	0x00000000
R32_DMA1_CFGR3	0x40020030	DMA1 channel 3 configuration register	0x00000000
R32_DMA1_CNTR3	0x40020034	DMA1 channel 3 transfer data number register	0x00000000
R32_DMA1_PADDR3	0x40020038	DMA1 channel 3 peripheral address register	0x00000000
R32_DMA1_MADDR3	0x4002003C	DMA1 channel 3 memory address register	0x00000000
R32_DMA1_M1ADDR3	0x40020040	DMA1 channel 3 memory address register 1	0x00000000
R32_DMA1_CFGR4	0x40020044	DMA1 channel 4 configuration register	0x00000000
R32_DMA1_CNTR4	0x40020048	DMA1 channel 4 transfer data number register	0x00000000
R32_DMA1_PADDR4	0x4002004C	DMA1 channel 4 peripheral address register	0x00000000
R32_DMA1_MADDR4	0x40020050	DMA1 channel 4 memory address register	0x00000000
R32_DMA1_M1ADDR4	0x40020054	DMA1 channel 4 memory address register 1	0x00000000
R32_DMA1_CFGR5	0x40020058	DMA1 channel 5 configuration register	0x00000000
R32_DMA1_CNTR5	0x4002005C	DMA1 channel 5 transfer data number register	0x00000000
R32_DMA1_PADDR5	0x40020060	DMA1 channel 5 peripheral address register	0x00000000
R32_DMA1_MADDR5	0x40020064	DMA1 channel 5 memory address register	0x00000000
R32_DMA1_M1ADDR5	0x40020068	DMA1 channel 5 memory address register 1	0x00000000
R32_DMA1_CFGR6	0x4002006C	DMA1 channel 6 configuration register	0x00000000
R32_DMA1_CNTR6	0x40020070	DMA1 channel 6 transfer data number register	0x00000000
R32_DMA1_PADDR6	0x40020074	DMA1 channel 6 peripheral address register	0x00000000
R32_DMA1_MADDR6	0x40020078	DMA1 channel 6 memory address register	0x00000000
R32_DMA1_M1ADDR6	0x4002007C	DMA1 channel 6 memory address register 1	0x00000000
R32_DMA1_CFGR7	0x40020080	DMA1 channel 7 configuration register	0x00000000
R32_DMA1_CNTR7	0x40020084	DMA1 channel 7 transfer data number register	0x00000000
R32_DMA1_PADDR7	0x40020088	DMA1 channel 7 peripheral address register	0x00000000
R32_DMA1_MADDR7	0x4002008C	DMA1 channel 7 memory address register	0x00000000
R32_DMA1_M1ADDR7	0x40020090	DMA1 channel 7 memory address register 1	0x00000000
R32_DMA1_CFGR8	0x40020094	DMA1 channel 8 configuration register	0x00000000
R32_DMA1_CNTR8	0x40020098	DMA1 channel 8 transfer data number register	0x00000000
R32_DMA1_PADDR8	0x4002009C	DMA1 channel 8 peripheral address register	0x00000000
R32_DMA1_MADDR8	0x400200A0	DMA1 channel 8 memory address register	0x00000000
R32_DMA1_M1ADDR8	0x400200A4	DMA1 channel 8 memory address register 1	0x00000000

Table 10-5 DMA2 registers

Name	Access address	Description	Reset value
R32_DMA2_INTFR	0x40020400	DMA2 interrupt status register	0x00000000
R32_DMA2_INTFCR	0x40020404	DMA2 interrupt flag clear register	0x00000000
R32_DMA2_CFGR1	0x40020408	DMA2 channel 1 configuration register	0x00000000
R32_DMA2_CNTR1	0x4002040C	DMA2 channel 1 transfer data number register	0x00000000
R32_DMA2_PADDR1	0x40020410	DMA2 channel 1 peripheral address register	0x00000000
R32_DMA2_MADDR1	0x40020414	DMA2 channel 1 memory address register	0x00000000
R32_DMA2_M1ADDR1	0x40020418	DMA2 channel 1 memory address register 1	0x00000000
R32_DMA2_CFGR2	0x4002041C	DMA2 channel 2 configuration register	0x00000000

R32_DMA2_CNTR2	0x40020420	DMA2 channel 2 transfer data number register	0x00000000
R32_DMA2_PADDR2	0x40020424	DMA2 channel 2 peripheral address register	0x00000000
R32_DMA2_MADDR2	0x40020428	DMA2 channel 2 memory address register	0x00000000
R32_DMA2_M1ADDR2	0x4002042C	DMA2 channel 2 memory address register 1	0x00000000
R32_DMA2_CFGR3	0x40020430	DMA2 channel 3 configuration register	0x00000000
R32_DMA2_CNTR3	0x40020434	DMA2 channel 3 transfer data number register	0x00000000
R32_DMA2_PADDR3	0x40020438	DMA2 channel 3 peripheral address register	0x00000000
R32_DMA2_MADDR3	0x4002043C	DMA2 channel 3 memory address register	0x00000000
R32_DMA2_M1ADDR3	0x40020440	DMA2 channel 3 memory address register 1	0x00000000
R32_DMA2_CFGR4	0x40020444	DMA2 channel 4 configuration register	0x00000000
R32_DMA2_CNTR4	0x40020448	DMA2 channel 4 transfer data number register	0x00000000
R32_DMA2_PADDR4	0x4002044C	DMA2 channel 4 peripheral address register	0x00000000
R32_DMA2_MADDR4	0x40020450	DMA2 channel 4 memory address register	0x00000000
R32_DMA2_M1ADDR4	0x40020454	DMA2 channel 4 memory address register 1	0x00000000
R32_DMA2_CFGR5	0x40020458	DMA2 channel 5 configuration register	0x00000000
R32_DMA2_CNTR5	0x4002045C	DMA2 channel 5 transfer data number register	0x00000000
R32_DMA2_PADDR5	0x40020460	DMA2 channel 5 peripheral address register	0x00000000
R32_DMA2_MADDR5	0x40020464	DMA2 channel 5 memory address register	0x00000000
R32_DMA2_M1ADDR5	0x40020468	DMA2 channel 5 memory address register 1	0x00000000
R32_DMA2_CFGR6	0x4002046C	DMA2 channel 6 configuration register	0x00000000
R32_DMA2_CNTR6	0x40020470	DMA2 channel 6 transfer data number register	0x00000000
R32_DMA2_PADDR6	0x40020474	DMA2 channel 6 peripheral address register	0x00000000
R32_DMA2_MADDR6	0x40020478	DMA2 channel 6 memory address register	0x00000000
R32_DMA2_M1ADDR6	0x4002047C	DMA2 channel 6 memory address register 1	0x00000000
R32_DMA2_CFGR7	0x40020480	DMA2 channel 7 configuration register	0x00000000
R32_DMA2_CNTR7	0x40020484	DMA2 channel 7 transfer data number register	0x00000000
R32_DMA2_PADDR7	0x40020488	DMA2 channel 7 peripheral address register	0x00000000
R32_DMA2_MADDR7	0x4002048C	DMA2 channel 7 memory address register	0x00000000
R32_DMA2_M1ADDR7	0x40020490	DMA2 channel 7 memory address register 1	0x00000000
R32_DMA2_CFGR8	0x40020494	DMA2 channel 8 configuration register	0x00000000
R32_DMA2_CNTR8	0x40020498	DMA2 channel 8 transfer data number register	0x00000000
R32_DMA2_PADDR8	0x4002049C	DMA2 channel 8 peripheral address register	0x00000000
R32_DMA2_MADDR8	0x400204A0	DMA2 channel 8 memory address register	0x00000000
R32_DMA2_M1ADDR8	0x400204A4	DMA2 channel 8 memory address register 1	0x00000000

Table 10-6 DMAMUX registers

Name	Access address	Description	Reset value
R32_DMAMUX1_4_CFGR	0x40020800	DMA request multiplexer channel 1-4 configuration register	0x00000000
R32_DMAMUX5_8_CFGR	0x40020804	DMA request multiplexer channel 5-8 configuration register	0x00000000
R32_DMAMUX9_12_CFGR	0x40020808	DMA request multiplexer channel 9-12 configuration register	0x00000000

R32_DMAMUX13_16_CFGR	0x4002080C	DMA request multiplexer channel 13-16 configuration register	0x00000000
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10.3.1 DMAx Interrupt Flag Register (DMAx_INTFR) (x=1/2)

Offset address: $0x00 + (x-1)*0x400$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TEIF 8	HTIF 8	TCIF 8	GIF8	TEIF 7	HTIF 7	TCIF 7	GIF7	TEIF 6	HTIF 6	TCIF 6	GIF6	TEIF 5	HTIF 5	TCIF 5	GIF5
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEIF 4	HTIF 4	TCIF 4	GIF4	TEIF 3	HTIF 3	TCIF 3	GIF3	TEIF 2	HTIF 2	TCIF 2	GIF2	TEIF 1	HTIF 1	TCIF 1	GIF1

Bit	Name	Access	Description	Reset value
31/27/23/19 /15/11/7/3	TEIFx	RO	Transmission error flag for channel x (x= 1/2/3/4/5/6/7/8): 1: A transmission error occurred on channel x. 0: There is no transmission error on channel x. The hardware is set, and the software writes the CTEIFx bit to clear this flag.	0
30/26/22/18 /14/10/6/2	HTIFx	RO	Transmission more than half flag of channel x (x= 1/2/3/4/5/6/7/8): 1: More than half of the transmission events occurred on channel x. 0: No more than half of the transmission is on channel x. The hardware is set, and the software writes the CHTIFx bit to clear this flag.	0
29/25/21/17 /13/9/5/1	TCIFx	RO	Transmission completion flags for channel x (x=1/2/3/4/5/6/7/8): 1: A transmission completion event was generated on channel x; 0: No transmission completion event on channel x. Hardware sets and software writes the CTCIFx bit to clear this flag.	0
28/24/20/16 /12/8/4/0	GIFx	RO	Global interrupt flag for channel x (x=1/2/3/4/5/6/7/8): 1: TEIFx or HTIFx or TCIFx was generated on channel x; 0: No TEIFx or HTIFx or TCIFx has been generated on channel x. Hardware sets and software writes the CGIFx bit to clear this flag.	0

10.3.2 DMA Interrupt Flag Clear Register (DMAx_INTFCR) (x=1/2)

Offset address: $0x04 + (x-1)*0x400$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

CTEIF	CHTIF	CTCIF	CGIF	CTEIF	CHTIF	CTCIF	CGIF	CTEIF	CHTIF	CTCIF	CGIF	CTEIF	CHTIF	CTCIF	CGIF
8	8	8	8	7	7	7	7	6	6	6	6	5	5	5	5
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CTEIF	CHTIF	CTCIF	CGIF	CTEIF	CHTIF	CTCIF	CGIF	CTEIF	CHTIF	CTCIF	CGIF	CTEIF	CHTIF	CTCIF	CGIF
4	4	4	4	3	3	3	3	2	2	2	2	1	1	1	1

Bit	Name	Access	Description	Reset value
31/27/23/19 /15/11/7/3	CTEIFx	WO	Clear the transmission error flag for channel x (x=1/2/3/4/5/6/8): 1: Clear the TEIFx flag in the DMA_INTFR register; 0: No effect.	0
30/26/22/18 /14/10/6/2	CHTIFx	WO	Clear the transmit half flag for channel x (x=1/2/3/4/5/6/8): 1: Clear the HTIFx flag in the DMA_INTFR register; 0: No effect.	0
29/25/21/17 /13/9/5/1	CTCIFx	WO	Clear the transmission completion flag for channel x (x=1/2/3/4/5/6/7/8): 1: Clear the TCIFx flag in the DMA_INTFR register; 0: No effect.	0
28/24/20/16 /12/8/4/0	CGIFx	WO	Clear the global interrupt flag for channel x (x=1/2/3/4/5/6/7/8): 1: Clears the TEIFx/HTIFx/TCIFx/ GIFx flags in the DMA_INTFR register; 0: No effect.	0

10.3.3 DMAy Channel x Configuration Register (DMAy_CFGRx) (x=1/2/3/4/5/6/7/8, y=1/2)

Offset address: $0x08 + (x-1)*20 + (y-1)*0x400$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															FLAG_CUR_MEM
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DOUBLE_MODE	MEM2MEM	PL[1:0]	MSIZE[1:0]	PSIZE[1:0]	MINC	PINC	CIRC	DIR	TEIE	HTIE	TCIE	EN			

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	FLAG_CUR_MEM	RW	Memory Address Selection Settings (Hardware automatically switches in dual-buffering mode): 1: M1ADDR (uses DMA_M1ADDRx pointer addressing); 0: MADDR (uses DMA_MADDRx pointer addressing). <i>Note: This bit is only used in dual-buffering mode. After</i>	0

			<i>channel enable, this bit functions as a status flag indicating the memory region serving as the current target.</i>	
15	DOUBLE_MODE	RW	Double-buffer mode enable: 1: Enables double-buffer mode, automatically entering a looping cycle. After one cycle completes, the memory address switches. 0: Disables double-buffer mode.	0
14	MEM2MEM	RW	Memory to memory mode enable: 1: Enable memory to memory mode; 0: Disable memory to memory mode.	0
[13:12]	PL[1:0]	RW	Channel priority level setting: 00: Low; 01: Medium; 10: High; 11: Very high.	00b
[11:10]	MSIZE[1:0]	RW	Memory address data width setting: 00: 8-bit; 01: 16-bit; 10: 32-bit; 11: 256-bit.	00b
[9:8]	PSIZE[1:0]	RW	Peripheral address data width setting: 00: 8-bit; 01: 16-bit; 10: 32 bits; 11: 256-bit.	00b
7	MINC	RW	Memory address increment mode enables: 1: Enable memory address increment operation; 0: Memory address remains unchanged operation.	0
6	PINC	RW	Peripheral address increment mode enables: 1: Enable memory address increment operation; 0: Peripheral address remains unchanged operation.	0
5	CIRC	RW	DMA channel cyclic mode enable: 1: Enable cyclic operation; 0: Perform a single operation.	0
4	DIR	RW	Data transmission direction: 1: Read from memory. 0: Read from the peripheral.	0
3	TEIE	RW	Transmission error interrupt enable control: 1: Enable transmission error interrupt. 0: Disable transmission error interrupt.	0
2	HTIE	RW	More than half of the transmission interruption enables control: 1: Enable transmission for more than half an interruption. 0: Disable transmission for more than half an interruption.	0
1	TCIE	RW	Transfer complete interrupt enable control: 1: Enable transmission completion interrupt; 0: Disable transmission completion interrupt.	0
0	EN	RW	Channel enable control: 1: Channel enabled; 0: Channel disabled. When a DMA transmission error occurs, it will be cleared	0

			to 0 automatically by hardware, and channel is disabled.	
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10.3.4 DMAy Channel x Transfer Data Number Register (DMAy_CNTRx) (x=1/2/3/4/5/6/7/8, y=1/2)

Offset address: $0x0C + (x-1)*20 + (y-1)*0x400$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDT[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	NDT[15:0]	RW	Number of data transfers, range 0-65535. Indicating the remaining number of transfers (the register contents are decremented after each DMA transfer). When the channel is in loop mode, the contents of the register are automatically reloaded to the previously configured value.	0

Note: Indicating the current number of transfers. This register can only be changed when EN=0, and cannot be written when EN=1. When the register content is 0, no data transfer occurs regardless of whether the channel is open or not.

10.3.5 DMAy Channel x Peripheral Address Register (DMAy_PADDRx) (x=1/2/3/4/5/6/7/8, y=1/2)

Offset address: $0x10 + (x-1)*20 + (y-1)*0x400$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PA[31:0]																															

Bit	Name	Access	Description	Reset value
[31:0]	PA[31:0]	RW	The base address of the peripheral, which is used as the source or destination address of the peripheral data transmission. When PSIZE [1:0] = '01' (16 bits), the module automatically ignores bit0, and the operation address automatically aligns with 2 bytes; when PSIZE [1:0] = '10' (32 bits), the module automatically ignores bit [1:0], and the operation address automatically aligns with 4 bytes; when PSIZE [1:0] = '11' (256 bits), the module automatically ignores bit [4:0], and the operation address automatically aligns with 32 bytes.	0

10.3.6 DMAy Channel x Memory Address Register (DMAy_MADDRx) (x=1/2/3/4/5/6/7/8, y=1/2)

Offset address: $0x14 + (x-1)*20 + (y-1)*0x400$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MA[31:0]																															

Bit	Name	Access	Description	Reset value
[31:0]	MA[31:0]	RW	Memory data address, as the source or destination address for data transmission. When MSIZE [1:0] = '01' (16 bits), the module automatically ignores bit0, and the operation address automatically aligns with 2 bytes; when MSIZE [1:0] = '10'(32 bits), the module automatically ignores bit [1:0], and the operation address automatically aligns with 4 bytes; when MSIZE [1:0] ='11' (256 bits), the module automatically ignores bit [4:0], and the operation address automatically aligns with 32 bytes.	0

10.3.7 DMAy Channel x Memory Address Register 1 (DMAy_M1ADDRx) (x=1/2/3/4/5/6/7/8, y=1/2)

Offset address: $0x18 + (x-1)*20 + (y-1)*0x400$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M1A[31:0]																															

Bit	Name	Access	Description	Reset value
[31:0]	M1A[31:0]	RW	Memory data address, as the source or destination address for data transmission. When MSIZE [1:0] = '01' (16-bit), the module automatically ignores bit0, and the operation address automatically aligns with 2 bytes; when MSIZE [1:0] = '10'(32-bit), the module automatically ignores bit [1:0], and the operation address automatically aligns with 4 bytes; when MSIZE [1:0] ='11' (256-bit), the module automatically ignores bit [4:0], and the operation address automatically aligns with 32 bytes.	0

10.3.8 DMA Request Multiplexer Channel 1-4 Configuration Register (DMAMUX1_4_CFGR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reser ved	CHANNEL4_MUX[6:0]						Reser ved	CHANNEL3_MUX[6:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	CHANNEL2_MUX[6:0]						Reser ved	CHANNEL1_MUX[6:0]							

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
[30:24]	CHANNEL4_MUX[6:0]	RW	DMAMUX multiplexer channel 4 input to resource allocation.	0
23	Reserved	RO	Reserved	0
[22:16]	CHANNEL3_MUX[6:0]	RW	DMAMUX multiplexer channel 3 input to resource allocation.	0
15	Reserved	RO	Reserved	0
[14:8]	CHANNEL2_MUX[6:0]	RW	DMAMUX multiplexer channel 2 input to resource allocation.	0
7	Reserved	RO	Reserved	0
[6:0]	CHANNEL1_MUX[6:0]	RW	DMAMUX multiplexer channel 1 input to resource allocation: 0000000: DMA request input 1; 0000001: DMA request input 2; 0000010: DMA request input 3; 1111010: DMA request input 123; Other: Reserved.	0

10.3.9 DMA Request Multiplexer Channel 5-8 Configuration Register (DMAMUX5_8_CFGR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	CHANNEL8_MUX[6:0]						Reserved	CHANNEL7_MUX[6:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CHANNEL6_MUX[6:0]						Reserved	CHANNEL5_MUX[6:0]							

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
[30:24]	CHANNEL8_MUX[6:0]	RW	DMAMUX multiplexer channel 8 input to resource allocation.	0
23	Reserved	RO	Reserved	0
[22:16]	CHANNEL7_MUX[6:0]	RW	DMAMUX multiplexer channel 7 input to resource allocation.	0
15	Reserved	RO	Reserved	0
[14:8]	CHANNEL6_MUX[6:0]	RW	DMAMUX multiplexer channel 6 input to resource allocation.	0
7	Reserved	RO	Reserved	0
[6:0]	CHANNEL5_MUX[6:0]	RW	DMAMUX multiplexer channel 5 input to resource allocation:	0

			0000000: DMA request input 1; 0000001: DMA request input 2; 0000010: DMA request input 3; 1111010: DMA request input 123; Other: Reserved.	
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10.3.10 DMA Request Multiplexer Channel 9-12 Configuration Register (DMAMUX9_12_CFGR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reser ved	CHANNEL12_MUX[6:0]						Reser ved	CHANNEL11_MUX[6:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	CHANNEL10_MUX[6:0]						Reser ved	CHANNEL9_MUX[6:0]							

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
[30:24]	CHANNEL12_MUX[6:0]	RW	DMAMUX multiplexer channel 12 input to resource allocation.	0
23	Reserved	RO	Reserved	0
[22:16]	CHANNEL11_MUX[6:0]	RW	DMAMUX multiplexer channel 11 input to resource allocation.	0
15	Reserved	RO	Reserved	0
[14:8]	CHANNEL10_MUX[6:0]	RW	DMAMUX multiplexer channel 10 input to resource allocation.	0
7	Reserved	RO	Reserved	0
[6:0]	CHANNEL9_MUX[6:0]	RW	DMAMUX multiplexer channel 9 input to resource allocation: 0000000: DMA request input 1; 0000001: DMA request input 2; 0000010: DMA request input 3; 1111010: DMA request input 123; Other: Reserved.	0

10.3.11 DMA Request Multiplexer Channel 13-16 Configuration Register (DMAMUX13_16_CFGR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reser ved	CHANNEL16_MUX[6:0]						Reser ved	CHANNEL15_MUX[6:0]							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CHANNEL14_MUX[6:0]							Reserved	CHANNEL13_MUX[6:0]						

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
[30:24]	CHANNEL16_MUX[6:0]	RW	DMAMUX multiplexer channel 16 input to resource allocation.	0
23	Reserved	RO	Reserved	0
[22:16]	CHANNEL15_MUX[6:0]	RW	DMAMUX multiplexer channel 15 input to resource allocation.	0
15	Reserved	RO	Reserved	0
[14:8]	CHANNEL14_MUX[6:0]	RW	DMAMUX multiplexer channel 14 input to resource allocation.	0
7	Reserved	RO	Reserved	0
[6:0]	CHANNEL13_MUX[6:0]	RW	DMAMUX multiplexer channel 13 input to resource allocation: 0000000: DMA request input 1; 0000001: DMA request input 2; 0000010: DMA request input 3; 1111010: DMA request input 123; Other: Reserved.	0

Chapter 11 Analog-to-digital Converter (ADC)

The ADC module contains two 12-bit n analog-to-digital converters with the highest 80MHz input clock. Support 16 external channels and 2 internal signal source sampling sources. It can complete the functions of single conversion, continuous conversion, automatic scanning mode, discontinuous mode, external trigger mode and so on. You can monitor whether the channel voltage is within the threshold range by simulating the watchdog function.

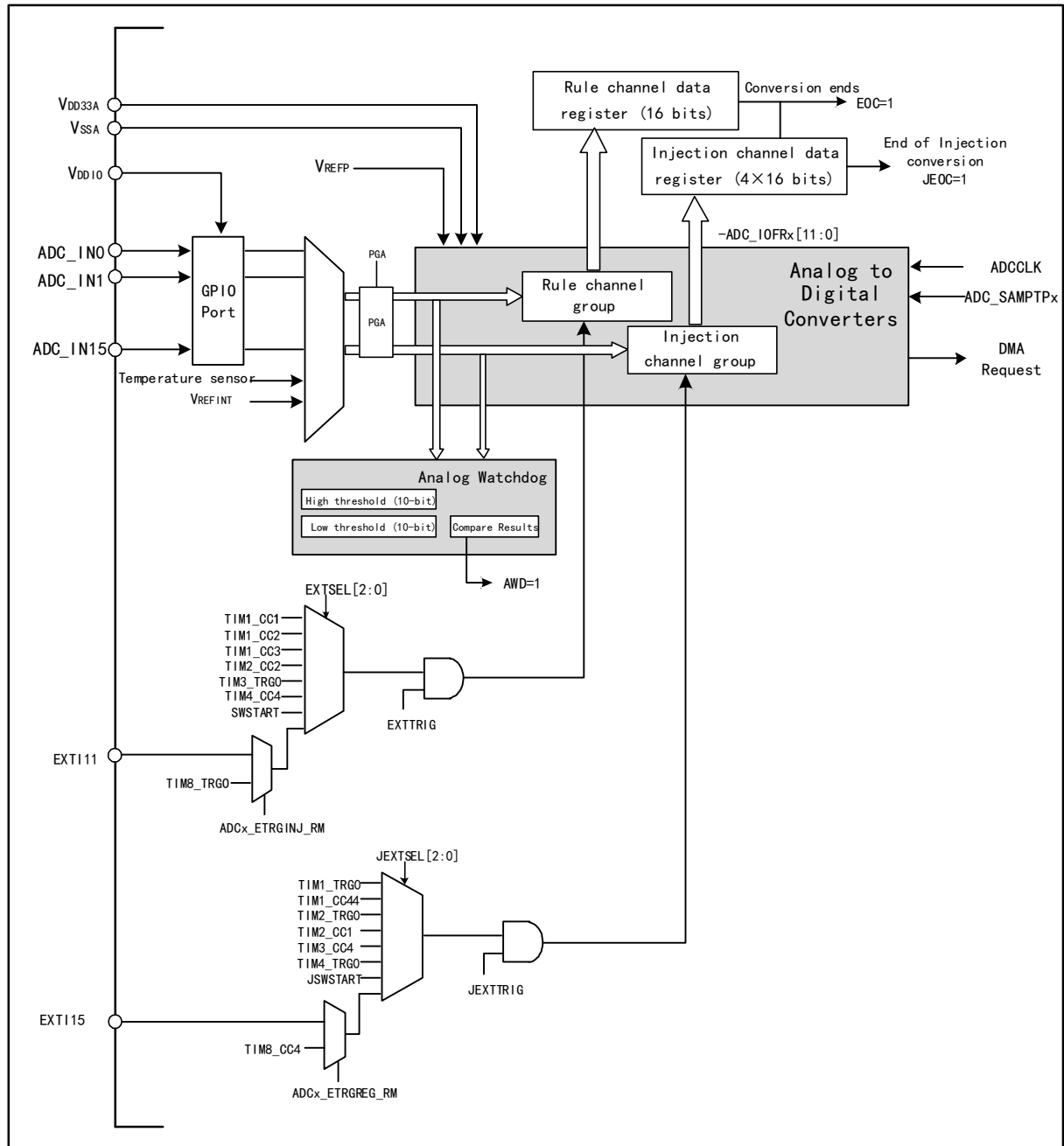
11.1 Main Features

- 12-bit resolution
- 16 external channels and 2 internal signal sources for sample
- Multiple sampling conversion modes for multiple channels: single, continuous, scan, trigger, discontinuous, etc.
- Data alignment mode: Left alignment, right alignment
- Sampling time can be programmed separately per channel
- Both regular conversion and injected conversion support external trigger
- Analog watchdog monitors the channel voltage
- Dual mode
- ADC channel input range: $V_{SS} \leq V_{IN} \leq V_{DDIO}$
- Adjustable input gain, to implement small signal amplification and sampling

11.2 Functional Description

11.2.1 Module Structure

Figure 11-1 ADC module block diagram



11.2.2 ADC Configuration

1) Module power on

An ADON bit of 1 in the ADCx_CTLR2 register indicates that the ADC module is powered on. When the ADC module enters the power-on state (ADON=1) from the power-off mode (ADON=0), it needs to be delayed for a period of time for t_{STAB} to stabilize the module. After that, the ADON bit is written again as 1, which is used as the startup signal for the software to start the ADC conversion. By clearing the ADON bit to 0, you can terminate the current conversion and put the ADC module in power-off mode, where ADC consumes almost no power.

2) Sample clock

The register operation of the module is based on the PCLK2 (HB bus) clock. The clock reference ADCCLK of the conversion unit is synchronized with the PCLK2. The frequency division is configured by the ADCPRE[1:0] domain of the RCC_CFGR0 register, and the maximum cannot exceed 80MHz.

3) Channel configuration

The ADC module provides 18 channel sampling sources, including 16 external channels and 2 internal channels. They can be configured into 2 translation groups: regular groups and injection groups. To realize the group conversion consisting of a series of conversions in any order on any number of channels.

Conversion group:

- Regular group: consists of up to 16 conversions. Regular channels and their conversion order are set in the ADCx_RSQRx register. The total number of translations in the regular group should be written to L [3:0] in the ADCx_RSQR1 register.
- Injection group: composed of up to 4 conversions. The injection channels and their conversion order are set in the ADCx_ISQR register. The total number of conversions in the injection group should be written to JL [1:0] in the ADCx_ISQR register.

Note: If the ADCx_RSQRx or ADCx_ISQR register is changed during the conversion, the current conversion is terminated and a new startup signal will be sent to ADC to convert the newly selected group.

3 internal channels:

- Temperature sensor: Connect the ADC_IN16 channel to measure the internal temperature(TA) of the chip.
- V_{REFINT} internal reference voltage: Connect the ADC_IN17 channel.

4) Calibration

Initialize the calibration register by writing bit 1 to the RSTCAL field of the ADCx_CTLR2 register. Wait for the RSTCAL bit to be cleared by hardware to indicate initialization completion. Set the CAL bit to initiate the calibration function. Once calibration finishes, the hardware automatically clears the CAL bit and stores the calibration code in ADCx_RDATAR. Normal conversion operations can then commence. It is recommended to perform ADC calibration once during module power-up.

The calibration voltage value can be configured by writing the CAL_VOL[1:0] bits in the ADCx_CTLR2 register. Set the CAL_AUTO bit to 1 to enable the hardware automatic calibration function.

Note: 1. Before starting the calibration, you must ensure that the ADC module is powered on (ADON=1) for at least 2 ADC clock cycles.

2. During ADC calibration, the sampling time must be at least 1μs.

3. When calibrating the ADC, the ADC channel gain must be set to ×1.

5) Programmable sample time

ADC uses several ADCCLK cycles to sample the input voltage, and the number of sampling cycles of the channels can be changed by the SMPx [2:0] bit in the ADCx_SAMPTR1 and ADCx_SAMPTR2 registers. Each channel can be sampled at different times.

The total conversion time is calculated as follows:

$$T_{\text{CONV}} = \text{Sampling time} + 12.5T_{\text{ADCCLK}}$$

The regular channel conversions of ADC supports the DMA feature. The value of regular channel conversion is stored in a single data register ADCx_RDATAR. In order to prevent the continuous conversion of multiple regular channels from not taking the data from the ADCx_RDATAR register in time, the DMA function of ADC can be turned on. The hardware generates an DMA request at the end of the conversion of the regular channel (EOC setting) and transfers the translated data from the ADCx_RDATAR register to the destination address specified by the user.

After the channel configuration of the DMA controller module is completed, write the DMA location 1 of the ADCx_CTLR2 register and turn on the DMA function of ADC.

Note: The injection group conversion does not support the DMA feature.

6) Data alignment

The ALIGN bit in the ADCx_CTLR2 register selects the data storage alignment after ADC conversion. 12-bit data supports left and right alignment modes.

The data register ADCx_RDATAR of the regular group channel stores the actual converted 12-bit digital value, while the data register ADCx_IDATARx of the injected group channel is the value written after the actual converted data minus the offset of the ADCx_IOFRx register, so there are positive and negative SIGNB.

Figure 11-2 Data left alignment

Regular group data register

D11	D10	D9	D8	D7	D6	D5	D4	D4	D2	D1	D0	0	0	0	0
-----	-----	----	----	----	----	----	----	----	----	----	----	---	---	---	---

Injected group data register

SIGNB	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	0	0	0
-------	-----	-----	----	----	----	----	----	----	----	----	----	----	---	---	---

Figure 11-3 Data right alignment

Regular group data register

0	0	0	0	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
---	---	---	---	-----	-----	----	----	----	----	----	----	----	----	----	----

Injected group data register

SIGNB	SIGNB	SIGNB	SIGNB	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
-------	-------	-------	-------	-----	-----	----	----	----	----	----	----	----	----	----	----

11.2.3 External Trigger Source

The start event of an ADC conversion can be triggered by an external event. If the EXTTRIG or JEXTTRIG bit of the ADCx_CTLR2 register is set, the conversion of the regular group or injection group channel can be triggered by external events, respectively. At this point, the configuration of the EXTSEL[2:0] and JEXTSEL[2:0] bits determines the external event sources of the regular group and the injection group.

Table 11-1 External trigger sources of regular group channel

EXTSEL[2:0]	Trigger source	Type
000	CC1 event of timer 1	Internal signal from on-chip timers
001	CC2 event of timer 1	
010	CC3 event of timer 1	
011	CC2 event of timer 2	
100	TRGO event of timer 3	
101	CC4 event of timer 4	

110	EXTI line 11/ TIM8_TRGO	From external pin/internal timer signal
111	RSWSTART position 1 software trigger	Software control bit

Table 11-2 External trigger sources of injected group channel

JEXTSEL[2:0]	Trigger source	Type
000	TRGO event of timer1	Internal signal from on-chip timers
001	CC4 event of timer1	
010	TRGO event of timer2	
011	CC1 event of timer2	
100	CC4 event of timer3	
101	TRGO event of timer4	
110	EXTI line 15/TIM8_CC4	From external pin/internal timer signal
111	JSWSTART position 1 software trigger	Software control bit

11.2.4 Conversion Mode

Table 11-3 Conversion mode combination

ADCx_CTLR1 and ADCx_CTLR2 register control bits					ADC conversion mode
CONT	SCAN	DISCEN/JDISCEN	JAUTO	Start event	
0	0	0	0	ADON bit set to 1	Single channel mode: A regular channel performs a single conversion.
				External trigger mode	Single channel mode: A regular channel or a channel of an injection channel performs a single conversion.
	1	0	0	ADON bit set to 1 or external trigger mode	Single scan mode: performs a single conversion on all selected regular group channels (ADCx_RSQRx) or all injection group channels (ADCx_ISQR) one by one in sequence. Trigger injection method: during the regular group channel conversion, all the conversions of the injection group channel can be inserted, and then the regular group channel conversion can be continued; however, the regular group channel conversion will not be inserted when the conversion is injected into the group channel.
			1	ADON bit set to 1 or external trigger mode	Single scan mode: performs a single conversion on all selected regular group channels (ADCx_RSQRx) or all injection group channels (ADCx_ISQR) one by one in sequence.

					Automatic injection mode: after the regular group channel is converted, the injection group channel is automatically converted. <i>Note: the external trigger signal of the injection channel is not allowed during the conversion process.</i>
	0	1 (DISCEN and JDISCEN cannot be 1 at the same time)	0	External trigger mode	Discontinuous break mode: each time an event is started, a short sequence of channel number conversions (the number defined by DISCNUM [2:0]) is performed until all selected channel conversions are completed. <i>Note: Regular group and injection group select this mode control bit as DISCEN and JDISCEN respectively. Discontinuous mode cannot be configured for both regular group and injection group. Discontinuous mode can only be used for one set of conversions.</i>
			1	-	Disable such mode.
	1	1	X	-	No such mode.
1	0	0	0	ADON bit set to 1 or external trigger mode	Continuous single channel/scan mode: repeat a new round of conversion at the end of each round until the CONT is cleared 0.
	1	0	1		

Note: The external trigger events of the regular group and the injection group are different, and the "ADON" bit can only start the regular group channel conversion, so the start events of the regular group and injection group channel conversion are independent.

1) Single channel conversion mode

In this mode, only one conversion is performed for the current 1 channel. This mode performs conversion on the channels sorted first in the regular group or injection group, where it can be started by setting ADON position 1 of the ADCx_CTLR2 register (for regular channels only) or by external triggering (for regular channels or injection channels). Once the conversion of the selected channel is completed:

If the regular group channel is converted, the conversion data is stored in the 16-bit ADCx_RDATAR register, the EOC flag is set, and if the EOCIE bit is set, the ADC interrupt will be triggered.

If the injection group channel is converted, the conversion data is stored in the 16-bit ADCx_IDATAR1 register, the EOC and JEOC flags are set, and if the JEOCIE or EOCIE bit is set, the ADC interrupt will be triggered.

2) Single scan mode conversion

Enter ADC scan mode by setting the scan bit of the ADCx_CTLR1 register to 1. This mode is used to scan a set of analog channels and perform a single conversion one by one for all channels selected by the ADCx_RSQRx register (for regular channels) or ADCx_ISQR (for injection channels). When the current channel conversion ends, the next channel of the same group is automatically converted.

In the scanning mode, according to the state of JAUTO bits, it can be divided into trigger injection mode and

automatic injection mode.

- Trigger injection

The JAUTO bit is 0, when the trigger event of the injection group channel conversion occurs in the process of scanning the regular group channel, the current conversion is reset, and the sequence of the injection channel is carried out in a single scan mode. After all the selected injection group channel scan conversion is completed, the last interrupted regular group channel conversion is restored.

If the start event of the regular channel occurs when scanning the channel sequence of the injection group, the conversion of the injection group will not be interrupted, but the conversion of the regular sequence will be performed after the conversion of the injection sequence is completed.

Note: when using triggered injection conversions, you must ensure that the interval between triggered events is longer than the injection sequence. For example, if it takes 28 ADCCLK to complete the conversion of the injection sequence, the minimum time between events to trigger the injection channel is 29 ADCCLK.

- Automatic injection

The JAUTO bit is 1, and after scanning all the channel conversions selected by the regular group, the channel selected by the injection group is converted automatically. This method can be used to convert up to 20 conversion sequences in ADCx_RSQRx and ADCx_ISQR registers.

In this mode, the external trigger (JEXTTRIG=0) of the injection channel must be disabled.

Note: When the ADC clock pre-division factor (ADCPRE [1:0]) is 4 to 8, 1 ADCCLK interval is automatically inserted when switching from regular conversion to injection sequence or from injection conversion to regular sequence; when the ADC clock pre-division factor is 2, there is a delay of 2 ADCCLK intervals.

3) Single discontinuous mode conversion

Enter the break mode of the regular group or injection group by setting the DISCEN or JDISCEN bit of the ADCx_CTLR1 register to 1. This mode distinguishes scanning a complete set of channels in the scan mode, but divides a group of channels into multiple short sequences, and each external trigger event will perform a scan conversion of a short sequence.

The length of the short sequence n ($n \leq 8$) is defined in the DISCNUM [2:0] of the ADCx_CTLR1 register. When DISCEN is 1, it is the discontinuous mode of the regular group, and the total length to be converted is defined in the L[3:0] of the ADCx_RSQR1 register. When JDISCEN is 1, it is the discontinuous mode of the injection group, and the total length to be converted is defined in the JL [1:0] of the ADCx_ISQR register. Both regular group and injection group cannot be set to discontinuous mode at the same time.

Example of regular group discontinuity mode:

DISCEN=1, DISCNUM [2:0] = 3, L[3:0] = 8, to be switched channel = 1, 3, 2, 5, 8, 4, 7, 6.

The first external trigger: the conversion sequence is: 1, 3, 2.

The second external trigger: the conversion sequence is: 5, 8, 4.

The third external trigger: the conversion sequence is: 10, 6, and EOC events are generated at the same time.

The 4th external trigger: the conversion sequence is: 1, 3, 2.

Example of injection group discontinuity mode:

JDISCEN=1, JL[1:0] = 3, Channel to be changed = 1, 3, 2.

The first external trigger: the conversion sequence is: 1.

The second external trigger: conversion sequence is: 3.

The third external trigger: the conversion sequence is: 2, and both EOC and JEOC events are generated.

The 4th external trigger: conversion sequence is: 1.

Note:

1. When a regular group or injection group is converted in discontinuous mode, the conversion sequence does not automatically start from scratch at the end of the conversion sequence. When all subgroups are converted, the next trigger event initiates the conversion of the first subgroup.
2. You cannot use both automatic injection (JAUTO=1) and discontinuous mode.
3. Discontinuous mode cannot be set for both regular groups and injection groups, and can only be used for one set of conversions.
4. In the intermittent mode of the injection group, the number of injection channels to be switched after an external trigger is 1.

4) Continuous conversion

In continuous conversion mode, another conversion is started as soon as the current ADC conversion is completed, and the conversion does not stop on the last channel of the selection group, but continues again from the first channel of the selection group. Boot events in this mode include external trigger events and ADON position 1. After setting boot, CONT position 1 needs to be set.

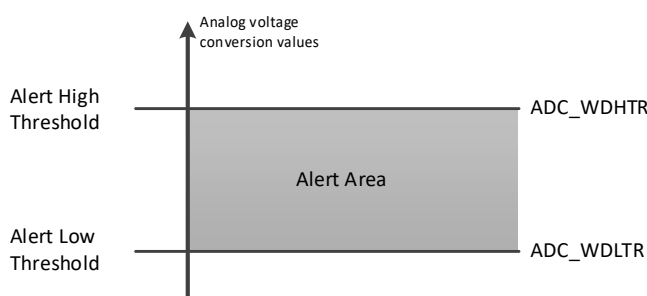
If a regular channel is converted, the converted data is stored in the ADCx_RDATAR register, the end of the conversion indicates that the EOC is set, and if EOCIE is set, an interrupt is generated.

If an injection channel is converted, the converted data is stored in the ADCx_IDATARx register, the end of injection conversion flag JEOC is set, and an interrupt is generated if JEOCIE is set.

11.2.5 Analog Watchdog

If the analog voltage being converted by the ADC is below the low threshold or above the high threshold, the AWD analog watchdog status bit is set. The threshold setting is located in the lowest 12 valid bits of the ADCx_WDHTR and ADCx_WDLTR registers. The corresponding interrupt is allowed to be generated by setting the AWDIE bit in the ADCx_CTLR1 register.

Figure 11-4 Analog watchdog threshold area



Configure the AWDSGL, AWDEN, JAWDEN, and AWDCH[4:0] bits of the ADCx_CTLR1 register to select the channel for analog watchdog alert, see the following table for the relationship:

Table 11-4 Analog Watchdog channel selection

Analog Watchdog alert channel	ADCx_CTLR1 register control bit			
	AWDSGL	AWDEN	JAWDEN	AWDCH[4:0]

No alert	Ignore	0	0	Ignore
All injection channels	0	0	1	Ignore
All regular channels	0	1	0	Ignore
All injection and regular channels	0	1	1	Ignore
Single injected channel	1	0	1	Determine channel No.
Single regular channel	1	1	0	Determine channel No.
Single injected and regular channel	1	1	1	Determine channel No.

11.2.6 Temperature Sensor

Chip built-in temperature sensor, connected to the ADC_INT16 channel, through ADC to convert the output voltage of the sensor into digital value to feedback the internal temperature of the chip, it is recommended to set the sampling time is 17.1μs. The output voltage of the temperature sensor varies linearly with the temperature. Due to the manufacturing discreteness, the slope and offset of the linear curve are different, so the internal temperature sensor is more suitable for detecting the change of temperature rather than measuring the absolute temperature. If you need to measure the temperature accurately, you should use an external temperature sensor.

By setting the TSVREFE position 1 of the ADCx_CTLR2 register, awakening the ADC internal sampling channel, software startup or external trigger starts the temperature sensor channel conversion of the ADC, and reads the data result (mV). Among them, the conversion formula of numerical value and temperature (°C) is as follows:

$$\text{Temperature (}^{\circ}\text{C)} = ((V_{\text{SENSE}} - V_{25}) / \text{Avg_Slope}) + 25$$

V₂₅: The voltage value of the temperature sensor at 25°C

Avg_Slope: Average slope of temperature and V_{SENSE} curve (mV/°C)

Refer to the actual values of V₂₅ and Avg_Slope in the electrical characteristics section of the datasheet.

Note: it takes a setup time for the internal temperature sensor to power on (the TSVREFE bit is changed from 0 to 1), and the ADC module also needs a setup time (the ADON bit is changed from 0 to 1), so in order to shorten the waiting time, both the ADON and TSVREFE bits can be set.

11.2.7 Dual ADC Mode

In ADC modules with 2 chips, both ADCs can be used together to enable dual ADC mode. In dual ADC mode, ADC1 serves as the primary ADC and ADC2 as the secondary ADC. By configuring DUALMOD[3:0] in ADC1_CTLR1 to select the mode, ADC1 and ADC2 can be triggered alternately or synchronously for conversion.

Note: In dual ADC mode, when selecting external trigger events for activation, users must enable external triggering on both the master and slave ADCs. The master ADC must be configured for the corresponding trigger, while the slave ADC should be set to software triggering. This prevents unnecessary triggering that could cause the slave ADC to convert.

The following possible modes can be achieved through configuration:

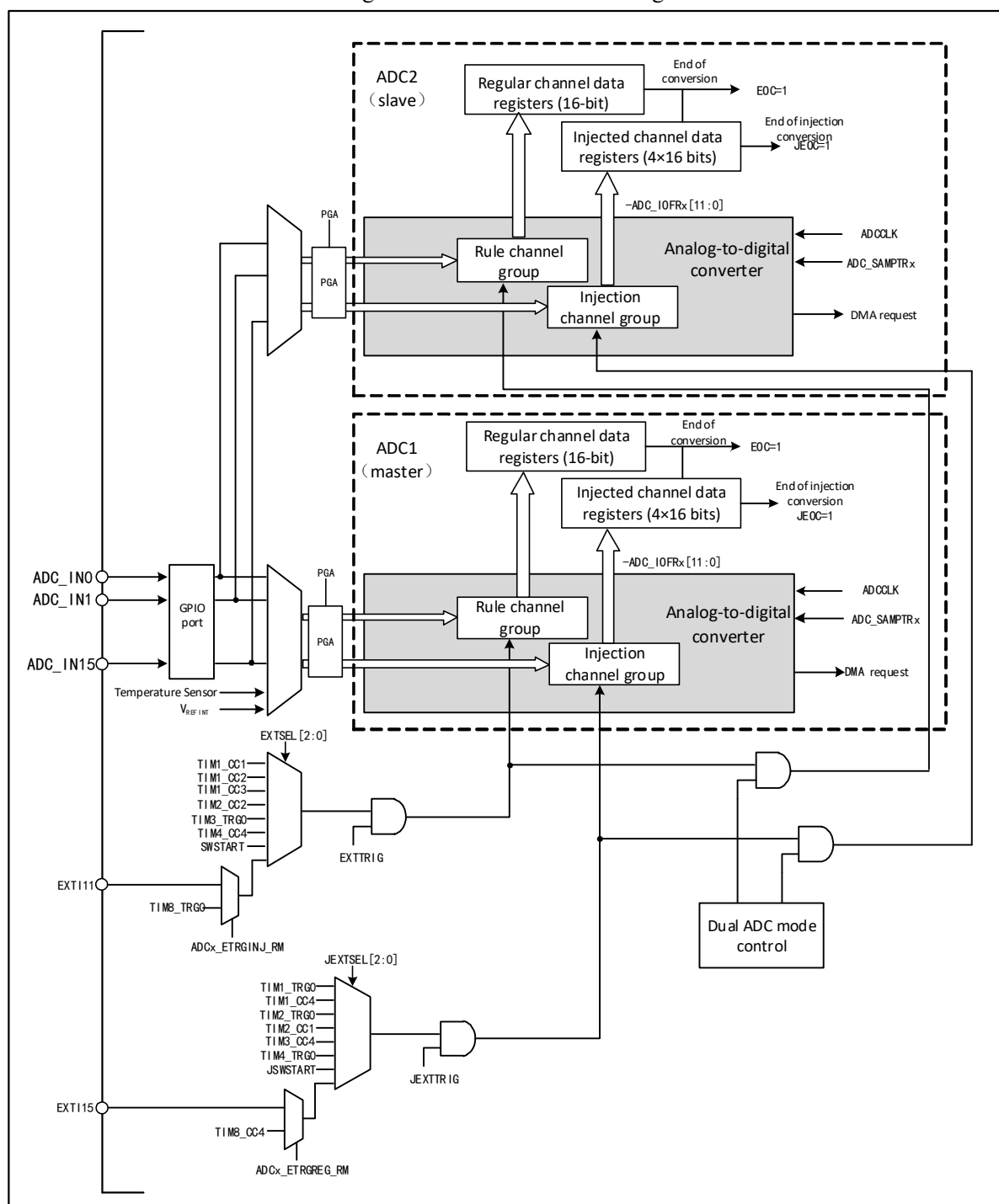
- Independent Mode
- Synchronous Injection Mode
- Synchronous Regular Mode
- Fast Alternating Mode
- Slow Alternating Mode
- Alternating Trigger Mode
- Synchronous Regular Mode + Synchronous Injection Mode

- Synchronous Regular Mode + Alternating Trigger Mode
- Synchronous Injection Mode + Fast Alternation Mode
- Synchronous Injection Mode + Slow Alternation Mode

Note: 1. In dual ADC mode, the DMA bit must be enabled to allow the main data register to read conversion data from the slave ADC.

2. Only ADC1 possesses DMA functionality. Data converted by ADC2 can only be transferred via dual ADC mode using ADC1's DMA capability.

Figure 11-5 Dual ADC block diagram



Note: 1. The conversion channels of ADC1 and ADC2 should not overlap at the same moment.

2. In synchronous mode, ADC1 and ADC2 should have conversion sequences of equal duration, or the duration of the longer sequence should be shorter than the trigger interval to ensure both sequences complete conversion with each trigger.

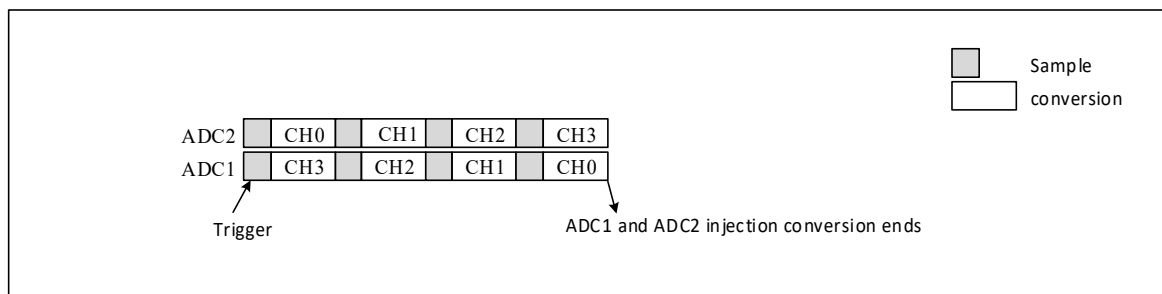
1) Independent Mode

In this mode, the dual ADCs work asynchronously and independently.

2) Synchronous injection mode

In this mode, it is used to convert a channel group. Set JEXTSEL[2:0] in ADC1_CTLR2 to select the trigger source, which will also synchronize the trigger for ADC2. Upon conversion completion, the converted data is stored in ADCx_IDATARx for each ADCx. If any ADC interrupt is enabled, a JEOC interrupt will be generated after conversion ends.

Figure 11-6 Channel synchronous injection conversion



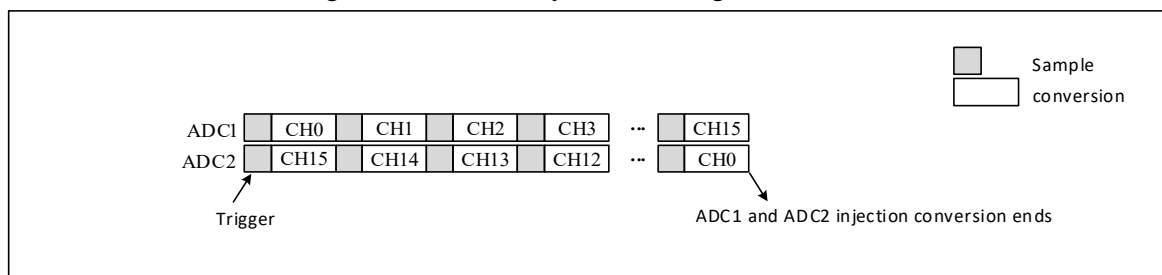
Note: 1. The conversion channels of ADC1 and ADC2 should not overlap at the same moment.

2. In synchronous mode, ADC1 and ADC2 should have conversion sequences of equal duration, or the duration of the longer sequence should be shorter than the trigger interval to ensure both sequences complete conversion with each trigger.

3) Synchronous regular mode

In this mode, the conversion regular channel sequence is used to configure EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source, which also synchronizes the trigger for ADC2. Upon conversion completion, a 32-bit DMA transfer request is generated to transfer the contents of the ADC1_RDATAR data register to SRAM. The upper 16 bits contain ADC2 conversion data, while the lower 16 bits contain ADC1 conversion data. Enabling interrupts for either ADC will generate an EOC interrupt.

Figure 11-7 Channel synchronous regular conversion



Note: 1. The conversion channels of ADC1 and ADC2 should not overlap at the same moment.

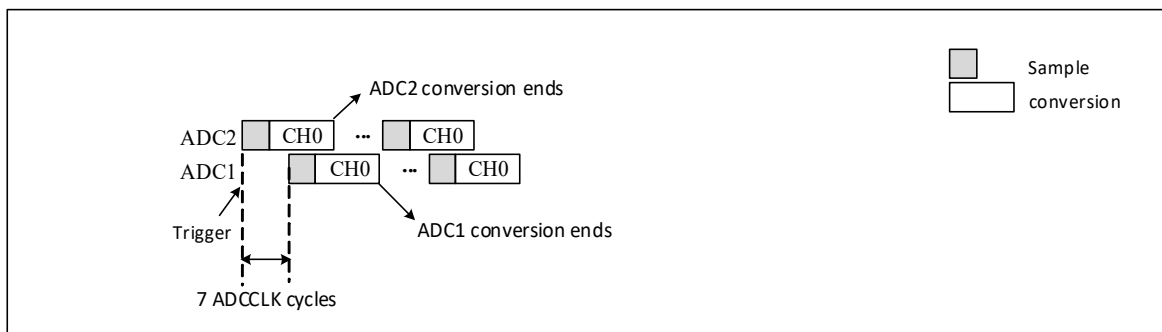
2. In synchronous mode, ADC1 and ADC2 should have conversion sequences of equal duration, or the duration of the longer sequence should be shorter than the trigger interval to ensure both sequences complete conversion with each trigger.

conversion with each trigger.

4) Fast alternating mode

This mode applies only to the regular channel (typically the sole channel). Configure EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. Upon trigger generation, ADC2 initiates conversion immediately, while ADC1 begins conversion after a delay of 7 ADC clock cycles. If both ADCs are enabled in continuous mode (CONT set), they will perform alternating conversions on the regular channel. If interrupts are enabled, ADC1 will generate an EOC interrupt. If DMA is also enabled, this will trigger a 32-bit DMA transfer request, transferring the contents of the ADC1_RDATAR data register to SRAM. The upper 16 bits contain ADC2 conversion data, while the lower 16 bits contain ADC1 conversion data.

Figure 11-8 Single channel fast alternating continuous conversion

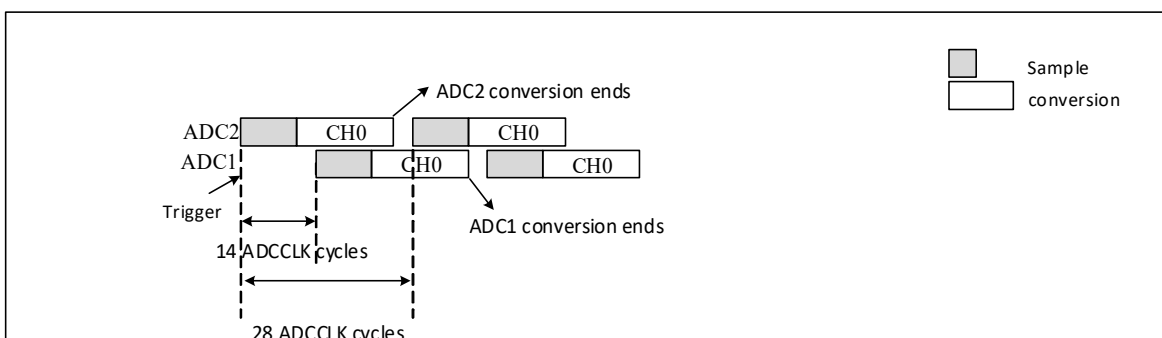


Note: The sampling time should be less than 7 ADC clock cycles, so as to avoid the problem of overlapping sampling periods when ADC1 and ADC2 sample the same channel.

5) Slow alternating mode

This mode is only applicable to the regular channel and supports only one channel. Configure EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. Upon trigger generation, ADC2 initiates conversion immediately, while ADC1 starts conversion after a delay of 14 ADC clock cycles. After another delay of 14 ADC clock cycles, ADC2 initiates conversion again, repeating this cycle. If interrupts are enabled, ADC1 will generate an EOC interrupt. If DMA is simultaneously enabled, a 32-bit DMA transfer request will be generated, transferring the contents of the ADC1_RDATAR data register to SRAM. The upper 16 bits contain the ADC2 conversion data, while the lower 16 bits contain the ADC1 conversion data.

Figure 11-9 Single channel slow alternating continuous conversion



Note: 1. The sampling time should be less than 14 ADC clock cycles to avoid overlapping with the next sampling cycle;

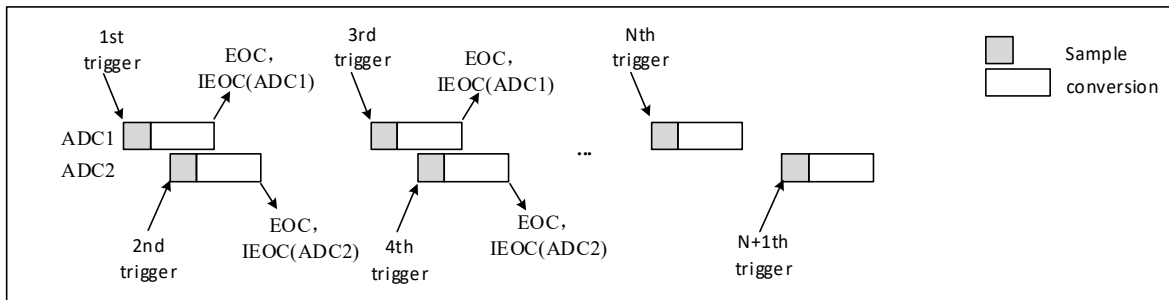
2. A new ADC2 conversion will automatically start after 28 ADC clock cycles;

3. The CONT bit does not need to be set.

6) Alternate trigger mode

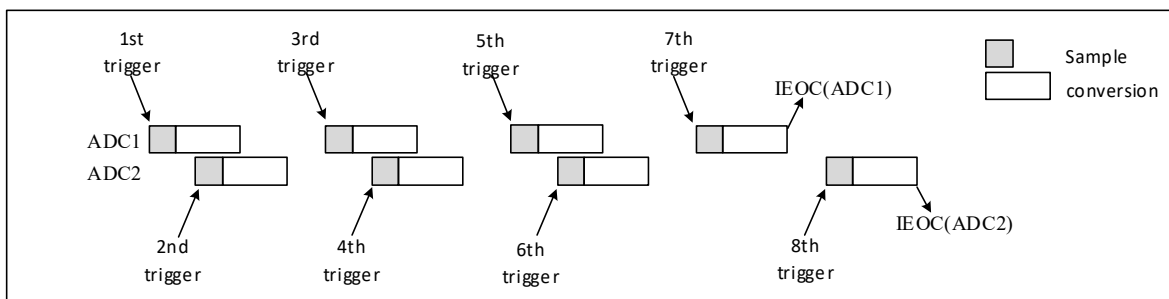
This mode applies only to injection channel groups. Configure JEXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. Upon the first trigger event, all injection channels on ADC1 undergo conversion. Upon the second trigger event, all injection channels on ADC2 undergo conversion. This cycle repeats sequentially. If interrupts are enabled, a JEOC interrupt is generated when all injection channels on ADC1 complete conversion, and another JEOC interrupt is generated when all injection channels on ADC2 complete conversion.

Figure 11-10 Alternate trigger conversion for each ADC injection channel group



If injection break mode is used simultaneously, the first injection channel on ADC1 is converted when the first trigger event occurs, the first injection channel on ADC2 is converted when the second trigger event occurs, and so on. If interrupt is enabled, a JEOC interrupt is generated when all injection channels on ADC1 are converted, and a JEOC interrupt is generated when all injection channels on ADC2 are converted.

Figure 11-11 Alternating trigger conversions for each ADC injection channel in discontinuous mode



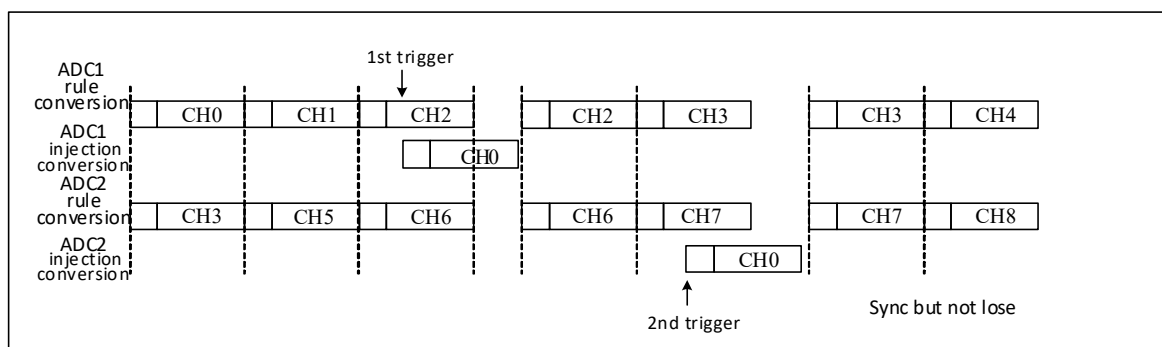
7) Synchronous regular mode + synchronous injection mode

In this mode, the synchronous conversion of a regular group can be interrupted to initiate the synchronous conversion of an injection group. At the same time, it is necessary to ensure that the conversions have the same length of time, or that the trigger time interval is longer than the longer of the two sequences.

8) Synchronous regular mode + alternating trigger mode

In this mode, synchronous conversion of regular groups can be interrupted to initiate alternate trigger conversion of injection groups. When an injection event occurs, alternate trigger conversion is initiated immediately. If synchronous regular conversion is ongoing, regular conversion of all ADCs is stopped, and synchronization resumes after injection conversion.

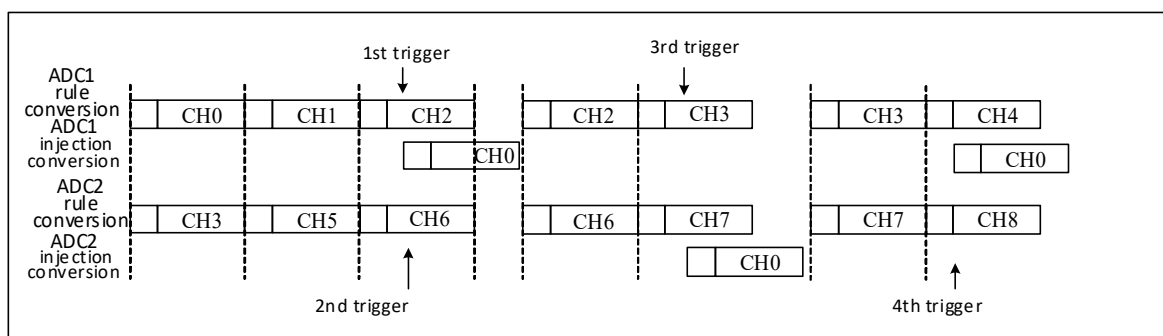
Figure 11-12 Alternating trigger injection channel switching in synchronous regular mode



Note: In this mode, you need to ensure that sequences with the same time length are converted, or that the trigger interval is longer than the longer of the two sequences.

If an injection trigger event occurs during an injection conversion that interrupts a regular conversion, the trigger event is ignored, such as in the case of the second trigger depicted in the figure below.

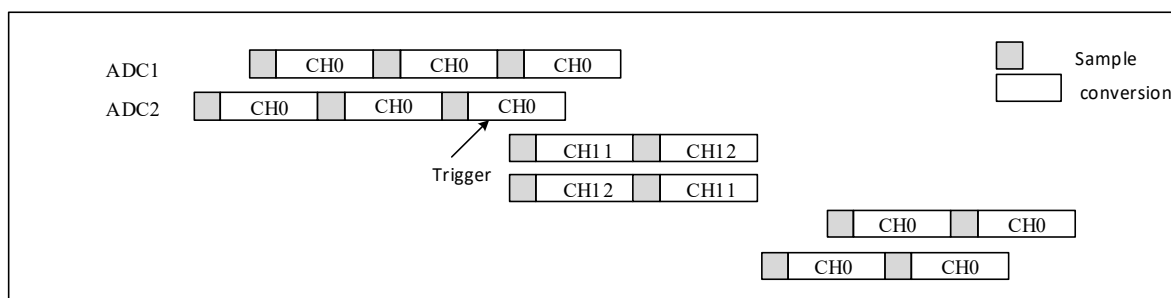
Figure 11-13 Trigger event occurs during injection conversion



9) Synchronous injection mode + alternating mode

In this mode, injection conversions can interrupt alternating conversions. When an injection event occurs, the alternating conversion is interrupted, the injection conversion begins, and after the injection conversion completes, the alternating conversion resumes.

Figure 11-14 Trigger injection group switching under alternate switching



Note: When the prescaler of the ADC is 4, the sampling interval after alternate conversion recovery is no longer uniform at 7 ADC clocks, but alternating between 8 and 6 clock cycles.

11.2.8 DFSDM Conversion

The ADC conversion result can be directly transferred to the digital filter with $\Sigma\Delta$ modulator (DFSDM). In this case,

the ADCx_AUX[31] bit must be set to 1, and the DMA bit must be cleared to 0. The ADC transfers the 12 least significant bits of the regular data register data to the DFSDM. Once the transfer is complete, the DFSDM resets the EOC flag.

11.2.9 ADC Low-power Mode

The ADC operating mode can be configured via the ADC_LP bit in the ADCx_CTLR2 register. When set to 0 (default), the mode consumes higher power and is suitable for sampling rates of 6M or higher, provided V_{DD33A} is above 3V. When set to 1, the low-power mode consumes lower power and is suitable for sampling rates below 2 Mbps.

11.3 Register Description

Table 11-5 ADC1-related registers

Name	Access address	Description	Reset value
R32_ADC1_STATR	0x40012400	ADC1 status register	0x00000000
R32_ADC1_CTLR1	0x40012404	ADC1 control register 1	0x00000000
R32_ADC1_CTLR2	0x40012408	ADC1 control register 2	0x000000D0
R32_ADC1_SAMPTR1	0x4001240C	ADC1 sample time configuration register 1	0x00000000
R32_ADC1_SAMPTR2	0x40012410	ADC1 sample time configuration register 2	0x00000000
R32_ADC1_IOFR1	0x40012414	ADC1 injected channel data offset register 1	0x00000000
R32_ADC1_IOFR2	0x40012418	ADC1 injected channel data offset register 2	0x00000000
R32_ADC1_IOFR3	0x4001241C	ADC1 injected channel data offset register 3	0x00000000
R32_ADC1_IOFR4	0x40012420	ADC1 injected channel data offset register 4	0x00000000
R32_ADC1_WDHTR	0x40012424	ADC1 watchdog high threshold register	0x00000FFF
R32_ADC1_WDLTR	0x40012428	ADC1 watchdog low threshold register	0x00000000
R32_ADC1_RSQR1	0x4001242C	ADC1 regular channel sequence register 1	0x00000000
R32_ADC1_RSQR2	0x40012430	ADC1 regular channel sequence register 2	0x00000000
R32_ADC1_RSQR3	0x40012434	ADC1 regular channel sequence register 3	0x00000000
R32_ADC1_ISQR	0x40012438	ADC1 injected channel sequence register	0x00000000
R32_ADC1_IDATAR1	0x4001243C	ADC1 injected data register 1	0x00000000
R32_ADC1_IDATAR2	0x40012440	ADC1 injected data register 2	0x00000000
R32_ADC1_IDATAR3	0x40012444	ADC1 injected data register 3	0x00000000
R32_ADC1_IDATAR4	0x40012448	ADC1 injected data register 4	0x00000000
R32_ADC1_RDATAR	0x4001244C	ADC1 regular data register	0x00000000
R32_ADC1_AUX	0x40012454	ADC1 sample time register	0x00000000

Table 11-5 ADC1-related registers

Name	Access address	Description	Reset value
R32_ADC2_STATR	0x40012800	ADC2 status register	0x00000000
R32_ADC2_CTLR1	0x40012804	ADC2 control register 1	0x00000000
R32_ADC2_CTLR2	0x40012808	ADC2 control register 2	0x00000000

R32_ADC2_SAMPTR1	0x4001280C	ADC2 sample time configuration register 1	0x00000000
R32_ADC2_SAMPTR2	0x40012810	ADC2 sample time configuration register 2	0x00000000
R32_ADC2_IOFR1	0x40012814	ADC2 injected channel data offset register 1	0x00000000
R32_ADC2_IOFR2	0x40012818	ADC2 injected channel data offset register 2	0x00000000
R32_ADC2_IOFR3	0x4001281C	ADC2 injected channel data offset register 3	0x00000000
R32_ADC2_IOFR4	0x40012820	ADC2 injected channel data offset register 4	0x00000000
R32_ADC2_WDHTR	0x40012824	ADC2 watchdog high threshold register	0x00000FFF
R32_ADC2_WDLTR	0x40012828	ADC2 watchdog high threshold register	0x00000000
R32_ADC2_RSQR1	0x4001282C	ADC2 regular channel sequence register 1	0x00000000
R32_ADC2_RSQR2	0x40012830	ADC2 regular channel sequence register 2	0x00000000
R32_ADC2_RSQR3	0x40012834	ADC2 regular channel sequence register 3	0x00000000
R32_ADC2_ISQR	0x40012838	ADC2 injected channel sequence register	0x00000000
R32_ADC2_IDATAR1	0x4001283C	ADC2 injected data register 1	0x00000000
R32_ADC2_IDATAR2	0x40012840	ADC2 injected data register 2	0x00000000
R32_ADC2_IDATAR3	0x40012844	ADC2 injected data register 3	0x00000000
R32_ADC2_IDATAR4	0x40012848	ADC2 injected data register 4	0x00000000
R32_ADC2_RDATAR	0x4001284C	ADC2 regular data register	0x00000000
R32_ADC2_AUX	0x40012854	ADC2 sample time register	0x00000000
R32_ADC2_DRV	0x40012858	ADC2_TOUCHKEY multiplex mask register	0x00000000

11.3.1 ADCx Status Register (ADCx_STATR) (x=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADC RSTF	Reserved										STRT	JSTR T	JEOC	EOC	AWD

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
15	ADCRSTF	RO	ADC reset status indicator: 1: Reset complete; 0: Reset in progress.	0
[14:5]	Reserved	RO	Reserved	0
4	STRT	RW0	Regular channel conversion start status: 1: The regular channel conversion has started; 0: The regular channel conversion has not started. This bit is set by hardware and cleared by software (invalid if writing 1).	0
3	JSTR	RW0	Injected channel conversion start status: 1: The injected channel conversion has started; 0: The injected channel conversion has not started;	0

			This bit is set by hardware and cleared by software (invalid if writing 1).	
2	JEOC	RW0	Injected channel group conversion completion status: 1: The conversion has completed; 0: The conversion has not completed. This bit is set to 1 by hardware (the conversion of all injected channels is completed), and cleared by software (invalid if writing 1).	0
1	EOC	RW0	Conversion completion status: 1: The conversion has completed; 0: The conversion has not completed. This bit is set to 1 by hardware (the regular or injected channel group conversion ends), and is cleared by software (invalid if writing 1) or clearing when ADCx_RDATAR is read.	0
0	AWD	RW0	Analog watchdog flag bit: 1: The analog watchdog event occurs; 0: No analog watchdog event occurs. This bit is set to 1 by hardware (the conversion value is out of the ADCx_WDHTR and ADCx_WDLTR register range), and is cleared by software (invalid if writing 1).	0

11.3.2 ADC Control Register 1 (ADCx_CTLR1) (x=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SW_PRE	ANA_RST	Reserved	PGA[1:0]	BUF_EN	TKITUNE	TKENABLE	AWDEN	JAWDEN	Reserved	DUALMOD[3:0]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DISCNUM[2:0]			JDISCEN	DISCEN	JAUTO	AWD_SGL	SCAN	JEOCIE	AWDIE	EOCIE	AWDCH[4:0]				

Bit	Name	Access	Description	Reset value
31	SW_PRE	RW	Channel Pre-Switching Off: 1: Channel pre-switching disabled; 0: Channel pre-switching enabled.	0
30	ANA_RST	RW	Analog module reset enable: 1: Enable; 0: Disable. <i>Note: Resetting the ADC analog circuit during an ADC conversion will corrupt the conversion result; this bit can be used in conjunction with the RCC's ADCxRST.</i>	0
29	Reserved	RO	Reserved	0

[28:27]	PGA[1:0]	RW	ADC channel gain configuration: 00: x1 01: x4 10: x16 11: x64 <i>Note: 1. The input gain can be configured, for amplifying small signals and sampling.</i> <i>2. ADC_Buffer needs to be enabled to use this function.</i>	00b
26	BUFEN	RW	ADC BUFFER enable. 0: Disable input Buffer. 1: Enable input Buffer. <i>Note: Conditions requiring buffer activation:</i> <i>1. When the TKENABLE bit or TSVREFE bit is set to 1, the buffer is enabled by default and cannot be disabled (thus, in these two scenarios, ADC calibration must occur before setting the TKENABLE bit or TSVREFE bit to 1, and the buffer must be disabled).</i> <i>2. When the external input impedance exceeds the maximum input impedance requirement, the buffer can be enabled to improve ADC data acquisition (in this case, the ADC sampling time should not be less than 7.5T). For details on external input impedance, refer to the DS0 datasheet.</i>	0
25	TKITUNE	RW	TKEY module charging current configuration: 1: Charging current is reduced half; 0: Charging current is 35μA.	0
24	TKENABLE	RW	TKEY module enable control, including TKEY_F and TKEY_V units: 1: Enable TKEY module; 0: Disable TKEY module.	0
23	AWDEN	RW	Analog watchdog enable bit on regular channels: 1: Enable analog watchdog on regular channels; 0: Disable analog watchdog on regular channels.	0
22	JAWDEN	RW	Analog watchdog enable bit on injected channels: 1: Enable analog watchdog on injected channels; 0: Disable analog watchdog on injected channels.	0
[21:20]	Reserved	RO	Reserved	0
[19:16]	DUALMOD[3:0]	RW	Dual mode selection: 0000: Independent mode; 0001: Synchronous regulars + synchronous injection mode; 0010: Synchronous regulars + alternating trigger mode; 0011: Synchronous injection + rapid alternating mode; 0100: Synchronous injection + slow alternating mode;	0000b

			0101: Synchronous injection mode; 0110: Synchronous regular mode; 0111: Fast alternating mode; 1000: Slow alternating mode; 1001: Alternating trigger mode. <i>Note: These bits in ADC2 are reserved. Any configuration bit modifications should be performed with dual mode disabled.</i>	
[15:13]	DISCNUM[2:0]	RW	In discontinuous mode, the number of regular channels to be converted after external trigger: 000: 1 channel; ... 111: 8 channels.	000b
12	JDISCEN	RW	Discontinuous mode enable bit on injected channel: 1: Enable discontinuous mode on the injected channel; 0: Disable discontinuous mode on the injected channel.	0
11	DISCEN	RW	Discontinuous mode enable level on the regular channel: 1: Enable discontinuous mode on regular channel. 0: Disable discontinuous mode on regular channel.	0
10	JAUTO	RW	After opening the regular channel, automatically convert the enable bit of the injected channel group: 1: Enable automatic injection channel group conversion. 0: Disable automatic injection channel group conversion. <i>Note: This mode needs to disable the external trigger function of the injection channel.</i>	0
9	AWDSGL	RW	In scan mode, analog watchdog enable bit on a single channel: 1: Enable analog watchdog on single channel (AWDCH[4:0] selection); 0: Disable analog watchdog on all channels.	0
8	SCAN	RW	Scan mode enable bit: 1: Enable scan mode (continuous conversion of all channels selected by ADCx_IOFRx and ADCx_RSQRx); 0: Disable scan mode.	0
7	JEOCIE	RW	Injected channel group conversion completion interrupt enable bit: 1: Enable injected channel group transfer completion interrupt (JEOC flag); 0: Disable injected channel group transfer completion interrupt.	0
6	AWDIE	RW	Analog watchdog interrupt enable bit: 1: Enable analog watchdog interrupt; 0: Disable analog watchdog interrupt.	0

			<i>Note: In scan mode, if this interrupt occurs, the scan will be aborted.</i>	
5	EOCIE	RW	Conversion completion (regular or injected channel group) interrupt enable bit; 1: Enable the transfer completion bit (EOC flag); 0: Disable the transfer completion interrupt.	0
[4:0]	AWDCH[4:0]	RW	Analog watchdog channel selection bit: 00000: Analog input channel 0; 00001: Analog input channel 1; ... 10000: Analog input channel 16; 10001: Analog input channel 17.	00000b

11.3.3 ADCx Control Register 2 (ADCx_CTLR2) (x=1/2)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TS VREFE	SW START	JSW START	EXT TRIG	EXTSEL[2:0]			Reser ved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JEXT TRIG	JEXTSEL[2:0]		ALIG N	Reserved		DMA	ADC _LP	CAL _AUT _O	CAL_VOL [1:0]		RST CAL	CAL	CON T	ADO N	

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23	TSVREFE	RW	Temperature sensor and internal voltage (V_{REFINT}) channel enable: 1: Enable the temperature sensor and V_{REFINT} channel; 0: Disable the temperature sensor and V_{REFINT} channel.	0
22	SWSTART	RW	To start a regular channel conversion, you need to set the software trigger: 1: Start regular channel conversion. 0: Reset state. This bit is set by the software, and the hardware is cleared 0 after the conversion starts.	0
21	JSWSTART	RW	To start an injection channel conversion, you need to set the software trigger: 1: Start the injection channel conversion. 0: Reset state. This bit is set by the software, and the hardware or software clears 0 after the conversion starts.	0

20	EXTTRIG	RW	External trigger conversion mode enable for regular channels: 1: Enable conversion on external event; 0: Disable conversion on external event.	0
[19:17]	EXTSEL[2:0]	RW	External trigger event selection for regular channel: 000: CC1 event of timer 1; 001: CC2 event of timer 1; 010: CC3 event of timer 1; 011: CC2 event of timer 2; 100: TRGO event of timer 3; 101: CC4 event of timer 4; 110: EXTI line11/TRGO event of Timer 8; 111: SWSTART software trigger.	000b
16	Reserved	RO	Reserved	0
15	JEXTTRIG	RW	External trigger conversion mode enable for injected channels: 1: Enable conversion on external event; 0: Disable conversion on external event.	0
[14:12]	JEXTSEL[2:0]	RW	External trigger event select for injected channels: 000: TRGO event of timer 1; 001: CC4 event of timer 1; 010: TRGO event of timer 2; 011: CC1 event of timer 2; 100: CC4 event of timer 3; 101: TRGO event of timer 4; 110: EXTI line 15/CC4 event of Timer 8; 111: JSWSTART software trigger.	000b
11	ALIGN	RW	Data alignment: 1: Left alignment; 0: Right alignment.	0
[10:9]	Reserved	RO	Reserved	0
8	DMA	RW	Direct memory access (DMA) mode enable: 1: Enable DMA mode; 0: Disable DMA mode.	0
7	ADC_LP	RW	ADC low-power mode control bit: 1: Low-power mode, lower power consumption, suitable for sampling rates below 2 MHz; 0: Higher power consumption, suitable for sampling rates of 6 MHz and above, and only applicable when VDD33A exceeds 3V.	1
6	CAL_AUTO	RW	Hardware auto-calibration function enabled: 1: ADC results will automatically subtract the offset value; 0: ADC results will not subtract the offset value.	1
[5:4]	CAL_VOL[1:0]	RW	Calibration voltage configuration:	01b

			00: 0.3*Vref (recommended); 01: 0.5*Vref; 10: 0.75*Vref; 11: Reserved.	
3	RSTCAL	RW	Reset calibration, this bit is set by software, and cleared by hardware after reset: 1: Initialize calibration register; 0: The calibration register initialized. <i>Note: If RSTCAL is set while the conversion is in progress, it takes extra cycles to clear the calibration register.</i>	0
2	CAL	RW	A/D calibration, set by software and cleared by hardware when the calibration is completed. 1: Enable the calibration; 0: Calibration completed.	0
1	CONT	RW	Continuous conversion enable: 1: Continuous conversion mode; 0: Single conversion mode. If this bit is set, the conversion will continue until the bit is cleared.	0
0	ADON	RW	A/D converter ON/OFF When this bit is 0, writing 1 will wake up the ADC from power-down mode; When this bit is 1, writing a 1 will start the conversion, and if any other bit changes, a new conversion will not be started. 1: Enable ADC and to start conversion; 0: Disable ADC conversion/calibration, and go to power down mode.	0

11.3.4 ADCx Sample Time Configuration Register 1 (ADCx_SAMPTR1) (x=1/2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								SMP17[2:0]			SMP16[2:0]			SMP15[2:1]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP15[0]]	SMP14[2:0]			SMP13[2:0]			SMP12[2:0]			SMP11[2:0]			SMP10[2:0]		

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:0]	SMPx[2:0]	RW	SMPx[2:0]: Sample time configuration of channel x: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 41.5 cycles; 101: 55.5 cycles;	000b

			110: 71.5 cycles; 111: 239.5 cycles; SMPx[2:0]: When the ADC_SMP_SELx bit is set, the sampling interval configuration of channel X is: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 2.5 cycles; 101: 3.5 cycles; 110: 4.5 cycles; 111: 5.5 cycles; These bits are used to independently select the sample time of each channel, and the channel configuration value must remain unchanged during the sampling period. <i>Note: When the sampling rate is higher than 1MHz, the sampling time is not recommended to be less than 3.5 cycles.</i>	
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11.3.5 ADCx Sample Time Configuration Register 2 (ADCx_SAMPTR2) (x=1/2)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	SMP9[2:0]			SMP8[2:0]			SMP7[2:0]			SMP6[2:0]			SMP5[2:1]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP5[0]	SMP4[2:0]			SMP3[2:0]			SMP2[2:0]			SMP1[2:0]			SMP0[2:0]		

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:0]	SMPx[2:0]	RW	SMPx [2:0]: sampling time configuration for channel x: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 41.5 cycles; 101: 55.5 cycles; 110: 71.5 cycles; 111: 239.5 cycles; SMPx[2:0]: When the ADC_SMP_SELx bit is set, the sampling interval configuration of channel X is: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 2.5 cycles; 101: 3.5 cycles; 110: 4.5 cycles; 111: 5.5 cycles; These bits are used to independently select the sample time of each channel, and the channel configuration value must remain unchanged during the sampling period. <i>Note: When the sampling rate is higher than 1MHz, the sampling time is not recommended to be less than 3.5 cycles.</i>	000b

11.3.6 ADCy Injected Channel Data Offset Register x (ADCy_IOFRx) (y=1/2, x=1/2/3/4)

Offset address: $0x14 + (x-1)*4$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				JOFFSETx[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	JOFFSETx[11:0]	RW	Data offset for injected channel x. When the injected channel is converted, these bits define the value to be subtracted from the original conversion data. The result of the conversion can be read in the ADCx_IDATARx register.	0

11.3.7 ADCx Watchdog High Threshold Register (ADCx_WDHTR) (x=1/2)

Offset address: $0x24$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				HT[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	HT[11:0]	RW	Analog watchdog high threshold set bits.	0xFFFF

Note: The values of WDHTR and LTR can be changed during the conversion, but they will take effect in the next conversion.

11.3.8 ADCx Watchdog Low Threshold Register (ADCx_WDLTR) (x=1/2)

Offset address: $0x28$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				LT[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	LT[11:0]	RW	Analog watchdog low threshold set bits.	0

Note: The values of WDHTR and LTR can be changed during the conversion, but they will take effect in the next conversion.

11.3.9 ADCx Regular Channel Sequence Register1 (ADCx_RSQR1) (x=1/2)

Offset address: $0x2C$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								L[3:0]				RSQ16[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ16[0]		SQ15[4:0]				SQ14[4:0]				SQ13[4:0]					

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[23:20]	L[3:0]	RW	The number of channels to be converted in a regular channel conversion sequence: 0000-1111: 1-16 conversions.	0
[19:15]	SQ16[4:0]	RW	Number of the 16th conversion channel in the regular sequence (0-17).	0
[14:10]	SQ15[4:0]	RW	Number of the 15th conversion channel in the regular sequence (0-17).	0
[9:5]	SQ14[4:0]	RW	Numbering of the 14th conversion channel in the regular sequence (0-17).	0
[4:0]	SQ13[4:0]	RW	Numbering of the 13th conversion channel in the regular sequence (0-17).	0

11.3.10 ADCx Regular Channel Sequence Register 2 (ADCx_RSQR2) (x=1/2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SQ12[4:0]				SQ11[4:0]				SQ10[4:1]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ10[0]		SQ9[4:0]				SQ8[4:0]				SQ7[4:0]					

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:25]	SQ12[4:0]	RW	Number of the 12th conversion channel in the regular sequence (0-17).	0
[24:20]	SQ11[4:0]	RW	Number of the 11th conversion channel in the regular sequence (0-17).	0
[19:15]	SQ10[4:0]	RW	Number of the 10th conversion channel in the regular sequence (0-17).	0
[14:10]	SQ9[4:0]	RW	Number of the 9th conversion channel in the regular sequence (0-17).	0
[9:5]	SQ8[4:0]	RW	Number of the 8th conversion channel in the regular sequence (0-17).	0
[4:0]	SQ7[4:0]	RW	Number of the 7th conversion channel in the regular sequence (0-7).	0

11.3.11 ADCx Regular Channel Sequence Register 3 (ADCx_RSQR3) (x=1/2)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SQ6[4:0]				SQ5[4:0]				SQ4[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ4[0]		SQ3[4:0]				SQ2[4:0]				SQ1[4:0]					

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:25]	SQ6[4:0]	RW	Number of the 6th conversion channel in the regular sequence (0-17).	0
[24:20]	SQ5[4:0]	RW	Number of the 5th conversion channel in the regular sequence (0-17).	0
[19:15]	SQ4[4:0]	RW	Number of the 4th conversion channel in the regular sequence (0-17).	0
[14:10]	SQ3[4:0]	RW	Number of the 3th conversion channel in the regular sequence (0-17).	0
[9:5]	SQ2[4:0]	RW	Number of the 2th conversion channel in the regular sequence (0-17).	0
[4:0]	SQ1[4:0]	RW	Number of the 1st conversion channel in the regular sequence (0-17).	0

11.3.12 ADCx Injected Channel Sequence Register (ADCx_ISQR) (x=1/2)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										JL[1:0]		JSQ4[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JSQ4[0]		JSQ3[4:0]				JSQ2[4:0]				JSQ1[4:0]					

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:20]	JL[1:0]	RW	Injects the number of channels to be converted in the channel conversion sequence: 00-11: 1-4 conversions.	0
[19:15]	JSQ4[4:0]	RW	Inject the number of the 4th conversion channel in the sequence (0-17). <i>Note: The software writes and assigns the channel number (0-17) as the 4th in the sequence to be converted.</i>	0
[14:10]	JSQ3[4:0]	RW	Inject the number of the 3rd conversion channel in the sequence (0-17).	0
[9:5]	JSQ2[4:0]	RW	Inject the number of the 2nd conversion channel in the sequence (0-17).	0
[4:0]	JSQ1[4:0]	RW	Inject the number of the 1st conversion channel in the	0

		sequence (0-17).	
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Note: Unlike regular conversion sequences, if the length of $JL[1:0]$ is less than 4, the sequence order of conversions starts with (4-JL).

For example, when $JL[1:0]=3$ (4 injected conversions in the sequencer), the ADC will convert channels in the following order: $JSQ1[4:0]$, $JSQ2[4:0]$, $JSQ3[4:0]$, and $JSQ4[4:0]$;

When $JL[1:0]=2$ (3 injected conversions in the sequencer), the ADC will convert the channels in the following order: $JSQ2[4:0]$, $JSQ3[4:0]$ and $JSQ4[4:0]$;

When $JL[1:0]=1$ (2 injected conversions in the sequencer), the ADC converts the channels in the following order: first $JSQ3[4:0]$, then $JSQ4[4:0]$;

When $JL[1:0] = 0$ (1 injection conversion in the sequencer), the ADC will convert only the $JSQ4[4:0]$ channels.

If $ADCx_ISQR[21:0]=10\ 0011\ 0001\ 0011\ 0001\ 0$, the ADC will convert channels in the following order: $JSQ2[4:0]$, $JSQ3[4:0]$, and $JSQ4[4:0]$, indicating that the scan conversions are performed in the following channel order: 7, 3, 7.

11.3.13 ADCy Injected Data Register (ADCy_IDATARx) (y=1/2, x=1/2/3/4)

Offset address: $0x3C + (x-1)*4$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JDATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved.	0
[15:0]	JDATA[15:0]	RO	Injected channel converted data (data left alignment or right alignment).	0

11.3.14 ADCx Regular Data Register (ADCx_RDATAR) (x=1/2)

Offset address: $0x4C$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADC2DATA[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	ADC2DATA[15:0]	RO	Data converted by ADC2: In ADC1: In dual-mode operation, these bits contain the regular channel data converted by ADC2. In ADC2: These bits are not used.	0
[15:0]	DATA[15:0]	RO	Regular channel converted data (data left alignment or right alignment).	0

11.3.15 ADCx Sampling Time Register (ADCx_AUX) (x=1/2)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADC_TO_DFSDM	Reserved													ADC_SMP_SEL_17	ADC_SMP_SEL_16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADC_SMP_SEL_15	ADC_SMP_SEL_14	ADC_SMP_SEL_13	ADC_SMP_SEL_12	ADC_SMP_SEL_11	ADC_SMP_SEL_10	ADC_SMP_SEL_9	ADC_SMP_SEL_8	ADC_SMP_SEL_7	ADC_SMP_SEL_6	ADC_SMP_SEL_5	ADC_SMP_SEL_4	ADC_SMP_SEL_3	ADC_SMP_SEL_2	ADC_SMP_SEL_1	ADC_SMP_SEL_0

Bit	Name	Access	Description	Reset value
31	ADC_TO_DFSDM	RW	ADC module sampling data sent to DFSDM module function enable: 1: Enable; 0: Disable.	0
[30:18]	Reserved	R0	Reserved	0
[17:0]	ADC_SMP_SELx	RW	x=0-17, optional enabling bits for sampling time of channel x (100-111 only in SMPx): 1: SMPx=100: The sampling time is 2.5 cycles; SMPx=101: The sampling time is 3.5 cycles; SMPx=110: The sampling time is 4.5 cycles; SMPx=111: The sampling time is 5.5 cycles. 0: Sampling time is the corresponding SMPx configuration cycle.	0

11.3.16 ADCx Multiplex Mask Register (ADCx_DRV) (x=1/2)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															TKEY_DRV_EN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TKEY_DRV_OUTEN															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	R0	Reserved	0
16	TKEY_DRV_EN	RW	TOUCHKEY multiplex mask enable: 1: Enable; 0: Disable.	0
[15:0]	TKEY_DRV_OUTEN	RW	TOUCHKEY multiplex mask enable each channel: 1: Enable;	0

			0: Disable.	
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Chapter 12 High-speed Analog/Digital Conversion (HSADC)

The chip incorporates a single 10-bit high-speed analog/digital converter (HSADC) that provides seven external channel sampling capabilities. With a sampling rate of up to 20Msps, it enables continuous conversion and supports DMA operations.

12.1 Functional Description

- Support sampling from 7 external channels
- Support receive FIFO
- Support burst mode
- Support DMA functionality
- Support ping-pong mode
- Support continuous conversion mode
- Provide multiple transfer interrupt flags and status
- Provides 1 interrupt vector

12.2 Functional Description

12.2.1 HSADC Module Power-up

The EN bit in the R32_HSADC_CFGR register set to 1 indicates the ADC module is powered on. When the ADC module conversions from power-down mode (EN=0) to power-up state (EN=1), a delay period tSTAB is required for module stabilization. Subsequently, writing a value of 1 to the START bit serves as the software trigger signal to initiate an ADC conversion. Clearing the EN bit to 0 terminates the current conversion and places the ADC module into power-down mode, where it consumes negligible power.

12.2.2 HSADC Clock

The HSADC clock source can be derived from the system clock (SYSCLK), PLL clock, USBHS_PLL clock, or ETH_PLL clock. It can be selected via the HSADCSRC[1:0] bits in the RCC_CFGR2 register. Additionally, the HSADC clock can be divided using the CLKDIV[5:0] bits in the R32_HSADC_CFGR register.

The total conversion time for the sampling cycle is calculated as follows:

$$T_{\text{CONV}} = \text{Sampling time} + 4T_{\text{HSADCCLK}};$$

The sampling time is fixed at $1T_{\text{HSADCCLK}}$.

12.2.3 HSADC Channel Configuration

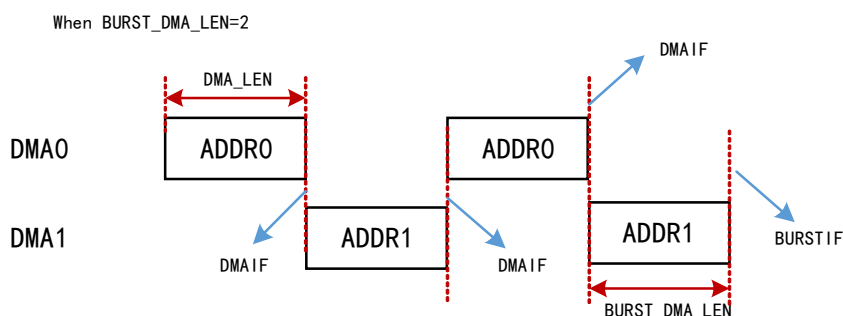
The HSADC module provides 7 external channel sampling sources.

12.2.4 HSADC DMA Function

The HSADC channel conversion supports DMA functionality. The channel conversion value is stored in a single data register, R32_HSADC_DATAR. Failing to retrieve data from the R32_HSADC_DATAR register in a timely manner enables the ADC's DMA capability (if ping-pong transfer is not enabled, only DMA0 can be configured). The hardware generates a DMA request upon completion of the channel conversion (when EOC is set) and transfers the converted data from the R32_HSADC_DATAR register to the user-specified destination address.

Supports burst transfer mode. When burst transfer is enabled, set the PPMODE bit to activate ping-pong memory mode. After the DMA transfer reaches the specified length, the destination address alternates between ADDR0 and ADDR1. Set the BURST_END bit to 1 in the R32_HSADC_CTLR register to enable termination of a burst transfer. Transfer stops upon reaching the configured burst length. The DMA transfer length for the final burst can be set as needed via the BURST_DMA_LEN bit in the R32_HSADC_CTLR2 register.

The following diagram illustrates the burst transfer functionality using BURST_DMA_LEN=2 as an example:



12.2.5 HSADC Conversion Mode

The HSADC supports only continuous conversion. In continuous conversion mode, a new conversion is initiated immediately upon completion of the previous ADC conversion. This mode is activated by setting the START bit to 1 in the R32_HSADC_CTLR1 register. Once conversion for the selected channel completes, the converted data is stored in the 10-bit R32_HSADC_DATAR register. The EOC flag is set, and if the EOCIE bit is enabled, an ADC interrupt is triggered.

12.3 Register Description

Table 12-1 HSADC-related registers

Name	Access address	Description	Reset value
R32_HSADC_CFGR	0x40017400	HSADC configuration register	0x00002400
R32_HSADC_CTLR1	0x40017404	HSADC control register 1	0x00000000
R32_HSADC_CTLR2	0x40017408	HSADC control register 2	0x00000000
R32_HSADC_STATR	0x4001740C	HSADC status register	0x00000000
R32_HSADC_DATAR	0x40017410	HSADC data register	0x00000000
R32_HSADC_ADDR0	0x40017414	HSADC DMA receive address register 0	0x00000000
R32_HSADC_ADDR1	0x40017418	HSADC DMA receive address register 1	0x00000000

12.3.1 HSADC Configuration Register (R32_HSADC_CFGR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_LEN[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BUR ST_E	PP MO	CLKDIV[5:0]						WI DT	SETUP [1:0]	CHSEL[2:0]			DM AE	EN	

N	DE		H			N	
---	----	--	---	--	--	---	--

Bit	Name	Access	Description	Reset value
[31:16]	DMA_LEN[15:0]	RW	DMA transfer length configuration bit.	0
15	BURST_EN	RW	Emergency transmission enable: 1: Enable; 0: Disable.	0
14	PPMODE	RW	Ping-pong storage mode enable: 1: Enable; 0: Disable.	0
[13:8]	CLKDIV[5:0]	RW	High-speed ADC clock frequency division configuration bit: 000000: No frequency division; 000001: Divided by 2; 000010: Divided by 3; 111111: Divided by 64.	0x9
7	WIDTH	RW	Memory bit width configuration: 1: 8-bit; 0: 16-bit.	0
[6:5]	SETUP[1:0]	RW	First conversion setup time configuration bit: 00: 8 cycles; 01: 9 cycles; 10: 10 cycles; 11: 11 cycles.	0
[4:2]	CHSEL[2:0]	RW	Channel selection bit: 000: Input channel 0; 110: Input channel 6; 111: Reserved.	0
1	DMAEN	RW	Direct Memory Access (DMA) mode enable: 1: Enable DMA mode; 0: Disable DMA mode.	0
0	EN	RW	High-speed ADC enable: 1: Enable; 0: Disable.	0

12.3.2 HSADC Control Register 1 (R32_HSADC_CTLR1)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						BUR	DM	EO	Reserved				BU	ST	

	STIE	AIE	CIE		RS T_E ND	AR T
--	------	-----	-----	--	-----------------	---------

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	BURSTIE	RW	Burst transmission complete interrupt enable: 1: Enable; 0: Disable.	0
9	DMAIE	RW	DMA transfer complete interrupt enable: 1: Enable; 0: Disable.	0
8	EOCIE	RW	Conversion completion interrupt enable: 1: Enable; 0: Disable.	0
[7:2]	Reserved	RO	Reserved	0
1	BURST_END	RW	Abort a burst transmission.	0
0	START	WO	Start high-speed ADC conversion: 1: Enable; 0: Disable.	0

12.3.3 HSADC Control Register 2 (R32_HSADC_CTLR2)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BURST_LEN[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BURST_DMA_LEN[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	BURST_LEN[15:0]	RW	Burst transmission length configuration.	0
[15:0]	BURST_DMA_LEN[15:0]	RW	Last secondary DMA transfer length configuration bit of burst transfer. When the data quantity of the final secondary burst transmission is not aligned with the DMA transmission length, the DMA transmission length configuration bit is set.	0

12.3.4 HSADC Status Register (R32_HSADC_STATR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FIFO_CNT[2:0]		FIF	FIF	FIF	Reserved		PP_	RX	BURS	DM	EO			

		O OV	O FU LL	O RD Y		AD DR	NE	TIF	AIF	CIF
--	--	---------	---------------	--------------	--	----------	----	-----	-----	-----

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:11]	FIFO_CNT[2:0]	RO	Receive FIFO current count value.	0
10	FIFO_OV	RO	Receive FIFO overflow status flag bit.	0
9	FIFO_FULL	RO	Receive FIFO full status flag.	0
8	FIFO_RDY	RO	Receive FIFO non-empty status flag.	0
[7:5]	Reserved	RO	Reserved	0
4	PP_ADDR	RO	Ping-pong storage mode cache address indicator: 1: Data stored at DMA receive address 1; 0: Data stored at DMA receive address 0.	0
3	RXNE	RO	Data register non-empty flag: When data conversion is complete and stored in the data register, this bit is set. Reading the data register clears this bit.	0
2	BURSTIF	RW1	Burst transfer completion interrupt flag.	0
1	DMAIF	RW1	DMA transfer completion interrupt flag.	0
0	EOCIF	RW1	Conversion c completion interrupt flag.	0

12.3.5 HSADC Data Register (R32_HSADC_DATAR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						DR[9:0]									

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
[9:0]	DR[9:0]	RW	Conversion data register.	0

12.3.6 HSADC DMA Receive Address Register 0 (R32_HSADC_ADDR0)

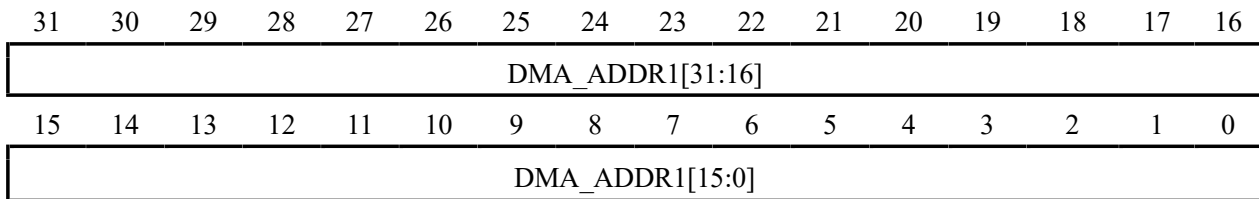
Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_ADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMA_ADDR0[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	DMA_ADDR0[31:0]	RW	DMA transfer address 0 configuration bit.	0

12.3.7 HSADC DMA Receive Address Register 1 (R32_HSADC_ADDR1)

Offset address: 0x18



Bit	Name	Access	Description	Reset value
[31:0]	DMA_ADDR1[31:0]	RW	DMA transfer address 1 configuration bit.	0

Chapter 13 Touch Key Detection (TKEY)

The touch detection control (TKEY) unit, with the help of the voltage conversion function of the ADC module, realizes the touch key detection function by converting the capacitance to the voltage for sampling. The detection channel reuses 16 external channels of the ADC, and the touch key detection is realized through the single conversion mode of the ADC module.

13.1 Functional Description

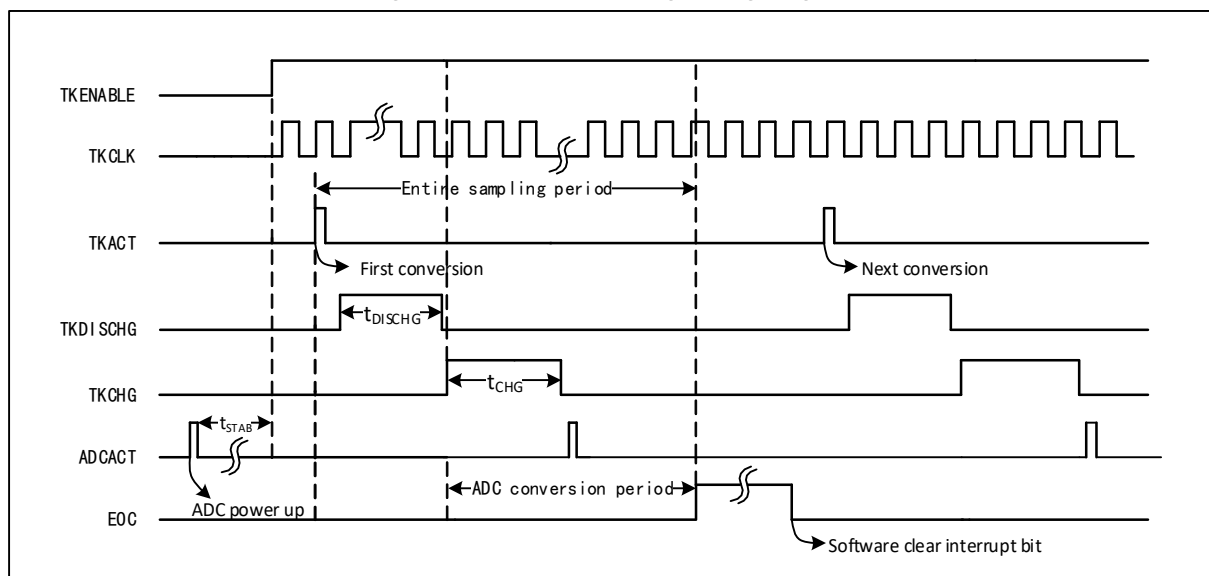
- Enable TKEY

For TKEY detection, the ADC module is needed. To enable TKEY, ensure that ADC is powered on (ADON=1), then set the TKENABLE bit in the ADC_CTLR1 register to 1. The charge current of the TKEY module can be adjusted by the TKITUNE bit.

TKEY only supports single 1-channel conversion mode, which configures the channel to be converted as the first channel in the regular sequence of ADC. And software starts conversion (write to TKEY_ACT_DCG).

Note: When the TKEY conversion is disabled, ADC channel configuration function can still be retained.

Figure 13-1 TKEY working timing diagram



- Programmable sampling time

TKEY cell conversion needs to use several HCLK clock cycles (t_{DISCHG}) to discharge first, and then charge the channels through several HCLK cycles (t_{CHG}) to sample the voltage. The number of charging cycles is TKEY_CHGOFFSET, and each channel can adjust the sampling voltage with different charging cycles.

13.2 TKEY Operations Steps

TKEY is an extended function of the ADC. Touch and release change the capacitance sensed by the channel; the programmable charge/discharge circuitry converts this capacitance change into a voltage change, which is then digitized by the ADC.

When sampling, you need to configure ADC to work in a single channel mode, and the "write operation" of the

TKEY_ACT_DCG register initiates a conversion. The specific process is as follows:

- 1) Initialize the ADC function, configure the ADC module as a single conversion module, set the ADON bit to 1, and wake the ADC module. Set the TKENABLE bit in ADC_CTLR1 to 1 to enable the TKEY unit.
- 2) Set the channel to be converted, write the channel number to the first conversion position in the ADC regular group sequence (ADC_RSQR3[4:0]), and set L[3:0] to 1.
- 3) Write the TKEY_CHGOFFSET register to set the channel's charge time offset (low 10 bits valid) to adjust the charge time.
- 4) Write the TKEY_ACT_DCG register, set the discharge time (valid for low 10 bits), and start a sampling and conversion of TKEY.
- 5) Wait until the EOC flag in the ADC status register is set, and read the ADC_DR register to get the conversion value.
- 6) Repeat steps 2-5 if you need to make the next conversion. If you do not need to modify the channel charging sampling time, you can omit step 3 or 4.

13.3 TKEY Register Description

Table 13-1 TKEY1-related registers

Name	Access address	Description	Reset value
R32_TKEY1_CHGOFFSET	0x4001243C	TKEY charge time offset register	0x00000000
R32_TKEY1_ACT_DCG	0x4001244C	TKEY activate and discharge time register	0x00000000
R32_TKEY1_DR	0x4001244C	TKEY data register	0x00000000

Table 13-2 TKEY2-related registers

Name	Access address	Description	Reset value
R32_TKEY2_CHGOFFSET	0x4001283C	TKEY charge time offset register	0x00000000
R32_TKEY2_ACT_DCG	0x4001284C	TKEY start-up and discharge time register	0x00000000
R32_TKEY2_DR	0x4001284C	TKEY data register	0x00000000

13.3.1 TKEYx Charge Time Offset Register (TKEY_CHGOFFSET) (x=1/2)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TKCGOFFSET[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	TKCGOFFSET[15:0]	WO	TKEY charge time configuration value. Total charge time TCHG = TKCGOFFSET	0

Note: This register maps the injected data register1 (ADC_IDATAR1) of the ADC module. So when write operation is performed on this address register, it serves as TKEY charge time (TKEY_CHGOFFSET). When 'read operation' is performed, it serves as the injected data register1 (ADC_IDATAR1) of the ADC module.

13.3.2 TKEY Activate and Discharge Time Register (TKEY_x_ACT_DCG) (x=1/2)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TKACT_DCG[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	TKACT_DCG[15:0]	WO	Write the discharge time and activate a TKEY channel detection.	0

Note: This register maps the regular data register (ADC_RDATAR) of the ADC module.

13.3.3 TKEY_x Data Register (TKEY_x_DR) (x=1/2)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	DATA[15:0]	RO	Converted data.	0

Note: This register maps the regular data register (ADC_RDATAR) of the ADC module.

Chapter 14 Advanced-control Timer (ADTM)

The advanced timer module includes two 16-bit auto-load up/down counters (TIM1 and TIM8) with 16-bit programmable prescalers. Beyond comprehensive general-purpose timer functionality, it can be configured as a three-phase PWM generator distributed across six channels. It supports complementary PWM outputs with dead-time insertion, enables timer refresh for repeated counting cycles after a specified number of counter cycles, and includes break functionality. It can measure pulse widths or generate pulses, PWM waves, and more. Applications include motor control and power supplies.

Many functions of the Advanced Timer are identical to those of the General-purpose Timer, and their internal structures are also the same. Therefore, the Advanced Timer can operate in coordination with other TIM timers through the timer link function, providing synchronization or event-linking capabilities.

14.1 Main Features

Advanced-control timer (TIM1/8) features include:

- 16-bit auto-reload counter, supports up count, down count and up/down count
- 16-bit prescaler; the frequency division factor is dynamically adjustable from 1 to 65536
- 4 independent compare/capture channels
- Each compare/capture channel supports multiple working modes, such as: input capture, output compare, PWM generation and single pulse output
- Complementary outputs with programmable dead-time
- External signal to control timer
- Repetition counter for updating the timer after a programmable number of counter cycles
- Break signal input to put the timer's output signals in reset status or in a known status
- DMA generation in multiple modes
- Incremental encoder
- Cascade connection and synchronization between timers

14.2 Principle and Structure

This section describes the internal structure of the advanced-control timer.

14.2.1 Overview

As shown in Figure 14-1, the structure of the advanced-control timer can be roughly divided into 3 parts: Input clock part, core counter block and compare/capture channel part.

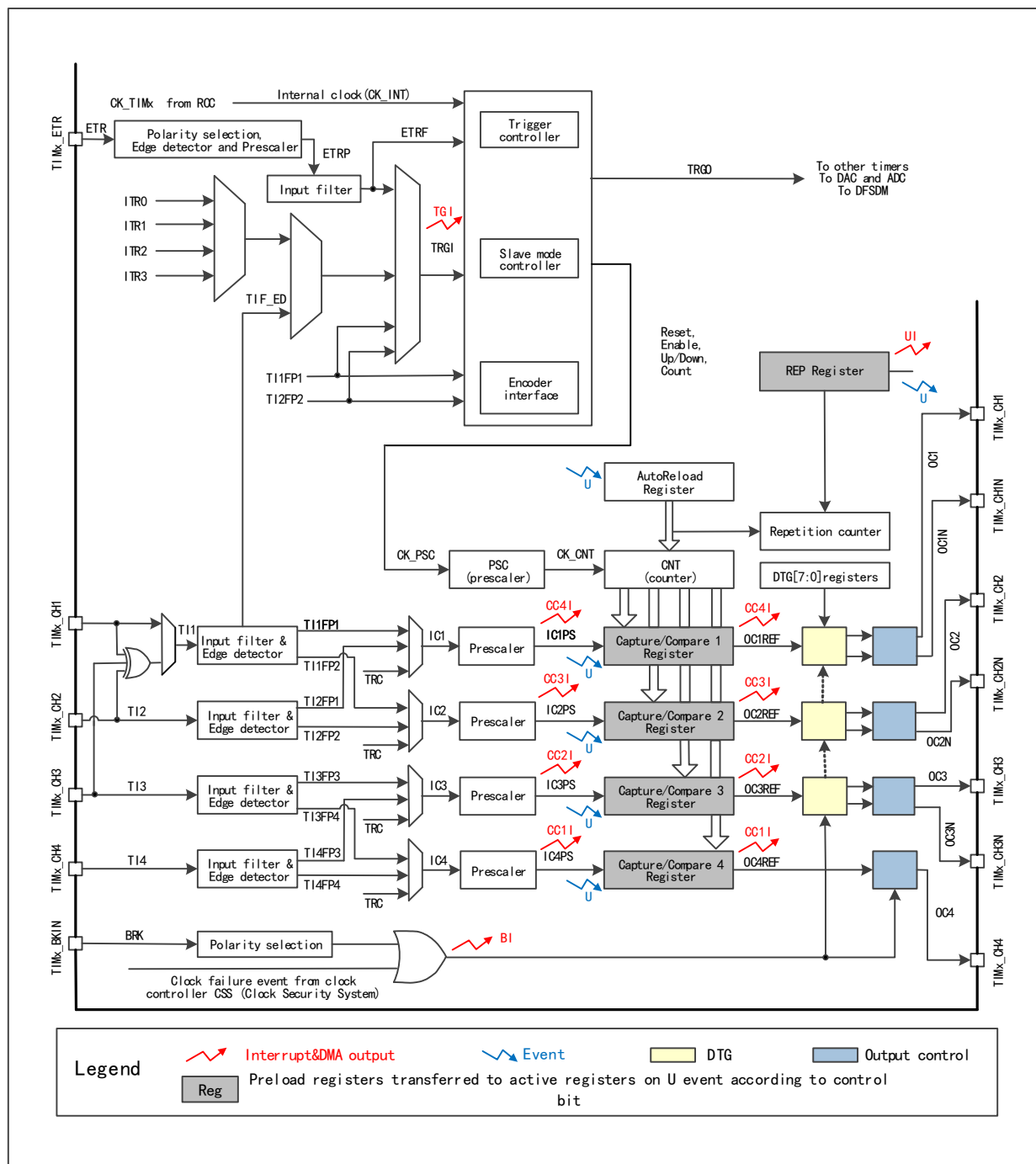
The advanced-control timer clock can come from HB bus clock (CK_INT), external clock input pin (TIMx_ETR), other timers with clock output function (ITRx), or the input end of compare/capture channel (TIMx_CHx). These input clock signals will become CK_PSC clocks after various set filtering and frequency division operations, and will output to the core counter block. In addition, these complex clock sources can also be output as TRGO to other timers, ADC, DAC and other peripherals.

The core of the advanced-control timer is a 16-bit counter (CNT). After frequency division by prescaler (PSC), CK_PSC becomes CK_CNT and is output to CNT, which supports counting up mode, counting down mode and counting up and down mode. In countdown mode, CNT is reloaded with initial value after each counting period. At

the end of each counting period, CNT counts from 0. In addition, there is an auxiliary counter that counts the number of times that ATRLR reloads the initial value for CNT. When the number of times reaches the number set in the repeat count register (RPTCR), a specific event can be generated.

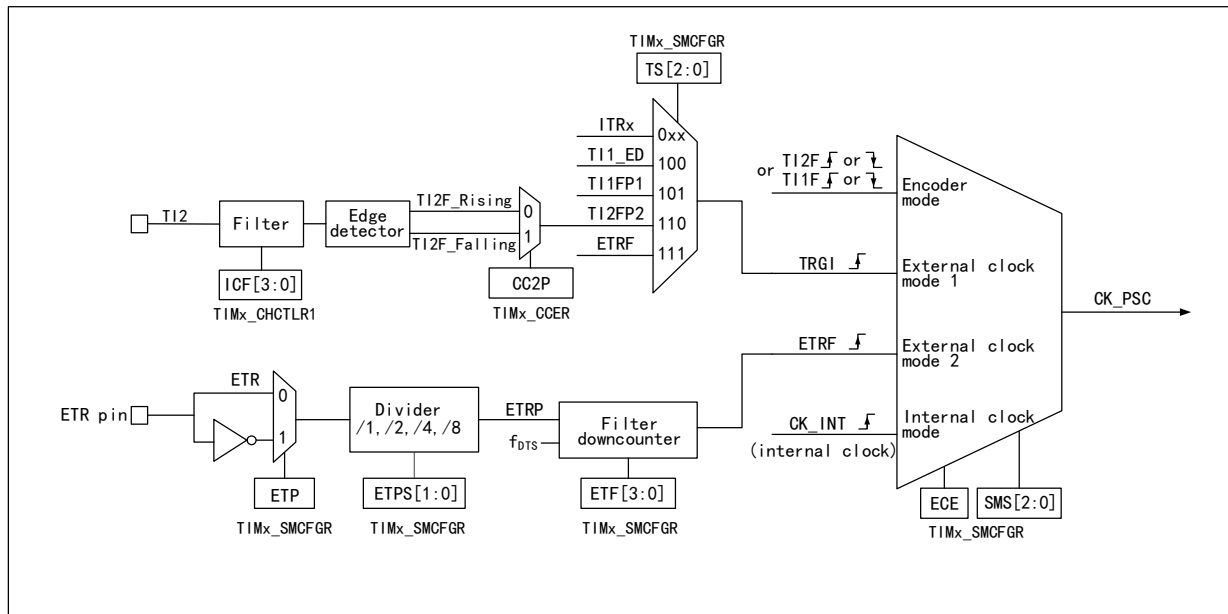
The advanced-control timer has 4 groups of compare/capture channels. On each group of compare/capture channel, pulses can be inputted from its dedicated pins or output waveforms to the pins, i.e., the compare/capture channels support input and output modes. The input of each channel of the compare/capture register supports operations such as filtering, frequency division and edge detection, and supports mutual triggering between channels, and can also provide a clock for the core counter CNT. Each compare/capture channel has a set of compare/capture register (CHxCVR), which supports comparison with the main counter (CNT) so as to output pulse.

Figure 14-1 Structure block diagram of advanced-control timer



14.2.2 Clock Input

Figure 14-2 CK_PSC source block diagram of advanced-control timer



There are many clock sources for advanced-control timer CK_PSC, which can be divided into 4 categories:

- 1) The route of external clock pin (ETR) input clock: ETR→ETRP→ETRF;
- 2) Internal HB clock input route: CK_INT;
- 3) The route from the compare/capture channel pin (TIMx_CHx): TIMx_CHx→TIx→TIxFPx; this route is also used in encoder mode;
- 4) Input from other internal timers: ITRx;

The actual operation can be divided into 4 categories by determining the input pulse selection of the SMS from the CK_PSC source:

- 1) Select the internal clock source (CK_INT);
- 2) External clock source mode 1;
- 3) External clock source mode 2;
- 4) Encoder mode;

The 4 clock sources mentioned above can be selected by these 4 operations.

14.2.2.1 Internal Clock Source (CK_INT)

If the advanced-control timer is started when the SMS field is kept at 000b, then the internal clock source (CK_INT) is selected as the clock. At this moment, CK_INT is CK_PSC.

14.2.2.2 External Clock Source Mode 1

If SMS is set to 111b, the external clock source mode1 is enabled. When external clock source 1 is enabled, TRGI is selected as the source of CK_PSC. It is worth noting that you need to configure TS to select the source of TRGI. For TS, the following pulses can be used as the clock sources:

- 1) Internal Trigger (ITRx, x is 0,1,2,3);
- 2) Signal of compare/capture 1 after passing through the edge detector (TI1F_ED);
- 3) Signals TI1FP1 and TI2FP2 of compare/capture channel;
- 4) Signal ETRF from external clock pin.

14.2.2.3 External Clock Source Mode 2

Use external trigger mode 2 to count on every rising or falling edge of the external clock pin input. When the ECE bit is set, the external clock source mode 2 is used. When the external clock source mode2 is used, ETRF is selected as CK_PSC. The ETR pin passes through the optional inverter (ETP) and frequency divider (ETPS) to become ETRP, and then passes through the filter (ETF) to become ETRF.

When ECE bit is set and the SMS is set to 111b, it means that the TS selects ETRF as the input.

14.2.2.4 Encoder Mode

Set SMS as 001b, 010b and 011b to enable the encoder mode. After enabling the encoder mode, you may choose to use another conversion edge as a signal for signal output at a certain level in TI1FP1 and TI2FP2. This mode is used when the external encoder is used. Refer to Section 14.3.9 for specific functions.

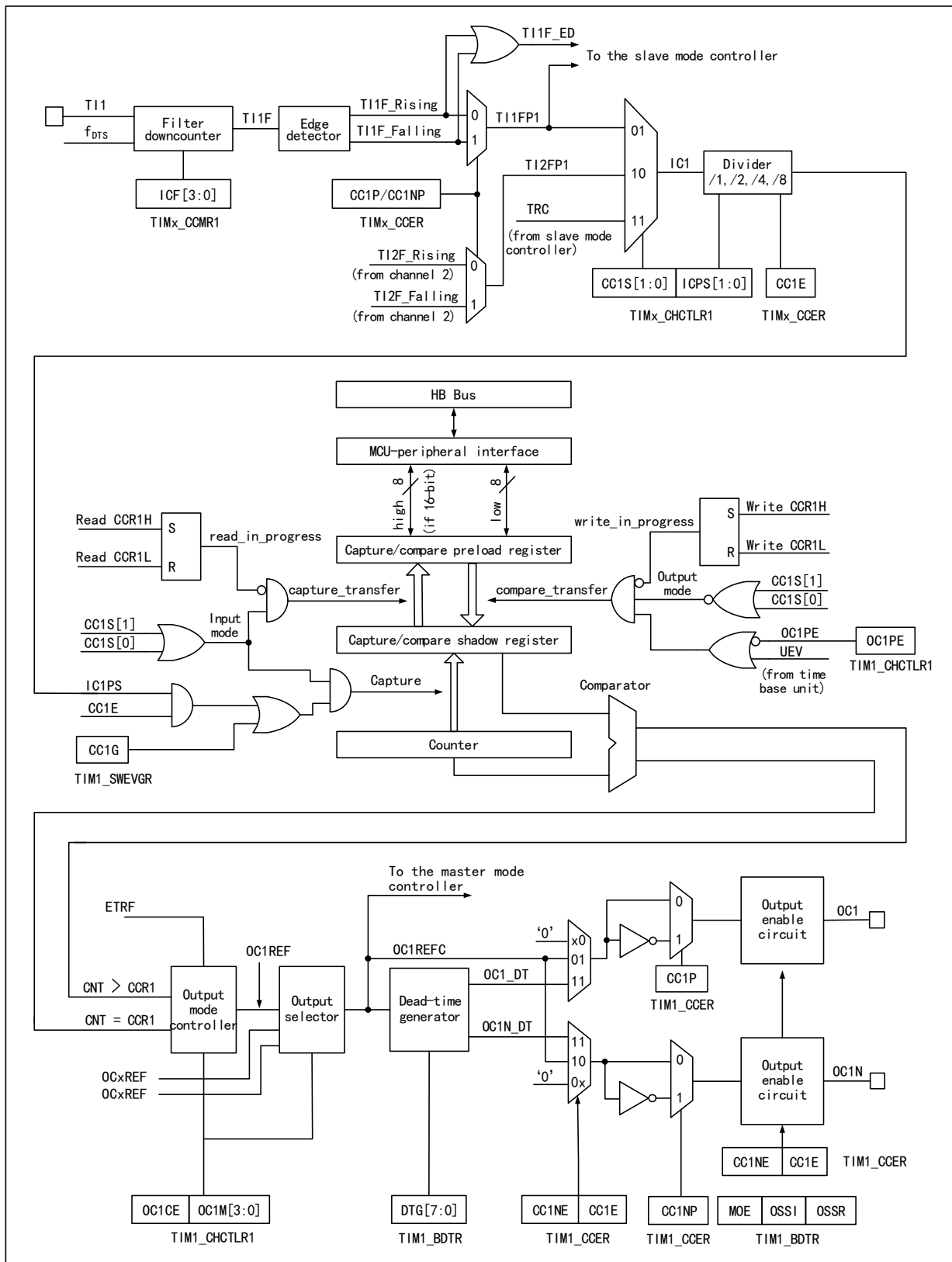
14.2.3 Counter and Periphery

CK_PSC inputs to the prescaler (PSC) for frequency division. PSC is 16 bits, and the actual frequency division factor is equivalent to the value of R16_TIMx_PSC+1. CK_PSC will become CK_CNT after PSC. The changed value of R16_TIM1_PSC will not take effect in real time, but will be updated to the PSC after the update event. Update events include clearing and resetting the UG bit. The core of the timer is a 16-bit counter (CNT). CK_CNT will eventually be inputted to CNT. CNT supports up-counting mode, down-counting mode and up/down counting mode, and there is an automatic reload value register (ATRLR) which re-loads the initial value for CNT after each counting cycle is completed. In addition, there is an auxiliary counter that records the number of times that ATRLR reloads the initial value for CNT. When the number of times reaches the number set in the repeat count register (RPTCR), a specific event can be generated.

14.2.4 Compare/Capture Channel and Periphery

The compare/capture channel is the main component of the timer to achieve complex functions. Its core is the compare/capture register, supplemented by the digital filtering of the peripheral input part, frequency division and channel multiplexing, the output part comparator and output control.

Figure 14-3 Structure block diagram of compare/capture channel



The block diagram of the compare/capture channel is as shown in Figure 14-3. After the signal is inputted from the channel x pin, it can be selected as T1x (the source of T11 may be more than CH1. See the timer structure block diagram 14-1). Take T11 as example, T11 passes through the filter (ICF[3:0]) to generate T11F, and then is divided into T11F_Rising and T11F_Falling after passing through the edge detector. These 2 signals are selected (CC1P) to

generate TI1FP1, TI1FP1 and TI2FP1 from channel 2 are sent to CC1S together to be selected as IC1, and then sent to the compare/capture register after going through the ICPS frequency division.

The compare/capture register is composed of a preload register and a shadow register, and only the preload register is operated during reading and writing. In the capture mode, the capture occurs on the shadow register, and then copied to the preload register; in the comparison mode, the content of the preload register is copied to the shadow register, and then the content of the shadow register is compared with the core counter (CNT).

14.3 Function and Implementation

The advanced-control timer complex functions are implemented by the operation of comparison & capture channel, clock input circuit, counter and peripheral parts of the timer. The timer's clock input can come from multiple clock sources including the input of the compare/capture channel. The operation of compare/capture channel and the clock source selection directly determines its function. The compare/capture channel is bidirectional and can work in input and output modes.

14.3.1 Input Capture Mode

The input capture mode is one of basic functions of timer. The principle of the input capture mode is that when a certain edge on the ICxPS signal is detected, a capture event will occur, and the current value of the counter will be latched into the compare/capture register (R16_TIMx_CHCTLRx). When a capture event occurs, CCxIF (In R16_TIMx_INTFR) bit will be set. If an interrupt or DMA is enabled, a corresponding interrupt or DMA will be generated. If CCxIF is already set when a capture event occurs, then the CCxOF bit will be set. CCxIF can be cleared by software or by hardware through reading the compare/capture register. CCxOF is cleared by the software.

Take an example of channel 1 to illustrate the steps to use the input capture mode, as follows:

- 1) Configure CC1S and select the source of IC1 signal. For example, it is set to 10b, and TI1FP1 is selected as the source of IC1, and the default setting cannot be used. CC1S defaults to use the compare/capture module as the output channel;
- 2) Configure IC1F domain, and these bits set the sampling frequency of TI1 input and the length of digital filter. The digital filter consists of an event counter, which will produce an output jump after recording n events;
- 3) Configure CC1P bit and set the polarity of TI1FP1. For example, maintain CC1P bit to be low and select the jump of rising edge;
- 4) Configure IC1PS and set IC1 signal as the frequency division factor between IC1PS. For example, maintain the IC1PS as 00b without frequency division;
- 5) Configure the CC1E bit to allow to capture the core counter (CNT) value to the compare/capture register.
- 6) The CC1IE and CC1DE bits are configured according to whether an interrupt or DMA is needed.

After these operations, the compare & capture channel configuration is completed.

When TI1 inputs a captured pulse, the value of the core counter (CNT) will be recorded in the compare/capture register, and CC1IF will be set. When CC1IF has been set before, the CCIOF bit will also be set. If CC1IE is set, then an interrupt will be generated; if CC1DE is set, a DMA request will be generated. An input capture event can be generated by software through writing the event generation register (TIMx_SWEVGR).

14.3.2 Compare Output Mode

The compare output mode is one of basic functions of timer. The principle of the compare output mode is to output a specific change or waveform when the value of the core counter (CNT) is consistent with the value of the compare/capture register. OCxM (in R16_TIMx_CHCTLRx) and the CCxP bit (in R16_TIMx_CCER) determine whether the output is determined high or low level or level inversion. When a comparison consistent event is generated, the CCxIF bit will be also set. If the CCxIE bit is preset, an interrupt will be generated; if the CCxDE bit is preset, a DMA request will be generated.

The procedure of compare output mode configuration is as follows:

- 1) Configure the clock source and auto-reload value of the core counter (CNT);
- 2) Set the count value to be compared to the compare/capture register (R16_TIMx_CHxCVR);
- 3) If an interrupt needs to be generated, set the CCxIE bit;
- 4) Keep OCxPE as 0 and disable the preload register of the compare register;
- 5) Set the output mode, and set OCxM and CCxP bit;
- 6) Enable the output and set the CCxE bit;
- 7) Set the CEN bit to start the timer.

14.3.3 Forced Output Mode

The output mode of the compare/capture channel of the timer can be forced to output a certain level by software, instead of relying on the shadow register and the core counter of the compare/capture register.

The specific means is to set OCxM to 100b, which means to force OCxREF to be low; or to set OCxM to 101b, which means setting OCxREF to a high value by force.

It should be noted that if OCxM is set to 100b or 101b by force, the comparison process between the internal core counter and the compare/capture register will be still in progress, the corresponding flag bit will be still set, and interrupts and DMA request will still be generated.

14.3.4 PWM Input Mode

The PWM input mode is used to measure the duty cycle and frequency of the PWM, which is a special case of the input capture mode. The operation is the same as the input capture mode except for the following differences: PWM occupies 2 compare/capture channels, and the input polarity of the 2 channels is set to opposite. One of the signals is set to trigger input, and SMS is set to reset mode.

For example, to measure the cycle and frequency of the PWM wave input from TI1, the following operations are required:

- 1) Set TI1 (TI1FP1) as the input of IC1 signal. Set CC1S to 01b;
- 2) Set TI1FP1 as the rising edge valid. Keep CC1P to 0;
- 3) Set TI1 (TI1FP2) as the input of IC2 signal. Set CC2S to 10b;
- 2) Set TI1FP2 as the falling edge valid. Set CC2P to 1;
- 5) The source of the clock source is TI1FP1. Set TS to 101b;
- 6) Set SMS to reset mode, i.e., 100b;
- 7) Enable the input capture. Set CC1E and CC2E bits;

In this way, the value of the compare/capture register 1 is the cycle of PWM, and the value of the compare/capture register 2 is its duty cycle.

14.3.5 PWM Output Mode

The PWM output mode is one of basic functions of timer. The most common method of PWM output mode is to use the reload value to determine the PWM frequency, and to use the capture comparison register to determine the duty cycle. Set 110b or 111b in OCxM to use PWM mode 1 or mode 2, set the OCxPE bit to enable the preload register, and finally set the ARPE bit. Since the value of the preload register can be sent to the shadow register when an update event occurs, it is necessary to set the UG bit to initialize all registers before the core counter starts counting. In the PWM mode, the core counter and the compare/capture register are always being compared. According to the CMS bit, the timer can output edge-aligned or center-aligned PWM signals.

- Edge alignment

When the edge alignment is used, the core counter counts up or down. In the scenario of PWM mode 1, when the value of the core counter is greater than that of the compare/capture register, OCxREF will be high; when the value of the core counter is less than the compare/capture register (such as When the core counter increases to the value of R16_TIMx_ATRLR and returns to all 0s), OCxREF drops to low.

- Central alignment

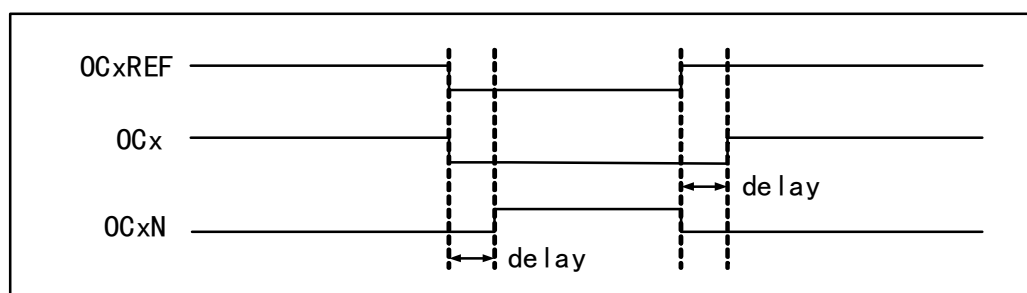
When the center-aligned mode is used, the core counter will run in a mode where up counting and down counting are performed alternately, and OCxREF performs rising and falling jumps when the values of the core counter and the compare/capture register are consistent. However, in 3 types of central alignment mode of comparison flag, the bit setting timing is different somewhat. When the center-alignment mode is used, it is the best to generate a software update flag (setting the UG bit) before starting the core counter.

14.3.6 Complementary Output and Deadband

The compare/capture channel generally has 2 output pins (compare/capture channel 4 has only one output pin), to output 2 complementary signals (OCx and OCxN). OCx and OCxN can be independently set by the CCxP and CCxNP bits. The output enable is set independently through CCxE and CCxNE, and the deadband and other controls are performed through the MOE, OIS, OISN, OSSI and OSSR bits. Meanwhile, OCx and OCxN outputs are enabled to insert into the deadband, each channel has a 10-bit deadband generator. If there is a break circuit, set the MOE bit. OCx and OCxN are generated by OCxREF in association. If OCx and OCxN are both high and effective, then OCx will be the same as OCxREF, but the rising edge of OCx is equivalent to OCxREF with a delay. OCxN is opposite to OCxREF, and its rising edge has a delay relative to the falling edge of the reference signal, and if the delay is greater than the effective output width, the corresponding pulse will not be generated.

Figure 14-4 shows the relationship between OCx, OCxN and OCxREF, and shows the deadband.

Figure 14-4 Complementary output and deadband



14.3.7 Break Signal

When the break signal is generated, the output enable signal and the invalid level will be modified according to the MOE, OIS, OISN, OSSI and OSSR bits. But OCx and OCxN will not be at the effective level at any time. The break event source can come from the break input pin, or it can be a clock failure event, and the clock failure event will be generated by CSS (Clock Security System).

After the system reset, the break function will be disabled by default (MOE bit is low). Setting the BKE bit can enable the break function. The polarity of the input break signal can be set by setting BKP. The BKE and BKP signals can be written at the same time. There will be a HB clock delay before the actual write, so you need to wait for a HB cycle to read the written value correctly.

When the selected level appears on the break pin, the system will generate the following actions:

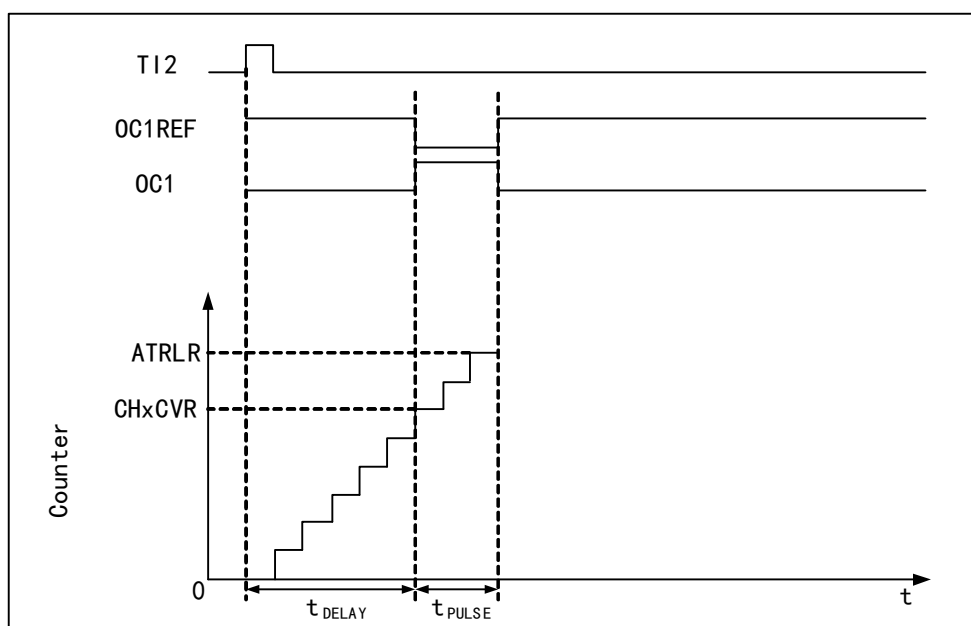
- 1) The MOE bit is asynchronously cleared, and the output is set to the invalid status, idle status or reset status according to the setting of the OSSI bit;
- 2) After MOE is cleared, each output channel will output the level determined by OISx;
- 3) During the supplementary output: the output will be in an invalid status, depending on the polarity;
- 4) If BIE is set, an interrupt will be generated when BIF is set; if the BDE bit is set, a DMA request will be generated;
- 5) If AOE is set, the MOE bit will be automatically set during the next update of event UEV.

14.3.8 One-Pulse Mode

The One-Pulse Mode can be used to allow the microcontroller to respond to a specific event to generate a pulse after a delay. The delay and pulse width are programmable. Setting the OPM bit can make the core counter stop when the next update event UEV is generated (the counter turns over to 0).

As shown in Figure 14-5, it is necessary to detect the beginning of a rising edge on the TI2 input pin. After delaying T_{delay}, a positive pulse of length T_{pulse} will be generated on OC1:

Figure 14-5 Single pulse generation



- 1) Set TI2 as trigger. Set CC2S to 01b and map TI2FP2 to TI2; set CC2P bit to 0b and set TI2FP2 to rising edge detection; set TS to 110b and set TI2FP2 as the trigger source; set SMS to 110b, and TI2FP2 is used to start the counter;

2) Tdelay is determined by the value of the compare/capture register, and Tpulse is determined by the value of the auto-reload value register and the value of the compare/capture register.

14.3.9 Encoder Mode

The encoder mode is a typical application of the timer. It can be used to access the dual-phase output of the encoder. The counting direction of the core counter is synchronized with the rotating shaft of the encoder. Each pulse outputted by the encoder will increase the core counter by adding one or subtracting one. The steps to use the encoder are: set the SMS field to 001b (counting only on TI2 edge), 010b (counting only on TI1 edge) or 011b (counting on both TI1 and TI2 edges), and connect the encoder to compare/capture channel 1, 2 input terminals, set a value for the reload value register and this value can be set to be greater. In the encoder mode, the internal compare/capture register of timer, prescaler, repeat count register, etc. all work normally. The following table shows the relationship between the counting direction and the encoder signal.

Table 14-1 Relationship between counting direction of timer encoder mode and encoder signal

Counting active edge	Relative signal level	TI1FP1 signal edge		TI2FP2 signal	
		Rising edge	Falling edge	Rising edge	Falling edge
Only count at TI1 edge	High	Downcount	Upcount	Not count	
	Low	Upcount	Downcount		
Only count at TI2 edge	High	Not count		Upcount	Downcount
	Low			Downcount	Upcount
Count on both edges of TI1 and TI2	High	Downcount	Upcount	Upcount	Downcount
	Low	Upcount	Downcount	Downcount	Upcount

14.3.10 Timer Synchronization Mode

The timer can output clock pulses (TRGO) and also receive inputs from other timers (ITRx). The source of ITRx for different timers (the TRGO from other timers) varies. The internal trigger connections within the timer are shown in Table 14-2.

Table 14-2 Relationship between counting direction of timer encoder mode and encoder signal

Slave timer	ITR0 (TS=000)	ITR1 (TS=001)	ITR2 (TS=010)	ITR3 (TS=011)
TIM1	TIM5	TIM2	TIM3	TIM4
TIM8	TIM1	TIM2	TIM4	TIM5

14.3.11 Dual Edge Capture Mode

Enable dual edge capture for pulse measurement on the corresponding channel via the CAP_ED_CHx bit in the TIMx_AUX register.

For example: To capture pulse width using Channel 2, configure the capture function by selecting the IC2 signal source (CC2S bit in TIMx_CHCTLR1 register set to 11b) and enabling Channel 2's dual-edge capture (CAP_ED_CH2 bit in TIMx_AUX register set to 1). This completes the dual-edge capture configuration.

The high/low-level width values of captured pulses can be read via the CH2CVR bit in the TIMx_CH2CVR register. Enabling the CAPLVL bit allows the corresponding level of the captured value to be indicated by bit[16] in the TIMx_CH2CVR register.

14.3.12 Dead-time Asymmetry

Based on complementary output configuration and dead time control, configure the DT_VLU2 bit (with a value greater than 0 and less than the number of Tdtg bits set in the DTG[7:0] field of the TIMx_BDTR register) to select whether the dead time configured by DT_VLU2 occurs on the rising or falling edge of OCxREF by setting the DTP_MODE or DTN_MODE bit. This method achieves asymmetric dead time.

14.3.13 Debug Mode

When the system enters debug mode, the timer continues to run or stops according to the settings of the DBG module.

14.4 Register Description

Table 14-3 TIM1-related registers list

Name	Access address	Description	Reset value
R16_TIM1_CTLR1	0x40012C00	Control register 1	0x0000
R16_TIM1_CTLR2	0x40012C04	Control register 2	0x0000
R16_TIM1_SMCFR	0x40012C08	Slave mode control register	0x0000
R16_TIM1_DMAINTENR	0x40012C0C	DMA/interrupt enable register	0x0000
R16_TIM1_INTFR	0x40012C10	Interrupt status register	0x0000
R16_TIM1_SWEVGR	0x40012C14	Event generation register	0x0000
R16_TIM1_CHCTLR1	0x40012C18	Compare/capture control register 1	0x0000
R16_TIM1_CHCTLR2	0x40012C1C	Compare/capture control register 2	0x0000
R16_TIM1_CCER	0x40012C20	Compare/capture enable register	0x0000
R16_TIM1_CNT	0x40012C24	Counters	0x0000
R16_TIM1_PSC	0x40012C28	Counting clock prescaler	0x0000
R16_TIM1_ATRLR	0x40012C2C	Auto-reload value register	0xFFFF
R16_TIM1_RPTCR	0x40012C30	Repeat Count Register	0x0000
R32_TIM1_CH1CVR	0x40012C34	Compare/capture register 1	0x0000
R32_TIM1_CH2CVR	0x40012C38	Compare/capture register 2	0x0000
R32_TIM1_CH3CVR	0x40012C3C	Compare/capture register 3	0x0000
R32_TIM1_CH4CVR	0x40012C40	Compare/capture register 4	0x0000
R16_TIM1_BDTR	0x40012C44	Break and deadband registers	0x0000
R16_TIM1_DMACFR	0x40012C48	DMA control register	0x0000
R16_TIM1_DMAADR	0x40012C4C	DMA address register for continuous mode	0x0000
R16_TIM1_AUX	0x40012C50	Dual edge capture register	0x0000

Table 14-4 TIM8-related registers list

Name	Access address	Description	Reset value
R16_TIM8_CTLR1	0x40013400	Control register 1	0x0000
R16_TIM8_CTLR2	0x40013404	Control register 2	0x0000
R16_TIM8_SMCFR	0x40013408	Slave mode control register	0x0000

R16_TIM8_DMAINTENR	0x4001340C	DMA/interrupt enable register	0x0000
R16_TIM8_INTFR	0x40013410	Interrupt status register	0x0000
R16_TIM8_SWEVGR	0x40013414	Event generation register	0x0000
R16_TIM8_CHCTLR1	0x40013418	Compare/capture control register 1	0x0000
R16_TIM8_CHCTLR2	0x4001341C	Compare/capture control register 2	0x0000
R16_TIM8_CCER	0x40013420	Compare/capture enable register	0x0000
R16_TIM8_CNT	0x40013424	Counters	0x0000
R16_TIM8_PSC	0x40013428	Counting clock prescaler	0x0000
R16_TIM8_ATRLR	0x4001342C	Auto-reload value register	0xFFFF
R16_TIM8_RPTCR	0x40013430	Repeat Count Register	0x0000
R32_TIM8_CH1CVR	0x40013434	Compare/capture register 1	0x0000
R32_TIM8_CH2CVR	0x40013438	Compare/capture register 2	0x0000
R32_TIM8_CH3CVR	0x4001343C	Compare/capture register 3	0x0000
R32_TIM8_CH4CVR	0x40013440	Compare/capture register 4	0x0000
R16_TIM8_BDTR	0x40013444	Break and deadband registers	0x0000
R16_TIM8_DMACFGR	0x40013448	DMA control register	0x0000
R16_TIM8_DMAADR	0x4001344C	DMA address register for continuous mode	0x0000
R16_TIM8_AUX	0x40013450	Dual edge capture register	0x0000

14.4.1 Control Register 1 (TIMx_CTLR1) (x=1/8)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
15	CAPLVL	RW	In double edge capture mode, capture level indication enables: 0: Disable the indication function. 1: Enable indicates function. <i>Note: after enabling, [16] of CHxCVR indicates the level corresponding to the capture value.</i>	0
14	CAPOV	RW	Capture value mode configuration. 0: The capture value is the value of the actual counter 1: The CHxCVR value is 0xFFFF when a counter overflow is generated before capture.	0
13	OE_MODE	RW	When the counter is disabled, the output pin status remains: 1: When the counter is disabled, the timer output pin is left floating by default; 0: When the counter is disabled, the timer output pin retains its current state. <i>Note: The current output pin state is retained when the advanced timer clock is enabled (TIMxEN=1) and the counter is disabled (CEN=0).</i>	0
[12:10]	Reserved	RO	Reserved	0
[9:8]	CKD[1:0]	RW	These 2 bits define the division ratio between the	00b

			timer clock (CK_INT) frequency, the dead time and the sampling clock used by the dead time generator and the digital filter (ETR,TIx). 00: Tdts=Tck_int 01: Tdts = 2 × Tck_int 10: Tdts = 4 × Tck_int 11: Reserved.	
7	ARPE	RW	Auto-reload preload enable bit. 0: Disable Auto-reload Value Register (ATRLR). 1: Enable Auto-reload Value Register (ATRLR).	0
[6:5]	CMS[1:0]	RW	Central alignment mode selection. 00: Edge-aligned mode. The counter counts up or down based on the direction bit (DIR). 01: Central alignment mode 1. The counter counts up and down alternately. The output compare interrupt flag bit of the channel configured as output (CCxS=00 in the CHCTLRx register) is set only when the counter counts down. 10: Central alignment mode 2. The counter counts up and down alternately. The output compare interrupt flag bit of the channel configured as output (CCxS=00 in the CHCTLRx register) is set only when the counter counts up. 11: Central alignment mode 3. The counter counts up and down alternately. The output compare interrupt flag bit of the channel configured as output (CCxS=00 in the CHCTLRx register) is set when the counter counts both up and down. <i>Note: When the counter is enabled (CEN=1), the conversion from edge-aligned mode to center-aligned mode is not allowed.</i>	00b
4	DIR	RW	Counter direction: 0: The counting mode of the counter is increment. 1: The counting mode of the counter is subtractive. <i>Note: This bit is not valid when the counter is configured in central alignment mode or encoder mode.</i>	0
3	OPM	RW	One-Pulse Mode. 0: The counter does not stop when the next update event occurs. 1: The counter stops (clear the CEN bit) when the next update event occurs.	0
2	URS	RW	Update request source, by which the software selects the source of the UEV event.	0

			0: If an update interrupt or DMA request is enabled, an update interrupt or DMA request is generated by any of the following events. -Counter overflow/underflow -Setting the UG position -Updates generated by the slave mode controller 1: If an update interrupt or DMA request is enabled, only an update interrupt or DMA request is generated if the counter overflows/underflows.	
1	UDIS	RW	Updates are prohibited, and the software allows / disables the generation of UEV events through this bit. 0: UEV is allowed. The UEV event is generated by any of the following events: -Counter overflow / underflow. -Set the UG bit. Updates generated from the mode controller with cached registers are loaded into their preload values. 1: UEV is prohibited. No update events are generated, and the registers (ARR, PSC, CCRx) hold their values. If the UG bit is set or a hardware reset is issued from the mode controller, the counter and prescaler are reinitialized.	0
0	CEN	RW	Enables the counter. 1: Enable the counter; <i>Note: The external clock, gated mode and encoder mode will not work until the CEN bit is set in software. Trigger mode can automatically set the CEN bit in hardware.</i> 0: Disable the counter.	0

14.4.2 Control Register 2 (TIMx_CTLR2) (x=1/8)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
15	Reserved	RO	Reserved	0
14	OIS4	RW	Output idle state 4. 1: When MOE=0, if OC4N is implemented, OC1=1 after deadband; 0: When MOE=0, if OC4N is implemented, OC1=0 after deadband. <i>Note: This bit cannot be modified after LOCK (TIMx_BDTR register) level 1, 2 or 3 has been set.</i>	0
13	OIS3N	RW	Output idle state 3. 1: OC1N = 1 after the dead zone when MOE = 0.	0

			0: When MOE=0, OC1N=0 after dead zone. <i>Note: This bit cannot be modified after the LOCK (TIMx_BDTR register) level 1, 2 or 3 has been set.</i>	
12	OIS3	RW	Output idle state 3, see OIS4.	0
11	OIS2N	RW	Output idle state 2, see OIS3N.	0
10	OIS2	RW	Output idle state 2, see OIS4.	0
9	OIS1N	RW	Output idle state 1, see OIS3N.	0
8	OIS1	RW	Output idle state 1, see OIS4.	0
7	TI1S	RW	TI1 selection. 1: TIMx_CH1, TIMx_CH2 and TIMx_CH3 pins connected to TI1 input after heterodyning. 0: TIMx_CH1 pin is connected directly to TI1 input.	0
[6:4]	MMS[2:0]	RW	Master mode selection: These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in master mode. The possible combinations are as follows. 000: The UG bit of the Reset-TIMx_EGR register is used as the trigger output (TRGO). In the case of a reset generated by a trigger input (from a mode controller in reset mode), there is a delay in the signal on TRGO relative to the actual reset. 001: Enable - The counter enable signal CNT_EN is used as a trigger output (TRGO). Sometimes it is necessary to start multiple timers at the same time or to control the enable from timers over a period of time. The counter enable signal is generated by the logical or of the trigger input signal in CEN control bit and gated mode. When the counter enable signal is controlled by a trigger input, there is a delay on TRGO unless master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFG register). 010: Update - The update event is selected as a trigger output (TRGO). For example, the clock of a master timer may be used as a prescaler for a slave timer. 011: Comparison pulse - on the occurrence of a capture or a successful comparison, when the CC1IF flag is to be set (even if it is already high), the trigger output sends a positive pulse (TRGO). 100: The comparison-OC1REF signal is used as a trigger output (TRGO). 101: The comparison-OC2REF signal is used as a trigger output (TRGO). 110: The comparison-OC3REF signal is used as a trigger output (TRGO).	000b

			111: The comparison -OC4REF signal is used as the trigger output (TRGO).	
3	CCDS	RW	Capture the DMA selection for comparison. 1: Send a DMA request for CHxCVR when an update event occurs. 0: Generate a DMA request for CHxCVR when CHxCVR occurs.	0
2	CCUS	RW	Compare/capture control update selection bits. 1: If CCPC is set, they can be updated by setting the COM bit or a rising edge on TRGI. 0: If the CCPC is set, they can only be updated by setting the COM bit. <i>Note: This bit only works for channels with complementary outputs.</i>	0
1	Reserved	RO	Reserved	0
0	CCPC	RW	Compare/capture preload control bits. 1: CCxE, CCxNE and OCxM bits are preloaded and when this bit is set they are only updated when the COM bit is set. 0: CCxE, CCxNE and OCxM bits are not preloaded. <i>Note: This bit only works for channels with complementary outputs.</i>	0

14.4.3 Slave Mode Control Register (TIMx_SMCFGR) (x=1/8)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
15	ETP	RO	ETR trigger polarity selection, this bit selects whether to input ETR directly or to input the inverse of ETR. 1: Invert ETR, low or falling edge active; 0: ETR, active high or rising edge.	0
14	ECE	RW	External clock mode 2 enable selection. 1: Enables external clock mode 2. 0: Disable external clock mode 2. <i>Note 1: Slave mode can be used simultaneously with external clock mode 2: reset mode, gated mode and trigger mode; however, TRGI cannot be connected to ETRF in this case (TS bit cannot be '111').</i> <i>Note 2: When both external clock mode 1 and external clock mode 2 are enabled, the external clock input is ETRF.</i>	0
[13:12]	ETPS[1:0]	RW	The external trigger signal (ETRP) divides the frequency of this signal, which cannot exceed a maximum of 1/4 of the TIMxCLK frequency, and can be downconverted through this domain.	00b

			00: Prescaler off. 01: ETRP frequency divided by 2. 10: ETRP frequency divided by 4. 11: ETRP frequency divided by 8.	
[11:8]	ETF[3:0]	RW	Externally triggered filtering, in fact, the digital filter is an event counter, which uses a certain sampling frequency to record up to N events and then produces a jump in the output. 0001: Sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, $N=2$. 0010: Sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, $N=4$. 0011: Sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, $N=8$. 0100: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/2$, $N=6$. 0101: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/2$, $N=8$. 0110: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/4$, $N=6$. 0111: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/4$, $N=8$. 1000: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/8$, $N=6$. 1001: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/8$, $N=8$. 1010: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N=5$. 1011: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N=6$. 1100: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N=8$. 1101: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N=5$. 1110: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N=6$. 1111: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N=8$.	0000b
7	MSM	RW	Master/slave mode selection. 1: The event on the trigger input (TRGI) is delayed to allow perfect synchronization between the current timer (via TRGO) and its slave timer. This is useful when the synchronization of several timers to a single external event is required. 0: Does not function.	0
[6:4]	TS[2:0]	RW	Trigger selection field, these 3 bits select the trigger input source used to synchronize the counter. 000: Internal trigger 0 (ITR0). 001: Internal trigger 1 (ITR1). 010: Internal trigger 2 (ITR2). 011: Internal trigger 3 (ITR3). 100: Edge detector of TI1 (TI1F_ED). 101: Filtered timer input 1 (TI1FP1). 110: Filtered timer input 2 (TI2FP2). 111: External trigger input (ETRF). The above only changes when SMS is 0. <i>Note: See Table 14-2 for details.</i>	000b
3	Reserved	RO	Reserved.	0

[2:0]	SMS[2:0]	RW	Input mode selection field. Selects the clock and trigger mode of the core counter. 000: driven by the internal clock CK_INT. 001: Encoder mode 1, where the core counter increments or decrements the count at the edge of TI2FP2 depending on the level of TI1FP1. 010: Encoder mode 2, where the core counter increments or decrements the count at the edge of TI1FP1, depending on the level of TI2FP2. 011: Encoder mode 3, where the core counter increments and decrements the count on the edges of TI1FP1 and TI2FP2 depending on the input level of another signal; 100: reset mode, where the rising edge of the trigger input (TRGI) will initialize the counter and generate a signal to update the registers. 101: Gated mode, when the trigger input (TRGI) is high, the counter clock is turned on; at the trigger input becomes low, the counter is stopped, and the counter starts and stops are controlled. 110: Trigger mode, where the counter is started on the rising edge of the trigger input TRGI and only the start of the counter is controlled. 111: External clock mode 1, rising edge of the selected trigger input (TRGI) drives the counter.	000b
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14.4.4 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=1/8)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
15	Reserved	RO	Reserved	0
14	TDE	RW	Trigger the DMA request enable bit. 1: Allow DMA requests to be triggered. 0: Triggering of DMA requests is disabled.	0
13	COMDE	RW	DMA request enable bit of COM. 1: Allow DMA requests for COM. 0: DMA request for COM is disabled.	0
12	CC4DE	RW	Compare the DMA request enable bit of capture channel 4. 1: Allow comparison of DMA requests for capture channel 4. 0: Disable comparison of DMA requests for capture channel 4.	0
11	CC3DE	RW	Compare the DMA request enable bit of capture channel 3. 1: Allow comparison of DMA requests for capture	0

			channel 3. 0: Disable comparison of DMA requests for capture channel 3.	
10	CC2DE	RW	Compare the DMA request enable bit of capture channel 2. 1: Allow comparison of DMA requests for capture channel 2. 0: Disable comparison of DMA requests for capture channel 2.	0
9	CC1DE	RW	Compare the DMA request enable bit of capture channel 1. 1: Allow comparison of DMA requests for capture channel 1. 0: Disable comparison of DMA requests for capture channel 1.	0
8	UDE	RW	Updated DMA request enable bit. 1: DMA requests that allow updates. 0: DMA requests for updates are disabled.	0
7	BIE	RW	Break interrupt enable bit. 1: Allow breaks to be interrupted. 0: Break interrupt is prohibited.	0
6	TIE	RW	Trigger the interrupt enable bit. 1: Enable triggering of interrupts. 0: Trigger interrupt is disabled.	0
5	COMIE	RW	COM interrupt allow bit. 1: Allow COM interrupts. 0: COM interrupt is disabled.	0
4	CC4IE	RW	Compare/capture channel 4 interrupt enable bit. 1: Allow comparison of capture channel 4 interrupts. 0: Disable compare/capture channel 4 interrupt.	0
3	CC3IE	RW	Compare/capture channel 3 interrupt enable bit. 1: Allow comparison of capture channel 3 interrupts. 0: Disable compare/capture channel 3 interrupt.	0
2	CC2IE	RW	Compare/capture channel 2 interrupt enable bit. 1: Allow comparison of capture channel 2 interrupts. 0: Disable compare/capture channel 2 interrupt.	0
1	CC1IE	RW	Compare/capture channel 1 interrupt enable bit. 1: Allow comparison of capture channel 1 interrupts. 0: Disable compare/capture channel 1 interrupt.	0
0	UIE	RW	Update the interrupt enable bit. 1: Allow updates to be interrupted. 0: Disable update interruption.	0

14.4.5 Interrupt Status Register (TIMx_INTFR) (x=1/8)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
12	CC4OF	RW0	Compare/capture channel 4 repeat capture flag bit.	0
11	CC3OF	RW0	Compare/capture channel 3 repeat capture flag bit.	0
10	CC2OF	RW0	Compare/capture channel 2 repeat capture flag bit.	0
9	CC1OF	RW0	Compare/capture channel 1 repeat capture flag bit is used only when the compare/capture channel is configured for input capture mode. This flag is set by hardware and a software write of 0 clears this bit. 1: The value of the counter is captured into the capture comparison register when the status of CC1IF has been set. 0: No duplicate captures are generated.	0
8	Reserved	RO	Reserved	0
7	BIF	RW0	Break interrupt flag bit, once the break input is valid, by hardware for this position bit, can be cleared by software. 1: A set valid level is detected on the break pin input. 0: No break event is generated.	0
6	TIF	RW0	Trigger interrupt flag bit, when a trigger event occurs by hardware to this location bit, by software to clear. Trigger events include the detection of a valid edge at the TRGI input from a mode other than gated, or any edge in gated mode. 1: Trigger event generation. 0: No trigger event is generated.	0
5	COMIF	RW0	COM interrupt flag bit, this bit is set by hardware and cleared by software once a COM event is generated. com events including CCxE, CCxNE, OCxM are updated. 1: COM event generation. 0: No COM event is generated.	0
4	CC4IF	RW0	Compare/capture channel 4 interrupt flag bit.	0
3	CC3IF	RW0	Compare/capture channel 3 interrupt flag bit.	0
2	CC2IF	RW0	Compare/capture channel 2 interrupt flag bit.	0
1	CC1IF	RW0	Compare/capture channel 1 interrupt flag bit. If the compare/capture channel is configured in output mode. This bit is set by hardware when the counter value matches the comparison value, except in center-aligned mode. This bit is cleared by software.	0

			1: The value of the core counter matches the value of compare/capture register 1; 0: No match occurs. If compare/capture channel 1 is configured as input mode. This bit is set by hardware when a capture event occurs, and it is cleared by software or by reading the compare/capture register. 1: The counter value has been captured compare/capture register 1. 0: No input capture is generated.	
0	UIF	RW0	Update interrupt flag bit, this bit is set by hardware when an update event is generated and cleared by software. 1: Update interrupt generation. 0: No update event is generated. The following scenarios generate update events. If UDIS = 0, when the repeat counter value overflows or underflows. If URS = 0, UDIS = 0, when the UG bit is set, or when the counter core counter is reinitialized by software. If URS = 0, UDIS = 0, when the counter CNT is reinitialized by a trigger event.	0

14.4.6 Event Generation Register (TIMx_SWEVGR) (x=1/8)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved.	0
7	BG	WO	The break event generation bit, which is set and cleared by software, is used to generate a break event. 1: Generate a break event. At this point, MOE=0, BIF=1, if the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated. 0: No action.	0
6	TG	WO	The trigger event generation bit, which is set by software and cleared by hardware, is used to generate a trigger event. 1: Generate a trigger event, TIF is set, and the corresponding interrupts and DMAs are generated if enabled. 0: No action.	0
5	COMG	WO	Compare/capture control update generation bit. Generates a compare/capture control update event. This bit is set by software and automatically cleared by	0

			hardware. 1: when CCPC = 1, allow updating of CCxE, CCxNE, OCxM bits. 0: No action. <i>Note: This bit is only valid for channels with complementary outputs (channels 1, 2, 3).</i>	
4	CC4G	WO	Compare/capture event generation bit 4. generates compare/capture event 4.	0
3	CC3G	WO	Compare/capture event generation bit 3. generates compare/capture event 3.	0
2	CC2G	WO	Compare/capture event generation bit 2. generates compare/capture event 2.	0
1	CC1G	WO	Compare/capture event generation bit 1. generates compare/capture event 1. This bit is set by software and cleared by hardware. It is used to generate a compare/capture event. 1: Generate a compare/capture event on compare/capture channel 1. If compare/capture channel 1 is configured as output. Set the CC1IF bit. Generate the corresponding interrupts and DMAs if they are enabled. If compare/capture channel 1 is configured as input. The current core counter value is captured to compare/capture register 1; set the CC1IF bit to generate the corresponding interrupts and DMAs if they are enabled; if CC1IF is already set, set the CC1OF bit. 0: No action.	0
0	UG	WO	Update event generation bit to generate an update event. This bit is set by software and is automatically cleared by hardware. 1: Initialize the counter and generate an update event. 0: No action. <i>Note: The prescaler counter is also cleared to zero, but the prescaler factor remains unchanged. The core counter is cleared if in center-aligned mode or incremental counting mode; if in decremental counting mode, the core counter takes the value of the reload value register.</i>	0

14.4.7 Compare/Capture Control Register 1 (TIMx_CHCTLR1) (x=1/8)

Offset address: 0x18

The channel can be used in input (capture mode) or output (compare mode), and the direction of the channel is defined by the corresponding CCxS bit. The other bits of this register have different roles in input and output modes.

OCxx describes the function of the channel in output mode and ICxx describes the function of the channel in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]		OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]		OC1PE	OC1FE	CC1S[1:0]			
IC2F[3:0]			IC2PSC[1:0]				IC1F[3:0]			IC1PSC[1:0]					

Compare mode (pin direction is output).

Bit	Name	Access	Description	Reset value
15	OC2CE	RW	Compare/capture channel 2 clear enable bit. 1: Clear OC2REF bit zero once ETRF input is detected high; 0: OC2REF is not affected by ETRF input.	0
[14:12]	OC2M[2:0]	RW	Compare/capture Channel 2 mode setting field. The 3 bits define the action of the output reference signal OC2REF, which determines the values of OC2, OC2N. OC2REF is active high, while the active levels of OC2 and OC2N depend on the CC2P, CC2NP bits. 000: Freeze. Comparison of the value of the capture register with the value of the comparison between the core counters does not work for OC2REF. 001: force to set to valid level. Forcing OC2REF high when the core counter has the same value as the comparison capture register 2. 010: Force to set to invalid level. Forcing OC2REF low when the value of the core counter is the same as the comparison capture register 2. 011: Flip. Flips the level of OC2REF when the core counter is the same as the value of compare/capture register 2. 100: Forced to invalid level. Forces OC2REF to low. 101: Forced to valid level. Force OC2REF to high. 110: PWM Mode 1: In up count, once the core counter is smaller than the value of the compare/capture register, channel 2 is valid, otherwise it is invalid; in downward counting, once the core counter is larger than the value of the compare/capture register, channel 2 is invalid (OC2REF=0), otherwise it is valid (OC2REF=1) 111: PWM mode 2: In up count, once the core counter is smaller than the value of the compare/capture register, channel 2 is invalid level, otherwise it is valid level; in down count, once the core counter is larger than the value of the compare/capture register, channel 2 is valid level (OC2REF=1), otherwise it is invalid	0

			level (OC2REF=0). <i>Note: This bit cannot be modified once the LOCK level is set to 3 and CC1S=00b. In PWM mode 1 or PWM mode 2, the OC2REF level is changed only when the comparison result is changed or when switching from freeze mode to PWM mode in the output comparison mode.</i>	
11	OC2PE	RW	Compare/capture Register 2 preload enable bit. 1: Enable the preload function of compare/capture register 2, read and write operations only operate on the preload registers, the preload value of compare/capture register 2 is loaded into the current shadow register when the update event comes; 0: Disable the preload function of compare/capture register 2, compare/capture register 2 can be written at any time, and the newly written value takes effect immediately. <i>Note: Once the LOCK level is set to 3 and CC2S=00, this bit cannot be modified; PWM mode can be used only in One-Pulse Mode (OPM=1) without confirming the pre-load register, otherwise its action is not determined.</i>	0
10	OC2FE	RW	Compare/capture Channel 2 fast enable bit, this bit is used to speed up the response of the compare/capture channel output to a trigger input event. 1: The active edge of the input to the flipflop acts as if a comparison match has occurred. Therefore, the OC is set to the comparison level independent of the comparison result. The delay between the valid edge of the sample trigger and the output of the compare/capture channel 2 is reduced to 3 clock cycles. 0: Based on the value of the counter and compare/capture register 1, compare/capture channel 2 operates normally, even if the flip-flop is open. The minimum delay to activate the compare/capture channel 2 output is 5 clock cycles when the input of the flipflop has a valid edge. OC2FE only works when the channel is configured to PWM1 or PWM2 mode.	0
[9:8]	CC2S[1:0]	RW	Compare/capture channel 2 input selection fields. 00: comparison capture channel 2 is configured as an output. 01: comparison capture channel 2 is configured as an input and IC2 is mapped on TI2. 10: comparison capture channel 2 is configured as an	0

			input and IC2 is mapped on TI1. 11: Compare/capture Channel 2 is configured as an input and IC2 is mapped on TRC. This mode works only when the internal trigger input is selected (by the TS bit). <i>Note: Compare/capture Channel 2 is writable only when the channel is off (when CC2E is zero).</i>	
7	OC1CE	RW	Compare/capture channel 1 clear enable bit.	0
[6:4]	OC1M[2:0]	RW	Compare/capture channel 1 mode setting field.	0
3	OC1PE	RW	Compare/capture register 1 preload enable bit.	0
2	OC1FE	RW	Compare/capture channel 1 fast enable bit.	0
[1:0]	CC1S[1:0]	RW	Compare/capture channel 1 input selection fields.	0

Capture mode (pin direction is input).

Bit	Name	Access	Description	Reset value
[15:12]	IC2F[3:0]	RW	The input capture filter 2 configuration field, these bits set the sampling frequency of the TI1 input and the digital filter length. The digital filter consists of an event counter, which records N events and then generates a jump in the output. 0000: no filter, sampled at fDTS. 1000: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/8$, $N = 6$. 0001: sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, $N = 2$. 1001: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/8$, $N = 8$. 0010: sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, $N = 4$. 1010: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N = 5$. 0011: sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, $N = 8$. 1011: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N = 6$. 0100: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/2$, $N = 6$. 1100: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N = 8$. 0101: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/2$, $N = 8$. 1101: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N = 5$. 0110: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/4$, $N = 6$. 1110: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N = 6$. 0111: sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/4$, $N = 8$. 1111: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N = 8$.	0
[11:10]	IC2PSC[1:0]	RW	Compare/capture channel 2 prescaler configuration field, these 2 bits define the prescaler coefficient for compare/capture channel 2. Once CC1E = 0, the prescaler is reset. 00: Without prescaler, one capture is triggered for each edge detected on the capture input. 01: Capture triggered every 2 events. 10: Capture triggered every 4 events.	0

			11: Capture is triggered every 8 events.	
[9:8]	CC2S[1:0]	RW	Compare the capture channel 2 input selection field, these 2 bits define the direction of the channel (input/output), and the selection of the input pin. 00: Compare/capture channel 1 channel is configured as an output. 01: Compare/capture channel 1 channel is configured as an input and IC1 is mapped on TI1. 10: Compare/capture channel 1 channel is configured as an input and IC1 is mapped on TI2. 11: Compare/capture channel 1 channel is configured as an input and IC1 is mapped on TRC. This mode works only when the internal trigger input is selected (by the TS bit). <i>Note: CC2S is writable only when the channel is off (CC2E is 0).</i>	0
[7:4]	IC1F[3:0]	RW	Input capture filter 1 configuration field.	0
[3:2]	IC1PSC[1:0]	RW	Compare the capture channel 1 prescaler configuration field.	0
[1:0]	CC1S[1:0]	RW	Compare/capture channel 1 input selection fields.	0

14.4.8 Compare/Capture Control Register 2 (TIMx_CHCTLR2) (x=1/8)

Offset address: 0x1C

The channel can be used in input (capture mode) or output (compare mode), and the direction of the channel is defined by the corresponding CCxS bit. The other bits of this register serve different purposes in input and output modes. OCxx describes the function of the channel in output mode and ICxx describes the function of the channel in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4CE	OC4M[2:0]		OC4PE	OC4FE	CC4S[1:0]		OC3CE	OC3M[2:0]		OC3PE	OC3FE	CC3S[1:0]			
IC4F[3:0]			IC4PSC[1:0]				IC3F[3:0]			IC3PSC[1:0]					

Compare mode (pin direction is output):

Bit	Name	Access	Description	Reset value
15	OC4CE	RW	Compare/capture channel 4 clear enable bit.	0
[14:12]	OC4M[2:0]	RW	Compare/capture channel 4 mode setting field.	0
11	OC4PE	RW	Compare/capture register 4 preload enable bit.	0
10	OC4FE	RW	Compare/capture channel 4 fast enable bit.	0
[9:8]	CC4S[1:0]	RW	Compare/capture channel 4 input selection fields.	0
7	OC3CE	RW	Compare/capture channel 3 clear enable bit.	0
[6:4]	OC3M[2:0]	RW	Compare/capture channel 3 mode setting field.	0
3	OC3PE	RW	Compare/capture register 3 preload enable bit.	0
2	OC3FE	RW	Compare/capture channel 3 fast enable bit.	0
[1:0]	CC3S[1:0]	RW	Compare/capture channel 3 input selection fields.	0

Capture mode (pin direction is input).

Bit	Name	Access	Description	Reset value
[15:12]	IC4F[3:0]	RW	Input capture filter 4 configuration field.	0
[11:10]	IC4PSC[1:0]	RW	Compare/capture channel 4 prescaler configuration field.	0
[9:8]	CC4S[1:0]	RW	Compare/capture channel 4 input selection fields.	0
[7:4]	IC3F[3:0]	RW	Input capture filter 3 configuration field.	0
[3:2]	IC3PSC[1:0]	RW	Compare/capture channel 3 prescaler configuration fields.	0
[1:0]	CC3S[1:0]	RW	Compare/capture channel 3 input selection fields.	0

14.4.9 Compare/Capture Enable Register (TIMx_CCER) (x=1/8)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[15:14]	Reserved	RO	Reserved	0
13	CC4P	RW	Compare the capture channel 4 output polarity setting bit.	0
12	CC4E	RW	Compare/capture channel 4 output enable bit.	0
11	CC3NP	RW	Compare/capture channel 3 complementary output polarity setting bit.	0
10	CC3NE	RW	Compare/capture channel 3 complementary output enable bits.	0
9	CC3P	RW	Compare/capture channel 3 output polarity setting bit.	0
8	CC3E	RW	Compare/capture channel 3 output enable bit.	0
7	CC2NP	RW	Compare/capture channel 2 complementary output polarity setting bit.	0
6	CC2NE	RW	Compare/capture channel 2 complementary output enable bits.	0
5	CC2P	RW	Compare/capture channel 2 output polarity setting bit.	0
4	CC2E	RW	Compare/capture channel 2 output enable bit.	0
3	CC1NP	RW	Compare/capture channel 1 complementary output polarity setting bit.	0
2	CC1NE	RW	Compare/capture channel 1 complementary output enable bit.	0
1	CC1P	RW	Compare the output polarity setting bits of capture channel 1. The CC1 channel is configured to output: 1: OC1 low level effective. 0: OC1 high level is effective. The CC1 channel is configured to enter: The bit selects whether the inverse signal of IC1 or IC1 is used as the trigger or capture signal. 1: inversion: capture occurs at the falling edge of the IC1; when used as an external trigger, the IC1 is inverted. 0: no inversion: capture occurs on the rising edge of the IC1; when used as an external trigger, the IC1 is not inverted. <i>Note: Once the LOCK level (the LOCK bit in the TIMx_BDTR</i>	0

			<i>register) is set to 3 or 2, the bit cannot be modified.</i>	
0	CC1E	RW	Compare the output enable bits of capture channel 1. The CC1 channel is configured to output: 1: On. The OC1 signal is output to the corresponding output pin, and its output level depends on the values of MOE, OSS1, OSSR, OIS1, OIS1N and CC1NE bits. 0: Off. OC1 forbids output, so the output level of OC1 depends on the values of MOE, OSS1, OSSR, OIS1, OIS1N, and CC1NE bits. The CC1 channel is configured to enter: This bit determines whether the value of the counter can be captured into the TIMx_CCR1 register. 1: Capture enable. 0: Capture is prohibited.	0

14.4.10 Counter of Advanced-control Timer (TIMx_CNT) (x=1/8)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	The real-time value of the timer's counter.	0

14.4.11 Counting Clock Prescaler (TIMx_PSC) (x=1/8)

Offset address: 0x28

Bit	Name	Access	Description	Reset value
[15:0]	PSC[15:0]	RW	The dividing factor of the prescaler of the timer; the clock frequency of the counter is equal to the input frequency of the divider/(PSC+1).	0

14.4.12 Auto-reload Value Register (TIMx_ATRLR) (x=1/8)

Offset address: 0x2C

Bit	Name	Access	Description	Reset value
[15:0]	ARR[15:0]	RW	The value of this field will be loaded into the counter, see section 14.2.3 for when the ATRLR acts and updates; the counter stops when the ATRLR is empty.	FFFFh

14.4.13 Repeat Count Value Register (TIMx_RPTCR) (x=1/8)

Offset address: 0x30

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:0]	REP[7:0]	RW	The value of the repeat counter.	0

14.4.14 Compare/Capture Register 1 (TIMx_CH1CVR) (x=1/8)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LEVEL1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15: 0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL1	RO	The level indicator bit corresponding to the capture value	0
[15:0]	CCR1[15:0]	RW	Compare/capture register channel 1.	0

14.4.15 Compare/Capture Register 2 (TIMx_CH2CVR) (x=1/8)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LEVEL2
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15: 0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL2	RO	The level indicator bit corresponding to the capture value	0
[15:0]	CCR1[15:0]	RW	Compare/capture register channel 2.	0

14.4.16 Compare/Capture Register 3 (TIMx_CH3CVR) (x=1/8)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LEVEL3
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15: 0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL3	RO	The level indicator bit corresponding to the capture value	0
[15:0]	CCR1[15:0]	RW	Compare/capture register channel 3.	0

14.4.17 Compare/Capture Register 4 (TIMx_CH4CVR) (x=1/8)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LEVEL4
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL4	RO	The level indicator bit corresponding to the capture value	0
[15:0]	CCR1[15:0]	RW	Compare/capture register channel 4.	0

14.4.18 Break and Dead-time Register (TIMx_BDTR) (x=1/8)

Offset address: 0x44

Bit	Name	Access	Description	Reset value
15	MOE	RW	Main output enable bit. Once the break signal is active, it will be cleared asynchronously. 1: Allow OCx and OCxN to be set as outputs. 0: Disable the output of OCx and OCxN or force to idle state.	0
14	AOE	RW	Auto output enable. 1: MOE can be set by software or set in the next update event. 0: MOE can only be set by software.	0
13	BKP	RW	The break input polarity setting bit. 1: Break input active high. 0: Break input is active low. <i>Note: When LOCK level 1 is set, this bit cannot be modified. A write to this bit requires an HB clock before it can take effect.</i>	0
12	BKE	RW	Break function enable bit. 1: Turn on the break input. 0: Break input is disabled. <i>Note: When LOCK level 1 is set, this bit cannot be modified. A write to this bit requires an HB clock before it can take effect.</i>	0
11	OSSR	RW	1: When the timer is not working, once CCxE=1 or CCxNE=1, first turn on OC/OCN and output invalid level, then set OCx, OCxN enable output signal=1. 0: When the timer is not operating, OC/OCN output is disabled.	0

			<i>Note: When LOCK level 1 is set, this bit cannot be modified.</i>	
10	OSSI	RW	1: when the timer is not operating, once CCxE = 1 or CCxNE = 1, OC/OCN first outputs its idle level, then OCx, OCxN enable output signal = 1. 0: When the timer is not operating, OC/OCN output is disabled. <i>Note: When LOCK level 1 is set, this bit cannot be modified.</i>	0
[9:8]	LOCK[1:0]	RW	Lock the function setting field. 00: Disable the locking function. 01: Lock level 1, no DTG, BKE, BKP, AOE, OISx and OISxN bits can be written. 10: Lock level 2, where the bits in lock level 1 cannot be written, nor the CC polarity bits, nor the OSSR and OSSI bits. 11: Lock level 3, cannot write to the bits in lock level 2, and cannot write to the CC control bits. <i>Note: After system reset, the LOCK bit can only be written once and cannot be modified again until reset.</i>	0
[7:0]	DTG[7:0]	RW	Deadband setting bits that define the duration of the deadband between complementary outputs. Assume that DT denotes its duration. DTG[7:5]=0xx=>DT=DTG[7:0]*Tdtg, Tdtg=TDTS; DTG[7:5]=10x=>DT=(64+DTG[5:0])*Tdtg, Tdtg=2*TDTS; DTG[7:5]=110=>DT=(32+DTG[4:0])*Tdtg, Tdtg=8*TDTS; DTG[7:5]=111=>DT=(32+DTG[4:0])*Tdtg, Tdtg=16*TDTS. <i>Note: Once the LOCK level (the LOCK[1:0] bit in the TIM1_BDTR register) is set to 1, 2, or 3, these bits cannot be modified.</i>	0

14.4.19 DMA Control Register (TIMx_DMCFGR) (x=1/8)

Offset address: 0x48

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
[12:8]	DBL[4:0]	RW	The length of the DMA continuous transmission, the actual value of which is the value of this field + 1. When reading or writing to the TIMx_DMAADR register, the timer performs a continuous transfer), that is, it defines the number of transfers, which can be half words (double bytes) or bytes:	0

			00000: 1 transmission; 00001: 2 transmissions; 00010: 3 transmissions; 10001: 18 transmissions. Suppose we perform the following transmission: DBL=7, DBA=TIM2_CTLR1. If DBL=7 and DBA=TIM2_CTLR1 represents the address of the data to be transferred, then the transfer address is given by the following formula: (TIMx_CTLR1 address) + DBA + (DMA index), where DMA index = DBL. where (TIMx_CTLR1 address) + DBA plus 7 provides the address where data will be written or read. Thus, data transfer occurs across 7 registers starting from the slave address (TIMx_CTLR1 address) + DBA. Depending on the configured DMA data length, the following scenarios may occur: 1. If the data is set as a half-word (16 bits), it will be transferred to all 7 registers. 2. If the data is set as a byte, it will still be transferred to all 7 registers: the first register contains the first MSB byte, the second register contains the first LSB byte, and so on. Therefore, for timers, the user must specify the data width transferred by the DMA.	
[7:5]	Reserved	RO	Reserved	0
[4:0]	DBA[4:0]	RW	These bits define the offset of the DMA in continuous mode from the address where control register 1 is located. 00000: TIMx_CTLR1; 00001: TIMx_CTLR2; 00010: TIMx_SMCFGFR;	0

14.4.20 DMA Address Register in Continuous Mode (TIMx_DMAADR) (x=1/8)

Offset address: 0x4C

Bit	Name	Access	Description	Reset value
[15:0]	DMAB [31:0]	RW	The address of the DMA in continuous mode. Reading or writing the TIMx_DMAADR register results in accessing the register located at the address: TIMx_CTLR1 Address + DBA + DMA Index, where: “TIMx_CTLR1 Address” is the address of Control Register 1 (TIMx_CTLR1); “DBA” is the base address defined in the TIMx_DMACFGR register; “DMA	0

			Index” is an offset automatically controlled by the DMA, determined by the DBL defined in the TIMx_DMCFGR register.	
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14.4.21 Dual-edge Capture Register (TIMx_AUX) (x=1/8)

Offset address: 0x50

Bit	Name	Access	Description	Reset value
[15:8]	DT_VLU2	RW	To enable asymmetric dead time functionality, set the duration to be less than the dead time duration specified in DTG. This requires coordination with the DTP_MODE and DTN_MODE bits. <i>Note: The set dead time duration is calculated as the set value multiplied by Tdtg specified in DTG. This value must be greater than 0 and less than the dead time duration set in DTG.</i>	0
7	DTN_MODE	RW	Enable DT_VLU2 to set the dead time duration to occur on the falling edge of OCxREF. 1: DT_VLU2 sets the dead time duration to occur on the falling edge of OCxREF; 0: Function disabled; the dead time duration on the falling edge of OCxREF remains at the original DTG setting.	0
6	DTP_MODE	RW	Enable DT_VLU2 to set the dead time duration to occur on the rising edge of OCxREF. 1: DT_VLU2 sets the dead time duration to occur on the rising edge of OCxREF; 0: Function disabled; the dead time duration on the rising edge of OCxREF remains at the original DTG setting.	0
[5:3]	BK_SEL	RW	Break source selection bit: 000: BKIN; 001: BKIN2; 01x: BKIN or BKIN2; 100: DFSDM_BK[0]; 101: DFSDM_BK[1]; Other: Reserved.	0
2	CAP_ED_CH4	RW	Dual edge capture enable for Channel 4: 1: Enable bidirectional edge capture for Channel 4; 0: Disable bidirectional edge capture for Channel 4.	0
1	CAP_ED_CH3	RW	Dual edge capture enable for Channel 3: 1: Enable bidirectional edge capture for Channel 3; 0: Disable bidirectional edge capture for Channel 3.	0
0	CAP_ED_CH2	RW	Dual edge capture enable for Channel 2: 1: Enable bidirectional edge capture for Channel 2;	0

			0: Disable bidirectional edge capture for Channel 2.	
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Chapter 15 General-Purpose Timer (GPTM)

The general timer module consists of 4 16-bit automatic reassembling timers (TIM2, TIM3, TIM4 and TIM5) and 4 32-bit automatic reinstalling timer (TIM9, TIM10, TIM11 and TIM12), which are used to measure pulse width or generate pulses and PWM waves with specific frequencies. It can be used in automatic control, power supply and other fields.

15.1 Main Features

The main features of a 16-bit general-purpose timer include:

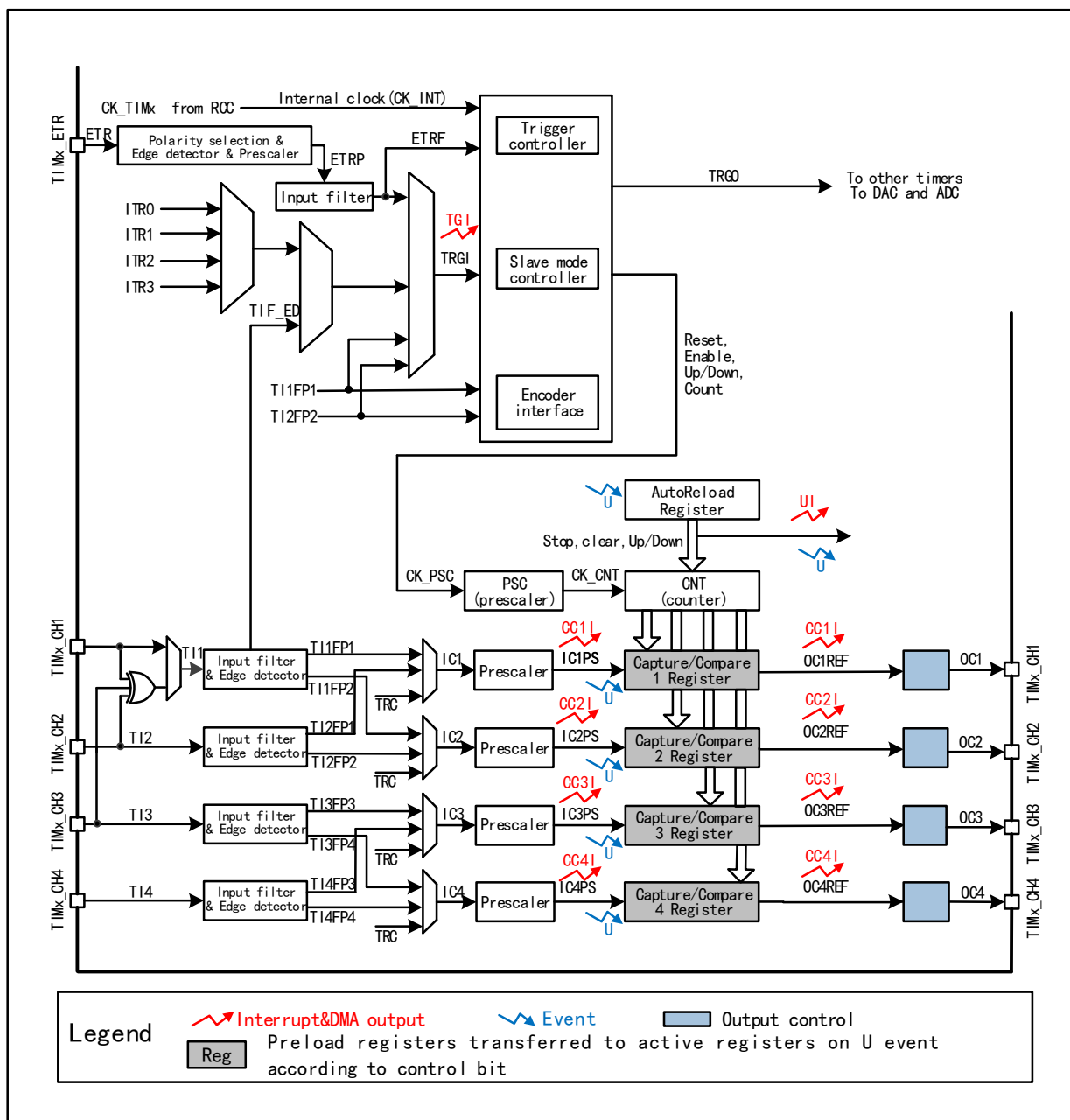
- 16-bit automatic reinstall counter, supporting increasing counting mode, decreasing counting mode and increasing and decreasing counting mode.
- 16-bit prescaler, the frequency division coefficient is dynamically adjustable from 1 to 65536.
- Support four independent comparison capture channels.
- Each comparison capture channel supports multiple operating modes, such as input capture, output comparison, PWM generation and single pulse output.
- Support external signal control timer.
- Support using DMA in multiple modes.
- Support incremental coding, cascading and synchronization between timers.

The main features of a 32-bit general-purpose timer include:

- 32-bit automatic reinstall counter, supporting increasing counting mode, decreasing counting mode and increasing and decreasing counting mode.
- 16-bit prescaler, the frequency division coefficient is dynamically adjustable from 1 to 65536.
- Support four independent comparison capture channels.
- Each comparison capture channel supports multiple operating modes, such as input capture, output comparison, PWM generation and single pulse output.
- Support external signal control timer.
- Support using DMA in multiple modes.
- Support incremental coding, cascading and synchronization between timers

15.2 Principle and Structure

Figure 15-1 Block diagram of the structure of the general-purpose timer



15.2.1 Overview

As shown in Figure 15-1, the structure of the general-purpose timer can be roughly divided into three parts, namely the input clock part, the core counter part and the compare/capture channel part.

The clock for the general-purpose timer can come from the HB bus clock (CK_INT), from the external clock input pin ($TIMx_ETR$), from other timers with clock output ($ITRx$), and from the input of the compare/capture channel ($TIMx_CHx$). These input clock signals become CK_PSC clocks after various set filtering and dividing operations, etc., and are output to the core counter section. In addition, these complex clock sources can also be output as $TRGO$ to other peripherals such as timers, ADCs and DACs.

The core of the general-purpose timer is a 16-bit or 32-bit counter (CNT). After frequency division by prescaler (PSC), CK_PSC becomes CK_CNT, which is finally exported to CNT. CNT supports counting up mode, counting down mode and counting up and down mode. In countdown mode, CNT is reloaded with initial value after each counting period. At the end of each counting period, CNT counts from 0.

The general-purpose timer has 4 sets of compare/capture channels, each of which can input pulses from exclusive pins or output waveforms to pins, i.e., the compare/capture channels support both input and output modes. The input of each channel of the compare/capture register supports filtering, dividing, edge detection, and other operations, and supports mutual triggering between channels, and can also provide clock for the core counter CNT. Each compare/capture channel has a set of compare/capture registers (CHxCVR) that support comparison with the main counter (CNT) to output pulses.

15.2.2 Difference between General-purpose Timer and Advanced-control Timer

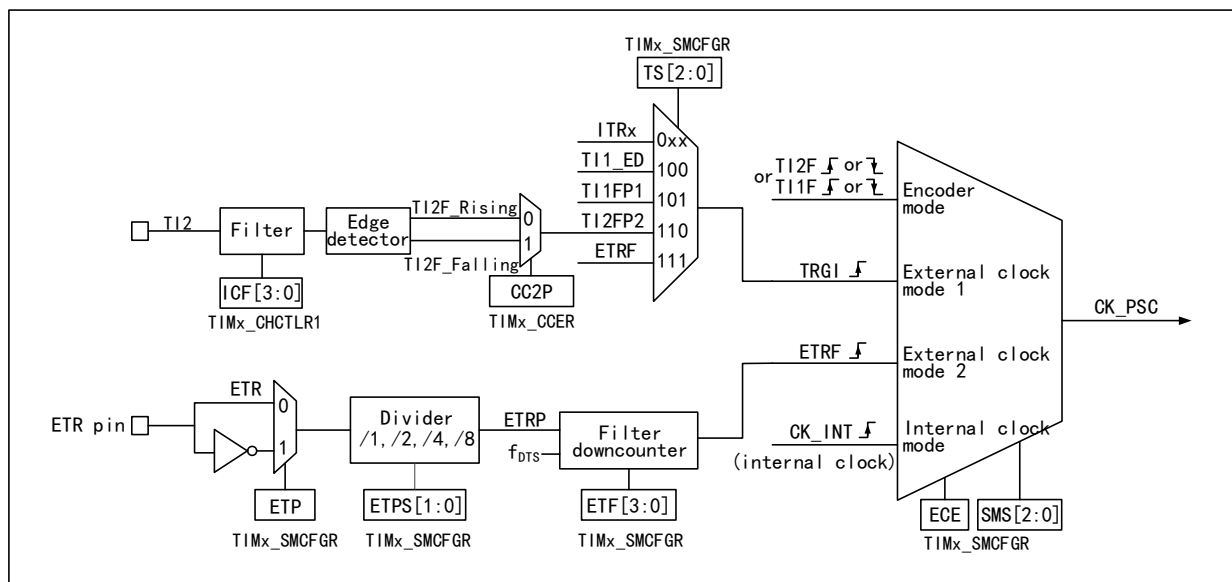
Compared with the advanced-control timer, the general-purpose timer is lack of the following functions:

- 1) The general-purpose timer lacks a repeated counting register that counts the count cycle of core counter.
- 2) The compare/capture of general-purpose timer lacks deadband generation and has no complementary output.
- 3) The general-purpose timer has no break signal mechanism.

15.2.3 Clock Input

This section describes the source of CK_PSC. The clock source part of the general structure block diagram of the general-purpose timer is abstracted here.

Figure 15-2 Block diagram of general-purpose timer source



The available input clocks can be divided into 4 categories:

- 1) External clock pin (ETR) input: ETR→ETRP→ETRF;
- 2) Internal HB clock input: CK_INT;
- 3) From the compare/capture pin (TIMx_CHx): TIMx_CHx→Tlx→TlxFPx; it is also used in encoder mode;
- 4) Input from other internal timers: ITRx.

The actual operation can be divided into 3 categories by determining the input pulse selection of the SMS from the CK_PSC source:

- 1) Select the internal clock source (CK_INT);
- 2) External clock source mode 1;
- 3) External clock source mode 2;
- 4) Encoder code.

The 4 clock sources mentioned above can be selected by these 4 operations.

15.2.3.1 Internal Clock Source (CK_INT)

If the general-purpose timer is started when the SMS domain is kept at 000b, then the internal clock source (CK_INT) is selected as the clock. At this moment, CK_INT is CK_PSC.

15.2.3.2 External Clock Source Mode 1

If SMS is set to 111b, the external clock source mode1 is enabled. When external clock source mode1 is enabled, TRGI is selected as the source of CK_PSC. It is worth noting that the user needs to configure TS to select the source of TRGI. For TS, the following pulses can be used as the clock source:

- 1) Internal Trigger (ITRx, x is 0, 1, 2, 3);
- 2) Signal of compare/capture1 after passing through the edge detector (TI1F_ED);
- 3) Signals TI1FP1 and TI2FP2 of compare/capture;
- 4) Signal ETRF from external clock pin.

15.2.3.3 External Clock Source Mode 2

Use external trigger mode2 to count on every rising or falling edge of the external clock pin input. When the ECE bit is set, the external clock source mode2 is enabled. When the external clock source mode2 is enabled, ETRF is selected as CK_PSC. The ETR pin passes through the optional inverter (ETP) and frequency divider (ETPS) to become ETRP, and then passes through the filter (ETF) to become ETRF.

When the ECE bit is set and the SMS is set to 111b, it means that the TS selects ETRF as the input.

15.2.3.4 Encoder Mode

Set SMS to 001b, 010b and 011b to enable the encoder mode. After enable the encoder mode, you may choose to use another conversion edge as a signal for signal output at a certain level in TI1FP1 and TI2FP2. This mode is used when the external encoder is used. Refer to Section 15.3.7 for the specific functions.

15.2.4 Counter and Periphery

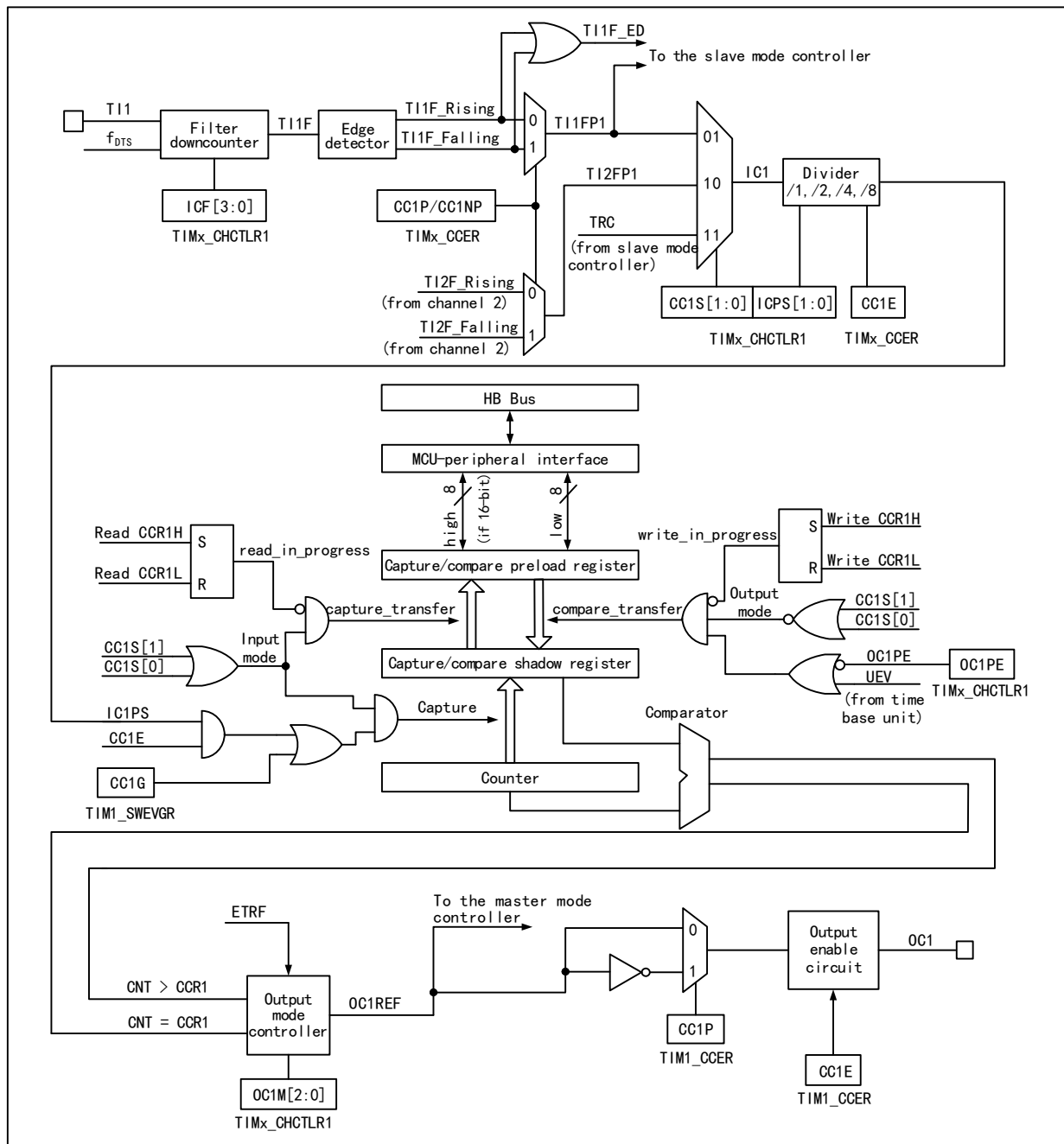
CK_PSC inputs to the prescaler (PSC) for frequency division. PSC has 16 bits, and the actual frequency division factor is equivalent to the value of R16_TIMx_PSC+1. CK_PSC becomes CK_CNT through PSC. The changed value of R16_TIM1_PSC does not take effect in real time, but can be updated to the PSC after the update event. Update events include clearing and resetting the UG bit. There is also an automatic reloading value register (ATRLR) that can set the counter threshold. When the CNT count value reaches the counter threshold set in ATRLR, CNT will be cleared to zero.

15.2.5 Compare/capture Channel

The compare/capture is the core of the timer to achieve complex functions. Its core is the compare/capture register, supplemented by the digital filtering of the peripheral input part, frequency division and channel multiplexing, the

output part comparator and output control. The block diagram of the compare/capture is as shown in Figure 15-3.

Figure 15-3 Structure block diagram of compare/capture channel



After the signal is input from the channel x pin, it can be selected as TI_x (the source of $TI1$ may be more than $CH1$. See Figure 15-1 timer block diagram). Take $T1$ as example, $TI1$ passes through the filter ($ICF[3:0]$) to generate $TI1F$, and then is divided into $TI1F_Rising$ and $TI1F_Falling$ after passing through the edge detector. These 2 signals are selected ($CC1P$) to generate $TI1FP1$, and $TI1FP1$ and $TI2FP1$ from channel 2 are sent to $CC1S$ together to be selected as $IC1$, and then sent to the compare/capture register after going through the $ICPS$ frequency division.

The compare/capture register is composed of preload register and shadow register, and only the preload register is operated during reading and writing. In the capture mode, the capture occurs on the shadow register, and then copied to the preload register; in the comparison mode, the content of the preload register is copied to the shadow register, and then the content of the shadow register is compared with the core counter (CNT).

15.3 Function and Implementation

The general-purpose timer complex functions are implemented by the operation of compare/capture channel, clock input circuit, counter and peripheral parts of the timer. The timer's clock input can come from multiple clock sources including the input of the compare/capture. The operation of compare/capture register channel and the clock source selection directly determines its function. The compare/capture is bidirectional and can work in input and output modes.

15.3.1 Input Capture Mode

Input capture mode is one of the basic functions of timer. The principle of the input capture mode is that when a determined edge on the ICxPS signal is detected, a capture event is generated and the current value of the counter is locked in the comparison capture register (R16_TIMx_CHCTLRx). When a capture event occurs, the CCxIF (in R16_TIMx_INTFR) is set, and if the interrupt or DMA is enabled, the corresponding interrupt or DMA is generated. If the CCxIF is already set when the capture event occurs, the CCxOF bit will be set. CCxIF can be cleared by software or by hardware by reading comparison capture registers. The CCxOF is cleared by the software.

Give an example of channel 1 to illustrate the steps to use input capture mode, as follows:

- 1) Configure the IC1 domain and select the source of the CC1S signal. For example, if you set it to 10b and select TI1FP1 as the source of the IC1, you cannot use the default setting. The CC1S domain defaults to using the comparison capture module as the output channel.
- 2) Configure the IC1F field. These bits set the sampling frequency for TI1 input and the length of the digital filter. The digital filter consists of an event counter that generates an output conversion after recording N events.
- 3) Configure the CC1P bit and set the polarity of TI1FP1. For example, keep the CC1P bit low and choose the rising edge jump.
- 4) Configure the IC1PS domain and set the IC1 signal to become the frequency division coefficient between IC1PS. For example, keep the IC1PS at 00b without frequency division.
- 5) Configure the CC1E bit to allow the value of the core counter (CNT) to be captured into the comparison capture register.
- 6) The CC1IE and CC1DE bits are configured according to whether an interrupt or DMA is needed.

At this point, the comparison capture channel configuration has been completed.

When the TI1 inputs a captured pulse, the value of the core counter (CNT) is recorded in the comparison capture register, the CC1IF is set, and the CCIOFbit is set when the CC1IF has been previously set. If the CC1IE bit, an interrupt is generated; if the CC1DE is set, an DMA request is generated. An input capture event can be generated by the software by writing events to generate registers (R16_TIMx_SWEVGR).

15.3.2 Compare Output Mode

The compare output mode is one of basic functions of timer. The principle of the compare output mode is to output a specific change or waveform when the value of the core counter (CNT) is consistent with the value of the compare/capture register. OCxM (in R16_TIMx_CHCTLRx) and the CCxP bit (in R16_TIMx_CCER) determine whether the output is determined high or low level or level inversion. When a compare consistent event is generated, the CCxIF bit will be also set. If the CCxIE bit is preset, an interrupt will be generated; if the CCxDE bit is preset, a DMA request will be generated.

The procedure of compare output mode configuration is as follows:

- 1) Configure the clock source and auto-reload value of the core counter (CNT);
- 2) Set the count value to be compared to the compare/capture register (R16_TIMx_CHxCVR);
- 3) If an interrupt needs to be generated, set the CCxIE bit;
- 4) Keep OCxPE as 0 and disable the preload register of the compare/capture register;
- 5) Set the output mode, and set OCxM and CCxP bit;
- 6) Enable the output and set the CCxE bit;
- 7) Set the CEN bit and start the timer;

15.3.3 Forced Output Mode

The output mode of the compare/capture of the timer can be forced to output a certain level by software, instead of relying on the shadow register and the core counter of the compare/capture register.

The specific means is to set OCxM to 100b, which means to force OCxREF to be low; or to set OCxM to 101b, which means setting OCxREF to a high value by force.

It shall be noted that if OCxM is set to 100b or 101b by force, the compare process between the internal main counter and the compare/capture register will be still in progress, the corresponding flag bit will be still set, and interrupts and DMA request will still be generated.

15.3.4 PWM Input Mode

The PWM input mode is used to measure the duty cycle and frequency of the PWM, which is a special case of the input capture mode. The operation is the same as the input capture mode except for the following differences: PWM occupies 2 compare/captures, and the input polarity of the 2 channels is set to opposite. One of the signals is set to trigger input, and SMS is set to reset mode.

For example, to measure the cycle and frequency of the PWM wave input from TI1, the following operations are required:

- 1) Set TI1 (TI1FP1) as the input of IC1 signal. Set CC1S to 01b;
- 2) Set TI1FP1 as the rising edge valid. Keep CC1P as 0;
- 3) Set TI1 (TI1FP2) as the input of IC2 signal. Set CC2S to 10b;
- 2) Set TI1FP2 as the falling edge valid. Set CC2P to 1;
- 5) The source of the clock source is TI1FP1. Set TS to 101b;
- 6) Set SMS to reset mode, i.e., 100b;
- 7) Enable the input capture. Set CC1E and CC2E bits.

15.3.5 PWM Output Mode

The PWM output mode is one of basic functions of timer. The most common method of PWM output mode is to use the reload value to determine the PWM frequency, and to use the capture comparison register to determine the duty cycle. Set 110b or 111b in OCxM to use PWM mode 1 or mode 2, set the OCxPE bit to enable the preload register, and finally set the ARPE bit. Since the value of the preload register can be sent to the shadow register when an update event occurs, it is necessary to set the UG bit to initialize all registers before the core counter starts counting. In the PWM mode, the core counter and the compare/capture register are always being compared. According to the CMS bit, the timer can output edge-aligned or center-aligned PWM signals.

- Edge alignment

When the edge alignment is used, the core counter counts up or down. In the scenario of PWM mode 1, when the value of the core counter is greater than that of the compare/capture register, OCxREF will rise to be high; when the value of the core counter is less than the compare/capture register (such as When the core counter increases to the value of R16_TIMx_ATRLR and returns to all 0s), OCxREF drops to low.

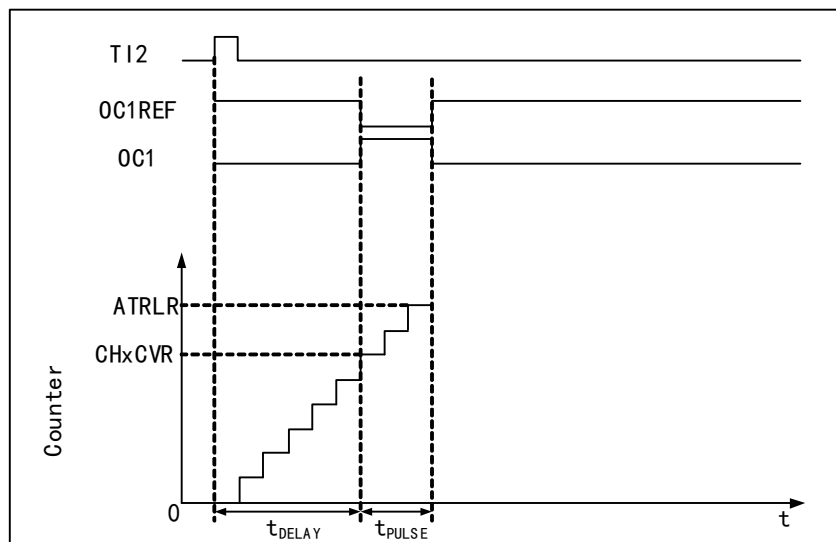
● Central alignment

When the center-aligned mode is used, the core counter will run in a mode where up counting and down counting are performed alternately, and OCxREF performs rising and falling jumps when the values of the core counter and the compare/capture register are consistent. However, in 3 types of central alignment mode, the bit setting timing of comparison flag is different somewhat. When the center-alignment mode is used, it is the best to generate a software update flag (set the UG bit) before starting the core counter.

15.3.6 One-Pulse Mode

The One-Pulse Mode can be used to respond to a specific event to generate a pulse after a delay. The delay and pulse width are programmable. Setting the OPM bit can make the core counter stop when the next update event UEV is generated (the counter turns over to 0).

Figure 15-4 Event generation and pulse response



As shown in Figure 15-4, it is necessary to detect the beginning of a rising edge on the TI2 input pin. After delaying t_{DELAY} , a positive pulse of length t_{PULSE} will be generated on OC1:

- 1) Set TI2 as trigger. Set the CC2S field to 01b and map TI2FP2 to TI2; set the CC2P bit to 0b and set TI2FP2 to rising edge detection; set the TS field to 110b and set TI2FP2 as the trigger source; set the SMS field to 110b, and TI2FP2 is used to start the counter;
- 2) t_{DELAY} is defined by the value of the compare/capture register, and t_{PULSE} is determined by the value of the auto-reload value register and the value of the compare/capture register.

15.3.7 Encoder Mode

The encoder mode is a typical application of the timer. It can be used to access the dual-phase output of the encoder. The count direction of the core counter is synchronized with the rotating shaft of the encoder. Each pulse output by the encoder will increase the core counter by adding one or subtracting one. The steps to use the encoder are: set the SMS field to 001b (count only on TI2 edge), 010b (count only on TI1 edge) or 011b (count on both TI1 and TI2

edges), and connect the encoder to compare/capture 1, 2 inputs, set a value for the reload value register and this value can be set to be greater. In the encoder mode, the internal compare/capture register of timer, prescaler, repeat count register and other registers all work normally. The following table shows the relationship between the counting direction and the encoder signal.

Table 15-1 Relationship between count direction of timer in encoder mode and encoder signal

Count active edge	Relative signal level	TI1FP1 signal edge		TI2FP2 signal edge	
		Rising edge	Falling edge	Rising edge	Falling edge
Only count at TI1 edge	High	Downcount	Upcount	Not count	
	Low	Upcount	Downcount		
Only count at TI2 edge	High	Not count		Upcount	Downcount
	Low			Downcount	Upcount
Count on both edges of TI1 and TI2	High	Downcount	Upcount	Upcount	Downcount
	Low	Upcount	Downcount	Downcount	Upcount

15.3.8 Timer Synchronous Mode

The timer can output clock pulses (TRGO) and can also receive input from other timers (ITRx). The sources of ITRx of different timers (TRGO of other timers) are different. Table 15-2 shows the internal trigger connection of timers.

Figure 15-2 GTPM internal trigger connection

Slave timer	ITR0(TS=000)	ITR1(TS=001)	ITR2(TS=010)	ITR3(TS=011)
TIM2	TIM1	TIM8/USB/ETH	TIM3	TIM4
TIM3	TIM1	TIM2	TIM5	TIM4
TIM4	TIM1	TIM2	TIM3	TIM8
TIM5	TIM2	TIM3	TIM4	TIM8
TIM9	TIM1	TIM2	TIM10	TIM11
TIM10	TIM1	TIM8	TIM2	TIM9
TIM11	TIM1	TIM2	TIM9	TIM12
TIM12	TIM1	TIM2	TIM9	TIM11

15.3.9 Dual Edge Capture Mode

Enable dual edge capture for pulse measurement on the corresponding channel via the CAP_ED_CHx bit in the TIMx_AUX register.

For example: To capture pulse width using Channel 2, configure the capture function by selecting the IC2 signal source (CC2S bit in TIMx_CHCTLR1 register set to 11b) and enabling Channel 2's dual-edge capture (CAP_ED_CH2 bit in TIMx_AUX register set to 1). This completes the dual-edge capture configuration.

The high/low-level width values of captured pulses can be read via the CH2CVR bit in the TIMx_CH2CVR register. Enabling the CAPLVL bit allows the corresponding level of the captured value to be indicated by bit[16] in the TIMx_CH2CVR register.

Note: This function is only available for TIM2/TIM3/TIM4/TIM5.

15.3.10 Debug Mode

When the system enters debug mode, the timer continues to run or stops according to the setting of the DBG module.

15.4 Register Description

Table 15-3 TIM2 registers

Name	Offset address	Description	Reset value
R16_TIM2_CTLR1	0x40000000	TIM2 control register1	0x0000
R16_TIM2_CTLR2	0x40000004	TIM2 control register2	0x0000
R16_TIM2_SMCFR	0x40000008	TIM2 slave mode configuration register	0x0000
R16_TIM2_DMAINTENR	0x4000000C	TIM2 DMA/interrupt enable register	0x0000
R16_TIM2_INTFR	0x40000010	TIM2 interrupt flag register	0x0000
R16_TIM2_SWEVGR	0x40000014	TIM2 event generation register	0x0000
R16_TIM2_CHCTLR1	0x40000018	TIM2 compare/capture control register 1	0x0000
R16_TIM2_CHCTLR2	0x4000001C	TIM2 compare/capture control register 2	0x0000
R16_TIM2_CCER	0x40000020	TIM2 compare/capture enable register	0x0000
R16_TIM2_CNT	0x40000024	TIM2 counter	0x0000
R16_TIM2_PSC	0x40000028	TIM2 prescaler	0x0000
R16_TIM2_ATRLR	0x4000002C	TIM2 auto-reload register	0xFFFF
R32_TIM2_CH1CVR	0x40000034	TIM2 compare/capture register 1	0x0000
R32_TIM2_CH2CVR	0x40000038	TIM2 compare/capture register 2	0x0000
R32_TIM2_CH3CVR	0x4000003C	TIM2 compare/capture register 3	0x0000
R32_TIM2_CH4CVR	0x40000040	TIM2 compare/capture register 4	0x0000
R16_TIM2_DMACFR	0x40000048	TIM2 DMA configuration register	0x0000
R16_TIM2_DMAADR	0x4000004C	TIM2 DMA address register in continuous mode	0x0000
R16_TIM2_AUX	0x40000050	TIM2 dual edge capture register	0x0000

Table 15-4 TIM3 registers

Name	Offset address	Description	Reset value
R16_TIM3_CTLR1	0x40000400	TIM3 control register 1	0x0000
R16_TIM3_CTLR2	0x40000404	TIM3 control register 2	0x0000
R16_TIM3_SMCFR	0x40000408	TIM3 slave mode configuration register	0x0000
R16_TIM3_DMAINTENR	0x4000040C	TIM3 DMA/interrupt enable register	0x0000
R16_TIM3_INTFR	0x40000410	TIM3 interrupt flag register	0x0000
R16_TIM3_SWEVGR	0x40000414	TIM3 event generation register	0x0000
R16_TIM3_CHCTLR1	0x40000418	TIM3 compare/capture control register 1	0x0000
R16_TIM3_CHCTLR2	0x4000041C	TIM3 compare/capture control register 2	0x0000
R16_TIM3_CCER	0x40000420	TIM3 compare/capture enable register	0x0000
R16_TIM3_CNT	0x40000424	TIM3 counter	0x0000
R16_TIM3_PSC	0x40000428	TIM3 prescaler	0x0000
R16_TIM3_ATRLR	0x4000042C	TIM3 auto reload register	0xFFFF
R32_TIM3_CH1CVR	0x40000434	TIM3 compare/capture register 1	0x0000

R32_TIM3_CH2CVR	0x40000438	TIM3 compare/capture register 2	0x0000
R32_TIM3_CH3CVR	0x4000043C	TIM3 compare/capture register 3	0x0000
R32_TIM3_CH4CVR	0x40000440	TIM3 compare/capture register 4	0x0000
R16_TIM3_DMACHFR	0x40000448	TIM3 DMA configuration register	0x0000
R16_TIM3_DMAADR	0x4000044C	TIM3 DMA address register in continuous mode	0x0000
R16_TIM3_AUX	0x40000450	TIM3 dual edge capture register	0x0000

Table 15-5 TIM4 registers

Name	Offset address	Description	Reset value
R16_TIM4_CTLR1	0x40000800	TIM4 control register1	0x0000
R16_TIM4_CTLR2	0x40000804	TIM4 control register2	0x0000
R16_TIM4_SMCFR	0x40000808	TIM4 slave mode configuration register	0x0000
R16_TIM4_DMAINTENR	0x4000080C	TIM4 DMA/interrupt enable register	0x0000
R16_TIM4_INTFR	0x40000810	TIM4 interrupt flag register	0x0000
R16_TIM4_SWEVGR	0x40000814	TIM4 event generation register	0x0000
R16_TIM4_CHCTL1	0x40000818	TIM4 compare/capture control register1	0x0000
R16_TIM4_CHCTL2	0x4000081C	TIM4 compare/capture control register2	0x0000
R16_TIM4_CCER	0x40000820	TIM4 compare/capture enable register	0x0000
R16_TIM4_CNT	0x40000824	TIM4 counter	0x0000
R16_TIM4_PSC	0x40000828	TIM4 prescaler	0x0000
R16_TIM4_ATRLR	0x4000082C	TIM4 auto-reload register	0xFFFF
R16_TIM4_CH1CVR	0x40000834	TIM4 compare/capture register1	0x0000
R16_TIM4_CH2CVR	0x40000838	TIM4 compare/capture register2	0x0000
R16_TIM4_CH3CVR	0x4000083C	TIM4 compare/capture register3	0x0000
R16_TIM4_CH4CVR	0x40000840	TIM4 compare/capture register4	0x0000
R16_TIM4_DMACHFR	0x40000848	TIM4 DMA configuration register	0x0000
R16_TIM4_DMAADR	0x4000084C	TIM4 DMA address register in continuous mode	0x0000
R16_TIM4_AUX	0x40000850	TIM4 dual edge capture register	0x0000

Table 15-6 TIM5 registers

Name	Offset address	Description	Reset value
R16_TIM5_CTLR1	0x40000C00	TIM5 control register1	0x0000
R16_TIM5_CTLR2	0x40000C04	TIM5 control register2	0x0000
R16_TIM5_SMCFR	0x40000C08	TIM5 slave mode configuration register	0x0000
R16_TIM5_DMAINTENR	0x40000C0C	TIM5 DMA/interrupt enable register	0x0000
R16_TIM5_INTFR	0x40000C10	TIM5 interrupt flag register	0x0000
R16_TIM5_SWEVGR	0x40000C14	TIM5 event generation register	0x0000
R16_TIM5_CHCTL1	0x40000C18	TIM5 compare/capture control register1	0x0000
R16_TIM5_CHCTL2	0x40000C1C	TIM5 compare/capture control register2	0x0000
R16_TIM5_CCER	0x40000C20	TIM5 compare/capture enable register	0x0000

R16_TIM5_CNT	0x40000C24	TIM5 counter	0x0000
R16_TIM5_PSC	0x40000C28	TIM5 prescaler	0x0000
R16_TIM5_ATRLR	0x40000C2C	TIM5 auto-reload register	0xFFFF
R32_TIM5_CH1CVR	0x40000C34	TIM5 compare/capture register1	0x0000
R32_TIM5_CH2CVR	0x40000C38	TIM5 compare/capture register2	0x0000
R32_TIM5_CH3CVR	0x40000C3C	TIM5 compare/capture register3	0x0000
R32_TIM5_CH4CVR	0x40000C40	TIM5 compare/capture register4	0x0000
R16_TIM5_DMCFGR	0x40000C48	TIM5 DMA configuration register	0x0000
R16_TIM5_DMAADR	0x40000C4C	TIM5 DMA address register in continuous mode	0x0000
R16_TIM5_AUX	0x40000C50	TIM5 dual edge capture register	0x0000

Table 15-7 TIM9 registers

Name	Offset address	Description	Reset value
R16_TIM9_CTLR1	0x40014C00	TIM9 control register1	0x0000
R16_TIM9_CTLR2	0x40014C04	TIM9 control register2	0x0000
R16_TIM9_SMCFR	0x40014C08	TIM9 slave mode configuration register	0x0000
R16_TIM9_DMAINTENR	0x40014C0C	TIM9 DMA/interrupt enable register	0x0000
R16_TIM9_INTFR	0x40014C10	TIM9 interrupt flag register	0x0000
R16_TIM9_SWEVGR	0x40014C14	TIM9 event generation register	0x0000
R16_TIM9_CHCTLR1	0x40014C18	TIM9 compare/capture control register1	0x0000
R16_TIM9_CHCTLR2	0x40014C1C	TIM9 compare/capture control register2	0x0000
R16_TIM9_CCER	0x40014C20	TIM9 compare/capture enable register	0x0000
R32_TIM9_CNT	0x40014C24	TIM9 counter	0x00000000
R16_TIM9_PSC	0x40014C28	TIM9 prescaler	0x0000
R32_TIM9_ATRLR	0x40014C2C	TIM9 auto-reload register	0x0000FFFF
R32_TIM9_CH1CVR	0x40014C34	TIM9 compare/capture register1	0x00000000
R32_TIM9_CH2CVR	0x40014C38	TIM9 compare/capture register2	0x00000000
R32_TIM9_CH3CVR	0x40014C3C	TIM9 compare/capture register3	0x00000000
R32_TIM9_CH4CVR	0x40014C40	TIM9 compare/capture register4	0x00000000
R16_TIM9_DMCFGR	0x40014C48	TIM9 DMA configuration register	0x0000
R16_TIM9_DMAADR	0x40014C4C	TIM9 DMA address register in continuous mode	0x0000

Table 15-8 TIM10 registers

Name	Offset address	Description	Reset value
R16_TIM10_CTLR1	0x40015000	TIM10 control register1	0x0000
R16_TIM10_CTLR2	0x40015004	TIM10 control register2	0x0000
R16_TIM10_SMCFR	0x40015008	TIM10 slave mode configuration register	0x0000

R16_TIM10_DMAINTENR	0x4001500C	TIM10 DMA/interrupt enable register	0x0000
R16_TIM10_INTFR	0x40015010	TIM10 interrupt flag register	0x0000
R16_TIM10_SWEVGR	0x40015014	TIM10 event generation register	0x0000
R16_TIM10_CHCTLR1	0x40015018	TIM10 compare/capture control register1	0x0000
R16_TIM10_CHCTLR2	0x4001501C	TIM10 compare/capture control register2	0x0000
R16_TIM10_CCER	0x40015020	TIM10 compare/capture enable register	0x0000
R32_TIM10_CNT	0x40015024	TIM10 counter	0x00000000
R16_TIM10_PSC	0x40015028	TIM10 prescaler	0x0000
R32_TIM10_ATRLR	0x4001502C	TIM10 auto-reload register	0x0000FFFF
R32_TIM10_CH1CVR	0x40015034	TIM10 compare/capture register1	0x00000000
R32_TIM10_CH2CVR	0x40015038	TIM10 compare/capture register2	0x00000000
R32_TIM10_CH3CVR	0x4001503C	TIM10 compare/capture register3	0x00000000
R32_TIM10_CH4CVR	0x40015040	TIM10 compare/capture register4	0x00000000
R16_TIM10_DMACHFR	0x40015048	TIM10 DMA configuration register	0x0000
R16_TIM10_DMAADR	0x4001504C	TIM10 DMA address register in continuous mode	0x0000

Table 15-9 TIM11 registers

Name	Offset address	Description	Reset value
R16_TIM11_CTLR1	0x40015400	TIM11 control register1	0x0000
R16_TIM11_CTLR2	0x40015404	TIM11 control register2	0x0000
R16_TIM11_SMCFR	0x40015408	TIM11 slave mode configuration register	0x0000
R16_TIM11_DMAINTENR	0x4001540C	TIM11 DMA/interrupt enable register	0x0000
R16_TIM11_INTFR	0x40015410	TIM11 interrupt flag register	0x0000
R16_TIM11_SWEVGR	0x40015414	TIM11 event generation register	0x0000
R16_TIM11_CHCTLR1	0x40015418	TIM11 compare/capture control register1	0x0000
R16_TIM11_CHCTLR2	0x4001541C	TIM11 compare/capture control register2	0x0000
R16_TIM11_CCER	0x40015420	TIM11 compare/capture enable register	0x0000
R32_TIM11_CNT	0x40015424	TIM11 counter	0x00000000
R16_TIM11_PSC	0x40015428	TIM11 prescaler	0x0000
R32_TIM11_ATRLR	0x4001542C	TIM11 auto-reload register	0x0000FFFF
R32_TIM11_CH1CVR	0x40015434	TIM11 compare/capture register1	0x00000000
R32_TIM11_CH2CVR	0x40015438	TIM11 compare/capture register2	0x00000000
R32_TIM11_CH3CVR	0x4001543C	TIM11 compare/capture register3	0x00000000
R32_TIM11_CH4CVR	0x40015440	TIM11 compare/capture register4	0x00000000
R16_TIM11_DMACHFR	0x40015448	TIM11 DMA configuration register	0x0000
R16_TIM11_DMAADR	0x4001544C	TIM11 DMA address register in continuous	0x0000

		mode	
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Table 15-10 TIM12 registers

Name	Offset address	Description	Reset value
R16_TIM12_CTLR1	0x40013C00	TIM12 control register1	0x0000
R16_TIM12_CTLR2	0x40013C04	TIM12 control register2	0x0000
R16_TIM12_SMCFGR	0x40013C08	TIM12 slave mode configuration register	0x0000
R16_TIM12_DMAINTENR	0x40013C0C	TIM12 DMA/interrupt enable register	0x0000
R16_TIM12_INTFR	0x40013C10	TIM12 interrupt flag register	0x0000
R16_TIM12_SWEVGR	0x40013C14	TIM12 event generation register	0x0000
R16_TIM12_CHCTLR1	0x40013C18	TIM12 compare/capture control register1	0x0000
R16_TIM12_CHCTLR2	0x40013C1C	TIM12 compare/capture control register2	0x0000
R16_TIM12_CCER	0x40013C20	TIM12 compare/capture enable register	0x0000
R32_TIM12_CNT	0x40013C24	TIM12 counter	0x00000000
R16_TIM12_PSC	0x40013C28	TIM12 prescaler	0x0000
R32_TIM12_ATRLR	0x40013C2C	TIM12 auto-reload register	0x0000FFFF
R32_TIM12_CH1CVR	0x40013C34	TIM12 compare/capture register1	0x00000000
R32_TIM12_CH2CVR	0x40013C38	TIM12 compare/capture register2	0x00000000
R32_TIM12_CH3CVR	0x40013C3C	TIM12 compare/capture register3	0x00000000
R32_TIM12_CH4CVR	0x40013C40	TIM12 compare/capture register4	0x00000000
R16_TIM12_DMACFGR	0x40013C48	TIM12 DMA configuration register	0x0000
R16_TIM12_DMAADR	0x40013C4C	TIM12 DMA address register in continuous mode	0x0000

15.4.1 Control Register 1 (TIMx_CTLR1) (x=2/3/4/5/9/10/11/12)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
15	CAPLVL	RW	In double edge capture mode, capture level indication enables: 1: Enable indicates function; 0: Disable indication function. <i>Note: After enabling, [16] of CHxCVR indicates the level corresponding to the capture value.</i>	0
14	CAPOV	RW	Capture value mode configuration: 1: When a counter overflow occurs before capture, the CHxCVR value is 0xFFFF; 0: Capture value is the value of the actual counter.	0
[13:10]	Reserved	RO	Reserved	0
[9:8]	CKD[1:0]	RW	These 2 bits define the frequency division ratio of timer	00b

			clock (CK_INT) frequency and sampling clock used for the digital filter: 00: Tdts=Tck_int; 01: Tdts= 2×Tck_int; 10: Tdts= 4×Tck_int; 11: Reserved.	
7	ARPE	RW	Auto-reload and preload enable: 1: Auto-reload value register (ATRLR) enabled; 0: Auto-reload value register (ATRLR) disabled.	0
[6:5]	CMS[1:0]	RW	Central alignment mode selection: 00: Edge alignment mode. The counter counts up or down according to the direction bit (DIR). 01: Center alignment mode 1. The counter counts up and down alternately. The output comparison interrupt flag bit of the channel configured as an output (CCxS=00 in the CHCTLRx register) is only set when the counter counts down. 10: Center alignment mode 2. The counter counts up and down alternately. The output comparison interrupt flag bit of the channel configured as an output (CCxS=00 in the CHCTLRx register) is only set when the counter counts up. 11: Center alignment mode 3. The counter counts up and down alternately. The output comparison interrupt flag bit of the channel configured as an output (CCxS=00 in the CHCTLRx register) is only set when the counter counts up and down. <i>Note: When the counter is enabled (CEN=1), it is not allowed to switch from edge alignment mode to center alignment mode.</i>	00b
4	DIR	RW	Counter direction: 1: Downcount; 0: Upcount. <i>Note: When the counter is configured in the center alignment mode or encoder mode, this bit will be invalid.</i>	0
3	OPM	RW	One-Pulse Mode 1: The counter stops when the next update event (clearing the CEN bit) occurs; 0: The counter does not stop when the next update event occurs.	0
2	URS	RW	Update Request Source: The software uses this bit to select the source of UEV events. 1: If update interrupts or DMA requests are enabled,	0

			only counter overflows/underflows generate update interrupts or DMA requests; 0: If update interrupts or DMA requests are enabled, any of the following events generate update interrupts or DMA requests: -Counter overflow/underflow. -Set the UG bit. -Updates generated from the mode controller.	
1	UDIS	RW	Updates are prohibited, and the software allows/disables the generation of UEV events through this bit. 1: UEV is prohibited. No update events are generated, and the registers (ATRLR, PSC, CHCTLRx) hold their values. If the UG bit is set or a hardware reset is issued from the mode controller, the counter and prescaler are reinitialized. 0: UEV is allowed. The UEV event is generated by any of the following events: -Counter overflow / underflow. -Set the UG bit. -Updates generated from the mode controller. Registers with caches are loaded into their preloaded values.	0
0	CEN	RW	Counter enable: 1: Enable counter. 0: Disables counters. <i>Note: the external clock, gating mode and encoder mode will not work until the CEN bit is set in the software. The trigger mode automatically sets the CEN bit through the hardware.</i>	0

15.4.2 Control Register 2 (TIMx_CTLR2) (x=2/3/4/5/9/10/11/12)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	TI1S	RW	TI1 selection: 1: TIMx_CH1, TIMx_CH2 and TIMx_CH3 pins are connected to TI1 input through XOR; 0: TIMx_CH1 pin is directly connected to TI1 input.	0
[6:4]	MMS[2:0]	RW	Master mode selection: These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in the master mode. The possible combination is as follows: 000: Reset – The UG bit is used as a trigger output (TRGO). If it is a reset generated by a trigger input (the	000b

			slave mode controller is in reset mode), the signal on TRGO will have a delay relative to the actual reset; 001: Enable-the counter enables signal CNT_EN to be used as a trigger output (TRGO). Sometimes, it is necessary to start multiple timers at the same time or control to enable slave timers within a period of time. The counter enable signal is generated by the logical OR of the CEN control bit and the trigger input signal in the gating mode. When the counter enable signal is controlled by the trigger input, there will be a delay on TRGO, unless the master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFGR register); 010: An update event is selected as the trigger output (TRGO). For example, the clock of a master timer can be used as a prescaler for a slave timer; 011: Comparison pulse, when a capture occurs or a comparison is successful, and the CC1IF flag is to be set (even if it is already high), the trigger output will send a positive pulse (TRGO); 100: OC1REF signal is used as trigger output (TRGO); 101: OC2REF signal is used as trigger output (TRGO); 110: OC3REF signal is used as trigger output (TRGO); 111: OC4REF signal is used as trigger output (TRGO).	
3	CCDS	RW	1: When an update event occurs, send a DMA request of CHxCVR; 0: When CHxCVR occurs, a DMA request of CHxCVR will be generated.	0
[2:1]	Reserved	RO	Reserved.	0
0	CCPC	RW	Compare/capture preload control bit: 1: The CCxE, CCxNE, and OCxM bits are preloaded. When this bit is set, they are updated only after the COM bit is set. 0: The CCxE, CCxNE, and OCxM bits are not preloaded. <i>Note: This bit affects only channels with complementary outputs.</i>	0

15.4.3 Slave Mode Control Register (TIMx_SMCFGR) (x=2/3/4/5/9/10/11/12)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
15	ETP	RO	ETR trigger polarity selection; this bit selects whether to directly input ETR or input inverted ETR. 1: ETR inverted, active at low level or falling edge;	0

			0: ETR, active at high level or rising edge.	
14	ECE	RW	External clock mode 2 enable. 1: External clock mode 2 enabled; 0: External clock mode 2 disabled. <i>Note 1: Slave mode can be used simultaneously with external clock mode 2: reset mode, gating mode and trigger mode; however, TRGI cannot be connected to ETRF at this time (TS bit cannot be 111b).</i> <i>Note 2: When both external clock mode 1 and external clock mode 2 are enabled at the same time, the input of the external clock will be ETRF.</i>	0
[13:12]	ETPS[1:0]	RW	External trigger prescaler (ETRP); the frequency must be at most 1/4 of TIMxCLK frequency, and the frequency can be reduced through this domain. 00: Prescale off; 01: ETRP frequency divided by 2; 10: ETRP frequency divided by 4; 11: ETRP frequency divided by 8.	00b
[11:8]	ETF[3:0]	RW	External trigger filter. In fact, the digital filter is an event counter. N events are needed to validate a conversion on the output. 0000: No filter, sampling is done at Fdts; 0001: Fsampling=Fck_int, N=2; 0010: Fsampling=Fck_int, N=4; 0011: Fsampling=Fck_int, N=8; 0100: Fsampling=Fdts/2, N=6; 0101: Fsampling=Fdts/2, N=8; 0110: Fsampling=Fdts/4, N=6; 0111: Fsampling=Fdts/4, N=8; 1000: Fsampling=Fdts/8, N=6; 1001: Fsampling=Fdts/8, N=8; 1010: Fsampling=Fdts/16, N=5; 1011: Fsampling=Fdts/16, N=6; 1100: Fsampling=Fdts/16, N=8; 1101: Fsampling=Fdts/32, N=5; 1110: Fsampling=Fdts/32, N=6; 1111: Fsampling=Fdts/32, N=8.	0000b
7	MSM	RW	Master/Slave mode selection: 1: The event on the trigger input (TRGI) is delayed to allow perfect synchronization between the current timer (via TRGO) and its slave timer. This is very useful when it is required to synchronize several timers to a single external event; 0: Not action.	0

[6:4]	TS[2:0]	RW	Trigger selection; these 3 bits select the trigger input source used to synchronize the counter. 000: Internal trigger 0 (ITR0); 100: Edge detector of TI1 (TI1F_ED); 001: Internal trigger 1 (ITR1); 101: Timer input 1 after filtering (TI1FP1); 010: Internal trigger 2 (ITR2); 110: Timer input 2 after filtering (TI12FP2); 011: Internal trigger 3 (ITR3); 111: External trigger input (ETRF); The values can be changed only when SMS is 0.	000b
3	Reserved	RO	Reserved	0
[2:0]	SMS[2:0]	RW	Input mode selection. Select the clock and trigger mode of the core counter. 000: Driven by the internal clock CK_INT; 001: Encoder mode1; depending on TI1FP1 level, the core counter counts up or down on edge of TI2FP2; 010: Encoder mode2; depending on TI2FP2 level, the core counter counts up or down on edge of TI1FP1; 011: Encoder mode3; depending on the input level of another signal, the core counter counts up and down on the edge of TI1FP1 and TI2FP2; 100: Reset mode; the rising edge of the trigger input (TRGI) will initialize the counter and generate a signal for updating the register; 101: Gating mode; when the trigger input (TRGI) is high, the clock of the counter will be turned on; when the trigger input becomes low, the counter will stop, and the start and stop of the counter will be controlled; 110: Trigger mode; the counter starts on the rising edge of the trigger input TRGI, and only the start of the counter is controlled; 111: External clock mode1; the rising edge of the selected trigger input (TRGI) drives the counter.	000b

15.4.4 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=2/3/4/5/9/10/11/12)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
15	Reserved	RO	Reserved	0
14	TDE	RW	Trigger DMA request enable. 1: Trigger DMA request enabled; 0: Trigger DMA request disabled.	0
13	Reserved	RW	Reserved	0
12	CC4DE	RW	DMA request enable of compare/capture4.	0

			1: DMA request of compare/capture4 enabled; 0: DMA request of compare/capture4 disabled.	
11	CC3DE	RW	DMA request enable of compare/capture3. 1: DMA request of compare/capture3 enabled; 0: DMA request of compare/capture3 disabled.	0
10	CC2DE	RW	DMA request enable of compare/capture2. 1: DMA request of compare/capture2 enabled; 0: DMA request of compare/capture2 disabled.	0
9	CC1DE	RW	DMA request enable of compare/capture1. 1: DMA request of compare/capture1 enabled; 0: DMA request of compare/capture1 disabled.	0
8	UDE	RW	Update DMA request enable. 1: Update DMA request enabled; 0: Update DMA request disabled.	0
7	Reserved	RO	Reserved	0
6	TIE	RW	Trigger interrupt enable. 1: Trigger interrupt enabled; 0: Trigger interrupt disabled.	0
5	Reserved	RO	Reserved	0
4	CC4IE	RW	Interrupt enable of compare/capture4. 1: Interrupt of compare/capture4 enabled; 0: Interrupt of compare/capture4 disabled.	0
3	CC3IE	RW	Interrupt enable of compare/capture3. 1: Interrupt of compare/capture3 enabled; 0: Interrupt of compare/capture3 disabled.	0
2	CC2IE	RW	Interrupt enable of compare/capture2. 1: Interrupt of compare/capture2 enabled; 0: Interrupt of compare/capture2 disabled.	0
1	CC1IE	RW	Interrupt enable of compare/capture1. 1: Interrupt of compare/capture1 enabled; 0: Interrupt of compare/capture1 disabled.	0
0	UIE	RW	Update interrupt enable. 1: Update interrupt enabled; 0: Update interrupt disabled.	0

15.4.5 Interrupt Flag Register (R16_TIMx_INTFR) (x=2/3/4/5/9/10/11/12)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
12	CC4OF	RW0	Compare/capture4 recapture flag.	0
11	CC3OF	RW0	Compare/capture3 recapture flag.	0
10	CC2OF	RW0	Compare/capture2 recapture flag.	0
9	CC1OF	RW0	Compare/capture1 recapture flag is only used when the compare/capture is configured in the input capture mode.	0

			This flag bit is set by the hardware, write 0 by software to clear the bit. 1: When the value of the counter is captured into the capture comparison register, the status of CC1IF has been set; 0: No recapture is generated.	
[8:7]	Reserved	RO	Reserved	0
6	TIF	RW0	Trigger interrupt flag. When a trigger event occurs, set by hardware and cleared by software. Trigger events include the detection of a valid edge at the TRGI input terminal from modes other than gating mode, or any edge in gating mode. 1: Trigger event occurs; 0: No trigger event occurs.	0
5	Reserved	RO	Reserved.	0
4	CC4IF	RW0	Compare/capture 4 interrupt flag.	0
3	CC3IF	RW0	Compare/capture 3 interrupt flag.	0
2	CC2IF	RW0	Compare/capture 2 interrupt flag.	0
1	CC1IF	RW0	Compare/capture 1 interrupt flag. If the compare/capture is configured as the output mode, this bit is set by hardware when the counter value matches the compare value, except in center-aligned mode. This bit is cleared by software. 1: The value of core counter matches the value of compare/capture register 1; 0: No. If the compare/capture is configured as the output mode, this bit is set by hardware when a capture event occurs, and it is cleared by software or cleared by reading the compare/capture register. 1: The counter value has been captured by the compare/capture register 1; 0: No input capture is generated.	0
0	UIF	RW0	Update interrupt flag. When an update event occurs, it is set by hardware and cleared by software. 1: Update interrupt generated; 0: No update interrupt generated. The update event generates in case of the following circumstances: For UDIS=0, when the repeated counter value overflows or underflows; For URS=0, UDIS=0, when the UG bit is set, or when the counter core is reinitialized by software;	0

			For URS=0, UDIS=0, when the counter CNT is reinitialized by a trigger event.	
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15.4.6 Event Generation Register (TIMx_SWEVGR) (x=2/3/4/5/9/10/11/12)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved.	0
6	TG	WO	Trigger event generation. Set by software, and cleared by hardware to generate a trigger event. 1: A trigger event generated; if TIF is set and the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA will be generated; 0: No effect.	0
5	Reserved	RO	Reserved.	0
4	CC4G	WO	Compare/capture 4 generation.	0
3	CC3G	WO	Compare/capture 3 generation.	0
2	CC2G	WO	Compare/capture 2 generation.	0
1	CC1G	WO	Compare/capture1 generation. This bit is set by software and cleared by hardware. It is used to generate a compare/capture event. 1: Generate compare/capture event on channel 1: If compare/capture 1 is configured as output: Set the CC1IF bit. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA will be generated; If compare/capture 1 is configured as input, the current core counter value is captured to compare/capture register 1; set the CC1IF bit, if the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA will be generated. If the CC1IF bit has been set, set the CC1OF bit. 0: No action.	0
0	UG	WO	Update event generation. This bit is set by software and cleared automatically by hardware. 1: Initialize the counter and generate an update event; 0: No action. <i>Note: The counter of the prescaler is also cleared, but the prescaler factor remains unchanged. In Centro symmetric mode or up-counting mode, the core counter will be cleared; in the down-counting mode, the core counter will take the value of the reload value register.</i>	0

15.4.7 Compare/Capture Control Register 1 (TIMx_CHCTLR1) (x=2/3/4/5/9/10/11/12)

Offset address: 0x18

The channel can be used for input (capture mode) or output (compare mode), and the direction of the channel is defined by the corresponding CCxS bit. The functions of other bits of this register are different in input and output modes. OCxx describes the function of the channel in output mode, and ICxx describes the function of the channel in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]		OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]		OC1PE	OC1FE	CC1S[1:0]			
IC2F[3:0]			IC2PSC[1:0]				IC1F[3:0]			IC1PSC[1:0]					

Compare mode (pin direction is output):

Bit	Name	Access	Description	Reset value
15	OC2CE	RW	Compare/capture 2 clear enable. 1: Once the ETRF input high level is detected, clear the OC2REF bit to zero; 0: OC2REF is not affected by the ETRF input.	0
[14:12]	OC2M[2:0]	RW	Compare/capture2 mode setting. These 3 bits define the action of the output reference signal OC2REF, and OC2REF determines the value of OC2 and OC2N. OC2REF is active at high level, while the active level of OC2 and OC2N depends on the CC2P and CC2NP bits. 000: Frozen. The comparison value between the value of the compare/capture register and the core counter has no effect on OC2REF; 001: Active by force. When the core counter and compare/capture register1 have the same value, force OC2REF to be high; 010: Set as inactive level by force. When the value of the core counter is the same as compare/capture register 1, force OC2REF to be low; 011: Overturn. When the core counter and compare/capture register1 have the same value, overturn the level of OC2REF; 100: Inactive by force. Force OC2REF to be low. 101: Force to be active level. Force OC2REF to be high. 110: PWM Mode 1: In up count, once the core counter is less than the value of the compare/capture register, channel 2 is valid level, otherwise it is invalid level; in downward counting, once the core counter is greater than the value of the compare/capture register, channel 2 is invalid level (OC2REF=0), otherwise it is valid level (OC2REF=1). 111: PWM mode 2: In up count, once the core counter	000b

			is smaller than the value of the compare/capture register, channel 2 is invalid level, otherwise it is valid level; in down count, once the core counter is larger than the value of the compare/capture register, channel 2 is valid level (OC2REF=1), otherwise it is invalid level (OC2REF=0). <i>Note: Once the LOCK level is set to 3 and CC2S=00b, this bit cannot be modified. In PWM mode1 or PWM mode2, the OC2REF level changes only when the comparison result changes or when switching from freezing mode to PWM mode in output compare mode.</i>	
11	OC2PE	RW	Compare/capture register 2 preload enable. 1: Enable the preload function of the compare/capture register. Read and write operations are only made on the preload register. The preload value of the compare/capture register 2 is loaded into the current shadow register when the update event arrives; 0: Disable the pre-loading function of compare/capture register 2. Compare/capture register 2 can be written at any time, and the newly written value takes effect immediately. <i>Note: Once the LOCK level is set to 3 and CC2S=00b, this bit cannot be modified; only in One-Pulse Mode (OPM=1) you can use PWM mode without confirming the preload register; otherwise its action is uncertain.</i>	0
10	OC2FE	RW	Compare/capture 2 fast enable. It is used to speed up the response of the compare/capture output to the trigger input event. 1: The effect of the inactive edge inputted to the trigger is like a comparison match. Therefore, OC is set to the comparison level regardless of the comparison result. The delay between the valid edge of the sampling trigger and the output of compare/capture 2 is shortened to 3 clock cycles; 0: According to the value of counter and compare/capture register 2, compare/capture 2 operates normally, even if the trigger is turned on. When the input of the trigger has a valid edge, the minimum delay for activating the output of the compare/capture 2 is 5 clock cycles. <i>Note: OC2FE only works when the channel is configured in PWM1 or PWM2 mode.</i>	0
[9:8]	CC2S[1:0]	RW	Compare/capture 2 input selection. 00: The compare/capture 2 is configured as output;	00b

			01: Compare/capture 2 is configured as input, and IC2 is mapped on TI2; 10: Compare/capture 2 is configured as input, and IC2 is mapped on TI1; 11: Compare/capture 2 is configured as an input, and IC2 is mapped on TRC. This mode only works when the internal trigger input is selected (selected by the TS bit). <i>Note: Compare/capture 2 is only writable when the channel is switched off (CC2E is zero).</i>	
7	OC1CE	RW	Compare/capture 1 clear enable.	0
[6:4]	OC1M[2:0]	RW	Compare/capture 1 mode setting.	0
3	OC1PE	RW	Compare/capture register 1 preload enable.	0
2	OC1FE	RW	Compare/capture 1 fast enable.	0
[1:0]	CC1S[1:0]	RW	Compare/capture 1 input selection.	0

Capture mode (pin direction is input):

Bit	Name	Access	Description	Reset value
[15:12]	IC2F[3:0]	RW	Input capture2 filter configuration. These bits set the sampling frequency and digital filter length of TI1 input. The digital filter is composed of an event counter, in which N events are needed to validate a conversion on the output. 0000: No filter, sampling is done at Fdts; 1000: Fsampling=Fdts/8, N=6; 0001: Fsampling=Fck_int, N=2; 1001: Fsampling=Fdts/8, N=8; 0010: Fsampling=Fck_int, N=4; 1010: Fsampling=Fdts/16, N=5; 0011: Fsampling=f=Fck_int, N=8; 1011: Fsampling=Fdts/16, N=6; 0100: Fsampling=Fdts/2, N=6; 1100: Fsampling=Fdts/16, N=8; 0101: Fsampling=Fdts/2, N=8; 1101: Fsampling=Fdts/32, N=5; 0110: Fsampling=Fdts/4, N=6; 1110: Fsampling=Fdts/32, N=6; 0111: Fsampling=Fdts/4, N=8; 1111: Fsampling=Fdts/32, N=8.	0000b
[11:10]	IC2PSC[1:0]	RW	Compare/capture 2 prescaler configuration. These 2 bits define the prescaler factor of compare/capture 2. Once CC1E=0, the prescaler will be reset. 00: Prescaler OFF, each edge detected on the capture input port triggers a capture;	00b

			01: Trigger a capture every 2 events; 10: Trigger a capture every 4 events; 11: Trigger a capture every 8 events.	
[9:8]	CC2S[1:0]	RW	Compare/capture2 input selection. These 2 bits define the direction of the channel (input/output) and selection of input pins. 00: Compare/capture1 is configured as output; 01: Compare/capture1 is configured as input, and IC1 is mapped on TI1; 10: Compare/capture1 is configured as input, and IC1 is mapped on TI2; 11: Compare/capture1 is configured as an input, and IC1 is mapped on TRC. This mode only works when the internal trigger input is selected (selected by the TS bit). <i>Note: CC1S can be written only when the channel is closed (CC1E is 0).</i>	00b
[7:4]	IC1F[3:0]	RW	Input capture1 filter configuration.	0
[3:2]	IC1PSC[1:0]	RW	Compare/capture1 prescaler configuration.	0
[1:0]	CC1S[1:0]	RW	Compare/capture1 input selection.	0

15.4.8 Compare/Capture Control Register 2 (TIMx_CHCTLR2) (x=2/3/4/5/9/10/11/12)

Offset address: 0x1C

The channel can be used for input (capture mode) or output (comparison mode), and the direction of the channel is defined by the corresponding CCxS bit. The functions of other bits of this register are different in input and output modes. OCxx describes the function of the channel in output mode, and ICxx describes the function of the channel in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4CE	OC4M[2:0]		OC4PE	OC4FE	CC4S[1:0]		OC3CE	OC3M[2:0]		OC3PE	OC3FE	CC3S[1:0]			
IC4F[3:0]			IC4PSC[1:0]				IC3F[3:0]			IC3PSC[1:0]					

Compare mode (pin direction is output):

Bit	Name	Access	Description	Reset value
15	OC4CE	RW	Compare/capture channel 4 clear enable bit.	0
[14:12]	OC4M[2:0]	RW	Compare/capture channel 4 mode setting fields.	0
11	OC4PE	RW	Compare/capture register 4 preload enable bit.	0
10	OC4FE	RW	Compare/capture channel 4 fast enable bit.	0
[9:8]	CC4S[1:0]	RW	Compare/capture channel 4 input selection fields.	0
7	OC3CE	RW	Compare/capture channel 3 clear enable bit.	0
[6:4]	OC3M[2:0]	RW	Compare/capture channel 3 mode setting fields.	0
3	OC3PE	RW	Compare/capture register 3 preload enable bit.	0
2	OC3FE	RW	Compare/capture channel 3 fast enable bit.	0

[1:0]	CC3S[1:0]	RW	Compare/capture 3 input selection fields.	0
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Capture mode (pin direction is input):

Bit	Name	Access	Description	Reset value
[15:12]	IC4F[3:0]	RW	Input capture 4 filter configuration fields.	0
[11:10]	IC4PSC[1:0]	RW	Compare/capture 4 prescale configuration fields.	0
[9:8]	CC4S[1:0]	RW	Compare/capture 4 input selection fields.	0
[7:4]	IC3F[3:0]	RW	Input capture 3 filter configuration fields.	0
[3:2]	IC3PSC[1:0]	RW	Compare/capture 3 prescale configuration fields.	0
[1:0]	CC3S[1:0]	RW	Compare/capture 3 input selection fields.	0

15.4.9 Compare/Capture Enable Register (TIMx_CCER) (x=2/3/4/5/9/10/11/12)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[15:14]	Reserved	RO	Reserved	0
13	CC4P	RW	Compare/capture 4 output polarity	0
12	CC4E	RW	Compare/capture 4 output enable	0
[11:10]	Reserved	RO	Reserved	0
9	CC3P	RW	Compare/capture 3 output polarity	0
8	CC3E	RW	Compare/capture 3 output enable	0
[7:6]	Reserved	RO	Reserved	0
5	CC2P	RW	Compare/capture 2 output polarity	0
4	CC2E	RW	Compare/capture 2 output enable	0
[3:2]	Reserved	RO	Reserved	0
1	CC1P	RW	Compare/capture channel 1 output polarity setting bit. CC1 channel is configured as output: 1: OC1 active low; 0: OC1 active high. CC1 channel is configured as input: This bit selects whether IC1 or the inverted signal of IC1 is used as the trigger or capture signal. 1: Inverting: Capture occurs on the falling edge of IC1; IC1 is inverted when used as an external trigger. 0: Non-inverting: Capture occurs on the rising edge of IC1; IC1 is non-inverted when used as an external trigger.	0
0	CC1E	RW	Compare/capture channel 1 output enable bit CC1 channel is configured as output: 1: Enable: OC1 signal output to the corresponding output pin; 0: Disable: OC1 output is inhibited. CC1 channel is configured as input: This bit determines whether the counter value can be	0

			captured into the TIMx_CH1CVR register. 1: Capture enabled; 0: Capture disabled.	
--	--	--	--	--

15.4.10 Counter of General-purpose Timer (TIMx_CNT) (x= 2/3/4/5)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	Real-time value of timer counter.	0

15.4.11 Counter of General-purpose Timer (TIMx_CNT) (x=9/10/11/12)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CNT[31:0]	RW	Real-time value of timer counter.	0

15.4.12 Count Clock Prescaler (TIMx_PSC) (x=2/3/4/5/9/10/11/12)

Offset address: 0x28

Bit	Name	Access	Description	Reset value
[15:0]	PSC[15:0]	RW	The frequency division factor of the timer's prescaler; the clock frequency of the counter is equal to the input frequency of the frequency divider/(PSC+1).	0

15.4.13 Auto-reload Value Register (TIMx_ATRLR) (x=2/3/4/5)

Offset address: 0x2C

Bit	Name	Access	Description	Reset value
[15:0]	ARR[15:0]	RW	Counter thresholds, when ATRLR is active and updated read Section 15.2.4; when ATRLR is empty, the counter stops.	0xFFFF

15.4.14 Auto-reload Value Register (TIMx_ATRLR) (x=9/10/11/12)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ARR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	ARR[31:0]	RW	Counter thresholds, when ATRLR is active and updated read Section 15.2.4; when ATRLR is empty, the counter stops.	0xFFFFFFFF F

15.4.15 Compare/Capture Register 1 (TIMx_CH1CVR) (x=2/3/4/5)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LEVEL1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CH1CVR[15:0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL1	RO	The level indicator bit corresponding to the capture value	0
[15:0]	CH1CVR[15:0]	RW	Compare/capture register channel 1.	0

15.4.16 Compare/Capture Register 1 (TIMx_CH1CVR) (x=9/10/11/12)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CH1CVR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CH1CVR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CH1CVR[31:0]	RW	Compare/capture register channel 1.	0

15.4.17 Compare/Capture Register 2 (TIMx_CH2CVR) (x=2/3/4/5)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LEVEL2
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CH2CVR[15:0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL2	RO	The level indicator bit corresponding to the capture value	0

[15:0]	CH2CVR[15:0]	RW	Compare/capture register channel 2.	0
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15.4.18 Compare/Capture Register 2 (TIMx_CH2CVR) (x=9/10/11/12)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CH2CVR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CH2CVR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CH2CVR[31:0]	RW	Compare/capture register channel 2.	0

15.4.19 Compare/Capture Register 3 (TIMx_CH3CVR) (x=2/3/4/5)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LEVEL3
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CH3CVR[15:0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL3	RO	The level indicator bit corresponding to the capture value	0
[15:0]	CH3CVR[15:0]	RW	Compare/capture register channel 3.	0

15.4.20 Compare/Capture Register3 (TIMx_CH3CVR) (x=9/10/11/12)

Offset address: 0x3C

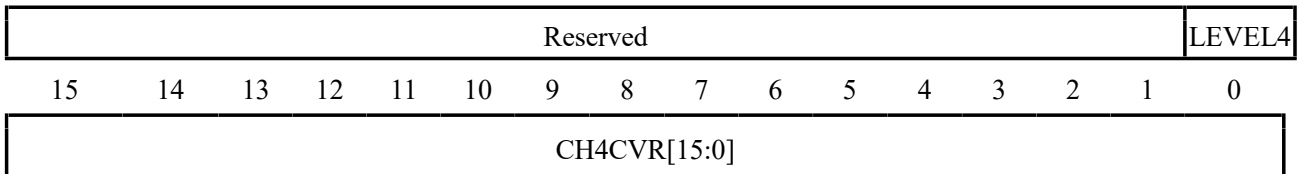
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CH3CVR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CH3CVR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CH3CVR[31:0]	RW	Compare/capture register channel 3.	0

15.4.21 Compare/Capture Register 4 (TIMx_CH4CVR) (x=2/3/4/5)

Offset address: 0x40

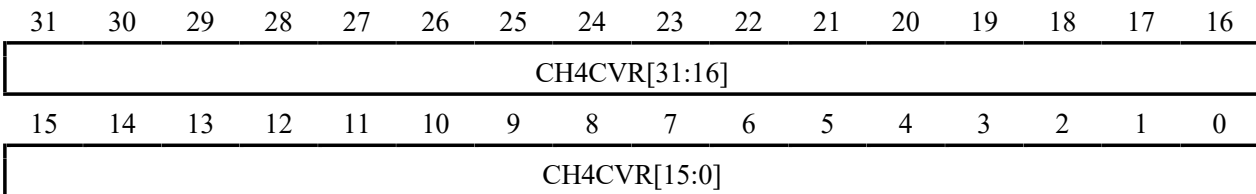
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----



Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	LEVEL4	RO	The level indicator bit corresponding to the capture value	0
[15:0]	CH4CVR[15:0]	RW	Compare/capture register channel 4.	0

15.4.22 Compare/Capture Register4 (TIMx_CH4CVR) (x=9/10/11/12)

Offset address: 0x40



Bit	Name	Access	Description	Reset value
[31:0]	CH4CVR[31:0]	RW	Compare/capture register channel 4.	0

15.4.23 DMA Control Register (TIMx_DMCFGR) (x=2/3/4/5/9/10/11/12)

Offset address: 0x48

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
[12:8]	DBL[4:0]	RW	Length of data that DMA continuously transfers; the actual value is the value of this domain + 1.	0
[7:5]	Reserved	RO	Reserved.	0
[4:0]	DBA[4:0]	RW	These bits define the offset of DMA from the address of control register1 in continuous mode.	0

15.4.24 DMA Address Register in Continuous Mode (TIMx_DMAADR) (x=2/3/4/5/9/10/11/12)

Offset address: 0x4C

Bit	Name	Access	Description	Reset value
[31:0]	DMAB[31:0]	RW	DMA address in continuous mode.	0

15.4.25 Dual-edge Capture Register (TIMx_AUX) (x=2/3/4/5)

Offset address: 0x50

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	RO	Reserved	0
2	CAP_ED_CH4	RW	Dual-edge capture enable for Channel 4:	0

			1: Enable dual-edge capture for Channel 4; 0: Disable dual-edge capture for Channel 4.	
1	CAP_ED_CH3	RW	Dual-edge capture enable for Channel 3: 1: Enable dual-edge capture for Channel 3; 0: Disable dual-edge capture for Channel 3.	0
0	CAP_ED_CH2	RW	Dual-edge capture enable for Channel 2: 1: Enable dual-edge capture for Channel 2; 0: Disable dual-edge capture for Channel 2.	0

Chapter 16 Basic Timer (BCTM)

The basic timer module comprises two 16-bit auto-reloadable timers (TIM6 and TIM7) for counting and generating interrupts upon update events or DMA requests. TIM6 and TIM7 support 16-bit programmable prescalers. They can provide clocks for digital-to-analog conversion (DAC) and trigger DAC synchronization circuits. The basic timers operate independently and do not share any resources.

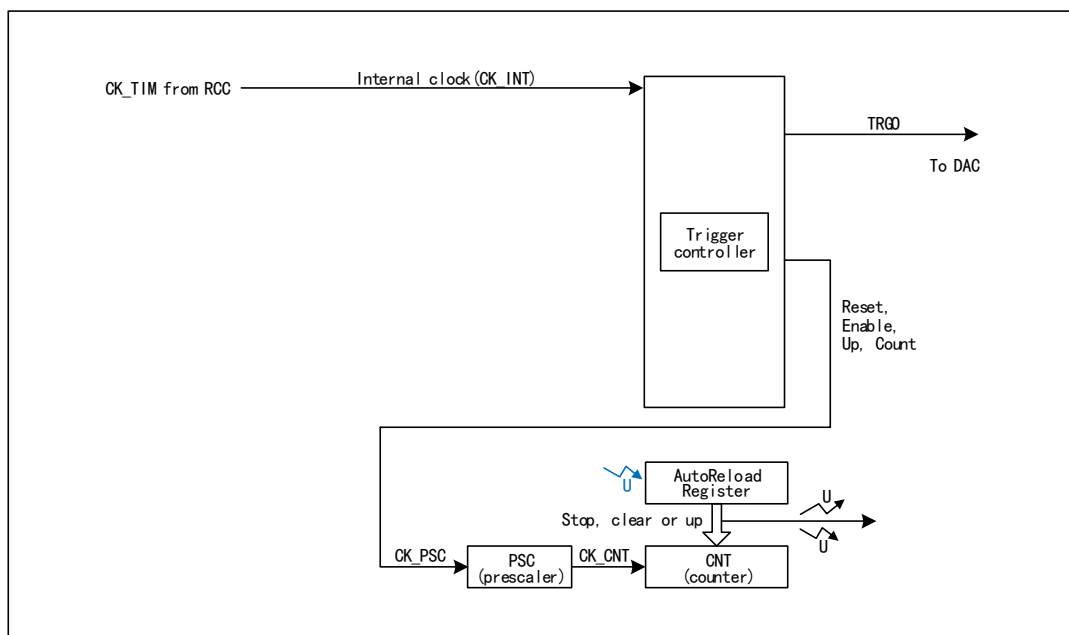
16.1 Main Features

Main features of the basic timer include:

- 16-bit auto-reload counter supporting increment mode
- 16-bit prescaler with dynamically adjustable division factor from 1 to 65536
- Triggered DAC synchronization circuit
- Generates interrupts or DMA requests upon update events

16.2 Principle and Structure

Figure 16-1 Block diagram of the structure of the basic timer



16.2.1 Overview

As shown in Figure 16-1, the structure of the basic timer can be broadly divided into two parts: the input clock section and the core counter section.

The clock for the basic timer originates from the HB bus clock (CK_INT). These input clock signals undergo various operations such as filtering and frequency division to become the CK_PSC clock, which is then output to the core counter section. Additionally, these complex clock sources can also be output as TRGO to the DAC peripheral.

The core of the basic timer is a 16-bit counter (CNT). After frequency division by prescaler (PSC), CK_PSC becomes CK_CNT, and finally it is lost to CNT, which supports counting up mode. At the end of each counting

period, CNT counts from 0.

16.2.2 Difference between Basic Timer and General-purpose Timer

Compared with the general timer, the basic timer lacks the following functions:

- 1) The basic timer lacks count-down mode and count-up mode.
- 2) The basic timer lacks 4-channel independent comparison capture channels.
- 3) The basic timer does not support external signal control timer.
- 4) The basic timer does not support incremental coding, cascade and synchronization between timers.

16.2.3 Clock Input

The clock of the basic timer is provided by the internal clock CK_INT.

16.2.4 Counter and Periphery

CK_PSC is input to the prescaler (PSC) for frequency division. The PSC is 16-bit, and the actual division factor corresponds to the value of R16_TIMx_PSC plus one. CK_PSC becomes CK_CNT after passing through the PSC. Changing the value of R16_TIM1_PSC does not take effect immediately but is updated to the PSC after an update event. Update events include clearing the UG bit and performing a reset. An Auto-Trigger Register (ATRLR) can be configured to set a counter threshold. When the CNT count value reaches the threshold stored in the ATRLR, the CNT is reset to zero.

16.3 Debug Mode

When the system enters debug mode, the timer continues to run or stops according to the setting of the DBG module.

16.4 Register Description

Table 16-1 TIM6-related registers

Name	Offset address	Description	Reset value
R16_TIM6_CTLR1	0x40001000	TIM6 control register 1	0x0000
R16_TIM6_CTLR2	0x40001004	TIM6 control register 2	0x0000
R16_TIM6_DMAINTENR	0x4000100C	TIM6 DMA/interrupt enable register	0x0000
R16_TIM6_INTFR	0x40001010	TIM6 interrupt status register	0x0000
R16_TIM6_SWEVGR	0x40001014	TIM6 event generation register	0x0000
R16_TIM6_CNT	0x40001024	TIM6 counter	0x0000
R16_TIM6_PSC	0x40001028	TIM6 count clock prescaler	0x0000
R16_TIM6_ATRLR	0x4000102C	TIM6 auto-reload value register	0xFFFF

Table 16-2 TIM7-related registers

Name	Offset address	Description	Reset value
R16_TIM7_CTLR1	0x40001400	TIM7 control register 1	0x0000
R16_TIM7_CTLR2	0x40001404	TIM7 control register 2	0x0000
R16_TIM7_DMAINTENR	0x4000140C	TIM7 DMA/interrupt enable register	0x0000

R16_TIM7_INTFR	0x40001410	TIM7 interrupt status register	0x0000
R16_TIM7_CNT	0x40001424	TIM7 counter	0x0000
R16_TIM7_PSC	0x40001428	TIM7 count clock prescaler	0x0000
R16_TIM7_ATRLR	0x4000142C	TIM7 auto-reload value register	0xFFFF

16.4.1 Control Register 1 (TIMx_CTLR1) (x=6/7)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	ARPE	RW	Auto-reload and preload enable: 1: Auto-reload value register (ATRLR) enabled; 0: Auto-reload value register (ATRLR) disabled.	0
[6:4]	Reserved	RO	Reserved	0
3	OPM	RW	One-Pulse Mode. 1: The counter stops when the next update event (clearing the CEN bit) occurs; 0: The counter does not stop when the next update event occurs.	0
2	URS	RW	Update Request Source: The software uses this bit to select the source of UEV events. 1: If update interrupts or DMA requests are enabled, only counter overflows/underflows generate update interrupts or DMA requests; 0: If update interrupts or DMA requests are enabled, any of the following events generate update interrupts or DMA requests: -Counter overflow/underflow. -Set the UG bit. -Updates generated from the mode controller.	0
1	UDIS	RW	Updates are prohibited, and the software allows/disables the generation of UEV events through this bit. 1: UEV is prohibited. No update events are generated, and the registers (ATRLR, PSC, CHCTLRx) hold their values. If the UG bit is set or a hardware reset is issued from the mode controller, the counter and prescaler are reinitialized. 0: UEV is allowed. The UEV event is generated by any of the following events: -Counter overflow / underflow. -Set the UG bit. -Updates generated from the mode controller. Registers with caches are loaded into their preloaded values.	0
0	CEN	RW	Counter enable:	0

			1: Enable counter. 0: Disables counters.	
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16.4.2 Control Register 2 (TIMx_CTLR2) (x=6/7)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved	0
[6:4]	MMS[2:0]	RW	Master mode selection: These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in the master mode. The possible combination is as follows: 000: Reset – The UG bit is used as a trigger output (TRGO). If it is a reset generated by a trigger input (the slave mode controller is in reset mode), the signal on TRGO will have a delay relative to the actual reset; 001: Enable-the counter enables signal CNT_EN to be used as a trigger output (TRGO). Sometimes, it is necessary to start multiple timers at the same time or control to enable slave timers within a period of time. The counter enable signal is generated by the logical OR of the CEN control bit and the trigger input signal in the gating mode. When the counter enable signal is controlled by the trigger input, there will be a delay on TRGO, unless the master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFGR register); 010: An update event is selected as the trigger output (TRGO). For example, the clock of a master timer can be used as a prescaler for a slave timer.	000b
[3:0]	Reserved	RO	Reserved.	0

16.4.3 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=6/7)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:9]	Reserved	RO	Reserved	0
8	UDE	RW	Update DMA request enable. 1: Update DMA request enabled; 0: Update DMA request disabled.	0
[7:1]	Reserved	RO	Reserved	0
0	UIE	RW	Update interrupt enable. 1: Update interrupt enabled; 0: Update interrupt disabled.	0

16.4.4 Interrupt Flag Register (R16_TIMx_INTFR) (x=6/7)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[15:1]	Reserved	RO	Reserved	0
0	UIF	RW0	Update interrupt flag. When an update event occurs, it is set by hardware and cleared by software. 1: Update interrupt generated; The update event generates in case of the following circumstances: For UDIS=0, when the repeated counter value overflows or underflows; For URS=0, UDIS=0, when the UG bit is set, or when the counter core is reinitialized by software; 0: No update interrupt generated.	0

16.4.5 Event Generation Register (TIMx_SWEVGR) (x=6/7)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:1]	Reserved	RO	Reserved.	0
0	UG	WO	Update event generation. This bit is set by software and cleared automatically by hardware. 1: Initialize the counter and generate an update event; 0: No action. <i>Note: The counter of the prescaler is also cleared, but the prescaler factor remains unchanged. In up-counting mode, the core counter will be cleared.</i>	0

16.4.6 Counter of General-purpose Register (TIMx_CNT) (x=6/7)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[15:0]	CNT[15:0]	RW	The real-time value of the counter of the timer.	0

16.4.7 Count Clock Prescaler (TIMx_PSC) (x=6/7)

Offset address: 0x28

Bit	Name	Access	Description	Reset value
[15:0]	PSC[15:0]	RW	The frequency division factor of the timer's prescaler; the clock frequency of the counter is equal to the input frequency of the frequency divider/(PSC+1).	0

16.4.8 Auto-reload Value Register (TIMx_ATRLR) (x=6/7)

Offset address: 0x2C

Bit	Name	Access	Description	Reset value
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[15:0]	ARR[15:0]	RW	Counter thresholds, when ATRLR is active and updated read Section 16.2.4; when ATRLR is empty, the counter stops.	0xFFFF
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Chapter 17 Low-power Timer (LPTIM)

LPTIM is two 16-bit upstream count timers. LPTIM has a variety of optional clock sources that allow LPTIM to operate in all power modes except standby mode. LPTIM can also run without an internal clock source, so LPTIM can be used as a "pulse counter". In addition, LPTIM can also wake up the system from low-power mode, so LPTIM is very suitable for implementing "timeout" at very low power consumption.

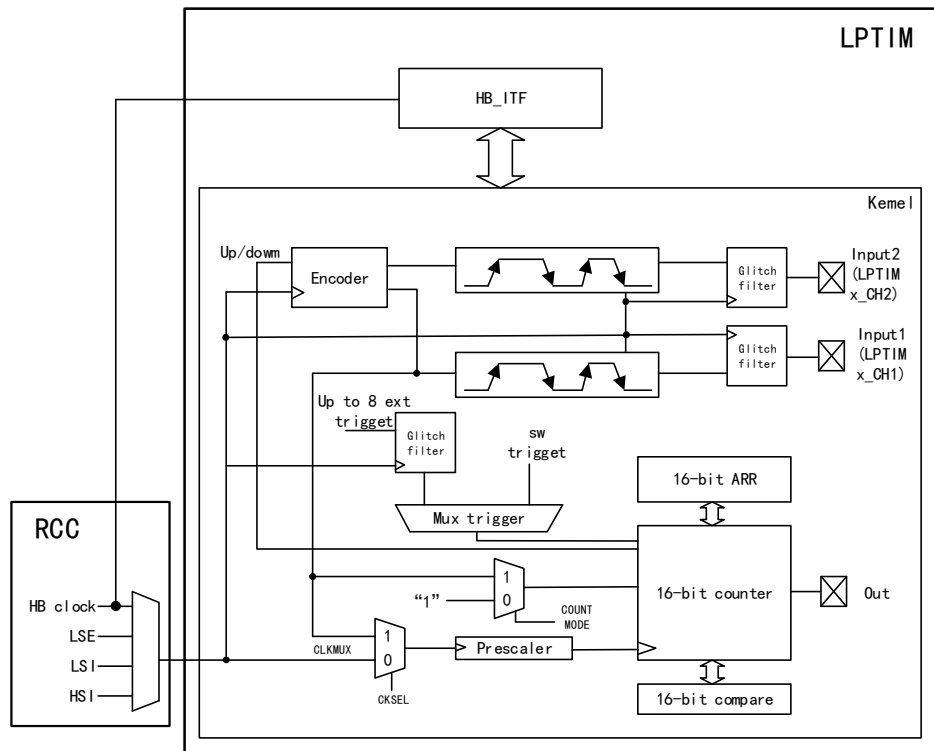
17.1 Main Features

- 16-bit upstream counter
- 3-bit prescaler, supporting 8 frequency division coefficients (1, 2, 4, 8, 16, 32, 64, 128)
- Optional clock source
 - Internal clock source: LSE, LSI, HSI or HB clock
 - External clock source: external clock on LPTIM input
- 16-bit ARR auto-reload register
- 16-bit comparison register
- Continuous/single trigger mode
- Optional software / hardware input trigger
- Programmable digital interference filter
- Configurable output PWM wave
- Configurable I/O polarity
- Encoder mode

17.2 Function Description

17.2.1 LPTIM Structure

Figure 17-1 Block diagram of LPTIM



17.3 LPTIM Trigger Mapping

The information for LPTIM external trigger connections is as follows:

Table 17-1 LPTIM external trigger connection

TRIGSEL[1:0]	External trigger
LPTIMx_TRG_00	LPTIMx_ETR
LPTIMx_TRG_01	RTC_ALARM

17.3.1 LPTIM Reset and Clock

The reset of the LPTIM module is controlled by the LPTIMRST bit of the RCC_HB1PRSTR register, which has no effect when setting 0 and resets the module when setting 1.

The LPTIM module clock is controlled by the LPTIMEN bit of the RCC_HB1PCENR register. The module clock is turned off at 0 and turned on at 1.

The counting clock of LPTIM is provided by multiple optional clock sources, which can be divided into internal clock sources and external clock sources.

When using the internal clock source count, the internal clock source can select the four HB, LSI, LSE, HIS clock sources through the CLKMX_SEL bit of the LPTIMx_CFGR register. In addition, the LPTIM can be timed using an external clock signal injected on the external input LPTIMx_CH1.

Table 17-2 LPTIM internal clock source

Register LPTIMx_CFGR[26:25]	Clock source
00	TIM_CLK (from HB_CLK)
01	HSI_CLK
10	LSE_CLK
11	LSI_CLK

When using external clock source counting, LPTIM may run in one of two configurations:

The first configuration is that the clock is provided to the LPTIM by the external signal, and the internal clock signal is provided by the configurable internal clock source (HB, LSI, LSE, HSI, etc.).

The second configuration is that the LPTIM only uses the external clock source through its external input channel 1, which realizes the pulse counter function or timeout function when all internal clock sources are turned off after entering the low power mode.

Programming the CKSEL and COUNTMODE bits controls whether the LPTIM clock uses an external clock source or an internal clock source. When configured to select an external clock source, the CKPOL bit is used to select the effective edge of the external clock signal. If the effective edge is set to the rising edge and the falling edge (double edge), an internal clock signal is also provided. In this case, the internal clock signal frequency should be four times higher than the external clock signal frequency.

17.3.2 Filter

LPTIM inputs, both external and internal, are protected by digital filters to prevent any burr and noise interference from propagating within the LPTIM, thereby avoiding accidental counting or triggering.

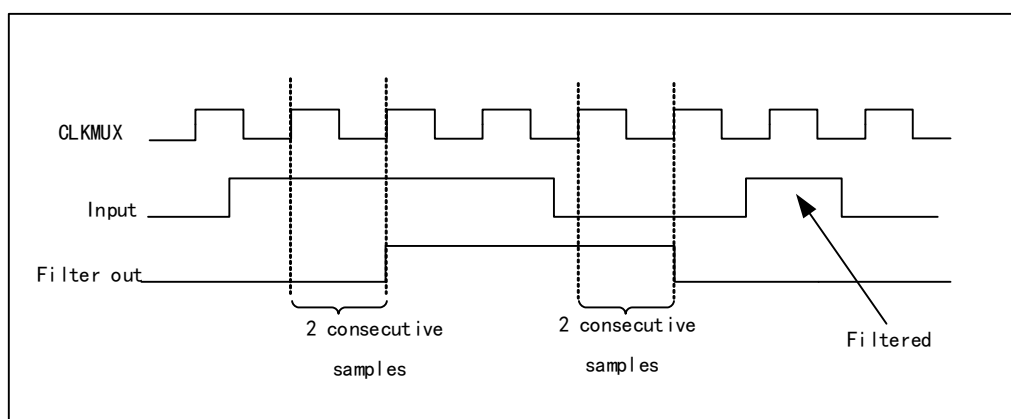
Before activating the digital filter, the internal clock source should be provided to the LPTIM to ensure the normal operation of the filter.

Digital filters are divided into 2 groups:

One group protects the LPTIM external input for the digital filter, and the sensitivity of the digital filter is controlled by the CKFLT bit; the other set of digital filter protects the internal trigger input of the LPTIM, and the sensitivity of the digital filter is controlled by the TRGFLT bit.

The sensitivity of the filter will affect the same number of continuous samples, and only when such continuous samples are detected on one of the LPTIM inputs can a signal level change be regarded as an effective switching. The following figure shows an example of interference filter behavior when programming two consecutive samples.

Figure 17-2 Timing block diagram of interference filter



Note: when the internal clock signal is not used, the digital filter must be disabled by zeroing the CKFLT and TRGFLT bits, and using an external analog filter to avoid interference caused by the external input of the LPTIM.

17.3.3 Prescaler

There should be a configurable 2^n prescaler in front of the LPTIM16 bit counter. The division ratio of the prescaler is controlled by the 3-bit field of PRESC[2:0]. All possible division ratios are listed in Table 17-3.

Table 17-3 Frequency division ratio of prescaler

PRESC[2:0]	Division ratio
000	1
001	1/2
010	1/4
011	1/8
100	1/16
101	1/32
110	1/64
111	1/128

17.3.4 Trigger Multiplexer

The LPTIM counter can be started in two ways, one is started by software, and the other is started after more than one valid edge in the trigger input is detected. The trigger mode of LPTIM is controlled by TRIGEN [1:0] and the trigger source is controlled by TRIGSEL[1:0] bit.

Table 17-4 Trigger mode

TRIGEN[1:0]	Trigger mode
00	Invalid
01	Rising edge
10	Falling edge
11	Dual edge

Table 17-5 Trigger source

TRIGSEL[1:0]	Trigger source
00	LPTIMx_ETR
01	RTC_ALARM
10	Invalid
11	Invalid

17.3.5 Operating Mode

LPTIM has 2 modes of operation:

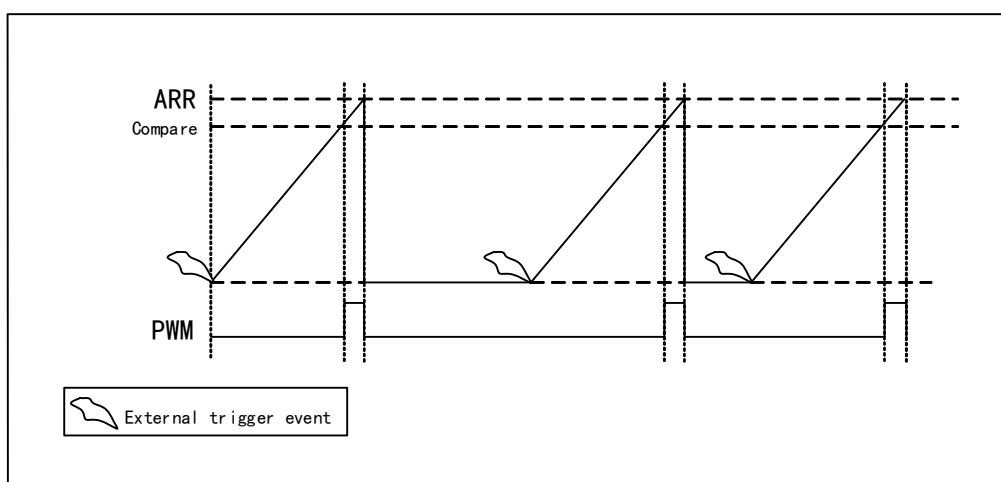
Continuous mode: the timer runs freely, starting from the trigger event and not stopping until the timer is disabled.

Single trigger mode: the timer starts from the trigger event and stops when the ARR value is reached.

In single trigger mode, to enable single counting, you must set SNGSTRT location 1, a new trigger event will restart the timer, and any trigger events that occur after the counter starts until it reaches ARR will be lost.

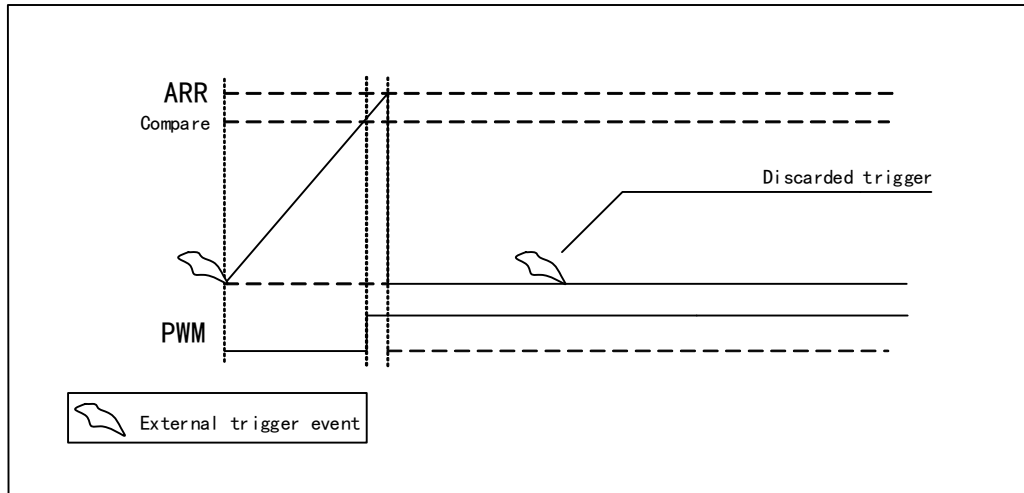
When an external trigger is selected, each external trigger event that arrives after the SNGSTRT bit is set and after the counter register stops starts the counter for a new count loop.

Figure 17-3 LPTIM output waveforms in single count mode configuration



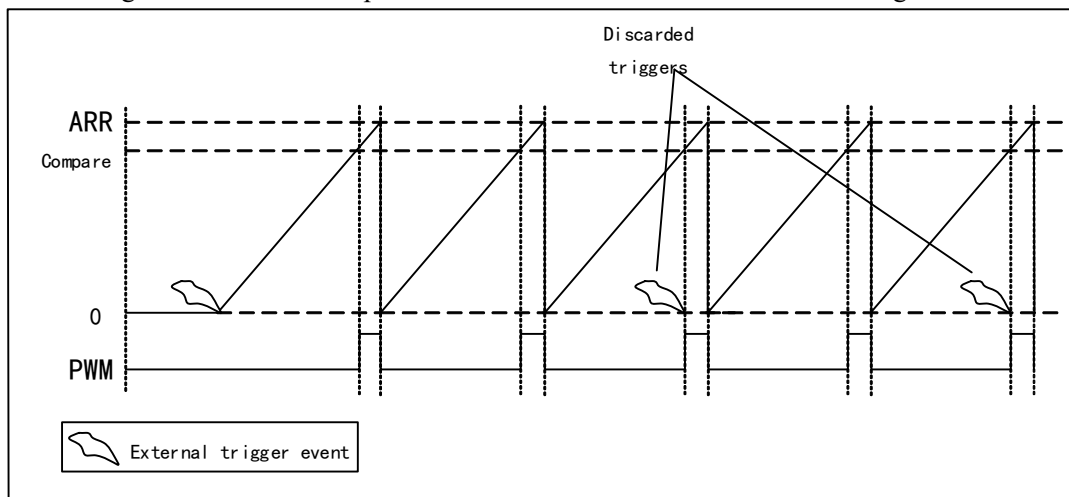
To activate one-trigger setup mode, it should be noted that in one-trigger mode, one-time setup mode is activated when the WAVE bit field in the LPTIMx_CFGR register is set. In this case, the counter is activated only once after the first trigger and any subsequent trigger events are discarded.

Figure 17-4 LPTIM output waveform for single count mode configuration with one setup mode activated



In continuous mode, to enable continuous counting, the CNTSTRT bit must be set, and if an external trigger is selected, external trigger events that arrive after setting CNTSTRT will start the counter for continuous counting. Any subsequent external trigger events will be discarded, and setting CNTSTRT in the case of software startup (TRIGEN=00) will start the counter to count continuously.

Figure 17-5 LPTIM output waveforms in continuous count mode configuration



The SNGSTRT and SNGSTRT bits can only be set when the timer is enabled (ENABLE bit is set to 1). You can change the LPTIM counter mode. If you previously selected continuous mode, setting SNGSTRT will switch LPTIM to single trigger mode, and the counter will stop counting when the ARR value is reached. If you have previously selected single trigger mode, setting CNTSTRT will switch CNTSTRT to continuous mode. Restart as soon as the counter reaches the ARR value.

17.3.6 Timeout Function

The detection of a valid edge on a selected trigger input can be used to reset the counter, which is controlled by the TIMOUT bit. The first trigger event will start the counter, any successive trigger event will reset the counter and the timer will restart, you can achieve low-power timeout function, if no trigger event occurs, the MCU will be awakened by the comparison match event.

17.3.7 Waveform Generation

Two 16-bit registers LPTIM_ARR and LPTIM_CMP are used to generate several different waveforms on the

LPTIM output.

The timer can generate the following waveforms:

- (1) PWM mode: once the counter value in LPTIMx_CNT exceeds the comparison value in LPTIMx_CMP, the LPTIM output is set. Once the LPTIMx_ARR and LPTIMx_CNT register values are equal, the LPTIM output is reset.
- (2) Mono-pulse mode: the output waveform is similar to the PWM mode of the first pulse, and then permanently reset.
- (3) one-time setting mode: the output waveform is similar to the mono-pulse mode, except that the output remains at the last signal level (depending on the polarity of the output configuration).

The above mode requires that the LPTIMx_ARR register value is strictly greater than the LPTIMx_CMP register value.

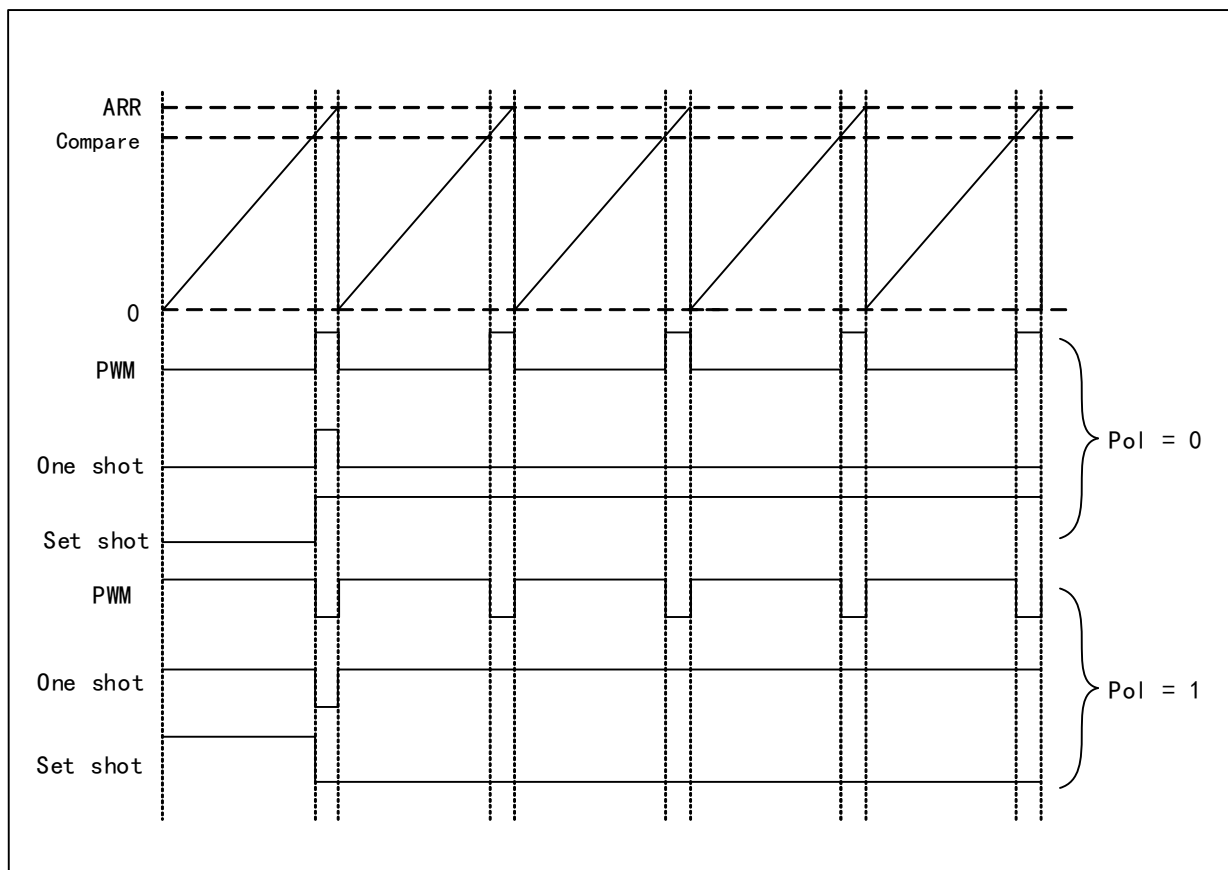
The LPTIM output waveform can be configured through the Wave bit as follows:

- (1) resetting the WAVE bit to 0 forces LPTIM to generate a PWM waveform or a mono-pulse waveform, depending on the bit set: CNTSTRT or SNGSTRT.
- (2) setting the WAVE bit to 1 forces LPTIM to generate a setting mode once.

The WAVPOL bit controls the polarity of the LPTIM output, and the change takes effect immediately, so the output default value changes immediately after the polarity is reconfigured, even before the timer is enabled.

The frequency of the generated signal is as high as the LPTIM clock frequency 2 division. Figure 17-6 shows three waveforms that may be generated on the LPTIM output. In addition, the figure shows the effect of changing polarity through the WAVPOL bit.

Figure 17-6 Waveform generation



17.3.8 Register Update

The LPTIM_ARR register and the LPTIMx_CMP register are updated immediately after the HB bus write operation, or if the timer has been started, at the end of the current cycle.

The PRELOAD bit controls how the LPTIMx_ARR and LPTIMx_CMP registers are updated:

When the PRELOAD bit is set to 0, the LPTIMx_ARR and LPTIMx_CMP registers are updated immediately after any write access.

When the PRELOAD bit is set to 1, the LPTIMx_ARR and LPTIMx_CMP registers are updated at the end of the current cycle if the timer has been started.

The LPTIM HB interface and LPTIM logic use different clocks, so there is some delay between HB writes and the values available to counter comparators, during which any additional writes to these registers must be avoided.

The ARROK flag and the CMPOK flag in the LPTIMx_ISR register indicate when to complete the writing operation to the LPTIMx_ARR register and the LPTIMx_CMP register, respectively.

After writing to the LPTIMx_ARR register or LPTIMx_CMP register, a new write operation to the same register can be performed only when the previous write operation is completed.

Any consecutive writes before setting the ARROK flag or the CMPOK flag will result in unpredictable results.

17.3.9 Counter Mode

LPTIM counters can be used to count external events on the LPTIM_CH1 as well as internal clock cycles. The CKSEL and count bits control which source will be used to update the counter.

If the LPTIM is configured to count external events on the LPTIM_CH1, the count can be updated after the rising, falling, or double edges based on the values written to the CKPOL [1:0] bit.

Depending on the CKSEL and COUNTMODE values, you can choose the following counting modes:

(1) CKSEL=0: LPTIM clocks are counted by internal clock sources.

The COUNTMODE=0, LPTIM is configured to count the internal clock source, and the LPTIM counter is configured to update after each internal clock pulse.

COUNTMODE=1, sampling the data on the LPTIM_CH1 using the internal clock provided to the LPTIM. Therefore, in order not to miss any events, the frequency of the signal on the LPTIM_CH1 should not exceed the frequency of the internal clock of the LPTIM.

(2) when CKSEL=1: LPTIM is counted by an external clock source, the value of COUNTMODE has no effect.

In this configuration, the LPTIM does not require an internal clock source (unless a digital filter is enabled), and the signal injected on the LPTIM_CH1 is used as the system clock of the LPTIM. This configuration applies to operating modes where the built-in oscillator is not enabled.

For this configuration, the LPTIM counter can be updated at the rising or falling edges of the LPTIM_CH1 clock signal, but not at the double edges. Because the signal injected by LPTIM_CH1 is also used for LPTIM counting, there is some initial delay before the counter is incremented (after LPTIM is enabled). To be exact, after enabling the LPTIM, the first five valid edges of the LPTIM external LPTIM_CH1 will be lost.

17.3.10 Timer Enabled

The ENABLE bit in the LPTIMx_CR register is used to enable / disable LPTIM logic, and after setting the ENABLE bit, a delay of two counter clocks is required before LPTIM is actually enabled.

17.3.11 Encoder Mode

This mode allows processing of signals from a quadrature encoder used to detect the position of the rotating element. The encoder interface mode is only used as an external clock with direction selection. This means that the counter counts only continuously between 0 and the automatic overload value in the LPTIMx_ARR register. Therefore, LPTIMx_ARR. Net must be configured before startup. A clock signal is generated based on two external input signals LPTIMx_CH1 and LPTIMx_CH2 to time the LPTIM counter. The phase between the two signals determines the counting direction.

Encoder mode is available only if the LPTIM is counted by an internal clock source. The signal frequency on the LPTIMx_CH1 and LPTIMx_CH2 must not exceed the frequency of the LPTIM internal clock by 4 divisions. The above conditions are met to ensure the normal operation of LPTIM.

The direction change is signaled by 2 up and down flags in the LPTIMx_ISR register. In addition, if enabled through the downside bit, interrupts can be generated for events that change in both directions.

To activate encoder mode, the ENC bit must be set to 1, and LPTIM must first be configured in continuous mode.

When the encoder mode is active, the LPTIM counter is automatically modified according to the speed and direction of the incremental encoder. Therefore, its content always represents the location of the encoder. The counting direction is indicated by the up and down marks, which corresponds to the rotation direction of the encoder rotor.

According to the edge sensitivity of CKPOL[1:0] bit configuration, the following possible combination schemes are obtained, in which LPTIM_CH1 and LPTIM_CH2 do not switch at the same time.

Table 17-6 Configuration of edge sensitivity combination scheme

Active edge (CKPOL[1:0])	The level of the opposite signal (The input for LPTIMx_CH1 is LPTIMx_CH2, and the input for LPTIMx_CH2 is LPTIMx_CH1.)	LPTIMx_CH1		LPTIMx_CH2	
		Rising	Falling	Rising	Falling
Rising edge (00)	High	decrement	non-counting	incremental	non-counting
	Low	incremental	non-counting	decrement	non-counting
Falling edge (01)	High	non-counting	incremental	non-counting	decrement
	Low	non-counting	decrement	non-counting	incremental
Dual edge (10)	High	decrement	incremental	incremental	decrement
	Low	incremental	decrement	decrement	Incremental
11	Invalid				

17.3.12 Debug Mode

When MCU enters modal mode (core stops), LPTIM continues to work properly.

17.3.13 LPTIM Low-power Mode

Table 17-7 Effect of low Power Mode on LPTIM

Mode	Description
Sleep mode	No effect, LPTIM interruption will cause the device to exit sleep mode
Stop mode	The LPTIM peripheral is active when it is timed by LSE or LSI, and the LPTIM interrupt causes the device to exit stop mode

17.4 LPTIM Interrupt

An interrupt / wake-up occurs if the following events are enabled through the LPTIM_IER register:

- Compare match.
- Automatic overload matching.
- External trigger event.
- Automatic overload register write complete.
- Comparison register write complete.
- Direction change (encoder mode)

Table 17-8 Interrupt event

Interrupt event	Description
Compare match	An interrupt flag is generated when the value of the counter register LPTIMx_CNT is equal to the value of the comparison register LPTIM_CMP
Automatic overload match	An interrupt flag occurs when the value of the counter register LPTIMx_CNT is equal to that of the automatic reload register LPTIMx_ARR
External trigger event	An interrupt flag is generated when an externally triggered event is detected
Automatic overload register write complete	An interrupt flag is generated when the write operation to the LPTIMx_ARR register is completed.
Comparison register write complete	An interrupt flag is generated when the write operation to the LPTIMx_CMP register is completed.
Change of counting direction	When used in encoder mode, two interrupt flags are embedded in the direction of the signal: Up sign: indicates a change in the direction of the upward count. Downward sign: indicates a change in the downward counting direction.

17.5 Register Description

Table 17-9 LPTIM1-related registers

Name	Access	Description	Reset value
------	--------	-------------	-------------

R32_LPTIM1_ISR	0x40002400	LPTIM1 interrupt status register	0x00000000
R32_LPTIM1_ICR	0x40002404	LPTIM1 interrupt clear register	0x00000000
R32_LPTIM1_IER	0x40002408	LPTIM1 interrupt enable register	0x00000000
R32_LPTIM1_CFGR	0x4000240C	LPTIM1 configuration register	0x00000000
R32_LPTIM1_CR	0x40002410	LPTIM1 control register	0x00000000
R16_LPTIM1_CMR	0x40002414	LPTIM1 compare register	0x00000000
R16_LPTIM1_ARR	0x40002418	LPTIM1 auto-reload register	0x00000001
R16_LPTIM1_CNT	0x4000241C	LPTIM1 counter register	0x00000000

Table 17-10 LPTIM2-related registers

Name	Access	Description	Reset value
R32_LPTIM2_ISR	0x40003400	LPTIM2 interrupt status register	0x00000000
R32_LPTIM2_ICR	0x40003404	LPTIM2 interrupt clear register	0x00000000
R32_LPTIM2_IER	0x40003408	LPTIM2 interrupt enable register	0x00000000
R32_LPTIM2_CFGR	0x4000340C	LPTIM2 configuration register	0x00000000
R32_LPTIM2_CR	0x40003410	LPTIM2 control register	0x00000000
R16_LPTIM2_CMR	0x40003414	LPTIM2 compare register	0x00000000
R16_LPTIM2_ARR	0x40003418	LPTIM2 auto-reload register	0x00000001
R16_LPTIM2_CNT	0x4000341C	LPTIM2 counter register	0x00000000

17.5.1 Interrupt Status Register (LPTIMx_ISR) (x=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DIR _S _Y N C	DO W N	UP	AR R O K	CM P O K	EX T T R I G	AR M	CM P M

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
7	DIR_SYNC	RO	Counter counting direction in encoder mode. 0: Count down. 1: Count up. This bit is only valid in encoder mode, that is, ENC position 1, and the DIR_EXTEN needs to be set to 1 to enable counting direction before use.	0
6	DOWN	RO	Count down. 1: Counting direction from top to bottom. 0: Invalid. In encoder mode, this bit is set by hardware to notify the application of a change in counter direction from top to	0

			bottom, which can be cleared by writing 1 to the LPTIMx_ICR[6] bit.	
5	UP	RO	Count up. 1: Counting direction from bottom to top. 0: Invalid. In encoder mode, this bit is set by hardware to notify the application of a change in counter direction from bottom to top, which can be cleared by writing 1 to the LPTIMx_ICR[5] bit.	0
4	ARROK	RO	Auto-reload register data updated successfully. 1: Data updated successfully. 0: Invalid. This bit is set by the hardware to inform the application that the write operation of the HB bus to the LPTIMx_ARR has completed successfully and can be cleared by writing 1 to the LPTIMx_ICR[4] bit.	0
3	CMPOK	RO	Compare register data updated successfully. 1: Data updated successfully. 0: Invalid. This bit is set by the hardware to inform the application that the write operation of the HB bus to the LPTIMx_CMR has completed successfully and can be cleared by writing 1 to the LPTIMx_ICR[3] bit.	0
2	EXTTRIG	RO	External trigger edge event. 1: Valid edge input has occurred. 0: Invalid. This bit is set by hardware to notify the application that valid edge input has occurred on the selected external trigger, and this flag is not set if the trigger is ignored because the timer has been started. 1 can be cleared by writing 1 to the LPTIMx_ICR[2] bit.	0
1	ARRM	RO	Automatic reload register data matching with LPTIMx_CNT register data. 1: Match successfully. 0: Invalid. This bit is set by the hardware to inform the application that the value of the LPTIMx_CNT register reaches the value of the LPTIMx_ARR register, which can be cleared by writing 1 to the LPTIMx_ICR[1] bit.	0
0	CMPM	RO	The data of the comparison register matches the data of the LPTIMx_CNT register. 1: Match successfully. 0: Invalid. This bit is set by the hardware to inform the application that	0

			the value of the LPTIMx_CNT register reaches the value of the LPTIMx_CMR register, which can be cleared by writing 1 to the LPTIMx_ICR[0] bit.	
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17.5.2 Interrupt Clear Register (LPTIMx_ICR) (x=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									DOWNCF	UPNCF	ARROKCF	CMPOKCF	EXTTRIGNCF	ARRMNCF	CMPMNCF

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
6	DOWNCF	W1	Clear downward flag bit. <i>Note: write 0 is invalid, write 1 clear 0 status register corresponding bit.</i>	x
5	UPCF	W1	Clear up flag bit. <i>Note: Write 0 is invalid, write 1 clear 0 status register corresponding bit.</i>	x
4	ARROKCF	W1	Clear auto-reload data update flag bit. <i>Note: Write 0 is invalid, write 1 clear 0 status register corresponding bit.</i>	x
3	CMPOKCF	W1	Clear comparator data update flag bit. <i>Note: Write 0 is invalid, write 1 clear 0 status register corresponding bit.</i>	x
2	EXTTRIGNCF	W1	Clear external trigger edge event flag bit. <i>Note: write 0 is invalid, write 1 clear 0 status register corresponding bit.</i>	x
1	ARRMCF	W1	Clear auto-reload register matching flag bit. <i>Note: write 0 is invalid, write 1 clear 0 status register corresponding bit.</i>	x
0	CMPMCF	W1	Clear the comparison register matching flag bit. <i>Note: write 0 is invalid, write 1 clear 0 status register corresponding bit.</i>	x

17.5.3 Interrupt Enable Register (LPTIMx_IER) (x=1/2)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	DOWNIE	UPIE	ARROKIE	CMPOKIE	EXTTRIGIE	ARRMIE	CMPMIE
----------	--------	------	---------	---------	-----------	--------	--------

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
6	DOWNIE	RW	Downwards interrupt enable: 0: Off 1: On	0
5	UPIE	RW	Upwards interrupt enable: 0: Off 1: On	0
4	ARROKIE	RW	Auto-reload register data update successfully interrupt enable. 0: Off 1: On	0
3	CMPOKIE	RW	Compare register data update successfully interrupt enable 0: Off 1: On	0
2	EXTTRIGIE	RW	External trigger edge event input interrupt enable 0: Off 1: On	0
1	ARRMIE	RW	Auto-reload register data matching successfully interrupt enable 0: Off 1: On	0
0	CMPMIE	RW	Compare register data matching successfully interrupt enable 0: Off 1: On	0

17.5.4 Configuration Register (LPTIMx_CFGR) (x=1/2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				FORCE_PWM	CLKMX_SEL	ENC	COUNTMODE	PRELOAD	WAVPOL	WAVE	TIMOUT	TRIGEN		Reserved	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TRIGSEL		Reserved	PRESC			Reserved	TRGFLT		Reserved	CKFLT		CKPOL		CKSEL

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
27	FORCE_PWM	RW	Force PWM output 1: Force PWM output high level;	0

			0: Invalid.	
[26:25]	CLKMX_SEL[1:0]	RW	LPTIM counter internal clock source selection 00: TIM_CLK (from HB_CLK) 01: HSI_CLK 10: LSE_CLK 11: LSI_CLK	0
24	ENC	RW	Encoder mode 1: On; 0: Off.	0
23	COUNTMODE	RW	Counter mode, select which clock source LPTIM uses to time the counter. 1: When each effective pulse is input outside the LPTIM, the counter increments; 0: The counter increments after each internal clock pulse.	0
22	PRELOAD	RW	Register update mode, control LPTIMx_ARR and LPTIMx_CMP register update mode. 1: The register is updated at the end of the current LPTIM cycle; 0: Update the register after each HB bus write access.	0
21	WAVPOL	RW	Polarity of PWM waveform. 1: The output reflects the inversion of the comparison between the LPTIMx_ARR and LPTIMx_CMP registers; 0: The output reflects the comparison between LPTIMx_ARR and LPTIMx_CMP registers. <i>Note: if the counter value is greater than the comparator value, the comparison result is 1, otherwise it is 0.</i>	0
20	WAVE	RW	PWM waveform. 1: Set the primary mode (one pulse waveform); 0: Turn off once mode.	0
19	TIMOUT	RW	Timeout is enabled to control the timeout function. 0: Trigger events that arrive when the timer has been started are ignored. 1: Trigger events that arrive when the timer has been started will reset and restart the counter.	0
[18:17]	TRIGEN[1:0]	RW	Trigger enable and polarity control whether the LPTIM counter is started by an external trigger, and if the external trigger option is selected, there are three configurations for the trigger active edge. 00: Software trigger (counter start by software). 01: Rising edge trigger. 10: Falling edge trigger. 11: Double edge trigger.	0
[16:15]	Reserved	RO	Reserved	0
[14:13]	TRIGSEL[1:0]	RW	Trigger source selection:	0

			00: LPTIM_ETR; 01: RTC_ALARM; 10: Invalid; 11: Invalid.	
12	Reserved	RO	Reserved	0
[11:9]	PRESC	RW	Clock prescaler, configured with prescaler 000: Divided by 1; 001: Divided by 2; 010: Divided by 4; 011: Divided by 8; 100: Divided by 16; 101: Divided by 32; 110: Divided by 64; 111: Divided by 128.	0
8	Reserved	RO	Reserved	0
[7:6]	TRGFLT[1:0]	RW	Configurable digital filter for flip-flop. The TRGFLT value sets the number of consecutive equal samples that should be detected when there is a change on the internal trigger, which is then treated as a valid level conversion. You must have an internal clock source to use this feature. 00: Any change to a trigger is considered a valid trigger. 01: Trigger activation level change must be stable for at least 2 clock cycles before it is considered a valid trigger. 10: Trigger activation level change must be stable for at least 4 clock cycles before it is considered a valid trigger. 11: Trigger activation level change must be stable for at least 8 clock cycles before it is considered a valid trigger.	0
5	Reserved	RO	Reserved	0
[4:3]	CKFLT[1:0]	RW	Configurable digital filter for external clock. The CKFLT value is used to set the number of consecutive equal samples. Such continuous samples should be detected when there is a level change in the external clock signal before the level change is regarded as an effective level conversion. You must have an internal clock source to use this feature. 00: Any change in the level of the external clock signal is considered an effective conversion. 01: External clock signal level changes must be stable for at least 2 clock cycles before they are considered valid. 10: External clock signal level changes must be stable for at least 4 clock cycles before they are considered valid. 11: External clock signal level changes must be stable for at least 8 clock cycles before they are considered valid.	0

[2:1]	CKPOL[1:0]	RW	If LPTIM chooses an external clock source count, the CKPOL bit is used to configure the valid edge: 00: Rising is still used in counting. 01: Falling is still used in counting. 10: Bilateral is still used for counting. When both sides of the external clock signal are valid, the LPTIM must also be timed by an internal clock source whose frequency is at least four times the external clock frequency. 11: Invalid. If LPTIM is configured in encoder mode (ENC position 1), the CKPOL bit is used to select the encoder mode: 00: Encoder mode 1 activation. 01: Encoder mode 2 activation. 10: Encoder mode 3 activated. 11: Invalid.	0
0	CKSEL	RW	Clock selector for selecting the clock used by LPTIM. 1: LPTIM uses an external clock; 0: LPTIM uses an internal clock.	0

17.5.5 Control Register (LPTIMx_CR) (x=1/2)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											DIR EXT EN	OUT EN	CNT STRT	SNG STRT	ENA BLE

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
4	DIR_EXTEN	RW	External trigger counting direction enable. 1: On; 0: Off. <i>Note: Use when encoder mode is turned on.</i>	0
3	OUTEN	RW	PWM wave output enable. 0: Off. 1: On. <i>Note: In non-encoder mode, the bit enable output PWM wave requires ENABLE to be enabled at the same time to be effective.</i>	0
2	CNTSTRT	RW	Start in continuous mode. This bit is set by the software and cleared by the hardware. In the case of software startup (TRIGEN==00), set this bit to start the LPTIM in continuous mode, and if the software	0

			startup is disabled (TRIGEN!=00), setting this bit will start the LPTIM in continuous mode immediately after an external trigger is detected. If this bit is set when LPTIM is in mono-pulse counting mode, LPTIM will not stop counting when the counter reaches the ARR value. <i>Note: you can only write when you ENABLE=1.</i>	
1	SNGSTRT	RW	LPTIM starts in single trigger mode. This bit is set by the software and cleared by the hardware. in the case of software startup (TRIGEN==00), set this bit to start the LPTIM in mono-pulse mode, and if the software startup is disabled (TRIGEN!=00), setting this bit will start the LPTIM in mono-pulse mode immediately after an external trigger is detected. If this bit is set when LPTIM is in continuous count mode, LPTIM will stop when the counter reaches the ARR value. <i>Note: write only when you ENABLE=1.</i>	0
0	ENABLE	RW	The timer enables, when the timer enable bit is cleared 0, the internal logic is reset and the LPT_CR[2:1] bit cannot be operated. 1: Low-power timer is turned on; 0: Low-power timer disables and resets internal logic.	0

17.5.6 Compare Register (LPTIMx_CMP) (x=1/2)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:0]	CMP	RW	Comparison value, when the value of the counter is equal to the value in the comparison register, the timer comparison flag will be set to 1, and if the corresponding enable signal is turned on before setting 1, it will produce an interrupt source and a low power wake-up signal.	0

17.5.7 Auto-reload Register (LPTIMx_ARR) (x=1/2)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[15:0]	ARR	RW	The count register reloaded count value. When counting up, if the count value is equal to the reloaded count value, the counter starts counting from 0; when counting down, if the counter value equals 0, the count starts from the reloaded count value.	1

17.5.8 Counter Register (LPTIMx_CNT) (x=1/2)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
-----	------	--------	-------------	-------------

[15:0]	COUNT	RO	Count value of the current timer counter register. When LPTIM is running with an asynchronous clock, reads to the LPTIMx_CNT register may return unreliable values; therefore, in this case, it is necessary to perform two consecutive read accesses and verify that the two returned values are the same. It should be noted that for reliable LPTIMx_CNT register read accesses, two consecutive reads must perform the access and be compared.	0
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Chapter 18 Digital-to-Analog Converter (DAC)

The digital-to-analog converter (DAC) contains 2 converters that convert 8/12-bit digital input to 2-channel analog voltage output. The DAC module has built-in triangular-wave generator and noise-wave generator, and it supports multiple event trigger conversions and supports DMA.

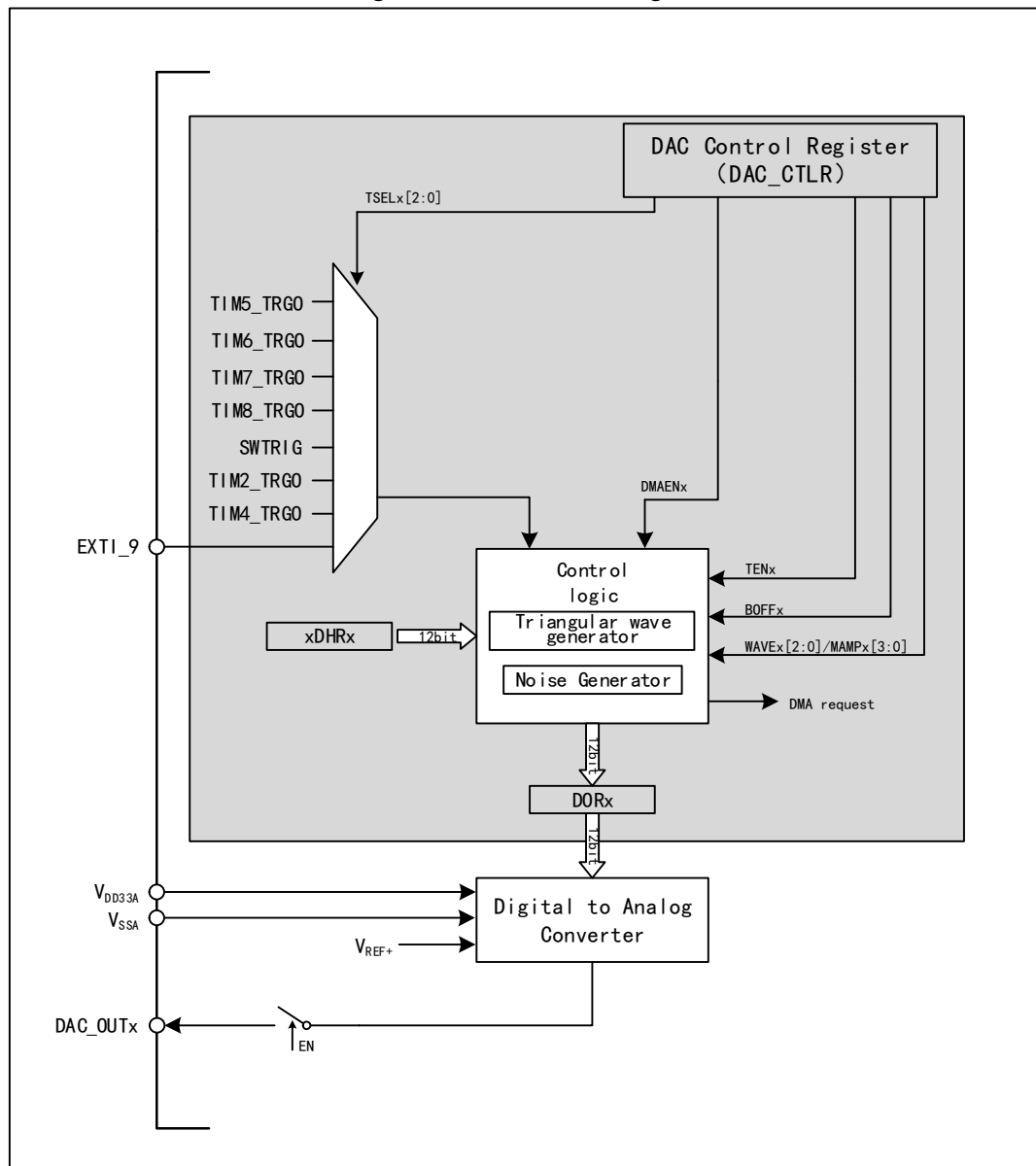
18.1 Main Features

- 2 DAC converters, and each has single output channel
- Triangular-wave generator and noise-wave generator
- Configurable 8-bit or 12-bit output
- Left or right data alignment in 12-bit mode
- Dual DAC channel for independent or simultaneous conversions
- DMA capability
- Multiple trigger events

18.2 Functional Description

18.2.1 DAC Structure

Figure 18-1 DAC block diagram



18.2.2 DAC Channel Configuration

18.2.2.1 Enable DAC:

Set the **ENx** bit in the **DAC_CTLR** register to 1, to enable the analog power to DAC channel x. After a period of start-up time, DAC channel x can be enabled. The DAC contains 2 analog output channels which can output independently or simultaneously.

Note: In order to avoid parasitic interference and additional power consumption, the corresponding pins of the DAC channel need to be set to analog input (AIN) mode in advance.

18.2.2.2 Enable Output Buffer:

DAC integrates output buffers, which can be used to reduce the output impedance and increase the drive capacity to directly drive the external loads. Each DAC channel output buffer can be enabled or disabled by setting the

BOFFx bit in the DAC_CTLR register.

18.2.2.3 Data Format:

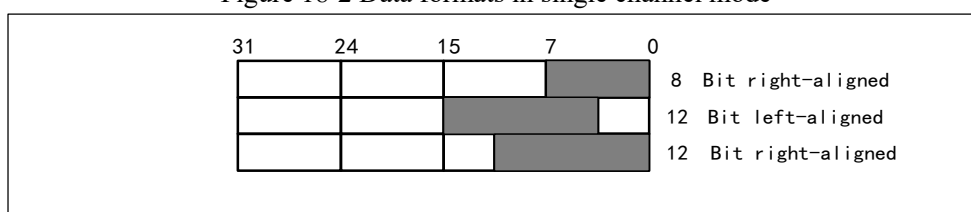
In single DAC channel mode, data formats include 8-bit right alignment, 12-bit left alignment and 12-bit right alignment.

When 8-bit data is right-aligned, write data to DAC_R8BDHRx[7:0], and the module will load (after 1 HB clock cycle) its left-shifted data to the data output register DAC_DORx[11:4].

When 12-bit data is right-aligned, write data to DAC_R12BDHRx[11:0], and the module will load (after 1 HB clock cycle) right-aligned data into the data output register DAC_DORx[11:0].

When 12 data are left-aligned, write the data to DAC_L12BDHRx[15:4]. After the corresponding shift, the module will load (after 1 HB clock cycle) the left-aligned data to the data output register DAC_DORx[11:0].

Figure 18-2 Data formats in single channel mode



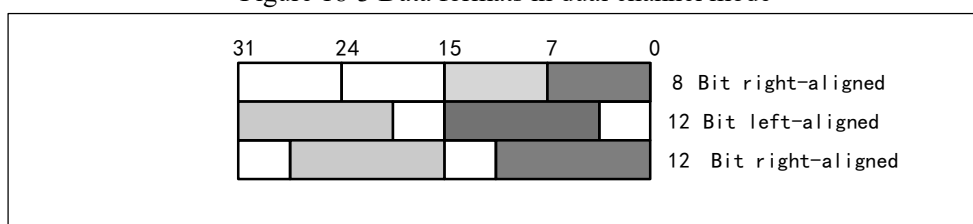
In dual DAC channel mode, there are also three methods: 8-bit data right alignment, 12-bit data left alignment, and 12-bit data right alignment.

For 8-bit data right-aligned, write data to DAC_RD8BDHR[7:0], the module will load (after 1 HB clock cycle) bits [7:0] shifted to DAC_DOR1[11:4] and bits [15:8] shifted to DAC_DOR2[11:4].

For 12 data left-aligned, write data to DAC_LD12BDHR[31:0], the module will load (after 1 HB clock cycle) bit[15:4] data shifted to DAC_DOR1[11:0] and bit[31:20] data shifted to DAC_DOR2[11:0].

12-bit data right-aligned, write data to DAC_RD12BDHR[31:0], the module will load (after 1 HB clock cycle) bit[11:0] data to DAC_DOR1[11:0] and bit[27:16] data to DAC_DOR2[11:0].

Figure 18-3 Data formats in dual channel mode



18.2.2.4 DMA Capability:

DAC channel has the DMA capability. Set the DMAENx bit in the DAC_CTLR register to 1, to enable DMA of the corresponding channel. When a trigger event occurs (Software trigger not included), a DMA request is generated. And then the data in the DAC_DORx register can be updated.

18.2.2.5 Trigger Event Selection:

DAC conversion can be triggered by the following events. When the TENx bit in the DAC_CTLR register is set to 1, a trigger event to trigger DAC conversion can be selected by configuring the TSELx[2:0] bits.

Table 18-1 Trigger events

Trigger source	Type	TSELx[2:0]
----------------	------	------------

Timer 6 TRGO event	Internal signal from on-chip timer	000
Timer 8 TRGO event		001
Timer 7 TRGO event		010
Timer 5 TRGO event		011
Timer 2 TRGO event		100
Timer 4 TRGO event		101
EXTI line9	External pin	110
SWTRIG (Software trigger)	Software control bit	111

Every time a DAC interface detects a rising edge on the selected timer TRGO output or on the selected external interrupt line 9, the DAC_DORx register is updated 3 HB clock cycles after the trigger occurs.

If the software trigger mode is configured, once the SWTRIG bit is set to 1, a conversion is started. The DAC_DORx register is updated single HB clock cycle after the trigger occurs, and the SWTRIG bit can be automatically cleared by hardware.

Note: The TSELx[2:0] bits cannot be changed when ENx is 1.

18.2.3 DAC Conversion

The data of the DAC channel comes from the DAC_DORx register, but data cannot be directly written to the register DAC_DORx. Any data output to the DAC channel x must be written into the DAC_R12BDHR1, DAC_L12BDHR1, DAC_R12BDHR2, DAC_L12BDHR2, DAC_RD12BDHR, DAC_LD12BDHR and DAC_RD8BDHR registers. The internal DAC_DHRx register obtains the value of above registers and transfers it to the DAC_DORx register after the corresponding time.

In the non-trigger mode, the data written into the DAC_xDHRx register is shifted into the DAC_DORx register in a HB clock cycle.

In software trigger mode, the DAC_DORx register is automatically updated in a HB clock cycle after the rising edge of the event trigger.

In hardware trigger mode (Timer TRGO event or external interrupt line 9 rising edge), the DAC_DORx register is automatically updated in 3 HB clock cycles after the trigger event.

Load the DAC_DORx register data. After the time of $t_{SETTLING}$, the output can be valid, and the length of this period of time will vary depending on the supply voltage and the analog output load.

The digital input is linearly converted to analog voltage output by the DAC, and it ranges from 0 to V_{DDA} . The output voltage on any DAC channel pin shall meet the following relationship:

DAC output voltage = $V_{DDA} * (DAC_DORx / 4096)$.

18.2.4 DAC Triangular-wave Generator

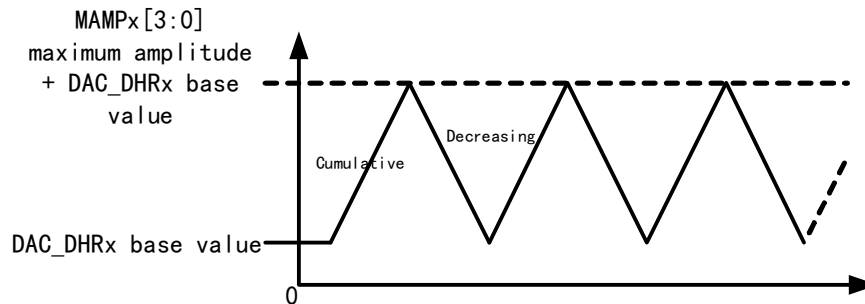
The module has a built-in triangular-wave generator, which can add a small amplitude triangle-wave to the reference signal. Set WAVEx[1:0] bit as '10b' and select the triangular-wave generation function of DAC. Set MAMPx[3:0] bit in the DAC_CTLR register to select the amplitude of triangular-wave.

The system contains a triangular-wave counter starting from 0, which accumulates by 1 in 3 HB clock cycles after each trigger event. The value of the counter is added to the value of the DAC_DHRx register and the overflow bit is discarded and then written to the DAC_DORx register. When the value transmitted into the DAC_DORx register is smaller than the maximum amplitude defined by the MAMPx[3:0] bits, the triangular-wave counter will gradually accumulate. Once it reaches the set maximum amplitude, the counter will begin to decrease progressively, and then start to accumulate after reaching 0. Repeat this cycle. Set WAVEx[1:0] bits to '00' to reset the generation of triangle-waves.

Note: 1. To generate a triangular-wave, DAC trigger must be enabled, i.e., setting the $TENx$ bit in the DAC_CTLR register to 1.

2. $MAMPx[3:0]$ bits must be set before enable the DAC. Otherwise, its value cannot be modified.

Figure 18-4 Triangular wave generation



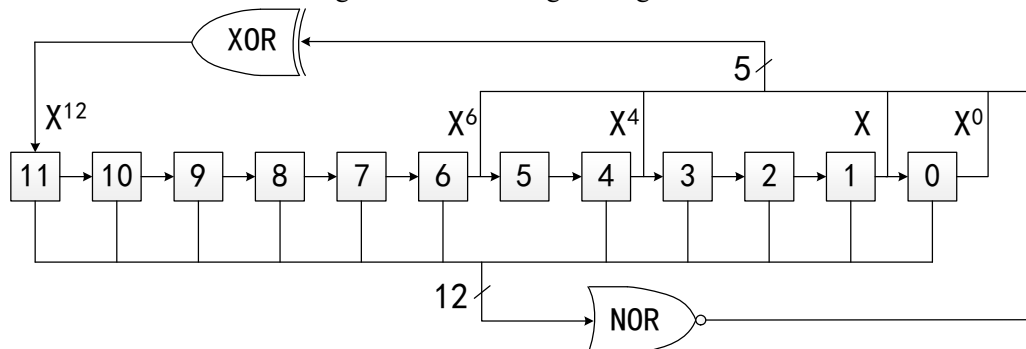
18.2.5 DAC Noise-wave Generator

The module has a built-in noise-generator that uses the Linear Feedback Shift Register (LFSR) to generate pseudo noise with varying amplitude. Set $WAVE[1:0]$ bits to '01b' to select the DAC noise generation function. Set the $MAMPx[3:0]$ bits in the DAC_CTLR register to select the data of the masked part of the LFSR.

The preload value of the register LFSR is 0xAAA. According to a specific algorithm, the value of this register is updated in 3 HB clock cycles after each trigger event. Setting the $MAMPx[3:0]$ bits in the DAC_CR register can mask part or all of the LFSR data, so that the LFSR value obtained is added to the value of DAC_DHRx , and the overflow bit is removed and then written into the DAC_DORx register. If the register LFSR value is 0x000, it will inject '1' (Anti-lock mechanism). Set $WAVEx[1:0]$ bit to '00b' to reset the generation algorithm of LFSR waveform.

Note: To generate a noise-wave, DAC trigger must be enabled, i.e., setting the $TENx$ bit in the DAC_CTLR register to 1.

Figure 18-5 LFSR register algorithm



18.3 Dual DAC Conversion

When the 2 DAC channels are required at the same time, the module integrates 3 data registers in dual DAC mode ($DAC_RD8BDHR$, $DAC_LD12BDHR$, $DAC_RD12BDHR$), in order to efficiently operate the DAC module. The conversion value of the 2 DACs can be updated when only one of the 3 registers is operated.

For dual DAC conversion, other registers of the module can be used to implement 11 types of conversion modes with different combinations. The value of data to be converted in the 2 channels needs to be written to one of the 3 data registers.

18.3.1 Independent Trigger with the Same LFSR Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b01. MAMPx is set as the same LFSR mask value. When a channel1 trigger event occurs, add the counter (LFSR1) with the same mask value to the channel1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 after delay of 3 HB clocks, for conversion. Then the LFSR1 counter is updated. When a channel2 trigger event occurs, add the counter (LFSR2) with the same mask value to the channel2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 HB clocks. Then the LFSR2 counter is updated.

18.3.2 Independent Trigger with Different LFSR Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b01. MAMPx can be set as different mask values of LFSR. When a channel1 trigger event occurs, add the LFSR1 counter, with the mask value configured by the MAMP1[3:0] bits, to the channel1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 for conversion after delay of 3 HB clocks. Then the LFSR1 counter is updated. When a channel2 trigger event occurs, add the LFSR2 counter, with the mask value configured by the MAMP2[3:0] bits, to the channel2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 HB clocks. Then the LFSR2 counter is updated.

18.3.3 Independent Trigger with the Same Triangle Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b1x. MAMPx is set as the same triangle amplitude value. When a channel1 trigger event occurs, add the channel1 triangle counter, with the same triangle amplitude, to the channel1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 for conversion after delay of 3 HB clocks. Then the channel1 triangle counter is updated. When a channel2 trigger event occurs, add the channel2 triangle counter, with the same triangle amplitude, to the channel2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 HB clocks. Then the channel2 triangle counter is updated.

18.3.4 Independent Trigger with Different Triangle Generation

Set the TENx bits. TSELx can be set as different values. WAVEx is set to 0b1x. MAMPx can be set as different triangle amplitude values. When a channel1 trigger event occurs, add the channel1 triangle counter, with a triangle amplitude configured by MAMP1, to the channel1 data register (DAC_DHR1), and the sum is transferred to DAC_DOR1 for conversion after delay of 3 HB clocks. Then the channel1 triangle counter is updated. When a channel2 trigger event occurs, add the channel2 triangle counter, with a triangle amplitude configured by MAMP2, to the channel2 data register (DAC_DHR2), and the sum is transferred to DAC_DOR2 for conversion after delay of 3 HB clocks. Then the channel2 triangle counter is updated.

18.3.5 Independent Trigger with No Wave Generation

Set the TENx bits. TSELx can be set as different values to select different trigger sources. When a channel1 trigger event occurs, the value of the channel1 data register (DAC_DHR1) delayed 3 HB clocks is transferred to DAC_DOR1 for conversion. When a channel2 trigger event occurs, the value of the channel2 data register (DAC_DHR2) delayed 3 HB clocks is transferred to DAC_DOR2 for conversion.

18.3.6 Simultaneous Software Trigger

In this conversion mode, write the dual DAC channel data to the desired register. A HB clock cycle later, the data

in DAC_DHR1 and DAC_DHR2 is respectively transferred to DAC_DOR1 and DAC_DOR2 for conversion.

18.3.7 Simultaneous Trigger with the Same LFSR Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b01. MAMPx is set as the same LFSR mask value. When a trigger event occurs, add the LFDR1 counter, with the same mask value, to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 HB clocks for conversion. Then the LFSR1 counter is updated. At the same time, add the LFDR2 counter, with the same mask value, to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 HB clocks for conversion. Then the LFSR2 counter is updated.

18.3.8 Simultaneous Trigger with Different LFSR Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b01. MAMPx can be set as different LFSR mask values. When a trigger event occurs, add the LFSR1 counter, with different mask values, to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 HB clocks for conversion. Then the LFSR1 counter is updated. At the same time, add the LFSR2 counter, with different mask values, to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 HB clocks for conversion. Then the LFSR2 counter is updated.

18.3.9 Simultaneous Trigger with the Same Triangle Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b1x. MAMPx is set as the same triangle amplitude value. When a trigger event occurs, add the counter, with the same triangle amplitude value, to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 HB clocks for conversion. Then the channel1 triangle counter is updated. At the same time, add the counter, with the same triangle amplitude value, to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 HB clocks for conversion. Then the channel2 triangle counter is updated.

18.3.10 Simultaneous Trigger with Different Triangle Generation

Set the TENx bits. TSELx is set as the same value. WAVEx is set to 0b1x. MAMPx can be set as different triangle amplitude values. When a trigger event occurs, add the counter, with a triangle amplitude configured by MAMP1[3:0], to the DAC_DHR1 register, and the sum is transferred to DAC_DOR1 after delay of 3 HB clocks for conversion. Then the channel1 triangle counter is updated. At the same time, add the counter, with a triangle amplitude configured by MAMP2[3:0], to the DAC_DHR2 register, and the sum is transferred to DAC_DOR2 after delay of 3 HB clocks for conversion. Then the channel2 triangle counter is updated.

18.3.11 Simultaneous Trigger with No Wave Generation

Set the TENx bits. TSELx is set as the same value. In this conversion mode, when a trigger event occurs, the values of the DAC_DHR1 register and the DAC_DHR2 register delayed 3 HB clocks are respectively transferred to DAC_DOR1 and DAC_DOR2 for DAC conversion.

18.4 Register Description

Table 18-2 DAC registers

Name	Access address	Description	Reset value
R32_DAC_CTLR	0x40007400	DAC control register	0x00000000
R32_DAC_SWTR	0x40007404	DAC software trigger register	0x00000000
R32_DAC_R12BDHR1	0x40007408	DAC channel1 right-aligned 12-bit data hold register	0x00000000
R32_DAC_L12BDHR1	0x4000740C	DAC channel1 left-aligned 12-bit data hold register	0x00000000
R32_DAC_R8BDHR1	0x40007410	DAC channel1 right-aligned 8-bit data hold register	0x00000000
R32_DAC_R12BDHR2	0x40007414	DAC channel2 right-aligned 12-bit data hold register	0x00000000
R32_DAC_L12BDHR2	0x40007418	DAC channel2 left-aligned 12-bit data hold register	0x00000000
R32_DAC_R8BDHR2	0x4000741C	DAC channel2 right-aligned 8-bit data hold register	0x00000000
R32_DAC_RD12BDHR	0x40007420	Dual channel right-aligned 12-bit data hold register	0x00000000
R32_DAC_LD12BDHR	0x40007424	Dual channel left-aligned 12-bit data hold register	0x00000000
R32_DAC_RD8BDHR	0x40007428	Dual channel right-aligned 8-bit data hold register	0x00000000
R32_DAC_DOR1	0x4000742C	DAC channel1 data output register	0x00000000
R32_DAC_DOR2	0x40007430	DAC channel2 data output register	0x00000000

18.4.1 DAC Control Register (DAC_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DMAEN2	MAMP2[3:0]				WAVE2[2:0]		TSEL2[2:0]		TEN2	BOFF2	EN2	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DMAEN1	MAMP1[3:0]				WAVE1[2:0]		TSEL1[2:0]		TEN1	BOFF1	EN1	

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved.	0
28	DMAEN2	RW	DMA enable of DAC channel2: 0: DMA of DAC channel2 disabled; 1: DMA of DAC channel2 enabled.	0
[27:24]	MAMP2[3:0]	RW	DAC channel2 mask/amplitude setting. The software sets this area to select the LFSR data mask bit in the noise generation mode, and	0000b

			select the waveform amplitude in the triangle waveform generation mode: 0000: Unmask bit0 of LFSR/Triangle amplitude equal to 1; 0001: Unmask bits[1:0] of LFSR / Triangle amplitude equal to 3; 0010: Unmask bits[2:0] of LFSR / Triangle amplitude equal to 7; 0011: Unmask bits[3:0] of LFSR / Triangle amplitude equal to 15; 0100: Unmask bits[4:0] of LFSR / Triangle amplitude equal to 31; 0101: Unmask bits[5:0] of LFSR / Triangle amplitude equal to 63; 0110: Unmask bits[6:0] of LFSR / Triangle amplitude equal to 127; 0111: Unmask bits[7:0] of LFSR / Triangle amplitude equal to 255; 1000: Unmask bits[8:0] of LFSR / Triangle amplitude equal to 511; 1001: Unmask bits[9:0] of LFSR / Triangle amplitude equal to 1023; 1010: Unmask bits[10:0] of LFSR / Triangle amplitude equal to 2047; ≥1011: Unmask bits[11:0] of LFSR / Triangle amplitude equal to 4095.	
[23:22]	WAVE2 [1:0]	RW	Noise/triangular wave generation enable of DAC channel2 00: Wave generator disabled; 01: Noise-wave generator enabled; 1x: Triangular-wave generator enabled.	00b
[21:19]	TSEL2[2: 0]	RW	Trigger event selection of DAC channel2: 000: TIM6 TRGO event; 001: TIM8 TRGO event; 010: TIM7 TRGO event; 011: TIM5 TRGO event; 100: TIM2 TRGO event; 101: TIM4 TRGO event; 110: External interrupt line9; 111: Software trigger; Others: Reserved.	000b
18	TEN2	RW	External trigger mode enable of DAC channel2: 0: Trigger of DAC channel2 disabled. The data written into the DAC_xDHR register is sent to the DAC_DOR2 register in 1 HB clock cycle. 1: Trigger of DAC channel2 enabled. The data written into the DAC_xDHR register is sent to the DAC_DOR2 register in 3 HB clock cycles. <i>Note: If software trigger is selected, the data in DAC_xDHR only needs to be sent to the DAC_DOR2 register in a HB clock cycle.</i>	0
17	BOFF2	RW	DAC channel2 output buffer disable control (recommended to be enabled): 0: DAC channel2 output buffer enabled; 1: DAC channel2 output buffer disabled.	0
16	EN2	RW	DAC channel2 enable: 0: DAC channel2 disabled; 1: DAC channel2 enabled.	0

[15:13]	Reserved	RO	Reserved.	0
12	DMAEN1	RW	DMA enable of DAC channel1: 0: DMA function of DAC channel1 disabled; 1: DMA function of DAC channel1 enabled.	0
[11:8]	MAMP1 [3:0]	RW	DAC channel 1 mask/amplitude setting. The software sets these bits to select the LFSR data mask bit in the noise generation mode, and select the wave amplitude in the triangle waveform generation mode: 0000: Unmask bit0 of LFSR / Triangle amplitude equal to 1; 0001: Unmask bits[1:0] of LFSR / Triangle amplitude equal to 3; 0010: Unmask bits[2:0] of LFSR / Triangle amplitude equal to 7; 0011: Unmask bits[3:0] of LFSR / Triangle amplitude equal to 15; 0100: Unmask bits[4:0] of LFSR / Triangle amplitude equal to 31; 0101: Unmask bits[5:0] of LFSR / Triangle amplitude equal to 63; 0110: Unmask bits[6:0] of LFSR / Triangle amplitude equal to 127; 0111: Unmask bits[7:0] of LFSR / Triangle amplitude equal to 255; 1000: Unmask bits[8:0] of LFSR / Triangle amplitude equal to 511; 1001: Unmask bits[9:0] of LFSR / Triangle amplitude equal to 1023; 1010: Unmask bits[10:0] of LFSR / Triangle amplitude equal to 2047; ≥1011: Unmask bits[11:0] of LFSR / Triangle amplitude equal to 4095.	0000b
[7:6]	WAVE1 [1:0]	RW	Noise/triangular-wave generation enable of DAC channel1. 00: Wave generator disabled; 01: Noise wave generator enabled; 1x: Triangular wave generator enabled.	00b
[5:3]	TSEL1 [2:0]	RW	Trigger event selection of DAC channel1: 000: TIM6 TRGO event; 001: TIM8 TRGO event; 010: TIM7 TRGO event; 011: TIM5 TRGO event; 100: TIM2 TRGO event; 101: TIM4 TRGO event; 110: External interrupt line9; 111: Software trigger; Others: Reserved.	000b
2	TEN1	RW	External trigger mode enable of DAC channel1: 0: Trigger of DAC channel1 disabled. The data written into the DAC_xDHR register is sent to the DAC_DOR1 register in 1 HB clock cycle. 1: Trigger of DAC channel1 enabled. The data written into the DAC_xDHR register is sent to the DAC_DOR1 register in 3 HB clock cycles. <i>Note: If software trigger is selected, the data in DAC_xDHR only needs to be sent to the DAC_DOR1 register in a HB clock cycle.</i>	0

1	BOFF1	RW	DAC channel1 output buffer disable control (Recommended to be enabled): 0: DAC channel1 output buffer enabled; 1: DAC channel1 output buffer disabled.	0
0	EN1	RW	DAC channel1 enable: 0: DAC channel1 disabled; 1: DAC channel1 enabled.	0

18.4.2 DAC Software Trigger Register (DAC_SWTR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														SW TRIG 2	SW TRIG 1

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved.	0
1	SWTRIG2	WO	Software trigger control bit of DAC channel2: 0: Software trigger of DAC channel2 disabled; 1: Software trigger of DAC channel2 enabled. <i>Note: Once the data in DAC_xDHR (After 1 HB clock cycle) is sent to the DAC_DOR2 register, this bit can be cleared by hardware.</i>	0
0	SWTRIG1	WO	Software trigger control bit of DAC channel1: 0: Software trigger of DAC channel1 disabled; 1: Software trigger of DAC channel1 enabled. <i>Note: Once the data in DAC_xDHR (After 1 HB clock cycle) is sent to the DAC_DOR1 register, this bit can be cleared by hardware.</i>	0

18.4.3 DAC Channel 1 Right-aligned 12-bit Data Hold Register (DAC_R12BDHR1)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DACC1DHR[11:0]										

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC1DHR[11:0]	RW	12-bit right-aligned data of DAC channel1.	0

18.4.4 DAC Channel 1 Left-aligned 12-bit Data Hold Register (DAC_L12BDHR1)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC1DHR[11:0]												Reserved			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved.	0
[15:4]	DACC1DHR[11:0]	RW	12-bit left-aligned data of DAC channel1.	0
[3:0]	Reserved	RO	Reserved.	0

18.4.5 DAC Channel 1 Right-aligned 8-bit Data Hold Register (DAC_R8BDHR1)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DACC1DHR[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	DACC1DHR[7:0]	RW	8-bit right-aligned data of DAC channel 1.	0

18.4.6 DAC Channel 2 Right-aligned 12-bit Data Hold Register (DAC_R12BDHR2)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC2DHR[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[11:0]	DACC2DHR[11:0]	RW	12-bit right-aligned data of DAC channel 2.	0

18.4.7 DAC Channel 2 Left-aligned 12-bit Data Hold Register (DAC_L12BDHR2)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

DACC2DHR[11:0]	Reserved
----------------	----------

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:4]	DACC2DHR[11:0]	RW	12-bit left-aligned data of DAC channel2.	0
[3:0]	Reserved	RO	Reserved	0

18.4.8 DAC Channel 2 Right-aligned 8-bit Data Hold Register (DAC_R8BDHR2)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DACC2DHR[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	DACC2DHR[7:0]	RW	8-bit right-aligned data of DAC channel2.	0

18.4.9 DAC Dual Channel Right-aligned 12-bit Data Hold Register (DAC_RD12BDHR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				DACC2DHR[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC1DHR[11:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	DACC2DHR[11:0]	RW	12-bit right-aligned data of DAC channel 2.	0
[15:12]	Reserved	RO	Reserved	0
[11:0]	DACC1DHR[11:0]	RW	12-bit right-aligned data of DAC channel 1.	0

18.4.10 DAC Dual Channel Left-aligned 12-bit Data Hold Register (DAC_LD12BDHR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DACC2DHR[11:0]												Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC1DHR[11:0]												Reserved			

Bit	Name	Access	Description	Reset
-----	------	--------	-------------	-------

				value
[31:20]	DACC2DHR[11:0]	RW	12-bit left-aligned data of DAC channel 2.	0
[19:16]	Reserved	RO	Reserved	0
[15:4]	DACC1DHR[11:0]	RW	12-bit left-aligned data of DAC channel 1.	0
[3:0]	Reserved	RO	Reserved	0

18.4.11 DAC Dual Channel Right-aligned 8-bit Data Hold Register (DAC_RD8BDHR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC2DHR[7:0]								DACC1DHR[7:0]							

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	DACC2DHR[7:0]	RW	8-bit right-aligned data of DAC channel 2.	0
[7:0]	DACC1DHR[7:0]	RW	8-bit right-aligned data of DAC channel 1.	0

18.4.12 DAC Channel 1 Data Output Register (DAC_DOR1)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC1DOR[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC1DOR[11:0]	RO	Output data of DAC channel 1	0

18.4.13 DAC Channel 2 Data Output Register (DAC_DOR2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC2DOR[11:0]											

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC2DOR[11:0]	RO	Output data of DAC channel 2	0

Chapter 19 Universal Synchronous Asynchronous Receiver Transmitter (USART)

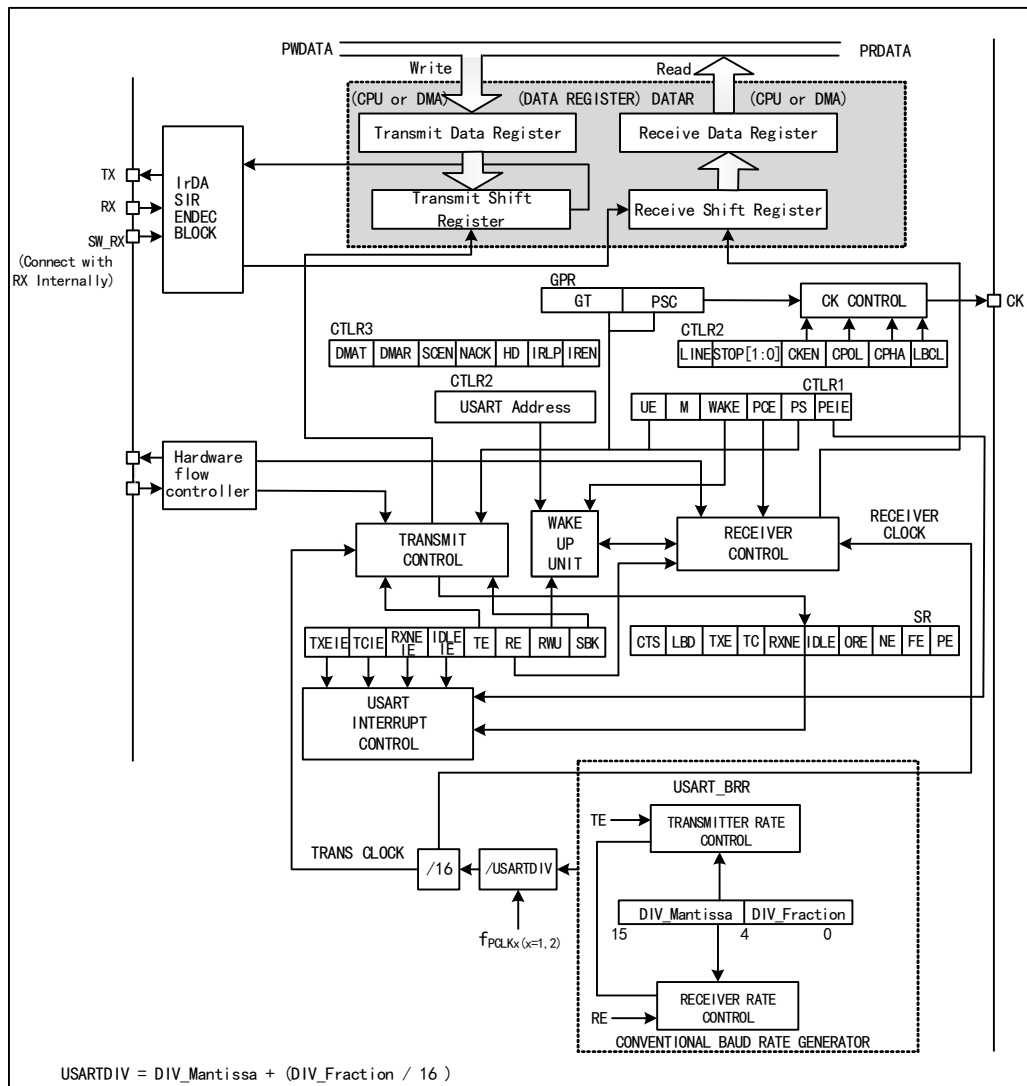
The module contains 8 universal synchronous asynchronous transceivers (USART1/2/3/4/5/6/7/8).

19.1 Main Features

- Full-duplex or half-duplex synchronous or asynchronous communication
- NRZ data format
- Fractional baud rate generator, highest 9Mbps
- Programmable data length
- Configurable stop bit
- Support LIN, IrDA encoders, smart cards
- Support DMA
- Multiple interrupt sources
- Support wake-up function in low power mode

19.2 Overview

Figure 19-1 USART block diagram



When TE (transmission enable bit) is set, the data in the transmitter shift register will be outputted on the TX pin, and the clock will be outputted on the CK pin. During transmission, the lowest significant bit is the first to be shifted out. Each data frame starts with a low-level start bit, and then the transmitter sends 8-bit data or 9-bit data according to the setting of the M (word length) bit, and finally a configurable number of stop bits. If there is a parity check bit, the last bit of the data word is the check bit. After TE is set, an idle frame is sent. The idle frame is 10-bit or 11-bit high level, including the stop bit. The break frame is a 10-bit or 11-bit low level, followed by a stop bit.

19.3 Baud Rate Generator

The baud rate of the transceiver = $F_{CLK} / (16 * USARTDIV)$. The value of USARTDIV is determined according to the 2 domains: DIV_M and DIV_F in USARTx_BRR. The specific calculation formula is:

$$USARTDIV = DIV_M + (DIV_F / 16)$$

It should be noted that the baud rate generated by the baud rate generator may not always generate just the baud rate required by the user, which may be biased. In addition to taking the value as close as possible, the method to reduce the deviation can also be to increase the HB clock. For example, when setting the baud rate to 9600bps, the

USARTDIV value is set to 78.125. At 12MHz, this yields a baud rate of 9600bps. However, if you require a baud rate of 115200bps, the calculated USARTDIV value is 6.51. but the closest value that can be entered into USART_BRR is 6.5. The resulting baud rate is 115384bps, resulting in an error of 0.16%.

When the serial port waveform sent by the transmitter is transmitted to the receiver, there is a certain error in the baud rate between the receiver and the sender. The error mainly comes from 3 aspects: the actual baud rate of the receiver and the sender are inconsistent; the clocks of the receiver and the sender have errors; the waveform changes in the circuit. The receiver of the peripheral module has a certain tolerance for receiving. When the sum of the total deviations generated in the above 3 aspects is less than the tolerance limit of the module, the total deviation will not affect the receiving and sending. The tolerance limit of the module is affected by the use of fractional baud rate and M bit (data field word length) or not. The use of fractional baud rate and the use of 9-bit data field length will reduce the tolerance limit, but it shall not be less than 3%.

19.4 Synchronous Mode

The synchronous mode enables the system to output clock signals when the USART module is used. When the synchronous mode is enabled to send data externally, the CK pin will output clock externally at the same time.

To enable synchronous mode, set CLKEN bit in the control register2 (R16_USARTx_CTLR2), but you need to switch off the LIN mode, smart card mode, infrared mode and half-duplex mode at the same time, i.e., to ensure that the SCEN, HDSEL and IREN bits are in the reset status. These 3 bits are in the control register3 (R16_USARTx_CTLR3).

The main point of the synchronous mode is the output control of the clock. Attention shall be paid to the following: The synchronous mode of the USART module only works in the master mode, i.e., the CK pin only outputs the clock and does not receive input;

The clock signal is outputted only when TX pin outputs data;

The LBCL bit determines whether the clock is outputted when the last data bit is sent. The CPOL bit determines the polarity of the clock, and the CPHA determines the phase position of the clock. These 3 bits are in the control register 2 (R16_USARTx_CTLR2). These 3 bits need to be set when TE and RE are not enabled. The specific difference is shown in Figure 19-2.

In the synchronous mode, the receiver will only sample when outputting the clock, and the slave needs to maintain a certain signal setup time and hold time, specifically as shown in Figure 19-3.

Figure 19-2 Example of USART clock timing (M=0)

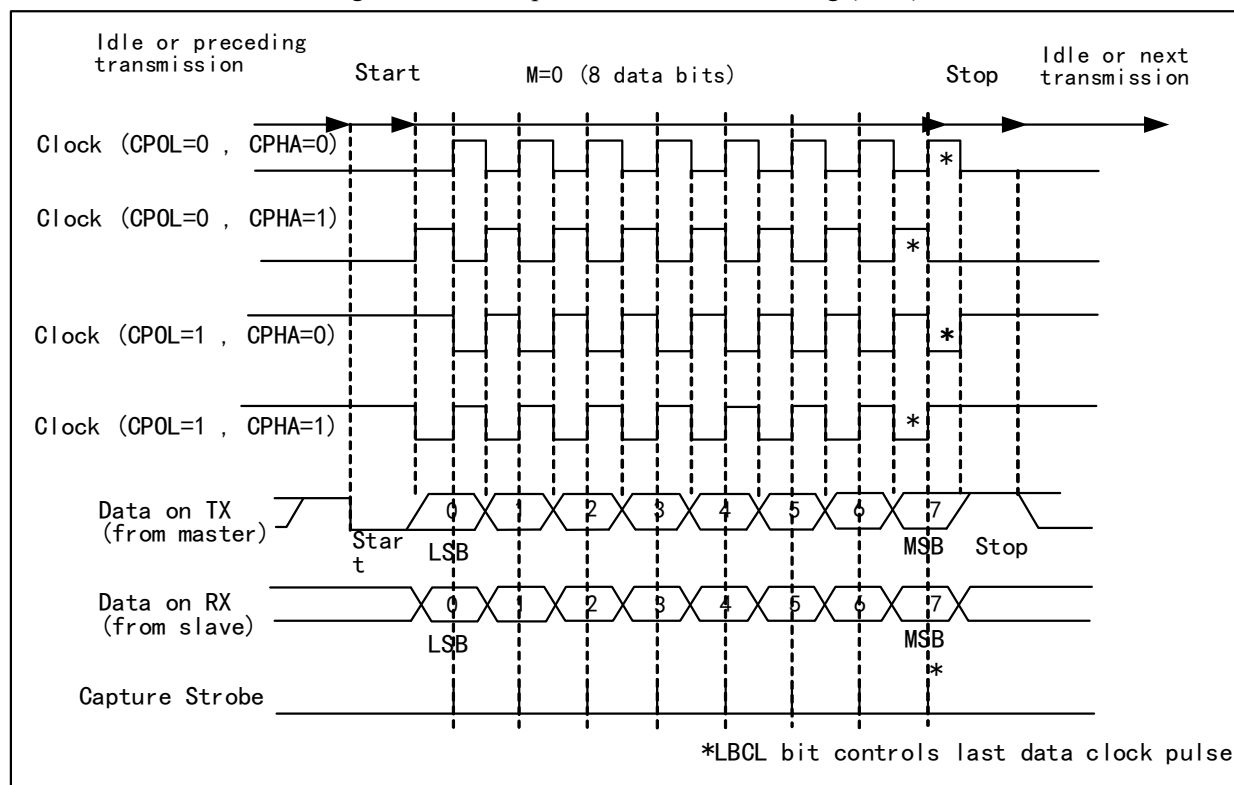
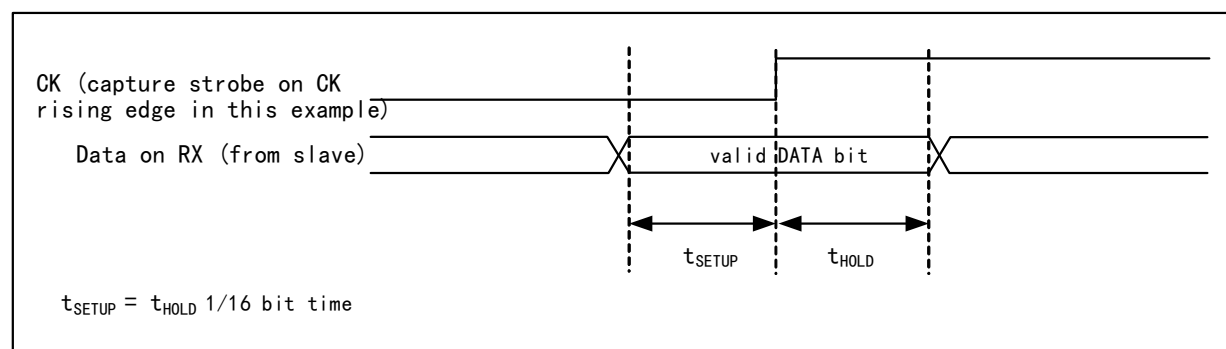


Figure 19-3 Data sample hold time



19.5 1-wire Half-duplex Mode

The half-duplex mode supports the use of a single pin (only TX pin) to receive and transmit, and the TX pin and RX pin are connected inside the chip.

To enable half-duplex mode, set HDSEL bit in the control register 3 (R16_USARTx_CTLR3), but you need to disable LIN mode, smartcard mode, infrared mode and synchronous mode at the same time, i.e., to ensure that the SCEN, CLKEN and IREN bits are in the reset status. These 3 bits are in the control register 2 and the control register 3 (R16_USARTx_CTLR2 and R16_USARTx_CTLR3).

After setting to half-duplex mode, it is needed to set the TX IO port to open-drain output high mode. When TE bit is set, the data will be sent out as long as the data is written to the data register. Special attention shall be paid to the fact that bus conflicts may occur when multiple devices use a single bus to transmit and receive in half-duplex mode. This requires users to avoid it by software.

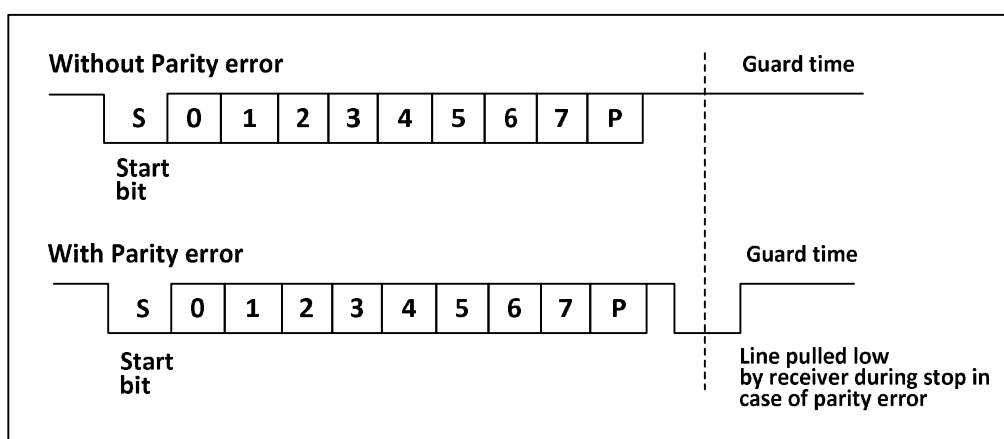
19.6 Smart Card

The smartcard mode supports ISO7816-3 protocol to access the smart card controller.

To enable smartcard mode, set the SCEN bit in the control register 3 (R16_USARTx_CTLR3), but it is needed to disable LIN mode, half-duplex mode and infrared mode at the same time, i.e., to ensure that the LINEN, HDSEL and IREN bits are in the reset status, but CLKEN can be switched on to output the clock. These 3 bits are in the control register 2 and the control register 3 (R16_USARTx_CTLR2 and R16_USARTx_CTLR3).

In order to support smartcard mode, USART shall be set to 8 data bits plus 1 check bit. It is recommended that the stop bit be configured to 1.5 bits for both sending and receiving. The smart card mode is a 1-wire half-duplex protocol, which uses TX line as the data communication and shall be configured as open drain output plus pull-up. When the receiver receives a frame of data and detects a parity check error, it will send a NACK signal at the stop bit, i.e., actively reducing a cycle of TX during the stop bit. After the sender detects the NACK signal, a frame error will be generated, and the application can resend accordingly. Figure 19-4 shows the waveforms on the TX pin under correct conditions and in the event of parity check errors. The TC flag (transmission completion flag) of the USART can delay the generation of GT (protection time) clocks, and the receiver will not recognize the NACK signal set by itself as the start bit.

Figure 19-4 (No) parity check error



In smartcard mode, the output waveform after the CK pin is enabled has nothing to do with the communication. It only provides the clock for the smart card. Its value is the HB clock and then the 5-bit settable clock frequency division (the frequency division value is double of PSC, and the highest is frequency division 62).

19.7 IrDA

USART module supports control IrDA infrared transceiver for physical layer communication. To use IrDA, the LINEN, STOP, CLKEN, SCEN and HDSEL bits must be cleared. NRZ (non-return-to-zero) coding is used between the USART module and the SIR physical layer (infrared transceiver), and the maximum support rate is 115200bps.

IrDA is a half-duplex protocol. If USART sends data to the SIR physical layer, the IrDA decoder ignores the newly sent infrared signal. If the USART receives data from SIR, then SIR does not accept USART signal. The level logic sent by USART to SIR and SIR to USART is different. In SIR receive logic, '1' represents high level and '0' represents low level. However, in the SIR transmit logic, '0' represents high level and '1' represents low level.

19.8 DMA

The USART module supports DMA, and can use DMA to implement fast continuous reception and transmission. When DMA is enabled and the TXE bit is set, DMA writes data to the transmit buffer from the set memory space. When DMA is used for reception, DMA transfers the data in the receive buffer to a specific memory space each time the RXNE bit is set.

19.9 Interrupt

The USART module supports multiple interrupt sources, including transmit data register empty (TXE), CTS, transmission complete (TC), received data ready (RXNE), data overrun error (ORE), idle line (IDLE), parity check error (PE), break flag (LBD), noise (NE), multi-buffer communication overrun (ORT) and framing error (FE).

Table 19-1 Interrupts and the corresponding enable bits

Interrupt source	Enable bit
Transmit data register empty (TXE)	TXEIE
Transmission allowed (CTS)	CTSIE
Transmission complete (TC)	TCIE
Received data ready (RXNE)	RXNEIE
Transmit data register empty (TXE)	
Idle line (IDLE)	IDLEIE
Parity error (PE)	PEIE
Break flag (LBD)	LBDIE
Noise (NE)	EIE
Overrun error in multi-buffer communication (ORT)	
Framing error (FE) in multi-buffer communication	

19.10 Interrupt

Enable the low-power wake-up function via the LPWKUP_EN bit in the USARTx_CTLR3 register, and read the low-power wake-up start indicator flag from the LPWKUP_ACT_FLAG bit in the USARTx_STATR register; The clock source for low-power wake-up is selected via the LPWKUP_CK_SRC bit in the USARTx_CTLR3 register, offering choices between LSE or LSI. The LPWKUP_DLY_CFG bit configures the clock source period for the low-power wake-up receive delay.

19.11 Register Description

Table 19-2 USART1-related registers

Name	Access address	Description	Reset value
R32_USART1_STATR	0x40013800	UASRT1 status register	0x000000C0
R32_USART1_DATAR	0x40013804	UASRT1 data register	0x00000000
R32_USART1_BRR	0x40013808	UASRT1 baud rate register	0x00000000

R32_USART1_CTLR1	0x4001380C	UASRT1 control register1	0x00000000
R32_USART1_CTLR2	0x40013810	UASRT1 control register2	0x00000000
R32_USART1_CTLR3	0x40013814	UASRT1 control register3	0x00000000
R32_USART1_GPR	0x40013818	UASRT1 guard time and prescaler register	0x00000000
R32_USART1_CTLR4	0x4001381C	UASRT1 control register4	0x00000000

Table 19-3 USART2-related registers

Name	Access address	Description	Reset value
R32_USART2_STATR	0x40004400	UASRT2 status register	0x000000C0
R32_USART2_DATAR	0x40004404	UASRT2 data register	0x00000000
R32_USART2_BRR	0x40004408	UASRT2 baud rate register	0x00000000
R32_USART2_CTLR1	0x4000440C	UASRT2 control register1	0x00000000
R32_USART2_CTLR2	0x40004410	UASRT2 control register2	0x00000000
R32_USART2_CTLR3	0x40004414	UASRT2 control register3	0x00000000
R32_USART2_GPR	0x40004418	UASRT2 guard time and prescaler register	0x00000000
R32_USART2_CTLR4	0x4000441C	UASRT2 control register4	0x00000000

Table 19-4 USART3-related registers

Name	Access address	Description	Reset value
R32_USART3_STATR	0x40004800	UASRT3 status register	0x000000C0
R32_USART3_DATAR	0x40004804	UASRT3 data register	0x00000000
R32_USART3_BRR	0x40004808	UASRT3 baud rate register	0x00000000
R32_USART3_CTLR1	0x4000480C	UASRT3 control register1	0x00000000
R32_USART3_CTLR2	0x40004810	UASRT3 control register2	0x00000000
R32_USART3_CTLR3	0x40004814	UASRT3 control register3	0x00000000
R32_USART3_GPR	0x40004818	UASRT3 guard time and prescaler register	0x00000000
R32_USART3_CTLR4	0x4000481C	UASRT3 control register4	0x00000000

Table 19-5 USART4-related registers

Name	Access address	Description	Reset value
R32_USART4_STATR	0x40004C00	UASRT4 status register	0x000000C0
R32_USART4_DATAR	0x40004C04	UASRT4 data register	0x000000XX
R32_USART4_BRR	0x40004C08	UASRT4 baud rate register	0x00000000
R32_USART4_CTLR1	0x40004C0C	UASRT4 control register1	0x00000000
R32_USART4_CTLR2	0x40004C10	UASRT4 control register2	0x00000000
R32_USART4_CTLR3	0x40004C14	UASRT4 control register3	0x00000000
R32_USART4_GPR	0x40004C18	UASRT4 guard time and prescaler register	0x00000000
R32_USART4_CTLR4	0x40004C1C	UASRT4 control register4	0x00000000

Table 19-6 USART5-related registers

Name	Access address	Description	Reset value
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R32_USART5_STATR	0x40005000	UASRT5 status register	0x000000C0
R32_USART5_DATAR	0x40005004	UASRT5 data register	0x00000000
R32_USART5_BRR	0x40005008	UASRT5 baud rate register	0x00000000
R32_USART5_CTLR1	0x4000500C	UASRT5 control register1	0x00000000
R32_USART5_CTLR2	0x40005010	UASRT5 control register2	0x00000000
R32_USART5_CTLR3	0x40005014	UASRT5 control register3	0x00000000
R32_USART5_GPR	0x40005018	UASRT5 guard time and prescaler register	0x00000000
R32_USART5_CTLR4	0x4000501C	UASRT5 control register4	0x00000000

Table 19-7 USART6-related registers

Name	Access address	Description	Reset value
R32_USART6_STATR	0x40001800	UASRT6 status register	0x000000C0
R32_USART6_DATAR	0x40001804	UASRT6 data register	0x00000000
R32_USART6_BRR	0x40001808	UASRT6 baud rate register	0x00000000
R32_USART6_CTLR1	0x4000180C	UASRT6 control register1	0x00000000
R32_USART6_CTLR2	0x40001810	UASRT6 control register2	0x00000000
R32_USART6_CTLR3	0x40001814	UASRT6 control register3	0x00000000
R32_USART6_GPR	0x40001818	UASRT6 guard time and prescaler register	0x00000000
R32_USART6_CTLR4	0x4000181C	UASRT6 control register4	0x00000000

Table 19-8 USART7-related registers

Name	Access address	Description	Reset value
R32_USART7_STATR	0x40001C00	UASRT7 status register	0x000000C0
R32_USART7_DATAR	0x40001C04	UASRT7 data register	0x00000000
R32_USART7_BRR	0x40001C08	UASRT7 baud rate register	0x00000000
R32_USART7_CTLR1	0x40001C0C	UASRT7 control register1	0x00000000
R32_USART7_CTLR2	0x40001C10	UASRT7 control register2	0x00000000
R32_USART7_CTLR3	0x40001C14	UASRT7 control register3	0x00000000
R32_USART7_GPR	0x40001C18	UASRT7 guard time and prescaler register	0x00000000
R32_USART7_CTLR4	0x40001C1C	UASRT7 control register4	0x00000000

Table 19-9 USART8-related registers

Name	Access address	Description	Reset value
R32_USART8_STATR	0x40002000	UASRT8 status register	0x000000C0
R32_USART8_DATAR	0x40002004	UASRT8 data register	0x00000000
R32_USART8_BRR	0x40002008	UASRT8 baud rate register	0x00000000
R32_USART8_CTLR1	0x4000200C	UASRT8 control register1	0x00000000
R32_USART8_CTLR2	0x40002010	UASRT8 control register2	0x00000000
R32_USART8_CTLR3	0x40002014	UASRT8 control register3	0x00000000
R32_USART8_GPR	0x40002018	UASRT8 guard time and prescaler register	0x00000000
R32_USART8_CTLR4	0x4000201C	UASRT8 control register4	0x00000000

19.11.1 USART Status Register (USARTx_STATR) (x=1/2/3/4/5/6/7/8)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LPWKUP_ACT_FLAG	Reserved			MS_ERR	RX_BUSY	CTS	LBD	TXE	TC	RXNE	IDLE	ORE	NE	FE	PE

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
15	LPWKUP_ACT_FLAG	RW1Z	Low power wake-up start indicator bit, cleared by writing 1.	0
[14:12]	Reserved	RO	Reserved	0
11	MS_ERR	RO	MARK or SPACE check error flag. In receive mode, if there is a MARK or SPACE check error, the hardware sets this bit. Reading this bit and then reading the data register will reset this bit. If MS_ERRIE has been set before, setting this bit will generate corresponding interrupt.	0
10	RX_BUSY	RO	Receive status indicator bit: 1: Set when operating in receive mode; 0: Unset when not in receive mode.	0
9	CTS	RW0	CTS status change flag. If the CTSE bit is set, when the nCTS output state changes, the bit will be set high by the hardware. Zero is cleared by the software. If the CTSIE bit has been set, an interrupt occurs. 1: There is a change in the nCTS state line. 0: There is no change on the nCTS state line.	0
8	LBD	RW0	LIN Break detection flag. When LIN Break is detected, this bit will be set by hardware. It is cleared by the software. If the LBDIE bit has been set, an interrupt will be generated. 1: LIN Break detected; 0: No LIN Break detected.	0
7	TXE	RO	Transmission data register empty flag. When data in TDR register is transferred to shift register by hardware, this bit will be set by hardware. If TXEIE bit has been set, an interrupt will be generated, the data register will be written and this bit will be reset. 1: Data is transferred to the shift register;	1

			0: Data is not transferred to the shift register.	
6	TC	RW0	Transmission complete flag. When a frame containing data is sent and TXE bit is set, the hardware will set this bit. If TCIE is set, a corresponding interrupt will be generated. The software will read this bit and then write the data register to clear this bit. You can also directly write 0 to clear this bit. 1: Transmission completed; 0: Transmission not completed.	1
5	RXNE	RW0	Read data register not empty flag. When the data in the shift register is transferred to the data register, this bit will be set by the hardware. If the RXNEIE bit has been set, the corresponding interrupt will be generated. This bit can be cleared by the write operation of the data register. This bit can be also cleared by directly writing 0. 1: The data is received and can be read; 0: The data is not received.	0
4	IDLE	RO	Idle line flag. When an idle line is detected, the bit will be set by hardware. If IDLEIE bit has been set, the corresponding interrupt will be generated. This bit can be cleared by reading the status register and then reading the data register. 1: The bus is idle now; 0: Idle bus is not detected. <i>Note: This bit will not be set again until RXNE is set.</i>	0
3	ORE	RO	Overrun error flag. When the receiving shift register has data that needs to be transferred to the data register, but this bit will be set when there is still data that has not been read in the receiving field of the data register. If the RXNEIE bit is set, the corresponding interrupt will be generated. 1: The overrun error has occurred; 0: No overrun error has occurred. <i>Note: When an overrun error occurs, the value of the data register will not be lost, but the value of the shift register will be overwritten. If the EIE bit is set, the ORE flag bit will generate an interrupt in the multi-buffer communication mode.</i>	0
2	NE	RO	Noise error flag. When the noise error flag is detected, it will be set by hardware. This bit can be reset by reading the status register and then reading the data register.	0

			1: The noise is detected; 0: No noise is detected. <i>Note: This bit will not generate the interrupt. If the EIE bit has been set, the FE flag bit will generate an interrupt in the multi-buffer communication mode.</i>	
1	FE	RO	Frame error flag. When a synchronization error, excessive noise or disconnection is detected, this bit will be set by hardware. This bit can be reset by reading the bit and then reading the data register. 1: A frame error is detected; 0: No frame error is detected. <i>Note: This bit will not generate interrupt. If the EIE bit has been set, the FE flag bit will generate an interrupt in the multi-buffer communication mode.</i>	0
0	PE	RO	Parity error flag. In the receiving mode, if a parity error occurs, this bit can be set by hardware. This bit can be reset by reading the bit and then reading the data register. Before this bit is cleared, the software must wait for the RXNE flag bit to be set. If PEIE bit has been set before, then the corresponding interrupt will be generated when this bit is set. 1: Parity check error occurs; 0: No parity check error occurs.	0

19.11.2 USART Data Register (USARTx_Datar) (x=1/2/3/4/5/6/7/8)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DR[8:0]							

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
[8:0]	DR[8:0]	RW	Data register. This register is actually composed of 2 registers: receive data register (RDR) and transmit data register (TDR). The start of the read and write operations of DR is to read the receive data register (RDR) and write to the transmit data register (TDR).	X

19.11.3 USART Baud Rate Register (USARTx_BRR) (x=1/2/3/4/5/6/7/8)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV_Mantissa[11:0]												DIV_Fraction[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:4]	DIV_Mantissa [11:0]	RW	These 12 bits define the integer portion of the divider division factor.	0
[3:0]	DIV_Fraction[3:0]	RW	These 4 bits define the decimal part of the divider division factor.	0

19.11.4 USART Control Register 1 (USARTx_CTLR1) (x=1/2/3/4/5/6/7/8)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M_EXT	UE	M	WAKE	PCE	PS	PEIE	TXEIE	TCIE	RXNEIE	IDLEIE	TE	RE	RWU	SBK	

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	M_EXT	RW	Word length extension bit: 00: Invalid, M bits determine the data length. 01: 7 data bits; 10: 6 data bits; 11: 5 data bits.	0
13	UE	RW	USART enable. 1: Enable USART; 0: Disable USART. When this bit is cleared, the frequency divider and output of USART both stop working after the current byte transmission is completed.	0
12	M	RW	Word length. 1: 9 data bits; 0: 8 data bits.	0
11	WAKE	RW	Wake-up. This bit decides the method to wake up USART: 1: Address flag; 0: Idle line.	0
10	PCE	RW	Parity control enable. 1: Enable parity bit; 0: Disable parity bit. For the receiver, the parity of the data is performed; for the	0

			transmitter, the check bit is inserted. Once this bit is set, the parity control enable takes effect only after the current byte transmission is completed.	
9	PS	RW	Parity selection. 1: Odd parity; 0: Even parity. After this bit is set, the parity control enable takes effect only after the current byte transmission is completed.	0
8	PEIE	RW	Parity check interrupt enable. 1: Allow generation of parity error interrupt; 0: Prohibit generation of parity error interrupt.	0
7	TXEIE	RW	Transmit buffer empty interrupt enable. 1: Allow generation of transmit buffer empty interrupt; 0: Prohibit generation of transmit buffer empty interrupt.	0
6	TCIE	RW	Transmission completion interrupt enable. 1: Allow generation of transmission complete interrupt; 0: Prohibit generation of transmission complete interrupt.	0
5	RXNEIE	RW	Receive buffer non-empty interrupt enable. 1: Allow generation of receive buffer not empty interrupt; 0: Prohibit generation of receive buffer not empty interrupt.	0
4	IDLEIE	RW	Idle line interrupt enable. 1: Allow generation of idle line interrupt; 0: Prohibit generation of idle line interrupt.	0
3	TE	RW	Transmit enable. 1: Enable transmitter; 0: Disable transmitter.	0
2	RE	RW	Receiver enable. 1: Enable receiver, and the receiver starts detecting the start bit on the RX pin. 0: Disable receiver.	0
1	RWU	RW	Receiver wake-up. This bit decides whether the USART is in mute mode: 1: The receiver is in mute mode; 0: The receiver is in active mode. <i>Note 1: Before the RWU bit is set, USART needs to receive a data byte firstly. Otherwise, it cannot be woken up by the idle bus in mute mode;</i> <i>Note 2: When configured to wake up from address flag, the RWU bit cannot be modified by software when RXNE is set.</i>	0
0	SBK	RW	Send break character control. This bit is set to transmit a frame break character. For the stop bit of break frame, the bit is set by hardware. 1: Break character transmitted; 0: No break character transmitted.	0

19.11.5 USART Control Register 2 (USARTx_CTLR2) (x=1/2/3/4/5/6/7/8)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	LINEN	STOP[1:0]	CLKEN	CPOL	CPHA	LBCL	Reserved	LBDIE	LBDL	Reserved	ADD[3:0]				

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	LINEN	RW	LIN mode enable bit: 1: Enables LIN mode; 0: Disables LIN mode. In LIN mode, the SBK bit can be used to transmit the LIN synchronization break character and to detect the LIN synchronization break character.	0
[13:12]	STOP[1:0]	RW	Stop bit setting. These bits are used to set the stop bits. 00: 1 stop bit; 01: 0.5 stop bit; 10: 2 stop bits; 11: 1.5 stop bit.	00b
11	CLKEN	RW	Clock enable. This bit is used to enable CK pin. 1: Enable; 0: Disable.	0
10	CPOL	RW	Clock polarity. In synchronous mode, this bit can be used to select the polarity of the clock output on the CK pin, and work with CPHA to generate the required clock/data sampling relationship. 1: High level is maintained on the CK pin when the bus is idle; 0: Low level is maintained on the CK pin when the bus is idle. <i>Note: This bit cannot be modified after enabling transmission.</i>	0
9	CPHA	RW	Clock phase position setting. In the synchronization mode, you can use this bit to select the phase position of the clock output on the CK pin, and work with CPOL bit to generate the required clock/data sampling relationship. 1: Data capture is performed on the second edge of the clock; 0: Data capture is performed on the first edge of the clock. <i>Note: This bit cannot be modified after enabling transmission.</i>	0

8	LBCL	RW	Last bit clock pulse control. In synchronous mode, it is used to control whether to output the clock pulse corresponding to the last data byte sent on the CK pin; 1: The clock pulse of the last bit of data is not output from CK; 0: The clock pulse of the last bit of data is output from CK. <i>Note: This bit cannot be modified after enabling transmission.</i>	0
7	Reserved	RW	Reserved.	0
6	LBDIE	RW	LIN Break detection interrupt enable. 1: Enable interrupt caused by LBD; 0: Disable interrupt caused by LBD.	0
5	LBDL	RW	LIN Break detection length, used to select 11-bit or 10-bit break character detection. 1: 11-bit break detection; 0: 10-bit break detection.	0
4	Reserved	RW	Reserved.	0
[3:0]	ADD[3:0]	RW	Address of the USART node, used to set the USART node address of the device. When the data is used during mute mode in multi-processor communication, the address flag is used to wake up a certain USART device.	0

19.11.6 USART Control Register 3 (USARTx_CTLR3) (x=1/2/3/4/5/6/7/8)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LPWKUP_DLY_CFG[2:0]		LPWKUP_CHK_SRC	LPWKUP_EN	CTSE	CTSE	RTSE	DMA_T	DMA_R	SCE_N	NACK	HDS_EL	IRLP	IREN	EIE	

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:13]	LPWKUP_DLY_CFG[2:0]	RW	Low-power wake-up receive delay configuration: 000: No delay; 001: Delay of 1 low-power wake-up clock source cycle; ... 111: Delay of 7 low-power wake-up clock source cycles.	0
12	LPWKUP_CHK_SRC	RW	Low power wake-up clock source selection: 1: LSE;	0

			0: LSI.	
11	LPWKUP_EN	RW	Low power wake-up enable: 1: Enable low-power wake-up; 0: Disable low-power wake-up.	0
10	CTSIE	RW	CTSIE interrupt enable. 1: An interrupt will be generated when CTS is set; 0: No interrupt is generated when CTS is set.	0
9	CTSE	RW	CTS enable. 1: Enable CTS flow control; 0: Disable CTS flow control.	0
8	RTSE	RW	RTS enable. 1: Enable RTS flow control; 0: Disable RTS flow control.	0
7	DMAT	RW	DMA transmission enable. 1: Use DMA when sending; 0: DMA is not used when sending.	0
6	DMAR	RW	DMA reception enable. 1: Use DMA when receiving; 0: DMA is not used when receiving.	0
5	SCEN	RW	Smart card mode enable. 1: Enable smartcard mode; 0: Disable smartcard mode.	0
4	NACK	RW	Smart card NACK enable. 1: NACK is transmitted when the check error occurs; 0: NACK is not transmitted when the check error occurs.	0
3	HDSEL	RW	Half-duplex mode selection bit: 1: Enable half-duplex mode; 0: Disable half-duplex mode.	0
2	IRLP	RW	Infrared low power selection bit: 1: Enable low-power mode when selecting infrared; 0: Disable low-power mode when selecting infrared.	0
1	IREN	RW	Infrared enable. 1: Enable infrared mode; 0: Disable infrared mode.	0
0	EIE	RW	Error interrupt bit enable: 1: Enable error interrupt; 0: Disable error interrupt. When this bit is set, and when DMAR is set, an interrupt is generated if the FE or ORE or NE bit is set.	0

19.11.7 USART Guard Time and Prescaler Register (USARTx_GPR) (x=1/2/3/4/5/6/7/8)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GT[7:0]								PSC[7:0]							

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	GT[7:0]	RW	Guard time domain. These bits specify the guard time in unit of baud rate clock. In smartcard mode, the transmission complete flag is set after the guard time has passed.	0
[7:0]	PSC[7:0]	RW	Prescaler domain. In infrared low-power mode, the source clock is divided by this value (all 8 bits are valid), and a value of 0 means reservation; In infrared normal mode, these bits can only be set to 1; In smartcard mode, the value (the lower 5 bits are valid) is multiplied by 2 to give the division factor of the source clock frequency, to provide the clock to the smart card. A value of 0 means reservation.	0

19.11.8 USART Control Register 4 (USARTx_CTLR4) (x=1/2/3/4/5/6/7/8)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CHECK_SE L [1:0]	MS_ ERRI E	Reser ved	

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
[3:2]	CHECK_SEL[1:0]	RW	Parity Function Selection Bit: 0x: Disable MARK and SPACE parity; 10: Enable MARK parity; 11: Enable SPACE parity.	0
1	MS_ERRIE	RW	SPACE or MARK Parity Enable Bit: 1: Enable SPACE or MARK parity error interrupts; 0: Disable SPACE or MARK parity error interrupts.	0
0	Reserved	RO	Reserved	0

Chapter 20 Serial-parallel Interconversion Controller and Transceiver (SerDes)

The module description in this chapter only applies to CH32H417 microcontroller products.

A set of SerDes modules supporting signal isolation and long-distance transmission are built into the system, which includes 2 independent SerDes controllers, supports 1.5Gbps high-speed differential signals (SERDES_RXP, SERDES_RXN, SERDES_TXP, SERDES_TXN pins), and can carry out long-distance data transmission through an optical fiber module or a differential peer-to-peer transmission medium in a network cable.

The SerDes module integrates a full-duplex PHY and 2 independent control units. Each control unit can be functionally paired with either the PHY's receive (Rx) channel or transmit (Tx) channel. Note: A single control unit must not interface with both the receive and transmit channels simultaneously. By configuring 2 controllers to 2 channels, the SerDes full-duplex communication functionality can be achieved. For the specific implementation approach and configuration process, please refer to the official EVT reference example provided.

The base addresses of the 2 controllers are configured as follows:

SerDes1 controller 1 base address: 0x40027C00

SerDes2 controller 2 base address: 0x40027C40

20.1 Main Features

- Programmable data transmission rates up to 1.5Gbps
- Support high-voltage signal isolation via network transformers and low-voltage capacitive isolation
- Built-in 8-bit/10-bit encoding/decoding with CRC checksum and serial number matching
- Integrated FIFO with dual-buffer mode for transmit/receive
- Support DMA functionality with byte-aligned address access
- Provide multiple transmission interrupt flags and status indicators for timely application layer feedback
- Differential transceiver capable of directly driving fiber optic modules

20.2 Function Description

20.2.1 Transmit Mode (Query Mode)

1. Set SERDESx_CTRL register
ALIGN_EN/PWR_UP/TX_PWR_UP/LINK_RESET/ALL_CLR is 1;
2. Wait PLL locked (PLL_LOCK=1);
3. Set SERDESx_CTRL register
ALIGN_EN/PWR_UP/TX_PWR_UP/INT_BUSY/DMA_EN/TX_EN is 1;
4. Set LINK_INIT to 1, start sending initialization packet, delay 100ms (software delay), and clear LINK_INIT;
5. Set transmit DMA address (Single buffer configuration DMA0 is enough);
6. Set SERDESx_DATA0 (Single buffer mode);
7. Set SERDESx_TX_LEN;
8. Check if LINK_FREE is set to 1 (Optional);

9. Set TX_VLD to 1 to initiate transmission (Steps 5, 6, and 7 may be performed in any order);
10. Check if the transmission complete interrupt flag is set to 1; disable transmission and clear the interrupt flag when TX_VLD is 0.

Note: The software delay in step 4 should ensure the receiver is ready to receive data.

20.2.2 Receive Mode (Interrupt Mode)

1. Set SERDESx_CTRL register
PWR_UP/RX_PWR_UP/LINK_RESET/ALL_CLR is 1;
2. Wait PLL locked (PLL_LOCK=1);
3. Set SERDESx_CTRL register
PWR_UP/RX_PWR_UP/DMA_EN/RX_EN is 1;
4. Set receive DMA address (Dual buffering mode is recommended.);
5. Enable interrupts RECV_DONE_IE/COMINIT_IE/FIFO_OV_IE;
6. After reception completes, check RECV_CRC_OK/RX_SEQ_MATCH. If in dual-buffered mode, query the least significant bit of RX_SEQ_NUM to determine which register is used for the current received packet: DATA0/DMA0/RX_LEN0 or DATA1/DMA1/RX_LEN1;

RX_SEQ_NUM[0] is 1, indicating the current received packet uses DATA0/DMA0/RX_LEN0;

RX_SEQ_NUM[0] is 0, indicating the current received packet uses DATA1/DMA1/RX_LEN1.

Note: RX_SEQ_NUM automatically adds 1 after receiving, so the RX_SEQ_NUM queried after receiving is subtracted from 1 to become the SEQ_NUM of the currently received packet.

20.2.3 Software Considerations:

1. The internal clock frequency SERDES_CLK within the controller is the PLL frequency divided by 40. For example, at 1.2GHz, SERDES_CLK is 30MHz.
2. The return values of LINK_RESET, RX_EN, and TX_VLD are synchronized by SERDES_CLK, resulting in read delays;
3. The length of transmitted or received data packets does not include the 4-byte length of DATA0/DATA1;
4. After transmission completes, TX_VLD must be cleared first. Only after verifying this bit returns 0 should the transmission complete interrupt flag be cleared; otherwise, asynchronous issues may occur.

20.3 Register Description

Table 20-1 SERDES1-related registers

Name	Access address	Description	Reset value
R32_SERDES1_CTRL	0x40027C00	SERDES1 control register	0x00041207
R32_SERDES1_INT_EN	0x40027C04	SERDES1 interrupt enable register	0x00000000
R32_SERDES1_INT_FS	0x40027C08	SERDES1 interrupt flag/status register	0x00000000
R32_SERDES1_RTX_CTRL	0x40027C0C	SERDES1 transceiver controller	0x00000000
R32_SERDES1_RX_LEN0	0x40027C10	SERDES1 receive length 0 register	0x00000000
R32_SERDES1_DATA0	0x40027C14	SERDES1 data 0 buffer register	0xFFFFFFFF
R32_SERDES1_DMA0	0x40027C18	Start address register for DMA0 buffer	0xFFFFFFFF
R32_SERDES1_RX_LEN1	0x40027C1C	SERDES1 receive length 1 register	0x00000000

R32_SERDES1_DATA1	0x40027C20	SERDES1 data 1 buffer register	0xFFFFFFFF
R32_SERDES1_DMA1	0x40027C24	Start address register for DMA1 buffer	0xFFFFFFFF

Table 20-2 SERDES1-related registers

Name	Access address	Description	Reset value
R32_SERDES2_CTRL	0x40027C40	SERDES2 control register	0x00041207
R32_SERDES2_INT_EN	0x40027C44	SERDES2 interrupt enable register	0x00000000
R32_SERDES2_INT_FS	0x40027C48	SERDES2 interrupt flag/status register	0x00000000
R32_SERDES2_RTX_CTRL	0x40027C4C	SERDES2 transceiver controller	0x00000000
R32_SERDES2_RX_LEN0	0x40027C50	SERDES2 receive length 0 register	0x00000000
R32_SERDES2_DATA0	0x40027C54	SERDES2 data 0 buffer register	0xFFFFFFFF
R32_SERDES2_DMA0	0x40027C58	Start address register for DMA0 buffer	0xFFFFFFFF
R32_SERDES2_RX_LEN1	0x40027C5C	SERDES2 receive length 1 register	0x00000000
R32_SERDES2_DATA1	0x40027C60	SERDES2 data 1 buffer register	0xFFFFFFFF
R32_SERDES2_DMA1	0x40027C64	Start address register for DMA1 buffer	0xFFFFFFFF

Table 20-3 SERDES1-related registers

Name	Access address	Description	Reset value
R32_SYS_CFGR	0x5003C010	System configuration register	0x0054422B

20.3.1 SERDESx Control Register (R32_SERDESx_CTRL) (x=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													ALIGN	CONT	Reserved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TX_PWR_UP	RX_PWR_UP	Reserved						DMA_EN	TX_EN	RX_EN	RX_POLARITY	INT_BUS_YEN	RESET_PHY	RESET_LINK	CLR_ALL

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
18	ALIGN_EN	RW	Insert ALIGN signal when transmitting, in order to solve the frequency offset, high active.	1
17	CONT_EN	RW	The SYNC when transmission is idle is replaced by CONT and scrambling code data to reduce EMI.	0
16	Reserved	RO	Reserved	0
15	TX_PWR_UP	RW	Transmit module power-on enable bit: 1: Enable;	0

			0: Disable.	
14	RX_PWR_UP	RW	Receive module power-on enable bit: 1: Enable; 0: Disable.	0
[13:8]	Reserved	RO	Reserved	0x12
7	DMA_EN	RW	DMA enable: 1: Enable; 0: Disable.	0
6	TX_EN	RW	Transmit enable: 1: Enable; 0: Disable.	0
5	RX_EN	RW	Receive enable: 1: Enable; 0: Disable.	0
4	RX_POLARITY	RW	RXN/RXP signal exchange.	0
3	INT_BUSY_EN	RW	Auto-pause transmission enable bit before SERDES transmission completion interrupt flag is cleared: 1: Enable; 0: Disable.	0
2	RESET_PHY	RW	SERDES physical layer software reset control bit: 1: Enable; 0: Disable.	1
1	RESET_LINK	RW	SERDES link layer software reset control bit: 1: Enable; 0: Disable.	1
0	CLR_ALL	RW	Clear SERDES interrupt flag and FIFO: 1: Enable; 0: Disable.	1

20.3.2 SERDESx Interrupt Enable Register (R32_SERDESx_INT_EN) (x=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										COM INIT_ IE	Reser ved	FIFO _OV_ IE	REC V_D _ONE_ IE	TRA N_D ONE_ IE _REC V_ER _IE	PHY RDY_ IE

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	COMINIT_IE	RW	COMINIT receive interrupt enable: 1: Enable; 0: Disable.	0
4	Reserved	RO	Reserved	0
3	FIFO_OV_IE	RW	FIFO overflow interrupt enable: 1: Enable; 0: Disable.	0
2	RECV_DONE_IE	RW	Receive complete interrupt enable: 1: Enable; 0: Disable.	0
1	TRAN_DONE_IE RECV_ERR_IE	RW	Transmit mode: Transmit complete interrupt enabled: 1: Enable; 0: Disable. Receive mode: receive CRC error interrupt enable: 1: Enable; 0: Disable.	0
0	PHYRDY_IE	RW	Physical layer READY interrupt enable: 1: Enable; 0: Disable.	0

20.3.3 SERDESx Interrupt Flag/Status Register (R32_SERDESx_INT_FS) (x=1/2)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TX_SEQ_NUM[3:0]				RX_SEQ_NUM[3:0]				Reserved		R_FI FO_R DY	LINK FRE E	PLL LOC K	REC V_C RC OK	RX_S EQ_ MAT CH	PHY RDY
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										COM INIT IF	Reser ved	FIFO OV IF	REC V_D ONE IF	TRA N_D ONE IF REC V_ER R_IF	PHY RDY IF

Bit	Name	Access	Description	Reset value
[31:28]	TX_SEQ_NUM[3:0]	RO	LINK_INIT automatically cleared when valid.	0
[27:24]	RX_SEQ_NUM[3:0]	RO	In receiving mode, the received COMINT is automatically cleared.	0

[23:22]	Reserved	RO	Reserved	0
21	R_FIFO_RDY	RO	SERDES receive FIFO data ready status: 1: Receive FIFO is not empty; 0: Receive FIFO is empty.	0
20	LINK_FREE	RO	SERDES link layer transmits idle status bits.	0
19	PLL_LOCK	RO	PLL lock status bit, which starts to transmit and receive after it is 1.	0
18	RECV_CRC_OK	RO	The check status bit of the received data frame is detected after the RECV_DONE interrupt, and updated after the packet reception is completed.	0
17	RX_SEQ_MATCH	RO	Receive that matching status bit of SEQ_NUM.	0
16	PHYRDY	RO	Physical layer READY status bit.	0
[15:6]	Reserved	RO	Reserved	0
5	COMINIT_IF	RW1Z	COMINIT receives the interrupt flag bit.	0
4	Reserved	RO	Reserved	0
3	FIFO_OV_IF	RW1Z	FIFO overflow interrupt flag bit.	0
2	RECV_DONE_IF	RW1Z	Reception completion interrupt flag bit.	0
1	TRAN_DONE_IF RECV_ERR_IF	RW1Z	Transmit Mode: Transmit Complete Interrupt Flag. Receive Mode: Receive CRC Error Interrupt Flag.	0
0	PHYRDY_IF	RW1Z	Physical layer READY interrupt flag bit.	0

20.3.4 SERDESx Transceiver Controller (R32_SERDESx_RTX_CTRL) (x=1/2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													BUF MOD E	TX_ VLD	LINK _INIT
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SERDES_TX_LEN[15:0]															

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
18	BUF_MODE	RW	Dual buffered control bits: 1: Enable; 0: Disable.	0
17	TX_VLD	RW	The transmitted data packet is valid and cleared by software, and this bit returns to be synchronized with PHY_CLK.	0
16	LINK_INIT	RW	This bit is 1, and the LINK initialization packet is sent.	0

[15:0]	SERDES_TX_LEN [15:0]	RW	Number of data bytes transmitted, with the lower 2 bits fixed at 0(4-byte alignment)	x
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20.3.5 SERDESx Receive Length 0 Register (R32_SERDESx_RX_LEN0) (x=1/2)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SERDES_RX_LEN0[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	SERDES_RX_LEN0[15:0]	RO	The packet length of the last received data, with the least significant two bits fixed as 0. In non-dual-buffered mode, this indicates the current received packet length; In dual-buffered mode, this represents the packet length where RX_SEQ_NUM is even.	0

20.3.6 SERDESx Data 0 Buffer Register (R32_SERDESx_DATA0) (x=1/2)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SERDES_DATA0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SERDES_DATA0[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	SERDES_DATA0[31:0]	RW	SERDES data buffer 0 Transmission mode: the first 4Byte data sent and received by LINK is stored in this register, and the upper 4 bits are replaced by TX_SEQ_NUM, so users can use the lower 28bit as self-defined information. Receiving mode: the first 4Bytes of data packet are stored in this register, and the subsequent data are stored in DMA.	X

20.3.7 Start Address Register for DMA0 Buffer (R32_SERDESx_DMA0) (x=1/2)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SERDES_DMA0[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SERDES_DMA0[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	SERDES_DMA0[31:0]	RW	Starting address of buffer 0. In receive mode, the software needs to fix the lower 4 bits to 0(16 bytes aligned), while in transmit mode, 16 bytes aligned is not required. <i>Note:</i> 1. In single-buffer mode, only DMA0 is used; 2. In dual-buffer mode, DMA0 is used when TX_SEQ_NUM/RX_SEQ_NUM is even.	X

20.3.8 SERDESx Receive Length 1 Register (R32_SERDESx_RX_LEN1) (x=1/2)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SERDES_RX_LEN1[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	SERDES_RX_LEN1[15:0]	RO	The packet length of the last received data, with the least 2 bits fixed at 0, is valid only in double buffering mode. Indicates that RX_SEQ_NUM is an odd packet length.	0

20.3.9 SERDESx Data 1 Buffer Register (R32_SERDESx_DATA1) (x=1/2)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SERDES_DATA1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SERDES_DATA1[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	SERDES_DATA1[31:0]	RW	SERDES data buffer 1 Transmit mode: The first 4B of data transmitted by LINK is stored in this register. The upper 4 bits are replaced with TX_SEQ_NUM, while the lower 28 bits can be used as custom information.	X

			Receive mode: The first 4 bytes of the data packet are stored in this register, with subsequent data stored in the DMA. <i>Note: This buffer is used in dual-buffer mode when RX_SEQ_NUM is an odd number.</i>	
--	--	--	---	--

20.3.10 Start Address Register for DMA1 Buffer (R32_SERDESx_DMA1) (x=1/2)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SERDES_DMA1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SERDES_DMA1[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	SERDES_DMA1	RW	Starting address of buffer 0. In receive mode, the software needs to fix the lower 4 bits to 0 (16 bytes aligned). While in transmit mode, 16 bytes aligned is not required. <i>Note: In dual-buffer mode, DMA0 is used when TX_SEQ_NUM/RX_SEQ_NUM is even.</i>	X

20.3.11 System Configuration Register (R32_SYS_CFGR)

Offset address: 0x5003C010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	RX_FBCAP_ADJ [1:0]		RX_FBRES_ADJ [1:0]		Reserved	TX_DE_EMP HASIS [1:0]		TX_OUTPUT_SWING [1:0]		Reserved					

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RW	Reserved, the original value must be kept when writing.	0x151
[13:11]	RX_FBCAP_ADJ [1:0]	RW	Receiver feedback capacitor adjustment bit. For detailed parameters, refer to CH32H417DS0.	000b
[10:9]	RX_FBRES_ADJ [1:0]	RW	Receiver feedback resistance adjustment bit. For detailed parameters, refer to CH32H417DS0.	01b
[8:7]	Reserved	RW	Reserved, the original value must be kept when writing.	00b
[6:5]	TX_DE_EMPHASI	RW	Transmit pre-emphasis amplitude adjustment	01b

	S[1:0]		bit. For detailed parameters, refer to CH32H417DS0.	
[4:3]	TX_OUTPUT_SWING[1:0]	RW	Transmit amplitude adjustment bit. For detailed parameters, refer to CH32H417DS0.	01b
[2:0]	Reserved	RW	Reserved, the original value must be kept when writing.	011b

Note: It is recommended to adjust the value of this register to 0x542E7B for long lines.

Chapter 21 Inter-integrated Circuit (I2C) Interface

The internal integrated circuit bus (I2C) is widely used in the communication between microcontrollers, sensors and other off-chip modules. It supports multi-master and multi-slave mode, and can communicate at both 100kHz (standard) and 400kHz (fast) speeds using only 2 wires (SDA and SCL). I2C bus is also compatible with SMBus protocol. It not only supports I2C timing, but also supports arbitration, timing and DMA, and has CRC check function.

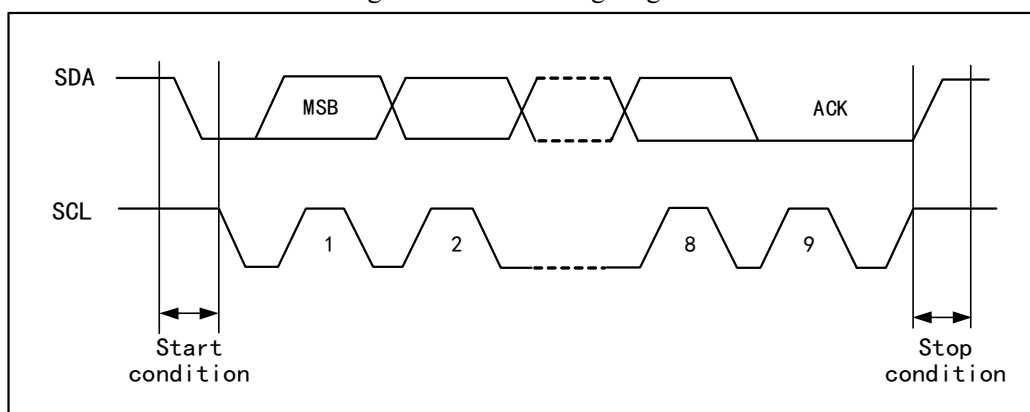
21.1 Main Features

- Master mode and slave mode
- 7-bit or 10-bit address
- Slave device supports dual 7-bit address.
- Two speed modes: 100kHz and 400kHz
- Multiple status modes, multiple error flags
- Support extended clock function
- 2 interrupt vectors
- Support DMA
- Support PEC
- Compatible with SMBus

21.2 Overview

I2C is a half-duplex bus, which can only run in one of the following four modes: master device sending mode, master device receiving mode, slave device sending mode and slave device receiving mode. The I2C module works in slave mode by default. After generating the starting condition, it will automatically switch to the master mode. When the arbitration is lost or a stop signal is generated, it will switch to the slave mode. The I2C module supports multi-host functions. When working in main mode, the I2C module will actively send out data and addresses. Data and addresses are transmitted in 8-bit units, with the high bit in front and the low bit in the back. After the start event is a byte (7-bit address mode) or two-byte (10-bit address mode) address. Every time the host sends 8-bit data or address, the slave needs to reply a reply ACK, that is, pull down the SDA bus, as shown in figure 21-1.

Figure 21-1 I2C timing diagram

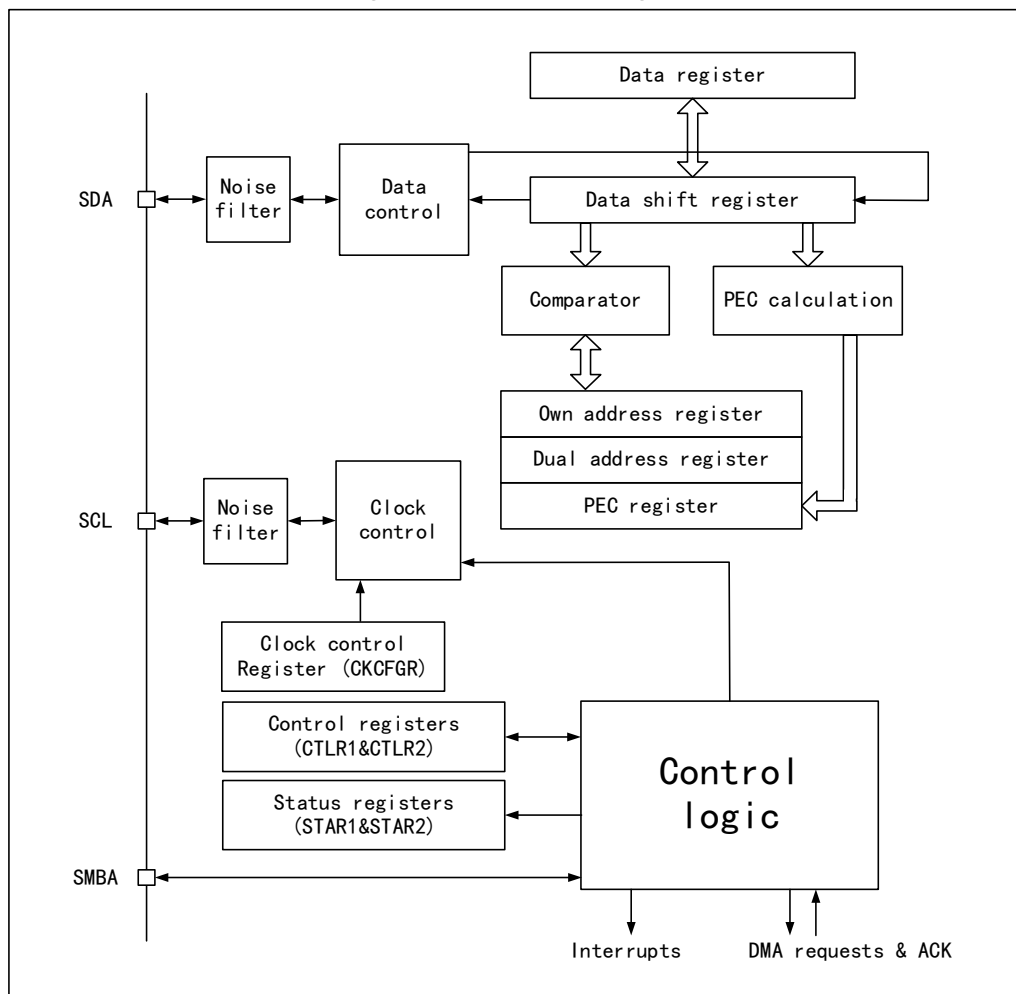


For normal use, the correct clock must be input to I2C. In the standard mode, the minimum input clock is 2MHz,

while the minimum input clock is 4MHz in the fast mode.

Figure 21-2 shows the block diagram of I2C.

Figure 21-2 I2C block diagram



21.3 Master Mode

In master mode, the I2C module leads the data transmission and outputs the clock signal. The data transfer starts with a Start event and ends with a Stop event. The following is the required operations in master mode:

Set the correct clock in the control register2 (R16_I2Cx_CTLR2) and the clock control register (R16_I2Cx_CKCFGR).

Set a proper rising edge in the rising edge register (R16_I2Cx_RTR).

Set the PE bit in R16_I2Cx_CTLR1 to start the peripheral.

Set the START bit in the control register (R16_I2Cx_CTLR1) to generate a start event.

After the START bit is set, the I2C module automatically switches to master mode, the MSL bit is set, and a Start event is generated. After the start event is generated, the SB bit is set. If the ITEVTEN bit (in R16_I2Cx_CTLR2) is set, an interrupt is generated. In this case, it is needed to read the R16_I2Cx_STAR1 register. After the slave address is written to the data register, the SB bit is automatically cleared.

If the 10-bit address mode is enabled, then write the data register to send the header sequence (the header sequence is 11110xx0b, of which the xx bits are the highest 2 bits of the 10-bit address).

After the header sequence is transmitted, the ADD10 bit in the status register is set. If the ITEVTEN bit is already set, an interrupt will be generated. At this time, read the R16_I2Cx_STAR1 register and write the second address byte to the data register. Then, clear the ADD10 bit.

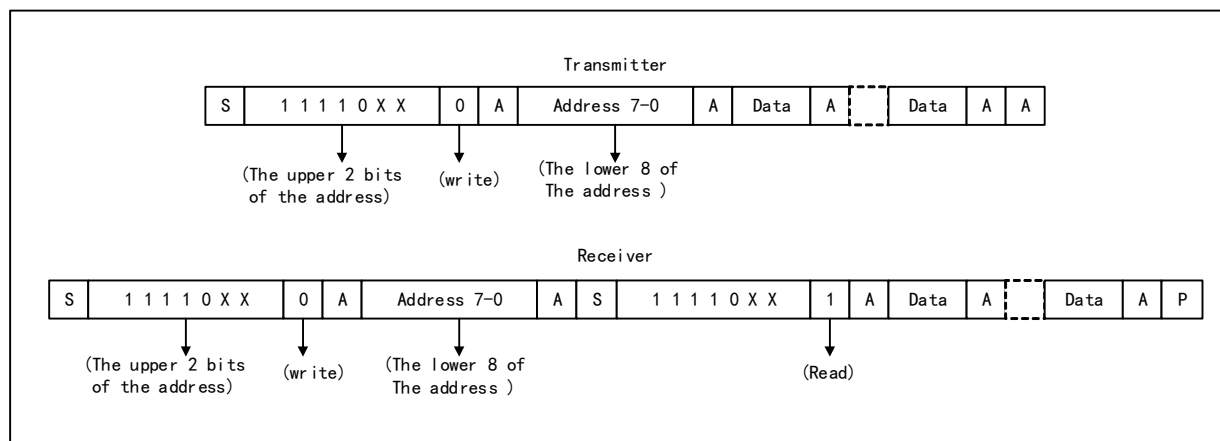
Then, write the data register to send the second address byte. After sending the second address byte, the ADDR bit in the status register is set. If the ITEVTEN bit has been set, an interrupt will be generated. Read the R16_I2Cx_STAR2 register at this time and then read R16_I2Cx_STAR1 register again to clear the ADDR bit;

If the 7-bit address mode is enabled, the write data register transmit the address byte. After the address byte is sent, the ADDR bit in the status register is set. If the ITEVTEN bit has been set, an interrupt will be generated. Read the R16_I2Cx_STAR1 register and then read the R16_I2Cx_STAR2 register again to clear the ADDR bit.

In the 7-bit address mode, the first byte sent is the address byte, the first 7 bits represent the address of the target slave device, the 8th bit determines the direction of the subsequent message, and '0' means the master device writes data to the slave device, '1' means that the master device reads information from the slave device.

In the 10-bit address mode, as shown in Figure 21-3, the first byte is 11110xx0, xx are the highest 2 bits of the 10-bit address, and the second byte is the lower 8 bits of the 10-bit address in the address transmission phase. If you subsequently enter the master transmitter mode, send data continuously. If the device enters the master receiver mode subsequently, a start event needs to be re-sent, and a byte of 11110xx1 will be sent together. Then, enter the master receiver mode.

Figure 21-3 Master receive/transmits data in 10-bit address mode

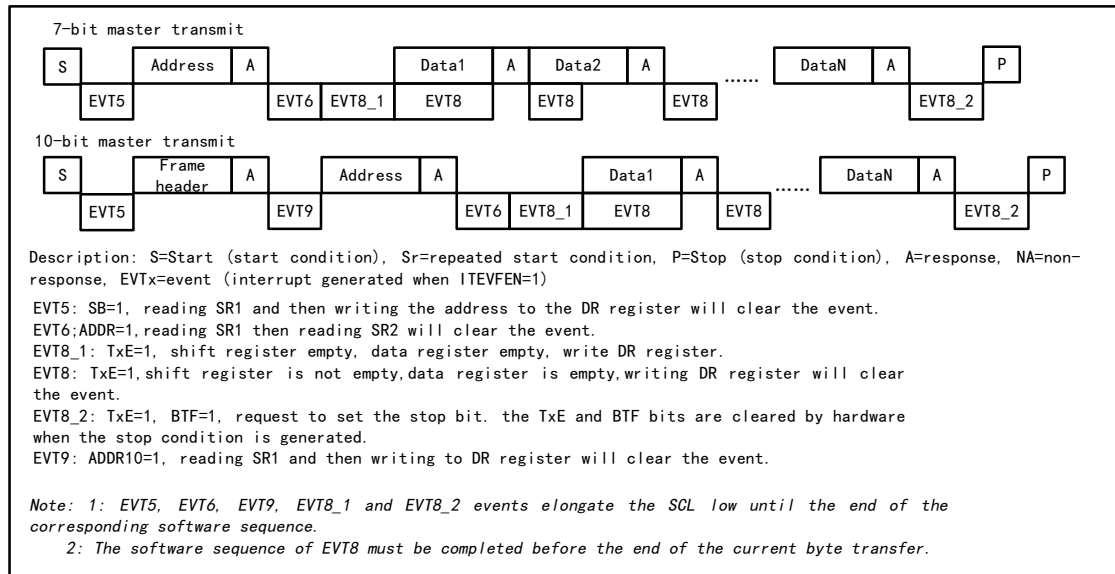


Master transmit mode:

The master device's internal shift register sends data from the data register to the SDA line. When the master device receives an ACK, TxE in status register 1 (R16_I2Cx_STAR1) is set, and an interrupt is also generated if ITEVTEN and ITBUFEN are set. Writing data to the data register will clear the TxE bit.

If the TxE bit is set and no new data was written to the data register before the last data was sent, then the BTF bit will be set and SCL will remain low until it is cleared, and writing data to the data register after reading R16_I2Cx_STAR1 will clear the BTF bit.

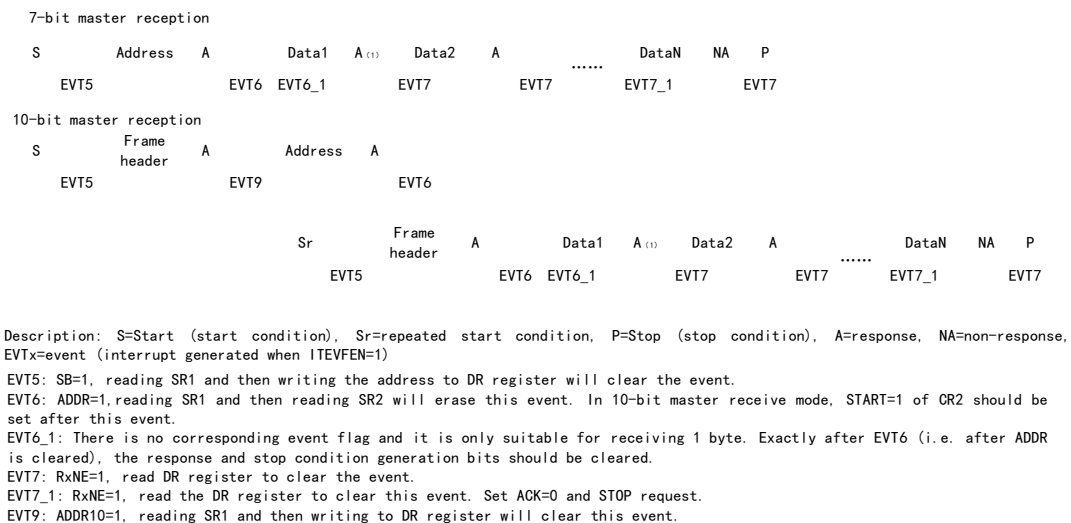
Figure 21-4 Master transmitter transmission sequence diagram



Master receive mode:

The I2C module will receive data from the SDA line and write it into the data register via a shift register. After each byte, if the ACK bit is set, then the I2C module will send an answer low, and the RxNE bit will be set, and an interrupt will be generated if ITEVTEN and ITBUFEN are set. If RxNE is set and the original data is not read before the new data is received, then the BTF bit will be set and SCL will remain low until the BTF is cleared. Reading R16_I2Cx_STAR1 and then reading the data register will clear the BTF bit.

Figure 21-5 Receiver transmission sequence diagram



When the master device ends sending data, it will actively send an end event, i.e. set the STOP bit, and the I2C will switch to slave mode. In receive mode, the master device needs to NAK at the answer position of the last data bit, and after receiving NACK, the slave device releases control of the SCL and SDA lines; the master device can then send a stop/restart condition. Note that the I2C module will automatically switch to slave mode after the stop condition is generated.

21.4 Slave Mode

In slave mode, the I2C module can recognize its own address and the broadcast call address. The software can

control the identification of broadcast call addresses on or off. Once the start event is detected, the I2C module compares the SDA data through the shift register with its own address (the number of bits depends on ENDUAL and ADDMODE) or the broadcast address (when ENGC is set). If there is a mismatch, it will be ignored until a new start event is generated. If it matches the header sequence, an ACK signal is generated and waits for the address of the second byte; if the address of the second byte also matches, or if the address of the whole segment matches in the case of a 7-bit address, then:

First an ACK reply is generated; the ADDR bit is set, and if the ITEVTEN bit is set, there will be a corresponding interrupt.

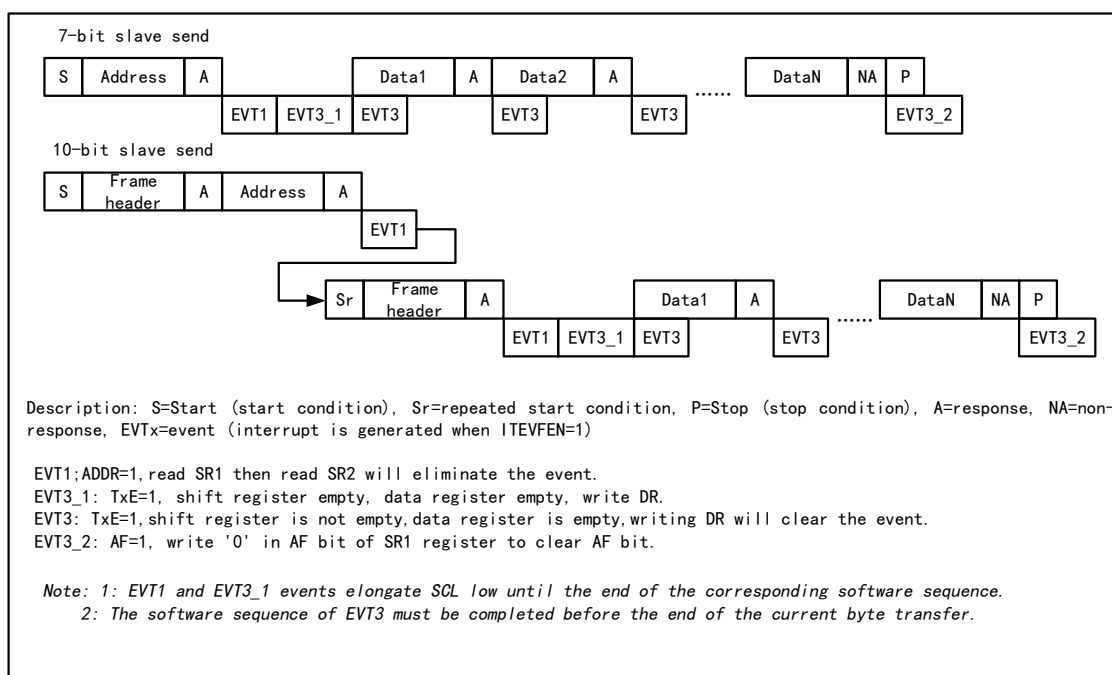
If you are using dual-address mode (the ENDUAL bit is set), you also need to read the DUALF bit to determine which address the host is calling.

The slave mode defaults to receive mode, and when the last bit of the received header sequence is 1, or the last bit of the 7-bit address is 1 (depending on whether the header sequence is received for the first time or a normal 7-bit address), the I2C module will enter sender mode, and the TRA bit will indicate whether it is currently in receiver or sender mode.

Slave transmit mode:

After the ADDR bit is cleared, the I2C module sends bytes from the data register to the SDA line via a shift register. The slave device holds SCL low until the ADDR bit is cleared and the data to be sent has been written to the data register. (See EVT1 and EVT3 in the figure below). Upon receipt of an answer ACK, the TxE bit will be set and an interrupt will be generated if ITEVTEN and ITBUFEN are set. The BTF bit will be set if TxE is set but no new data is written to the data register before the end of the next data transmission. SCL will remain low until the BTF is cleared. Reading status register 1 (R16_I2Cx_STAR1) and then writing data to the data register will clear the BTF bit.

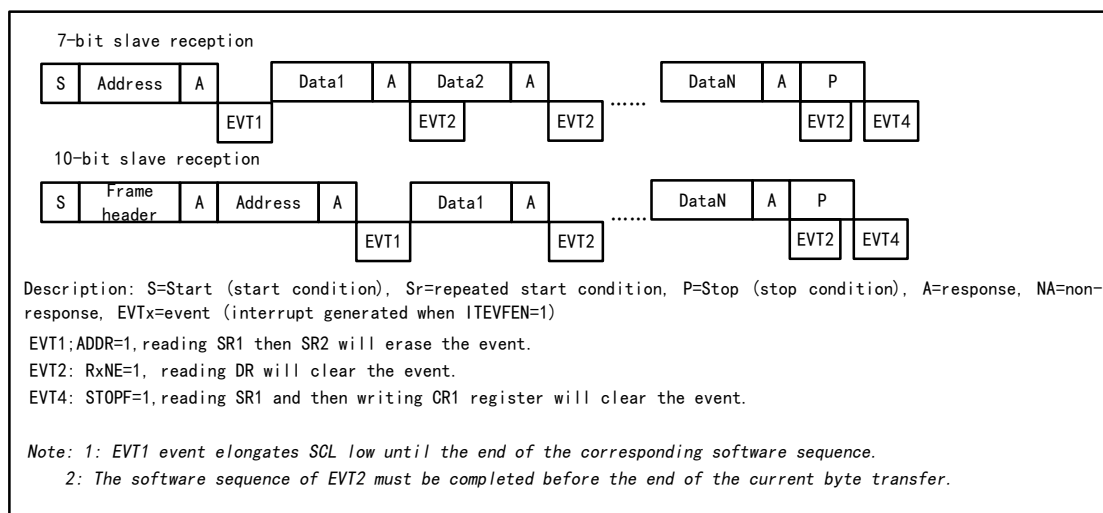
Figure 21-6 Slave transmitter transmission sequence diagram



Slave receive mode:

After ADDR is cleared, the I2C module stores the data on SDA into the data register via a shift register. After every byte received, the I2C module sets an ACK bit and sets the RxNE bit. If ITEVTEN and ITBUFEN are set, an interrupt is also generated. If RxNE is set and the old data is not read before the new data is received, then BTF is set. SCL will remain low until the BTF bit is cleared. Reading status register 1 (R16_I2Cx_STAR1) and reading the data in the data register will clear the BTF bit.

Figure 21-7 Receiver transmission sequence diagram



The master device will generate a stop condition after the last data byte is transferred. When the I2C module detects a stop event, it will set the STOPF bit, and if the ITEVFEN bit is set, it will also generate an interrupt. The user needs to read the status register (R16_I2Cx_STAR1) and then write the control register (e.g. reset control word SWRST) to clear it. (See EVT4 in the figure above).

21.5 Error

21.5.1 Bus Error (BERR)

When the I2C module detects an external start or stop event during address or data transmission, a bus error occurs. When a bus error is generated, the BERR bit is set, and an interrupt occurs if ITERREN is set. In slave mode, the data is discarded and the hardware releases the bus. If it is a start signal, the hardware will consider it as a restart signal and start waiting for the address or stop signal; if it is a stop signal, it will operate according to the normal stop condition in advance. In main mode, the hardware does not release the bus and does not affect the current transmission, and it is up to the user code to decide whether to abort the transmission.

21.5.2 Acknowledge Failure (AF)

When the I2C module detects no response after a byte, a reply error occurs. When a reply error is generated: AF will be set, and an interrupt will be generated if ITERREN is set; if an AF error is encountered, if the I2C module works in slave mode, the hardware must release the bus, and if it is in master mode, the software must generate a stop event.

21.5.3 Arbitration Lost (ARLO)

When the I2C module detects the arbitration loss, an arbitration loss error occurs. When an arbitration loss error occurs: the ARLO bit is set and an interrupt is generated if ITERREN is set; the I2C module switches to slave mode

and no longer responds to slave-initiated transfers against it unless a host initiates a new start event; the hardware releases the bus.

21.5.4 Overrun/Underrun Error (OVR)

- Overload error:

In slave mode, if clock extension is prohibited, the I2C module is receiving data. If one byte of data has been received, but the last received data has not been read out, an overload error will occur. When an overload error occurs, the last received byte is discarded and the sender should resend the last sent byte.

- Underload error:

In slave mode, if clock extension is prohibited, the I2C module is sending data, and if new data is not written to the data register before the next byte clock arrives, an underload error will occur. When an underload error occurs, the data in the previous data register will be sent twice. If an underload error occurs, the receiver should discard the repeatedly received data. In order not to generate underload errors, the I2C module should write data to the data register before the first rising edge of the next byte.

21.6 Clock Extension

If clock extension is prohibited, there is a possibility of overload / underload errors. But if the clock is extended when enabled:

- In transmit mode, if TxE is set and BTF is set, SCL will always be low, waiting for the user to read the status register and write the data to be sent to the data register.
- In receive mode, if RxNE is set and BTF is set, the SCL will remain low after receiving the data until the user reads the status register and reads the data register.

Thus it can be seen that the extension of the enable clock can avoid overload/underload errors.

21.7 SMBus

SMBus is also a 2-wire interface, which is generally used between the system and power management. SMBus and I2C have many similarities. For example, SMBus uses the same 7-bit address mode as I2C.

Similarities between SMBus and I2C:

- 1) Master-slave communication mode; the host provides the clock and supports multiple masters and multiple slaves;
- 2) 2-wire communication structure, of which a warning line can be selected for SMBus;
- 3) Support 7-bit address format.

Differences between SMBus and I2C:

- 1) I2C supports the maximum speed of 400 KHz, while SMBus supports the maximum speed of 100 KHz, and SMBus has the minimum speed limit of 10 KHz;
- 2) When the SMBus clock is lower than 35mS, it will report a timeout, but I2C has no such limitation;
- 3) SMBus has a fixed logic level, but I2C does not, depending on VDD;
- 4) SMBus has a bus protocol, but I2C does not.

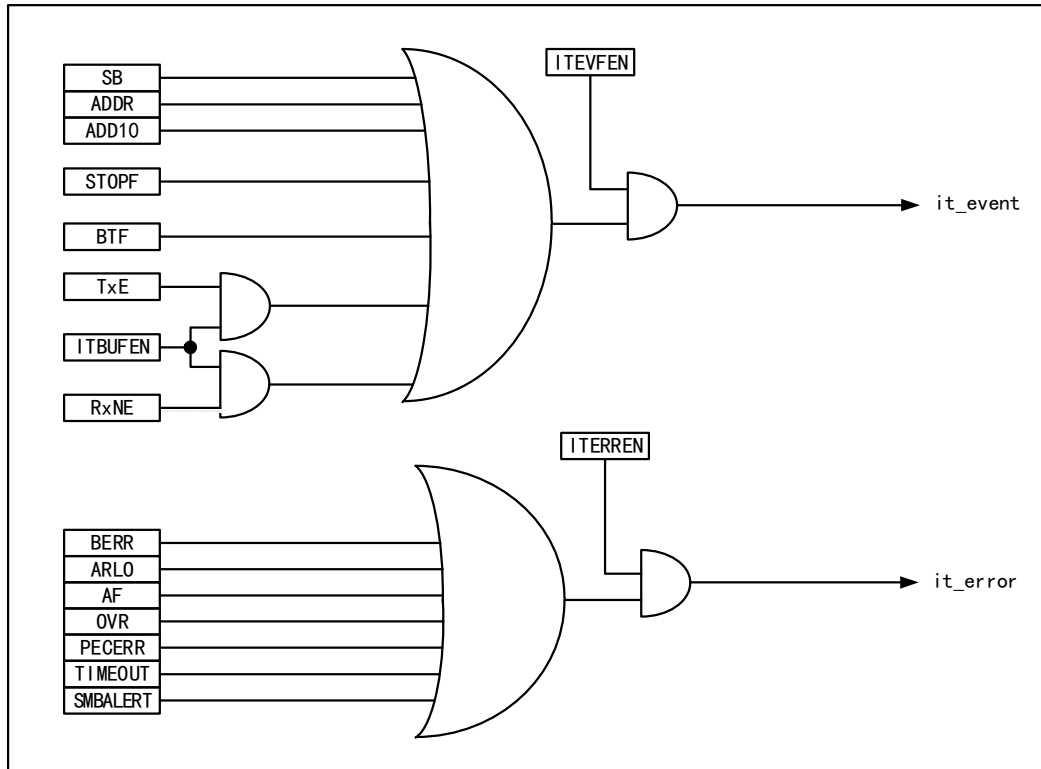
SMBus also includes device identification, address resolution protocol, unique device identifier, SMBus reminder, and various bus protocols. For details, please refer to SMBus specification version 2.0. When SMBus is used, only the SMBus bit of the control register needs to be set, and the SMBTYPE and ENAARP bits need to be configured

as needed.

21.8 Interrupt

Each I2C module has 2 interrupt vectors: event interrupt and error interrupt. 2 types of interrupts support the interrupt sources as shown in Figure 21-8.

Figure 21-8 I2C interrupt request



21.9 DMA

DMA can be used to receive/transmit bulk data. When DMA is used, the ITBUFEN bit of the control register cannot be set.

● DMA is used for transmission

The DMA mode can be activated by setting the DMAEN bit in the CTLR2 register. As long as the TxE bit is set, the data can be loaded into the I2C data register from the set memory by DMA. The following settings are required to allocate channels for I2C.

- 1) Set the I2Cx_DATAR register address to the DMA_PADDRx register, and set the memory address in the DMA_MADDRx register, so that the data will be sent from the memory to the I2Cx_DATAR register after each TxE event.
- 2) Set the required number of transferred bytes in the DMA_CNTRx register. After each TxE event, this value will be reduced progressively.
- 3) The channel priority is configured using the PL[0:1] bits in the DMA_CFGRx register.
- 4) Set the DIR bit in the DMA_CFGRx register, and it can be configured to issue an interrupt request according to application requirements when the entire transmission is half or wholly completed.

5) Activate the channel by setting the EN bit in the DMA_CFGRx register.

When the number of data transfer set in the DMA controller has been completed, the DMA controller will send an EOT/EOT_1 signal indicating the end of the transmission to the I2C interface. When the interrupt is allowed, a DMA interrupt will be generated.

- DMA is used for reception

After the DMAEN bit in the CTLR2 register is set, DMA receiver mode can be started. When DMA is used for reception, DMA transfers the data in the data register to the preset memory area. The following steps are required to allocate channels for I2C.

- 1) Set the I2Cx_DATAR register address to the DMA_PADDRx register, and set the memory address in the DMA_MADDRx register, so that the data will be written into the memory from the I2Cx_DATAR register after each RxNE event.
- 2) Set the required number of transferred bytes in the DMA_CNTRx register. After each RxNE event, this value will be reduced progressively.
- 3) The channel priority is configured by the PL[0:1] bits in the DMA_CFGRx register.
- 4) Clear the DIR bit in the DMA_CFGRx register, and it can be configured to issue an interrupt request according to application requirements when the data transmission is half or wholly completed.
- 5) Activate the channel by setting the EN bit in the DMA_CFGRx register.

When the number of data transfer set in the DMA controller has been completed, the DMA controller will send an EOT/EOT_1 signal indicating the end of the transmission to the I2C interface. When the interrupt is allowed, a DMA interrupt will be generated.

21.10 Packet Error Checking

Packet error checking (PEC) is a CRC8 check step added to provide the transmission reliability. Each bit of serial data can be calculated through the following polynomial:

$$C=X^8+X^2+X+1$$

PEC calculation is activated by the ENPEC bit in the control register, and all information bytes are calculated, including address and read/write bits. During transmission, enabling PEC will add a byte of CRC8 calculation result after the last byte of data. While in receiver mode, the last byte is considered to be the CRC8 check result. If it does not match the internal calculation result, it will reply with a NAK. For the master receiver, it will reply with a NAK regardless of whether the check result is correct or not.

21.11 Debug Mode

After the system enters the debug mode, the DBG_I2Cx_SMBUS_TIMEOUT bit in the DEBUG module can be used to determine whether to continue operating or stop the time-out control of I2CSMBus.

21.12 Register Description

Table 21-1 I2C1 registers

Name	Access address	Description	Reset value
R16_I2C1_CTLR1	0x40005400	I2C1 control register 1	0x0000
R16_I2C1_CTLR2	0x40005404	I2C1 control register 2	0x0000
R16_I2C1_OADDR1	0x40005408	I2C1 address register 1	0x0000
R16_I2C1_OADDR2	0x4000540C	I2C1 address register 2	0x0000
R16_I2C1_DATAR	0x40005410	I2C1 data register	0x0000
R16_I2C1_STAR1	0x40005414	I2C1 status register 1	0x0000
R16_I2C1_STAR2	0x40005418	I2C1 status register 2	0x0000
R16_I2C1_CKCFGR	0x4000541C	I2C1 clock register	0x0000
R16_I2C1_RTR	0x40005420	I2C1 rise time register	0x0002

Table 21-2 I2C2 registers

Name	Access address	Description	Reset value
R16_I2C2_CTLR1	0x40005800	I2C2 control register 1	0x0000
R16_I2C2_CTLR2	0x40005804	I2C2 control register 2	0x0000
R16_I2C2_OADDR1	0x40005808	I2C2 address register 1	0x0000
R16_I2C2_OADDR2	0x4000580C	I2C2 address register 2	0x0000
R16_I2C2_DATAR	0x40005810	I2C2 data register	0x0000
R16_I2C2_STAR1	0x40005814	I2C2 status register 1	0x0000
R16_I2C2_STAR2	0x40005818	I2C2 status register 2	0x0000
R16_I2C2_CKCFGR	0x4000581C	I2C2 clock register	0x0000
R16_I2C2_RTR	0x40005820	I2C2 rise time register	0x0002

Table 21-3 I2C3 registers

Name	Access address	Description	Reset value
R16_I2C3_CTLR1	0x40005C00	I2C3 control register 1	0x0000
R16_I2C3_CTLR2	0x40005C04	I2C3 control register 2	0x0000
R16_I2C3_OADDR1	0x40005C08	I2C3 address register 1	0x0000
R16_I2C3_OADDR2	0x40005C0C	I2C3 address register 2	0x0000
R16_I2C3_DATAR	0x40005C10	I2C3 data register	0x0000
R16_I2C3_STAR1	0x40005C14	I2C3 status register 1	0x0000
R16_I2C3_STAR2	0x40005C18	I2C3 status register 2	0x0000
R16_I2C3_CKCFGR	0x40005C1C	I2C3 clock register	0x0000
R16_I2C3_RTR	0x40005C20	I2C3 rise time register	0x0002

Table 21-4 I2C4 registers

Name	Access address	Description	Reset value
R16_I2C4_CTLR1	0x40014000	I2C4 control register 1	0x0000

R16_I2C4_CTLR2	0x40014004	I2C4 control register 2	0x0000
R16_I2C4_OADDR1	0x40014008	I2C4 address register 1	0x0000
R16_I2C4_OADDR2	0x4001400C	I2C4 address register 2	0x0000
R16_I2C4_DATAR	0x40014010	I2C4 data register	0x0000
R16_I2C4_STAR1	0x40014014	I2C4 status register 1	0x0000
R16_I2C4_STAR2	0x40014018	I2C4 status register 2	0x0000
R16_I2C4_CKCFGR	0x4001401C	I2C4 clock register	0x0000
R16_I2C4_RTR	0x40014020	I2C3 rise time register	0x0002

21.12.1 I2C Control Register (I2Cx_CTLR1) (x=1/2/3/4)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
15	SWRST	RW	Software reset. Setting this bit by user code will reset the I2C peripheral. Before reset, make sure that the pins of the I2C bus are released and the bus is idle. <i>Note: This bit can reset the I2C module when no stop condition is detected on the bus but the busy bit is 1.</i>	0
14	Reserved	RO	Reserved.	0
13	ALERT	RW	SMBus alert. This bit can be set or cleared by the user code. When PE is set, this bit can be cleared by hardware. 1: Drive the SMBusALERT pin to make it low, and the response address header shall closely follow the ACK signal; 0: Release the SMBusALERT pin to make it high, and the response address header shall closely follow the NACK signal.	0
12	PEC	RW	Packet error check enable. Set this bit to enable data packet error detection. This bit can be set or cleared by the user code. When the PEC is transmitted, or a start or end signal is generated, or the PE bit is cleared to 0, the bit can be cleared by hardware. 1: Provided with PEC; 0: Not provided with PEC. <i>Note: PEC will fail when the arbitration is lost.</i>	0
11	POS	RW	ACK and PEC position setting. This bit can be set and cleared by user code, and it can be cleared by hardware after PE is cleared; 1: The ACK bit controls the ACK or NAK of the next byte received in the shift register. The next byte received in the PEC shift register is PEC; 0: The ACK bit controls the ACK or NAK of the byte currently being received in the shift register. The PEC bit indicates that the byte of the current shift register	0

			is PEC. <i>Note: The usage of POS bit in 2-byte data reception is as follows: It must be configured before receiving. For the second byte of NACK, the ACK bit must be cleared immediately after the ADDR bit is cleared; in order to detect the PEC of the second byte, the PEC bit must be set after the ADDR event occurs following the POS bit.</i>	
10	ACK	RW	Acknowledge enable. This bit can be set or cleared by user code. When PE bit is set, this bit can be cleared by hardware; 1: Acknowledge returned after a byte is received; 0: No acknowledge is returned.	0
9	STOP	RW	Stop event generation. It can be set or cleared by user code, or cleared by hardware when a stop event is detected, or set by hardware when a timeout error is detected. In master mode: 1: A stop event is generated after the current byte transfer or the current start condition is issued; 0: No stop event occurs. In slave mode: 1: Release the SCL and SDA lines after the current byte transfer; 0: No stop event occurs.	0
8	START	RW	Start event generation. This bit can be set or cleared by the user code. When the start condition is issued or PE is cleared, it can be cleared by hardware. In master mode: 1: A start event is generated repeatedly; 0: No start event is generated. In slave mode: 1: When the bus is idle, a start event is generated; 0: No start event is generated.	0
7	NOSTRETCH	RW	Clock stretching disable. This bit is used to disable clock stretching in slave mode when the ADDR or BTF flag bit is set until it is cleared by software. 1: Clock stretching disabled; 0: Clock stretching enabled.	0
6	ENGCG	RW	General call enable. Set this bit to enable the general call, and respond to general address 00h.	0
5	ENPEC	RW	PEC enable. 1: Enable PEC calculation; 0: Disable PEC calculation.	0

4	ENARP	RW	ARP enable. 1: Enable ARP; 0: Disable ARP. If SMBTYPE=0, the default address of the SMBus device is used. If SMBTYPE=1, the main address of the SMBus is used.	0
3	SMBTYPE	RW	SMBus device type: 1: SMBus master device; 0: SMBus slave device.	0
2	Reserved	RO	Reserved.	0
1	SMBUS	RW	SMBus mode selection. 1: SMBus mode; 0: I2C mode.	0
0	PE	RW	I2C peripheral enable. 1: I2C module enabled; 0: I2C module disabled.	0

21.12.2 I2C Control Register 2 (I2Cx_CTLR2) (x=1/2/3/4)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:13]	Reserved	RO	Reserved	0
12	LAST	RW	Last transfer setting of DMA. 1: Next DMA EOT is the last transfer; 0: Next DMA EOT is not the last transfer. <i>Note: This bit is used in master receiver mode and can generate a NAK when the data is received at the last time.</i>	0
11	DMAEN	RW	DMA request enable. 1: Enable DMA request when TxEN or RxEN bit is set; 0: Disable DMA request when TxEN or RxEN bit is set.	0
10	ITBUFEN	RW	Buffer interrupt enable. 1: When the TxEN bit is set, or when the RxEN bit is set, an event interrupt is generated; 0: When the TxEN bit is set, or when the RxEN bit is set, no interrupt is generated.	0
9	ITEVTEN	RW	Event interrupt enable. 1: Enable event interrupt; 0: Disable event interrupt. Under the following conditions, the interrupt can be generated: SB=1 (master mode); ADDR=1(master and slave modes);	0

			ADDR10=1 (master mode); STOPF=1 (slave mode); BTF=1, but no TxE or RxEN event occurs; If ITBUFEN=1, TxE event is 1; If ITBUFEN=1, RxNE event is 1.	
8	ITERREN	RW	Error interrupt enable. 1: The error interrupt is enabled; 0: The error interrupt is disabled Under the following conditions, the interrupt can be generated: BERR=1; ARLO=1; AF=1; OVR=1; PECERR=1; TIMEOUT=1; SMBAlert=1.	0
[7:6]	Reserved	RO	Reserved.	0
[5:0]	FREQ[5:0]	RW	I2C module clock frequency domain, must input the correct clock frequency to produce the correct timing, the allowable range is between 4-60MHz. Must be set between 000100b and 111100b in MHz.	0

21.12.3 I2C Address Register 1 (I2Cx_OADDR1) (x=1/2/3/4)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
15	ADDMODE	RW	Addressing mode. 1: 10-bit slave address (7-bit address not acknowledged); 0: 7-bit slave address (10-bit address not acknowledged).	0
[14:10]	Reserved	RO	Reserved.	0
[9:8]	ADD[9:8]	RW	Interface address, which is 9-8 bits when using 10-bit addresses and ignored when 7-bit addresses are used.	0
[7:1]	ADD[7:1]	RW	Interface address, bit 7-1.	0
0	ADD0	RW	Interface address. Bit0 when a 10-bit address is used. It is ignored when a 7-bit address is used.	0

21.12.4 I2C Address Register 2 (I2Cx_OADDR2) (x=1/2/3/4)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:1]	ADD2[7:1]	RW	Interface address. Bit7 to bit1 of address in dual-address mode.	0
0	ENDUAL	RW	Dual addressing mode enable. When this bit is set, the ADD2 can be identified.	0

21.12.5 I2C Data Register (I2Cx_DATAR) (x=1/2/3/4)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
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[15:8]	Reserved	RO	Reserved.	0
[7:0]	DR[7:0]	RW	Data register. This domain is used to store received data or store data to be transmitted to the bus.	0

21.12.6 I2C Status Register 1 (I2Cx_STAR1) (x=1/2/3/4)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
15	SMBALERT	RW0	SMBus alert. It can be reset by user writing 0, or reset by hardware when PE becomes low. In master mode of SMBus: 1: SMBus alert is generated on the pin; 0: No SMBus alert. In slave mode of SMBus: 1: SMBAlert response address header to SMBAlert LOW received; 0: No SMBAlert response address header.	0
14	TIMEOUT	RW0	Timeout or Tlow error flag. It can be reset by user writing 0, or reset by hardware when PE becomes low. 1: SCL has been at low level for 25ms, or the accumulated time of host low level exceeds 10ms, or the accumulated time of device low level exceeds 25ms; 0: No timeout error. <i>Note: When this bit is set in slave mode, the slave device resets the communication and the hardware releases the bus. When this bit is set in master mode, hardware issues a stop condition.</i>	0
13	Reserved	RO	Reserved	0
12	PECERR	RW0	PEC error flag occurs during reception. This bit can be reset by user writing 0, or reset by hardware when PE becomes low. 1: PEC error. After PEC is received, NAK is returned; 0: No PEC error.	0
11	OVR	RW0	Overrun and underrun flag. 1: Overrun or underrun event occurs: In case of NOSTRETCH=1, when a new byte is received in the receiver mode and the content in the data register has not been read, the newly received byte will be lost. In the transmitter mode, no new data is written into the data register, and the same byte will be sent twice; 0: No overrun and underrun event.	0

10	AF	RW0	Acknowledge failure flag. This bit can be reset by user writing 0, or reset by hardware when PE becomes low. 1: Acknowledge error; 0: Normal acknowledge.	0
9	ARLO	RW0	Arbitration lost flag. It can be reset by user writing 0, or reset by hardware when PE becomes low. 1: Arbitration lost is detected and the module loses control of the bus; 0: Normal arbitration.	0
8	BERR	RW0	Bus error flag. It can be reset by user writing 0, or reset by hardware when PE becomes low. 1: Start or stop condition error; 0: Normal.	0
7	TxE	RO	Data register empty flag, which can be cleared by writing data to the data register, or it is automatically cleared by hardware after a start or stop bit is generated, or when PE is 0. 1: When the data is transmitted, the transmit data register is empty; 0: The data register is non-empty.	0
6	RxNE	RO	The data register is not an empty log bit, which will be cleared by reading and writing to the data register, or by the hardware when PE is 0. 1: When receiving data, the data register is not empty. 0: Normal.	0
5	Reserved	RO	Reserved.	0
4	STOPF	RO	Stop event flag. After the user reads the status register1, writing to the control register1 will clear this bit, or when PE is 0, the hardware will clear this bit. 1: After the response, the slave device will detect a stop event on the bus; 0: No stop event is detected.	0
3	ADD10	RO	10-bit address header sent flag. After the user reads the status register1, writing to the control register1 will clear this bit, or when PE is 0, the hardware will clear this bit. 1: In 10-bit address mode, the master device has sent the first address byte; 0: None	0
2	BTF	RO	Byte transmission end flag. After the user reads the status register1, reading and writing to the data	0

			register will clear this bit. During transmission, after a start or stop event is initiated, or when PE is 0, this bit will be cleared by hardware. 1: Byte transmission completed. In case of NOSTRETCH=0: when a new data is sent and the data register has not been written with new data during transmission; when a new byte is received but the data register has not been read; 0: None	
1	ADDR	RW0	Address transmitted/matched flag. After the user reads the status register 1, the read operation of the status register2 will clear this bit, or when PE is 0, the hardware will clear this bit. Master mode: 1: End of address transmission: In the 10-bit address mode, it is set after receiving the ACK of the second byte of the address; in 7-bit address mode, the bit will be set after the ACK of the address is received; 0: The address transmission is not finished. Slave mode: 1: The received address matches; 0: The address does not match or no address is received.	0
0	SB	RO	Start bit transmission flag. After reading the status register 1, the operation of writing the data register will clear this bit, or when PE is 0, the hardware will clear this bit. 1: The start bit has been transmitted; 0: The start bit has not been transmitted.	0

21.12.7 I2C Status Register 2 (I2Cx_STAR2) (x=1/2/3/4)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[15:8]	PEC[7:0]	RO	Packet error checking. When PEC is enabled (ENPEC is set), this domain stores the value of PEC.	0
7	DUALF	RO	Matched detection flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit. 1: The received address matched with OAR2; 0: The received address matched with OAR1.	0
6	SMBHOST	RO	SMBus host header flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit.	0

			1: When SMBTYPE=1 and ENARP=1, the SMBus host address will be received; 0: SMBus host address is not received.	
5	SMBDEFAULT	RO	SMBus device default address flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit. 1: When ENARP=1, the default address of the SMBus device is received; 0: No address is received.	0
4	GENCALL	RO	General call address flag. When the stop bit or start bit is generated, or when PE=0, the hardware will clear this bit. 1: When ENGC=1, the address of general call is received; 0: No general call address is received.	0
3	Reserved	RO	Reserved.	0
2	TRA	RO	Transmitter/receiver flag, cleared by hardware when a stop event (STOPF=1) is detected, repeated start condition or bus arbitration is lost (ARLO=1) or PE=0. 1: Data has been sent; 0: Data is received. This bit is determined by R/W bit of address byte.	0
1	BUSY	RO	Bus busy flag. This bit is cleared when a stop bit is detected. When the interface is disabled (PE=0), the information is still updated. 1: Busy bus: SDA or SCL has a low level; 0: The bus is idle and does not have communication.	0
0	MSL	RO	Master/slave mode indication. When the interface is in master mode (SB=1), the hardware will set this bit. When the bus detects a stop bit and the arbitration is lost, or PE=0, the hardware will clear this bit.	0

21.12.8 I2C Clock Register (I2Cx_CKCFGR) (x=1/2/3/4)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
15	F/S	RW	Master mode selection. 1: Fast mode; 0: Standard mode.	0
14	DUTY	RW	Duty cycle in the fast mode: 1: $T_{Low\ level}/T_{High\ level}=16/9$; 0: $T_{Low\ level}/T_{High\ level}=2$.	0
[13:12]	Reserved	RO	Reserved.	0
[11:0]	CCR[11:0]	RW	Clock frequency division factor. These bits	0

			determine the frequency waveform of the SCL clock.	
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21.12.9 I2C Rise Time Register (I2Cx_RTR) (x=1/2/3/4)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[15:6]	Reserved	RO	Reserved	0
[5:0]	TRISE[5:0]	RW	Maximum rise time domain. The rise time of SCL in master mode is set at this bit. The maximum rising edge time is equal to TRISE-1 clock cycle. This bit can only be set when PE is cleared. If the input clock cycle of the IIC module is 125nS and the value of TRISE is 9h, then the maximum rising edge time is (9-1)*125nS, i.e., 1000nS.	000010b

Chapter 22 I3C Bus (I3C)

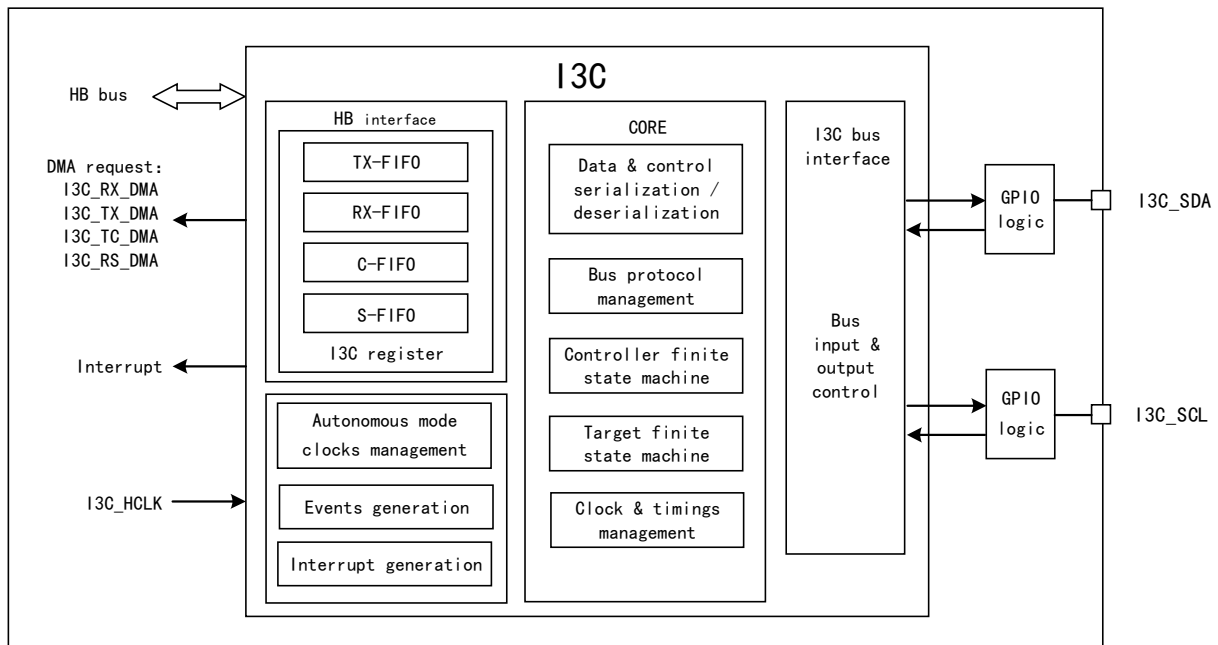
The I3C bus is a 2-wire serial single-ended multi-drop bus designed to enhance the traditional I2C bus. The I3C interface handles communication between this device and other devices connected to the I3C bus. It supports operation as both a master and a slave device. When functioning as a master, it extends the capabilities of the I2C interface while maintaining a degree of backward compatibility.

22.1 Main Features

- Support master and slave devices
- Support MIPI I3C specification v1.1
- Support multi-host functionality
- Support DMA
- In-band interrupt (IBI) capability
- Built-in error detection and recovery
- I3C SCL bus clock up to 12.5MHz
- Support dynamic address allocation, direct and broadcast common command codes (CCC), and private read/write transfers

22.2 Overview

Figure 22-1 I3C timing diagram



22.3 Function Description

22.3.1 I3C Peripheral Status

I3C peripherals can be used as both I3C master devices and I3C slave devices. In any case, the peripheral is in one of the following states:

● Forbidden state

Operating Conditions: After the I3C peripheral is reset (by setting bit 1 of I3CRST in the RCC module), the peripheral enters a disabled state.

Conversion from disabled to idle state: When the software sets bit 1 of the R32_I3C_CFGR register's EN field, the peripheral conversions from the disabled state to the idle state after completing verification of its internal configuration parameters.

● Forbidden state

In the idle state, the software can partially update the I3C peripheral configuration.

Conversion from idle state to working state:

1) Software-controlled triggering:

In master mode, triggered when frame transmission starts via software write operation setting TSFSET bit 1 in the R32_I3C_CFGR register;

In slave mode, triggered when software issues an IBI/CR/HJ request;

2) Hardware Event Triggering:

The master device triggers upon receiving a start request and maximum TCAS time from the slave device during bus arbitration;

The slave device triggers upon detecting broadcast/direct CCC or private read/write operations via the SDA/SCL lines.

● Working state

In the working state, the peripheral performs transmission on the bus.

In the working state, when the requested transmission is completed, the software will be notified by the event in the R32_I3C_EVR register, and the corresponding interrupt will be enabled through the R32_I3C_INTENR register. After the peripheral is switched from the working state to the idle state, it can still communicate on the bus and can be partially reconfigured.

- 1) As master device: in the working state, the event/flag generated after the requested transmission is completed can be Frame Complete (FCF), IBI/ Master Device Role/Hot Join Request Complete (IBIF/CRF/HJF) or Transmission Error (ERRF);
- 2) As slave device: During operation, events/flags generated upon completion of a requested transmission may include Dynamic Address Allocation Complete (DAUPDF), IBI Complete (IBIENDF), Controller Role Acquisition Complete (CRUPDF), Broadcast/Direct CCC Complete (xxUPDF/RSTF/GETF/STAF), Private Read/Write Complete (FCF), or Transmission Error (ERRF).

22.3.2 I3C Master Status and Programming Sequence

In this section, we provide a complete programming sequence when I3C peripheral is the master device, covering state conversion, main subtasks and conditions.

Master device initialization

When the master device is in the forbidden state (the EN bit in R32_I3C_CFGR register is 0), the software must follow the following steps to initialize:

- Configure the R32_I3C_CFGR register, set CRINIT to 1 to mark the master device role
 - Clear the HST_SIE_RST bit in the R32_I3C_RESET register
 - Configure I3C bus timing:
- 1) Configure the R32_I3C_TIMINGR0 register to set the SCL clock high-level duration and low-level duration

for the open-drain/push-pull stages of traditional I2C and I3C.

- 2) Configure the R32_I3C_TIMINGR1 register to set the SDA hold time (tHD_PP) for the push-pull phase, the bus idle condition time (I3C tCAS, traditional I2C tBUF), the I3C repeat start bit timing (tCASr, tCBSr), I3C stop timing (tCBP), and maximum SCL clock low stall time (tSTALLDAA for ENTDAACCC, ACK/NACK for traditional I2C, the address phase of I3C transfers, the parity bit of write data transfers, the ACK/NACK data phase of traditional I2C write transfers, the conversion bit of I3C read transfers, or the ACK/NACK phase of traditional I2C write transfers (tSTALL). Peripherals acting as master devices may adjust the SCL clock low-level stall time and tNEWCRLock according to their requirements to accommodate the master role conversion process (following the GETACCCR CCC command).
- 3) Configure the R32_I3C_TIMINGR2 register to set the SCL clock low-level dwell time. Each phase can be individually enabled or disabled to adjust the SCL clock low-level dwell time (to meet the requirements of the addressed I3C slave device or traditional I2C slave device during SDA switching).
 - Configure the DA[6:0] bits of the R32_I3C_DEVR0 register to set the device's dynamic address
 - Configure the R32_I3C_DEVRx register to set management parameters for any slave device x (x = 1/2/3/4)
 - Configure the R32_I3C_CFGR register to set the execution mode for frame transfer or target request transfer
 - Configure the R32_I3C_INTENR register to set interrupt generation or polling mode for any event
 - After initialization completes, set EN to 1 to enable the I3C peripheral

Start the frame transmission initiated by the master device

When the master device is in the enabled state (the EN bit in R32_I3C_CFGR register is 1), the software can start frame transmission by any of the following configuration methods:

1. Software trigger: set the TSFSET bit of R32_I3C_CFGR register to 1 by performing a write operation;
This operation will cause the hardware to set the CFNFF flag in the R32_I3C_EVR register to request writing the first control word to the R32_I3C_CTLR register.
2. No trigger: directly write the first control word to R32_I3C_CTLR register through software.

Regardless of the initialization method used, the I3C peripheral subsequently conversions to operational status. When the control word is not the final message in an I3C frame (i.e., the MEND bit in the R32_I3C_CTLR register is not set) and no transmission errors have occurred (i.e., the ERRF bit in the R32_I3C_EVR register is not set), the hardware will continue to request the next control word and proceed with frame transmission:

- 1) If the C-FIFO is in non-DMA mode (CDMAEN bit in the R32_I3C_CFGR register is 0), the software will write the next control word after the CFNFF flag in the R32_I3C_EVR register is set or the corresponding interrupt is triggered (CFNFIE bit in the R32_I3C_INTENR register is 1);
- 2) If the C-FIFO is in DMA mode (CDMAEN bit in the R32_I3C_CFGR register is set to 1), the corresponding I3C DMA request will be automatically processed by the assigned DMA channel, and the next control word will be written.

Initiate and receive transmissions initiated from devices

When the master device is in the enabled state (the EN bit in the R32_I3C_CFGR register is 1), the master device can initiate the transmission, and the slave device can also initiate the transmission by issuing a start request (driving SDA to a low level), but only if the master device has allowed the hot join request, IBI request or master role request through the R32_I3C_DEVR0 register.

In this case, even if the master software does not plan to start the frame transmission, the hardware will automatically enter the working state (activate the SCL clock before the maximum tCAS time (which can be defined as 1μs, 100μs, 2ms or 50ms, corresponding to the bus working state 0, 1, 2 or 3 respectively) in order to receive the hot join/in-

band interrupt/master role request from the slave.

Perform frame transmission initiated by the master device

On the bus, the master device continuously executes frame transfers until the final message is transmitted (R32_I3C_EVR register FCF bit is 1) or an error occurs during transmission (R32_I3C_EVR register ERRF bit is 1), triggering the corresponding interrupt (if enabled). This process relies on the R32_I3C_CTLR and R32_I3C_TDW/TDR registers (written explicitly by software or automatically pushed by assigned DMA channels) and the R32_I3C_RDW/RDR registers (read explicitly by software or by assigned DMA channels). After completing the transmission, the I3C master device returns to idle state.

When the S-FIFO is disabled (the SMODE bit in the R32_I3C_CFGR register is 0), the hardware will update the I3C status register (R32_I3C_STATR) for each message that has been completed without transmission errors, indicating that the transmission of an exchange message has been completed on the I3C bus, and the software can choose to read or ignore this register.

When the S-FIFO is disabled in the R32_I3C_STATR register (SMODE bit in the R32_I3C_CFGR register is 0), except for the completion of the last message transmission (FCF bit in the R32_I3C_EVR register is 1) or a transmission error occurs (ERRF bit in R32_I3C_EVR register is 1) triggering a corresponding interrupt (if enabled), the software is also notified when the slave device prematurely terminates a read transfer during a direct CCC read or private read transfer. This notification occurs via the RXTGTENDF bit in the R32_I3C_EVR register being set to 1 and triggering a corresponding interrupt (if enabled). Software can subsequently read the R32_I3C_STATR register to obtain additional information about the executed transfer.

If the S-FIFO is enabled (bit 1 in the SMODE field of the R32_I3C_CFGR register), the status register R32_I3C_STATR must be read once after each word message transfer completes, regardless of whether the device prematurely terminates the read transfer. This can be done via software (notified by the SFNEF bit in the R32_I3C_EVR register being set to 1 and the corresponding interrupt, if enabled) or via DMA (if the SDMAEN bit in the R32_I3C_CFGR register is set to 1). Frame transfer completion is only reported after reading the status of the last message (S-FIFO is empty) (R32_I3C_EVR register FCF bit set to 1).

Update master device transmission configuration

After the software returns to the idle state, it can update the configuration of the I3C peripheral before the next transmission:

- Configure the R32_I3C_TIMINGR2 register to modify the SCL clock pause;
- Configure the R32_I3C_INTENR register to modify the interrupt/polling mode strategy;
- Configure the TXTHRES, RXTHRES, TMODE, SMODE, TXDMAEN, CDMAEN, RXDMAEN, SDMAEN, EXITPTRN, RSTPTRN, and NOARBH bits in the R32_I3C_CFGR register;

Modify/prepare the control word, status word and read/write data of the next frame transmission to be executed by software and/or DMA.

Generally, after broadcast/direct DISEC/ENEC CCC is issued and completed:

- Modify the hot-join acknowledgment strategy via the HJACK bit in the R32_I3C_CFGR register;
- Modify the IBI/CR acknowledgment strategy for any slave device x via the R32_I3C_DEVRx register.

22.3.3 I3C Slave Status and Programming Sequence

In this section, we provide a complete programming sequence when I3C peripheral is the slave device, covering

state conversion, main subtasks and conditions.

Slave device initialization

When the slave device is in the forbidden state (the EN bit in R32_I3C_CFGR register is 0), the software must follow the following steps to initialize:

- Configure R32_I3C_CFGR register, set CRINIT to 0, and set I3C peripheral as slave.
- Clear DEV_SIE_RST bit in R32_I3C_RESET register.
- Set I3C bus timing through I3C timing register 1(R32_I3C_TIMINGR1): Write AVAL[7:0] to set:
 - Bus Available Condition Time (tAVAL), for IBI or master role requests
 - Bus Idle Condition Time (tIDLE), for hot join requests
 - tNEWCRLOCK, for master role conversion process (after GETACCCRCCC)
- Configure the request initiated by the device: write the following bits in I3C's own device characteristic register (R32_I3C_DEVR0):
 - IBIEN: In-band interrupt (also known as IBI) request enable/disable
 - CREN: Controller role enable/disable
 - HJEN: Hot join enable/disable
- Initialize slave device features and functions
- Configure the execution mode of the transmission:

In the R32_I3C_CFGR register, TXDMAEN and RXDMAEN enable/disable DMA mode for the TX-FIFO and RX-FIFO, respectively; TXTHRES and RXTHRES represent the byte/word thresholds for the TX-FIFO and RX-FIFO, respectively; SMODE bit 0 disables the S-FIFO.

- Configure the R32_I3C_INTENR register to set interrupt generation or polling mode for any event
- After initialization completes, set EN to 1 to enable the I3C peripheral

Receive a message (broadcast CCC, direct read/write CCC or private read/write) from the master device

When the slave device is in an idle state (the EN bit in the R32_I3C_CFGR register is 1), it means that it is ready to receive the communication message from the master device on the I3C bus and ready to switch to the working state.

In the active state, the slave device first receives the broadcast ENTDAACCC (possibly after receiving the optional broadcast ENEC/DISEC CCC), then is assigned a dynamic address. The DAUPF event in the R32_I3C_EVR register is set to 1, and a corresponding interrupt is generated (if enabled). The slave device subsequently returns to the idle state.

Thereafter, the slave device in idle state is ready to receive any additional broadcast CCC messages, direct read/write CCC messages, or private read/write messages from the master device.

Read message status register

For each message received without errors, the hardware will update the I3C status register (R32_I3C_STATR) to report that an exchange message has been transmitted on the I3C bus, and the software will be notified by the corresponding flag in the I3C event register (R32_I3C_EVR) or the corresponding interrupt in the I3C interrupt enable register (R32_I3C_INTENR).

If the following message appears, the software must read the I3C status register (R32_I3C_STATR):

- Private Read: Retrieve the number of bytes exchanged, as the master device may terminate transmission earlier than the slave device anticipates (if R32_I3C_STATR register XDCNT[15:0] is less than R32_I3C_TGTTDR register TGTTDCNT[15:0]).
- At this point, the software must set TXFLUSH in the R32_I3C_CFGR register to 1 to clear the TX-FIFO.

- DEFTGTS CCC or DEFGRPA CCC: Retrieves the number of data bytes received in the RX-FIFO.

Initiates a transfer initiated from the device

When a device conversions from the disabled state to the idle state (software writes 1 to the EN bit in the R32_I3C_CFGR register) and is simultaneously capable of receiving the master device's broadcast CCC, the software may issue a hot join request (software writes 1000b to the MTYPE[3:0] bits in the R32_I3C_CTLR register) to apply for eligibility to join the next ENTDAACCC. This is contingent upon the hot join request being enabled (set HJEN bit to 1 in the R32_I3C_DEVR0 register).

After dynamic address allocation (R32_I3C_EVR register DAUPF bit 1 set) – typically concurrent with frame transmission initiated by the master device – the software may issue an IBI (in-band interrupt request) or master role request to the master device by writing the relevant control word to the R32_I3C_CTLR register.

Update the configuration of I3C peripherals (as slaves)

After returning to the idle state, the software can update the configuration of the I3C slave device before the next transmission:

- Configure the R32_I3C_INTENR register to modify the interrupt/polling mode strategy
- Configure the TXTHRES, RXTHRES, TXDMAEN, and RXDMAEN bits in the R32_I3C_CFGR register
- Modify/prepare the I3CIBI payload data register (R32_I3C_IBIDR). If any payload exists (when bit BCR2 in the R32_I3C_BCR register is set to 1), this operation should be performed before initiating the IBI transfer (writing the MTYPE[3:0] bits to 1010b in the I3C Message Control Register [Multiplex] (R32_I3C_CTLR) with MTYPE[3:0] set to 1010b)
- Modify/prepare the I3C slave device transmission configuration register (R32_I3C_TGTTDR). Before receiving a private read or direct CCC read message (excluding GETSTATUS CCC) from the master device, disable or enable the TX-FIFO to preload the specified number of bytes of data to be transmitted.

22.3.4 I3C Bus Transmission and Programming

22.3.4.1 I3C Command Set (CCC) (as master/slave)

Table 22-1 provides a list of supported I3C command sets (such as CCC list and general command code) and an overview of how to handle this command set when peripherals are used as master or slave devices.

Table 22-1 Supported I3C CCC list (as Master/Slave)

CCC name	CCC value	Read/Write	With/without interpreted bytes With/without subcommand bytes	With/without optional data bytes	As master device	Used as a slave device to generate an I3C_EVR event.	As a slave device: specific operations
ENEC	0x00	Write	No interpretation byte/subcommand byte	There is one data byte (enable slave event byte)	X	X, INTUPDF	Update and enable R32_I3C_DEVR0 register: HJEN, CREN, IBIEN (if any)
DISEC	0x01			There is one data byte (disable slave event byte)	X	X, INTUPDF	Update and disable

				event byte)			R32_I3C_DEVR 0 register: HJEN, CREN IBIEN (if any)
ENTAS _x (x=0...3)	0x02 ... 0x05			No data byte	X	X, ASUPDF	Update R32_I3C_DEV R0 register AS[1:0] bit
RSTDAA	0x06			-	X	X, DAUPDF	Clear DAVAL bit of R32_I3C_DEV R0 register
ENTDAA	0x07			-	X	X, DAUPDF	Update R32_I3C_DEV R0 register DA[6:0] bit and set DAVAL to 1
DEFTGTS	0x08			There are [1+4x (1+n)] data bytes (n-number of slaves)	X	X, DEFF	Update R32_I3C_RDR/ R32_I3C_RDW R register
SETMWL	0x09			There are 2 data bytes	X	X, MWLUPDF	Update R32_I3C_MAX WLR register
SETMRL	0x0A			There are 2 or 3 data bytes	X	X, MRLUPDF	Update R32_I3C_MAX RLR register
ENTTM	0x0B			There is 1 data byte	X	-	-
RSTACT	0x2A		Interpreted byte (0x00, 0x01 or 0x02)	No data byte	X	X, RSTF after reset timing is detected	Update R32_I3C_DEVR 0 register RSTACT[1:0]bit and set RSTVAL bit to 1
DEFGRPA	0x2B		No interpretation byte/subcommand byte	There are multiple data bytes.	X	X, GRPF	Update R32_I3C_RDR/ RDWR register
RSTGRPA	0x2C			No data byte	X	-	-
Direct CCC							Execute operation upon ACK (if I3C slave device address matches R32_I3C_DEV R0 register's

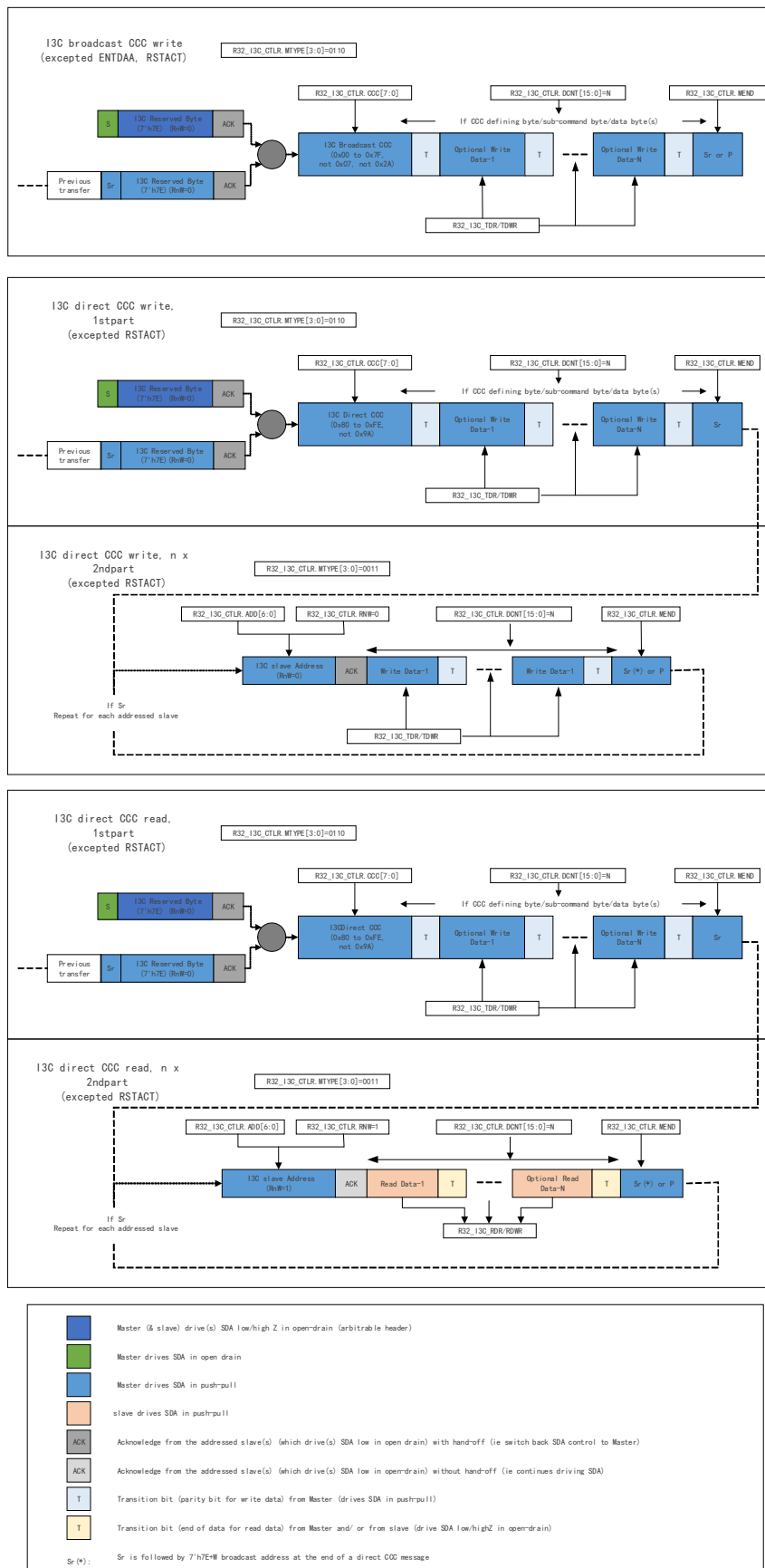
							DA[6:0] and R32_I3C_DEV R0 register's DAVAL=1) (otherwise NACK)
ENEC	0x80	Write	No interpretation byte/subcommand byte	There is a data byte (enable slave event byte)	X	X, INTUPDF	Update and enable R32_I3C_DEV R0 register: HJEN, CREN, IBIEN (if any)
DISEC	0x81			There is a data byte (slave event byte is prohibited).	X	X, INTUPDF	Update and disable R32_I3C_DEV R0 register: HJEN, CREN, IBIEN (if any)
ENTAS _x (x=0...3)	0x82 ... 0x85			No data byte	X	X, ASUPDF	Update R32_I3C_DEV R0 register AS[1:0] bit
SETDASA	0x87			No data byte	X	-	-
SETNEWD A	0x88			There is 1 data byte.	X	X, DAUPDF	Update R32_I3C_DEV R0 register DA[6:0] bit, set DAVAL to 1
SETMWL	0x89			There are 2 data bytes	X	X, MWLUPDF	Update R32_I3C_MAX WLR register
SETMRL	0x8A			There are 2 or 3 data bytes	X	X, MRLUPDF	Update R32_I3C_MAX RLR register
GETMWL	0x8B	Read		There are 2 data bytes	X	X, GETF	Data bytes are returned from R32_I3C_MAX WL register RMWL[15:0].
GETMRL	0x8C			There are 2 or 3 data bytes	X	X, GETF	Return the data byte from the MRL[15:0] field of the R32_I3C_MAX RLR register. If BCR2=1 in the R32_I3C_BCR register, return the third byte from the IBIP[2:0] field of the

							R32_I3C_MA XRLR register.
GETPID	0x8D			There are 6 data bytes	X	X, GETF	Return data byte from the R32_I3C_EPID R register
GETBCR	0x8E				X	X, GETF	Return the data byte from BCR[7:0] of the R32_I3C_BCR register
GETDCR	0x8F			There is 1 data byte.	X	X, GETF	Return R32_I3C_DCR register BCR[7:0]
GETSTATUS	0x90		With/without interpreted bytes (TGTSTAT, PRECR)	There are 2 data bytes (format 1 or format 2 (with PRECR))	X	X, STAF (in format 1) X, GETF (in format 2)	Returns 2 data bytes.
GETACCCR	0x91		No interpretation byte/subcommand byte	There is 1 data byte.	X	X, CRUPDF	Return the data byte with parity bit from the R32_I3C_DEV R0 register's DA[6:0] field.
GETMXDS	0x94		With/without interpreted bytes (WRRDTURN, CRHDLY)	2 data bytes (Format 1), 5 data bytes (Format 2 or Format 3 when WRRDTURN is present), or 1 data byte (Format 3 when CRHDLY is present)	X	X, GETF	Return data bytes from the R32_I3C_GET MXDSR register
GETCAPS	0x95	Read	With/without interpreted bytes (TGTSTAT, CRCAPS)	3 data bytes (Format 1 or Format 2 when TGTSTAT is present) or 2 data bytes (Format 2 when CRCAPS is present)	X	X, GETF	Return 3 GETCAPx data bytes from the R32_I3C_GET CAPR register or 2 CRCAPx data bytes from the R32_I3C_CRC APR register.
RSTACT	0x9A	Read/ Write	With interpreted bytes 0x00, 0x01 or 0x02)	With interpreted bytes (0x00, 0x01 or 0x02)	X	X, set RSTF upon detection of reset timing sequence	Read: Return data bytes from RSTACT[1:0] in the R32_I3C_DE VR0 register Write: Update RSTACT[1:0] in the R32_I3C_DE

							VR0 register and set RSTVAL bit 1
SETGRPA	0x9B	Write	No interpretation byte/subcommand byte	No interpretation byte/subcommand byte	X	-	-
RSTGRPA	0x9C				X		

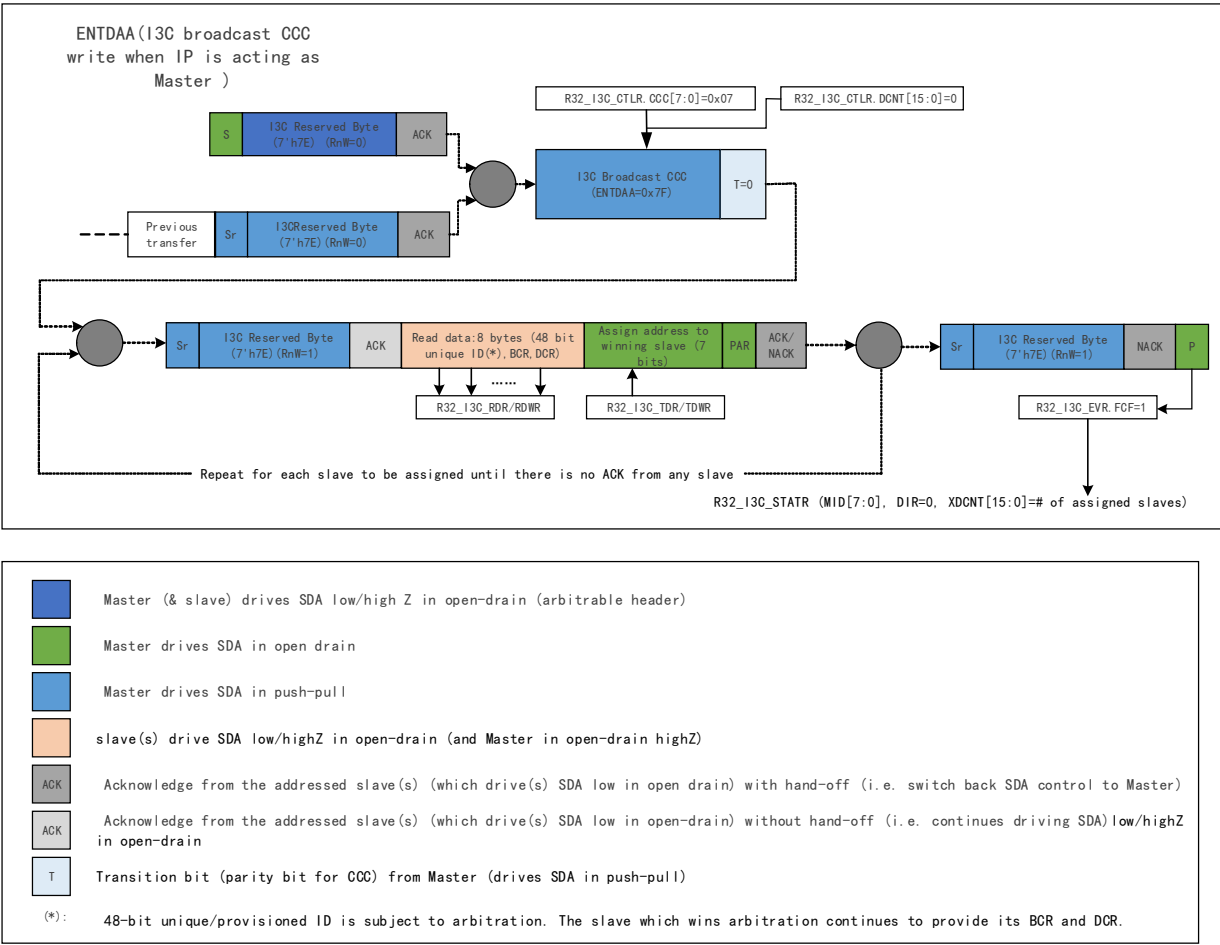
22.3.4.2 I3C Broadcast/Direct CCC Transfer (except ENTDA and RSTACT) (as master device)

Table 22-2 I3C CCC message



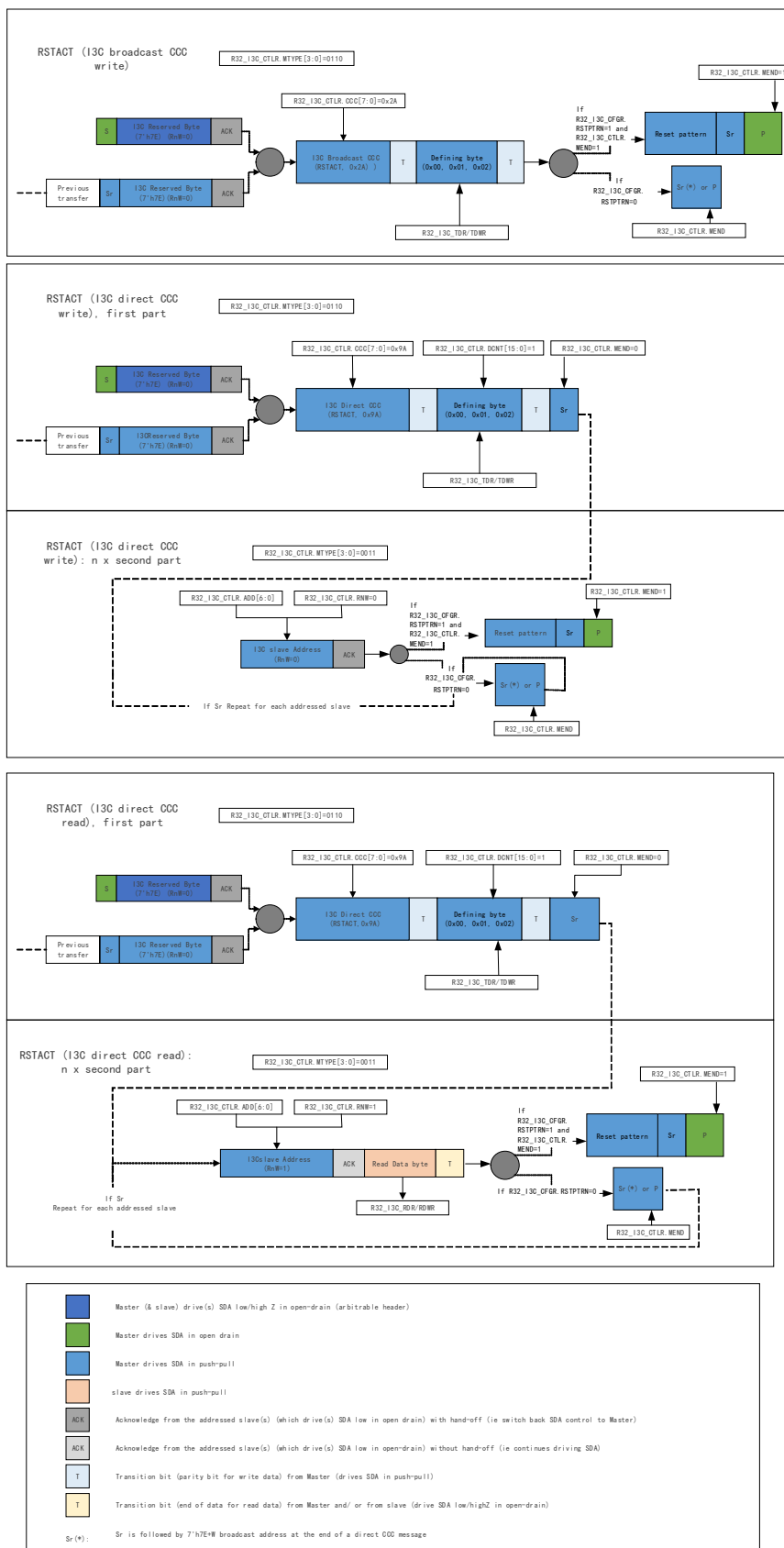
22.3.4.3 I3C Broadcast ENTDA A CCC Transfer (as master device)

Table 22-3 I3C broadcast ENTDA A CCC transmission (as master device)



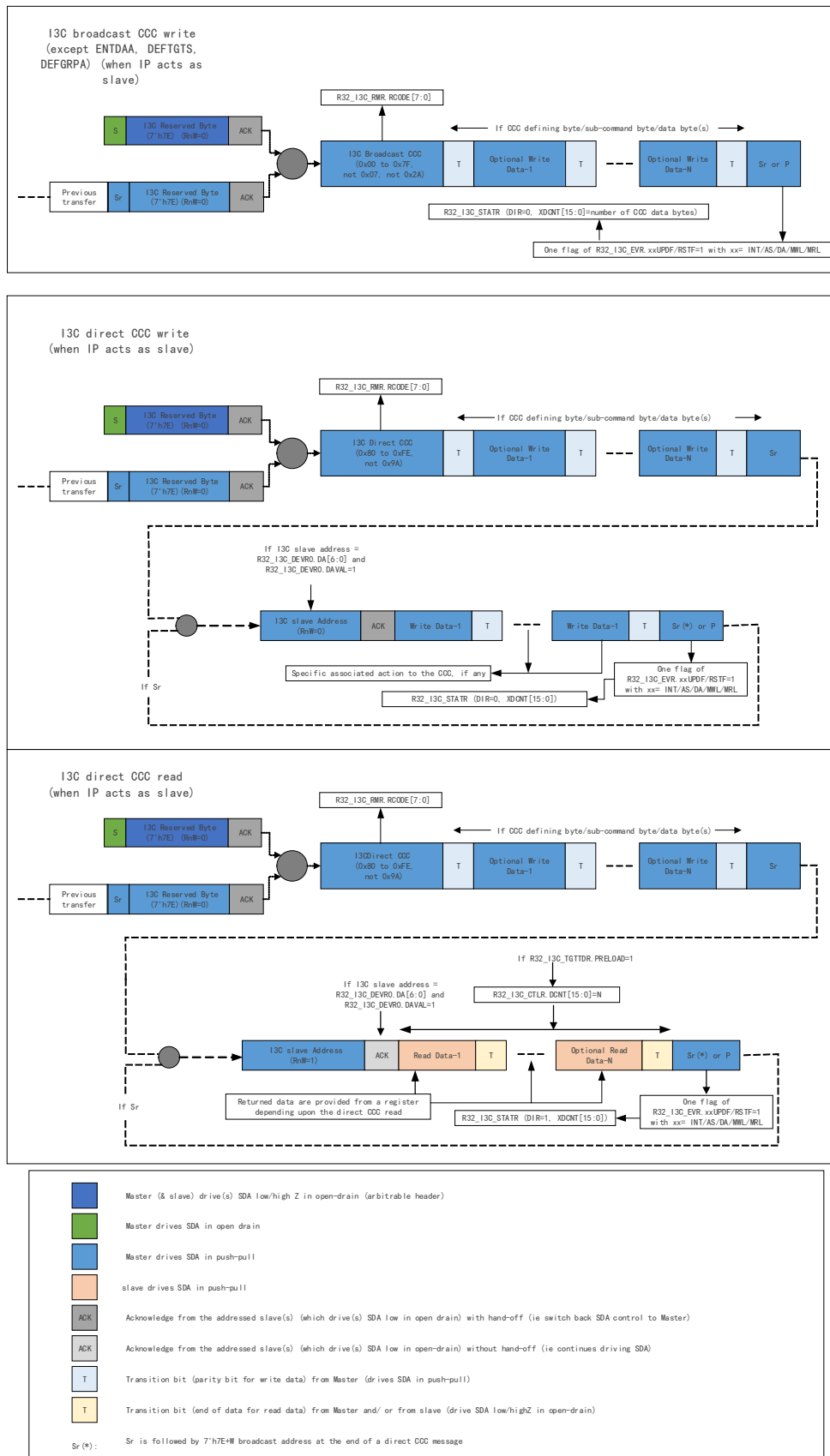
22.3.4.4 I3C Broadcast/Direct RSTACT CCC Transfer (as master device)

Table 22-4 I3C broadcast, direct read/write RSTACT CCC (as master device)



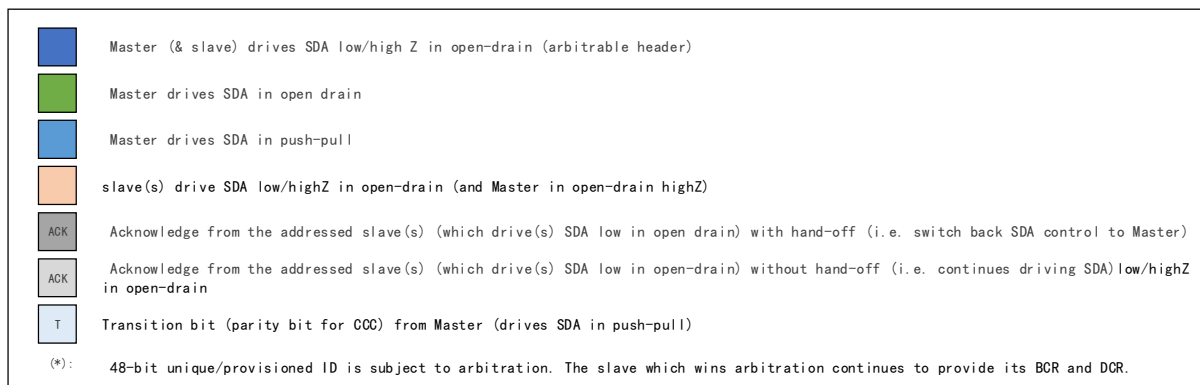
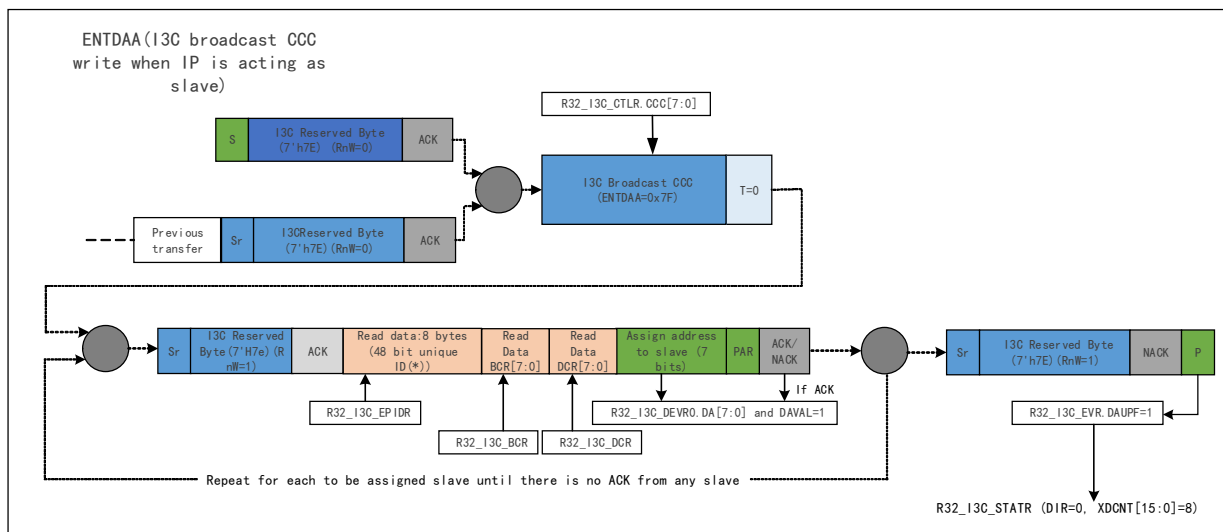
22.3.4.5 I3C Broadcast/Direct CCC Transfer (Except ENTDA, DEFTGTS and DEFGRPA) (as slave devices)

Table 22-5 I3C CCC message (as slave device)



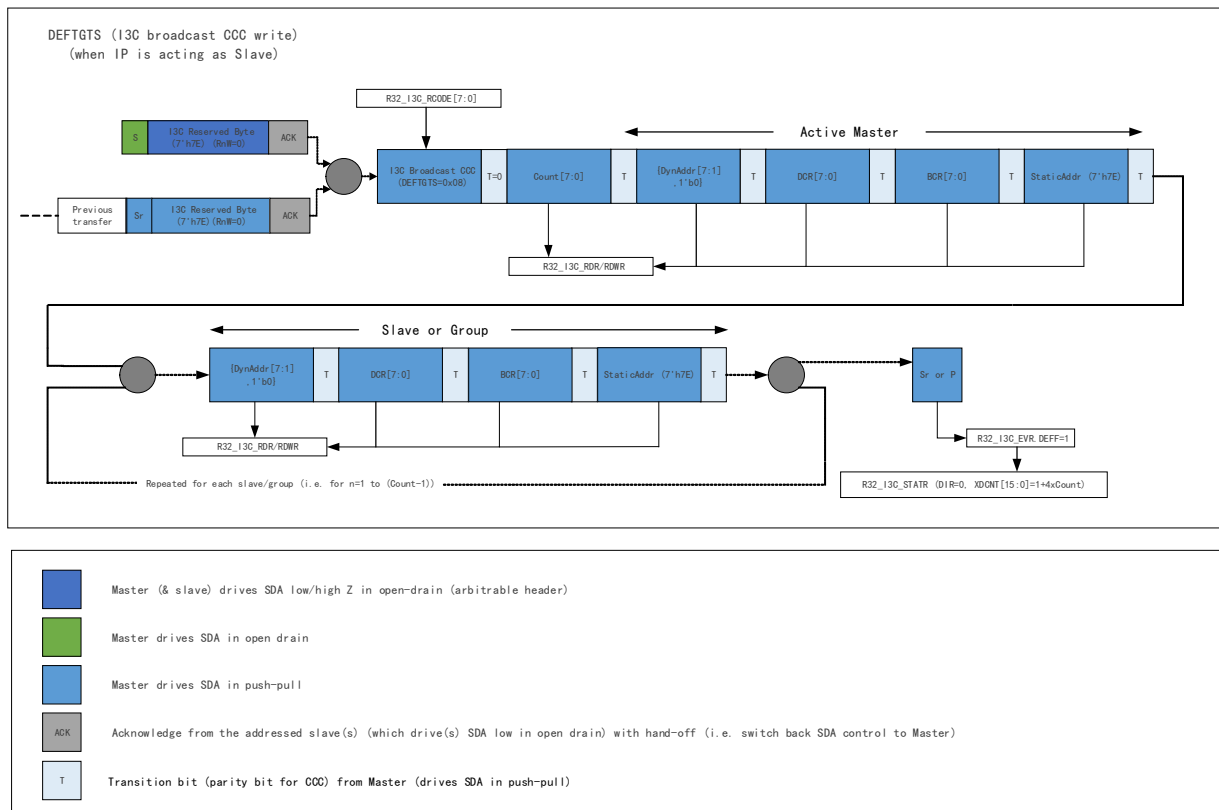
22.3.4.6 I3C Broadcast ENTDAACCC Transfer (as slave devices)

Table 22-6 I3C broadcast ENTDAACCC transmission (as slave device)



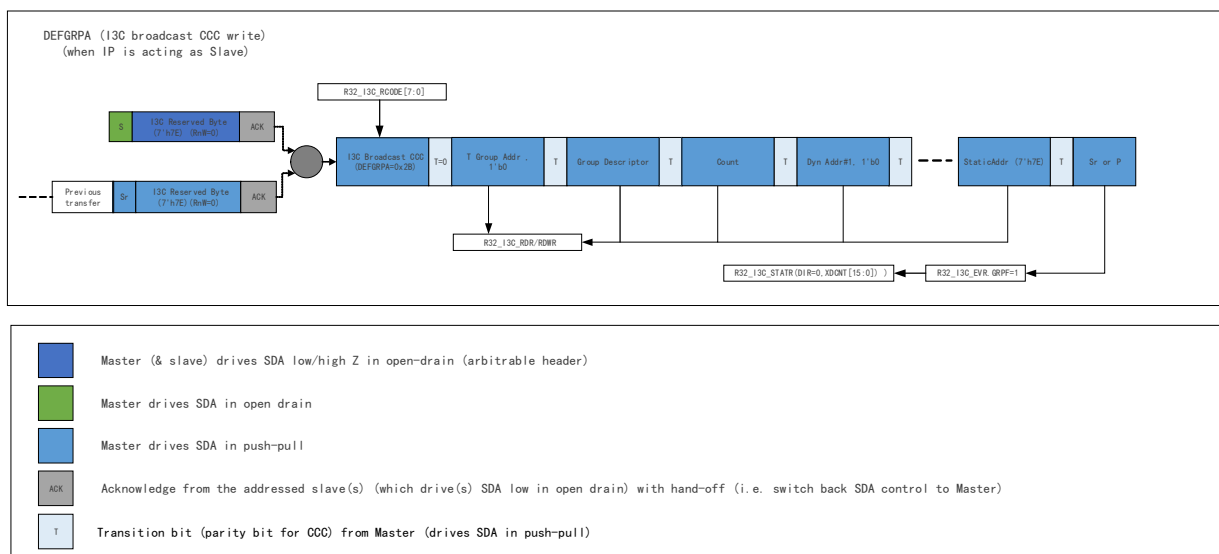
22.3.4.7 I3C Broadcast DEFTGTS CCC Transfer (as slave devices)

Table 22-7 I3C broadcast DEFTGTS CCC transmission (as slave device)



22.3.4.8 I3C Broadcast DEFGRPA CCC Transfer (as slave devices)

Table 22-8 I3C broadcast DEFGRPA CCC transmission (as slave device)



22.3.4.9 I3C Direct GETSTATUS CCC Transfer (as slave devices)

When I3C operates as a slave device, the hardware returns two data bytes upon receiving a GETSTATUS CCC

command, following either Format 1 (without an interpretation byte or with an interpretation byte TGTSTAT = 0x00) or Format 2 (with an interpretation byte PRECR = 0x91).

On the I3C bus, the 2-byte STATUS[15:0] returned in Format 1 is as follows:

STATUS[15:14] = 00 (unused);

STATUS[13] = 1 (if a missing start bit is detected since the previous GETSTATUS CCC reception) or 0 (otherwise);

STATUS[12] = 1 (if an overflow/underflow error is detected since the previous GETSTATUS CCC reception) or 0 (otherwise);

STATUS[11] = 1 (if SCL stabilization time exceeds 125μs during SDR read transmission since the previous GETSTATUS CCC reception) or 0 (otherwise);

STATUS[10:8] = 000 to 110 (if a protocol error is detected since the previous GETSTATUS CCC reception (provided STATUS[5] = 1), encoded value x = 0 to 6, corresponding to slave device error TEx) or 000 (otherwise);

STATUS[7:6] = 00 (ready to execute the switching process);

STATUS[5] = 1 (if a protocol error was detected since the previous GETSTATUS CCC reception) or 0 (otherwise);

STATUS[4] = 0 (reserved);

STATUS[3:1] = 000 (unused);

STATUS[0] = 1 (if a pending interrupt exists—provided IBI is configured in the R32_I3C_CTLR register, IBIEN = 1 and DAVAL = 1 in the R32_I3C_DEVR0 register, and IBI has not been acknowledged by the master device or disabled via DISEC) or 0 (otherwise).

On the I3C bus, the 2-byte STATUS[15:0] returned in format 2 is as follows:

STATUS[15:8] = 00000000 (unused);

STATUS[7:2] = 00000 (unused);

STATUS[1] = 1 (if software is still processing a received DEFTGTS or DEFGRPACCC and has not yet cleared the associated event (DEFF = 1 or GRPF = 1 in the R32_I3C_EVR register)); the master device must wait for software to complete these operations before issuing a GETACCR CCC (otherwise it will not be acknowledged);

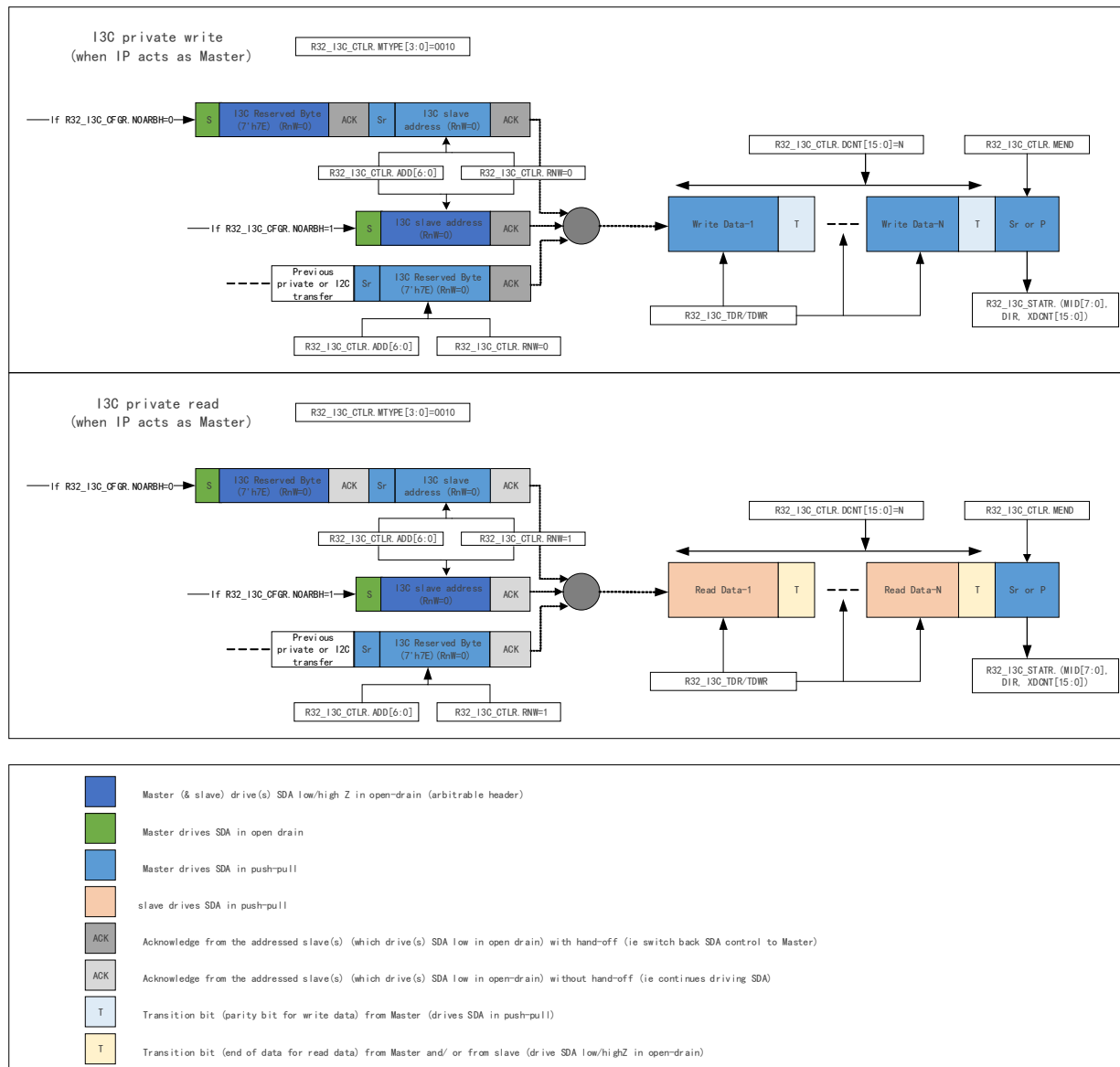
STATUS[0] = 1 (if the DEFTSGTS or DEFGRPA CCC may have been lost). If a missing start bit is detected (WKPF = 1 in the R32_I3C_EVR register), this bit will be set to 1; if DEFF = 1 or GRPF = 1 in the R32_I3C_EVR register, this bit will be cleared.

Upon completion of Format 1 GETSTATUS CCC, it is reported via STAF = 1 in the R32_I3C_EVR register and the corresponding interrupt (if enabled, i.e., STAIE = 1 in the R32_I3C_INTENR register).

Upon completion of a GETSTATUS CCC in Format 2, it is reported via GETF = 1 in the R32_I3C_EVR register and the corresponding interrupt (if enabled, i.e., GETIE = 1 in the R32_I3C_INTENR register).

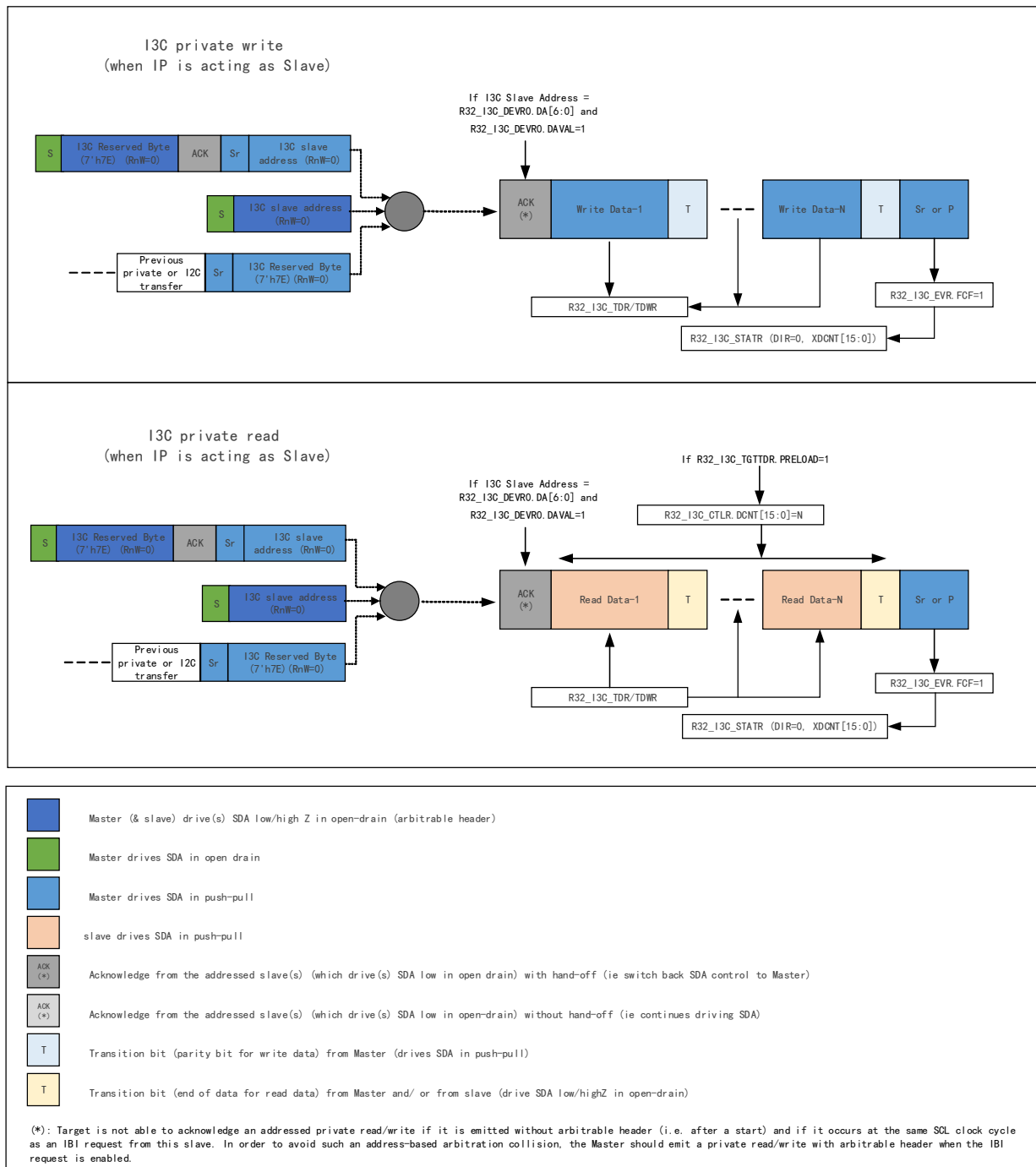
22.3.4.10 I3C Private Read/Write Transfer (as master devices)

Figure 22-9 I3C private read/write message (as master devices)



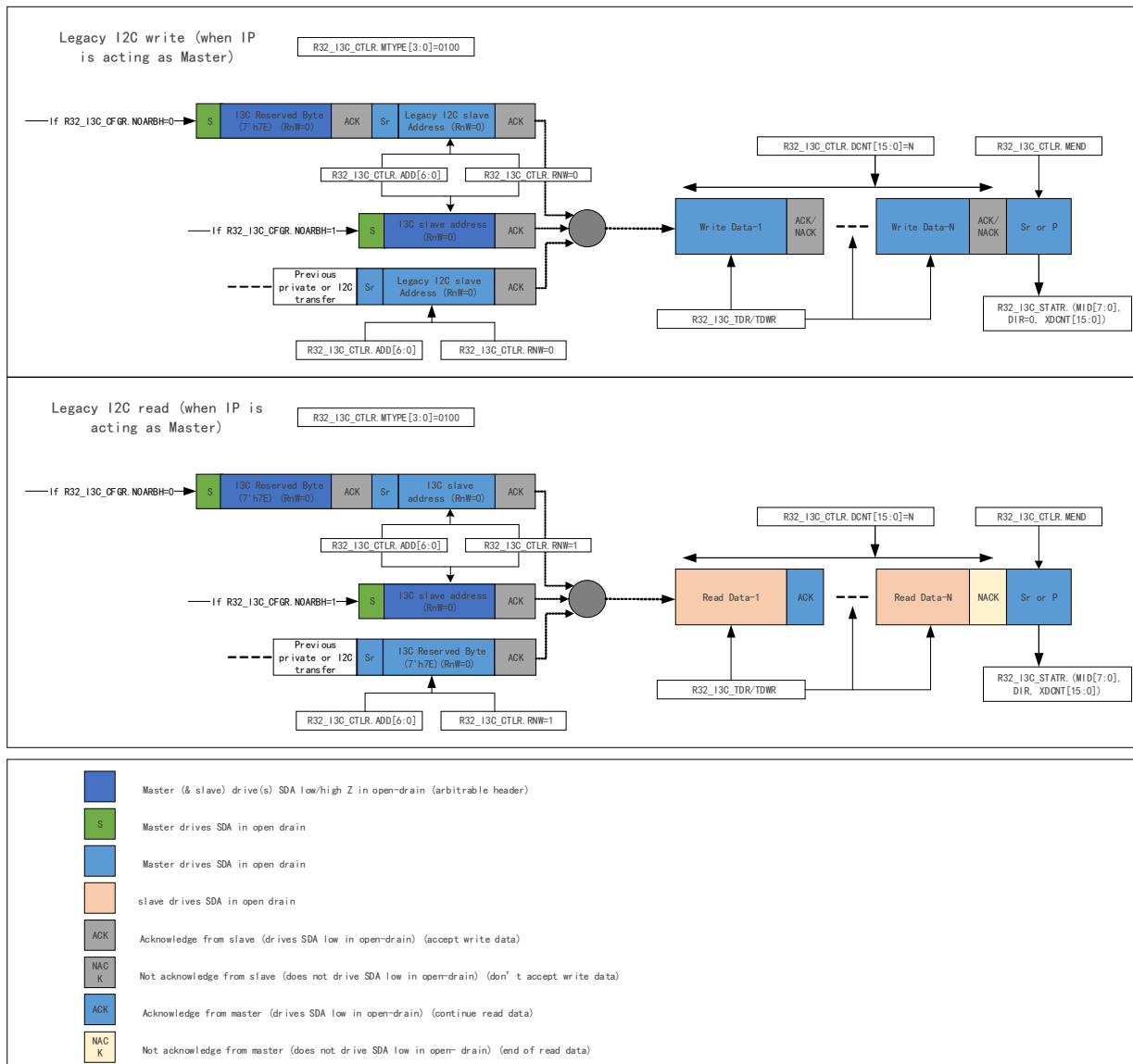
22.3.4.11 I3C Private Read/Write Transfer (as slave devices)

Figure 22-10 I3C private read/write message (as slave devices)



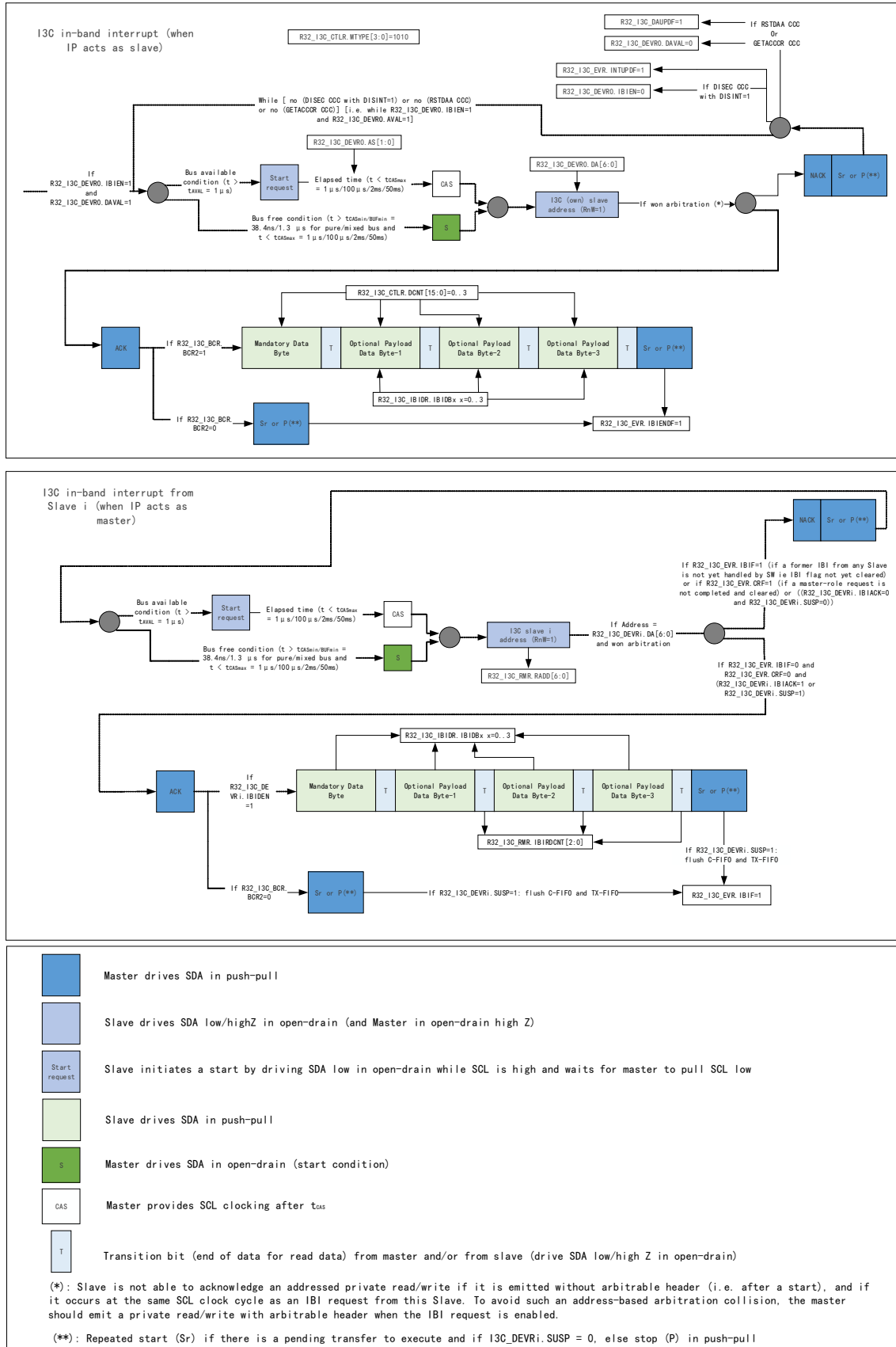
22.3.4.12 Traditional I2C Read/Write Transfer (as master devices)

Figure 22-11 Traditional I2C read/write message (as master devices)



22.3.4.13 I3C IBI Transfer (as master/slave devices)

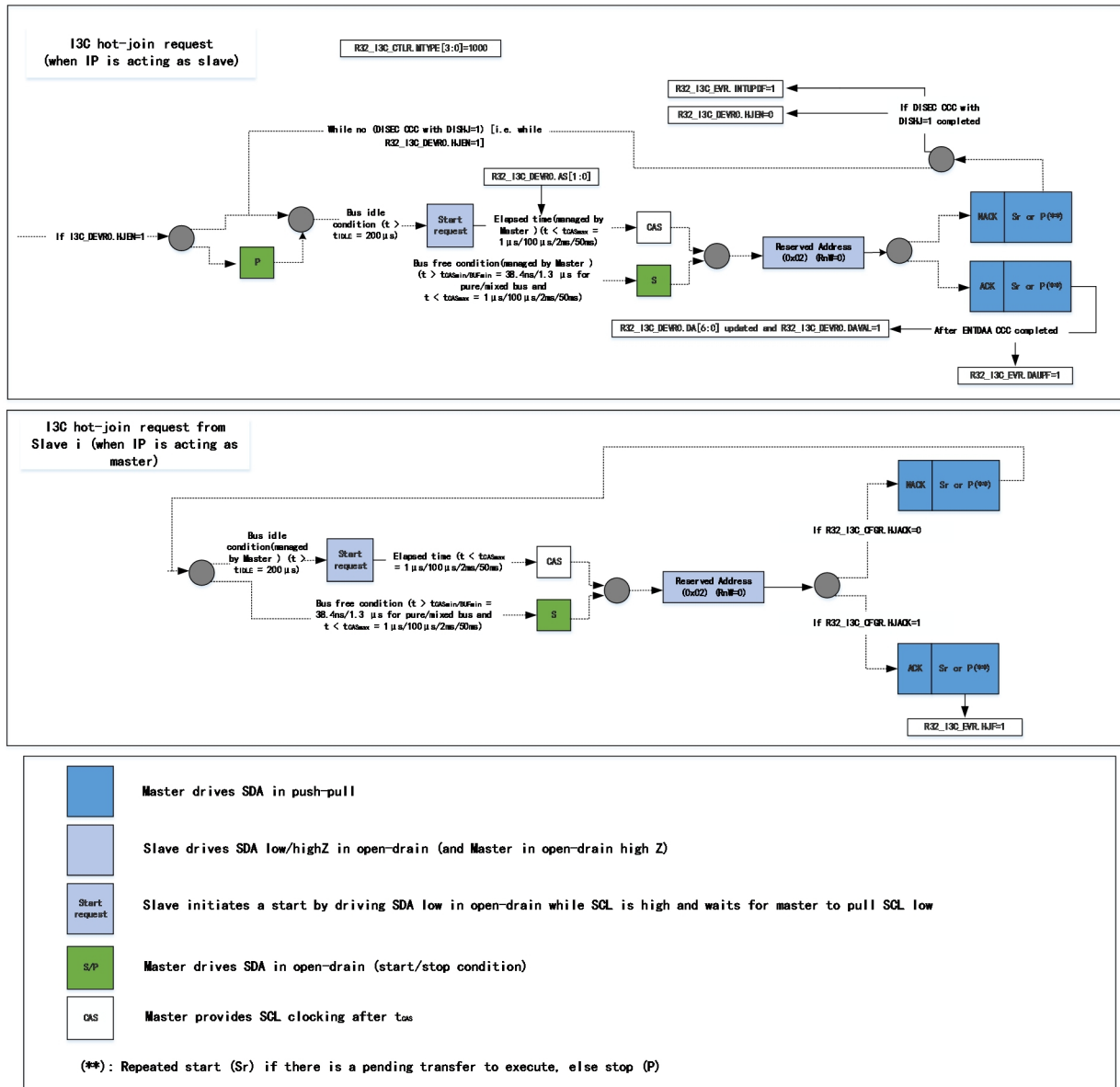
Figure 22-12 I3C IBI Transfer (as master/slave devices)



When the peripheral operates as the master device, the R32_I3C_IBIDR register is used to receive the IBI data payload. Therefore, IBI requests from the slave device must not exceed a 4-byte data payload. If more information needs to be exchanged within the context of this in-band interrupt, the master device software must issue a private read control word.

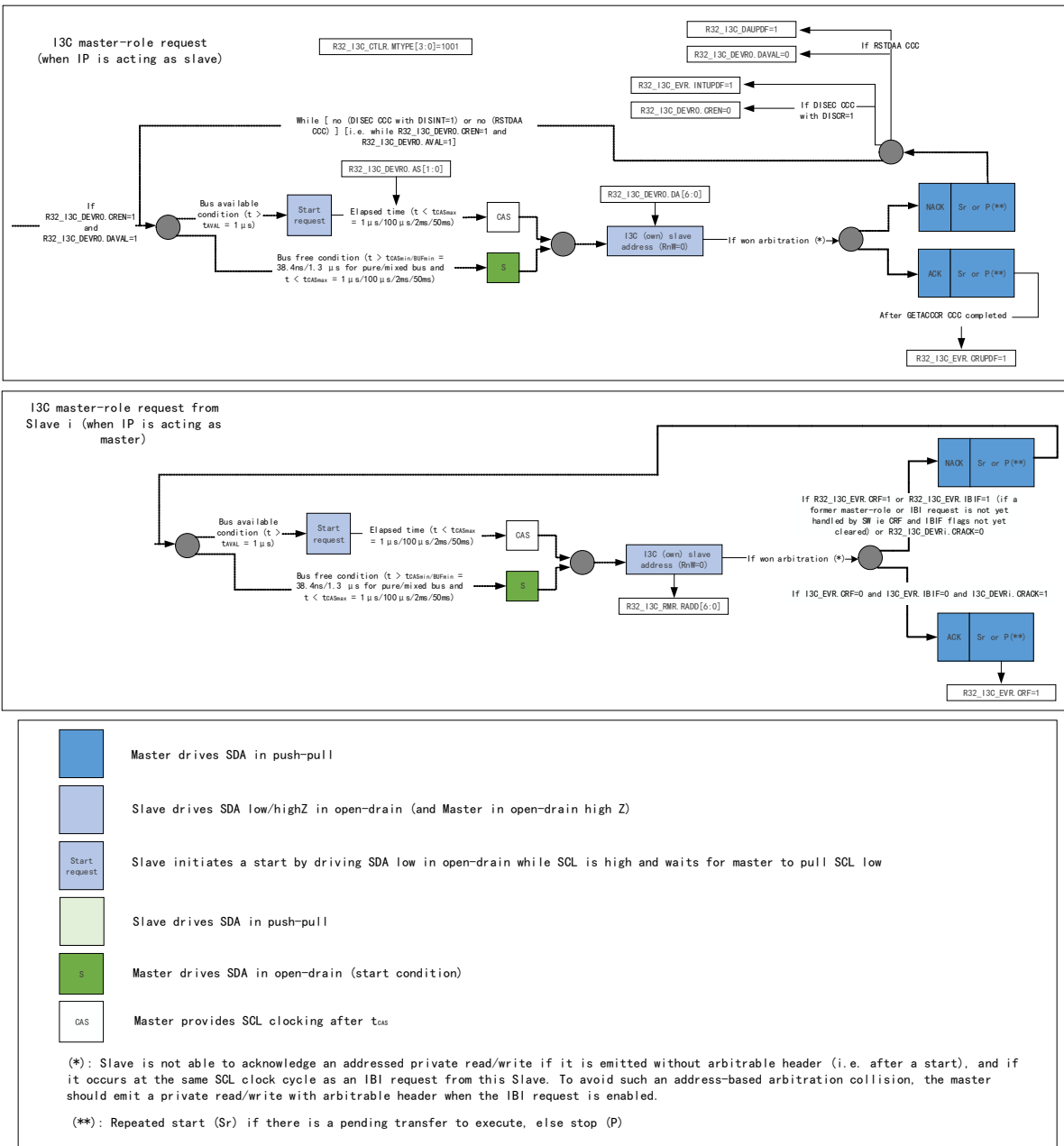
22.3.4.14 I3C Hot Join Request Transfer (as master/slave devices)

Figure 22-13 I3C hot join request transfer (as master/slave devices)



22.3.4.15 I3C Master Role Requests Transfer (as master/slave devices)

Figure 22-14 I3C master role requests transfer (as master/slave devices)



22.3.5 I3C FIFO Management

Table 22-2 I3C FIFO implementation

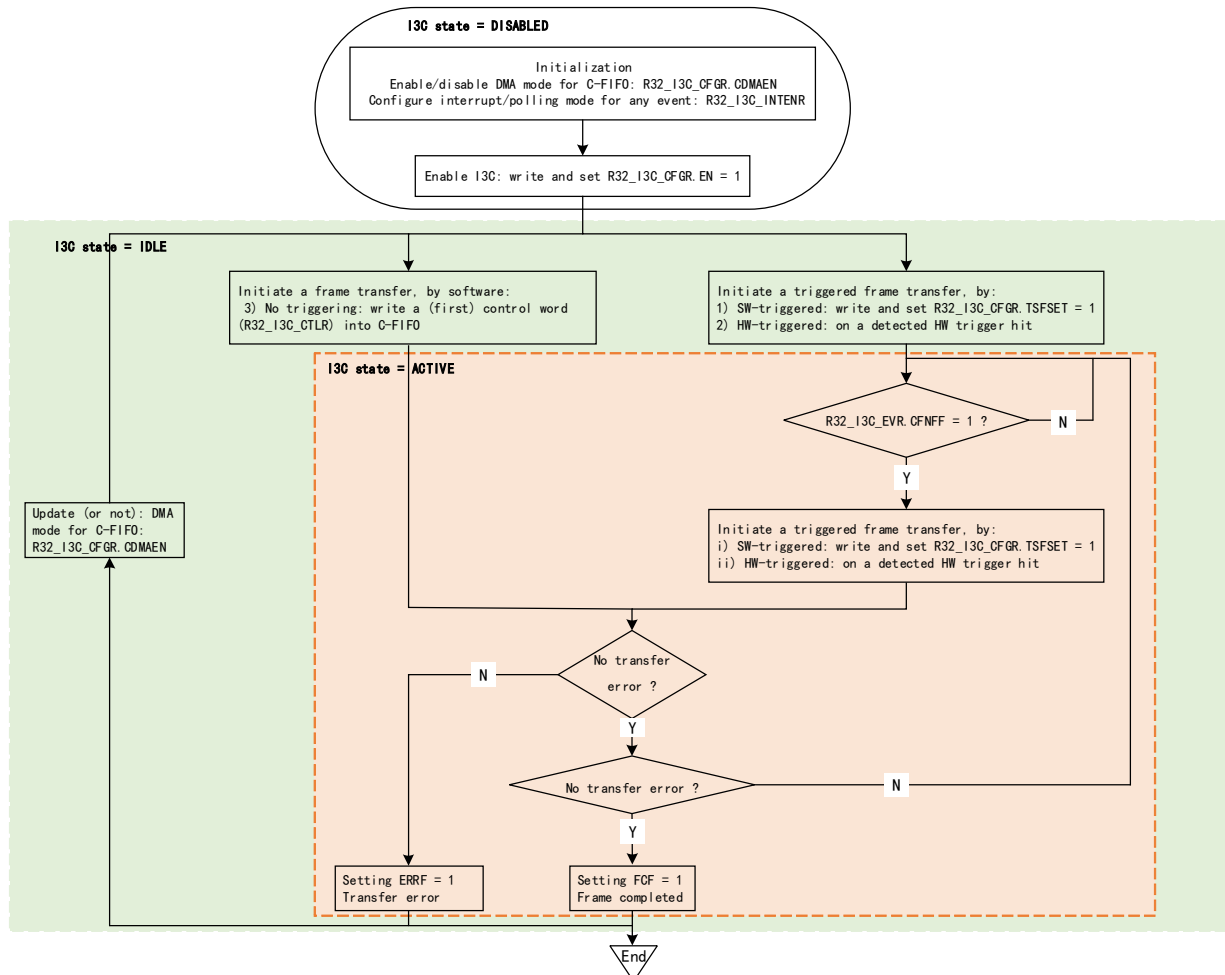
FIFO	Content	Unit	Size	Act as a master/slave (basic principle)
C-FIFO	32-bit control word	Word	2	Master device (a frame can be based on multiple control words; But not when used as a slave device)
S-FIFO	32-bit status word			Master device (slave device: status in register mode only)
TX-FIFO	Transmitted data	Byte	8	Master and slave devices
RX-FIFO	Received data			

22.3.5.1 C-FIFO Management (as master)

When it is used as a master device, it can be used during C-FIFO broadcast CCC, direct read/write CCC, private read/write, traditional I2C read/write, and software-initiated error recovery.

Figure 22-15 illustrates how to manage the C-FIFO to queue the control words on the I3C bus when the I3C peripheral is used as the master device.

Figure 22-15 C-FIFO management (as master)



First, the software must initialize the C-FIFO management through the R32_I3C_CFGR register CDMAEN, and the specific steps are as follows:

- The software writes directly according to the control word (when CDMAEN=0):
 - By polling mode (setting CFNFIE=0 in R32_I3C_INTENR register): Wait for the hardware to request the next control word (CFNFF=1 in R32_I3C_INTENR register) before explicitly writing into R32_I3C_CTLR register.
 - Through enabled interrupt notification (when CFNFIE=1):
- Written by the assigned DMA channel (when CDMAEN=1) to enable the corresponding I3C peripheral DMA request:
 - According to the configuration at the module level, DMA will automatically push/write the control word from its memory source buffer to the R32_I3C_CTLR register until the frame is completed (after the last message of the frame, a stop bit will be issued on the I3C bus), except when a transmission error occurs.
- In any case, as long as the C-FIFO is empty and the repeated start bit of a new control word must be issued, the C-FIFO underflow will be reported (ERRF=1 in the R32_I3C_EVR register and COVR=1 in the

R32_I3C_STATR register). If the ERRIE bit in the R32_I3C_INTENR register is enabled, an interrupt is generated.

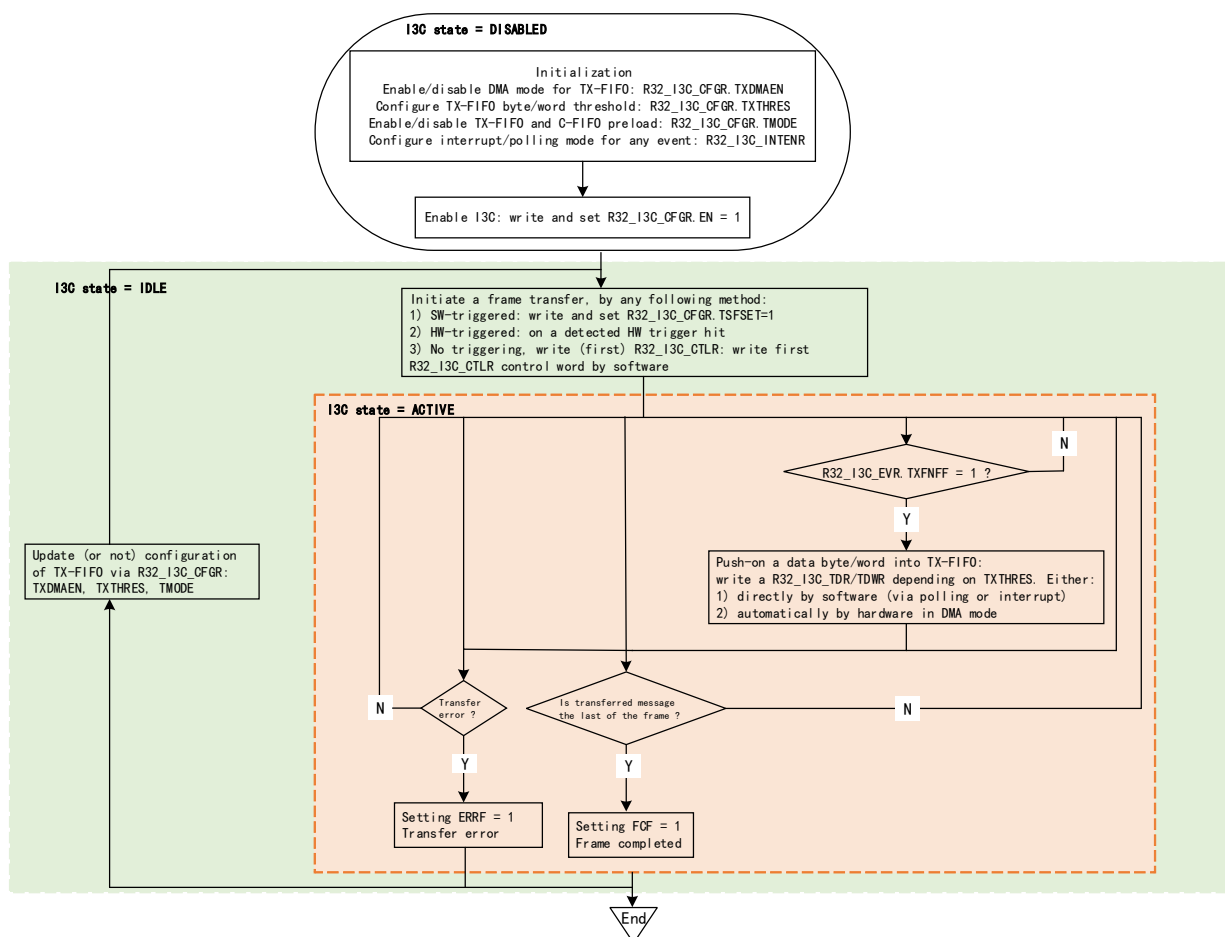
- When the I3C peripheral is not working, the DMA mode configuration of C-FIFO management can be modified.
- When used as a master device, if there is a transmission error (ERRF=1 in the R32_I3C_EVR register), the hardware will automatically empty the CFIFO.

22.3.5.2 TX-FIFO Management (as master)

When operating as a master device, the TX-FIFO may be used during broadcasts or direct CCCs (including ENTDA and RSTACT), when parsing bytes/subcommand bytes or data bytes, during private writes, or during traditional I2C writes.

Figure 22-16 illustrates how the TX-FIFO is managed to queue data bytes or words to be transmitted on the I²C bus when the I²C peripheral operates as a master device.

Figure 22-16 TX-FIFO management (as master)



First, TX-FIFO management must be initialized through the following fields in the R32_I3C_CFGR register:

- TXDMAEN: Enable/disable DMA mode for the TX-FIFO
- TXTHRES: Push data bytes or words into the TX-FIFO
- TMODE: Enable/disable TX-FIFO and C-FIFO preload

Then, according to the TXDMAEN bit in R32_I3C_CFGR register, there are two ways to write data into the TX-FIFO:

- Software writes directly by byte/word (when TXDMAEN=0):

- Through polling mode (R32_I3C_INTENR register TXFNFIE=0): Wait for the hardware to request the next data byte/word (R32_I3C_INTENR register TXFNFF = 1) before explicitly writing to R32_I3C_TDR or R32_I3C_TDWR register, depending on R32_I3C_CGFR register TXTHRES.
- Through enabled interrupt notification (when TXFNFIE = 1)
- Written by the assigned DMA channel (when TXDMAEN = 1) to enable the corresponding I3C peripheral DMA request:
 - According to the configuration at DMA module level, DMA will automatically push/write data bytes/words from its memory source buffer into R32_I3C_TDR or R32_I3C_TDWR register (depending on the TXTHRES bit in R32_I3C_CGFR register) until the frame is completed (after the last message of the frame, a stop bit will be issued on I3C bus), except when a transmission error occurs.

I3C messages start with a start bit or a repeat start bit and end with a stop bit or a repeat start bit. At the message level, the last data byte/word to be sent is indicated by the R32_I3C_EVR register TXLASTF=1. When the I3C frame contains multiple messages (separated by repeated start bits), the software can use this event to update the corresponding pointer to the buffer for storing the byte/word of the next message.

When the report frame completes (R32_I3C_EVR register FCF = 1 and the corresponding interrupt is enabled), the TX-FIFO is empty. If the TX-FIFO is empty and a data byte must be transmitted on the I3C bus, a TX-FIFO underflow is reported (R32_I3C_EVR register ERRF = 1 and R32_I3C_STATER register DOVR = 1). If the corresponding interrupt is enabled (R32_I3C_INTENR register ERRIE = 1), an interrupt is generated. When the I3C peripheral is not active, the configuration managing the TX-FIFO can be modified.

When operating as a master device, the hardware automatically clears the TX-FIFO upon a transmission error (ERRF = 1).

No C-FIFO/TX-FIFO preload

C-FIFO size is 2 words and TX-FIFO size is 8 bytes.

When the preload function of the C-FIFO/TX-FIFO is not enabled, that is, the R32_I3C_CFGR register TMODE=0, once the C-FIFO receives the first control word, the I3C peripheral will immediately send out the start bit on the I3C bus. Subsequently, the peripheral will decode the R32_I3C_CTLR register and write the next data byte or word as needed. Once it is detected that the hardware requires to send the next control word on the I3C bus (for example, it is necessary to send a repeated start bit on the I3C bus or the C-FIFO needs to obtain available space), the peripheral will immediately request to write the control word into the C-FIFO until the last message (that is, R32_I3C_CTLR register MEND=1). Similarly, once it is detected that the hardware requires another data byte or word to be sent on the I3C bus (for example, a repeated start bit needs to be sent on the I3C bus, the TX-FIFO is not full, or the data byte or word must be sent during the I3C message), the data byte/word must be written into the TX-FIFO immediately.

C-FIFO/TX-FIFO preload

When C-FIFO/TX-FIFO preload is configured (R32_I3C_CFGR register TMODE = 1), I3C peripherals need to wait for C-FIFO and TX-FIFO to load as much as possible before sending out the start bit on the bus, as follows:

Wait for the first control word to be written to the C-FIFO. Wait for data bytes/words to be written to the TX-FIFO (if any), as defined by the first control word (if RNW=0 in the R32_I3C_CTLR register and DCNT[15:0]=0), up to the size of the TX-FIFO. If TX-FIFO is not full and the first control word is not the final message in the frame (MEND=0 in the R32_I3C_CTLR register):

- Wait for the second control word to be written into the C-FIFO, after which the C-FIFO becomes full.
- If the TX-FIFO is not full, wait for data bytes/words to be written into the TX-FIFO (if any), as defined by the second control word (RNW=0 in the R32_I3C_CTLR register and DCNT[15:0]), up to the maximum size of the TX-FIFO.

Then, whenever the hardware detects a requirement to transmit the next control word on the I3C bus (if a duplicate start bit must be sent on the I3C bus), it immediately requests that control word be written into the C-FIFO, continuing until the final message (MEND=1 in the R32_I3C_CTLR register).

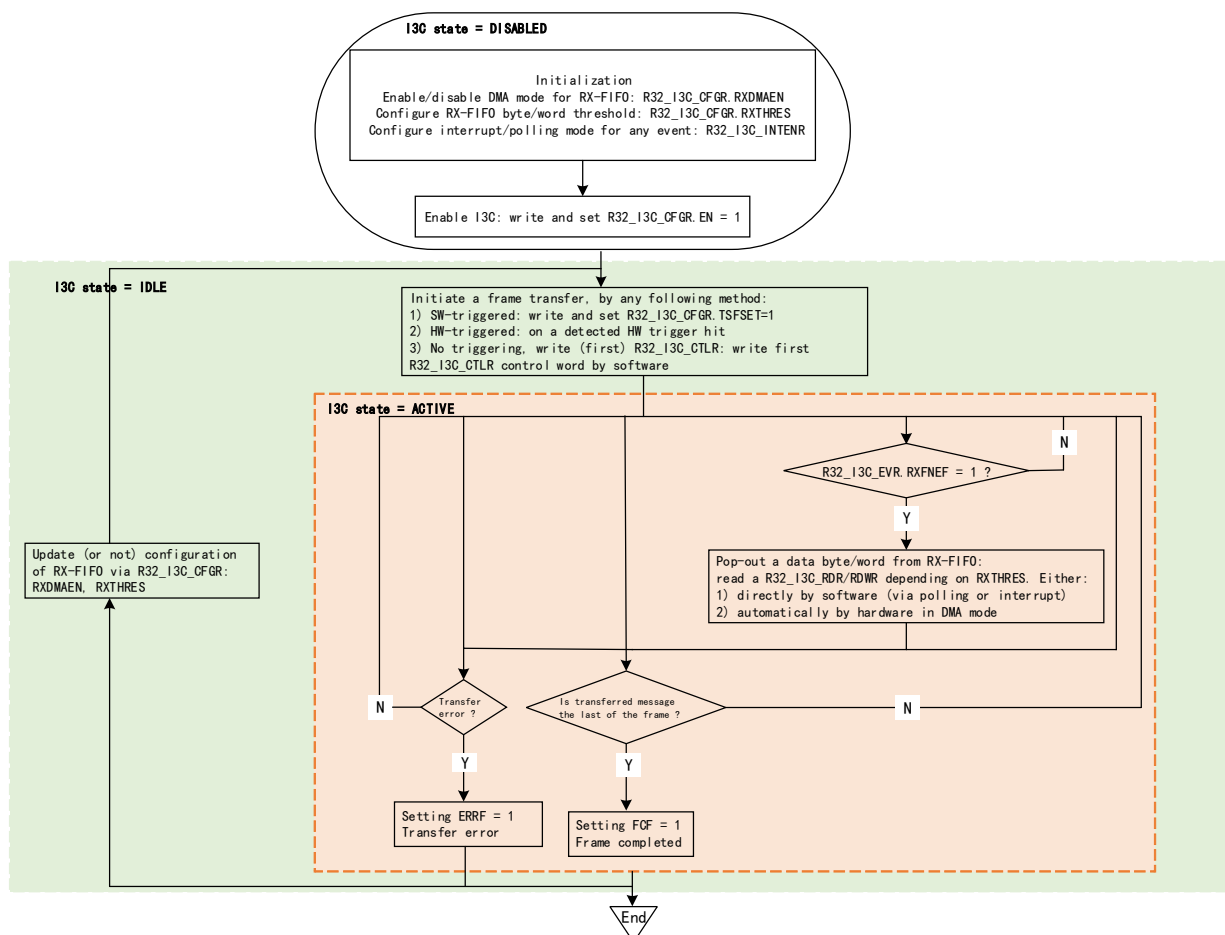
Similarly, whenever a hardware request is detected to send the next data byte/word on the I3C bus (if a repeat start bit must be issued on the I3C bus, the TX-FIFO is not full, or data bytes/words will be sent during that I3C message), that data byte/word must be written to the TX-FIFO immediately.

22.3.5.3 RX-FIFO Management (as master)

When used as a master device, RX-FIFO can be used during broadcasting ENTDAACCC, direct CCC reading (including direct RSTACT CCC reading), private reading and traditional I2C reading.

Figure 22-17 illustrates how to manage the RX-FIFO to queue and eject data bytes or words to be received on the I3C bus when the I3C peripheral is used as the master device.

Figure 22-17 RX-FIFO management (as master)



First, the software must initialize the RX-FIFO management through the following fields in the R32_I3C_CFGR register:

- RXDMAEN: Enable/disable DMA mode for the RX-FIFO
- RXTHRES: Pop data bytes or words from the RX-FIFO

Then, read the RX-FIFO according to the RXDMAEN bit:

- Software reads directly by byte/word (RXDMAEN=0):
 - Through polling mode (RXFNEIE=0 in R32_I3C_INTENR register): Wait for the hardware to request the next data byte/word (RXFNEF=1 in R32_I3C_INTENR register) before explicitly reading R32_I3C_RDR or R32_I3C_RDWR register, depending on R32_I3C_CGFR register.
 - Notify via enabled interrupt (if R32_I3C_INTENR register RXFNEIE=1)
- Read by the assigned DMA channel (if RXDMAEN=1) to enable the corresponding I3C peripheral DMA request:
 - According to the configuration at DMA module level, DMA will automatically pop up/read data bytes/words from R32_I3C_RDR or R32_I3C_RDWR register (depending on the RXTHRES bit) and write them into its memory slave buffer until the frame is completed (after the last message of the frame, a stop bit will be issued on the I3C bus), except when a transmission error occurs.

I3C messages start with a start bit or a repeat start bit and end with a stop bit or a repeat start bit. At the message level, the last data byte/word received from the I3C bus is indicated by IRXLASTF=1 in the I3C_EVR register. When the I3C frame contains multiple messages (separated by repeated start bits), the software can use this event to update the corresponding pointer to the buffer for storing the data byte/word of the next message.

If the RX-FIFO is full and data bytes are received on the I3C bus, the RX-FIFO overflow will be reported (ERRF=1 in the R32_I3C_EVR register and DOVR=1 in the R32_I3C_STATR register). If the corresponding interrupt is enabled (ERRIE=1 in the R32_I3C_INTENR register), an interrupt is generated.

When the I3C peripheral is not working, the configuration of RX-FIFO management can be modified.

Device early termination of read transfer

Disabling the S-FIFO (SMODE=0 in the R32_I3C_CFGR register) will notify the software of a read transfer premature termination via RXTGTENDF=1 in the R32_I3C_EVR register and the corresponding interrupt (if enabled). The software can then read the status register R32_I3C_STATR to examine information related to the last message and obtain the number of data bytes received during the prematurely terminated read transfer (XDCNT[15:0] in the R32_I3C_STATR register).

In any case, if the S-FIFO is enabled (SMODE=1 in the R32_I3C_CFGR register), the software or DMA (depending on SDMAEN in the R32_I3C_CFGR register) must read the status register R32_I3C_STATR once for each message. The number of valid data bytes received in a preemptively terminated read message is reported by XDCNT[15:0] in the R32_I3C_STATR register (followed by ABT=1).

22.3.5.4 S-FIFO Management (as master)

When used as a master device, the software can read the I3C status register (R32_I3C_STATR) for each transmitted message using the S-FIFO.

Reading status register when S-FIFO is disabled

When the SMODE bit of the R32_I3C_CFGR register is set to 0, the S-FIFO is disabled, and the reading mode of the status register is the same as that of a normal register:

- When transmitting a new message, the register contents are overwritten by hardware.

- R32_I3C_STATR contains the status of the last transmitted message.
- The SCL clock does not pause if the status register is not read.

When SMODE=0, that is, the slave device ends the private read transmission in advance under certain circumstances:

- Notify the software by setting the RXTGTENDF bit in the R32_I3C_EVR register to 1 and triggering the corresponding interrupt (if enabled).
 - Then, before the software clears the event flag by writing a 0 to the CRXTGTENDF bit in the R32_I3C_CEV register:
- No further data bytes are received on the I3C bus and are written to the R32_I3C_RDR/R32_I3C_RDWR registers by hardware
 - The R32_I3C_STATR register cannot be updated
 - Pause the SCL clock (if necessary)

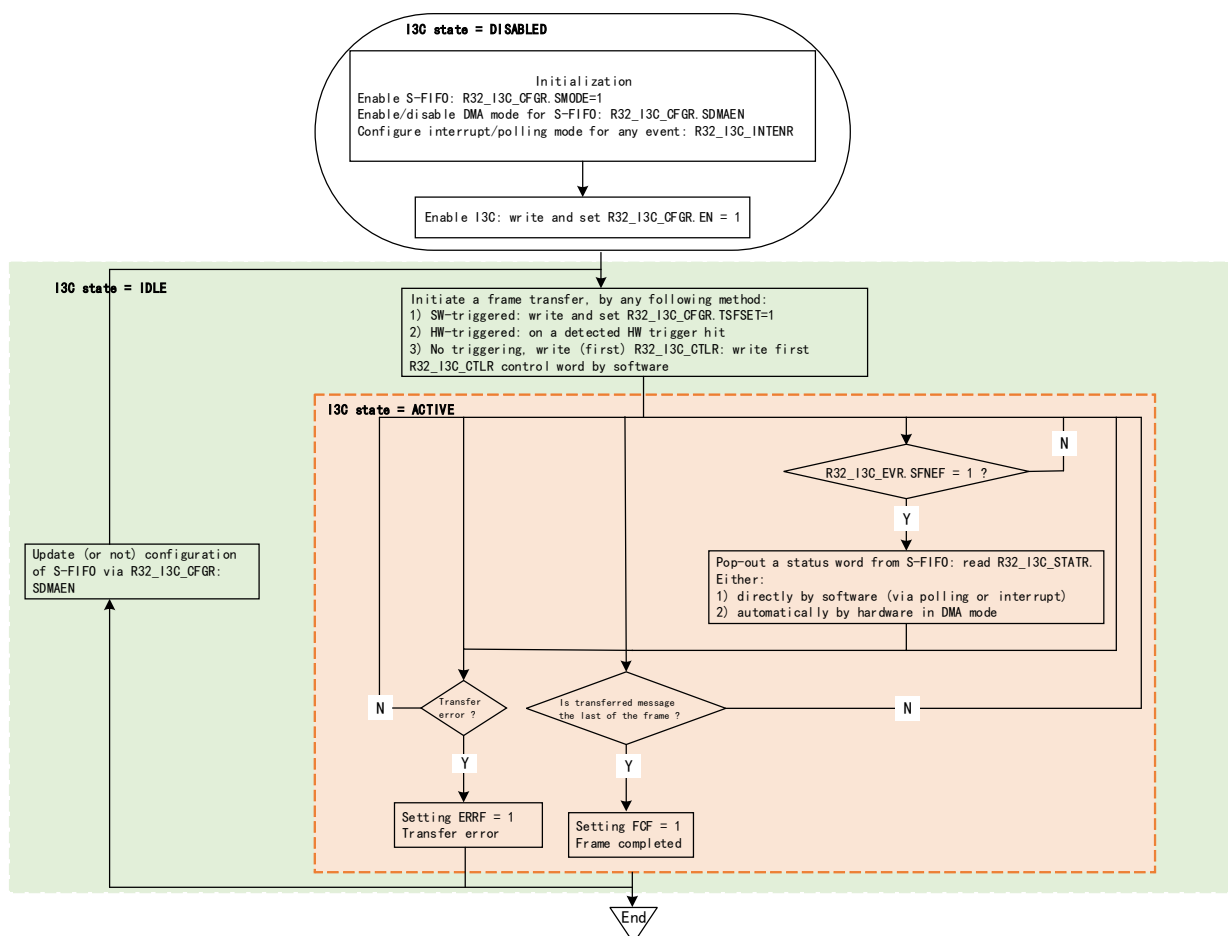
Under normal circumstances, the R32_I3C_STATR register can be read when FCF=1, ERRF=1, or RXTGTENDF=1 in the R32_I3C_EVR register. After the slave device prematurely terminates the read transfer, the number of valid data bytes received during the private read transfer can be obtained by reading XDCNT[15:0] in the R32_I3C_STATR register.

Read status register when S-FIFO is enabled

If the SMODE bit in the R32_I3C_CFGR register is set to 1, the S-FIFO is enabled.

Figure 22-18 illustrates how to manage the S-FIFO to queue and pop up the status word of each message executed on the I3C bus when the I3C peripheral is used as the master device.

Figure 22-18 S-FIFO management (as master)



First, the software must initialize the S-FIFO management through the SDMAEN bit field in R32_I3C_CFGR register (DMA mode of enabling/disabling S-FIFO). Then, read the S-FIFO according to the SDMAEN bit:

- Read directly by software (if SDMAEN=0):
 - Through polling mode (SFNEIE=0 in R32_I3C_INTENR register): Wait for the hardware to request the next status word before explicitly reading I3C_STATR register (SFNEF=1 in R32_I3C_INTENR register).
 - Through enabled interrupt notification (SFNEIE=1)
- Read by the assigned DMA channel (if SDMAEN=1) to enable the corresponding I3C peripheral DMA request (i3c_rs_dma):
 - According to the configuration at DMA module level, DMA will automatically pop up/read status words from R32_I3C_STATR register and write them into its memory slave buffer until the frame is completed (after the last message of the frame, a stop bit will be issued on I3C bus), except in case of transmission error.

Every message status must be read, otherwise when the S-FIFO is full and the next message status must be written, the hardware will set the overflow error flag to 1 (ERRF=1 in the R32_I3C_EVR register and COVR=1 in the R32_I3C_STATR register). If the corresponding interrupt is enabled (ERRIE=1 in the R32_I3C_INTENR register), an interrupt is generated.

Frame completion is reported only when the S-FIFO is empty (FCF=1 in R32_I3C_EVR register).

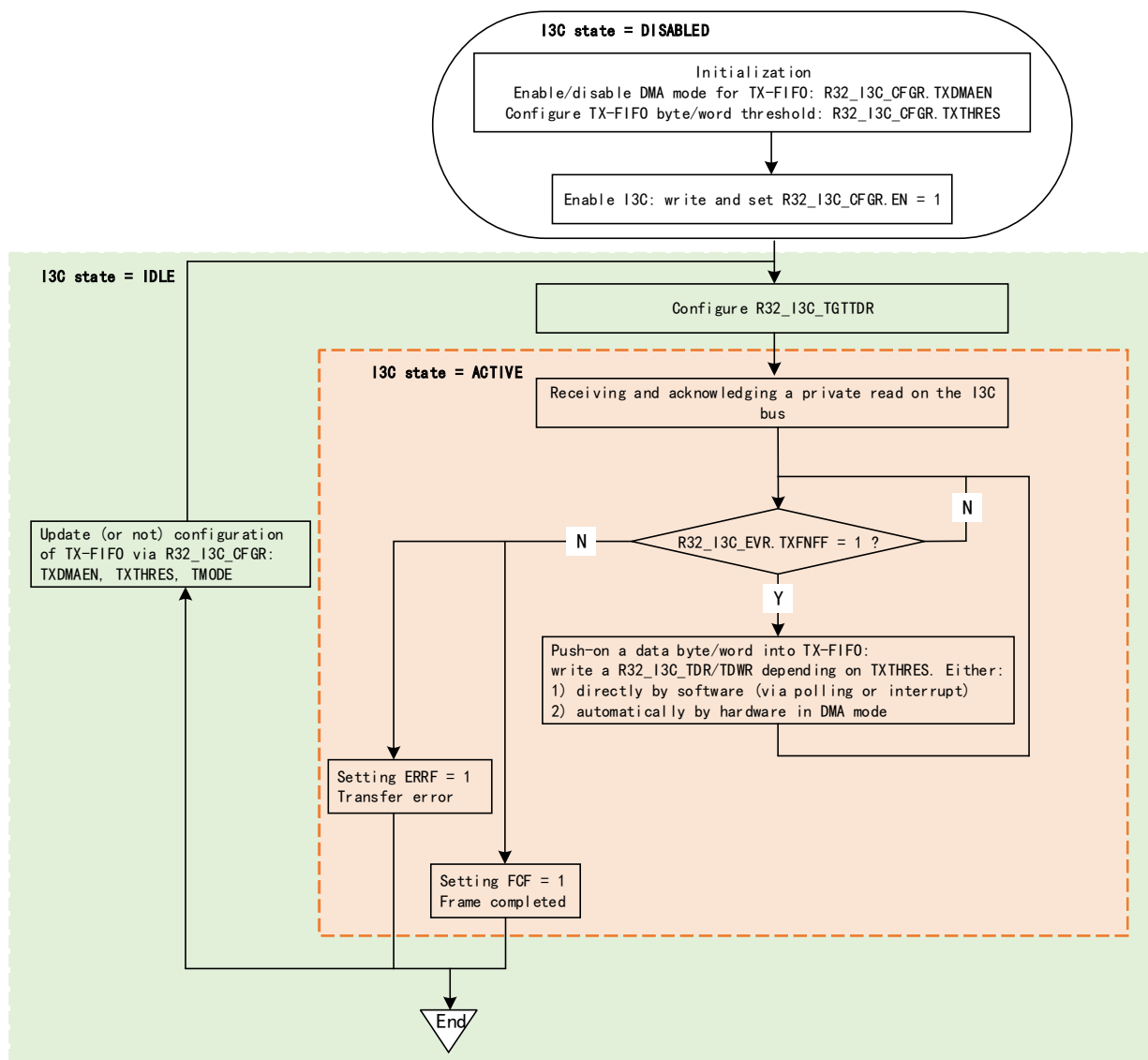
When the I3C peripheral is not working, the configuration of S-FIFO management can be modified.

22.3.5.5 TX-FIFO Management (as slave)

When used as a slave device, the TX-FIFO can only be used during private read transmission.

Figure 22-19 illustrates how to manage the TX-FIFO to queue and push data bytes or words to be sent on the I3C bus when the I3C peripheral is used as a slave device.

Figure 22-19 TX-FIFO management (as slave)



First, the software must initialize the TX-FIFO management and write the following bits in R32_I3C_CFGR:

- TXDMAEN: Enable/disable DMA mode for the TX-FIFO
- TXTHRES: Push data bytes or words into the TX-FIFO

Then, before receiving a private read transfer on the I3C bus, the software must configure the I3C slave device to send the configuration register (R32_I3C_TGTTDR) to preload a specified number of data bytes into the TX-FIFO (by writing TGTTDCNT[15:0] and setting PRELOAD=1 in a single access). This prepares the I3C bus to transmit data bytes from the slave device:

- If PRELOAD=1 and TGTTDCNT[15:0] > TX-FIFO size, the TX-FIFO will first preload up to the FIFO size.
- If PRELOAD=1 and TGTTDCNT[15:0] ≤ TX-FIFO size, the TX-FIFO will preload to TGTTDCNT[15:0].

Pre-load TX-FIFO according to TXDMAEN bit:

- Preloading by byte/word directly by software (TXDMAEN=0):
 - Through polling mode (R32_I3C_INTENR register TXFNFIE=0): Wait for the hardware to request the next data byte/word (R32_I3C_INTENR register TXFNF=1) before explicitly writing to R32_I3C_TDR or R32_I3C_TDWR register, and the specific byte or word depends on the R32_I3C_CFGR register.
 - Via enabled interrupt notification (if TXFNFIE=1)

- Preloaded by the allocated DMA channel (TXDMAEN=1) to enable the corresponding I3C peripheral DMA request (i3c_tx_dma):
 - According to the configuration of DMA module level, DMA will automatically push/write data bytes/words from its memory source buffer into R32_I3C_TDR or R32_I3C_TDWR register (depending on TXTHRES) until the transmission is completed (R32_I3C_EVR register FCF=1), except in case of transmission error.

Then, when the remaining data in TGTDCNT[15:0] of R32_I3C_TGTTDR register is to be loaded into the TX-FIFO to continue private read transmission (PRELOAD is set to 1 and TGTDCNT[15:0]>TX-FIFO size), if the master device has not completed private read transmission, it will be preloaded in the same way (directly through software or through the allocated DMA channel).

- If TXTHRES=0: When one byte is transmitted on the I3C bus, the next byte is preloaded into the TX-FIFO.
- If TXTHRES=1: When four bytes are transmitted on the I3C bus, the next byte is preloaded into the TX-FIFO.

When the slave device or the master device terminates the data byte transmission first, the private read transmission is completed (FCF=1 in the R32_I3C_EVR register). After the transmission is completed, the software can:

- Read XDCNT[15:0] in the R32_I3C_STATR register: Number of valid data bytes transmitted
- Read TGTDCNT[15:0] in the R32_I3C_TGTTDR register: Number of bytes remaining to be loaded and transmitted on the I3C bus
- Clear TX-FIFO: Whether to set TXFLUSH in the R32_I3C_CFGR register to 1 via a write operation and proceed with another private read transfer depends on the user application.

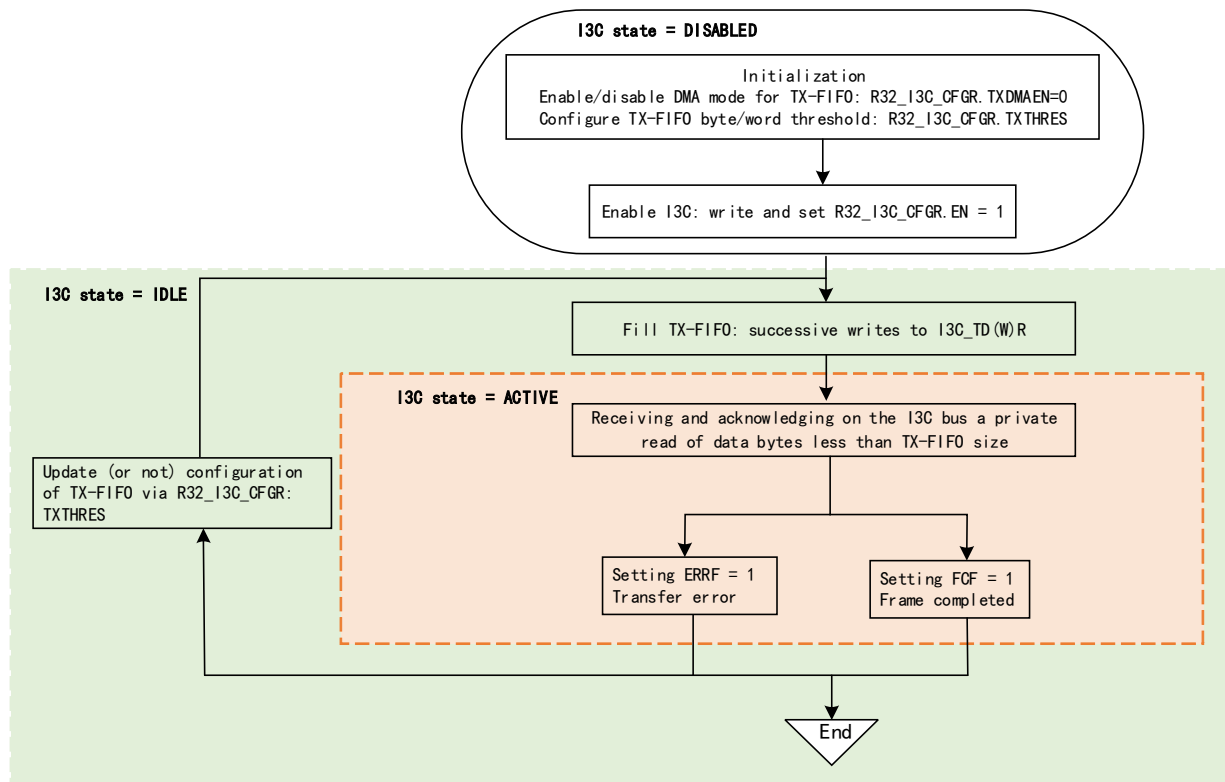
If the TX-FIFO is empty and a data byte must be transmitted on the I3C bus, a TX-FIFO underflow is reported (R32_I3C_EVR register ERRF=1 and R32_I3C_STATR register DOVR=1). If the corresponding interrupt is enabled (R32_I3C_EVR register ERRIE=1), an interrupt is generated.

Configuration managed by the TX-FIFO can be modified when the I3C peripheral is not active.

If the number of bytes is less than the TX-FIFO size, you can choose not to use R32_I3C_TGTTDR.

In addition to using the R32_I3C_TGTTDR register, as shown in Figure 22-20, when DMA is disabled (TXDMAEN=0), if the number of data bytes to be read from the I3C bus is less than the TX-FIFO size, the software can prepare and fill the TX-FIFO directly with the number of bytes read by writing to R32_I3C_TDR or R32_I3C_TDWR consecutively (depending on TXTHRES).

Figure 22-20 If the number of bytes read is smaller than the size of the TX-FIFO, the TX-FIFO (as a slave device) is managed by software without using I3C_TGTTDR

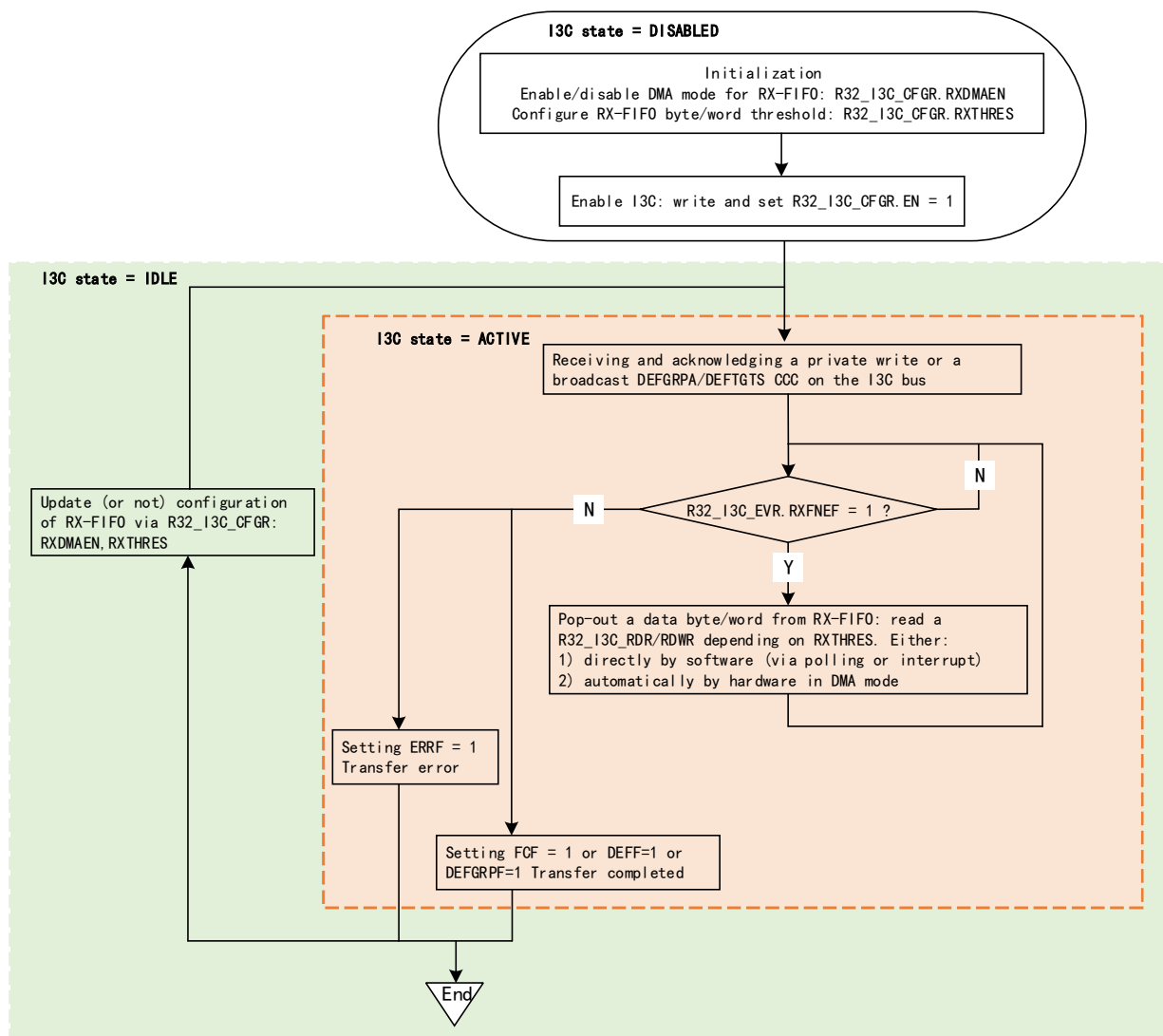


22.3.5.6 RX-FIFO Management (as slave)

When used as a slave device, the RX-FIFO can be used during the received and acknowledged transmission of broadcast DEFTGTS CCC, broadcast DEFGRPA CCC and private writing:

Figure 22-21 illustrates how to manage the RX-FIFO to queue and eject data bytes/words received from the I3C bus when the I3C peripheral is used as a slave device.

Figure 22-21 RX-FIFO management (as slave)



First, the software must initialize the RX-FIFO management through the following R32_I3C_CFGR register bit fields:

- RXDMAEN: Enable/disable DMA mode for the RX-FIFO;
- RXTHRES: Pop data bytes or words from the RX-FIFO;

Then, read the RX-FIFO based on the RXDMAEN bit:

- Read directly by software in bytes/words (if RXDMAEN=0):
 - Through polling mode (RXFNEIE=0 in the R32_I3C_INTENR register): Wait for the hardware to request the next data byte/word (RXFNEF=1 in the R32_I3C_INTENR register) before explicitly reading the R32_I3C_RDR or R32_I3C_RDWR register. Whether it is a byte or a word depends on RXTHRES in the R32_I3C_CFGR register.
 - Via enabled interrupt notification (if RXFNEIE=1)
- Read by the assigned DMA channel (if RXDMAEN=1) to enable the corresponding I3C peripheral DMA request (I3C_RX_DMA):
 - Based on the configuration at the DMA module level, the DMA automatically pops/reads data bytes/words from the R32_I3C_RDR or R32_I3C_RDWR register (depending on RXTHRES) and writes them into its memory slave device buffer until the transfer completes (when FCF=1 in the R32_I3C_EVR register (private write), GRPF=1 (DEFGRPA CCC), or DEFF=1 (DEFTGTS CCC)), except when a transfer error occurs.

If the RX-FIFO is full and a new data byte is received on the I3C bus, an overflow of the RX-FIFO will be reported (ERRF=1 in the R32_I3C_EVR register and DOVR=1 in the R32_I3C_STATR register) and a corresponding interrupt will be generated (if enabled).

When the transmission is completed (FCF=1, GRPF=1 or DEFF=1 in the R32_I3C_EVR register):

- RX-FIFO is empty
- Before the software has finished processing the RX data buffer corresponding to completed private write/DEFTGTS/DEFGRPA transfers (i.e., before the software has cleared the corresponding flags by writing CFCF/CDEFF/CGRPF to the R32_I3C_CEVR register), receiving the next private write/ DEFTGTS CCC/DEFGRPA CCC is received and the hardware must write data bytes to the RX-FIFO, an RX-FIFO overflow error will be reported (ERRF=1 in the R32_I3C_EVR register and DOVR=1 in the R32_I3C_STATR register) and a corresponding interrupt will be generated (if enabled).

When the I3C peripheral is not working, the configuration of RX-FIFO management can be modified.

22.3.6 I3C Error Management

22.3.6.1 Master Device Error Management

Table 22-3 Master device error management

Error type	Description	Error detection	Master device operation	Report errors
CE0	Illegally formatted CCC (for example, fewer data bytes are returned)	When direct CCC read transmission, the slave device ends reading data ahead of schedule.	Hardware issue stop bit	ERRF=1, PERR=1 and CODERR[3:0]=0000b
CE1	Monitoring error	An incorrect ACK was detected at the end of the traditional I2C read transmission.	The hardware keeps SCL running for nine clock cycles, and another byte can be exchanged, and then a NACK is issued again, followed by a stop bit.	ERRF=1, PERR=1 and CODERR[3:0]=0001b
		When I3C SDR read transmission ends, repeated start bit or stop bit cannot be generated.	SCL remains running. Software can stop SCL by writing the control word MTYPE[3:0]=0000 (header message) to the R32_I3C_CTLR register, and then it usually has to wait at least 150μs before sending another message.	
CE2	No response to broadcast address (1111110b)	In the process of message transmission (except for upgrade failure or reset timing message), it is detected that the header	The hardware issues HDR exit timing followed by a stop bit.	ERRF=1, PERR=1 and CODERR[3:0]=0010b

		(1111110b+RnW=0) is NACK.		
CE3	Master device switching failed.	The new master device has not driven SCL low after the SDA signal went low (triggered by a test header or a start request from a slave device) and after the delay time defined in its operating state has elapsed.	The hardware issues the start bit +1111110b+RnW=0, followed by the ACK/NACK from the slave device, and finally the stop bit.	ERRF=1, PERR=1 and CODERR[3:0]=0011b
-	The 7-bit slave address with odd parity returned in GETACCCR CCC is incorrect.	An incorrect dynamic address and/or parity bit was detected.	The hardware cancels the GETACCCR CCC by issuing the restart bit +1111110b+RnW=0, followed by the ACK/NACK from the slave device, and finally the stop bit.	ERRF=1 and DERR=1
-	The addressed slave device was NACK for the first time in the direct CCC read transmission.	NACK detected	The hardware performs a single retry by issuing a duplicate start bit +7 bits of the same slave address +RnW=1.	-
-	The addressed slave device is NACK for the second time in direct CCC write transmission, I3C private read/write transmission, traditional I2C transmission or direct CCC read transmission.	NACK detected	The hardware issues a stop bit.	ERRF=1 and ANACK=1
-	The allocated/sent dynamic address with parity bit was NACK in ENTDAACCC for the first time.	NACK detected	The hardware executes a single retry allocation cycle by transmitting a repeated start bit followed by 1111110b + RnW = 1, then receiving an ACK and 8 bytes of read data from the (highest priority) slave device, followed by	-

			the allocated address and parity bit.	
-	The allocated/transmitted dynamic address with parity bit was NACK in ENTDAACCC for the second time.	NACK detected	The hardware issues a stop bit.	ERRF=1 and DNACK=1
-	Write data is NACK in traditional I2C write transmission.			
-	Not writing/reading control words, status words, sending data or reading data in time relative to I3C bus timing.	SCL pause timeout	The hardware issues a stop bit.	ERRF=1 and (COVR=1 or DOVR=1)

22.3.6.2 Slave Device Error Management

Table 22-4 Slave device error management

Error type	Description	Error detection	Master device operation	Report errors
TE0	The broadcast address (1111110b+RnW=0) allocated by DAA is invalid or the 7-bit dynamic address +RnW=1 is invalid.	A forbidden address was detected after the start bit or repeated start bit.	Hardware waiting HDR exit timing	ERRF=1, PERR=1 and ODERR[3:0]=1000b
TE1	CCC code	A parity error was detected in CCC code.		ERRF=1, PERR=1 and ODERR[3:0]=1001b
TE2	Write data	A parity error was detected in the I3C private write message in the write data byte.	Hardware waits for stop bit or repeat start bit.	ERRF=1, PERR=1 and ODERR[3:0]=1010b
TE3	Addresses assigned during dynamic address arbitration	A parity error was detected in the assigned dynamic address in ENTDAACCC.		ERRF=1, PERR=1 and ODERR[3:0]=1011b
TE4	Missing 1111110b+RnW=1	Detected missing {1111110b+RnW=1}		ERRF=1, PERR=1 and

	after Sr bit during dynamic address arbitration.	bit after Sr bit during dynamic address arbitration.		ODERR[3:0]=1100b
TE5	Transactions after CCC detection	In CCC direct read/write transmission, an invalid CCC direction (relative to the direction provided by the address phase) was detected.		ERRF=1, PERR=1 and ODERR[3:0]=1101b
TE6	Monitoring error	During SDR data read transmission (during direct CCC reading, private reading or IBI transmission), it is detected that SDA is driven by an abnormal value.	The hardware releases the SDA and then waits for the stop bit or repeats the start bit.	ERRF=1, PERR=1 and ODERR[3:0]=1110b
-	Monitor SCL during SDR data reading and transmission	During SDR data read transmission (during direct CCC reads, private reads, or IBI transfers), the SCL stabilization time exceeds 125μs.		ERRF=1 and STALL=1
-	Data to be sent is not written/available in time (relative to I3C bus timing)	In private read transmission, the software or DMA did not write the data to be sent in time.	The hardware detected a stop bit or repeated start bit.	ERRF=1 and DOVR=1
-	Unable to read the received data in time (relative to I3C bus timing)	In private writing, DEFTGTS or DEFGRPA CCC, the software or DMA did not read the received data in time.		

22.3.7 I3C Low-power Mode

I3C in low-power mode is not affected when it is in sleep mode, and I3C interrupt can make the device exit sleep mode.

When in low power mode, changes on the SCL and SDA buses wake up the system and exit low power mode.

22.3.8 I3C Interrupt

Table 22-5 I3C interrupt request

Interrupt event	Interrupt division	As a master device	As a slave device	Interrupt enable control	Event flag	Event clearing control
Request control word	Request	×	-	CFNFIE	CFNFF	-
Status word available		×	-	SFNEIE	SFNEF	-
Request data to be transmitted		×	×	TXFNFIE	TXFNFF	-
Request data to be received		×	×	RXFNEIE	RXFNEF	-
Master: Frame transmission completed Slave: Private transmission completed		×	×	FCIE	FCF	CFCF
The slave device ends the private read transmission ahead of schedule (and the bit of the R32_I3C_CFGR register SMODE is 0).		×	-	RXTGTENDIE	RXTGTENDF	CRXTGTENDF
Received IBI request		×	-	IBIIE	IBIF	CIBIF
Received a request from the master device.		×	-	CRIE	CRF	CCRF
Hot join request received		×	-	HJIE	HJF	CHJF
IBI request completed		-	×	IBIENDIE	IBIENDF	CIBIENDF
I3C bus start bit lost		-	×	WKPIE	WKPF	CWKPF
Received direct GETACCR CCC		-	×	CRUPDIE	CRUPDF	CCRUPDF
Received direct GETSTATUS CCC		-	×	STAIE	STAF	CSTAF
Receive any direct connection to GET xxx CCC (except GETSTATUS)		-	×	GETIE	GETF	CGETF
Dynamic address update (received broadcast ENTDAAs, broadcast RSTDAA or direct SETNEWDA)		-	×	DAUPDIE	DAUPDF	CDAUPDF
Received direct SETMWL CCC		-	×	MWLUPDIE	MWLUPDF	CMWLUPDF
Received direct SETMRL CCC		-	×	MRLUPDIE	MRLUPDF	CMRLUPDF
Reset timing detected		-	×	RSTIE	RSTF	CRSTF
Bus status update (direct/broadcast ENTASx CCC received)		-	×	RSTIE	ASUPDF	CASUPDF
Received broadcast/direct		-	×	INTUPDIE	INTUPDF	CINTUPDF

ENEC/DISEC CCC						
Received broadcast DEFTGTS CCC		-	×	DEFIE	DEFF	CDEFF
Received broadcast DEFGRPA CCC		-	×	GRPIE	GRPF	CGRPF
An error occurred	Error	×	×	ERRIE	ERRF	CERRF

22.4 Register Description

Table 22-6 I3C-related registers

Name	Access address	Description	Reset value
R32_I3C_CTLR	0x40014400	I3C control register	0x00000000
R32_I3C_CFGR	0x40014404	I3C configuration register	0x00000000
R32_I3C_RDR	0x40014410	I3C receive data byte register	0x00000000
R32_I3C_RDWR	0x40014414	I3C received data word register	0x00000000
R32_I3C_TDR	0x40014418	I3C transmit data byte register	0x00000000
R32_I3C_TDWR	0x4001441C	I3C transmit data word register	0x00000000
R32_I3C_IBIDR	0x40014420	I3C IBI payload data register	0x00000000
R32_I3C_TGTTDR	0x40014424	I3C slave device transmits configuration register	0x00000000
R32_I3C_RESET	0x4001442C	I3C reset register	0x03000000
R32_I3C_STATR	0x40014430	I3C state register	0x00000000
R32_I3C_STATER	0x40014434	I3C status error register	0x00000000
R32_I3C_RMR	0x40014440	I3C received message register	0x00000000
R32_I3C_EVR	0x40014450	I3C event register	0x00000003
R32_I3C_INTENR	0x40014454	I3C interrupt enable register	0x00000000
R32_I3C_CEV	0x40014458	I3C clear event register	0x00000000
R32_I3C_DEVR0	0x40014460	I3C device register 0	0x00000000
R32_I3C_DEVR1	0x40014464	I3C device register 1	0x00000000
R32_I3C_DEVR2	0x40014468	I3C device register 2	0x00000000
R32_I3C_DEVR3	0x4001446C	I3C device register 3	0x00000000
R32_I3C_DEVR4	0x40014470	I3C device register 4	0x00000000
R32_I3C_MAXRLR	0x40014490	I3C maximum read length register	0x00000000
R32_I3C_MAXWLR	0x40014494	I3C maximum write length register	0x00000000
R32_I3C_TIMINGR0	0x400144A0	I3C timing register 0	0x00000000
R32_I3C_TIMINGR1	0x400144A4	I3C timing register 1	0x00000000
R32_I3C_TIMINGR2	0x400144A8	I3C timing register 2	0x00000000
R32_I3C_BCR	0x400144C0	I3C bus feature register	0x00000000
R32_I3C_DCR	0x400144C4	I3C device feature register	0x00000000
R32_I3C_GETCAPR	0x400144C8	I3C get capability register	0x00000000
R32_I3C_CRCAPR	0x400144CC	I3C master role capability register	0x00000000
R32_I3C_GETMDSR	0x400144D0	I3C get maximum data speed register	0x00000000
R32_I3C_EPIDR	0x400144D4	I3C extended configuration ID register	0x00000000

22.4.1 I3C Control Register (R32_I3C_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEN D	MTYPE[3:0]				Reserved				ADD[6:0]						RN W
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCNT[15:0]															

Bit	Name	Access	Description	Reset value
31	MEND	WO	Message end type/last message in frame: 1: This message from the master device ends with a stop bit and is the last message in the frame; 0: This message from the master device is followed by a repeat start bit before another message can be sent. <i>Note: (1) This bit applies only to the master device.</i> <i>(2) When MTYPE[3:0] = 0100b, this bit must not be 0.</i>	0
[30:27]	MTYPE[3:0]	WO	Message type When I3C is used as the master device: 0000: SCL clock forced stop until the next control word is executed; 0001: Header message; 0010: Private message; 0011: Direct message (second part of I3C SDR direct CCC commands); 0100: Traditional I2C message; Others: Reserved. When I3C is used as a slave device: 1000: Hot-add request; 1001: Master role request; 1010: IBI (in-band interrupt) request; Others: Reserved.	0
[26:24]	Reserved	RO	Reserved	0
[23:17]	ADD[6:0]	WO	7-bit I3C dynamic/I2C static slave device address. <i>Note: When I3C operates as a master device, this bit field is used if MTYPE[3:0] is 0010b, 0011b, or 0100b.</i>	0
16	RNW	WO	Read/Non-Write Message. 1: Read message; 0: Write message.	0

			<i>Note: When I3C operates as a master device, this bit field is used if MTYPE[3:0] is 0010b, 0011b, or 0100b.</i>	
[15:0]	DCNT[15:0]	WO	Data count to be transmitted during read/write operations, in bytes: 0x0000: No data to transmit; 0x0001: 1 byte; 0x0002: 2 bytes;... 0xFFFF: 64KB-1 byte. <i>When I3C operates as a master device, this bit field is used to set the number of data bytes exchanged on the bus if MTYPE[3:0] is 0010b, 0011b, or 0100b. For private or traditional I2C read/write messages, this bit field must be non-empty.</i> <i>When I3C operates as a slave device, this bit field is used to set the number of bytes for the IBI data payload (1, 2, 3, or 4) if MTYPE[3:0] is 1010b and any IBI data payload exists (data to be sent when the BCR2 bit in the I3C_BCR register is set to 1).</i>	0

22.4.2 I3C Control Register[multiplex] (R32_I3C_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEND	MTYPE[3:0]				Reserved				CCC[7:0]						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCNT[15:0]															

Bit	Name	Access	Description	Reset value
31	MEND	WO	Message end type/last message in frame: 1: This message from the master device ends with a stop bit and is the last message in the frame; 0: This message from the master device is followed by a repeat start bit before another message can be sent. <i>Note: (1) This bit applies only to the master device. (2) When MTYPE[3:0] = 0100b, this bit must not be 0.</i>	0
[30:27]	MTYPE[3:0]	WO	Message type. When I3C is used as master: 0110: Broadcast/direct CCC commands;	0

			Bit [23:16] (CCC[7:0]) is the transmitted 8-bit CCC code; Bit [15:0] (DCNT[15:0]) indicates the number of CCC interpretation bytes, CCC subcommand bytes, or CCC data bytes. If bit [23] = CCC[7] = 1: The message is a direct CCC command transmitted as the first part of an I3C SDR direct CCC command. The direct CCC command (first part) message is: - {S/S + 0b111_1110 + RnW = 0/Sr+*} + (Direct CCC + T) + (8-bit data + T)* + Sr; - After S (start byte), the NOARBH bit in the R32_I3C_CFGR register determines whether to insert an arbitration header (0b111_1110 + RnW = 0). - Sr+*: After Sr (repeat start byte), the hardware automatically inserts it (0b111_1110 + R/W). If bit [23] = CCC[7] = 0: The message is an I3C SDR broadcast CCC command; the transmitted broadcast CCC command message is: - {S/S + 0b111_1110 + RnW = 0/Sr + *} + (Broadcast CCC + T) + (8-bit data + T)* + Sr/P - After S (Start bit), the NOARBH bit determines whether to insert an arbitrable header (0b111_1110 + RnW = 0). - Sr+*: After Sr (Repeat Start bit), the hardware automatically inserts (0b111_1110 + R/W). Other values: Reserved.	
[26:24]	Reserved	RO	Reserved	0
[23:16]	CCC[7:0]	WO	8-bit CCC code: If bit [23] = CCC[7] = 1, the message is the first part of an I3C SDR direct CCC command. If bit [23] = CCC[7] = 0, the message is an I3C SDR broadcast CCC command (including ENTDAAs). <i>Note: This bit applies only to the master device.</i>	0
[15:0]	DCNT[15:0]	WO	The count (in bytes) of CCC command-related data to be transmitted as CCC interpretation bytes, CCC subcommand bytes, or CCC data bytes: Linear encoding up to 64KB - 1. 0x0000: No data to transmit; <i>Note: Mandatory value when issuing an ENTDAAs broadcast CCC.</i> 0x0001: 1 byte; <i>Note: Mandatory value when issuing a direct/broadcast CCC for RSTACT.</i> 0x0002: 2 bytes;	0

		...	
		0xFFFF: 64KB - 1 byte.	

22.4.3 I3C Configuration Register (R32_I3C_CFGR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	TSFSET	Reserved								CFLUSH	CDMAEN	TMODE	SMODE	SFLUSH	SDMAEN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TXTHRESH	TXFLUSH	TXDMAEN	Reserved	RXTHRESH	RXFLUSH	RXDMAEN	HJACK	Reserved	EXIPTRN	RSTPTRN	NOARBH	CRIINIT	EN	

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
30	TSFSET	W0	Frame transfer settings: 1: Hardware sets the CFNFF bit to initiate frame transfer; 0: No action is taken. <i>Note: (1) This bit applies only to the master device.</i> <i>(2) If this bit is not set to 1, frame transfer can be initiated via software by directly writing the first control word register (R32_I3C_CTLR) when the C-FIFO is empty (R32_I3C_EVR register's CFEF bit is 1). Then, if the first control word written is not marked as message end (MEND bit in R32_I3C_CTLR register is 0), it will cause the hardware to set CFNFF to 1.</i> <i>Note: This bit applies only to the master device.</i>	0
[29:22]	Reserved	RO	Reserved	0
21	CFLUSH	W0	C-FIFO clear: 1: Clear C-FIFO; 0: Do not perform any operation. <i>Note: This bit applies only to the master device.</i>	0
20	CDMAEN	RW	C-FIFO DMA request enable: 1: Enables DMA mode for C-FIFO; 0: Disables DMA mode for C-FIFO. <i>Note: This bit applies only to the master device.</i>	0
19	TMODE	RW	Frames transmitted on the I3C bus manage the C-FIFO and TX-FIFO: 1: Preload the C-FIFO and TX-FIFO before initiating frame transmission.	0

			0: Do not preload the C-FIFO and TX-FIFO before initiating frame transmission. Frame transmission begins immediately when the first control word appears in the C-FIFO. <i>Note: This bit applies only to the master device.</i>	
18	SMODE	RW	S-FIFO enable/status receive mode: When I3C operates as a master device: 1: Enable S-FIFO; 0: Disable S-FIFO. When I3C operates as a slave device, this bit must be cleared.	0
17	SFLUSH	WO	S-FIFO clear: 1: Clear S-FIFO; 0: Do not perform any operation. <i>Note: This bit applies only to the master device.</i>	0
16	SDMAEN	RW	S-FIFO DMA request enable: 1: Enables reading the status register R32_I3C_STATER via DMA mode; 0: Disables reading the status register R32_I3C_STATER via DMA mode. <i>Note: This bit applies only to the master device.</i>	0
15	Reserved	RO	Reserved	0
14	TXTHRES	RW	TX-FIFO threshold: 1: 1-word/4-byte threshold; 0: 1-byte threshold.	0
13	TXFLUSH	WO	TX-FIFO clear: 1: Clear the TX-FIFO; 0: Do not perform any operation.	0
12	TXDMAEN	RW	TX-FIFO DMA request enable: 1: Enables DMA mode for the TX-FIFO; 0: Disables DMA mode for the TX-FIFO.	0
11	Reserved	RO	Reserved	0
10	RXTHRES	RW	RX-FIFO threshold: 1: 1-word/4-byte threshold; 0: 1-byte threshold.	0
9	RXFLUSH	WO	RX-FIFO clear: 1: Clear the TX-FIFO; 0: Do not perform any operation.	0
8	RXDMAEN	RW	RX-FIFO DMA request enable: 1: Enables DMA mode for the RX-FIFO; 0: Disables DMA mode for the RX-FIFO.	0
7	HJACK	RW	Hot join request acknowledgment: 1: Hot join request acknowledged; 0: Hot join request not acknowledged.	0

			<i>Note: This bit applies only to the master device.</i>	
[6:5]	Reserved	RO	Reserved	0
4	EXITPTRN	RW	HDR exit timing enable: 1: After issuing the message header (R32_I3C_CTLR register MTYPE[3:0] set to 0001b), transmit the HDR exit timing. This is used for master device error detection and escalation when the slave device fails to respond to a private message or a direct CCC read. 0: Do not transmit HDR exit timing after issuing the message header (R32_I3C_CTLR register MTYPE[3:0] set to 0001b). This is used to send the header and test bus ownership if a problem is suspected after a master role switch (where the new master fails to enable its master role by accessing the previous master within the delay defined in the operating state). <i>Note: This bit applies only to the master device.</i>	0
3	RSTPTRN	RW	HDR reset timing enable: 1: Insert HDR reset timing before the stop bit of any transmitted frame (including RSTACT CCC commands); 0: Transmit standard stop bit at frame end. <i>Note: This bit applies only to the master device.</i>	0
2	NOARBH	RW	No arbitration header after the start bit: 1: No arbitration header - For traditional I2C messages or I3C SDR proprietary read/write messages, the slave device address is sent immediately after the start bit. - This is a more efficient option (when sending the 0x7E arbitrable header is unnecessary), but it can only be used when the master is certain that the addressed slave device cannot simultaneously send an IBI or Master Role Request (to ensure that for IBIs or MRs, there is no misinterpretation or potential conflict between the address sent by the master in open-drain mode and the same address that the slave device could send after the start bit). 0: Arbitrated header (0b111_1110 + RnW = 0) transmitted after the start bit, preceding traditional I2C messages or I3C SDR proprietary read/write messages (default). <i>Note: This bit applies only to the master device.</i>	0
1	CRINIT	RW	Mark primary/secondary device roles: 1: Master device role;	0

			0: Slave device role.	
0	EN	RW	I3C enable: 1: Enable I3C; 0: Disable I3C.	0

22.4.4 I3C Receive Data Byte Register (R32_I3C_RDR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								RDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	RDB0[7:0]	RO	8-bit receive data on the I3C bus.	0

22.4.5 I3C Receive Data Word Register (R32_I3C_RDWR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDB3[7:0]								RDB2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDB1[7:0]								RDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	RDB3[7:0]	RO	8-bit receive data (latest byte on I3C bus).	0
[23:16]	RDB2[7:0]	RO	8-bit receive data (bytes following RDB1 on the I3C bus).	0
[15:8]	RDB1[7:0]	RO	8-bit receive data (bytes following RDB0 on the I3C bus).	0
[7:0]	RDB0[7:0]	RO	8-bit receive data (earliest byte on the I3C bus).	0

22.4.6 I3C Transmit Data Byte Register (R32_I3C_TDR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0

[7:0]	TDB0[7:0]	RO	8-bit transmit data on the I3C bus.	0
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22.4.7 I3C Transmit Data Byte Register (R32_I3C_TDWR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TDB3[7:0]								TDB2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDB1[7:0]								TDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	TDB3[7:0]	WO	8-bit transmit data (latest byte on I3C bus).	0
[23:16]	TDB2[7:0]	WO	8-bit transmit data (bytes following RDB1 on the I3C bus).	0
[15:8]	TDB1[7:0]	WO	8-bit transmit data (bytes following RDB0 on the I3C bus).	0
[7:0]	TDB0[7:0]	WO	8-bit transmit data (earliest byte on the I3C bus).	0

22.4.8 I3C IBI Payload Data Register (R32_I3C_IBIDR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IBIDB3[7:0]								IBIDB2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIDB1[7:0]								IBIDB0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	IBIDB3[7:0]	RW	8-bit IBI payload data (latest byte on I3C bus).	0
[23:16]	IBIDB2[7:0]	RW	8-bit IBI payload data (byte following IBIDB1[7:0] on I3C bus).	0
[15:8]	IBIDB1[7:0]	RW	8-bit IBI payload data (bytes following IBIDB0[7:0] on I3C bus).	0
[7:0]	IBIDB0[7:0]	RW	8-bit IBI payload data (the earliest byte on the I3C bus, MDB[7:0] forced data bytes).	0

22.4.9 I3C Slave Transmit Configuration Register (R32_I3C_TGTTDR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														PRELOAD	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGTTDCNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	PRELOAD	RW	TX-FIFO Preload: 1: Execute TX-FIFO preload; 0: Do not execute TX-FIFO preload. <i>Note: This bit applies only to slave devices.</i>	0
[15:0]	TGTTDCNT[15:0]	RW	Data transmission counter, in bytes When PRELOAD is set to 1, this bit field must be written by software during the same access to define the number of bytes to be preloaded and transmitted. This bit field is updated by hardware; reading it reports the number of bytes remaining to be loaded into the TX-FIFO. <i>Note: This bit applies only to slave devices.</i>	0

22.4.10 I3C Reset Register (R32_I3C_RESET)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						TG T_S IE_ RS T	HS T_S IE_ RS T	Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bit	Name	Access	Description	Reset value
[31:26]	Reserved	RO	Reserved	0
25	TGT_SIE_RST	RW	Reset status of the slave device: 1: The protocol layer of the slave device is in reset state; 0: The protocol layer of the slave device is in operational state;	1
24	HST_SIE_RST	RW	Reset status of the master device: 1: The protocol layer of the master device is in reset state; 0: The protocol layer of the master device is in operational state;	1
[23:0]	Reserved	RO	Reserved. Must be set to 0x1000 during initialization.	0

22.4.11 I3C Status Register (R32_I3C_STATR)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

MID[7:0]								Reserved					DIR	AB T	Res erve d
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XDCNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:24]	MID[7:0]	RO	Message Identifier/Counter for a Given Frame: When I3C operates as a master device, this bit field identifies the control word message referenced by the R32_I3C_STATR status register (R32_I3C_CTLR). The first message in a frame is identified with MID[7:0] = 0. This bit field is incremented (by hardware) when a new message control word (R32_I3C_CTLR) on the I3C bus is completed. It is reset each time a new frame begins. <i>Note: This bit applies only to the master device.</i>	0
[23:19]	Reserved	RO	Reserved	0
18	DIR	RO	Message Direction: 1: Read; 0: Write. <i>Note: ENTDAACCC is treated as a write command.</i>	0
17	ABT	RO	Early termination of private read messages from the slave device: 1: Slave device completed early; 0: Slave device did not complete early. <i>Note: This bit applies only to the master device.</i>	0
16	Reserved	RO	Reserved	0
[15:0]	XDCNT[15:0]	RO	Data counters: Condition: During dynamic address allocation (ENTDAACCC) - When I3C is used as a master device: The number of slave devices detected on the bus; - When I3C is used as a slave device: The number of bytes transmitted. Condition: For other transfers during the message period - Regardless of whether I3C is used as a master or slave device: The number of data bytes read from the I3C bus or transmitted on the I3C bus during the message period.	0

22.4.12 I3C Status Error Register (R32_I3C_STATER)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DE RR	DN ACK	AN ACK	CO VR	DO VR	ST AL L	PER R	CODERR[3:0]			

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	DERR	RO	Data error: 1: The master detects a data error during the master role switch (GETACC RCCC, formerly GETACC MST) when the received slave address or/and parity bits do not match. The active master retains the master role. 0: No errors detected. <i>Note: This bit applies only to the master device.</i>	0
9	DNACK	RO	Data not confirmed: 1: The master device detected that the slave device did not acknowledge data bytes during the following processes: i) Traditional I2C write transfer; ii) Second attempt to send a dynamic address during the ENTDA process. 0: No error detected. <i>Note: This bit applies only to the master device.</i>	0
8	ANACK	RO	Address not confirmed: 1: The master detects that the slave does not acknowledge static/dynamic addresses in the following procedure: i) Traditional I2C read/write transfer; ii) Direct CCC write transfer; iii) Try direct CCC read transfer for the second time; iv) Private read/write transfer; 0: No errors were detected. <i>Note: This bit applies only to the master device.</i>	0
7	COVR	RO	C-FIFO underflow or S-FIFO overflow: 1: Master device detected: i) C-FIFO underflow: Control FIFO is empty; a repeat start bit must be issued; ii) S-FIFO overflow: S-FIFO is full; new message ends; 0: No error detected.	0

			<i>Note: This bit applies only to the master device.</i>	
6	DOVR	RO	RX-FIFO overflow or TX-FIFO underflow: 1: Regardless of whether acting as master or slave, the hardware detects: i) TX-FIFO underflow: TX-FIFO is empty and must transmit a write data byte; ii) RX-FIFO overflow: RX-FIFO is full and receives a new data byte; 0: No error detected.	0
5	STALL	RO	SCL pause error: 1: During I3C SDR data read transmission (including direct CCC read, private read, or IB transmission), the device detected an SCL stabilization time exceeding 125µs; 0: No error detected. <i>Note: This bit applies only to the slave device.</i>	0
4	PERR	RO	Protocol error: 1: Protocol error detected by hardware, whether acting as master or slave device; 0: No error detected.	0
[3:0]	CODERR[3:0]	RO	0000: CE0 Error (Transaction After CCC Transmission): The master device detected an illegally formatted CCC; 0001: CE1 Error (Monitoring Error): The master device detected data transmitted on the bus that did not match expectations; 0010: CE2 Error (No Response to Broadcast Address): The master device detected an unacknowledged broadcast address (0b111_1110); 0011: CE3 Error (Master Role Switch Failure): The master device detected that the new master device failed to drive the bus after the master role switch; 1000: TE0 Error (Invalid Broadcast Address 0b111_1110+W): The slave device detected an invalid broadcast address 0b111_1110+W; 1001: TE1 Error (CCC Code): The slave device detected a parity error in the CCC code via parity check (relative to the T bit); 1010: TE2 Error (Write Data): The slave device detected a parity error in the write data (relative to the T bit) through parity checking; 011: TE3 Error (Address Assigned During Dynamic Address Arbitration): The slave device detected a parity error in the assigned address during dynamic	0

		address arbitration (relative to the PAR bit). 1100: TE4 Error (Missing 0b111_1110+R after Sr bit during dynamic address arbitration): The slave device detected a missing 0b111_1110 + R sequence after the Sr bit during dynamic address arbitration. 1101: TE5 Error (Transaction After CCC Detected): Device detected an invalidly formatted CCC; 1110: TE6 Error (Monitoring Error): Device detected data transmitted on the bus that did not match expectations; Other values: Reserved.	
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22.4.13 I3C Receive Message Register (R32_I3C_RMR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								RADD[6:0]							Reserved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RCODE[7:0]								Reserved				IBIRDCNT[2:0]			

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:17]	RADD[6:0]	RO	Received Slave Address: When I3C is configured as a master device, this bit field records the dynamic address received from the slave device during an acknowledged IBI or master role request. <i>Note: This bit applies only to master devices.</i>	0
16	Reserved	RO	Reserved	0
[15:8]	RCODE[7:0]	RO	Received CCC Code: When I3C is configured as a slave device, this bit field is used to record the received CCC code. <i>Note: This bit applies only to slave devices.</i>	0
[7:3]	Reserved	RO	Reserved	0
[2:0]	IBIRDCNT[2:0]	RO	IBI Received Payload Data Count: When I3C is configured as the master device, this bit field records the number of valid data bytes received in the R32_I3C_IBIDR register. <i>Note: This bit applies only to master devices.</i>	0

22.4.14 I3C Event Register (R32_I3C_EVR)

Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
GRPFF	DEFF	INTUPDF	ASUPDF	RS TF	MRLUPDF	MWLUPDF	DAUPDF	STAF	GETF	WKPFF	Reserved	HJF	CRUPDF	CRF	IBIENDF
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIF	Reserved			ERRF	RXTGTENDF	FCF	Reserved	RXLA STF	TXLA STF	RXFN EF	TXFN FF	SFNEF	CFNFF	TXFEF	CFEF

Bit	Name	Access	Description	Reset value
31	GRPFF	RO	Group Addressing Flag: When the I3C operates as a slave device (typically supporting the master role), this flag is set to 1 by hardware to indicate that a broadcast DEFGRPA CCC (Define Group Address List) has been received. Subsequently, software can store the received data for use when assuming the master role. This flag is cleared by writing a 1 to the CGRPFF bit in the R32_I3C_CTLR register via software. <i>Note: This bit applies only to slave devices.</i>	0
30	DEFF	RO	DEFTGTS Flag: When I3C operates as a slave device (typically supporting the master role), this flag is set to 1 by hardware to indicate reception of a broadcast DEFTGTS CCC (Define Enslaved Device List). Software can then store the received data for use when assuming the master role. This flag is cleared by writing a 1 to the CDEFF bit of the R32_I3C_CEVF register via software. <i>Note: This bit applies only to slave devices.</i>	0
29	INTUPDF	RO	Interrupt/Master Role/Hot-Join Update Flag: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate receipt of a direct or broadcast ENEC/DISEC CCC (Enable/Disable Slave Event), where the slave event is an interrupt/IBI request, master role request, or hot-join request. Software must then read IBIEN, CREN, or HJEN from the R32_I3C_DEVR0 register, respectively. This flag is cleared by writing a 1 to the CINTUPDF bit in the R32_I3C_CEVF register via software. <i>Note: This bit applies only to slave devices.</i>	0
28	ASUPDF	RO	Working Status Update Flag:	0

			When I3C operates as a slave device, this flag is set to 1 by hardware to indicate receipt of a direct or broadcast ENTASx CCC (where x = 0...3). Subsequently, software must read the R32_I3C_DEVR0 register AS[1:0]. This flag is cleared by writing a 1 to the CASUPDF bit in the R32_I3C_CEVR register via software. <i>Note: This bit applies only to slave devices.</i>	
27	RSTF	RO	Reset Timing Flag: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate detection of a reset sequence (14 SDA conversions occurring while SCL is low, followed by repeated start and stop bits). Then, when not in stop mode, software must read RSTACT[1:0] and RSTVAL in the I3C_DEVR0 register to determine the required reset level. - If RSTVAL = 1: When RSTF is set (and/or the corresponding interrupt is generated if enabled), RSTACT[1:0] in the R32_I3C_DEVR0 register indicates the reset operation (if any) that software must perform. - If RSTVAL = 0: When RSTF is set to 1 (and/or the corresponding interrupt is generated if enabled), the software must issue an I3C reset after the first detected reset sequence and a system reset after the second detected reset sequence. When in stop mode, the device can be woken up using the corresponding interrupt. This flag is cleared by writing a 1 to the corresponding CRSTF bit in the R32_I3C_CEVR register via software. <i>Note: This bit applies only to slave devices.</i>	0
26	MRLUPDF	RO	Maximum Read Length Update Flag: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate that a direct SETMRL CCC (Set Maximum Read Length) command has been received. Subsequently, software must read the MRL[15:0] bits of the R32_I3C_MAXRLR register to obtain the maximum read length value. This flag is cleared by writing a 1 to the CMRLUPDF bit in the R32_I3C_CEVR register via software. <i>Note: This bit applies only to slave devices.</i>	0
25	MWLUPDF	RO	Maximum Write Length Update Flag: When I3C operates as a slave device, this flag is set	0

			to 1 by hardware to indicate that a direct SETMWL CCC (Set Maximum Write Length) command has been received. Subsequently, software must read the MWL[15:0] bits of the R32_I3C_MAXRLR register to obtain the maximum write length value. This flag is cleared by writing a 1 to the CMWLUPDF bit in the R32_I3C_CEVr register via software. <i>Note: This bit applies only to slave devices.</i>	
24	DAUPDF	RO	Dynamic Address Update Flag: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate that a dynamic address update has been received via broadcasting ENTDA, RSTDA, or direct SETNEWDA CCC. Subsequently, software must read the R32_I3C_DEVR0 register's DA[6:0] and DAVAL to obtain the dynamic address update. This flag is cleared by writing a 1 to the CDAUPDF bit in the R32_I3C_CEVr register via software. <i>Note: This bit applies only to slave devices.</i>	0
23	STAF	RO	Status Flag Acquisition: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate that a Format 1 direct GETSTATUS CCC (with or without a TGTSTAT interpretation byte) has been received. This flag is cleared by writing a 1 to the CSTAF bit of the R32_I3C_CEVr register via software. <i>Note: This bit applies only to slave devices.</i>	0
22	GETF	RO	Get Flag: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate that a direct CCC (GET*** CCC) of any get type—excluding GETSTATUS in format 1 but including GETSTATUS in format 2—has been received. This flag is cleared by writing a 1 to the CGETF bit in the R32_I3C_CEVr register via software. <i>Note: This bit applies only to slave devices.</i>	0
21	WKPF	RO	Wake-up/Lost Start Bit Flag: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate that a start bit (SDA falling edge followed by SCL falling edge) has been detected, but the I3C core clock remains gated at the next SCL falling edge. Consequently, the slave device may have missed the I3C bus transaction. The corresponding interrupt can be used to wake the	0

			device from low-power (sleep or stop) modes. This flag is cleared by writing a 1 to the CWKPF bit in the R32_I3C_CEVr register via software. <i>Note: This bit applies only to slave devices.</i>	
20	Reserved	RO	Reserved	0
19	HJF	RO	Hot-joined Flag: When I3C operates as the master device, this flag is set to 1 by hardware to indicate a received hot-joined request. This flag is cleared by writing a 1 to the CHJF bit of the R32_I3C_CEVr register via software. <i>Note: This bit applies only to master devices.</i>	0
18	CRUPDF	RO	Master Role Update Flag: When the I3C operates as a slave device, this flag is set to 1 by hardware to indicate that the I3C has acquired the master role following completion of the master role conversion process. This flag is cleared by writing a value of 1 to the CCRUPDF bit of the R32_I3C_CEVr register via software. <i>Note: This bit applies only to slave devices.</i>	0
17	CRF	RO	Master Role Request Flag: When the I3C operates as the master device, this flag is set to 1 by hardware to indicate that the hardware has acknowledged and completed the master role request. Subsequently, the software must issue a GETACCCR CCC (Get Accept Master Role) command to initiate the master role switching process. This flag is cleared by writing a value of 1 to the CCRF bit in the R32_I3C_CEVr register via software. <i>Note: This bit applies only to master devices.</i>	0
16	IBIENDF	RO	IBI End Flag: When I3C operates as a slave device, this flag is set to 1 by hardware to indicate that an IBI transmission has been received and completed (the master device acknowledged the IBI and read the IBI data byte, if present). This flag is cleared by writing a 1 to the CIBIENDF bit in the R32_I3C_CEVr register via software. <i>Note: This bit applies only to slave devices.</i>	0
15	IBIF	RO	IBI Flag: When I3C operates as a master device, this flag is set	0

			to 1 by hardware to indicate an IBI request has been received. This flag is cleared by writing a 1 to the CIBIF bit in the R32_I3C_CEVN register via software. <i>Note:</i> (1) This bit applies only to master devices; (2) For chips where the fifth digit of the lot number is less than 3, this bit is disabled when receiving an IBI request without data from the slave device. For specific details on handling such packets, please refer to the EVT example code on the official website.	
[14:12]	Reserved	RO	Reserved	0
11	ERRF	RO	Flag: This flag is set by hardware to indicate an error has occurred. Subsequently, software must read the R32_I3C_STATR register to determine the error type. This flag is cleared by writing a 1 to the CERRF bit in the R32_I3C_CEVN register via software.	0
10	RXTGTENDF	RO	Read End Flag Initiated by Slave Device: When I3C operates as a master device, and only when the S-FIFO is disabled (SMODE bit in the R32_I3C_CFGR register is 0), this flag is set to 1 by hardware to indicate that the slave device has prematurely terminated the read transfer. Software must then read the status register R32_I3C_STATR to examine information related to the last message and retrieve the number of data bytes received during the prematurely terminated read transfer (XDCNT in the R32_I3C_STATR register). This flag is cleared by writing a 1 to the CRXTGTENDF bit in the R32_I3C_CEVN register via software. <i>Note: This bit applies only to master devices.</i>	0
9	FCF	RO	Frame Complete Flag: When I3C is used as a master device, this flag is set to 1 by hardware to indicate that a frame on the I3C bus has completed (normally), such as when a stop bit is issued. When I3C operates as a slave device, this flag is set to 1 by hardware to indicate that a message sent to or addressed by this slave device on the I3C bus has completed (normally), such as when the master device issues the next stop bit or repeats the start bit.	0

			This flag is cleared by writing a 1 to the CFCF bit of the R32_I3C_CEV register via software.	
8	Reserved	RO	Reserved	0
7	RXLASTF	RO	Last Data Byte/Word Flag: When I3C operates as a master device, this flag is set by hardware to indicate the last data byte/word of a message that must be read from the RX-FIFO (depending on the RXTHRES bit in the I3C_CFGR register). This flag is cleared by hardware when the last data byte/word of the message is read. <i>Note: This bit applies only to master devices.</i>	0
6	TXLASTF	RO	Last Data Byte/Word Written Flag: This flag is set by hardware to indicate the last data byte/word of a message that must be written to the TX-FIFO (depending on the TXTHRES bit in the R32_I3C_CFGR register). When the last data byte/word of the message is written, this flag is cleared by hardware.	0
5	RXFNEF	RO	RX-FIFO Not Empty Flag: This flag is set/cleared by hardware to indicate whether data bytes must/must not be read from the RX-FIFO. <i>Note: Before reading data bytes from the RX-FIFO (by executing R32_I3C_RDR or R32_I3C_RDWR depending on RXTHRES), the software must wait for RXFNEF = 1 (via polling or an enabled interrupt).</i>	0
4	TXFNFF	RO	TX-FIFO Not Empty Flag: This flag is set/cleared by hardware to indicate whether data bytes must/must not be written to the TX-FIFO. <i>Note: (1) Before writing to the TX-FIFO (writing to R32_I3C_TDR or R32_I3C_TDWR depending on TXTHRES), software must wait for TXFNFF = 1 (via polling or an enabled interrupt).</i> <i>(2) When I3C operates as a slave device, if software intends to write to R32_I3C_TDR/R32_I3C_TDWR using the TXFNFF flag, the TX-FIFO must be preloaded (set the PRELOAD bit in the I3C_TGTTDR register to 1).</i>	0
3	SFNEF	RO	S-FIFO Not Empty Flag: When I3C operates as a master device, if S-FIFO is enabled (R32_I3C_CFGR register SMODE bit set to 1), this flag is set to 1 by hardware to indicate that the status word must be read from S-FIFO. This flag is	0

			cleared by hardware to indicate that the status word is not read from S-FIFO. <i>Note: This bit applies only to master devices.</i>	
2	CFNFF	RO	C-FIFO Not Full Flag: When I3C operates as a master device, this flag is set by hardware to indicate that a control word must be written to the C-FIFO. This flag is cleared by hardware to indicate that no control word is written to the C-FIFO. <i>Note: Before writing to the C-FIFO (writing to R32_I3C_CTLR), the software must wait for CFNFF = 1 (either through polling or via an enabled interrupt).</i> <i>Note: This bit applies only to master devices.</i>	0
1	TXFEF	RO	TX-FIFO Empty Flag: This flag is set to 1 by hardware to indicate the TX-FIFO is empty; This flag is cleared by hardware to indicate the TX-FIFO is not empty.	1
0	CFEF	RO	C-FIFO Empty Flag: When acting as the primary device: This flag is set to 1 by hardware to indicate the C-FIFO is empty; This flag is cleared by hardware to indicate the C-FIFO is not empty. When acting as a slave device: This flag is set by hardware when the R32_I3C_CTLR register does not contain a control word (no IBI/CR/HJ requests pending); This flag is cleared by hardware when the R32_I3C_CTLR register contains a control word (IBI/CR/HJ requests pending). <i>Note: When I3C is used as the master device, if C-FIFO and TX-FIFO preloading are configured (R32_I3C_CFGR register TMODE bit set to 1), the software must wait for TXFEF = 1 and CFEF = 1 before initiating a new frame transmission.</i>	1

22.4.15 I3C Interrupt Enable Register (R32_I3C_INTENR)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
GRPI	DE	INT	AS	RS	MR	M	DA	ST	GE	WK	Res	HJI	CR	CRI	IBI

E	FIE	UPDIE	UPDIE	TIE	LUPDIE	WLUPDIE	UPDIE	AIE	TIE	PIE	erved	E	UPDIE	E	ENDIE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIIE	Reserved			ERIE	RXTGTENDIE	FCIE	Reserved			RXFN EIE	TXFN EIE	SFNEIE	CFN FIE	Reserved	

Bit	Name	Access	Description	Reset value
31	GRPIE	RW	DEFGRPA CCC interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
30	DEFIE	RW	DEFTGTS CCC interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
29	INTUPDIE	RW	ENEC/DISEC CCC interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
28	ASUPDIE	RW	ENTASx CCC interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
27	RSTIE	RW	Reset timing interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
26	MRLUPDIE	RW	SETMRL CCC interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
25	MWLUPDIE	RW	SETMWL CCC interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
24	DAUPDIE	RW	ENTDAA/RSTDAA/SETNEWDA CCC interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
23	STAIE	RW	Format 1 GETSTATUS CCC interrupt enable:	0

			1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	
22	GETIE	RW	GETxxx CCC interrupt enable (except GETSTATUS in format 1): 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
21	WKPIE	RW	Wakeup interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
20	Reserved	RO	Reserved.	0
19	HJIE	RW	Hot join interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
18	CRUPDIE	RW	Master role update interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
17	CRIE	RW	Master role request interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to master devices.</i>	0
16	IBIENDIE	RW	IBI end interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
15	IBIIE	RW	IBI request interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to master devices.</i>	0
[14:12]	Reserved	RO	Reserved	0
11	ERRIE	RW	Error interrupt enable: 1: Enable; 0: Disable.	0
10	RXTGTENDIE	RW	Read end interrupt enable for slave device: 1: Enable; 0: Disable. <i>Note: This bit applies only to master devices.</i>	0
9	FCIE	RW	Frame completion interrupt enable: 1: Enable; 0: Disable.	0

[8:6]	Reserved	RO	Reserved	0
5	RXFNEIE	RW	RX-FIFO non-empty interrupt enable: 1: Enable; 0: Disable.	0
4	RXFNEIE	RW	TX-FIFO non-full interrupt enable: 1: Enable; 0: Disable.	0
3	SFNEIE	RW	S-FIFO non-empty interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to master devices.</i>	0
2	CFNFIE	RW	C-FIFO non-full interrupt enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to master devices.</i>	0
[1:0]	Reserved	RO	Reserved	0

22.4.16 I3C Clear Event Register (R32_I3C_CEVF)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CGR PF	CD EFF	CIN TU PDF	CASU PDF	CR STF	CM RL UP DF	CM WL UP DF	CD AU PDF	CS TAF	CG ETF	CW KPF	Res erve d	CH JF	CC RU PDF	CC RF	CIB IEN DF
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CIBI F	Reserved			CE RR F	CR XT GT EN DF	CF CF	Reserved								

Bit	Name	Access	Description	Reset value
31	CGRPF	WO	Clear DEFGRPA CCC flag: 1: Clear GRPF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
30	CDEFF	WO	Clear DEFTGTS CCC flag: 1: Clear DEFF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
29	CINTUPDF	WO	Clear ENEC/DISEC CCC flag: 1: Clear INTUPDF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
28	CASUPDF	WO	Clear ENTASx CCC flag:	0

			1: Clear ASUPDF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	
27	CRSTF	WO	Clear reset timing flag: 1: Clear RSTF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
26	CMRLUPDF	WO	Clear SETMRL CCC flag: 1: Clear MRLUPDF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
25	CMWLUPDF	WO	Clear SETMWL CCC flag: 1: Clear MWLUPDF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
24	CDAUPDF	WO	Clear ENTDA A/RSTDAA/SETNEWDA CCC flag: 1: Clear DAUPDF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
23	CSTAF	WO	Clear format 1 GETSTATUS CCC flag: 1: Clear STAF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
22	CGETF	WO	Clear GETxxx CCC flag (except GETSTATUS in format 1): 1: Clear GETF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
21	CWKPF	WO	Clear the wake-up flag: 1: Clear WKPF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
20	Reserved	RO	Reserved	0
19	CHJF	WO	Clear hot join flag: 1: Clear HJF; 0: No effect. <i>Note: This bit applies only to master devices.</i>	0
18	CCRUPDF	WO	Clear the master role update flag: 1: Clear CRUPDF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
17	CCRF	WO	Clear master role request flag: 1: Clear CRF; 0: No effect.	0

			<i>Note: This bit applies only to master devices.</i>	
16	CIBIENDF	WO	Clear IBI end flag: 1: Clear IBIENDF; 0: No effect. <i>Note: This bit applies only to slave devices.</i>	0
15	CIBIF	WO	Clear IBI request flag: 1: Clear IBIF; 0: No effect. <i>Note: This bit applies only to master devices.</i>	0
[14:12]	Reserved	RO	Reserved	0
11	CERRF	WO	Clear error flag: 1: Clear ERRF; 0: No effect.	0
10	CRXTGTENDF	WO	Clear read end flag for slave device: 1: Clear RXTGTENDF; 0: No effect. <i>Note: This bit applies only to master devices.</i>	0
9	CFCF	WO	Clear frame completion flag: 1: Clear FCF; 0: No effect.	0
[8:0]	Reserved	RO	Reserved	0

22.4.17 I3C Device Register 0 (R32_I3C_DEVR0)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							RSTVAL	RSTACT[1:0]	AS[1:0]	HJEN	Reserved	CR EN	IBI EN		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DA[6:0]						DAVAL	

Bit	Name	Access	Description	Reset value
[31:25]	Reserved	RO	Reserved	0
24	RSTVAL	RO	Reset Operation Valid: This bit is set to 1 by hardware to indicate that the RSTACT[1:0] bit field has been updated and is active upon receiving a broadcast or direct write to RSTACT CCC (slave device reset operation). This bit is cleared by hardware when the slave device receives a frame start. When the device is not in stop mode: - If RSTVAL = 1: When the R32_I3C_EVR	0

			register's RSTF bit is set to 1 (and/or the corresponding interrupt is generated if enabled), the R32_I3C_DEVR0 register's RSTACT[1:0] bits indicate the reset operation (if any) to be performed by software. - If RSTVAL = 0: When RSTF is set to 1 (and/or the corresponding interrupt is generated if enabled), the software must issue an I3C reset after the first detected reset timing and a system reset after the second detected reset timing. When in stop mode, the corresponding interrupt can be used to wake up the device. <i>Note: This bit applies only to slave devices.</i>	
[23:22]	RSTACT[1:0]	RO	Reset Operation/Level Upon Receiving Reset Signal: 00: No reset operation performed 01: First-level reset: Application software must: a) Perform a partial reset of the peripheral by clearing the enable bit in the I3C configuration register (write EN = 0). This resets the I3C bus interface and part of the I3C core, but does not modify the contents of the I3C HB register (except for the EN bit). b) A write operation to the I3C reset control bit in the RCC (Reset and Clock Controller) register performs a full reset of the peripherals (including all their registers). 10: Second-level reset: Application software must initiate a warm reset, also known as a system reset. This functions identically to pin reset (NRST = 0): 11: No reset operation is performed. <i>Note: This bit applies only to slave devices.</i>	0
[21:20]	AS[1:0]	RO	Operating Status: 00: Operating Status 0; 01: Operating Status 1; 10: Operating Status 2; 11: Operating Status 3. <i>Note: This bit applies only to slave devices.</i>	0
19	HJEN	RW	Hot join request enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
18	Reserved	RO	Reserved	0
17	CREN	RW	Master role request enable: 1: Enable; 0: Disable.	0

			<i>Note: This bit applies only to slave devices.</i>	
16	IBIEN	RW	IBI request enable: 1: Enable; 0: Disable. <i>Note: This bit applies only to slave devices.</i>	0
[15:8]	Reserved	RO	Reserved	0
[7:1]	DA[6:0]	RW	7-bit dynamic address: When I3C operates as a master device, this bit field can be written by software to define its own dynamic address. When I3C operates as a slave device, this bit field is updated by hardware upon receiving a broadcast ENTDAACCC or a direct SETNEWDACCC command.	0
0	DAVAL	RW	Dynamic address valid: When I3C operates as a master device, this bit may be written by software to validate its own dynamic address, such as prior to switching master roles. When I3C operates as a slave device, this bit is set to 1 by hardware upon acknowledging the broadcast ENTDAACCC or direct SETNEWDACCC command, and cleared upon acknowledging the broadcast RSTDACCC command.	0

22.4.18 I3C Device Register x (R32_I3C_DEVR_x) (x=1/2/3/4)

Offset address: 0x64, 0x68, 0x6C, 0x70

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DIS	Reserved											SUSP	IBIDEN	CRACK	IBIACK
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												DA[6:0]		Reserved	

Bit	Name	Access	Description	Reset value
31	DIS	RO	DA[6:0] write disable: 1: Disables/locks write operations to DA[7:0] and IBIDEN; 0: Enables write operations to the R32_I3C_DEVR _x register's DA[7:0] and IBIDEN. <i>Note: This bit applies only to the master device.</i>	0
[30:20]	Reserved	RO	Reserved	0
19	SUSP	RW	Suspending/Stopping I3C transmission upon receiving IBI:	0

			1: Upon receiving an IBI request from slave device x, stop the I3C transmission and clear the C-FIFO and TX-FIFO; 0: Upon acknowledging and completing the IBI request from slave device x, do not clear the C-FIFO and TX-FIFO, and issue a repeat start bit or stop bit based on whether the next control word exists. <i>Note: This bit applies only to the master device.</i>	
18	IBIDEN	RW	IBI data enabled: 1: Confirmed IBI from slave device x followed by mandatory data bytes MDB[7:0]; 0: Confirmed IBI from slave device x with no data bytes. <i>Note: This bit applies only to the master device.</i>	0
17	CRACK	RW	Master role request acknowledgment: 1: Must acknowledge the master role request from slave device x (via 7-bit dynamic address DA[6:0]) - The -DIS bit field is set to 1 by hardware to protect DA[6:0] from software modification, while the hardware internally stores the current DA[6:0] into the core clock domain. After ACK, messages continue transmitting according to the initial programming. The software is notified via the master role request flag (bit 1 of the CRF field in the R32_I3C_EVR register) and/or the corresponding interrupt (if enabled). To effectively grant the master role to the requesting slave device, the software must issue a GETACCCR (formerly known as GETACCMST) followed by a stop bit. -Regardless of the CRACK configuration on this or other devices, all other master role requests will be NACKed until both the master role request flag (CRF) and the IBI flag (IBIF) in the I3C_EVR register are cleared. 0: Must NACK After the master device role request from slave device x receives a NACK, message transmission continues according to the initial programming (the slave device has received the NACK and may issue another master device role request later). <i>Note: This bit applies only to the master device.</i>	0
16	IBIACK	RW	IBI request acknowledgment: 1: Must acknowledge the IBI request originating from slave device x (via 7-bit dynamic address DA[6:0])	0

			- The -DIS bit field is set to 1 by hardware to protect DA[6:0] from software modification, while the hardware internally stores the current DA[6:0] into the core clock domain. - After ACK, the master device records the IBI payload data (if present) based on I3C_DEVRx.IBIDEN. - Software is notified via the IBI flag (IBIF=1) and/or the corresponding interrupt (if enabled); - Regardless of the IBIACK configuration on this or other devices, all subsequent IBI requests will be NACKed until both the IBI request flag (IBIF) and the master role request flag (CRF) are cleared. 0: After the NACK for the IBI request from slave device x is received, message transmission resumes according to the initial programming (the slave device has received the NACK and may issue another IBI request later). <i>Note: This bit applies only to the master device.</i>	
[15:8]	Reserved	RO	Reserved	0
[7:1]	DA[6:0]	RW	I3C dynamic address assigned from slave device x: When I3C operates as the master device, this bit field must be written by software to store the 7-bit dynamic address sent by the master device via a broadcast ENTDAACK acknowledged by slave device x or a direct SETNEWDA CCC command.	0
0	Reserved	RO	Reserved	0

22.4.19 I3C Maximum Read Length Register (R32_I3C_MAXRLR)

Offset address: 0x90

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													IBIP[2:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MRL[15:0]															

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
[18:16]	IBIP[2:0]	RW	Maximum IBI load data size in bytes: 000: Empty load data size (allowed only when BCR2 bit in R32_IC3_BCR register is 0); 001: 1 byte (forced data byte MDB[7:0]); 010: 2 bytes (includes first MDB[7:0]);	0

			011: 3 bytes (includes first MDB[7:0]); Others: 4 bytes (includes first MDB[7:0]). <i>Note: This bit applies only to slave devices.</i>	
[15:0]	MRL[15:0]	RW	Maximum read length: Notifies software of MRL updates via MRLUPF and corresponding interrupts (if enabled). When the device receives a GETMRL CCC request, the hardware returns this bit field on the I3C bus. <i>Note: This bit applies only to slave devices.</i>	0

22.4.20 I3C Maximum Write Length Register (R32_I3C_MAXWLR)

Offset address: 0x94

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MWL[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	MWL[15:0]	RW	Maximum data write length: This bit field is initially written by software when EN = 0 and updated by hardware upon receiving the SETMWL command. MWL updates are notified to software via the MWLUPF field in the R32_I3C_EVR register and the corresponding interrupt (if enabled). When the slave device receives GETMWL CCC, hardware uses this bit field to return the value on the I3C bus. <i>Note: This bit is applicable only to slave devices.</i>	0

22.4.21 I3C Timing Register 0 (R32_I3C_TIMINGR0)

Offset address: 0xA0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SCLH_I2C[7:0]								SCLL_OD[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SCLH_I3C[7:0]								SCLL_PP[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	SCLH_I2C[7:0]	RW	SCL high duration for traditional I2C messages, in microsecond units: $t_{SCLH_I2C} = (SCLH_I2C + 1) \times t_{HCLK}$	0

[23:16]	SCLL_OD[7:0]	RW	The duration of the low level of the SCL pin during the open-drain phase, measured in core clock cycles, for traditional I2C messages and I3C open-drain phases (address phase following the start bit, ACK phase during master-initiated messages, and T-bit phase during direct/private/IBI payloads): $t_{SCLL_OD} = (SCLL_OD + 1) \times t_{HCLK}$	
[15:8]	SCLH_I3C[7:0]	RW	SCL high level duration for I3C messages (in push-pull and open-drain phases), measured in core clock cycles: $t_{SCLH_I3C} = (SCLH_I3C + 1) \times t_{HCLK}$	
[7:0]	SCLL_PP[7:0]	RW	The low-level duration of SCL in the I3C push-pull stage, measured in core clock cycles: $t_{SCLL_PP} = (SCLL_PP + 1) \times t_{HCLK}$	

22.4.22 I3C Timing Register 1 (R32_I3C_TIMINGR1)

Offset address: 0xA4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			SDA_HD	Reserved					FREE[6:0]						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						ASNCR[1:0]		AVAL[7:0]							

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved	0
28	SDA_HD	RW	SDA hold time in core clock cycles. $SDA_hold\ time = (SDA_HD + 0.5) \times t_{HCLK}$	0
[27:23]	Reserved	RO	Reserved	0
[22:16]	FREE[6:0]	RW	The number of core clock cycles used to set certain MIPI timing parameters (e.g., bus idle condition time). <i>Note: This bit applies only to master devices.</i>	0
[15:10]	Reserved	RO	Reserved	0
[9:8]	ASNCR[1:0]	RW	Operating status of new master device: The time required to wait before accessing as a new slave device. Refer to AVAL[7:0]. <i>Note: This bit applies only to the master device.</i>	0
[7:0]	AVAL[7:0]	RW	Number of core clock cycles used to set the 1μs time unit: The hardware then uses this time unit to construct several internal timers corresponding to the following MIPI I3C timing:	0

		When I3C is used as a slave device: 1. For the bus available condition time: After the stop bit and before issuing an IBI start request or a master role request, a (bus available condition) time period must be observed (the bus idle condition must persist for at least t_{AVAL}). Refer to MIPI timing where $t_{AVAL} = 1\mu s$. This timing is defined as: $t_{AVAL} = (AVAL[7:0] + 2) \times t_{HCLK}$ 2. For Bus Idle Condition Time: After both SDA and SCL are high and stable, and before issuing a hot-join event, a Bus Idle Condition time period must be observed. Refer to MIPI v1.1 timing $t_{IDLE} = 200\mu s$. This timing is defined as: $t_{IDLE} = (AVAL[7:0] + 2) \times 200 \times t_{HCLK}$ When the I3C is used as a master, its clock pause time must not exceed the maximum pause time (pause SCL clock low), as follows: 1. For the first bit of the address allocated during dynamic address allocation: the pause clock shall not exceed the MIPI timing $t_{STALLDAA} = 15ms$. The timing is defined as: $t_{STALLDAAmax} = (AVAL[7:0] + 1) \times 15000 \times t_{HCLK}$ 2. ACK/NACK phase for I3C/I2C transmission, parity bit for write data transmission, conversion bit for I3C read transmission: pause clock time must not exceed MIPI timing $t_{STALL} = 100\mu s$. The timing is defined as: $t_{STALLmax} = (AVAL[7:0] + 1) \times 100 \times t_{HCLK}$ Whether I3C is used as a master or slave (supporting the master role), during the master role switch: 1. The new master device must wait for $t_{NEWCRLOCK}$ before pulling SDA low (emitting the start bit) after completing the GETACCR CCC. The new master device may then pull SCL low within t_{CAS} to activate the SCL clock. The active master device must wait for the same $t_{NEWCRLOCK}$ duration, or at least $100\mu s$, before testing whether the new master has acquired bus control by pulling SDA low. The wait time depends on the value of ANSCR[1:0] in the I3C_TIMINGR1 register: – ANSCR[1:0] = 00: $t_{NEWCRLOCK} = (AVAL[7:0] + 1) \times t_{HCLK}$ – ANSCR[1:0] = 01: $t_{NEWCRLOCK} = (AVAL[7:0] + 1) \times 100 \times t_{HCLK}$ – ANSCR[1:0] = 10: $t_{NEWCRLOCK} = (AVAL[7:0] + 1) \times 2000 \times t_{HCLK}$	
--	--	---	--

			– ASNCR[1:0] = 11: $t_{\text{NEWCRLock}} = (\text{AVAL}[7:0] + 1) \times 50000 \times t_{\text{HCLK}}$	
--	--	--	--	--

22.4.23 I3C Timing Register 2 (R32_I3C_TIMINGR2)

Offset address: 0xA8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STALL[7:0]								Reserved				ST AL LA	ST AL LC	ST AL LD	ST AL LT

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	STALL[7:0]	RW	Master clock stall time in core clock cycles: $t_{\text{SCLL_STALL}} = \text{STALL} \times t_{\text{HCLK}}$	0
[7:4]	Reserved	RO	Reserved	0
3	STALLA	RW	ACK phase master clock stall enable: 1: Enable stall; 0: Disable stall.	0
2	STALLC	RW	CCCPAR phase master clock stall enable: 1: Enable stall; 0: Disable stall.	0
1	STALLD	RW	Data PAR phase master clock stall enable: 1: Enable stall; 0: Disable stall.	0
0	STALLT	RW	Master clock stall enable for data T-bit phase (and ACK/NACK phase for data bytes of legacy I2C read messages): 1: Enable stall; 0: Disable stall.	0

22.4.24 I3C Bus Characteristic Register (R32_I3C_BCR)

Offset address: 0xC0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								BC R6	Reserved			BC R2	Res erve d	BC R0	

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0

6	BCR6	RW	Support master device role: 1: Supports I3C master device role; 0: I3C slave device (does not support master device role).	0
[5:3]	Reserved	RO	Reserved	0
2	BCR2	RW	In-Band Interrupt (IBI) payload: 1: At least one mandatory data byte (maximum of 4 data bytes) follows the accepted IBI; 0: No data bytes follow the accepted IBI.	0
1	Reserved	RO	Reserved	0
0	BCR0	RW	Maximum data speed limit: 1: Limited; 0: Unlimited.	0

22.4.25 I3C Device Characteristic Register (R32_I3C_DCR)

Offset address: 0xC4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DCR[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	DCR[7:0]	RW	Device characteristic ID: 0x00: General Device (for v1.0 devices); Other values: Describe the ID for I3C sensor/device types.	0

22.4.26 I3C Get Capacity Register (R32_I3C_GETCAPR)

Offset address: 0xC8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CAPPEND	Reserved													

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	CAPPEND	RW	IBI MDB supports suspended read notifications: 1: When acting as a slave device, the I3C will send an IBI request with a forced data byte value	0

			(MDB[7:5]=101) indicating a suspended read notification; 0: When acting as a slave device, the I3C will send an IBI request without a forced data byte value indicating a suspended read notification.	
[13:0]	Reserved	RO	Reserved	0

22.4.27 I3C Master Role Capacity Register (R32_I3C_CRCAPR)

Offset address: 0xCC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CAPGRP	Reserved						CAPDHOFF	Reserved	

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
9	CAPGRP	RW	Group management support: 1: Support group address functionality (after becoming the master device); 0: Does not support group address functionality. <i>Note: This bit applies only to the master device.</i>	0
[8:4]	Reserved	RO	Reserved	0
3	CAPDHOFF	RW	Delay primary device role switch: 1: Additional time is required to process the primary device role switch; 0: No additional time is required to process the primary device role switch.	0
[2:0]	Reserved	RO	Reserved	0

22.4.28 I3C Get Maximum Data Speed Register (R32_I3C_GETMDSR)

Offset address: 0XD0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							TS CO	RDTURN[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						FMT[1:0]		Reserved					HOFFAS[1:0]		

Bit	Name	Access	Description	Reset value
[31:25]	Reserved	RO	Reserved	0

24	TSCO	RW	Clock to data turn time (t_{SCO}): 1: $t_{SCO} > 12ns$; 0: $t_{SCO} \leq 12ns$.	0
[23:16]	RDTURN[7:0]	RW	Write the selected byte value of the 3-byte MaxRdTurn (via the FMT[1:0] bit field), returning a value that encodes the maximum read switch turn time in response to the GETMXDS CCC.	0
[15:10]	Reserved	RO	Reserved	0
[9:8]	FMT[1:0]	RW	GETMXDS CCC Format: 00: Format 1 (2 bytes: MaxWr (no interpretation byte), MaxRd); 01: Format 2 (5 bytes: MaxWr (no interpretation byte), MaxRd, MaxRdTurn); - Returns 3 bytes MaxRdTurn, where MSB=0, middle byte =0, LSB=RDTURN[7:0]; - Maximum read turn time is less than 256 μs . 10: Format 2 (5 bytes: MaxWr (no interpretation byte), middle byte of MaxRd and MaxRdTurn); - Return 3 bytes of MaxRdTurn, where MSB=0, middle byte = RDTURN[7:0], LSB=0; - Maximum read turn time is between 256 and 65535 μs . 11: Format 2 (5 bytes: MaxWr (no interpretation byte), MaxRd, MaxRdTurn MSB); - Returns 3 bytes of MaxRdTurn, where MSB = RDTURN[7:0], middle byte = 0, LSB = 0; - Maximum read turn time ranges from 65535 μs to 16s.	0
[7:2]	Reserved	RO	Reserved	0
[1:0]	HOFFAS[1:0]	RW	Master device switching operating states: 00: Operating state 0 is the initial operating state of the I3C before becoming the master device and when becoming the master device; 01: Operating state 1 is the initial operating state of the I3C when becoming the master device; 10: Operating state 2 is the initial operating state of the I3C when becoming the master device; 11: Operating state 3 is the initial operating state of the I3C when becoming the master device.	0

22.4.29 I3C Extended Configuration ID Register (R32_I3C_EPIDR)

Offset address: 0XD4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MIPIMID[14:0]															IDT

															SE
															L
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MIPIID[3:0]				Reserved											

Bit	Name	Access	Description	Reset value
[31:17]	MIPIMID[14:0]	RO	15 MIPI vendor IDs.	0x0104
16	IDTSEL	RO	Configuration ID type selector: This bit field is set to 0. This bit field represents bit [32] of the 48-bit configuration ID. Note: Bits [31:16] of the configuration ID may be set to 0.	0
[15:12]	MIPIID[3:0]	RW	4-bit MIPI instance ID: This bit field is written by software to individually configure and identify each instance of this I3C IP on a single I3C bus using a specific number. This bit field represents bits [15:12] of the 48-bit Configuration ID. Note: Bits [11:0] of the Configuration ID may be set to 0.	0
[11:0]	Reserved	RO	Reserved	0

Chapter 23 Serial Peripheral Interface (SPI/I2S)

The chip features four built-in SPI interfaces (SPI1/2/3/4). SPI supports data exchange in 3-wire synchronous serial mode, and the chip select line supports hardware switching between master and slave modes, and supports communication with a single data line.

I2S is also a 3-wire synchronous serial interface communication protocol, which supports 4 audio standards, including Philips I2S standard, MSB alignment standard, LSB alignment standard and PCM standard.

23.1 Main Features

23.1.1 SPI Features

- Support full-duplex synchronous serial mode
- Support 1-wire half-duplex mode
- Support master mode and slave mode, multi-slave mode
- Support 8-bit or 16-bit data structures
- The highest clock frequency is supported to half of F_{HCLK}
- Data ordering supports MSB or LSB first
- Support hardware or software control of the NSS pin
- Transceiver supports hardware CRC
- Transceiver buffer supports DMA transfer
- Support modifying clock phase and polarity

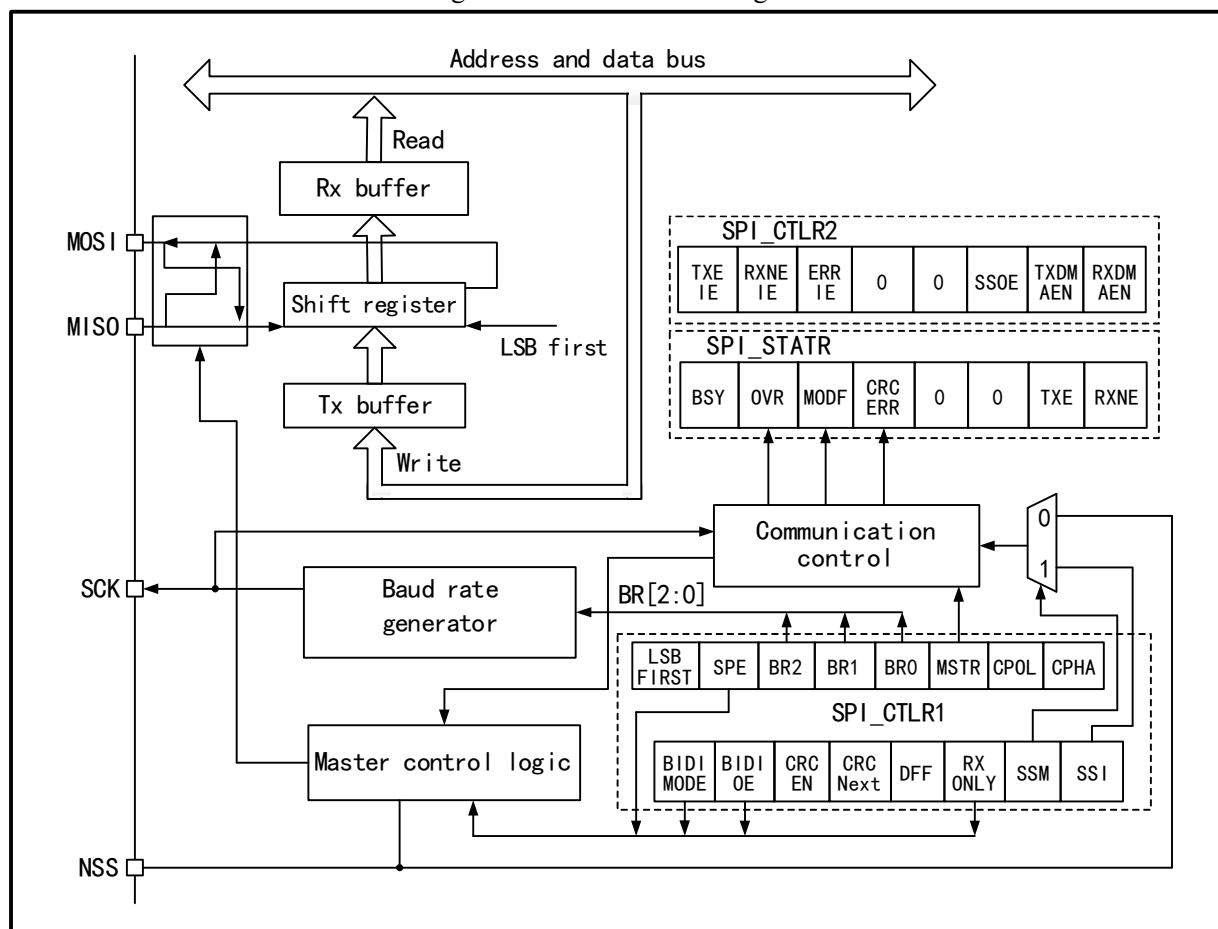
23.1.2 I2S Features

- Support simplex communication
- Support master mode and slave mode
- Support 16-bit, 24-bit and 32-bit data formats
- Audio sampling frequency support range 8kHz-562.2kHz
- Support clock polarity programmable
- Support common I2S protocols: Philips standard, MSB alignment standard, LSB alignment standard and PCM standard
- Transceiver buffer supports DMA transfer
- Support master clock output to external audio equipment

23.2 SPI functional Description

23.2.1 Overview

Figure 23-1 SPI structure diagram



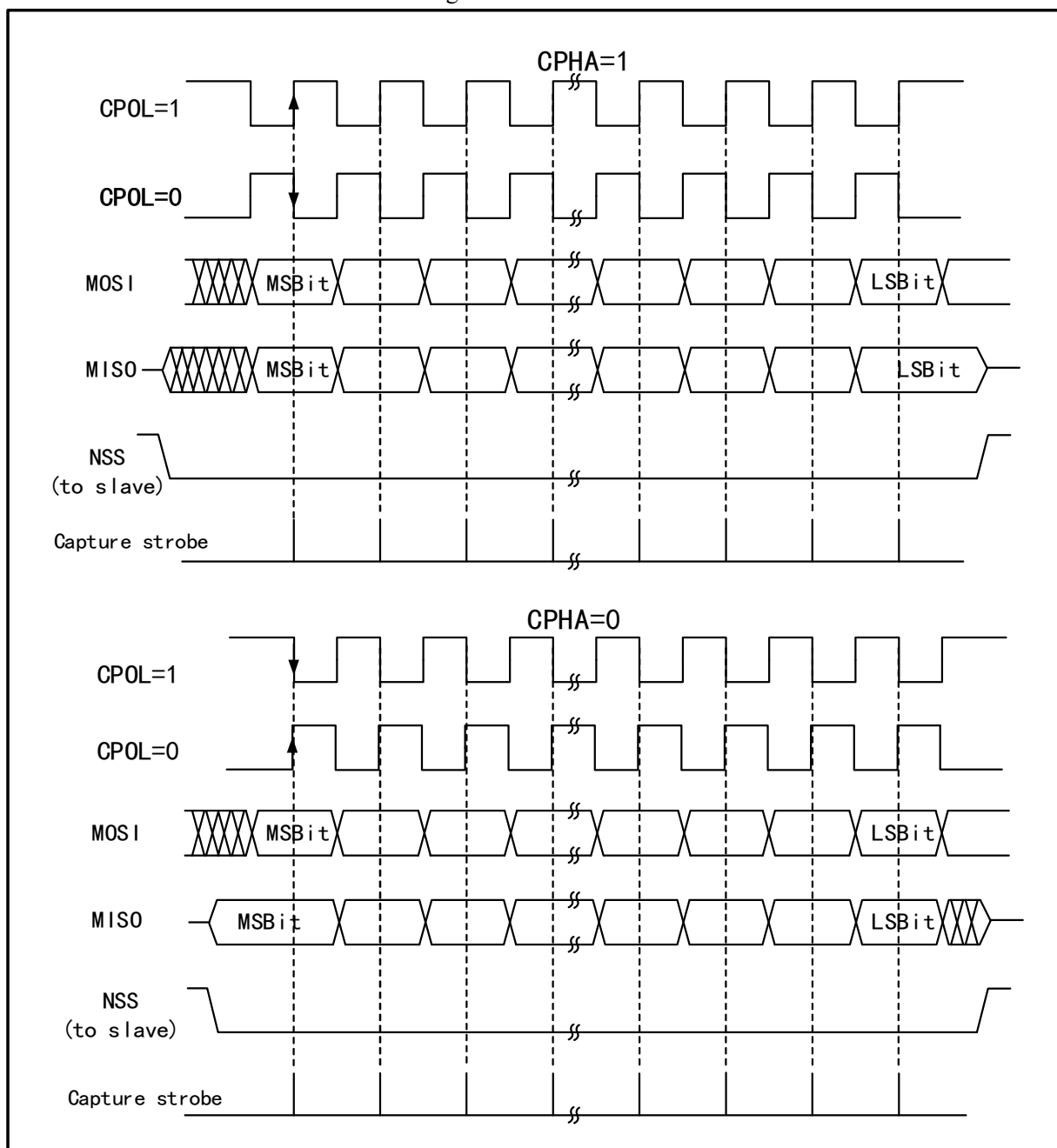
As can be seen from Figure 23-1, the 4 pins related to SPI are MISO, MOSI, SCK and NSS. The MISO pin is the data input pin when the SPI module works in the master mode; it is the data output pin when it works in the slave mode. When the MOSI pin works in master mode, it is a data output pin; when it works in slave mode, it is a data input pin. SCK is the clock pin, the clock signal is always output by the master, and the slave receives the clock signal and synchronizes data transmission and reception. The NSS pin is a chip select pin and has the following uses:

- 1) NSS is controlled by software: At this time, SSM is set, and the internal NSS signal is determined by SSI to output high or low. This situation is generally used in SPI master mode;
- 2) NSS is controlled by hardware: When the NSS output is enabled, i.e., when SSOE is set, the NSS pin will be actively pulled down when the SPI host sends the output outward, and a hardware error will be generated if the NSS pin cannot be successfully pulled down and pulled down, indicating that there are other host devices on the main line that are communicating; SSOE If it is not set, it can be used in multi-master mode. If it is pulled low, it will be forced into slave mode, and the MSTR bit will be automatically cleared.

The working mode of the SPI can be configured through CPHA and CPOL. If CPHA is set, it means that the module samples the data on the second edge of the clock, and the data is latched. If CPHA is not set, it means that the SPI module samples on the first edge of the clock, and the data is latched. CPOL indicates whether the clock remains

high or low when there is no data. See Figure 23-2 below for details.

Figure 23-2 SPI model



The host and device need to be set to the same SPI mode, and the SPE bit needs to be cleared before configuring the SPI mode. The DEF bit can decide whether the single data length of SP is 8 bits or 16 bits. LSBFIRST can control whether a single data word is big-endian or little-endian.

23.2.2 Master Mode

When the SPI module works in master mode, the serial clock is generated by SCK. To configure into master mode do the following steps:

- Configure the BR[2:0] bits in the control register to determine the clock;
- Configure the CPOL and CPHA bits to determine the SPI mode;

Configure DEF to determine the data word length;

Configure LSBFIRST to determine the frame format;

Configure the NSS pin, such as setting the SSOE bit to let the hardware reset NSS. You can also set the SSM bit and set the SSI bit high;

To set the MSTR bit and the SPE bit, it is necessary to ensure that NSS is already high at this time.

When you need to send data, you only need to write the data to be sent to the data register. SPI will send data from the transmit buffer to the shift register in parallel, and then send the data from the shift register according to the setting of LSBFIRST. When the data has reached the shift register, the TXE flag will be set. If it has been set TXEIE, then an interrupt will be generated. If the TXE flag bit is set, data needs to be filled into the data register to maintain a complete data flow.

When the receiver receives data, when the last sampling clock edge of the data word arrives, the data is transferred from the shift register to the receive buffer in parallel, the RXNE bit is set, and if the RXNEIE bit was previously set, it will also generate interrupt. At this time, the data register should be read as soon as possible to remove the data.

23.2.3 Slave Mode

When the SPI module works in slave mode, SCK is used to receive the clock sent by the host, and its own baud rate setting is invalid. The steps to configure into slave mode are as follows:

Configure the DEF bit to set the data bit length;

Configure the CPOL and CPHA bits to match the host mode;

Configure LSBFIRST to match the host data frame format;

In hardware management mode, the NSS pin needs to be kept at a low level. If NSS is set to software management (SSM is set), then please keep SSI not set;

Clear the MSTR bit and set the SPE bit to enable SPI mode.

When transmitting, when the first slave receive sampling edge occurs on SCK, the slave starts to transmit. The process of sending is to move the data in the send buffer to the send shift register. When the data in the send buffer is moved to the shift register, the TXE flag will be set. If the TXEIE bit was previously set, an interrupt will be generated.

When receiving, after the last clock sampling edge, the RXNE bit is set, the byte received by the shift register is transferred to the receive buffer, and the read operation of the read data register can obtain the data in the receive buffer. An interrupt will be generated if RXNEIE is set before RXNE is set.

23.2.4 Simplex Mode

The SPI interface can work in half-duplex mode, that is, the master device uses the MOSI pin, and the slave device uses the MISO pin for communication. When using half-duplex communication, BIDIMODE needs to be set, and BIDIOE is used to control the transmission direction.

Setting the RXONLY bit in normal full-duplex mode can set the SPI module to receive-only simplex mode. After RXONLY is set, a data pin will be released. The pins released in master mode and slave mode are not the same. It is also possible to ignore the received data and set the SPI to transmit-only mode.

23.2.5 CRC

The SPI module uses CRC to ensure the reliability of full-duplex communication, and separate CRC calculators are used for data transmission and reception. The polynomial of the CRC calculation is determined by the polynomial register. For 8-bit data width and 16-bit data width, different calculation methods are used respectively.

Setting the CRCEN bit enables CRC and resets the CRC calculator. After the last data byte is sent, setting the CRCNEXT bit will send the calculation result of the TXCRCR calculator after the current byte is sent. At the same time, if the last received value of the receive shift register is not the same as the locally calculated value of RXCRCR, then the CRCERR bit will be set. To use the CRC, you need to set the polynomial calculator and set the CRCEN bit when configuring the SPI working mode, and set the CRCNEXT bit in the last word or half word to send the CRC and check the received CRC. Note that the CRC calculation polynomials of the sender and receiver should be unified.

23.2.6 DMA

The SPI module supports the use of DMA to speed up data communication. You can use DMA to fill in data into the send buffer, or use DMA to take data from the receive buffer in time. DMA will take or send data in time with RXNE and TXE as signals. DMA can also work in simplex or CRC mode.

23.2.7 Error

- Master mode failure error

When the SPI works in the NSS pin hardware management mode, an external operation of pulling down the NSS pin occurs; or in the NSS pin software management mode, the SSI bit is cleared; or the SPE bit is cleared, causing the SPI to be turned off; or the MSTR bit is cleared, the SPI enters slave mode. An interrupt will also be generated if the ERRIE bit has been set. Clear MODF bit steps: first perform a read or write operation to R16_SPI1_STATR, and then write to R16_SPI1_CTLR1.

- Overflow error

If the master sends data and there is unread data in the slave's receive buffer, an overrun error occurs, the OVR bit is set, and an interrupt is generated if ERRIE is set. Sending an overflow error should restart the current transfer. Reading the data register followed by the status register will clear this bit.

- CRC error

When the received CRC word does not match the value of RXCRCR, a CRC error will occur, and the CRCERR bit will be set.

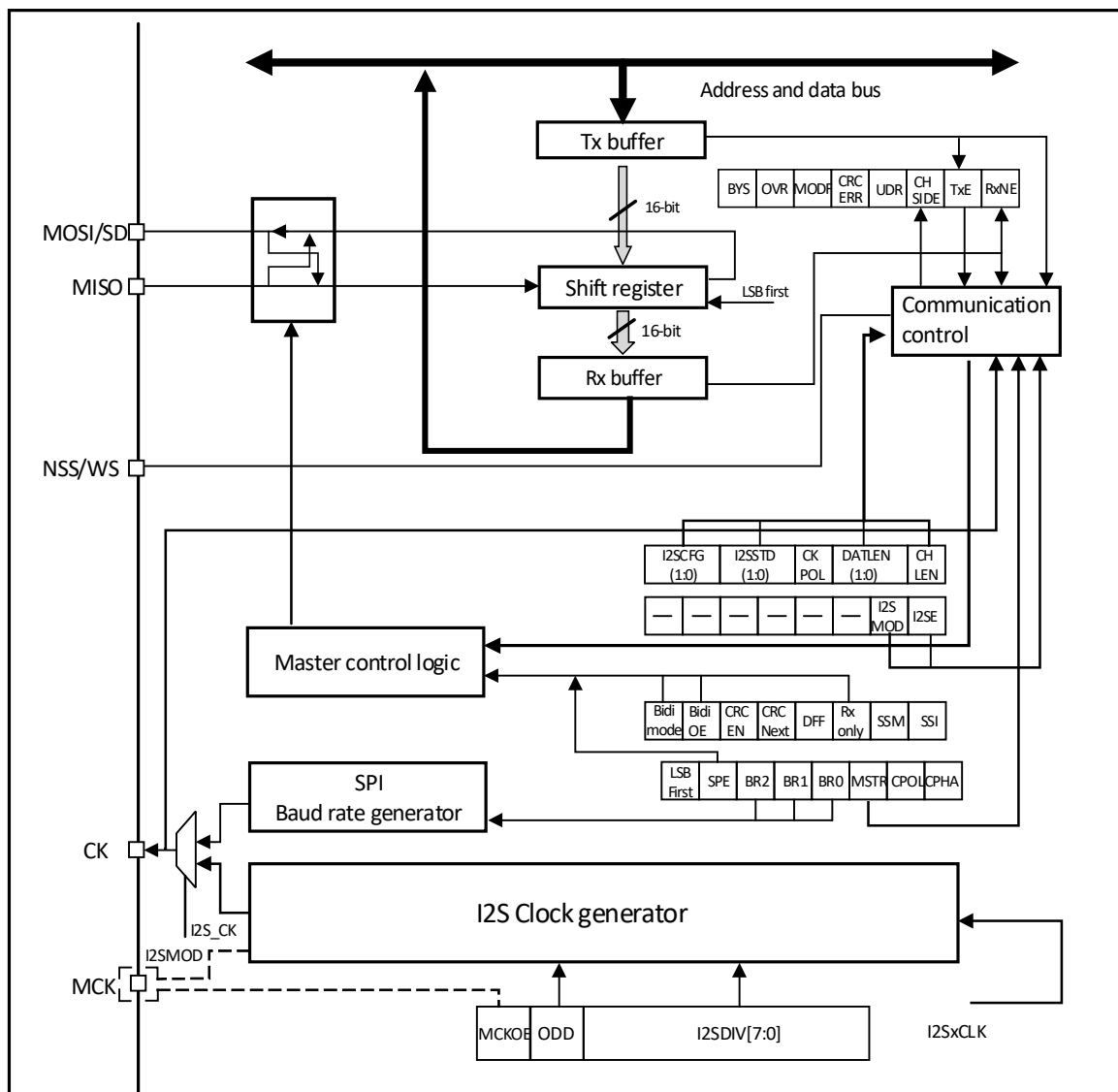
23.2.8 Interrupt

The interrupt of the SPI module supports 5 interrupt sources. The 2 events of the send buffer empty and the receive buffer non-empty will set TXE and RXNE respectively, and an interrupt will be generated when the TXEIE and RXNEIE bits are set respectively. In addition, the 3 errors mentioned above will also generate interrupts, namely MODF, OVR and CRCERR. After the ERRIE bit is enabled, these 3 errors will also generate error interrupts.

23.3 I2S Functional Description

23.3.1 I2S Overview

Figure 23-3 I2S structure diagram



The I2S function is enabled by setting the I2SMOD bit in register I2SCFGR. At this point, the SPI module can be used as an I2S audio interface. I2S and SPI share 3 pins:

- SD: Serial data (Mapped to MOSI pin), used to send and receive 2 time-multiplexed data channels;
- WS: word selection (Mapped to NSS pin), output as data control signal in master mode, and input in slave mode;
- CK: Serial clock (Mapped to the SCK pin), output as a clock signal in master mode, and input in slave mode.

When the main clock is required for some external audio devices, there is an additional pin to output the clock:

- **MCK**: main clock (Independent mapping), when the I2S is configured as the main mode and the MCKOE bit of the register I2SPR is 1, it is used as an output additional clock signal pin. The frequency of the output clock signal is preset to $256 \times F_s$, where F_s is the sampling frequency of the audio signal.

When set to master mode, I2S uses its own clock generator to generate the clock signal for communication. This clock generator is also the clock source for the master clock output. There are 2 additional registers in I2S mode,

one is the clock generator configuration related register I2SPR, the other is the I2S general configuration register I2SCFGR (Can set the audio standard, slave/master mode, data format, packet frame, clock pole parameters, etc.). Register CTLR1 and all CRCR registers are not used in I2S mode. Similarly, the SSOE bit in the CTLR2 register, and the MODF bit and the CRCERR bit in the register STATR are not used in the I2S mode. I2S uses the same register DATAR as SPI for 16-bit wide mode data transfer.

23.3.2 Supported Audio Protocols

The 3-wire bus supports time division multiplexing of audio data on 2 channels: left and right, but only one 16-bit register is used for transmit or receive. Therefore, when writing data to the data register, the software must write the corresponding data according to the channel currently being transmitted; similarly, when reading the register data, check the CHSIDE bit of the register STATR to determine which channel the received data belongs to. The left channel always sends data before the right channel (The CHSIDE bit has no meaning under the PCM protocol). There are 4 available data and packet frame combinations. Data can be sent in the following 4 data formats:

- 16-bit data is packed into 16-bit frames
- 16-bit data is packed into 32-bit frames
- 24-bit data is packed into 32-bit frames
- 32-bit data is packed into 32-bit frames

When using 16-bit data to expand to a 32-bit frame, the first 16 bits (MSB) are meaningful data, and the last 16 bits (LSB) are forced to 0. This operation requires no software intervention and no DMA request (Only requires a read or write operation). 24-bit and 32-bit data frames require the CPU to perform 2 read or write operations on the register DATAR, and 2 DMA transfers are required when using DMA. For 24-bit data, after expansion to 32 bits, the lowest 8 bits are set to 0 by hardware. For all data formats and communication standards, the most significant bit (MSB) is always sent first. The I2S interface supports 4 audio standards, which can be selected by setting the I2SSTD[1:0] bits and PCMSYNC bits in the I2SCFGR register.

23.3.2.1 I2S Philips Standard

Under this standard, pin WS is used to indicate which channel the data being sent belongs to. The pin is valid 1 clock cycle before the first data bit (MSB) is sent. The sender changes the data on the falling edge of the clock signal (CK) and the receiver reads the data on the rising edge. The WS signal also changes on the falling edge of the clock signal.

Figure 23-4 Philips protocol waveform (16/32 full precision, CPOL=0)

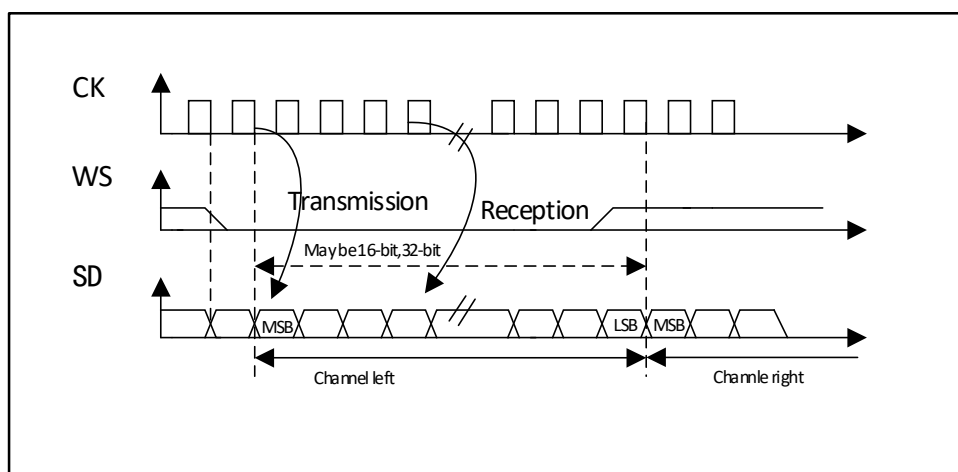
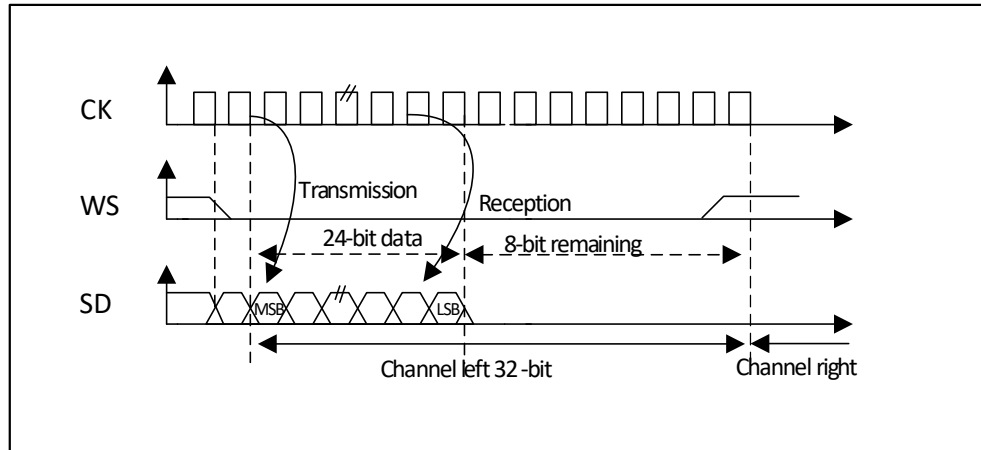
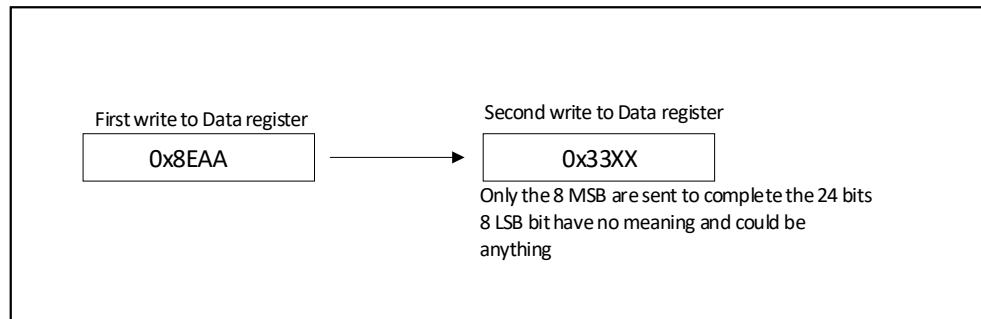


Figure 23-5 Philips protocol waveform (24-bit frame, CPOL=0)



This mode requires 2 read or write operations to the SPI_Datar register. In transmit mode: if you need to send 0x8EAA33 (24-bits):



In receive mode: if 0x8EAA33 is received:

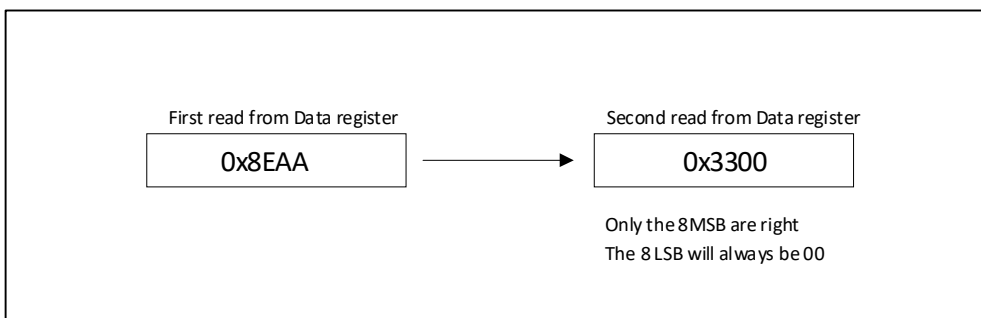
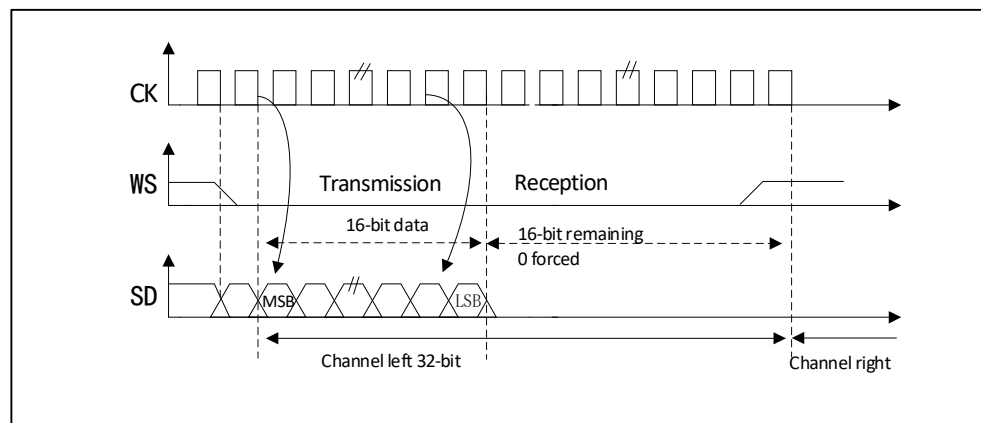


Figure 23-6 Philips protocol standard waveforms (16-bit extended to 32-bit packet frame, CPOL=0)

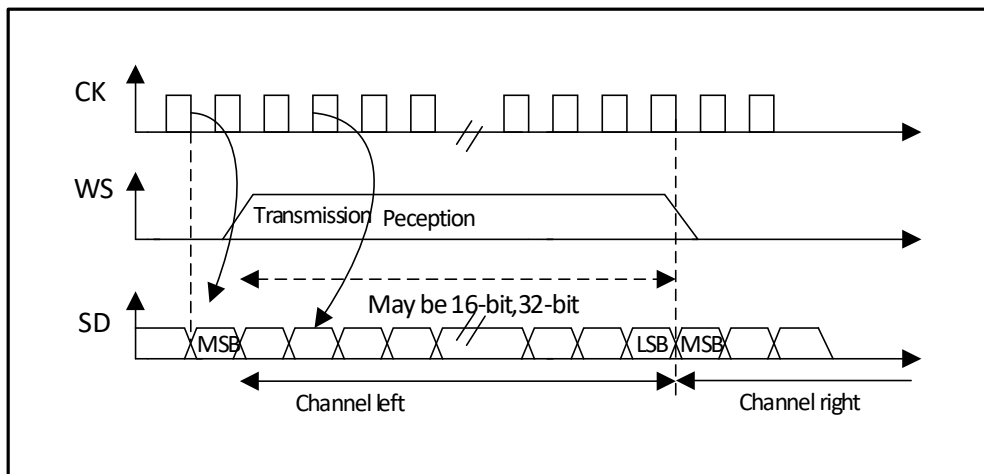


In the I2S configuration stage, if you choose to extend the 16-bit data to a 32-bit channel frame, you only need to access the register DATAR once to extend to 32-bit. The lower 16 bits are set to 0x0000 by hardware. If the data to be transmitted or received is 0x76A3 (Extended to 32 bits is 0x76A30000), only one operation of DATAR is required. When sending, the MSB needs to be written into the register DATAR; if the flag bit TXE is 1, it means that new data can be written, and if the corresponding interrupt is allowed, an interrupt can be generated. The transmission is done by hardware, even if the last 16 bits of 0x0000 have not been sent, the TXE will be set and the corresponding interrupt will be generated. When receiving, each time the upper 16-bit half-word (MSB) is received, the flag bit RXNE is set to 1, and if the corresponding interrupt is enabled, an interrupt can be generated. This way, there is more time between the 2 reads and writes, preventing underflow or overflow conditions.

23.3.2.2 MSB alignment standard

Under this standard, the WS signal and the first data bit, the most significant bit (MSB), are generated simultaneously.

Figure 23-7 MSB aligned 16-bit or 32-bit full precision (CPOL = 0)



The sender changes data on the falling edge of the clock signal; the receiver reads data on the rising edge.

Figure 23-8 MSB aligned 24-bit data, CPOL = 0

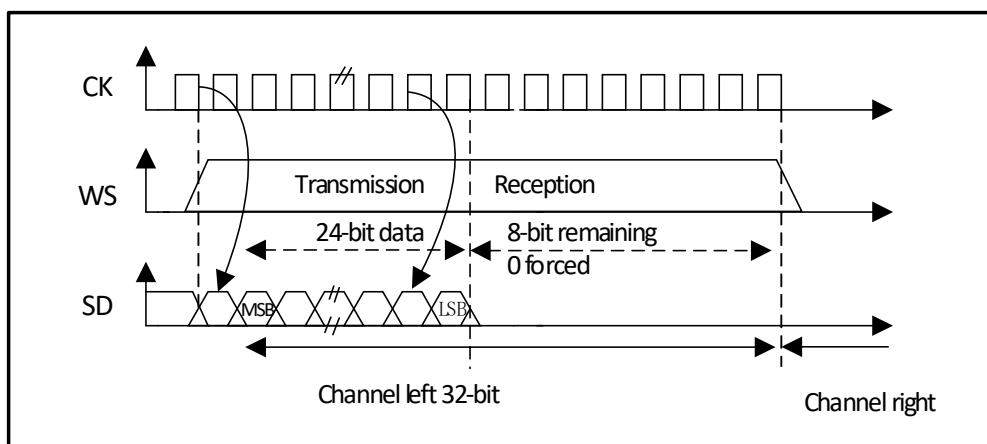
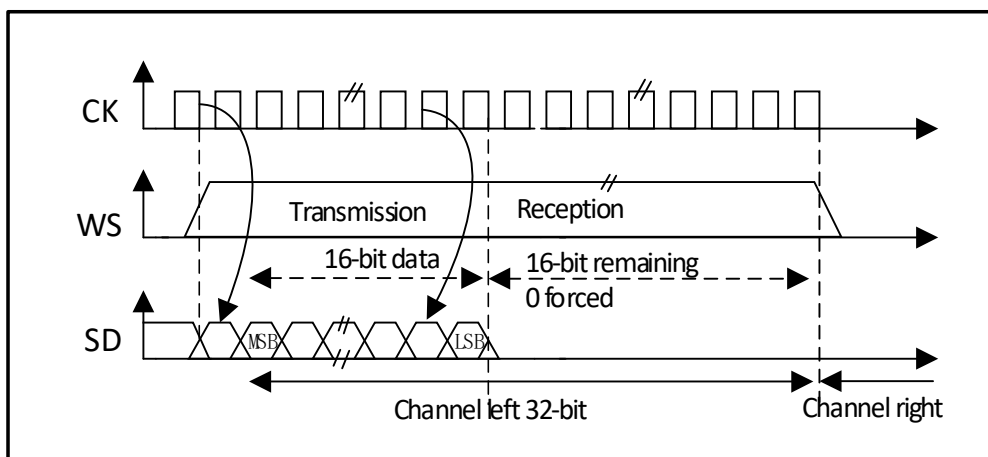


Figure 23-9 MSB-aligned 16-bit data extended to 32-bit packet frame, CPOL = 0



23.3.2.3 LSB alignment standard

This standard is similar to the MSB alignment standard (No difference in 16-bit or 32-bit full-precision frame format).

Figure 23-10 LSB-aligned 16- or 32-bit full precision, CPOL = 0

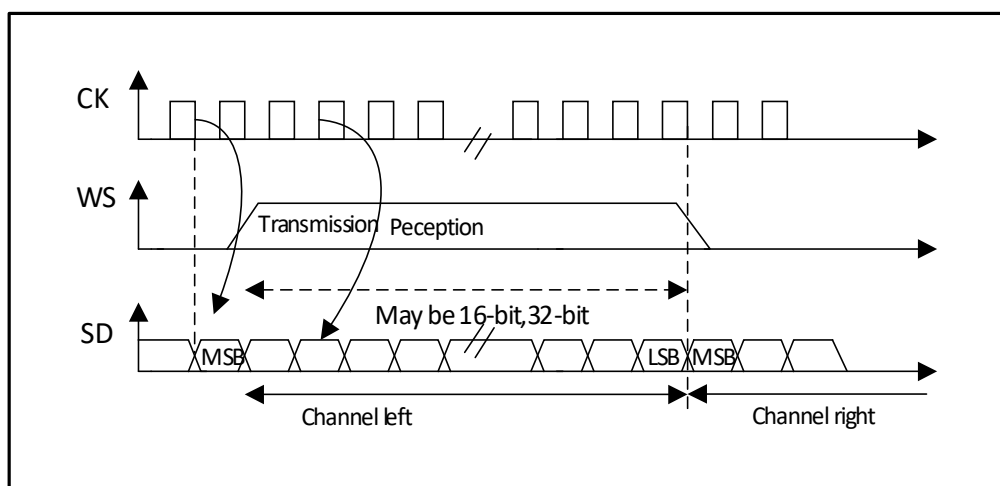
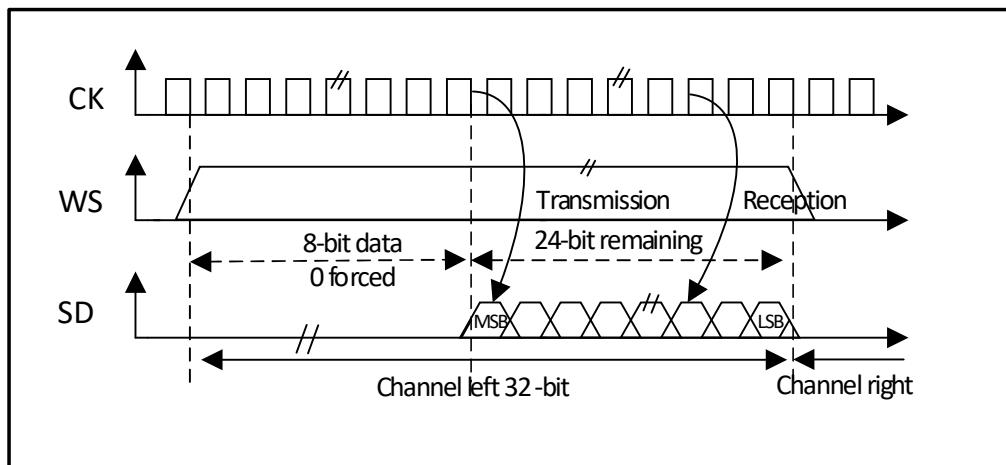
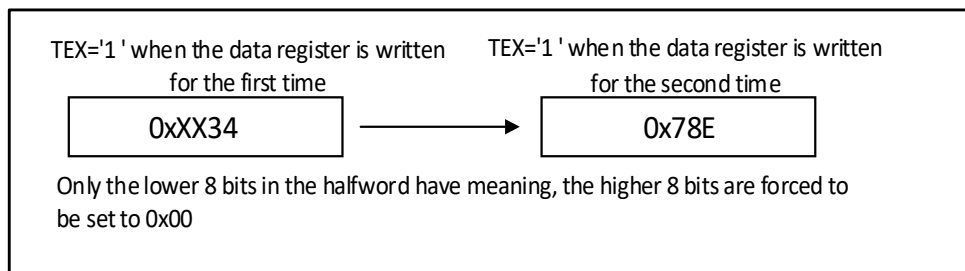


Figure 23-11 LSB-aligned 24-bit data, CPOL = 0



In transmit mode, if you want to send data 0x3478AE, you need to write the register DATAR twice by software or

DMA.



In receive mode, if you want to receive data 0x3478AE, you need to read the register DATAR once when 2 consecutive RXNE events occur.

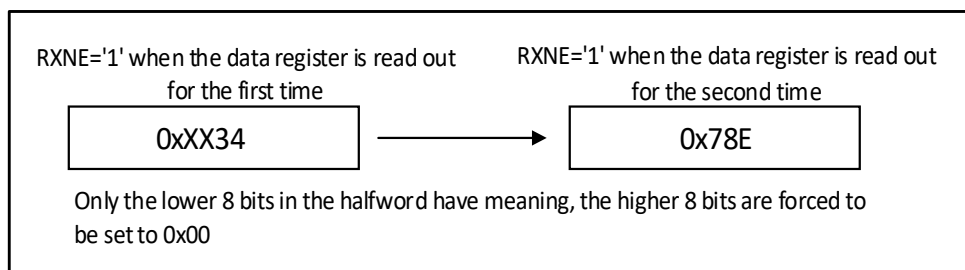
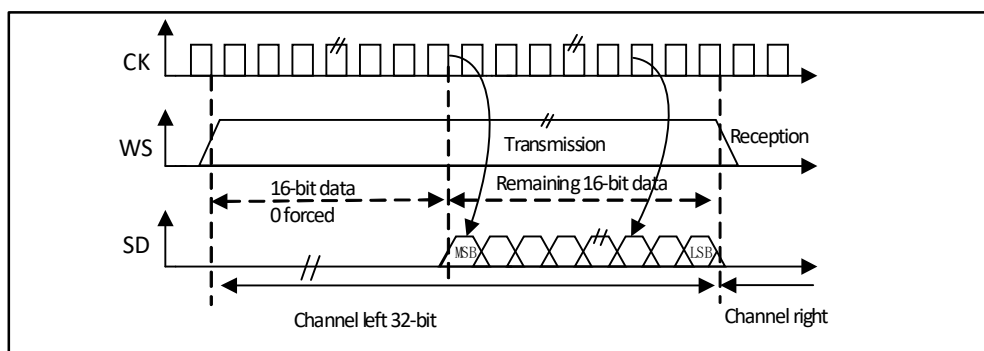


Figure 23-12 LSB aligned 16-bit data extended to 32-bit packet frame, CPOL = 0



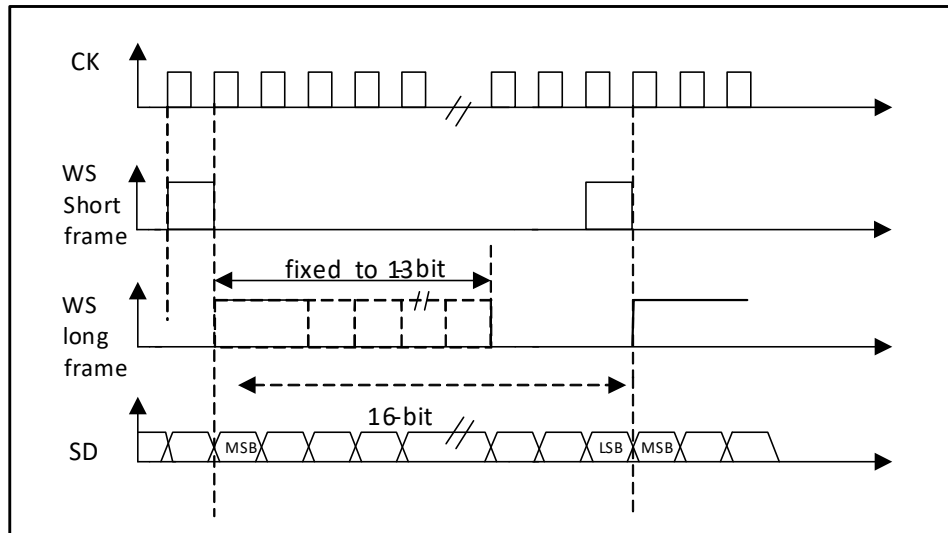
During the I2S configuration stage, if you choose to extend the 16-bit data to a 32-channel frame, you only need to access the DATAR register once. At this point, the high half word (16-bit MSB) after extending to 32 bits is set to 0x0000 by hardware.

If the data to be transmitted or received is 0x76A3 (Extended to 32 bits is 0x000076A3), only one operation of DATAR is required. When sending, if TXE is 1, the user needs to write the data to be sent (i.e. 0x76A3). The 0x0000 part, which is used to extend to 32 bits, is first sent by the hardware, and once the valid data starts to be sent from the SD pin, the next TXE event occurs. On reception, the RXNE event occurs as soon as valid data is received (Instead of the 0x0000 part). This way, there is more time between the 2 reads and writes, preventing underflow or overflow conditions.

23.3.2.4 PCM standard

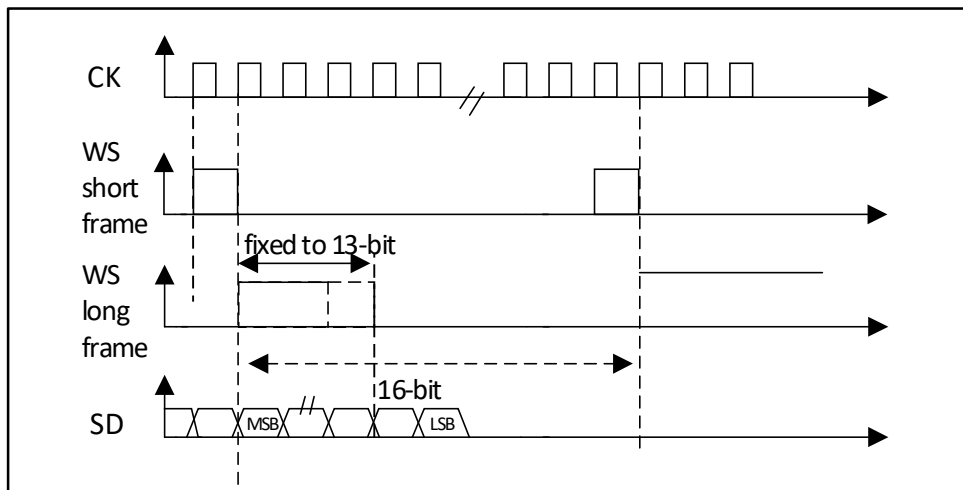
Under the PCM standard, there is no information for channel selection. The PCM standard has 2 available frame structures, short or long, which can be selected by setting the PCMSYNC bit in register I2SCFGR.

Figure 23-13 PCM standard waveform (16-bit)



For the long frame, in the main mode, the valid time of the WS signal used for synchronization is fixed to 13 bits. For short frames, the length of the WS signal used for synchronization is only 1 bit.

Figure 23-14 PCM standard waveform (16-bit)



No matter which mode (Master or slave), which synchronization method (Short frame or long frame), the time difference between 2 consecutive frames of data and between 2 synchronization signals, (Even in slave mode) needs to be set by setting the I2SCFGR register determined by the DATLEN bit and the CHLEN bit.

23.3.3 Clock Generator

The bit rate of I2S determines the data flow on the I2S data line and the frequency of the I2S clock signal. I2S bit rate = number of bits per channel × number of channels × audio sampling frequency.

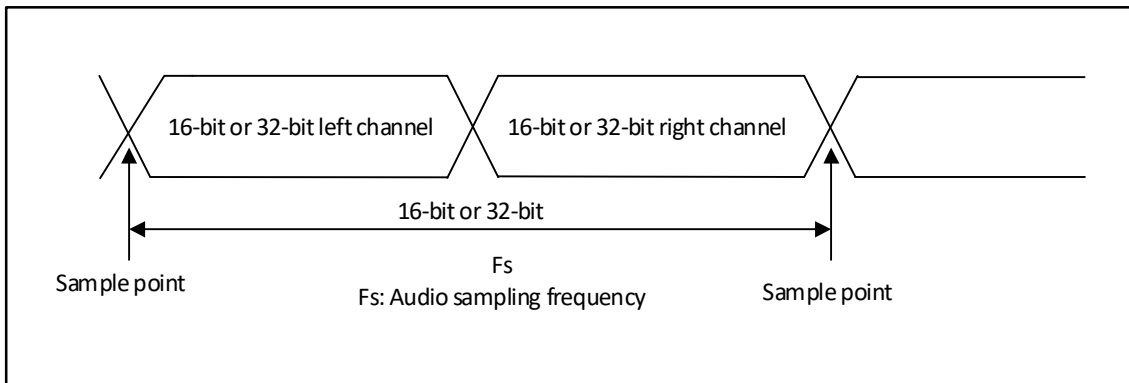
For a signal with left and right channels and 16-bit audio, the I2S bit rate is calculated as follows:

$$\text{I2S bit rate} = 16 \times 2 \times F_s$$

If the packet length is 32 bits, the I2S bit rate is calculated as follows:

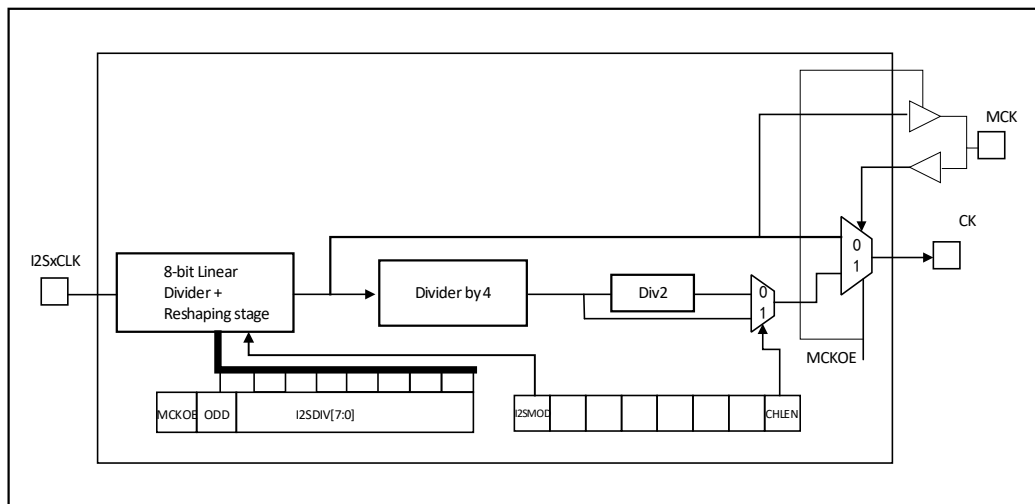
$$\text{I2S bit rate} = 32 \times 2 \times F_s$$

Figure 23-15 Audio sampling frequency definition



In master mode, the linear divider needs to be set correctly in order to obtain the desired audio frequency.

Figure 23-16 I2S clock generator structure



The clock source of I2SxCLK in the figure is the system clock (i.e., the HSI, HSE, or PLL that drives the HB clock). I2SxCLK can come from SYSCLK, or the PLL3 VCO (2xPLL3CLK) clock, which can be selected by the I2S2SRC and I2S3SRC bits in the RCC_CFGR2 register. Audio can be sampled at 96kHz, 48kHz, 44.1kHz, 32kHz, 22.05kHz, 16kHz, 11.025kHz, or 8kHz (Or any value within this range). To obtain the desired frequency, set the linear divider according to the following formula:

When the master clock needs to be generated (The MCKOE bit of the register SPI_I2SPR is 1):

When the frame length of the channel is 16 bits, $F_s = \text{I2SxCLK} / [(16 \times 2) \times ((2 \times \text{I2SDIV}) + \text{ODD}) \times 8]$

When the frame length of the channel is 32 bits, $F_s = \text{I2SxCLK} / [(32 \times 2) \times ((2 \times \text{I2SDIV}) + \text{ODD}) \times 4]$

When the main clock is turned off (MCKOE bit is '0'):

When the frame length of the channel is 16 bits, $F_s = \text{I2SxCLK} / [(16 \times 2) \times ((2 \times \text{I2SDIV}) + \text{ODD})]$

When the frame length of the channel is 32 bits, $F_s = \text{I2SxCLK} / [(32 \times 2) \times ((2 \times \text{I2SDIV}) + \text{ODD})]$

23.3.4 I2S Master Mode

Set I2S to work in master mode, the serial clock is output by pin CK, and the word selection signal is generated by pin WS. The master clock (MCK) can be selected to be output or not output by setting the MCKOE bit in register I2SPR.

23.3.4.1 Configuration Process

- Set I2SDIV[7:0] in the I2SPR register to define the serial clock baud rate corresponding to the audio sampling frequency. Also define the ODD bit of the register I2SPR.
- Set the CKPOL bit to define the level state of the communication clock when it is idle. If the main clock MCK needs to be provided to the external DAC/ADC audio device, the MCKOE position of the register I2SPR is required.
- Set the I2SMOD bit in the I2SCFGR register to 1 to activate the I2S function, set the I2SSTD[1:0] and PCMSYNC bits to select the I2S standard used, and set CHLEN to select the number of data bits per channel. Also set the I2SCFG[1:0] in the SPI_I2SCFGR register to select the I2S master mode and direction (transmitter or receiver).
- If required, the required interrupt function and DMA function can be turned on by setting register CR2.
- The I2SE bit in the I2SCFGR register must be set to 1.
- Pins WS and CK need to be configured for output mode. If the MCKOE bit in the SPI_I2SPR register is 1, the pin MCK is also configured to output mode.

23.3.4.2 Transmission Process

The transmission process starts when 1 half word (16 bits) of data is written to the transmit buffer. It is assumed that the first data written to the transmit buffer corresponds to the left channel data. When the data is moved from the transmit buffer to the shift register, the flag bit TXE is set to 1. At this time, the data corresponding to the right channel should be written into the transmit buffer. The flag bit CHSIDE indicates which channel the data to be transmitted currently corresponds to. The value of the flag bit CHSIDE is updated when TXE is 1, so it has meaning when TXE is 1. After the data of the left channel and then the right channel have been transmitted, it can be regarded as a complete data frame. It is not possible to transmit only part of the data frame, such as data only for the left channel.

When the first bit of data is sent out, the half-word data is transferred to the 16-bit shift register in parallel, and then the following bits are sent out from pins MOSI/SD in high-order first. Each time data is moved from the transmit buffer to the shift register, the flag bit TXE is set to 1, and an interrupt is generated if the TXEIE bit in the CR2 register is 1.

In order to ensure continuous audio data transmission, it is recommended to write the next data to be transmitted to the register DATAR before the current transmission is completed. It is recommended to wait for the flag bit TXE=1 and BSY=0 when the I2S function is to be turned off, and then clear the I2SE bit to '0'.

23.3.4.3 Reception Process

Except for point 3, the configuration steps of the reception process are consistent with the transmission process (See the aforementioned "transmission process"). It is necessary to select the main reception mode by configuring I2SCFG[1:0]. Regardless of the data and channel length, audio data is always received in 16-bit packets. That is, after the receive buffer is filled up each time, the flag bit RXNE is set to 1, and if the RXNEIE bit of the register CR2 is 1, an interrupt is generated. Depending on the configured data and channel length, receiving data from the left or right channel will require 1/2 times to transmit the data to the receive buffer. The RXNE flag is cleared by reading the DATAR register. CHSIDE is updated after each reception. Its value depends on the WS signal generated by the I2S unit. If the previous received data has not been read, and new data is received, that is, an overflow occurs,

and the flag bit OVR is set to 1. If the ERRIE bit in the CR2 register is 1, an interrupt is generated, indicating that an error has occurred. To disable I2S, special operations need to be performed to ensure that the I2S module can normally complete the transfer cycle without starting a new data transfer. The operation process is related to the data configuration and channel length, and the mode of the audio protocol:

- 16-bit data extended to 32-bit channel length (DATLEN=00 and CHLEN=1), using LSB (Low bit) alignment mode (I2SSTD=10)
 - a) Wait for the penultimate (n-1) RXNE=1;
 - b) Wait for 17 I2S clock cycles (Using software delay);
 - c) Disable I2S (I2SE=0).
- 16-bit data extended to 32-bit channel length (DATLEN=00 and CHLEN=1), using MSB (High bit) alignment, I2S or PCM mode (I2SSTD=00, I2SSTD=01 or I2SSTD=11 respectively)
 - a) Wait for the last RXNE=1;
 - b) Wait for 1 I2S clock cycle (Using software delay);
 - c) Disable I2S (I2SE=0).
- For all other combinations of DATLEN and CHLEN, any audio mode selected by I2SSTD, use the following method to turn off I2S:
 - a) Wait for the penultimate (n-1) RXNE=1;
 - b) Wait for one I2S clock cycle (Using software delay);
 - c) Disable I2S (I2SE=0).

Note: The BSY flag is always low during transmission.

23.3.5 I2S Slave Mode

In slave mode, I2S can be set to transmission and reception mode. The configuration method of the slave mode basically follows the same process as the configuration of the master mode. In slave mode, no clock is required from the I2S interface. Both the clock signal and the WS signal are provided by an external master I2S device connected to the corresponding pins. So the user does not need to configure the clock.

The configuration steps are as follows:

- Set the I2SMOD bit in the I2SCFGR register to activate the I2S function; set I2SSTD[1:0] to select the I2S standard used; set DATLEN[1:0] to select the number of data bits; set CHLEN to select the number of data bits per channel. Set the I2SCFG[1:0] in the I2SCFGR register to select the data direction (transmitter or receiver) of the I2S slave mode.
- As needed, set register CR2 to turn on the desired interrupt function and DMA function.
- The I2SE bit in the I2SCFGR register must be set to 1.

23.3.5.1 Transmission Process

The transmission process starts when the external master sends a clock signal, and when the NSS_WS signal requests data transmission. The slave device must be enabled and the I2S data register must be written before the external master device can start communication. For the MSB-aligned and LSB-aligned modes of I2S, the first data item written to the data register corresponds to the data of the left channel. When the communication starts, the data is transferred from the transmit buffer to the shift register, and then the flag bit TXE is set to 1; at this time, the data item corresponding to the right channel should be written into the I2S data register. The flag bit CHSIDE indicates which channel the data to be transmitted currently corresponds to. Compared to the transmit flow in master mode, in slave mode, CHSIDE depends on the WS signal from the external master I2S. This means that the slave I2S has to prepare the first data to be sent before receiving the clock signal generated by the master. A WS signal of 1 means

that the left channel is sent first.

23.3.5.2 Reception Process

Except for point 1, the configuration steps are the same as the transmission process. The master reception mode needs to be selected by configuring I2SCFG[1:0]. Regardless of the data and channel length, audio data is always received in the form of 16-bit packets, that is, every time the receive buffer is filled, the flag bit RXNE is set to 1, and if the RXNEIE bit in the I2S_CTLR2 register is 1, an interrupt is generated. According to different data and channel length settings, receiving left channel or right channel data will require 1 or 2 times to transmit data to the receive buffer. CHSIDE is updated every time data is received (to be read out from DATAR), which corresponds to the WS signal generated by the I2S unit. Reading the SPI_DATAR register will clear the RXNE bit. When the previous received data has not been read out and new data is received, an overflow occurs, and the flag bit OVR is set to 1; if the ERRIE bit in the register I2S_CTLR2 is 1, an interrupt is generated, indicating that an error has occurred. To turn off the I2S function, it is necessary to clear the I2SE bit to 0 when the last RXNE=1 is received.

23.3.6 Status Flag

There are 3 status flags for the user to monitor the status of the I2S bus.

23.3.6.1 Busy Flag (BSY)

The BSY flag is set and cleared by hardware (writing this bit has no effect). This flag bit indicates the status of the I2S communication layer. When this bit is 1, it indicates that I2S communication is in progress, with one exception: in master receive mode (I2SCFG=11), the BSY flag is always low during reception. Before the software closes the SPI module, the BSY flag can be used to detect whether the transfer is over, so as to avoid destroying the last transfer, so it is necessary to strictly follow the procedure below. When a transfer starts, the BSY flag is set to 1, unless the I2S module is in master receive mode. This flag is cleared when the transfer ends or when the I2S module is turned off. When communication is continuous, in master transmit mode, the BSY flag is always high during the entire transfer; in slave mode, between each data item transmission, the BSY flag goes low within 1 I2S clock cycle.

23.3.6.2 TX Buffer Empty Flag (TXE)

When the flag bit is 1, it indicates that the sending buffer is empty, and new data to be sent can be written into the sending buffer. When there is data in the transmit buffer, the flag bit is cleared to 0. When I2S is turned off (I2SE bit is 0), this flag is also 0.

23.3.6.3 RX Buffer Not Empty Flag (RXNE)

Setting this flag bit indicates that there is valid data received in the receive buffer. This bit is cleared to 0 when the DATAR register is read.

23.3.6.4 Channel Side Flag (CHSIDE)

In transmit mode, this flag is refreshed when TXE is high, indicating which channel the data is sent from the SD pin. If an underflow error occurs in the slave sending mode, the value of this flag is invalid, and the I2S needs to be turned off and on before restarting the communication. In receive mode, this flag bit is refreshed when the register DATAR receives data, indicating the channel where the received data is located. Note that if an error occurs, this flag is meaningless, and I2S needs to be turned off and on again. Under the PCM standard, no matter the short frame format or the long frame format, this flag has no meaning. If the flag bit OVR or UDR in the STATR register is 1,

and the ERRIE bit in the CR2 register is 1, an interrupt will be generated. (After the interrupt source has been cleared) The interrupt flag can be cleared by reading the STATR register.

23.3.7 Error Flag

23.3.7.1 Underrun Flag (UDR)

In the slave transmit mode, if the new data has not been written to the DATAR register when the first clock edge of the data transfer arrives, this flag will be set to 1. This flag is valid only after the I2SMOD bit of the register I2SCFGR is set. An interrupt will be generated if the ERRIE bit in register CR2 is 1. This flag is cleared by reading the STATR register.

23.3.7.2 Overrun Flag (OVR)

If the previous received data has not been read, and new data is received, that is, an overflow occurs, and the flag is set to 1. If the ERRIE bit of the register CTLR2 is 1, an interrupt is generated to indicate that an error has occurred. At this time, the content of the receiving buffer will not be refreshed with the new data sent from the sending device. A read of register DATAR returns the last correctly received data. All other 16-bit data sent by the sending device after the overflow occurs are lost. This flag is cleared by first reading the DATAR register and then reading the STATR register.

23.3.8 I2S Interrupt

I2S has 4 interrupt sources, of which the send buffer is empty and the receive buffer is not empty. These 2 events will set TXE and RXNE respectively, and an interrupt will be generated when the TXEIE and RXNEIE bits are set respectively. If the previous received data has not been read out, new data is received, that is, an overflow occurs. If ERRIE is set, an overflow interrupt will be generated; in slave transmit mode, if the first clock of data transmission When the edge arrives, the new data is still not written to the DATAR register. If ERRIE is set, an underflow interrupt will be generated.

23.3.9 DMA Features

The way DMA works in I2S mode is exactly the same as in SPI mode except that the CRC function is not available. Because there is no data transmission protection system in I2S mode.

23.4 Register Description

Table 23-1 SPI1 registers

Name	Access address	Description	Reset value
R16_SPI1_CTLR1	0x40013000	SPI1 control register1	0x0000
R16_SPI1_CTLR2	0x40013004	SPI1 control register2	0x0000
R16_SPI1_STATR	0x40013008	SPI1 status register	0x0002
R16_SPI1_DATAR	0x4001300C	SPI1 data register	0x0000
R16_SPI1_CRCCR	0x40013010	SPI1 polynomial register	0x0007
R16_SPI1_RCRCR	0x40013014	SPI1 receive CRC register	0x0000
R16_SPI1_TRCRCR	0x40013018	SPI1 transmit CRC register	0x0000
R16_SPI1_I2S_CFGR	0x4001301C	SPI1 I2S configuration register	0x0000
R16_SPI1_HSCR	0x40013024	SPI1 high-speed control register	0x0000

Table 23-2 SPI2 registers

Name	Access address	Description	Reset value
R16_SPI2_CTLR1	0x40003800	SPI2 control register1	0x0000
R16_SPI2_CTLR2	0x40003804	SPI2 control register2	0x0000
R16_SPI2_STATR	0x40003808	SPI2 status register	0x0002
R16_SPI2_DATAR	0x4000380C	SPI2 data register	0x0000
R16_SPI2_CRCCR	0x40003810	SPI2 polynomial register	0x0007
R16_SPI2_RCRCR	0x40003814	SPI2 receive CRC register	0x0000
R16_SPI2_TCRCR	0x40003818	SPI2 transmit CRC register	0x0000
R16_SPI2_I2S_CFGR	0x4000381C	SPI2 I2S configuration register	0x0000
R16_SPI2_I2SPR	0x40003820	SPI2 I2S prescaler register	0x0000
R16_SPI2_HSCR	0x40003824	SPI2 high-speed control register	0x0000

Table 23-3 SPI3 registers

Name	Access address	Description	Reset value
R16_SPI3_CTLR1	0x40003C00	SPI3 control register1	0x0000
R16_SPI3_CTLR2	0x40003C04	SPI3 control register2	0x0000
R16_SPI3_STATR	0x40003C08	SPI3 status register	0x0002
R16_SPI3_DATAR	0x40003C0C	SPI3 data register	0x0000
R16_SPI3_CRCCR	0x40003C10	SPI3 polynomial register	0x0007
R16_SPI3_RCRCR	0x40003C14	SPI3 receive CRC register	0x0000
R16_SPI3_TCRCR	0x40003C18	SPI3 transmit CRC register	0x0000
R16_SPI3_I2S_CFGR	0x40003C1C	SPI3 I2S configuration register	0x0000
R16_SPI3_I2SPR	0x40003C20	SPI3 I2S prescaler register	0x0000
R16_SPI3_HSCR	0x40003C24	SPI3 high-speed control register	0x0000

Table 23-4 SPI4 registers

Name	Access address	Description	Reset value
R16_SPI4_CTLR1	0x40004000	SPI4 control register1	0x0000
R16_SPI4_CTLR2	0x40004004	SPI4 control register2	0x0000
R16_SPI4_STATR	0x40004008	SPI4 status register	0x0002
R16_SPI4_DATAR	0x4000400C	SPI4 data register	0x0000
R16_SPI4_CRCCR	0x40004010	SPI4 polynomial register	0x0007
R16_SPI4_RCRCR	0x40004014	SPI4 receive CRC register	0x0000
R16_SPI4_TCRCR	0x40004018	SPI4 transmit CRC register	0x0000
R16_SPI4_HSCR	0x40004024	SPI4 high-speed control register	0x0000

23.4.1 SPI Control Register 1 (SPIx_CTLR1) (x=1/2/3/4)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
15	BIDIMODE	RW	Unidirectional data mode enable. 0: 2-wire bidirectional mode; 1: 1-wire bidirectional mode.	0
14	BIDIOE	RW	Single-line output enable, used in conjunction	0

			with BIDIMODE. 0: Disable output, only receive; 1: Enable output, only transmit.	
13	CRCEN	RW	Hardware CRC check enable, this bit can only be written when SPE is 0, this bit can only be used in full duplex mode. 0: Disable CRC calculation; 1: Enable CRC calculation.	0
12	CRCNEXT	RW	Value of the transmit CRC register after next data transfer. This bit should be set immediately after writing the last data to the data register. 0: Continue to transmit the data of the data register; 1: Transmit the CRC result.	0
11	DFF	RW	Data frame length, this bit can only be written when SPE is 0. 0: Use 8-bit data length for transmission and reception; 1: Use 16-bit data length to transmit and receive.	0
10	RXONLY	RW	Rx only in 2-wire mode. It is used in conjunction with BIDIMODE. Setting this bit allows the device to only receive and not transmit. 0: Full-duplex mode; 1: Receive only, simplex mode.	0
9	SSM	RW	CS pin management, this bit determines whether the level of the NSS pin is controlled by hardware or software. 0: Hardware controls NSS pin; 1: Software control NSS pin.	0
8	SSI	RW	CS pin control. When SSM is set, this bit determines the level of the NSS pin. 0: NSS is low level; 1: NSS is high level.	0
7	LSBFIRST	RW	Frame format control. This bit cannot be modified during communication. 0: Transmit MSB first; 1: Transmit LSB first.	0
6	SPE	RW	SPI enable. 0: Disable SPI; 1: Enable SPI.	0
[5:3]	BR[2:0]	RW	Baud rate setting domain, this domain cannot be modified during communication. When the HSRXEN bit is 0, the SCK frequency is:	0

			000: FPCLK/2; 001: FPCLK/4; 010: FPCLK/8; 011: FPCLK/16; 100: FPCLK/32; 101: FPCLK/64; 110: FPCLK/128; 111: FPCLK/256. When the HSRXEN bit is 1, the SCK frequency is: 000: FPCLK/2; 001: FPCLK/3; 010: FPCLK/4; 011: FPCLK/5; 100: FPCLK/6; 101: FPCLK/7; 110: FPCLK/8; 111: FPCLK/9.	
2	MSTR	RW	Master/slave setting. It cannot be modified during communication. 0: Configured as the slave device; 1: Configure as the master device.	0b
1	CPOL	RW	Clock polarity selection. It cannot be modified during communication. 0: In idle state, SCK remains low; 1: In idle state, SCK remains high.	0
0	CPHA	RW	Clock phase setting. It cannot be modified during communication. 0: Data sampling starts from the first clock edge; 1: Data sampling starts from the second clock edge.	0

23.4.2 SPI Control Register 2 (SPIx_CTLR2) (x=1/2/3/4)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	TXEIE	RW	Transmit buffer empty interrupt enable. 0: Disable interrupt generation when TXE is set; 1: Enable interrupt generation when TXE is set.	0
6	RXNEIE	RW	Receive buffer not empty interrupt enable. 0: Disable interrupt generation when RXNE is set; 1: Enable interrupt generation when RXNE is set.	0
5	ERRIE	RW	Error interrupt enable. 0: Disable interrupt generation when an error (CRCERR, OVR, MODF) is generated; 1: Enable interrupt generation when an error (CRCERR, OVR, MODF) is generated.	0
[4:3]	Reserved	RO	Reserved	0
2	SSOE	RW	SS output enable. Disable SS output to work in multi-master mode. 0: Disable SS output in master mode; 1: Enable SS output.	0

1	TXDMAEN	RW	Transmit buffer DMA enable. 0: Disable transmit buffer DMA; 1: Enable transmit buffer DMA.	0
0	RXDMAEN	RW	Receive buffer DMA enable. 0: Disable receive buffer DMA; 1: Enable receive buffer DMA.	0

23.4.3 SPI Status Register (SPIx_STATR) (x=1/2/3/4)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
7	BSY	RO	Busy flag. Set and reset by hardware. 0: SPI is not communicating; 1: SPI is communicating, or the transmit buffer is not empty.	0
6	OVR	RO	Overflow flag. Set by hardware, and reset by software. 0: No overflow error occurred; 1: An overflow error occurred.	0
5	MODF	RO	Mode error flag. Set by hardware, and reset by software. 0: No mode error occurred; 1: A mode error occurred.	0
4	CRCERR	RW0	CRC error flag. Set by hardware, and reset by software. 0: The received CRC value is the same as the RCRCR value; 1: The received CRC value is inconsistent with the RCRCR value.	0
3	UDR	RO	Underflow flag. Set by hardware, and reset by software. 0: Underflow has not occurred; 1: Underflow occurs.	0
2	CHSIDE	RO	Channel. Set by hardware and reset by software. 0: The left channel needs to be transmitted or received; 1: The right channel needs to be transmitted or received.	0
1	TXE	RO	Transmit buffer empty flag. 0: The transmit buffer is not empty; 1: The transmit buffer is empty.	1
0	RXNE	RO	Receive buffer not empty flag. 0: The receive buffer is empty; 1: The receive buffer is not empty.	0

			Note: Read DATAR and auto-zero.	
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23.4.4 SPI Data Register (SPIx_DATAR) (x=1/2/3/4)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:0]	DATAR	RW	Data register. It is used to store the received data or pre-store the data to be sent, so the read and write of the data register is actually a different area for operations, among which the read pair uses the receive buffer, and the write corresponds to the send buffer. Data reception and transmission can be 8-bit or 16-bit, and it is necessary to determine how many bits of data to use before transmission. When using 8 bits for data transmission, only the lower 8 bits of the data register are used, and the upper 8 bits are forced to 0 when receiving. Using the 16-bit data structure will cause all 16-bit data registers to be used.	0

23.4.5 SPI Polynomial Register (SPIx_CRCCR) (x=1/2/3/4)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CRCPOLY[15:0]															

Bit	Name	Access	Description	Reset Value
[15:0]	CRCPOLY[15:0]	RW	CRC polynomial. This domain defines the polynomial used for the CRC calculation.	0x0007

23.4.6 SPI Receive CRC Register (SPIx_RCRCCR) (x=1/2/3/4)

Offset address: 0x14

Bit	Name	Access	Description	Reset Value
[15:0]	RXCRC[15:0]	RO	Rx CRC value. Stores the result of the calculated CRC check of the received bytes. Setting CRCEN will reset this register. The calculation method uses the polynomial used by CRCPOLY. In 8-bit mode, only the lower 8 bits participate in the calculation, and in 16-bit mode, all 16 bits will participate in the calculation. This register needs to be read when BSY is 0.	0

23.4.7 Transmit CRC Register (SPIx_TCR) (x=1/2/3/4)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[15:0]	TXCRC[15:0]	RO	Tx CRC value. Stores the calculated result of the CRC check of the bytes that have been sent. Setting CRCEN will reset this register. The calculation method uses the polynomial used by CRCPOLY. In 8-bit mode, only the lower 8 bits participate in the calculation, and in 16-bit mode, all 16 bits will participate in the calculation. This register needs to be read when BSY is 0.	0

23.4.8 SPI_I2S Configuration Register (SPIx_I2S_CFG) (x=2/3)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
[15:12]	Reserved	RO	Reserved	0
11	I2SMOD	RW	I2S mode selection, this bit can only be set when SPI or I2S is disabled. 0: Select SPI mode; 1: Select I2S mode.	0
10	I2SE	RW	I2S enable, not used in SPI mode. 0: I2S disabled; 1: I2S enabled.	0
[9:8]	I2SCFG[1:0]	RW	I2S mode selection, this bit can only be set when I2S is disabled: 00: Slave device transmits; 01: Slave device receives; 10: Master device transmit; 11: Master device receives.	00b
7	PCMSYNC	RW	PCM frame synchronization. This bit is only meaningful when I2SSTD = 11 (Using the PCM standard). 0: Short frame sync; 1: Long frame synchronization.	0
6	Reserved	RO	Reserved	0
[5:4]	I2SSTD[1:0]	RW	I2S standard selection, this bit can only be set when I2S is disabled. 00: I2S Philips standard; 01: High byte alignment standard (Left alignment); 10: Low byte alignment standard (Right justified); 11: PCM standard.	00b
3	CKPOL	RW	Quiescent clock polarity, for correct operation,	0

			this bit should only be set when I2S is turned off. 0: I2S clock quiescent state is low; 1: I2S clock quiescent state is high level.	
[2:1]	DATLEN[1:0]	RW	Length of the data to be transmitted. For correct operation, this bit can only be set when I2S is turned off. 00: 16-bit data length; 01: 24-bit data length; 10: 32-bit data length; 11: Not allowed.	00b
0	CHLEN	RW	Channel length. Only when DATLEN = 00, the write operation of this bit is meaningful, otherwise the channel length is fixed to 32 bits by hardware. 0: 16 bits wide; 1: 32 bits wide.	0

23.4.9 SPI_I2S Prescaler Register (SPIx_I2SPR) (x=2/3)

Offset Address: 0x20

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0
9	MCKOE	RW	Master clock output enable, for proper operation, this bit can only be set when I2S is turned off. This bit is only used in I2S master mode. 0: Master clock output disabled; 1: Master device clock output enabled.	0
8	ODD	RW	Odd coefficient prescaler, for correct operation, this bit should only be set when I2S is turned off. This bit is only used in I2S master mode. 0: Actual division factor = I2SDIV*2; 1: Actual division factor = (I2SDIV*2)+1.	0
[7:0]	I2SDIV[7:0]	RW	I2S linear prescaler. For proper operation, this bit should only be set when I2S is turned off. This bit is only used in I2S master mode. Disable setting I2SDIV[7:0] = 0 or I2SDIV[7:0] = 1	0

23.4.10 SPI High-speed Control Register (SPIx_HSCR) (x=1/2/3/4)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[15:3]	Reserved	RO	Reserved	0
2	HSRXEN2	RW	SPI High Speed Read Mode 2 This mode is used only when HSRXEN is turned on and SPI PCLK is greater than or equal to 120M; this bit is not recommended for less than 120M. A	0

			higher host read speed can be obtained than if only HSRXEN is turned on. 0: Disables high-speed read mode 2; 1: Enable high-speed read mode 2.	
1	Reserved	RO	Reserved	0
0	HSRXEN	W0	High-speed read mode enable bit: 0: High-speed read mode disabled; 1: High-speed read mode enabled.	0

Chapter 24 USB PD Controller (USBPD)

24.1 USB PD Controller Introduction

The chip has built-in USB Power Delivery controller and PD transceiver PHY, supports USB Type-C master-slave detection, automatic BMC codec and CRC, hardware edge control.

Supports USB PD 2.0, PD 3.0, and PD 3.2 Power Delivery control, supports SPR and EPR, supports 100W or 240W fast charging, supports PD sink and PD source applications, and DRP applications. Supports PDUSB, supports UFP and DFP, and DRD applications.

Supports PD packets including SOP, SOP', and SOP'', supports USB PD reset signal frame hardware reset, supports a maximum packet length of 510 bytes, and supports DMA.

24.2 Register Description

Table 24-1 USBPD-related registers

Name	Access address	Description	Reset value
R32_USBPD_CONFIG	0x40027000	PD configuration register	0x00000002
R16_CONFIG	0x40027000	PD interrupt enable register	0x0002
R16_BMC_CLK_CNT	0x40027002	BMC sampling clock counter	0x0000
R32_USBPD_CONTROL	0x40027004	PD control register	0x00000000
R16_CONTROL	0x40027004	PD transceiver control register	0x0000
R8_CONTROL	0x40027004	PD transceiver enable register	0x00
R8_TX_SEL	0x40027005	PD transmit SOP selection register	0x00
R16_BMC_TX_SZ	0x40027006	PD transmit length register	0x0000
R32_USBPD_STATUS	0x40027008	PD status register	0x000000XX
R16_STATUS	0x40027008	PD interrupt and data register	0x00XX
R8_DATA_BUF	0x40027008	DMA cache data register	0xXX
R8_STATUS	0x40027009	PD interrupt flag register	0x00
R16_BMC_BYTE_CNT	0x4002700A	Byte counter	0x0000
R32_USBPD_PORT	0x4002700C	Port control register	0x00030003
R16_PORT_CC1	0x4002700C	CC1 port control register	0x0003
R16_PORT_CC2	0x4002700E	CC2 port control register	0x0003
R32_USBPD_DMA	0x40027010	DMA cache address register	0x00XXXXXX

24.2.1 PD Configuration Register (R16_CONFIG)

Offset address: 0x00

Bit	Name
[31:16]	R16_BMC_CLK_CNT
[15:0]	R16_CONFIG

24.2.2 PD Interrupt Enable Register (R16_CONFIG)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
15	IE_TX_END	RW	End-of-transmit interrupt enable.	0
14	IE_RX_RESET	RW	Receive reset interrupt enable.	0
13	IE_RX_ACT	RW	Receive completion interrupt enable.	0
12	IE_RX_BYTE	RW	Receive byte interrupt enable.	0
11	IE_RX_BIT	RW	Receive bit interrupt enable.	0
10	IE_PD_IO	RW	PD IO interrupt enable.	0
9	Reserved	RO	Reserved	0
8	RX_MULTI_0	RO	Continuous reception of multiple zero indicator bits: 1: Continuous reception of multiple zeros; 0: No continuous reception of multiple zeros.	0
7	CC_INC	RW	CC bias current adjustment position: 1: Bias current +10%; 0: Default setting.	0
6	CC_TR	RW	Tr adjustment for CC LV waveform: 1: Increase Tr by 5%; 0: Keep Tr unchanged.	0
5	WAKE_POLAR	RW	PD port wake-up level: 1: Active high; 0: Active low.	0
4	PD_RST_EN	RW	The PD mode reset command enable: 1: Reset; 0: Invalid.	0
3	PD_DMA_EN	RW	Enable the DMA of USBPD, which must be set to 1 in normal transfer mode: 1: Enable DMA function and DMA interrupt; 0: Disable DMA.	0
2	CC_SEL	RW	Select the current PD communication port: 1: Use CC2 port to communicate; 0: Use CC1 port to communicate.	0
1	PD_ALL_CLR	RW	PD mode clears all interrupt flag bits: 1: Clear the interrupt flag bit; 0: Invalid.	1
0	PD_FILT_EN	RW	Control the input filter enable for the PD pin: 1: Enable filtering; 0: Disable filtering.	0

24.2.3 BMC Sampling Clock Counter (R16_BMC_CLK_CNT)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
[15:9]	Reserved	RO	Reserved	0
[8:0]	BMC_CLK_CNT	RW	BMC transmits or receives sampling clock	0

			counters.	
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Note: During transmission, $BMC_CLK_CNT = (HCLK / CC \text{ communication frequency}) / 2$

During reception, $BMC_CLK_CNT = (HCLK / CC \text{ communication frequency}) / 4 * 3$

24.2.4 PD Control Register (R32_USBPD_CONTROL)

Offset address: 0x04

Bit	Name
[31:16]	R16_BMC_TX_SZ
[15:0]	R16_CONTROL

24.2.5 PD Transceiver Control Register (R16_CONTROL)

Offset address: 0x04

Bit	Name
[15:8]	R8_TX_SEL
[7:0]	R8_CONTROL

24.2.6 PD Transceiver Enable Register (R8_CONTROL)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
7	BMC_BYTE_HI	RO	Indicates the current half-byte status during PD data transmission/reception: 1: Processing upper 4 bits; 0: Processing lower 4 bits.	0
6	TX_BIT_BACK	RO	Indicates the bit status when the current BMC is transmitting code: 1: Indicates BMC byte transmission in progress; 0: Idle.	0
5	DATA_FLAG	RO	Cache Data Validity Flag.	0
[4:2]	RX_STATE[2:0]	RO	PD Receive Status Indicator 000: Initial receive state; 001: Start receiving SOP; 010: Receive Reset; 011: Receive SOP 100: End of receive; 101: Receive not used; 110: Receive EOP; 111: Receive byte.	0
1	BMC_START	RW	BMC transmits a start signal.	0
0	PD_TX_EN	RW	USBPD transceiver mode and transmit enable: 0: PD receive enable. 1: PD transmit enable.	0

24.2.7 PD Transmit SOP Selection Register (R8_TX_SEL)

Offset address: 0x05

Bit	Name	Access	Description	Reset value
[7:6]	TX_SEL4[1:0]	RW	Select the K-CODE4 type in PD transmitting mode:	0

			00: SYNC2; 01: SYNC3; 1x: RST2.	
[5:4]	TX_SEL3[1:0]	RW	Select the K-CODE3 type in PD transmitting mode: 00: SYNC1; 01: SYNC3; 1x: RST1.	0
[3:2]	TX_SEL2[1:0]	RW	Select the K-CODE2 type in PD transmitting mode: 00: SYNC1; 01: SYNC3; 1x: RST1.	0
1	Reserved	RO	Reserved	0
0	TX_SEL1	RW	Select the K-CODE1 type in PD transmitting mode: 1: RST1; 0: SYNC1.	0

24.2.8 PD Transmit Length Register (R16_BMC_TX_SZ)

Offset address: 0x06

Bit	Name	Access	Description	Reset value
[15:9]	Reserved	RO	Reserved	0
[8:0]	BMC_TX_SZ	RW	The total length transmitted in PD mode.	0

24.2.9 PD Status Register (R32_USBPD_STATUS)

Offset address: 0x08

Bit	Name
[31:16]	R16_BMC_BYTE_CNT
[15:0]	R16_STATUS

24.2.10 PD Interrupt and Data Register (R16_STATUS)

Offset address: 0x08

Bit	Name
[15:8]	R8_STATUS
[7:0]	R8_DATA_BUF

24.2.11 DMA Cache Data Register (R8_DATA_BUF)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[7:0]	DATA_BUF	RO	DMA cache data	X

24.2.12 PD Interrupt Flag Register (R8_STATUS)

Offset address: 0x09

Bit	Name	Access	Description	Reset value
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7	IF_TX_END	RW1	Transfer completed interrupt flag, write 1 clear 0, write 0 invalid.	0
6	IF_RX_RESET	RW1	Receive reset interrupt flag, write 1 clear 0, write 0 invalid.	0
5	IF_RX_ACT	RW1	Transfer completed interrupt flag, write 1 clear 0, write 0 invalid.	0
4	IF_RX_BYTE	RW1	Receive byte or SOP interrupt flag, write 1 clear 0, write 0 is invalid.	0
3	IF_RX_BIT	RW1	Receive bit or 5bit interrupt flag, write 1 clear 0, write 0 invalid.	0
2	BUF_ERR	RW1	BUFFER or DMA error interrupt flag, write 1 clear 0, write 0 invalid.	0
[1:0]	BMC_AUX[1:0]	RO	Indicates the current PD status: When the PD is received or after the reception is completed, the status is as follows: 00: Receiving idle or no valid packet received; 01: SOP received i.e. SOP0; 10: SOP' received i.e. SOP1 or Hard Reset; 11: SOP' received i.e. SOP2 or Cable Reset. When PD is transmitting, the status is as follows: 00: CRC32[7:0] is being transmitted; 01: CRC32[15:8] is being transmitted; 10: CRC32[23:16] is being transmitted; 11: CRC32[31:24] is being transmitted.	0

24.2.13 Byte Counter (R16_BMC_BYTE_CNT)

Offset address: 0x0A

Bit	Name	Access	Description	Reset value
[15:9]	Reserved	RO	Reserved	0
[8:0]	BMC_BYTE_CNT	RO	Byte counter.	0

24.2.14 Port Control Register (R32_USBDPD_PORT)

Offset address: 0x0C

Bit	Name
[31:16]	R16_PORT_CC2
[15:0]	R16_PORT_CC1

24.2.15 CC1 Port Control Register (R16_PORT_CC1)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:5]	CC1_CE	RW	Enable the CC1 port voltage comparator: 000: Off.	000

			001: Reserved 010: 0.22V; 011: 0.43V; 100: 0.55V; 101: 0.66V; 110: 0.96V; 111: 1.23V.	
4	CC1_LVE	RW	The CC1 port outputs low voltage enable: 1: Low voltage drive output; 0: Normal V _{DD33} voltage drive output.	0
[3:2]	CC1_PU	RW	CC1 port pull-up current selection: 00: No pull-up current 01: 330uA; 10: 180uA; 11: 80uA.	0
1	CC1_PD	RW	CC1 port pull-down resistor Rd enable: 1: Enable Rd pull-down resistor, about 5.1KΩ; 0: Disable the pull-down resistor. <i>Note: Approximately 600kΩ of weak pull-down remains after shutdown.</i>	1
0	PA_CC1_AI	RO	The CC1 port comparator simulates input.	1

24.2.16 CC2 Port Control Register (R16_PORT_CC2)

Offset address: 0x0E

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:5]	CC2_CE	RW	Enable the CC2 port voltage comparator: 000: Off. 001: Reserved 010: 0.22V; 011: 0.43V; 100: 0.55V; 101: 0.66V; 110: 0.96V; 111: 1.23V.	0
4	CC2_LVE	RW	The CC2 port outputs low voltage enable: 1: Low voltage drive output; 0: Normal V _{DD33} voltage drive output.	0
[3:2]	CC2_PU	RW	CC2 port pull-up current selection: 00: No pull-up current 01: 330uA; 10: 180uA; 11: 80uA.	0
1	CC2_PD	RW	CC2 port pull-down resistor Rd enable:	1

			1: Enable Rd pull-down resistor, about 5.1K Ω ; 0: Disable pull-down resistor. <i>Note: about 600kΩ weak pull-down remains after disabling</i>	
0	PA_CC2_AI	RO	The CC2 port comparator Analog input.	1

24.2.17 DMA Cache Address Register (R32_USBPD_DMA)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	USBPD_DMA_AD DR	RW	USBPD_DMA cache address. The low 21-bit is valid and the address must be 4-byte aligned.	X

Chapter 25 USB High-speed Host/Device Controller (USBHS)

The USB 2.0 high-speed controller functions as both a host controller and a device controller, featuring an integrated 480Mbps USB-PHY physical layer transceiver. When operating as a host controller, it supports low-speed, full-speed, and high-speed USB devices. When functioning as a device controller, it can be flexibly configured for low-speed, full-speed, or high-speed modes to accommodate diverse applications.

25.1 USB Controller Introduction

USB Controller features include:

- Support USB 2.1, USB 2.0, USB 1.1, and USB 1.0 protocol specifications
- Support USB Host function and USB Device function.
- Support control transfer, bulk transfer, interrupt transfer and isochronous/real-time transfer.
- Provide bus reset, suspend, wake-up and recovery functions.
- Host supports USB HUB
- Non-zero endpoints all support the maximum packet size of 1024 bytes, with built-in FIFO, interrupt and DMA support.
- Support USART or I2C pin mapping, and is also used for 2-wire debugging.

25.2 Register Description

The USB registers are divided into 2 parts:

- USB high-speed device control registers
- USB high-speed host control registers

25.2.1 Device Register Description

Table 25-1 Device-related registers

Name	Access address	Description	Reset value
R8_USB_CTRL	0x40030000	USBHS control register	0x07
R8_USB_BASE_MODE	0x40030001	USBHS mode control register	0x00
R8_USB_INT_EN	0x40030002	USBHS interrupt enable register	0x00
R8_USB_DEV_AD	0x40030003	USBHS device address register	0x00
R8_USB_WAKE_CTRL	0x40030004	USBHS remote wakeup register	0x00
R8_USB_TEST_MODE	0x40030005	USBHS test mode register	0x00
R16_USB_LPM_DATA	0x40030006	USBHS power management register	0x8000
R8_USB_INT_FG	0x40030008	USBHS interrupt flag register	0x00
R8_USB_INT_ST	0x40030009	USBHS interrupt status register	0x00
R8_USB_MIS_ST	0x4003000A	USBHS miscellaneous status register	0x00
R16_USB_FRAME_NO	0x4003000C	USBHS frame number register	0x0000
R16_USB_BUS	0x4003000E	USBHS bus status register	0x0000
R16_UEP_TX_EN	0x40030010	USBHS endpoint transmit enable register	0x0000
R16_UEP_RX_EN	0x40030012	USBHS endpoint receive enable register	0x0000
R16_UEP_T_TOG_AUTO	0x40030014	USBHS endpoint transmit auto-flip enable	0x0000

		register	
R16_UEP_R_TOG_AUTO	0x40030016	USBHS endpoint receive auto-flip enable register	0x0000
R8_UEP_T_BURST	0x40030018	USBHS endpoint transmit burst register	0x00
R8_UEP_T_BURST_MODE	0x40030019	USBHS endpoint transmit burst mode register	0x00
R8_UEP_R_BURST	0x4003001A	USBHS endpoint receive burst register	0x00
R8_UEP_R_RES_MODE	0x4003001B	USBHS endpoint receive burst mode register	0x00
R32_UEP_AF_MODE	0x4003001C	USBHS endpoint alter Register	0x00000000
R32_UEP0_DMA	0x40030020	Start address register of endpoint 0 receive buffer	0x000XXXXX
R32_UEP1_RX_DMA	0x40030024	Start address register of endpoint 1 receive buffer	0x000XXXXX
R32_UEP2_RX_DMA	0x40030028	Start address register of endpoint 2 receive buffer	0x000XXXXX
R32_UEP3_RX_DMA	0x4003002C	Start address register of endpoint 3 receive buffer	0x000XXXXX
R32_UEP4_RX_DMA	0x40030030	Start address register of endpoint 4 receive buffer	0x000XXXXX
R32_UEP5_RX_DMA	0x40030034	Start address register of endpoint 5 receive buffer	0x000XXXXX
R32_UEP6_RX_DMA	0x40030038	Start address register of endpoint 6 receive buffer	0x000XXXXX
R32_UEP7_RX_DMA	0x4003003C	Start address register of endpoint 7 receive buffer	0x000XXXXX
R32_UEP1_TX_DMA	0x40030040	Start address register of endpoint 1 transmit buffer	0x000XXXXX
R32_UEP2_TX_DMA	0x40030044	Start address register of endpoint 2 transmit buffer	0x000XXXXX
R32_UEP3_TX_DMA	0x40030048	Start address register of endpoint 3 transmit buffer	0x000XXXXX
R32_UEP4_TX_DMA	0x4003004C	Start address register of endpoint 4 transmit buffer	0x000XXXXX
R32_UEP5_TX_DMA	0x40030050	Start address register of endpoint 5 transmit buffer	0x000XXXXX
R32_UEP6_TX_DMA	0x40030054	Start address register of endpoint 6 transmit buffer	0x000XXXXX
R32_UEP7_TX_DMA	0x40030058	Start address register of endpoint 7 transmit buffer	0x000XXXXX
R32_UEP0_MAX_LEN	0x4003005C	Endpoint 0 maximum length packet register	0x000000XX
R32_UEP1_MAX_LEN	0x40030060	Endpoint 1 maximum length packet register	0x000000XX
R32_UEP2_MAX_LEN	0x40030064	Endpoint 2 maximum length packet register	0x000000XX
R32_UEP3_MAX_LEN	0x40030068	Endpoint 3 maximum length packet register	0x000000XX
R32_UEP4_MAX_LEN	0x4003006C	Endpoint 4 maximum length packet register	0x000000XX
R32_UEP5_MAX_LEN	0x40030070	Endpoint 5 maximum length packet register	0x000000XX
R32_UEP6_MAX_LEN	0x40030074	Endpoint 6 maximum length packet register	0x000000XX
R32_UEP7_MAX_LEN	0x40030078	Endpoint 7 maximum length packet register	0x000000XX
R16_UEP0_RX_LEN	0x4003007C	Endpoint 0 receive length register	0x00XX
R16_UEP1_RX_LEN	0x40030080	Endpoint 1 single receive length register	0XXXXX
R16_UEP1_R_SIZE	0x40030082	Endpoint 1 total receive data length register	0XXXXX
R16_UEP2_RX_LEN	0x40030084	Endpoint 2 single receive length register	0XXXXX
R16_UEP2_R_SIZE	0x40030086	Endpoint 2 total receive data length register	0XXXXX
R16_UEP3_RX_LEN	0x40030088	Endpoint 3 single receive length register	0XXXXX
R16_UEP3_R_SIZE	0x4003008A	Endpoint 3 total receive data length register	0XXXXX
R16_UEP4_RX_LEN	0x4003008C	Endpoint 4 single receive length register	0XXXXX
R16_UEP4_R_SIZE	0x4003008E	Endpoint 4 total receive data length register	0XXXXX
R16_UEP5_RX_LEN	0x40030090	Endpoint 5 single receive length register	0XXXXX
R16_UEP5_R_SIZE	0x40030092	Endpoint 5 total receive data length register	0XXXXX
R16_UEP6_RX_LEN	0x40030094	Endpoint 6 single receive length register	0XXXXX
R16_UEP6_R_SIZE	0x40030096	Endpoint 6 total receive data length register	0XXXXX

R16_UEP7_RX_LEN	0x40030098	Endpoint 7 single receive length register	0xFFFF
R16_UEP7_R_SIZE	0x4003009A	Endpoint 7 total receive data length register	0xFFFF
R16_UEP0_T_LEN	0x4003009C	Endpoint 0 transmit length register	0x00XX
R8_UEP0_TX_CTRL	0x4003009E	Endpoint 0 transmit control register	0x00
R8_UEP0_RX_CTRL	0x4003009F	Endpoint 0 receive control register	0x00
R16_UEP1_T_LEN	0x400300A0	Endpoint 1 transmit length register	0x0000
R8_UEP1_TX_CTRL	0x400300A2	Endpoint 1 transmit control register	0x00
R8_UEP1_RX_CTRL	0x400300A3	Endpoint 1 receive control register	0x00
R16_UEP2_T_LEN	0x400300A4	Endpoint 2 transmit length register	0x0000
R8_UEP2_TX_CTRL	0x400300A6	Endpoint 2 transmit control register	0x00
R8_UEP2_RX_CTRL	0x400300A7	Endpoint 2 receive control register	0x00
R16_UEP3_T_LEN	0x400300A8	Endpoint 3 transmit length register	0x0000
R8_UEP3_TX_CTRL	0x400300AA	Endpoint 3 transmit control register	0x00
R8_UEP3_RX_CTRL	0x400300AB	Endpoint 3 receive control register	0x00
R16_UEP4_T_LEN	0x400300AC	Endpoint 4 transmit length register	0x0000
R8_UEP4_TX_CTRL	0x400300AE	Endpoint 4 transmit control register	0x00
R8_UEP4_RX_CTRL	0x400300AF	Endpoint 4 receive control register	0x00
R16_UEP5_T_LEN	0x400300B0	Endpoint 5 transmit length register	0x0000
R8_UEP5_TX_CTRL	0x400300B2	Endpoint 5 transmit control register	0x00
R8_UEP5_RX_CTRL	0x400300B3	Endpoint 5 receive control register	0x00
R16_UEP6_T_LEN	0x400300B4	Endpoint 6 transmit length register	0x0000
R8_UEP6_TX_CTRL	0x400300B6	Endpoint 6 transmit control register	0x00
R8_UEP6_RX_CTRL	0x400300B7	Endpoint 6 receive control register	0x00
R16_UEP7_T_LEN	0x400300B8	Endpoint 7 transmit length register	0x0000
R8_UEP7_TX_CTRL	0x400300BA	Endpoint 7 transmit control register	0x00
R8_UEP7_RX_CTRL	0x400300BB	Endpoint 7 receive control register	0x00
R16_UEP_T_ISO	0x400300BC	USBHS endpoint transmit synchronization mode enable register	0x0000
R16_UEP_R_ISO	0x400300BE	USBHS endpoint receive synchronization mode enable register	0x0000
R32_UEP1_RX_FIFO	0x400300C0	Receive FIFO address of USBHS endpoint 1	0x00000000
R32_UEP2_RX_FIFO	0x400300C4	Receive FIFO address of USBHS endpoint 2	0x00000000
R32_UEP3_RX_FIFO	0x400300C8	Receive FIFO address of USBHS endpoint 3	0x00000000
R32_UEP4_RX_FIFO	0x400300CC	Receive FIFO address of USBHS endpoint 4	0x00000000
R32_UEP5_RX_FIFO	0x400300D0	Receive FIFO address of USBHS endpoint 5	0x00000000
R32_UEP6_RX_FIFO	0x400300D4	Receive FIFO address of USBHS endpoint 6	0x00000000
R32_UEP7_RX_FIFO	0x400300D8	Receive FIFO address of USBHS endpoint 7	0x00000000
R32_UEP1_TX_FIFO	0x400300DC	Transmit FIFO address of USBHS endpoint 1	0x00000000
R32_UEP2_TX_FIFO	0x400300E0	Transmit FIFO address of USBHS endpoint 2	0x00000000
R32_UEP3_TX_FIFO	0x400300E4	Transmit FIFO address of USBHS endpoint 3	0x00000000
R32_UEP4_TX_FIFO	0x400300E8	Transmit FIFO address of USBHS endpoint 4	0x00000000
R32_UEP5_TX_FIFO	0x400300EC	Transmit FIFO address of USBHS endpoint 5	0x00000000
R32_UEP6_TX_FIFO	0x400300F0	Transmit FIFO address of USBHS endpoint 6	0x00000000

R32_UEP7_TX_FIFO	0x400300F4	Transmit FIFO address of USBHS endpoint 7	0x00000000
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25.2.1.1 USBHS Control Register (R8_USB_CTRL)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
7	RB_UD_LPM_EN	RW	LPM enable: 1: Enable; 0: Disable.	0
6	Reserved	RO	Reserved	0
5	RB_UD_DEV_EN	RW	USB device enable: 1: Enable; 0: Disable.	0
4	RB_UD_DMA_EN	RW	DMA transfer enable: 1: Enable; 0: Disable.	0
3	RB_UD_PHY_SUSPENDM	RW	USB PHY suspended: 1: Normal operation; 0: Suspended.	0
2	RB_UD_CLR_ALL	RW	Clear all interrupt flags: 1: Clear USB interrupt flag and FIFO; requires software reset; 0: Do not clear.	1
1	RB_UD_RST_SIE	RW	USB protocol processor reset: 1: Force reset the USB Protocol Processor (SIE), including endpoint-related registers, requiring software clearing; 0: Do not reset.	1
0	RB_UD_RST_LINK	RW	LINK layer reset: 1: USB Link layer reset; 0: No reset.	1

25.2.1.2 USBHS Mode Control Register (R8_USB_BASE_MODE)

Offset address: 0x01

Bit	Name	Access	Description	Reset value
[7:2]	Reserved	RO	Reserved	0x0
[1:0]	RB_UD_SPEED_TY PE[1:0]	RW	Desired speed mode for device: 00: Full-Speed device; 01: High-Speed device; 10: Low-Speed device; 11: Reserved.	0

25.2.1.3 USBHS Interrupt Enable Register (R8_USB_INT_EN)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
7	RB_UDIE_FIFO_OVER	RW	FIFO overflow interrupt. 1: Enable interrupt; 0: Disable interrupt.	0
6	RB_UDIE_LINK_RDY	RW	USB connection interrupt enabled. 1: Enable interrupt; 0: Disable interrupt.	0
5	RB_UDIE_SOF_ACT	RW	SOF packet interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
4	RB_UDIE_TRANSFER	RW	USB transfer complete interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
3	RB_UDIE_LPM_ACT	RW	LPM transfer end interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
2	RB_UDIE_BUS_SLEEP	RW	USB bus sleep interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
1	RB_UDIE_SUSPEND	RW	USB bus interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0
0	RB_UDIE_BUS_RST	RW	USB bus reset interrupt enable. 1: Enable interrupt; 0: Disable interrupt.	0

25.2.1.4 USBHS Device Address Register (R8_USB_DEV_AD)

Offset address: 0x03

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved	0
[6:0]	RB_UD_DEV_ADDR[6:0]	RW	USB device address.	0

25.2.1.5 USBHS Remote Wakeup Register (R8_USB_WAKE_CTRL)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[7:1]	Reserved	RO	Reserved	0
0	RB_UD_REMOTE_WAKEUP	RW1Z	Remote wake-up, hardware automatically reset.	0

25.2.1.6 USBHS Test Mode Register (R8_USB_TEST_MODE)

Offset address: 0x05

Bit	Name	Access	Description	Reset value
7	RB_UD_TEST_EN	RW	Test mode enable.	0
[6:4]	Reserved	RO	Reserved	0
3	RB_UD_TEST_SE0NAK	RW	Test mode, output SE0.	0
2	RB_UD_TEST_PKT	RW	Test mode, output one packet. The packet uses the data address and length from endpoint 4, with TOG set to DATA0.	0
1	RB_UD_TEST_K	RW	Test mode, output K.	0
0	RB_UD_TEST_J	RW	Test mode, output J.	0

25.2.1.7 USBHS Power Management Register (R16_USB_LPM_DATA)

Offset address: 0x06

Bit	Name	Access	Description	Reset value
15	RB_UD_LPM_BUSY	RW	Power management busy.	1
[14:11]	Reserved	RO	Reserved	0
[10:0]	RB_UD_LPM_DATA[10:0]	RO	Power management data.	0

25.2.1.8 USB Interrupt Flag Register (R8_USB_INT_FG)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
7	RB_UDIF_FIFO_OV	RW1Z	FIFO overflow interrupt flag bit; write 1 to clear. 1: FIFO overflow triggered; 0: No event.	0
6	RB_UDIF_LINK_RDY	RW1Z	USB connection interrupt flag bit; write 1 to clear. 1: USB connection event triggered; 0: No event.	0
5	RB_UDIF_RX_SOF	RW1Z	Receive SOF packet interrupt flag bit; write 1 to clear. 1: SOF packet event triggered; 0: No event.	0
4	RB_UDIF_RTX_ACT	RO	USB transfer complete interrupt flag: Receive cleared by RB_UEP_R_DONE; Transmit cleared by RB_UEP_T_DONE.	0
3	RB_UDIF_LPM_ACT	RW1Z	LPM transmission end interrupt flag, write 1 to clear. 1: LPM transmission end event triggered; 0: No event.	0
2	RB_UDIF_BUS_SLEEP	RW1Z	USB Bus Sleep Interrupt Flag When LPM is enabled, entering L1 (Sleep)	0

			state triggers this flag, writing 1 to clear. 1: USB bus sleep event triggered; 0: No event.	
1	RB_UDIF_SUSPEND	RW1Z	USB bus suspend interrupt flag, write 1 to clear. 1: USB suspend event triggered; 0: No event.	0
0	RB_UDIF_BUS_RST	RW1Z	USB bus reset interrupt flag, write 1 to clear. 1: USB bus reset event triggered; 0: No event.	0

25.2.1.9 USB Interrupt Status Register (R8_USB_INT_ST)

Offset address: 0x09

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	0
4	RB_UDIS_EP_DIR	RO	Endpoint data transmission direction: 1: Endpoint IN data; 0: Endpoint OUT/SETUP data.	0
3	Reserved	RO	Reserved	0
[2:0]	RB_UDIS_EP_ID_MASK[2:0]	RO	The endpoint number where data transmission occurs.	0

25.2.1.10 USB Miscellaneous Status Register (R8_USB_MIS_ST)

Offset address: 0x0A

Bit	Name	Access	Description	Reset value
7	RB_UDMS_HS_MOD	RO	Whether the host is at high speed.	0
[6:5]	Reserved	RO	Reserved	0
4	RB_UDMS_SUSP_REQ	RO	USB suspend request.	0
3	RB_UDMS_SIE_FREE	RO	USB idle state.	0
2	RB_UDMS_SLEEP	RO	USB sleep state.	0
1	RB_UDMS_SUSPEND	RO	USB suspend state. 1: The USB bus is in suspend state, with no USB activity for a period of time; 0: The USB bus is in non-suspend state.	0
0	RB_UDMS_READY	RO	USB connection status.	0

25.2.1.11 USB Frame Register (R16_USB_FRAME_NO)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[15:13]	RB_UD_MFRAME_NO[2:0]	RO	The received micro-frame number.	0
[12:11]	Reserved	RO	Reserved	0

[10:0]	RB_UD_FRAME_NO[10:0]	RO	The received frame number.	0
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25.2.1.12 USB Bus Status Register (R16_USB_BUS)

Offset address: 0x0E

Bit	Name	Access	Description	Reset value
[15:4]	Reserved	RO	Reserved	0
3	RB_USB_DM_ST	RO	UDM status.	0
2	RB_USB_DP_ST	RO	UDP status.	0
1	Reserved	RO	Reserved	0
0	RB_USB_WAKEUP	RO	USB wakeup (active high).	0

25.2.1.13 USBHS Endpoint Transmit Enable Register (R16_UEP_TX_EN)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[15:0]	RB_UEP_TX_EN[15:0]	RW	0~15 endpoint transmit enable register: 1: Enable; 0: Disable.	0

25.2.1.14 USBHS Endpoint Receive Enable Register (R16_UEP_RX_EN)

Offset address: 0x12

Bit	Name	Access	Description	Reset value
[15:0]	RB_UEP_RX_EN[15:0]	RW	0~15 endpoint receive enable register: 1: Enable; 0: Disable.	0

25.2.1.15 USBHS Endpoint Transmit Auto-flip Enable Register (R16_UEP_T_TOG_AUTO)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:1]	RB_UEP_T_TOG_AUTO[7:1]	RW	Auto flip enable for synchronous trigger bits on endpoints 1~7: 1: Automatically flips after successful data transmission; 0: Manually controls flipping.	0
0	Reserved	RO	Reserved	0

25.2.1.16 USBHS Endpoint Transmit Auto-flip Enable Register (R16_UEP_R_TOG_AUTO)

Offset address: 0x16

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:0]	RB_UEP_R_TOG_	RW	Auto flip enable for synchronous trigger	0

	AUTO[7:1]		bits on endpoints 1~7: 1: Automatically flips after successful data reception; 0: Manually controls flipping.	
0	Reserved	RO	Reserved	0

25.2.1.17 USBHS Endpoint Transmit Burst Register (R8_UEP_T_BURST)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[7:1]	RB_UEP_T_BURST_EN[6:0]	RW	1~7 endpoint burst transmit enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved	0

25.2.1.18 USBHS Endpoint Transmit Mode Register (R8_UEP_T_BURST_MODE)

Offset address: 0x19

Bit	Name	Access	Description	Reset value
[7:1]	RB_UEP_T_BURST_MODE[6:0]	RW	1~7 endpoint burst transmit mode: 1: Do not transmit zero-length data in burst mode; 0: Transmit zero-length data in burst mode.	0
0	Reserved	RO	Reserved	0

25.2.1.19 USBHS Endpoint Receive Burst Register (R8_UEP_R_BURST)

Offset address: 0x1A

Bit	Name	Access	Description	Reset value
[7:1]	RB_UEP_R_BURST_EN[6:0]	RW	1~7 endpoint burst receive enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved	0

25.2.1.20 USBHS Endpoint Reply Mode Register (R8_UEP_R_RES_MODE)

Offset address: 0x1B

Bit	Name	Access	Description	Reset value
[7:1]	RB_UEP_R_RES_MODE[6:0]	RW	1~7 endpoint receive return mode: 1: NYET; 0: ACK.	0
0	Reserved	RO	Reserved	0

25.2.1.21 USBHS Endpoint Alternate Register (R32_UEP_AF_MODE)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:1]	RB_UEP_T_AF[6:0]	RW	Endpoint multiplexing enable for 1–7: 1: Multiplex 9–15; 0: Multiplex 1–7.	0
0	Reserved	RO	Reserved	0

25.2.1.22 Start Address Register of Endpoint 0 Buffer (R32_UEP0_DMA)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:0]	UEP0_DMA[23:0]	RW	Start address register of endpoint 0 buffer.	X

25.2.1.23 Start Address Register of Endpoint n Receive Buffer (R32_UEPn_RX_DMA) (n=1~7)

Offset address: 0x24 + 4*(n-1)

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:0]	UEPn_RX_DMA[23:0]	RW	Start address register of endpoint n receive buffer. The address must be 4 bytes aligned.	X

25.2.1.24 Start Address Register of Endpoint n Transmit Buffer (R32_UEPn_TX_DMA) (n=1~7)

Offset address: 0x40 + 4*(n-1)

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:0]	UEPn_TX_DMA[23:0]	RW	The start address of endpoint n transmit buffer. The address must be 4-byte aligned.	X

25.2.1.25 Endpoint 0 Maximum Length Package Register (R32_UEPn_MAX_LEN)

Offset address: 0x5C

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
[6:0]	UEP0_MAX_LEN[6:0]	RW	The maximum offset length allowed for endpoint 0 DMA.	X

25.2.1.26 Endpoint n Maximum Length Package Register (R32_UEPn_MAX_LEN) (n=1~7)

Offset address: 0x60 + 4 (n-1)

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[6:0]	UEPn_MAX_LEN[10:0]	RW	The maximum offset length allowed for endpoint n DMA.	X

25.2.1.27 Endpoint 0 Receive Length Register (R16_UEP0_RX_LEN)

Offset address: 0x7C

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved	0
[6:0]	UEP0_RX_LEN[6:0]	RW	Endpoint 0 receive data length.	X

25.2.1.28 Endpoint n Single Receive Length Register (R16_UEPn_RX_LEN) (n=1~7)

Offset address: 0x80 + 4 (n-1)

Bit	Name	Access	Description	Reset value
[15:0]	UEPn_RX_LEN[15:0]	RW	Endpoint n single receive data length.	X

25.2.1.29 Endpoint n Total Receive Length Register (R16_UEPn_R_SIZE) (n=1~7)

Offset address: 0x82 + 4 (n-1)

Bit	Name	Access	Description	Reset value
[15:0]	UEPn_R_SIZE[15:0]	RW	Endpoint n total receive data length.	X

25.2.1.30 Endpoint 0 Transmit Length Register (R16_UEP0_T_LEN)

Offset address: 0x9C

Bit	Name	Access	Description	Reset value
[15:7]	Reserved	RO	Reserved	0
[6:0]	UEP0_T_LEN[6:0]	RW	Endpoint 0 transmit data length.	X

25.2.1.31 Endpoint 0 Transmit Control Register (R8_UEP0_TX_CTRL)

Offset address: 0x9E

Bit	Name	Access	Description	Reset value
7	RB_UEP_T_DONE	RW0	Endpoint 0 transmits the end flag, writes 0 to clear it.	0
6	RB_UEP_T_NAK_AC T	RW0	Endpoint 0 transmits NAK end flag, writes 0 to clear it.	0
[5:3]	Reserved	RO	Reserved	0
2	RB_UEP_T_TOG_MA SK	RW	Synchronous trigger bit expected to be sent by endpoint 0: 1: DATA1; 0: DATA0.	0
[1:0]	RB_UEP_T_RES_MA SK[1:0]	RW	Endpoint 0 response control for transmission: 00: Respond with NAK or busy; 01: Respond with STALL or error; 10: Respond with ACK; 11: Reserved.	0

25.2.1.32 Endpoint 0 Receive Control Register (R8_UEP0_RX_CTRL)

Offset address: 0x9F

Bit	Name	Access	Description	Reset value
7	RB_UEP_R_DONE	RW0	Endpoint 0 receives the end flag and writes 0 to clear it.	0
6	RB_UEP_R_NAK_ACT	RW0	Endpoint 0 receives the NAK end flag and writes 0 to clear it.	0
5	RB_UEP_R_NAK_TOG	RO	Endpoint 0 For the received return NAK, packet type: 1: DATA1; 0: DATA0.	0
4	RB_UEP_R_TOG_MAT CH	RO	Received synchronous trigger bit matching status with expected synchronous trigger bit: 1: Synchronized; 0: Not synchronized.	0
3	RB_UEP_R_SETUP_IS	RO	Whether endpoint 0 receives a SETUP transaction.	0
2	RB_UEP_R_TOG_MAS K	RW	Expected synchronous trigger bits for endpoint 0 reception: 1: DATA1; 0: DATA0.	0
[1:0]	RB_UEP_R_RES_MAS K[1:0]	RW	Endpoint 0 response control for received responses: 00: Respond with NAK or busy; 01: Respond with STALL or error; 10: Respond with ACK; 11: Reserved.	0

25.2.1.33 Endpoint n Transmit Length Register (R16_UEPn_T_LEN) (n=1~7)

Offset address: 0xA0 + 4*(n-1)

Bit	Name	Access	Description	Reset value
[15:0]	UEPn_T_LEN[15:0]	RW	The length of data sent by endpoint n.	0

25.2.1.34 Endpoint n Transmit Control Register (R8_UEPn_TX_CTRL) (n=1~7)

Offset address: 0xA2 + 4*(n-1)

Bit	Name	Access	Description	Reset value
7	RB_UEP_T_DONE	RW0	Endpoint n sends the end flag and writes 0 to clear it.	0
6	RB_UEP_T_NAK_AC T	RW0	Endpoint n sends the NAK end flag and writes 0 to clear it.	0
[5:4]	Reserved	RO	Reserved	0
[3:2]	RB_UEP_T_TOG_MA	RW	Expected synchronization trigger bit for	0

	SK[1:0]		transmission of endpoint n: 00: DATA0; 01: DATA1; 10: DATA2; 11: MDATA. <i>Note: This bit can be configured in manual mode and not in automatic mode.</i>	
[1:0]	RB_UEP_T_RES_MASK[1:0]	RW	Endpoint n response control for transmission: 00: Respond with NAK or busy; 01: Respond with STALL or error; 10: Respond with ACK; 11: Reserved.	0

25.2.1.35 Endpoint n Receive Control Register (R8_UEPn_RX_CTRL) (n=1~7)

Offset address: 0xA3 + 4*(n-1)

Bit	Name	Access	Description	Reset value
7	RB_UEP_R_DONE	RW0	Endpoint n receives the end flag and writes 0 to clear it.	0
6	RB_UEP_R_NAK_ACT	RW0	Endpoint n receives the NAK end flag and writes 0 to clear it.	0
5	RB_UEP_R_NAK_TOG	RO	Endpoint n for the received return NAK, packet type: 1: DATA1; 0: DATA0.	0
4	RB_UEP_R_TOG_MATCH	RO	The synchronization trigger bit received by endpoint n matches the expected synchronization trigger bit: 1: Synchronized; 0: Not synchronized.	0
[3:2]	RB_UEP_R_TOG_MASK[1:0]	RW	Receiving expected synchronization trigger bit of endpoint n: 00: DATA0; 01: DATA1; 10: DATA2; 11: MDATA. <i>Note: This bit can be configured in manual mode and not in automatic mode.</i>	0
[1:0]	RB_UEP_R_RES_MASK[1:0]	RW	Endpoint n controls responses received: 00: Responds with NAK or busy; 01: Responds with STALL or error; 10: Responds with ACK; 11: Reserved.	0

25.2.1.36 Endpoint Transmit Synchronization Mode Enable Register (R16_UEP_T_ISO)

Offset address: 0xBC

Bit	Name	Access	Description	Reset value
[15:9]	RB_UEP_T_FIFO_EN [6:0]	RO	FIFO mode enable of Endpoint TX	0
8	Reserved	RO	Reserved	0
[7:1]	RB_UEPn_T_ISO_EN[6:0]	RW	The upload endpoint (IN) synchronization mode is enabled.	0
0	Reserved	RO	Reserved	0

25.2.1.37 USBHS Endpoint Receive Synchronization Mode Enable Register (R16_UEP_R_ISO)

Offset address: 0xBE

Bit	Name	Access	Description	Reset value
[15:9]	RB_UEP_R_FIFO_EN [6:0]	RO	The FIFO mode of the RX of the endpoint is enabled.	0
8	Reserved	RO	Reserved	0
[7:1]	RB_UEPn_R_ISO_EN [6:0]	RW	Downstream endpoint (OUT) synchronization mode is enabled.	0
0	Reserved	RO	Reserved	0

25.2.1.38 Received FIFO Address of USBHS Endpoint n (R32_UEPn_RX_FIFO) (n=1~7)

Offset address: 0xC0 + 4*(n-1)

Bit	Name	Access	Description	Reset value
[31:16]	RB_UEP_RX_FIFO_E	RW	FIFO end address of endpoint received.	0
[15:0]	RB_UEP_RX_FIFO_S	RW	FIFO start address of endpoint received.	0

25.2.1.39 Transmitted FIFO Address of USBHS Endpoint n (R32_UEPn_TX_FIFO) (n=1~7)

Offset address: 0xDC + 4*(n-1)

Bit	Name	Access	Description	Reset value
[31:16]	RB_UEP_TX_FIFO_E	RW	FIFO end address of endpoint transmitted.	0
[15:0]	RB_UEP_TX_FIFO_S	RW	FIFO start address of endpoint transmitted.	0

25.2.2 Host Register Description

Table 25-2 Host-related registers

Name	Access address	Description	Reset value
R8_UH_CFG	0x40030100	USBHS host configuration register	0x07
R8_UH_INT_EN	0x40030102	USBHS host interrupt enable register	0x00
R8_UH_DEV_AD	0x40030103	USBHS host device address register	0x00
R32_UH_CONTROL	0x40030104	USBHS host control register	0x00XXXXXX
R8_UH_INT_FLAG	0x40030108	USBHS host interrupt flag register	0x00
R8_UH_INT_ST	0x40030109	USBHS host interrupt status register	0xFF

R8_UH_MIS_ST	0x4003010A	USBHS host miscellaneous status register	0xXX
R32_UH_LPM_DATA	0x4003010C	USBHS host power management data register	0x0000XXX
R32_UH_SPLIT_DATA	0x40030110	USBHS host SPLIT data register	0x000XXXXX
R32_UH_FRAME	0x40030114	USBHS host frame register	0x00000000
R32_UH_TX_LEN	0x40030118	USBHS host transmit length register	0x0000XXX
R32_UH_RX_LEN	0x4003011C	USBHS host receive length register	0x0000XXX
R32_UH_RX_MAX_LEN	0x40030120	USBHS host receive maximum length register	0x0000XXX
R32_UH_RX_DMA	0x40030124	DMA receive address register	0x000XXXXX
R32_UH_TX_DMA	0x40030128	DMA transmit address register	0x000XXXXX
R32_UH_PORT_CTRL	0x4003012C	USBHS host port control register	0x0000X000
R8_UH_PORT_CFG	0x40030130	USBHS host port configuration register	0x00
R8_UH_PORT_INT_EN	0x40030132	USBHS host port interrupt enable register	0x00
R8_UH_PORT_TEST_CT	0x40030133	USBHS host port test mode register	0x00
R16_UH_PORT_ST	0x40030134	USBHS host port status register	0x0010
R8_UH_PORT_CHG	0x40030136	USBHS host port status change register	0x00
R32_UH_BC_CTRL	0x4003013C	USBHS host BC charging control register	0x00000000

25.2.2.1 USBHS Host Control Register (R8_UH_CFG)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
7	RB_UH_LPM_EN	RW	LPM enable.	0
6	RB_UH_FORCE_FS	RW	Forced to use USB_FS.	0
5	RB_UH_SOF_EN	RW	Enable SOF package transmitted.	0
4	RB_UH_DMA_EN	RW	Enable DMA transmitted.	0
3	RB_UH_PHY_SUSPENDM	RW	PHY suspended.	0
2	RB_UH_CLR_ALL	RW	Clearing the interrupt flag and FIFO requires software clearing.	1
1	RB_UH_RST_SIE	RW	The USB protocol processor is reset, which requires software clearing.	1
0	RB_UH_RST_LINK	RW	The USB connection control module is reset.	1

25.2.2.2 USBHS Host Interrupt Enable Register (R8_UH_INT_EN)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
7	RB_UHIE_FIFO_OVER	RW	FIFO overflow interrupt enable	0
6	RB_UHIE_TX_HALT	RW	Transmit pause interrupt enable	0
5	RB_UHIE_SOF_ACT	RW	SOF packet transmission interrupt enable	0
4	RB_UHIE_TRANSFER	RW	USB end or transfer complete interrupt enable	0
3	RB_UHIE_RESUME_ACT	RW	Bus recovery interrupt enable	0

2	RB_UHIE_WKUP_ACT	RW	Wake-up interrupt enable	0
[1:0]	Reserved	RO	Reserved	0

25.2.2.3 USBHS Host Device Address Register (R8_UH_DEV_AD)

Offset address: 0x03

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved	0
[6:0]	RB_UH_DEV_ADDR[6:0]	RW	The address of the currently operating USB device.	0

25.2.2.4 USBHS Host Control Register (R32_UH_CONTROL)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
23	RB_UH_RX_NO_RES	RW	IN-DATA no response, used for synchronous transmission or high-speed SPLIT packets.	X
22	RB_UH_TX_NO_RES	RW	OUT/SETUP-DATA does not expect a response and is used for synchronous transmission or high-speed SPLIT packets.	X
21	RB_UH_RX_NO_DATA	RW	IN token packet followed by no expected data packet, used for high-speed SPLIT packets.	X
20	RB_UH_TX_NO_DATA	RW	No data packets follow the OUT/SETUP token packet, intended for high-speed SPLIT packets.	X
19	RB_UH_PRE_PID_EN	RW	Low-speed preamble packet PRE PID enable bit: When the port operates at full speed and requires transmission of low-speed packets (PRE), this bit must be enabled. 1: Enabled, used for communication with low-speed USB devices via an external full-speed HUB; 0: Disables low-speed preamble packets.	X
18	RB_UH_SPLIT_VALID	RW	Transmit SPLIT packet is valid.	X
17	RB_UH_LPM_VALID	RW	Transmit LPM packet is valid.	X
16	RB_UH_HOST_ACTION	RW	HOST performs transaction enabling, and this bit is cleared automatically after the transaction is completed.	0
[15:11]	Reserved	RO	Reserved	0
10	RB_UH_BUF_MODE	RW	Data cache control bit.	X

			1: Use R32_UH_RX_DMA for transmission and R32_UH_TX_DMA for reception; 0: Use R32_UH_TX_DMA for transmission and R32_UH_RX_DMA for reception.	
[9:8]	RB_UH_T_TOG_MASK [1:0]	RW	Transmit data PID: 00: PID_DATA0; 01: PID_DATA1; 10: PID_DATA2; 11: PID_MDAT0.	X
[7:4]	RB_UH_T_ENDP_MASK [3:0]	RW	The endpoint number of the transaction token packet sent by the host.	X
[3:0]	RB_UH_T_TOKEN_MASK [3:0]	RW	The PID of the transaction token packet sent by the host.	X

25.2.2.5 USBHS Host Interrupt Flag Register (R8_UH_INT_FLAG)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
7	RB_UHIF_FIFO_OVERFLOW	RW1Z	FIFO overflow interrupt flag.	0
6	RB_UHIF_TX_HALT	RW1Z	The transmission stop interrupt flag is used in the host mode for reading data overflow during downloading, or when the transmission is not finished at the time point of EOF2, the transmission is stopped, and the write 1 is cleared.	0
5	RB_UHIF_SOF_ACT	RW1Z	SOF packet transmission completion interrupt flag, write 1 to clear.	0
4	RB_UHIF_TRANSACTION_COMPLETE	RW1Z	USB transaction transmission complete interrupt flag, write 1 to clear.	0
3	RB_UHIF_RESUME_ACT_IF	RW1Z	Bus recovery interrupt flag, write 1 to clear.	0
2	RB_UHIF_WKUP_ACT	RW1Z	Wake-up interrupt flag, write 1 to clear.	0
[1:0]	Reserved	RO	Reserved	0

25.2.2.6 USBHS Host Interrupt Status Register (R8_UH_INT_ST)

Offset address: 0x09

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	X
4	RB_UHIS_PORT_RX_RESUME	RW	A bit of 1 indicates that the port has received a wake-up signal.	X

[3:0]	RB_UH_R_TOKEN_MASK [3:0]	RO	The received PID.	X
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25.2.2.7 USBHS Host Miscellaneous Status Register (R8_UH_MIS_ST)

Offset address: 0x0A

Bit	Name	Access	Description	Reset value
7	RB_UHMS_BUS_SE0	RO	SE0 on USB bus.	X
6	RB_UHMS_BUS_J	RO	J on USB bus.	X
[5:4]	RB_UHMS_LINESTATE[1:0]	RO	PHY Linestate signal.	X
3	RB_UHMS_USB_WAKEUP	RO	USB bus wake status: 1: Bus wake in progress; 0: No bus wake in progress.	X
2	RB_UHMS_SOF_ACT	RO	USB bus SOF packet transmission status bit: 1: SOF packet being transmitted; 0: Transmission complete or idle.	X
1	RB_UHMS_SOF_PRE	RO	The USB bus SOF packet status indicates: 1: An SOF packet is about to be sent; 0: No SOF packet is being sent.	X
0	RB_UHMS_SOF_FREE	RO	Port enable status: 1: Port enabled; 0: Port disabled.	X

25.2.2.8 USBHS Host Power Management Data Register (R32_UH_LPM_DATA)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_LPM_DATA[10:0]	RW	Link power management packet data content.	X

25.2.2.9 USBHS Host SPLIT Register (R32_UH_SPLIT_DATA)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
[18:0]	RB_UH_SPLIT_DATA[18:0]	RW	The data content of the SPLIT packet sent by the host endpoint, from high to low bits, is: ET, E, S, Port, SC, Hub Addr.	X

25.2.2.10 USBHS Host Frame Register (R32_UH_FRAME)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[31:26]	Reserved	RO	Reserved	0
25	RB_UH_SOF_CNT_CLR	RW	SOF count clear.	0
24	RB_UH_SOF_CNT_EN	RW	SOF count enable.	0
[23:19]	Reserved	RO	Reserved	0
[18:16]	RB_UH_MFRAME_NO[2:0]	RO	The micro-frame number of the upcoming SOF packet.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_FRAME_NO[10:0]	RW	The frame number of the upcoming SOF packet.	0

25.2.2.11 USBHS Host Transmit Length Register (R32_UH_TX_LEN)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_TX_LEN[10:0]	RW	Data transmission length.	X

25.2.2.12 USBHS Host Receive Length Register (R32_UH_RX_LEN)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_RX_LEN[10:0]	RW	Received data length.	X

25.2.2.13 USBHS Host Receive Maximum Length Register (R32_UH_RX_MAX_LEN)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:0]	RB_UH_RX_MAX_LEN[10:0]	RW	Maximum length of reception.	X

25.2.2.14 DMA Receive Address Register (R32_UH_RX_DMA)

Offset address: 0x24

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:0]	UH_RX_DMA[23:0]	RW	Receiving address, the lower two digits are invalid, and 4 bytes alignment is required.	X

25.2.2.15 DMA Transmit Address Register (R32_UH_TX_DMA)

Offset address: 0x28

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:0]	UH_TX_DMA[23:0]	RW	Transmitting address, the lower two digits are invalid, and 4 bytes alignment is required.	X

25.2.2.16 USBHS Host Port Control Register (R32_UH_PORT_CTRL)

Offset address: 0x2C

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	RB_UH_BUS_RST_LONG	RW	Bus reset time selection: 1: Extended reset; 0: Normal reset.	0
[15:12]	RB_UH_PORT_SLEEP_BESL[3:0]	RW	Wake-up time control: 0000: 125us; 1000: 3ms; 0001: 150us; 1001: 4ms; 0010: 200us; 1010: 5ms; 0011: 300us; 1011: 6ms; 0100: 400us; 1100: 7ms; 0101: 500us; 1101: 8ms; 0110: 1ms; 1110: 9ms; 0111: 2ms; 1111: 10ms.	X
[11:9]	Reserved	RO	Reserved	0
8	RB_UH_CLR_PORT_SLEEP	WO	PORT exit SLEEP status (LPM).	0
[7:6]	Reserved	RO	Reserved	0
5	RB_UH_CLR_PORT_CONNECT	WO	Put the port into the port status.	0
4	RB_UH_CLR_PORT_EN	WO	PORT exits the enabled state and enters the DISABLED status.	0
3	RB_UH_SET_PORT_SLEEP	WO	PORT enter SLEEP status (LPM).	0
2	RB_UH_CLR_PORT_SUSP	WO	PORT exits the suspended status.	0
1	RB_UH_SET_PORT_SUSP	WO	PORT enters the suspended status.	0
0	RB_UH_SET_PORT_RESET	WO	Port transmit reset.	0

25.2.2.17 USBHS Host Port Configuration Register (R8_UH_PORT_CFG)

Offset address: 0x30

Bit	Name	Access	Description	Reset value
7	RB_UH_PD_EN	RW	15K resistor pull-down is enabled in host	0

			mode.	
[6:1]	Reserved	RO	Reserved	0
0	RB_UH_HOST_EN	RW	USB port mode selection: 1: Port operates in host mode; 0: Port operates in device mode.	0

25.2.2.18 USBHS Host Port Interrupt Enable Register (R8_UH_PORT_INT_EN)

Offset address: 0x32

Bit	Name	Access	Description	Reset value
[7:6]	Reserved	RO	Reserved	0
5	RB_UHIE_PORT_SLP	RW	Port sleep state change interrupt enabled.	0
4	RB_UHIE_PORT_RESET	RW	Port reset state change interrupt enabled.	0
3	Reserved	RO	Reserved	0
2	RB_UHIE_PORT_SUSP	RW	Port suspended state change interrupt enabled.	0
1	RB_UHIE_PORT_EN	RW	Port enable state change interrupt enabled.	0
0	RB_UHIE_PORT_CONNECT	RW	Port connection state change interrupt enabled.	0

25.2.2.19 USBHS Host Port Test Mode Register (R8_UH_PORT_TEST_CT)

Offset address: 0x33

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	0
4	RB_UH_TEST_SE0_NAK	RW	Test TEST_SE0_NAK.	0
3	RB_UH_TEST_PACKET	RW	Test TEST_PKT.	0
2	RB_UH_TEST_FORCE_EN	RW	Test mode enabled.	0
1	RB_UH_TEST_K	RW	Test output K.	0
0	RB_UH_TEST_J	RW	Test output J.	0

25.2.2.20 USBHS Host Port Status Register (R16_UH_PORT_ST)

Offset address: 0x34

Bit	Name	Access	Description	Reset value
[15:12]	Reserved	RO	Reserved	0
11	RB_UHIS_PORT_TEST	RO	Whether the port is in test mode.	0
10	RB_UHIS_PORT_HS	RO	Whether the port connection speed is high.	0
9	RB_UHIS_PORT_LS	RO	Whether the port connection speed is low.	0
[8:6]	Reserved	RO	Reserved	0
5	RB_UHIS_PORT_SLP	RO	Port sleep.	0

4	RB_UHIS_PORT_RST	RO	Port reset status.	1
3	Reserved	RO	Reserved	0
2	RB_UHIS_PORT_SUSP	RO	Port suspended status.	0
1	RB_UHIS_PORT_EN	RO	Port enabled.	0
0	RB_UHIS_PORT_CONN CT	RO	Port connection status.	0

25.2.2.21 USBHS Host Port Status Change Register (R8_UH_PORT_CHG)

Offset address: 0x36

Bit	Name	Access	Description	Reset value
[7:6]	Reserved	RO	Reserved	0
5	RB_UHIF_PORT_SLP	RW1Z	Port sleep state changes, write 1 to clear.	0
4	RB_UHIF_PORT_RESET	RW1Z	Port reset state changes, write 1 to clear.	0
3	Reserved	RO	Reserved	0
2	RB_UHIF_PORT_SUSP	RW1Z	Port suspend state changes, write 1 to clear.	0
1	RB_UHIF_PORT_EN	RW1Z	Port enable state changes, write 1 to clear.	0
0	RB_UHIF_PORT_CONN NECT	RW1Z	Port connection status changes, write 1 to clear.	0

25.2.2.22 USBHS Host BC Charging Control Register (R32_UH_BC_CTRL)

Offset address: 0x3C

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	RB_UDM_VSRC_A CT	RO	In automatic mode, this bit indicates UDM output V_{BC_SRC} ; otherwise, it is controlled by RB_UDM_BC_CMPE. <i>Note: This bit is set to 1 when UDP is higher than V_{BC_SRC} but UDP is not high; otherwise, it is 0.</i>	0
9	RB_UDM_BC_VSR C	RW	UDM pin BC protocol source voltage enable: 1: UDM pin outputs BC protocol source voltage V_{BC_SRC} ; 0: Output disabled.	0
8	RB_UDP_BC_VSRC	RW	UDP Pin BC Protocol Source Voltage Enable: 1: Outputs BC protocol source voltage V_{BC_SRC} via UDP pin; 0: Disables output.	0
7	Reserved	RO	Reserved	0
6	RB_BC_AUTO_MO	RW	Auto mode enable:	0

	DE		1: Enabled; 0: Disabled.	
5	RB_UDM_BC_CMPE	RW	UDM pin BC protocol comparator enable: 1: Enable; 0: Disable.	0
4	RB_UDP_BC_CMPE	RW	UDP pin BC protocol comparator enable: 1: Enable; 0: Disable.	0
[3:2]	Reserved	RO	Reserved	0
1	RB_UDM_BC_CMPO	RO	UDM pin BC protocol comparator state: 1: UDM pin voltage is higher than the BC protocol reference value V_{BC_REF} ; 0: UDM pin voltage is lower than the BC protocol reference value V_{BC_REF} .	0
0	RB_UDP_BC_CMPO	RO	UDP pin BC protocol comparator state: 1: UDP pin voltage is higher than the BC protocol reference value V_{BC_REF} ; 0: UDP pin voltage is lower than the BC protocol reference value V_{BC_REF} .	0

Chapter 26 USB Full-speed Host/Device Controller (USBFS/OTG_FS)

26.1 USB Controller Introduction

USB 2.0 Full-Speed Host Controller and Device Controller (USBFS) conforms to the USB 2.0 Full-Speed standard. It provides 16 configurable USB device endpoints and a set of host endpoints. Supports control/bulk/synchronous/interrupt transfers, dual-buffer mechanism, USB bus suspend/resume operations, and offers standby/wake-up functionality.

OTG_FS is a dual-role USB controller supporting both host and device functionality, compliant with the On-The-Go Supplement to the USB 2.0 specification. Additionally, this controller can be configured as a host-only or device-only controller, compliant with the USB 2.0 Full-Speed specification.

The characteristics are as follows:

- Support the USB On-The-Go Supplement, Revision 1.3 specification, defined as an optional feature in the OTG_FS controller's physical layer
- Software-configurable as USB full-speed host, USB full-speed/low-speed device, or USB dual-role device
- Provide power-saving functionality
- Support control transfer, lot transfer, interrupt transmission and synchronous/real-time transmission.
- Provide bus reset, suspend, wake-up and recovery functions.
- Support packets up to 64 bytes, built-in FIFO, interrupt and DMA.

26.2 Register Description

The USB registers are divided into 3 parts, and some registers are multiplexed in the host and device modes.

- USB global registers
- USB device control registers
- USB host control registers

26.2.1 Global Register Description

Table 26-1 USBFS-related registers

Name	Access address	Description	Reset value
R8_USB_CTRL	0x40023400	USB control register	0x06
R8_USB_INT_EN	0x40023402	USB interrupt enable register	0x00
R8_USB_DEV_AD	0x40023403	USB device address register	0x00
R8_USB_MIS_ST	0x40023405	USB miscellaneous status register	0xXX
R8_USB_INT_FG	0x40023406	USB interrupt flag register	0x20
R8_USB_INT_ST	0x40023407	USB interrupt status register	0xXX
R16_USB_RX_LEN	0x40023408	USB receive length register	0x0XXX
R32_USB_OTG_CR	0x40023454	USB OTG control register	0x00000000
R32_USB_OTG_SR	0x40023458	USB OTG status register	0x0000000X

26.2.1.1 USB Control Register (R8_USB_CTRL)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
7	RB_UC_HOST_MODE	RW	USB operating mode selection: 1: Host mode; 0: Device mode.	0
6	RB_UC_LOW_SPEED	RW	USB bus signal transfer rate selection: 1: 1.5Mbps; 0: 12Mbps.	0
5	RB_UC_DEV_PU_EN	RW	The USB device enable and internal pull-up resistor control bits in USB Device mode, a value of 1 enables USB device transfer and enables the internal pull-up resistor.	
[5:4]	MASK_UC_SYS_CTRL	RW	See the table below to configure USB system.	0
3	RB_UC_INT_BUSY	RW	The USB transmission completion interrupt flag automatically pauses the enable bit before zero is cleared: 1: Automatically pause before the interrupt flag UIF_TRANSFER is cleared, answer busy NAK in device mode, and automatically pause subsequent transmission in host mode. 0: No pause.	0
2	RB_UC_RST_SIE	RW	USB protocol processor software reset control bit: 1: Forces reset of the USB Protocol Processor (SIE), requiring software reset; 0: No reset.	1
1	RB_UC_CLR_ALL	RW	USB FIFO and interrupt flags clear: 1: Forced clearing and clearing; 0: Not clear.	1
0	RB_UC_DMA_EN	RW	DMA and DMA interrupt control for USB: 1: Enable DMA function and DMA interrupt; 0: Disable DMA.	0

USB system control combination is composed of RB_UC_HOST_MODE and MASK_UC_SYS_CTRL:

Table 26-2 USB system control combination

RB_UC_HOST_MODE	MASK_UC_SYS_CTRL	USB system control description
0	00	Disable the USB device function, turn off the internal pull-up resistor.
0	01	Enable the USB device function, close the internal pull-up resistor, and need to add an external pull-up.
0	1x	Enable USB device function, enable internal 1.5K pull-up resistor. This pull-up resistor takes precedence over the pull-down resistor and can also

		be used in GPIO mode.
1	00	USB host mode, normal working state.
1	01	USB host mode, forcing DP/DM to output SE0 state.
1	10	USB host mode, forces DP/DM to output J state.
1	11	USB host mode, force DP/DM to output K state/wake up.

26.2.1.2 USB Interrupt Enable Register (R8_USB_INT_EN)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved	0
6	RB_UIE_DEV_NAK	RW	USB device mode, receive NAK interrupt: 1: Enable interrupt; 0: Disable interrupt.	0
5	RB_UID_1_WIRE	RW	USB 1-wire mode enable 1: Enable 0: Disable	0
4	RB_UIE_FIFO_OV	RW	FIFO overflow interrupt: 1: Enable interrupt; 0: Disable interrupt.	0
3	RB_UIE_HST_SOF	RW	USB host mode, SOF timing interrupt: 1: Enable interrupt; 0: Disable interrupt.	0
2	RB_UIE_SUSPEND	RW	USB bus suspend or wakeup event interrupt: 1: Enable interrupt; 0: Disable interrupt.	0
1	RB_UIE_TRANSFER	RW	USB transfer complete interrupt: 1: Enable interrupt; 0: Disable interrupt.	0
0	RB_UIE_DETECT	RW	USB host mode, USB device connect or disconnect event interrupt: 1: Enable interrupt; 0: Disable interrupt.	0
	RB_UIE_BUS_RST	RW	USB device mode, USB bus reset event interrupt: 1: Enable interrupt; 0: Disable interrupt.	0

26.2.1.3 USB Device Address Register (R8_USB_DEV_AD)

Offset address: 0x03

Bit	Name	Access	Description	Reset value
7	RB_UDA_GP_BIT	RW	USB general-purpose flag, user-defined.	0
[6:0]	MASK_USB_ADDR	RW	Host mode: current operated USB device address; Device Mode: USB own address.	0

26.2.1.4 USB Miscellaneous Status Register (R8_USB_MIS_ST)

Offset address: 0x05

Bit	Name	Access	Description	Reset value
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7	RB_UMS_SOF_PRES	RO	SOF packet presage status in USB host mode: 1: SOF packet is about to be transmitted, if there are other USB packets at this time, it will be automatically delayed; 0: No SOF packet is transmitted.	x
6	RB_UMS_SOF_ACT	RO	SOF packet transfer status in USB host mode: 1: SOF packet is being sent; 0: Send complete or idle.	x
5	RB_UMS_SIE_FREE	RO	USB protocol handler free: 1: The protocol device is free; 0: Busy, USB transfer in progress.	1
4	RB_UMS_R_FIFO_RDY	RO	USB receive FIFO data ready: 1: The receive FIFO is not empty; 0: The receive FIFO is empty.	0
3	RB_UMS_BUS_RST	RO	USB bus reset: 1: The current USB bus is in reset state; 0: The current USB bus is in a non-reset state.	x
2	RB_UMS_SUSPEND	RO	USB suspend: 1: The USB bus is in a suspended state, and there is no USB activity for a period of time; 0: The USB bus is in a non-suspend state.	0
1	RB_UMS_DM_LEVEL	RO	In USB host mode, level state of the DM pin when the device is just connected to the USB port is used to judge the speed: 1: High level/low-speed; 0: Low level/full-speed.	0
0	RB_UMS_DEV_ATTACH	RO	USB device attach status for the port in USB host mode: 1: The port has been connected to a USB device; 0: The port has no USB device connected.	0

26.2.1.5 USB Interrupt Flag Register (R8_USB_INT_FG)

Offset address: 0x06

Bit	Name	Access	Description	Reset value
7	RB_U_IS_NAK	RO	In USB device mode, NAK response status: 1: Respond to NAK during the current USB transfer process; 0: No NAK response.	0
6	RB_U_TOG_OK	RO	Current USB transfer DATA0/1 sync flag	0

			match status: 1: Synchronized; 0: Not synchronized.	
5	RB_U_SIE_FREE	RO	USB Protocol Handler free: 1: USB idle; 0: Busy, USB transfer in progress.	1
4	RB_UIF_FIFO_OV	RW1Z	SOF timer interrupt flag in USB host mode, write 1 to clear: 1: SOF packet transmission completion trigger; 0: No event.	0
3	RB_UIF_HST_SOF	RW1Z	USB bus suspend or wake-up event interrupt flag, write 1 to clear: 1: Triggered by USB suspend event or wake- up event; 0: No event.	0
2	RB_UIF_SUSPEND	RW1Z	USB bus suspend or wake-up event interrupt flag, write 1 to clear: 1: Triggered by USB suspend event or wake- up event; 0: No event.	0
1	RB_UIF_TRANSFER	RW1Z	USB transfer completion interrupt flag, write 1 to clear: 1: Trigger when a USB transfer is completed; 0: No event.	0
0	RB_UIF_DETECT	RW1Z	In USB host mode, USB device connect or disconnect event interrupt flag, write 1 to clear: 1: Detected USB device connection or disconnection trigger; 0: No event.	0
	RB_UIF_BUS_RST	RW1Z	In USB device mode, USB bus reset event interrupt flag bit, write 1 to clear: 1: USB bus reset event trigger; 0: No event.	0

26.2.1.6 USB Interrupt Status Register (R8_USB_INT_ST)

Offset address: 0x07

Bit	Name	Access	Description	Reset value
7	RB_UIS_IS_NAK	RO	In USB device mode, the NAK response status bit is identical to RB_U_IS_NAK: 1: A NAK response was issued during the current USB transfer; 0: No NAK response.	0
6	RB_UIS_TOG_OK	RO	The current USB transport DATA0/1	0

			synchronization flag matches the status bit, which is the same as RB_U_TOG_OK: 1: Synchronous; 0: Out of sync.	
[5:4]	MASK_UIS_TOKEN	RO	Token PID identity of the current USB transport transaction in device mode.	X
[3:0]	MASK_UIS_ENDP	RO	The endpoint number of the current USB transport transaction in device mode.	X
	MASK_UIS_H_RES	RO	In host mode, the reply PID identification of the current USB transport transaction, 0000 indicates that the device does not answer or times out; other values indicate the reply PID.	X

MASK_UIS_TOKEN is used to identify the token PID of the current USB transfer transaction in the USB device mode: 00 means OUT packet; 01 reserved; 10 means IN packet; 11 means SETUP packet.

MASK_UIS_H_RES is only valid in host mode. In the host mode, if the host sends the OUT/SETUP token packet, the PID is the handshake packet ACK/NAK/STALL, or the device has no response/timeout. If the host sends an IN token packet, the PID is the PID of the data packet (DATA0/DATA1) or the PID of the handshake packet.

26.2.1.7 USB Receive Length Register (R16_USB_RX_LEN)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0
[9:0]	R16_USB_RX_LEN	RO	The current number of data bytes received by the USB endpoint	x

26.2.1.8 USB OTG Control Register (R32_USB_OTG_CR)

Offset address: 0x54

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	RB_CR_SESS_VTH	RW	OTG session valid threshold voltage settings: 1: SESS_VLD level is 1.4V; 0: SESS_VLD level is 0.8V.	0
4	RB_CR_VBUS_VTH	RW	OTG VBUS threshold voltage settings: 1: VBUS_VLD level is 4.4V; 0: VBUS_VLD level is 4.8V.	0
3	RB_CR_OTG_EN	RW	OTG function enable: 1: Enable; 0: Disable.	0
2	RB_CR_IDPU	RW	USB_OTG_ID pin pull-up enable: 1: Enable; 0: Disable.	0

1	RB_CR_CHARGE_VBUS	RW	OTG VBUS charging enable: 1: Enable; 0: Disable.	0
0	RB_CR_DISCHAR_VBUS	RW	OTG VBUS discharging enable: 1: Enable; 0: Disable.	0

26.2.1.9 USB OTG Status Register (R32_USB_OTG_SR)

Offset address: 0x58

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	RB_SR_ID_DIG	RO	OTG ID flag: 1: B device; 0: A device.	0
2	RB_SR_SESS_END	RO	OTG session end status indicator: 1: Valid; 0: Invalid.	0
1	RB_SR_SESS_VLD	RO	OTG session valid flag: 1: Valid, session active level exceeds threshold voltage; 0: Invalid, session active level falls below threshold voltage.	0
0	RB_SR_VBUS_VLD	RO	OTG VBUS input level: 1: VBUS voltage is greater than the threshold voltage; 0: VBUS voltage is less than the threshold voltage.	X

26.2.2 Device Register Description

In USB device mode, the USB OTG module provides eight sets of bidirectional endpoint configuration registers numbered 0 through 7. These can map configurations for endpoints numbered 8 through 15. Except for endpoint 3, all endpoints support a maximum packet length of 64 bytes. Endpoint 3 supports a maximum packet length of 1023 bytes (synchronous transfers).

- Endpoint 0 is the default endpoint that supports controlling transmission, transmitting and receiving sharing a 64-byte data buffer.
- Endpoints 1-15 support configurable independent 64-byte transmit and receive buffers or dual 64-byte data buffers, enabling lot transmission, interrupt transmission, and real-time/synchronous transmission. Each endpoint pair possesses a control register R8_UEPn_CTRL and a transmission length register R16_UEPn_T_LEN, which are used to configure the endpoint's synchronous trigger bit, response to OUT and IN transactions, and the length of transmitted data.

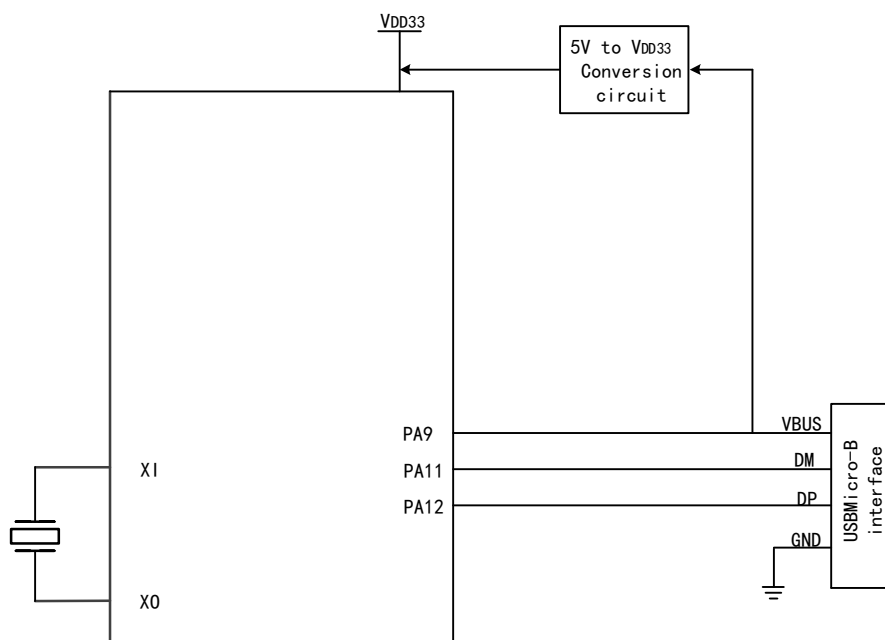
The host and device roles of the USB OTG module are determined by the state of the OTG_FS_ID pin. When the OTG_FS_ID pin is left floating, its internal pull-up resistor sets the RB_SR_ID_DIG bit to 1 in the USB OTG status register R32_USB_OTG_SR. At this point, the controller should initialize as a B device. When the OTG_FS_ID

pin is grounded, the RB_SR_ID_DIG bit in the USB OTG status register R32_USB_OTG_SR is set to 0. The controller should initialize as a B device.

Device B ensures the VBUS level remains below $V_{\text{SESS_VLD.min}}$ during session initiation based on the RB_SR_SESS_VLD bit in the OTG status register R32_USB_OTG_SR. If this bit is 0, a new session may commence. If this bit is set to 1, the device discharges the VBUS line by setting the RB_CR_DISCHAR_VBUS bit in the OTG control register R32_USB_OTG_CR to 1, thereby reducing VBUS below the session threshold voltage.

As a class B device, if it needs to get power from VBUS, it needs a conversion circuit externally, as shown in Figure 26-1.

Figure 26-1 Connection Diagram of OTG Class B Equipment



The USB bus pull-up resistor necessary for the USB device can be set by the software at any time. When the RB_UC_DEV_PU_EN in the USB control register R8_USB_CTRL is set to 1, the controller connects the pull-up resistor for the DP/DM pin of the USB bus internally according to the speed setting of the RB_UC_SPEED_TYPE, and enables the USB device function.

When a USB bus reset, a USB bus hang or wake-up event is detected, or when the USB successfully processes data transmission or data reception, the USB protocol processor will set the corresponding interrupt flag, and if the interrupt is enabled, a corresponding interrupt request will be generated. The application program can query directly or query and analyze the interrupt flag register R8_USB_INT_FG in the USB interrupt service program, and deal with it according to RB_UIF_BUS_RST and RB_UIF_SUSPEND. And, if the RB_UIF_TRANSFER is valid, then you need to continue to analyze the USB interrupt status register R8_USB_INT_ST and process it accordingly according to the current endpoint number MASK_UIS_ENDP and the current transaction token PID identification MASK_UIS_TOKEN. If the synchronization trigger bit RB_UEP_R_TOG of the OUT transaction of each endpoint is set in advance, then whether the synchronization trigger bit of the currently received data packet matches the synchronization trigger bit of the endpoint can be determined by RB_U_TOG_OK or RB_UIS_TOG_OK. If the data is synchronized, the data is valid; if the data is not synchronized, the data should be discarded. After each interrupt of USB transmission or reception, the synchronization trigger bit of the corresponding endpoint should be modified correctly to detect whether the data packet sent or received next time is synchronized; in addition, setting

RB_UEP_T_AUTO_TOG or RB_UEP_R_AUTO_TOG can automatically modify the corresponding synchronization trigger bit (flip or self-subtraction) after successful transmission or reception.

The data to be sent by each endpoint is in its own buffer, and the length of the data to be sent is independently set in the R8_UEPn_T_LEN; the data received by each endpoint is in the respective buffer, but the length of the received data is in the USB receiving length register R8_USB_RX_LEN, which can be distinguished according to the current endpoint number when the USB receives the interrupt.

Table 26-3 Device-related registers

Name	Access address	Description	Reset value
R8_UDEV_CTRL	0x40023401	USB device physical port control register	0xX0
R8_UEP4_1_MOD	0x4002340C	Endpoint 1(9)/4(8/12) mode control register	0x00
R8_UEP2_3_MOD	0x4002340D	Endpoint 2(10)/3(11) mode control register	0x00
R8_UEP5_6_MOD	0x4002340E	Endpoint 5(13)/6(14) mode control register	0x00
R8_UEP7_MOD	0x4002340F	Endpoint 7(15) mode control register	0x00
R32_UEP0_DMA	0x40023410	Endpoint 0 buffer start address	0xFFFFFFFF
R32_UEP1_DMA	0x40023414	Endpoint 1(9) buffer start address	0xFFFFFFFF
R32_UEP2_DMA	0x40023418	Endpoint 2(10) buffer start address	0xFFFFFFFF
R32_UEP3_DMA	0x4002341C	Endpoint 3(11) buffer start address	0xFFFFFFFF
R32_UEP4_DMA	0x40023420	Endpoint 4(8/12) buffer start address	0xFFFFFFFF
R32_UEP5_DMA	0x40023424	Endpoint 5(13) buffer start address	0xFFFFFFFF
R32_UEP6_DMA	0x40023428	Endpoint 6(14) buffer start address	0xFFFFFFFF
R32_UEP7_DMA	0x4002342C	Endpoint 7(15) buffer start address	0xFFFFFFFF
R32_USB_EP0_CTRL	0x40023430	Endpoint 0 transmit length and control register	0xFF
R16_UEP0_T_LEN	0x40023432	Endpoint 0 transmit length register	0x00
R16_UEP0_CTRL	0x40023433	Endpoint 0 control register	0x00
R32_USB_EP1_CTRL	0x40023434	Endpoint 1(9) transmit length and control register	0xFF
R16_UEP1_T_LEN	0x40023436	Endpoint 1(9) transmit length register	0x00
R16_UEP1_CTRL	0x40023437	Endpoint 1(9) control register	0x00
R32_USB_EP2_CTRL	0x40023438	Endpoint 2(10) transmit length and control register	0xFF
R16_UEP2_T_LEN	0x4002343A	Endpoint 2(10) transmit length register	0x00
R16_UEP2_CTRL	0x4002343B	Endpoint 2(10) control register	0x00
R32_USB_EP3_CTRL	0x4002343C	Endpoint 3(11) transmit length and control register	0xFF
R16_UEP3_T_LEN	0x4002343E	Endpoint 3(11) transmit length register	0x00
R16_UEP3_CTRL	0x4002343F	Endpoint 3(11) control register	0x00
R32_USB_EP4_CTRL	0x40023440	Endpoint 4(8/12) transmit length and control register	0xFF
R16_UEP4_T_LEN	0x40023442	Endpoint 4(8/12) transmit length register	0x00
R16_UEP4_CTRL	0x40023443	Endpoint 4(8/12) control register	0x00
R32_USB_EP5_CTRL	0x40023444	Endpoint 5(13) transmit length and control	0xFF

		register	
R16_UEP5_T_LEN	0x40023446	Endpoint 5(13) transmit length register	0x00
R16_UEP5_CTRL	0x40023447	Endpoint 5(13) control register	0x00
R32_USB_EP6_CTRL	0x40023448	Endpoint 6(14) transmit length and control register	0xXX
R16_UEP6_T_LEN	0x4002344A	Endpoint 6(14) transmit length register	0x00
R16_UEP6_CTRL	0x4002344B	Endpoint 6(14) control register	0x00
R32_USB_EP7_CTRL	0x4002344C	Endpoint 7(15) transmit length and control register	0xXX
R16_UEP7_T_LEN	0x4002344E	Endpoint 7(15) transmit length register	0x00
R16_UEP7_CTRL	0x4002344F	Endpoint 7(15) control register	0x00

26.2.2.1 USB Device Physical Port Control Register (R8_UDEV_CTRL)

Offset address: 0x01

Bit	Name	Access	Description	Reset value
7	RB_UD_PD_DIS	RW	USB device port UDP/UDM pin internal pull-down resistance control bit: 1: Disable internal drop-down. 0: Enable internal pull-down Can be used in GPIO mode to provide pull-down resistors.	1
6	Reserved	RO	Reserved	0
5	RB_UD_DP_PIN	RO	Current UDP pin status: 1: High level. 0: Low level.	X
4	RB_UD_DM_PIN	RO	Current UDM pin status: 1: High level. 0: Low level.	X
3	Reserved	RO	Reserved	0
2	RB_UD_LOW_SPEED	RW	USB device physical port low-speed mode enable bit: 1: Select 1.5Mbps low-speed mode. 0: Select 12Mbps full-speed mode.	0
1	RB_UD_GP_BIT	RW	USB device mode common flag bit, user-defined.	0
0	RB_UD_PORT_EN	RW	USB device physical port enable bit: 1: Enable physical port. 0: Disable physical port.	0

26.2.2.2 Endpoint 1(9)/4(8/12) Mode Control Register (R8_UEP4_1_MOD)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
7	RB_UEP1_RX_EN	RW	1: Enable endpoint 1(9) to receive (OUT).	0

			0: Disable endpoint 1(9) to receive.	
6	RB_UEP1_TX_EN	RW	1: Enable endpoint 1(9) to transmit (IN). 0: Disable endpoint 1(9) to transmit.	0
5	Reserved	RO	Reserved	0
4	RB_UEP1_BUF_MOD	RW	Endpoint 1(9) data buffer mode control bit.	0
3	RB_UEP4_RX_EN	RW	1: Enable endpoint 4(8/12) to receive (OUT). 0: Disable endpoint 4(8/12) to receive.	0
2	RB_UEP4_TX_EN	RW	1: Enable endpoint 4(8/12) to transmit (IN). 0: Disable endpoint 4(8/12) to transmit.	0
1	Reserved	RO	Reserved	0
0	RB_UEP4_BUF_MOD	RW	Endpoint 4(8/12) data buffer mode control bit.	0

Note: Endpoint 1 configures option mapping Endpoint 9, and Endpoint 4 configures option mapping Endpoints 8 and 12.

26.2.2.3 Endpoint 2(10)/3(11) Mode Control Register (R8_UEP2_3_MOD)

Offset address: 0x0D

Bit	Name	Access	Description	Reset value
7	RB_UEP3_RX_EN	RW	1: Enable endpoint 3(11) to receive (OUT). 0: Disable endpoint 3(11) to receive.	0
6	RB_UEP3_TX_EN	RW	1: Enable endpoint 3(11) to transmit (IN). 0: Disable endpoint 3(11) to transmit.	0
5	Reserved	RO	Reserved	0
4	RB_UEP3_BUF_MOD	RW	Endpoint 3(11) data buffer mode control bit.	0
3	RB_UEP2_RX_EN	RW	1: Enable endpoint 2(10) to receive (OUT). 0: Disable endpoint 2(10) to receive.	0
2	RB_UEP2_TX_EN	RW	1: Enable endpoint 2(10) to transmit (IN). 0: Disable endpoint 2(10) to transmit.	0
1	Reserved	RO	Reserved	0
0	RB_UEP2_BUF_MOD	RW	Endpoint 2(10) data buffer mode control bit.	0

Note: Endpoint 2 configures option mapping Endpoint 10, and Endpoint 3 configures option mapping Endpoint 11.

26.2.2.4 Endpoint 5(13)/6(14) Mode Control Register (R8_UEP5_6_MOD)

Offset address: 0x0E

Bit	Name	Access	Description	Reset value
7	RB_UEP6_RX_EN	RW	1: Enable endpoint 6(14) to receive (OUT). 0: Disable endpoint 6(14) to receive.	0
6	RB_UEP6_TX_EN	RW	1: Enable endpoint 6(14) to transmit (IN). 0: Disable endpoint 6(14) to transmit.	0
5	Reserved	RO	Reserved	0
4	RB_UEP6_BUF_MOD	RW	Endpoint 6(14) data buffer mode control bit.	0
3	RB_UEP5_RX_EN	RW	1: Enable endpoint 5(13) to receive (OUT). 0: Disable endpoint 5(13) to receive.	0
2	RB_UEP5_TX_EN	RW	1: Enable endpoint 5(13) to transmit (IN).	0

			0: Disable endpoint 5(13) to transmit.	
1	Reserved	RO	Reserved	0
0	RB_UEP5_BUF_MOD	RW	Endpoint 5(13) data buffer mode control bit.	0

Note: Endpoint 5 configures option mapping Endpoint 13, and Endpoint 6 configures option mapping Endpoint 14.

20.2.2.5 Endpoint 7 Mode Control Register (R8_UEP7_MOD)

Offset address: 0x0F

Bit	Name	Access	Description	Reset value
[7:4]	Reserved	RO	Reserved	0
3	RB_UEP7_RX_EN	RW	1: Enable endpoint 7(15) to receive (OUT). 0: Disable endpoint 7(15) to receive.	0
2	RB_UEP7_TX_EN	RW	1: Enable endpoint 7(15) to transmit (IN). 0: Disable endpoint 7(15) to transmit.	0
1	Reserved	RO	Reserved	0
0	RB_UEP7_BUF_MOD	RW	Endpoint 7(15) data buffer mode control bit.	0

Note: Endpoint 7 configures option mapping Endpoint 15.

The data buffer modes of USB endpoints 1-15 are configured by the combination of RB_UEPn_RX_EN and RB_UEPn_TX_EN and RB_UEPn_BUF_MOD, respectively, with specific reference to Table 26-4. Among them, in the double-64-byte buffer mode, the USB data transfer will select the first 64-byte buffer according to RB_UEP*_TOG=0. In the dual 64-byte buffer mode, USB data transfer will select the back 64-byte buffer according to RB_UEP*_TOG=1, and setting RB_UEP_AUTO_TOG=1 can realize automatic switching. Note: During synchronous transmission, when (RB_UEPn_RX_EN, RB_UEPn_TX_EN, RB_UEPn_BUF_MOD) are only (1,0,0) or (0,1,0), endpoint 3 supports a maximum of 1023 bytes.

Table 26-4 Endpoint n buffer mode (n=1-7)

RB_UEPn_RX_EN	RB_UEPn_TX_EN	RB_UEPn_BUF_MOD	Description: Arranged from low to high with R16_UEPn_DMA as the starting address
0	0	X	The endpoint is disabled and the R16_UEPn_DMA buffer is not used.
1	0	0	Single 64-byte receive buffer (OUT).
1	0	1	Dual 64-byte receive buffer (OUT), selected by RB_UEP_R_TOG.
0	1	0	Single 64-byte transmit buffer (IN).
0	1	1	Dual 64-byte transmit buffer (IN), selected by RB_UEP_T_TOG.
1	1	0	Single 64-byte receive buffer (OUT), single 64-byte transmit buffer (IN).
1	1	1	Dual 64-byte receive buffer (OUT), selected via RB_UEP_R_TOG. Dual 64-byte transmit buffer (IN), selected via RB_UEP_T_TOG. All 256 bytes are arranged as follows: UEPn_DMA+0 address: endpoint receive address with RB_UEP_R_TOG=0; UEPn_DMA+64 address: endpoint receive address when RB_UEP_R_TOG=1; UEPn_DMA+128 address: endpoint transmit address with RB_UEP_T_TOG=0; UEPn_DMA+192 address: endpoint transmit address when RB_UEP_T_TOG=1.

Note: The configuration selection in Table 21-4 supports n=1-7, and the configuration of endpoints 8-15 maps the configuration of endpoints 1-7.

26.2.2.6 Endpoint n Buffer start address (R32_UEPn_DMA) (n=0-7)

Offset address: 0x10+(4*n)

Bit	Name	Access	Description	Reset value
[31:0]	R32_UEPn_DMA	RW	Start address of endpoint n's buffer. The address must be 4-byte aligned.	X

Note: 1. The length of the buffer for receiving data $> = \min$ (the maximum possible packet length + 2 bytes, 64 bytes).
 2. The endpoint DMA configuration supports endpoints 0-7 and can be mapped to configure endpoints 8-15.

26.2.2.7 Endpoint n Transmit Length Register (R8_UEPn_T_LEN) (n=0-2, 4-7)

Offset address: 0x30+(4*n)

Bit	Name	Access	Description	Reset value
[7:0]	R8_UEPn_T_LEN	RW	Set the number of data bytes n=0, 1, 2, 4, 5, 6, 7 that USB endpoint n is ready to transmit.	X

Note 1. Endpoint transmission length configuration supports endpoints 0-7, which can be mapped to configure transmission for endpoints 8-15.
 2. Host transmission supports a maximum of 1023 bytes (for synchronous endpoints).

26.2.2.8 Endpoint n Transmit Length Register (R16_UEPn_T_LEN) (n=3)

Offset address: 0x3C

Bit	Name	Access	Description	Reset value
[15:10]	Reserved	RO	Reserved	0
[9:0]	R16_UEP3_T_LEN	RW	Set the number of data bytes n=3 that USB endpoint n is ready to send.	X

Note 1. Endpoint transmission length configuration supports endpoints 0-7, which can be mapped to configure transmission for endpoints 8-15.
 2. Host transmission supports a maximum of 1023 bytes (for synchronous endpoints).

26.2.2.9 Endpoint n Control Register (R8_UEPn_TX_CTRL) (n=0-7)

Offset address: 0x32+(4*n)

Bit	Name	Access	Description	Reset value
[7:4]	Reserved	RO	Reserved	0
3	RB_UEP_T_AUTO_TOG	RW	Synchronous trigger bit auto flip enable control bit: 1: Automatically flip the corresponding synchronous trigger bit after the data is sent successfully. 0: Do not flip automatically, you can switch manually. <i>Note: this bit of endpoint 0 is reserved.</i>	0
2	RB_UEP_T_TOG	RW	Synchronous trigger bits prepared by the sender of USB endpoint n (processing IN transactions): 1: Transmit DATA1; 0: Transmit DATA0.	0

[1:0]	MASK_UEP_T_RES	RW	Response control of the sender of endpoint n to the IN transaction: 00: DATA0/DATA1 data is ready and expects ACK. 01: Reply DATA0/DATA1 and expect no response for real-time/synchronous transmission of non-endpoint 0. 10: Answer NAK or busy. 11: Answer STALL or error.	00b
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Note 1. Endpoint configuration supports endpoints 0-7, which can be mapped to configure transmission for endpoints 8-15.

26.2.2.10 Endpoint n Control Register (R8_UEPn_RX_CTRL) (n=0-7)

Offset address: 0x33+(4*n)

Bit	Name	Access	Description	Reset value
[7:4]	Reserved	RO	Reserved	0
3	RB_UEP_R_AUTO_TOG	RW	Synchronous trigger bit auto flip enable control bit: 1: Automatically flip the corresponding synchronous trigger bit after successful data reception. 0: Do not flip automatically, you can switch manually. <i>Note: this bit of endpoint 0 is reserved.</i>	0
2	MASK_UEP_R_TOG	RW	The desired synchronous trigger bit of the receiver of the USB endpoint n (processing OUT transactions): 1: Expected DATA1; 0: expected DATA0.	0
[1:0]	MASK_UEP_R_RES	RW	Response control of the receiver of endpoint n to the OUT transaction: 00: Reply ACK. 01: Timeout / no response for real-time / synchronous transmission of non-endpoint 0. 10: Answer NAK or busy. 11: Answer STALL or error.	00b

Note 1. Endpoint configuration supports endpoints 0-7, which can be mapped to configure transmission for endpoints 8-15.

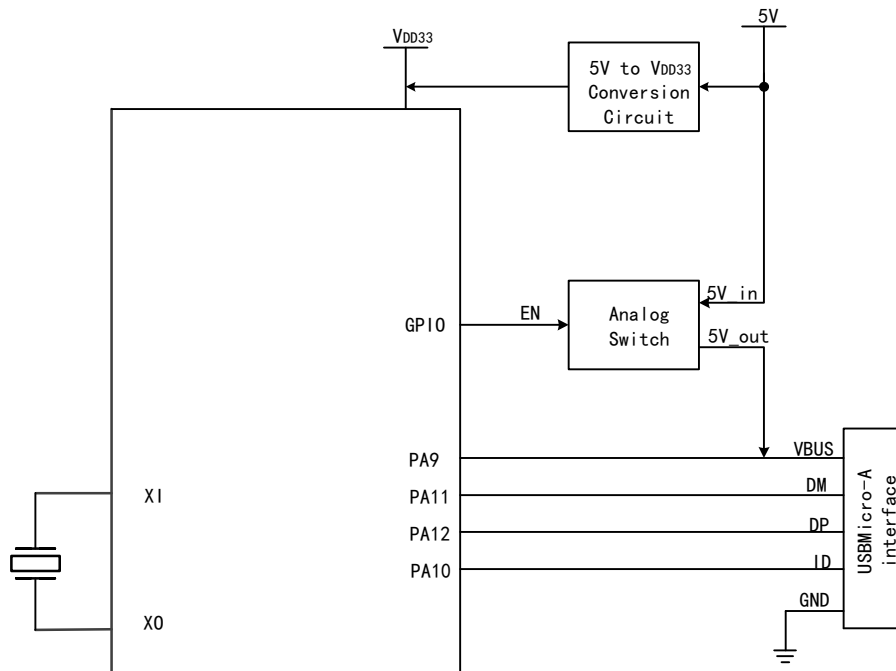
26.2.3 USB Host Register

In USB host mode, the chip provides a set of bi-directional host endpoints, including a sending endpoint OUT and a receiving endpoint IN. The maximum length of a packet is 1023 bytes (synchronous transmission), supporting control transmission, interrupt transmission, lot transmission and real-time/synchronous transmission.

In USB OTG host mode, if the controller cannot provide 5V power to VUBS, an external charge pump must be

used. If the application board can supply 5V power, an analog switch can be employed to control the enable and disable of VBUS, as shown in Figure 26-2.

Figure 26-2 OTG class A device connection



Every USB transaction initiated by the host endpoint automatically sets the RB_UIF_TRANSFER interrupt flag at the end of the processing. The application program can directly query or query and analyze the interrupt flag register R8_USB_INT_FG in the USB interrupt service program, and carry out corresponding processing according to each interrupt flag; moreover, if the RB_UIF_TRANSFER is valid, then it is necessary to continue to analyze the USB interrupt status register R8_USB_INT_ST and carry out corresponding processing according to the reply PID identification MASK_UIS_H_RES of the current USB transmission transaction.

If the synchronization trigger bit (RB_UH_R_TOG) of the IN transaction of the host receiving endpoint is set in advance, then whether the synchronization trigger bit of the currently received data packet matches the synchronization trigger bit of the host receiving endpoint can be determined by RB_U_TOG_OK or RB_UIS_TOG_OK. If the data is synchronized, the data is valid; if the data is not synchronized, the data should be discarded. After each interrupt of USB transmission or reception, the synchronization trigger bit of the corresponding host endpoint should be correctly modified to synchronize the next sent data packet and detect whether the next received data packet is synchronized; in addition, by setting RB_UH_T_AUTO_TOG and RB_UH_R_AUTO_TOG, the corresponding synchronization trigger bit can be flipped automatically after successful transmission or reception.

The USB host token setup register R8_UH_EP_PID is used to configure the endpoint number of the target device being operated on and the token PID packet identifier for the current USB transfer transaction. Data corresponding to the SETUP token and OUT token is provided by the host transmit endpoint. Data ready for transmission resides in the R32_UH_TX_DMA buffer, with its length set in R16_UH_TX_LEN. Data corresponding to the IN token is returned by the target device to the host's receiving endpoint. Received data is stored in the R32_UH_RX_DMA buffer, with the received data length stored in R16_USB_RX_LEN. The maximum packet length receivable by the host endpoint must be pre-written into the R16_UH_RX_MAX_LEN register.

Table 26-5 Host-related registers

Name	Access address	Description	Reset value
R8_UHOST_CTRL	0x40023401	USB host physical port control register	0xX0
R32_UH_EP_MOD	0x4002340D	USB host endpoint mode control register	0x00000000
R32_UH_RX_DMA	0x40023418	USB host receive buffer start address	X
R32_UH_TX_DMA	0x4002341C	USB host transmit buffer start address	X
R16_UH_SETUP	0x40023436	USB host auxiliary setting register	0x0000
R8_UH_EP_PID	0x40023438	USB host token setting register	0x00
R8_UH_RX_CTRL	0x4002343B	USB host receive endpoint control register	0x00
R16_UH_TX_LEN	0x4002343C	USB host transmit length register	X
R8_UH_TX_CTRL	0x4002343E	USB host transmit endpoint control register	0x00

26.2.3.1 USB Host Physical Port Control Register (R8_UHOST_CTRL)

Offset address: 0x01

Bit	Name	Access	Description	Reset value
7	RB_UH_PD_DIS	RW	USB host port UD+/UD- pin internal pull-down resistance control bit: 1: Disable internal pull-down. 0: Enable internal pull-down. Can be used in GPIO mode to provide pull-down resistors.	1
6	Reserved	RO	Reserved	0
5	RB_UH_DP_PIN	RO	Current UD+ pin status: 1: High; 0: Low.	x
4	RB_UH_DM_PIN	RO	Current UD- pin status: 1: High; 0: Low.	x
3	Reserved	RO	Reserved	0
2	RB_UH_LOW_SPEED	RW	USB host port low-speed mode enable bit: 1: Select 1.5Mbps low-speed mode. 0: Select 12Mbps full-speed mode.	0
1	RB_UH_BUS_RESET	RW	USB host mode bus reset control bit: 1: Force output USB bus reset. 0: End the output.	0
0	RB_UH_PORT_EN	RW	USB host port enable bit: 1: Enable host port. 0: Disable host port. This bit is automatically cleared 0 when the USB device is disconnected.	0

26.2.3.2 USB Host Endpoint Mode Control Register (R32_UH_EP_MOD)

Offset address: 0x0D

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved	0

6	RB_UH_EP_TX_EN	RW	Host send endpoint transmit (SETUP/OUT) enable bit: 1: Enable endpoint to transmit. 0: Enable endpoint to transmit.	0
5	Reserved	RO	Reserved	0
4	RB_UH_EP_TBUF_MOD	RW	Host transmitting endpoint transmits data buffer mode control bits.	0
3	RB_UH_EP_RX_EN	RW	Host receive endpoint receive (IN) enable bit: 1: Enable endpoint to receive. 0: Disable endpoint to receive.	0
[2:1]	Reserved	RO	Reserved	00b
0	RB_UH_EP_RBUF_MOD	RW	Host receiving endpoint receives the data buffer mode control bits.	0

The host transmit endpoint data buffer mode is controlled by the combination of RB_UH_EP_TX_EN and RB_UH_EP_TBUF_MOD, refer to the following table.

Table 26-6 Host transmit buffer mode

RB_UH_EP_TX_EN	RB_UH_EP_TBUF_MOD	Description: start address with R16_UH_TX_DMA
0	X	Endpoints are disabled and the R16_UH_TX_DMA buffer is not used.
1	0	Single 64-byte transmit buffer (SETUP/OUT).
1	1	Dual 64-byte transmit buffer, selected by RB_UH_T_TOG: The first 64-byte buffer is selected when RB_UH_T_TOG=0; Back 64-byte buffer selected when RB_UH_T_TOG=1.

The host receives endpoint data buffer mode by the combination of RB_UH_EP_RX_EN and RB_UH_EP_RBUF_MOD, refer to the following table.

Table 26-7 Host receive buffer mode

RB_UH_EP_RX_EN	RB_UH_EP_RBUF_MOD	Description: start address with R16_UH_TX_DMA
0	X	Endpoints are disabled and the R16_UH_RX_DMA buffer is not used.
1	0	Single 64-byte receive buffer (IN).
1	1	Dual 64-byte receive buffer, selected by RB_UH_R_TOG: The first 64-byte buffer is selected when RB_UH_R_TOG=0; Back 64-byte buffer selected when RB_UH_R_TOG=1.

26.2.3.3 USB Host Receive Buffer Start Address (R32_UH_RX_DMA)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	UH_RX_DMA	RW	Host endpoint data receive buffer starting address. The lower 20 bits are valid and the address must be 4 bytes aligned.	x

26.2.3.4 USB Host Transmit Buffer Start Address (R32_UH_TX_DMA)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:0]	UH_TX_DMA	RW	Host endpoint data transmit buffer starting address. The lower 20 bits are valid and the address must be 4 bytes aligned.	x

26.2.3.5 USB Host Auxiliary Setting Register (R16_UH_SETUP)

Offset address: 0x36

Bit	Name	Access	Description	Reset value
[15:11]	Reserved	RO	Reserved	0
10	RB_UH_PRE_PID_EN	RW	PREPID enable level of low-speed pilot package: 1: Enable, used to communicate with low-speed USB devices through external HUB. 0: Disable the low-speed pilot package.	0
[9:3]	Reserved	RO	Reserved	0
2	RB_UH_SOF_EN	RW	Automatically generate SOF packet enable bit: 1: The host automatically generates SOF packets. 0: Disable automatic SOF function.	0
[1:0]	Reserved	RO	Reserved	0

26.2.3.6 USB Host Token Setting Register (R8_UH_EP_PID)

Offset address: 0x38

Bit	Name	Access	Description	Reset value
[7:4]	MASK_UH_TOKEN	RW	Set the token PID package identity for this USB transport transaction.	0
[3:0]	MASK_UH_ENDP	RW	Set the endpoint number of the target device to be operated this time.	0

26.2.3.7 USB Host Receive Endpoint Control Register (R8_UH_RX_CTRL)

Offset address: 0x3B

Bit	Name	Access	Description	Reset value
[7:4]	Reserved	RO	Reserved	0
3	RB_UH_R_AUTO_TOG	RW	Synchronous trigger bit auto flip enable control bit: 1: Automatically flip the corresponding expected synchronous trigger bit (RB_UH_R_TOG) after successful data reception. 0: Do not flip automatically, you can switch manually (RB_UH_R_TOG).	0
2	RB_UH_R_TOG	RW	The synchronization trigger bits expected by the USB host receiver (processing IN transactions): 1: No response, used for real-time/synchronous transmission on non-zero endpoints; 0: ACK response.	0
1	Reserved	RO	Reserved	0
0	RB_UH_R_RES	RW	Host receiver response control bits for IN transactions: 1: No response for real-time / synchronous transmission of non-zero endpoints. 0: Reply ACK.	0

26.2.3.8 USB Host Transmit Length Register (R16_UH_TX_LEN)

Offset address: 0x3C

Bit	Name	Access	Description	Reset value
[15:0]	R8_UH_TX_LEN	RW	Sets the number of bytes of data that the USB host transmitting endpoint is ready to transmit.	x

26.2.3.9 USB Host Transmit Endpoint Control Register (R8_UH_TX_CTRL)

Offset address: 0x3E

Bit	Name	Access	Description	Reset value
[7:4]	Reserved	RO	Reserved	0
3	RB_UH_T_AUTO_TOG	RW	Synchronous trigger bit auto flip enable control bit: 1: Automatically flip the corresponding synchronous trigger bit (RB_UH_T_TOG) after the data is sent successfully. 0: Do not flip automatically, you can switch manually (RB_UH_T_TOG).	0
2	RB_UH_T_TOG	RW	Synchronous trigger bits prepared by the USB host transmitter (processing SETUP/OUT	0

			transactions): 1: Transit DATA1. 0: Transit DATA0.	
1	Reserved	RO	Reserved	0
0	RB_UH_T_RES	RW	The response control bits of the USB host transmitter to the SETUP/OUT transaction: 1: No response is expected for real-time/synchronous transmission of non-zero endpoints. 0: Expect to reply ACK.	0

Chapter 27 USB3.0 SuperSpeed Host/Device Controller (USBSS)

The module description in this chapter is only applicable to CH32H417 and CH32H416 microcontroller products.

The USB 3.0 SuperSpeed controller functions as both a host controller and a device controller. It incorporates a built-in SuperSpeed USB PHY physical layer transceiver, enabling USB 3.0 interface functionality and supporting 5Gbps USB SuperSpeed signals.

The USBSS module provides application code with a link-layer register access interface for managing device connection/disconnection, bus status, and power modes. It offers separate host (HOST) and device (DEVICE) function access interfaces to implement various data transfers and upper-layer protocols defined by the USB 3.0 specification.

USBSS module peripheral base address: 0x40034000

27.1 Main Features

- Support USB 3.0 protocol specifications and USB 3.2 Gen1
- Support USB Host functionality and USB Device functionality
- Power management modes support U1/U2/U3 low-power states
- Support driving USB 3.0 hubs
- Support control transfers, bulk transfers, interrupt transfers, real-time/synchronous transfers
- All non-zero endpoints support maximum 1024-byte packets with burst mode capability
- Support direct memory access (DMA) to endpoint buffer data
- Proprietary controller and transceiver with high-speed integrated design, tested at 450 Mbytes/s

27.2 Register Description

The USBSS registers are divided into 4 parts:

- USBSS global registers
- USBSS device control registers
- USBSS host control registers
- USBSS LINK control registers

27.2.1 Global Register Description

Table 27-1 USBSS-related registers

Name	Access address	Description	Reset value
R32_USBSS_CTRL	0x40034070	USBSS control register	0x00000006
R32_USBSS_STATUS	0x40034074	USBSS status register	0x00000000
R32_UH_TX_DMA R32_UEP0_TX_DMA	0x4003408C	Transmit buffer address register	0xFFFFFFFF
R32_UH_RX_DMA R32_UEP0_RX_DMA	0x40034090	Receive buffer address register	0xFFFFFFFF

27.2.1.1 USBSS Control Register (R32_USBSS_CTRL)

Offset address: 0x70

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
[30:24]	RB_DEV_ADDR	RW	Host mode: The address of the currently operated USB device; Device mode: The address of this USB device.	0
23	RB_UIE_FIFO_RXOV	RW	Receive FIFO overflow interrupt enable: 1: Enabled; 0: Disabled.	0
22	RB_UIE_FIFO_TXOV	RW	Transmit FIFO overflow interrupt enable: 1: Enabled; 0: Disabled.	0
21	Reserved	RO	Reserved	0
20	RB_UIE_ITP	RW	Transmit ITP complete interrupt enable: 1: Enabled; 0: Disabled.	0
19	RB_UIE_RX_PING	RW	Receive PING-TP complete interrupt enable: 1: Enabled; 0: Disabled.	0
18	RB_UDIE_STATUS	RW	Device mode: Enable interrupt upon completion of STATUS transaction: 1: Enabled; 0: Disabled.	0
18	RB_UHIE_NOTIF	RW	Host mode: Enable interrupt upon completion of DEV_NOTIF-TP transaction: 1: Enabled; 0: Disabled.	0
17	RB_UDIE_SETUP	RW	Device mode: Enable interrupt upon completion of SETUP transaction: 1: Enabled; 0: Disabled.	0
17	RB_UHIE_ERDY	RW	Host mode: Enable interrupt upon completion of SETUP transaction: 1: Enabled; 0: Disabled.	0
16	RB_UIE_TRANSFER	RW	USB transaction completion interrupt enable: 1: Enabled; 0: Disabled.	0
15	Reserved	RO	Reserved	0
14	RB_TX_ERDY_MODE	RW	Usage in device mode: 1: Prohibit ERDY-TP insertion between two DPH packets; 0: Allow ERDY-TP insertion between two	0

			DPH packets.	
[13:10]	Reserved	RO	Reserved	0
[9:8]	RB_REG_HP_PENDING[1:0]	RW	Packet pending control bit for sending TP/DP packets: In burst mode, the low bit indicates the PP bit status of the last packet, while the high bit indicates the sPP bit status of non-last packets. In non-burst mode, only the low bit is valid.	0
7	RB_HOST_MODE	RW	USB operating mode selection bit: 1: Host mode (HOST); 0: Device mode (DEVICE).	0
6	RB_ITP_EN	RW	In host mode, send ITP enable: 1: Enable; 0: Disable.	0
5	RB_SETUP_FLOW	RW	SETUP flow control 1: Upon receiving a SETUP transaction, enter flow control; device required. 0: Upon receiving a SETUP transaction, flow control is not required.	0
4	Reserved	RO	Reserved	0
3	RB_DMA_MODE	RW	1: During burst transmission DPH, do not prefetch the next packet into the FIFO; 0: During burst transmission DPH, prefetch the next packet into the FIFO. <i>Note: Enabling prefetching minimizes the interval between burst DPH packets, increasing transmission bandwidth. Prefetching is enabled by default.</i>	0
2	RB_FORCE_RST	RW	Protocol layer and FIFO module reset requires software clearing: 1: Reset; 0: Do not reset.	1
1	RB_USB_CLR_ALL	RW	Reset all software configuration registers, high-order bit is valid, requires software clearing; Clear all interrupt flags, clear device addresses, clear all endpoint configurations. <i>Note: In device mode, this bit should be set after receiving HOT_RESET/WARM_RESET, then cleared.</i>	1
0	RB_DMA_EN	RW	Enable DMA: 1: Enabled, the controller can access data in SRAM; 0: Disabled, the controller cannot access data in	0

			SRAM.	
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27.2.1.2 USBSS Status Register (R32_USBSS_STATUS)

Offset address: 0x74

Bit	Name	Access	Description	Reset value
[31:30]	RB_HRX_RES [1:0]	RW	Received the reply TP returned from the device. 00: ACK-TP; 01: RX_FAILED; 10: NRDY-TP; 11: STALL-TP. <i>Note: TX_FAILED/RX_FAILED indicates that when the host initiates a transaction, the HUB's downstream port is in low-power mode. The HUB will mark the packet as deferred and return this status to the host. The host of the asynchronous transaction should wait until it receives the device's ERDY-TP before resuming the transaction.</i>	0
[29:24]	Reserved	RO	Reserved	0
[23:22]	RB_HTX_RES [1:0]	RW	Received a TP sent from the host: 00: ACK-TP; 01: TX_FAILED; 10: NRDY-TP; 11: STALL-TP. <i>Note: TX_FAILED/RX_FAILED means that when the host initiates a transaction, the downlink port of the HUB is in a low power consumption state, and the HUB will change the label to deferred and return it to the host at the same time. The asynchronous transaction host should wait until it receives the ERDY-TP of the device to restart the transaction.</i>	0
21	Reserved	RO	Reserved	0
[20:16]	RB_HOST_ACK_N UMP	RW	This bit is invalid under synchronous transmission, and has the following definitions in control transmission, block transmission and interrupt transmission: After sending DPH+DPP(OUT/SETUP transaction), the NUMP field in the ACK-TP is received, which indicates the number of packets that the endpoint of the device can receive next. If NUMP=0, it indicates that the endpoint cannot continue to receive DPP.	0

			When the endpoint of the device can receive DPH+DPP, the device will send ERDY-TP packets to inform the host. But it does not affect the host sending data to other endpoints at this time. In particular, if the NUMP of the reply ACK-TP of the SETUP transaction is 0, indicating that the flow control state is entered, the host should pause the data phase or the state phase until receiving the ERDY-TP from the device. In addition, the host can send the SETUP transaction to release the flow control status. Of course, when in the flow control state, the host can also carry out the data phase and the state phase, without generating an error, and only returns NRDY-TP when the endpoint of the device is not ready.	
[15:13]	Reserved	RO	Reserved	0
12	RB_EP_DIR	RW	At present, the direction of the endpoint with the transmission completion interrupt flag. If there is an interrupt flag at the same time, then this bit is 1, that is, the transmission is given priority. (SETUP/STATUS independent interrupt) 1: Transmit (IN); 0: Receive (OUT).	0
11	Reserved	RO	Reserved	0
[10:8]	RB_EP_ID	RW	The endpoint number currently has a transmission completion interrupt flag. If multiple endpoints simultaneously have interrupt flags, the priority order for endpoint types is as follows (highest to lowest): synchronous transmission > control transmission > block transmission/interrupt transmission. Within the same endpoint type, lower endpoint numbers have higher priority.	0
7	RB_UIF_FIFO_RX OV	RW1Z	Receive the FIFO overflow interrupt flag; writing 1 to clear it: 1: Receiving FIFO overflow interrupt trigger; 0: No event.	0
6	RB_UIF_FIFO_TX OV	RW1Z	Transmit the FIFO overflow interrupt flag; writing 1 to clear it: 1: Transmitting FIFO overflow interrupt triggered;	0

			0: No event.	
5	Reserved	RO	Reserved	0
4	RB_UIF_ITP	RW1Z	Transmit the ITP completion interrupt flag; writing 1 to clear it: 1: Transmitting ITP completion interrupt triggered; 0: No event.	0
3	RB_UIF_RX_PING	RW1Z	Receive the PING-TP completion interrupt flag; writing 1 to clear it: 1: Receiving PING-TP completion interrupt triggered; 0: No event.	0
2	RB_UHIF_NOTIF	RW1Z	Host Mode: Receive the DEV_NOTIF-TP completion interrupt flag; writing 1 to clear it: 1: Receiving DEV_NOTIF-TP completion interrupt triggered; 0: No event.	0
2	RB_UDIF_STATUS	RW1Z	Device mode: Receive the STATUS completion interrupt flag; writing 1 to clear it: 1: Receiving STATUS completion interrupt triggered; 0: No event.	0
1	RB_UHIF_ERDY	RW1Z	Host mode: Receive the ERDY-TP completion interrupt flag; writing 1 to clear it: 1: Receiving ERDY-TP completion interrupt triggered; 0: No event.	0
1	RB_UDIF_SETUP	RW1Z	Device mode: Receive the SETUP transaction completion interrupt flag; writing 1 to clear it: 1: Receiving SETUP completion interrupt triggered; 0: No event.	0
0	RB_UIF_TRANSFE R	RW1Z	USB transaction completion interrupt flag; writing 1 to clear it: 1: USB transaction completion interrupt triggered; 0: No event.	0

27.2.1.3 Transmit Buffer Address Register (R32_UH_TX_DMA/U3EP0_TX_DMA)

Offset address: 0x8C

Bit	Name	Access	Description	Reset value
[31:0]	U3EP0_TX_DMA	RW	Host mode: Host transmits buffer start address.	X

			Device mode: Endpoint 0 transmits buffer start address.	
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27.2.1.4 Receive Buffer Address Register (R32_UH_RX_DMA/U3EP0_RX_DMA)

Offset address: 0x90

Bit	Name	Access	Description	Reset value
[31:0]	U3EP0_RX_DMA	RW	Host mode: Start address of the host receive buffer. Device mode: Start address of endpoint 0 receive buffer.	X

27.2.2 Device Register Description

Table 27-2 USBSS device registers

Name	Access address	Description	Reset value
R32_USBSS_ITP	0x40034078	Interval register in the ITP package of USB	0x00000000
R32_USBSS_ITP_ADJ	0x4003407C	Interval adaptive register in the ITP package of USB	0x00000000
R16_UEP_TX_EN	0x40034080	Endpoint transmit enable register	0x00000000
R16_UEP_RX_EN	0x40034082	Endpoint receive enable register	0x00000000
R32_UEP0_TX_CTRL	0x40034084	Endpoint 0 transmit control register	0x00000000
R32_UEP0_RX_CTRL	0x40034088	Endpoint 0 receive control register.	0x00000000

27.2.2.1 Interval Register in the ITP Package of USB (R32_USBSS_ITP)

Offset address: 0x78

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	REG_ITP_INTERVAL	RO	Receive Bus Interval Counter field in the ITP. <i>Note: Bus Interval Counter increments every 125 microseconds.</i>	0

27.2.2.2 Interval Adaptive register in the ITP package of USB (R32_USBSS_ITP_ADJ)

Offset address: 0x7C

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:8]	ITP_DELTA	RO	The upper 13 bits (Delta) of the ITS within an ITP received in device mode represent the time difference from the start of the current ITP packet to the boundary of the preceding bus interval. If the host transmits ITPs precisely every 125 microseconds, the host should set	0

			this bit to 0.	
7	ITP_DELAYED	RO	In device mode, the Delayed bit in the Link Control Word of the received ITP is set to 1 when the ITP is transmitted with delay.	0
[6:0]	ITP_ADJ_CR	RO	In device mode, the Bus Interval Adjustment Control field in the received ITP should be set to 0 in the ITP sent by the host after a power-on reset or device disconnection.	0

27.2.2.3 Endpoint Transmit Enable Register (R16_UEP_TX_EN)

Offset address: 0x80

Bit	Name	Access	Description	Reset value
[15:1]	RB_EP_TX_EN	RW	Endpoints 1–15 upload enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved	1

27.2.2.4 Endpoint Receive Enable Register (R16_UEP_RX_EN)

Offset address: 0x82

Bit	Name	Access	Description	Reset value
[15:1]	RB_EP_RX_EN	RW	Endpoints 1–15 download enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved	1

27.2.2.5 Endpoint 0 Transmit Control Register (R32_UEP0_TX_CTRL)

Offset address: 0x84

Bit	Name	Access	Description	Reset value
31	RB_UIF_EP0_TX_ACT	RW	Upload the transaction completion interrupt flag, the software writes 0 to clear it, and the hardware sets it to 1.	0
[20:26]	Reserved	RO	Reserved	0
25	RB_EP0_TX_FLOW	RO	Sign to complete NRDY-TP transmission (response).	0
24	RB_EP0_TX_PP	RO	PP bit in the received ACK-TP.	0
23	RB_EP0_TX_ERDY	RW	Write 1 to this bit to send ERDY. After ERDY is sent, this bit is automatically cleared by hardware.	0
[22:21]	RB_EP0_TX_RES[1:0]	RW	Response to ACK-TP (nump $\neq$ 0): 00: NRDY-TP; 01: DPH; 10: STALL-TP; 11: No response (invalid).	0

			Automatically clears after IN transaction completion.	
[20:16]	RB_EP0_TX_SEQ[4:0]	RW	The current serial number of the endpoint, which is automatically cleared after receiving the SETUP transaction.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	RB_EP0_TX_LEN[10:0]	RW	Endpoint send length register, with a maximum value of 512B.	x

27.2.2.6 Endpoint 0 Receive Control Register (R32_UEP0_RX_CTRL)

Offset address: 0x88

Bit	Name	Access	Description	Reset value
31	RB_UIF_EP0_RX_ACT	RW	Downloading transaction completion interrupt flag, which is cleared by software writing 0 and set by hardware.	0
[30:25]	Reserved	RO	Reserved	0
24	RB_EP0_RX_PP	RO	PP bit in the received DPH.	0
23	RB_EP0_RX_ERDY	RW	Write 1 to this bit to send ERDY. After ERDY is sent, this bit is automatically cleared by hardware.	0
[22:21]	RB_EP0_RX_RES[1:0]	RW	Response to DPH: 00: NRDY-TP; 01: ACK-TP; 10: STALL-TP; 11: No response (invalid). EPn_R_RES automatically clears after SETUP/OUT transaction completion; additionally, the SETUP packet directly responds with ACK-TP, which is unrelated to this.	0
[20:16]	RB_EP0_RX_SEQ[4:0]	RW	The sequence number that the endpoint expects to receive.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	RB_EP0_RX_LEN[10:0]	RO	Endpoint receive length register.	x

27.2.3 Device Register-Endpoints 1 ~ 7 Transmit-related Registers.

Table 27-3 Endpoint 1 transmit-related registers

Name	Access address	Description	Reset value
R8_UEP1_TX_CFG	0x400340C0	Endpoint 1 configuration register	0x86
R8_UEP1_TX_CR	0x400340C1	Endpoint 1 control register	0x10
R8_UEP1_TX_SEQ	0x400340C2	Endpoint 1 serial number register	0x00
R8_UEP1_TX_ST	0x400340C3	Endpoint 1 status register	0x00

R8_UEP1_TX_CHAIN_CR	0x400340C4	Endpoint 1 CHAIN control register	0x00
R8_UEP1_TX_CHAIN_ST	0x400340C5	Endpoint 1 CHAIN status register	0x00
R16_UEP1_TX_CHAIN_LEN	0x400340C6	Endpoint 1 CHAIN transmit last packet length	0x00
R8_UEP1_TX_CHAIN_EXP_NUMP	0x400340C8	Endpoint 1 expects to transmit NUMP number	0x00
R8_UEP1_TX_CHAIN_NUMP	0x400340C9	Endpoint 1 has transmitted NUMP number	0x00
R16_UEP1_TX_DMA_OFS	0x400340CA	DMA offset length of endpoint 1	0x00
R32_UEP1_TX_DMA	0x400340CC	DMA start address of endpoint 1	0x0400

Table 27-4 Endpoint 2 transmit-related registers

Name	Access address	Description	Reset value
R8_UEP2_TX_CFG	0x400340E0	Endpoint 2 configuration register	0x86
R8_UEP2_TX_CR	0x400340E1	Endpoint 2 control register	0x10
R8_UEP2_TX_SEQ	0x400340E2	Endpoint 2 serial number register	0x00
R8_UEP2_TX_ST	0x400340E3	Endpoint 2 status register	0x00
R8_UEP2_TX_CHAIN_CR	0x400340E4	Endpoint 2 CHAIN control register	0x00
R8_UEP2_TX_CHAIN_ST	0x400340E5	Endpoint 2 CHAIN status register	0x00
R16_UEP2_TX_CHAIN_LEN	0x400340E6	Endpoint 2 CHAIN transmit last packet length	0x00
R8_UEP2_TX_CHAIN_EXP_NUMP	0x400340E8	Endpoint 2 expects to send NUMP number	0x00
R8_UEP2_TX_CHAIN_NUMP	0x400340E9	Endpoint 2 has transmitted NUMP number	0x00
R16_UEP2_TX_DMA_OFS	0x400340EA	DMA offset length of endpoint 2	0x00
R32_UEP2_TX_DMA	0x400340EC	DMA start address of endpoint 2	0x0400

Table 27-5 Endpoint 3 transmit-related registers

Name	Access address	Description	Reset value
R8_UEP3_TX_CFG	0x40034100	Endpoint 3 configuration register	0x86
R8_UEP3_TX_CR	0x40034101	Endpoint 3 control register	0x10
R8_UEP3_TX_SEQ	0x40034102	Endpoint 3 serial number register	0x00
R8_UEP3_TX_ST	0x40034103	Endpoint 3 status register	0x00
R8_UEP3_TX_CHAIN_CR	0x40034104	Endpoint 3 CHAIN control register	0x00
R8_UEP3_TX_CHAIN_ST	0x40034105	Endpoint 3 CHAIN status register	0x00
R16_UEP3_TX_CHAIN_LEN	0x40034106	Endpoint 3 CHAIN transmit last packet length	0x00
R8_UEP3_TX_CHAIN_EXP_NUMP	0x40034108	Endpoint 3 expects to send NUMP number	0x00
R8_UEP3_TX_CHAIN_NUMP	0x40034109	Endpoint 3 has transmitted NUMP number	0x00

R16_UEP3_TX_DMA_OFS	0x4003410A	DMA offset length of endpoint 3	0x00
R32_UEP3_TX_DMA	0x4003410C	DMA start address of endpoint 3	0x0400

Table 27-6 Endpoint 4 transmit-related registers

Name	Access address	Description	Reset value
R8_UEP4_TX_CFG	0x40034120	Endpoint 4 configuration register	0x86
R8_UEP4_TX_CR	0x40034121	Endpoint 4 control register	0x10
R8_UEP4_TX_SEQ	0x40034122	Endpoint 4 serial number register	0x00
R8_UEP4_TX_ST	0x40034123	Endpoint 4 status register	0x00
R8_UEP4_TX_CHAIN_CR	0x40034124	Endpoint 4 CHAIN control register	0x00
R8_UEP4_TX_CHAIN_ST	0x40034125	Endpoint 4 CHAIN status register	0x00
R16_UEP4_TX_CHAIN_LEN	0x40034126	Endpoint 4 CHAIN transmit last packet length	0x00
R8_UEP4_TX_CHAIN_EXP_NUMP	0x40034128	Endpoint 4 expects to send NUMP number	0x00
R8_UEP4_TX_CHAIN_NUMP	0x40034129	Endpoint 4 has transmitted NUMP number	0x00
R16_UEP4_TX_DMA_OFS	0x4003412A	DMA offset length of endpoint 4	0x00
R32_UEP4_TX_DMA	0x4003412C	DMA start address of endpoint 4	0x0400

Table 27-7 Endpoint 5 transmit-related registers

Name	Access address	Description	Reset value
R8_UEP5_TX_CFG	0x40034140	Endpoint 5 configuration register	0x86
R8_UEP5_TX_CR	0x40034141	Endpoint 5 control register	0x10
R8_UEP5_TX_SEQ	0x40034142	Endpoint 5 serial number register	0x00
R8_UEP5_TX_ST	0x40034143	Endpoint 5 status register	0x00
R8_UEP5_TX_CHAIN_CR	0x40034144	Endpoint 5 CHAIN control register	0x00
R8_UEP5_TX_CHAIN_ST	0x40034145	Endpoint 5 CHAIN status register	0x00
R16_UEP5_TX_CHAIN_LEN	0x40034146	Endpoint 5 CHAIN transmit last packet length	0x00
R8_UEP5_TX_CHAIN_EXP_NUMP	0x40034148	Endpoint 5 expects to send NUMP number	0x00
R8_UEP5_TX_CHAIN_NUMP	0x40034149	Endpoint 5 has transmitted NUMP number	0x00
R16_UEP5_TX_DMA_OFS	0x4003414A	DMA offset length of endpoint 5	0x00
R32_UEP5_TX_DMA	0x4003414C	DMA start address of endpoint 5	0x0400

Table 27-8 Endpoint 6 transmit-related registers

Name	Access address	Description	Reset value
R8_UEP6_TX_CFG	0x40034160	Endpoint 6 configuration register	0x86
R8_UEP6_TX_CR	0x40034161	Endpoint 6 control register	0x10

R8_UEP6_TX_SEQ	0x40034162	Endpoint 6 serial number register	0x00
R8_UEP6_TX_ST	0x40034163	Endpoint 6 status register	0x00
R8_UEP6_TX_CHAIN_CR	0x40034164	Endpoint 6 CHAIN control register	0x00
R8_UEP6_TX_CHAIN_ST	0x40034165	Endpoint 6 CHAIN status register	0x00
R16_UEP6_TX_CHAIN_LEN	0x40034166	Endpoint 6 CHAIN transmit last packet length	0x00
R8_UEP6_TX_CHAIN_EXP_NUMP	0x40034168	Endpoint 6 expects to send NUMP number	0x00
R8_UEP6_TX_CHAIN_NUMP	0x40034169	Endpoint 6 has transmitted NUMP number	0x00
R16_UEP6_TX_DMA_OFS	0x4003416A	DMA offset length of endpoint 6	0x00
R32_UEP6_TX_DMA	0x4003416C	DMA start address of endpoint 6	0x0400

Table 27-9 Endpoint 7 transmit-related registers

Name	Access address	Description	Reset value
R8_UEP7_TX_CFG	0x40034180	Endpoint 7 configuration register	0x86
R8_UEP7_TX_CR	0x40034181	Endpoint 7 control register	0x10
R8_UEP7_TX_SEQ	0x40034182	Endpoint 7 serial number register	0x00
R8_UEP7_TX_ST	0x40034183	Endpoint 7 status register	0x00
R8_UEP7_TX_CHAIN_CR	0x40034184	Endpoint 7 CHAIN control register	0x00
R8_UEP7_TX_CHAIN_ST	0x40034185	Endpoint 7 CHAIN status register	0x00
R16_UEP7_TX_CHAIN_LEN	0x40034186	Endpoint 7 CHAIN transmit last packet length	0x00
R8_UEP7_TX_CHAIN_EXP_NUMP	0x40034188	Endpoint 7 expects to send NUMP number	0x00
R8_UEP7_TX_CHAIN_NUMP	0x40034189	Endpoint 7 has transmitted NUMP number	0x00
R16_UEP7_TX_DMA_OFS	0x4003418A	DMA offset length of endpoint 7	0x00
R32_UEP7_TX_DMA	0x4003418C	DMA start address of endpoint 7	0x0400

27.2.3.1 Endpoint n Configuration Register (R8_UEPn_TX_CFG) (n=1-7)

Offset address: $0xC0 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
7	RB_EP_TX_CHAIN_AUTO	RW	CHAIN automatically switches mode, which is recommended.	1
6	RB_EP_TX_FIFO_MODE	RW	A bit of 1 indicates that the current endpoint uses FIFO mode. Before this bit is 1, the start address and end address of FIFO should be configured first.	0
5	RB_EP_TX_FIFO_CFG	RW	1: Access offset addresses 0xC to 0xF, with the operation target being UEP_DMA[31:16]/UEP_DMA[15:0]; 0: Access offset addresses 0xC to 0xF, with the	0

			operation target being UEP_DMA.	
4	Reserved	RO	Reserved	0
3	RB_EP_TX_EOB_MODE	RW	1: When sending a short packet, EOB/LPF in ACK-TP is 0; 0: When sending a short packet, EOB/LPF in ACK-TP is 1; <i>Note: The EOB_LPF of CHAIN_ST is ORed with the EOB/LPF set in this mode; this bit applies only to short packets, whereas CHAIN_ST->EOB_LPF applies to the last packet of the CHAIN and is independent of packet length.</i>	0
2	RB_EP_TX_ERDY_AUTO	RW	ERDY automatic mode, hardware will send ERDY without software control; The synchronization endpoint recommends clearing this bit.	1
1	RB_EP_TX_SEQ_AUTO	RW	1: Software is prohibited from writing to R16_EPn_ST->EP_SEQ_NUM; 0: Software is permitted to write to R16_EPn_ST->EP_SEQ_NUM.	1
0	RB_EP_TX_ISO_MODE	RW	A value of 1 indicates that the current endpoint is a synchronous endpoint.	0

27.2.3.2 Endpoint n Control Register (R8_UEPn_TX_CR) (n=1-7)

Offset address: $0xC1 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
7	RB_EP_TX_HALT	RW	Endpoint stop, active high, endpoint stop responding to DPH STALL.	0
6	RB_EP_TX_CLR	RW1	Write 1 to clear all configuration values and states of endpoints except UEP_CFG.	0
5	RB_EP_TX_CHAIN_CLR	RW1	Write 1 to clear all CHAIN configuration values and status.	0
[4:0]	RB_EP_TX_ERDY_NUMP	RW	The hardware sends the nump field of ERDY-TP, and the general configuration value is the number of supported bursts; For example, it supports 16-level burst, and the configuration value of this field is 16.	0x10

27.2.3.3 Endpoint n Serial Number Register (R8_UEPn_TX_SEQ) (n=1-7)

Offset address: $0xC2 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	0
[4:0]	RB_TX_EP_SEQ	RW	Current serial number of endpoint, writable in	0

	NUM		non-SEQ_AUTO mode and read-only in SEQ_AUTO mode.	
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27.2.3.4 Endpoint n Status Register (R8_UEPn_TX_ST) (n=1-7)

Offset address: $0xC3 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
7	RB_TX_EP_INT_F LAG	RO	Endpoint current interrupt flag. This bit is read-only. If all CHAIN_IF are 0, this bit is 0, otherwise it is 1.	0
6	RB_TX_EP_FC_ST	RW1Z	Current flow control status of endpoint, write 1 to clear.	0
5	RB_TX_EP_ERDY _REQ	RW1	Indicates that it is currently in the sending ERDY; Writing a 1 to this bit will send ERDY for use in non-ERDY_AUTO mode.	0
4	RB_TX_CHAIN_R ES	RO	CHAIN response status, corresponding to 4 independent CHAINS: 1: Responds with ACK_TP; 0: Responds with NRDY-TP. <i>Note: In endpoint halt mode (EP_HALT=1), responds with STALL; this bit is inactive.</i>	0
[3:0]	RB_TX_CHAIN_E N	RO	CHAIN enabled state, corresponding to 4 independent CHAIN.	0

27.2.3.5 Endpoint n CHAIN Control Register (R8_UEPn_TX_CHAIN_CR) (n=1-7)

Offset address: $0xC4 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[7:6]	RB_TX_CUR_USE	RO	CHAIN serial number currently in use.	0
[5:4]	RB_TX_CUR_CFG	RO	CHAIN serial number currently configured.	0
3	Reserved	RO	Reserved	0
2	RB_TX_FORCE_R ET	RW	This bit effectively forces the return of the selected CHAIN state machine configuration.	0
[1:0]	RB_TX_RET_SEL	RW	When FORCE_RET is valid, this bit indicates the returned CHAIN status and configuration.	0

27.2.3.6 Endpoint n CHAIN Status Register (R8_UEPn_TX_CHAIN_ST) (n=1-7)

Offset address: $0xC5 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
7	RB_TX_CHAIN_E N	RO	When the previously enabled CHAIN is enabled, it is automatically set by hardware after configuring the UEP_CHAIN_NUMP register, and the CHAIN is automatically cleared after the transmission.	0

6	RB_TX_CHAIN_IF	WO	This bit is written only to release the current CHAIN_IF; The current CHAIN's NUMP_EMPTY being 1 generates this interrupt.	0
5	RB_TX_EOB_LPF	RW	Current EOB/LPF bit in the last DPH packet of CHAIN: For lot transmission, when this bit is set to 1, CHAIN transmission ends and enters flow control (End Of Block), preventing it from being linked with the next CHAIN for transmission. For synchronous transmission, when this bit is set to 1, CHAIN transmission ends, and the microframe does not send another DPH, indicating this is the last packet for this microframe (LAST PACKET FLAG).	0
4	Reserved	RO	Reserved	0
3	RB_TX_NUMP_EMPTY	RO	NUMP in the current CHAIN is 0, then this bit is set to 1.	0
2	RB_TX_DPH_PP	RW	Status of the PP bit in the currently received DPH.	0
[1:0]	RB_TX_CHAIN_NUMP_O	RW	CHAIN sequence number that currently generates the interrupt.	0

27.2.3.7 Endpoint n CHAIN Transmit Packet Length (R16_UEPn_TX_CHAIN_LEN) (n=1-7)

Offset address: $0xC6 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[15:11]	Reserved	RO	Reserved	0
[10:0]	CHAIN_TX_LEN	RW	The length of the last packet sent by the CHAIN.	0

27.2.3.8 Endpoint n Expects to Transmit NUMP Number (R8_UEPn_TX_CHAIN_EXP_NUMP) (n=1-7)

Offset address: $0xC8 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[7:0]	TX_CHAIN_EXP_NUMP	RW	The number of DPP packets that can be sent by the currently completed CHAIN.	0

27.2.3.9 Endpoint n has Transmitted NUMP Number (R8_UEPn_TX_CHAIN_NUMP) (n=1-7)

Offset address: $0xC9 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[7:0]	TX_CHAIN_NUMP	RW	Number of DPP packets currently transmitted; when CHIN_EN is 0, writing	0

			UEP_CHAIN_NUMP will automatically clear this register.	
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27.2.3.10 DMA Offset Length of Endpoint n (R16_UEPn_TX_DMA_OFS) (n=1-7)

Offset address: $0xCA + (n-1) \times 0x20$

Bit	Name	Access	Description	Reset value
[15:0]	CHAIN_TX_DMA_OFS	RW	The offset address of the DPP in this CHAIN.	0x0400

27.2.3.11 DMA Start Address of Endpoint n (R32_UEPn_TX_DMA) (n=1-7)

Offset address: $0xCC + (n-1) \times 0x20$

Bit	Name	Access	Description	Reset value
[31:0]	CHAIN_TX_DMA	RW	Normal mode: The DMA start address for data transmission by this CHAIN. FIFO mode: The FIFO start address corresponds to bits 16–23 of the mapped address in SRAM; The FIFO end address corresponds to bits 16–23 of the mapped address in SRAM.	0

27.2.4 Device Register-Endpoints 1 ~ 7 Receive-related Registers.

Table 27-10 Endpoint 1 receive-related registers

Name	Access address	Description	Reset value
R8_UEP1_RX_CFG	0x400340D0	Endpoint 1 configuration register	0x86
R8_UEP1_RX_CR	0x400340D1	Endpoint 1 control register	0x10
R8_UEP1_RX_SEQ	0x400340D2	Endpoint 1 serial number register	0x00
R8_UEP1_RX_ST	0x400340D3	Endpoint 1 status register	0x00
R8_UEP1_RX_CHAIN_CR	0x400340D4	Endpoint 1 CHAIN control register	0x00
R8_UEP1_RX_CHAIN_ST	0x400340D5	Endpoint 1 CHAIN status register	0x00
R16_UEP1_RX_CHAIN_LEN	0x400340D6	Endpoint 1 CHAIN receive last packet length	0x00
R8_UEP1_RX_CHAIN_MAX_NUMP	0x400340D8	Endpoint 1 expects to receive NUMP number	0x00
R8_UEP1_RX_CHAIN_NUMP	0x400340D9	Endpoint 1 has received NUMP number	0x00
R16_UEP1_RX_DMA_OFS	0x400340DA	DMA offset length of endpoint 1	0x0400
R32_UEP1_RX_DMA	0x400340DC	DMA start address of endpoint 1	0x00

Table 27-11 Endpoint 2 receive-related registers

Name	Access address	Description	Reset value
R8_UEP2_RX_CFG	0x400340F0	Endpoint 2 configuration register	0x86
R8_UEP2_RX_CR	0x400340F1	Endpoint 2 control register	0x10
R8_UEP2_RX_SEQ	0x400340F2	Endpoint 2 serial number register	0x00
R8_UEP2_RX_ST	0x400340F3	Endpoint 2 status register	0x00
R8_UEP2_RX_CHAIN_CR	0x400340F4	Endpoint 2 CHAIN control register	0x00
R8_UEP2_RX_CHAIN_ST	0x400340F5	Endpoint 2 CHAIN status register	0x00
R16_UEP2_RX_CHAIN_LEN	0x400340F6	Endpoint 2 CHAIN receive last packet length	0x00
R8_UEP2_RX_CHAIN_MAX_NUMP	0x400340F8	Endpoint 2 expects to receive NUMP number	0x00
R8_UEP2_RX_CHAIN_NUMP	0x400340F9	Endpoint 2 has received NUMP number	0x00
R16_UEP2_RX_DMA_OFS	0x400340FA	DMA offset length of endpoint 2	0x0400
R32_UEP2_RX_DMA	0x400340FC	DMA start address of endpoint 2	0x00

Table 27-12 Endpoint 3 receive-related registers

Name	Access address	Description	Reset value
R8_UEP3_RX_CFG	0x40034110	Endpoint 3 configuration register	0x86
R8_UEP3_RX_CR	0x40034111	Endpoint 3 control register	0x10
R8_UEP3_RX_SEQ	0x40034112	Endpoint 3 serial number register	0x00
R8_UEP3_RX_ST	0x40034113	Endpoint 3 status register	0x00
R8_UEP3_RX_CHAIN_CR	0x40034114	Endpoint 3 CHAIN control register	0x00
R8_UEP3_RX_CHAIN_ST	0x40034115	Endpoint 3 CHAIN status register	0x00
R16_UEP3_RX_CHAIN_LEN	0x40034116	Endpoint 3 CHAIN receive last packet length	0x00
R8_UEP3_RX_CHAIN_MAX_NUMP	0x40034118	Endpoint 3 expects to receive NUMP number	0x00
R8_UEP3_RX_CHAIN_NUMP	0x40034119	Endpoint 3 has received NUMP number	0x00
R16_UEP3_RX_DMA_OFS	0x4003411A	DMA offset length of endpoint 3	0x0400
R32_UEP3_RX_DMA	0x4003411C	DMA start address of endpoint 3	0x00

Table 27-13 Endpoint 4 receive-related registers

Name	Access address	Description	Reset value
R8_UEP4_RX_CFG	0x40034130	Endpoint 4 configuration register	0x86
R8_UEP4_RX_CR	0x40034131	Endpoint 4 control register	0x10
R8_UEP4_RX_SEQ	0x40034132	Endpoint 4 serial number register	0x00

R8_UEP4_RX_ST	0x40034133	Endpoint 4 status register	0x00
R8_UEP4_RX_CHAIN_CR	0x40034134	Endpoint 4 CHAIN control register	0x00
R8_UEP4_RX_CHAIN_ST	0x40034135	Endpoint 4 CHAIN status register	0x00
R16_UEP4_RX_CHAIN_LEN	0x40034136	Endpoint 4 CHAIN receive last packet length	0x00
R8_UEP4_RX_CHAIN_MAX_NUMP	0x40034138	Endpoint 4 expects to receive NUMP number	0x00
R8_UEP4_RX_CHAIN_NUMP	0x40034139	Endpoint 4 has received NUMP number	0x00
R16_UEP4_RX_DMA_OFS	0x4003413A	DMA offset length of endpoint 4	0x0400
R32_UEP4_RX_DMA	0x4003413C	DMA start address of endpoint 4	0x00

Table 27-14 Endpoint 5 receive-related registers

Name	Access address	Description	Reset value
R8_UEP5_RX_CFG	0x40034150	Endpoint 5 configuration register	0x86
R8_UEP5_RX_CR	0x40034151	Endpoint 5 control register	0x10
R8_UEP5_RX_SEQ	0x40034152	Endpoint 5 serial number register	0x00
R8_UEP5_RX_ST	0x40034153	Endpoint 5 status register	0x00
R8_UEP5_RX_CHAIN_CR	0x40034154	Endpoint 5 CHAIN control register	0x00
R8_UEP5_RX_CHAIN_ST	0x40034155	Endpoint 5 CHAIN status register	0x00
R16_UEP5_RX_CHAIN_LEN	0x40034156	Endpoint 5 CHAIN receive last packet length	0x00
R8_UEP5_RX_CHAIN_MAX_NUMP	0x40034158	Endpoint 5 expects to receive NUMP number	0x00
R8_UEP5_RX_CHAIN_NUMP	0x40034159	Endpoint 5 has received NUMP number	0x00
R16_UEP5_RX_DMA_OFS	0x4003415A	DMA offset length of endpoint 5	0x0400
R32_UEP5_RX_DMA	0x4003415C	DMA start address of endpoint 5	0x00

Table 27-15 Endpoint 6 receive-related registers

Name	Access address	Description	Reset value
R8_UEP6_RX_CFG	0x40034170	Endpoint 6 configuration register	0x86
R8_UEP6_RX_CR	0x40034171	Endpoint 6 control register	0x10
R8_UEP6_RX_SEQ	0x40034172	Endpoint 6 serial number register	0x00
R8_UEP6_RX_ST	0x40034173	Endpoint 6 status register	0x00
R8_UEP6_RX_CHAIN_CR	0x40034174	Endpoint 6 CHAIN control register	0x00
R8_UEP6_RX_CHAIN_ST	0x40034175	Endpoint 6 CHAIN status register	0x00
R16_UEP6_RX_CHAIN_LEN	0x40034176	Endpoint 6 CHAIN receive last packet length	0x00

R8_UEP6_RX_CHAIN_MAX_NUMP	0x40034178	Endpoint 6 expects to receive NUMP number	0x00
R8_UEP6_RX_CHAIN_NUMP	0x40034179	Endpoint 6 has received NUMP number	0x00
R16_UEP6_RX_DMA_OFS	0x4003417A	DMA offset length of endpoint 6	0x0400
R32_UEP6_RX_DMA	0x4003417C	DMA start address of endpoint 6	0x00

Table 27-16 Endpoint 7 receive-related registers

Name	Access address	Description	Reset value
R8_UEP7_RX_CFG	0x40034190	Endpoint 7 configuration register	0x86
R8_UEP7_RX_CR	0x40034191	Endpoint 7 control register	0x10
R8_UEP7_RX_SEQ	0x40034192	Endpoint 7 serial number register	0x00
R8_UEP7_RX_ST	0x40034193	Endpoint 7 status register	0x00
R8_UEP7_RX_CHAIN_CR	0x40034194	Endpoint 7 CHAIN control register	0x00
R8_UEP7_RX_CHAIN_ST	0x40034195	Endpoint 7 CHAIN status register	0x00
R16_UEP7_RX_CHAIN_LEN	0x40034196	Endpoint 7 CHAIN receive last packet length	0x00
R8_UEP7_RX_CHAIN_MAX_NUMP	0x40034198	Endpoint 7 expects to receive NUMP number	0x00
R8_UEP7_RX_CHAIN_NUMP	0x40034199	Endpoint 7 has received NUMP number	0x00
R16_UEP7_RX_DMA_OFS	0x4003419A	DMA offset length of endpoint 7	0x0400
R32_UEP7_RX_DMA	0x4003419C	DMA start address of endpoint 7	0x00

27.2.4.1 Endpoint n Configuration Register (R8_UEPn_RX_CFG) (n=1-7)

Offset address: $0xD0 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
7	RB_EP_RX_CHAIN_AUTO	RW	CHAIN automatically switches mode, which is recommended.	1
6	RB_EP_RX_FIFO_MODE	RW	A bit of 1 indicates that the current endpoint uses FIFO mode. Before this bit is 1, the start address and end address of FIFO should be configured first.	0
5	RB_EP_RX_FIFO_CFG	RW	1: Access offset addresses 0xC to 0xF, with the operation target being UEP_DMA; 0: Access offset addresses 0xC to 0xF, with the operation target being R16_EPn_FIFO_START/R16_EPn_FIFO_DEST.	0
4	RB_EP_RX_TOUT_MODE	RO	When this bit is set to 1, after receiving consecutive burst packets, if PP=1 and the packet is not a short packet, and DPH is not	0

			received within the timeout period, a TOUT_IF interrupt is generated.	
3	RB_EP_RX_NRDY_MODE	RW	When this bit is set to 1, upon receiving a short packet or DPH (PP=0), clearing all CHAIN_EN flags will cause subsequent DPH responses to be NRDY; software reconfiguration is required.	0
2	RB_EP_RX_ERDY_AUTO	RW	ERDY automatic mode, hardware will send ERDY without software control; The synchronization endpoint recommends clearing this bit.	1
1	RB_EP_RX_SEQ_AUTO	RW	1: Software is prohibited from writing to R16_EPn_ST->EP_SEQ_NUM; 0: Software is permitted to write to R16_EPn_ST->EP_SEQ_NUM.	1
0	RB_EP_RX_ISO_MODE	RW	A value of 1 indicates that the current endpoint is a synchronous endpoint.	0

27.2.4.2 Endpoint n Control Register (R8_UEPn_RX_CR) (n=1-7)

Offset address: $0xD1 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
7	RB_EP_RX_HALT	RW	Endpoint stop, active high, endpoint stop responding to DPH STALL.	0
6	RB_EP_RX_CLR	RW1Z	Write 1 to clear all configuration values and states of endpoints except UEP_CFG.	0
5	RB_EP_RX_CHAIN_CLR	RW1Z	Write 1 to clear all CHAIN configuration values and status.	0
[4:0]	RB_EP_RX_ERDY_NUMP	RW	The hardware sends the nump field of ERDY-TP, and the general configuration value is the number of supported bursts; For example, it supports 16-level burst, and the configuration value of this field is 16.	0x10

27.2.4.3 Endpoint n Serial Number Register (R8_UEPn_RX_SEQ) (n=1-7)

Offset address: $0xD2 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	0
[4:0]	RB_EP_RX_SEQ_NUM	RW	Current serial number of endpoint, writable in non-SEQ_AUTO mode and read-only in SEQ_AUTO mode. <i>Note: In general, this field is automatically controlled by hardware, and the software does not need to write EP_SEQ_NUM.</i>	0

27.2.4.4 Endpoint n Status Register (R8_UEPn_RX_ST) (n=1-7)

Offset address: 0xD3 + (n-1)*0x20

Bit	Name	Access	Description	Reset value
7	RB_EP_RX_INT_F LAG	RO	Endpoint current interrupt flag. This bit is read-only. If all CHAIN_IF are 0, this bit is 0, otherwise it is 1.	0
6	RB_EP_RX_FC_ST	RW1Z	Current flow control status of endpoint, write 1 to clear.	0
5	RB_EP_RX_ERDY _REQ	RW1	Indicates that it is currently in the sending ERDY; Writing a 1 to this bit will send ERDY for use in non-ERDY_AUTO mode.	0
4	RB_EP_RX_CHAI N_RES	RO	CHAIN response status, corresponding to 4 independent CHAINS: 1: Responds with ACK_TP; 0: Responds with NRDY-TP. <i>Note: In endpoint halt mode (EP_HALT=1), responds with STALL; this bit is inactive.</i>	0
[3:0]	RB_EP_RX_CHAI N_EN	RO	CHAIN enabled state, corresponding to 4 independent CHAIN.	0

27.2.4.5 Endpoint n CHAIN Control Register (R8_UEPn_RX_CHAIN_CR) (n=1-7)

Offset address: 0xD4 + (n-1)*0x20

Bit	Name	Access	Description	Reset value
[7:6]	RB_EP_RX_CUR_ USE	RO	CHAIN serial number currently in use.	0
[5:4]	RB_EP_RX_CUR_ CFG	RO	CHAIN serial number currently configured.	0
3	Reserved	RO	Reserved	0
2	RB_EP_RX_FORC E_RET	RW	This bit effectively forces the return of the selected CHAIN state machine configuration.	0
[1:0]	RB_EP_RX_RET_S EL	RW	When FORCE_RET is valid, this bit indicates the returned CHAIN status and configuration.	0

27.2.4.6 Endpoint n CHAIN Status Register (R8_UEPn_RX_CHAIN_ST) (n=1-7)

Offset address: 0xD5 + (n-1)*0x20

Bit	Name	Access	Description	Reset value
7	RB_EP_RX_CHAI N_EN	RO	When the previously enabled CHAIN is enabled, it is automatically set by hardware after configuring the UEP_CHAIN_NUMP register, and the CHAIN is automatically cleared after the transmission.	0
6	RB_EP_RX_CHAI	RW1	This bit is written only to release the current	0

	N_IF		CHAIN_IF; 1. The length of the currently received packet is less than 1024; 2. The PP bit is set to 0 in the DPH received by the asynchronous endpoint; 3. The LPF bit is set to 1 in the DPH received by the synchronous endpoint; 4. The NUMP_EMPTY bit for the current CHAIN is set to 1; 5. No DPH interrupt is received within 4 microseconds after DPH completion (optional). When receiving a short packet, the hardware may optionally trigger an automatic NRDY or disable all chains; When receiving PP=0, the hardware may optionally trigger an automatic NRDY or disable all chains; If no DPP is received within the timeout period, the hardware may optionally trigger an automatic NRDY or disable all chains. In synchronous mode, if the expected number of packets is not received (CRC error)	
5	RB_EP_RX_LPF_F LAG	RO	Only the downstream endpoint is used synchronously, and the LPF status in the currently received DPH.	0
4	RB_EP_RX_ISO_P KT_ERR	RO	Only the downstream endpoint is used synchronously, and there is a CRC32 error in the currently received DPP.	0
3	RB_EP_RX_NUMP _EMPTY	RO	NUMP in the current CHAIN is 0, then this bit is set to 1.	0
2	RB_EP_RX_DPH_ PP	RO	Status of the PP bit in the currently received DPH.	0
[1:0]	RB_EP_RX_CHAI N_NO	RO	CHAIN sequence number that currently generates the interrupt.	0

27.2.4.7 Endpoint n CHAIN Receive Packet Length (R16_UEPn_RX_CHAIN_LEN) (n=1-7)

Offset address: $0xD6 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[15:11]	Reserved	RO	Reserved	0
[10:0]	RB_EP_RX_CHAI N_RX_LEN	RO	The length of the last packet received by the CHAIN.	0

27.2.4.8 Endpoint n Expects to Receive NUMP Number (R8_UEPn_RX_CHAIN_MAX_NUMP) (n=1-7)

Offset address: $0xD8 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[7:0]	RX_CHAIN_MAX_NUMP	RW	The number of DPP packets that can be received by the currently completed CHAIN.	0

27.2.4.9 Endpoint n has Received NUMP Number (R8_UEPn_RX_CHAIN_NUMP) (n=1-7)

Offset address: $0xD9 + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[7:0]	RX_CHAIN_NUMP	RO	Number of DPP packets that have been received at present	0

27.2.4.10 DMA Offset Length of Endpoint n (R16_UEPn_RX_DMA_OFS) (n=1-7)

Offset address: $0xDA + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[15:0]	CHAIN_DMA_OFS	RW	The offset address of the DPP in this CHAIN.	0x0400

27.2.4.11 DMA Start Address of Endpoint n (R32_UEPn_RX_DMA) (n=1-7)

Offset address: $0xDC + (n-1)*0x20$

Bit	Name	Access	Description	Reset value
[31:0]	CHAIN_DMA_BA	RW	Normal mode: The DMA start address for data transmission by this CHAIN. FIFO mode: The FIFO start address corresponds to bits 16–23 of the mapped address in SRAM; The FIFO end address corresponds to bits 16–23 of the mapped address in SRAM.	0

27.2.5 Host Register Description

Table 27-17 USBSS host registers

Name	Access address	Description	Reset value
R32_UH_TX_CTRL	0x40034084	Host transmit control register	0x0000XXXX
R32_UH_RX_CTRL	0x40034088	Host receive control register	0xFF00XXXX
R32_UH_TX_DMA_OFS	0x40034094	Host transmit address offset register	0x00000400
R32_UH_RX_DMA_OFS	0x40034098	Host receive address offset register	0x00000400
R16_HOST_TX_NUMP	0x4003409C	Host transmit NUMP register	0x0000
R16_HOST_RX_NUMP	0x4003409E	Host receive NUMP register	0xFF00
R32_HOST_STATUS	0x400340A0	Host status register	0x00000100
R16_HOST_TX_FC_STATUS	0x400340A4	Host endpoint transmit flow control register	0x0000
R16_HOST_RX_FC_STATUS	0x400340A6	Host endpoint receive flow control register	0x0000

R32_TP_RX_DATA0	0x400340A8	DATA0 register of DEV_NOTIF	0xFFFFFFFF
R32_TP_RX_DATA1	0x400340AC	DATA1 register of DEV_NOTIF	0xFFFFFFFF
R32_TP_RX_DATA2	0x400340B0	DATA2 register of DEV_NOTIF	0xFFFFFFFF

27.2.5.1 Host Transmit Control Register (R32_UH_TX_CTRL)

Offset address: 0x84

Bit	Name	Access	Description	Reset value
31	RB_UH_TX_ACT	RW	IN transaction completion interrupt flag, which is cleared by software writing 0 and set by hardware.	0
30	RB_UH_TX_ISO	RW	The host is ready to send an ISO packet.	0
29	RB_UH_TX_SETUP	RW	Indicates that the data packet sent by the host is a Setup data packet, and sets the setup flag.	0
28	RB_UH_TX_STATUS	RW	Indicates that the packet sent by the host is STATUS TP.	0
[27:24]	Reserved	RO	Reserved	0
23	RB_UH_TX_LPF	RW	For burst transmission, this bit only represents the LPF/EOB of the last packet, and the LPF/EOB of the previous packet uses a fixed value of 0. The synchronization endpoint is defined as follows: Last Packet Flag-- The LPF in the DPH packet is returned in the -IN transaction, and a bit of 1 indicates that the packet is the last packet in this frame interval. For asynchronous endpoints, there are the following definitions: End Of Block--The EOB IN the DPH packet is returned IN the IN transaction. When this bit is 1, the endpoint enters the flow control state, and the host waits for the endpoint of the device to receive ERDY-TP and then restarts the IN transaction.	0
[22:21]	RB_UH_TX_RES	RW	Response to DPH+DPP: 00: NRDY-TP; 01: ACK-TP; 10: STALL-TP; 11: No response (invalid). After SETUP/OUT transaction completion, EPn_R_RES automatically clears itself.	0
[20:16]	RB_UH_TX_SEQ	RW	The SEQ_NUM field in received packets is incremented by hardware automatically, except for endpoint 0: For synchronous endpoints, the SEQ field is automatically reset to zero upon receiving an	0

			ITP packet, as protocol requirements mandate that the first packet in a frame interval must have a SEQ value of 0.	
[15:12]	RB_UH_TX_EP	RW	Indicates the destination (target endpoint number of the device) of the packet sent in host mode.	x
11	Reserved	RO	Reserved	0
[10:0]	RB_UH_TX_LEN	RW	Endpoint receiving length register. For burst transmission, this length indicates the data packet length of the last packet of burst transmission, and other packets must be 1024B according to the protocol.	x

27.2.5.2 Host Receive Control Register (R32_UH_RX_CTRL)

Offset address: 0x88

Bit	Name	Access	Description	Reset value
31	RB_UH_RX_ACT	RW	OUT transaction completion interrupt flag, software writes 0 to clear, and hardware sets 1.	0
30	RB_UH_RX_ISO	RO	The received data packet (DPP) is synchronous transmission.	0
29	Reserved	RO	Reserved	0
[28:24]	RB_UH_RX_NUM P	RW	The number of data packets (DPP) that an endpoint can receive (burst transmission).	x
23	Reserved	RO	Reserved	0
[22:21]	RB_UH_RX_RES	RO	Response to DPH+DPP or STATUS-TP: 00: NRDY-TP; 01: ACK-TP; 10: STALL-TP; 11: No response (invalid). The SETUP/OUT/STATUS transaction automatically resets upon completion.	0
[20:16]	RB_UH_RX_SEQ	RW	The SEQ_NUM expected by the endpoint is incremented automatically by the hardware, except for endpoint 0. For synchronous endpoints, the hardware automatically resets the SEQ upon receiving an ITP packet, as the protocol requires the SEQ of the first packet in a frame interval to be 0.	0
[15:12]	RB_UH_RX_EP	RW	Indicates the source (device endpoint number) of the received packet in host mode.	x
11	Reserved	RO	Reserved	0
[10:0]	RB_UH_RX_LEN	RO	Endpoint receiving length register. For burst transmission, this length indicates the data packet length of the last packet of burst	x

			transmission, and other packets must be 1024B according to the protocol.	
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27.2.5.3 Host Transmit Address Offset Register (R32_UH_TX_DMA_OFS)

Offset address: 0x94

Bit	Name	Access	Description	Reset value
[31:0]	RB_UH_TX_DMA_OFS	RW	The size of the DMA address offset after the host sends it.	0x00000400

27.2.5.4 Host Receive Address Offset Register (R32_UH_RX_DMA_OFS)

Offset address: 0x98

Bit	Name	Access	Description	Reset value
[31:0]	RB_UH_RX_DMA_OFS	RW	The size of the DMA address offset after the host receives it.	0x00000400

27.2.5.5 Host Transmit NUMP Register (R16_HOST_TX_NUMP)

Offset address: 0x9C

Bit	Name	Access	Description	Reset value
[15:8]	Reserved	RO	Reserved	0
[7:0]	RB_UH_TX_NUMP	RW	The number of packets that the host expects to send is automatically reduced by 1 for each packet sent.	0

27.2.5.6 Host Receive NUMP Register (R16_HOST_RX_NUMP)

Offset address: 0x9E

Bit	Name	Access	Description	Reset value
[15:8]	RB_UH_RX_DPP_NUM	RW	The number of DPP that the host has accepted.	x
[7:0]	RB_UH_RX_NUMP	RW	The number of packets that the host expects to receive. If it is synchronous transmission, the hardware will automatically subtract 1 for each packet received.	0

27.2.5.7 Host Status Register (R32_HOST_STATUS)

Offset address: 0xA0

Bit	Name	Access	Description	Reset value
[31:20]	Reserved	RO	Reserved	0
[19:18]	RB_UH_ITP_PRESAGE	RO	In host mode, this bit indicates the time for sending ITP packets: 00: Host does not send ITP packets within 9us; 01: Host sends ITP packets within 9us; 11: Host sends ITP packets within 6us;	0

			10: Host sends ITP packets within 3us.	
17	RB_UH_RX_ISO_PKT_ERR	RO	CRC error of data packet (DPP) was received during synchronous transmission.	0
16	RB_UH_RX_EOB_LPF	RO	This bit indicates the EOB/LPF status of the received packet. For burst transmission, this bit only indicates the LPF/EOB of the last packet; the LPF/EOB of preceding packets use a fixed value of 0. For synchronous endpoints, the following definition applies: Last Packet Flag (LPF) — The LPF bit in the DPH packet returned during an IN transaction. When this bit is set to 1, it indicates that this packet is the last packet in the current frame interval. For asynchronous endpoints, the following definition applies: End Of Block (EOB) — The EOB bit in the DPH packet returned during an IN transaction. When this bit is set to 1, the endpoint enters flow control state. The host waits for the device's endpoint to send an ERDY-TP before resuming the IN transaction.	0
15	RB_UH_RX_ERDY_DIR	RO	ERDY is received from the device, which indicates the direction of the endpoint.	0
[14:13]	Reserved	RO	Reserved	0
[12:8]	RB_UH_RX_ERDY_NUMP	RO	Received ERDY from the device, which indicates the number of data packets that can be sent/received by the corresponding endpoint of the device.	0x1
[7:4]	RB_UH_RX_ERDY_EP	RO	Received ERDY from the device. This segment indicates the endpoint number of the device ERDY.	0
[3:0]	Reserved	RO	Reserved	0

27.2.5.8 Host Endpoint Transmit Flow Control Register (R16_HOST_TX_FC_STATUS)

Offset address: 0xA4

Bit	Name	Access	Description	Reset value
[15:1]	EPx_TX_FC	RW1Z	Flow control status of host to sending endpoints 1-15.	0
0	Reserved	RO	Reserved	0

27.2.5.9 Host Endpoint Receive Flow Control Register (R16_HOST_RX_FC_STATUS)

Offset address: 0xA6

Bit	Name	Access	Description	Reset value
[15:1]	EPx_RX_FC	RW1Z	Flow control status of host to receiving	0

			endpoints 1-15.	
0	Reserved	RO	Reserved	0

27.2.5.10 DEV_NOTIF-TP Data 0 Register (R32_TP_RX_DATA0)

Offset address: 0xA8

Bit	Name	Access	Description	Reset value
[31:0]	USB3_NOTIF_DATA0	RO	After receiving DEV_NOTIF-TP, HP data is stored in the register.	x

27.2.5.11 DEV_NOTIF-TP Data 1 Register (R32_TP_RX_DATA1)

Offset address: 0xAC

Bit	Name	Access	Description	Reset value
[31:0]	USB3_NOTIF_DATA1	RO	After receiving DEV_NOTIF-TP, HP data is stored in the register.	x

27.2.5.12 DEV_NOTIF-TP Data 2 Register (R32_TP_RX_DATA2)

Offset address: 0xB0

Bit	Name	Access	Description	Reset value
[31:0]	USB3_NOTIF_DATA2	RO	After receiving DEV_NOTIF-TP, HP data is stored in the register.	x

27.2.6 USBSS LINK Register Description

Table 27-18 USBSS LINK registers

Name	Access address	Description	Reset value
R32_LINK_CFG	0x40034000	LINK configuration register	0xC0000128
R32_LINK_CTRL	0x40034004	LINK control	0xXX000013
R32_LINK_INT_CTRL	0x40034008	LINK interrupt enable register	0x00000000
R32_LINK_INT_FLAG	0x4003400C	LINK interrupt flag register	0x00000000
R32_LINK_STATUS	0x40034010	LINK status register	0xX0000430
R8_LINK_ITP_PRE	0x40034017	LINK ITP timeout mode register	0x00
R8_LINK_U2_INACT_TIME	0x4003401D	LINK U2 inactivity timeout counter threshold register	0x00
R8_LINK_U1_WKUP_FILTER	0x40034028	U1 wake-up LFPS duration configuration register	0x4A
R8_LINK_U2_WKUP_FILTER	0x4003402C	U2 wake-up LFPS duration configuration register	0x02
R8_LINK_U3_WKUP_FILTER	0x40034030	U3 wake-up LFPS duration configuration register	0x64
R16_LINK_ISO_DLY	0x40034040	LINK synchronization delay register	0x0028
R16_LINK_LPM_CR	0x40034050	Link power management register	0x0180
R32_LINK_LMP_PORT_CAP	0x40034054	PORT_CAP register	0XXXXX0000
R32_LINK_LMP_RX_DATA0	0x40034058	LMP receive data 0 register	0XXXXXXXXX

R32_LINK_LMP_RX_DATA1	0x4003405C	LMP receive data 1 register	0xFFFFFFFF
R32_LINK_LMP_RX_DATA2	0x40034060	LMP receive data 2 register	0xFFFFFFFF
R32_LINK_LMP_TX_DATA0	0x40034064	USB custom HP data 0 register	0xFFFFFFFF
R32_LINK_LMP_TX_DATA1	0x40034068	USB custom HP data 1 register	0xFFFFFFFF
R32_LINK_LMP_TX_DATA2	0x4003406C	USB custom HP data 2 register	0xFFFFFFFF

27.2.6.1 LINK Configuration Register (R32_LINK_CFG)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
31	RB_LINK_RESET	RW	LINK reset, including resetting the state machine and all interrupt flags, is active high.	1
[30:22]	Reserved	RO	Reserved	0
21	RB_LINK_TOUT_MODE	RW	1: 3.2 SPEC; 0: 3.0 SPEC. <i>Note: Different protocol versions affect the timer duration in flow control:</i> <i>PM_LC_TIMER: 3.2_SPEC: 4us/3.0_SPEC: 3us;</i> <i>PM_ENTRY_TIMER: 3.2_SPEC: 8us/3.0_SPEC: 6us;</i> <i>PEND_HP_TIMER: 3.2_SPEC: 10us/3.0_SPEC: 3us.</i>	0
20	RB_LINK_U1_PING_EN	RW	U1 PING_FDFS enable: 1: Enable transmission; 0: Disable transmission.	0
[19:18]	RB_LINK_SKP_MODE	RW	SKP discard control for descrambling: 00: If the lower byte of the lower two bytes is SKP, disable descrambling for the lower two bytes; if the lower byte of the upper two bytes is SKP, disable descrambling for the upper two bytes; 01/11: Both low bytes are SKP, disabling descrambling enable for the low two bytes; both high bytes are SKP, disabling descrambling enable for the high two bytes; 10: Either byte in the low two bytes is SKP, disabling descrambling enable for the low two bytes; either byte in the high two bytes is SKP, disabling descrambling enable for the high two bytes.	0
17	RB_LINK_U2_ALLOWED	RW	Active high: Upon receiving LGO_U2, it responds with LAU to permit entry into U2 state; otherwise, upon receiving LGO_U2, it responds with LXU to deny entry into U2 state.	0

			<i>Note: For the downstream port, this enable has a fixed relationship with the U2_TIMEOUT value. When U2_TIMEOUT > 0, configuring U2_ALLOW is effective; otherwise, it is ineffective.</i>	
16	RB_LINK_U1_ALLOW	RW	Active high: Upon receiving LGO_U1, it responds with LAU to permit entry into U1 state; otherwise, upon receiving LGO_U1, it responds with LXU to deny entry into U1 state. <i>Note: For the downstream port, this enable has a fixed relationship with the U1_TIMEOUT value. When U1_TIMEOUT > 0, configuring U1_ALLOW is effective; otherwise, it is ineffective.</i>	0
15	RB_LINK_LTSSM_MODE	RW	The link state machine enters DISABLED mode: 1: Enter DISABLED mode according to protocol specifications upon timeout, detection count, etc.; 0: Enter DISABLED mode without adhering to protocol specifications upon timeout, detection count, etc.	0
14	RB_LINK_LOOPBACK_ACT	RW	Used in LOOPBACK mode for LOOPBACK master control pattern transmission (LOOPBACK slave maintains 0), ACT is a high-level signal sustained for a period of time. 1. When this bit is pulled high, the LOOPBACK master must send BRST to the LOOPBACK slave to reset the LOOPBACK slave's error counter (slave loopback BRST). 2. While ACT is high, the LOOPBACK master sends BDAT to the slave. The slave compares the received BDAT with a scrambled 0 to count errors (slave loopback BDAT). 3. When ACT goes low, the LOOPBACK master sends BERC to the slave (the slave returns the error count BCNT to the master).	0
13	RB_LINK_LOOPBACK_EN	RW	Enable the LOOPBACK enable bit, active high. This must be used in conjunction with the LOOPBACK enable bit in TX/RX_TS_CFG[3]. Both must be active for the LINK to enter LOOPBACK mode (used to enable data loopback and error counting for the LOOPBACK slave; the LOOPBACK master remains at 0).	0

12	RB_LINK_U2_DET_EN	RW	Device detection method in U2 state: 1: Software actively initiates detection of connected devices in U2 state; 0: Hardware automatically times detection of connected devices in U2 state.	0
[11:10]	RB_LINK_CP78_SEL	RW	When using Compliance Pattern 7/8, the length of consecutive zeros or consecutive ones sent: 00: 190 bits; 01: 120 bits; 10: 50 bits; 11: 250 bits.	0
[9:8]	RB_LINK_TX_DEMPH	RW	Transmitter de-emphasis control: 00: -6.0 dB de-emphasis; 01: -3.5 dB de-emphasis (default); 10: No de-emphasis; 11: Reserved. <i>Note: Normal operating mode must be 0/1 only; other values are reserved for COMPLIANCE.</i>	01b
7	RB_LINK_TX_SWING	RW	Transmitter signal swing control: Low swing reduces power consumption but affects transmission distance. 1: Low Swing (400mv-1200mv); 0: Full Swing (800mv-1200mv)。	0
6	RB_LINK_RX_EQ_EN	RW	Receiver equalization enables control, and protocol provisions are optional.	0
5	RB_LINK_LFPS_RX_PD	RW	LFPS reception control, this bit is 1 to disable LFPS reception.	1
4	RB_LINK_COMPLIANCE_EN	RW	POLLING_LFPS timeout entry into COMPLIANCE mode: 1: Enter upon timeout; 0: Do not enter upon timeout.	0
3	RB_LINK_PHY_RESET	RW	PIPE interface reset 1: Reset; 0: Clear reset.	1
2	RB_LINK_SS_PLR_SWAP	RW	Swap SSTX and SSRX polarities as follows: 1: SSTXA(SSTXM), SSTXB(SSTXP); SSRXA(SSRXM), SSRXB(SSRXP)。 0: SSTXA(SSTXP), SSTXB(SSTXM), SSRXA(SSRXP), SSRXB(SSRXM)。 <i>Note: The difference of this adaptive rx polarity is that rx polarity only exchanges SSRXM and SSRXP.</i>	0
1	RB_LINK_RX_TE	RW	Terminal resistor control:	0

	RM_EN		1: Connect termination resistor; 0: Remove termination resistor.	
0	RB_LINK_DOWN_MODE	RW	Peripheral type: 1: downstream; 0: upstream. In host mode, it should be configured as downstream, and in device modeling, it should be configured as upstream. Specifically: When OTG mode support is required, the port should be configured as downstream. The Tiebreaker in Port_Capability(LMP) should be set to 0, indicating that when connected to a host, the port will act as a device, and when connected to a device, the port will act as a host. The protocol specifies that when both ports are downstream or upstream, the port with a Tiebreaker of 1 will become the downstream (host) port. Subsequently, both parties must exchange Port_Capability(LMP) again.	0

27.2.6.2 LINK Control Register (R32_LINK_CTRL)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[31:24]	LINK_RX_TS_CFG	RO	Received Link Control: BIT0: 1: Reset; 0: Normal transmission. BIT1: Reserved. BIT2: 1: Enable Loopback; 0: Disable Loopback. BIT3: 1: Disable scrambling; 0: Enable scrambling. BIT4~7: Reserved.	0
[23:16]	LINK_TX_TS_CFG	RW	Link configuration for sending TS1/TS2 training sequences: BIT0: 1: HOT_RESET; 0: Normal training mode. BIT1: Reserved. BIT2: 1: Enable Loopback; 0: Disable Loopback. BIT3: 1: Disable scrambling; 0: Enable scrambling. BIT4~7: Reserved. Specifically, reserved bits must be set to 0; BIT0 and BIT2 cannot be both 1	0

			simultaneously. In device mode, if a valid HOT_RESET is received, it generates a HOT_RESET interrupt flag while BIT0 is automatically set to 1 by hardware to respond to the host's HOT_RESET.	
15	LINK_TX_LGO_U3	RW	When this bit is active, send LGO_U3 (high active) to trigger cleared automatically by hardware. The protocol mandates that the receiver cannot refuse to enter U3 state. Querying the LINK status as U3 indicates entry into the U3 low-power state. At this point, the software should change the current power mode to P3.	0
14	LINK_TX_LGO_U2	RW	When this bit is active, send LGO_U2 (high active) to trigger cleared automatically by hardware. If the peer responds with LXU, it indicates refusal to enter U2 state. If the LINK status query returns U2, it confirms the peer accepts entering U2 low-power state. At this point, the software should change the current power mode to P2.	0
13	LINK_TX_LGO_U1	RW	When this bit is active, send LGO_U1 (high active) to trigger cleared automatically by hardware. If the peer responds with LXU, it indicates refusal to enter U1 mode. If the LINK status query returns U1, it confirms the peer accepts entering U1 low-power mode. At this point, the software should change the current power mode to P1.	0
12	LINK_POLLING_EN	RW	High active. If SS.RX_DETECT detects TERM, the POLLING handshake will proceed after this bit is set active in software. Note: When AUTO_MODE[9] is 1, if AUTO_MODE[8] is 1 and the current power state is P2, initiate POLLING_LFPS handshake; When AUTO_MODE[9] is 1, if AUTO_MODE[7] is 1 and the current power supply is P0, initiate POLLING_LFPS handshake.	0
11	LINK_REG_ROUT	RW	Enable the HUB's routing function, high for	0

	_EN		enabled, low for disabled: 1: Uplink port can route header packets to downlink port normally; 0: Uplink port cannot route header packets to downlink port.	
10	LINK_LUP_LDN_EN	RW	U0 state: When no data is present, whether to transmit LUP/LDN packets every 10μs: 1: Enable transmission; 0: Disable transmission.	0
9	LINK_TX_UX_EXIT	RW1Z	Active high, cleared automatically by hardware: 1. If LINK is in U3 state, it is equivalent to executing U3_WAKEUP. After U3_WAKEUP handshake failure, the software should re-initiate U3_WAKEUP after at least 100ms. In host mode, if three consecutive handshakes fail, the software should set LINK to INACTIVE and check TERM; the device may have disconnected. If a U3_WAKEUP is received but the handshake signal cannot be returned for any reason, the device should proactively send a U3_WAKEUP when ready to enter U0. 2. If LINK is in U2 state, it is equivalent to executing U2_EXIT (LFPS). 3. If LINK is in U1 state, this is equivalent to executing U1_EXIT (LFPS). 4. If the U1_EXIT/U2_EXIT handshake fails, LINK enters INACTIVE.	0
8	LINK_TX_WARM_RST	RW1Z	Active high, cleared by software. This bit is valid and will send warm-reset. Host Mode: Pulled high by software, with optional software or hardware pull-down 1. The duration of the warm_reset is determined by the software. The host should send the warm_reset for 80ms to 120ms (typically 100ms). After the device detects the warm_reset, it should send a warm_reset response within the tResetDelay timeframe (18ms to 50ms) until it detects the host has ceased sending the warm_reset, at which point the device also stops sending the warm_reset. 2. After the software pulls the signal high, the hardware automatically pulls it low after timing 100ms (requires enabling the	0

			corresponding hardware timeout via reg_auto_mode[13]). In device mode: Upon detecting a warm reset initiated by the host after 18ms of continuous transmission, the device controller will automatically acknowledge the warm reset and proceed with the reset. <i>Specifically, the protocol stipulates that only the host has the capability to actively initiate a warm reset; device mode can only passively respond to a warm reset.</i>	
7	LINK_GO_RX_DETECT	RW1Z	Before setting this bit, configure PD_MODE to P2 mode. Set LINK to enter SS.RX_DETECT. Software polls TERM_PRESENT to determine if a connection exists or has been lost. High active, auto-reset.	0
6	LINK_GO_RECOVERY	RW1Z	Set LINK to enter SS.RECOVERY mode, active high, cleared automatically by hardware.	0
5	LINK_GO_INACTIVE	RW1Z	Set LINK to SS.INACTIVE, active high, cleared automatically by hardware.	0
4	LINK_GO_DISABLED	RW	Set LINK to SS.DISABLED, high active, requiring software reset. If hardware automatically enters SS.DISABLED, this bit is automatically set to 1 by hardware. In device mode, upon receiving a USB 2.0 bus reset, the software should first set PD_MODE to P2 mode, then clear this bit. LTSSM will automatically enter SS.RX_DETECT, where the device performs 8 detections (every 12ms). If TERM is not detected, it enters DISABLE state while communicating in USB 2.0 mode. In Host Mode, the BUS_RESET signal should be sent via the USB 2.0 controller. The software should first set PD_MODE to P2 mode, then clear this bit. If TERM is detected, a POLLING_LFPS handshake will occur. Otherwise, the software should set the power mode to P3 and communicate in USB 2.0 mode.	1
[3:2]	Reserved	RO	Reserved	0
[1:0]	LINK_PD_MODE	RW	Configure the PHY's current power mode, corresponding to PO/P1/P2/P3 in the PIPE: 00: PO: Normal operating mode;	11b

			01: P1: Low-power mode. In this mode, both ends of the port periodically send PING-LFPS to maintain the connection; 10: P2: Low-power mode. This mode consumes less power than P1 mode, but the time required to exit this mode also increases; 11: P3: Ultra-low power mode. In this mode, the PHY clock (125MHz) is disabled. Changing this configuration value will cause LINK_BUSY to go high. The software should check that LINK_BUSY is 0 to indicate the switchover is complete. Query PD_MODE_CURRENT in LINK_STATUS to obtain the PHY's current power mode (optional operation).	
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27.2.6.3 LINK Interrupt Enable Register (R32_LINK_INT_CTRL)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
31	LINK_IE_STATE_C HG	RW	Link state change flag interrupt enable: 1: Enable; 0: Disable.	0
30	LINK_IE_U1_TOU T	RW	U1 timeout interrupt enable: 1: Enable; 0: Disable.	0
29	LINK_IE_U2_TOU T	RW	U2 timeout interrupt enable: 1: Enable; 0: Disable.	0
28	LINK_IE_UX_FAI L	RW	UX conversion failure interrupt enable: 1: Enable; 0: Disable.	0
27	LINK_IE_TX_WAR MRST	RW	Transmit warm_reset to enable/disable interrupts: 1: Enable; 0: Disable.	0
26	LINK_IE_UX_EXI T_FAIL	RW	Ux exit failure interrupt enable: 1: Enable; 0: Disable.	0
[25:24]	Reserved	RO	Reserved	0
23	LINK_IE_RX_LMP _TOUT	RW	Receive LMP timeout interrupt enable: 1: Enabled; 0: Disabled.	0
22	LINK_IE_TX_LMP	RW	Successful entry into U0 allows sending HP packet interrupt enable:	0

			1: Enable; 0: Disable.	
21	LINK_IE_RX_LMP	RW	Received link command flag interrupt enable: 1: Enable; 0: Disable.	0
20	LINK_IE_RX_DET	RW	Interrupt enable for link entering Rx.Detect state: 1: Enabled; 0: Disabled.	0
19	LINK_IE_LOOPBACK	RW	The link enters loopback mode for testing and fault isolation. Interrupt enable: 1: Enabled; 0: Disabled.	0
18	LINK_IE_COMPLIANCE	RW	Link enter compliance test, enabled via compatibility test or physical layer consistency test interrupt: 1: Enabled; 0: Disabled.	0
17	LINK_IE_HPBUF_FULL	RW	HP_BUF transmit FIFO full interrupt enable: 1: Enable; 0: Disable.	0
16	LINK_IE_HPBUF_EMPTY	RW	HP_BUF transmit FIFO empty interrupt enable: 1: Enable; 0: Disable.	0
15	LINK_IE_HOT_RESET	RW	Hot RESET, using reset interrupts from the TS1/TS2 ordered set: 1: Enable; 0: Disable.	0
14	LINK_IE_U3_WAKEUP	RW	In U3 state, upon receiving a low-frequency periodic signal (LPFS) wake-up interrupt enable: 1: Enable; 0: Disable.	0
13	LINK_IE_WARM_RST	RW	Enable warm reset (unconnected) interrupts using LPFS: 1: Enable; 0: Disable.	0
12	LINK_IE_UX_EXIT	RW	Interrupt enable for Ux exit request received: 1: Enable; 0: Disable.	0
11	LINK_IE_TXEQ	RW	LINK enters POLLING_RXEQ for receiver equalization training interrupt enable: 1: Enable;	0

			0: Disable.	
10	LINK_IE_TERM_P RES	RW	LINK enters RX_DETECT mode for detecting remote receiver tracking segment impedance detection interrupt enable: 1: Enable; 0: Disable.	0
9	LINK_IE_UX_REJ	RW	Transmit LGO_Ux, receive an LXU interrupt enable sent by Ux: 1: Enable; 0: Disable.	0
8	LINK_IE_U3_WK_ TOUT	RW	Enable U3 command timeout interrupt request: 1: Enable; 0: Disable.	0
7	LINK_IE_GO_U0	RW	LINK is in U0 interrupt enable mode: 1: Enabled; 0: Disabled.	0
6	LINK_IE_GO_U1	RW	LINK is in U1 interrupt enable mode: 1: Enabled; 0: Disabled.	0
5	LINK_IE_GO_U2	RW	LINK is in U2 interrupt enable mode: 1: Enabled; 0: Disabled.	0
4	LINK_IE_GO_U3	RW	LINK is in U3 interrupt enable mode: 1: Enabled; 0: Disabled.	0
3	LINK_IE_DISABL E	RW	LINK is in DISABLED interrupt enable mode: 1: Enabled; 0: Disabled.	0
2	LINK_IE_INACTI VE	RW	LINK is in INACTIVE interrupt enable mode: 1: Enabled; 0: Disabled.	0
1	LINK_IE_RECOVE RY	RW	Error LINK is in RECOVERY interrupt enable mode: 1: Enable; 0: Disable.	0
0	LINK_IE_READY	RW	LINK initialization completed, including two ports prior to (Header Sequence Number Advertisement) and (RX Header Buffer Credit Advertisement). Interrupt enable: 1: Enabled; 0: Disabled.	0

27.2.6.4 LINK Interrupt Flag Register (R32_LINK_INT_FLAG)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
31	LINK_IF_STATE_CHANGE	RW1Z	LINK_RESET is reset to 0 (write 1 to clear it): 1: Link state machine change flag; 0: Link state machine has not changed.	0
30	LINK_IF_U1_TIMEOUT	RW1Z	U1 timeout interrupt; (Write 1 to clear): 1: Timeout occurred; 0: No timeout occurred;	0
29	LINK_IF_U2_TIMEOUT	RW1Z	U2 timeout interrupt; (Write 1 to clear): 1: Timeout occurred; 0: No timeout occurred;	0
28	LINK_IF_UX_FAIL	RW1Z	Ux entry failure interrupt; (Write 1 to clear): 1: Ux entry failed; 0: No action.	0
27	LINK_IF_TX_WARM_RST	RW1Z	Transmit WARM_RESET to terminate interrupt; (Write 1 to clear): 1: Transmit WARM_RESET to terminate; 0: No action.	0
26	LINK_IF_UX_EXIT_FAIL	RW1Z	Ux exit failed interrupt; (Write 1 to clear): 1: Ux exit failed; 0: No action.	0
[25:24]	Reserved	RO	Reserved	0
23	LINK_IF_RX_LMP_TOUT	RW1Z	After LINK initialization completes, exchange Port Capabilities/Configuration LMPs timeout (20us) interrupt flags.	0
22	LINK_IF_TX_LMP	RW1Z	LINK Initialization Complete Interrupt Flag: When this signal goes high, software configures the transmission of the Port Capabilities LMP for port capability exchange.	0
21	LINK_IF_RX_LMP	RW1Z	Receive LMP interrupt flag: Control related LMP transmission during the port capabilities exchange phase: When the downstream port receives a Port Capabilities LMP, software configures and transmits a Port Configuration LMP; When the upstream port receives a Port Capabilities LMP, software configures and transmits a Port Configuration Response LMP.	0
20	LINK_IF_RX_DET	RW1Z	The RX_DETECT status interrupt flag is set. Once this flag is set to 1, the software configures PWR_MODE to P2, preparing for RX_DETECT.	0

19	LINK_IF_LOOPBACK	RW1Z	LINK enters LOOPBACK status interrupt flag.	0
18	LINK_IF_COMPLIANCE	RW1Z	LINK enters COMPLIANCE status interrupt flag.	0
17	LINK_IF_HPBUF_FULL	RW1Z	Header Packet buffer full interrupt flag.	0
16	LINK_IF_HPBUF_EMPTY	RW1Z	Header Packet buffer empty interrupt flag.	0
15	LINK_IF_HOT_RESET	RW1Z	The device receives the HOT RESET interrupt flag: After this signal goes high, the software should clear TX_TS_CFG[0] to complete the HOT RESET process.	0
14	LINK_IF_WAKEUP	RW1Z	The power supply is in P3 mode, and the LFPS signal interrupt flag is detected.	0
13	LINK_IF_WARM_RST	RW1Z	Interrupt flag of WARMRESET state change (valid-> invalid or invalid-> valid) received by the device.	0
12	LINK_IF_UX_EXIT	RW1Z	Upon receiving the LFPS ready signal, LINK prepares to exit the U1/U2/U3 request interrupt flag. Once this bit is set to 1, the software should set PWR_MODE to P0. After setting PWR_MODE to P0, the Ux exit timer begins counting. Upon reaching U1_EXIT_TIMER, U2_EXIT_TIMER, or U3_WAKE_TIMER, LINK exits the Ux state and enters Recovery mode.	0
11	LINK_IF_TXEQ	RW1Z	LINK entering TXEQ State Interrupt Flag: Indicates completion of the POLLING handshake, awaiting software to set P0 to P0_MODE to enter the TXEQ phase.	0
10	LINK_IF_TERM_PRES	RW1Z	Detected TERM disconnect or connection interruption flag.	0
9	LINK_IF_UX_REJ	RW1Z	LINK refused to enter low power consumption mode (U1/U2) interrupt flag.	0
8	LINK_IF_U3_WK_TOUT	RW1Z	Wake up timeout interrupt flag from U3 (10ms).	0
7	LINK_IF_GO_U0	RW1Z	LINK enters U0 status interrupt flag.	0
6	LINK_IF_GO_U1	RW1Z	LINK enters U1 status interrupt flag.	0
5	LINK_IF_GO_U2	RW1Z	LINK enters U2 status interrupt flag.	0
4	LINK_IF_GO_U3	RW1Z	LINK enters U3 status interrupt flag.	0
3	LINK_IF_DISABLE	RW1Z	LINK enters SS.DISABLE status interrupt flag: <i>Note: The software is directly connected with</i>	0

			<i>GO_DISABLE enable (LINK_CTRL), which causes LINK to directly enter DISABLE, and the interrupt will not be pulled high.</i>	
2	LINK_IF_INACTIVE	RW1Z	LINK enters SS.INACTIVE status interrupt flag. <i>Note: Software directly configures GO_INACTIVE enable (LINK_CTRL), causing LINK to enter INACTIVE directly without interrupt flag being pulled high.</i>	0
1	LINK_IF_RECOVERY	RW1Z	LINK enters SS.RECOVERY state interrupt flag. After this bit is set, query LINK_ERR_STATUS to determine the cause of LINK entering SS.RECOVERY. <i>Note: Direct software configuration of GO_RECOVERY enable (LINK_CTRL) causes LINK to enter RECOVERY directly without raising the interrupt flag.</i>	0
0	LINK_IF_READY	RW1Z	LINK enters the U0 state and clears the LINK initialization interrupt flag.	0

27.2.6.5 LINK Status Register (R32_LINK_STATUS)

Offset address: 0x10

Bit	Name	Access	Description	Reset value
31	LINK_HPBUF_EMPTY	RO	HP buffer is empty.	x
30	LINK_HPBUF_FULL	RO	HP buffer is full.	x
29	LINK_HPBUF_IDLE	RO	HP_BUF transmit state of FIFO idle: 1: HP_BUF transmit FIFO has data; 0: HP_BUF transmit FIFO is empty.	x
[28:23]	RESERVE	RO	Reserved	0
22	LINK_U3_SLEEP_ALLOW	RO	The link is in U3 state, allowing sleep.	x
21	LINK_U2_SLEEP_ALLOW	RO	The link is in U2 state, allowing sleep.	x
20	LINK_RXDET_SLEEP_ALLOW	RO	The link is in RXDET state, allowing sleep.	x
19	LINK_WAKUP	RO	Link wake-up signal received.	x
18	LINK_RX_LFPS	RO	Link receives LFPS signal.	x
17	LINK_RX_DETECT	RO	Link is at P2, and RX_DETECT is in progress.	x
16	LINK_RX_UX_EXIT_REQ	RO	Received Ux exit request: 1: LINK received a valid Ux exit LFPS	0

			signal, preparing to exit U1/U2/U3. Software should configure the power mode to P0 (after the power mode conversions to P0, the Ux exit timer starts counting and LINK enters RECOVERY upon reaching the preset value); 0: No change.	
[15:12]	Reserved	RO	Reserved	0
[11:8]	LINK_STATE	RO	Link status: 0000: STATE_U0; 0001: STATE_U1; 0010: STATE_U2; 0011: STATE_U3; 0100: STATE_DISABLED; 0101: STATE_RXDET; 0110: STATE_INACTIVE; 0111: STATE_POLLING; 1000: STATE_RECOVERY; 1001: STATE_HOTRST; 1010: STATE_COMPLIANC; 1011: STATE_LOOPBACK; Others: Reserved.	4h
7	Reserved	RO	Reserved	0
6	LINK_TXEQ	RO	The link is in POLLING_RXEQ.	0
[5:4]	LINK_PD_MODE_MASK	RO	Link power status: 00: P0 status; 01: P1 status; 10: P2 status; 11: Default status P3.	10b
3	LINK_READY	RO	When LINK enters U0 state, this bit is set after initialization (broadcast) is completed, and this bit is automatically cleared when it exits U0 state.	0
2	LINK_BUSY	RO	LINK busy status. When switching PD_MODE, this bit is 1, and it is automatically cleared by hardware after switching.	0
1	LINK_RX_WARM_RST	RO	Received a valid warm-reset signal from the host (hardware automatically powers up 18ms after receiving the host's WARM_RESET signal).	0
0	LINK_RX_TERM_PRES	RO	After RX_DETECT, if a receive terminal resistor is present, this bit is set to 1.	0

27.2.6.6 LINK ITP Timeout Mode Register (R8_LINK_ITP_PRE)

Offset address: 0x17

Bit	Name	Access	Description	Reset value
[7:0]	ITP_PRE	RO	ITP timeout mode: 1: Ignore the impact of ITP transmission/reception on the link idle state for both downlink and uplink ports; 0: The link is considered non-idle when the downstream port transmits non-delayed ITP or the upstream port receives non-delayed ITP.	0x80

27.2.6.7 LINK U2 Inactivity Timeout Counter Threshold Register (R8_LINK_U2_INACT_TIMER)

Offset address: 0x1D

Bit	Name	Access	Description	Reset value
[7:0]	U2_INACTIVE_TIMER	RW	Value of U2's inactivity timeout counter threshold: U2 Timeout value (this value represents the maximum allowed time in U0/U1 inactive state, and enters U2 state after timeout), value range: 0x00~0xff, 256us/LSB. <i>Note: When the U1/U2Timeout value is 0xff, it means that the timeout time of the link inactive state is infinite, and the conversion request to U1/U2 will not be actively initiated when the timer reaches 0xff.</i>	0

27.2.6.8 U1 Wake-up LFPS Duration Configuration Register (R8_LINK_U1_WKUP_FILTER)

Offset address: 0x28

Bit	Name	Access	Description	Reset value
[7:0]	U1_WKUP_FILTER	RW	The valid duration of the LFPS received during U1 exit. The port sends LFPS for handshaking. When the received LFPS reaches this duration, the received U1 EXIT is considered valid. Pull high the sent LFPS_last to indicate successful handshaking. Default is 600ns. U1_WKUP_FILTER[7]: Controls the time unit of U1_WKUP_FILTER[6:0]; 1: U1_WKUP_FILTER[6:0] unit is in us, duration is (n)us; 0: U1_WKUP_FILTER[6:0] unit is 8ns, duration is (8n)ns, where n ranges from 0 to 124; <i>Note: When the duration exceeds 1ns, the configuration method with</i>	0x4A

			<i>U1_WKUP_FILTER[7] set to 1 must be used; otherwise, U1 cannot be awakened.</i>	
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27.2.6.9 U2 Wake-up LFPS Duration Configuration Register (R8_LINK_U2_WKUP_FILTER)

Offset address: 0x2C

Bit	Name	Access	Description	Reset value
[7:0]	U2_WKUP_FILTER	RW	The valid determination duration for LFPS received during U2 wake-up and loopback exit. The port sends LFPS for handshaking. When the received LFPS reaches this duration, the LFPS_LAST signal is pulled high, indicating successful handshaking. The received U2 EXIT can then be considered valid, with a duration of (n)us, defaulting to 2us.	0x02

27.2.6.10 U3 Wake-up LFPS Duration Configuration Register (R8_LINK_U3_WKUP_FILTER)

Offset address: 0x30

Bit	Name	Access	Description	Reset value
[7:0]	U3_WKUP_FILTER	RW	The valid determination duration for LFPS received during U3 wake-up. The port sends LFPS for handshaking. When the received LFPS reaches this duration, the LFPS_LAST signal is pulled high, indicating successful handshaking. The received U3 EXIT signal is then considered valid. The duration is (n)us, with a default value of 100us.	0x64

27.2.6.11 LINK Synchronous Delay Register (R16_LINK_ISO_DLY)

Offset address: 0x40

Bit	Name	Access	Description	Reset value
[15:0]	LINK_ISOCH_DLY	RW	The delay time for serializing bitstreams into parallel data defaults to 40ns. As the device, write the delay value to this register. Note that the lower three bits of this register are invalid (since the clock frequency is 125MHz, the delay time must be set to 8*n).	0x28

27.2.6.12 Link Power Management Register (R16_LINK_LPM_CR)

Offset address: 0x50

Bit	Name	Access	Description	Reset value
[15:14]	Reserved	RO	Reserved	0
13	PHY_CHSEL_AUTO	RW	PHY layer transceiver channel type auto-Selection:	0

			1: After device detection completes, if the connection status flag remains unchanged, switch to another channel type. The default channel type is standard USB interface. 0: PHY layer disables automatic transceiver channel type switching. Only software-configured transceiver types can be used, with standard USB interface as default.	
12	LPM_RXDET_EN	RO	When the lpm counter reaches the expected value RXDET_EXP, confirming a device is connected to the PHY, this bit is pulled high.	0
11	LPM_TERM_PRESENT	RO	The PHY layer detected a device connection.	0
10	LPM_TERM_CHG	RO	The connected device changes, and the device is inserted or pulled out.	0
9	LPM_EN	RW	CLKLPM is fixed at 12MHz. When LINK_PD_MODE=3 & LPM_EN=0 & OSCOUTEN=0 & PLLALIV=0, CLKOSCO stops and the crystal oscillator ceases operation. 1: Enables CLKLPM in L1 mode. 0: Disables. <i>Note: This bit is unrelated to the USBSS controller. To enter low-power mode, the crystal oscillator must be disabled by clearing this bit.</i>	0
8	LPM_RST	RW	Reset signal for LPM-related registers: 1: Resets LPM counter and LPM terminal detection; 0: Does not reset.	1
[7:0]	RXDET_EXP	RW	Detect external device counter register.	0x80

27.2.6.13 PORT_CAP Register (R32_LINK_LMP_PORT_CAP)

Offset address: 0x54

Bit	Name	Access	Description	Reset value
31	LINK_LMP_RX_CAP_VLD	RO	Upon receiving a valid PORT_CAP_LMP, the protocol specifies that both LINK parties exchange PORT_CAP_LMP within 20 microseconds after entering the U0 state. <i>Note: This flag signal is set to 0 when the LINK state changes and enters Port_State_Polling.</i>	0
30	LINK_LMP_TX_CAP_VLD	RO	PORT Capability configuration complete indicator (software operation, write 1 to clear 0):	0

			1: Indicates PORT Capability configuration is incomplete; 0: Indicates PORT Capability configuration is complete (distinguished between host and device scenarios). The host clears the indicator to 0 upon receiving the device's PORT Configuration Response. The device clears the indicator to 0 upon receiving the host's PORT Configuration and replying with a PORT Configuration Response.	
[29:24]	LINK_SPEED	RO	[24] Position 1 indicates that the device supports USB 3.2 Gen 1 (5Gbps).	x
[23:0]	LINK_REG_PORT_CAP	RO	1. LINK_REG_PORT_CAP[23:20]: Tiebreaker. Used for arbitrating port types when connecting two ports simultaneously capable of both uplink and downlink (LINK_REG_PORT_CAP[17:16]=11). The port with the higher value becomes the downlink port. If both ports have equal Tiebreaker values, PORT_CAP will be re-exchanged. 2. LINK_REG_PORT_CAP[17:16]: Direction LINK_REG_PORT_CAP[17]: Set to 1 indicates the port can be configured as an uplink port; LINK_REG_PORT_CAP[16]: Set to 1 indicates the port can be configured as a downlink port; 3. LINK_REG_PORT_CAP[7:0]: Number of Header Packet Buffers, indicating the number of header packet buffers supported by the device; 4. Other bits are reserved.	0

27.2.6.14 LMP Receive Data 0 Register (R32_LINK_LMP_RX_DATA0)

Offset address: 0x58

Bit	Name	Access	Description	Reset value
[31:0]	LINK_LMP_RX_DATA0	RO	After LMP reception is complete, HP data is stored in this register [31:0].	x

27.2.6.15 LMP Receive Data 1 Register (R32_LINK_LMP_RX_DATA1)

Offset address: 0x5C

Bit	Name	Access	Description	Reset value
[31:0]	LINK_LMP_RX_DATA1	RO	After LMP reception is complete, HP data is	x

	ATA1		stored in this register [63:32].	
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27.2.6.16 LMP Receive Data 2 Register (R32_LINK_LMP_RX_DATA2)

Offset address: 0x60

Bit	Name	Access	Description	Reset value
[31:0]	LINK_LMP_RX_D ATA2	RO	After LMP reception is complete, HP data is stored in this register [95:64].	x

27.2.6.17 USB Custom HP Data 0 Register (R32_LINK_LMP_TX_DATA0)

Offset address: 0x64

Bit	Name	Access	Description	Reset value
[31:0]	LINK_LMP_TX_D ATA0	RW	User-defined data sent to HP [31:0].	x

27.2.6.18 USB Custom HP Data 1 Register (R32_LINK_LMP_TX_DATA1)

Offset address: 0x68

Bit	Name	Access	Description	Reset value
[31:0]	LINK_LMP_TX_D ATA1	RW	User-defined data sent to HP [63:32].	x

27.2.6.19 USB Custom HP Data 2 Register (R32_LINK_LMP_TX_DATA2)

Offset address: 0x6C

Bit	Name	Access	Description	Reset value
[31:0]	LINK_LMP_TX_D ATA2	RW	User-defined data sent to HP [95:64].	x

Note:

1. After writing to LINK_LMP_TX_DATA2, the hardware automatically generates a trigger signal to transmit the information from DATA0/DATA1/DATA2.
2. Custom HP transmission is used for sending undefined packets from registers such as LMP (TX_PORT_CAP/TX_PORT_CFG, etc.) and PING-TP.

Chapter 28 Controller Area Network (CAN)

Controller Area Network is a high-performance communication protocol for serial data communication. The CAN controller provides a complete implementation of the CAN protocol, supporting CAN protocols 2.0A and 2.0B. The CAN controller can be used to construct a powerful local area network to realize safe distributed real-time control, process a large number of data packets with a small CPU load, and has a wide range of applications in the industrial and automotive fields.

28.1 Main Features

- Compatible with CAN specification 2.0A and 2.0B (active)
- Programmable transfer rate up to 1Mbit/s
- Support time-triggered communication function to avoid low-priority message blocking
- Support 3 sending mailboxes, the priority of sending messages can be determined by the message identifier or the order of sending requests, and can record the time stamp of the SOF moment of sending the message
- Each CAN group supports 2 receive FIFOs with 3 levels of mailbox depth. A total of 42 filter groups are available for configuration across three CANs. Each filter group can be configured in 32-bit or 16-bit mode, using mask bits or identifier list mode. This minimizes software intervention in message filtering. FIFO overflow handling is flexible and can record timestamps for the SOF moment of received messages.
- Each group of CAN occupies 4 interrupt vectors, and each interrupt source can be configured independently.

28.2 CAN Controller Operating Mode

The CAN controller can operate the SLEEP or INRQ bits in the register CAN_CTLR to switch between the initialization mode, sleep mode and normal mode.

28.2.1 Initialization Mode

After reset, CAN works in sleep mode by default to reduce power consumption. At this time, the sending and receiving of messages is prohibited, the internal pull-up resistor of the TX pin is enabled, and the TX pin outputs a recessive bit. The INRQ bit in the register CANx_CTLR is set to 1, and the CAN controller is requested to enter the initialization mode. When the INAK bit of the register CANx_STATR is automatically set to 1, the initialization state is successfully entered. Similarly, clear the INRQ bit in the register CANx_CTLR to request the CAN controller to exit the initialization mode. When the INAK bit of the register CANx_STATR is automatically cleared to 0, the initialization state is successfully exited.

The filter group can be initialized in the non-initialization mode, but the FINIT bit of the register CANx_FCTLR must be set to 1, and the reception of the message is prohibited at this time.

28.2.2 Sleep Mode

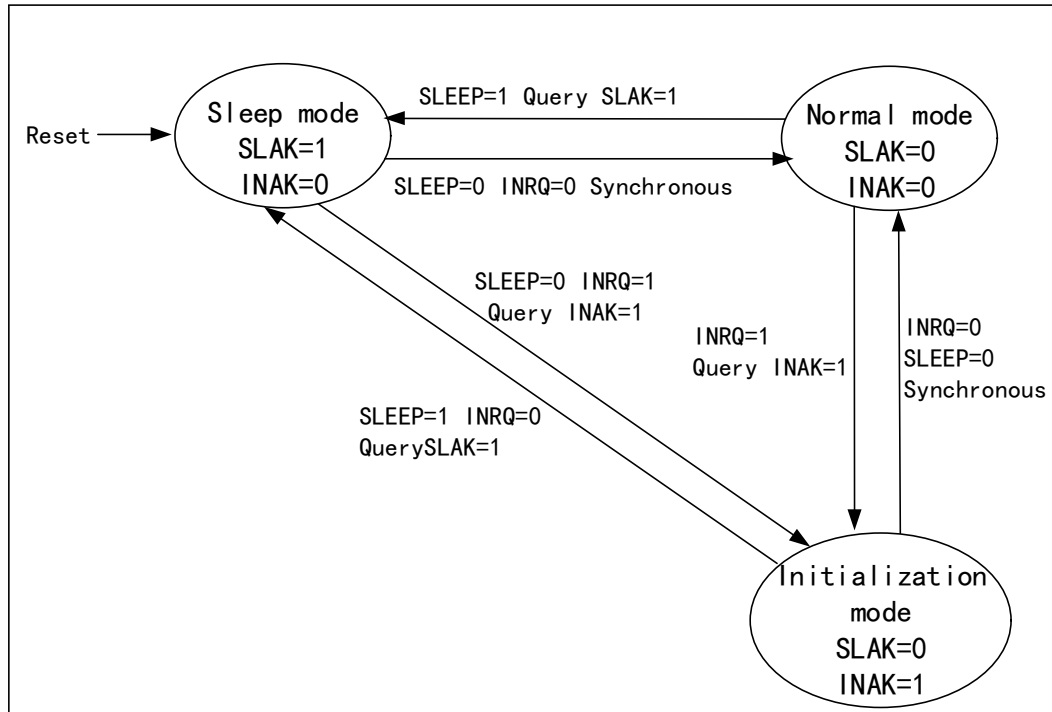
For the SLEEP position 1 in the register CANx_CTLR, the CAN controller is requested to enter the sleep mode. When the SNAK bit of the register CANx_STATR is automatically set to 1, the CAN successfully enters the sleep mode, and the clock of the CAN controller stops, but the mailbox register can still be accessed.

From sleep mode to initialization mode, the SLEEP bit of CANx_CTLR must be cleared to position 1, and when

the INAK bit of register CANx_STATR is automatically set to 1, it will be switched to initialization state.

To enter the normal mode from sleep mode, the SLEEP bit of CANx_CTLR must be cleared 0, and when the SNAK bit of register CANx_STATR is automatically cleared 0, it will enter normal mode.

Figure 28-1 CAN operating mode switch



28.3 CAN Controller Test Mode

In the initialization mode, operate the SILM and LBKM bits of the register CANx_BTMR to select a test mode, and then exit the initialization mode and enter the test mode by clearing the INRQ bit of the register CANx_CTLR. There are 3 test modes: silent mode, loopback mode and silent loopback mode.

28.3.1 Silent Mode

Setting the SILM bit in register CANx_BTMR to 1 can optionally enter silent mode. In this mode, the CAN controller can receive, but cannot send messages to the outside world. It is always in a recessive bit to the outside world, which can avoid affecting the bus, but the message can be received by the controller of the node where it is located. Usually, silent mode is used for the status analysis of the CAN bus.

28.3.2 Loopback Mode

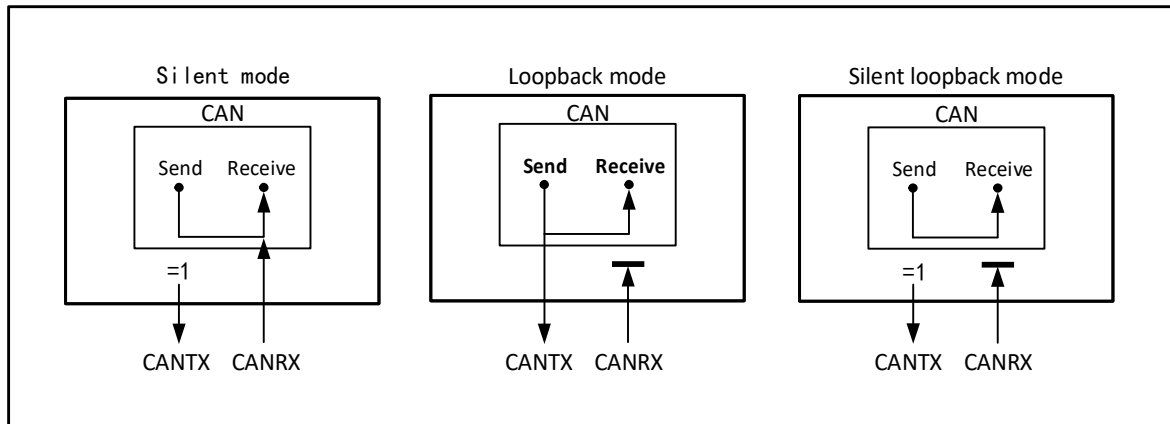
By setting the LBKM bit of the register CANx_BTMR to 1, the loopback mode can be selected. In this mode, the CAN controller can send external messages, but cannot receive external messages, but the sent messages can be received by the controller of the node where it is located, and the reception filtering mechanism is effective. Usually, loopback mode is used for transceiver testing of CAN controllers.

28.3.3 Silent Loopback Mode

Setting the SILM and LBKM bits in register CANx_BTMR to 1 can optionally enter silent loopback mode. This

mode is usually used for the closed self-test of the CAN controller. In this mode, it has no effect on the CAN bus, the RX pin is disconnected from the bus, and the TX pin is set to a recessive bit.

Figure 28-2 CAN 3 test modes of the bus



28.4 MCU Operating State of CAN Controller in Debug Mode

When the MCU enters the debug mode, the core is in a suspended state, but it can be determined whether the CAN controller is in a normal operation or a stop state through the configuration bits in the debug module.

28.5 CAN Controller Functional Description

28.5.1 Transmit Processing Flow

The transmit processing flow is as follows: If there are vacant mailboxes among the 3 sending mailboxes, the application layer software only has write access to the registers of the vacant mailboxes, and operates the registers `CANx_TXMIyR`, `CANx_TXMDTyR`, `CANx_TXMDLyR` and `CANx_TXMDHyR`, and can set the message identifier, message length, time stamp, and message data. After the data is ready, the `TXRQ` bit of the register `CANx_TXMIyR` is set to 1 to request transmission, the mailbox enters the registered state, and the priority is queued; once it becomes the highest priority mailbox, it becomes the scheduled transmission state and waits for the CAN bus to be idle; when the CAN bus is idle When the message is scheduled to be sent, the message of the mailbox will enter the sending state immediately; after the message is sent, the mailbox will become a vacant mailbox again, and the `RQCP` and `TXOK` bits of the register `CANx_TSTATR` are set to 1 to indicate that the sending is successful; if the arbitration fails during sending, the `ALST` of the register `CANx_TSTATR` Set to 1, `TERR` set to 1 if an error is sent.

28.5.2 Transmit Priority

The transmit priority can be determined by the identifier or the order of sending requests. The `TXFP` position of the register `CANx_CTLR` is 1 and sent according to the order of sending requests. According to the order of sending requests, it is mainly used for segmented sending; the `TXFP` bit is cleared to 0 and sent according to the priority of the identifier. In order, the smaller the identifier is, the higher the priority is. In the case of the same identifier, the mailbox with the lower number has a higher priority.

28.5.3 Transmit Abort Processing

If the ABRQ bit in the register CANx_TSTATR is set, the transmission request can be aborted. When the mailbox status is registered or scheduled to send, the sending request is directly aborted; when the mailbox is in the sending state, the abort request may succeed (stop sending) or fail (sending complete), and the result can be queried by the TXOK bit in the CANx_TSTATR register.

28.5.4 Time-based Trigger Mode

When the traditional CAN communication bus is busy, it is easy to cause low-priority messages to be blocked for a long time, and even cannot meet the requirements of its time limit. In order to solve the bottleneck, related protocols based on time-triggered mode have been introduced. Such protocols have a certain scale of application in the industry, and the functions based on time-triggered mode are the application of such protocols.

There are 2 modes to choose from in the time-triggered mode. To use this mode, the automatic retransmission function needs to be turned off. The default mode and enhanced mode are selected by configuring the MODE bit of the CANx_TTCTLR register. Set the TTCM and NART bits of the register CANx_CTLR to 1, enable the time-triggered mode and disable automatic retransmission. The MODE bit of the CANx_TTCTLR register defaults to 0. At this time, it works in the default mode, and the internal timer is activated to generate timestamps of the transmit and receive mailboxes. The timer accumulates at the CAN bit time, and the internal timer is sampled and generates a timestamp at the sample point position of the received and transmitted frame start bits. If the enhanced mode is used, the MODE bit of the CANx_TTCTLR register needs to be set to 1 to enable the enhanced mode. Using this mode, there must be 3 or more nodes in the entire CAN network, one of which sends the time reference, and the other nodes receive the time stamp of the reference node, they reset the internal counter by writing 1 to the TIMRST bit of the CANx_TTCTLR register to synchronize the internal counter. So that in addition to the node that sends the time reference, the rest of the CAN nodes achieve time synchronization. Afterwards, write the data to be sent to the sending mailbox, configure the time trigger count value (TIMCNT of the CANx_TTCNT register) and the internal counter count end value (TIMCMV of the CANx_TTCTLR register) of each node in turn. The time-triggered count value and the final count value of the internal counter are determined by the CAN nodes, the CAN communication rate and the number of data bits in a frame. After the configuration is completed, each node waits for the internal counter to count to the time-triggered count value, and then triggers the transmitting action.

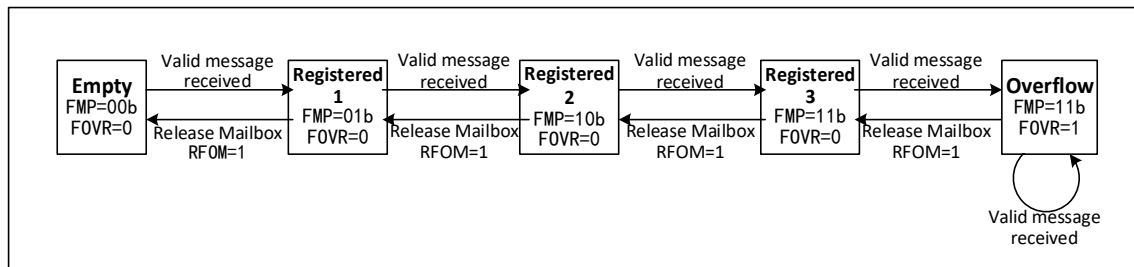
28.5.5 Receive Processing Flow

The reception of CAN bus messages is completed by the controller hardware without the intervention of the MCU, which reduces the processing load of the MCU. The received messages are stored in 2 FIFOs with 3-level mailbox depth according to the setting of the register CANx_FAFIFOR. If the application layer needs to obtain the message, it can only read the valid received message through the receiving FIFO mailbox.

Initially, the receiving FIFO is empty, and the value of FMR[1:0] in the receiving FIFO register CAN_RFIFOR is binary 00b. After receiving a valid receiving message, it becomes the registered 1 state, and the controller automatically sets the FMR[1:0] in the receiving FIFO register CAN_RFIFOR to binary 01b; if the mailbox data registers CANx_RXMDLyR and CANx_RXMDHyR are read at this time, the mailbox is released by setting the RFOM bit of the receiving FIFO register CAN_RFIFOR to 1, and the receiving FIFO state becomes empty again; if the mailbox is not released in the registered 1 state, after the next valid receiving message is received, the receiving FIFO state switches to the registered 2 state; at this time, the FMR[1:0] of the receiving FIFO register CAN_RFIFOR is automatically set to binary 10b. If the mailbox data register is read and the mailbox is released, then the state returns to registered 1; if the mailbox is not released in the registered 2 state, the receiving FIFO enters the registered

3 state; also in the registered 3 state, the message is read and the mailbox is released, then the registered 2 state is returned; if the registered 3 state is not If the mailbox is released, when the next valid packet is received, packet loss will inevitably occur.

Figure 28-3 Receive FIFO state switching diagram



In the case of message loss above, that is, the receiving FIFO is full, and the message overflow causes the message to be lost. The FOVR bit of the receiving FIFO register CANx_RFIFOx will be automatically set to 1 by hardware for overflow query. When the RFLM bit of the register CANx_CTLR is set to 1, the receiving FIFO locking function is enabled, and the discarded message is a new received message; when the RFLM bit of the register CANx_CTLR is cleared to 0, the receiving FIFO locking function is disabled. Among the 3 original messages of the receiving FIFO, the last received message will be overwritten by the new message.

When the relevant bit of the register CANx_INTENR is set, an interrupt can be generated when the state of the receiving FIFO is switched, so as to process the received message more efficiently, see section 28.6 CAN interrupt for details.

28.5.6 Received Message Identifier Filtering

There are as many as 42 filter groups in the module. By setting the filter groups, each CAN node can receive messages that meet the filtering regulars, and messages that do not meet the filtering regulars are discarded by hardware without software intervention.

Each filter group consists of two 32-bit registers: CANx_FxR0 and CANx_FxR1. The bit width of the filter group can be independently configured as either one 32-bit filter or two 16-bit filters by setting individual bits in the CANx_FSCFGR register. Each filter group can be configured as a mask bit or identifier list mode by setting individual bits in the CANx_FMCFGR register. Individual filter groups can be enabled or disabled by setting corresponding bits in the CANx_FWR register. Bits in the CANx_FAFIFOR register determine which receive FIFO stores messages passing through the filter.

As shown in Table 28-1, in mask bit mode, the two registers are the identifier register and the mask register, which must be used in conjunction. Each bit in the identifier register indicates whether the corresponding bit should be set to a dominant or recessive state. Each bit in the mask register indicates whether the corresponding bit should match the desired state specified by the identifier register.

Table 28-1 32-bit mask bit patterns

Identifier register	CANx_FxR1[31:24]	CANx_FxR1[23:16]	CANx_FxR1[15:8]	CANx_FxR1[7:0]
Mask bit register	CANx_FxR2[31:24]	CANx_FxR2[23:16]	CANx_FxR2[15:8]	CANx_FxR2[7:0]
Mapping	STID[10:3]	STID[2:0] EXID[17:13]	EXID[12:5]	EXID[4:0] IDE RTR 0

In identifier list mode, both registers are used as identifier registers, and the identifier of the received message must be consistent with one of the registers before it can pass the screening.

Table 28-2 32-bit identifier list mode

Identifier register	CANx_FxR1[31:24]	CANx_FxR1[23:16]		CANx_FxR1[15:8]	CANx_FxR1[7:0]			
Mask bit register	CANx_FxR2[31:24]	CANx_FxR2[23:16]		CANx_FxR2[15:8]	CANx_FxR2[7:0]			
Mapping	STID[10:3]	STID[2:0]	EXID[17:13]	EXID[12:5]	EXID[4:0]	IDE	RTR	0

In the 16-bit mode, the register group is divided into four registers, and the mask bit mode of each group of filters can have two filters, each of which contains a 16-bit identifier register and a 16-bit mask register; All four registers in identifier list mode are used as identifier registers.

Table 28-3 16-bit mask bit patterns

Identifier register n	CANx_FxR1[15:8]	CANx_FxR1[7:0]			
Mask bit register n	CANx_FxR1[31:24]	CANx_FxR1[23:16]			
Identifier register n+1	CANx_FxR2[15:8]	CANx_FxR2[7:0]			
Mask bit register n+1	CANx_FxR2[31:24]	CANx_FxR2[23:16]			
Mapping	STID[10:3]	STID[2:0]	RTR	IDE	EXID[17:15]

Table 28-4 16-bit identifier list mode

Identifier register n	CANx_FxR1[15:8]	CANx_FxR1[7:0]			
Mask bit register n	CANx_FxR1[31:24]	CANx_FxR1[23:16]			
Identifier register n+1	CANx_FxR2[15:8]	CANx_FxR2[7:0]			
Mask bit register n+1	CANx_FxR2[31:24]	CANx_FxR2[23:16]			
Mapping	STID[10:3]	STID[2:0]	RTR	IDE	EXID[17:15]

Messages entering the FIFO mailbox are read and stored by the application, which typically distinguishes message data based on message identifiers. The CAN controller provides a filter number for messages filtered through different filters in the receive FIFO. This number is stored in the FMI[7:0] field of the CANx_RXMDTyR register, regardless of whether the filter group is enabled. Refer to Figure 28-4 for the numbering regulars.

When a message passes multiple filters, the stored filter number in the receive buffer is determined by the filter priority regulars. The filter priority regulars are as follows:

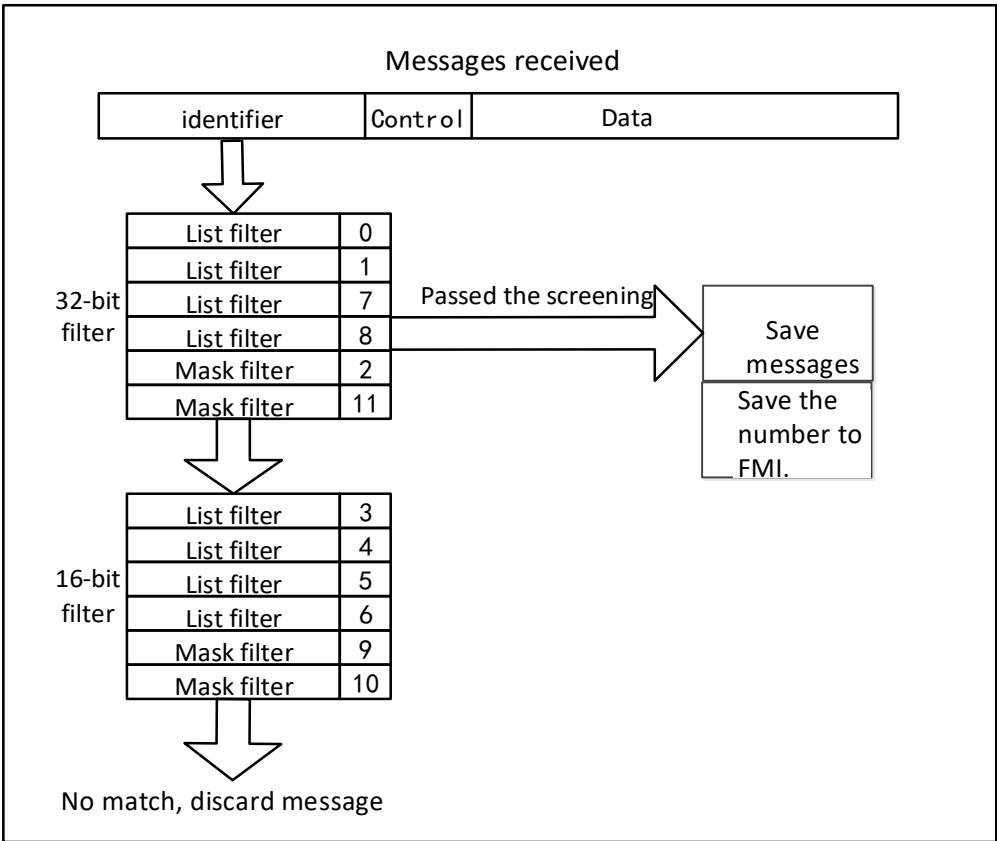
- All 32-bit filters have higher priority than 16-bit filters.
- For filters of the same width, those using an identifier list have higher priority than those using a mask bit pattern.
- For filters with identical width and pattern, the one with the lower number has higher priority.

As shown in Figure 28-5: When receiving a message, the identifier is first matched against the 32-bit identifier list pattern filter. If no match is found, it is then matched against the 32-bit mask pattern filter. If still no match occurs, it proceeds to match against the 16-bit identifier list pattern filter. If no match is found, it finally matches against the 16-bit mask pattern filter. If no match is found at any stage, the message is discarded. If a match occurs, the message is stored in the mailbox of the receive FIFO, and the identifier number is stored in the FMI field of the CANx_RXMDTyR register.

Figure 28-4 Example of Filter Number

Filter group number	FIFO0	Filter number	Filter group number	FIFO1	Filter number
0	32-bit mask mode	0	1	16-bit list mode	0 1 2 3
2	16-bit list mode	1 2 3 4	4	16-bit mask mode	4 5
3	32-bit list mode	5 6	6	32-bit list mode	6 7
5	Disabled 16-bit mask Mode	7 8	9	32-bit mask mode	8
7	32-bit mask mode	9	11	Disabled 16-bit mask Mode	9 10
8	32-bit list mode	10 11	12	32-bit list mode	11 12
10	16-bit mask mode	12 13	13	32-bit list mode	13 14

Figure 28-5 Filter filtering example



28.5.7 Error Handling

The CAN controller relies on the state error register CANx_ERRSR to manage errors on the bus. The TEC and REC

in the state error register `CANx_ERRSR` represent the sending and receiving error count values respectively. According to the increase of the sending and receiving errors and the decrease of the sending and receiving success, the stability of the CAN bus can be judged according to their values.

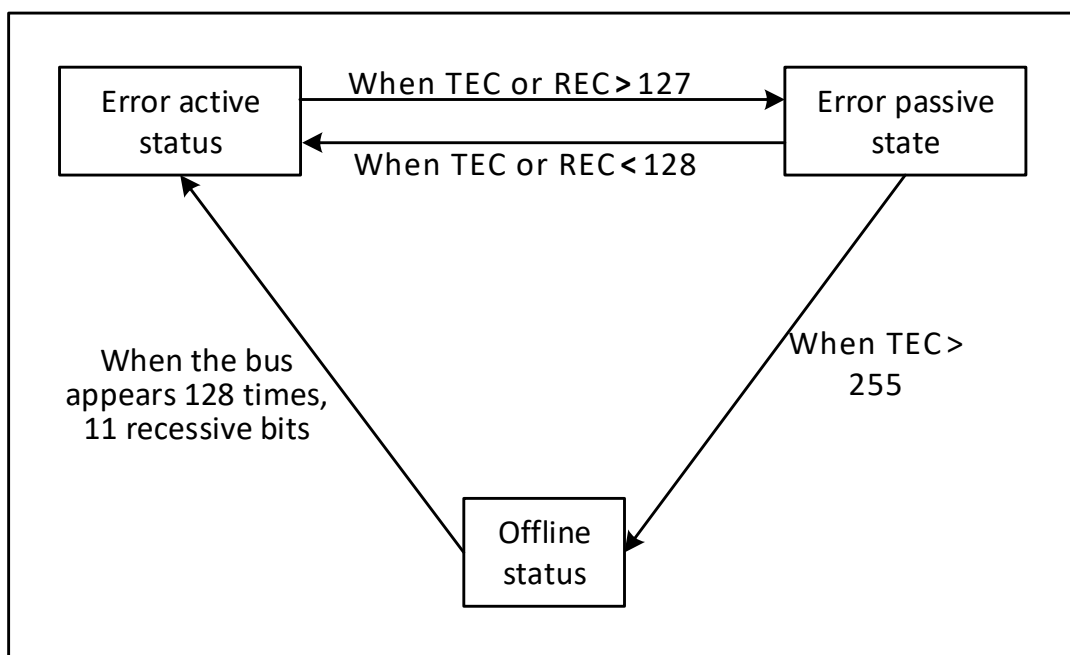
When the TEC and REC in the state error register `CANx_ERRSR` are less than 128, the current CAN node is in an error active state, can normally participate in bus communication, and issue an active error flag when an error is detected.

When the TEC and REC in the state error register `CANx_ERRSR` are greater than 127, the current CAN node is in an error passive state, and when an error is detected, it is not allowed to issue an active error flag, but can only issue a passive error flag.

When the TEC in the status error register `CANx_ERRSR` is greater than 255, the current CAN node goes offline.

When the bus monitors 11 consecutive hidden bits for 128 times, it returns to the error active state, and the recovery mode is affected by the ABOM bit in the main control register `CANx_CTLR`. If ABOM is set to 1, the hardware automatically exits the offline state. If ABOM is 0, the software needs to operate the INRQ bit to enter the initialization mode, and then exit the initialization before exiting the offline state.

Figure 28-6 CAN error state switch diagram



28.5.8 Bit Timing

According to the standard of CAN bus, each bit time is divided into four segments: synchronization segment, propagation time segment, phase buffer segment 1 and phase buffer segment 2. These segments consist of the minimum time unit T_q . The CAN controller monitors the CAN bus change by sampling and synchronizes through the edge of the start bit of the frame.

The CAN controller re-divides the above four segments into 3 segments, which are:

- Synchronization segment (SS): the synchronization segment in the CAN standard is fixed as a minimum time unit, and the expected bit jump normally occurs within this period of time.
- Time period 1 (BS1): contains the propagation time period and phase buffer section 1 in the CAN standard, which can be set to contain 1 to 16 minimum time units and can be automatically extended to compensate for

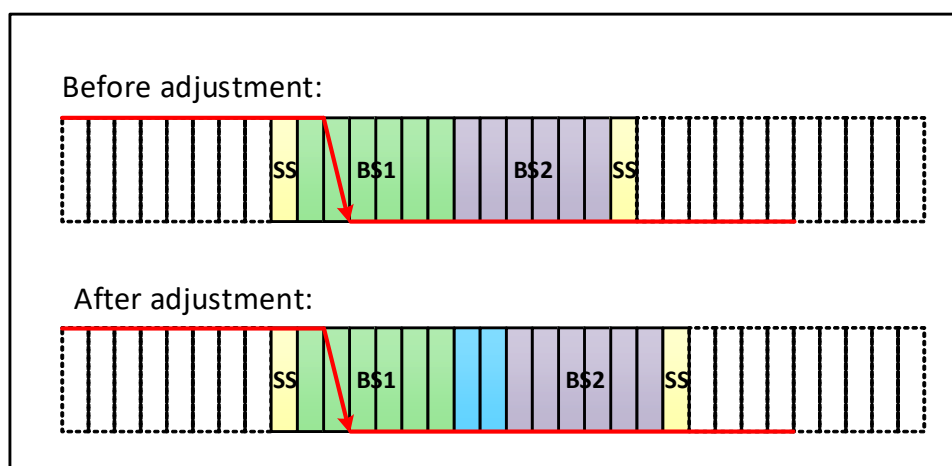
the positive phase drift caused by frequency accuracy errors of different nodes on the CAN bus. The end of the time period is the location of the sampling point.

- Time period 2 (BS2): phase buffer section 2 in the CAN standard can be set to 1 to 8 minimum time units and can be automatically shortened to compensate for the negative phase drift caused by frequency accuracy errors of different nodes on the CAN bus.

Resynchronization jump width (SJW) is the upper limit of the minimum number of time units that can be extended and reduced per person, and the range can be set to 1 to 4 minimum time units.

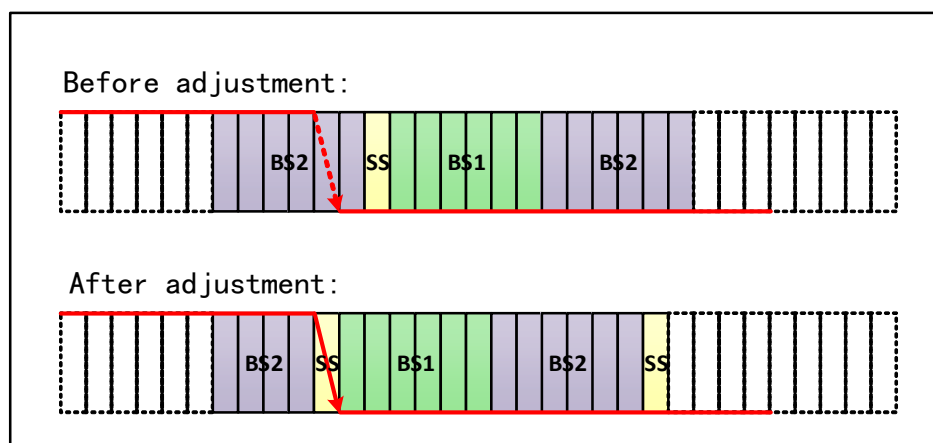
The above parameters can be configured in the CAN bus timing register CANx_BTMR.

Figure 28-7 The jump appears in BS1



If the SJW of figure 28-7 is 2, and the bus level jump is detected in time period 1, the length of time period 1 needs to be extended, with a maximum extension of SJW, thus delaying the position of the sampling point.

Figure 28-8 The jump appears in BS2



If the SJW of figure 28-8 is 2 and the bus level jump is detected in time period 2, it is necessary to reduce the length of time period 2 and the maximum SJW, so as to advance the position of the sampling point.

28.6 CAN Interrupt

The CAN controller has 4 interrupt vectors, which are send interrupt, FIFO_0 interrupt, FIFO_1 interrupt, error and state change interrupt.

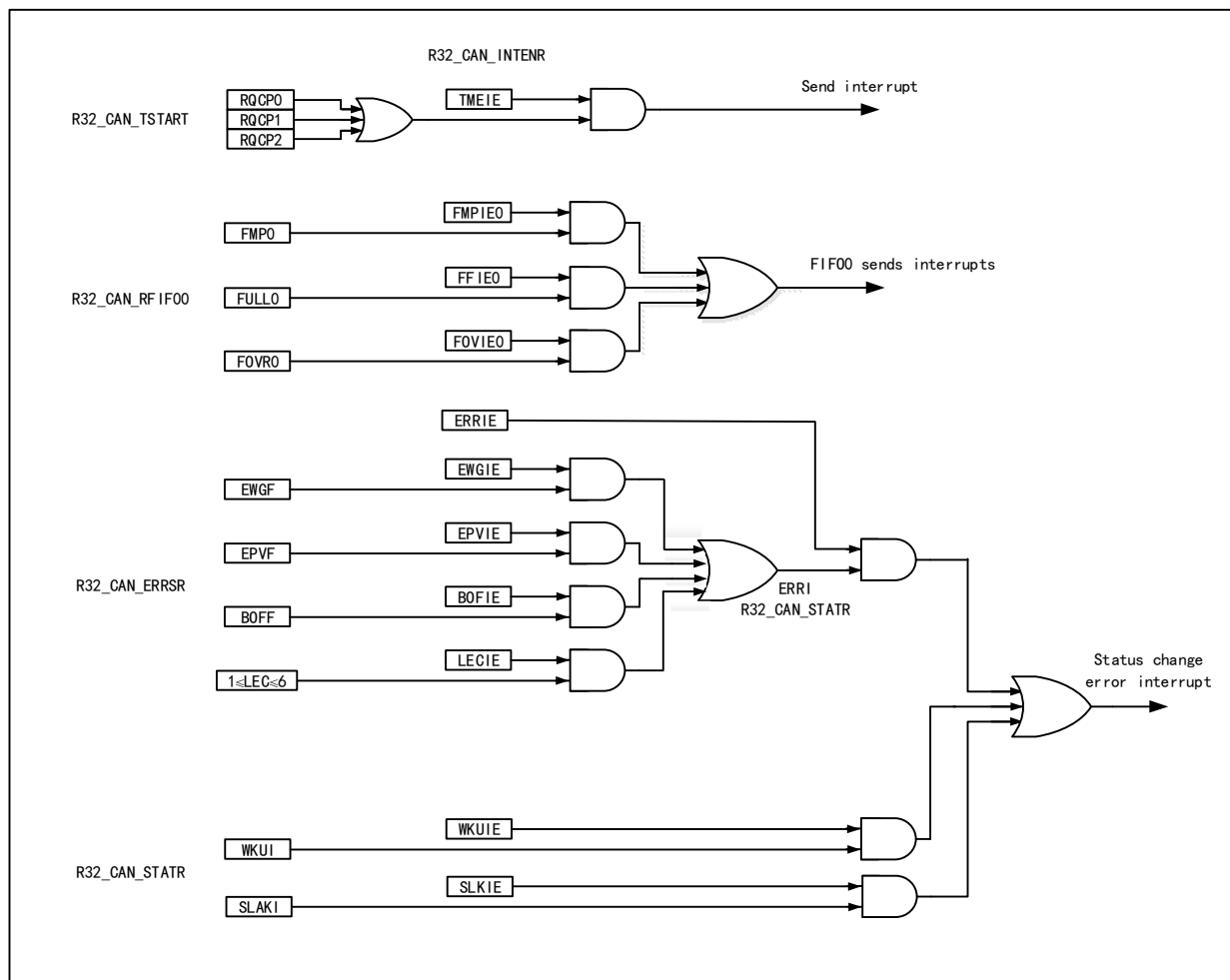
Set the CAN interrupt allow register CANx_INTENR to allow or disable each interrupt source.

The sending interrupt is caused by the sending mailbox empty event. After the interrupt is generated, the RQCP0, RQCP1 and RQCP2 bits of the register CANx_TSTATR are queried to determine which mailbox empty event is generated.

FIFO0 interrupts are caused by receiving new messages, receiving mailbox fullness and overflow events. After the interrupts are generated, the FMP0, FULL0 and FOVER0 bits of register CANx_RFIFO0 are queried to determine which mailbox emptiness event is generated.

Errors and state change interruptions are caused by errors, arousal, and sleep events.

Figure 28-9 CAN interrupt logic diagram



28.7 Register Description

The registers related to the CAN controller must be manipulated in 32-bit words. In order to avoid the influence of the current node on the entire CAN bus, the application software can only modify the bit timing register CAN_BTMR in the initialization mode.

Table 28-6 CAN1-related registers

Name	Access address	Description	Reset value
R32_CAN1_CTLR	0x40006400	CAN1 main control register	0x00010002

R32_CAN1_STATR	0x40006404	CAN1 main status register	0x00000C02
R32_CAN1_TSTATR	0x40006408	CAN1 transmit status register	0x1C000000
R32_CAN1_RFIFO0	0x4000640C	CAN1 receive FIFO0 control and status registers	0x00000000
R32_CAN1_INTENR	0x40006414	CAN1 interrupt enable register	0x00000000
R32_CAN1_ERRSR	0x40006418	CAN1 error status register	0x00000000
R32_CAN1_BTMR	0x4000641C	CAN1 bit timing register	0x01230000
R32_CAN1_TTCTLR	0x40006420	CAN1 time trigger control register	0x0000FFFF
R32_CAN1_TTCNT	0x40006424	CAN1 time trigger count value register	0x00000000
R32_CAN1_TERR_CNT	0x40006428	CAN1 offline recovery error counter	0x00000000

Table 28-7 CAN2-related registers

Name	Access address	Description	Reset value
R32_CAN2_CTLR	0x40006800	CAN1 main control register	0x00010002
R32_CAN2_STATR	0x40006804	CAN1 main status register	0x00000C02
R32_CAN2_TSTATR	0x40006808	CAN1 transmit status register	0x1C000000
R32_CAN2_RFIFO0	0x4000680C	CAN1 receive FIFO0 control and status registers	0x00000000
R32_CAN2_INTENR	0x40006810	CAN1 interrupt enable register	0x00000000
R32_CAN2_ERRSR	0x40006814	CAN1 error status register	0x00000000
R32_CAN2_BTMR	0x40006818	CAN1 bit timing register	0x01230000
R32_CAN2_TTCTLR	0x4000681C	CAN1 time trigger control register	0x0000FFFF
R32_CAN2_TTCNT	0x40006820	CAN1 time trigger count value register	0x00000000
R32_CAN2_TERR_CNT	0x40006824	CAN1 offline recovery error counter	0x00000000

Table 28-8 CAN3-related registers

Name	Access address	Description	Reset value
R32_CAN3_CTLR	0x40006800	CAN3 main control register	0x00010002
R32_CAN3_STATR	0x40006804	CAN3 main status register	0x00000C02
R32_CAN3_TSTATR	0x40006808	CAN3 transmit status register	0x1C000000
R32_CAN3_RFIFO0	0x4000680C	CAN3 receive FIFO0 control and status registers	0x00000000
R32_CAN3_INTENR	0x40006810	CAN3 interrupt enable register	0x00000000
R32_CAN3_ERRSR	0x40006814	CAN3 error status register	0x00000000
R32_CAN3_BTMR	0x40006818	CAN3 bit timing register	0x01230000
R32_CAN3_TTCTLR	0x4000681C	CAN3 time trigger control register	0x0000FFFF
R32_CAN3_TTCNT	0x40006820	CAN3 time trigger count value register	0x00000000
R32_CAN3_TERR_CNT	0x40006824	CAN3 offline recovery error counter	0x00000000

Table 28-9 CAN1 mailbox-related registers

Name	Access address	Description	Reset value
R32_CAN1_TXMIR0	0x40006580	CAN1 Tx mailbox 0 identifier register	X
R32_CAN1_TXMDTR0	0x40006584	CAN1 Tx mailbox 0 data length and timestamp register	X
R32_CAN1_TXMDLR0	0x40006588	CAN1 Tx mailbox 0 data low register	X

R32_CAN1_TXMDHR0	0x4000658C	CAN1 Tx mailbox 0 data high register	X
R32_CAN1_TXMIR1	0x40006590	CAN1 Tx mailbox 1 identifier register	X
R32_CAN1_TXMDTR1	0x40006594	CAN1 Tx mailbox 1 data length and timestamp register	X
R32_CAN1_TXMDLR1	0x40006598	CAN1 Tx mailbox 1 data low register	X
R32_CAN1_TXMDHR1	0x4000659C	CAN1 Tx mailbox 1 data high register	X
R32_CAN1_TXMIR2	0x400065A0	CAN1 Tx mailbox 2 identifier register	X
R32_CAN1_TXMDTR2	0x400065A4	CAN1 Tx mailbox 2 data length and timestamp register	X
R32_CAN1_TXMDLR2	0x400065A8	CAN1 Tx mailbox 2 data low register	X
R32_CAN1_TXMDH2R	0x400065AC	CAN1 Tx mailbox 2 data high register	X
R32_CAN1_RXMI0R	0x400065B0	CAN1 Rx FIFO0 mailbox identifier register	X
R32_CAN1_RXMDT0R	0x400065B4	CAN1 Rx FIFO0 mailbox data length and timestamp register	X
R32_CAN1_RXMDL0R	0x400065B8	CAN1 Rx FIFO0 mailbox data low register	X
R32_CAN1_RXMDH0R	0x400065BC	CAN1 Rx FIFO0 mailbox data high register	X
R32_CAN1_RXMI1R	0x400065C0	CAN1 Rx FIFO1 mailbox identifier register	X
R32_CAN1_RXMDT1R	0x400065C4	CAN1 Rx FIFO1 mailbox data length and timestamp register	X
R32_CAN1_RXMDL1R	0x400065C8	CAN1 Rx FIFO1 mailbox data low register	X
R32_CAN1_RXMDH1R	0x400065CC	CAN1 Rx FIFO1 mailbox data high register	X

Table 28-10 CAN2 mailbox-related registers

Name	Access address	Description	Reset value
R32_CAN2_TXMIR0	0x40006580	CAN2 Tx mailbox 0 identifier register	X
R32_CAN2_TXMDTR0	0x40006584	CAN2 Tx mailbox 0 data length and timestamp register	X
R32_CAN2_TXMDLR0	0x40006588	CAN2 Tx mailbox 0 data low register	X
R32_CAN2_TXMDHR0	0x4000658C	CAN2 Tx mailbox 0 data high register	X
R32_CAN2_TXMIR1	0x40006590	CAN2 Tx mailbox 1 identifier register	X
R32_CAN2_TXMDTR1	0x40006594	CAN2 Tx mailbox 1 data length and timestamp register	X
R32_CAN2_TXMDLR1	0x40006598	CAN2 Tx mailbox 1 data low register	X
R32_CAN2_TXMDHR1	0x4000659C	CAN2 Tx mailbox 1 data high register	X
R32_CAN2_TXMIR2	0x400065A0	CAN2 Tx mailbox 2 identifier register	X
R32_CAN2_TXMDTR2	0x400065A4	CAN2 Tx mailbox 2 data length and timestamp register	X
R32_CAN2_TXMDLR2	0x400065A8	CAN2 Tx mailbox 2 data low register	X
R32_CAN2_TXMDH2R	0x400065AC	CAN2 Tx mailbox 2 data high register	X
R32_CAN2_RXMI0R	0x400065B0	CAN2 Rx FIFO0 mailbox identifier register	X
R32_CAN2_RXMDT0R	0x400065B4	CAN2 Rx FIFO0 mailbox data length and timestamp register	X
R32_CAN2_RXMDL0R	0x400065B8	CAN2 Rx FIFO0 mailbox data low register	X
R32_CAN2_RXMDH0R	0x400065BC	CAN2 Rx FIFO0 mailbox data high register	X
R32_CAN2_RXMI1R	0x400065C0	CAN2 Rx FIFO1 mailbox identifier register	X
R32_CAN2_RXMDT1R	0x400065C4	CAN2 Rx FIFO1 mailbox data length and timestamp register	X
R32_CAN2_RXMDL1R	0x400065C8	CAN2 Rx FIFO1 mailbox data low register	X

R32_CAN2_RXMDH1R	0x400065CC	CAN2 Rx FIFO1 mailbox data high register	X
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Table 28-11 CAN3 mailbox-related registers

Name	Access address	Description	Reset value
R32_CAN3_TXMIR0	0x40006580	CAN3 Tx mailbox 0 identifier register	X
R32_CAN3_TXMDTR0	0x40006584	CAN3 Tx mailbox 0 data length and timestamp register	X
R32_CAN3_TXMDLR0	0x40006588	CAN3 Tx mailbox 0 data low register	X
R32_CAN3_TXMDHR0	0x4000658C	CAN3 Tx mailbox 0 data high register	X
R32_CAN3_TXMIR1	0x40006590	CAN3 Tx mailbox 1 identifier register	X
R32_CAN3_TXMDTR1	0x40006594	CAN3 Tx mailbox 1 data length and timestamp register	X
R32_CAN3_TXMDLR1	0x40006598	CAN3 Tx mailbox 1 data low register	X
R32_CAN3_TXMDHR1	0x4000659C	CAN3 Tx mailbox 1 data high register	X
R32_CAN3_TXMIR2	0x400065A0	CAN3 Tx mailbox 2 identifier register	X
R32_CAN3_TXMDTR2	0x400065A4	CAN3 Tx mailbox 2 data length and timestamp register	X
R32_CAN3_TXMDLR2	0x400065A8	CAN3 Tx mailbox 2 data low register	X
R32_CAN3_TXMDH2R	0x400065AC	CAN3 Tx mailbox 2 data high register	X
R32_CAN3_RXMI0R	0x400065B0	CAN3 Rx FIFO0 mailbox identifier register	X
R32_CAN3_RXMDT0R	0x400065B4	CAN3 Rx FIFO0 mailbox data length and timestamp register	X
R32_CAN3_RXMDL0R	0x400065B8	CAN3 Rx FIFO0 mailbox data low register	X
R32_CAN3_RXMDH0R	0x400065BC	CAN3 Rx FIFO0 mailbox data high register	X
R32_CAN3_RXMI1R	0x400065C0	CAN3 Rx FIFO1 mailbox identifier register	X
R32_CAN3_RXMDT1R	0x400065C4	CAN3 Rx FIFO1 mailbox data length and timestamp register	X
R32_CAN3_RXMDL1R	0x400065C8	CAN3 Rx FIFO1 mailbox data low register	X
R32_CAN3_RXMDH1R	0x400065CC	CAN3 Rx FIFO1 mailbox data high register	X

Table 28-12 CAN filter-related registers

Name	Access address	Description	Reset value
R32_CAN1_FCTLR	0x40006600	CAN1 filter main control register	0x2A1C0E01
R32_CAN1_FMCFGR	0x40006604	CAN1 filter mode register	0x00000000
R32_CAN1_FMCFGR1	0x40006608	CAN1 filter mode register 1	0x00000000
R32_CAN1_FSCFGR	0x4000660C	CAN1 filter bit width register	0x00000000
R32_CAN1_FSCFGR1	0x40006610	CAN1 filter bit width register 1	0x00000000
R32_CAN1_FAFIFOR	0x40006614	CAN1 filter FIFO association register	0x00000000
R32_CAN1_FAFIFOR1	0x40006618	CAN1 filter FIFO association register 1	0x00000000
R32_CAN1_FWR	0x4000661C	CAN1 filter activation register	0x00000000
R32_CAN1_FWR1	0x40006620	CAN1 filter activation register 1	0x00000000
R32_CAN1_F0R1	0x40006640	CAN1 filter group 0 register 1	X
R32_CAN1_F0R2	0x40006644	CAN1 filter group 0 register 2	X
R32_CAN1_F1R1	0x40006648	CAN1 filter group 1 register 1	X
R32_CAN1_F1R2	0x4000664C	CAN1 filter group 1 register 2	X

R32_CAN1_F2R1	0x40006650	CAN1 filter group 2 register 1	X
R32_CAN1_F2R2	0x40006654	CAN1 filter group 2 register 2	X
R32_CAN1_F3R1	0x40006658	CAN1 filter group 3 register 1	X
R32_CAN1_F3R2	0x4000665C	CAN1 filter group 3 register 2	X
R32_CAN1_F4R1	0x40006660	CAN1 filter group 4 register 1	X
R32_CAN1_F4R2	0x40006664	CAN1 filter group 4 register 2	X
R32_CAN1_F5R1	0x40006668	CAN1 filter group 5 register 1	X
R32_CAN1_F5R2	0x4000666C	CAN1 filter group 5 register 2	X
R32_CAN1_F6R1	0x40006670	CAN1 filter group 6 register 1	X
R32_CAN1_F6R2	0x40006674	CAN1 filter group 6 register 2	X
R32_CAN1_F7R1	0x40006678	CAN1 filter group 7 register 1	X
R32_CAN1_F7R2	0x4000667C	CAN1 filter group 7 register 2	X
R32_CAN1_F8R1	0x40006680	CAN1 filter group 8 register 1	X
R32_CAN1_F8R2	0x40006684	CAN1 filter group 8 register 2	X
R32_CAN1_F9R1	0x40006688	CAN1 filter group 9 register 1	X
R32_CAN1_F9R2	0x4000668C	CAN1 filter group 9 register 2	X
R32_CAN1_F10R1	0x40006690	CAN1 filter group 10 register 1	X
R32_CAN1_F10R2	0x40006694	CAN1 filter group 10 register 2	X
R32_CAN1_F11R1	0x40006698	CAN1 filter group 11 register 1	X
R32_CAN1_F11R2	0x4000669C	CAN1 filter group 11 register 2	X
R32_CAN1_F12R1	0x400066A0	CAN1 filter group 12 register 1	X
R32_CAN1_F12R2	0x400066A4	CAN1 filter group 12 register 2	X
R32_CAN1_F13R1	0x400066A8	CAN1 filter group 13 register 1	X
R32_CAN1_F13R2	0x400066AC	CAN1 filter group 13 register 2	X
R32_CAN1_F14R1	0x400066B0	CAN1 filter group 14 register 1	X
R32_CAN1_F14R2	0x400066B4	CAN1 filter group 14 register 2	X
R32_CAN1_F15R1	0x400066B8	CAN1 filter group 15 register 1	X
R32_CAN1_F15R2	0x400066BC	CAN1 filter group 15 register 2	X
R32_CAN1_F16R1	0x400066C0	CAN1 filter group 16 register 1	X
R32_CAN1_F16R2	0x400066C4	CAN1 filter group 16 register 2	X
R32_CAN1_F17R1	0x400066C8	CAN1 filter group 17 register 1	X
R32_CAN1_F17R2	0x400066CC	CAN1 filter group 17 register 2	X
R32_CAN1_F18R1	0x400066D0	CAN1 filter group 18 register 1	X
R32_CAN1_F18R2	0x400066D4	CAN1 filter group 18 register 2	X
R32_CAN1_F19R1	0x400066D8	CAN1 filter group 19 register 1	X
R32_CAN1_F19R2	0x400066DC	CAN1 filter group 19 register 2	X
R32_CAN1_F20R1	0x400066E0	CAN1 filter group 20 register 1	X
R32_CAN1_F20R2	0x400066E4	CAN1 filter group 20 register 2	X
R32_CAN1_F21R1	0x400066E8	CAN1 filter group 21 register 1	X
R32_CAN1_F21R2	0x400066EC	CAN1 filter group 21 register 2	X
R32_CAN1_F22R1	0x400066F0	CAN1 filter group 22 register 1	X
R32_CAN1_F22R2	0x400066F4	CAN1 filter group 22 register 2	X
R32_CAN1_F23R1	0x400066F8	CAN1 filter group 23 register 1	X

R32_CAN1_F23R2	0x400066FC	CAN1 filter group 23 register 2	X
R32_CAN1_F24R1	0x40006700	CAN1 filter group 24 register 1	X
R32_CAN1_F24R2	0x40006704	CAN1 filter group 24 register 2	X
R32_CAN1_F25R1	0x40006708	CAN1 filter group 25 register 1	X
R32_CAN1_F25R2	0x4000670C	CAN1 filter group 25 register 2	X
R32_CAN1_F26R1	0x40006710	CAN1 filter group 26 register 1	X
R32_CAN1_F26R2	0x40006714	CAN1 filter group 26 register 2	X
R32_CAN1_F27R1	0x40006718	CAN1 filter group 27 register 1	X
R32_CAN1_F27R2	0x4000671C	CAN1 filter group 27 register 2	X
R32_CAN1_F28R1	0x40006720	CAN1 filter group 28 register 1	X
R32_CAN1_F28R2	0x40006724	CAN1 filter group 28 register 2	X
R32_CAN1_F29R1	0x40006728	CAN1 filter group 29 register 1	X
R32_CAN1_F29R2	0x4000672C	CAN1 filter group 29 register 2	X
R32_CAN1_F30R1	0x40006730	CAN1 filter group 30 register 1	X
R32_CAN1_F30R2	0x40006734	CAN1 filter group 30 register 2	X
R32_CAN1_F31R1	0x40006738	CAN1 filter group 31 register 1	X
R32_CAN1_F31R2	0x4000673C	CAN1 filter group 31 register 2	X
R32_CAN1_F32R1	0x40006740	CAN1 filter group 32 register 1	X
R32_CAN1_F32R2	0x40006744	CAN1 filter group 32 register 2	X
R32_CAN1_F33R1	0x40006748	CAN1 filter group 33 register 1	X
R32_CAN1_F33R2	0x4000674C	CAN1 filter group 33 register 2	X
R32_CAN1_F34R1	0x40006750	CAN1 filter group 34 register 1	X
R32_CAN1_F34R2	0x40006754	CAN1 filter group 34 register 2	X
R32_CAN1_F35R1	0x40006758	CAN1 filter group 35 register 1	X
R32_CAN1_F35R2	0x4000675C	CAN1 filter group 35 register 2	X
R32_CAN1_F36R1	0x40006760	CAN1 filter group 36 register 1	X
R32_CAN1_F36R2	0x40006764	CAN1 filter group 36 register 2	X
R32_CAN1_F37R1	0x40006768	CAN1 filter group 37 register 1	X
R32_CAN1_F37R2	0x4000676C	CAN1 filter group 37 register 2	X
R32_CAN1_F38R1	0x40006770	CAN1 filter group 38 register 1	X
R32_CAN1_F38R2	0x40006774	CAN1 filter group 38 register 2	X
R32_CAN1_F39R1	0x40006778	CAN1 filter group 39 register 1	X
R32_CAN1_F39R2	0x4000677C	CAN1 filter group 39 register 2	X
R32_CAN1_F40R1	0x40006780	CAN1 filter group 40 register 1	X
R32_CAN1_F40R2	0x40006784	CAN1 filter group 40 register 2	X
R32_CAN1_F41R1	0x40006788	CAN1 filter group 41 register 1	X
R32_CAN1_F41R2	0x4000678C	CAN1 filter group 41 register 2	X

Note: The clock of CAN filter table is provided by CAN1. Therefore, to configure or use the filter table function, the clock of CAN1 must be enabled.

28.7.1 CANx Main Control Register (CANx_CTLR) (x=1/2/3)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved													CFG CAN M	DBF	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RST	Reserved							TTC M	ABO M	AWU M	NAR T	RFL M	TXFP	SLEE P	INRQ

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0
17	CFGCANM	RW	Configure CAN offline recovery time: 1: Recover from offline to normal state in compliance with CAN protocol; 0: Recover from offline to normal state more quickly.	0
16	DBF	RW	Debug whether to disable CAN bus from working 1: During debugging, the CAN transceiver is prohibited, but the control and read and write operations of the receiving FIFO are normal; 0: When debugging, the CAN controller works normally.	1
15	RST	RW1	CAN controller software reset request. Writing 0 to this bit is invalid 1: Reset the CAN controller. After reset, the controller enters sleep mode, and then the hardware automatically clears it to 0; 0: The CAN controller is in normal state.	0
[14:8]	Reserved	RO	Reserved	0
7	TTCM	RW	Whether to allow time trigger mode 1: Enable time trigger mode; 0: Disable time trigger mode. The time trigger mode is mainly used with the TTCAN protocol.	0
6	ABOM	RW	Offline automatic exit control 1: The hardware detects 11 consecutive implicit bits 128 times and automatically exits the offline state; 0: The INRQ bit of the software operation register CAN_CTLR is required to be set to 1 and then cleared to 0. After 11 consecutive implicit bits are detected 128 times, the offline state is exited.	0
5	AWUM	RW	CAN controller automatic wake-up enable 1: When a message is detected, the hardware automatically wakes up, and the SLEEP and SLAK bits of the register CAN_STATR are automatically cleared to 0; 0: Software operation is required to clear the SLEEP	0

			bit of the CAN_CTLR register to wake up the CAN controller.	
4	NART	RW	The automatic packet retransmission function is disabled 1: No matter whether the transmission is successful or not, the message can only be sent once; 0: The CAN controller keeps retransmitting until the transmission is successful.	0
3	RFLM	RW	Receive FIFO message lock mode enable. 1: When the received FIFO overflows, the received mailbox message is not read, and the mailbox is not released, the newly received message is discarded. 0: When the received FIFO overflows, the received mailbox message is not read, and when the mailbox is not released, the newly received message will overwrite the original message. <i>Note: this bit is only used for traditional CAN.</i>	0
2	TXFP	RW	Transmit mailbox priority method selection 1: The priority is determined by the order in which the requests are sent; 0: The priority is determined by the message identifier.	0
1	SLEEP	RW	Sleep mode request 1: Set to 1 to request the CAN controller to enter sleep mode. After the current activity is completed, the controller enters sleep mode. If the AWUM bit is set to 1, the controller clears the SLEEP bit to 0 when a message is received; 0: After the software clears to 0, the controller exits the sleep mode.	1
0	INRQ	RW	Initialize mode request 1: Set to 1 to request the CAN controller to enter the initialization mode. After the current activity is completed, the controller enters the initialization mode, and the hardware sets the INAK bit of the register CAN_STATR to 1; 0: Set to 0 to request the CAN controller to exit the initialization mode and enter the normal mode, and the hardware clears the INAK bit of the CAN_STATR register to 0.	0

28.7.2 CANx Main Status Register (CANx_STATR) (x=1/2/3)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				RX	SAMP	RXM	TXM	Reserved			SLAKI	WKUI	ERRI	SLAK	INAK

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
11	RX	RO	Current actual level of CAN controller Rx pin.	1
10	SAMP	RO	Level of a receive bit on the CAN controller RX pin	1
9	RXM	RO	Rx mode query 1: CAN controller is in Rx mode currently; 0: CAN controller is not in Rx mode currently.	0
8	TXM	RO	Tx mode query 1: CAN controller is in Tx mode currently; 0: CAN controller is not in Tx mode currently.	0
[7:5]	Reserved	RO	Reserved	0
4	SLAKI	RW1	Sleep interrupt enable, that is, interrupt generation flag when the SLKIE bit in the CAN_INTENR register is set to 1, write 1 to clear it, and writing 0 is invalid. 1: When entering sleep mode, an interrupt is generated and the hardware is set to 1; 0: When exiting sleep mode, it can be cleared by hardware or by software.	0
3	WKUI	RW1	Wake-up interrupt flag. When the WKUI bit in the CAN_INTENR register is set to 1, if the SOF bit is detected when the CAN controller is in sleep mode, the hardware will set it to 1. Set to 1 by software to clear to 0, and set to 0 is invalid.	0
2	ERRI	RW1	Error interrupt. When the ERRIE bit in the CAN_INTENR register is set to 1, an error and status change interrupt is generated. This bit is set to 1 and cleared to 0 by software, and set to 0 is invalid.	0
1	SLAK	RO	Sleep mode indication. 1: CAN controller is in sleep mode; 0: CAN controller is not in sleep mode.	1
0	INAK	RO	Initialization mode indication. 1: CAN controller is in initialization mode; 0: CAN controller is not in initialization mode.	0

28.7.3 CANx Transmit Status Register (CANx_TSTATR) (x=1/2/3)

Offset Address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOW 2	LOW 1	LOW 0	TME 2	TME 1	TME 0	CODE[1:0]		ABR Q2	Reserved			TER R2	ALST 2	TXO K2	RQC P2
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABR Q1	Reserved			TER R1	ALST 1	TXO K1	RQC P1	ABR Q0	Reserved			TER R0	ALST 0	TXO K0	RQC P0

Bit	Name	Access	Description	Reset value
31	LOW2	RO	Lowest priority flag for Tx mailbox2 1: The priority of Tx mailbox2 is the lowest; 0: The priority of Tx mailbox2 is not the lowest.	0
30	LOW1	RO	Lowest priority flag for Tx mailbox1 1: The priority of Tx mailbox1 is the lowest; 0: The priority of Tx mailbox1 is not the lowest.	0
29	LOW0	RO	Lowest priority flag for Tx mailbox0 1: The priority of Tx mailbox0 is the lowest; 0: The priority of Tx mailbox0 is not the lowest.	0
28	TME2	RO	Indicates the empty flag bit of sending mailbox 2. 1: Indicates that mailbox 2 does not have a message waiting to be sent. 0: Indicates that mailbox 2 is waiting to send a message.	1
27	TME1	RO	Indicates the empty flag bit of sending mailbox 1. 1: Indicates that mailbox 1 does not have a message waiting to be sent. 0: Indicates that mailbox 1 is waiting to send a message.	1
26	TME0	RO	Indicates the empty flag bit of sending mailbox 0. 1: Indicates that mailbox 0 does not have a message waiting to be sent. 0: Indicates that mailbox 0 is waiting to send a message.	1
[25:24]	CODE[1:0]	RO	Mailbox number. When more than one mailbox is empty, the next mailbox number is empty; when the mailbox is empty, it means the mailbox number with the lowest priority.	00
23	ABRQ2	RW1	Transmit a send abort request for mailbox 2. Software setting 1 can abort the transmitting request of mailbox 2, and the hardware is cleared when transmitting messages are cleared. If mailbox 2 is empty, software setting 1 is invalid.	0
[22:20]	Reserved	RO	Reserved	0
19	TERR2	RW1	Tx mailbox2 transmit error. When Tx mailbox2	0

			fails, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	
18	ALST2	RW1	Tx mailbox 2 arbitration failure flag. When the Tx mailbox 2 has a low arbitration priority and fails to send, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
17	TXOK2	RW1	Tx mailbox 2 transmit OK. 1: The last transmission was successful; 0: The last transmission failed. Set to 1 by software to clear, software write 0 is invalid.	0
16	RQCP2	RW1	Tx mailbox 2 request completion flag, this bit is automatically set to 1 when the sending or aborting request of sending mailbox 2 is completed. Set to 1 by software to clear, software write 0 is invalid.	0
15	ABRQ1	RW1	Tx a transmit abort request for mailbox 1. When the software is set to 1, the transmitting request of mailbox 1 can be aborted and the hardware will clear 0 when the message is cleared. The software write 0 is invalid.	0
[14:12]	Reserved	RO	Reserved	0
11	TERR1	RW1	Tx mailbox1 transmit error. When Tx mailbox1 fails, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
10	ALST1	RW1	Tx mailbox1 arbitration failure flag. When the Tx mailbox1 has a low arbitration priority and fails to send, this bit is automatically set to 1.	0
9	TXOK1	RW1	Tx mailbox1 transmit OK. 1: The last transmission was successful; 0: The last transmission failed. Set to 1 by software to clear, software write 0 is invalid.	0
8	RQCP1	RW1	Tx mailbox1 request completion flag, this bit is automatically set to 1 when the sending or aborting request of sending mailbox 1 is completed. Set to 1 by software to clear, software write 0 is invalid.	0
7	ABRQ0	RW1	Tx a transmit abort request for mailbox 0. Set to 1 by software to abort the transmitting request of mailbox 0, and reset to 0 by hardware when the sent message is cleared. Software write 0 is	0

			invalid.	
[6:4]	Reserved	RO	Reserved	0
3	TERR0	RW1	Tx mailbox0 transmit error. When Tx mailbox0 fails, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
2	ALST0	RW1	Tx mailbox0 arbitration failure flag. When Tx mailbox0 has a low arbitration priority and fails to send, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0
1	TXOK0	RW1	Tx mailbox0 transmit OK. 1: The last transmission was successful; 0: The last transmission failed. Set to 1 by software to clear, software write 0 is invalid.	0
0	RQCP0	RW1	Tx mailbox0 request completion flag. When the send or abort request of Tx mailbox0 is completed, this bit is automatically set to 1. Set to 1 by software to clear, software write 0 is invalid.	0

28.7.4 CANx Receive FIFO 0 Status Register (CANx_RFIFO0) (x=1/2/3)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RFO M0	FOV R0	FULL 0	Reser ved	FMP0[1:0]	

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	RFOM0	RW1	When the software sets this bit to 1, it releases the current mailbox message of the receiving FIFO_0, and automatically clears it to 0 after the release, and software write 0 is invalid.	0
4	FOVR0	RW1	Receive FIFO_0 overflow flag. When there are 3 messages in FIFO_0, a new message is received, and the hardware is set to 1. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
3	FULL0	RW1	Receive FIFO_0 full flag. Set by hardware when there are 3 messages in FIFO_0. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
2	Reserved	RO	Reserved	0

[1:0]	FMP0[1:0]	RO	Number of received FIFO_0 messages.	0
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28.7.5 CANx Receive FIFO 1 Status Register (CANx_RFIFO1) (x=1/2/3)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RFO M1	FOV R1	FULL 1	Reser ved	FMP1[1:0]	

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	RFOM1	RW1	When the software sets this bit to 1, it releases the current mailbox message of the receiving FIFO_1, and automatically clears it to 0 after the release, and software write 0 is invalid.	0
4	FOVR1	RW1	Receive FIFO_1 overflow flag. When there are 3 messages in FIFO_1, a new message is received, and the hardware is set to 1. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
3	FULL1	RW1	Receive FIFO_1 full flag. Set by hardware when there are 3 messages in FIFO_1. This bit needs software to be set to 1 and cleared to 0, and software write 0 is invalid.	0
2	Reserved	RF	Reserved	0
[1:0]	FMP1[1:0]	RO	Number of received FIFO_1 messages.	0

28.7.6 CANx Interrupt Enable Register (CANx_INTENR) (x=1/2/3)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														SLKI E	WKU IE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ERRI E	Reserved			LECI E	BOFI E	EPVI E	EWG IE	Reser ved	FOV IE1	FFIE 1	FMP IE1	FOV IE0	FFIE 0	FMP IE0	TMEI E

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0
17	SLKIE	RW	Sleep interrupt enable. 1: When entering the sleep state, an interrupt is generated;	0

			0: No interrupt is generated when entering sleep state.	
16	WKUIE	RW	Wake-up interrupt enable. 1: When the CAN controller is woken up, an interrupt is generated; 0: No interrupt is generated when the CAN controller wakes up.	0
15	ERRIE	RW	Error interrupt enable, CAN error interrupt always enable. 1: When the CAN controller generates an error, an interrupt is generated; 0: No interrupt is generated when the CAN controller generates an error.	0
[14:12]	Reserved	RF	Reserved.	0
11	LECIE	RW	Last error number interrupt enable. 1: When an error is detected, the hardware updates LEC[2:0], updates the ERRI bit to 1, and triggers an error interrupt; 0: When an error is detected, the hardware updates LEC[2:0], does not update the ERRI bit, and does not trigger an error interrupt.	0
10	BOFIE	RW	Offline interrupt enable. 1: When entering the offline state, update the ERRI bit to 1, triggering an error interrupt; 0: When entering the offline state, the ERRI bit will not be updated, and the error interrupt will not be triggered.	0
9	EPVIE	RW	Error passive interrupt enable. 1: When entering the error passive state, update the ERRI bit to 1, triggering an error interrupt; 0: When entering the error passive state, the ERRI bit is not updated and the error interrupt is not triggered.	0
8	EWGIE	RW	Error warning interrupt enable. 1: When the number of errors reaches the warning threshold, update the ERRI bit to 1, triggering an error interrupt; 0: When the number of errors reaches the warning threshold, the ERRI bit will not be updated, and the error interrupt will not be triggered.	0
7	Reserved	RF	Reserved	0
6	FOVIE1	RW	Receive FIFO_1 overflow interrupt enable. 1: When FIFO_1 overflows, trigger FIFO_1 interrupt;	0

			0: When FIFO_1 overflows, not trigger FIFO_1 interrupt.	
5	FFIE1	RW	Receive FIFO_1 full interrupt enable. 1: When FIFO_1 is full, trigger FIFO_1 interrupt; 0: When FIFO_1 is full, not trigger FIFO_1 interrupt.	0
3	FOVIE0	RW	Receive FIFO_0 overflow interrupt enable. 1: When FIFO_0 overflows, trigger FIFO_0 interrupt; 0: When FIFO_0 overflows, not trigger FIFO_0 interrupt.	0
2	FFIE0	RW	Receive FIFO_0 full interrupt enable. 1: When FIFO_0 is full, trigger FIFO_0 interrupt; 0: When FIFO_0 is full, not trigger FIFO_0 interrupt.	0
1	FMPIE0	RW	Receive FIFO_0 message registration interrupt enable. 1: When FIFO_0 updates the FMP bit and is not 0, trigger FIFO_0 interrupt; 0: When FIFO_0 updates the FMP bit, and it is not 0, the FIFO_0 interrupt is not triggered.	0
0	TMEIE	RW	Tx mailbox empty interrupt. 1: When the Tx mailbox is empty, an interrupt is generated; 0: No interrupt is generated when the Tx mailbox is empty.	0

28.7.6 CANx Error Status Register (CANx_ERRSR) (x=1/2/3)

Offset Address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REC[7:0]								TEC[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								LEC[2:0]			Reserved	BOFF	EPVF	EWGF	

Bit	Name	Access	Description	Reset value
[31:24]	REC[7:0]	RO	Receive error counter. When the CAN receives an error, according to the error condition, the counter is incremented by 1 or 8; after successful reception, the counter is decreased by 1 or set to 120 (the error count value is greater than 127). When the counter value exceeds 127, the CAN enters the error passive state.	0

[23:16]	TEC[7:0]	RO	Transmit error counter. When the CAN sends an error, according to the error condition, the counter is incremented by 1 or 8; after the transmission is successful, the counter is decremented by 1 or set to 120 (the error count value is greater than 127). When the counter value exceeds 127, the CAN enters the error passive state.	0
[15:7]	Reserved	RO	Reserved	0
[6:4]	LEC{2:0}	RW	Last error code. When detecting the sending error on the CAN bus, the controller will set according to the error condition, and set 000b when sending and receiving the message correctly. 000: no error; 001: Bit stuffing error; 010: FORM format error; 011: ACK confirmation error; 100: recessive bit error; 101: Dominant bit error; 110: CRC error; 111: Software settings. Usually when the application software reads the error, the code name is set to 111b, and the code name update can be detected.	0
3	Reserved	RO	Reserved	0
2	BOFF	RO	Offline status flag. When the CAN controller enters the offline state, the hardware automatically sets it to 1; when it exits the offline state, the hardware automatically clears it to 0.	0
1	EPVF	RO	Error passive flag. When the transceiver error counter reaches the error passive threshold, that is, greater than 127, the hardware is set to 1.	0
0	EWGF	RO	Error warning flag bit. When the sending and receiving error counter reaches the warning threshold, that is, greater than or equal to 96, the hardware is set to 1.	0

28.7.8 CANx Bit Timing Register (CANx_BTMR) (x=1/2/3)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SILM	LBK M	Reserved	SJW[3:0]			TS2[3:0]			TS1[3:0]						

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BTR_TS1_T					Reserved		BRP[9:0]								

Bit	Name	Access	Description	Reset value
31	SILM	RW	Silent mode setting. 1: Enter silent mode; 0: Exit silent mode.	0
30	LBKM	RW	Loopback mode setting. 1: Enter loopback mode; 0: Exit loopback mode.	0
[29:28]	Reserved	RO	Reserved	0
[27:24]	SJW[3:0]	RW	Defines the resync jump width setting value. When implementing resynchronization, the upper limit of the minimum number of time units that can be extended and reduced in the bit, the actual value is (SJW[3:0]+1), and the range can be set to 1 to 16 minimum time units.	0001b
[23:20]	TS2[3:0]	RW	Time period 2 set value. It defines how many minimum time units are occupied by time period 2, and the actual value is (TS2[3:0]+1).	0010b
[19:16]	TS1[3:0]	RW	Time period 1 set value. It defines how many minimum time units are occupied by time period 1, and the actual value is (TS1[3:0]+1).	0011b
[15:12]	BTR_TS1_T	RW	For legacy CAN, CLAS_LONG_TS1=0, then TS1 is TS[3:0] (4bit); CLAS_LONG_TS1=1, then TS1 is (TS[1:0]<<4)+BTR_TS1_T[15:12] (6bit).	0
[11:10]	Reserved	RO	Reserved	0
[9:0]	BRP[9:0]	RW	Minimum time unit length setting value $T_q = (BRP[9:0]+1) \times t_{HCLKS}$	0

28.7.9 CANx Time Trigger Control Register (CANx_TTCTLR) (x=1/2/3)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													MOD E	TIMR ST	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMCMV [15:0]															

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0

17	MODE	RW	Time-triggered mode selection. 1: Enhanced mode; 0: Default mode.	0
16	TIMRST	WZ	Internal counter reset control. Write 1 to reset the internal counter, the hardware will automatically clear 0	0
[15:0]	TIMCMV[15:0]	RW	Internal counter count end value	0xFFFF

28.7.10 CANx Time Trigger Count Value Register (CANx_TTCNT) (x=1/2/3)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMCNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	TIMCNT[15:0]	RW	Time-triggered count value	0

28.7.11 CANx Offline Recovery Error Counter (CANx_TERR_CNT) (x=1/2/3)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TX_ERR_CNT							

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
[8:0]	TX_ERR_CNT	RW	Currently, the error count value is recovered offline. If you modify the count value, you can recover it immediately from offline.	0

28.7.12 CANx Tx Mailbox Identifier Register (CANx_TXMIDyR) (x=1/2/3, y=0/1/2)

Offset address: 0x180,0x190,0x1A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STID[10:0]/EXID[28:18]											EXID[17:13]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXID[12:0]												IDE	RTR	TXR Q	

Bit	Name	Access	Description	Reset value
[31:21]	STID[10:0] /EXID[28:18]	RW	The upper 11 bits of a standard or extended identifier.	X
[20:3]	EXID[17:0]	RW	The lower 18 bits of the extended identifier.	X
2	IDE	RW	Identifier selection flag. 1: Extended identifier; 0: Standard identifiers.	X
1	RTR	RW	Remote frame selection flag. 1: Remote frame; 0: Data frame.	X
0	TXRQ	RW	Data transmission request flag. When the software is set to 1, the data in the mailbox is requested to be sent. When the mailbox is empty after sending, the hardware is cleared to 0.	0

28.7.13 CAN Tx Mailbox Data Length and Timestamp Register (CANx_TXMDTyR) (x=1/2/3, y=0/1/2)

Offset address: 0x184, 0x194, 0x1A4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TIME[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TGT	Reserved				DLC[3:0]		

Bit	Name	Access	Description	Reset value
[31:16]	TIME[15:0]	RW	The 16-bit timer value used to send the message SOF time.	X
[15:9]	Reserved	RO	Reserved	0
8	TGT	RW	Message timestamp transmission selection flag. This bit is valid when TTCM is set to 1 and the message length is 8. 1: Transmit timestamp, the value is the immediate value of TIME[15:0], replacing the last 2 bytes of the 8-byte message; 0: No timestamp is transmitted.	X
[7:4]	Reserved	RO	Reserved	0
[3:0]	DLC[3:0]	RW	The data length of the data frame or the data length of the remote frame request. The data length can be set from 0 to 64.	0

28.7.14 CANx Tx Mailbox Data Low Register (CANx_TXMDLyR) (x=1/2/3, y=0/1/2)

Offset address: 0x188, 0x198, 0x1A8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA3[7:0]								DATA2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA1[7:0]								DATA0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA3[7:0]	RW	Tx the content of data byte 3.	X
[23:16]	DATA2[7:0]	RW	Tx the content of data byte 2.	X
[15:8]	DATA1[7:0]	RW	Tx the content of data byte 1.	X
[7:0]	DATA0[7:0]	RW	Tx the contents of data byte 0.	X

28.7.15 CANx Tx Mailbox Data High Register (CANx_TXMDHyR) (x=1/2/3, y=0/1/2)

Offset address: 0x18C, 0x19C, 0x1AC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA7[7:0]								DATA6[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA5[7:0]								DATA4[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA7[7:0]	RW	Tx the content of data byte 7.	X
[23:16]	DATA6[7:0]	RW	Tx the content of data byte 6.	X
[15:8]	DATA5[7:0]	RW	Tx the content of data byte 5.	X
[7:0]	DATA4[7:0]	RW	Tx the contents of data byte 4.	X

28.7.16 CANx Rx Mailbox Identifier Register (CANx_RXMIyR) (x=1/2/3, y=0/1/2)

Offset address: 0x1B0, 0x1C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STID[10:0]/EXID[28:18]											EXID[17:13]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXID[12:0]												IDE	RTR	FDF	

Bit	Name	Access	Description	Reset value
[31:21]	STID[10:0] /EXIDH[28:18]	RO	The upper 11 bits of a standard or extended identifier.	X
[20:3]	EXIDL[17:0]	RO	The lower 18 bits of the extended identifier.	X
2	IDE	RO	Identifier selection flag. 1: Select extended identifier;	X

			0: Use standard identifiers.	
1	RTR	RO	Remote frame selection flag. 1: Remote frame; 0: Data frame.	X
0	FDF	RO	Reserved	0

28.7.17 CANx Rx Mailbox Data Length and Timestamp Register (CANx_RXMDTyR) (x=1/2/3, y=0/1/2)

Offset address: 0x1B4, 0x1C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TIME[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FMI[7:0]								Reserved	RES	ESI	BRS	DLC[3:0]			

Bit	Name	Access	Description	Reset value
[31:16]	TIME[15:0]	RO	16-bit timer value used to receive the SOF time of the message.	0
[15:7]	FMI[7:0]	RO	The filter number to which the message matches.	X
6	RES	RO	The RES bit of the currently received frame	0
5	ESI	RO	The ESI bit of the currently received frame	0
4	BRS	RO	The BRS bit of the currently received frame	0
[3:0]	DLC[3:0]	RO	Received message data length. DLC=0-8: Indicating data frame lengths of 0 through 8 and 0 for remote frames.	X

28.7.18 CANx Rx Mailbox Data Low Register (CANx_RXMDLyR) (x=1/2/3, y=0/1/2)

Offset address: 0x1B8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA3[7:0]								DATA2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA1[7:0]								DATA0[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA3[7:0]	RO	The data byte 3 that receives the message.	X
[23:16]	DATA2[7:0]	RO	The data byte 2 that receives the message.	X
[15:8]	DATA1[7:0]	RO	The data byte 1 that receives the message.	X
[7:0]	DATA0[7:0]	RO	The data byte 0 that receives the message.	X

28.7.19 CANx Rx Mailbox Data High Register (CANx_RXMDHyR) (x=1/2/3, y=0/1/2)

Offset address: 0x1BC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA7[7:0]								DATA6[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA5[7:0]								DATA4[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DATA7[7:0]	RO	The data byte 7 that receives the message.	X
[23:16]	DATA6[7:0]	RO	The data byte 6 that receives the message.	X
[15:8]	DATA5[7:0]	RO	The data byte 5 that receives the message.	X
[7:0]	DATA4[7:0]	RO	The data byte 4 that receives the message.	X

28.7.20 CANx Filter Main Control Register (CANx_FCTLR) (x=1)

Offset address: 0x200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										CAN3SB[5:0]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				CAN2SB[4:0]				Reserved						FINI T	

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0x2A
[21:16]	CAN3SB[5:0]	RW	CAN3 filter start group (CAN3SB[5:0]+1). 000000: Reserved; 000001: CAN3 Filter Groups 2–41; 000010: CAN3 filter group 3–41; ... 100111: CAN3 filter group 40–41; 101000: CAN3 filter group 41; 101001: Reserved; 101010: CAN3 not participating in filter group allocation; Others: Reserved.	0x1C
[15:13]	Reserved	RO	Reserved	0
[12:8]	CAN2SB[4:0]	RW	CAN2 filter start group (CAN2SB[4:0]). 00000: CAN1 not participating in filter group allocation; CAN2 filter group 0~CAN3SB; 00001: CAN1 filter group 0; CAN2 filter group 1~CAN2EB; 00010: CAN1 filter group 0~1; CAN2 filter group 2~CAN2EB;	0xE

			00011: CAN1 filter group 0~2; CAN2 filter group 3~CAN2EB; 11011: CAN1 filter group 0~26; CAN2 filter group 27~CAN2EB; 11100: CAN1 filter group 0~27; CAN2 not participating in filter group allocation; 11101: CAN1 filter group 0~27; CAN2 filter group 29~CAN2EB; 11111: CAN1 filter group 0~27; CAN2 filter group 31~CAN2EB. <i>Note: When CAN3SB[5:0]=42, CAN2EB=41; When CAN3SB[5:0] is not equal to 42, CAN2EB=CAN3SB[5:0].</i>	
[7:1]	Reserved	RO	Reserved	0
0	FINIT	RW	Filter initialization mode enable flag. 1: Filter group is in initialization mode; 0: Filter group is in normal mode.	1

Note:

1. When CAN2SB[4:0] equals 28, CAN3SB[5:0] can take values from 27 to 40 and 42;
2. When CAN2SB[4:0] does not equal 28, the value of CAN2SB[4:0] must be less than or equal to the value of CAN3SB[5:0].

28.7.21 CANx Filter Mode Register (CANx_FMCFGR) (x=1)

Offset address: 0x204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				FBM 27	FBM 26	FBM 25	FBM 24	FBM 23	FBM 22	FBM 21	FBM 20	FBM 19	FBM 18	FBM 17	FBM 16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FBM 15	FBM 14	FBM 13	FBM 12	FBM 11	FBM 10	FBM 9	FBM 8	FBM 7	FBM 6	FBM 5	FBM 4	FBM 3	FBM 2	FBM 1	FBM 0

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:0]	FBMx	RW	The FINIT bit must be set to 1 to write to the operating mode control bit of filter group x. 1: Filter group x registers operate in identifier list mode; 0: Filter group x registers operate in mask bit mode.	0

28.7.22 CANx Filter Mode Register (CANx_FMCFGR1) (x=1)

Offset address: 0x208

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FBM41	FBM40	FBM39	FBM38	FBM37	FBM36	FBM35	FBM34	FBM33	FBM32	FBM31	FBM30	FBM29	FBM28	

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	FBMx	RW	The FINIT bit must be set to 1 to write to the operating mode control bit of filter group x. 1: Filter group x registers operate in identifier list mode; 0: Filter group x registers operate in mask bit mode.	0

28.7.23 CANx Filter Bit Width Register (CANx_FSCFGR) (x=1)

Offset address: 0x20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				FSC27	FSC26	FSC25	FSC24	FSC23	FSC22	FSC21	FSC20	FSC19	FSC18	FSC17	FSC16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FSC15	FSC14	FSC13	FSC12	FSC11	FSC10	FSC9	FSC8	FSC7	FSC6	FSC5	FSC4	FSC3	FSC2	FSC1	FSC0

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:0]	FSCx	RW	Bit width control bit for filter group x; FINIT must be 1 to write. 1: Filter group x registers are a single 32-bit register; 0: Filter group x registers are two 16-bit registers.	0

28.7.24 CANx Filter Bit Width Register (CANx_FSCFGR1) (x=1)

Offset address: 0x210

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FSC41	FSC40	FSC39	FSC38	FSC37	FSC36	FSC35	FSC34	FSC33	FSC32	FSC31	FSC30	FSC29	FSC28	

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	FSCx	RW	Bit width control bit for filter group x; FINIT must be 1 to write. 1: Filter group x registers are a single 32-bit register; 0: Filter group x registers are two 16-bit registers.	0

28.7.25 CANx Filter FIFO Association Register (CANx_FAFIFOR) (x=1)

Offset address: 0x214

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				FFA2 7	FFA2 6	FFA2 5	FFA2 4	FFA2 3	FFA2 2	FFA2 1	FFA2 0	FFA1 9	FFA1 8	FFA1 7	FFA1 6
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FFA1 5	FFA1 4	FFA1 3	FFA1 2	FFA1 1	FFA1 0	FFA9	FFA8	FFA7	FFA6	FFA5	FFA4	FFA3	FFA2	FFA1	FFA0

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:0]	FFAx	RW	The associated FIFO control bit for filter group x must be FINIT=1 to enable writing. 1: Filter group x is associated with FIFO_1; 0: Filter group x is associated with FIFO_0.	0

28.7.26 CANx Filter FIFO Association Register (CANx_FAFIFOR1) (x=1)

Offset address: 0x218

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FFA4 1	FFA4 0	FFA3 9	FFA3 8	FFA3 7	FFA3 6	FFA3 5	FFA3 4	FFA3 3	FFA3 2	FFA3 1	FFA3 0	FFA2 9	FFA2 8	

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	FFAx	RW	The associated FIFO control bit for filter group x must be FINIT=1 to enable writing. 1: Filter group x is associated with FIFO_1; 0: Filter group x is associated with FIFO_0.	0

28.7.27 CANx Filter Activation Register (CANx_FWR) (x=1)

Offset address: 0x21C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved				FACT 27	FACT 26	FACT 25	FACT 24	FACT 23	FACT 22	FACT 21	FACT 20	FACT 19	FACT 18	FACT 17	FACT 16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FACT 15	FACT 14	FACT 13	FACT 12	FACT 11	FACT 10	FACT 9	FACT 8	FACT 7	FACT 6	FACT 5	FACT 4	FACT 3	FACT 2	FACT 1	FACT 0

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:0]	FACTx	RW	Activation control bit for filter group x. 1: Filter group x activated; 0: Filter group x disabled.	0

28.7.28 CANx Filter Activation Register (CANx_FWR1) (x=1)

Offset address: 0x220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FACT 41	FACT 40	FACT 39	FACT 38	FACT 37	FACT 36	FACT 35	FACT 34	FACT 33	FACT 32	FACT 31	FACT 30	FACT 29	FACT 28	

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	FACTx	RW	Activation control bit for filter group x. 1: Filter group x activated; 0: Filter group x disabled.	0

28.7.29 CANx Filter Group Register (CANx_FiRy) (x=1, i=0-41, y=1/2)

Offset address: 0x240-0x38C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FB31	FB30	FB29	FB28	FB27	FB26	FB25	FB24	FB23	FB22	FB21	FB20	FB19	FB18	FB17	FB16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FB15	FB14	FB13	FB12	FB11	FB10	FB9	FB8	FB7	FB6	FB5	FB4	FB3	FB2	FB1	FB0

Bit	Name	Access	Description	Reset value
[31:0]	FB	RW	The FINIT flag in the filter group register must be set to 1 before writing. Identifier mode: 1: The corresponding bit expects a low level; 0: The corresponding bit expects a high level. Mask bit mode: 1: Must match the corresponding identifier register bit;	0

			0: Does not need to match the corresponding identifier register bit.	
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Chapter 29 Digital Video Port (DVP)

Digital Video Port (DVP) is used to connect the camera module to obtain the image data stream. 8/10/12-bit parallel interface communication is provided, which supports the maximum input frequency of 150MHz pixel clock. It supports image data organized in original line and frame formats, such as YUV and RGB, and also supports compressed image data in JPEG format, and can receive high-speed parallel data streams output by external 8-bit, 10-bit and 12-bit camera modules. When receiving, it mainly depends on the synchronization of VSYNC and HSYNC signals. Support image cropping function.

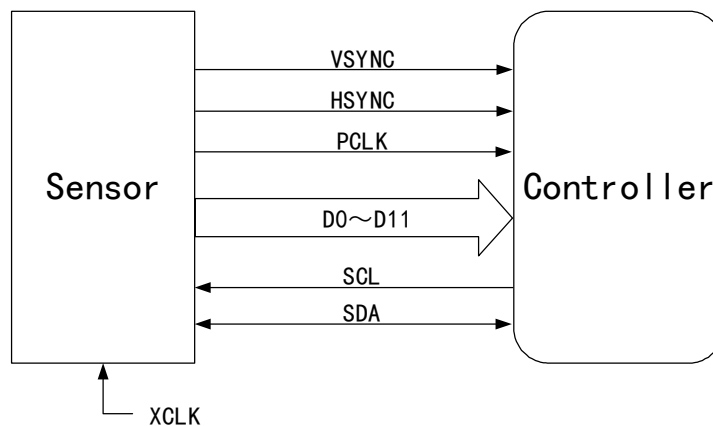
29.1 Main Features

- Configurable 8/10/12-bit data width modes
- Support YUV and RGB format data
- Support JPEG compressed format data
- Built-in FIFO with DMA transfer support
- Support dual-buffered reception
- Support cropping functionality
- Support continuous mode and snapshot mode

29.2 Register Description

29.2.1 Connected to the Sensor

Figure 29-1 DVP interface connection



- PCLK (Pixel clk): pixel clock, each clock corresponds to single pixel data (Uncompressed data). The PCLK clock output by the external DVP interface sensor supports a maximum of 96MHz.
- HSYNC (Horizontal synchronization): horizontal synchronization signal.
- VSYNC (Vertical synchronization): Frame synchronization signal.
- DATA: pixel data or compressed data, the bit width supports 8/10/12 bits.
- XCLK: The reference clock of the Sensor, which can be provided by the microcontroller or externally, generally using a crystal oscillator.
- I2C interface: The register used to configure the sensor, which can simulate the I2C timing communication through the controlled common GPIO port, or use the hardware I2C interface to operate.

Table 29-1 DVP pins

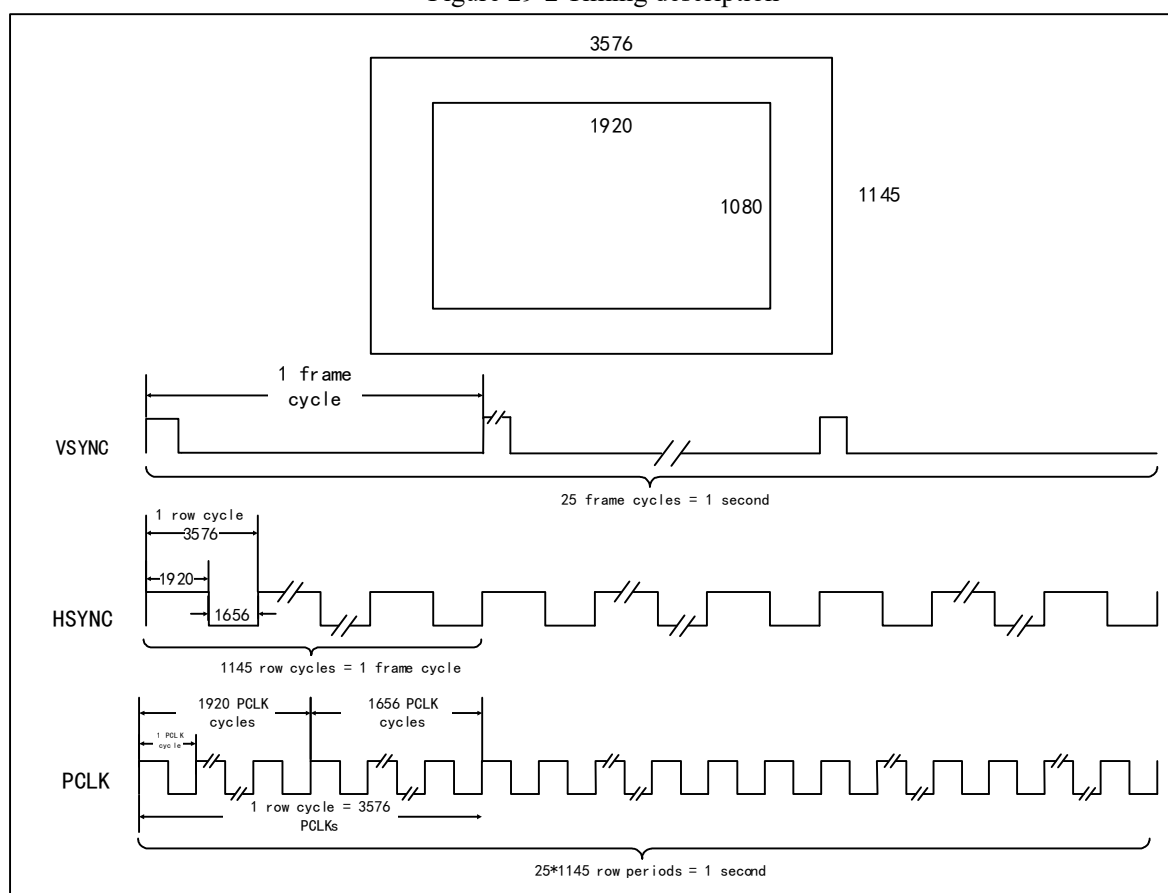
Name	Signal Type Description
PLCK	Pixel clock input
DATA[11:0]	Pixel data input
VSYNC	Frame sync signal input
HSYNC	Line sync signal input

29.2.2 Working Timing

There is a certain relationship between the digital signal data stream output by the commonly used DVP interface sensor module and the image size. The following is an example of image data.

As shown in Figure 29-2, the inside of the sensor is a complete image size of 3576*1145. After internal scaling process, the final image data size output from the interface is 1920*1080, and the image refresh rate is 25.

Figure 29-2 Timing description



PCLK is the transmission time of a pixel, so HSYNC is 3576 times more than PCLK. Among the 3576 pixels, only 1920 pixels are valid, and the sensor does not transmit data during the remaining 1656 pixels. VSYNC is a frame synchronization signal, so the VSYNC time is 3576*1145 times of PCLK. Similarly, the sensor is transmitting data only during the 1920*1080 effective pixel time. If the sensor is transmitting JPEG compressed data, the HSYNC signal may not be needed.

The relationship between DVP interface signal and image data is mainly based on the description of the selected sensor data sheet.

29.2.3 RGB/YUV/JPEG Compressed Data Format Description

- RGB
3 primary colors: Red, green, blue
- YUV
The luminance signal Y, the chrominance signal U and V, Pb and Pr, Cb and Cr.
- JPEG (Joint Photographic Experts Group)

There is lossy compression, but the lost part is the part that is not easily perceptible by human vision, and the human eye is insensitive to the high-frequency information part of the computer color. Remove the visual redundant information (Spatial redundancy) and remove the redundant information of the data itself (Structural redundancy).

29.3 Application of Digital Video Port

29.3.1 Digital Video Port Configuration Description

- 1) In the data reception of DVP, each frame of data is stored alternately by BUF0 and BUF1, starting from BUF0. For RGB and YUV data streams, the hardware resets and selects BUF0 every time the frame signal changes from an inactive level to an active level. When a line of data is full, BUF1 will be switched to realize alternate storage; for JPEG compressed data, the hardware will set the switching threshold of BUF0 and BUF1 according to the set DMA receive length.
- 2) When the data bus width is 10-bit or 12-bit, each time a data is received, the system will automatically unsigned-extend the data to 16-bit and then store it.
- 3) The R16_DVP_ROW_NUM and R16_DVP_COL_NUM registers must match the image size actually output by the sensor.
- 4) In video stream RGB mode, R16_DVP_COL_NUM represents the number of valid PCLK cycles of 1-line data, and R16_DVP_ROW_NUM represents the number of lines contained in a frame of image data; in image JPEG mode, R16_DVP_COL_NUM is used to configure the DMA length, and in this mode, the R16_DVP_ROW_NUM register cannot be used effectively.

29.3.2 Digital Video Port Application Note

When using the digital image interface to receive image data, the DVP control registers must be correctly configured to match the mode of the image sensor. The specific operation steps are as follows:

- 1) Clear RB_DVP_ALL_CLR and RB_DVP_RCV_CLR using the R8_DVP_CR1 register.
- 2) Configure the image mode, data bit width, PCLK polarity, HSYNC polarity and VSYNC polarity through the R8_DVP_CR0 register to match the output of the sensor.
- 3) According to the effective image pixels output by the configured image sensor, configure the R16_DVP_ROW_NUM and R16_DVP_COL_NUM registers to match the output of the sensor. In the image JPEG mode, only configure the R16_DVP_COL_NUM register.
- 4) Configure the DMA receive address through the R32_DVP_DMA_BUF0/1 register.
- 5) If the snapshot mode is used, the RB_DVP_CM field needs to be configured through the R8_DVP_CR1 register to enable the snapshot mode.
- 6) If the cropping mode is used, the RB_DVP_CROP and RB_DVP_FCRC fields need to be configured through the R8_DVP_CR1 register to enable the cropping function and control the frame capture rate. At the same time,

configure the R16_DVP_HOFFCNT, R16_DVP_VST, R16_DVP_CAPCNT and R16_DVP_VLINE to set the size of the cropped image.

- 7) According to the requirements, enable the corresponding interrupt through the R8_DVP_IER register, configure the interrupt priority through the interrupt controller NVIC or PFIC, and enable the DVP interrupt.
- 8) The DMA is enabled through the R8_DVP_CR1 register, and the DVP interface is enabled through the R8_DVP_CR0 register.
- 9) Wait for the generation of the relevant receiving interrupt, and process the received data in time.

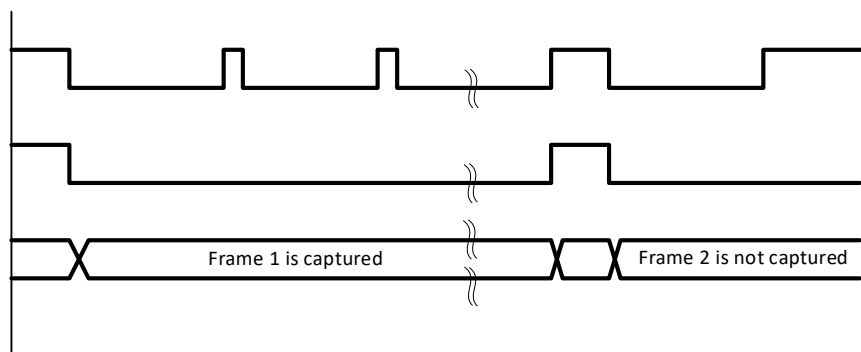
29.3.3 Capture Mode Description

The DVP interface supports 2 capture modes: snapshot (Single frame) mode and continuous mode.

29.3.3.1 Snapshot Mode

In this mode, only a single frame is captured (RB_DVP_CM is set to 1 in the R8_DVP_CR1 register). When the DVP interface is enabled, wait for the system to detect the start of the frame, and then start to sample the image data. After receiving a complete frame of data, the DVP interface will be turned off. (The RB_DVP_ENABLE field of the R8_DVP_CR0 register is cleared to 0). Figure 29-3 shows the single frame capture waveform in snapshot mode.

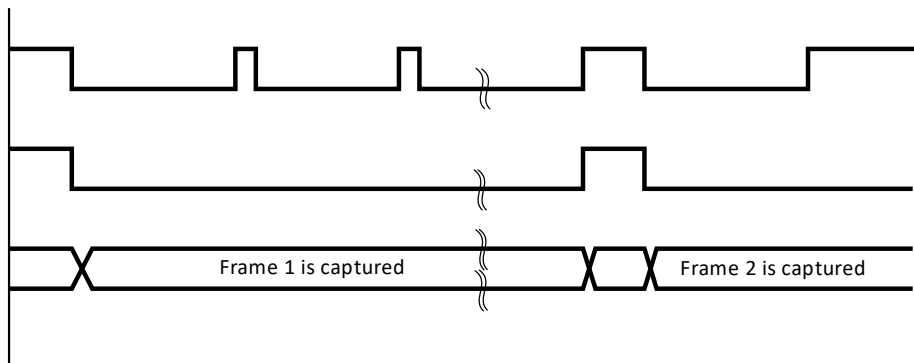
Figure 29-3 Single frame capture waveform in Snapshot mode



29.3.3.2 Continuous Mode

In this mode (RB_DVP_CM is cleared to 0 in the R8_DVP_CR1 register), after the DVP interface is enabled, each frame of data will be continuously sampled before the RB_DVP_ENABLE field of the R8_DVP_CR0 register is cleared to 0. The frame capture waveform in continuous mode is shown in Figure 29-4.

Figure 29-4 Frame capture waveform in continuous mode



29.3.4 Cropping Function Description

DVP can use the crop function to cut a rectangular window from the received image. The starting coordinates of the rectangular window (X coordinate of the upper left corner of the rectangle R16_DVP_HOFFCNT, Y coordinate R16_DVP_VST) and window size (R16_DVP_CAPCNT represents the horizontal size, R16_DVP_VLINE represents the vertical size) are configurable. The coordinates and size of the cropping window are shown in Figure 29-5. The cropped window data capture waveform is shown in Figure 29-6.

Figure 29-5 Crop window coordinates and size

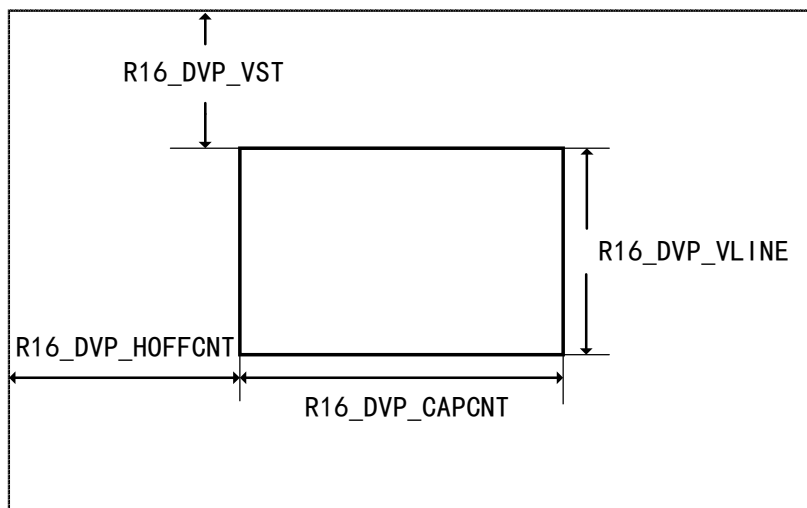
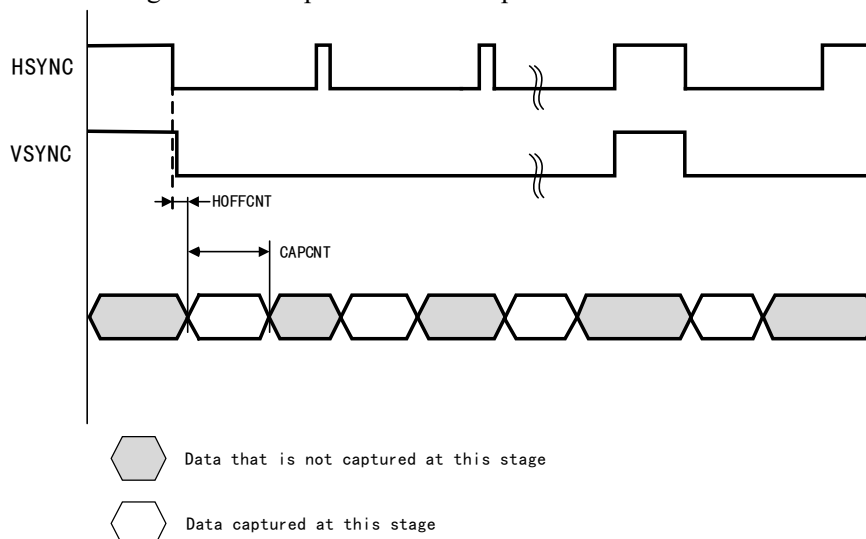


Figure 29-6 Crop window data capture waveform



29.4 Register Description

Table 29-2 DVP registers

Name	Access address	Description	Reset value
R8_DVP_CR0	0x40025800	DVP control register0	0x00
R8_DVP_CR1	0x40025801	DVP control register1	0x06
R8_DVP_IER	0x40025802	DVP interrupt enable register	0x00

R16_DVP_ROW_NUM	0x40025804	DVP row number configuration register	0x0000
R16_DVP_COL_NUM	0x40025806	DVP column number configuration register	0x0000
R32_DVP_DMA_BUF0	0x40025808	DVP DMA address 0 register	0x00000000
R32_DVP_DMA_BUF1	0x4002580C	DVP DMA address 1 register	0x00000000
R8_DVP_IFR	0x40025810	DVP interrupt flag register	0x00
R8_DVP_STATUS	0x40025811	DVP receive FIFO status register	0x00
R16_DVP_ROW_CNT	0x40025814	DVP row count register	0x0000
R16_DVP_HOFFCNT	0x40025818	Horizontal displacement register for the start of the window	0x0000
R16_DVP_VST	0x4002581A	Window start line number register	0x0000
R16_DVP_CAPCNT	0x4002581C	DVP capture count register	0x0000
R16_DVP_VLINE	0x4002581E	DVP vertical line count register	0x0000
R32_DVP_DR	0x40025820	DVP data register	0x00000000

29.4.1 DVP Control Register (R8_DVP_CR0)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
6	RB_DVP_JPEG	RW	JPEG mode enable. 0: Raw data format; 1: JPEG compression format.	0
[5:4]	RB_DVP_MSK_DAT_MOD[1:0]	RW	DVP data bit width configuration. 00: 8-bit mode; 01: 10-bit mode; 1x: 12-bit mode.	00b
3	RB_DVP_P_POLAR	RW	PCLK polarity configuration. 0: Sample data on the rising edge of PCLK; 1: Sample data on the falling edge of PCLK.	0
2	RB_DVP_H_POLAR	RW	HSYNC polarity configuration. 0: HSYNC high level data is valid; 1: HSYNC low level data is valid.	0
1	RB_DVP_V_POLAR	RW	VSYNC polarity configuration. 0: VSYNC low level data is valid; 1: VSYNC high level data is valid.	0
0	RB_DVP_ENABLE	RW	DVP enable. 0: Disable DVP; 1: Enable DVP.	0

29.4.2 DVP Control Register (R8_DVP_CR1)

Offset address: 0x01

Bit	Name	Access	Description	Reset value
[7:6]	RB_DVP_FCRC	RW	DVP frame capture rate control. 00: Capture all frames; 01: Capture every other frame;	00b

			10: Capture every third frame. 11: Reserved	
5	RB_DVP_CROP	RW	Crop function control. 0: Capture full image; 1: Capture only the data in the window specified by the clipping register.	0
4	RB_DVP_CM	RW	Capture mode. 0: Continuous mode; 1: Snapshot mode.	0
3	RB_DVP_BUF_TOG	RWT	Buffer address flag. Hardware control toggle, the software sets 1 to flip this bit, and writing 0 is invalid. 0: Data is stored at receive address 0; 1: Data is stored at receive address 1.	0
2	RB_DVP_RCV_CLR	RW	Receive logic reset control. 0: Cancel the reset operation; 1: Reset the receiving logic circuit.	1
1	RB_DVP_ALL_CLR	RW	Flag and FIFO clear control, write 1 or write 0 by software: 0: Cancel the reset operation; 1: Reset flag and FIFO.	1
0	RB_DVP_DMA_ENABLE	RW	DMA enable: 0: Disable DMA; 1: Enable DMA.	0

29.4.3 DVP Interrupt Enable Register (R8_DVP_IER)

Offset address: 0x02

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved	0
4	RB_DVP_IE_STP_FRM	RW	Frame stop interrupt enable. (Interrupt is generated when VSYNC changes from an active level to an inactive level.) 0: Disable frame stop interrupt; 1: Enable frame stop interrupt.	0
3	RB_DVP_IE_FIFO_OV	RW	Receive FIFO overflow interrupt enable. 0: Disable FIFO overflow interrupt; 1: Enable FIFO overflow interrupt.	0
2	RB_DVP_IE_FRM_DONE	RW	Frame reception done interrupt enable. (Interrupt is generated when the counter reaches the ROW/COL_NUM configuration value, indicating that the last data has been written to RAM) 0: Disable frame reception done interrupt; 1: Enable frame reception done interrupt.	0
1	RB_DVP_IE_ROW_DONE	RW	Row done interrupt enable. (Interrupt is	0

			generated when the counter reaches the COL_NUM configuration value) 0: Disable row done interrupt; 1: Enable row done interrupt.	
0	RB_DVP_IE_STR_FRM	RW	New frame start interrupt enable. (Interrupt is generated when VSYNC changes from an inactive level to an active level, indicating that a new frame starts and data is coming.) 0: Disable new frame start interrupt; 1: Enable new frame start interrupt.	0

29.4.4 DVP Row Number Configuration Register (R16_DVP_ROW_NUM)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_ROW_NUM[15:0]	RW	In RGB mode, it indicates the number of rows contained in a frame of image data. In JPEG mode, this register has no practical meaning.	0

29.4.5 DVP Column Number Configuration Register (R16_DVP_COL_NUM)

Offset address: 0x06

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_COL_NUM[15:0]	RW	In RGB mode, indicates the number of PCLK cycles contained within a line of data. In JPEG mode, it is used to configure the DMA receive length.	0

29.4.6 DVP DMA Address 0 Register (R32_DVP_DMA_BUF0)

Offset address: 0x08

Bit	Name	Access	Description	Reset value
[31:0]	RB_DVP_DMA_BUF0[31:0]	RW	DMA receive address 0 (32-byte alignment).	0

29.4.7 DVP DMA Address 1 Register (R32_DVP_DMA_BUF1)

Offset address: 0x0C

Bit	Name	Access	Description	Reset value
[31:0]	RB_DVP_DMA_BUF1[31:0]	RW	DMA receive address 1 (32-byte alignment).	0

29.4.8 DVP Interrupt Flag Register (R8_DVP_IFR)

Offset Address: 0x10

Bit	Name	Access	Description	Reset value
[7:5]	Reserved	RO	Reserved.	0
4	RB_DVP_IF_STP_FRM	RW1Z	Frame stop interrupt flag, active high, clear by writing 1.	0

3	RB_DVP_IF_FIFO_OV	RW1Z	Receive FIFO overflow interrupt flag, active high, clear by writing 1.	0
2	RB_DVP_IF_FRM_DONE	RW1Z	Frame receive completion interrupt flag, valid high, clear by writing 1.	0
1	RB_DVP_IF_ROW_DONE	RW1Z	Row done interrupt flag, active high, clear by writing 1.	0
0	RB_DVP_IF_STR_FRM	RW1Z	Frame start interrupt flag, active high, clear by writing 1.	0

29.4.9 DVP Receive FIFO Status Register (R8_DVP_STATUS)

Offset address: 0x11

Bit	Name	Access	Description	Reset value
7	Reserved	RO	Reserved.	0
[6:4]	RB_DVP_FIFO_CNT[2:0]	RO	FIFO counter.	0
3	Reserved	RO	Reserved.	0
2	RB_DVP_FIFO_OV	RO	FIFO overflow status. 0: FIFO not overflow; 1: FIFO overflow.	0
1	RB_DVP_FIFO_FULL	RO	FIFO full status. 0: The FIFO is not full; 1: The buffer is full.	0
0	RB_DVP_FIFO_RDY	RO	FIFO ready status. 0: No data in the FIFO; 1: There is data in the FIFO.	0

29.4.10 DVP Row Count Register (R16_DVP_ROW_CNT)

Offset address: 0x14

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_ROW_CNT[15:0]	RO	In actual reception, the number of rows contained in a frame of image data, this register is updated at the end of the frame. In JPEG format, the value of this register has no meaning.	0

29.4.11 DVP Horizontal Shift Register for the Start of the Window (R16_DVP_HOFFCNT)

Offset address: 0x18

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_HOFFCNT[15:0]	RW	Within the window line, each row needs to be blanked for the number of PCLK cycles before capturing data. The lower 14 bits are valid.	0

29.4.12 DVP Window Start Line Number Register (R16_DVP_VST)

Offset address: 0x1A

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_VST[15:0]	RW	The number of lines the image to start capturing, data before this line is not captured. The lower 13 bits are valid.	0

29.4.13 DVP Capture Count Register (R16_DVP_CAPCNT)

Offset address: 0x1C

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_CAPCNT[15:0]	RW	The number of PCLK cycles to capture within the clipping window. The lower 14 bits are valid.	0

29.4.14 DVP Vertical Line Count Register (R16_DVP_VLINE)

Offset address: 0x1E

Bit	Name	Access	Description	Reset value
[15:0]	RB_DVP_VLINE[15:0]	RW	The number of lines to capture within the crop window. The lower 14 bits are valid.	0

29.4.15 DVP Data Register (R32_DVP_DR)

Offset address: 0x20

Bit	Name	Access	Description	Reset value
[31:0]	RB_DVP_DR[31:0]	RO	Each time the DVP interface receives 4 bytes of data, a DMA request is triggered, and the 4-byte deep FIFO can leave sufficient time for DMA transfers. It can effectively prevent the occurrence of DMA overflow.	0

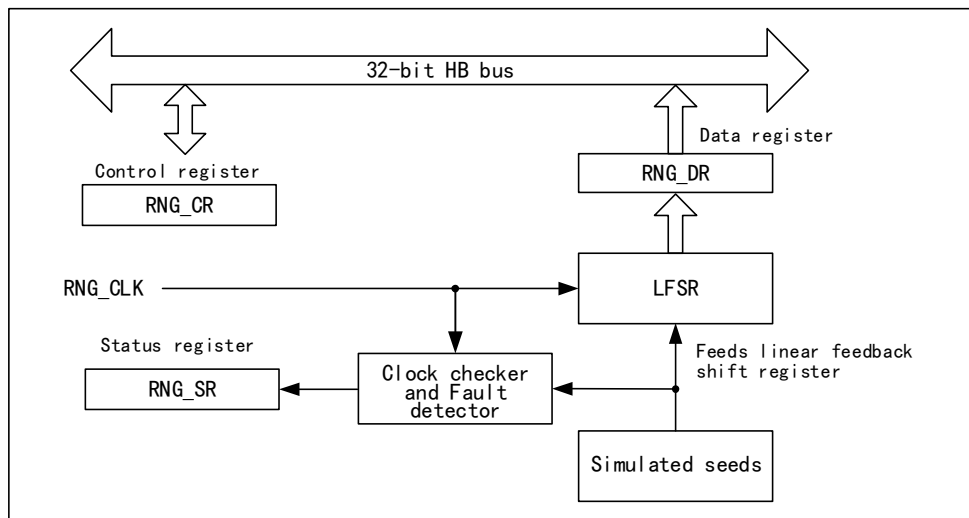
Chapter 30 Random Number Generator (RNG)

The chip has a built-in hardware random number generator, which provides a 32-bit random number through an internal analog circuit based on continuous analog noise.

30.1 Main Features

- Can generate 32-bit random numbers
- Error management is possible
- Can be disabled individually to reduce power consumption

Figure 30-1 RNG block diagram



30.2 Functional Description

The random transmitter is implemented using an analog circuit that generates a seed for a linear feedback shift register (RNG_LFSR) for generating 32-bit random numbers. RNG_LFSR provides clock information at a constant frequency by a dedicated clock (PLL48CLK), so the quality of random numbers is related to the HCLK clock. When a large number of seeds are introduced into RNG_LFSR, the content of RNG_LFSR will be transferred to the data register (R32_RNG_DR).

30.2.1 RNG Operations

RNG operation procedure:

- 1) If the interrupt is enabled, by setting the IE bit in the R32_RNG_CR register (The interrupt is generated when a random number is ready or an error occurs).
- 2) Random number generation is enabled by configuring the RNGEN bit in the R32_RNG_CR register, while the analog section, RNG_LFSR and error detector are activated.
- 3) If the interrupt is enabled, every time an interrupt is generated, the SEIS and CEIS bits in the R32_RNG_SR register are 0 to confirm that no error occurs and that DRDY is 1 to confirm that the random number is ready. The contents of the RNG_DR register can then be read.

30.2.2 Error Management

RNG errors include clock errors and seed errors. When a clock error occurs, the RNG can no longer generate random numbers. At this time, it is necessary to check whether the clock controller is configured correctly and whether the RNG clock can be provided, and then clear the CEIS bit. When the CECS bit is 0, the RNG can work normally. When a clock error occurs, it has no effect on the last random number and can be used normally. When a seed error occurs, the value in the R32_RNG_DR register cannot use the random number at this time. If the RNG is to be reused, the SEIS bit needs to be cleared first, then the RNGEN bit is cleared and set to 1, and the RNG is reinitialized and restarted.

30.3 Register Description

Table 30-1 RNG registers

Name	Address	Description	Reset value
R32_RNG_CR	0x40023C00	RNG control register	0x00000000
R32_RNG_SR	0x40023C04	RNG status register	0x00000000
R32_RNG_DR	0x40023C08	RNG data register	0x00000000

30.3.1 RNG Control Register (R32_RNG_CR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												IE	RNG EN	Reserved	

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	IE	RW	Interrupt enable: 0: Disable RNG interrupt. 1: Enable RNG interrupt.	0
2	RNGEN	RW	Random number generator enable: 0: Disable random number generator. 1: Enable random number generator.	0
[1:0]	Reserved	RW	Reserved	0

30.3.2 RNG Status Register (R32_RNG_SR)

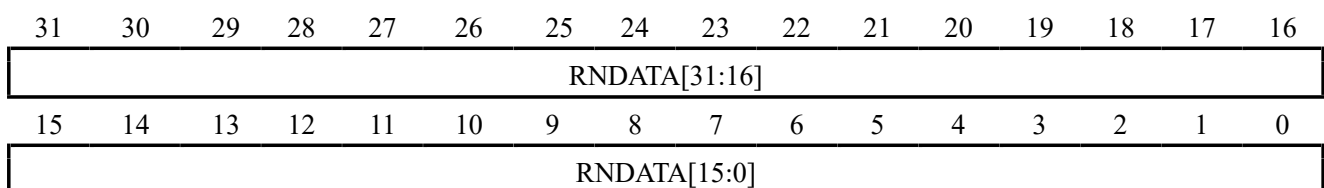
Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									SEIS	CEIS	Reserved	SECS	CECS	DRDY	

Bit	Name	Access	Description	Reset Value
[31:7]	Reserved	RO	Reserved	0
6	SEIS	RW	Seed error interrupt status (This bit is set at the same time as SECS) 1: One of the following error sequences is detected: - more than 64 identical consecutive bits; - more than 32 consecutive alternating 0's and 1's. 0: No inverted error sequence detected.	0
5	CEIS	RW	Clock error interrupt status (This bit is set at the same time as CECS) 1: PLL48CLK clock not detected correctly; 0: PLL48CLK clock detected correctly.	0
[4:3]	MODE4	RW	Reserved	0
2	SECS	RO	Seed error current status 1: One of the following error sequences was detected: - more than 64 identical consecutive bits; - more than 32 consecutive alternating 0s and 1s. 0: No error sequence detected.	0
1	CECS	RO	Clock error current status 1: PLL48CLK clock not detected correctly; 0: PLL48CLK clock detected correctly.	0
0	DRDY	RO	Data ready (This bit is cleared after reading RNG_DR) 1: RNG_DR register is valid, this random number is available; 0: RNG_DR register is invalid, this random number is not available.	0

30.3.3 RNG Data Register (R32_RNG_DR)

Offset address: 0x08



Bit	Name	Access	Description	Reset Value
[31:0]	RNDATA	RO	32-bit random number	0

Chapter 31 Ethernet Transceiver (ETH)

This chapter's module description applies only to the CH32H417 microcontroller product.

The "Ethernet transceiver" mentioned in this chapter is a proper term, which means the Ethernet (Ethernet) data link layer transceiver inside the microcontroller, the communication rate is up to gigabit per second (1Gbps), it is a communication peripheral. The "MAC" mentioned in this chapter refers to the role name of the Ethernet transceiver at the data link layer and is a component of the Ethernet transceiver.

Ethernet transceiver (Ethernet Transceiver MAC) is an important high-speed communication component of microcontroller, which allows microcontroller to access Ethernet at Gigabit connection speed to achieve extremely fast data communication.

31.1 Main Features

The Ethernet transceiver in a microcontroller is a critical high-speed communication peripheral. It integrates a Gigabit MAC (Media Access Controller), a 256-bit-wide DMA controller, a Management Counter (MMC), a Precision Time Protocol (PTP) controller, and a 10/100 Mbps Ethernet Physical Layer (10M/100M PHY). When paired with an external Gigabit Ethernet PHY, it enables Gigabit Ethernet (1Gbps) data transmission. Operating at the data link layer, the Ethernet transceiver requires software implementation of the TCP/IP protocol stack and interfaces. The Ethernet transceiver supports a standard RGMII interface for connecting to the PHY. If users require gigabit access speeds, they must use the RGMII interface. Ethernet MAC controls PHY through SMI interface, and the timing of the interface is automatically realized by MAC, without the need for users to generate it through software. The RGMII interface supports the phase inversion of the transmission clock and the relative data delay, with a maximum delay of 4 nanoseconds.

Ethernet MAC supports Ethernet of standard IEEE802.3 protocol, magic frame and specific wake-up frame. The DMA controller matched with Ethernet transceiver manages data receiving and dispatching and carries memory in the form of descriptors. The number of descriptors is determined by the user according to the communication density. The DMA controller can write the received data into the memory space specified by descriptors or retrieve the data to be sent at a speed of 32 bits. In addition, the MAC of Ethernet transceiver also supports IEEE1588 precise time protocol.

31.1.1 MAC Features

- Support RGMII interface
- RGMII supports transmit clock delay and inversion
- Support full-duplex and half-duplex, support 10M/100M/1000Mbps data transfer rate
- Support automatic insertion of frame header sequence and SFD
- Support automatic insertion and check CRC
- Support automatic calculation and verification of the checksum of IP/ICMP/TCP/UDP protocol packets
- Support sending frame length control
- Support sending gap adjustment
- Support VLAN frame
- Support perfect address filtering and HSAH filtering of frame receiving addresses

- Support frame sending address filtering
- Support multicast broadcast frame reception control
- Support promiscuous mode
- Support SMI management interface
- Support magic frame and custom wake-up frame to wake up microcontroller
- Support dedicated Ethernet wake-up interrupt entry
- Support data loopback at the link layer
- Built-in pre-allocated globally unique MAC address
- Application-selectable Ethernet controller MAC + external 1Gbps PHY, or Ethernet controller MAC + built-in 10Mbps/100Mbps PHY

31.1.2 DMA Features

- 256-bit wide MAC dedicated DMA
- Minimize CPU operations
- Support access to RAM in a byte-aligned manner
- Manage send and receive buffers in the form of descriptors
- Part of the status feedback sent by the receiver is in the descriptor
- Can dynamically modify descriptors and buffers that are not in use
- Support manual stop or start
- Support chained or ringed formal connection descriptors
- Support multiple interrupt sources such as transceiver completion interrupt

31.1.3 MMC Module Features

- Support manual reset, stop or freeze
- Multiple counters supporting multiple counting modes for sending and receiving
- Support multiple interrupts

31.1.4 PTP Module Features

- Support IEEE 1588 protocol
- Support to automatically save the current moment when sending and receiving
- Come with a 32-bit second timer and a 32-bit signed sub-second timer
- Support to adjust the time in 2 ways: fine adjustment and coarse adjustment
- Single interrupt source
- Support PPS output

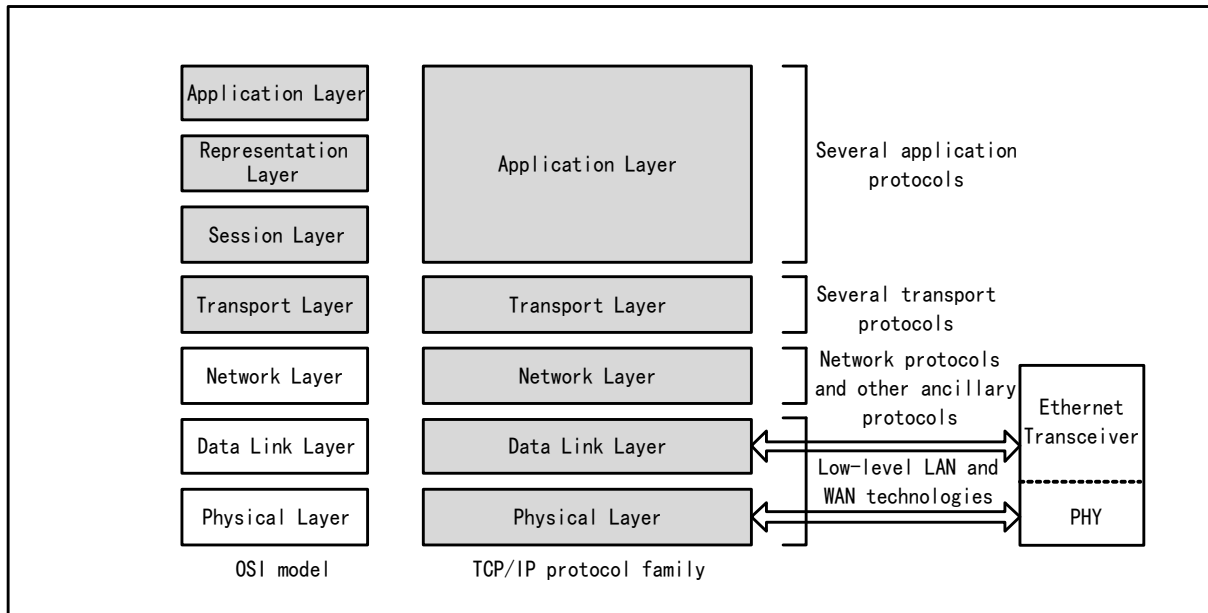
31.1.5 Internal 10M/100M Physical Layer Features

- Support 10BASE-T and 100BASE-TX and auto-negotiation
- Support Auto-MDIX switching TX/RX and automatic identification of positive and negative signal lines
- Support WOL network wake-up support
- Support interrupt function
- Support half-duplex, full-duplex
- 5 network status LEDs

31.2 Overview

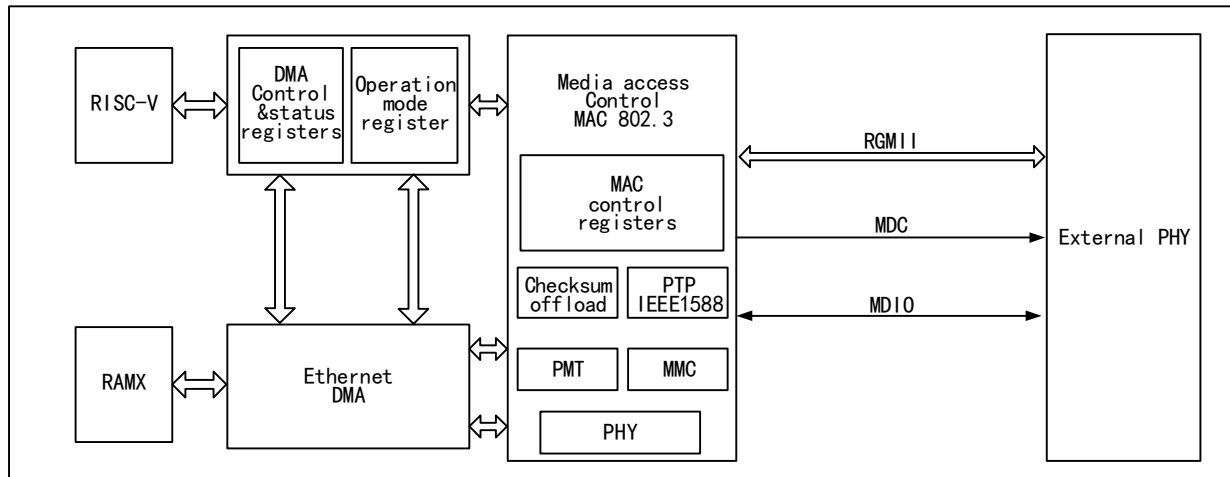
Ethernet transceivers work at the data link layer and physical layer in the OSI seven-layer model. In order to establish the communication of protocols such as IP, TCP and UDP in the widely used Ethernet, the user also needs to implement the TCP/IP protocol stack in software. The Ethernet transceiver is composed of a media access control layer (MAC), a matching DMA and their control registers, and a 10M physical layer.

Figure 31-1 Position of Ethernet Transceivers in OSI Model and TCP/IP Model



The MAC of the Ethernet transceiver is designed according to the specifications of the IEEE802.3 protocol, and is equipped with a 256-bit wide DMA to ensure that the data can be quickly forwarded from the network cable to the memory of the microcontroller. The Ethernet transceiver has powerful and complete DMA control registers, MAC control registers and mode control registers. The microcontroller's CPU operates the registers of the Ethernet transceiver through the HB bus. The Ethernet transceiver is connected to the gigabit Ethernet physical layer through the RGMII interface. If only the speed of 100M Ethernet is required, it can be connected to the 100M Ethernet physical layer through the MII or RMII interface. The pins of the MII, RMII and RGMII interfaces of the Ethernet transceiver are multiplexed, see section 31.3 for details. The Ethernet transceiver manages the Ethernet physical layer through the SMI interface. When using an Ethernet transceiver, the clock of the HB bus cannot be lower than 50MHz.

Figure 31-2 Block diagram of Ethernet Transceiver



In addition, the Ethernet transceiver also supports the IEEE1588 Precision Time Protocol (PTP) to provide precise time data for microcontroller systems or off-chip.

31.3 Ethernet Transceiver Pinouts and Configuration

The microcontroller supports RGMII interface. The RGMII interface can be used for 10M, 100M and Gigabit Ethernet. The following table shows the distribution of standard RGMII interfaces and internal physical layers on package pins.

Table 31-1 Media independent interface of microcontroller, media dependent interface with built-in physical layer and other related pinouts and required configuration

Pins	RGMII	RGMII pin configuration
PD15	GTXC	Push-pull multiplexed output
PA14	RXDV	Floating input
PA15	GRXC	Floating input
PC6	RXD3	Floating input
PD13	TXD0	Push-pull multiplexed output
PD12	TXD1	Push-pull multiplexed output
PD11	TXD2	Push-pull multiplexed output
PD10	TXD3	Push-pull multiplexed output
PC8	RXD1	Floating input
PC0	MDC	Push-pull multiplexed output
PC1	MDIO	Push-pull multiplexed output
PC9	RXD0	Floating input
PC7	RXD2	Floating input
PD14	TXEN	Push-pull multiplexed output
PC2	PPS	Push-pull multiplexed output
-	MDITP	No IO configuration required
-	MDITN	No IO configuration required
-	MDIRP	No IO configuration required
-	MDIRN	No IO configuration required

31.4 Physical Layer (PHY) Management and Data Interaction

The MAC of the Ethernet transceiver manages the PHY through the site management interface (SMI interface). The microcontroller uses different pins to bring out the SMI interface in RGMII mode.

31.4.1 SMI Interface

The SMI interface is a serial communication interface. It uses MDC (Clock line) and MDIO (Data line) to access the registers of the PHY to manage the PHY, and can manage up to 32 PHY chips. Where MDC is the clock line, which is kept low when idle, and MDIO is the data line. The read and write operations of SMI and the composition of the frame are dominated by the MAC, and the user only needs to write the address and data. The relevant registers are the MII address register (R32_ETH_MACMIAR) and the MII data register (R32_ETH_MACMIIDR).

31.4.1.1 Frame Format

The format of the SMI interface management frame is shown in Table 31-2 below.

Table 31-2 SMI Frame Format

	Preamble	STR	OP	PHY ADR	REG ADR	T	DATA	P
Read	32 "1"	01	10	PPPPP	RRRRR	Z0	DDDDDDDDDDDDDDDDDD	Z
Write	32 "1"	01	01	PPPPP	RRRRR	10	DDDDDDDDDDDDDDDDDD	Z

The fields of the management frame are defined as follows:

- 1) Preamble: Preamble, consisting of 32 "1"s, used for MAC and PHY synchronization;
- 2) STR: start character, fixed to "01";
- 3) OP: operator, read as "10", written as "01";
- 4) PHY ADR: physical layer address, 5 bits;
- 5) REG ADR: register address, 5 bits;
- 6) T: Converter, 2 bits, used to switch the control of the MDIO line between the MAC and the PHY. During the read operation, the MAC maintains a high impedance to the MDIO line, and the PHY maintains a high impedance to the first bit, pulls down on the second bit, and obtains control of the MDIO; during a write operation, the MAC first sets the MDIO line high and then pulls it down. PHY maintains a high-impedance state for MDIO, and MAC maintains control over MDIO;
- 7) DATA: data field, 16 bits, data read and written by MAC to PHY, MSB first;
- 8) P: Both MAC and PHY maintain a high-impedance state for MDIO, but the pull-up resistor of PHY will pull MDIO high.

31.4.1.2 Read and Write Timing

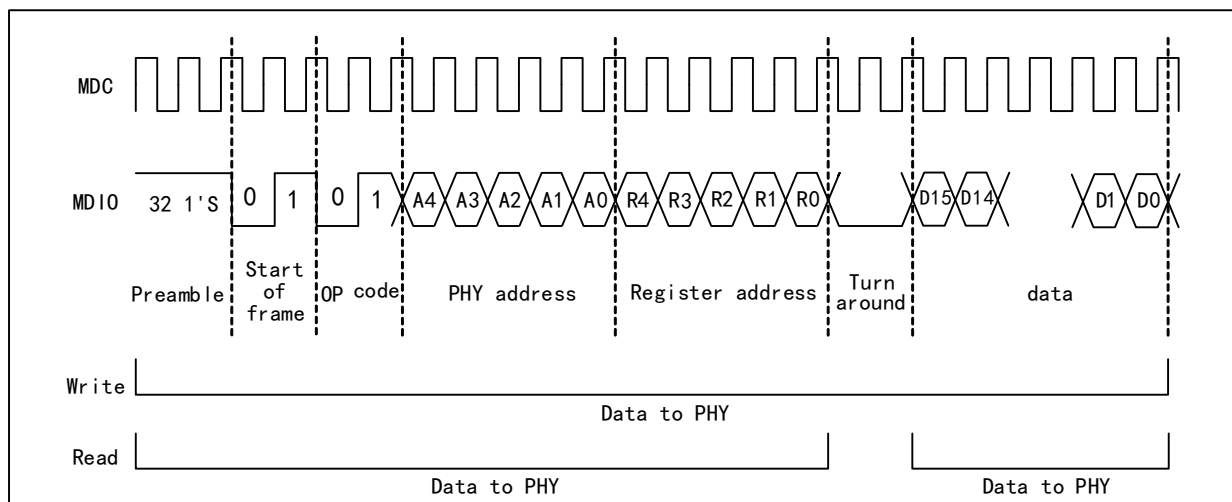
Writing to the PHY register operates as follows:

When the user sets the MII write bit (ETH_MACMIAR: MW) and the busy bit (ETH_MACMIAR: MB), the SMI interface will send the PHY address and register address to the PHY, and then send the data (ETH_MACMIIDR). In the process of sending data through the SMI interface, the values of the MII address register and MII data register cannot be modified; during this process, the busy bit remains high, and the write operation to the MII address register or MII data register will be ignored, and the entire transmission is affected. When the write operation is completed, the SMI interface will clear the busy bit, and the user can judge the end of the write operation according to the busy bit. The write part of Figure 31-3.

The operation of reading the PHY register is as follows:

When the user sets the MII busy bit of the MII address register (ETH_MACMIAR) of the Ethernet MAC and keeps the MII write bit reset, the SMI interface sends the PHY address and register address, and performs the operation of reading the PHY register. During the entire transfer process, the user cannot modify the contents of the MII address register and the MII data register. During the transfer, the busy bit remains high, and writes to the MII address register or the MII data register will be ignored and will not affect the correct completion of the entire transfer. After the read operation is completed, the SMI interface will clear the busy bit and write the data read back from the PHY into the MII data register. The read section of Figure 31-3.

Figure 31-3 SMI interface read and write time diagram



31.4.1.3 SMI Clock

Generally speaking, the SMI clock needs to be kept in a fixed range to ensure that the SMI clock actually needs to be received by the PHY. For details, refer to the manual of the PHY chip selected by the user.

The SMI clock frequency is divided from the HB clock by the CR field of the MII address register (ETH_MACMIAR). The following table shows the frequency division relationship between SMI clock and HB clock. By default, divide by 42 is selected.

Table 31-3 Frequency division relationship between SMI clock and HB clock

	Range of AHB clocks suitable for matching	SMI clock
0000b	Above 60MHz	HB clock/42
0010b	20-35MHz	HB clock/16
0011b	35-60MHz	HB clock/26
other values	meaningless	

31.4.2 RGMII interface

31.4.2.1 Pins

The pins and functions of the RGMII are as follows

Table 31-4 RGMII pins and functions

Pins	Features
TXC	Transmit clock
TXCTL	Transmit data control
TXD0	Transmit data lines [0:3]
TXD1	
TXD2	
TXD3	
RXC	Receive clock
RXCTL	Receive data control
RXD0	Receive data lines[0:3]
RXD1	
RXD2	
RXD3	

Note: 1. TXCTL/RXCTL is used in the Ethernet transceiver of this microcontroller as a valid signal for sending and receiving data;

2. RGMII has an independent SMI interface.

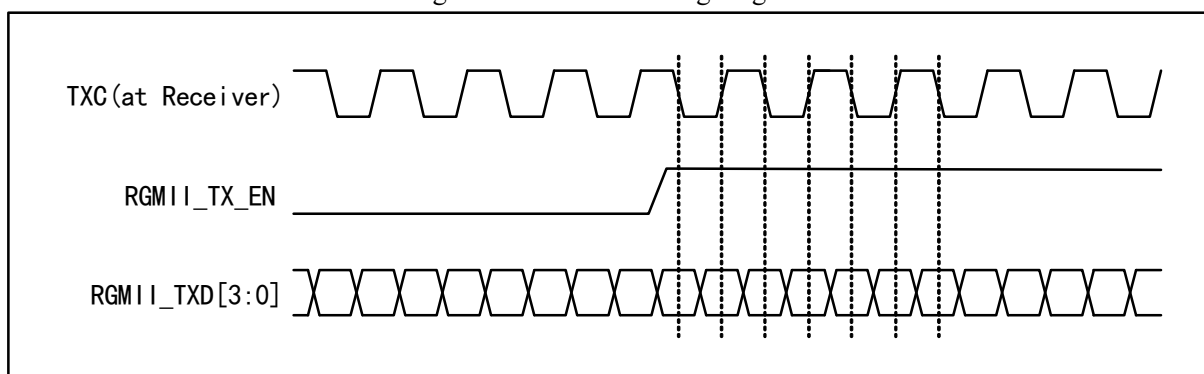
31.4.2.2 Timing

RGMII works in 10M and 100M rate modes, and the timing is similar to MII; when working in gigabit, the clock of RGMII is 125MHz, and the double-edge sampling method is adopted. RGMII does not support half-duplex.

Since RGMII uses double-edge sampling and operates at a clock frequency of 125MHz, circuit designers should be aware of signal integrity issues.

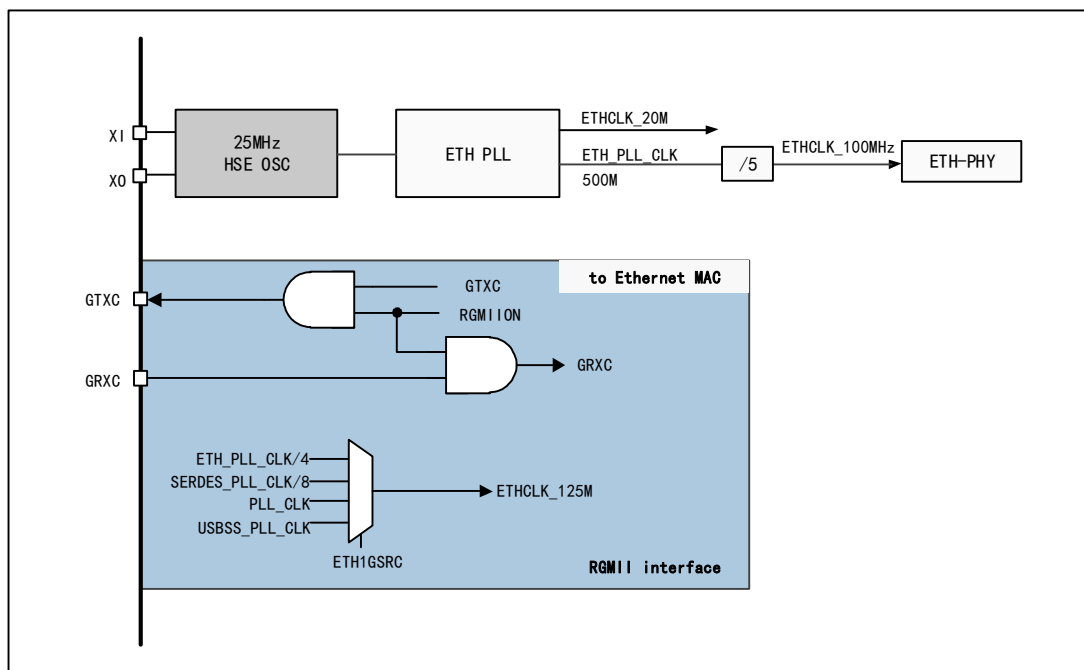
The clock received by the receiver of the RGMII should lag the data by 90° to ensure correct sampling. Ethernet transceivers are designed to transmit clock delay output and phase inversion.

Figure 31-4 RGMII timing diagram



31.4.3 Peripheral clock source configuration

Figure 31-5 Ethernet peripheral clock tree



When using RGMII, set `RCC_CFGR0[21]` to 1 to enable RGMII mode, and configure `RCC_CFG2[31:30]` to select the appropriate 125MHz clock source for the MAC. The RGMII transmit clock GTXC is generated by the MAC, while the receive clock GRXC is generated by the PHY. For details, see EVT routines on the official website.

31.5 IEEE802.3 and IEEE1588

The IEEE802.3 protocol and its supplementary protocols constitute the official standard of the current Ethernet, which defines all aspects of the currently used Ethernet in detail. We can think of Ethernet as part of the physical layer and data link layer in the OSI model. This section discusses the parts that users need to use in the IEEE802.3 protocol from the application point of view, that is, the frame format and the content related to the MAC address. At the same time, the Ethernet transceiver of the microcontroller also supports the IEEE1588 Precision Time Protocol, which will also be discussed in part in this article.

In the Ethernet model constructed by the IEEE802.3 protocol, the data transmission unit is an Ethernet frame. Ethernet frames have different encoding forms when transmitted on different media at different rates. After receiving, they are decoded by the physical layer and sent to the MAC through the MII interface. After the MAC checksum is filtered, it is extracted by the TCP/IP protocol stack. Application information is sent to different application processes.

31.5.1 Frame Format

The frame format of an Ethernet frame is shown in Table 31-5.

Table 31-5 Normal Ethernet frame format

Preamble	SFD	Target address	Source address	Length/Type	Data field	CRC
7 Bytes	1Byte	6 Bytes	6 Bytes	2 Bytes	46-1500 Bytes	4 Bytes

Total length: 64 to 1518 bytes plus 8 bytes of physical layer header (Preamble and SFD)

Preamble: 56-bit (7 bytes) alternating low-level and high-level jumps, the value is fixed, which is 0xAA-0xAA-0xAA-0xAA-0xAA in hexadecimal. This field is used for clock synchronization. This field is added/removed automatically by the hardware and the user does not need to care about it.

SFD: Frame start delimiter, 8 bits (1 byte), the value is 10101011b, SFD is used to remind the receiver that this is the last opportunity to synchronize the clock, followed by the target address. This byte is automatically added/removed by the hardware, and the user does not need to care about it.

Destination address: The device that sends this frame wants to receive the device address of this frame, and the address here refers to the hardware address, also known as the MAC address, which is assigned by the IEEE for the vendor, globally unique, 6-byte (48-bit). The MAC addresses of WCH microcontrollers have been programmed into the devices when they left the factory. The transmission of the address follows the principle of the least significant bit first.

Source Address: The hardware address of the device that sent this frame. The source address must be a unicast address.

Length/Type: 2 bytes, Ethernet is generally used as a type, and is also used as a length in the IEEE802.3 standard. It is demarcated by 1536 (i.e. 0x0600). The protocol above 1536 indicates which upper-layer protocol the data part is organized according to, for example, 0x0806 means ARP, 0x0800 means IPv4, 0x86dd means IPv6; below 1536 means data length.

Data field: Minimum 46 bytes, maximum 1500 bytes. When the data field is less than 46 bytes, you need to add padding to 46 bytes, and if it exceeds 1500 bytes, please create another frame. The data field is loaded with the data that needs to be actually sent.

CRC: Cyclic Redundancy Check, the CRC32 check is used here.

31.5.2 Frame Transmitting

When the Ethernet transceiver sends a data frame, its dedicated DMA controller takes out the data to be sent from the RAM specified by the transmit descriptor (Trans descript in Section 31.5.1) and pushes it into the MAC through the dedicated data bus. The filling level of the FIFO will be returned to the DMA controller. When all the data to be sent is sent, the DMA controller will send a data start signal to the MAC, and the MAC will start sending; read the data from the FIFO and send pilot, SFD, and sends actual data to MII interface (RMII/RGMII); after the DMA controller sends the data end signal to the MAC, the MAC adds the automatically calculated CRC.

The MAC will calculate the checksum for the checksum field according to the type of the frame sent by the configured automatic device, and replace the value of the original checksum field. This feature can be turned off.

The MAC will automatically add the CRC32 check to the padding, or it can be set to not add the CRC check bit. According to the provisions of the IEEE802.3 protocol, the length of the data field is not less than 46 bytes. When the MAC detects that the data field to be sent is shorter than 46 bytes, it will automatically add padding after the number field. If automatic padding is selected, then The CRC32 checksum must be added, ignoring the register configuration. The CRC32 verification formula used by the MAC is as follows:

$$G(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$$

When IEEE 1588 (PTP) functionality is enabled, enhanced descriptors must be used for storing timestamps.

31.5.3 Frame Receiving

When the Ethernet transceiver receives the Ethernet frame, the Ethernet frame enters the MAC from MII, RMII or RGMII, first enters the FIFO in the MAC, and then is forwarded to the buffer in the RAM by the DMA. The MAC will filter and check the Ethernet frame. The filtering includes perfect address filtering and HASH filtering. The inspection then typically performs frame length, IP/ICMP/TCP/UDP checks and CRC checks on frames that do not pass the filtering or checks are marked out by the descriptor or dropped, and packets that exceed the length may be choked off.

When IEEE 1588 (PTP) functionality is enabled, enhanced descriptors must be used for storing timestamps.

31.5.4 Frame Filtering

Use the MAC frame filter to perform perfect filtering or HASH filtering on the destination MAC address and source MAC address of the received frame, and can set the frame that does not pass the filtering to be discarded, received, or marked in the receive descriptor. Please read the description of the MAC frame filter register (R32_ETH_MACFFR) for this function. The MAC has 4 built-in MAC address registers, of which MAC address register 0 is used by default to store its own MAC address, and the remaining 3 MAC address registers can be used for perfect filtering, compared with the source address or destination address of the received frame. After setting R32_ETH_MACFFR: RA, all received frames will be received, but the corresponding status will be marked in the status field of the first word of the descriptor after that, if R32_ETH_MACFFR:PM is set, there will have a similar effect, but the state will not be marked. After setting R32_ETH_MACFFR: DAIF/SAIF, the result will be flipped. Those that pass the filtering will be discarded or marked, and those that do not pass the filtering will be transferred to the buffer in RAM.

31.5.4.1 Unicast Filtering

The MAC uses the HPF bit and the HU bit to determine whether the unicast frame is HASH filtering or perfect address filtering.

31.5.4.2 Multicast Filtering

The MAC uses the HPF bit and the HM bit to determine whether the multicast frame is HASH filtered or perfect address filtering. When the PAM bit is set, all multicast packets can pass through the filter.

31.5.4.3 Broadcast Filtering

By setting the BFD bit, the MAC can block all broadcast packets.

31.5.4.4 Source Address Filtering Selection

The MAC address register can be enabled by setting the AE bit in the MAC address register, while setting the SA bit determines whether the MAC address register is to be compared as a source or target address sample.

31.5.4.5 Summary

The filter settings for target address and source address are shown in Table 31-6 and Table 31-7 respectively.

Table 31-6 Acceptance of the setting of each bit of R32_ETH_MACFFR to the target MAC address of the

received frame							
Frame type	M	PF	U	DAIF	M	PAM	Effect
Broadcast frame	1	-	-	-	-	-	Pass
	0	-	-	-	-	-	Fail (BFD set)
Unicast frame	1	-	-	-	-	-	All frames pass
	0	-	0	0	-	-	Pass when perfect filter matches
	0	-	0	1	-	-	Fail when perfect filter matches
	0	0	1	0	-	-	Pass when HSAH filter matches
	0	0	1	1	-	-	Fail when HSAH filter matches
	0	1	1	0	-	-	Pass when perfect filter or HSAH filter matches
	0	1	1	1	-	-	Fail when perfect filter or HSAH filter matches
Multicast frame	1	-	-	-	-	-	Pass
	-	-	-	-	-	1	Pass
	0	-	-	0	0	0	Pass when perfect filter matches
	0	0	-	0	1	0	Pass when HSAH filter matches
	0	1	-	0	1	0	Pass when perfect filter or HSAH filter matches
	0	-	-	1	0	0	Fail when perfect filter matches
	0	0	-	1	1	0	Fail when HSAH filter matches
	0	1	-	1	1	0	Fail when perfect filter or HSAH filter matches

Table 31-7 Acceptance of the setting of each bit of R32_ETH_MACFFR to the source MAC address of the received frame

Frame Type	RA	SAIF	SAF	Effect
Unicast frame	1	-	-	All frames pass
	0	0	0	Passes on perfect filter match, marks but does not discard failed frames
	0	1	0	Do not pass on perfect filter match, mark but do not discard failed frames
	0	0	1	Pass when perfect filter matches, discard failed frames
	0	1	1	Fail when perfect filter matches, discard failed frames

Note: "-" indicates that the setting of this bit is of no concern.

31.5.5 MMC

The function of the management counter (MMC) is mainly to count the receiving and sending status of various instruction frames and the running status of the MAC. It can generate settings and generate interrupts. In general, we can get the number of currently received good frames by receiving the "good" frame counter (MMCRGUFCR), get the number of successfully sent frames by sending the "good" frame counter (MMCTGFCR), and get the number of frames successfully sent by receiving the CRC error frame counter (MMCRFCECR) to check whether there is a

frame with CRC error received, in general, if the data is correct but the CRC is wrong, it can be considered that there is a problem with the RGMII line layout.

Users need to be clear about a concept: what kind of frame is a "good" frame. Under normal circumstances, as long as a frame starts to send, its frame length meets the requirements (Automatic padding is turned on), and automatic CRC calculation is turned on, then it will be a "good" frame. And a frame is received, as long as its CRC is correct, the frame length is within the Ethernet frame length range, the frame length and the value of the length field are the same (If the length type field indicates the length), or there is no misalignment, then it will be considered a "good" received frame.

31.5.6 PMT

31.5.6.1 Overview

The function of the PMT (Power management) part is to wake up the microcontroller from a low-power mode via Ethernet. The Gigabit Ethernet controller supports 2 types of frames to wake up the system, magic frames and (Remote) wake-up frames. When Ethernet receives these 2 kinds of frames, because the microcontroller is in low-power mode, it does not necessarily generate a receive interrupt, even if it is legal, but if it is recognized by the MAC's magic frame and wake-up frame, it will generate PMT interrupt, PMT interrupt is a separate interrupt from the Ethernet interrupt. By querying the PMT control and status registers, you can find out which Ethernet frame generated the interrupt.

31.5.6.2 Magic Frame

The magic frame (Magic package, also called magic data packet) defined by AMD is a commonly used Ethernet frame to wake up the microcontroller. It has a fixed and special frame format, that is, the target network card can receive it through frame filtering. It can be encapsulated into a broadcast frame, and there are 6 consecutive bytes of full height (OXFF) in the frame, followed by the MAC address of the target network card repeated 16 times. This combination can exist anywhere in the frame, so magic frames can be encapsulated into Mac frames or IP packets or even UDP packets. Below is the format of the magic frame.

xx xx xx xx xx xx xx xx(Unlimited previous data, even no previous data)—ff ff ff ff ff 84 c2 e4 01 02 02 84 c2
e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4
01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01
02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 xx xx xx xx xx xx (Some network cards require the magic frame to be
attached with a password at the end).

31.5.6.3 Wakeup Frame

Due to the limitation of the format of the magic frame, the format of the (Remote) wakeup frame can be defined by the user. As long as the Ethernet frame passes through the remote wakeup frame filter register (ETH_MACRWUFR) group, it is regarded as a wakeup frame. Users can define the wakeup frame that meets their needs by setting the remote wakeup frame filter register group. It should be noted that the remote wakeup frame filter register group has only single address entry, the user can write data to it through 8 consecutive write operations, and then read the data originally written into the remote wakeup frame filter register by reading data 8 consecutive times.

The structure of the remote wakeup frame filter register group is as follows:

Table 31-10 Structure of the remote wakeup frame filtering register bank

Filter register 0	Filter byte mask of register 0
-------------------	--------------------------------

Filter register 1	Filter byte mask of register 1							
Filter register 2	Filter byte mask of register 2							
Filter register 3	Filter byte mask of register 3							
Filter register 4	Reserved	Filter 3 command	Reserved	Filter 2 command	Reserved	Filter 1 command	Reserved	Filter 0 command
Filter register 5	Filter 3 offset		Filter 2 offset		Filter 1 offset		Filter 0 offset	
Filter register 6	Filter 1 CRC-16				Filter 0 CRC-16			
Filter register 7	Filter 3 CRC-16				Filter 2 CRC-16			

As can be seen from the above table, the four fields of the filter register, byte mask, command, offset, and filter, work together to determine whether a frame is a remote wake-up frame. In fact, four different frames that meet the requirements can be set.

The highest bit of the 32-bit byte mask must be 0. Bit[30:0] corresponds to the first 31 bytes of data defined by the offset field. If any bit is set to 1, it indicates that the corresponding byte participates in CRC-16 parity, and up to 31 bytes participate in calculation;

The highest bit of the 4-bit command field indicates what kind of frame it works on. 1 is only valid for multicast addresses, 0 is only valid for unicast addresses; the 2nd and 1st bits of the command field are reserved, and bit 0 is Enable bit, set high to enable this set of filters;

The offset field indicates how many bytes are offset from the beginning of the frame to calculate the CRC16 value. The minimum value is 12. If the offset field is 12, the CRC16 value is calculated starting from the 13th byte at the beginning. (Parametric model using CRC-16-IBM with default parameters)

The filter stores the value of the CRC result that the user expects to calculate. The MAC will compare the value of the CRC16 calculated by itself with the value in this field. If they are consistent, it will be regarded as a (remote) wake-up frame. If used If the wake-up frame interrupt and PMT interrupt are disabled, a PMT interrupt will also be generated.

In addition, according to the bit definition of the PMT control status register, if the GU bit is set, the unicast frame that passes the frame filter will also be considered a wake-up frame.

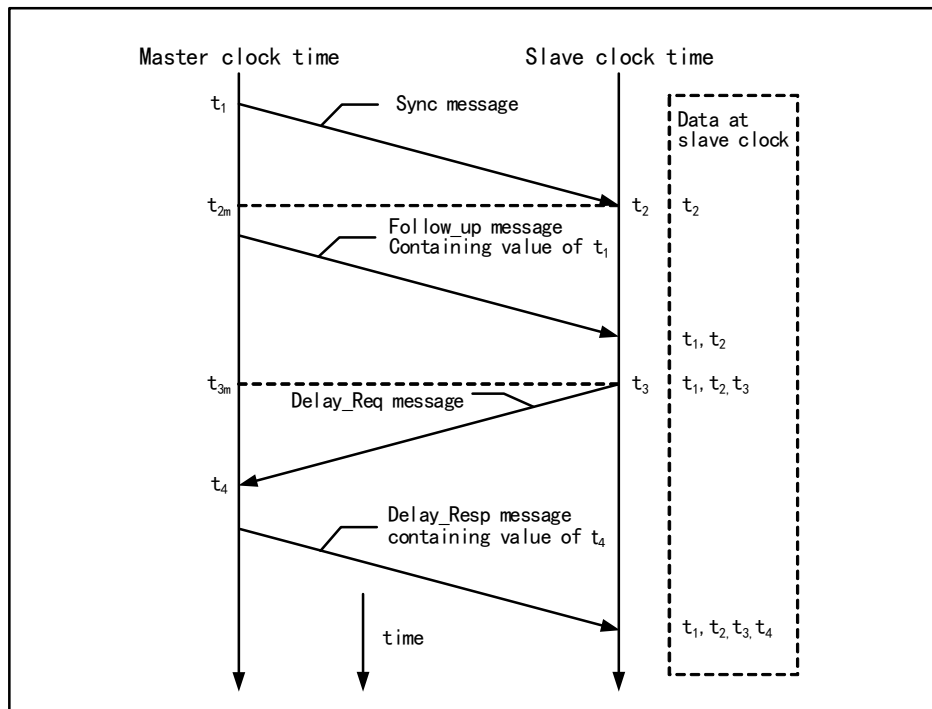
31.5.7 IEEE1588 PTP

31.5.7.1 PTP Principle and Implementation

The IEEE1588 standard defines a set of precise protocols for acquiring time, and its goal is to achieve time synchronization within 10 microseconds of error. The original NTP protocol can already achieve time synchronization at the level of 200 microseconds, but in fact this level cannot meet the time synchronization requirements in the field of industrial automation, so the Network Precision Clock Synchronization Committee drafted the PTP protocol, which was approved by the IEEE Standards Committee at the end of 2002 as the IEEE1588 standard.

The implementation of the PTP protocol requires both the host and the slave to accurately record the time when the MAC receives and sends Ethernet frames, which requires both the host and the slave to have their own set of high-precision time counters. Then, the master and slave supporting PTP perform time synchronization through a set of procedures, and the slave can obtain the time difference between itself and the master and correct it. The following figure shows the process of master-slave time synchronization:

Figure 31-6 IEEE1588 PTP protocol synchronization message timing diagram



Step-by-step description:

- The master sends a syn message to the slave, and the slave receives this message and records the local time t_2 when the syn message is received;
- The master sends a follow_up message to the slave, including the master time t_1 when the master sends the syn message;
- The slave sends a delay_req message to the master, and the slave records the sending time t_3 ;
- The master sends a delay_resq message to the slave, including the reception time t_4 of the delay_req message;

In actual use, the host sends out a syn message every 2 seconds, and every other syn message can be regarded as the follow_up message of the previous syn message, and they will all be accompanied by the sending time of the last syn message. Through this process, the slave can know the delay time T_{delay} from the master network to the slave network, and then calculate the time offset between the master time and the slave time.

$$T_{\text{delay}} = \frac{(t_2 - t_1) + (t_4 - t_3)}{2}$$

The time sent by either host minus the offset is the host's time.

The synchronization process of PTP is generally implemented through the UDP protocol, of course, the user can also implement a custom protocol. PTP is highly dependent on the latency stability of the intranet.

31.5.7.2 Local Time Update and Correction

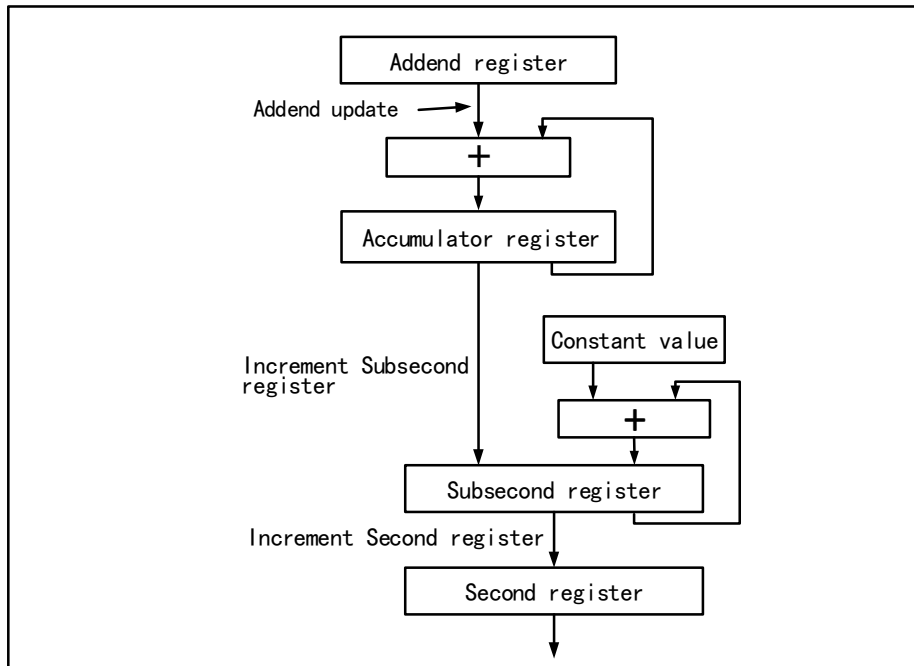
In order to implement the PTP host, the device needs to have a high-precision time counter locally, at least to the nanosecond level. The PTP module of the microcontroller's Gigabit Ethernet counter has a 32-bit counter in seconds, and a 31-bit sub-second counter. When the sub-second counter overflows, it will cause the second counter to increment automatically, so the resolution of the local time. The rate can be done around 0.46 nanoseconds.

There are 2 ways to update the local time, namely coarse correction and fine correction. The update timing of the

coarse correction mode is determined externally. When the time update is required, the time update bit (PTPTSCR: TSSTU) of the time stamp control register is set, and the PTP module of the microcontroller subtracts or adds the value of the timestamp update registers (TSHUR, TSLUR) to the second counter and sub-second counter. Coarse correction is a simpler and more convenient time update mechanism, but less precise.

The fine correction time update method is more commonly used. The update process is as follows:

Figure 31-7 Process for updating time using fine correction



The way to use the fine correction mode requires a clear understanding of the system frequency and the fine correction process. Different from the coarse correction mode, the timing of the update event in the fine correction mode is the overflow of the accumulator (32 bits, not listed in the register list. The Accumulator register in the figure) overflows, and the accumulator will automatically increment the number register in each system clock cycle. The value of (PTPTSAR, Addend register in the figure) will generate a time update event once it overflows, that is, the value of the sub-second auto-increment register (PTPSSIR, Constant Value in the figure) will be added to the sub-second counter (Sub-second register), complete time update. The second counter is incremented when the sub-second counter overflows. In fact, the time for the sub-second counter to increment by one bit is $1/(2^{31})=0.46566128730\dots$ nanoseconds. The user needs to ensure that the time spent on the accumulator overflow is exactly equal to the value of the sub-second auto-increment register multiplied by the sub-second the time for the counter to increment by one bit.

The correction of local time is relatively simple. Set the time correction bit (PTPTSCR: TSSTI) of the time stamp control register, and the value of the second counter and sub-second counter will be replaced by the value of the time stamp update counter (TSHUR, TSLUR).

31.6 DMA Operation

31.6.1 Overview

In an Ethernet transceiver, data enters the FIFO from the MII interface and is then transferred into RAM by DMA.

Even for the largest ordinary Ethernet frame, the data part reaches the maximum 1500 bytes, and it only takes about tens of microseconds to receive and transmit. Even with the receiving detection of the MAC, the DMA transfer time and the frame interval time are counted, a frame needs to be processed by the CPU in 100us. The advantages of Ethernet transceivers are fast speed and large throughput. In order to maintain this advantage, it is necessary to minimize the intervention of the CPU in the process of receiving and sending Ethernet frames. Here, a dedicated DMA for Ethernet transceivers is used.

The DMA used by Ethernet is 32-bit, and it is managed by the CPU through 2 data structures: traditional control and status registers, and receive and transmit descriptors. Since the DMA is 32 bits wide, the base address of the transceiver descriptor queue in RAM is required to be aligned with 4 bytes. There is no alignment requirement for the receive/transmit buffer.

31.6.2 DMA Descriptor

Users use traditional peripherals mainly by writing the control bits of the register, and by reading the status register to obtain the status and return information of the peripheral. These registers are independent of the core, space outside of SRAM and non-volatile memory, and actually exist, and can be called "hardware registers". Traditional communication peripherals, such as USART or SPI, have a data register to temporarily store the data sent and received, and a set of DMA to store all received data in a specific address space.

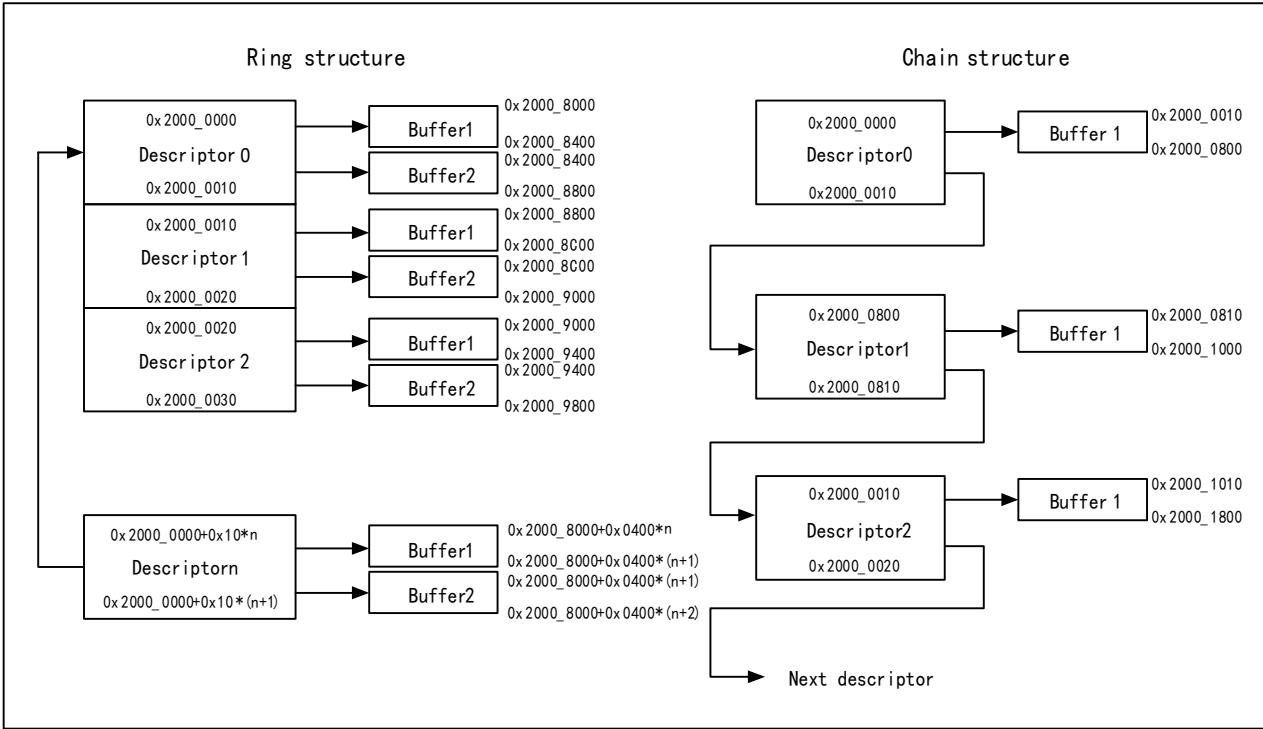
The Ethernet transceiver is different from the traditional communication transceiver because of its extremely high data transmission speed, great data throughput and unique data frame organization. The Ethernet transceiver receives a large amount of data as densely as possible. It must store the received data stream in a separate memory space in units of frames, and complete the receiving and sending operations by itself, so that the CPU can read it with a minimum of operations. And dispose of these data, release the corresponding memory space, and ensure that the DMA controller and the CPU will not conflict with the memory usage rights. The single size of the buffer that is opened up to temporarily store the Ethernet frame in the memory is generally set to the maximum packet length of the Ethernet frame. IEEE802.3 stipulates that it is 1518 bytes (Including the source-end hardware address, length or type field and CRC32 calibration). The number of buffers for sending and receiving Ethernet frames is determined by the user according to the frequency of actual interaction and the memory resources of the microcontroller.

The buffer of the Ethernet frame is organized in the form of a queue. In the receiving direction, the Ethernet frame is written into the memory by DMA through the MAC, and is read and dequeued by the CPU; in the sending direction, the Ethernet frame is written to the memory and queued by the CPU, read by DMA and pushed into the transmit FIFO and out of the queue. The depth of the queue is the number of buffers. Different from ordinary communication peripherals, the starting address and number of the buffer of the Ethernet frame are not fixed in a certain register, but are managed by a special type of data structure, which is stored in the memory. A single such data structure unit manages a buffer, which holds the start address, length of the buffer, the settings that the DMA controller needs to make when calling this buffer, the send status of the write-back when the DMA send is completed, and the next such data structure the address of. This data structure plays the role of the traditional communication peripheral control register and status register, but the actual location is in the memory, so it can be called "software register", and the official name is "DMA descriptor".

DMA descriptors are divided into 2 types: sending and receiving, and each format is fixed. The storage structure of DMA descriptors is divided into 2 types, one is linked list form, that is, the fourth word of each descriptor is the address of the next descriptor, and the DMA controller will directly read the next descriptor from TDes3/RDes3. The other is a ring structure, where all descriptors must be closely arranged, DMA control takes a descriptor from

the end of the current descriptor, when the DMA controller detects the TER/RER bit of TDes4/RDes4, i.e., from the beginning of the descriptor list. The address of the descriptor array must be 4-byte aligned, and the size of a single descriptor is 16 bytes.

Figure 31-8 A buffer allocation scheme which describes the ring structure and the chain structure for reference when users allocate space.



31.6.2.1 Transmit DMA Descriptor

Table 31-9, 31-10 shows the structure of the transmit descriptor.

Table 31-9 Structure of the transmit descriptor

	31							0
TDes0	OWM (31)	CTRL (30:26)	TTSE (25)	Reserved	Control (23:20)	Reserved (19:18)	TTSS (17)	State (16:0)
TDes1	Reserved (31:29)		Buffer 2-byte count (28:16)			Reserved (15:13)		Buffer 1-byte count (12:0)
TDes2	Buffer 1 address, timestamp low							
TDes3	Buffer 2 address, address of next descriptor, timestamp high							

Table 31-10 Structure of the enhanced transmit descriptor

	31							0
TDes0	OWM (31)	CTRL (30:26)	TTSE (25)	Reserved	Control (23:20)	Reserved (19:18)	TTSS (17)	Status (16:0)

TDes1	Reserved (31:29)	Buffer 2-byte count (28:16)	Reserved (15:13)	Buffer 1 byte count (12:0)
TDes2	Buffer 1 address			
TDes3	Buffer 2 address/address of next descriptor			
TDes4	Reserved			
TDes5	Reserved			
TDes6	Low timestamp			
TDes7	High timestamp			

It can be seen from the above table that the send descriptor is composed of 4 32-bit words, namely TDes0, TDes1, TDes2 and TDes3, of which TDes0 is used for control and return transmission status, TDes1 is used to indicate the transmission length, and TDes2 is used to indicate the transmission buffer. The position of the zone or the low bit of the return transmit timestamp, TDes3 is used to indicate the address for the second transmit buffer (When TCH is not set).

Compared with the conventional transmission descriptor, the enhanced transmission descriptor adds 4 32-bit words: TDes4, TDes5, TDes6 and TDes7. Among them, when the IEEE1588 function is enabled, TDes6 is used to represent the low timestamp and TDes7 is used to represent the high timestamp.

Each 32-bit word is described below.

Table 31-11 Definitions of TDes0 bits

31	30	29	28	27	26	25	24	23:22	21	19	19:18	17	16
OWN	IC	LS	FS	DC	DP	TTE	Res	CIC	TER	TCH	Res	TSS	IHE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ES	JT	FF	IPE	LCA	NC	LCO	EC	VF	CC				ED	UF	DB

Bit	Name	Description
31	OWN	Descriptor own This bit indicates who occupies this descriptor. 0: This descriptor is owned by the CPU, and the CPU can modify the value of this descriptor; 1: This descriptor is owned by the DMA, and the CPU cannot modify this descriptor. When this bit is 0, it is set to 1 by the CPU after the CPU completes the operation on the descriptor and buffer. When this bit is 1, 0 is automatically written by the DMA after the DMA completes the operation on the descriptor and the buffer. In this way, the user software and hardware complete the operation

		handover of the descriptor and the receive/transmit buffer.
30	IC	Transmission completed interrupt enable When this bit is set, the transmission interrupt bit flag (ETH_DMASR: TS) will be set after the current frame is sent.
29	LS	Last segment This bit is set to indicate that the buffer indicated by this descriptor contains the end of the frame.
28	FS	First segment This bit is set to indicate that the buffer indicated by this descriptor contains the beginning of the frame.
27	DC	Disable automatic CRC calculation When this bit is set, the DMA controller will not calculate the CRC32 value of the Ethernet frame, and no value will be appended to the end of the frame. This bit is only valid when the FS bit is set. Also, the setting of the DP bit takes precedence over this bit.
26	DP	Disable auto-padding When this bit is set, the DMA controller will not add auto-padding for Ethernet frames less than 64 bytes. When this bit is 0, DMA will automatically add padding and CRC value to Ethernet frames less than 64 bytes, regardless of whether DC is set.
25	TTE	Timestamp transmission enable On the premise that ETH_PTPTSCR: TSE is set, the DMA controller will enable the IEEE1588 timestamp function in the Ethernet frame indicated by the frame descriptor after this bit is set. This bit is only valid when FS is set.
24	Reserved	Reserved
23:22	CIC	Checksum and insert control 00: Prohibit insertion of checksum; 01: Only enable the calculation and insertion of IP header checksum; 10: Reserved; 11: Enable calculation and insertion of IP header checksum and payload checksum, enable calculation of pseudo-header checksum.
21	TER	Transmit descriptor end flag setting Users set this bit to indicate to the DMA controller that the current transmit descriptor is the last descriptor of the transmit descriptor array. The DMA controller will next read the first descriptor in the transmit descriptor array.
20	TCH	Next descriptor addresses valid This bit is set to indicate that the second address is the address of the next descriptor and not the address of the next buffer. When

		this bit is set, the value of TBS2 has no effect. This bit is only valid when the FS bit is set. The TER bit has higher priority than this bit.
19:18	Reserved	Reserved
17	TSS	Transmit timestamp status DMA controller sets this bit to indicate that the transmit timestamp is captured and stored in TDes2 and TDes3.
16	IHE	IP header error DMA controller checks the IP header according to the received data: for IPv4, the DMA controller checks whether the length field of the header is correct; for IPv6, the DMA controller checks whether the header is 40 bytes. In addition, the IP protocol type must be the same as the type/length field in the Ethernet frame.
15	ES	Error summary Set by the DMA controller if an error occurs while sending a frame. The ES bit is set when one of the following bits is set: UF[TDES0:1] data underflow error bits; IPE[TDES0:12] IP data error bits; FF[TDES0:13] frame empty bits; JT[TDES0:14] Jabber timeout bits; IHE[TDES0:16] IP header error bits.
14	JT	Jabber timeout When this bit is set, a jabber timeout error occurs at the MAC transmitter. This bit is only set when the JD bit (ETH_MACCR:22) is not set.
13	FF	Frame flushed This bit is set to indicate that the DMA controller clears the frame from FIFO due to the command issued by the CPU.
12	IPE	IP payload error The MAC compares the total packet length in the IPv4 or IPv6 header of the received TCP/UDP/ICMP packet with the actual packet length, and set this bit if they are inconsistent.
11	LCA	Lose carrier This bit is set to indicate that a carrier loss occurs when the frame is sent, that is, the CSR signal is invalid. This bit only works in half-duplex mode.
10	NC	No carrier This bit indicates that the carrier sense signal of the physical layer is invalid when the frame is sent. This bit only works in half-duplex mode.
9	LCO	Late collision This bit indicates that the frame collided after sending the preamble. This bit only works in half-duplex mode.
8	EC	Excessive collisions

		This bit indicates that a collision of more than 16 bits occurs when the frame is sent. If the RD bit in MACCR is set, this bit indicates that only one collision is sent. This bit only works in half-duplex mode.
7	VF	VLAN frame. Set when sending VLAN frames.
6:3	CC	Collision counter These bits indicate how many collisions the frame had when it was sent. No effect when EC is set. These bits only works in half-duplex mode.
2	EC	Excessive deferral This bit is set to indicate that when the DC in MACCR is set, the transmission of the frame fails due to the delay of more than 24288 bits. This bit only works in half-duplex mode.
1	UF	UF data underflow error When the DMA controller fetches data from the specified RAM and finds that the buffer is empty, the transmission enters the suspended state, and sets this bit and the relevant bit of the DMASR register.
0	DB	Deferred bit This bit is set to indicate that the transmission fails due to carrier occupancy. This bit only works in half-duplex mode.

Table 27-12 Definitions of TDes1 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TBS2											

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				TBS1											

Bit	Name	Description
31:29	Reserved	Reserved
28:16	TBS2	Size of the transmit buffer 2.
15:13	Reserved	Reserved
12:0	TBS1	Size of the transmit buffer 1.

Table 27-13 Definitions of TDes2 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TBAD1															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TBAD1															

Bit	Name	Description
31:0	TBAD1	All 32 bits of TDes2 are used as a whole unit to store the address of the transmit buffer 1. (Buffer start address needs to be 32 bytes aligned)

Table 27-14 Definitions of TDes3 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TDAD2															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDAD2															

Bit	Name	Description
31:0	TDAD2	This field is used to store the address of the next descriptor. When TCH is not set, this field stores the address of the transmit buffer 2. (Buffer start address needs to be 32 bytes aligned)

31.6.2.2 Receive DMA Descriptor

Table 31-15 Receive descriptor structure

	31						0
RDes0	OWM(31)	Status (30:0)					
RDes1	Control (31)	Reserved (30:29)	Buffer 2-byte count (28:16)	RER (15:14)	Reserved 13	Buffer 1 byte count (12:0)	
RDes2	Buffer 1 address, timestamp low						
RDes3	Buffer 2 address, address of next descriptor, timestamp high						

Table 31-16 Enhanced Receive descriptor structure

	31						0
RDes0	OWM (31)	Status (30:0)					
RDes1	Control (31)	Reserved (30:29)	Buffer 2-byte count (28:16)	RER (15:14)	Reserved 13	Buffer 1 byte count (12:0)	
RDes2	Buffer 1 address						
RDes3	Buffer 2 address, address of next descriptor						

RDes4	Reserved
RDes5	Reserved
RDes6	Timestamp low
RDes7	Timestamp high

It can be seen from the above figure that the receiving descriptor is also composed of 4 32-bit words, of which the first 32-bit word mainly returns the state when receiving, the second 32-bit word contains the received data length, and the third 32-bit word defines the address of the receive buffer or the low order of the return timestamp, the fourth 32-bit word is the address of the second buffer (RCH is not set), the address of the next buffer (RCH is set) or Returns the high order bit of the timestamp.

The enhanced receiver descriptor adds 4 32-bit words—RDes4, RDes5, RDes6, and RDes7—compared to the standard receiver descriptor. When IEEE 1588 functionality is enabled, RDes6 represents the low-order bits of the timestamp, while RDes7 represents the high-order bits of the timestamp.

The meanings of bits of the receive descriptor are as follows:

Table 31-17 Definitions of RDes0

31	30	29:16
OWN	AFM	FL

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ES	DE	SAF	LE	OE	VLAN	FS	LS	IPHCE	LCO	PT	RWT	RE	DE	CE	PCE

Bit	Name	Description
31	OWN	Descriptor own This bit indicates who occupies this descriptor. 0: This descriptor is owned by the CPU, and the CPU can modify the value of this descriptor; 1: This descriptor is owned by the DMA, and the CPU cannot modify this descriptor. When this bit is 0, it is set to 1 by the CPU after the CPU completes the operations on the descriptor and buffer. When this bit is 1, 0 is automatically written by the DMA after the DMA completes the operations on the descriptor and buffer. In this way, the operation handover of the descriptor and the receive/transmit buffer between the user software and the hardware is completed.
30	AFM	Destination address filter fail This flag is set if the received frame does not pass the MAC's target address filter.
29:16	FL	Frame length This field is valid when ES (RDes0:15) is 0. When LS (RDes0:8) is 1, this field

		indicates the length of the frame received by the DMA controller, including the CRC. When LS is 0, this field indicates the cumulative length sent to the memory by the DMA controller so far, the unit is in bytes.
15	ES	Error summary Set when the MAC detects any of the following errors: CE[RDes0:1]: CRC error; RE[RDes0:3]: Receive error; RWT[RDes0:4]: Watchdog timeout; IPHCE[RDes0:7]: Jumbo frame (note that it is necessary to distinguish whether it is a jumbo frame or an IP header error when reporting IPHCE); OE[RDes0:11]: Overflow error; DE[RDes0:14]: Descriptor error.
14	DE	Descriptor error This bit is set to indicate that the buffer indicated by the descriptor is cut off as the buffer indicated by the descriptor cannot fit the current frame, and the DMA does not occupy the next descriptor. This bit is only valid when the LS bit (RDes0 [8]) is set.
13	SAF	Source address filter fail This bit is set to indicate that the frame does not pass the source address filter of the MAC.
12	LE	Length error Set to indicate that the actual received frame length does not match the length indicated in the Ethernet Type/Length field. This bit is only valid when FT (RDes0 [5]) is set.
11	OE	Overflow error Set to indicate that the received frame is corrupted due to an overflow of the receive FIFO.
10	VLAN	VLAN tag Set to indicate that a VLAN frame is received.
9	FS	First descriptor This bit indicates that this descriptor contains the beginning of the frame.
8	LS	Last descriptor It indicates that this descriptor contains the end of the frame.
7	IPHCE	IP header checksum error This bit is set to indicate that there is an error in the IPv4 or IPv6 header. The specific reasons may be: 1. The protocol indicated by the Ethernet frame type/length field is inconsistent with the actual IP version; 2. The IP header checksum is incorrect; 3. The length indicated by the IP header is incorrect.
6	LCO	Late collision Set to indicate that a late collision occurs. This bit only works in half-duplex mode.
5	FT	Frame type indication bit.

		1 indicates that the received frame is an Ethernet type encapsulated frame (RFC 894). 0 indicates that the received frame is an IEEE802.3 type encapsulated frame (RFC1042). This bit is invalid when the frame length is less than 14 bytes. <i>Note: Refer to Table 31-16 for the special meaning of FT.</i>
4	RWT	Receive watchdog timeout Set to indicate that when the current frame is received, the watchdog times out and the current frame is truncated.
3	RE	Receive error Set to indicate that the RX_ERR signal is valid when RX_DV is valid in the process of receiving a frame.
2	DE	Dribble bit error Set to indicate that the length of the frame received by the MAC is not multiples of the 8-bit frame, and the period may be missed.
1	CE	CRC error This bit is set to indicate that the received frame has a CRC check error. This bit is valid only when the LS bit (RDes0 [8]) is set.
0	PCE	Payload checksum error This bit is set to indicate that the TCP/UDP/ICMP packet received by the MAC does not match the value indicated in the checksum field.

It can be seen that the verification mechanism is implemented in the receiving process of the MAC. In fact, the data length of this layer is described at the Ethernet frame layer (Data link layer), network layer (IPv4/IPv6) and transport layer (TCP/UDP/SCTP), and some check measures are taken to ensure the correctness of the data content. The 0th, 5th and 7th bits of Des0 are related to the checksum of the data, which is summarized in the following table.

Table 31-18 Relationship between the value of Des0:7/5/0 and the status of the received frame

RDes0:5	RDes0:7	RDes0:0	Frame status
Frame Type Indicator Bit (FT)	IP header checksum error flag bit (IPHCE)	Load checksum error (PCE)	
1	0	0	IP type frame, IP header not detected and load checksum error
1	0	1	IP type frame, load checksum error
1	1	0	IP type frame, IP header checksum error
1	1	1	IP type frame, IP header checksum error, load checksum error
0	0	0	IEEE802.3 Type Encapsulated Frame (RFC1042)
0	0	1	IP type frame, IP header checksum error not detected, not detected because load checksum is not supported
0	1	1	Not an IP type frame (e.g. ARP frame)
0	1	0	Reserved

Table 31-19 Definitions of RDes1 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DIC	Reserved	RBS2													

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RER	RCH	Res	RBS1												

Bit	Name	Description
31	DIC	Disable reception completion interrupt setting bit.
30:29	Reserved	Unused.
28:16	RBS2	Receive buffer 2 size. (The length is required to be a multiple of 32)
15	RER	End receive descriptor flag. It indicates that the current descriptor is the last descriptor. The DMA controller will go back to the Descriptor Pair Column Base Address Register (ETH_DMARDLAR) to fetch the next descriptor.
14	RCH	Next receive descriptor address valid bit. This bit is set to indicate that in the last 32-bit word is the address of the next receive descriptor, otherwise the address of the second buffer.
13	Reserved	Unused.
12:0	RBS1	Receive buffer 1 size. (The length is required to be a multiple of 32)

Table 31-20 Definitions of RDes2 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RBAD1															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RBAD1															

Bit	Name	Description
31:0	RBAD	All 32 bits of RDes2 are used as a whole unit to store the address of the receive buffer. (Buffer start address needs to be 32 bytes aligned)

Table 31-21 Definitions of RDes3 bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDAD2															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDAD2															

Bit	Name	Description
31:0	RDAD	Used to store the address of the next buffer when RCR is set, otherwise used to store the address of the second buffer. (Buffer start address needs to be 32 bytes aligned)

31.6.3 Data Cache Alignment

Since both the transceiver buffer and the transceiver descriptor are called by the DMA controller and exist in the RAM space, and the DMA is 256-bit, it is necessary to ensure that their start addresses are aligned in 32 bytes when setting the transceiver descriptor queue, but setting the buffer does not require this forcibly.

In order to improve efficiency, it is most appropriate that an Ethernet frame is received by a buffer. For example, setting the buffer to 1518 bytes can accommodate the largest Ethernet ordinary frame, including source-end address field, length type field, Data padding field and CRC field. It should be noted that the maximum frame length of the tagged VLAN frame is 4 bytes longer than the general Ethernet frame, that is, 1522 bytes. Because the length of the buffer is required to be a multiple of 32, a buffer can be set to 1536 bytes.

31.6.4 DMA Transceiver Configuration

The data received and sent is automatically dumped and pushed by the DMA, but if a fatal error is encountered, the DMA will stop and update the DMA status register. The user needs to manually start the DMA after initializing the DMA to continue running.

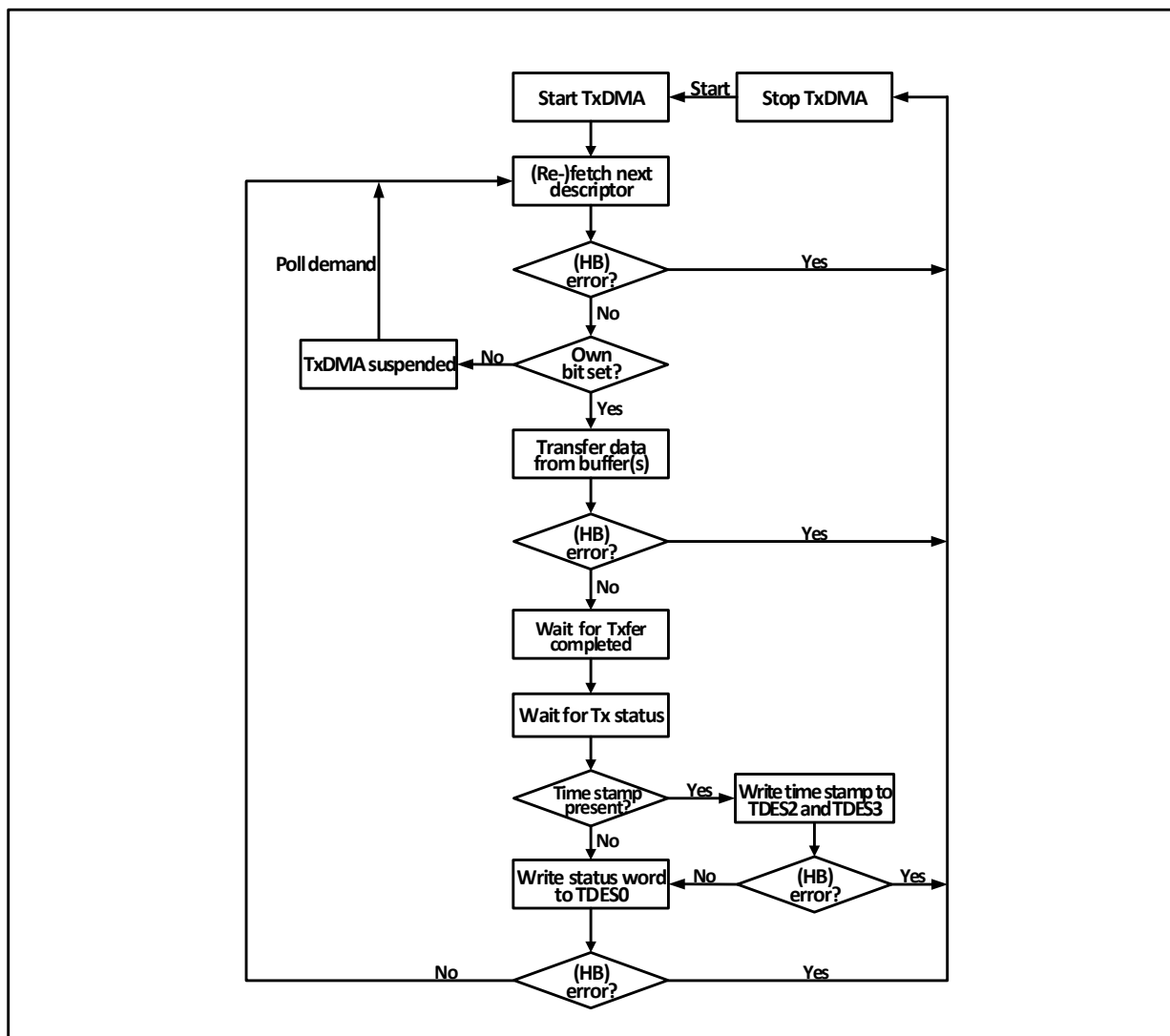
31.6.4.1 Transmit DMA Configuration

The steps for the DMA controller to establish the operating mechanism are as follows:

- 1) Set up the transmit buffer queue and fill the transmit buffer with the content to be sent. Set up the transmit descriptor queue, fill in the fields of the transmit descriptor, set OWN, and deliver the management right of the transmit descriptor to the DMA controller; register the initial address of the descriptor in the DMATDLAR register;
- 2) Set the ST bit (DMAOMR: 13) to enable DMA;
- 3) In Run mode, the DMA descriptor will automatically read the content of the transmit descriptor, and push the data to the transmit FIFO according to the address and length indicated by it. After finishing the DMA, it will read the next transmit descriptor immediately following the chain structure for the next transmission. When the DMA controller detects that the OWM bit is not set, resulting in no access to the transmit descriptor or other normal errors, it will terminate the transfer and set the TBUS bit (DMASR: 2) or other bits (Other than errors caused by OWN being 0) and the NIS bit (DMASR:16) is set.
- 4) A single frame is not allowed to span multiple descriptors; a frame must be clearly associated with a descriptor and a buffer.
- 5) If the MAC turns on the IEEE1588 PTP mode, after the DMA controller pushes the data to the FIFO, the MAC will write the transmit timestamp to TDes2 and TDes3, and reset the OWN bit.
- 6) After sending a frame, if the transmit descriptor enables the transmit completion interrupt (Set TDes1:31), the DMA controller will set the transmit completion interrupt flag (DMASR:0), and then continue to fetch the next transmit descriptor.

The following figure shows the default process of transmission.

Figure 31-9 Process of transmission



31.6.4.2 Receive DMA Configuration

The steps to establish a DMA receive flow mechanism are as follows:

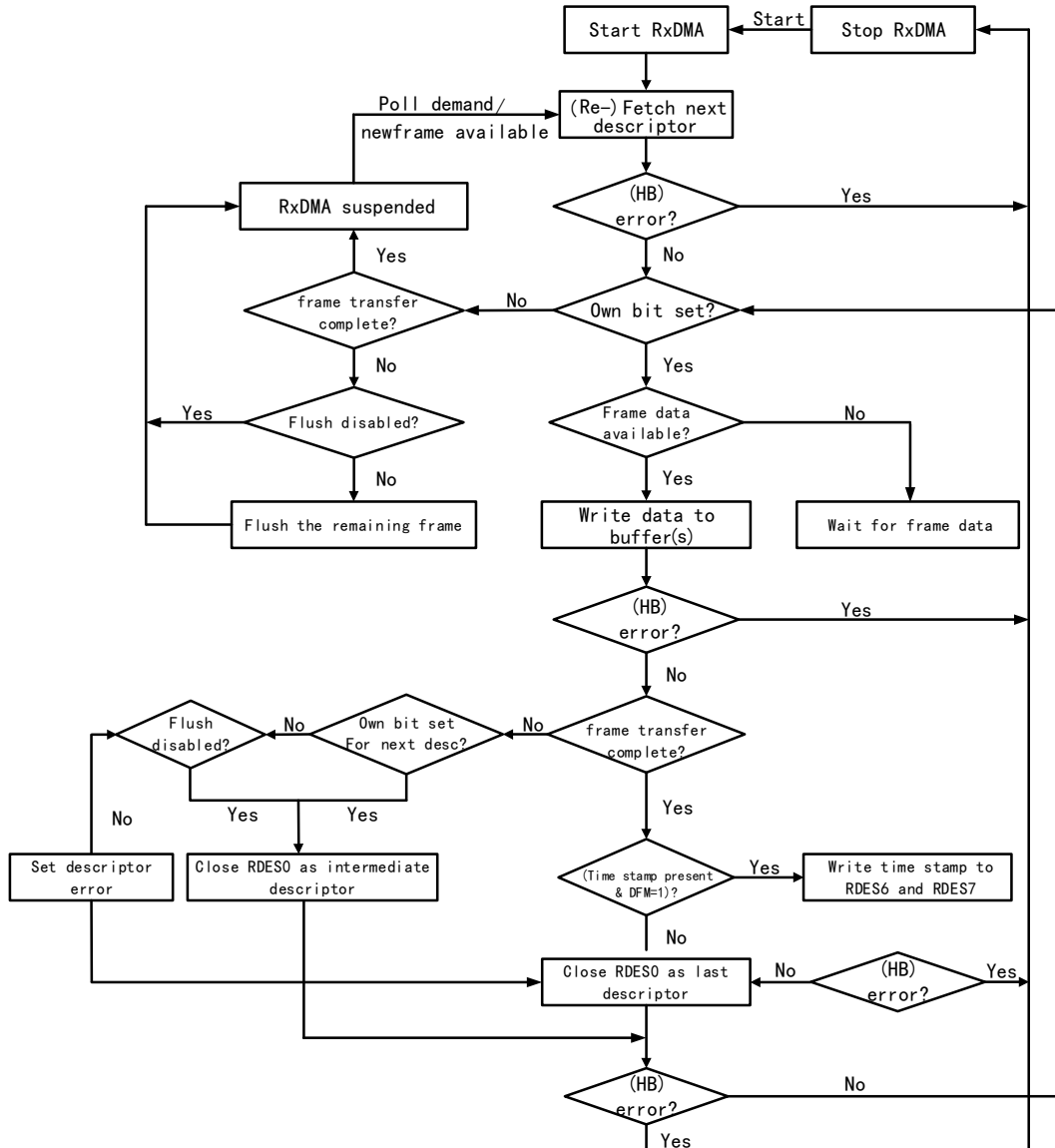
- 1) Set the transmit buffer queue and descriptor queue, and set up the bits of each field of the receive descriptor; register the initial address of the descriptor in the DMARDLAR register; set the OWN bit to give the descriptor use permission to the DMA controller;
- 2) Set the SR bit to start the receiving process;
- 3) When the receive flow mechanism is running, the DMA controller obtains the next descriptor, checks the receiving descriptor configuration, and when the FIFO receives the next frame, performs multiple checks such as filtering and identifying the frame content, and forwards the frame data to the buffer specified by the symbol, writes the descriptor status field and obtains the next receiving descriptor, and reports the receiving completion interrupt. If the frame does not pass the filtering, it will be marked or discarded. If the frame has a checksum error, CRC error or the frame is too short, it will be marked. If the frame is too long, it may be cut off or receive a watchdog timeout error. The DMA controller will stop the receiving process when it encounters a fatal error such as the unavailability of the receive descriptor, and the user needs to pay special attention;
- 4) The user needs to enable at least one receive completion interrupt, and restore the used receive descriptor to the standby state in the Ethernet interrupt function to ensure that the receive process can run uninterrupted. The

user can pass the address of the data buffer to be processed in the interrupt function, or handle some abnormal events that interrupt the receiving process.

- 5) If the MAC opens the IEEE1588PTP mode and enables the enhanced descriptor, when the DMA controller dumps data and writes the descriptor status field, the current timestamp will also be written into the last two words of the descriptor.

The following figure shows the default flow mechanism of reception:

Figure 31-10 Process of reception



31.7 Interrupt

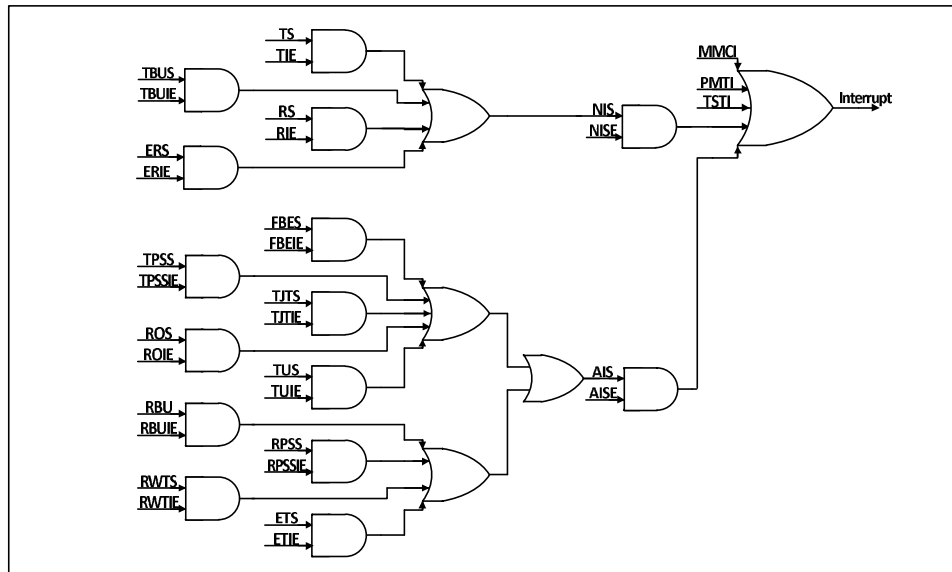
The Ethernet transceiver has 2 interrupt vectors, one is the Ethernet wake-up event and the other is the normal transceiver event. When a wake-up frame or magic frame is detected, an Ethernet wake-up event is triggered. Common Ethernet transceiver interrupt events include DMA interrupt and ETH interrupt.

31.7.1 DMA Interrupt

DMA interrupts can be roughly divided into 2 groups, namely normal interrupts (NIS) and abnormal interrupts

(AIS). Abnormal interrupts generally mean abnormal data transmission and reception, which requires special attention. When processing interrupts, users need to retrieve all interrupt flag bits and process all generated interrupt sources. The following figure is a schematic diagram of the interrupt of the Ethernet transceiver.

Figure 31-11 Interrupt schematic



The DMA interrupt is the most important interrupt in the Ethernet transceiver. Generally, user logic needs to rely on interrupts to receive frames, confirm that the frames are sent, or deal with the interrupted transceiver logic in time.

31.7.2 ETH Interrupt

The ETH interrupt mainly includes the time alarm trigger of PTP, the sending and receiving count to a certain set value of the MMC register, and the PMT. The usefulness of ETH is mainly functional, not very complex and important relative to DMA interrupts. Users can use PTP interrupt for alarm clock, MMC interrupt for speed measurement, or PMT interrupt for remote wake-up. In addition, ETH also supports interrupts when the state of the built-in physical layer connection changes.

31.7.3 PMT Interrupt

The reason why the PMT interrupt is raised separately and reiterated is that this interrupt has an independent interrupt vector and needs to be paid attention to when using it.

31.8 Register Description

Table 31-22 Ethernet registers

MAC control register address mapping

Name	Offset Address	Description	Reset Value
R32_ETH_MACCR	0x40028000	MAC control register	0x00000000
R32_ETH_MACFFR	0x40028004	Frame filter register	0x00000000
R32_ETH_MACHTHR	0x40028008	Hash value list register high	0x00000000
R32_ETH_MACHTLR	0x4002800C	Hash value list register low	0x00000000

R32_ETH_MACMIAR	0x40028010	MII address register	0x00000000
R32_ETH_MACMIIDR	0x40028014	MII data register	0x00000000
R32_ETH_MACFCR	0x40028018	MAC flow control register	0x00000000
R32_ETH_MACVLAN	0x4002801C	VLAN tag register	0x00000000
R32_ETH_MACRWUFR	0x40028028	Wake-up frame filter register	0x00000000
R32_ETH_MACPMTCSR	0x4002802C	PMT control and status Register	0x00000000
R32_ETH_MACSR	0x40028038	MAC interrupt status register	0x00000000
R32_ETH_MACIMR	0x4002803C	MAC interrupt mask register	0x00000000
R32_ETH_MACA0HR	0x40028040	MAC address register 0 high 32 bits	0x8000FFFF
R32_ETH_MACA0LR	0x40028044	MAC address register 0 low 32 bits	0xFFFFFFFF
R32_ETH_MACA1HR	0x40028048	MAC address register 1 high 32 bits	0x0000FFFF
R32_ETH_MACA1LR	0x4002804C	MAC address register 1 low 32 bits	0xFFFFFFFF
R32_ETH_MACA2HR	0x40028050	MAC address register 2 high 32 bits	0x0000FFFF
R32_ETH_MACA2LR	0x40028054	MAC address register 2 low 32 bits	0xFFFFFFFF
R32_ETH_MACA3HR	0x40028058	MAC address register 3 high 32 bits	0x0000FFFF
R32_ETH_MACA3LR	0x4002805C	MAC address register 3 low 32 bits	0xFFFFFFFF
R32_ETH_PHY_CFGR	0x40028080	MAC_PHY configuration register	0x40000001

MMC control register address mapping, note: the addresses are not consecutive

Name	Address	Description	Reset Value
R32_ETH_MMCCR	0x40028100	MMC control register	0x00000000
R32_ETH_MMCRIR	0x40028104	MMC receive register	0x00000000
R32_ETH_MMCTIR	0x40028108	MMC transmit interrupt register	0x00000000
R32_ETH_MMCRIMR	0x4002810C	MMC receive interrupt mask register	0x00000000
R32_ETH_MMCTIMR	0x40028110	MMC transmit interrupt mask register	0x00000000
R32_ETH_MMCTGFSCCR	0x4002814C	MMC transmit good frame counter after single conflict	0x00000000
R32_ETH_MMCTGFMSCCR	0x40028150	MMC transmit good frame counter after multiple conflicts	0x00000000
R32_ETH_MMCTGFCR	0x40028168	MMC transmit good frame count register	0x00000000
R32_ETH_MMCRFCECR	0x40028194	MMC receive CRC error frame count register	0x00000000
R32_ETH_MMCRFAECR	0x40028198	MMC receive alignment error frame counter register	0x00000000
R32_ETH_MMCRAFCR	0x4002819C	MMC receives all frame count registers	0x00000000
R32_ETH_MMCRGUFCR	0x400281C4	MMC receive good unicast frame count register	0x00000000

IEEE1588 (PTP) register address mapping

Name	Address	Description	Reset Value
R32_ETH_PTPTSCR	0x40028700	PTP timestamp control register	0x00000000
R32_ETH_PTPSSIR	0x40028704	PTP subsecond increment register	0x00000000
R32_ETH_PTPTSHR	0x40028708	PTP timestamp register high	0x00000000

R32_ETH_PTPTSLR	0x4002870C	PTP timestamp register low	0x00000000
R32_ETH_PTPTSHUR	0x40028710	PTP timestamp update register high	0x00000000
R32_ETH_PTPTSLUR	0x40028714	PTP timestamp update register low	0x00000000
R32_ETH_PTPTSAR	0x40028718	PTP timestamp adder register	0x00000000
R32_ETH_PTPTTHR	0x4002871C	PTP target register high	0x00000000
R32_ETH_PTPTTLR	0x40028720	PTP target register low	0x00000000

DMA register address mapping. Note: The addresses are not consecutive

Name	Address	Description	Reset Value
R32_ETH_DMABMR	0x40029000	DMA bus mode register	0x00000001
R32_ETH_DMATPDR	0x40029004	DMA transmit poll demand register	0x00000000
R32_ETH_DMARPTR	0x40029008	DMA receive query register	0x00000000
R32_ETH_DMARDLAR	0x4002900C	DMA receive descriptor address register	0x00000000
R32_ETH_DMATDLAR	0x40029010	DMA transmit descriptor address register	0x00000000
R32_ETH_DMASR	0x40029014	DMA status register	0x00000000
R32_ETH_DMAOMR	0x40029018	DMA operation mode register	0x00000000
R32_ETH_DMAIER	0x4002901C	DMA interrupt enable register	0x00000000
R32_ETH_DMAMFBOCR	0x40029020	DMA lost frame register	0x00000000
R32_ETH_DMACHTDR	0x40029048	DMA current transmit descriptor register	0x00000000
R32_ETH_DMACHRDR	0x4002904C	DMA current receive descriptor register	0x00000000
R32_ETH_DMACHTBAR	0x40029050	DMA current transmit buffer register	0x00000000
R32_ETH_DMACHRBAR	0x40029054	DMA current receive buffer register	0x00000000

Internal 10M/100M physical layer register address

Name	Offset Address	Description	Reset Value
R16_BMCR	0x00	Basic control register	0x3100
R16_BMSR	0x01	Basic status register	0x7849
R16_ANAR	0x04	Auto-negotiation message register	0x01E1
R16_ANALPAR	0x05	Auto-negotiation linker capability register	0x0001
R16_PAGE_SEL	0x1F	Page selection register	0x0000

Note: the offset address of the internal physical layer registers is used in the SMI interface

PAGE0-related register address

Name	Offset Address	Description	Reset Value
R16_PHY_STATUS	0x1A	PHY status register	0x0020
R16_INTERRUPT_IND	0x1E	Interrupt indication register	0x0000

PAGE7-related register address

Name	Offset Address	Description	Reset Value
R16_INTERRUPT_MASK	0x13	Interrupt /LED function register	0x0030

31.8.1 MAC Control Registers

31.8.1.1 MAC Control Register (R32_ETH_MACCR)

Offset Address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TCD[2:0]			Reserved					WD	JD	PI	PR	IFG[2:0]			CSD
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FES[1:0]		ROD	LM	DM	IPCO	RD	Res	APCS	BL		DC	TE	RE	TCF	Res

Name	Name	Access	Description	Reset Value
[31:29]	TCD[2:0]	RW	Transmit clock delay field, this field is used to delay the transmit clock. The MAC controller delays the output of the transmit clock selected by the CES bit (ETH_MACCR[1]). Delay time calculation formula: $T_{delay} = TCD(ETH_MACCR[31:29]) * 0.5ns$.	0
[28:24]	Reserved	RO	Reserved	0
23	WD	RW	Watchdog setting. 0: The MAC watchdog is enabled, and can only accept Ethernet frames with a maximum size of 2048 bytes, and the excess data is truncated; 1: The MAC watchdog is disabled, and can receive Ethernet frames with a maximum size of 16384 bytes.	0
22	JD	RW	Jabber setting. 0: If the user tries to send an Ethernet frame with a length of more than 2048 bytes, the MAC will close the transmitter; 1: The MAC closes the Jabber timer, and can send up to 16384 bytes of Ethernet frames.	0
21	PI	RW	Built-in 10MPHY transmit drive bias current setting bits. 0: Rated drive; 1: Low-power transmission.	0
20	PR	RW	Built-in 10MPHY on-chip 50Ω pull-up resistor enable. 0: On-chip 50Ω resistor disconnected; 1: On-chip 50Ω resistor connected.	0
[19:17]	IFG[2:0]	RW	Frame gap setting, used to set the shortest time gap between sending 2 frames. 000: 96-bit time; 001: 88-bit time; 010: 80-bit time; 011: 72-bit time; 100: 64-bit time; 101: 56-bit time;	0

			110: 48-bit time; 111: 40-bit time.	
16	Reserved	RO	Reserved	0
[15:14]	FES[1:0]	RW	Ethernet speed setting. 00: 10Mbit/s; 01: 100Mbit/s; 10: 1Gbit/s; 11: Reserved, unused.	0
13	Reserved	RO	Reserved	0
12	LM	RW	Loopback mode enable. Setting this bit enables loopback mode.	0
11	DM	RW	Duplex mode enable. Setting this bit enables full duplex mode.	0
10	IPCO	RW	IPv4 checksum check enable. 0: Disable the checksum verification function of IPv4 at the receiving end, and the corresponding PCE and PHCE flags are always 0. See the definition of each bit of the receive descriptor; 1: Enable IPv4 checksum verification, the MAC controller will check the TCP, UDP and ICMP headers.	0
[9:8]	Reserved	RO	Unused.	0
7	APCS	RW	Padding & CRC auto-stripping enable. 0: The MAC does not modify the contents of the frame; 1: The MAC removes the Pad/FCS field from the frame only if the length/type field value is less than or equal to 1500 bytes. If the Length/Type field is greater than or equal to 1501 bytes, the frame contents are not changed.	0
[6:4]	Reserved	RO	Unused	0
3	TE	RW	Transmit enable. 0: Disable the transmitter, and no frame will be transmitted after the MAC finishes sending the current frame; 1: Enable the MAC transmitter.	0
2	RE	RW	Receive enable. 0: Disable the receiver, and no frame will be received after the MAC finishes receiving the current frame; 1: Enable the MAC receiver.	0
1	TCF	RW	Transmit clock toggle setting. 0: TXC selected by TCES is directly used as the GTX_CLK output by the chip; 1: Use the inversion of TXC selected by TCES as the GTX_CLK output by the chip.	0

0	Reserved	RO	Reserved	0
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31.8.1.2 MAC Frame Filter Register (R32_ETH_MACFFR)

Offset Address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RA	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						HPF	SAF	SAIF	PCF[1:0]	BFD	PAM	DAIF	HM	HU	PM

Bit	Name	Access	Description	Reset Value
31	RA	RW	Receive all 0: MAC only forwards frames that have passed the filter to the receive queue; 1: The MAC forwards all received frames to the receive queue, regardless of whether they pass the filter or not.	0
[30:11]	Reserved	RO	Reserved	0
10	HPF	RW	HASH filter or perfect filter 0: On the premise that HM or HU is set, as long as it meets the HASH filter, it can pass the address filter; 1: Determine which filtering method is used in unicast/multicast mode according to the value of HM or HU.	0
9	SAF	RW	Source address filter 0: MAC marks frames that do not pass the source MAC address filtering; 1: MAC will directly discard frames that do not pass the source MAC address filtering.	0
8	SAIF	RW	Source address filter result inverted. 0: If the source address of the received frame is inconsistent with the source address enabled in the MAC address register, it is considered to have failed the source address filter; 1: If the source address of the received frame is consistent with the source address enabled in the MAC address register, it is considered to have failed the source address filter.	0
[7:6]	PCF[1:0]	RW	Flow control frames pass control. 00/01: The MAC does not forward any flow control frames to the application; 10: MAC forwards all flow control frames to the application, including flow control frames that fail the address filter;	00b

			11: MAC only forwards flow control frames that pass the address filter.	
5	BFD	RW	Broadcast frame reception control 0: Receive all broadcast frames; 1: Discard all broadcast frames.	0
4	PAM	RW	Pass all multicast frame control. 0: Whether the multicast frame can pass filtering depends on the value of HM; 1: All multicast frames can be filtered by address.	0
3	DAIF	RW	Destination MAC address filter results inverted control. 0: The filter result takes effect normally; 1: For multicast frames and unicast frames, whether the result of the filter passing is reversed before it takes effect.	0
2	HM	RW	Multicast frame filter mode 0: Perfect address filtering; 1: HSAH address filtering.	0
1	HU	RW	Unicast frame filter mode 0: Perfect address filtering; 1: HSAH address filtering.	0
0	PM	RW	Promiscuous mode enable All frames pass the address filter without marking the result of the filter.	0

31.8.1.3 Hash List Register (R32_ETH_MACHTHR, R32_ETH_MACHTLR)

Offset Address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTH[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTH[15:0]															

Offset Address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTL[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTL[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HTH[31:0]	RW	Hash list high 32 bits	0
[31:0]	HTL[31:0]	RW	Hash list low 32 bits	0

31.8.1.4 MII Address Register (R32_ETH_MACMIAR)

Offset Address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PA[4:0]					MR[4:0]					Reserved	CR[2:0]		MW	MB	

Bit	Name	Access	Description	Reset Value
[31:16]	Reserved	RO	Unused.	0
[15:11]	PA[4:0]	RW	Physical layer address. The user writes the physical layer address to be operated into this field.	0
[10:6]	MR[4:0]	RW	Physical layer register address field. The user writes the register address to be operated into this field.	0
5	Reserved	RO	Unused.	0
[4:2]	CR[2:0]	RW	Clock range setting. Users, please keep it as 000b, so that the MII frequency is divided by 42 of the main frequency.	0
1	MW	RW	Read and write setting. 0: Read the physical layer; 1: Write to the physical layer.	0
0	MB	RW1Z	MII busy flag. This bit is set by the user, indicating that the hardware is instructed to start reading or writing. The physical address, register address and data field should remain unchanged during reading and writing. After the hardware clears this bit, the operation is completed.	0

31.8.1.5 MII Data Register (R32_ETH_MACMIIDR)

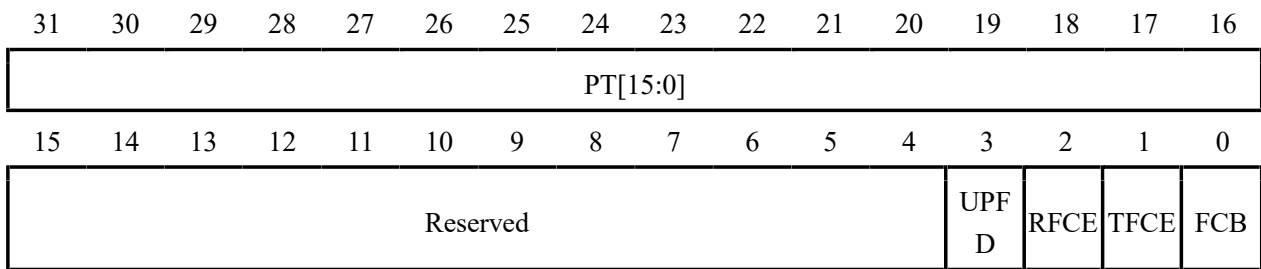
Offset Address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MD[15:0]															

Bit	Name	Access	Description	Reset Value
[31:16]	Reserved	RO	Reserved.	0
[15:0]	MD[15:0]	RW	MII operation data. This field is used to store the data to be read from the physical layer through the MII interface, or to store the data written to the physical layer.	0

31.8.1.6 MAC Flow Control Register (R32_ETH_MACFCR)

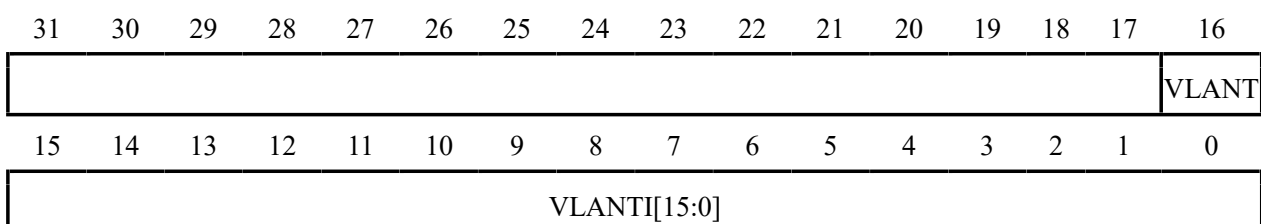
Offset address: 0x18



Bit	Name	Access	Description	Reset Value
[31:16]	PT[15:0]	RW	Pause interval. This field is used to control the value of the Pause time field, and the unit is the time it takes for the current MII interface to send 64 bytes.	0
[15:4]	Reserved	RO	Reserved	0
3	UPFD	RW	Unicast Pause frame detection. 0: The MAC only receives Pause frames with the unique address defined by the protocol specification; 1: The MAC also detects whether the Pause frame is a unicast address defined in the MAC address register 0.	0
2	RFCE	RW	Receive flow control enable. 0: The MAC does not parse the Pause frame; 1: The MAC parses the Pause frame and turns off the transmitter for a while.	0
1	TFCE	RW	Transmit flow control enable. 0: MAC closes sending flow control and does not send Pause frames; 1: MAC enables sending flow control and can send Pause frames.	0
0	FCB	RW1Z	Flow control busy flag. This bit can be set to send a Pause frame, which is cleared by hardware after the transmission is completed. When operating on the entire MACFCR register, it is necessary to ensure that the FCB bit is 0.	0

31.8.1.7 VLAN Tag Register (MACVLAN)

Offset address: 0x1C



Bit	Name	Access	Description	Reset Value
[31:17]	Reserved	RO	Reserved	0

16	VLANT[15:0]	RW	Tag compare control. 0: Compare all 16-bit data of the 15th and 16th bytes of the VLAN frame with the VLANTI field; 1: Only use the [11:0] bits of the 15th and 16th bytes of the VLAN frame and compare them with the corresponding bits of the VLANTI field.	0
[15:0]	VLANTI[15:0]	RW	Label vs. sample domain. According to the IEEE 802.1 protocol, bits [15:13] of the VLAN frame are the user priority, [12] is the canonical format indicator, and bits [11:0] of the VLAN identifier field. If VLANTI is all 0, then the MAC will no longer care about the 15th and 16th bytes of the VLAN frame, and when the 13th and 14th bytes are 0x8100 (pay attention to the size end), it will be judged as a VLAN frame.	0

31.8.1.8 Wake-up Frame Filter Register (R32_MACRWUFR)

Offset Address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RWUFR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RWUFR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RWUFR[31:0]	RW	The wake-up frame filter register is actually 8 different registers, and 8 consecutive read operations can read out all registers, and 8 consecutive write operations can be written into all 8 registers. The description of each bit of these 8 registers is described in Section 31.5.6.3.	0

31.8.1.9 PMT Control and Status Register (R32_ETH_MACPMTCSR)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WFF RP	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						GU	Reser ved	PHY_ PME BR	WFR	MPR	Reser ved	PHY_ PME B_EN	WFE	MPE	PD

Bit	Name	Access	Description	Reset Value
31	WFFRP	RW1Z	Wake-up frame filter register pointer reset. Setting this	0

			bit clears the MACRWUFR register all the way to zero. This bit is automatically cleared on a clock cycle.	
[30:10]	Reserved	RO	Reserved.	0
9	GU	RW	Global unicast. Setting this bit causes the MAC to consider all unicast frames that pass the filter to be wake-up frames.	0
8	Reserved	RO	Reserved.	0
7	PHY_PMEBR	RZ	When a wake-up frame is received from the PHY, this bit is set. It is automatically cleared upon read.	0
6	WFR	RZ	Wake-up frame received flag. This bit is set when a wake-up frame is received. Read it and it will be cleared automatically.	0
5	MPR	RZ	Magic frame received flag. This bit is set when a magic frame is received. Read it and it will be cleared automatically.	0
4	Reserved	RO	Reserved.	0
3	PHY_PMEB_EN	RW	PHY_PMEB enable bit. Setting this bit allows the PHY to generate a PMT event upon receiving a wake-up frame.	0
2	WFE	RW	Wake-up frame enable. Setting this bit allows a PMT event to be generated when a wake-up frame is received.	0
1	MPE	RW	Magic frame enable. Setting this bit allows a PMT event to be generated when a magic frame is received.	0
0	PD	RW1Z	Power-down control. Setting this bit will put the MAC into power-down mode: all other frames are discarded until it receives a wake-up frame or magic frame, and the MAC is automatically cleared after wake-up. Before setting this bit, WFE or MPE needs to be set.	0

31.8.1.10 MAC Interrupt Status Register (R32_ETH_MACSR)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TSTS	Reserved	MMCTS	MMCRS	MMCS	PMTS	Reserved			

Bit	Name	Access	Description	Reset Value
[31:10]	Reserved	RO	Reserved	0
9	TSTS	RC	Timestamp trigger interrupt flag. When the time set by the PTP system clock is reached, this flag will be set.	0
[8:7]	Reserved	RO	Reserved	0

6	MMCTS	RO	MMC transmit interrupt flag. This bit is set when any interrupt is generated in the MMCTIR register in the MMC register bank. This bit is also cleared when all the MMCTIR registers in the MMC register bank are cleared.	0
5	MMCRS	RO	MMC receive interrupt flag. This bit is set when any interrupt is generated in the MMCRIR register in the MMC register bank. This bit is also cleared when the MMCRIR registers in the MMC register bank are all cleared.	0
4	MMCS	RO	MMC status flag. When MMCTS or MMCRS is set, this bit is triggered. When MMCTS and MMCRS are all cleared, this bit is cleared.	0
3	PMTS	RO	PMT status flag. In power-down mode, if the MAC receives a magic frame or wake-up frame wakes up the MAC, this bit will be set. After clearing the WFR and MPR bits, this bit is cleared.	0
[2:0]	Reserved	RO	Reserved	0

31.8.1.11 MAC Interrupt Mask Register (R32_ETH_MACIMR)

Offset Address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TSTIM	Reserved					PMTIM	Reserved		

Bit	Name	Access	Description	Reset Value
[31:10]	Reserved	RO	Reserved	0
9	TSTIM	RW	Timestamp interrupt mask. Setting this bit disables generation of timestamp interrupts.	0
[8:4]	Reserved	RO	Reserved	0
3	PMTIM	RW	PMT interrupt mask. Setting this bit will mask the PMT interrupt.	0
[2:0]	Reserved	RO	Reserved	0

31.8.1.12 MAC Address Register 0 High 32 Bits (R32_ETH_MACA0HR)

Offset Address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MO	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

MACA0H[15:0]

Bit	Name	Access	Description	Reset Value
31	MO	RO	Always 1.	1
[30:16]	Reserved	RO	Reserved	0
[15:0]	MACA0H[15:0]	RW	High 16 bits of the MAC address, that is, 47:32 bits.	0xFFFF

31.8.1.13 MAC Address Register 0 Low 32 Bits (R32_ETH_MACA0LR)

Offset Address: 0x44

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

MACA0L[31:16]

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

MACA0L[15:0]

Bit	Name	Access	Description	Reset Value
[31:0]	MACA0L[31:0]	RW	Low 32 bits of the MAC address, namely 31:0 bits. In general, the address of the MAC address register 0 is the address of the MAC itself.	0xFFFFFFFF F

31.8.1.14 MAC Address Register 1 High 32 Bits (R32_ETH_MACA1HR)

Offset address: 0x48

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

AE	SA	MBC[5:0]	Reserved
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15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

MACA1H[15:0]

Bit	Name	Access	Description	Reset Value
31	AE	RW	Address filter enable. 0: The address filter ignores MAC address 1. 1: The address filter uses MAC address 1 for perfect filtering;	0
30	SA	RW	Address role selection. 0: MAC address 1 is used to compare with the destination address of the received frame; 1: MAC address 1 is used to compare with the source address of the received frame.	0
[29:24]	MBC[5:0]	RW	Mask word control. Each bit of the MBC is used to correspond to a certain byte of the masked MAC address 1, and is used to prohibit a certain byte of the MAC address 1 from participating in the comparison of	0

			the source address or the destination address of the received frame. R32_ETH_MACA1HR[29] is set to mask MACA1H[15:8]; R32_ETH_MACA1HR[28] is set to mask MACA1H[7:0]; R32_ETH_MACA1HR[27] is set to mask MACA1L[31:24]; R32_ETH_MACA1HR[26] is set to mask MACA1L[23:16]; R32_ETH_MACA1HR[25] is set to mask MACA1L[15:8]; R32_ETH_MACA1HR[24] is set to mask MACA1L[7:0].	
[23:16]	Reserved	RO	Reserved	0
[15:0]	MACA1H	RW	High 16 bits of the MAC address, that is, 47:32 bits.	0xFFFF

31.8.1.15 MAC Address Register 1 Low 32 Bits (R32_ETH_MACA1LR)

Offset Address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA1L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA1L[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	MACA1L[31:0]	RW	Low 32 bits of the MAC address, namely 31:0 bits.	0xFFFFFFFF F

31.8.1.16 MAC Address Register 2 High 32 Bits (R32_ETH_MACA2HR)

Offset Address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA2H[15:0]															

Bit	Name	Access	Description	Reset value
31	AE	RW	Address filter enable. 0: The address filter ignores the MAC address 2. 1: The address filter uses MAC address 2 for perfect filtering.	0
30	SA	RW	Address role selection.	0

			0: MAC address 2 is used to compare with the destination address of the received frame; 1: MAC address 2 is used to compare with the source address of the received frame.	
[29:24]	MBC[5:0]	RW	Mask word control. Each bit of the MBC is used to correspond to a certain byte of the masked MAC address 2, and is used to prohibit a certain byte of the MAC address 2 from participating in the comparison of the source address or the destination address of the received frame. R32_ETH_MACA2HR[29] is set to mask MACA2H[15:8]; R32_ETH_MACA2HR[28] is set to mask MACA2H[7:0]; R32_ETH_MACA2HR[27] is set to mask MACA2L[31:24]; R32_ETH_MACA2HR[26] is set to mask MACA2L[23:16]; R32_ETH_MACA2HR[25] is set to mask MACA2L[15:8]; R32_ETH_MACA2HR[24] is set to mask MACA2L[7:0].	0
[23:16]	Reserved	RO	Reserved	0
[15:0]	MACA2H[15:0]	RW	High 16 bits of the MAC address, that is, 47:32 bits.	0xFFFF

31.8.1.17 MAC Address Register 2 Low 32 Bits (R32_ETH_MACA2LR)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA2L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA2L[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MACA2L[31:0]	RW	Low 32 bits of the MAC address, that is, 31:0 bits.	0xFFFFFFFF

31.8.1.18 MAC Address Register 3 High 32 Bits (R32_ETH_MACA3HR)

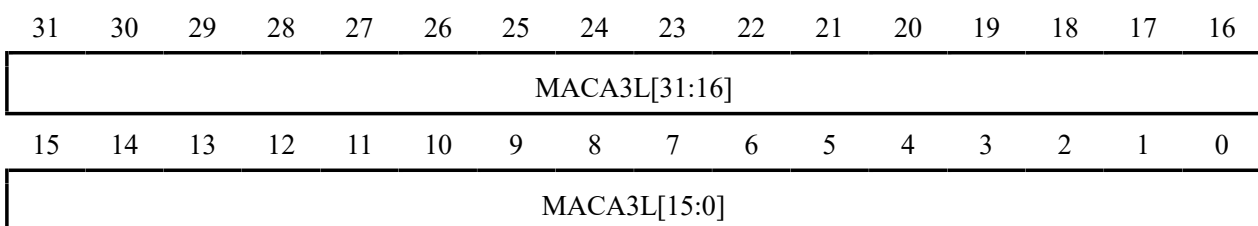
Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA3H[15:0]															

Bit	Name	Access	Description	Reset Value
31	AE	RW	Address filter enable. 0: The address filter ignores the MAC address 3; 1: The address filter uses MAC address 3 for perfect filtering.	0
30	SA	RW	Address role selection. 0: MAC address 3 is used to compare with the destination address of the received frame; 1: MAC address 3 is used to compare with the source address of the received frame.	0
[29:24]	MBC[5:0]	RW	Mask word control. Each bit of the MBC is used to correspond to a certain byte of the masked MAC address 3, and is used to prohibit a certain byte of the MAC address 3 from participating in the comparison of the source address or the destination address of the received frame. R32_ETH_MACA3HR[29] is set to mask MACA3H[15:8]; R32_ETH_MACA3HR[28] is set to mask MACA3H[7:0]; R32_ETH_MACA3HR[27] is set to mask MACA3L[31:24]; R32_ETH_MACA3HR[26] is set to mask MACA3L[23:16]; R32_ETH_MACA3HR[25] is set to mask MACA3L[15:8]; R32_ETH_MACA3HR[24] is set to mask MACA3L[7:0].	0
[23:16]	Reserved	RO	Reserved	0
[15:0]	MACA3H[15:0]	RW	High 16 bits of the MAC address, that is, 47:32 bits.	0xFFFFFFFF F

31.8.1.19 MAC Address Register 3 Low 32 Bits (R32_ETH_MACA3LR)

Offset address: 0x5C



Bit	Name	Access	Description	Reset Value
[31:0]	MACA3L[31:0]	RW	Low 32 bits of the MAC address, that is, 31:0 bits.	0xFFFFFFFF

				F
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31.8.1.20 ETH_PHY Configuration Register (R32_ETH_PHY_CFGR)

Offset address: 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PHY_RSTN	PHY_PD	Reserved													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DUPIEX	SPEED	LINK_STATUS	PHYADDR_EN	Reserved		REPHYADDR[4:0]				

Bit	Name	Access	Description	Reset Value
31	PHY_RSTN	RW	Ethernet PHY Global Reset Signal: 1: Normal operating mode; 0: Reset the Ethernet PHY.	0
30	PHY_PD	RW	In power-down mode, the MAC can access the PHY's registers via the MDIO interface: 1: Power-down mode; 0: Normal operating mode.	1
[29:11]	Reserved	RO	Reserved	0
10	DUPIEX	RO	1: Connection is full-duplex; 0: Connection is half-duplex.	0
9	SPEED	RO	1: Connection is 100 Mbps; 0: Connection is 10 Mbps.	0
8	LINK_STATUS	RO	1: PHY connection successful; 0: Not connected.	0
7	PHYADDR_EN	RW	Reconfigure PHY Address: 1: Bits 4-0 serve as the PHY address. 0: PHY address defaults to 1.	0
[6:5]	Reserved	RO	Reserved	0
[4:0]	REPHYADDR[4:0]	RW	When bit7 is 1, bit[4:0] is used as the address of PHY.	0x1

31.8.2 MMC registers

31.8.2.1 MMC Control Register (R32_ETH_MMCCR)

Offset address: 0x0100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	MCF	ROR	CSR	CR
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Bit	Name	Access	Description	Reset Value
[31:4]	Reserved	RO	Reserved	0
3	MCF	RW	Counter freeze control. Setting this bit will freeze all MMC counter values. Resetting this bit will restore the count of each counter. Setting ROR while frozen and then reading any counter will cause that counter to be cleared.	0
2	ROR	RW	Reset control on read. Setting this bit will clear the value of the counter after reading either counter.	0
1	CSR	RW	Counter roll stop. Setting this bit causes the counter to increment to its maximum value and stop without auto-zeroing.	0
0	CR	RW1Z	Counter reset control. Setting this bit will reset all counters. This bit is automatically cleared after single system cycle.	0

31.8.2.2 MMC Receive Interrupt Register (R32_ETH_MMCRIR)

Offset address: 0x0104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														RGU FS	Reser ved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RFCES	Reserved				

Bit	Name	Access	Name	Reset Value
[31:18]	Reserved	RO	Reserved	0
17	RGUFS	RC	This bit is set when more than half of the good frames are received.	0
[16:6]	Reserved	RO	Reserved	0
5	RFCES	RC	This bit is set when more than half of the frames with CRC errors are received.	0
4:0	Reserved	RO	Reserved	0

31.8.2.3 MMC Transmit Interrupt Register (R32_ETH_MMCTIR)

Offset address: 0x0108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TGFS	Reserved				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved

Bit	Name	Access	Description	Reset Value
[31:22]	Reserved	RO	Reserved	0
21	TGFS	RC	This bit is set when more than half of the frames are sent.	0
[20:0]	Reserved	RO	Reserved	0

31.8.2.4 MMC Receive Interrupt Mask Register (R32_ETH_MMCRIMR)

Offset address: 0x010C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														RGU FM	Reser ved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RFCM	Reserved				

Bit	Name	Access	Description	Reset Value
[31:18]	Reserved	RO	Reserved	0
17	RGUFM	RW	Receive a good frame half interrupt mask. Setting this bit will mask the interrupt generated when the received frame counter reaches half the value.	0
[16:6]	Reserved	RO	Reserved	0
5	RFCM	RW	Receive CRC error frame half interrupt mask. Setting this bit will mask the interrupt that occurs when the frame counter value reaches half the value of the frame counter that receives a CRC error.	0
[4:0]	Reserved	RO	Reserved	0

31.8.2.5 MMC Transmit Interrupt Mask Register (R32_ETH_MMCTIMR)

Offset address: 0x0110

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TGFM	Reserved				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bit	Name	Access	Description	Reset Value
[31:22]	Reserved	RO	Reserved	0
21	TGFM	RW	Transmit good frame half interrupt mask. Setting this bit will mask the interrupt generated when the sent frame counter reaches half the value.	0
[20:0]	Reserved	RO	Reserved	0

31.8.2.6 MMC Transmit Good Frame after Single Conflict Counter (R32_ETH_MMCTGFSCCR)

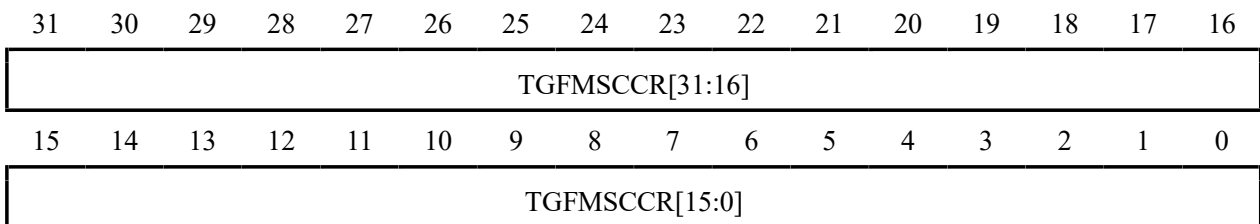
Offset address: 0x014C



Bit	Name	Access	Description	Reset Value
[31:0]	TGFSCCR[31:0]	RO	This field contains the number of frames successfully sent after a single conflict in half-duplex mode, i.e., the counter of good frames sent after a single conflict.	0

31.8.2.7 MMC Transmit Good Frame Counter after Multiple Conflicts (R32_ETH_MMCTGFMSCCR)

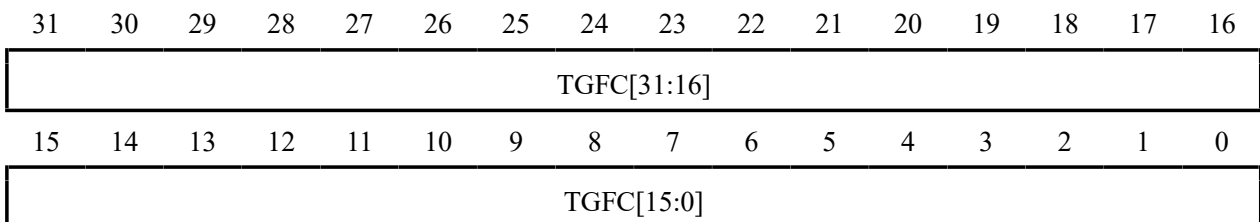
Offset address: 0x0150



Bit	Name	Access	Description	Reset Value
[31:0]	TGFMSCCR[31:0]	RO	This field contains the number of frames successfully sent after more than one conflict in half-duplex mode, i.e., a counter of good frames sent after multiple conflicts.	0

31.8.2.8 MMC Transmit Good Frame Count Register (R32_ETH_MMCTGFCR)

Offset address: 0x0168



Bit	Name	Access	Description	Reset Value
[31:0]	TGFC[31:0]	RO	Count of correct frames sent by the MAC.	0

31.8.2.9 MMC Receive Frame with CRC Error Count Register (R32_ETH_MMCRFCECR)

Offset address: 0x0194

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RFCECR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RFCECR[15:0]															
Bit	Name	Access	Description	Reset Value											
[31:0]	RFCECR[31:0]	RO	Counter of frames received by the MAC with CRC errors.	0											

31.8.2.10 MMC Receive Frame with Alignment Error Count Register (R32_ETH_MMCRFAECR)

Offset address: 0x0198

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RFAECR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RFAECR[15:0]															
Bit	Name	Access	Description	Reset Value											
[31:0]	RFAECR[31:0]	RO	Counter of frames received by the MAC with alignment errors.	0											

31.8.2.11 MMC Receives All Frame Count Registers (R32_ETH_MMCRAFCR)

Offset address: 0x019C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CRAFCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RAFCR[15:0]															
Bit	Name	Access	Description	Reset Value											
[31:0]	RAFCR[31:0]	RO	The number of all frames received by the MAC, whether there is an error or not.	0											

31.8.2.12 MMC Receive Good Frame Count Register (R32_ETH_MMCRGUFCR)

Offset address: 0x01C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RGUFCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RGUFCR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RGUFCR[31:0]	RO	The number of normal frames received by the MAC.	0

31.8.3 IEEE 1588 (PTP) Registers

31.8.3.1 PTP Timestamp Control Register (R32_ETH_PTPTSCR)

Offset address: 0x0700

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										TSARU	TSITE	TSSTU	TSSTI	TSFCU	TSE

Bit	Name	Access	Description	Reset Value
[31:6]	Reserved	RO	Reserved	0
5	TSARU	RW	Addend register update control. When this bit is set, the value of the addend register will be added to the accumulator. This bit is used in fine tuning mode. This bit is automatically cleared after the accumulator is incremented. This bit can only be set when it is 0.	0
4	TSITE	RW	Timestamp interrupt trigger enable. When this bit is set, an interrupt will be generated when the PTP system time reaches the value set in the target time register.	0
3	TSSTU	RW	System time update control. When this bit is set, the PTP system time will be added to the value in the update register. This bit is automatically cleared after the update is complete.	0
2	TSSTI	RW	Timestamp initialization control. When this bit is set, the system time of the PTP will be replaced with the value in the update register. This bit is automatically cleared after the update is complete.	0
1	TSFCU	RW	Update mode selection. 0: Coarse mode; 1: Fine adjustment mode.	0
0	TSE	RW	Additional timestamp enable. 0: No timestamp is added to the descriptor; 1: Add a timestamp to the descriptor at the end of reception or transmission.	0

31.8.3.2 PTP Subsecond Increment Register (R32_ETH_PTPSSIR)

Offset address: 0x0704

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								STSSI[7:0]							

Bit	Name	Access	Description	Reset Value
[31:8]	Reserved	RO	Reserved	0
[7:0]	STSSI[7:0]	RW	Sub-second step value. In coarse mode, the PTP system time will automatically increase by this value every main frequency cycle. In the fine adjustment mode, when the accumulator overflows, the PTP system time will automatically increase by this value.	0

31.8.3.3 PTP Timestamp Register High (R32_ETH_PTPTSHR)

Offset address: 0x0708

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STS[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STS[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TSHR[31:0]	RO	PTP system time value, real-time value, in seconds.	0

31.8.3.4 PTP Timestamp Register Low (R32_ETH_PTPTSLR)

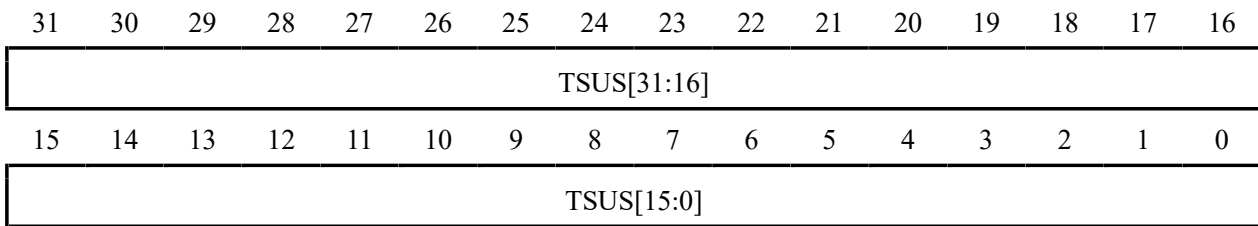
Offset address: 0x070C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STPNS	STSS[30:16]														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STSS[15:0]															

Bit	Name	Access	Description	Reset Value
31	STPNS	RO	System time positive/negative flag. 1: Negative; 0: Positive.	0
[30:0]	STSS[30:0]	RO	PTP system time value, real-time value, in sub-seconds, i.e., about 0.46ns. When STSS overflows, STS increments by 1s.	0

31.8.3.5 PTP Timestamp Update Register High 32 Bits (R32_ETH_PTPTSHUR)

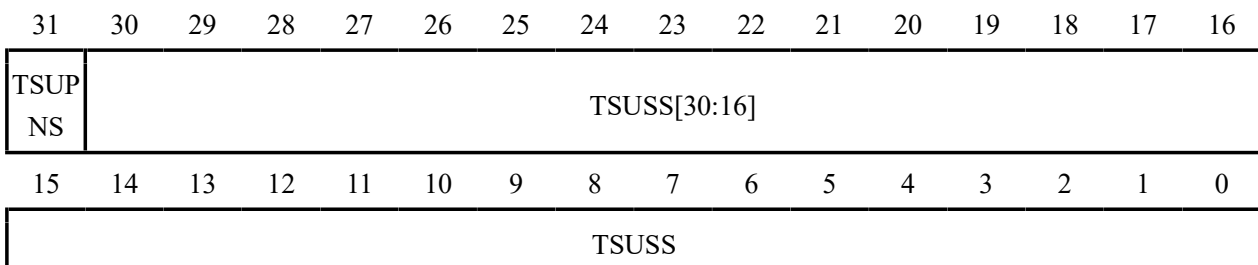
Offset address: 0x0710



Bit	Name	Access	Description	Reset Value
[31:0]	TSUS	RW	Timestamp update second value, which is used to replace the high bit of the PTP system time or the second value representing its addition or subtraction to the system time.	0

31.8.3.6 PTP Timestamp Update Register Low 32 Bits (R32_ETH_PTPTSLUR)

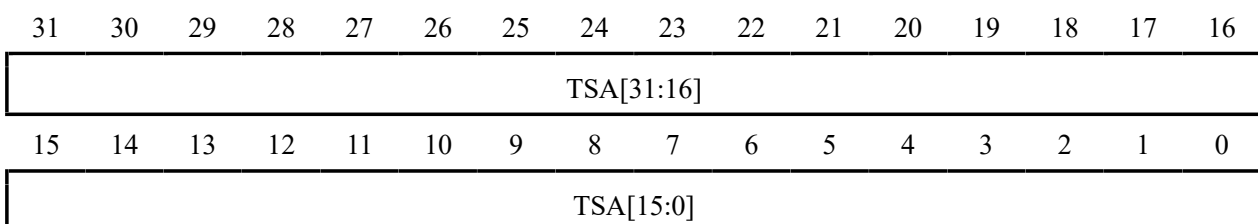
Offset address: 0x0714



Bit	Name	Access	Description	Reset Value
31	TSUPNS	RW	Timestamp update positive/negative flag. When the update time method is to use TSUS and TSUSS to directly replace the PTP system time, TSUPNS should be 0; when the update time method is to use TSUS and TSUSS to add or subtract the original PTP system time value, TSUPNS is 1, indicating that the PTP system Subtract TSUS and TSUSS from the time base, and 0 means add TSUS and TSUSS to the PTP system time.	0
[30:0]	TSUSS[30:0]	RW	Timestamp update sub-second value, which is used to replace the high bit of the PTP system time or a sub-second value that represents its addition or subtraction from the system time.	0

31.8.3.7 PTP Timestamp Adder Register (R32_ETH_PTPTSAR)

Offset address: 0x0718



Bit	Name	Access	Description	Reset Value
[31:0]	TSA[31:0]	RW	Timestamp adder value. This register is only used in fine tuning mode. The value of TSA is added to the accumulator every system cycle, and if the accumulator overflows, it will trigger a system time update in fine tuning mode.	0

31.8.3.8 PTP Target Time Register High 32 Bits (R32_ETH_PTPTTHR)

Offset address: 0x071C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TTSH[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TTSH[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TTSH[31:0]	RW	Target time in seconds. If the PTP system time reaches or exceeds this value and the associated interrupt is enabled, an interrupt will be generated.	0

31.8.3.9 PTP Target Time Register Low 32 Bits (R32_ETH_PTPTTLR)

Offset address: 0x0720

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TTSL[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TTSL[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	TTSL[31:0]	RW	Target time sub-second value. An interrupt is generated if the PTP system time reaches or exceeds this value and the associated interrupt is enabled.	0

31.8.4 DMA Control Registers

31.8.4.1 DMA Bus Mode Register (R32_ETH_DMABMR)

Offset address: 0x1000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ERX_RST	ETX_RST	DMA_RX_RST	DMA_TX_RST	Reserved											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	DFM	DSL[4:0]	Reserved	SR
----------	-----	----------	----------	----

Bit	Name	Access	Description	Reset Value
31	ERX_RST	RW	Write 1 to reset the MAC receiving module, and write 0 to cancel the reset.	0
30	ETX_RST	RW	Write 1 to reset the MAC sending module, and write 0 to cancel the reset.	0
29	DMARX_RST	RW	Write 1 to reset the DMA receiving module, and write 0 to cancel the reset.	0
28	DMATX_RST	RW	Write 1 to reset the DMA sending module, and write 0 to cancel the reset.	0
[27:8]	Reserved	RO	Reserved	0
7	DFM	RW	Descriptor mode: 1: Enhanced descriptor mode; 0: Standard descriptor mode.	0
[6:2]	DSL[4:0]	RW	Descriptor skip length These bits define the jump distance in words (32 bits) between 2 descriptors that are not connected in a chained structure. The address skip is the address difference from the end of the current descriptor to the beginning of the next descriptor. When the DSL field is 0, the DMA considers the descriptors to be contiguous under a ring structure.	0
1	Reserved	RO	Reserved	0
0	SR	RW	Software Reset When set to 1, the MAC's DMA controller resets the internal registers and logic circuits of all subsystems of the MAC. This bit is automatically cleared after the reset operation is completed for the different clock domain modules inside the MAC. Before rewriting the MAC's registers, you should ensure that this bit is 0.	1

31.8.4.2 DMA Transmit Poll Demand Register (R32_ETH_DMATPDR)

Offset address: 0x1004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TPDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

TPDR[15:0]

Bit	Name	Access	Description	Reset Value
[31:0]	TPDR[31:0]	RW	Transmit poll demand. The user initiates a suspended transmit process by writing any value to this register. After restarting the transmission process, this register will be automatically cleared.	0

31.8.4.3 DMA Receive Query Register (R32_ETH_DMARPDR)

Offset address: 0x1008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RPDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RPDR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RPDR[31:0]	RW	Receive query commands. The receiving process may be interrupted by various accidents, and the user needs to write any value to this register to restart the receiving process. After restarting the receive process, this register will be automatically cleared.	0

31.8.4.4 DMA Receive Descriptor Address Register (R32_ETH_DMARDLAR)

Offset address: 0x100C

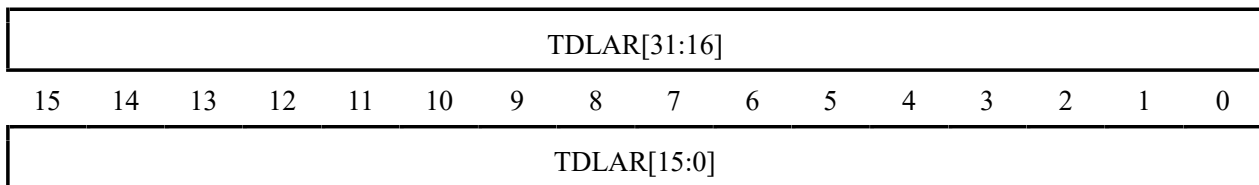
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDLAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDLAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RDLAR[31:0]	RW	This register is used to store the address of the first received DMA descriptor. Note that descriptors require 16-byte alignment, so the last 4 bits of this register should be 0.	0

31.8.4.5 DMA Transmit Descriptor Address Register (R32_ETH_DMATDLAR)

Offset address: 0x1010

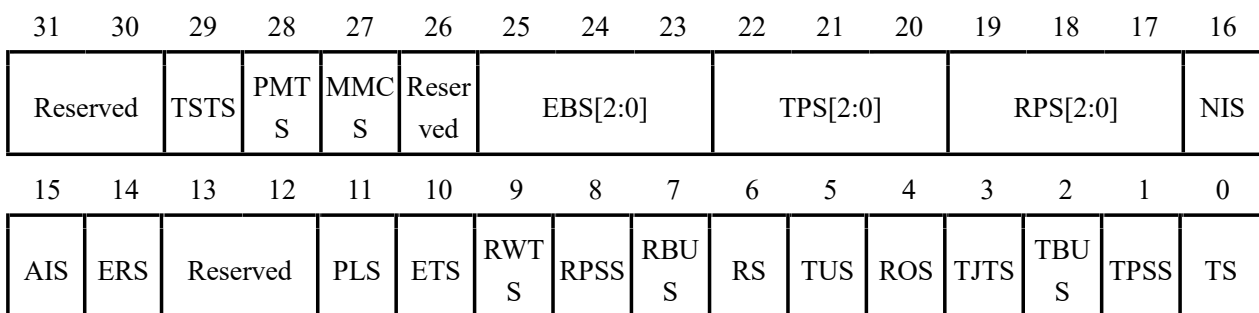
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----



Bit	Name	Access	Description	Reset Value
[31:0]	TDLAR[31:0]	RW	This register is used to store the address of the first transmit DMA descriptor. Note that descriptors require 32-byte alignment.	0

31.8.4.6 DMA Status Register (R32_ETH_DMASR)

Offset address: 0x1014



Bit	Name	Access	Description	Reset Value
[31:30]	Reserved	RO	Reserved	0
29	TSTS	RO	Timestamp trigger status. This bit will be set when an interrupt event occurs in the PTP section, or an interrupt will be generated if the PTP interrupt is enabled. This bit is automatically cleared after clearing all flag bits in the PTP section.	0
28	PMTS	RO	PMT trigger status. This bit will be set when an interrupt event is generated in the PMT part, and an interrupt will be generated if the PMT interrupt is enabled. This bit is automatically cleared after clearing all flag bits in the PMT section.	0
27	MMCS	RO	MMC trigger status. This bit will be set when an interrupt event is generated in the MMC part, and an interrupt will be generated if the MMC interrupt is enabled. This bit is automatically cleared after clearing all flag bits in the MMC section.	0
26	Reserved	RO	Reserved	0
[25:23]	EBS[2:0]	RO	Error status. This field indicates the type of bus error that caused it. This field is only valid when the DMASR[13] bit is set. DMASR[23]:0: Error sending DMA forwarding data; 1: Error receiving DMA forwarding data;	0

			DMASR[24]:0: Error reading data forwarding; RO 1: Error writing data forwarding; DMASR[25]:0: Error accessing descriptor; 1: Error accessing data cache.	
[22:20]	TPS[2:0]	RO	Transmit process status. This field is used to indicate the status of the current transmit DMA. 000: Stop. Receive reset or stop sending command; 001: Run. Fetching the send descriptor; 010: Run. Waiting for status information; 011: Run. Reading the transmit buffer data and pushing it into the FIFO; 100,101: Reserved; 110: Pause. The transmit descriptor is not available or the transmit buffer data underflows; 111: Run. Closing the transmit descriptor.	000b
[19:17]	RPS[2:0]	RO	Receive process status. This field is used to indicate the current state of receiving DMA. 000: Stop. Receive reset or stop sending command; 001: Run. Fetching the receive descriptor; 010: Reserved; 011: Run. Waiting to receive data packets; 100: Pause. The receive descriptor is not available; 101: Run. Closing the receive descriptor; 110: Reserved; 111: Run. Pushing the received data from the FIFO into the memory.	000b
16	NIS	RW1Z	Normal interrupt summary. With interrupts enabled in the DMAIER register, if any of the following bits are set, the NIS bit will also be set. -DMASR[0]: Send interrupt; -DMASR[2]: The transmit buffer is not available; -DMASR[6]: Receive interrupt; -DMASR[14]: Early receive interrupt; -DMASR[31]: Internal 10M physical layer connection state change The NIS bit is sticky, when the corresponding interrupt status bit that causes NIS location 1 is cleared, the NIS bit must also be cleared by writing 1.	0
15	AIS	RW1Z	Abort interrupt summary. With interrupts enabled in the DMAIER register, the AIS bit will also be set if any of the following bits are set. -DMASR[1]: The transmission is stopped; -DMASR[3]: Transmit Jabber timeout; -DMASR[4]: Receive FIFO overflow; -DMASR[5]: Transmit data underflow;	0

			-DMASR[7]: The receive buffer is not available; -DMASR[8]: The reception is stopped; -DMASR[9]: Receive watchdog timeout; -DMASR[10]: Send early; -DMASR[13]: Bus error; The AIS bit is a sticky bit, and when the corresponding interrupt status bit that caused AIS location 1 is cleared to zero, the AIS bit must also be cleared to zero by writing a 1.	
14	ERS	RW1Z	Early receive status. When this bit is set, it means that the DMA has filled the first buffer when the data frame is received, but the complete frame has not been received yet. After RS is set, the ERS bit is automatically cleared.	0
[13:12]	Reserved	RO	Reserved	0
11	PLS	RW1Z	Internal physical layer connection state change flag bit. This bit indicates that the physical layer is connected or disconnected. Or write 1 to clear.	0
10	ETS	RW1Z	Early transmit status. When set, it indicates that the transmit frame has been fully pushed into the FIFO.	0
9	RWTS	RW1Z	Watchdog time-out status. When set, it indicates that the frame length has exceeded 2048 bytes.	0
8	RPSS	RW1Z	Receive process stop status. This bit is set to indicate that the receive process has stopped.	0
7	RBUS	RW1Z	Receive buffer unavailable status. This bit is set to indicate that the permission to receive the descriptor belongs to the CPU, the DMA cannot be obtained, and the receiving process has been suspended. The user needs to release the descriptor and fill the RPDR register with a value to resume the receive flow. The DMA will retry to acquire the descriptor when the next frame is received.	0
6	RS	RW1Z	Receive completed status. This bit is set to indicate that the reception of a frame has been completed, and the frame information is also updated into the descriptor.	0
5	TUS	RW1Z	Transmit data underflow status. This bit is set to indicate that the transmit buffer under flowed the transmit data while transmitting the frame. At this point the sending process is halted and the data underflow error bit is set.	0
4	ROS	RW1Z	Receive status overflow. This bit is set to indicate that a data overflow has occurred in the receive buffer.	0
3	TJTS	RW1Z	Transmit Jabber timer timeout status. This bit is set to indicate that the transmitter is too busy, transmission	0

			has stopped, and the descriptor's Jabber timeout bit has been set.	
2	TBUS	RW1Z	Transmit buffer unavailable status bit. This bit is set to indicate that the transmit descriptor is occupied by the CPU, the DMA cannot be acquired, and the transmit process has been suspended. Bits [22:20] show the current transmit status. The application needs to release the send descriptor.	0
1	TPSS	RW1Z	Transmit process stop status. This bit is set to indicate that the transmission process has stopped.	0
0	TS	RW1Z	Transmit completed status. This bit is set to indicate that the transmission of a frame has been completed, and the ownership of the descriptor has been returned to the CPU.	0

31.8.4.7 DMA Operation Mode Register (R32_ETH_DMAOMR)

Offset address: 0x1018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					DTC EFD	Reserved				TSF	FTF	Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		ST	Reserved				FEF	FUGF	Reserved				SR	Res	

Bit	Name	Access	Description	Reset Value
[31:27]	Reserved	RO	Reserved	0
26	DTCEFD	RW	Discard TCP/IP checksum error frame disable 0: If the FEF bit is 0, the MAC will discard all frames with errors; 1: When an error is found in the checksum of protocols such as TCP/IP/ICMP/UDP, the frame will not be discarded.	0
[25:22]	Reserved	RO	Reserved	0
21	TSF	RW	Transmit store and forward. 0: After the data written into the FIFO in the sending process reaches a certain value, the sending will be started; 1: The sending process will start sending after the complete frame is written into the FIFO.	0
20	FTF	RW	Transmit FIFO empty control. Setting this bit will reset the transmit FIFO.	0
[19:14]	Reserved	RO	Reserved	0
13	ST	RW	Start or stop transmission.	0

			0: After sending the current frame, the sending process enters the stop state; 1: Put the sending process in the running state.	
[12:8]	Reserved	RO	Reserved	0
7	FEF	RW	Forward error frame. 0: Receive FIFO discards frames with errors. 1: All frames are forwarded to the DMA except for frames that are too short.	0
6	FUGF	RW	Forward too short frame. 0: The receive FIFO discards all frames less than 64 bytes in length; 1: The receive FIFO forwards a frame whose length is too short.	0
[5:2]	Reserved	RO	Reserved	0
1	SR	RW	Start or stop reception. 0: After forwarding the currently received frame, the receiving DMA enters the stop mode, and the next transmission starts from the current receiving descriptor position; 1: Open the receiving process, DMA fetches the receiving descriptor from the current position or fetches the receiving descriptor from the identifier header position.	0
0	Reserved	RO	Reserved	0

31.8.4.8 DMA Interrupt Enable Register (R32_ETH_DMAIER)

Offset address: 0x101C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															NISE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AISE	ERS	Reserved	PLE	ETIE	RWTI E	RPSI E	RBUI E	RIE	TUIE	ROIE	TJTI E	TBUI E	TPSI E	TIE	

Bit	Name	Access	Description	Reset Value
[31:17]	Reserved	RO	Reserved	0
16	NISE	RW	Normal interrupt enable. Enabling this bit will enable the following interrupts. -DMASR[0]: Transmit interrupt; -DMASR[2]: Transmit buffer unavailable; -DMASR[6]: Receive interrupt; -DMASR[14]: Early receive interrupt.	0
15	AISE	RW	Abnormal interrupt. Enabling this bit will enable the	0

			following interrupts. -DMASR[1]: The sending process is stopped; -DMASR[3]: Send Jabber timeout; -DMASR[4]: Receive FIFO overflow; -DMASR[5]: transmit data underflow; -DMASR[7]: Send buffer unavailable; -DMASR[8]: The receiving process is stopped; -DMASR[9]: Receive watchdog timeout; -DMASR[10]: Early transmit; -DMASR[13]: Bus error.	
14	ERS	RW	Early receive interrupt enable. Enable this bit to generate an early receive interrupt. NISE must be set. AISE must be set.	0
[13:12]	Reserved	RO	Reserved	0
11	PLE	RW	Internal 100M physical layer connection state change interrupt enable bit.	0
10	ETIE	RW	Early transmit interrupt enable. Enable this bit to generate an early transmit interrupt. AISE must be set.	0
9	RWTIE	RW	Receive watchdog interrupt enable. Enable this bit to generate a receive watchdog timeout interrupt. AISE must be set.	0
8	RPSIE	RW	Receive process stop interrupt enable. Enable this bit to generate a receive process stop interrupt.	0
7	RBUIE	RW	Receive buffer unavailable interrupt enable. Enable this bit to generate a receive buffer unavailable interrupt. AISE must be set.	0
6	RIE	RW	Receive completed interrupt enable bit. Enable this bit to generate receive completed interrupt. NISE must be set.	0
5	TUIE	RW	Transmit underflow interrupt enable. Enable this bit to generate a transmit underflow interrupt. AISE must be set.	0
4	ROIE	RW	Receive overflow interrupt. Enable this bit to generate a receive overflow interrupt. AISE must be set.	0
3	TJTIE	RW	Transmit Jabber timeout interrupt enable. Enable this bit to generate transmit Jabber timeout interrupt. AISE must be set.	0
2	TBUIE	RW	Transmit buffer unavailable interrupt enable. Enable this bit to generate transmit buffer unavailable interrupt. NISE must be set	0
1	TPSIE	RW	Transmit process stop interrupt enable. Enable this bit to generate transmit process stop interrupt. AISE must be set.	0
0	TIE	RW	Transmit completed interrupt enable. Enable this bit to	0

			generate transmit completed interrupt. NISE must be set.	
--	--	--	--	--

31.8.4.9 DMA Lost Frame Register (R32_ETH_DMAMFBOCR)

Offset address: 0x1020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			OFOC	MFA[10:0]											OMFC
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MFC[15:0]															

Bit	Name	Access	Description	Reset Value
[31:29]	Reserved	RO	Reserved	0
28	OFOC	RZ	The overflow bit of the FIFO overflow counter.	0
[27:17]	MFA[10:0]	RZ	Frames lost by the application.	0
16	OMFC	RZ	Missing frame counter overflow.	0
[15:0]	MFC[15:0]	RZ	Lost frame counter. This field indicates the number of frames lost due to unavailability of receive buffers.	0

31.8.4.10 DMA Current Transmit Descriptor Register (R32_ETH_DMACHTDR)

Offset address: 0x1048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTDAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTDAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HTDAR[31:0]	RO	This register value points to the transmit descriptor currently in use. This register is updated in real time by DMA.	0

31.8.4.11 DMA Current Receive Descriptor Register (R32_ETH_DMACHRDR)

Offset address: 0x104C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HRDAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HRDAR[15:0]															

Bit	Name	Access	Description	Reset Value
-----	------	--------	-------------	-------------

[31:0]	HRDAR[31:0]	RO	This register value points to the receive descriptor currently in use. This register is updated in real time by DMA.	0
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31.8.4.12 DMA Current Transmit Buffer Register (R32_ETH_DMACHTBAR)

Offset address: 0x1050

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTBAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTBAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HTBAR[31:0]	RO	This register value points to the transmit buffer currently in use. This register is updated in real time by DMA.	0

31.8.4.13 DMA Current Receive Buffer Register (R32_ETH_DMACHRBAR)

Offset address: 0x1054

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HRBAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HRBAR[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	HRBAR[31:0]	RO	This register value points to the receive buffer currently in use. This register is updated in real time by DMA.	0

31.8.5 Internal 10M/100M Physical Layer Register Address

31.8.5.1 Basic Control Register (R16_BMCR)

Offset address: 0x00

Bit	Name	Access	Description	Reset Value
15	RST	RW/SC	1: PHY reset; 0: Normal operation.	0
14	LOOPBACK	RW	1: Enable loopback; 0: Normal operation.	0
13	SPEED_SELECTION	RW	1: 100Mb/s; 0: 10Mb/s; After auto-negotiation, the status of negotiation working speed is displayed.	0

12	AUTO_NEGOTIATION	RW	0: Disable auto-negotiation; 1: Enable auto-negotiation.	1
11	POWER_DOWN	RW	1: Turn off the power supply; 0: Normal operation.	0
10	ISOLATE	RW	1. MII/RMII interface is isolated from PHY, and PHY can still respond to MDC/MDIO; 0: It works normally.	
9	RESTART_AUTO_NEGOTIATION	RW/SC	0: Normal operation; 1: Restart auto-negotiation.	0
8	DUPLEX_MODE	RW	0: Half-duplex; 1: Full-duplex.	1
7	COLLISION_TEST	RW	Collision test. 1: Conflict test enabled; 0: Normal operation.	0
[6:0]	Reserved	RO	Reserved	0

31.8.5.2 Basic Status Register (R16_BMSR)

Offset address: 0x01

Bit	Name	Access	Description	Reset Value
15	100BASE_T4	RO	1: Supports enabling 100BASE-T4; 0: Does not support 100BASE-T4.	0
14	100BASE_TX_FULL_DUPLEX	RO	1: Supports enabling 100BASE-TX full-duplex; 0: Does not support 100BASE-TX full-duplex.	1
13	100BASE_TX_HALF_DUPLEX	RO	1: Supports enabling 100BASE-TX half-duplex; 0: Does not support 100BASE-TX half-duplex.	1
12	10BASE_T_FULL_DUPLEX	RO	1: Supports enabling 10BASE-TX full-duplex; 0: Does not support 10BASE-TX full-duplex.	1
11	10BASE_T_HALF_DUPLEX	RO	1: Supports enabling 10BASE-TX half-duplex; 0: Does not support 10BASE-TX half-duplex.	1
[10:7]	Reserved	RO	Reserved	0
6	MF_PREAMBLE_SUPPRESSION	RO	Management frames with preamble suppression are allowed to be received.	1
5	AUTO_NEGOTIATION_COMPLETE	RO	1: Auto-negotiation is completed; 0: Auto-negotiation is not completed.	0
4	REMOTE_FAULT	RO	1: Remote error condition detected (read clear); 0: No remote error condition was detected.	0
3	AUTO_NEGOTIATION_ABILITY	RO	1: PHY can perform self-negotiation; 0: PHY cannot perform self-negotiation.	1
2	LINK_STATUS	RO	1: A valid link has been established; 0: No valid link has been established.	0
1	JABBER_DETECT	RO	1: Jabber condition detected; 0: Jabber condition not detected.	0
0	EXTENDED_CAPABILITY	RO	1: Extended register functionality; 0: No extended register functionality.	1

31.8.5.3 Auto-Negotiation Message Register(R16_ANAR)

Offset address: 0x04

Bit	Name	Access	Description	Reset Value
15	NP	RW	Next page position: 1: Send protocol regular data page; 0: Send primary capability data page.	0
14	ACK	RO	1: Acknowledged receipt of the link partner capability data word; 0: No acknowledgment signal received.	0
13	RF	RW	1: Notify remote error detection capabilities; 0: Do not notify remote error detection capabilities.	0
12	Reserved	RO	Reserved	0
11	ASYPAUSE	RW	1: Supports asymmetric flow control; 0: Does not support asymmetric flow control.	0
10	PAUSE	RW	1: Supports flow control; 0: Does not support flow control.	0
9	100BASE_T4	RO	1: Local node supports 100Base-T4; 0: Local node does not support 100Base-T4.	0
8	100BASE_TX_FD	RW	1: Local node supports 100Base-TX full-duplex; 0: Local node supports 100Base-TX full-duplex.	1
7	100BASE_TX	RW	1: Local node supports 100Base-TX; 0: Local node supports 100Base-TX.	1
6	10BASE_T_FD	RW	1: Local node supports 10Base-TX full-duplex; 0: Local node supports 10Base-TX full-duplex.	1
5	10BASE_T	RW	1: Local node supports 10Base-TX half-duplex; 0: Local node supports 10Base-TX half-duplex.	1
[4:0]	SELECT[4:0]	RO	The binary code selector supported by this node is currently only CSMA/CD00001, and no other protocols support it.	1

31.8.5.4 Auto-Negotiation Linker Capability Register(R16_ANLPAR)

Offset address: 0x05

Bit	Name	Access	Description	Reset Value
15	NP	RW	Next page bit. 1: Transmit protocol details data page; 0: Transmit basic capability data page;	0
14	ACK	RO	1: Acknowledgment received for peer auto-negotiation	0

			feature register; 0: No acknowledgment received.	
13	RF	RW	1: Enable remote error detection notification; 0: Disable remote error detection notification.	0
12	Reserved	RO	Reserved	0
11	ASYPAUSE	RO	1: Supports asymmetric pause notifications; 0: Does not support asymmetric pause. When auto-negotiation is enabled, this bit reflects the link partner's capability.	0
10	PAUSE	RO	1: Supports symmetric notification suspension; 0: Does not support symmetric suspension.	0
9	100BASE_T4	RO	1: PHY supports 100BASE-T4; 0: PHY does not support 100BASE-T4.	0
8	100BASE_TX_FD	RO	1: PHY supports 100BASE-T full-duplex; 0: PHY does not support 10BASE-T full-duplex.	0
7	100BASE_TX	RO	1: PHY supports 100BASE-T; 0: PHY does not support 10BASE-T.	0
6	10BASE_T_FD	RO	1: PHY supports 10BASE-T full-duplex; 0: PHY does not support 10BASE-T full-duplex.	0
5	10BASE_T	RO	1: PHY supports 10BASE-T; 0: PHY does not support 10BASE-T.	0
[4:0]	SELECT[4:0]	RO	The PHY supports a binary encoding selector. Currently, only CSMA/CD 00001 is special; no other protocols are supported.	0x01

31.8.5.5 Page Selection Register (R16_PAGE_SEL)

Offset address: 0x1F

Bit	Name	Access	Description	Reset Value
[15:8]	Reserved	RO	Reserved	0
[7:0]	SEL_PAGE_ADDR[7:0]	RW	Select page address: 00000000: 11111111.	0

31.8.6 PAGE0-related Registers

31.8.6.1 PHY Status Register (R16_PHY_STATUS)

Offset address: 0x1A

Bit	Name	Access	Description	Reset Value
15	INCOMPATIBLE_LINK	RO	After auto-negotiation priority resolution, the compatibility status between the local device and the	0

			peer device's capabilities: 1: Incompatible; 0: Compatible.	
[14:7]	Reserved	RO	Reserved	0
6	SIGNAL_STATUS	RO	100M signal status: 1: Signal status is good; 0: No signal.	0
5	MDI_MDIX_STATUS	RO	MDI/MDIX status: 1: MDI; 0: MDIX.	1
4	POLARITY_STATUS	RO	Polarity status: 1: Polarity reversed; 0: Polarity normal.	0
[3:0]	PHY_SPEED_DUP[3:0]	RO	PHY operating rate and full/half-duplex operating mode.	0

PHY_SPEED_DUP detail table				
3	PHY100_INDICATE	RO	Operation rate 100M.	0
2	PHY10_INDICATE	RO	Operation rate 10M.	0
1	FULL_INDICATE	RO	Full-duplex.	0
0	HALFFULL_INDICATE	RO	Half-duplex.	0

31.8.6.2 Interrupt Indicator Register (R16_INTERRUPT_IND)

Offset address: 0x1E

Bit	Name	Access	Description	Reset Value
[15:12]	Reserved	RO	Reserved	0
11	LINKSTATUS_CHG	RZ	Link state change notification.	0
[10:0]	Reserved	RO	Reserved	0

31.8.7 PAGE7-related Registers

31.8.7.1 Interrupt/LED Flag Register (R16_INTERRUPT_MASK)

Offset address: 0x13

Bit	Name	Access	Description	Reset Value
[15:14]	Reserved	RO	Reserved	0
13	INT_LINKCHG	RW	Link change interrupt Mask.	0
[12:10]	Reserved	RO	Reserved	0
9	LED_EN	RW	LED function enabled.	0

[8:0]	Reserved	RO	Reserved	0x30
-------	----------	----	----------	------

Chapter 32 SDIO Interface (SDIO)

This chapter's module description applies only to the CH32H417, CH32H415 microcontroller product.

The "SDIO" mentioned separately in this chapter refers to a communication interface on the microcontroller designed to operate external memory cards such as SD cards or other devices, and is a peripheral of the microcontroller. The SDIO of the microcontroller is directly mounted on the HB bus, and the clock is directly provided by HCLK, which can achieve higher communication speed. The SDIO of the microcontroller is used as the SDIO host, and the controlled devices are also collectively referred to as SDIO devices. In applications, SDIO is generally used to read and write SD cards, TF cards or eMMC particles, or to control other devices that use SDIO as a communication interface, such as Wi-Fi/4G modules.

32.1 Main Features

32.1.1 Features

- Support SD card, SDIO card and MMC card
- 1-bit, 4-bit and 8-bit bus support
- The highest communication clock can reach 100MHz.
- Compatible with MMC Specification 4.5 (Forward compatibility)
- Compatible with SD card specification 2.0, SDIO card specification 2.0
- Not compatible with SPI or QSPI

32.1.2 Overview

The SDIO of the microcontroller supports communication with memory such as SD cards or MMC cards. It needs to be clear that SDIO only provides a set of clocks, data and command control sequences required to realize the single command transmission of SD cards and MMC cards. The sequence and combination of commands need to be determined by the user through the program. In addition, for various memory cards, SDIO can only realize the function of reading and writing, and the function of the file system provided by the file system needs to be realized by the user to build the file system through the program.

SDIO is different from the QSPI interface. It has no chip select pin and has an extra CMD pin. CMD can be regarded as a special data line, which is specially used to transmit commands and responses; SDIO is available in 1-bit, 4-bit and 8-bit data line widths; the SDIO clock generally works at a frequency below 400kHz during configuration. When the data transmission is officially performed, it can be configured as the maximum clock supported by the SDIO device. The maximum SDIO clock output supported by the microcontroller is Half of HCLK, according to the protocol, when the clock received by the SDIO device is greater than a certain threshold, the waveform peaks of the clock line, data line and command line need to be reduced to save the time-consuming of waveform rising and falling.

Unlike SD cards, SDIO cards often refer to non-storage devices such as WIFI/Bluetooth modules and 4G modules that use SDIO interfaces. If there is no special instruction, the content described in this chapter must be applicable to SD card, and the content only applicable to SDIO card or MMC card will be specially pointed out. In the description of this chapter, the SD card is regarded as the potential operation object first, followed by the SDIO card, and finally the MMC card.

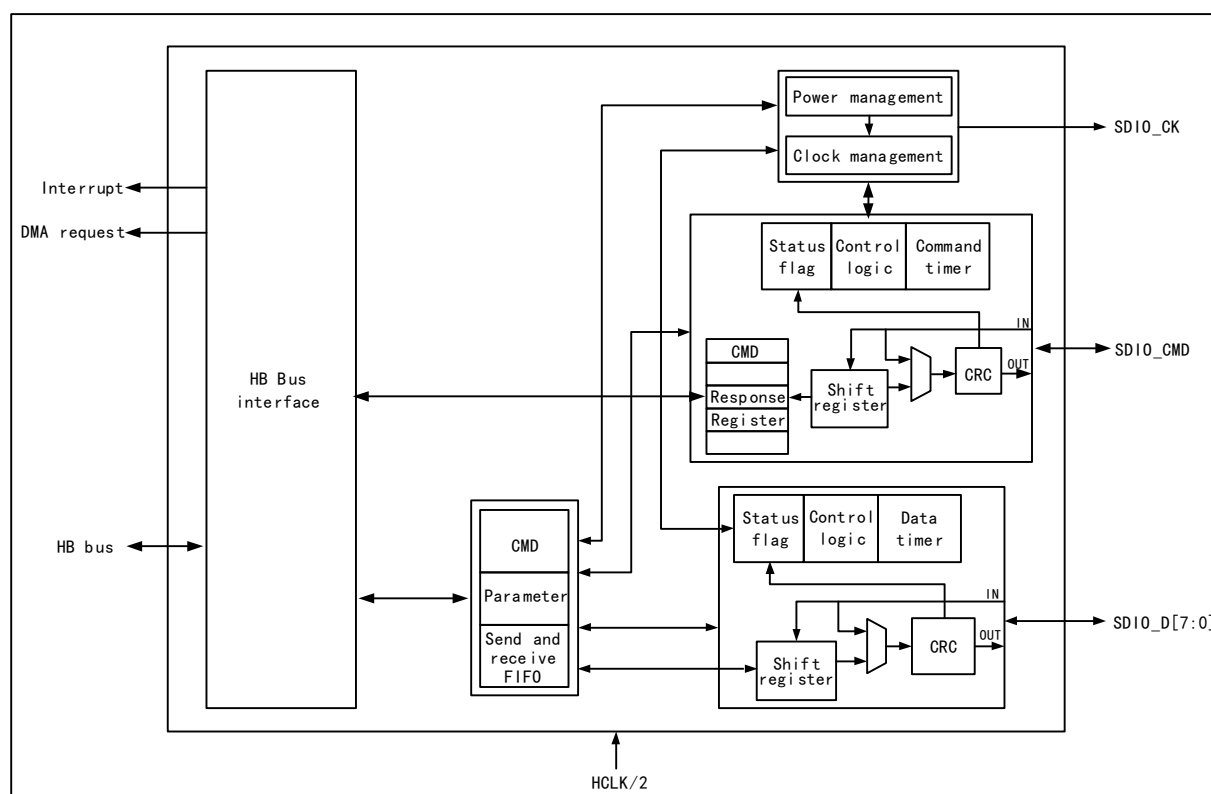
32.2 Interface and Clock

SDIO receives the control of the CPU through the HB bus. There is a FIFO interface in the SDIO register. The CPU or DMA obtains or generates data by reading and writing the FIFO. SDIO is directly clocked by HCLK, and has an interrupt entry that supports multiple interrupt sources. The pins directly controlled by SDIO are SDIO_CK, SDIO_CMD, SDIO_D [7:0], which are connected to SDIO devices through several pins. SDIO is a host-dominated communication interface, and all transfers must be initiated by the microcontroller.

32.2.1 Peripheral Structure

The structure of SDIO is shown in Figure 32-1.

Figure 32-1 SDIO structure diagram



SDIO is a peripheral that the microcontroller directly operates the SDIO device as a host. It is roughly composed of 5 modules: HB interface, clock control part, CMD line control part, data line control part and control register part. SDIO is a half-duplex for peripherals, the CPU writes the commands and data to be sent to the control register, and the command line and data line control module is responsible for pushing the command or data to the IO and adding CRC. The data flow of SDIO is responsible for the transfer from RAM to SDIO FIFO by general-purpose DMA. SDIO FIFO has 32 32-bit sizes.

32.2.2 Pins and Configuration

The pins that need to be configured for SDIO and their modes are shown in Table 32-1.

Table 32-1 SDIO pin configuration

SDIO alternate function	Required pin mode
SDIO_D[0:3]	Push-pull alternate output

SDIO_CK	Push-pull alternate output
SDIO_CMD	Push-pull alternate output
SDIO_D4	Push-pull alternate output
SDIO_D5	Push-pull alternate output
SDIO_D6	Push-pull alternate output
SDIO_D7	Push-pull alternate output

Note: The SDIO_CK pin should be configured as a float input when in slave mode.

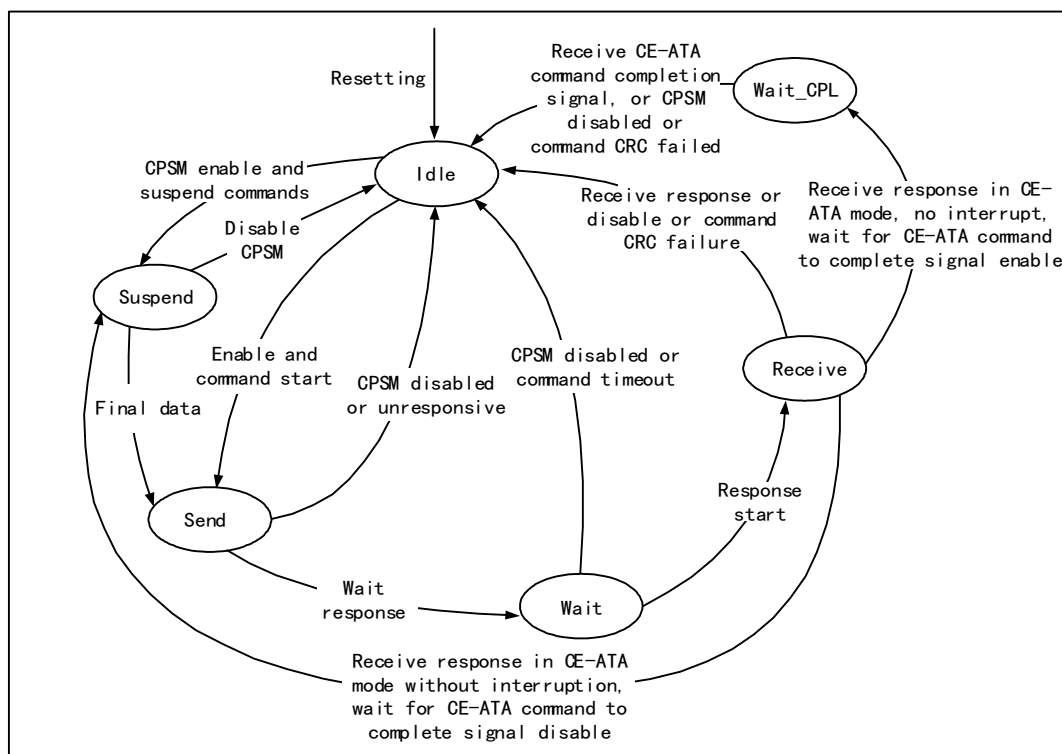
32.2.3 Clock

The clock pin of SDIO is SDIO_CK, and its output clock is obtained by dividing the frequency of HCLK, and the frequency dividing coefficient can be configured as any integer value between 2 and 261. SDIO devices generally only support a single bus mode with a clock of up to 400kHz during initialization. After initialization, the host generally initiates a switch to a low voltage operation, and at the same time increases the clock to the maximum clock that both the microcontroller and the SDIO device can accept. Different versions and speed grades of SD cards support different clock speeds and switching processes, and users need to understand for themselves.

32.2.4 Command Path State Machine

The command workflow of SDIO follows the state machine shown in the figure below.

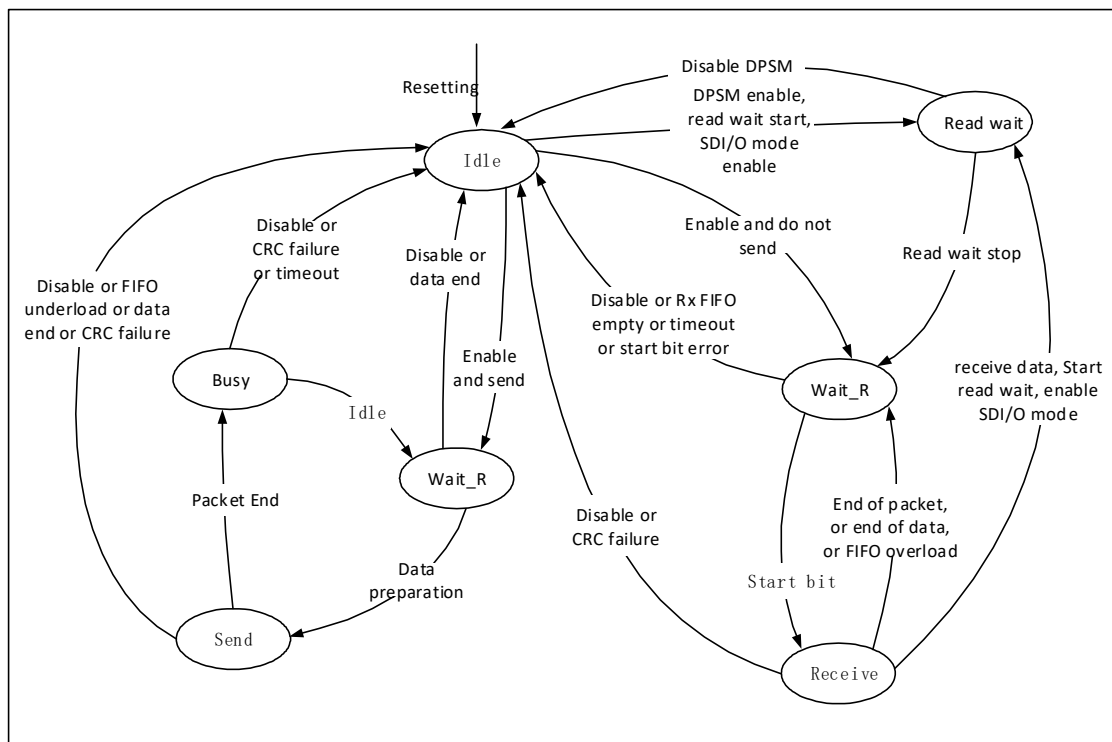
Figure 32-2 Command path state machine



32.2.5 Data Path State Machine

The data workflow of SDIO follows the state machine shown in the figure below.

Figure 32-3 Data path state machine



32.3 SDIO Protocol Description

The communication on SDIO is based on transmission as the smallest unit. Each transmission always starts with the host sending a command on the CMD line. After some commands are sent, the SDIO device will also send a piece of data on the CMD line to reply to the host, which is called 'Response', sometimes accompanied by the transmission of data, the transmission of data is on the D line. The format of commands and responses is fixed, and the definition of each field is determined according to different commands or responses. Responses and data transfers need to be issued or stopped within a specified time after the end of the command or response, otherwise a timeout error will be generated.

The purpose of this section is to let users have a preliminary understanding of the details of some specifications necessary to use SDIO in a relatively small space, and it does not guarantee detailed and timely updates. The SDIO of the microcontroller only guarantees that it implements the hardware operation basis of the SD 2.0, SDIO 2.0 and MMC 4.5 specifications, and does not necessarily support the functions defined in the higher version specifications, such as double-edge sampling. Users should carry out SDIO interactive programming based on SD specification, SDIO specification and MMC specification when doing specific development.

32.3.1 Bus Timing

The transmission of the SD card is initiated by the host initiating the CMD. The SD card may not reply with a response, but may reply with a short response and a long response, and some responses will be accompanied by data transmission. In the communication of the SDIO card or MMC card, there may also be a situation where the SDIO device actively reports an interruption. The timing is shown in the group of figures below.

Figure 32-4 SDIO no response timing (Taking SD card as an example)

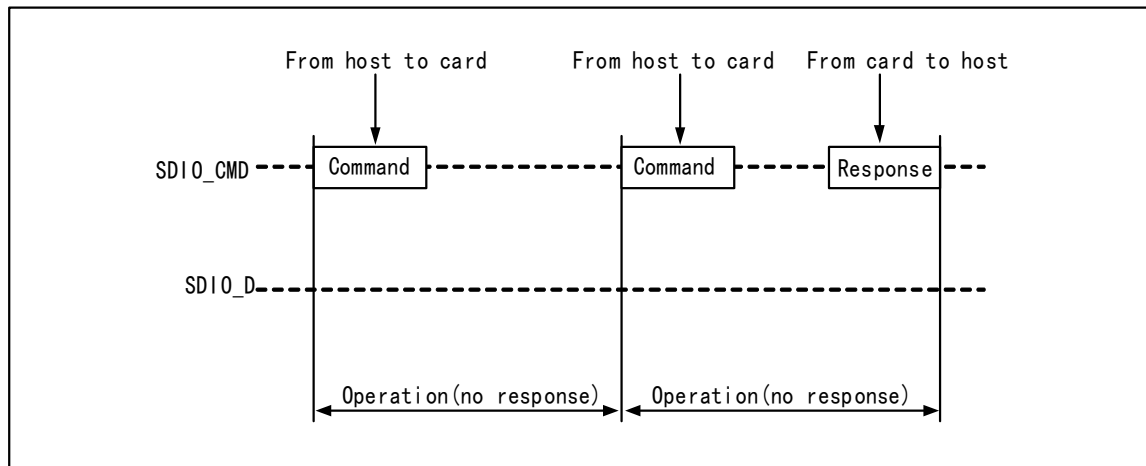


Figure 32-5 SDIO multi-block read timing (Taking SD card as an example)

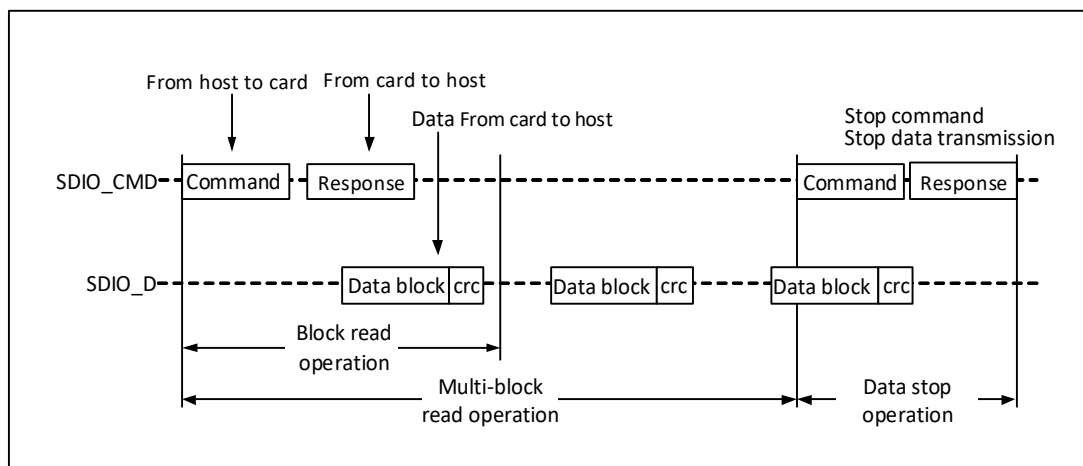


Figure 32-6 SDIO multi-block write timing (Taking SD card as an example)

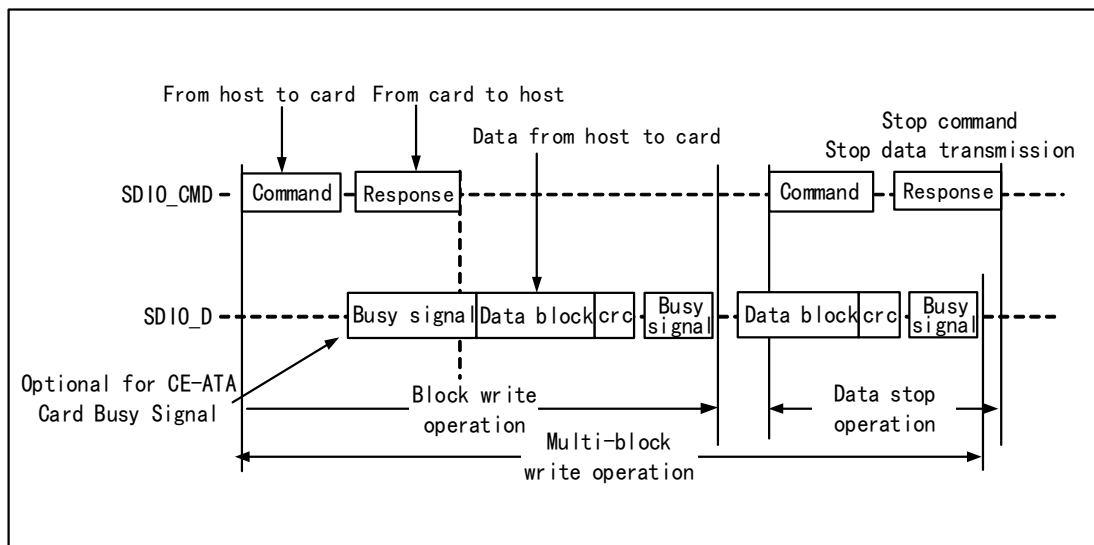


Figure 32-7 SDIO data stream read timing (Taking SDIO card as an example)

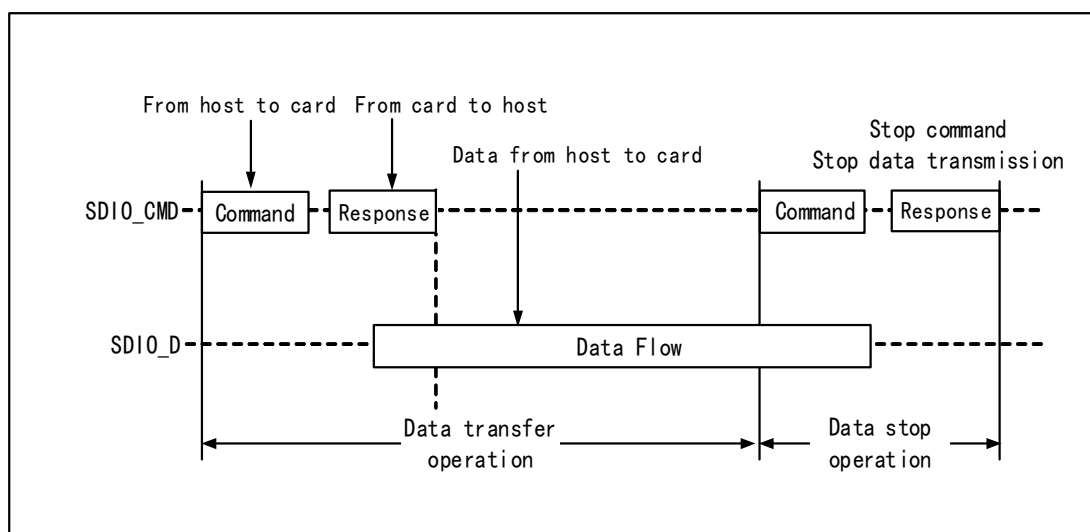
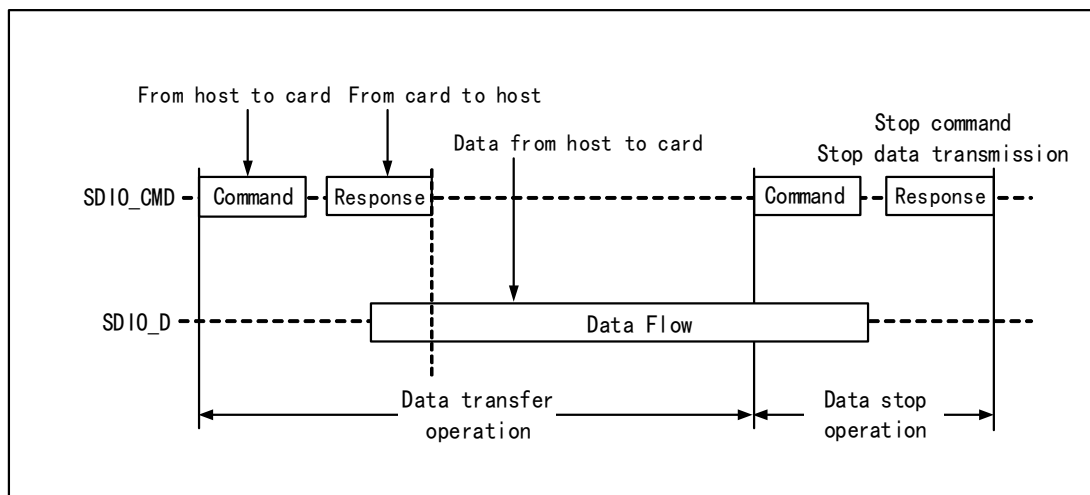


Figure 32-8 SDIO data stream write timing (Taking SDIO card as an example)



32.3.2 Command

Most SDIO transfers start with a command (CMD), through which the SDIO host informs the device of its intentions. The format of the command is as follows.

Table 32-2 Command format

Bit/Field name	Start bit	Transmission bit	Command index	Command parameters	CRC7	End bit
Rank/Width	47	46	[45:40] / 6 bits	[39:8] / 32 bits	[7:1] / 7 bits	0
Value	0b	1b	X	X	X	1b

4 types of commands:

- 1) Broadcast command (bc): sent to all cards on the bus, no response returned;
- 2) Broadcast command with response (bcr): sent to all cards on the bus, and a response is returned;
- 3) Point-to-point command (ac): issued to a specific card, but no data transmission;

4) Point-to-point command with data transmission (adtc): issued to a specific card with data transmission attached.

Below are some commonly used commands.

32.3.2.1 Basic Command

Basic commands are some of the more basic functions supported by the SD card.

Table 32-3 SD Card Basic Commands

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD0	bc	[31:0] stuff bits are meaningless.	None	GO_IDLE_STATE	Reset all cards to IDLE state
CMD2	bcr	[31:0] stuff bits are meaningless.	R2	ALL_SEND_CID	All cards reply to CID
CMD3	bcr	[31:0] stuff bits are meaningless.	R6	SEND_RELATIVE_ADDR	Reply to new RCA
CMD7	ac	[31:16] RCA; [15:0] stuff bits are meaningless.	R1b Selected Card	SELECT/DESELECT_CARD	Check or uncheck a card
CMD8	bcr	[31:12] reserved bits; [11:8] powered; [7:0] check mode.	R7	SEND_IF_COND	Send SD card interface conditions.
CMD9	ac	[31:16] RCA; [15:0] stuff bits are meaningless.	R2	SEND_CSD	Require CSD.
CMD10	ac	[31:16] RCA; [15:0] stuff bits are meaningless.	R2	SEND_CID	Require CID.
CMD11	ac	[31:0] stuff bits are meaningless.	R1	VOLTAGE_SWITCH	Switch to 1.8V.
CMD12	ac	[31:0] stuff bits are meaningless.	R1b	STOP_TRANSMISSION	Force the card to stop transmission;
CMD13	ac	[31:16] RCA; [15] stuff bits are meaningless; [14:0] stuff bits.	R1	SEND_STATUS/SEND_TASK_STATUS	Transmit Status or Task Status Register
CMD15	ac	[31:16] RCA, [15:0] stuff bits are meaningless.	None	GO_INACTIVE_STATE	The card is required to enter the INACTIVE mode;

32.3.2.2 Erase Command

The SD card is also a FLASH structure, and the FLASH needs to be erased before writing, but the SD card integrates erasing logic. If it is found that there is no erasure before executing the write command, it will automatically supplement the erasing operation. In many cases, especially before large lot writes, it can help to improve efficiency

if the erase is actively performed.

Table 32-4 SD Card Erase Command ^[1]

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD32	ac	[31:0] address to start erasing ^[2]	R1	ERASE_WR_BLK_START	Set the first address for erasing
CMD33	ac	[31:0] address to start erasing ^[2]	R1	ERASE_WR_BLK_END	Set the first address for erasing
CMD38	ac	[31:0] erase mode	R1b	ERASE	The parameter is 0, that is, ordinary wipe

Note 1: The erase command here is the erase command of SD card defined by SD protocol specification, and the erase command of SDIO card and MMC card is different from this.

Note 2: Currently commonly used SDHC/SDXC (2GB to 2TB) level erase address written by the card must be a block address, that is, 512-byte alignment.

32.3.2.3 Block Read Command

Table 32-5 SD Card block read command ^[1]

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD16	ac	[31:0] block length	R1	SET_BLOCKLEN	write block length, 512
CMD17	adtc	[31:0] address of a single block ^[1]	R1	READ_SINGLE_BLOCK	Set the start address of a single read
CMD18	adtc	[31:0] address of a single block ^[1]	R1	READ_MULTIPLE_BLOCK	Data blocks are continuously transmitted from the card to the host until interrupted by the STOP_TRANSMISSION command.
CMD19	adtc	[31:0] reserved bits	R1	SEND_TUNING_BLOCK	Send a 64-byte sequence representing the mode change
CMD20	ac	[31:28] speed control bits [27:0] see command description for details.	R1b	SPEED_CLASS_CONTROL	Speed level control command: [27:0] speed level [27:0] UHS speed level [27:0] video speed level
CMD22	ac	[31:6] reserved bits	R1	ADDRESS_EXTENSION	SDUC will be used.

		[5:0] extended address			
CMD23	ac	[31:0] block counter	R1	SET_BLOCK_COUNT	block counter

Note: The erasing address written by the SDHC/SDXC (2GB to 2TB) level cards commonly used at present must be a block address, that is, 512-byte alignment.

32.3.2.4 Write Command for Block Transfer

Table 32-6 SD Card block write command ^[1]

Command index	Type	Parameter	Response format	Abbreviation	Description
CMD16	ac	[31:0] block length	R1	SET_BLOCKLEN	write block length, 512
CMD24	adtc	[31:0] block address ^[1]	R1	WRITE_BLOCK	Set the start address of a single block write
CMD25	adtc	[31:0] addresses of multiple blocks ^[1]	R1	WRITE_MULTIPLE_BLOCK	Set the start address of multiple blocks write
CMD27	adtc	[31:0] stuff bits	R1	PROGRAM_CSD	Programmable word programming for CSD

Note: The erasing address written by the SDHC/SDXC (2GB to 2TB) level cards commonly used at present must be a block address, that is, 512-byte alignment.

32.3.3 Response

As a necessary reply from the SDIO device to the host, the response is also transmitted on the CMD line and must be replied within a specified time. The response transmission is high-order first and low-order last. The length of the response and the definition of each bit and each field are determined by the type of response, but all responses start with a fixed 0 start bit, followed by a fixed 0 transmission Direction bit ^[1]. All responses have a stop bit at the end, fixed at 1 ^[1]. There are roughly 7 kinds of responses, SD card supports R1/R1b/R2/R3/R6/R7, SDIO card also supports R4 / R5, see the format below.

32.3.3.1 R1 Response

Normal response, the total length is 48 bits, with CRC7, and the card status field is 32 bits. The format is as follows.

Table 32-7 R1 response

Bit/Field name	Start bit	Transmission bit	Response index	Card Status	CRC7	End bit
Rank/Width	47	46	[45:40] / 6 bits	[39:8]/32 bits	[7:1]/7 bits	0
value	0b	0b	Follow CMD Index	X	X	1b

32.3.3.2 R1b Response

The format of R1b is the same as that of R1, but a busy signal can be added after the response, that is, the data line

D2 is clamped. The host needs to deal with it after receiving the busy signal (Detecting that SDIO_D2 is low).

32.3.3.3 R2 Response

Responses to specific commands, the total length is 136 bits, the CRC7 is included in the card status field, and the card status field is 128 bits. The card status field stores the value of the CID register or CSD register. The CID register is generally used as the reply to CMD2/CMD10, and the CSD register is generally used as the reply to CMD9. For the specific meaning of the CID/CSD register, see the section 32.4.2 Device Registers. It should be noted that R2 will only reply to the [127:1] segment of the CID/CSD register. The reserved bit [2] of the CID/CSD [0] fixed to 1 is occupied by the end bit, and the end bit is also fixed to 1. The format is as follows.

Table 32-8 R2 response

Bit/Field name	Start bit	Transmission bit	Command index	Card status	End bit
Rank/Width	135	134	[133:128]/6 bits	[127:1]/127 bits	0
Value	0b	0b	111111b	CID/CSD	1b

32.3.3.4 R3 Response

Reply to the dedicated response of the OCR register, the total length is 48 bits, and there is no CRC7. The OCR register is generally used as a reply to ACMD41. The format is as follows.

Table 32-9 R3 response

Bit/Field name	Start bit	Transmission bit	Command index	Card Status	Reserved	End bit
Rank/Width	47	46	[45:40]/6 bits	[39:8]/32 bits	[7:1]/7 bits	0
Value	0b	0b	111111b	OCR	1111111b	1b

32.3.3.5 R4 Response

In response to the response of CMD5, reply to the special response of OCR and related registers, with a total length of 48 bits and a CRC7. R4 is used in SDIO card, the format is as follows.

Table 32-10 R4 response

Bit/Field name	Bit position	Width (bit)	value
Start bit	47	1	0b
Transmission bit	46	1	0b
Reserve	[45:40]	6	111111b
Card ready	39	1	X
IO number of functions	[38:36]	3	X
Current register	[35]	1	X
Stuff bits	[34:33]	2	00b
S18A	32	1	X
IO OCR	[31:8]	24	OCR
CRC field	[7:1]	7	1111111b
End bit	0	1	1b

Note: The format of MMC card R4 is different from SDIO card.

32.3.3.6 R5 Response

A dedicated response to CMD5, with a total length of 48 bits and a CRC7. R5 is used in SDIO card, the format is as follows.

Table 32-11 R5 response

Bit/Field name	Start bit	Transmission bit	Command index	Stuff bit	Response format	Read and write data	CRC7	End bit
Rank	47	46	[45:40]	[39:24]	[23:16]	[15:8]	[7:1]	0
width	1	1	6	16	8	8	7	1
Value	0b	0b	110100b	0000h	X	X	X	1b

Note: The format of MMC card R5 is different from SDIO card.

32.3.3.7 R6 Response

Reply to the special response of RCA, with a total length of 48 bits and a CRC7. The format is as follows.

Table 32-12 R6 response

Bit/Field name	Start bit	Transmission bit	Command index	Card RCA	Card status bit	CRC7	End bit
rank	47	46	[45:40]	[39:24]	[23:8]	[7:1]	0
Width	1	1	6	16	16	7	1
Value	0b	0b	000011b	X	X	X	1b

32.3.3.8 R7 Response

In response to the dedicated response of CMD8, it indicates the information of the supported voltage, with a total length of 48 bits, with CRC7, and the format is as follows.

Table 32-13 R7 response

Bit/Field name	Start bit	Transmission bit	Command index	Reserved bit	PCIe 1V2 support	PCIe response	Accepted Voltage	Check Feedback	CRC7	End bit
Bit position	47	46	[45:40]	[39:22]	21	20	[19:16]	[15:8]	[7:1]	0
Width	1	1	6	18	1	1	4	8	7	1
Value	0b	0b	001000b	00000h	X	X	X	X	X	1b

32.3.4 Data Transfer

Data transfer is carried out on the data line SDIO_D, which has 3 widths of 1/4/8 bits. During data transfer, the start bit of a clock is generally at the front, the high bit of each byte is in the front, and the low bit is in the back. For the block transfer mode only supported by the SD card, the data transfer of each block is followed by a CRC.

Enable the RANDOM_LEN_EN bit, configure the byte length indicated by DBLOCKSIZE2, configure the total length of the transferred data indicated by R32_SDIO_DLEN, enable the DTEN bit, and fill in the data to the R32_SDIO_FIFO, then you can complete the transfer of the block of any byte length with CRC checksum.

The MMC card also supports the data streaming transfer mode, at this time no CRC is attached. the following figure

shows the format of the data transfer.

Figure 32-9 SD card single bus data transfer byte format

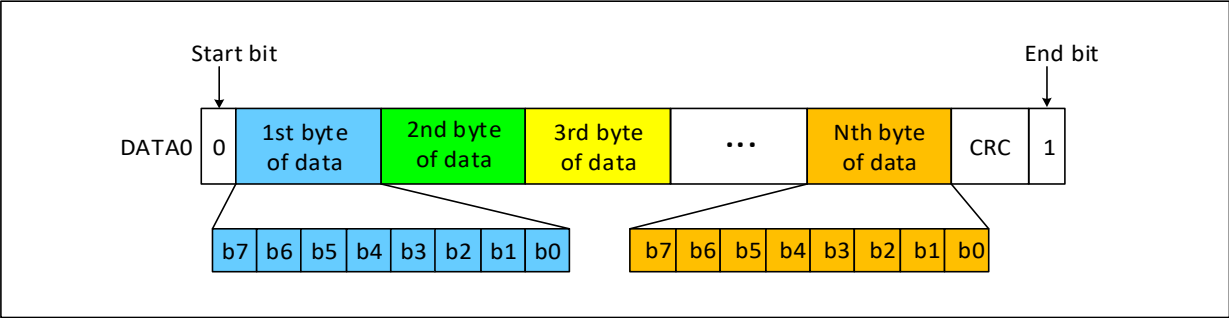


Figure 32-10 SD card 4 bus data transfer byte format

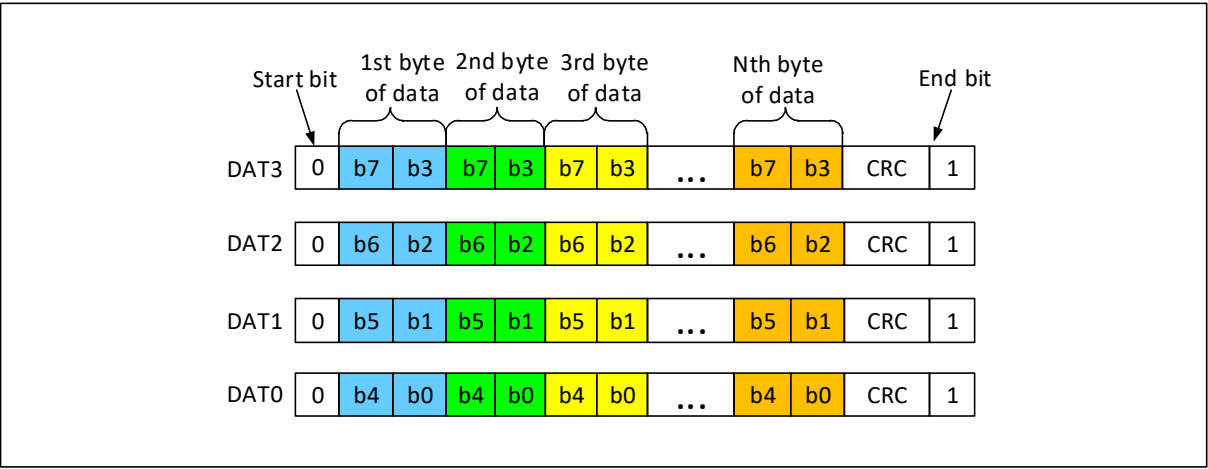


Figure 32-11 SD card single bus data transfer in a 512-bit word format

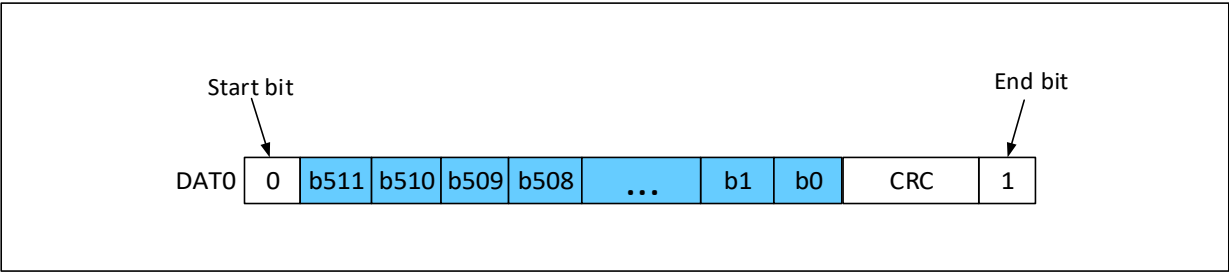
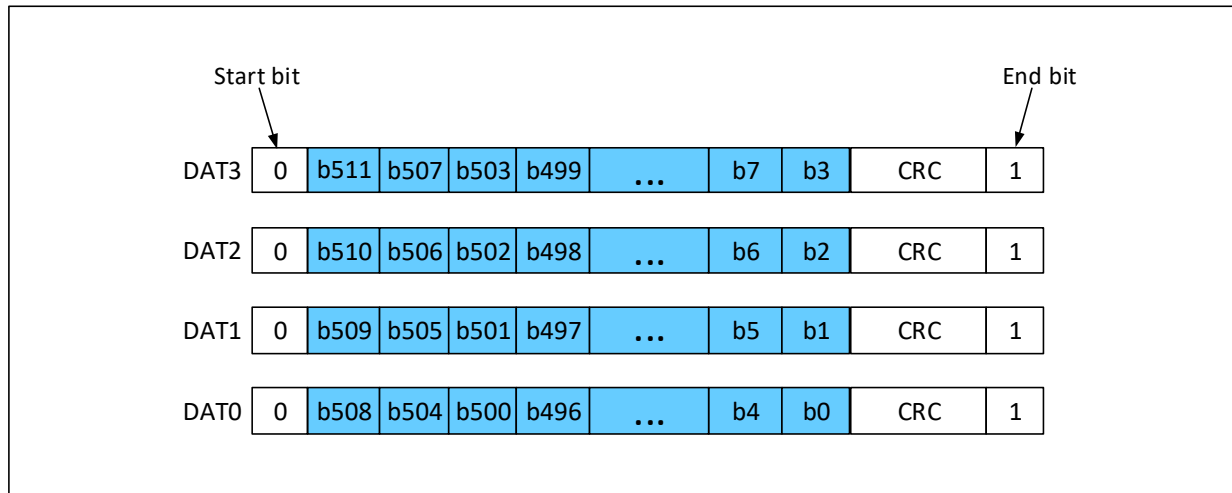


Figure 32-12 SD card 4 bus data transfer in a 512-bit word format



32.3.5 Slave Mode

Turning on the SLV_MODE bit means that it is in the state of waiting for commands from the host, with received command parameters placed in RESP1 and received command indexes placed in RESP2, while automatically answering to the host with an R1 type of answer, with the answer parameter being the value of CMDARG.

Data sent and received in slave mode refer to master mode, but the slave can use the SLV_FORCE_ERR bit to force a data block CRC error, thus indicating an undesired data read or write.

Note: The slave supports only the R1 type of answer, and the command indexes and parameter meanings used for master-slave communication are defined by the software;

32.4 Application

32.4.1 Device Initialization and Device Registers

32.4.1.1 OCR

The Operation Conditions Register stores some information about the power supply voltage and related status bits of the SD card. The relevant bits are defined in the following table.

Table 32-14 OCR bit definitions

Bit	Bit definition	Description
[0-3]	Reserved	Supported V _{DD33} voltage range in volts.
4	Reserved	
5	Reserved	
6	Reserved	
7	Reserved for low voltage range	
8	Reserved	
9	Reserved	
10	Reserved	
11	Reserved	
12	Reserved	
13	Reserved	

14	Reserved	
15	2.7-2.8	
16	2.8-2.9	
17	2.9-3.0	
18	3.0-3.1	
19	3.1-3.2	
20	3.2-3.3	
21	3.3-3.4	
22	3.4-3.5	
23	3.5-3.6	
24	accept switching to 1.8V	
[25:26]	Reserved	
27	Over 2TB support status bit	
28	Reserved	
29	UHS-II card status bits	This bit is set to indicate that the card supports the UHS-II interface.
30	Card capacity status (CCS)	This bit is set to indicate that the card capacity is greater than 2GB.
31	Card power-on status bit (busy)	This bit is set when the card is powered up. After power-on, other bits have meaning.

32.4.1.2 CID

CID stores some identifying information.

Table 32-15 CID bit definitions

Bit/Field name	Abbreviation	Width	Bit position
Vendor ID	MID	8	[127:120]
App ID	OID	16	[119:104]
Product Name	PNM	40	[103:64]
Product Revision	PRV	8	[63:56]
Product Serial Number	PSN	32	[55:24]
Reserved	None	4	[23:20]
Date of Manufacture	MDT	12	[19:8]
CRC7	CRC	7	[7:1]
Fixed bit, always 1	None	1	[0]

32.4.1.3 CSD

The CSD register stores the characteristic data of the SD card. Taking the most commonly used SDHC and SDXC cards of the second version of the CSD as an example, the definitions of each field are as follows.

Table 32-16 CSD bit definitions

Name	Abbreviation	Width	Value	Read and write	Bit position
CSD version	CSD_STRUCTURE	2	01b	RO	[127:126]
Reserved	None	6	00_0000b	RO	[125:120]

Reading access time	TAAC	8	0Eh	RO	[119:112]
Reading access time in clock cycles	NSAC	8	00h	RO	[111:104]
Maximum data transmission speed	TRAN_SPEED	8	32h5Ah0Bh2Bh	RO	[103:96]
Card command class	CCC	12	X1X1101101X1b	RO	[95:84]
Maximum length of read data block	READ_BL_LEN	4	9	RO	[83:80]
Allow block partial read	READ_BL_PARTIAL	1	0	RO	[79]
block write misalignment	WRITE_BLK_MISALIGN	1	0	RO	[78]
block read misaligned	READ_BLK_MISALIGN	1	0	RO	[77]
Executed DSR	DSR_IMP	1	X	RO	[76]
Reserved	None	6	00_0000b	RO	[75:70]
Device size	C_SIZE	22	XXXXXXh	RO	[69:48]
Reserved	None	1	0	RO	[47]
Single block erase enable	ERASE_BLK_EN	1	1	RO	[46]
Erase sector size	SECTOR_SIZE	7	7Fh	RO	[45:39]
Write-protect group size	WP_GRP_SIZE	7	0000000b	RO	[38:32]
Write Protect Group Enable	WP_GRP_ENABLE	1	0	RO	[31]
Reserved	None	2	00b	RO	[30:29]
Write speed factor	R2W_FACTOR	3	010b	RO	[28:26]
Maximum write data block length	WRITE_BL_LEN	4	9	RO	[25:22]
Allow block partial writes	WRITE_BL_PARTIAL	1	0	RO	[21]
Reserved	None	5	00000b	RO	[20:16]
file format group	FILE_FORMAT_GRP	1	0	RO	[15]

Copy sign	TMP_WRITE_PROTECT	1	X	RW OTP	[14]
Permanent write protection	PERM_WRITE_PROTECT	1	X	RW OTP	[13]
Temporary write protection	TMP_WRITE_PROTECT	1	X	RW	[12]
File format	FILE_FOMAT	2	00b	RO	[11:10]
Reservef	None	2	00b	RO	[9:8]
CRC	CRC	7	0000000b	RW	[7:1]
Not used, must use 1	None	1	1b	RO	[0]

32.4.1.4 RCA

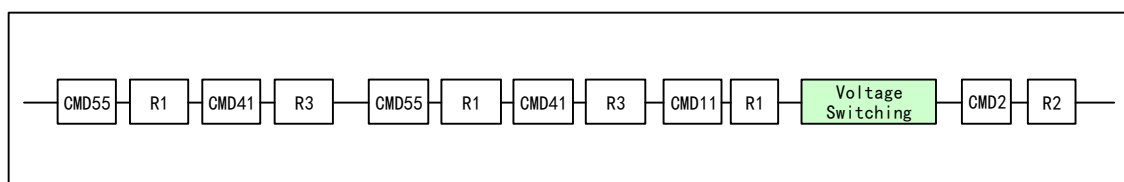
The relative card address register stores the address of the card, which is 16 bits, and the default value is 0.

32.4.2 Voltage Switch

In the later stage of SD card initialization, it is necessary to switch the interface level, and switch the IO level of the clock line, data line and command line of the SD card to the 1.8V level. For devices whose slew rate is not good enough, using a lower-level standard can help increase the frequency. However, the power supply voltage of SD does not necessarily change, and only SD cards with low voltage power supply appear in the newer version of the protocol.

The steps for switching the voltage are shown in the figure below.

Figure 32-13 Voltage switch sequence



32.4.3 Clock Switch

During initialization, the clock of the SD card is only 400KHz. After the voltage switch is completed, the clock can be increased to a higher level. For example, the first speed of the SDHC card UHS-I mode, the bus clock can reach 80MHz. In view of the IO output capability of the microcontroller, the clock should be limited to 50MHz.

32.5 Interrupt

32.5.1 SDIO Interrupt

SDIO supports a variety of interrupt sources. As shown in the interrupt enable register (R32_SDIO_IER), there are 24 situations that can trigger interrupts, and users can enable them as appropriate.

32.5.2 SDIO Device Interrupt

It should be noted that not only SDIO peripherals can report interrupts to the CPU, but also external SDIO cards and MMC cards can report interrupts to SDIO peripherals. In 4-bit bus mode, the interrupt line is D1, and in 8-bit bus mode, the interrupt line is D7, active low. If SDIO detects that D1 or D7 is low level in the idle state, it should read the status register or interrupt flag register of the SDIO device to respond to the interrupt in time. The CPU can get whether the SDIO host receives an interrupt through the SDIOIT bit of the R32_SDIO_STA register.

32.6 Register Description

Table 32-17 SDIO registers

Name	Address	Description	Reset Value
R32_SDIO_POWER	0x40018000	Power register	0x00000000
R32_SDIO_CLKCR	0x40018004	Clock register	0x00000000
R32_SDIO_ARG	0x40018008	Command parameter register	0x00000000
R32_SDIO_CMD	0x4001800C	Command register	0x00000000
R32_SDIO_RESPCMD	0x40018010	Response register	0x00000000
R128_SDIO_RESPX	0x40018014	Response parameter register	0x00000000
R32_SDIO_DTIMER	0x40018024	Data timing register	0x00000000
R32_SDIO_DLEN	0x40018028	Data length register	0x00000000
R32_SDIO_DCTRL	0x4001802C	Data control register	0x00000000
R32_SDIO_DCOUNT	0x40018030	Data count register	0x00000000
R32_SDIO_STA	0x40018034	Status register	0x00000000
R32_SDIO_ICR	0x40018038	Interrupt clear register	0x00000000
R32_SDIO_MASK	0x4001803C	Interrupt enable register	0x00000000
R32_SDIO_FIFOCNT	0x40018048	FIFO counter	0x00000000
R32_SDIO_DCTRL2	0x40018060	Data control register 2	0x00000000
R32_SDIO_FIFO	0x40018080	FIFO register	0x00000000

32.6.1 Power Register (R32_SDIO_POWER)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														PWRCTRL[1:0]	

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved	0
[1:0]	PWRCTRL[1:0]	RW	Power control: 00: Power down, clock stops; 01: Reserved; 10: Reserved power-on status; 11: Power-on, the card clock is turned on;	0

32.6.2 Clock Register (R32_SDIO_CLKCR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	HWFC_EN	NEGEDGE	WIDBUS[1:0]	BYPASS	PWRSAPV	CLKEN	CLKDIV[7:0]								

Bit	Name	Access	Description	Reset Value
[31:15]	Reserved	RO	Reserved	0
14	HWFC_EN	RW	Hardware flow control enable. TXFIFOE and RXFIFOE signals have effect after this bit is set. 0: Disable hardware flow control; 1: Enable hardware flow control.	0
13	NEGEDGE	RW	SDIO_CLK phase selection. 0: SDIO_CLK is generated on the rising edge of HCLK; 1: SDIO_CLK is generated on the falling edge of HCLK.	0
[12:11]	WIDBUS[1:0]	RW	Bus width configuration. 00: 1-bit bus mode, use SDIO_D0; 01: 4-bit bus mode, use SDIO_D[3:0]; 10: 8-bit bus mode, use SDIO_D[7:0]; 11: Not used.	0
[10]	BYPASS	RW	Clock bypass enable. 0: SDIO_CLK is obtained after divider; 1: SDIO_CLK is directly connected to HCLK. <i>Note: The HWFC_EN bit needs to be enabled when using this bit.</i>	0
[9]	PWRSAPV	RW	Free clock state configuration. After this bit is set, SDIO_CLK output is disabled when bus is free, to save power. 0: SDIO_CLK is always output; 1: SDIO_CLK is only output when needed.	0
[8]	CLKEN	RW	Clock enable. 0: SDIO_CLK output is disabled; 1: SDIO_CLK output is enabled.	0
[7:0]	CLKDIV[7:0]	RW	Clock frequency division factor. This field represents the relationship between SDIO_CLK and HCLK. $SDIO_CLK = HCLK / (CLKDIV + 2)$. Note that SDIO_CLK should be lower than 400kHz during initialization.	0

Note: (1) The clock configuration register is used to control SDIO_CLK related parameters. It should be noted that

this register cannot be changed during the period of reading and writing data to 7 HCLK cycles.

(2) No need to configure clock register in slave mode.

32.6.3 Command Parameter Register (R32_SDIO_ARG)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CMDARG[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMDARG[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	CMDARG	RW	Master mode: Command parameter field. This field stores the parameters in the command, which will be sent to the CMD line as part of the command. Slave mode: 32-bit response parameter register to which the slave replies to the answer.	0

32.6.4 Command Register (R32_SDIO_CMD)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	ATACMD	NIE N	ENCMDCompl	SDIOSuspend	CPSMEN	WAITPEND	WAITINT	WAITRESP[1:0]	CMDINDEX[5:0]						

Bit	Name	Access	Description	Reset Value
[31:15]	Reserved	RO	Reserved	0
[14]	ATACMD	RW	Carry out CE-ATA command. If this bit is set, CPSM will go to CMD61.	0
[13]	NIEN	RW	Not enable CE-ATA interrupt setting. If this bit is set, CE-ATA will not generate an interrupt.	0
[12]	ENCMDCompl	RW	Enable CMD completed signal enable. If this bit is set, command completion will be signaled.	0
[11]	SDIOSuspend	RW	Suspend command. If this bit is set, a suspend signal will be sent. Applies to SDIO cards only.	0
[10]	CPSMEN	RW	CPSM (Command Path State Machine) enable. If set, CPSM is enabled.	0
[9]	WAITPEND	RW	Command wait control. If this bit is set, the CPSM will wait for the data transfer to complete before sending the command.	0
[8]	WAITINT	RW	Command wait interrupt control. If this bit is set,	0

			the CPSM will turn off the timeout control and wait for an interrupt to be generated.	
[7:6]	WAITRESP[1:0]	RW	Response type. This field indicates the type of response the CPSM expects to receive. 00: No response, wait for the CMDSENT flag; 01: Short response, wait for CMDREND or CCRCFAIL flag; 10: No response, wait for the CMDSENT flag; 11: Long response, wait for CMDREND or CCRCFAIL flag;	0
[5:0]	CMDINDEX[5:0]	RW	Command index. This field indicates the specific command value.	0

32.6.5 Response Register (R32_SDIO_RESPCMD)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RESPCMD					

Bit	Name	Access	Description	Reset Value
[31:6]	Reserved	RO	Reserved	0
[5:0]	RESPCMD[5:0]	RO	This field records the index value of the received response.	0

32.6.6 Response Parameter Register (R128_SDIO_RESPX)

Bit	Name	Access	Description	Reset Value
[127:0]	CARDSTATUSx	RO	Master mode: When the response is a long response, the whole 128 bits indicate the card status; when the response is a short response, the lower 32 bits indicate the card status. the SDIO peripheral receives the highest bit of the card status first and stores it starting from the lowest bit of R128_SDIO_RESPX.	0

32.6.6.1 Response Parameter Register High 32 Bits (R128_SDIO_RESP1[127:96])

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS1															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS1															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS1	RO	Master mode: Long response: Card status [127:96]; Short response: Card status [31:0]. Slave mode: Command parameters received by the receive command.	0

32.6.6.2 Response Parameter Register Second High 32 Bits (R128_SDIO_RESP2[95:64])

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS2[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS2[79:64]															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS2	RO	Master mode: Card status [95:64]; Slave mode: Index of the command received by the receive command.	0

32.6.6.3 Response Parameter Register Second Low 32 Bits (R128_SDIO_RESP3[63:32])

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS3[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS3[48:32]															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS3	RO	Master mode: Card status [63:32].	0

32.6.6.4 Response Parameter Register Low 32 Bits (R128_SDIO_RESP4[31:0])

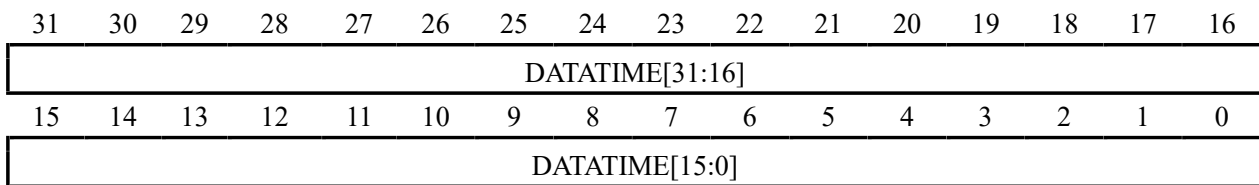
Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS4[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS4[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	CARDSTATUS4	RO	Master mode: Card status [31:0];	0

32.6.7 Data Timing Register (R32_SDIO_DTIMER)

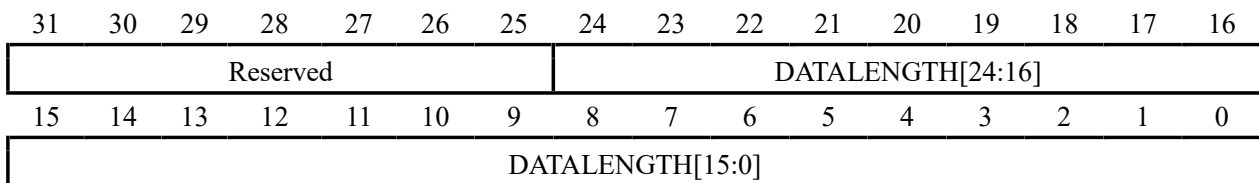
Offset address: 0x24



Bit	Name	Access	Description	Reset Value
[31:0]	DATETIME	RW	Data timeout period. SDIO_CK cycle as the unit.	0

32.6.8 Data Length Register (R32_SDIO_DLEN)

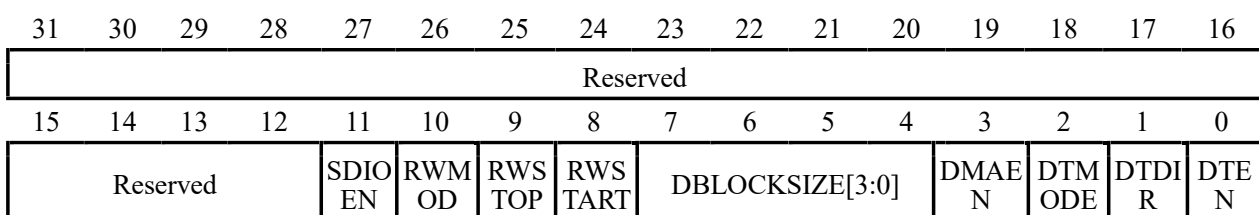
Offset address: 0x28



Bit	Name	Access	Description	Reset Value
[31:25]	Reserved	RO	Reserved	0
[24:0]	DATALENGTH[24:0]	RW	Data length. The value of this field is loaded into the transfer counter when a transfer is started. For block transfers, the value of this field must be an integer multiple of the block size. The block size is defined by the SDIO device and stored in R32_SDIO_DCTLR[7:4]. Common values are 512B, etc.	0

32.6.9 Data Control Register (R32_SDIO_DCTRL)

Offset address: 0x2C



Bit	Name	Access	Description	Reset Value
[31:12]	Reserved	RO	Reserved	0
11	SDIOEN	RW	SDIO enable. When this bit is set, the DPSM can perform some operations specific to the SDIO card.	0
10	RWMOD	RW	Read wait mode. 0: Stop SDIO_CK control read wait; 1: Use SDIO_D2 to control read wait.	0
9	RWSTOP	RW	Read wait stop. If the RWSTART bit is set, the read wait will be stopped.	0

8	RWSTART	RW	Read wait start. Setting this bit will perform a read wait operation.	0
[7:4]	DBLOCKSIZE[3:0]	RW	Data block size. This field stores the size of the data block, and the block transfer length must be defined before using the block transfer. The value that can be written in this field is between 0 and 1110b, which indicates that the block transfer length is 2^{BLKLEN} (Between 0 and 16384 bytes).	0
3	DMAEN	RW	DMA enable. 0: Disable DMA; 1: Enable DMA.	0
2	DTMODE	RW	Data mode. Set this bit to configure the transfer mode. 0: Block transfer; 1: Stream transfer. <i>Note: The 4-bit and 8-bit bus mode of streaming transmission only supports products with the last six digits of lot numbers of CH32F20x_D8, CH32F20x_D8C, CH32V30x_D8C, and CH32V31x_D8C not being 0.</i>	0
1	DTDIR	RW	Data direction. Set this bit to configure the transfer direction. 0: From controller to card; 1: From card to controller.	0
0	DTEN	RW	Data enable. Set this bit to start data transfer. Specific process: after setting this bit, DPSM enters the process of Wait_S or Wait_R (Depending on the data direction).	0

32.6.10 Data Count Register (R32_SDIO_DCOUNT)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							DATACOUNT[24:16]								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATACOUNT[15:0]															

Bit	Name	Access	Description	Reset Value
[31:25]	Reserved	RO	Reserved	0
[24:0]	DATACOUNT[24:0]	RO	Data count. The value of the transmission length register will be loaded into this counter when a transfer is initiated and decremented as the transfer progresses.	0

32.6.11 Status Register (R32_SDIO_STA)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CEATAEND	SDIOIT	RXDAVL	TXDAVL	RXFIFOE	TXFIFOE	RXFIFO	TXFIFO
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXFIFOHF	TXFIFOHE	RXACT	TXACT	CMDACT	DBCKEND	STBITERR	DATAEND	CMDSENT	CMDREND	RXOVERR	TXUNDERR	DTIMEOUT	CTIMEOUT	DCRCFAIL	CCRCFAIL

Bit	Name	Access	Description	Reset Value
[31:24]	Reserved	RO	Reserved	0
23	CEATAEND	RO	When this bit is set, CMD61 receives the CE-ATA completed signal.	0
22	SDIOIT	RO	When this bit is set, SDIO receives a device interrupt.	0
21	RXDAVL	RO	When this bit is set, data in the Rx FIFO is valid.	0
20	TXDAVL	RO	When this bit is set, data in the Tx FIFO is valid.	0
19	RXFIFOE	RO	When this bit is set, the Rx FIFO is empty.	0
18	TXFIFOE	RO	When this bit is set, the Tx FIFO is empty.	0
17	RXFIFO	RO	When this bit is set, the Rx FIFO is full.	0
16	TXFIFO	RO	When this bit is set, the Tx FIFO is full.	0
15	RXFIFOHF	RO	When this bit is set, the Rx FIFO is half full.	0
14	TXFIFOHE	RO	When this bit is set, the Tx FIFO is half empty.	0
13	RXACT	RO	When this bit is set, data is being received.	0
12	TXACT	RO	When this bit is set, data is being transmitted.	0
11	CMDACT	RO	When this bit is set, a command is being transmitted.	0
10	DBCKEND	RO	When this bit is set, the data block has been sent or received and the CRC has passed.	0
9	STBITERR	RO	When this bit is set, no start signal is detected on all data lines in wide bus mode.	0
8	DATAEND	RO	When this bit is set, the data transfer is complete (The transfer counter is zero).	0
7	CMDSENT	RO	When this bit is set, the command has been sent.	0
6	CMDREND	RO	When this bit is set, the response has been received and the CRC is successful.	0
5	RXOVERR	RO	When this bit is set, the Rx FIFO overflows.	0
4	TXUNDERR	RO	When this bit is set, the transmit FIFO underflows.	0
3	DTIMEOUT	RO	When this bit is set, the data times out.	0
2	CTIMEOUT	RO	When this bit is set, the command timeout exceeds 64 SDIO_CLK cycles.	0
1	DCRCFAIL	RO	When this bit is set, a data block has been sent or received but the CRC failed.	0
0	CCRCFAIL	RO	When this bit is set, the response has been received but the CRC has failed.	0

32.6.12 Interrupt Clear Register (R32_SDIO_ICR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CEATAENDC	SDIOITC	Reserved					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DBCKENDC	STBITERRC	DAT AENDC	CMDSENTC	CMDREND C	RXOVERRC	TXUNDERRC	DTIMEOUTC	CTIMEOUTC	DCRCFAILC	CCRCFAILC

Bit	Name	Access	Description	Reset Value
[31:24]	Reserved	RO	Reserved	0
23	CEATAENDC	RW	Setting this bit clears the CEATAEND bit in the status register.	0
22	SDIOITC	RW	Setting this bit clears the SDIOIT bit in the status register.	0
[21:11]	Reserved	RW	Reserved	0
10	DBCKENDC	RW	Setting this bit clears the DBCKEND bit in the status register.	0
9	STBITERRC	RW	Setting this bit clears the STBITERR bit in the status register.	0
8	DATAENDC	RW	Setting this bit clears the DATAEND bit in the status register.	0
7	CMDSENTC	RW	Setting this bit clears the CMDSENT bit in the status register.	0
6	CMDREND C	RW	Setting this bit clears the CMDREND bit in the status register.	0
5	RXOVERRC	RW	Setting this bit clears the RXOVERR bit in the status register.	0
4	TXUNDERRC	RW	Setting this bit clears the TXUNDERR bit in the status register.	0
3	DTIMEOUTC	RW	Setting this bit clears the DTIMEOUT bit in the status register.	0
2	CTIMEOUTC	RW	Setting this bit clears the CTIMEOUT bit in the status register.	0
1	DCRCFAILC	RW	Setting this bit clears the DCRCFAIL bit in the status register.	0
0	CCRCFAILC	RW	Setting this bit clears the CCRCFAIL bit in the status register.	0

32.6.13 Interrupt Enable Register (R32_SDIO_MASK)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CEATAENDIE	SDIOITIE	RXDAVLIE	TXDAVLIE	RXFIFOEIE	TXFIFOEIE	RXFIFOEIE	TXFIFOEIE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXFIFOHFIE	TXFIFOHEIE	RXACTIE	TXACTIE	CMDACTIE	DBCKENDIE	STBITERRIE	DATAENDIE	CMDSENTIE	CMDRENDIE	RXOVERIE	TXUNDERRIE	DTIMEOUTIE	CTIMEOUTIE	DCRCFAILIE	CCRCFAILIE

Bit	Name	Access	Description	Reset Value
[31:24]	Reserved	RO	Reserved	0
23	CEATAENDIE	RW	After this bit is set, the status register will generate an interrupt when the CEATAEND bit is set.	0
22	SDIOITIE	RW	After this bit is set, the status register will generate an interrupt when the SDIOIT bit is set.	0
21	RXDAVLIE	RW	After this bit is set, the status register will generate an interrupt when the RXDAVL bit is set.	0
20	TXDAVLIE	RW	After this bit is set, the status register will generate an interrupt when the TXDAVL bit is set.	0
19	RXFIFOEIE	RW	After this bit is set, the status register will generate an interrupt when the RXFIFOE bit is set.	0
18	TXFIFOEIE	RW	After this bit is set, the status register will generate an interrupt when the TXFIFOE bit is set.	0
17	RXFIFOEIE	RW	After this bit is set, the status register will generate an interrupt when the RXFIFOE bit is set.	0
16	TXFIFOEIE	RW	After this bit is set, the status register will generate an interrupt when the TXFIFOE bit is set.	0
15	RXFIFOHFIE	RW	After this bit is set, the status register will generate an interrupt when the RXFIFOHF bit is set.	0
14	TXFIFOHEIE	RW	After this bit is set, the status register will generate an interrupt when the TXFIFOHE bit is set.	0
13	RXACTIE	RW	After this bit is set, the status register will generate an interrupt when the RXACT bit is set.	0
12	TXACTIE	RW	After this bit is set, the status register will generate an interrupt when the TXACT bit is set.	0
11	CMDACTIE	RW	After this bit is set, the status register will generate an interrupt when the CMDACT bit is set.	0
10	DBCKENDIE	RW	After this bit is set, the status register will generate an interrupt when the DBCKEND bit is set.	0
9	STBITERRIE	RW	After this bit is set, the status register will generate an interrupt when the STBITERR bit is set.	0
8	DATAENDIE	RW	After this bit is set, the status register will generate an interrupt when the DATAEND bit is set.	0
7	CMDSENTIE	RW	After this bit is set, the status register will generate an interrupt when the CMDSENT bit is set.	0

			interrupt when the CMDSENT bit is set.	
6	CMDRENDIE	RW	After this bit is set, the status register will generate an interrupt when the CMDREND bit is set.	0
5	RXOVERRIE	RW	After this bit is set, the status register will generate an interrupt when the RXOVERR bit is set.	0
4	TXUNDERRIE	RW	After this bit is set, the status register will generate an interrupt when the TXUNDERR bit is set.	0
3	DTIMEOUTIE	RW	After this bit is set, the status register will generate an interrupt when the DTIMEOUT bit is set.	0
2	CTIMEOUTIE	RW	After this bit is set, the status register will generate an interrupt when the CTIMEOUT bit is set.	0
1	DCRCFAILIE	RW	After this bit is set, the status register will generate an interrupt when the DCRCFAIL bit is set.	0
0	CCRCFAILIE	RW	After this bit is set, the status register will generate an interrupt when the CCRCFAIL bit is set.	0

32.6.14 FIFO Count Register (R32_SDIO_FIFOCNT)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FIFOCOUNT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FIFOCOUNT[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	FIFOCOUNT[31:0]	RO	The FIFO contains the number of words (32 bits) that have not been written to or read from the FIFO. When setting R32_SDIO_DCTLR: R32_SDIO_DCTLR, if the DPSM is idle, the FIFO counter will load the transfer length value from R32_SDIO_TLEN, if this value is not divisible by 4, the last 1 to 3 bytes will be treated as a word.	0

32.6.15 FIFO Data Control Register 2 (R32_SDIO_DCTRL2)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SLV CK_P HAS E	SLV FOR CE_E RR	SLV MOD E	Reserved							RAN DOM LEN _EN
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DBLOCKSIZE2										

Bit	Name	Access	Description	Reset Value
[31:27]	Reserved	R0	Reserved	0
26	SLV_CK_PHASE	RW	Phase selection bit when outputting DATA from mode when SLV_MODE is 1: 0: Outputs DATA on the rising edge of the internal SDCK; 1: outputs DATA on the falling edge of the internal SDCK. When SLV_MODE is 0: 0: DTEN bit writes 0 to clear the FIFO; 1: DTEN bit writes 0 without clearing the FIFO.	0
[25]	SLV_FORCE_ERR	RW	Software forced data block CRC error in slave mode 0: Use hardware correct CRC value; 1: Software forced data block CRC is the wrong value.	0
[24]	SLV_MODE	RW	Slave mode enable bit: 0: The controller operates in host mode; 1: The controller CPSM and DPSM work in slave mode.	0
[23:17]	Reserved	R0	Reserved	0
[16]	RANDOM_LEN_EN	RW	Data block arbitrary byte length enable bit: 0: The data block can only be 2 DBLOCKSIZE bytes; 1: The data block can be any length from 0 to 2048 in DBLOCKSIZE2.	0
[15:12]	Reserved	R0	Reserved	0
[11:0]	DBLOCKSIZE2	RW	The data block length field for arbitrary byte length mode.	0

32.6.16 FIFO Register (R32_SDIO_FIFO)

Offset address: 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FIFODATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FIFODATA[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	FIFODATA[31:0]	RW	FIFO data field. Reading and writing this field will read the received data or send the pending data. The SDIO FIFO has a total of 32 words (One word is 32 bits).	0

Chapter 33 SD/EMMC Controller (SDMMC)

This chapter's module description applies only to the CH32H417, CH32H416 microcontroller product.

The system provides one SDMMC controller host/slave interface with a maximum transmission clock of 200MHz. It supports 1-wire, 4-wire, and 8-wire communication modes, as well as dual-edge sampling. External devices such as SD/TF cards and eMMC cards can be connected. Application code can flexibly configure various parameters including data transmission/reception commands, response packet formats, valid data packet modes and lengths, and dual-buffer length switching thresholds.

Note: When using 1-wire or 4-wire mode, unused data lines corresponding to GPIO pins cannot be repurposed for output. They may be used for input multiplexing or as general-purpose GPIO outputs.

33.1 Main Features

- Support SD physical layer 1.0 and 2.0 specifications, as well as UHS-I SDR50, DDR50, and SDR104 modes compliant with SD 3.0 specifications
- Compliant with EMMC Card Specifications 4.4 and 4.5.1; supports HS200 and HS400 modes per EMMC Card Specification 5.0
- Communication modes support single-wire, four-wire, and eight-wire configurations
- Maximum communication clock single-edge up to 200MHz, dual-edge up to 180MHz
- Support dual-edge sampling
- Flexible configurable packet length, command format, and response status
- Provide hardware-automated clock stop function during data block intervals
- Support SD cards, SDIO cards, eMMC cards, and other devices compliant with SD interface protocols
- Support SDIO slave interface, enabling data exchange with chips featuring SDIO host interfaces
- DMA dual-buffering capability

33.2 Register Description

33.2.1 Communication Clock

- **Frequency configuration**

The controller interface provides two clock modes: low-speed mode and high-speed mode. Generally, the SD protocol stipulates that the communication clock of about 400kHz should be used for the early initialization of the equipment with SD interface to ensure better communication compatibility. When the parameter information of the external equipment is obtained, it can be switched to a higher clock frequency for communication according to its internally supported configuration.

Set the bit9 of R16_EMMC_CLK_DIV register to select the clock mode, and set the bit[4:0] frequency division value to get the final SDIO interface output clock. Bit8 controls whether the EMMC peripheral outputs a clock signal to the MSDCK pin, and it is necessary to ensure that the IO mode of the MSDCK pin is configured as push-pull output.

- **Phase configuration**

When communicating with external SDIO interface devices, due to high clock frequency, hardware wiring, device characteristics and other factors, signal sampling errors will occur.

The system SDIO controller's host interface provides an output clock phase inversion (180° offset) function. Setting

bit 10 of register R16_EMMC_CLK_DIV to 1 physically inverts the output clock signal, while the controller's internal clock remains unchanged. This method allows adjustment of communication timing.

Note: At higher SDCLK frequency, if the recommended timing adjustment register value can't communicate stably, users need to initiate a bus sampling tuning sequence to find a better sampling point.

33.2.2 Command Sending and Response

The 48-bit data packet with fixed command length sent by SDIO host interface to slave machine is transmitted serially on MCMD line. Some commands don't need the slave to respond, and some commands need the slave to respond to reply packets or data packets with different lengths.

- General command format: Start bit (0)+ Transmission bit (1)+ Command index+ Parameter +CRC7+ End bit (1)

The start bit, data bits, CRC7, and stop bit are automatically filled by the hardware. The command index and parameters vary depending on the command and must be passed to the hardware controller by the application code. Configure the parameter field in the SD command by writing to the R32_EMMC_ARGUMENT register; Writing bits [5:0] of the R16_EMMC_CMD_SET register populates the command index, while bits [11:8] of this register configure the hardware's expected response packet detection for this command. Writing to the R16_EMMC_CMD_SET register triggers the hardware to transmit the command packet sequence on the MCMD signal line.

- Short Response Format: Start bit (0) + Transmission bit (0) + Command Index + Parameters + CRC7 + Stop bit (1)
- Long Response Format: Start bit (0) + Transmission bit (0) + Reserved field (111111b) + Data + Stop bit (1)

Slave response packets are categorized by length into 48-bit and 136-bit packets, with 32 bits and 128 bits of user-valid data respectively. After sending a host command, the R16_EMMC_INT_FG register can be queried. If the RB_EMMC_IF_CMDDONE flag is set to 1, the R32_EMMC_RESPONSE register can be read to obtain the valid data from the slave interface response packet. If the response packet is abnormal, corresponding flags in bits [2:0] of the R16_EMMC_INT_FG register will be set.

33.2.3 Read Multiple Data Blocks Consecutively

For the operation of reading data (transmitted through data lines) in the SD protocol command, the following configuration is required:

- 1) Initialization: Configure the R16_EMMC_CONTROL register to set the sampling edges for commands and data, the data bus width for transmission and reception, and enable DMA functionality. Configure the RB_EMMC_TOCNT_MASK bit in the R8_EMMC_TIMEOUT register to set the command timeout and data timeout durations. Configure the R32_EMMC_DMA_BEG1 register to set the DMA start address for storing read data. Configure the R32_EMMC_BLOCK_CFG register to set the single data block length during transfer and the total number of data blocks to be read. Configure the R32_EMMC_TRAN_MODE register to set the DMA transfer direction to "SD to Controller." Optionally configure whether to automatically stop the clock during block transfer intervals, allowing sufficient time for application code to perform processing tasks before software resumes the transfer.
- 2) Send commands to the SDIO slave interface device. These commands instruct the slave device to transmit data from the data bus to the host interface, such as CMD17 and CMD18 in the SD protocol.
- 3) The controller interface reads data from the data line: After the slave interface device responds to the host command, if the device is ready for data transfer, it outputs data from the data line based on the host clock signal, with the start signal being a low level. The host controller samples the signal and transfers it via internal DMA to the user-configured SRAM region. Each transfer of data set by RB_EMMC_BKSIZE_MASK indicates

completion of one data block. Simultaneously, the internal total number of transferred data blocks is decremented by one. Once all data blocks are transferred, the internal DMA ceases further data transfers. During this process, the hardware sets the RB_EMMC_IF_BKGAP flag upon completing each data block read. When all data blocks are read, the hardware sets the RB_EMMC_IF_TRANDONE flag. If a data timeout occurs during the controller's receive wait period or a CRC error is detected in the received data, the RB_EMMC_IF_DATTMO and RB_EMMC_IF_TRANERR flags are set respectively. Application code can configure the corresponding interrupt control bits in the interrupt enable register R16_EMMC_INT_EN. When the corresponding interrupt flag is set, it triggers the EMMC interrupt service routine.

- 4) Detection Status: The R32_EMMC_STATUS register enables real-time monitoring of the logic levels on data line MD0 and command line MCMD. It evaluates these levels according to protocol-defined timing specifications. The MASK_BLOCK_NUM field records the number of data blocks successfully transferred via DMA during the current command transmission.

33.2.4 Write Multiple Data Blocks Consecutively

- 1) Configure the R16_EMMC_CONTROL register to set the sampling edges for commands and data, the data bus width for transmission and reception, and enable DMA functionality. Configure the RB_EMMC_TOCNT_MASK bit in the R8_EMMC_TIMEOUT register to set the command timeout and data timeout durations. After successful command transmission, configure the R32_EMMC_BLOCK_CFG register to set the single data block length during transfer and the total number of data blocks to be read, thereby initiating data transfer. Configure the R32_EMMC_TRAN_MODE register to set the DMA transfer direction to "Controller to SD".
- 2) Send commands to the SDIO slave interface device. These commands require the host interface to transmit data to the slave device via the data cable, such as CMD24 and CMD25 in the SD protocol.
- 3) The controller interface writes data to the data line: After the slave interface device responds to the host command, it generates a response packet and releases the data line state (non-low level). At this point, the host can drive the data line to output a data block. Writing to the DMA register R32_EMMC_DMA_BEG1 or performing a write operation to the R32_EMMC_WRITE_CONT register both trigger the host interface to drive the data line. The difference lies in the behavior when manipulating register R32_EMMC_DMA_BEG1: the hardware loads the current value of the R32_EMMC_DMA_BEG1 register internally and fetches data from this address to send to the controller. whereas writing to the R32_EMMC_WRITE_CONT register instructs the hardware to continue fetching SRAM data from the position currently occupied by the DMA internally. Therefore, to change the DMA address (making it non-contiguous with previous data), the R32_EMMC_DMA_BEG1 register must be rewritten to resume data line output. If the new address is contiguous with previously written data, the hardware automatically fetches data from the address and moves it. You can directly write to the R32_EMMC_WRITE_CONT register to instruct the controller to continue data line output. Each time data of the size specified by RB_EMMC_BKSIZE_MASK is output, it indicates the completion of one data block transfer. Simultaneously, the internal total number of transferred data blocks is decremented by one. Once all data blocks have been transferred, the internal DMA will cease further data output. During this process, upon completing the read of each data block, the hardware sets the RB_EMMC_IF_BKGAP flag. When all data blocks are read, the hardware sets the RB_EMMC_IF_TRANDONE flag. If a data timeout occurs during the controller's wait for reception or a CRC error is detected in the received data, the RB_EMMC_IF_DATTMO and RB_EMMC_IF_TRANERR flags are set respectively. Application code can configure the corresponding interrupt control bits in the interrupt enable register R16_EMMC_INT_EN. When the corresponding interrupt flag is set, it triggers the EMMC interrupt service routine.

Note 1: When the controller interface writes data blocks to external devices, it must wait for command transmission and acknowledgment completion before setting the DMA address. This is because setting the DMA address notifies the hardware to drive the data lines; if the command has not yet finished transmitting at this point, the device will not accept the data.

Note 2: During a transmission process (including data read and write) for a given command execution, if a CRC error (RB_EMMC_IF_TRANERR) occurs, the hardware will not count it as a successful transmission and will not decrement the total transmission block count. It will proceed to transmit the next block. However, DMA will still place the data block containing this CRC error into the corresponding SRAM address area. Application code must handle this situation appropriately.

33.2.5 DMA Dual-buffer Function

The system provides DMA dual-buffering functionality, featuring two DMA address registers: R32_EMMC_DMA_BEG1 and R32_EMMC_DMA_BEG2, along with the data block length threshold RB_EMMC_DMATN_CNT for DMA switching. When the data block length threshold starting from the R32_EMMC_DMA_BEG1 register address completes transmission, the hardware switches to the address range set by R32_EMMC_DMA_BEG2 for data transfer. Upon reaching the transfer threshold, it cycles back to the R32_EMMC_DMA_BEG1 range. This process repeats until all data blocks configured by RB_EMMC_BKNUM_MASK are fully transferred.

Each time non-zero data is written to the RB_EMMC_BKNUM_MASK field of the R32_EMMC_BLOCK_CFG register, a data transfer is initiated. Concurrently, the hardware directs DMA operations to the address range starting at R32_EMMC_DMA_BEG1. The flag-setting conditions during the transfer process remain consistent with the previously described behavior. Application code may optionally enable the corresponding interrupt flag to trigger the EMMC interrupt service routine.

33.2.6 Slave Mode

Slave mode operation

The hardware in slave mode only supports R1-type responses, so the R16_EMMC_CMD_SET register has no effect in slave mode.

1) Host queries slave data status:

The slave device pre-populates the ARGUMENT register with its desired data reception/transmission status. The host sends CMD13 to query the slave's status and determines whether to read or write to the slave based on the response.

Note: The use of CMD13 and the bit definitions in the slave's response are software-defined.

2) Host sends CMD18 to read multiple blocks:

The CMD18 command parameter from the host no longer specifies a sector address but instead indicates the number of blocks to be read in this operation. Upon receiving the command interrupt, the slave configures the DMA address, sets the R32_EMMC_TRAN_MODE, reads the block count from the command parameter register, and writes it into the R32_EMMC_BLOCK_CFG register.

2) Host sends CMD25 to write multiple blocks:

The CMD25 command parameter from the host no longer specifies a sector address but instead indicates the number of blocks to be written in this operation. Upon receiving the command interrupt, the slave configures the DMA address and sets R32_EMMC_TRAN_MODE. It then reads the block count from the command parameter register and writes it into the R32_EMMC_BLOCK_CFG register.

Slave error processing

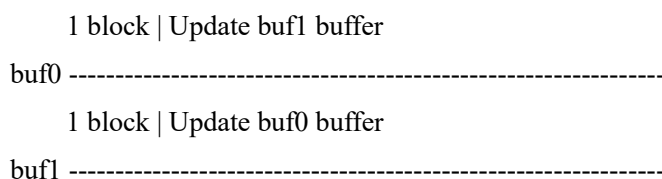
The slave responds to host commands with an R1-type acknowledgment, where the response parameter is the value of the slave's ARGUMENT register. If the slave fails to acknowledge the host due to a CRC error during command reception, the host should retry the current command.

If the slave experiences an internal FIFO overflow during transmission or reception, the hardware automatically forces the current data block to be CRC-erroneous, signaling transmission issues to the host. Alternatively, software can set `RB_SLV_FORCE_ERR` to force transmission of the erroneous CRC block.

The slave hardware parses a special command, CMD63, to simplify the slave software's handling of transmission errors. If a CRC error occurs during host transmission or reception, the host immediately sends CMD63. Upon receiving CMD63, the slave software must reset the data state machine. The hardware automatically sets the DMA address to the starting address of the previous erroneous packet. Once master-slave negotiation resumes transmission, for read operations, the slave software only needs to set the block count in `R32_EMMC_BLOCK_CFG`. For write operations, the software configures the block count in `R32_EMMC_BLOCK_CFG` and writes to the `RESPONSE3` register to complete retransmission and resume data transfer.

Dual buffer mode

If the slave operates in dual-buffered mode, and an error occurs while the master reads the last block of data from `buf0`, the master will send CMD63 during the transmission of the first block of `buf1` to indicate a CRC error in the previous packet. Therefore, the slave should only update the data in `buf0` after the first block of `buf1` has been successfully transmitted. The newly added interrupt bit `RB_SIF_SLV_BUF_RELEASE` is generated when the first sector of the new buffer is successfully transmitted.



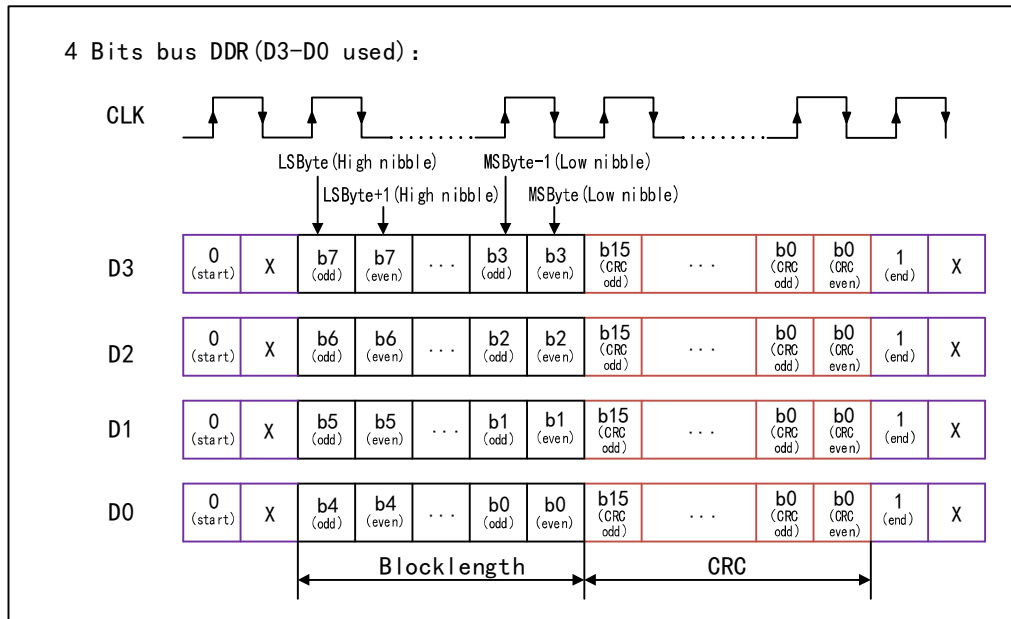
DDR mode

Support DDR mode under 4-wire and 8-wire.

1) 4-wire DDR mode:

That is, through the rising edge and falling edge of the clock, data is output and sampled to improve the data transmission rate. When applied, 4-wire mode register and DDR mode register should be enabled.

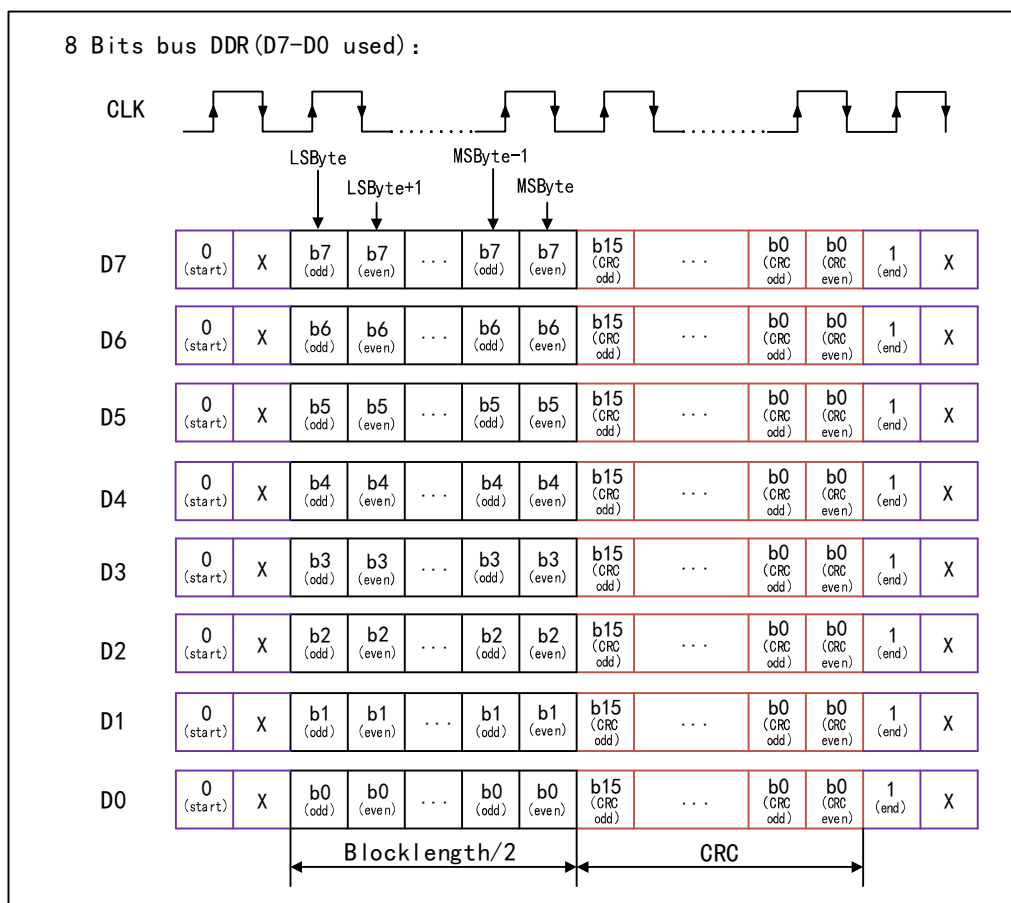
Figure 33-1 4-wire DDR mode



2) 8-wire DDR mode:

That is, through the rising edge and falling edge of the clock, data is output and sampled to improve the data transmission rate. When applied, 8-wire mode register and DDR mode register should be enabled.

Figure 33-2 8-wire DDR mode



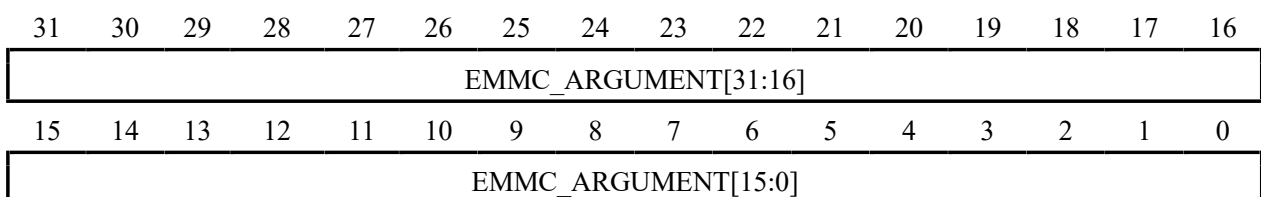
33.3 Register Description

Table 33-1 EMMC-related registers

Name	Address	Description	Reset value
R32_EMMC_ARGUMENT	0x40024000	Command parameter register	0x00000000
R16_EMMC_CMD_SET	0x40024004	Command setting register	0x0000
R32_EMMC_RESPONSE0	0x40024008	Response parameter register 0	0x00000000
R32_EMMC_RESPONSE1	0x4002400C	Response parameter register 1	0x00000000
R32_EMMC_RESPONSE2	0x40024010	Response parameter register 2	0x00000000
R32_EMMC_RESPONSE3	0x40024014	Response parameter register 3	0x00000000
R32_EMMC_WRITE_CONT	0x40024014	Continue writing start register.	0x00000000
R16_EMMC_CONTROL	0x40024018	Control register	0x0015
R8_EMMC_TIMEOUT	0x4002401C	Timeout count register	0x0C
R32_EMMC_STATUS	0x40024020	Status register	0x00000000
R16_EMMC_INT_FG	0x40024024	Interrupt flag register	0x0000
R16_EMMC_INT_EN	0x40024028	Interrupt enable register	0x0000
R32_EMMC_DMA_BEG1	0x4002402C	DMA start address 1 register	0x0000XXXX
R32_EMMC_BLOCK_CFG	0x40024030	Transmission block configuration register	0x00000000
R32_EMMC_TRAN_MODE	0x40024034	Transmission mode register	0x00001040
R16_EMMC_CLK_DIV	0x40024038	Clock configuration register	0x0213
R32_EMMC_DMA_BEG2	0x4002403C	DMA start address 2 register	0x00000000
R32_EMMC_TUNE_DAT0	0x40024040	Data output delay register	0x00000000
R32_EMMC_TUNE_DAT1	0x40024044	Data input delay register	0x00000000
R32_EMMC_TUNE_CLK_CMD	0x40024048	Clock and command delay register	0x00000000

33.3.1 Command Parameter Register (R32_EMMC_ARGUMENT)

Offset address: 0x00



Bit	Name	Access	Description	Reset Value
[31:0]	EMMC_ARGUMENT	RW	Host mode: 32-bit command parameter register for SD/eMMC. Slave mode: 32-bit response parameter register for slave acknowledgment.	0

33.3.2 Command Setting Register (R16_EMMC_CMD_SET)

Offset address: 0x04

Bit	Name	Access	Description	Reset Value
[15:12]	Reserved	RO	Reserved	0

11	RB_EMMC_CKIDX	RW	Command index for check response: 1: Required; 0: Not required.	0
10	RB_EMMC_CKCRC	RW	CRC for response verification: 1: Required; 0: Not required.	0
[9:8]	RB_EMMC_RPTY_MASK	RW	Expected Response Type: 00: No response; 01: Response length is 136 bits; 10: Response length is 48 bits; 11: Response length is 48 bits and is an R1b type response.	0
[7:6]	Reserved	RO	Reserved	0
[5:0]	RB_EMMC_CMDIDX_MASK	RW	The index number of the currently sent command.	0

33.3.3 Response Parameter Register (R32_EMMC_RESPONSE)

Offset address: 0x08, 0x0C, 0x10, 0x14, 0x14

Bit	Name	Access	Description	Reset Value
[31:0]	R32_EMMC_RESPONSE0	RO	Response parameter register 0.	0
[63:32]	R32_EMMC_RESPONSE1	RO	Response parameter register 1.	0
[95:64]	R32_EMMC_RESPONSE2	RO	Host Mode: Response parameter register 2; Slave Mode: Receive command parameter register.	0
[127:96]	R32_EMMC_RESPONSE3	RO	Host Mode: Response parameter register 3; Slave Mode: Receive command parameter register.	0
[127:96]	R32_EMMC_WRITE_CONT	WO	Multiplex the EMMC_RESPONSE3 register, which is used to start the write operation in the multi-block write process.	0

Note: When the response length is 136 bits, the valid data is 128 bits; when the response length is 48 bits, the valid data length is 32 bits. The R32_EMMC_RESPONSE3 register stores the valid data parameters of the response.

The R32_EMMC_RESPONSE3 register is multiplexed. This multiplexing occurs as follows: When using the CMD25 command to write multiple blocks of data to the card consecutively, writing to this register upon completion of a block transfer causes the controller to automatically offset the address by the block length and continue writing data. If the DMA address needs to be changed, writing the DMA address register will initiate the write operation, eliminating the need to use register writes for initiation.

33.3.4 Control Register (R16_EMMC_CONTROL)

Offset address: 0x18

Bit	Name	Access	Description	Reset Value
[15:10]	Reserved	RO	Reserved	0
9	RB_SLV_FORCE_ERR	RW	Software-forced data block CRC error in machine mode: 1: Software-forced data block CRC is an invalid	0

			value; 0: Use the hardware-correct CRC value.	
8	RB_SLV_MODE	RW	Slave mode enable bit: 1: Slave mode, SDCK input; 0: Master mode, SDCK output.	0
[7:6]	Reserved	RO	Reserved	0
5	RB_EMMC_NEGSM P	RW	CMD and data signal line sampling mode selection Bit: 1: Falling edge sampling; 0: Rising edge sampling.	0
4	RB_EMMC_RST_LG C	RW	Module internal logic transmit/receive reset: 1: Execute reset; 0: Normal operation;	1
3	RB_EMMC_DMAEN	RW	Controller DMA enable: 1: Enable DMA; 0: Disable DMA.	0
2	RB_EMMC_ALL_CL R	RW	SD controller logic reset: 1: Reset SD controller logic; requires software reset; 0: Normal operation.	1
[1:0]	RB_EMMC_LW_MA SK	RW	Valid data width for transmission/reception: 00: Transceiver uses only D[0], single data line; 01: Transceiver uses D[3:0], 4 data lines; 10: Transceiver uses D[7:0], 8 data lines.	01b

33.3.5 Timeout Control Register (R8_EMMC_TIMEOUT)

Offset address: 0x1C

Bit	Name	Access	Description	Reset Value
[7:4]	Reserved	RO	Reserved	0
[3:0]	RB_EMMC_TOCNT_ MASK	RW	Response/Data Timeout Configuration. Non-zero: Sets the timeout duration, valid values 1-15; Calculation: Module clock cycle * 4194304 * RB_EMMC_TOCNT_MASK. Example: For an SDCLK cycle of 10ns and a value of 12 entered, the timeout duration is 10ns * (4194304) * (12) = 503ms. 0: Not available.	0xC

Note 1: The aforementioned data timeouts encompass the following four scenarios:

- 1) D[0] busy timeout following the R1b response;
- 2) D[0] busy timeout after the CRC status during data block write;
- 3) CRC status wait timeout during data block write;
- 4) Start bit wait timeout during data block read.

Note 2: Command responses also support a timeout mechanism. If a response times out, it is indicated by the R16_EMMC_INT_FG interrupt in the interrupt register. Command timeouts use the maximum timeout value

specified in the protocol: 64Tsdclk.

33.3.6 Status Indication Register (R32_EMMC_STATUS)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														RB_E MMC _DAT 0STA	RB_E MMC _CM DSTA
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MASK_BLOCK_NUM															

Bit	Name	Access	Description	Reset Value
[32:18]	Reserved	RO	Reserved	0
17	RB_EMMC_DAT0STA	RO	Current level status of D0 data line: 1: High level; 0: Low level.	0
16	RB_EMMC_CMDSTA	RO	Current level status of CMD signal line: 1: High level; 0: Low level.	0
[15:0]	MASK_BLOCK_NUM	RO	In the current multi-block transfer operation, the number of blocks that have been successfully transferred.	0

33.3.7 Interrupt Flag Register (R16_EMMC_INT_FG)

Offset address: 0x24

Bit	Name	Access	Description	Reset Value
[15:11]	Reserved	RO	Reserved	0
10	RB_SIF_SLV_BUF_RELEASE	RW1Z	In dual-buffer mode, the BUF release flag is cleared by writing 1: 1: Software can update (write/read) the corresponding buffer data; 0: The corresponding buffer data is still in use.	0
9	RB_EMMC_IF_SDIOINT	RW1Z	SDIO Card Interrupt Flag, write 1 to clear: 1: SDIO card has generated a card interrupt; 0: No event.	0
8	RB_EMMC_IF_FIFO_OV	RW1Z	FIFO overflow flag, write 1 to clear: 1: FIFO overflow triggered; 0: No event.	0
7	RB_EMMC_IF_BKGAP	RW1Z	Single-block transfer complete flag, Write 1 to clear: 1: Triggered by single-block transmission/reception completion; 0: No event occurred.	0
6	RB_EMMC_IF_TRAN_DONE	RW1Z	All-in-transmission-complete flag, Write 1 to clear: 1: Triggered when all requested blocks are	0

			transmitted; 0: No event.	
5	RB_EMMC_IF_TRANERR	RW1Z	Data transmission CRC error flag, Write 1 to clear: 1: CRC error triggered; 0: No event.	0
4	RB_EMMC_IF_DAT_TMO	RW1Z	Data timeout flag, write 1 to clear: 1: Data timeout triggered; 0: No event.	0
3	RB_EMMC_IF_CMD_DONE	RW1Z	Command completion flag/Slave receives command completion flag, Write 1 to clear: 1: Host sends command and receives acknowledgment; slave completes command reception; 0: No event.	0
2	RB_EMMC_IF_REIDX_ER	RW1Z	Response index number check error flag, Write 1 to clear: 1: Response index number check error triggered; 0: No event.	0
1	RB_EMMC_IF_RECRC_WR	RW1Z	CRC error acknowledgment flag, Write 1 to clear: 1: Acknowledges CRC error trigger; 0: No event.	0
0	RB_EMMC_IF_RE_TMOU	RW1Z	Receive acknowledgment timeout flag, Write 1 to clear: 1: Acknowledgment timeout triggered; 0: No event.	0

33.3.8 Interrupt Enable Register (R16_EMMC_INT_EN)

Offset address: 0x28

Bit	Name	Access	Description	Reset Value
[15:11]	Reserved	RO	Reserved	0
10	RB_SIE_SLV_BUF_RELEASE	RW	In dual-buffer mode, BUF release interrupt enable: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
9	RB_EMMC_IE_SDIO_INT	RW	SDIO card interrupt enable: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
8	RB_EMMC_IE_FIFO_OV	RW	FIFO overflow interrupt enable: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
7	RB_EMMC_IE_BKGAP	RW	Single block transmission completion interrupt enable: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
6	RB_EMMC_IE_TRAN	RW	All block transmission completion interrupt	0

	DONE		enabled: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	
5	RB_EMMC_IE_TRANERR	RW	Block transmission CRC error interrupt: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
4	RB_EMMC_IE_DAT_TMO	RW	Data timeout interrupt: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
3	RB_EMMC_IE_CMD_DONE	RW	Command completion interrupt: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
2	RB_EMMC_IE_REIDX_ER	RW	Answer index check error interrupt: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
1	RB_EMMC_IE_RECR_C_WR	RW	Reply to CRC check error interrupt: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0
0	RB_EMMC_IE_RE_TMO	RW	Command response timeout interrupt: 1: Enable the corresponding interrupt; 0: Disable the corresponding interrupt.	0

33.3.9 Data Block DMA Start Address Register (R32_EMMC_DMA_BEG1)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RB_EMMC_DMAAD1_MASK[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RB_EMMC_DMAAD1_MASK[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RB_EMMC_DMAAD1_MASK[31:0]	RW	Start address of reading and writing data buffer, with the lower 4 bits fixed at 0(16 bytes aligned). After this register is set, reception or transmission is officially started.	0

Note: This register stores the starting address in SRAM for data read from the SD card during read operations. During write operations to the SD card, it stores the starting address in SRAM for data to be written. Addressing occurs within the SRAM region.

When performing consecutive multi-block read/write operations to SD, after a single block transfer completes (RB_EMMC_IF_BKGAP), the user may write the DMA_BEG1 register to change the DMA address as needed. Do not change the DMA address during a transfer, as this may cause data counting errors. For consecutive multi-block writes, after a single block transfer completes, initiate the next write operation by writing to either the R32_EMMC_WRITE_CONT or DMA_BEG1 register. This is not required for multi-block reads.

33.3.10 Transmission Block Configuration Register (R32_EMMC_BLOCK_CFG)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				RB_EMMC_BKSIZE_MASK											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RB_EMMC_BKNUM_MASK															

Bit	Name	Access	Description	Reset Value
[31:28]	Reserved	RO	Reserved	0
[27:16]	RB_EMMC_BKSIZE_MASK	RW	Single block transmission size (1-2048-byte)	0
[15:0]	RB_EMMC_BKNUM_MASK	RW	The block count (1 ~ 65535 blocks) to be transmitted in this DMA is automatically cleared internally. If the number of blocks is not zero, a receiving or transmitting signal will be generated.	0

33.3.11 Transmission Mode Register (R32_EMMC_TRAN_MODE)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										RB_SW		RB_CARE_NEG	RB_DDR_MODE	RB_EMMC_DULEDM_A_EN	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	RB_EMMC_DMATN_CNT					Reserved					RB_EMMC_AU_TOGAP_STOP	Reserved	RB_EMMC_MODE_BOOT	RB_EMMC_GAP_STOP	RB_EMMC_DMA_DIR

Bit	Name	Access	Description	Reset Value
[31:21]	Reserved	RO	Reserved	0
[20:19]	RB_SW	RW	In DDR mode, clock switching: 00: Clock does not switch; 01: Automatic switching; 10: Manual switching; 11: Reserved.	0
18	RB_CARE_NEG	RW	In DDR mode, data check on the falling edge of the clock: 1: Check; 0: Do not check.	0
17	RB_DDR_MODE	RW	Enable DDR mode: 1: Enable; 0: Disable.	0
16	RB_EMMC_DULED	RW	Enable DMA double buffering:	0

	MA_EN		1: Enable; 0: Disable.	
15	Reserved	RO	Reserved	0
[14:8]	RB_EMMC_DMATN_CNT	RW	In double buffer mode, set the block count value of buffer switching.	0x10
[7:5]	Reserved	RW	Reserved, Bit5 must write 0.	010b
4	RB_EMMC_AUTOGAPSTOP	RW	Hardware automatic data block stop clock enable: 1: Enabled; 0: Disabled.	0
3	Reserved	RW	Must write 0.	0
2	RB_EMMC_MODE_BOOT	RW	Set eMMC card transfer mode: 1: Boot mode; 0: Normal mode. Note: For eMMC cards only.	0
1	RB_EMMC_GAP_STOP	RW	Block completion clock stop mode. When reading block data, setting this bit to 1 enables hardware automatic clock output shutdown after sampling one data block. To initiate a new block transfer, software must clear this bit to 0. After a delay of one SCLK cycle, resetting it to 1 resumes counting. If the RB_EMMC_AUTOGAPSTOP function is enabled, this bit is set to 1 by hardware. Software only needs to clear it to 0.	0
0	RB_EMMC_DMA_DIRECTION	RW	Direction of DMA Transfer: 1: Controller to SD; 0: SD to Controller.	0

33.3.12 Clock Configuration Register (R16_EMMC_CLK_DIV)

Offset address: 0x38

Bit	Name	Access	Description	Reset Value
[15:11]	Reserved	RO	Reserved	0
10	RB_EMMC_PHASEINVERT	RW	SDCK clock output phase inversion.	0
9	RB_EMMC_CLKMode	RW	Clock Frequency Mode Selection Bit: 1: High-speed mode, 25 MHz to 200 MHz; 0: Low-speed mode, below 5 MHz.	1
8	RB_EMMC_CLKOE	RW	SD Physical clock signal line output control bit: 1: Enable, output communication clock; 0: Disable.	0
[7:5]	Reserved	RO	Reserved	0
[4:0]	RB_EMMC_DIV_MASK[4:0]	RW	SD Controller Clock (SDCLK) Division Factor: When RB_EMMC_CLKMode = 1, then SDCLK = SYSPLL_SEL / RB_EMMC_DIV_MASK; When RB_EMMC_CLKMode = 0, then SDCLK = SYSPLL_SEL /	10011b

			RB_EMMC_DIV_MASK / 64.	
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Note: RB_EMMC_DIV_MASK needs to be configured to a minimum value of 2 in slave mode, which does not affect the input clock value.

33.3.13 Data Block DMA Start Address Register (R32_EMMC_DMA_BEG2)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RB_EMMC_DMAAD2_MASK[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RB_EMMC_DMAAD2_MASK[15:0]															

Bit	Name	Access	Description	Reset Value
[31:0]	RB_EMMC_DMAAD2_MASK[31:0]	RW	Start address of reading and writing data buffer, with the lower 4 bits fixed at 0(16 bytes aligned).	0

Note: When reading data from SD, this register holds the starting address of the read data in SRAM. When writing data to SD card, the starting address of the data to be written in SRAM is saved. Addressed in SRAM area.

33.3.14 Data Output Delay Register (R32_EMMC_TUNE_DATO)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RB_TUNNE_DAT7_O[3:0]				RB_TUNNE_DAT6_O[3:0]				RB_TUNNE_DAT5_O[3:0]				RB_TUNNE_DAT4_O[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RB_TUNNE_DAT3_O[3:0]				RB_TUNNE_DAT2_O[3:0]				RB_TUNNE_DAT1_O[3:0]				RB_TUNNE_DAT0_O[3:0]			

Bit	Name	Access	Description	Reset Value
[31:28]	RB_TUNNE_DAT7_O[3:0]	RW	Delay of D7 output.	0
[27:24]	RB_TUNNE_DAT6_O[3:0]	RW	Delay of D6 output.	0
[23:20]	RB_TUNNE_DAT5_O[3:0]	RW	Delay of D5 output.	0
[19:16]	RB_TUNNE_DAT4_O[3:0]	RW	Delay of D4 output.	0
[15:12]	RB_TUNNE_DAT3_O[3:0]	RW	Delay of D3 output.	0
[11:8]	RB_TUNNE_DAT2_O[3:0]	RW	Delay of D2 output.	0
[7:4]	RB_TUNNE_DAT1_O[3:0]	RW	Delay of D1 output.	0

[3:0]	RB_TUNNE_DAT0_O [3:0]	RW	Delay of D0 output.	0
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Note: Output delay = 0.1ns * RB_TUNNE_DATx_O.

33.3.15 Data Input Delay Register (R32_EMMC_TUNE_DATI)

Offset address: 0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RB_TUNNE_DAT7_I[3:0]]				RB_TUNNE_DAT6_I[3:0]]				RB_TUNNE_DAT5_I[3:0]]				RB_TUNNE_DAT4_I[3:0]]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RB_TUNNE_DAT3_I[3:0]]				RB_TUNNE_DAT2_I[3:0]]				RB_TUNNE_DAT1_I[3:0]]				RB_TUNNE_DAT0_I[3:0]]			

Bit	Name	Access	Description	Reset Value
[31:28]	RB_TUNNE_DAT7_I [3:0]	RW	Delay of D7 input.	0
[27:24]	RB_TUNNE_DAT6_I [3:0]	RW	Delay of D6 input.	0
[23:20]	RB_TUNNE_DAT5_I [3:0]	RW	Delay of D5 input.	0
[19:16]	RB_TUNNE_DAT4_I [3:0]	RW	Delay of D4 input.	0
[15:12]	RB_TUNNE_DAT3_I [3:0]	RW	Delay of D3 input.	0
[11:8]	RB_TUNNE_DAT2_I [3:0]	RW	Delay of D2 input.	0
[7:4]	RB_TUNNE_DAT1_I [3:0]	RW	Delay of D1 input t.	0
[3:0]	RB_TUNNE_DAT0_I [3:0]	RW	Delay of D0 input.	0

Note: Input delay = 0.1ns * RB_TUNNE_DATx_O.

33.3.16 Clock and Command Delay Register (R32_EMMC_TUNE_CLK_CMD)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								RB_TUNNE_CMD_I[3:0]				RB_TUNNE_CMD_O[3:0]]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								RB_TUNNE_CLK_I[3:0]				RB_TUNNE_CLK_O[3:0]			

Bit	Name	Access	Description	Reset Value
[31:24]	Reserved	RO	Reserved	0

[23:20]	RB_TUNNE_CMD_I [3:0]	RW	CMD input delay: Delay value = $0.2\text{ns} \times \text{RB_TUNNE_CMD_I}$.	0
[19:16]	RB_TUNNE_CMD_O [3:0]	RW	CMD output delay: Delay value = $0.2\text{ns} * \text{RB_TUNNE_CMD_O}$.	0
[15:8]	Reserved	RO	Reserved	0
[7:4]	RB_TUNNE_CLK_I [3:0]	RW	CLK input delay: Delay value = $0.2\text{ns} * \text{RB_TUNNE_CLK_I}$. <i>Note: The delay applied to this clock input only affects the strobe signal of the eMMC card. When using cards without this signal, this field is invalid.</i>	0
[3:0]	RB_TUNNE_CLK_O [3:0]	RW	CLK output delay: Delay value = $0.2\text{ns} * \text{RB_TUNNE_CLK_O}$.	0

Chapter 34 Programmable Protocol I/O Microcontroller (PIOC)

A programmable protocol I/O microcontroller is embedded in the chip. The microcontroller is based on the RISC core with a single clock cycle, runs at the main frequency of the system, has a program ROM with 2K instructions, 49 SFR registers and PWM timers/counters, and supports the protocol control of two I/O pins.

34.1 Main Features

- RISC core, optimized single cycle bit manipulation instruction set, fully static design.
- Multiplex 4K bytes of system SRAM as 2K capacity program ROM, supports program pause and dynamic loading.
- Provide 33 bytes of 1 register each in bidirectional and unidirectional mode, providing 6 levels of independent stacking.
- Support 2 general-purpose bi-directional I/O protocol controls and input level change detection.
- Support 1- and 2-wire interfaces with multiple protocol specifications by dynamically loading different protocol programs.
- Support 1-wire ARGB chip 1024-stage series connection

34.2 Instruction, Program Space and Stack

34.2.1 Instruction Set

eMCU adopts RISC8B core with reduced instruction set, with data width of 8 bits and instruction width of 16 bits. There are 66 instructions, all of which are single clock cycles, except that the jump instruction is double clock cycle and the program space reading and writing instruction is double cycle. For more information, please refer to RISC8B core instruction set and assembly tool document CHRISC8B.PDF.

34.2.2 Program ROM

EMCU's program ROM comes from 4-kbyte system SRAM multiplexing. When SB_MST_CLK_GATE is 0, it is dedicated to the system SRAM on the host side, and when SB_MST_CLK_GATE is 1, it is dedicated to eMCU's program ROM.

The program space of eMCU is 2048 words, and the address of the program space ranges from 0x0000 to 0x07FF. The instruction at address 0 in the program ROM has a special purpose and will not be executed.

eMCU chip supports reading program space. The steps are as follows: first, write the lower 8 bits of the target address into SFR_INDIR_ADDR, and the upper 3 bits of the target address into the A register, and then execute the RDCODE instruction, so that the 16-bit data in the program space can be read out at one time, which is divided into high and low bytes, with the low byte returned to the A register and the high byte returned to the SFR_INDIR_ADDR. This operation can be used for double-byte table lookup.

eMCU chip does not support writing to program space.

34.2.3 Stack

The eMCU features a stack memory with a depth of 6 levels and an 11-bit data width. It is used to store the program return address during subroutine calls and can also be employed with the PUSHAS instruction to save variables, status data, and other information from the running program onto the stack.

34.2.4 Sleep and Awakening

When SB_MST_CLK_GATE is 0, the eMCU program will suspend execution. After executing the SLEEP or SLEEPX instruction, the eMCU enters sleep mode. The eMCU is designed as fully static, where program suspension and sleep mode have equivalent effects.

After entering sleep, the eMCU supports two wake-up methods: - Wake-up via level change on a pin with level change detection enabled. - Wake-up via the SB_MST_CLK_GATE signal conversion from 0 to 1.

34.2.5 Event Wait Instruction

The WAITB instruction supports waiting for 8 types of events. When the event is invalid, WAITB continues to wait indefinitely, with the program counter (PC) remaining unchanged. Upon detecting the target event, the wait is terminated, and the next instruction is executed. Timer timeout reset can be optionally enabled during the wait.

The instruction parameter WB_DATA_SW_MR_0 indicates waiting until SB_DATA_SW_MR becomes 0 to exit; if it is already 0, exit immediately.

The instruction parameter WB_BIT_CYC_TAIL_1 indicates waiting until SB_BIT_CYC_TAIL becomes 1 to exit; if it is already 1, it exits immediately.

The instruction parameter WB_PORT_I0_FALL indicates waiting until SB_PORT_IN0 detects a falling edge to exit; if a falling edge was previously detected and recorded, it also exits immediately, clearing the falling edge detection record upon exit.

The instruction parameter WB_PORT_I0_RISE indicates waiting until SB_PORT_IN0 detects a rising edge to exit. If a rising edge has already been detected as recorded, it exits directly and clears the rising edge detection record upon exit.

The instruction parameter WB_DATA_MW_SR_1 indicates waiting until SB_DATA_MW_SR is 1 to exit. If it is already 1, it exits directly.

The instruction parameter WB_PORT_XOR1_1 indicates waiting until SB_PORT_XOR1 is 1 before exiting. If it is already 1, exit directly.

The instruction parameter WB_PORT_XOR0_0 indicates waiting until SB_PORT_XOR0 is 0 before exiting. If it is already 0, exit directly.

The instruction parameter WB_PORT_XOR0_1 indicates waiting until SB_PORT_XOR0 is 1 before exiting; if it is already 1, exit immediately.

34.2.6 Bit Transfer Instruction

Bit transfer instructions support two bit operation registers: BP1F and BG1F correspond to register SFR_INDIR_ADDR, while BP2F and BG2F correspond to register SFR_DATA_EXCH.

Bit transfer instructions BP1F and BP2F support four independent bit outputs: Parameter 0# BIO_FLAG_C corresponds to SB_FLAG_C bit, parameter 1# BO_BIT_TX_O0 corresponds to SB_BIT_TX_O0, parameter 2# BO_PORT_OUT0 corresponds to SB_PORT_OUT0, and parameter 3# BO_PORT_OUT1 corresponds to SB_PORT_OUT1.

Bit transfer instructions BG1F and BG2F support four independent bit inputs: Parameter 0# (BIO_FLAG_C) corresponds to bit SB_FLAG_C, parameter 1# (BI_BIT_RX_I0) corresponds to bit SB_BIT_RX_I0, parameter 2# (BI_PORT_IN0) corresponds to bit SB_PORT_IN0, and parameter 3# (BI_PORT_IN1) corresponds to bit SB_PORT_IN1.

The bit transfer instruction BCTC supports four independent bit inputs: Parameter 0# BI_C_XOR_IN0 updates bit SB_FLAG_C to the XOR result of SB_FLAG_C and SB_PORT_IN0; Parameter 1# BI_BIT_RX_I0 corresponds to SB_BIT_RX_I0; Parameter 2# BI_PORT_IN0 corresponds to SB_PORT_IN0; Parameter 3# BI_PORT_IN1

corresponds to SB_PORT_IN1.

34.3 Data Space and Registers and Addressing

The data space of the eMCU includes 49 dedicated special function registers (SFRs) addressed via 8-bit addresses within the range 0x00 to 0x3F. All registers have a data width of 1 byte, equivalent to 8 bits.

Some SFRs are accessible for read/write or read-only operations from the host side, which supports 8-bit, 16-bit, or 32-bit width read/write operations. The host side has write priority. When both the eMCU and host attempt to write to the same register simultaneously, the eMCU's write operation is automatically discarded.

The eMCU's addressing modes include: immediate address, immediate fast address, standard direct address, extended direct address, indirect address, and bit address. The latter four are used for register addressing.

The indirect addressing range is 0x000-0x0FF, covering all registers of the eMCU. The eMCU provides two sets of indirect address registers, each consisting of an address register and a data read/write port. First, write the target register's address into the address register, then read or write the target register via the data read/write port.

The address range for extended direct addressing is 0x000-0x1FF, where the instruction directly provides a 9-bit register address. This is only applicable to the following two direct register read/write instructions: MOV register,A or F, and MOVA register.

The addressing range for standard direct addressing is 0x000-0x0FF. The instruction directly provides an 8-bit register address, applicable to all direct address instructions except those covered by extended direct addressing, such as CLR register, ADD register, BS register, bit, etc. When addressing a larger range is required, indirect addressing should be employed.

Immediate number fast addressing enables rapid writing of operands from instruction codes to target registers without requiring an A register intermediary. MOVA1F is used for fast setting of register #1 (SFR_PORT_DIR), while MOVA2F is used for fast setting of register #2 (SFR_PORT_IO). MOVA1P and MOVA2P employ indirect addressing for rapid setting of all registers, equivalent to first performing immediate addressing followed by indirect addressing within a single cycle.

Bit addressing utilizes a 3-bit address provided directly by the instruction. Register addressing can employ either standard direct addressing or indirect addressing, covering address ranges 0x000 to 0x0FF respectively, thereby enabling access to any bit within any register.

34.4 Special Function Register SFR

Some SFR or some bits are not really realized, and they are reserved bits. Reading is 0, and writing must keep the original value or write 0.

Name	Access address	Description	Host-side reading and writing	Reset value
R8_SFR_INDIR_PORT	0x40025C00	Indirect addressed data read-write port	UUUUUUUU	XXXXXXXX
R8_SFR_INDIR_PORT2	0x40025C01	Data read-write port of indirect addressing	UUUUUUUU	XXXXXXXX

		2		
R8_SFR_PRG_COUNT	0x40025C02	Low byte of program counter PC	UUUUUUUU	00000000
R8_SFR_STATUS_REG	0x40025C03	State register	UUUUSUSU	0000-0-0
R8_SFR_INDIR_ADDR	0x40025C04	Indirectly addressed address register	RRRRRRRR	XXXXXXXX
R8_SFR_TMR0_COUNT	0x40025C05	Count register of timer 0	RRSSRRRR	00000000
R8_SFR_TIMER_CTRL	0x40025C06	Control register of timer	RRRRRRRR	00000000
R8_SFR_TMR0_INIT	0x40025C07	Initial value register of timer 0	RRRRRRRR	00000000
R8_SFR_BIT_CYCLE	0x40025C08	Coded bit period register	WWWWWWWW	00000000
R8_SFR_INDIR_ADDR2	0x40025C09	Address register of indirect addressing 2	RRRRRRRR	00000000
R8_SFR_PORT_DIR	0x40025C0A	Port direction setting register	RRRRRRRR	00000000
R8_SFR_PORT_IO	0x40025C0B	Port input output register	RRRRRRRR	XXXXXX00
R8_SFR_BIT_CONFIG	0x40025C0C	Coded bit configuration register	WWWRRRRR	00010000
R8_SFR_SYS_CFG	0x40025C1C	System configuration register	RRRWWWWW	00000000
R8_SFR_CTRL_RD	0x40025C1D	EMCU read-write and host read-only register	RRRRRRRR	00000000
R8_SFR_CTRL_WR	0x40025C1E	Host reads and writes and eMCU read-only register	WWWWWWWW	00000000
R8_SFR_DATA_EXCH	0x40025C1F	Data exchange register	WWWWWWWW	00000000
R8_SFR_DATA_REG0	0x40025C20	Data register 0	WWWWWWWW	00000000
R8_SFR_DATA_REG1	0x40025C21	Data register 1	WWWWWWWW	00000000
R8_SFR_DATA_REG2	0x40025C22	Data register 2	WWWWWWWW	00000000
R8_SFR_DATA_REG3	0x40025C23	Data register 3	WWWWWWWW	00000000
R8_SFR_DATA_REG4	0x40025C24	Data register 4	WWWWWWWW	00000000
R8_SFR_DATA_REG5	0x40025C25	Data register 5	WWWWWWWW	00000000
R8_SFR_DATA_REG6	0x40025C26	Data register 6	WWWWWWWW	00000000
R8_SFR_DATA_REG7	0x40025C27	Data register 7	WWWWWWWW	00000000
R8_SFR_DATA_REG8	0x40025C28	Data register 8	WWWWWWWW	00000000
R8_SFR_DATA_REG9	0x40025C29	Data register 9	WWWWWWWW	00000000
R8_SFR_DATA_REG10	0x40025C2A	Data register 10	WWWWWWWW	00000000
R8_SFR_DATA_REG11	0x40025C2B	Data register 11	WWWWWWWW	00000000

R8_SFR_DATA_REG12	0x40025C2C	Data register 12	WWWWWWWW	00000000
R8_SFR_DATA_REG13	0x40025C2D	Data register 13	WWWWWWWW	00000000
R8_SFR_DATA_REG14	0x40025C2E	Data register 14	WWWWWWWW	00000000
R8_SFR_DATA_REG15	0x40025C2F	Data register 15	WWWWWWWW	00000000
R8_SFR_DATA_REG16	0x40025C30	Data register 16	WWWWWWWW	00000000
R8_SFR_DATA_REG17	0x40025C31	Data register 17	WWWWWWWW	00000000
R8_SFR_DATA_REG18	0x40025C32	Data register 18	WWWWWWWW	00000000
R8_SFR_DATA_REG19	0x40025C33	Data register 19	WWWWWWWW	00000000
R8_SFR_DATA_REG20	0x40025C34	Data register 20	WWWWWWWW	00000000
R8_SFR_DATA_REG21	0x40025C35	Data register 21	WWWWWWWW	00000000
R8_SFR_DATA_REG22	0x40025C36	Data register 22	WWWWWWWW	00000000
R8_SFR_DATA_REG23	0x40025C37	Data register 23	WWWWWWWW	00000000
R8_SFR_DATA_REG24	0x40025C38	Data register 24	WWWWWWWW	00000000
R8_SFR_DATA_REG25	0x40025C39	Data register 25	WWWWWWWW	00000000
R8_SFR_DATA_REG26	0x40025C3A	Data register 26	WWWWWWWW	00000000
R8_SFR_DATA_REG27	0x40025C3B	Data register 27	WWWWWWWW	00000000
R8_SFR_DATA_REG28	0x40025C3C	Data register 28	WWWWWWWW	00000000
R8_SFR_DATA_REG29	0x40025C3D	Data register 29	WWWWWWWW	00000000
R8_SFR_DATA_REG30	0x40025C3E	Data register 30	WWWWWWWW	00000000
R8_SFR_DATA_REG31	0x40025C3F	Data register 31	WWWWWWWW	00000000

The default values after reset are all represented by binary numbers, and the bit values are described as follows:

0: Always 0 after reset;

1: Always 1 after reset;

X: Reset does not affect data; initial data value is undefined;

-: Always clears to 0 after power-on reset; system reset or host forced reset does not affect data.

The read/write specifications for the host side are as follows:

W: Bits that can be read or written;

R: Read-only bits;

U: Invisible bit, neither readable nor writable;

S: Read-only switch bit. The host-side visible SFR_TIMER_CTRL differs from the eMCU side in that the SB_TMR0_ENABLE and SB_TMR0_OUT_EN bits are replaced by SB_GP_BIT_Y and SB_GP_BIT_X respectively.

34.4.1 Indirect Addressed Data Read-write Port (SFR_INDIR_PORT)

34.4.2 Address Register of Indirect Addressed (SFR_INDIR_ADDR)

34.4.3 Indirect Addressed 2 Data Read-write Port (SFR_INDIR_PORT2)

34.4.4 Address Register of Indirect Addressed 2 (SFR_INDIR_ADDR2)

These are 2 sets of indirect address registers, each consisting of an address register and a data read/write port. First, write the destination register's address into the address register. Then, read or write the destination register by

accessing the data read/write port.

The data read/write port SFR_INDIR_PORT for indirect addressing reads/writes the destination register specified by the indirectly addressed address register SFR_INDIR_ADDR. The data read/write port SFR_INDIR_PORT2 for indirect addressing reads/writes the destination register specified by the indirectly addressed address register SFR_INDIR_ADDR2.

The dedicated instruction MOVIP is used to load an 8-bit destination address into SFR_INDIR_ADDR. Additionally, SFR_INDIR_ADDR can be modified independently by other non-dedicated instructions.

The dedicated instruction MOVIA is used to load an 8-bit target address into SFR_INDIR_ADDR2, and SFR_INDIR_ADDR2 can also be independently modified by other non-dedicated instructions.

The address register SFR_INDIR_ADDR2 for Indirect Addressing Mode 2 features automatic incrementing. When executing the “MOV SFR_INDIR_PORT2,A” read instruction or the “MOVA SFR_INDIR_PORT2” write instruction to RAM, SFR_INDIR_ADDR2 will automatically increment by 1 upon instruction completion.

The dedicated instructions MOVA1P and MOVA2P are used to directly and rapidly write operands from the instruction code into target registers, equivalent to performing immediate addressing followed by indirect addressing within a single cycle.

34.4.5 Status Register (SFR_STATUS_REG)

The status register contains the result status of the ALU, general-purpose bit variables, and stack usage flags. After executing arithmetic or logical operations, the eMCU sets corresponding status bits based on the results, enabling the program to make judgments and perform further processing. General-purpose bit variables can be defined for specific purposes by the application.

The timer timeout reset monitors the timeout of WAITB wait instructions. After disabling SB_TMR0_ENABLE and enabling SB_EN_TOUT_RST, the timer counts during WAITB execution. When overflow occurs and SB_TMR0_CYCLE is 1, a timeout reset is triggered. The timer initial value is automatically reloaded each time the WAITB instruction exits the WAITB wait state upon condition fulfillment. Counting does not occur during execution of non-WAITB instructions when SB_TMR0_ENABLE is 0, meaning the timer timeout reset applies exclusively to the WAITB instruction.

Bit	Name	Description	Reset value
7	Reserved	Reserved	0
6	Reserved	Reserved	0
5	SB_EN_TOUT_RST	Timer timeout reset enable: 1: Enables the eMCU to be reset upon timer timeout overflow; 0: Disabled.	0
4	SB_STACK_USED	Current stack usage flags: 1: Stacked; 0: Not yet used or completely off the stack.	0
3	SB_GP_BIT_Y	The universal bit variable Y is defined and used by the application. It is initialized to 0 after power-on reset and is unaffected by system reset or host forced reset.	-

2	SB_FLAG_Z	ALU zero flag, is the result 00H: 1: Result is 0; 0: Result is not 0.	0
1	SB_GP_BIT_X	The universal bit variable X is defined and used by the application. It is initialized to 0 after power-on reset and is unaffected by system reset or host forced reset.	-
0	SB_FLAG_C	ALU carry flag: Indicates whether the result generated a carry or was caused by a shift: 1: The result generated a carry; 0: The result did not generate a carry.	0

34.4.6 Low Byte of Program Counter PC (SFR_PRG_COUNT)

The program counter (PC) of the eMCU is 11 bits wide. The lower 8 bits of the PC can be modified by reading or writing SFR_PRG_COUNT, but this does not affect the upper 3 bits. This allows for short jumps and lookup table operations within a 256-byte range.

34.4.7 Port Direction Setting Register (SFR_PORT_DIR)

The following table is a description of the port direction setting register SFR_PORT_DIR.

Bit	Name	Description	Reset value
7	SB_PORT_MOD3	Pin mode control, the host side defines the purpose.	0
6	SB_PORT_MOD2		0
5	SB_PORT_MOD1		0
4	SB_PORT_MOD0		0
3	SB_PORT_PU1	IO1 pin pull-up enable: 1: Enabled; 0: Disabled.	0
2	SB_PORT_PU0	IO0 pin pull-up enable: 1: Enabled; 0: Disabled.	0
1	SB_PORT_DIR1	IO1 pin direction control: 1: Output; 0: Input.	0
0	SB_PORT_DIR0	IO0 pin direction control: 1: Output; 0: Input.	0

34.4.8 Port Input Output Register (SFR_PORT_IO)

The port contains two bidirectional input/output pins, each independently configurable as input or output via its respective direction bit SB_PORT_DIR. When set as output, data from SB_PORT_OUT is output to the pin. The default direction after reset is input. SB_PORT_IN is used to acquire the current pin input level (when SB_PORT_IN_EDGE=0) or the pin input level from the previous cycle (when SB_PORT_IN_EDGE=1), with the latter being approximately half a clock cycle ahead of the former.

Both pins feature level change detection. When the input level differs from the data in the output data register

SB_PORT_OUT, the corresponding SB_PORT_XOR bit becomes 1.

The IO0 pin offers more available input instructions and methods, making it more suitable than IO1 for input or bidirectional data.

The following table is a description of the port input and output register SFR_PORT_IO.

Bit	Name	Description	Reset value
7	SB_PORT_IN_XOR	The XOR result of the input states at IO1 and IO0 pins: 1: IO1 and IO0 have different levels; 0: IO1 and IO0 have the same level.	X
6	SB_BIT_RX_IO	The data received by the IO0 pin is decoded into a bit-by-bit state.	X
5	SB_PORT_IN1	IO1 pin input state (refer to SB_PORT_IN_EDGE).	X
4	SB_PORT_IN0	IO0 pin input state (refer to SB_PORT_IN_EDGE).	X
3	SB_PORT_XOR1	The XOR result between the input state of IO1 pin and SB_PORT_OUT1: 1: IO1 level differs from SB_PORT_OUT1, used to detect level changes; 0: IO1 level is the same as SB_PORT_OUT1.	X
2	SB_PORT_XOR0	The XOR result between the IO0 pin input state and SB_PORT_OUT0: 1: IO0 level differs from SB_PORT_OUT0, used to detect level changes; 0: IO0 level is the same as SB_PORT_OUT0.	X
1	SB_PORT_OUT1	IO1 pin output data: 1: Output high; 0: Output low.	0
0	SB_PORT_OUT0	IO0 pin output data: 1: Output high; 0: Output low. The timer output signal takes precedence. When SB_TMR0_OUT_EN is set to 1, it outputs either SB_TMR0_CYCLE or its divided-by-two signal.	0

34.4.9 Count Register of Timer 0 (SFR_TMR0_COUNT)

34.4.10 Initial Value Register of Timer 0 (SFR_TMR0_INIT)

34.4.11 Control Register of Timer (SFR_TIMER_CTRL)

Timer 0 is an 8-bit counter. The count value is stored in the SFR_TMR0_COUNT register, which supports direct data write to load new values. The clock frequency for Timer 0 is selected via SB_TMR0_FREQ. Timer 0 can operate in either timer mode or pulse width modulator (PWM) mode.

In timer mode, Timer 0's initial count value is stored in advance in the initial value register SFR_TMR0_INIT. Timer 0 always counts from this initial value up to 0FFH. Upon overflowing from 0FFH, Timer 0 automatically reloads

the initial value from the initial value register SFR_TMR0_INIT and begins counting from the initial value toward 0FFH again. When counting from 0FEH to 0FFH, Timer 0 automatically sets the cycle signal SB_TMR0_CYCLE to 1. When counting overflows from 0FFH back to the initial value, Timer 0 automatically clears the cycle signal SB_TMR0_CYCLE to 0. The frequency of SB_TMR0_CYCLE is $(\text{SB_TMR0_FREQ clock frequency}) / (256 - \text{SFR_TMR0_INIT})$, with a duty cycle of $1 / (256 - \text{SFR_TMR0_INIT})$. This means SB_TMR0_CYCLE is active for one clock cycle per period.

In pulse width modulator mode, Timer 0 always counts from 00H to 0FFH, then overflows from 0FFH and resets to 00H to begin a new count cycle. Its period is always 256 clock cycles. When the count value exceeds the value stored in the Timer 0 initialization register (SFR_TMR0_INIT), Timer 0 automatically sets the cycle signal SB_TMR0_CYCLE to 1. When the count value overflows from 0FFH back to 00H, Timer 0 automatically clears the cycle signal SB_TMR0_CYCLE to 0. The duty cycle of SB_TMR0_CYCLE is $(255 - \text{SFR_TMR0_INIT}) / 256$.

SB_TMR0_OUT_EN determines whether the internal signal SB_TMR0_CYCLE or its 2-division signal is output to the IO0 pin. When the internal signal is high, the pin outputs a high level; otherwise, it outputs a low level. If the cycle signal output is enabled, then in timer mode, the 2-division signal of SB_TMR0_CYCLE (reduced to half frequency with a 50% duty cycle) will be output through the IO0 pin; in pulse width modulator mode, the SB_TMR0_CYCLE signal will be output through the IO0 pin.

The following table describes the timer control register SFR_TIMER_CTRL.

Bit	Name	Description	Reset value
7	SB_EN_LEVEL1	IO1 pin level change activates interrupt flag and wake-up enable: 1: Enables interrupt flag activation and wake-up; 0: Does not activate interrupt flag or wake-up.	0
6	SB_EN_LEVEL0	IO0 pin level change activates interrupt flag and wake-up enable: 1: Enables interrupt flag activation and wake-up; 0: Does not activate interrupt flag or wake-up.	0
5	SB_TMR0_ENABLE	Count enable for timer 0: 1: Counting enabled; 0: Counting disabled.	0
4	SB_TMR0_OUT_EN	Timer 0 count cycle signal output enable: 1: Enable output to IO0 pin; 0: Disable output.	0
3	SB_TMR0_MODE	Timer 0 operating modes: 1: Pulse width modulator mode; 0: Timer mode.	0
[2:0]	SB_TMR0_FREQ	Select the clock frequency for Timer 0 (as a fraction of the system clock frequency): 000: 1024X; 001: 256X; 010: 64X; 011: 16X; 100: 8X; 101: 4X; 110: 2X; 111: 1X.	000

34.4.12 System Configuration Register (SFR_SYS_CFG)

34.4.13 Host Reads and Writes and eMCU Read-only Register (SFR_CTRL_WR)

34.4.14 System Configuration Register (SFR_CTRL_RD)

The System Configuration Register provides control bits for the host and status bits for bidirectional data exchange. SB_INT_REQ is fully controlled by the eMCU to request interrupts or cancel requests to the host. However, the host can clear SB_INT_REQ to 0 by performing a dummy write operation to the read-only register SFR_CTRL_RD. Writing to SFR_CTRL_RD has no practical effect on its data.

After the eMCU writes to SFR_CTRL_RD, SB_DATA_SW_MR is automatically set to 1. After the host reads SFR_CTRL_RD, SB_DATA_SW_MR is automatically cleared to 0.

After the host writes to SFR_CTRL_WR, SB_DATA_MW_SR is automatically set to 1. After the eMCU reads SFR_CTRL_WR, SB_DATA_MW_SR is automatically cleared to 0.

SB_MST_CLK_GATE also controls the multiplexing of the eMCU program ROM. When SB_MST_CLK_GATE is 0, it is dedicated to the host-side system SRAM, totaling 4K bytes, enabling dynamic loading of new programs for the eMCU. When SB_MST_CLK_GATE is 1, it is dedicated to the eMCU program ROM, containing 2K instructions.

Bits 0 to 4 of the system configuration register are fully controlled by the host and are read-only on the eMCU side. The following table describes the system configuration register SFR_SYS_CFG.

Bit	Name	Description	Reset value
7	SB_INT_REQ	Interrupt request enable bit: 1: Request an interrupt to the host; 0: Cancel the interrupt request.	0
6	SB_DATA_SW_MR	SFR_CTRL_RD read status bit: 1: eMCU has written data but the host has not yet read it; 0: eMCU has not written data or the host has already read it.	0
5	SB_DATA_MW_SR	SFR_CTRL_WR read status bit: 1: Written by host but not yet read by eMCU; 0: Not yet written by host or already read by eMCU.	0
4	SB_MST_CFG_B4	Configuration information bit, software-defined purpose	0
3	SB_MST_IO_EN1	IO1 pin mode and output control switch: 1: Controlled by eMCU; 0: Controlled by host.	0
2	SB_MST_IO_EN0	IO0 pin mode and output control switch: 1: Controlled by eMCU; 0: Controlled by host.	0
1	SB_MST_RESET	Forced eMCU reset: 1: Host initiates an additional forced reset specifically for the eMCU; 0: No forced reset; the eMCU resets only when the host resets.	0
0	SB_MST_CLK_GATE	Global clock control for eMCU: 1: Enable the eMCU clock; 0: Disable the eMCU clock, suspending its program equivalent to sleep mode.	0

34.4.15 Coded Bit Period Register (SFR_BIT_CYCLE)

34.4.16 Coded Bit Configuration Register (SFR_BIT_CONFIG)

The eMCU supports 2 bit-modulation methods: PWM duty cycle modulation and Manchester modulation.

Setting SB_BIT_CYCLE greater than 3 enables encoding/decoding. When SB_BIT_TX_EN=0, the system receives data. Input status is received from IO0 based on the preset width, and decoded data is stored in SB_BIT_RX_I0. When SB_BIT_TX_EN=1, the system transmits data. Raw data to be transmitted is placed in SB_BIT_TX_O0. After encoding, it is automatically delivered to SB_PORT_OUT0 for pin output.

SB_BIT_CYC_CNT and SB_BIT_CYC_TAIL indicate the cycle status of the encoded bit. When SB_BIT_CYC_TAIL=1, it signifies the current phase is within the last 25% of the bit cycle, indicating that SB_BIT_RX_I0 has completed reception and decoding, and the transmission encoding is finished and has reached SB_PORT_OUT0.

Setting SB_BIT_CYCLE=0 forces SB_BIT_CYC_TAIL to 1, facilitating Manchester encoding/decoding initiation. Setting SFR_BIT_CYCLE clears the rising and falling edge detection records for IO0, enabling the WAITB instruction to restart waiting.

The following table describes the encoding bit cycle register SFR_BIT_CYCLE.

Bit	Name	Description	Reset value
7	SB_BIT_TX_O0	The original bit data of the coded bit to be sent, which has a double buffer structure. When a new bit period starts, this bit can load the next bit data in advance.	0
[6:0]	SB_BIT_CYCLE	Sets the width of the coded bit in clock period, which is the actual clock number of the bit minus 1. Its high 5 bits are 0 to turn off the encoding and decoding, and they are not 0 to turn on.	0000000

The following table is a description of the encoding bit configuration register SFR_BIT_CONFIG.

Bit	Name	Description	Reset value
7	SB_BIT_TX_EN	Transmission enable for encoding bit: 1: Enable transmission. Send the encoded SB_BIT_TX_O0 to SB_PORT_OUT0. 0: Disable transmission. If SB_BIT_CYCLE is active, reception is permitted.	0
6	SB_BIT_CODE_MOD	Modulation methods for encoding bits: 1: Manchester modulation, where 0 remains unchanged within a cycle and 1 flips within a cycle; 0: PWM duty cycle modulation, with 0 having a 25% duty cycle and 1 having a 75% duty cycle.	0
5	SB_PORT_IN_EDGE	Pin input level sampling timing selection: 1: Sample at the edge of the previous cycle, equivalent to sampling approximately half a cycle early; 0: Sample at the midpoint of the current clock cycle, offering relatively higher real-time performance.	0

4	SB_BIT_CYC_TAIL	Periodic state of the encoding bit: 1: Located in the last 25% of the bit period, indicating bit encoding/decoding is complete; 0: Located in the first 75% of the bit period.	1
3	SB_BIT_CYC_CNT6	The cycle timing status of the encoding bit. Bit cycle timing operates in clock cycles and is a 7-bit counter. SB_BIT_CYC_CNT6 to 3 represent bits [6:3] of the counter.	0
2	SB_BIT_CYC_CNT5		0
1	SB_BIT_CYC_CNT4		0
0	SB_BIT_CYC_CNT3		0

34.4.17 Data Register (SFR_DATA_x) (x=0-31)

SFR_DATA_x are all bidirectional read/write data registers. Write operations from the host take precedence, and their functions are software-defined.

Among these, SFR_DATA_EXCH also supports single-cycle bit transfer instructions.

34.5 Application

The eMCU operates at the same clock frequency as the host side, facilitating dynamic data exchange. It features a rich set of bit manipulation instructions, enabling single-cycle I/O configuration, single-cycle I/O status polling, or bit data copying. It supports hardware encoding/decoding of PWM or Manchester-modulated bit data, making it suitable for implementing I/O for various low-to-medium speed communication protocols and for I/O control requiring precise timing.

Chapter 35 Operational Amplifier (OPA) and Comparator (CMP)

The module consists of 3 independently configurable operational amplifier (OPA1/OPA2/OPA3) and 1 independently configurable voltage comparator (CMP). The operational amplifier (OPA) supports gain selection or can be used as a voltage comparator.

The input and output of each operational amplifier are connected to the I/O port, and the input pin or gain is optional, and the output pin can be optionally configured to the universal I/O port or multiplexed as an ADC sampling channel. The external analog small signal is amplified into the ADC to achieve small-signal ADC conversion.

The input and output of each voltage comparator are connected to the I/O port, and the input pins are optional, and the output pins can be optionally configured to the universal I/O port or multiplexed as an TIM internal sampling channel (without using the I/O pin).

35.1 Main Features

- OPA input pin/output channel can be selected.
- OPA1 output channel directly connected to comparator input
- OPA supports voltage buffer (BUFFER) and programmable gain amplifier (PGA)
- CMP input pin selectable
- CMP positive input supports selectable DAC output; negative input supports selectable internal self-bias voltage
- CMP supports selectable digital filtering and hysteresis
- CMP output pin selectable as general-purpose I/O port or TIM internal sampling channel

35.2 Function Description

35.2.1 OPA1/OPA2/OPA3

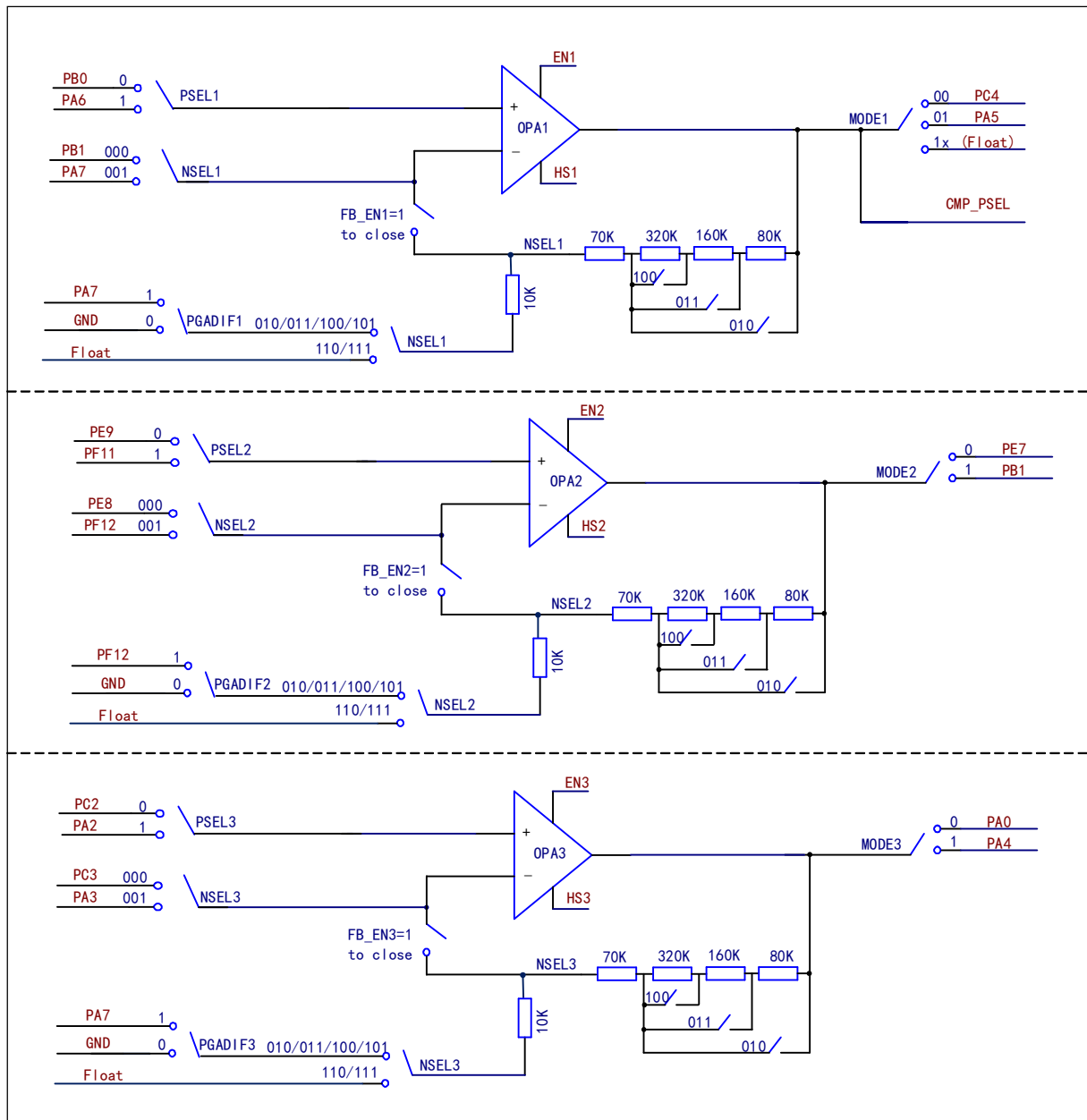
OPA1, OPA2, and OPA3 support voltage buffers (BUFFER) and programmable gain amplifiers (PGA). As shown in Figure 35-1, these 3 OPAs share essentially identical functionality. Compared to OPA2 and OPA3, OPA1 features an additional output channel connected to the CMP_P2 input of the voltage comparator.

In the R32_OPA_CTLR1 register: Setting the EN1 bit enables OPA1; Setting the PSEL1 bit selects the positive input channel for OPA1; Setting the NSEL1 bit selects the negative input channel for OPA1 or enables PGA mode to configure the internal gain of OPA1. Configuring the FB_EN1 bit enables the PGA mode feedback for OPA1. Configuring the PGADIF1 bit enables the differential input PGA mode, allowing OPA1 to function as a PGA in conjunction with NSEL1, with the N terminal connected to PA7. Configuring the HS1 bit enables high-speed mode, which improves slew rate in this mode. Configuring the MODE1[1:0] bits selects the output channel as either a GPIO or the positive input of a comparator.

In the R32_OPA_CTLR2 register: Setting the EN2 bit enables OPA2; Setting the PSEL2 bit selects the positive input channel for OPA2; Setting the NSEL2 bit selects the negative input channel for OPA2 or enables PGA mode to configure the internal gain of OPA2. Configure the FB_EN2 bit to enable OPA2's PGA mode feedback; configure the PGADIF2 bit to enable differential input PGA mode, allowing OPA2 to function as a PGA with NSEL2 set, with the N terminal connected to PF12; configure the HS2 bit to enable high-speed mode, which improves slew rate in this mode; configure the MODE2 bit to select the output channel.

In the R32_OPA_CTLR3 register: Configure the EN3 bit to enable OPA3; Configure the PSEL3 bit to select the positive input channel for OPA3; Configure the NSEL3 bit to select the negative input channel for OPA3 or enable PGA mode, setting the internal gain of OPA3. Configure the FB_EN3 bit to enable OPA3's PGA mode feedback. Configure the PGADIF3 bit to enable differential input PGA mode, allowing OPA3 to function as a PGA with NSEL3 set, with the N terminal connected to PA3. Configure the HS3 bit to enable high-speed mode, which improves slew rate in this mode. Configure the MODE3 bit to select the output channel.

Figure 35-1 OPA1/OPA2/OPA3 Structure Diagram



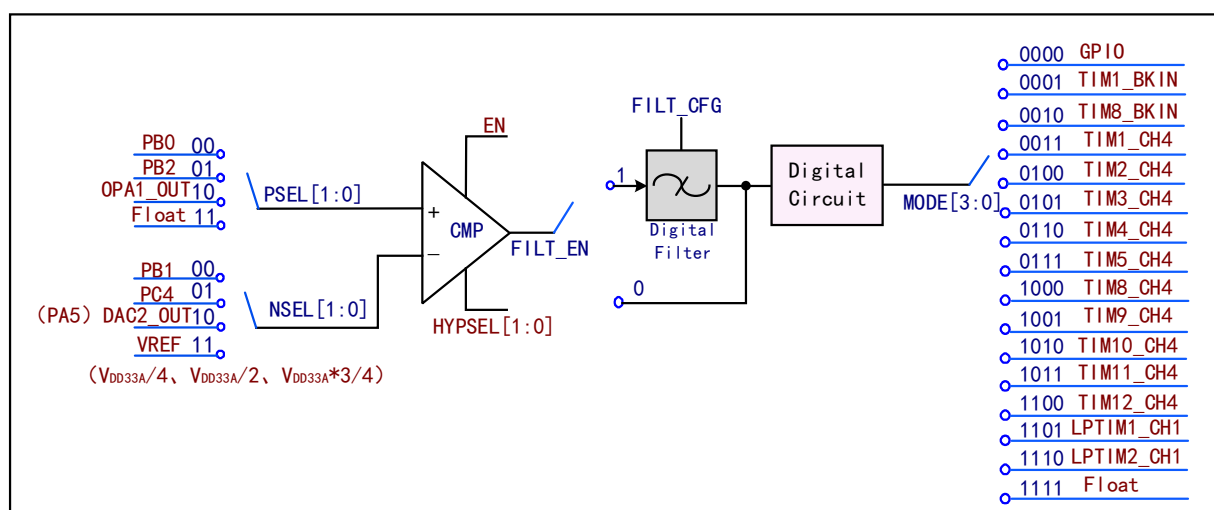
35.2.2 CMP

CMP supports optional hysteresis characteristics, digital filtering at the output, and optional output filtering. The structure diagram is shown in Figure 35-2.

In the R32_CMP_CTLR register: Configure the EN bit to enable the CMP module; Configure the PSEL[1:0] bits to select whether the P-channel is connected to a GPIO input or internally linked to the OPA operational amplifier; Configure the NSEL[1:0] bits to select whether the N-channel is connected to a GPIO input, internally linked to the DAC output, or connected to an optional internal self-bias voltage. Configure the HYPSEL bit to select the comparator hysteresis voltage; configure the VREF bit to select the CMP internal bias voltage magnitude; configure the FILT_EN bit to enable the CMP output digital filtering function; configure the FILT_CFG[8:0] bits to select the CMP output digital filtering sampling interval, which must be greater than the noise width and less than twice the noise width; Configure the FILT_BASE[2:0] bits to set the time reference for the CMP output digital filter sampling. Configure the MODE[3:0] bits to select whether the CMP2 output connects internally to the I/O or connects to the timer break input/internal sampling channel of the timer.

In the R32_CMP_STATR register, the output result of CMP filtering function closing/opening can be read through the OUT_FILT bit.

Figure 35-2 CMP Structure Diagram



35.3 Register Description

Table 35-4 OPA-related registers

Name	Access address	Description	Reset value
R32_OPA_CTLR1	0x40017800	OPA control register 1	0x00000076
R32_OPA_CTLR2	0x40017804	OPA control register 2	0x00000070
R32_OPA_CTLR3	0x40017808	OPA control register 3	0x00000070
R32_CMP_CTLR	0x4001780C	CMP control register	0x000000FF
R32_CMP_STATR	0x40017810	CMP status register	0x00000000

35.3.1 OPA Control Register 1 (R32_OPA_CTLR1)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	HS1	PGA DIF1	FB_E N1	Reser ved	NSEL1[2:0]	PSEL 1	MODE1[1:0]	EN1
----------	-----	-------------	------------	--------------	------------	-----------	----------------	-----

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	HS1	RW	OPA1 high-speed mode enable: 1: Enabled; 0: Disabled.	0
9	PGADIF1	RW	OPA1 is used as a PGA in conjunction with NSEL1, with the N terminal connected to OPA1_CHN1 (PA7): 1: Enable differential input PGA mode; 0: Disable differential input PGA mode.	0
8	FB_EN1	RW	OPA1 PGA mode feedback enable: 1: Enabled; 0: Disabled.	0
7	Reserved	RO	Reserved	0
[6:4]	NSEL1[2:0]	RW	OPA1 negative channel selection and PGA gain selection: 000: OPA1_CHN0 (PB1); 001: OPA1_CHN1 (PA7); 010: PGA mode, no negative input channel, internal gain 8; 011: PGA mode, no negative input channel, internal gain 16; 100: PGA mode, no negative input channel, internal gain 32; 101: PGA mode, no negative input channel, internal gain 64; Others: Disabled.	111b
3	PSEL1	RW	OPA1 positive terminal channel selection: 1: OPA1_CHP1 (PA6); 0: OPA1_CHP0 (PB0).	0
[2:1]	MODE1[1:0]	RW	OPA1 output channel selection: 00: Output signal via PC4; 01: Output signal via PA5; 1X: Output signal not connected to I/O but still connected to the comparator's positive input.	11b
0	EN1	RW	OPA1 enable: 1: Enabled; 0: Disabled.	0

35.3.2 OPA Control Register 2 (R32_OPA_CTLR2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	HS2	PGA DIF2	FB_E N2	Reser ved	NSEL2[2:0]	PSEL 2	Reser ved	MOD E2	EN2						

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	HS2	RW	OPA2 high-speed mode enable: 1: Enabled; 0: Disabled.	0
9	PGADIF2	RW	OPA2 is used as a PGA in conjunction with NSEL2, with the N terminal connected to OPA2_CHN1 (PF14): 1: Enable differential input PGA mode; 0: Disable differential input PGA mode.	0
8	FB_EN2	RW	OPA2 PGA mode feedback enable: 1: Enabled; 0: Disabled.	0
7	Reserved	RO	Reserved	0
[6:4]	NSEL2[2:0]	RW	OPA2 negative channel selection and PGA gain selection: 000: OPA2_CHN0 (PE8); 001: OPA2_CHN1 (PF12); 010: PGA mode, no negative input channel, internal gain 8; 011: PGA mode, no negative input channel, internal gain 16; 100: PGA mode, no negative input channel, internal gain 32; 101: PGA mode, no negative input channel, internal gain 64; Others: Disabled.	111b
3	PSEL2	RW	OPA2 positive terminal channel selection: 1: OPA2_CHP1 (PF11); 0: OPA2_CHP0 (PE9).	0
2	Reserved	RO	Reserved	0
1	MODE2	RW	OPA2 output channel selection: 00: Output signal via PB1; 01: Output signal via PE7;	0
0	EN2	RW	OPA2 enable: 1: Enabled; 0: Disabled.	0

35.3.3 OPA Control Register 3 (R32_OPA_CTLR3)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					HS3	PGA	FB_E	Reser	NSEL3[2:0]			PSEL	Reser	MOD	EN3

		DIF3	N3	ved		3	ved	E3	
--	--	------	----	-----	--	---	-----	----	--

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
10	HS3	RW	OPA3 high-speed mode enable: 1: Enabled; 0: Disabled.	0
9	PGADIF3	RW	OPA3 is used as a PGA in conjunction with NSEL3, with the N terminal connected to OPA3_CHN1 (PA3): 1: Enable differential input PGA mode; 0: Disable differential input PGA mode.	0
8	FB_EN3	RW	OPA3 PGA mode feedback enable: 1: Enabled; 0: Disabled.	0
7	Reserved	RO	Reserved	0
[6:4]	NSEL3[2:0]	RW	OPA3 negative channel selection and PGA gain selection: 000: OPA3_CHN0 (PC3); 001: OPA3_CHN1 (PA3); 010: PGA mode, no negative input channel, internal gain 8; 011: PGA mode, no negative input channel, internal gain 16; 100: PGA mode, no negative input channel, internal gain 32; 101: PGA mode, no negative input channel, internal gain 64; Others: Disabled.	111b
3	PSEL3	RW	OPA3 positive terminal channel selection: 1: OPA3_CHP1 (PA2); 0: OPA3_CHP0 (PC2).	0
2	Reserved	RO	Reserved	0
1	MODE3	RW	OPA3 output channel selection: 00: Output signal via PA4; 01: Output signal via PA0;	0
0	EN3	RW	OPA3 enable: 1: Enabled; 0: Disabled.	0

35.3.4 CMP Control Register 3 (R32_CMP_CTLR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	FILT_BASE[2:0]			Reserved			FILT_CFG[8:0]								

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FILT_EN	VREF[1:0]	HYPSEL[1:0]	EN	MODE[3:0]	NSSEL[1:0]	PSEL[1:0]								

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
[30:28]	FILT_BASE[2:0]	RW	CMP filter sampling timing configuration.	0
[27:25]	Reserved	RO	Reserved	0
[24:16]	FILT_CFG[8:0]	RW	CMP filter sampling interval configuration.	0
[15:14]	Reserved	RO	Reserved	0
13	FILT_EN	RW	CMP digital filter enable: 1: Enabled; 0: Disabled.	0
[12:11]	VREF[1:0]	RW	CMP internal bias voltage selection bits: 00: Disabled; 01: 1/4 V_{DD33A} ; 10: 1/2 V_{DD33A} ; 11: 3/4 V_{DD33A} .	0
[10:9]	HYPSEL[1:0]	RW	CMP hysteresis voltage selection bits: 00: 0mV; 01: 10mV; 10: 20mV; 11: 30mV.	0
8	EN	RW	CMP enable: 1: Enabled; 0: Disabled.	0
[7:4]	MODE[3:0]	RW	CMP output channel selection: 0000: Connect to GPIO (PA6: AF10; PA8: AF12; PB12: AF13; PC5: AF13; PE6: AF11; PE12: AF13; PE15: AF13); 0001: TIM1_BKIN; 0010: TIM8_BKIN; 0011: TIM1_CH4; 0100: TIM2_CH4; 0101: TIM3_CH4; 0110: TIM4_CH4; 0111: TIM5_CH4; 1000: TIM8_CH4; 1001: TIM9_CH4; 1010: TIM10_CH4; 1011: TIM11_CH4; 1100: TIM12_CH4; 1101: LPTIM1_CH1; 1110: LPTIM2_CH1;	1111b

			1111: Turn off the output function.	
[3:2]	NSEL[1:0]	RW	CMP negative channel selection bit: 00: CHN0 (PB1); 01: CHN1 (PC4); 10: DAC2 output (PA5); 11: CMP internal bias voltage.	11b
[1:0]	PSEL[1:0]	RW	CMP positive channel selection bits: 00: CHP0 (PB0); 01: CHP1 (PB2); 10: OPA1 output; 11: Reserved.	11b

35.3.5 CMP Status Register (R32_CMP_STATR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														OUT_FILT	

Bit	Name	Access	Description	Reset value
[31:1]	Reserved	RO	Reserved	0
0	OUT_FILT	RO	CMP output: When CMP digital filtering is enabled (FILT_EN = 1), the output is the filtered value; When CMP digital filtering is disabled (FILT_EN = 0), the output is the normal value.	0

Chapter 36 Serial Audio Interface (SAI)

Serial Audio Interface (SAI) supports various common audio protocols, such as I2S, PCM/DSP, AC'97, LSB or MSB alignment and TDM, which is suitable for mono and stereo applications. In order to realize the flexibility and configurability of SAI interface, it adopts two completely independent audio submodules, each of which contains up to four IO pins (SD, SCK, FS and MCLK).

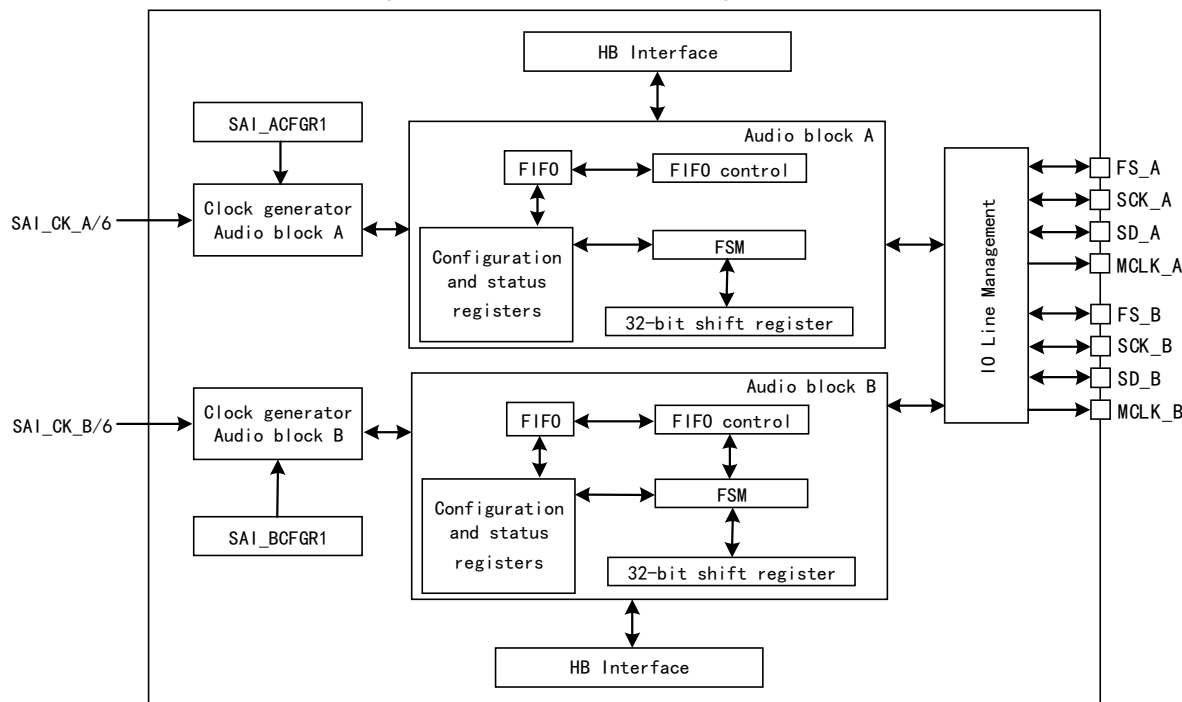
SAI can be configured as any combination of master/slave and sending/receiving, and its operation mode can be set as full duplex/simplex according to the synchronous/asynchronous configuration of audio submodule.

36.1 Main Features

- Provide 2 independent control modules, with each audio submodule configurable in any combination of master/slave or transmit/receive roles, each featuring an 8-word FIFO
- Audio submodules can operate in synchronous or asynchronous modes, with master/slave configurations independent of each other
- Flexible slot length configuration, with slots configurable as valid or invalid
- Frame synchronization configuration (valid level, valid length, and offset)
- Each audio frame contains up to 16 configurable slots
- Up to 16 slots, each supporting 8-bit, 10-bit, 16-bit, 20-bit, 24-bit, or 32-bit data
- Support LSB or MSB data transmission
- Support multiple audio protocols including I2S standard, LSB or MSB alignment, PCM/DSP, TDM, and AC'97
- Features mono/stereo audio functionality with mute settings
- Serial clock strobe edge selection (SCK)
- Each audio submodule has 2 independent DMA interfaces
- Error flag bits and interrupt sources:
 - FIFO overflow and underflow;
 - Frame sync advance detection in slave mode;
 - Frame sync lag detection in slave mode;
 - AC'97 codec not ready;
 - Clock configuration error.

36.2 Overview

Figure 36-1 Structural block diagram of SAI



The SAI primarily consists of 2 independent audio submodules connected to the output via an I/O management module. The I/O line controller manages four dedicated pins (SD, SCK, FS, MCLK) for the specified audio module within the SAI. If both submodules are declared as synchronous modules, certain pins can be shared, freeing others for use as general-purpose I/O. Whether the MCLK pin can function as an output pin depends on the specific application and decoding requirements, as well as whether the audio module is configured in master mode.

Each audio module integrates a 32-bit shift register controlled by the module's own functional state machine. Data storage and retrieval are performed via dedicated FIFOs. These FIFOs can be accessed by the CPU or via DMA to reduce the CPU's communication load.

The audio submodule can function as either a transmitter or receiver in both master and slave modes. Master mode indicates that the SAI generates the SCK bit clock and frame synchronizing signal, while slave mode indicates that the SCK bit clock and frame synchronizing signal originate from another external or internal master device. In specific cases, the direction of the FS signal is not directly related to the master or slave mode definition. Within the AC'97 protocol, even when the SAI (link controller) is configured to consume the SCK clock, the FS signal remains an SAI output (this also applies in slave mode).

36.3 Function Description

36.3.1 Clock Generator

Both independent audio submodules of the SAI have their own clock generators, but these clock generators function identically.

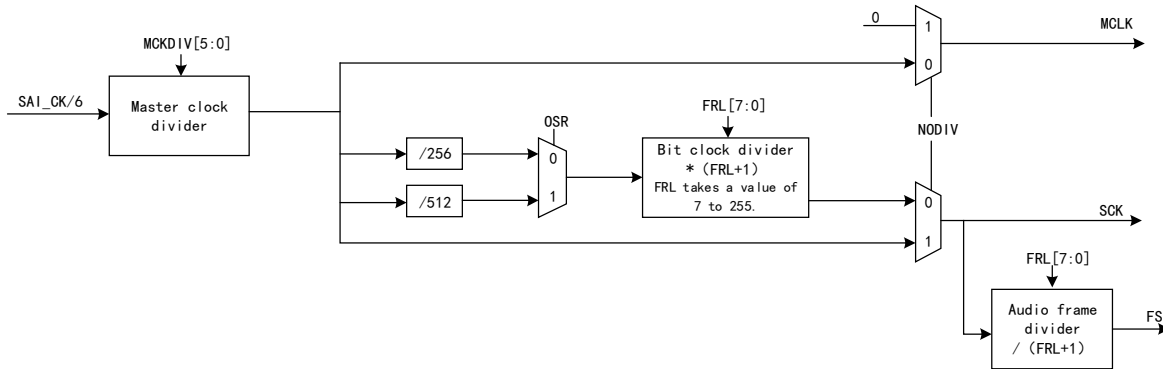
When the audio module is configured in master mode, the clock generator provides the bit clock (SCK) and the master clock (MCLK) for the external decoder. When the audio module is defined as slave mode, the clock generator is disabled.

The bit clock strobe edge on SCK can be configured via the CKSTR bit in the R32_SAI_xCFGR1 register. This bit applies to both master and slave modes.

It is recommended to select the PLL output of 295MHz as the input clock for the SAI.

The architecture of the audio module clock generator is shown in Figure 36-2.

Figure 36-2 Architecture of clock generator for audio module



Note: When NODIV is set to 1, if the MCLK pin is configured as the SAI pin in the GPIO peripheral, its signal level will be set to 0.

The clock source of the clock generator comes from the product clock controller. SAI_CK_x clock is equivalent to the master clock, and can divide the frequency for the external decoder through MCKDIV[5:0] bits:

If MCKDIV[5:0] is not equal to 0000b, then $MCLK_x = SAI_CK_x / (MCKDIV[5:0] * 2)$

If MCKDIV[5:0] is equal to 0000b, then $MCLK_x = SAI_CK_x$

The MCLK signal is used only in TDM.

Frequency division must be uniform to maintain a 50% duty cycle on both the MCLK output and the SCK clock. If the MCKDIV[5:0] bits are set to 0000b, a single division ratio is used to make MCLK equal to SAI_CK_x.

In the SAI, a single ratio $MCLK/FS = 256$ is used. In most cases, three frequency ranges will be encountered, as shown in Table 36-1.

Table 36-1 Examples of possible audio sampling range

Input SAI_CK_x clock frequency	Available common audio sampling frequencies	MCKDIV[5:0]
192kHz * 256	192kHz	000000
	96kHz	000001
	48kHz	000010
	16kHz	000110
	8kHz	001100
44.1kHz * 256	44.1kHz	000000
	22.05kHz	000001
	11.025kHz	000010
SAI_CK_x = MCLK	MCLK	000000

Note: This happens when the product clock controller selects an external clock source instead of the PLL clock.

If the corresponding audio module is configured as the master module and the NODIV bit in the R32_SAI_xCFGR1 register is set to 0, the master clock can be generated for an external decoder via the I/O port. When the audio module is defined as slave mode, the clock generator is disabled. The values of the NODIV and MCKDIV bits are

ignored. Additionally, the MCLK I/O pin is released and can be used as a general-purpose I/O.

The bit clock is derived from the master clock. The bit clock divider sets the division factor between the bit clock (SCK) and the master clock (MCLK) according to the following formula:

$$SCK = MCLK * (FRL[7:0] + 1) / 256$$

Among them:

256 is the fixed ratio between MCLK and the audio sampling frequency;

FRL[7:0] represents the number of bit clock cycles within an audio frame minus one, configured in the R32_SAI_xFRCR register.

In Master Mode, (FRL[7:0]+1) must equal a power of 2 to yield an even number of MCLK pulses per bit clock cycle. A 50% duty cycle is guaranteed on the bit clock (SCK).

The SAI_CK_x clock may also equal the bit clock frequency. In this case, the NODIV bit in the R32_SAI_xCFGR1 register should be set to 1, and the values within the MCKDIV divider and the bit clock divider will be ignored. At this time, the number of bits in each frame is completely configurable, and it is not necessary to be equal to several powers of 2.

36.3.2 Operation Mode

The SAI audio submodule can be configured via the MODE[1:0] bits of the R32_SAI_xCFGR1 register to operate in any combination of master/slave, transmit/receive modes.

Each audio module within the SAI is enabled via the SAIEN bit in the R32_SAI_xCFGR1 register. In slave mode, once this bit is activated, the transmitter or receiver becomes sensitive to activity on the clock line, data line, and sync line.

In master TX mode, the enabled audio module immediately generates bit clocks for the external slave module even if no data is present in the FIFO, but the generation of the FS signal is controlled by the presence of data in the FIFO. Upon receiving the first data to be transmitted, the FIFO outputs this data to the external slave module. If no data is present in the FIFO for transmission, subsequent values of zero are transmitted within the audio frame, and an underflow flag is generated.

In slave mode, audio frames commence when the audio module is enabled and the SOF bit is detected.

In slave TX mode, an underflow event cannot occur on the first frame after enabling the audio module, as the forced operation sequence is as follows:

- 1) Write to R32_SAI_xDATAR via software or DMA;
- 2) Wait until the FIFO threshold (FLTH) flag differs from 000b (indicating FIFO is empty);
- 3) Enable the audio module for slave transmit mode.

Master mode

In master mode, the SAI provides clock signals to connected external devices:

- 1) Bit clock and frame sync are output on pins SCK and FS, respectively;
- 2) If required, the SAI can also generate the master clock on the MCLK pin.

FS is generated when the FIFO is not empty and at the start of a frame. CK and MCLK generation, however, are not dependent on the FIFO status and are produced whenever the audio submodule is enabled.

Slave mode

In slave mode, the SAI receives clock signals from external devices:

- 1) When the SAI submodule is configured for asynchronous mode, the SCK and FS pins are configured as inputs.

2) If the SAI submodule is configured to operate synchronously with the second audio submodule, the corresponding SCK and FS pins are released for use as general-purpose I/O.

In slave mode, the MCLK pin is not used and can be assigned to other functions. It is recommended to enable the slave device before enabling the master device.

36.3.3 Internal Synchronization Mode

The audio submodule can run synchronously with another audio submodule in the same SAI, which is also called internal synchronization mode. In this case, they will share the bit clock and frame synchronization signal to reduce the number of external pins occupied during communication. An audio module configured in synchronous mode will release its SCK, FS and MCLK pins to be used as GPIO, while an audio module configured in asynchronous mode will use its SCK, FS and MCLKI/O pins (if the audio module is configured as a master module).

Typically, the audio module in synchronous mode can be used to configure the SAI in full-duplex mode. One of the two audio modules can be configured as the master module and the other as the slave module; alternatively, both can be configured as slave modules. One module can be set as the asynchronous module (with the SYNCEN[1:0] bits in the R32_SAI_xCFGR1 register set to 00b), and the other as the synchronous module (with the SYNCEN[1:0] bits in the R32_SAI_xCFGR1 register set to 01b).

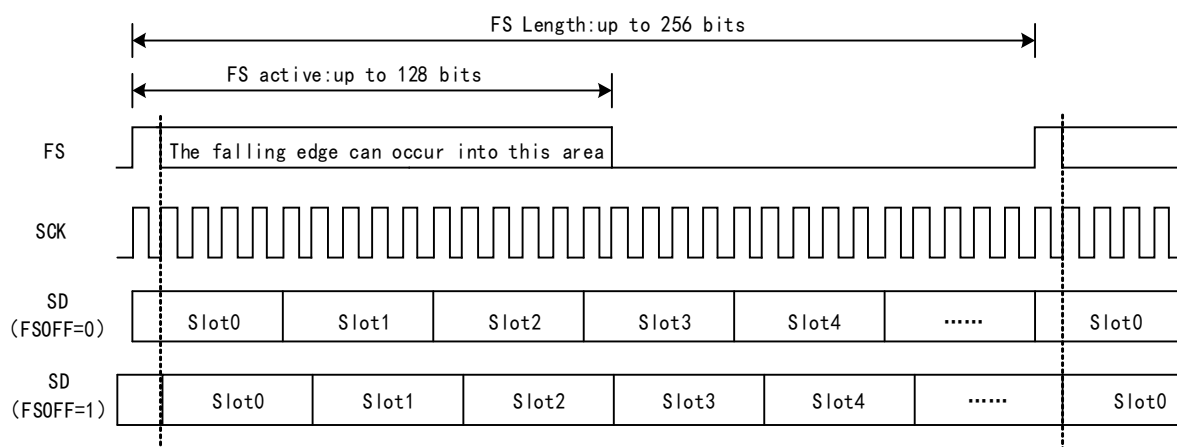
Note: Due to the internal resynchronization phase, the HCLK frequency must be greater than twice the bit rate clock frequency.

36.3.4 Frame Configuration

36.3.4.1 Frame Synchronization

The FS signal is used as a frame synchronization signal (SOF) in audio. The waveform of this signal can be completely configurable by configuring R32_SAI_xFRCR register, which can support various audio protocols with special specifications during frame synchronization. The following figure shows the configurability of audio frames.

Figure 36-3 Audio frame



In AC'97 mode or SPDIF mode (when the PRTCFG[1:0] bits of the R32_SAI_xCFGR1 register are set to 10b or 01b), the frame sync signal waveform is forced to conform to the AC'97 protocol. The value of the R32_SAI_xFRCR register is ignored.

Each audio module operates independently and therefore requires specific configuration.

36.3.4.2 Frame Length

Master mode

Setting the FRL[7:0] field of the R32_SAI_xFRCR register to 1 configures the audio frame length to a maximum of 256 bit clock cycles.

If the frame length exceeds the number of slots declared for that frame, the remaining bits to be transmitted will be zero-padded or the SD line will be released to a high configuration, depending on the state of the TRIS bit in the R32_SAI_xCFGR2 register. In receive mode, the remaining bits are ignored.

If the NODIV bit is cleared, (FRL+1) must equal a power of 2 (from 8 to 256) to ensure each bit clock cycle of the audio frame contains an integer number of MCLK pulses. If the NODIV bit is set to 1, the (FRL+1) field can take any value from 8 to 256.

Slave mode

The audio frame length is primarily used to specify the number of bit clock cycles for each audio frame transmitted from the external master module to the slave module. It is mainly employed to detect frame synchronization signals that occur ahead of schedule or behind schedule during audio frame transmission from the master module. Errors occur in such cases.

In slave mode, the configuration of the FRL[7:0] bits in the R32_SAI_xFRCR register is unrestricted.

36.3.4.3 Frame Synchronization Polarity

Set the effective polarity of the FS pin through the FSPOL bit in R32_SAI_xFRCR register, and start the frame through this polarity. SOF signal is sensitive to edge.

In the slave mode, the audio module is in a waiting state until a valid frame signal is received to perform the sending or receiving operation, and the SOF signal is synchronized with the frame signal. The frame synchronization polarity is valid only when the SOF signal is not detected during communication and the SOF signal is consistent with the expected SOF signal.

In the main mode, every time an audio frame is completed, a frame synchronization signal is sent until the SAIEN bit in the R32_SAI_xCTLR1 register is cleared. If there is no data in the FIFO at the end of the previous audio frame, the underflow will be handled as described in section 36.3.12.1, but the audio communication flow will not be interrupted.

36.3.4.4 Frame Synchronization Effective Level Length

The FSALL[6:0] bits of the R32_SAI_xFRCR register configure the length of the active level for the frame sync signal. This length can be set to between 1 and 128 bit clock cycles.

In I2S, LSB, or MSB alignment modes, the active length is set to half the frame length; in PCM/DSP or TDM modes, the active length is set to 1 bit.

36.3.4.5 Frame Synchronization Offset

According to the audio protocols supported in the application (such as I2S standard protocol and MSB alignment protocol), setting the audio frame synchronization offset can make the frame synchronization signal valid when the last bit or the first bit is transmitted. You can select one of two configurations by using the FSOFF bit in the R32_SAI_xFRCR register.

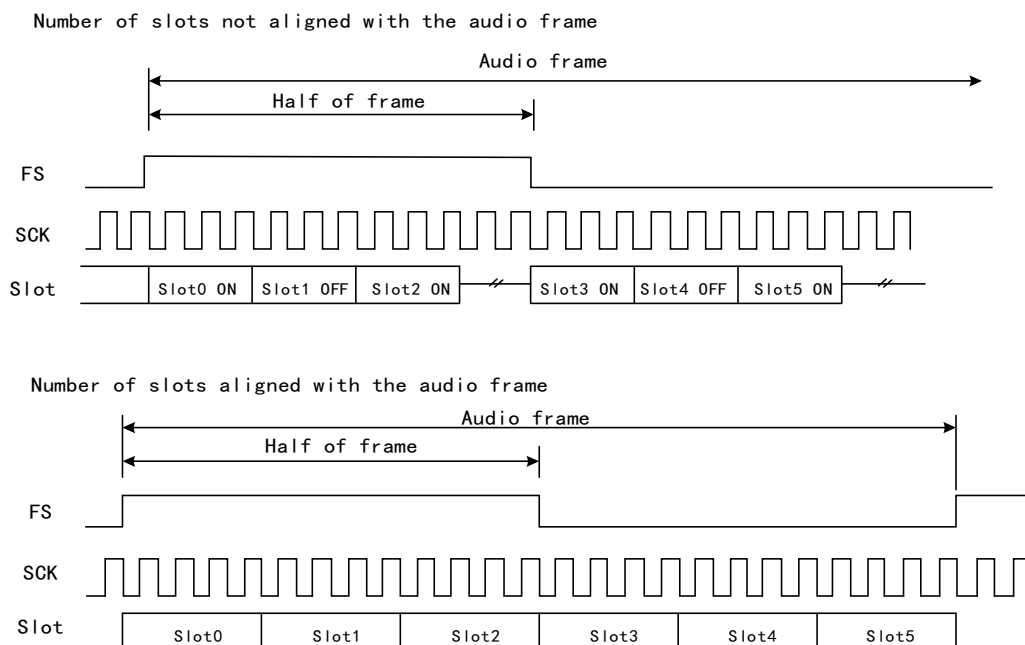
36.3.4.6 Function of FS Signal

The meaning of FS signal varies with the function of FS, and the meaning of FS signal can be configured in the FSDEF bit of R32_SAI_xFRCR register.

- When the FSDEF bit is set to 1, the FS signal simultaneously carries both the SOF signal and channel identification signal within the frame, suitable for audio frame structures such as I2S, MSB-aligned, or LSB-aligned protocols;
- When the FSDEF bit is set to 0, the FS signal functions as the SOF signal, suitable for PCM/DSP, TDM, AC'97, and audio protocols.

When interpreting the FS signal as the SOF signal and channel identification signal within a frame, the declared number of slots should be evenly distributed between the left and right channels. If the number of bit clock cycles on a half audio frame exceeds the dedicated slot count for a channel and the TRIS bit is set to 0, the remaining bit clock cycles in the R32_SAI_xCFGR2 register will output 0. Conversely, if the TRIS bit is set to 1, the SD line will enter a high-impedance state. During reception, the remaining bit clock cycles are only considered when a channel change occurs.

Figure 36-4 FS is used for SOF signal+channel identification signal



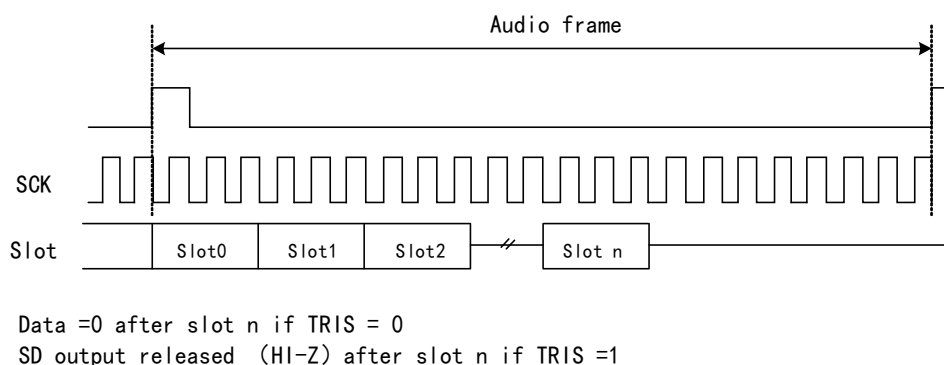
Note: The frame length should be even.

If the FSDEF bit in the R32_SAI_xFRCR register remains cleared, the FS signal is equivalent to the SOF signal. If the result of multiplying the number of slots defined by the NBSLOT[3:0] bits in the R32_SAI_xSLOTR register by the number of slots configured in the SLOTSZ[1:0] bits of the R32_SAI_xSLOTR register is less than the frame size (as indicated by the FRL[7:0] bits in the R32_SAI_xFRCR register), then: is less than the frame size (R32_SAI_xFRCR register FRL[7:0] bits), then:

If the TRIS bit in the R32_SAI_xCFGR2 register is set to 0, the remaining bits after the last Slot will be forced to 0 until the transmitter frame terminates.

If the TRIS bit in the R32_SAI_xCFGR2 register is set to 1, the data lines are released to high-impedance state during transmission of these remaining bits. In receive mode, these bits are discarded.

Figure 36-5 FS is used for SOF signal



In transmit mode, when the audio module is configured to capture SPDIF output on the SD line, the FS signal will not be utilized. The corresponding FS I/O will be released for other purposes.

36.3.5 Slot Configuration

The basic unit of an audio frame is a Slot, with each audio frame capable of being divided into up to 16 Slots. The number of Slots within an audio frame is determined by $NBSLOT[3:0]+1$.

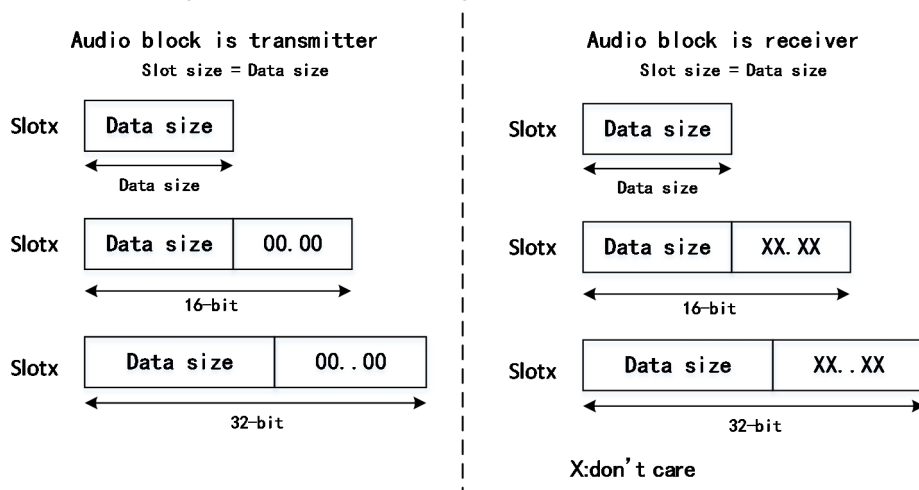
For AC'97 or SPDIF protocols (when $PRTCFCF[1:0]$ bits are set to 10b or 01b), the number of Slots is automatically configured according to the respective protocol specifications, and the value of $NBSLOT[3:0]$ is ignored.

By setting the $SLOTEN[15:0]$ bits in the $R32_SAI_xSLOTR$ register, individual slots can be enabled or disabled.

When transmitting an invalid slot, the SD data line is forcibly set to 0 or placed in a high-impedance state, depending on the configuration of the TRIS bit in transmit mode (see the section on invalid slot output data line management for details). In receive mode, data received after the end of this invalid slot is ignored. Consequently, there are no FIFO accesses, nor any FIFO read/write requests associated with the state of this invalid slot.

Additionally, the slot size can be selected by setting bits $SLOTSZ[1:0]$ in the $R32_SAI_xSLOTR$ register to 1, as shown in Figure 36-6.

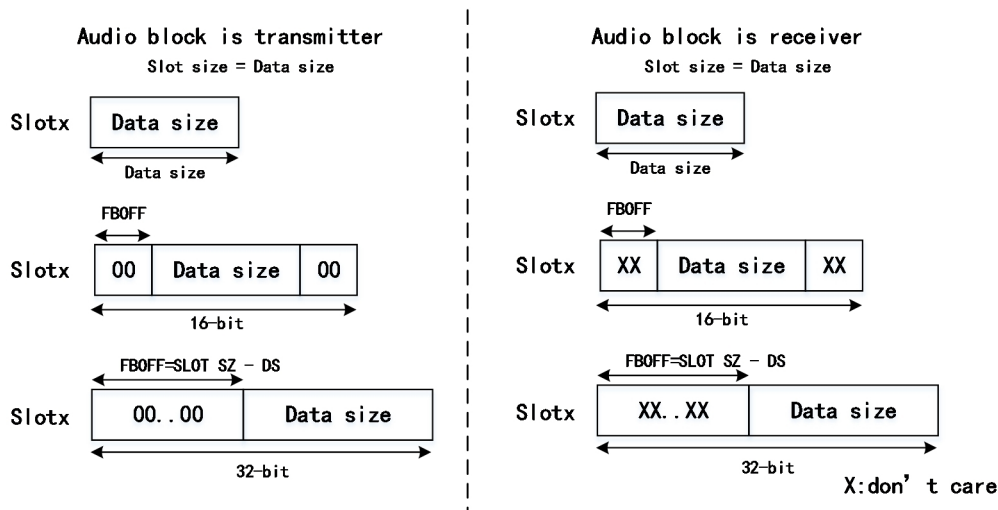
Figure 36-6 Slot size configuration when FBOFF bit 0



The position of the first data bit to be transmitted within the slot, i.e., the offset value of the first bit, can be configured via the $FBOFF[4:0]$ bits of the $R32_SAI_xSLOTR$ register. As shown in Figure 36-7, in transmit mode, a value of 0 is injected starting from the Slot's beginning position until the preset offset position is reached. In

receive mode, bits within the offset phase are ignored. This feature applies to LSB-aligned protocols (if the offset equals the Slot size minus the data size).

Figure 36-7 First bit offset



To avoid SAI malfunction, the following conditions must be met:

$$FBOFF \leq (SLOTSZ - DS),$$

$$DS \leq SLOTSZ,$$

$$NBSLOT \times SLOTSZ \leq FRL \text{ (Frame Length)},$$

When bit 1 of the R32_SAI_xFRCR register's FSDEF field is set, the number of slots must be even.

In AC'97 and SPDIF protocols (PRTCFCFG[1:0] bits set to 10b or 01b), slot size is automatically configured according to Section 36.3.7: AC'97 Link Controller.

36.3.6 Internal FIFO

Each SAI audio submodule contains an independent 8-word-wide synchronous FIFO to enhance transmission efficiency. These FIFOs can be accessed by either the CPU or DMA, with FIFO request interrupts used to request CPU and DMA access.

Depending on whether the audio module is defined as a transmitter or receiver, corresponding reads or writes to the FIFO are executed. Regardless of the access size, each read or write operation to the FIFO operates on a single word-sized FIFO cell. Each FIFO word contains one audio slot. Following each access to the R32_SAI_xDATAR register, the FIFO pointer increments by one word. Consequently, only one FIFO request is associated with the FREQ bit (FIFO request flag) in the R32_SAI_xSR register.

When the FREQIE bit in the R32_SAI_xINTENR register is enabled, an interrupt will be generated. The triggering conditions for this interrupt depend on the operating mode, FIFO threshold, FIFO status, and DMA burst transfer size.

The specific interrupt triggering mechanism is detailed in Table 36-2 and the subsequent sections 'Generating Interrupts in Transmit Mode' and 'Generating Interrupts in Receive Mode'.

Table 36-2 Generation conditions of FIFO request interrupt

Transmit: MODE[1:0]=x0				Receive: MODE[0]=x1			
FIFO threshold	FTH[2:0]	FIFO status	FLTH[2:0]	FIFO threshold	FTH[2:0]	FIFO status	FLTH[2:0]

Null	=000	Null	=000	Null	=000	Not null	≥001
1/4 full	=001	<1/4 full	<010	1/4 full	=001	≥1/4 full	≥010
1/2 full	=010	<1/2 full	<011	1/2 full	=010	≥1/2 full	≥011
3/4 full	=011	<3/4 full	<100	3/4 full	=011	≥3/4 full	≥100
Full	=100	Not full	<101	Full	=100	Full	=101

Note: (1) If the fundamental conditions are not met, the interrupt request will be cleared by hardware.

(2) After disabling SAI by setting bit 1 of the FFLUSH field in the R32_SAI_xCFGR2 register, the FIFO pointer can be reinitialized. If FFLUSH is set to 1 while SAI is enabled, the data in the FIFO will be automatically cleared, and the read/write pointer will be reset to 0.

Generate an interrupt in send mode

According to the FIFO configuration in transmit mode, an interrupt is generated:

When the FIFO threshold bits in the R32_SAI_xCFGR2 register are configured for an empty FIFO (FTH[2:0] set to 000b), an interrupt is generated if the register contains no data (FLTH[2:0] bits in the R32_SAI_xSR register are less than 001b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When the FIFO is no longer empty (FLTH[2:0] in R32_SAI_xSR is not 000b), i.e., when one or more data items are stored in the FIFO, this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bit in the R32_SAI_xCFGR2 register is configured to trigger on one-quarter FIFO fill (FTH[2:0] set to 001b), an interrupt is generated if less than one-quarter of the FIFO contains data (FLTH[2:0] bits in R32_SAI_xSR are less than 010b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When at least one-quarter of the FIFO contains data (FLTH[2:0] bits in the R32_SAI_xSR register are greater than or equal to 010b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bits in the R32_SAI_xCFGR2 register are configured for half-full FIFO (FTH[2:0] set to 010b), an interrupt is generated if less than half of the FIFO contains data (FLTH[2:0] bits in the R32_SAI_xSR register are less than 011b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When at least half of the FIFO contains data (FLTH[2:0] bits in R32_SAI_xSR are greater than or equal to 011b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bits in the R32_SAI_xCFGR2 register are configured to three-quarters full (FTH[2:0] set to 011b), an interrupt is generated if less than three-quarters of the FIFOs contain data (FLTH[2:0] bits in the R32_SAI_xSR register are less than 100b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When at least three-quarters of the FIFO contains data (the FLTH[2:0] bits in the R32_SAI_xSR register are greater than or equal to 100b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bits in the R32_SAI_xCFGR2 register are configured for a full FIFO (FTH[2:0] set to 100b), an interrupt is generated if the FIFO is not full (R32_SAI_xSR register FLTH[2:0] bits less than 101b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When the FIFO is full (FLTH[2:0] bits in the R32_SAI_xSR register equal 101b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

Generate an interrupt in receive mode

According to the FIFO configuration in receiving mode, an interrupt is generated:

When the FIFO threshold bits in the R32_SAI_xCFGR2 register are configured for an empty FIFO (FTH[2:0] set to 000b), an interrupt is generated if at least one data bit is present in the R32_SAI_xDATAR register (FLTH[2:0] bits in the R32_SAI_xSR register are greater than or equal to 001b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When the FIFO becomes empty (FLTH[2:0] in the R32_SAI_xSR register equals 000b), i.e., when no data is stored in the FIFO, this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bit in the R32_SAI_xCFGR2 register is configured for one-quarter FIFO fill (FTH[2:0] set to 001b), an interrupt is generated if at least one-quarter of the FIFO data cells are available (R32_SAI_xSR register FLTH[2:0] bits greater than or equal to 010b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When less than one-quarter of the FIFO data cells are available (FLTH[2:0] bits in R32_SAI_xSR are less than 010b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bits in the R32_SAI_xCFGR2 register are configured for half-full FIFO (FTH[2:0] set to 010b), an interrupt is generated if at least half of the FIFO data cells are available (FLTH[2:0] bits in the R32_SAI_xSR register are greater than or equal to 011b). (The FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When less than half of the FIFO data cells are available (FLTH[2:0] bits in the R32_SAI_xSR register are less than 011b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bit in the R32_SAI_xCFGR2 register is configured to three-quarters full (FTH[2:0] set to 011b), an interrupt is generated if at least three-quarters of the FIFO data cells are available (FLTH[2:0] bits in SAI_xSR are greater than or equal to 100b). (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When less than three-quarters of the FIFO data cells are available (the FLTH[2:0] bits in the R32_SAI_xSR register are less than 100b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

When the FIFO threshold bits in the R32_SAI_xCFGR2 register are configured for FIFO full (FTH[2:0] set to 100b), an interrupt is generated if the FIFO is full (R32_SAI_xSR register FLTH[2:0] bits equal 101b) (the FREQ bit in the R32_SAI_xSR register is set to 1 by hardware). When the FIFO is not full (FLTH[2:0] bits in the R32_SAI_xSR register are less than 101b), this interrupt (the FREQ bit in the R32_SAI_xSR register) is cleared by hardware.

Note: Similar to interrupt generation, if the DMAEN bit in the R32_SAI_xCFGR1 register is set to 1, the SAI can utilize DMA.

36.3.7 AC'97 Link Controller

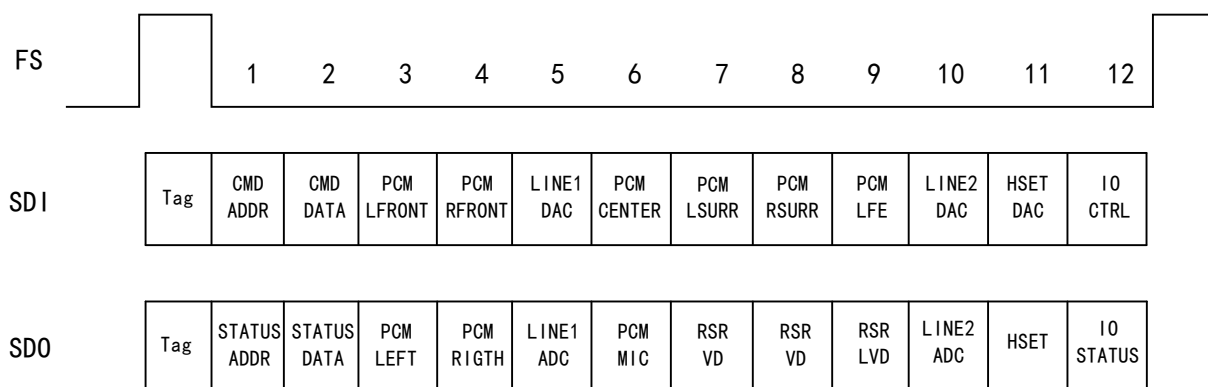
SAI can function as an AC'97 link controller. This mode is configured via the PRTCFG[1:0] bits in the R32_SAI_xCFGR1 register. When this protocol is selected, numerous configuration fields are ignored. Consequently, the number of slots and slot size are fixed, and the frame synchronization signal must be fully defined with a fixed waveform. Failure to meet these requirements will prevent SAI from operating correctly.

The NBSLOT[3:0] bits and SLOTSZ[1:0] bits will be ignored;

- The number of slots is fixed at 13. The first slot is 16 bits wide, while all other slots are 20 bits wide (data slots);
- The FBOFF[4:0] bits in the SAI_xSLOTR register are ignored;
- The SAI_xFRCCR register is ignored;
- MCLK is not used.

Regardless of whether the master or slave configuration is adopted, since the AC'97 link controller drives the FS signal, the FS signal generated by the asynchronous module will be automatically configured as an output.

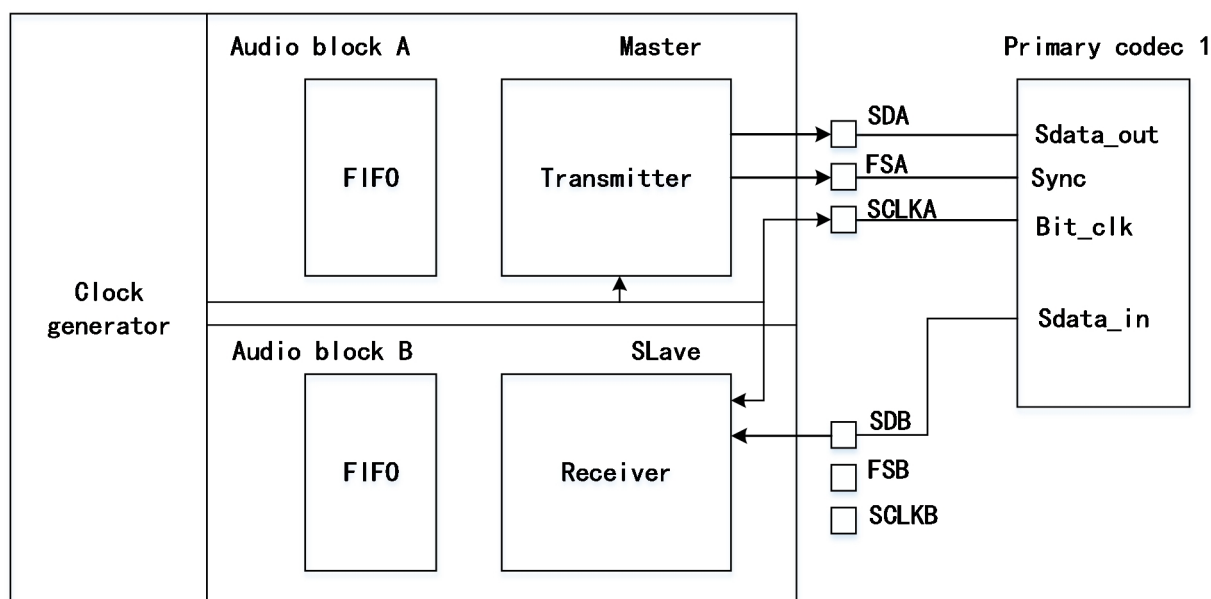
Figure 36-8 AC'97 audio frame



Note: In the AC'97 protocol, bit 2 of the TAG is reserved (always set to 0). Therefore, regardless of the value written to the SAI FIFO, bit 2 of the TAG is forced to 0. For detailed information on TAG representation, refer to the AC'97 protocol standard.

One SAI can be used for AC'97 point-to-point communication, as shown in Figure 36-9. In SAI, audio module A must be declared as an asynchronous master transmitter, while audio module B must be defined as a slave receiver and synchronized with audio module A internally.

Figure 36-9 SAI for AC'97 point-to-point communication



In receiver mode, SAI, which is used as AC'97 link controller, does not need FIFO request, so when the codec ready bit in Slot0 is decoded to low level, no data is stored in FIFO. If the CNRDYIE bit in the R32_SAI_xINTENR register is enabled, the flag CNRDY in the R32_SAI_xSR register will be set to 1 and an interrupt will be generated. This flag is dedicated to AC'97 protocol.

Clock generator programming in AC'97 mode

In AC'97 mode, the frame length is fixed at 256 bits, and its frequency should be set to 48 kHz. The registers specified in Section 36.3.1: SAI Clock Generator should be applied with FRL=255 to generate the appropriate frame rate (FFS).

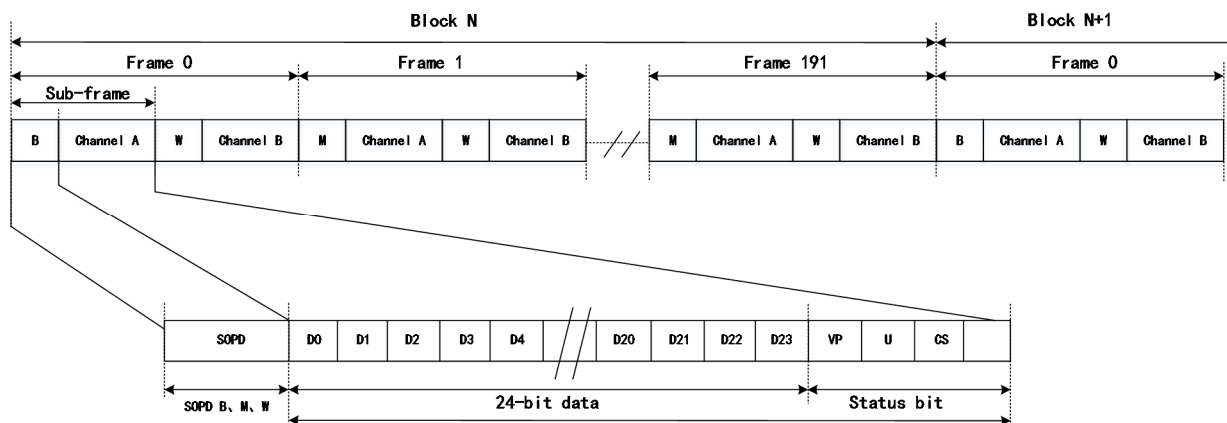
36.3.8 SPDIF Output

SPDIF (Sony/Philips Digital Interface) is a digital audio interconnect for consumer audio equipment, designed to transmit audio over reasonably short distances. The SPDIF interface is available only in transmit mode and supports the IEC 60958 standard. SPDIF mode can be selected by setting the PRTCFG[1:0] bits in the R32_SAI_xCFGR1 register to 01b.

For the SPDIF protocol:

- 1) Enable only the SD data line;
- 2) Release the FS, SCK, and MCLK I/O pins;
- 3) To enable the SAI clock generator and manage the data rate on the SD line, force the MODE[1] bit to 0 to select master mode;
- 4) Force the data size to 24 bits. Values set in DS[2:0] bits of the R32_SAI_xCFGR1 register are ignored;
- 5) The clock generator must be configured to define the symbol rate, where the known-bit clock should be twice the symbol rate. Data is encoded using the Manchester protocol;
- 6) Ignore the SAI_xFRCR and SAI_xSLOTR registers. Internally configure the SAI to comply with SPDIF protocol requirements, as shown in Figure 36-0.

Figure 36-10 SPDIF format



The SPDIF block contains 192 frames. Each frame consists of two 32-bit subframes, typically one for the left channel and one for the right channel. Each subframe begins with a 4-bit SOPD mode bit, which specifies whether the subframe marks the start of the block (and identifies Channel A), identifies Channel A within the block, or refers to Channel B (see Table 36-3). The subsequent 28 bits contain channel information, comprising 24 data bits and 4 status bits.

Table 36-3 SOPD mode

SOPD	Header coding		Description
	The last digit is 0	The last digit is 1	
B	11101000	00010111	Channel A data, and is the start subframe of a block.
W	11100100	00011011	Channel B data
M	11100010	00011101	Channel A data

The data transmission of SPDIF should be filled in the R32_SAI_xDATAR register according to the following:

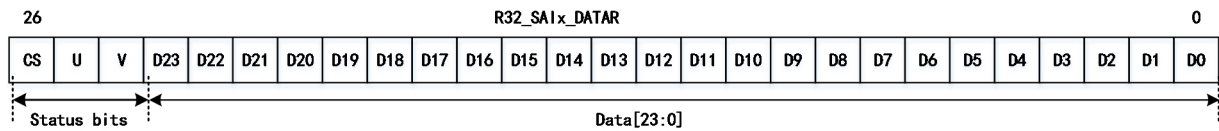
- R32_SAI_xDATAR[26:24] contains channel status bits, user bits, and validity bits;

- R32_SAI_xDATAR[23:0] contains the 24-bit data for the channel under consideration.

Note: (1) If the data size is 20 bits, map the data to R32_SAI_xDATAR[23:4].

(2) If the data size is 16 bits, map the data to R32_SAI_xDATAR[23:8].

Figure 36-11 R32_SAI_xDATAR register data schematic diagram



Note: R32_SAI_xDATAR[23] always stands for MSB. When performing transmission, LSB always takes precedence.

SAI first sends the appropriate header of each subframe in the block, and then sends R32_SAI_xDATAR (encoded in Manchester protocol) on the SD line. SAI ends the subframe by transmitting parity bits calculated as described in Table 36-4.

Table 36-4 Odd parity bits

R32_SAI_xDATAR[26:0]	The value of the transmission parity bit P
Odd number of 0	0
Odd number of 1	1

In SPDIF mode, underflow is the only error flag used in R32_SAI_xSR register. Since SAI can only work in the send mode, to recover from the underflow error detected by the underflow interrupt or underflow status bit, the following steps should be performed in sequence:

- 1) If DMA is already in use, disable the DMA stream via the DMA peripheral.
- 2) Disable SAI operation and verify physical disablement by polling the SAIEN bit in the R32_SAI_xCFGR1 register.
- 3) Clear the flag by writing a 1 to the OVRUDR bit in the R32_SAI_xSR register.
- 4) Flush the FIFO by setting the FFLUSH bit to 1 in the R32_SAI_xCFGR2 register. The software must point to the subsequent data address corresponding to the start of the new data block (data in header B). If DMA is in use, update the DMA source start address pointer accordingly.
- 5) Re-enable the DMA stream if the DMA controller resumes data transfer based on the new source start address pointer.
- 6) Re-enable the SAI by setting the SAIEN bit in the R32_SAI_xCFGR1 register to 1.

Programming of clock generator in SPDIF generator mode

For SPDIF generator, SAI shall provide a bit clock with equal symbol rate:

Table 36-5 Audio sampling frequency and symbol rate (SHARK)

Audio sampling frequency (Fs)	Symbolic rate
44.1kHz	2.8224MHz
48kHz	3.072MHz
96kHz	6.144MHz
192kHz	12.288MHz

Generally, the relationship between the audio sampling rate (F_S) and the bit clock rate (F_{SCK}) is given by the following formula:

$$F_S = \frac{F_{SAI_CK_x}}{64}$$

36.3.9 Stereo/Mono

In transmit mode, if the number of slots equals 2 (with the NBSLOT[3:0] bits of the R32_SAI_xSLOTR register set to 0001b), mono mode can be addressed without any data preprocessing in memory. In this case, since the data from Slot 0 is copied to data Slot 1 during transmission, the FIFO access time is reduced by half.

To enable mono mode, the following steps are required:

- 1) Set the MONO bit in the R32_SAI_xCFGR1 register to 1;
- 2) Set the NBSLOT bit in the R32_SAI_xSLOTR register to 1 and configure the SLOTEN bit to 3.

In receive mode, the MONO bit setting to 1 is only meaningful when the Slot count equals 2 (same as in transmit mode). When this bit is set to 1, only data from Slot 0 will be stored in the FIFO. Data from Slot 1 is discarded as it is considered identical to the previous slot's data. If the data stream in receive mode is a true stereo audio stream where the left and right channel data are distinctly different, the MONO bit is irrelevant. The conversion from the output stereo file to an equivalent mono file is handled by software.

36.3.10 Mute Mode

When the audio sub-module is used as a transmitter or receiver, the mute mode can be used.

Audio sub-module in transmission mode

In transmit mode, mute mode can be selected at any time. Mute mode is effective for all audio frames. During frame transmission, mute mode can be enabled by setting the MUTE bit in the R32_SAI_xCFGR2 register to 1. This mute mode bit is latched only at the end of a frame. If the MUTE bit is set to 1 at the end of a frame, mute mode activates at the start of the next audio frame and persists throughout its entire duration until the next frame ends. The bit is then latched to determine whether the subsequent frame remains a mute frame.

If the number of slots set via the NBSLOT[3:0] bits in the R32_SAI_xSLOTR register is less than or equal to 2, you can specify whether the value sent in mute mode is 0 or the last value for each slot. This is selected via the MUTEVAL bit in the R32_SAI_xCFGR2 register.

If the number of slots set in the NBSLOT[3:0] bits of the R32_SAI_xSLOTR register exceeds 2, the MUTEVAL bit in the R32_SAI_xCFGR2 register becomes meaningless since a value of 0 is sent on every bit of each slot.

In mute mode, the FIFO pointer continues to increment. This means that data requested for transmission in the FIFO during mute mode will be discarded.

Audio sub-module in receiving mode

In receive mode, for a given number of consecutive audio frames (R32_SAI_xCFGR2 register MUTECONT[5:0] bits), if zero is received in all declared active slots of the audio frame, a mute pattern from an external transmitter is detected. Upon detecting the corresponding number of mute frames, the MUTEDDET flag in the R32_SAI_xSR register is set to 1 and triggers an interrupt if the MUTEDDETIE bit in the R32_SAI_xCFGR2 register is set to 1.

The mute frame counter is reset when the audio submodule is disabled or when at least one data frame is received within a valid slot. An interrupt is generated only once when the counter reaches the value specified in the MUTECONT[5:0] bits. Subsequent interrupt events are reinitialized when the counter is reset.

Note: Mute mode is not available for SPDIF audio module.

36.3.11 Compressed Expansion Mode

In the field of mobile communications, pre-processing of data for transmission often involves the application of data compression and decompression algorithms.

The application can select whether to process data before transmission on the SD serial output line (compression) and whether to decompress data after reception on the SD serial input line (expansion) based on the COMP[1:0] bits of the R32_SAI_xCFGR2 register (used only when TDM mode is selected), as shown in Figure 36-11. The two supported compression/decompression modes are μ -Law and A-Law, both of which are standards recommended by CCITT G.711.

The μ -Law standard, adopted in the United States and Japan, permits a 14-bit dynamic range (R32_SAI_xCFGR2 register COMP[1:0] bits set to 10b). The European A-Law standard supports a 13-bit dynamic range (R32_SAI_xCFGR2 register COMP[1:0] bits set to 11b).

Regarding the calculation method for μ -Law or A-Law compression/expansion standards, based on the setting of the CPL bit in the R32_SAI_xCFGR2 register, either 1's complement or 2's complement may be used for computation.

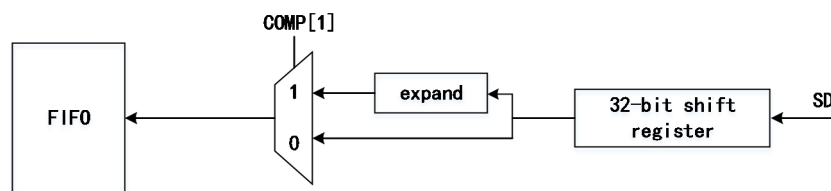
In both μ -Law and A-Law standards, data is encoded in an 8-bit format aligned to the most significant bit (MSB), and the width of data after compression/expansion processing remains consistently 8 bits.

Therefore, when the SAI audio module is enabled (R32_SAI_xCFGR1 register SAIEN bit set to 1) and either of these two compression/expansion modes is selected via the COMP[1:0] bits, the DS[2:0] bits of the R32_SAI_xCFGR1 register are forced to 010b. If no compression/expansion processing is required, the COMP[1:0] bits should remain cleared.

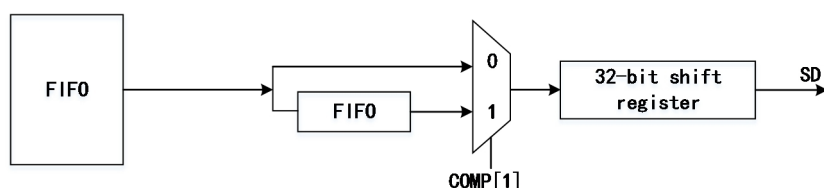
Specifically, if the SAI audio module is configured as a transmitter and the COMP[1] bit in the R32_SAI_xCFGR2 register is set to 1, the compression mode is employed. If the SAI audio module is declared as a receiver, the expansion algorithm is used.

Figure 36-12 Data compressed expansion hardware in audio module of SAI

Receiver mode (bit MODE[0]=1 in R32_SAI_xCFGR1)



Transmitter mode (bit MODE[0]=0 in R32_SAI_xCFGR1)



Note: Not applicable when AC'97 is selected.

Output data line management on invalid Slot

In the transmission mode, when an invalid Slot is transmitted on the data line, the behavior of SD line output can be controlled by TRIS bit.

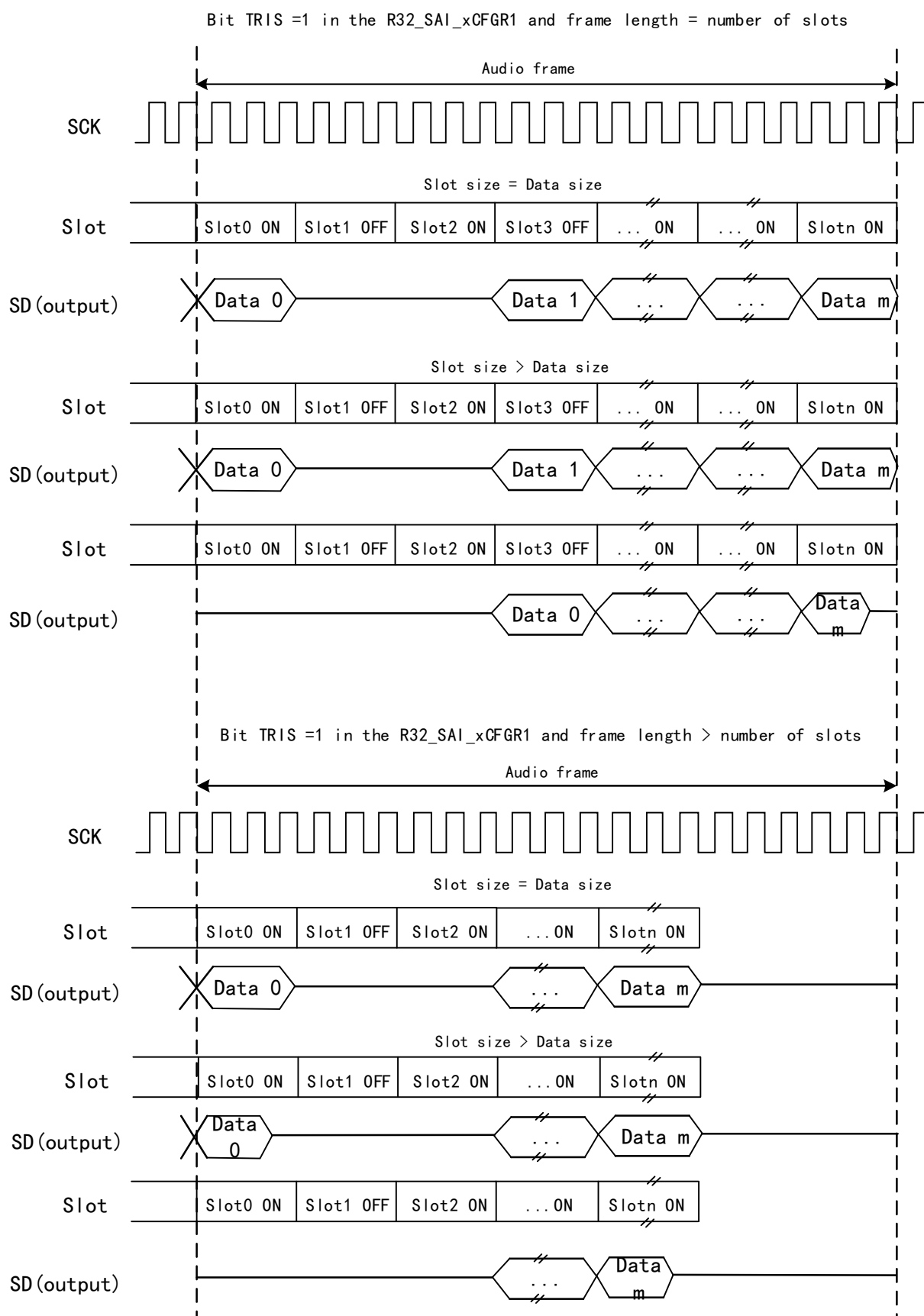
- When an invalid slot is transmitted, the SAI forces the SD output line to 0.
- After the last data bit is transmitted, the SD output line is placed in a high-impedance state to release the data line for use by other transmitters connected to this node.

2 transmitters must not simultaneously drive the same SD output pin to prevent short-circuit conditions. To ensure a gap exists during transmission, if the data is less than 32 bits, set the SLOTSZ[1:0] bits in the R32_SAI_xSLOTR register to 10b to expand the data to 32 bits. Subsequently, if the next slot is declared invalid, the SD output pin will be tri-stated at the end of the LSD of the valid slot (during the zero-padding phase to extend data to 32 bits).

Additionally, if the product of the number of slots and the slot size is less than the frame length, the SD output line will be tri-stated when supplementing the audio frame end with zero-padding.

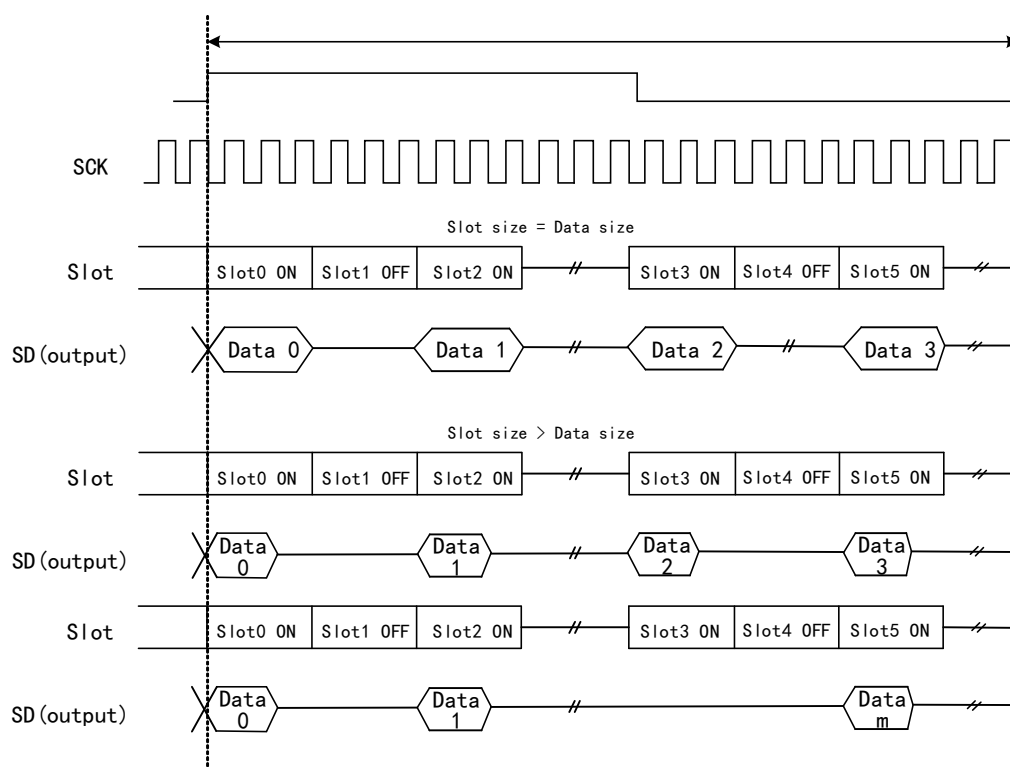
Figure 36-13 illustrates these behaviors.

Figure 36-13 Three-state strategy on SD output line when sending invalid Slot



When the selected audio protocol uses the FS signal as the SOF signal or channel identification signal (R32_SAI_xFRCR register FSDEF bit 1), the tri-state mode is managed according to Figure 36-14 (where R32_SAI_xCFGR1 register TRIS bit 1, FSDEF bit 1, half-frame length greater than Slot count/2, and NBSLOT=6).

Figure 36-14 Three-State strategy on output data line with I2S and other protocols



If the TRIS bit in the R32_SAI_xCFGR2 register is cleared, all high impedance state on the SD output line in Figure 36-13 and Figure 36-14 will be replaced by driving with a value of 0.

36.3.12 Error Flag

SAI uses the following error flags:

- FIFO overflow/underflow;
- Frame sync advance detection;
- Frame sync delay detection;
- Codec not ready (AC'97 only);
- Master mode clock configuration error.

36.3.12.1 FIFO Overflow/Underflow (OVRUDR)

The FIFO overflow/underflow bit is the OVRUDR bit in R32_SAI_xSR register.

Since audio modules can be used as both receivers and transmitters, and each audio module in the specified SAI has its own R32_SAI_xSR register, overflow or underflow errors share the same bit.

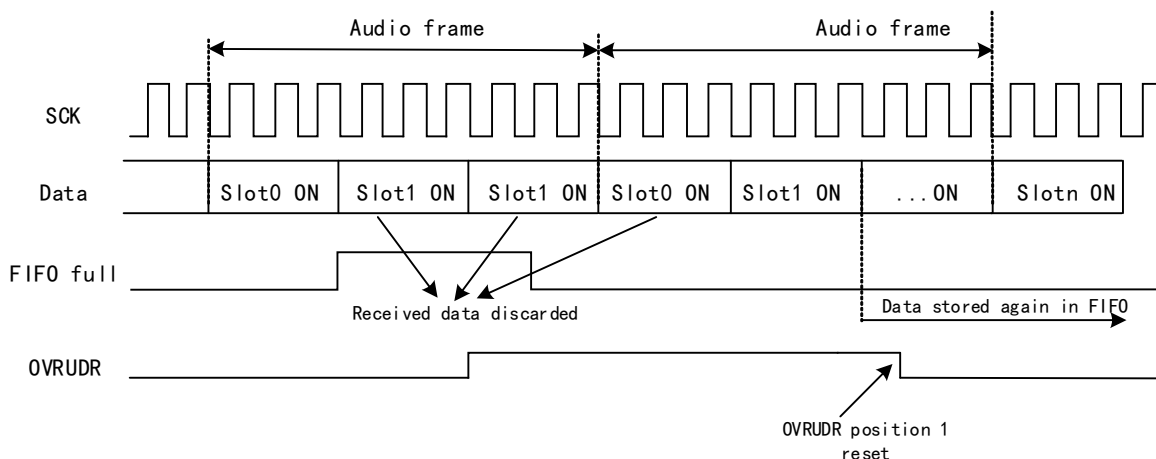
Overflow

If the audio module is configured as a receiver, an overflow occurs when audio frame data is received while the FIFO is full and cannot store any more received data. In this case, the received data is lost, and the OVRUDR flag in the R32_SAI_xSR register is set to 1. If the OVRUDRIE bit in the R32_SAI_xINTENR register is set to 1, an

interrupt is also generated. The internal system records the slot number at the time of overflow. The FIFO cannot store additional data until space is freed to accommodate new data. Once the FIFO has freed space for at least one data item, the SAI audio module receiver begins receiving new data from the new audio frame starting from the slot number internally recorded after the overflow detection. This prevents data slot misalignment in the target memory. Refer to Figure 36-15 for details.

To clear the OVRUDR flag, set the OVRUDR bit in the R32_SAI_xSR register to 1.

Figure 36-15 Overflow error detection



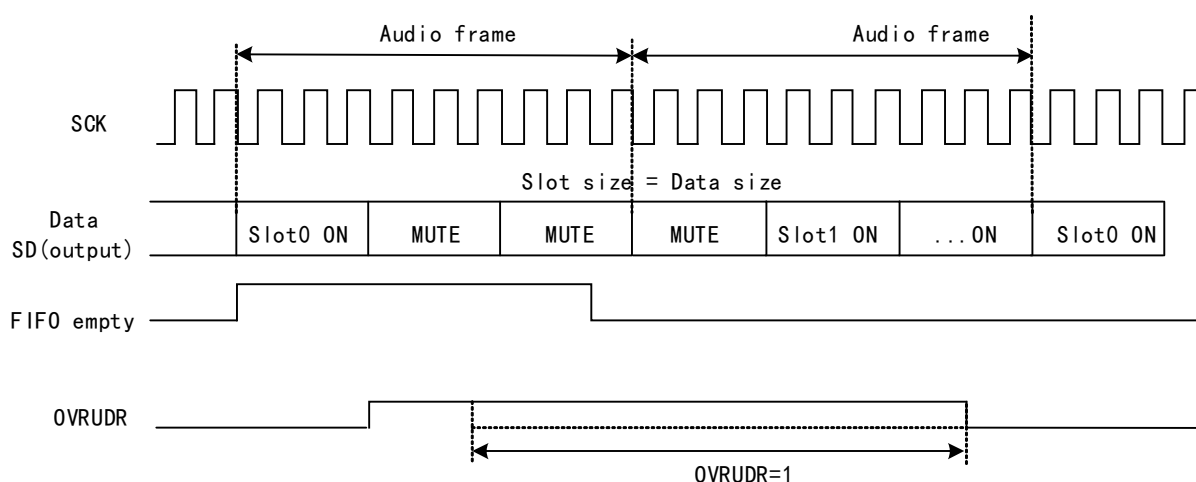
Underflow

When the audio module in SAI is used as a transmitter, if the FIFO is empty when data needs to be sent, underflow may occur. If underflow is detected, the Slot number of the event will be stored and the MUTE value (00b) will be sent until the FIFO is ready to send the data corresponding to the Slot where underflow is detected. See Figure 36-16 for details. This can avoid synchronization failure between memory pointer and Slot in audio frame.

The underflow event will set the OVRUDR flag in the R32_SAI_xSR register to 1, and if the OVRUDRIE bit in the R32_SAI_xINTENR register is set to 1, an interrupt will be generated. To clear this flag, just write 1 to bit OVRUDR in R32_SAI_xSR register.

When the audio submodule is configured as master mode or slave mode, an underflow event will occur.

Figure 36-16 FIFO underflow event



36.3.12.2 Frame Synchronization Advance Detection (AFSDET)

AFSDET flag is only used when SAI audio module is used as slave mode, and it will be disabled in master mode. In R32_SAI_xFRCR register, the frame length, frame polarity and frame offset configuration are all known, and this flag is used to indicate whether the frame synchronization (FS) signal is detected earlier than expected. When the frame detection advance occurs, the AFSDET flag in R32_SAI_xSR register will be set.

For the current audio frame which is insensitive to FS advance, the secondary detection will not affect it. In other words, the "parasitic" event of the FS signal will be marked, but it will not interfere with the processing of the current audio frame.

If the AFSDETIE bit in the R32_SAI_xINTENR register is set to 1, an interrupt will be generated. Clearing the AFSDET flag requires writing a 1 to this bit.

To resynchronize with the main module after a Frame Sense Detect Early error occurs, the following steps must be followed:

- 1) Disable the SAI module by resetting the SAIEN bit in the R32_SAI_xCFGR1 register. To ensure SAI is disabled, read back the SAIEN bit and verify its value is set to 0.
- 2) Flush the FIFO by setting the FFLUSH bit in the R32_SAI_xCFGR2 register to 1.
- 3) Re-enable the SAI peripheral by setting the SAIEN bit in the R32_SAI_xCFGR1 register to 1.
- 4) The SAI module will wait for the FS enable signal to restart synchronization with the main module.

Note: The SAIEN flag is not enabled in AC'97 mode because the SAI audio module, acting as a link controller, generates the FS signal even when declared as a slave module. Since the AC'97 protocol does not utilize the FS signal, this feature serves no practical purpose in SPDIF mode.

36.3.12.3 Frame Synchronization Lag Detection (LFSDET)

The LFSDET flag is only used when the SAI audio module operates as a slave module; it is disabled in master mode. In the R32_SAI_xFRCR register, the frame length, frame polarity, and frame offset configurations are known. If the external master module fails to send the FS signal within the scheduled time or generates the FS signal too late, the LFSDET flag will be set to 1. An interrupt will also be generated if the LFSDETIE bit in the R32_SAI_xINTENR register is set to 1. Writing a value of 1 to the LFSDET bit in the R32_SAI_xSR register clears the LFSDET flag.

The Frame Synchronization Lag Detection flag is set to 1 when the corresponding error is detected, indicating that the SAI requires resynchronization with the master module. Refer to Section 36.3.12.2: Frame Synchronization Advance Detection (AFSDET) for the specific sequence described.

In noisy environments, the audio module's state machine may incorrectly detect interference on the SCK clock and shift SAI data to an erroneous frame position. The SAI detects this event and reports it as a frame synchronization delay detection error.

Frame corruption does not occur if the external master module manages audio data frame transmission in continuous mode, which is the case for most applications. If this situation occurs, the LFSDET flag will be set to 1.

Note: The LFSDET flag is not enabled in AC'97 mode because the SAI audio module, acting as a link controller, generates the FS signal even when declared as a slave module. Since the AC'97 protocol does not utilize the FS signal, this functionality is not relevant in SPDIF mode.

36.3.12.4 Codec Not Ready (CNRDY AC'97)

When the SAI audio module is configured to operate in AC'97 mode (i.e., when the PRTCFCFG[1:0] bits in the R32_SAI_xCFGR1 register are set to 10b), the CNRDY flag in the R32_SAI_xSR register becomes meaningful. If the CNRDYIE bit in the R32_SAI_xINTENR register is set to 1, an interrupt will be generated when the CNRDY flag is set to 1.

During reception of the TAG0 (slot0) AC'97 audio frame, if the codec is not yet ready for communication, the CNRDY flag will be set to 1. In this case, data will not be automatically stored into the FIFO until TAG0 indicates the codec is ready, because the codec is not yet prepared. Once the codec is ready, all valid slots defined in the R32_SAI_xSLOTR register will be captured.

To clear the CNRDY flag, the CNRDY bit in the R32_SAI_xSR register must be written to 1.

Master mode clock configuration error (NODIV=0)

When the audio module works in the main mode (MODE[1:0]=0x) and the NODIV position is 0, if the following conditions are met, the WCKCFG flag will be set to 1 as long as SAI is enabled:

- (FRL+1) is not a power of 2;
- (FRL+1) is not between 8 and 256.

The MODE bit, the NODIV bit and the SAIEN bit belong to the R32_SAI_xCFGR1 register, and the FRL bit belongs to the R32_SAI_xFRCR register.

If the WCKCFGIE bit is set, an interrupt will be generated when the WCKCFG flag in the R32_SAI_xSR register is set. To clear this flag, write a 1 to bit WCKCFG in R32_SAI_xSR register.

When WCKCFG is set to 1, the audio module is automatically disabled, so the SAIEN bit will be hardware cleared.

36.3.13 DMA Interface

To reduce CPU load and optimize bus bandwidth, each SAI audio module features an independent DMA interface for read/write operations on the R32_SAI_xDATAR register (accessing the internal FIFO). Each audio submodule is equipped with a DMA channel supporting the basic DMA request/response protocol.

When configuring DMA transfers to serve the audio submodule, set bit 1 of the DMAEN field in the R32_SAI_xCFGR1 register. DMA requests are directly managed by the FIFO controller, and their generation depends on the FIFO threshold. When the audio submodule is configured in transmit mode, the FIFO threshold must be set to a specific value to ensure sufficient remaining space for a complete DMA burst write operation under worst-case conditions. Failure to do so may result in a FIFO overflow error. When the audio submodule is configured in receive mode, the FIFO threshold must be set to a specific value to ensure sufficient data in the FIFO for a complete DMA burst read operation, thereby preventing FIFO underflow errors.

The DMA transfer direction is related to the configuration of the SAI audio submodule:

- If the audio module functions as a transmitter, the audio module's FIFO controller will issue DMA requests to load data written into the SAI_xDATAR register into the FIFO.
- If the audio module functions as a receiver, DMA requests are associated with read operations from the R32_SAI_xDATAR register.

Configure the SAI interface for DMA mode in the following sequence:

- 1) Configure the SAI and FIFO thresholds to specify when to initiate DMA requests.
- 2) Configure the SAI DMA channel.
- 3) Enable DMA.
- 4) Enable the SAI interface.

Note: Before configuring the SAI module, the SAI DMA channel must be disabled.

36.3.14 SAI Interrupt

Table 36-6 SAI interrupt request

Interrupt source	Interrupt division	Interrupt occurrence condition	Interrupt enable control	Interrupt clearing control
FREQ	Request	MODE[1:0] is any mode	FREQIE	Depends on: - FIFO threshold settings - Communication transmission direction See Section 36.3.6 for details.
MUTEDET	Mute	MODE[1:0] is receiver	MUTEDETIE	CMUTEDET = 1
OVRUDR	Error	MODE[1:0] is any mode	OVRUDRIE	COVRUDR = 1
AFSDDET	Error	MODE[1:0] is the slave mode (not applicable to AC'97 mode and SPDIF mode).	AFSDETIE	CAFSDET = 1
LFSDET	Error	MODE[1:0] is the slave mode (not applicable to AC'97 mode and SPDIF mode).	LFSDETIE	CLFSDET = 1
CNRDY	Error	MODE[1:0] is the slave mode (AC'97 mode only)	CNRDYIE	CCNRDY = 1
WCKCFG	Error	MODE[1:0] is the master mode and NODIV = 0.	WCKCFGIE	CWCKCFG = 1

Enable interrupts in the following sequence:

- 1) Disable SAI interrupts;
- 2) Configure the SAI function register;
- 3) Enable SAI interrupt sources;
- 4) Enable SAI.

36.4 Register Description

Table 35-4 SAI-related registers

Name	Access address	Description	Reset value
R32_SAI_ACFGR1	0x40015804	SAI module A configuration register 1	0x00000040
R32_SAI_ACFGR2	0x40015808	SAI module A configuration register 2	0x00000000
R32_SAI_AFRCCR	0x4001580C	SAI module A frame configuration register	0x00000007
R32_SAI_ASLOTR	0x40015810	SAI module A Slot register	0x00000000
R32_SAI_AINTENR	0x40015814	SAI module A interrupt enable register	0x00000000
R32_SAI_ASR	0x40015818	SAI module A status register	0x00000008
R32_SAI_ADATAR	0x40015820	SAI module A data register	0x00000000
R32_SAI_BCFGR1	0x40015824	SAI module B configuration register 1	0x00000040
R32_SAI_BCFGR2	0x40015828	SAI module B configuration register 2	0x00000000
R32_SAI_BFRCCR	0x4001582C	SAI module B frame configuration register	0x00000007

R32_SAI_BSLOTR	0x40015830	SAI module B Slot register	0x00000000
R32_SAI_BINTENR	0x40015834	SAI module B interrupt enable register	0x00000000
R32_SAI_BSR	0x40015838	SAI module B status register	0x00000008
R32_SAI_BDATAR	0x40015840	SAI module B data register	0x00000000

36.4.1 SAI Module x Configuration Register 1 (R32_SAI_xCFGR1) (x=A/B)

Offset address: 0x04, 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					OSR	MCKDIV[5:0]					NODIV	Reserved	DMAEN	SAIEN	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				MONO	SYNCEN[1:0]	CKSTR	LSBFIRST	DS[2:0]		Reserved	PRTCFCG[1:0]		MODE[1:0]		

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
26	OSR	RW	Oversampling rate of the master clock: 1: FFS × 512; 0: FFS × 256.	0
[25:20]	MCKDIV[5:0]	RW	Master clock divider: 000000: Master clock input divided by 1; Others: The master clock frequency will be calculated according to the following formula: $F_{SCK_x} = \frac{F_{SAI_CK_x}}{MCKDIV}$ <i>Note: This bit is only configured if the audio module prohibits it.</i>	0
19	NODIV	RW	No divider: 1: No divider is used in the clock generator (the main clock divider bit is inactive at this time); 0: Enables the main clock generator.	0
18	Reserved	RO	Reserved	0
17	DMAEN	RW	DMA enable: 1: Enable; 0: Disable. <i>Note: In the receiving mode, the MODE[1:0] bits must be configured before the DMAEN bit is set to avoid DMA request, because the audio module will work in the sending mode by default after reset.</i>	0
16	SAIEN	RW	Audio module enable: 1: Enable SAIx audio module; 0: Disable SAIx audio module.	0

			<i>Note: When the SAI module (A or B) is configured in the main mode, there must be a clock in the input of the SAI module before the SAIEN can be set to 1.</i>	
[15:13]	Reserved	RO	Reserved	0
12	MONO	RW	Mono Mode Selection: 1: Mono mode; 0: Stereo mode. <i>Note: This bit is meaningful only when the Slot number is 2.</i>	0
[11:10]	SYNCEN[1:0]	RW	Synchronous enable: 00: Audio submodule operates in asynchronous mode; 01: Audio submodule synchronizes with another internal audio submodule. In this case, the audio submodule must be configured as slave mode; Others: Reserved. <i>Note: (1) After enabling SPDIF mode, the audio submodule should be configured as asynchronous;</i> <i>(2) Configure this bit only when the audio module is disabled.</i>	0
9	CKSTR	RW	Clock strobe edge: 1: Modify the signal generated by SAI on the falling edge of SCK, and sample the signal received by SAI on the rising edge of SCK; 0: Modify the signal generated by SAI on the rising edge of SCK, and sample the signal received by SAI on the falling edge of SCK. <i>Note: Configure this bit only when the audio module is disabled.</i>	0
8	LSBFIRST	RW	Least Significant Bit Priority: 1: Transmit the LSB of the data first; 0: Transmit the MSB of the data first. <i>Note: (1) Configure this bit only when the audio module is disabled.</i> <i>(2) When using LSB mode, the received data is in high-byte order: For example, if the received data size is 16 bits and the data content is 0x1234, the actual received data is 0x12340000.</i>	0
[7:5]	DS[2:0]	RW	Data size selection: 000/001: Reserved; 010: 8 bits; 011: 10 bits; 100: 16 bits; 101: 20 bits;	010b

			110: 24 bits; 111: 32 bits. <i>Note: Configure this bit only when the audio module is disabled.</i>	
4	Reserved	RO	Reserved	0
[3:2]	PRTCFCG[1:0]	RW	Protocol Configuration: 00: Free Protocol. By setting most configuration register bits and the frame configuration register, the free protocol allows users to leverage the audio module's powerful configuration capabilities to handle specific audio protocols (such as I2S, LSB/MSB alignment, TDM, PCM/DSP...). 01: SPDIF Protocol; 10: AC'97 Protocol; 11: Reserved. <i>Note: Configure this bit only when the audio module is disabled.</i>	0
[1:0]	MODE[1:0]	RW	SAIx audio module modes: 00: Master Transmitter; 01: Master Receiver; 10: Slave Transmitter; 11: Slave Receiver. <i>Notes:</i> (1) If the audio module is configured in SPDIF mode, the Master Transmit mode will be forced (MODE[1:0] = 00). In Master Transmit mode, the audio module will immediately begin generating FS and clock signals; (2) Configure this bit only when the audio module is disabled.	0

36.4.2 SAI Module x Configuration Register 2 (R32_SAI_xCFGR2) (x=A/B)

Offset address: 0x08, 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP[1:0]		CP L	MUTECNT[5:0]						MUT EVAL	MUT E	TRIS	FFLU SH	FTH[2:0]		

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:14]	COMP[1:0]	RW	Compression/Expansion Mode: 00: Compression/expansion algorithm not supported;	0

			01: Reserved; 10: μ -Law algorithm; 11: A-Law algorithm. <i>Note: Compression/expansion mode is only available when the TDM protocol is selected.</i>	
13	CPL	RW	Patch bit: 1: 2's complement representation; 0: 1's complement representation. <i>Note: This bit is valid only when the compression/expansion mode is set to μ-Law or A-Law algorithm.</i>	0
[12:7]	MUTECONT[5:0]	RW	Mute counter: The value set in these bits is compared with the number of consecutive silent frames detected in receive mode. When the number of silent frames equals this value, the MUTEDET flag is set to 1, and an interrupt is also generated if the MUTEDETIE bit is set to 1. <i>Note: This field is used only in receive mode.</i>	0
6	MUTEVAL	RW	Mute value setting: 1: Send the previous value during mute mode; 0: Send a bit value of 0 during mute mode. <i>Note: This bit is irrelevant to the SPDIF audio module and is therefore not used.</i>	0
5	MUTE	RW	Mute mode enable: 1: Enable mute mode; 0: Disable mute mode. <i>Note: This bit has no effect on the SPDIF audio module and is therefore not used.</i>	0
4	TRIS	RW	Three-state management of data lines: 1: The SD output line will be released (high-impedance state) upon completion of the last data bit transmission in the previous valid slot (the next slot is invalid). 0: When the slot is invalid, the SD output line remains driven by the SAI.	0
3	FFLUSH	RW	FIFO refresh: 1: FIFO refresh. Programming this bit to 1 triggers a FIFO refresh. All internal FIFO pointers (read and write) will be cleared to zero. Data remaining in the FIFO at this time will be lost (transmitted or received data will not continue to be lost). Before refreshing the SAI, DMA data flow/interrupts must be disabled. 0: Disable FIFO refresh.	0

[2:0]	FTH[2:0]	RW	FIFO threshold setting: 000: FIFO is empty; 001: $\frac{1}{4}$ FIFO; 010: $\frac{1}{2}$ FIFO; 011: $\frac{3}{4}$ FIFO; 100: FIFO is full; Others: Reserved.	0
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36.4.3 SAI Module x Frame Configuration Register 2 (R32_SAI_xFRCR) (x=A/B)

Offset address: 0x0C, 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													FS OF F	FSP OL	FS DE F
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	FSALL[6:0]							FRL[7:0]							

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
18	FSOFF	RW	Frame synchronization offset: 1: Enable FS on the bit preceding the first bit of Slot 0; 0: Enable FS on the first bit of Slot 0.	0
17	FSPOL	RW	Frame synchronization polarity: 1: FS is active high (rising edge); 0: FS is active low (falling edge). <i>Note: Chips with the fifth digit of the lot number less than 3 require an external pull-up resistor when FSPOL=0.</i>	0
16	FSDEF	RO	Frame synchronization definition: 1: FS signal is SOF signal + channel identification signal; 0: FS signal is the start-of-frame signal. When this bit is set to 1, the number of slots defined in the R32_SAI_xSLOTR register must be even. This means half the slots will be used for the left channel, and the remaining slots for the right channel (e.g., for protocols like I2S or MSB/LSB alignment, this bit must be set to 1).	0
15	Reserved	RO	Reserved	0

[14:8]	FSALL[6:0]	RW	Frame Synchronization Active Level Duration: Specifies the active level duration of the FS signal within an audio frame, expressed in bit clock cycles (SCK) + 1 (FSALL[6:0] + 1).	0
[7:0]	FRL[7:0]	RW	Define the audio frame length in terms of SCK clock cycles: The number of bits in a frame equals FRL[7:0] + 1. The number of bits transmitted within an audio frame must be greater than or equal to 8; otherwise, the audio module will exhibit operational anomalies. This condition occurs when the data size is 8 bits and only one slot is defined in NBSLOT[4:0] of the R32_SAI_xSLOTR register (NBSLOT[3:0] = 0000). In master mode, if the master clock (provided on the MCLK_x pin) is used, the frame length should be a power of 2 between 8 and 256. When the master clock is not used (NODIV = 1), it is recommended to program the frame length to a value between 8 and 256.	0x07

Note: The R32_SAI_AFRCCR and R32_SAI_BFRCCR registers must be configured when the audio module is disabled and are irrelevant for AC'97 and SPDIF audio protocols.

36.4.4 SAI Module x Slot Register (R32_SAI_xSLOTR) (x=A/B)

Offset address: 0x10, 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SLOTEN[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				NBSLOT[3:0]				SLOTSZ [1:0]	Reserved	FBOFF[4:0]					

Bit	Name	Access	Description	Reset value
[31:16]	SLOTEN[15:0]	RW	Slot enable: 1: Enabled Slot; 0: Disabled Slot. Each SLOTEN bit corresponds to a slot position from 0 to 15 (maximum of 16 slots).	0
[15:12]	Reserved	RO	Reserved	0
[11:8]	NBSLOT[3:0]	RW	Number of slots in an audio frame: The set value indicates the number of slots in the audio frame plus one (including invalid slots). The maximum number of slots is 16. When bit 1 of the FSDEF field in the R32_SAI_xFRCCR register is set, the number of slots must be even.	0

[7:6]	SLOTSZ[1:0]	RW	Slot size: 00: Slot size is equivalent to data size (specified in DS[3:0] bits of the R32_SAI_xCFGR1 register); 01: 16 bits; 10: 32 bits; 11: Reserved. The slot size must be greater than or equal to the data size. If this condition is not met, the SAI's behavior is undefined.	0
5	Reserved	RO	Reserved	0
[4:0]	FBOFF[4:0]	RW	First bit offset: The set value defines the position of the first data transmission bit within the Slot. It represents an offset value. In transmit mode, bits beyond this data field will be forced to zero. In receive mode, any additional received bits will be discarded.	0

Note: The R32_SAI_ASLOTR and R32_SAI_BSLOTR registers must be configured when the audio module is disabled and are irrelevant for AC'97 and SPDIF audio protocols.

36.4.5 SAI Module x Interrupt Enable Register (R32_SAI_xINTENR) (x=A/B)

Offset address: 0x14, 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									LFSD ETIE	AFSD ETIE	CNRD YIE	FREQ IE	WCK CFG IE	MUT EDET IE	OVR UDRI E

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
6	LFSDDETIE	RW	Frame synchronization latency detection interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
5	AFSDDETIE	RW	Frame synchronization early detection interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
4	CNRDYIE	RW	Codec not ready interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
3	FREQIE	RW	FIFO request interrupt enable: 1: Enable interrupts;	0

			0: Disable interrupts.	
2	WCKCFGIE	RW	Clock configuration error interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
1	MUTEDETIE	RW	Silent detection interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
0	OVRUDRIE	RW	Overflow/Underflow interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0

36.4.6 SAI Module x Status Register (R32_SAI_xSR) (x=A/B)

Offset address: 0x18, 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													FLTH[2:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									LFSD ET	AFSD ET	CNRD Y	FREQ	WCK CFG	MUT EDET	OVR UDR

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
[18:16]	FLTH[2:0]	RO	FIFO threshold flags: When the SAI module is configured as a transmitter: 000: FIFO is empty; 001: $FIFO \leq 1/4$ but not empty; 010: $1/4 < FIFO \leq 1/2$; 011: $1/2 < FIFO \leq 3/4$; 100: $3/4 < FIFO$ but not full; 101: FIFO full; Others: Reserved. If the SAI module is configured as a receiver: 000: FIFO is empty; 001: $FIFO < 1/4$, but not empty; 010: $1/4 \leq FIFO < 1/2$; 011: $1/2 \leq FIFO < 3/4$; 100: $3/4 \leq FIFO$, but not full; 101: FIFO is full; Others: Reserved.	0
[15:7]	Reserved	RO	Reserved	0
6	LFSDET	RW1Z	Frame synchronization latency detection: 1: Frame synchronization signal did not appear at the correct time;	0

			0: No error. <i>Note: Writing 1 clears 0; writing 0 has no effect.</i>	
5	AFSDDET	RW1Z	Frame synchronization early detection: 1: Frame synchronization signal detected early; 0: No error. <i>Note: Writing 1 clears 0; writing 0 has no effect.</i>	0
4	CNRDY	RW1Z	Codec not ready: 1: External AC'97 codec not ready; 0: External AC'97 codec ready. <i>Note: (1) This bit is used only when the AC'97 audio module is selected in the R32_SAI_xCFGR1 register and the audio module is configured in receiver mode. (2) Writing 1 clears the bit; writing 0 has no effect.</i>	0
3	FREQ	RW1Z	FIFO request flag: 1: FIFO request to read or write R32_SAI_xDATAR; 0: No FIFO request. The request content depends on the audio module configuration: - If the module is configured in transmit mode, the FIFO request relates to writing to R32_SAI_xDATAR. - If the audio module is configured in transmit mode, the FIFO request relates to writing to R32_SAI_xDATAR. <i>Note: Writing 1 clears the flag; writing 0 has no effect.</i>	1
2	WCKCFG	RW1Z	Clock Configuration Error Flag: 1: Clock configuration does not comply with the frame length specification defined in frame synchronization (configuration of FRL[7:0] bits in the R32_SAI_xFRCR register); 0: Clock configuration is correct. <i>Note: Writing 1 clears the flag; writing 0 has no effect.</i>	0
1	MUTEDET	RW1Z	Mute detection indicator: 1: Detected MUTE value (0 value) in a specified number of consecutive audio frames on the SD input line; 0: No MUTE value detected on the SD input line. <i>Note: Writing 1 clears 0; writing 0 has no effect.</i>	0
0	OVRUDR	RW1Z	Overflow/Underflow error flag: 1: Overflow/underflow error detected; 0: No overflow/underflow error.	0

			<i>Note: Writing 1 clears the flag; writing 0 has no effect.</i>	
--	--	--	--	--

36.4.7 SAI Module x Data Register (R32_SAI_xDATAR) (x=A/B)

Offset address: 0x20, 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	DATA[31:0]	RW	Data: If the FIFO is not full, writing to this register loads data into the FIFO. If the FIFO is not empty, reading this register clears the FIFO.	0

Chapter 37 QSPI Interface (QuadSPI)

This chapter's module description applies only to the CH32H417, CH32H416 microcontroller product.

The chip incorporates 2 dedicated QSPI communication interfaces (QuadSPI) for connecting to single-, dual-, or quad-wire (data-line) SPI flash storage media. The interface primarily supports the following operating modes:

- Indirect Mode: In this mode, the QSPI interface performs all operations through the QSPI register. This mode is suitable for applications requiring frequent read/write operations.
- Status Polling Mode: In this mode, the system periodically reads the external FLASH status register. When the flag is set to 1 (e.g., erase or program complete), the system generates an interrupt. This mode is suitable for applications requiring real-time monitoring of the external Flash status.

Memory-mapped mode: In this mode, the external Flash is mapped into the microcontroller's address space, allowing the system to access it as internal memory. For chips where the fifth digit of the lot number is less than 3, the memory-mapped address may only be used for data access.

Dual-Flash Mode: When operating in dual-Flash mode, the system can simultaneously access 2 Quad-SPI FLASH devices, doubling both throughput and capacity. This mode is suitable for applications requiring extensive data processing. Memory mapping function is not supported in dual flash mode.

37.1 Main Features

- 3 functional modes: Indirect mode, Status polling mode, and Memory-mapped mode
- Dual flash mode via parallel access to two FLASH devices
- Integrated FIFOs for transmission and reception
- Supports SDR mode
- Fully programmable frame format for indirect and memory-mapped modes
- Fully programmable opcode for indirect and memory-mapped modes
- Generates DMA trigger signals upon reaching FIFO threshold and transmission completion
- Generates interrupts upon reaching FIFO threshold, timeout, operation completion, and access errors

37.2 Overview

Figure 37-1 Structure block diagram of QSPI (Dual flash mode disabled)

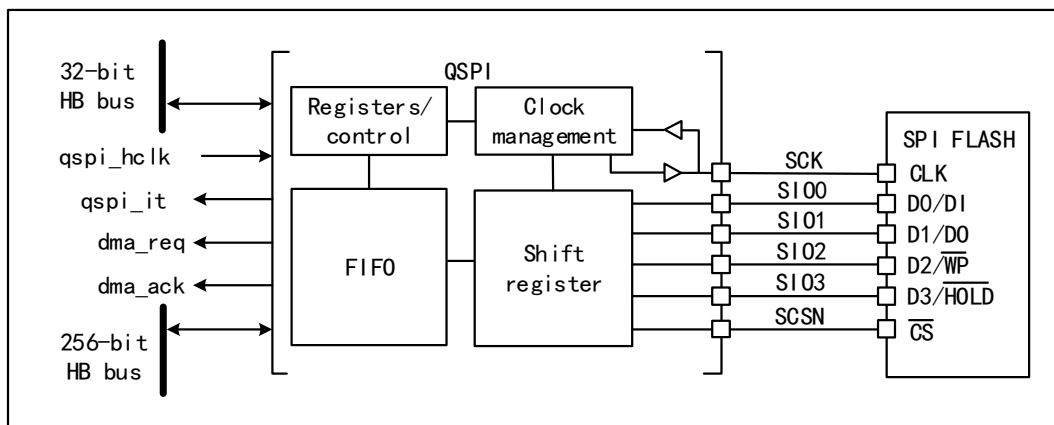
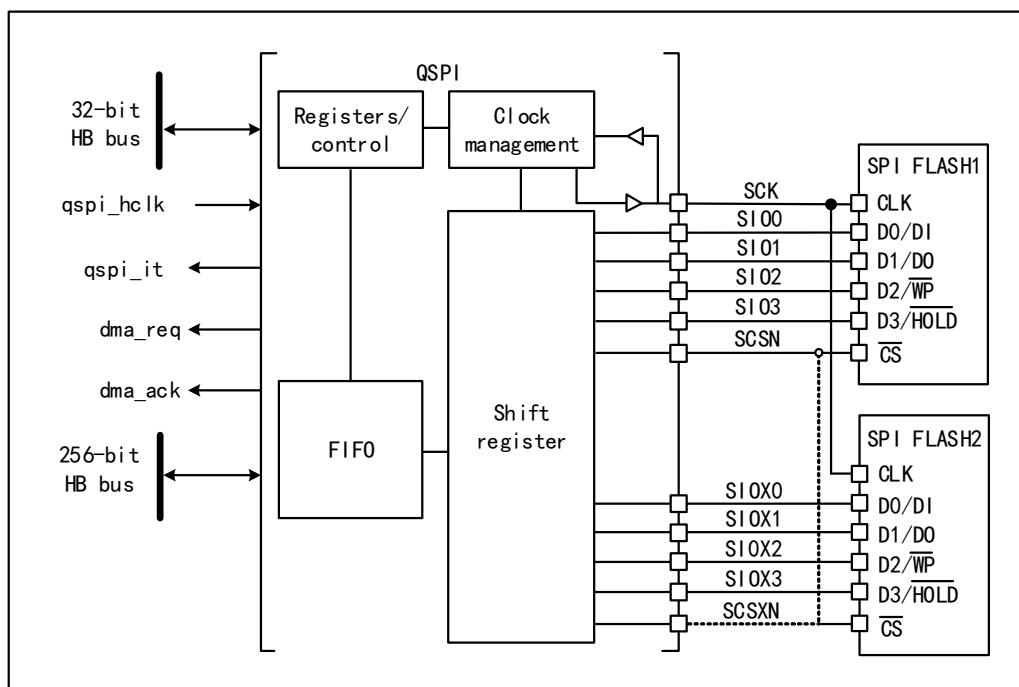


Figure 37-2 Structure block diagram of QSPI (Dual flash mode enabled)

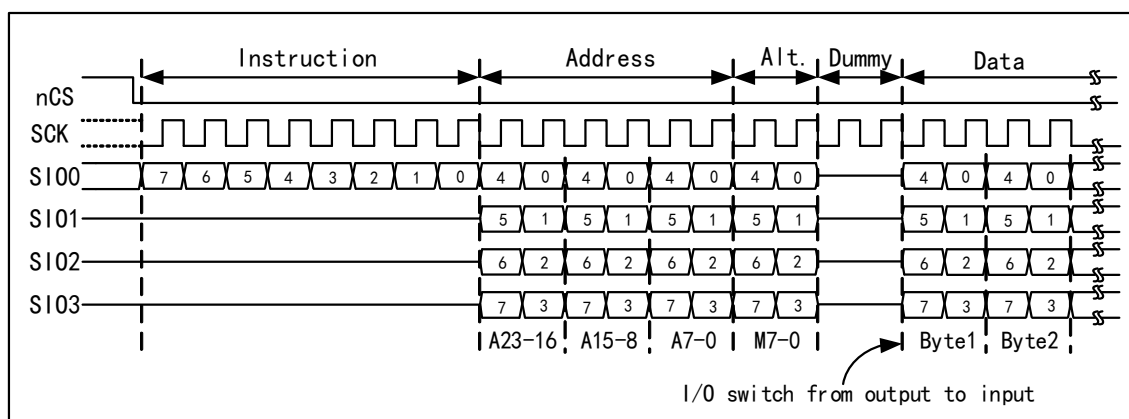


37.3 Function Description

37.3.1 QSPI Command Sequence

QSPI communicates with FLASH through commands, and each command includes five stages: instruction, address, alternate byte, null instruction and data. Any phase can be skipped, but at least one of instruction, address, alternate byte or data phase should be reserved. NCS drops before each instruction starts and rises again after each instruction is completed.

Figure 37-3 Example of read command in 4-wire mode



Instruction Phase

During the instruction phase, an 8-bit instruction configured in the INSTRUCTION[7:0] field of the R32_QSPIx_CCR register is sent to the FLASH to specify the type of operation to be executed. Most FLASH devices can only receive instructions one bit at a time via the SIO0 signal (single-wire SPI mode). However, by

configuring the IMODE[1:0] field in the R32_QSPIx_CCR register, selective transmission during the instruction phase can be enabled. This allows sending 2 bits at a time (via SIO0/SIO1 in dual-wire SPI mode) or 4 bits at a time (via SIO0/SIO1/SIO2/SIO3 in quad-wire SPI mode). If IMODE = 00b, the instruction phase is skipped, and the command sequence begins with the address phase (if present).

Address Phase

During the address phase, 1 to 4 bytes may be sent to the FLASH to indicate the operation address. The number of address bytes to be sent is configured in the ADSIZE[1:0] field of the R32_QSPIx_CCR register. In Indirect Mode and Auto Polling Mode, the address bytes to be sent are specified in the ADDRESS[31:0] field of the R32_QSPIx_AR register. In Memory-Mapped Mode, the address is provided directly by the system.

The address phase can be configured to transmit 1 bit (via SIO0 in single-wire SPI mode), 2 bits (via SIO0/SIO1 in dual-wire SPI mode), or 4 bits (via SIO0/SIO1/SIO2/SIO3 in quad-wire SPI mode) by setting the ADMODE[1:0] field in the R32_QSPIx_CCR register. If ADMODE = 00b, the address phase is skipped, and the command sequence proceeds directly to the next phase (if present).

Alternate Byte Phase

During the alternate byte phase, 1-4 bytes of data are sent to the FLASH, typically used to control operating modes. The number of alternate bytes to be sent is configured in the ABSIZE[1:0] field of the R32_QSPIx_CCR register. The specification of the bytes to be sent is completed in the R32_QSPIx_ABR register.

Configured via the ABMODE[1:0] field in the R32_QSPIx_CCR register, the Alternate Byte Phase can transmit 1 bit at a time (via SIO0 in single-wire SPI mode), 2 bits at a time (via SIO0/SIO1 in dual-wire SPI mode), or 4 bits at a time (via SIO0/SIO1/SIO2/SIO3 in quad-wire SPI mode). If ABMODE=00b, the alternate byte phase is skipped, and the command sequence proceeds directly to the next phase (if present).

During the alternate byte phase, it is sometimes sufficient to transmit a single half-byte rather than a full byte. For example: When transmitting alternate bytes using only 2 cycles in dual-wire mode. In this case, the firmware can adopt four-wire mode (ABMODE=11b) and transmit a byte by setting bits [7] and [3] of ALTERNATE to 1 (SIO3 remains high) and bits [6] and [2] to 0 (SIO2 line remains low). The upper 2 bits of the half-byte are stored in bits [4:3] of ALTERNATE, while the lower 2 bits reside in bits [1] and [0]. For instance, if half-byte 2 (0010) is transmitted via SIO0/SIO1, ALTERNATE should be set to 0x8A (1000_1010).

Null Instruction Cycle Phase

During the null instruction cycle phase, when operating at higher clock frequencies, no data transmission or reception occurs within a specified 1-31 cycle window to ensure sufficient time for the FLASH to prepare data. The number of cycles in this phase is defined by the DCYC[4:0] field in the R32_QSPIx_CCR[22:18] register.

In SDR mode, the duration is defined by a fixed number of full clock cycles. If DCYC is 0, the empty instruction cycle phase is skipped, and the command sequence proceeds directly to the data phase (if present).

The operating mode during the empty instruction cycle phase is fixed to single-wire mode.

Data Phase

During the data phase, any number of bytes can be received from or transmitted to the FLASH. In indirect mode and automatic polling mode, the number of bytes to be received/transmitted is specified in the R32_QSPIx_DLR register. In indirect write mode, data to be sent to the FLASH is written into the R32_QSPIx_DR register. In indirect read mode, data received from the FLASH is obtained by reading the R32_QSPIx_DR register.

In memory-mapped mode, read data is returned directly to the system.

By configuring the ABMODE[1:0] field in the R32_QSPIx_CCR register, the data phase can transmit/receive 1 bit at a time (via SIO0 in single-wire SPI mode), 2 bits at a time (via SIO0/SIO1 in dual-wire SPI mode), or 4 bits at a time (via SIO0/SIO1/SIO2/SIO3 in quad-wire SPI mode). If DMODE = 00b, the data phase is skipped, and the command sequence completes immediately upon pulling nCS high. This configuration applies only to indirect write mode.

37.3.2 QSPI Signal Interface Protocol Mode

Single-wire SPI mode

Traditional SPI mode permits serial transmission/reception of individual 1-bit data. In this mode, data is sent to the FLASH via SIO0. Data received from the FLASH is delivered via SIO1.

To utilize this single-bit mode, configure different command phases by setting the IMODE/ADMODE/ABMODE/DMODE field in R32_QSPIx_CCR to 01b.

Within each phase configured for single-wire mode:

- SIO0 is in output mode
- SIO1 is in input mode (high impedance)
- SIO2 is in output mode and forced to 0 (to disable the “write protection” function)
- SIO3 is in output mode and forced to 1 (to disable the “hold” function)

If DMODE=01b, this also applies to the empty instruction phase as described above.

Dual-Wire SPI Mode

In dual-wire mode, 2 bits are transmitted/received simultaneously via the SIO0/SIO1 signals. By configuring the IMODE/ADMODE/ABMODE/DMODE field of the R32_QSPIx_CCR register to 10b, different command phases can be set accordingly.

In each phase configured for dual-wire mode:

- SIO0/SIO1 are in high-impedance state (input) during data phase read operations and in output mode otherwise.
- SIO2 is in output mode and forced to 0.
- SIO3 is in output mode and forced to 1.

During the empty instruction phase, if DMODE=01b, SIO0/SIO1 remain in high-impedance state at all times.

Quad-wire SPI Mode

In quad-wire mode, four bits of data are transmitted/received simultaneously via the SIO0/SIO1/SIO2/SIO3 signals. Configuring the IMODE/ADMODE/ABMODE/DMODE fields in the R32_QSPIx_CCR register to 11b enables corresponding settings for different command phases.

In each phase configured for quad-wire mode:

- SIO0/SIO1/SIO2/SIO3 are in high-impedance state (input) during the data phase of read operations and function as outputs in other cases.
- SIO2 and SIO3 are used exclusively for QSPI mode. If no phase is configured for quad-wire SPI mode, the corresponding SIO2 and SIO3 pins can be used for other functions even when QSPI is active.

During the empty command phase, if DMODE=11, SIO0/SIO1/SIO2/SIO3 are all in high-impedance state.

SDR Mode

QSPI operates in Single Data Rate (SDR) mode.

In SDR mode, when QSPI drives the SIO0, SIO1, SIO2, and SIO3 signals, these signals conversion only on the falling edge of SCK.

When receiving data in SDR mode, the QSPI assumes the FLASH transmits data on the falling edge of SCK and samples the signal on the subsequent edge (rising edge) of SCK.

Dual Flash Mode

When the DFM bit (R32_QSPIx_CR[6]) is set to 1, QSPI enters dual flash mode. The QSPI utilizes 2 external four-wire SPIFLASHes (FLASH1 and FLASH2), transmitting/receiving 8 bits of data per cycle, effectively doubling throughput and capacity. Each FLASH uses the same SCK signal and may optionally share the same nCS signal, but their SIO0, SIO1, SIO2, and SIO3 signals are independent.

Dual Flash mode can be combined with Single-bit mode, Dual-bit mode, Quad-bit mode, and SDR mode.

Specify the FLASH size in FSIZE[4:0] (R32_QSPIx_DCR[20:16]). The specified parameter should reflect the total FLASH capacity, i.e., twice the capacity of a single component. When address X is even, the byte assigned by QSPI to address X is the byte stored at address X/2 in FLASH1. The byte assigned by QSPI to address X+1 is the byte stored at address X/2 in FLASH2. That is, bytes at even addresses are stored in FLASH1, while bytes at odd addresses are stored in FLASH2.

In dual-flash mode, the number of bytes required to read the FLASH status register is twice that of single-flash mode. Specifically, since each FLASH provides 8 valid bits after executing the status register read instruction, QSPI must be configured for a data length of 2 bytes (16 bits), with QSPI receiving 1 byte from each FLASH. If each FLASH provides a 16-bit status, the QSPI must be configured to read 4 bytes to capture all status bits from both FLASHes in dual-flash mode. Consequently, the least significant byte in the data register is the least significant byte of FLASH1's status register, while the next byte is the least significant byte of FLASH2's status register. The third byte in the data register is the second byte of FLASH1, and the fourth byte is the second byte of FLASH2 (when the FLASH has a 16-bit status register).

In dual flash mode, accesses must always be to an even number of bytes. Therefore, when DFM=1, bit 0 of the data length field (R32_QSPIx_DLR[0]) is always maintained as 1.

In dual flash mode, the behavior of the FLASH1 interface signals is essentially identical to that in normal mode. During instruction, address, alternate byte, and empty instruction cycle phases, the waveforms of the FLASH2 interface signals are identical to those of the FLASH1 interface signals. That is, each FLASH always receives the same instruction and address. Subsequently, during the data phase, the SIOx and SIOx buses transmit data in parallel, but the data sent to (or received from) FLASH1 differs from that in FLASH2.

37.3.3 QSPI Indirect Mode

In indirect mode, the command is triggered by writing to the QSPI register; Data is transmitted by reading and writing data registers in the same way as other communication peripherals.

When FMODE=00b (R32_QSPIx_CCR[27:26]), the QSPI operates in indirect write mode, transmitting bytes to the FLASH during the data phase. Data is provided by writing to the data register (R32_QSPIx_DR). When FMODE=01b, QSPI operates in indirect read mode, receiving bytes from FLASH during the data phase. Data is acquired by reading from the data register (R32_QSPIx_DR).

Specify the number of bytes to read/write in the Data Length Register (R32_QSPIx_DLR). If R32_QSPIx_DLR = 0xFFFF_FFFF (all bits set to 1), the data length is considered undefined, and QSPI will continue transferring data until reaching the end of the FLASH (as defined by FSIZE). If no bytes are transferred, the DMODE bit in the R32_QSPIx_CCR register is set to 00b. If R32_QSPIx_DLR = 0xFFFF_FFFF and FSIZE = 0x1F (data length of

4G bytes), during J-mode flash access, the configuration must ensure flash address + DLR $\leq$ FSIZE. In this case, the transfer will continue indefinitely, stopping only when QSPI is disabled or a terminate request occurs. After reading the last memory address (address 0xFFFF_FFFF), reading will resume starting from address 0x0000_0000.

When the programmed number of bytes to transmit or receive is reached, if TCIE=1, TCF is set to 1 and an interrupt is generated. When the data quantity is uncertain, the system will also set TCF to 1 when the external SPI limit is reached, based on the FLASH size defined in the R32_QSPIx_CR register.

QSPI supports write flow control. This means that when the length filled in the DLR has not been fully transmitted and the FIFO is empty, the shift register transmission is paused. In this mode, for efficiency, the software should first write to the DR register to fill data into the FIFO, or set DMAEN to 1 to automatically request a certain amount of data from the DMA before writing the START bit to initiate transmission.

Trigger Command Start

Essentially, once the firmware provides the last information required by the command, the command will start. According to the configuration of QSPI, in indirect mode, there are three ways to trigger the command to start:

- 1) Perform a write operation to the INSTRUCTION[7:0] field of the R32_QSPIx_CCR register. This is required when ADMODE=00b (address is mandatory) and when no data is provided by the firmware (when FMODE=01b or DMODE=00b);
- 2) Write to the ADDRESS[31:0] field of the R32_QSPIx_AR register if an address is required (when ADMODE=00b) and no data is provided by the firmware (when FMODE=01b or DMODE=00b);
- 3) Write to the DATA[31:0] field of the R32_QSPIx_DR register if an address is required (when ADMODE $\neq$ 00b) and data is provided by the firmware (when FMODE = 00b and DMODE $\neq$ 00b).

Writing to the Alternate Byte Register (R32_QSPIx_ABR) never triggers a command start. If alternate bytes are required, they must be programmed in advance. Once a command starts, the BUSY bit in the R32_QSPIx_SR register is automatically set to 1.

FIFO and Data Management

In indirect mode, data is transferred through the 32-byte FIFO internal to the QSPI. The FLEVEL[5:0] bits of the R32_QSPIx_SR register indicate how many bytes are currently stored in the FIFO.

In indirect write mode (FMODE=00b), when the firmware writes data to the R32_QSPIx_DR register, the data is appended to the FIFO. A word write adds 4 bytes to the FIFO, a half-word write adds 2 bytes, and a byte write adds only 1 byte. If the firmware writes data to the FIFO exceeding the value indicated by DL[31:0], any excess bytes are cleared from the FIFO upon completion of the write operation (when TCF is set to 1). Access to bytes/half-words in the R32_QSPIx_DR register must be strictly confined to the least significant byte/half-word of this 32-bit register.

FTHRES[4:0] defines the FIFO threshold. When the threshold is reached, the FTF (FIFO Threshold Flag) is set to 1. In indirect read mode, the FTF is set to 1 when the number of valid bytes read from the FIFO exceeds the threshold. After reading the last byte from FLASH, if data remains in the FIFO, FTF will be set to 1 regardless of the FTHRES setting. In indirect write mode, FTF is set to 1 when the number of empty bytes in the FIFO exceeds the threshold. If FTIE=1, an interrupt is generated when FTF is set to 1. Once the threshold condition is no longer met, that is, after the CPU or DMA has transmitted enough data, the HW will clear the FTF.

In indirect mode, QSPI will pause byte reads from Flash when the FIFO is full to prevent overflow. Reading from FLASH resumes only when four bytes in the FIFO are empty (FLEVEL $\leq$ 11). Therefore, if FTHRES $\geq$ 13, the

application must read a sufficient number of bytes to ensure QSPI can continue retrieving data from FLASH. Conversely, whenever $11 < \text{FLEVEL} < \text{FTHRES}$, the FTF flag will remain at 0.

37.3.4 QSPI Status Flag Polling Mode

In automatic polling mode, QSPI periodically starts commands to read a certain number of status bytes (up to 4). By masking the received bytes, some status bits can be isolated. When the selected bit has a preset value, an interrupt can be generated.

Access to FLASH starts in the same way as in indirect reading mode: if an address (AMODE=00b) is not needed, access will start when R32_QSPIx_CCR is written. Conversely, if an address is required, the first access will start when R32_QSPIx_AR is written. At this time, BUSY becomes high level and remains unchanged even during periodic access.

In automatic polling mode, the contents of the R32_QSPIx_PSMAR register's MASK[31:0] bits are used to mask data from the FLASH. If MASK[n]=0, bit n in the result is masked and ignored. If MASK[n]=1 and the content of bit [n] matches the content of bit [n] in the R32_QSPIx_PSMAR register's MATCH[n] field, bit n is considered a match.

If the Polling Match Mode bit (PMM, bit 23 of R32_QSPIx_CR) is 0, the “AND” match mode is activated. This means the Status Match Flag (SMF) is set only when all unmasked bits match. If PMM=1, the “OR” match mode is activated, meaning SMF is set to 1 if a match occurs on any unmasked bit. If SMIE=1, an interrupt is triggered when SMF is set to 1. If the automatic polling mode is stopped (APMS=1), the operation halts and the BUSY bit is cleared upon detecting a match. Otherwise, the BUSY bit remains set, and periodic accesses continue until an abort occurs or QSPI is disabled (EN=0).

The data register (R32_QSPIx_DR) contains the most recently received status byte (FIFO disabled). The contents of the data register are unaffected by the masking method used by the matching logic. The FTF status bit is set to 1 after a new status read completes; once data is read, FTF is cleared.

37.3.5 QSPI Memory Mapping Mode

When configured in memory-mapped mode, the external SPI device is treated as internal memory. If the QSPI peripheral is not properly configured and enabled, the QSPI Flash memory area cannot be accessed. Even with larger FLASH capacity, its address space is limited to a maximum of 256MB.

Note: For chips where the fifth digit of the lot number is less than 3, the memory-mapped addresses may only be used for data access, not for instruction access. Memory mapping function is not supported in dual flash mode.

Byte, half-word, and word access types are supported. The QSPI anticipates subsequent microcontroller accesses and preloads bytes from later addresses. Subsequent accesses to contiguous addresses are accelerated because the data has already been fetched.

By default, the QSPI never interrupts its pre-fetch operation. Even during extended periods without access to the FLASH, it maintains the previous read operation, and nCS remains at a low level. However, when nCS stays low, FLASH power consumption increases. Therefore, the application may activate the timeout counter (TCEN=1, bit 3 of R32_QSPIx_CR). Consequently, after the FIFO is filled with prefetched data, nCS is released if no access occurs within TIMEOUT[15:0] (R32_QSPIx_LPTR) cycles. Upon the first memory-mapped access, the BUSY signal conversions high. Due to the prefetch operation, the BUSY signal remains high until a timeout occurs, an abort is

triggered, or the peripheral is disabled.

37.3.6 QSPI FLASH Configuration

The device configuration register (R32_QSPIx_DCR) is used to define the characteristic parameters of the external SPI FLASH. The FSIZE[4:0] field determines the capacity of the external memory according to the following formula:

$$\text{Number of bytes in FLASH} = 2^{[\text{FSIZE}+1]}$$

FSIZE+1 represents the number of address bits required for FLASH addressing. In indirect mode, the maximum FLASH capacity reaches 4GB (using 32-bit addressing), but the addressable space in memory-mapped mode is limited to 256MB. If DFM=1, FSIZE denotes the combined capacity of 2 FLASH chips.

When QSPI executes two consecutive commands, it holds the chip select signal (nCS) high between commands for a default duration of one SCK cycle. If the FLASH requires a longer interval between commands, the Chip Select High Time (CSHT) field can be used to specify the minimum number of SCK cycles (maximum 8) that nCS must remain high. The clock mode (CKMODE) bit indicates the logic level of the SCK signal between commands (when nCS=1).

37.3.7 QSPI Configuration

QSPI configuration occurs in 2 stages: QSPI IP configuration and QSPI FLASH configuration. Once configured and enabled, the QSPI can operate in one of three modes: Indirect Mode, Status Polling Mode, and Memory-Mapped Mode.

QSPI IP Configuration: Configured via R32_QSPIx_CR. The user must configure the clock prescaler division factor for incoming data and the sampling shift settings. The FIFO level for generating DMA triggers or interrupts is programmed in the FTHRES bit. If a timeout counter is required, set the TCEN bit to 1 and program the timeout value in the R32_QSPIx_LPTR register. Dual flash mode can be activated by setting DFM to 1.

QSPI FLASH Configuration: Configure parameters related to the external target FLASH via the R32_QSPIx_DCR register. Users must program the FLASH size in the FSIZE bit, program the minimum hold time for the chip select high level in the CSHT bit, and set the functional mode (Mode 0 or Mode 3) in the MODE bit.

37.3.8 Usage of QSPI

Select the operation mode through R32_QSPIx_CCR register FMODE[1:0].

Indirect Mode

When FMODE=00b, the indirect write mode is selected, enabling data transmission to the FLASH. When FMODE=01b, the indirect read mode is selected, allowing data retrieval from the FLASH.

When QSPI operates in indirect mode, frames are constructed as follows:

- 1) Specify the target address in the R32_QSPIx_AR register;
- 2) Specify the operating mode in the R32_QSPIx_CR register;
- 3) Specify the number of bytes to read or write in the R32_QSPIx_DLR register;
- 4) Specify the frame format, mode, and command code in the R32_QSPIx_CCR register;
- 5) Read data from/write data to the FIFO via the R32_QSPIx_DR register;
- 6) Specify optional alternate bytes to be sent immediately after the address phase in the R32_QSPIx_ABR register.

When writing to the control register (R32_QSPIx_CR), the user can specify the following settings:

- Enable bit (EN) set to 1

- Timeout counter enable bit (TCEN)
- FIFO threshold (FTRHES), indicating when the FTF flag should be set to 1
- Interrupt enable
- Auto-polling mode parameters: Match mode and Stop mode (effective when FMODE=11b)
- Clock prescaler

When writing to the communication configuration register (R32_QSPIx_CCR), the user specifies the following parameters:

- DCYC bit specifies the number of empty instruction bytes
- DBMODE bit specifies whether empty instruction bytes exist
- INSTRUCTION bit specifies the instruction byte
- Specify the number of alternate bytes (1/2/3/4) via the ABSIZE bit
- Specify the instruction transmission mode (1/2/4-wire) via the IMODE bit
- Specify the address length (8/16/24/32 bits) via the ADSIZE bit
- Specify the address transmission mode (none/1/2/4-wire) via the ADMODE bit
- Specify alternate byte transmission mode (None/1-wire/2-wire/4-wire) via the ABMODE bit
- Specify data transmission/reception mode (None/1-wire/2-wire/4-wire) via the DMODE bit

If no data register (R32_QSPIx_DR) or address register (R32_QSPIx_AR) needs updating for a command, the command sequence initiates immediately upon writing the R32_QSPIx_CCR register. This occurs when both ADMODE and DMODE are 00b, or when ADMODE=00b alone in indirect read mode (FMODE=01b). When an address is required (ADMODE ≠ 00) and no data register write is needed (FMODE = 01b or DMODE = 00b), the command sequence starts immediately upon updating the address via writing to the R32_QSPIx_AR register. In data transfer scenarios (FMODE=00b and DMODE!=00b), writing to the FIFO via the R32_QSPIx_DR register triggers communication initiation.

Status Flag Polling Mode

Enable Status Flag Polling Mode: Set the FMODE field in the R32_QSPIx_CCR register to 10b. In this mode, programmed frames are sent and data is periodically retrieved. The maximum data read per frame is 4 bytes. If the data requested in the R32_QSPIx_DLR register exceeds 4 bytes, the excess is ignored and only 4 bytes are read. The periodicity is specified by the R32_QSPIx_PISR register.

After retrieving status data, it can be processed internally to achieve the following purposes:

- Automatically stops periodic status byte retrieval
- Sets the status match flag to 1 and generates an interrupt when enabled

The received value can be masked using the value stored in the R32_QSPIx_PSMKR register, and then ORed or ANDed with the value stored in the R32_QSPIx_PSMAR register. If a match exists, the status match flag is set to 1 and an interrupt is generated if enabled; if AMPS is set to 1, the QSPI automatically stops. In all cases, the latest read value is available in the R32_QSPIx_DR register.

Memory-Mapped Mode

In memory-mapped mode, the external FLASH is treated as internal memory despite access latency. Only read operations to the external FLASH are supported in this mode. Memory-mapped mode can be activated by setting FMODE to 11b in the R32_QSPIx_CCR register. When the host device accesses the memory-mapped space, it transmits the configured frames and commands.

The FIFO functions as a prefetch buffer to support continuous linear read operations. In this mode, any access to the R32_QSPIx_DR register returns zero. The data length register (R32_QSPIx_DLR) is invalid in memory-mapped mode.

37.3.9 Instruction Sent Only Once

Certain FLASH devices (e.g., Winbound) support a mode where commands must be sent using only the first command sequence, with subsequent commands executed directly based on the address. Users enable this feature by setting the SIOO bit in the R32_QSPIx_CCR register.

The SIOO bit applies to all functional modes: Indirect Mode, Status Polling Mode, and Memory-Mapped Mode. When SIOO is set to 1, only the first command is sent as an instruction, and a write operation is performed on the R32_QSPIx_CCR register. Subsequent command sequences skip the instruction phase until the R32_QSPIx_CCR register is written again. However, the SIOO bit is ineffective when IMODE=00b (no instruction).

37.3.10 QSPI Error Management

The possible error situations are:

- In indirect mode or status flag polling mode, if an incorrect address is set in the R32_QSPIx_AR register, TEF will be set to 1 based on the FLASH size defined by FSIZE[4:0] in the R32_QSPIx_DCR register. If enabled, an interrupt will also be generated.
- In indirect mode, if the sum of the address and data length exceeds the FLASH size, TEF will be set to 1 when the access is triggered.

37.3.11 Busy Bit and Abort Function of QSPI

When QSPI begins operating on the FLASH, the BUSY bit in the R32_QSPIx_SR register is automatically set to 1. In indirect mode, the BUSY bit is cleared when QSPI completes the requested command sequence and the FIFO is empty; In automatic polling mode, the BUSY bit goes low only after the last periodic access completes, either due to a match when APMS=1 or due to an abort operation. In memory-mapped mode, after the first access, the BUSY bit goes low only upon a timeout event or an abort.

Any operation can be aborted by setting the ABORT bit to 1 in the R32_QSPIx_CR register. Upon completion of the abort, the BUSY and ABORT bits automatically reset, and the FIFO is cleared.

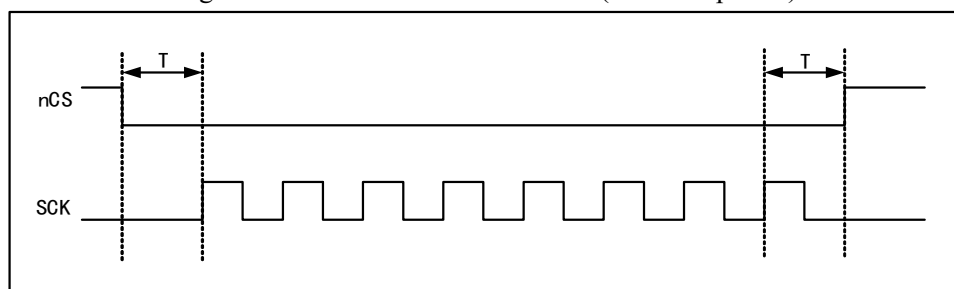
Note: If the writing operation to the status register is stopped, some FLASH may have abnormal behavior.

37.3.12 NCS Behavior

By default, nCS is high, disabling external FLASH. nCS goes low before the operation begins and immediately goes high after the operation completes.

When CKMODE=0 ("Mode 0", where SCK remains low when no operation is in progress), nCS is pulled low one SCK cycle before the first rising edge of SCK during an operation and pulled high one SCK cycle after the last rising edge of SCK during an operation, as shown in the figure below.

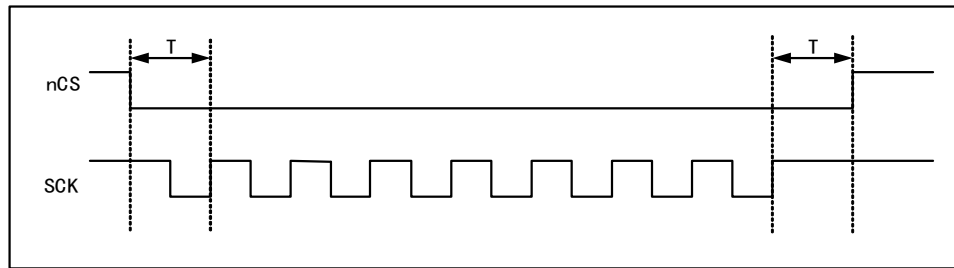
Figure 37-4 nCS with CKMODE = 0 (T = SCK period)



When CKMODE=1 ('Mode 3', where SCK rises to high level without any operation), nCS still falls to low level

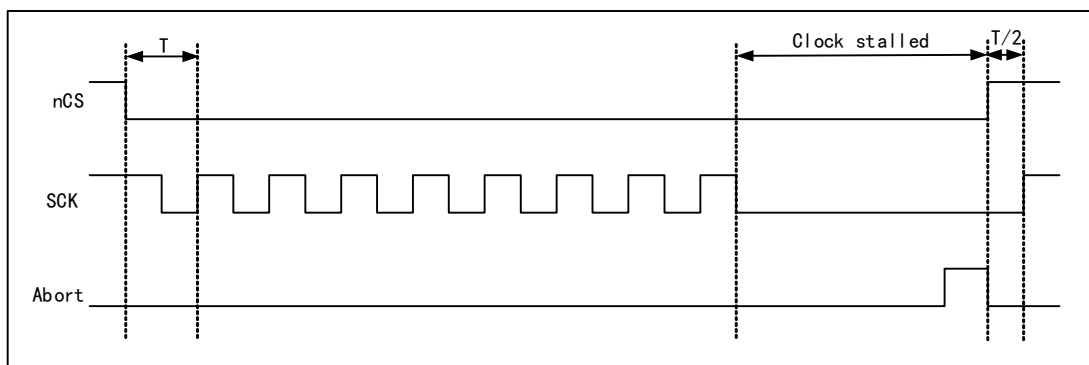
one SCK cycle before the first rising edge of SCK during operation, and rises to high level one SCK cycle after the last rising edge of SCK during operation, as shown in the figure below.

Figure 37-5 nCS with CKMODE = 1 ($T = \text{SCK period}$)



If the FIFO remains full in a read operation or empty in a write operation, the operation stops and SCK remains low before the firmware intervenes in the FIFO. If the operation stops, the nCS will rise to the high level after requesting the suspension, and the SCK will rise to the high level after half a cycle, as shown in the following figure.

Figure 37-6 nCS($T = \text{SCK period}$) when CKMODE = 1 and termination occurs



When not in Dual Flash Mode ($\text{DFM}=0$) and $\text{FSEL}=0$ (default), only FLASH1 is accessed. Therefore, if $\text{FSEL}=1$, only FLASH1 is accessed, and nCS remains high. In Dual Flash Mode, the nCS behavior for both FLASH1 and FLASH2 is identical. Consequently, if FLASH2 is present and the application always operates in Dual Flash Mode, the nCS signal for FLASH1 can also be used for FLASH2, while the nCS pin output for FLASH2 can be repurposed for other functions.

37.3.13 Interrupt

Table 37-1 Interrupt request

Interrupt event	Event flag	Enable control bit
Overtime	TOF	TOIE
State match	SMF	SMIE
FIFO threshold	FTF	FTIE
Transmission complete	TCF	TCIE
Transmission error	TEF	TEIE

37.4 Register Description

Table 37-2 QSPI1-related register

Name	Access address	Description	Reset value
R32_QSPI1_CR	0x40024C00	QSPI1 control register	0x00000000
R32_QSPI1_DCR	0x40024C04	QSPI1 device configuration register	0x00000000
R32_QSPI1_SR	0x40024C08	QSPI1 status register	0x00000000
R32_QSPI1_FCR	0x40024C0C	QSPI1 flag clear register	0x00000000
R32_QSPI1_DLR	0x40024C10	QSPI1 data length register	0x00000000
R32_QSPI1_CCR	0x40024C14	QSPI1 communication configuration register	0x00000000
R32_QSPI1_AR	0x40024C18	QSPI1 address register	0x00000000
R32_QSPI1_ABR	0x40024C1C	QSPI1 alternate byte register	0x00000000
R32_QSPI1_DR	0x40024C20	QSPI1 data register	0x00000000
R32_QSPI1_PSMKR	0x40024C24	QSPI1 polling status mask register	0x00000000
R32_QSPI1_PSMAR	0x40024C28	QSPI1 polling status match register	0x00000000
R32_QSPI1_PIR	0x40024C2C	QSPI1 polling interval register	0x00000000
R32_QSPI1_LPTR	0x40024C30	QSPI1 low-power timeout register	0x00000000

Table 37-3 QSPI2-related register

Name	Access address	Description	Reset value
R32_QSPI2_CR	0x40025000	QSPI2 control register	0x00000000
R32_QSPI2_DCR	0x40025004	QSPI2 device configuration register	0x00000000
R32_QSPI2_SR	0x40025008	QSPI2 status register	0x00000000
R32_QSPI2_FCR	0x4002500C	QSPI2 flag clear register	0x00000000
R32_QSPI2_DLR	0x40025010	QSPI2 data length register	0x00000000
R32_QSPI2_CCR	0x40025014	QSPI2 communication configuration register	0x00000000
R32_QSPI2_AR	0x40025018	QSPI2 address register	0x00000000
R32_QSPI2_ABR	0x4002501C	QSPI2 alternate byte register	0x00000000
R32_QSPI2_DR	0x40025020	QSPI2 data register	0x00000000
R32_QSPI2_PSMKR	0x40025024	QSPI2 polling status mask register	0x00000000
R32_QSPI2_PSMAR	0x40025028	QSPI2 polling status match register	0x00000000
R32_QSPI2_PIR	0x4002502C	QSPI2 polling interval register	0x00000000
R32_QSPI2_LPTR	0x40025030	QSPI2 low-power timeout register	0x00000000

37.4.1 QSPI Control Register (R32_QSPIx_CR) (x=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRESCALER[7:0]								PMM	APMS	Reserved	TOIE	SMIE	FTIE	TCIE	TEIE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		SIO XE	FTHRES[4:0]					FSEL	DFM	START	Reserved	TCEN	DMA EN	ABOR T	EN

	N									
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Bit	Name	Access	Description	Reset value
[31:24]	PRESCALER[7:0]	RW	Clock prescaler: This field defines the prescaler factor (value + 1) used to generate SCK from the HB bus clock. 0x0: F_{SCK} = F_{HCLK}; HCLK clock is directly used as QSPI SCK (prescaler bypassed); 0x1: F_{SCK} = F_{HCLK}/2; 0x2: F_{SCK} = F_{HCLK}/3; ... 0xFF: F_{SCK} = F_{HCLK}/256. For odd clock division factors, the SCK duty cycle is not 50%. The low level duration of the clock signal is one cycle longer than the high level duration. This field can only be modified when BUSY=0.	0
23	PMM	RW	Polling matching mode: 1: OR matching mode. If any unmasked bit received from FLASH matches the corresponding bit in the matching register, the SMF is set to 1; 0: AND matching mode. If all unmasked bits received from FLASH match the corresponding bits in the matching register, the SMF is set to 1. This bit can only be modified when BUSY=0.	0
22	APMS	RW	Automatic polling mode stop: 1: Automatic polling mode stops when a match occurs; 0: Automatic polling mode stops only by terminating or disabling QSPI. This bit can only be modified when BUSY=0.	0
21	Reserved	RO	Reserved	0
20	TOIE	RW	Timeout interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
19	SMIE	RW	Status matching interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
18	FTIE	RW	FIFO threshold interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
17	TCIE	RW	Transmission completion interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
16	TEIE	RW	Transmission error interrupt enable:	0

			1: Enable interrupt; 0: Disable interrupt.	
[15:14]	Reserved	RO	Reserved	0
13	SIOXEN	RW	SIOX enable: 1: Enable 4-wire IO output; 0: Disable 4-wire IO output.	0
[12:8]	FTHRES[4:0]	RW	FIFO Threshold Level: Defines the byte threshold in indirect mode that will cause the FIFO threshold flag (FTF, R32_QSPIx_SR[2]) to be set to 1. In Indirect Write Mode (FMODE=00): 00000: FTF is set to 1 if 1 or more free bytes are available for writing in the FIFO; 00001: FTF is set to 1 if 2 or more free bytes are available for writing in the FIFO; ... 11111: FTF is set to 1 if 32 free bytes are available for writing in the FIFO; In indirect read mode (FMODE=01): 00000: If one or more valid bytes are available for reading in the FIFO, FTF is set to 1; 00001: If two or more valid bytes are available for reading in the FIFO, FTF is set to 1; ... 11111: If 32 valid bytes are available for reading in the FIFO, FTF is set to 1.	0
7	FSEL	RW	FLASH selection: 1: Select FLASH2; 0: Select FLASH1. This bit can only be modified when BUSY=0. It is ignored when DFM=1.	0
6	DFM	RW	Dual Flash mode: 1: Enable dual flash mode; 0: Disable dual flash mode. This bit can only be modified when BUSY=0.	0
5	START	WZ	In indirect mode and status polling mode, after the required registers have been configured, writing a 1 to the START bit initiates the flash command sequence. This bit is automatically cleared by hardware.	0
4	Reserved	RO	Reserved	0
3	TCEN	RW	Timeout counter enable: This bit is active only in memory-mapped mode (FMODE=11). When enabled, the chip select (nCS)	0

			is released after a defined period of inactivity (specified by TIMEOUT[15:0] (R32_QSPIx_LPTR)), thereby reducing power consumption. Enables the timeout counter. By default, even if the FLASH remains unaccessed for extended periods, the QSPI does not cease its pre-fetch operations. Previous read operations remain active, and nCS stays at a low level. Since power consumption increases when nCS remains low, the application may activate the timeout counter (TCEN=1, bit 3 of R32_QSPIx_CR). Consequently, after the FIFO is filled with prefetched data, nCS is released if no access occurs within TIMEOUT[15:0] (R32_QSPIx_LPTR) cycles. 1: Enable the timeout counter. In memory-mapped mode, the chip select (nCS) is released after TIMEOUT[15:0] cycles of continuous inactivity in the FLASH. 0: Disable the timeout counter. In memory-mapped mode, the chip select (nCS) remains active after an access operation. This bit can only be modified when BUSY=0.	
2	DMAEN	RW	DMA mode enable: 1: Enable DMA; 0: Disable DMA.	0
1	ABORT	RW	Abort request/Clear FIFO: 1: Request abort/clear FIFO; 0: Do not request abort/clear FIFO. Abort Request: This bit aborts the currently executing command sequence. It resets automatically upon completion of the abort. This bit can halt the current transfer. In polled mode or memory-mapped mode, this bit is also used to reset the APMS bit or DMAEN bit. Clear FIFO: Programming this bit to 1 clears the FIFO. All internal FIFO pointers (read and write) will be reset to zero. Data remaining in the FIFO at this time will be lost (transmitted or received data will not continue to be lost). When the QSPI read/write direction is switched, the FIFO must be cleared once. When the IDLEF bit in	0

			the R32_QSPIx_SR register is set to 1 and the ABORT bit is enabled, a transfer complete flag is generated, indicated by the TCF bit in the R32_QSPIx_SR register.	
0	EN	RW	Enable: 1: Enable QSPI; 0: Disable QSPI.	0

37.4.2 QSPI Device Configuration Register (R32_QSPIx_DCR) (x=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved											FSIZE[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CSHT[2:0]			Reserved					CKMODE		

Bit	Name	Access	Description	Reset value
[31:21]	Reserved	RO	Reserved	0
[20:16]	FSIZE[4:0]	RW	FLASH Size: Number of bytes in FLASH = $2^{[FSIZE+1]}$, where FSIZE+1 represents the number of address bits required for FLASH addressing. In indirect mode, the maximum FLASH capacity reaches 4GB (using 32-bit addressing), but the addressable space in memory-mapped mode is limited to 256MB. If DFM=1, FSIZE represents the combined capacity of two FLASH banks. This field can only be modified when BUSY=0.	0
[15:11]	Reserved	RO	Reserved	0
[10:8]	CSHT[2:0]	RW	Chip Select High Duration: CSHT+1 defines the minimum number of SCK cycles that the chip select (nCS) must remain high between commands sent to the FLASH. When PRESCALER[7:0] bits are configured to non-0x2: 000: nCS remains high for at least 1 cycle between FLASH commands; 001: nCS remains high for at least 2 cycles between FLASH commands; ... 111: nCS remains high for at least 8 cycles between FLASH commands.	0

			When PRESCALER[7:0] bits are configured as 0x2: 000: nCS remains high for at least one clock cycle between FLASH commands; 001: nCS remains high for at least two clock cycles between FLASH commands; ... 110: nCS must remain high for at least 7 cycles between FLASH commands; 111: Reserved. This field can only be modified when BUSY=0.	
[7:1]	Reserved	RO	Reserved	0
0	CKMODE	RW	Mode 0/Mode 3: 1: When nCS is high (chip select released), SCK must remain high. This is referred to as Mode 3; 0: When nCS is high (chip select released), SCK must remain low. This is referred to as Mode 0. This field can only be modified when BUSY=0.	0

37.4.3 QSPI Status Register (R32_QSPIx_SR) (x=1/2)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															IDL EF
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		FLEVEL[5:0]					Reserved		BUS Y	TO F	SM F	FTF	TC F	TE F	

Bit	Name	Access	Description	Reset value
[31:17]	Reserved	RO	Reserved	0
16	IDLEF	RO	Idle flag: When this bit is set to 1, the QSPI bus is in idle state.	0
[15:14]	Reserved	RO	Reserved	0
[13:8]	FLEVEL[5:0]	RO	FIFO level: This field indicates the number of valid bytes in the FIFO. When the FIFO is empty, FLEVEL=0; when full, FLEVEL=32. In memory-mapped mode and automatic status polling mode, FLEVEL is zero.	0
[7:6]	Reserved	RO	Reserved	0
5	BUSY	RO	Busy: This bit is set during operation. It automatically clears	0

			when the FLASH operation completes and the FIFO is empty.	
4	TOF	RO	Timeout flag: When a timeout occurs, this bit is set to 1. Writing a 1 to CTOF clears this bit.	0
3	SMF	RO	Status match flag: In automatic polling mode, if unmasked received data matches the corresponding phase in the match register (R32_QSPIx_PSMAR), this bit is set to 1. Writing a 1 to CSMF clears this bit.	0
2	FTF	RO	FIFO threshold flag: In indirect mode, this bit is set to 1 when the FIFO threshold is reached or when data remains in the FIFO after a read operation from FLASH completes. It automatically clears when the threshold condition is no longer true. In automatic polling mode, this bit is set to 1 each time the status register is read; it is cleared when the data register is read.	0
1	TCF	RO	Transmission complete flag: In indirect mode, this bit is set to 1 when the transmitted data volume reaches the programmed set value, or when transmission is terminated in any mode. Writing a 1 to CTCF clears this bit.	0
0	TEF	RO	Transmission error flag: This bit is set to 1 when accessing an invalid address in indirect mode. Writing a 1 to CTEF clears this bit.	0

37.4.4 QSPI Flag Clear Register (R32_QSPIx_FCR) (x=1/2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											CT OF	CS MF	Reser ved	CT CF	CT EF

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
4	CTOF	RW1Z	Clear timeout flag: Writing a 1 to the R32_QSPIx_SR register clears the TOF flag.	0
3	CSMF	RW1Z	Clear status match flag:	0

			Writing a 1 clears the SMF flag in the R32_QSPIx_SR register.	
2	Reserved	RO	Reserved	0
1	CTCF	RW1Z	Clear transmission complete flag: Writing a 1 clears the TCF flag in the R32_QSPIx_SR register.	0
0	CTEF	RW1Z	Clear transmission error flag: Writing a 1 to the R32_QSPIx_SR register clears the TEF flag.	0

37.4.5 QSPI Data Length Register (R32_QSPIx_DLR) (x=1/2)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DL[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DL[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	DL[31:0]	RW	Data length: The number of data items to be retrieved (value + 1) in Indirect Mode and Status Polling Mode. For Status Polling Mode, use a value no greater than 3 (indicating 4 bytes). In Indirect Mode, all position 1s indicate undefined length; QSPI will continue transferring data until reaching the end of memory defined by FSIZE. 0x0000_0000: Transfer 1 byte; 0x0000_0001: Transfer 2 bytes; 0x0000_0002: Transfer 3 bytes; 0x0000_0003: Transfer 4 bytes; ... 0xFFFF_FFDD: Transfers 4,294,967,294 bytes (4G-2); 0xFFFF_FFDE: Transfers 4,294,967,295 bytes (4G-1); 0xFFFF_FFFF: Undefined length—transfers all bytes until reaching the end of FLASH defined by FSIZE. If FSIZE=0x1F, reading continues indefinitely. In Dual Flash Mode (DFM=1), DL[0] remains 1 even when this bit is written as 0, ensuring each access transfers an even number of bytes. This field is inactive in Memory-Mapped Mode	0

			(FMODE=10); It can only be written when BUSY=0.	
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37.4.6 QSPI Communication Configuration Register (R32_QSPIx_CCR) (x=1/2)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			SIO O	FMODE [1:0]	DMODE [1:0]	Rese rved	DCYC[4:0]				ABSIZE [1:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABMODE [1:0]		ADSIZE [1:0]		ADMODE [1:0]		IMODE [1:0]		INSTRUCTION[7:0]							

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved	0
28	SIOO	RW	Single command transmission mode: When IMODE=00, this bit is inactive. 1: Transmit commands only for the first command; 0: Transmit commands for every transaction. This field can only be written when BUSY=0.	0
[27:26]	FMODE[1:0]	RW	Function mode: 00: Indirect Write Mode; 01: Indirect Read Mode; 10: Auto Polling Mode; 11: Memory-Mapped Mode. This field can only be written when BUSY=0.	0
[25:24]	DMODE[1:0]	RW	Data mode: 00: No data; 01: Single-wire data transmission; 10: Dual-wire data transmission; 11: Quad-wire data transmission. This field also defines the empty command phase. This field may only be written when BUSY=0.	0
23	Reserved	RO	Reserved	0
[22:18]	DCYC[4:0]	RW	Empty instruction cycle count: This field defines the duration of the empty instruction phase. In SDR mode, it specifies the number of SCK cycles (0-31). This field can only be written when BUSY=0.	0
[17:16]	ABSIZE[1:0]	RW	Alternate byte length: 00: 8-bit alternate byte; 01: 16-bit alternate byte; 10: 24-bit alternate byte;	0

			11: 32-bit alternate byte. This field can only be written when BUSY=0.	
[15:14]	ABMODE[1:0]	RW	Alternate byte mode: 00: No alternate byte; 01: Single-wire transmission of alternate byte; 10: Dual-wire transmission of alternate byte; 11: Quad-wire transmission of alternate byte. This field can only be written when BUSY=0.	0
[13:12]	ADSIZE[1:0]	RW	Address length: 00: 8-bit address; 01: 16-bit address; 10: 24-bit address; 11: 32-bit address. This field can only be written when BUSY=0.	0
[11:10]	ADMODE[1:0]	RW	Address mode: 00: No address; 01: Single-wire transmission address; 10: Dual-wire transmission address; 11: Quad-wire transmission address. This field can only be written when BUSY=0.	0
[9:8]	IMODE[1:0]	RW	Command mode: 00: No command; 01: Single-wire transmission command; 10: Dual-wire transmission command; 11: Quad-wire transmission command; This field can only be written when BUSY=0.	0
[7:0]	INSTRUCTION[7:0]	RW	Command: Specifies the command to be sent to the external SPI device. This field may only be written when BUSY=0.	0

37.4.7 QSPI Address Register (R32_QSPIx_AR) (x=1/2)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDRESS[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDRESS[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	ADDRESS[31:0]	RW	Address: Specifies the address to be written to external FLASH.	0

			When BUSY=0 or FMODE=11 (memory-mapped mode), writes to this field are ignored. In dual-flash mode, since addresses are always even, ADDRESS[0] is automatically set to 0.	
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37.4.8 QSPI Alternate Byte Register (R32_QSPIx_ABR) (x=1/2)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ALTERNATE[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALTERNATE[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	ALTERNATE[31:0]	RW	Alternate byte: Specifies optional data to be sent to the external SPI device immediately following the address. This field may only be written when BUSY=0.	0

37.4.9 QSPI Data Register (R32_QSPIx_DR) (x=1/2)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	DATA[31:0]	RW	Data: Specifies the data to be exchanged with external SPI devices. In indirect write mode, data written to this register is sent to the FLASH during the data phase, having been stored in the FIFO beforehand. If the FIFO is too full, writing is paused until sufficient space becomes available to accept the data to be written. In indirect mode, reading this register retrieves data (via the FIFO) that has been received from FLASH. If the number of bytes in the FIFO is less than the number of bytes requested by the read operation and BUSY=1, the read operation will pause until sufficient data is available or the transfer completes	0

			(whichever occurs first). In automatic polling mode, this register contains the last data read from FLASH (unmasked). Word, half-word, and byte accesses are supported for this register. In indirect write mode, a byte write adds 1 byte to the FIFO, a half-word write adds 2 bytes, and a word write adds 4 bytes. Similarly, in indirect read mode, a byte read erases one byte from the FIFO, a half-word read erases two bytes, and a word read erases four bytes. Accesses in indirect mode must align with the least significant bits of this register: byte reads must access DATA[7:0], while half-word reads must access DATA[15:0].	
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37.4.10 QSPI Polling Status Mask Register (R32_QSPIx_PSMKR) (x=1/2)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MASK[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MASK[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MASK[31:0]	RW	Status masking: Masks the status byte received in polling mode. For bit n: 1: Does not mask bit n in data received during automatic polling mode; its value is considered in matching logic. 0: Masks bit n in data received during automatic polling mode; its value is ignored in matching logic. This field can only be written when BUSY=0.	0

37.4.11 QSPI Polling Status Match Register (R32_QSPIx_PSMAR) (x=1/2)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MATCH[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MATCH[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MATCH[31:0]	RW	Status matching: This value will be compared with the mask status register for matching. This field can only be written when BUSY=0.	0

37.4.12 QSPI Polling Interval Register (R32_QSPIx_PIR) (x=1/2)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTERVAL[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	MATCH[31:0]	RW	Polling Interval: The number of SCK cycles between read operations during the automatic polling phase. This field can only be written when BUSY=0.	0

37.4.13 QSPI Low-power Timeout Register (R32_QSPIx_LPTR) (x=1/2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMEOUT[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	TIMEOUT[15:0]	RW	Timeout duration: In memory-mapped mode, after each access completes, the QSPI pre-fetches subsequent bytes and stores them in the FIFO. This field specifies how many SCK clock cycles the QSPI waits after the FIFO fills before raising nCS high to place the FLASH into low-power mode. This field can only be written when BUSY=0.	0

Chapter 38 Single-wire Protocol Main Interface (SWPMI)

The Single Wire Protocol Master Interface (SWPMI) is a full-duplex single-wire communication technology that implements Single Wire Protocol (SWP) communication based on the ETSI TS 102 613 standard specification.

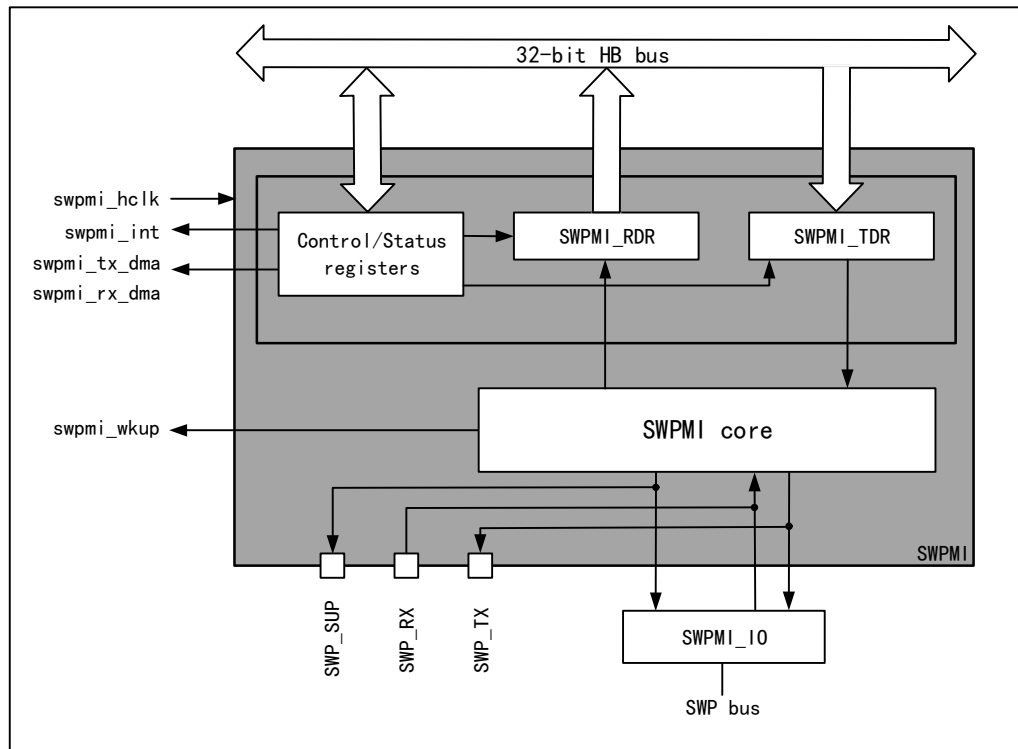
In SWPMI, data transmission occurs via two physical methods: First, data flows from the master device to the slave device through the voltage domain (S1 signal). Second, data flows from the slave device to the master device through the current domain (S2 signal). The S1 signal employs pulse width modulation (PWM) for digital transmission, while the S2 signal conveys data through current variations.

38.1 Main Features

- Support full-duplex communication mode
- Support automatic processing of fill bits
- Support automatic SWP bus state management
- Provide loopback mode for testing
- Support automatic processing of Start of Frame (SOF) and End of Frame (EOF)
- Support configurable bit rate up to 2Mbit/s with configurable interrupts
- Provide CRC error, underflow, overflow, and slave device recovery monitoring flags
- Support CRC-16 calculation, generation, and verification

38.2 Overview

Figure 38-1 Structure block diagram of single-wire protocol main interface



38.3 Register Description

38.3.1 SWP Initialization and Activation

SWP initialization and activation conversion the SWPMI_IO state from low to high. When using the internal transceiver, the software operation follows this sequence:

- 1) Enable the SWPMI_IO transceiver via bit 1 of the R32_SWPMI_CR register (SWPTEN) and set SWPMI_IO to low (SWP bus “disabled”);
- 2) Wait for bit 1 of the RDYF field in the R32_SWPMI_SR register (polling flag) or enable interrupts via the RDYIE bit in the R32_SWPMI_IER register;
- 3) Set bit 1 of the SWPACT field in the R32_SWPMI_CR register to conversion the SWP bus from “Disabled” to “Suspended” state, thereby activating SWP.

38.3.2 SWP Bus Status

The SWP bus has three states: “Disabled,” “Suspended,” and “Activated.”

State conversions include:

- Activate: Conversion from “Disabled” state to “Suspended” state;
- Suspend: Conversion from “Activated” state to “Suspended” state;
- Disable: Conversion from “Suspended” state to “Disabled” state;
- Master Device Recovery: Conversion initiated by the master device from “Suspended” state to “Activated” state;
- Slave Device Recovery: Conversion initiated by the slave device from “Suspended” state to “Activated” state.

Activation

During the reset phase and immediately after reset completion, SWPMI_IO is configured for emulation mode. Refer to Section 38.3.1 for the SWP bus activation procedure.

Suspended

If communication exists between the SWP bus and a slave device (regardless of whether in transmit or receive mode), the SWP bus remains in the “active” state. If no subsequent transmit or receive requests occur, the SWP bus will conversion back to the “suspended” state after seven idle bits.

Disable

1) Disabling Request

When the SWP is in the “Suspended” state, if communication is no longer required, the user may request switching the SWP to the ‘Disabled’ state by disabling the SWPMI peripheral. Setting the DEACT bit in the R32_SWPMI_CR register to 1 enables switching to the “Disabled” state. If the SWPMI does not detect a “Slave Device Recovery” state, the DEACTF bit in the R32_SWPMI_ISR register will be set to 1, and the SWPACT bit in the R32_SWPMI_ICR register will be cleared. If SWPMI detects a “Slave Device Recovery” state after DEACT is set to 1, the SRF flag in the R32_SWPMI_ISR register will be set to 1, DEACTF remains cleared, SWPACT remains set to 1, and the DEACT bit is cleared. Clear the DEACT bit in the R32_SWPMI_CR register, then set SWPACT to 1 to reactivate SWP.

2) Disabled Mode

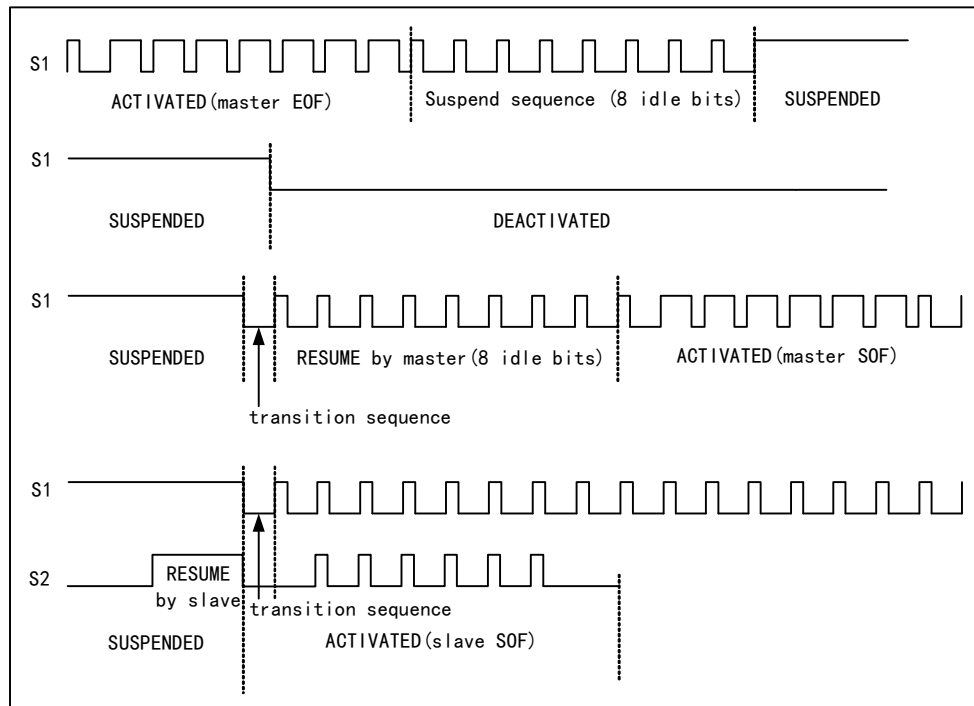
The user must clear the SWPACT bit in the R32_SWPMI_CR register for the SWP to immediately conversion to the “Disabled” mode and ignore any incoming “Slave Device Recovery” state.

Note: To further reduce current consumption, configure the SWPMI_IO port in the GPIO controller to push-pull output mode, set it to low, and finally clear the SWPTEN bit in the R32_SWPMI_CR register.

Master device recovery

After SWPMI is enabled, the user may request SWPMI frame transmission. First, SWPMI transmits a conversion sequence and eight idle bits, indicating master device recovery, then commences frame transmission. Following “Master Device Recovery,” the SWP conversions from the ‘Suspended’ state to the “Activated” state. Refer to Figure 38-2: SWP Bus States.

Figure 38-2 SWP bus state diagram



Slave device recovery

When SWPMI is enabled, if SWPMI receives a slave device recovery status signal, the SWP conversions from the “suspended” state to the ‘activated’ state. “Slave device recovery” triggers the SRF flag bit in the R32_SWPMI_ISR register to be set to 1.

38.3.3 SWPMI_IO (Internal Transceiver) Bypass

The microcontroller integrates an SWPMI_IO (internal transceiver) compliant with the ETSI TS 102 613 technical specification. To bypass this transceiver, set the SWP_TBYP bit in the R32_SWPMI_OR register to 1. In bypass mode, the SWPMI_IO is disabled, and the SWP_RX, SWP_TX, and SWPMI_SUP signals become available as multiplexed functions on these three GPIO pins. This configuration can be used to connect an external transceiver. Note: In SWPMI_IO bypass mode, the SWPTEN bit in the R32_SWPMI_CR register must be cleared.

38.3.4 SWPMI Baud Rate

In the R32_SWPMI_BRR register, set the bit rate according to the following formula:

$$F_{SWP} = F_{HCLK} / ((BR[7:0] + 1) \times 4)$$

Note: Maximum bit rate is 2 Mbit/s.

38.3.5 SWPMI Frame Processing

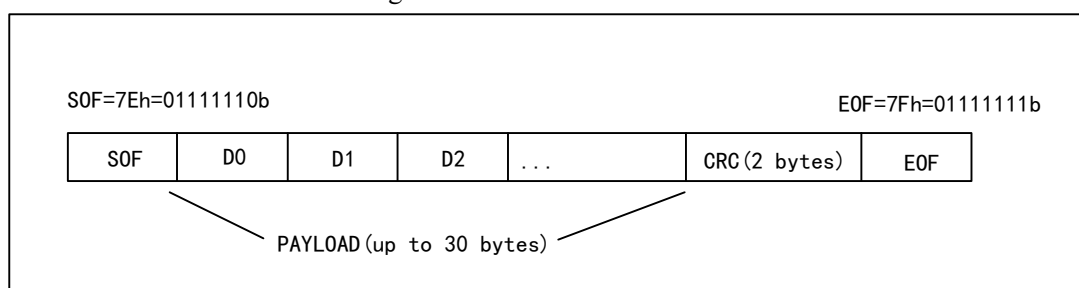
The SWP frame consists of the following components: Start of Frame (SOF), a payload of 1 to 30 bytes, a 16-bit CRC (Cyclic Redundancy Check), and End of Frame (EOF). See Figure 38-3.

The SWPMI integrates a 32-bit data transmit register (R32_SWPMI_TDR) and a 32-bit data receive register (R32_SWPMI_RDR).

In data transmission mode, SOF insertion, EOF insertion, CRC calculation, and insertion are all automatically managed by SWPMI. The user only needs to provide the payload size and content. After writing data to the R32_SWPMI_TDR register, frame transmission occurs immediately. Specific flag bits indicate when the transmission data register is empty and when the data frame transmission is complete.

In data reception mode, SOF deletion, EOF deletion, CRC calculation, and verification are all automatically managed by SWPMI. Users need only read the payload size and content. Specific flag bits will indicate frame reception completion, register full status, and potential CRC error events.

Figure 38-3 SWP frame structure



Insertion (in transmit mode) and deletion (in receive mode) of padding bits are automatically managed by SWPMI core. These operations are transparent to users.

38.3.6 Transmit Processing

Before transmitting a frame, the user must first activate the SWP; refer to Section 38.3.1. Software implementations for frame transmission include multiple modes: no software buffer mode, single software buffer mode, and multiple software buffer mode. When using software buffers, data is transferred via a DMA channel from the software buffer in RAM memory to the SWPMI peripheral's transmit data register.

No software buffer mode

In No Software Buffer Mode, DMA does not need to be enabled. Polling the status flag within the main loop or SWPMI interrupt routine handles SWP frame transmission. The SWPMI includes a 32-bit transmit data register (R32_SWPMI_TDR); writing to this register triggers transmission of up to 4 bytes.

Software-Bufferless Mode can be selected by clearing the TXDMA bit in the R32_SWPMI_CR register.

The first write to the R32_SWPMI_TDR register triggers frame transmission. Within the first word written to the R32_SWPMI_TDR register, bits [7:0] indicate the number of data bytes in the payload. Bits [15:8], bits [23:16], and bits [31:24] contain the first three bytes of the payload, where bits [15:8] contain the first byte, bits [23:16] contain the second byte, and bits [31:24] contain the third byte. Subsequent writes to the R32_SWPMI_TDR register will only contain additional payload data bytes (up to four bytes per write operation).

Note: The least significant byte of the first 32-bit word written to the R32_SWPMI_TDR register encodes the number of data bytes in the payload. This value must fall within the range of 1 to 30. If the least significant byte exceeds this range, it will be ignored and will not trigger data transmission.

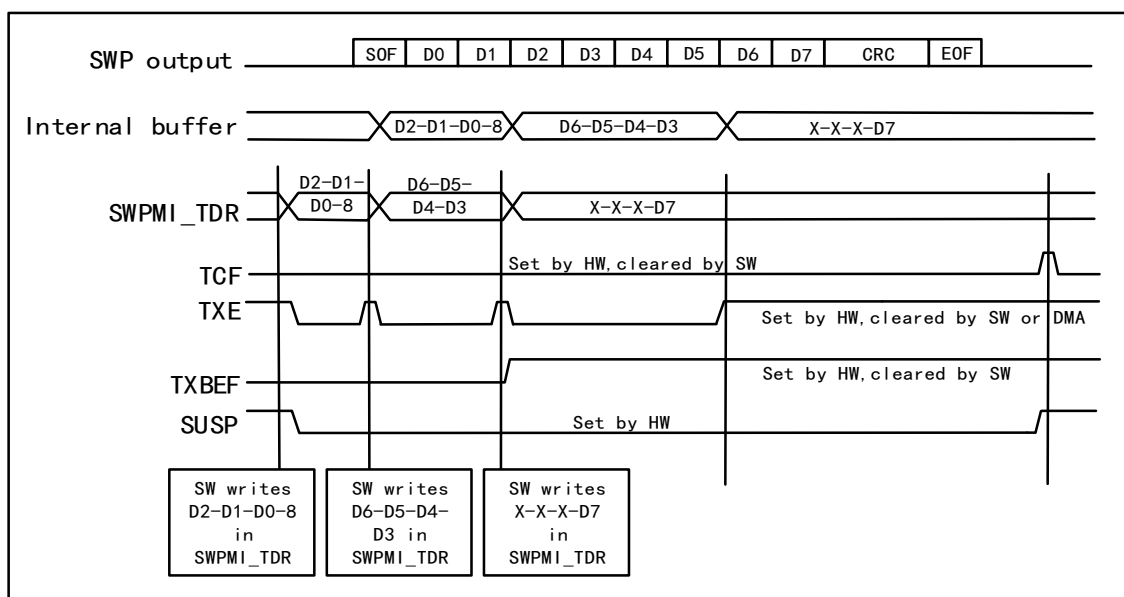
Writing to the R32_SWPMI_TDR register triggers the following sequence of operations:

- 1) When the SWP bus is in the “Suspended” state, the system sends a conversion sequence followed by 8 idle bits (for host recovery). Conversely, if the SWP bus is in the “Active” state, this operation is not triggered;
- 2) Frame start transmission;
- 3) Transmit the payload based on the contents of the R32_SWPMI_TRD register. If the payload contains more than 3 bytes, the R32_SWPMI_TDR must be refilled by software whenever the TXE flag in the R32_SWPMI_ISR register is set to 1, provided the TXBEF flag in the R32_SWPMI_ISR register is not set to 1;
- 4) Transmit the 16-bit CRC (automatically computed by the SWPMI core);
- 5) Transmit the end-of-frame (EOF) signal.

When the software performs a write operation to the R32_SWPMI_TDR register, it will automatically clear the TXE flag.

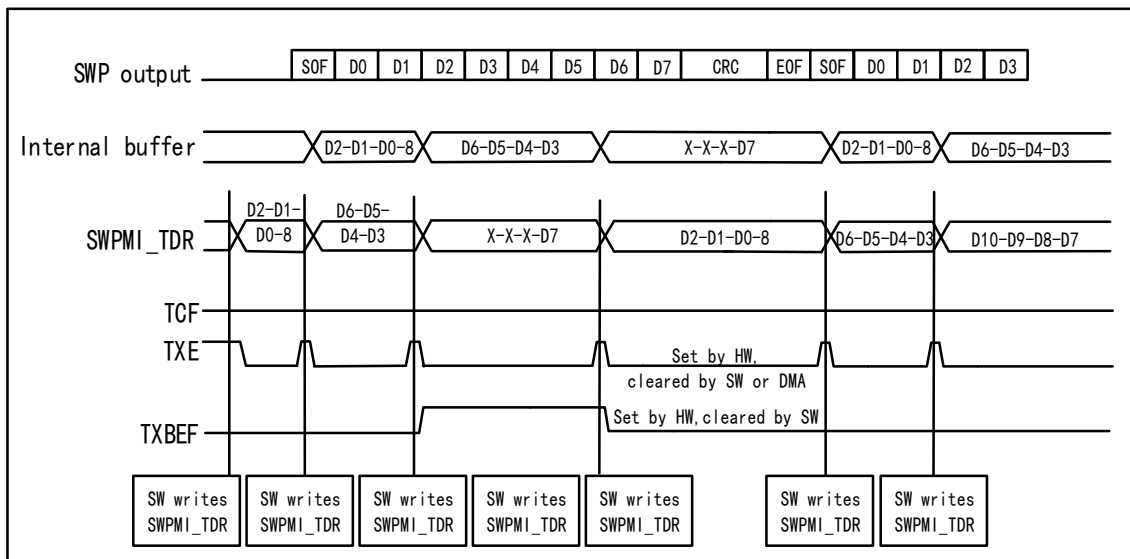
After a complete frame transmission succeeds, if no other frame transmission is requested (i.e., no subsequent write operation to the R32_SWPMI_TDR register occurs after the TXBEF flag is set to 1), the TCF and SUSP flags in the R32_SWPMI_ISR register will be set to 1 after 7 idle bits following the completion of the EOF transmission. Additionally, setting the TCIE bit in the R32_SWPMI_IER register triggers an interrupt. See Figure 38-4.

Figure 38-4 SWPMI no software buffer mode transmission



If the frame transmission request is initiated again before the EOF transmission ends, the TCF flag bit will not be set, and the frame will be connected with the previous frame (there is only an idle bit between them, please refer to Figure 38-5).

Figure 38-5 SWPMI no software buffer mode transmission, continuous frames



Single software buffer mode

In Single Software Buffer Mode, complete SWP frames can be transmitted via DMA without CPU intervention. DMA will refill the 32-bit R32_SWPMI_TDR register. Software can poll the SWPMI_TXBEF flag to determine if frame transmission has completed. To select Single Software Buffer Mode: Clear the TXDMA bit 1 and TXMODE bit in the R32_SWPMI_CR register.

The DMA channel or data stream must be configured in one of the following modes (refer to the DMA section):

- Memory size set to 32 bits;
- Peripheral size set to 32 bits;
- Enable memory increment mode;
- Disable cycle mode;
- Disable peripheral increment mode;
- Disable memory-to-memory mode;
- Number of words to transfer must be set according to SWP frame length;
- Target address is the R32_SWPMI_TDR register;
- Data transfer direction is set to read from memory.
- Source address is the SWP frame buffer in RAM.

Subsequently, the user performs the following operations:

- 1) Set the TXDMA bit in the R32_SWPMI_CR register to 1;
- 2) Set the TXBEIE bit in the R32_SWPMI_IER register to 1;
- 3) Populate the buffer in RAM memory (the number of data bytes in the payload is indicated by the least significant byte of the first word);
- 4) Trigger the DMA transfer and frame transmission by enabling the data stream or channel within the DMA module.

When the TXE flag in the R32_SWPMI_ISR is set to 1, the SWPMI will initiate a DMA request. During the DMA write operation to the R32_SWPMI_TDR register, the TXE flag bit will be automatically cleared.

Within the SWPMI interrupt service routine, the user must check the TXBEF bit in the R32_SWPMI_ISR register. If this bit is set and additional frames remain to be transmitted, the user must perform the following actions:

- 1) Disable the data stream or channel within the DMA module;

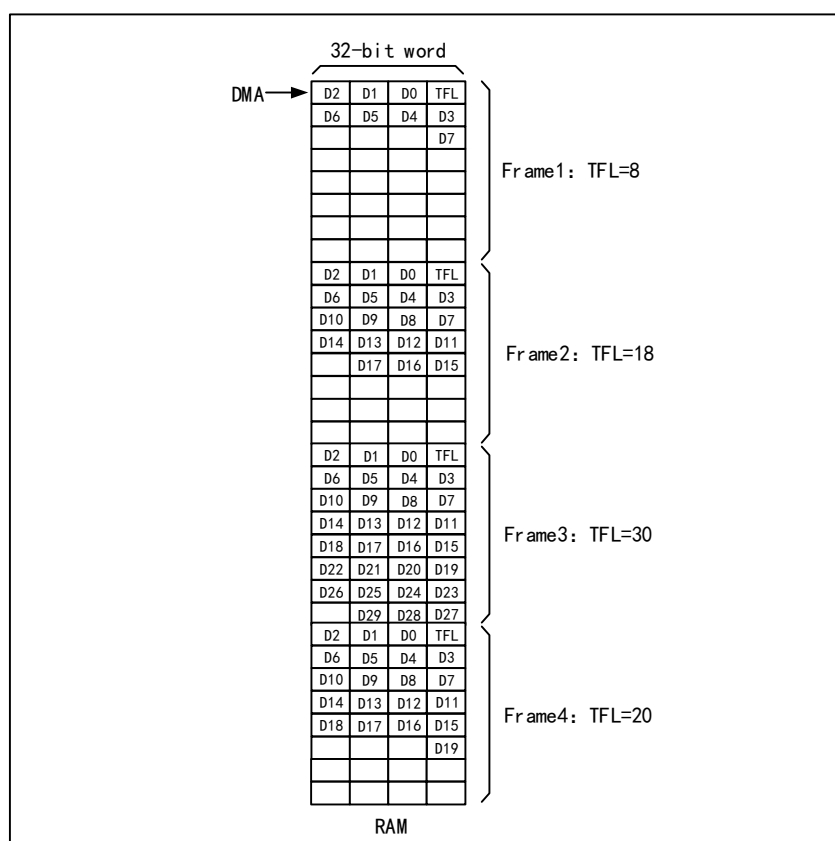
- 2) Update the buffer in RAM with the contents of the next frame to be transmitted;
- 3) Set the total number of words to transfer in the DMA module;
- 4) Enable the data stream or channel in the DMA module to trigger the next frame transmission;
- 5) Clear the TXBEF flag by setting the CTXBEF bit in the R32_SWPMI_ICR register to 1.

Multi-Software buffer mode

Multi-Software buffer mode supports configuring multiple frame buffers in RAM memory to ensure continuous data transmission, maintain extremely low CPU load, and provide software with additional latency to execute buffer updates (using DMA). Software can monitor the DMA counter at any time and update the SWP frame in RAM memory accordingly.

For optimal performance, Multi-Software Buffer Mode should be used in conjunction with DMA in cyclic mode. Within the SWP frame payload, each transmit buffer in RMA memory must have a fixed length of eight 32-bit words, regardless of the actual byte count. These transmit buffers require software filling, with an 8-byte offset maintained between consecutive buffers. The first data byte of the buffer indicates the total number of bytes in the frame payload, as illustrated below.

Figure 38-6 SWPMI multi-software buffer mode transmission



Select the multi-software buffer mode by setting the TXDMA and TXMODE bits in the R32_SWPMI_CR register to 1. To use 4 transmit buffers, the user must configure the DMA as follows:

The DMA channel or data stream must be configured in the following modes (refer to the DMA section):

- Memory size set to 32 bits;
- Peripheral size set to 32 bits;

- The number of words to transfer must be set to 32 (8 words per buffer);
- Data transfer direction set to read from memory;
- Disable memory-to-memory mode;
- Disable peripheral increment mode;
- Enable memory increment mode;
- Enable cycle mode;
- Source address is buffer 1 in RAM;
- Destination address is the R32_SWPMI_TDR register.

Subsequently, the user performs the following operations:

- 1) Set the TXDMA bit in the R32_SWPMI_CR register to 1;
- 2) Set the TXBEIE bit in the R32_SWPMI_IER register to 1;
- 3) Populate buffers 1, 2, 3, and 4 in RAM (the number of data bytes in the payload is indicated by the least significant byte of the first word);
- 4) Trigger the DMA transfer and frame transmission by enabling the data stream or channel within the DMA module.

Within the SWPMI interrupt routine, the user must check the TXBEF bit in the R32_SWPMI_ISR register. If this bit is set to 1, the user should set the CTXBEF bit in the R32_SWPMI_ICR register to 1 to clear the TXBEF flag and update Buffer 1 in RAM memory. Upon the next SWPMI interrupt, the user updates Buffer 2, repeating this cycle.

Software can also read the DMA counter (data volume to be transferred) in the DMA register to retrieve frames already transferred from RAM and sent. For example, if software uses 4 transmit buffers and the DMA counter value equals 17, this indicates two buffers in the RAM region are ready for update. This feature effectively addresses scenarios where the software must transmit multiple frames before handling the SWPMI interrupt. In such cases, the software must update multiple buffers.

When no frames require transmission, the user must disable the DMA module's cyclic mode. Transmission will cease upon completion of buffer 4.

If transmission is terminated prematurely, such as upon completion of buffer 2, the user must set the least significant byte of the first word in both buffer 3 and buffer 4 to 0.

When the data byte count in the least significant byte of the first word read is 0, the TXDMA bit in the R32_SWPMI_CR register will be cleared by hardware.

38.3.7 Receive Processing

Before frame transmission, the user must activate the SWP. Refer to Section 38.3.1. After setting bit 1 of the SWPACT field in the R32_SWPMI_CR register, the "slave device recovery" state will set the SRF flag in the R32_SWPMI_ISR register to 1 and automatically enable the SWPMI for frame reception.

If the SWP bus is in an "active" state, such as during frame transmission, the SWPMI core does not require a "slave device recovery" state and can immediately proceed with reception.

Software implementation for frame reception includes multiple modes: no software buffer mode, single software buffer mode, and multiple software buffer mode. Utilizing software buffers requires a DMA channel to transfer data from the receive data register within the SWPMI peripheral to the software buffer in RAM memory.

No software buffer mode

In No Software Buffer Mode, DMA does not need to be enabled. Polling the status flag within the main loop or SWPMI interrupt routine handles SWP frame transmission. SWPMI includes a 32-bit receive data register (R32_SWPMI_RDR), allowing reception of up to 4 bytes before reading this register.

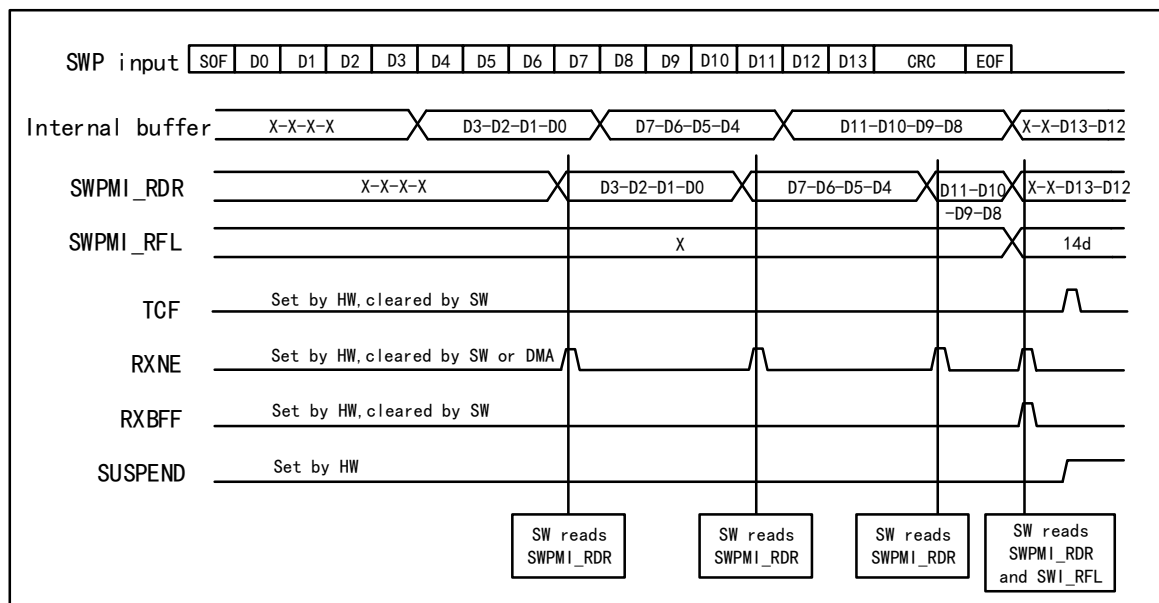
No Software-buffer mode is selected by clearing the RXDMA bit in the R32_SWPMI_CR register.

After receiving the Start of Frame (SOF), subsequent payload bytes are stored in the R32_SWPMI_RDR register. When R32_SWPMI_RDR is full, the RXNE flag in R32_SWPMI_ISR is set to 1. Additionally, an interrupt is generated if the RIE bit in the R32_SWPMI_IER register is set to 1. Users can read the R32_SWPMI_RDR register to retrieve data. When software reads the R32_SWPMI_RDR register, the RXNE flag is automatically cleared.

Upon receiving a complete frame (CRC, EOF), the RXNE and RXBFF flags in the R32_SWPMI_ISR register are set to 1. The user must read the last byte(s) of the payload from the R32_SWPMI_RDR register and set the CRXBFF flag in the R32_SWPMI_ICR register to clear the RXBFF flag. The number of data bytes in the payload can be obtained from the R32_SWPMI_RFL register. When software reads the R32_SWPMI_RDR register, the RXNE flag is automatically reset again. As shown in the figure below.

When RXNE is cleared, reading the R32_SWPMI_RDR register returns 0.

Figure 38-7 SWPMI no software buffer mode reception



Single software buffer mode

In Single Software Buffer Mode, complete SWP frames can be transmitted via DMA without CPU intervention. DMA transfers received data from the 32-bit R32_SWPMI_RDR register to RAM storage. Software can poll the SWPMI_RBFF flag to determine if frame reception has completed. Single Software Buffer Mode: Clear the RXDMA bit position 1 and RXMODE bit in the R32_SWPMI_CR register.

The DMA channel or data stream must be configured in one of the following modes (refer to the DMA section):

- Memory size set to 32 bits;
- Peripheral size set to 32 bits;
- Enable memory increment mode;
- Disable cycle mode;

- Disable peripheral increment mode;
- Disable memory-to-memory mode;
- Set the number of words transferred to 8;
- The destination address is the SWP frame buffer in RAM;
- Set the data transfer direction to read from memory;
- The source address is the R32_SWPMI_RDR register.

Subsequently, the user performs the following actions:

- 1) Set bit 1 in the RXDMA field of the R32_SWPMI_CR register;
- 2) Set bit 1 in the RXBEIE field of the R32_SWPMI_IER register;
- 3) Enable the data stream or channel within the DMA module.

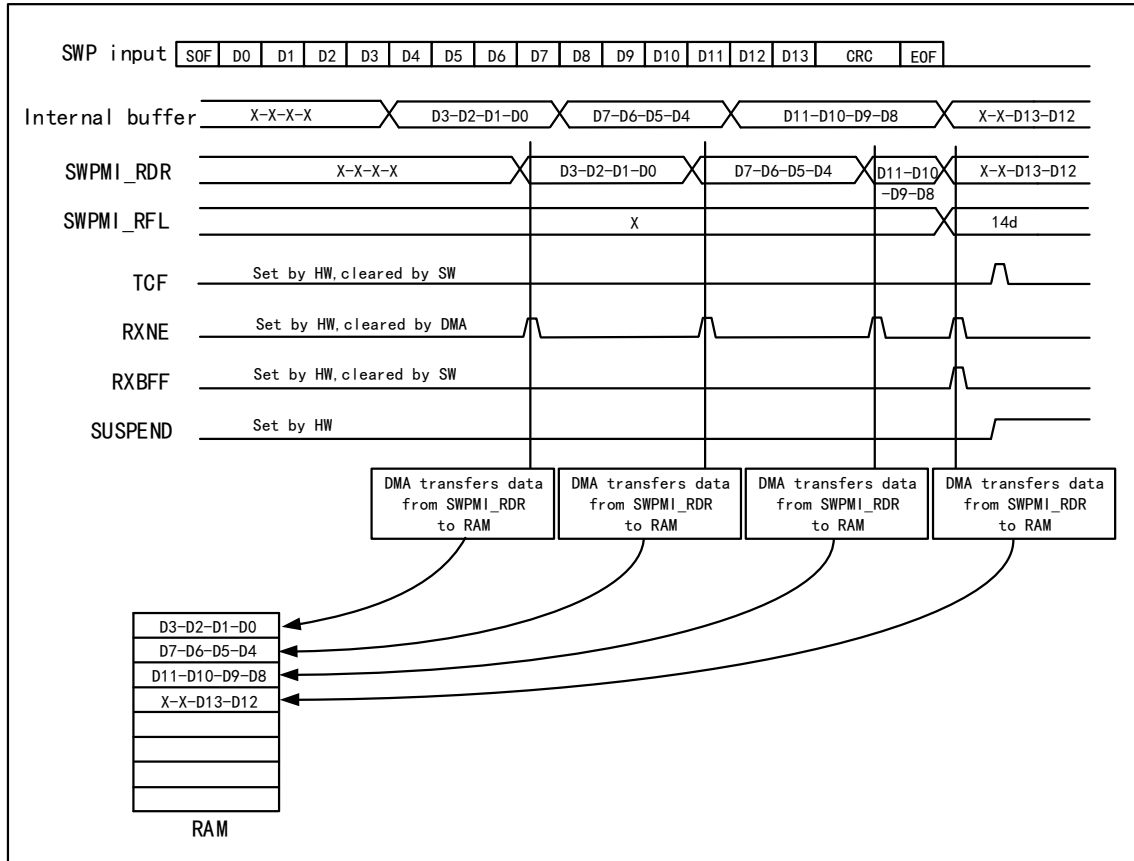
When the RXNE flag in the R32_SWPMI_ISR is set to 1, the SWPMI initiates a DMA request. During the DMA write operation to the R32_SWPMI_RDR register, the RXNE flag is automatically cleared.

Within the SWPMI interrupt service routine, the user must check the RXBEF bit in the R32_SWPMI_ISR register. If this bit is set and additional frames remain to be transmitted, the user must perform the following actions:

Disable data streams or channels in the DMA module;

- 2) Read the number of bytes of frame payload received from the R32_SWPMI_RFL register;
- 3) Read the frame payload from the RAM buffer;
- 4) Enable the data stream or channel in the DMA module;
- 5) Clear the RXBEF flag by setting bit CTXBEF in the R32_SWPMI_ICR register to 1. Refer to Figure 38-8 for details.

Figure 38-8 SWPMI single software buffer mode reception



Multi-Software Buffer Mode

Multi-Software Buffer Mode supports configuring multiple frame buffers in RAM memory to ensure continuous data transmission while maintaining extremely low CPU load. Frame payloads and frame status flags are stored together in RAM memory. Software can check the DMA counter and status flags at any time to process SWP frames received in RAM memory.

For optimal performance, Multi-Software Buffer Mode should be used in conjunction with DMA in cyclic mode. Select Multi-Software Buffer Mode: Set the RXDMA and RXMODE bits in the R32_SWPMI_CR register to 1.

To implement n receive buffers in RAM, the DMA channel or data stream must be configured in one of the following modes (refer to the DMA section):

- Memory size set to 32 bits;
- Peripheral size set to 32 bits;
- Number of words to transfer must be set to $8 \times n$ (8 words per buffer);
- Data transfer direction set to read from peripheral;
- Disable memory-to-memory mode;
- Disable peripheral increment mode;
- Enable memory increment mode;
- Enable cyclic mode;
- Source address is the R32_SWPMI_TDR register;
- Destination address is buffer 1 in RAM.

Subsequently, the user performs the following operations:

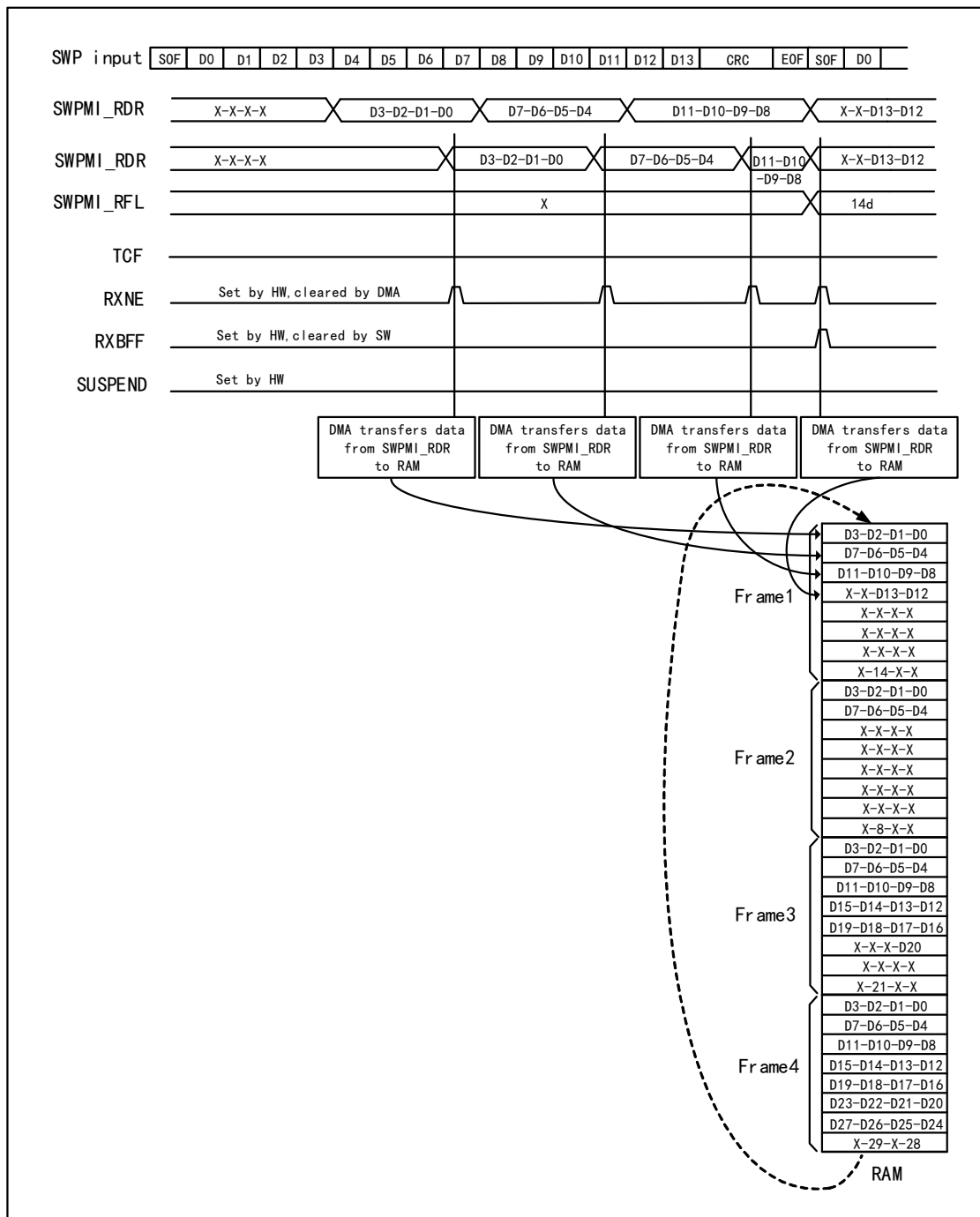
- 1) Set bit 1 in the RXDMA field of the R32_SWPMI_CR register;

- 2) Set bit 1 in the RXBEIE field of the R32_SWPMI_IER register;
- 3) Enable the data stream or channel within the DMA module.

In the SWPMI interrupt routine, the user must check the RXBEF bit in the R32_SWPMI_ISR register. If this bit is set to 1, the user should set the CTXBEF bit in the R32_SWPMI_ICR register to 1 to clear the RXBEF flag. The first frame payload received in the first buffer (the RAM address set in DMA2_CMAR1) can then be read. The number of data bytes in the payload can be obtained from bits [23:16] of the 8th word from the end.

Upon the next SWPMI interrupt routine, the user reads the second frame received in the second buffer (address set in DMA2_CMAR1 + 8), repeating this process cyclically. See Figure 38-9.

Figure 38-9 SWPMI multi-software buffer mode reception



If the application software cannot guarantee processing the SMPMI interrupt before receiving the next frame, each buffer status will be presented in the most significant byte of the 8th buffer word:

- **Receive Overflow Flag:** The RXOVRF flag in the R32_SWPMI_ISR register, indicated by bit 25 of the eighth word. For details on overflow errors, refer to Section 38.3.8.
- **Receive Buffer Full Flag:** The RXBFF flag in the R32_SWPMI_ISR register, indicated by bit 26 of the eighth word.
- **CRC error flag:** The RXBERF flag in the R32_SWPMI_ISR register, indicated by bit 24 of the 8th word. For details on CRC errors, see Section 38.3.8.

When a CRC error occurs, both the RXBFF and RXBERF flags are set to 1, meaning bits 24 and 26 will be set to 1. When an overflow occurs, the overflow flag bit is set to 1, i.e., bit 25 is set to 1. The receive buffer full flag is only set to 1 if the overflow occurs during the reception of the last byte; Subsequently, for both the current and next frame reception, bits 25 and 26 will be set to 1.

Software can read the DMA counter in the DMA register to retrieve frames transferred via DMA and stored in RAM. For example, if software uses 4 receive buffers and the DMA counter value equals 17, this indicates two buffers are ready for software readout in the RAM region. In multi-software buffer reception mode, if the software is reading bits 24, 25, and 26 of the eighth word, the RXBERF, RXOVRF, and RXBFF flags need not be cleared after each received frame.

38.3.8 Error Management

Underflow occurred during payload transmission

In the process of transmitting the frame payload, if the transmission underflow occurs, the TXUNRF flag in R32_SWPMI_ISR register will be triggered as a warning. If the TXBUNREIE bit in R32_SWPMI_IER register is set, an interrupt will be generated. When transmission underflow occurs, SWPMI will stop sending payload and send damaged CRC (the first bit of the first CRC byte sent is inverted), followed by EOF flag. If DMA is used, the TXDMA bit in R32_SWPMI_CR register will be cleared automatically.

When TXUNRF is set to 1, any writing to the R32_SWPMI_TDR register is ignored. The user must set the CTXUNRF bit of R32_SWPMI_ICR register to 1 to clear the TXUNRF flag.

Overflow occurred during payload reception

In the process of receiving the frame payload, if there is reception overflow, the RXOVRF flag in R32_SWPMI_ISR register will be triggered as a warning. When receiving overflow occurs, the incoming data will not be used by SWPMI to update R32_SWPMI_RDR, so the incoming data will be lost.

Data reception continues until EOF. Once the overflow condition ceases, the RXBFF flag will be set to 1. When the RXBFF flag is set, the user may check the RXOVRF flag. To clear the RXOVRF flag, the user must set the CRXOVRF bit in the R32_SWPMI_ICR register to 1. If immediate overflow detection is required, set the RXBOVREIE bit in the R32_SWPMI_IER register to ensure an immediate interrupt trigger upon overflow occurrence.

When both the RXOVRF and RXNE flags are set, two R32_SWPMI_RDR read operations will execute following an overflow event. This indicates that at least one received byte was lost, and the byte loaded into R32_SWPMI_RDR contains the byte received immediately before the overflow occurred.

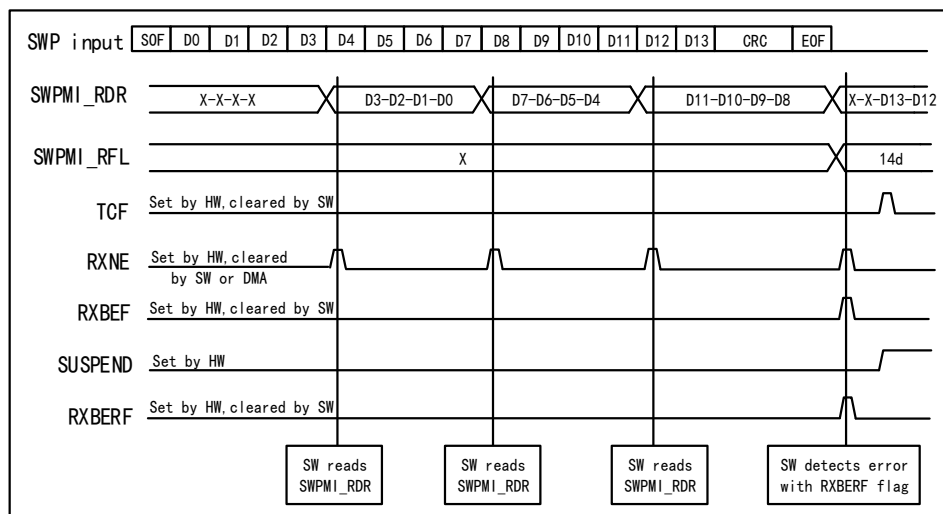
In multi-software buffer mode, if the last byte of a received frame sets the RXOVRF flag, both the current frame

and the next frame will set the overflow bit (bit 25 of the 8th byte) to 1.

A CRC error occurred during payload reception

Upon receiving two CRC bytes, if a CRC error occurs, the RXBERF flag in the R32_SWPMI_ISR register will be set to 1 after completing EOF reception. At this point, if the RXBEIE bit in the R32_SWPMI_IER register is set to 1, an interrupt will be triggered. Refer to Figure 38-10. To clear the RXBERF flag, the user must set the CRXBERF bit in the R32_SWPMI_ICR register to 1.

Figure 38-10 SWPMI single buffer mode reception (with CRC error)



Loss or corruption of padding bits during payload reception

When the padding bits in the payload are lost or damaged, the RXBERF and RXBFF flags in R32_SWPMI_ISR will be set to 1 after receiving EOF.

Receiving damaged EOF

After receiving SOF, SWPMI will start to accumulate the received bytes until EOF is received (ignoring any possible SOF). After receiving EOF, SWPMI will be ready to start a new round of frame reception and wait for the next SOF. If the EOF is damaged, the RXBERF and RXBFF flags in the SWPMI_ISR register will be set after the next EOF is received.

Note: When the damaged EOF is received, the payload will continue to be received. Therefore, if the number of bytes received exceeds 30, the number of bytes in the payload can take the value of 31. The number of bytes in the payload is read in the R32_SWPMI_RFL register, or in bits [23:16] of the 8th word in the RAM memory buffer according to different working modes.

38.3.9 Loopback Mode

You can use loopback mode for testing. The user must set the LPBK bit in the R32_SWPMI_CR register to enable loopback mode. When loopback mode is enabled, SWP_TX and SWP_RX signals will be connected. Therefore, all frames sent through SWPMI will be retracted.

38.3.10 SWPMI Interrupt

All SWPMI interrupts are connected to PFIC. To enable SWPMI interrupts, the following process is required:

- 1) Configure the SWPMI interrupt channel in PFIC and enable it;

- 2) Configure SWPMI to generate SWPMI interrupts (refer to R32_SWPMI_IER register).

Table 38-2 Interrupt requests

Interrupt event	Interrupt event flag	Interrupt enable control bit
Receive buffer is full	RXBFF	RXBFIE
Transmit buffer is empty	TXBEF	TXBEIE
Receive buffer error (CRC error)	RXBERF	RXBEIE
Receive buffer overflow	RXOVRF	RXBOVEREIE
Transmit buffer underflow	TXUNRF	TXBUNREIE
Receive data register is not empty	RXNE	RIE
Transmit data register is full	TXE	TIE
Transmission completion flag	TCF	TCIE
Recovery flag from device	SRF	SRIE
Transceiver ready flag	RDYF	RDYIE

38.4 Register Description

Table 38-3 SWPMI-related register

Name	Access address	Description	Reset value
R32_SWPMI_CR	0x40008400	SWPMI configuration/control register	0x00000000
R32_SWPMI_BRR	0x40008404	SWPMI bit rate register	0x00000001
R32_SWPMI_ISR	0x4000840C	SWPMI interrupt and status register	0x000002C2
R32_SWPMI_ICR	0x40008410	SWPMI interrupt flag clear register	0x00000000
R32_SWPMI_IER	0x40008414	SWPMI interrupt enable register	0x00000000
R32_SWPMI_RFL	0x40008418	SWPMI receive frame length register	0x00000000
R32_SWPMI_TDR	0x4000841C	SWPMI transmit data register	0x00000000
R32_SWPMI_RDR	0x40008420	SWPMI receive data register	0x00000000
R32_SWPMI_OR	0x40008424	SWPMI option register	0x00000000

38.4.1 SWPMI Configuration/Control Register (R32_SWPMI_CR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				SWP TEN	DE AC T	Reserved				SW PA CT	LP BK	TX MO DE	RX MO DE	TX DM A	RX DM A

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0

11	SWPTEN	RW	Single-wire protocol master transceiver enable: 1: SWPMI_IO transceiver controlled by SWPMI; 0: SWPMI_IO pin controlled by GPIO controller.	0
10	DEACT	RW	The single-wire protocol master interface is disabled to request the SWP “disabled” state. Except when the “slave device recovery” state is potentially received, which keeps the SWP in the “activated” state, setting this bit to 1 has the same effect as clearing SWPACT.	0
[9:6]	Reserved	RO	Reserved	0
5	SWPACT	RW	Single-wire protocol master interface activation: 1: SWPMI_IO is released, SWP bus switches to “suspended” state; 0: SWPMI_IO is pulled low to ground, SWP bus switches to “disabled” state. To set SWPACT bit 1, the DEACT bit must be cleared first.	0
4	LPBK	RW	Return mode enable: 1: Enable return mode; 0: Disable return mode.	0
3	TXMODE	RW	Transmit buffer mode: 1: Configures SWPMI for multiple software buffer mode for transmission; 0: Configures SWPMI for single software buffer mode for transmission. Note: This bit cannot be written when SWPACT is set to 1.	0
2	RXMODE	RW	Receive buffer mode: 1: Configures SWPMI for multiple software buffers during reception; 0: Configures SWPMI for a single software buffer during reception. Note: This bit cannot be written when SWPACT is set to 1.	0
1	TXDMA	RW	Transmit DMA enable: 1: Enable transmit DMA; 0: Disable transmit DMA. <i>Note: If the payload size of a transmitted frame is specified as 0x00 (in the TDR LSB of the frame's first word), TXDMA will automatically reset. TXDMA also resets automatically upon an underflow event (when the TXUNRF flag in the R32_SWP_ISR register is set to 1).</i>	0
0	RXDMA	RW	Receive DMA enable: 1: Enables receive DMA;	0

			0: Disables receive DMA.	
--	--	--	--------------------------	--

Table 38-4 Select buffer mode for transmit/receive

Buffer mode	No software buffer	Single software buffer	Multiple software buffer
RXMODE/TXMODE	X	0	1
RXDMA/TXDMA	0	1	1

38.4.2 SWPMI Bit Rate Register (R32_SWPMI_BRR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								BR[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	BR[7:0]	RW	Bit rate prescaler: This bit field must be programmed according to the following formula, taking into account the FHCLK programmed in the RCC (Reset and Clock Control) to set the SWP bus bit rate: $FSWP = F_{HCLK} / ((BR[7:0] + 1) \times 4)$ <i>Note: The programming bit rate must be within the following range: 100 kbit/s to 2 Mbit/s.</i> <i>When the SWPACT bit in the R32_SWPMI_CR register is set to 1, writing to BR[7:0] is not permitted.</i>	0x01

38.4.3 SWPMI Interrupt and Status Register (R32_SWPMI_ISR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				RD YF	DE AC TF	SUSP	SR F	TC F	TX E	RXNE	TX UN RF	RX OV RF	RX BE RF	TX BE F	RX BF F

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
11	RDYF	RO	Transceiver ready indicator: This bit is immediately set to 1 by hardware when the transceiver is ready. After enabling the SWPMI_IO transceiver by setting the SWPTEN bit in the R32_SWPMI_CR register to	0

			1, software must wait for this indicator to be set to 1 before setting the SWPACT bit to 1 to activate the SWP bus. 1: Transceiver is ready; 0: Transceiver is not ready.	
10	DEACTF	RO	Disabled indicator: 1: SWP bus is in the “Disabled” state; 0: SWP bus is in the “Enabled” or “Suspended” state.	0
9	SUSP	RO	Suspended indicator: 1: SWP bus is in “Suspended” or “Disabled” state; 0: SWP bus is in “Enabled” state.	1
8	SRF	RO	Slave device recovery flag: This bit is set to 1 by hardware to indicate detection of a “slave device recovery” state. It is cleared by software by writing a 1 to the CSRF bit in the R32_SWPMI_ICR register. 1: “Slave device recovery” state detected while the SWP bus is in a “suspended” state; 0: No “slave device recovery” state detected.	0
7	TCF	RO	Transmission complete flag: This flag is immediately set to 1 by hardware when both transmission and reception are complete and the SWP conversions to the “Suspended” state. This bit is cleared by software by writing a 1 to the CTCF bit in the R32_SWPMI_ICR register. 1: Both transmission and reception are complete, and the SWP has conversioned to the “Suspended” state; 0: Transmission or reception is incomplete.	1
6	TXE	RO	Transmit data register is empty: 1: Data written to transmit data register R32_SWPMI_TDR has been transmitted; R32_SWPMI_TDR can be written to again. 0: Data written to transmit data register R32_SWPMI_TDR has not yet been transmitted.	1
5	RXNE	RO	Receive data register not empty: 1: Data received in the R32_SWPMI_RDR register is ready for reading; 0: No data received in the R32_SWPMI_RDR register.	0
4	TXUNRF	RO	Transmit underflow error flag: This flag is set to 1 by hardware to indicate an underflow occurred during payload transmission, meaning software or DMA failed to write to R32_SWPMI_TDR in time. Software clears this flag by writing 1 to the CTXUNRF bit in the R32_SWPMI_ICR register.	0

			1: Underflow error detected during transmission; 0: No underflow error during transmission.	
3	RXOVRF	RO	Receive overflow error flag: This flag is set to 1 by hardware to indicate an overflow occurred during payload reception, meaning software or DMA failed to read R32_SWPMI_RDR in time. Software clears this flag by writing a 1 to the CRXOVRF bit in the R32_SWPMI_ICR register. 1: Overflow error detected during reception; 0: No overflow error during reception.	0
2	RXBERF	RO	Received CRC error flag: This flag is set to 1 by hardware to indicate a CRC error in the received frame. It is synchronized with the RXBFF flag when set. This bit is cleared by software by writing a 1 to the CRXBERF bit in the R32_SWPMI_ICR register. 1: CRC error detected during reception; 0: No CRC error during reception.	0
1	TXBEF	RO	Transmit buffer empty flag: This flag is set to 1 by hardware to indicate that no further R32_SWPMI_TDR updates are required to complete the current frame transmission. This bit is cleared by software by writing 1 to the CTXBEF bit in the R32_SWPMI_ICR register. 1: Frame transmission buffer has been cleared; 0: Frame transmission buffer has not yet been cleared.	1
0	RXBFF	RO	Transmit buffer empty flag: This flag is set to 1 by hardware to indicate that no further R32_SWPMI_TDR updates are required to complete the current frame transmission. This bit is cleared by software by writing 1 to the CTXBEF bit in the R32_SWPMI_ICR register. 1: Frame transmission buffer has been cleared; 0: Frame transmission buffer has not yet been cleared.	0

38.4.4 SWPMI Interrupt Flag Clear Register (R32_SWPMI_ICR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				CRDYF	Reserved		CSRF	CTCF	Reserved		CTX UNR F	CRX OVR F	CRXB ERF	CT XB EF	CR XB FF

Bit	Name	Access	Description	Reset value
-----	------	--------	-------------	-------------

[31:12]	Reserved	RO	Reserved	0
11	CRDYF	RW1Z	Trigger clear transceiver ready flag: Writing a 1 to this bit clears the RDYF flag in the R32_SWPMI_ISR register to this bit. Writing a 0 has no effect.	0
[10:9]	Reserved	RO	Reserved	0
8	CSRF	RW1Z	Clear device recovery flag: Writing a 1 to this bit clears the SRF flag in the R32_SWPMI_ISR register to this bit. Writing a 0 has no effect.	0
7	CTCF	RW1Z	Transmission clear completion flag: Writing a 1 to this bit clears the TCF flag in the R32_SWPMI_ISR register to this bit. Writing a 0 has no effect.	0
[6:5]	Reserved	RO	Reserved	0
4	CTXUNRF	RW1Z	Clear transmit underflow error flag: Writing a 1 to this bit clears the TXUNRF flag in the R32_SWPMI_ISR register to this bit. Writing a 0 has no effect.	0
3	CRXOVRF	RW1Z	Clear receive overflow error flag: Writing a 1 to this bit clears the RXBOCREF flag in the R32_SWPMI_ISR register to this bit. Writing a 0 has no effect.	0
2	CRXBERF	RW1Z	Clear receive CRC error flag: Writing a 1 to this bit clears the RXBERF flag in the R32_SWPMI_ISR register for this bit. Writing a 0 has no effect.	0
1	CTXBEF	RW1Z	Clear transmit buffer empty flag: Writing a 1 to this bit clears the TXBEF flag in the R32_SWPMI_ISR register to this bit. Writing a 0 has no effect.	0
0	CRXBFF	RW1Z	Clear receive buffer full flag: Writing a 1 to this bit clears the RXBFF flag in the R32_SWPMI_ISR register to this bit. Writing a 0 has no effect.	0

38.4.5 SWPMI Interrupt Enable Register (R32_SWPMI_IER)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				RDYIE	Reserved		SRIE	TCIE	TIE	RIE	TXUNRIE	RXOVRIE	RXBERIE	TXBEIE	RXBFFIE

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
11	RDYIE	RW	Transceiver ready interrupt enable: 1: Generates SWPMI interrupt when the RDYF flag in the R32_SWPMI_ISR register is set to 1;	0

			0: Disables interrupts.	
[10:9]	Reserved	RO	Reserved	0
8	SRIE	RW	Enabling interrupts from the device: 1: Generates a SWPMI interrupt when the SRF flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupts.	0
7	TCIE	RW	Enable transmit complete interrupt: 1: SWPMI interrupt is generated when the TCF flag in the R32_SWPMI_ISR register is set to 1; 0: Disable interrupt.	0
6	TIE	RW	Enable transmission interrupt: 1: Generates a SWPMI interrupt when the TXE flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupts.	0
5	RIE	RW	Receive interrupt enable: 1: Generates a SWPMI interrupt when the RXNE flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupts.	0
4	TXUNRIE	RW	Enable transmit underflow error interrupt: 1: Generates SWPMI interrupt when the TXBUNRF flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupt.	0
3	RXOVRIE	RW	Enable receive overflow error interrupt: 1: Generates SWPMI interrupt when the RXBOVRF flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupt.	0
2	RXBERIE	RW	Receive CRC error interrupt enable: 1: Generates a SWPMI interrupt when the RXBERF flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupts.	0
1	TXBEIE	RW	Transmit buffer empty interrupt enable: 1: Generates SWPMI interrupt when the TXBEF flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupts.	0
0	RXBFIE	RW	Receive buffer full interrupt enable: 1: Generates SWPMI interrupt when the RXBFF flag in the R32_SWPMI_ISR register is set to 1; 0: Disables interrupt.	0

38.4.6 SWPMI Receive Frame Length Register (R32_SWPMI_RFL)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											RFL[4:0]				

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
[4:0]	RFL[4:0]	RO	Received frame length: The number of data bytes in the payload of the received frame. The two least significant bits RFL[1:0] indicate the byte count associated with the last R32_SWPMI_RDR register read operation.	0

38.4.7 SWPMI Transmit Data Register (R32_SWPMI_TDR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TD[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TD[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	TD[31:0]	WO	Data transmission: Writing to this register triggers SOF transmission or the next payload data transmission, and clears the TXE flag.	0

38.4.8 SWPMI Receive Data Register (R32_SWPMI_RDR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RD[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RD[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	RD[31:0]	RO	Data received: Contains the received data. Reading this register clears the RXNE flag.	0

38.4.9 SWPMI Option Register (R32_SWPMI_OR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	Reserved	SWP_TBYP
----------	----------	----------

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved	0
1	Reserved	RO	Reserved	0
0	SWP_TBYP	RW	Disable internal transceiver: 1: Disable the internal transceiver. The SWP_RX, SWP_TX, and SWPMI_SUP signals are available as multiplexed functions on GPIO pins. Selecting this configuration allows connection of an external transceiver; 0: Enable internal transceivers. The external interface for SWPMI is SWPMI_IO (SWP_RX, SWP_TX, and SWP_SUP signals are unavailable on GPIO).	0

Chapter 39 Electronic Signature (ESIG)

Electronic Signature contains chip identification information: Flash memory area capacity and unique identification. It is burned into the system storage area of the memory module by the vendor when leaving the factory, and can be read through SWD (SDI) or application code.

39.1 Function Description

Flash memory area capacity: Indicates the size that the current chip user application can use.

Unique identification: 96-bit binary code, unique to any microcontroller, users can only read and access it and cannot modify it. This unique identification information can be used as the security password, decryption key, product serial number, etc. of the microcontroller (product) to improve the system security mechanism or indicate identity information.

Users of the above content can perform read access in 8/16/32 bits.

39.2 Register Description

Table 39-1 ESIG-related register

Name	Access address	Description	Reset value
R16_ESIG_FLACAP	0x1FFFF7E0	Flash capacity register	0xFFFF
R32_ESIG_UNIID1	0x1FFFF7E8	UID register 1	0xFFFFFFFF
R32_ESIG_UNIID2	0x1FFFF7EC	UID register 2	0xFFFFFFFF
R32_ESIG_UNIID3	0x1FFFF7F0	UID register 3	0xFFFFFFFF

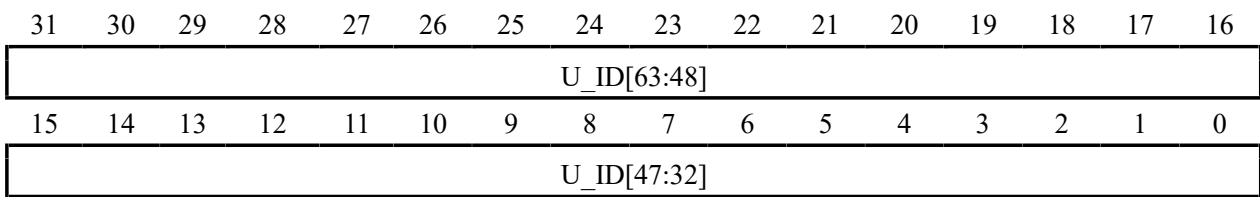
39.2.1 Flash Capacity Register (ESIG_FLACAP)

Bit	Name	Access	Description	Reset value
[15:0]	F_SIZE[15:0]	RO	Flash memory capacity in Kbyte. Example: 0x0080 = 128 Kbytes	X

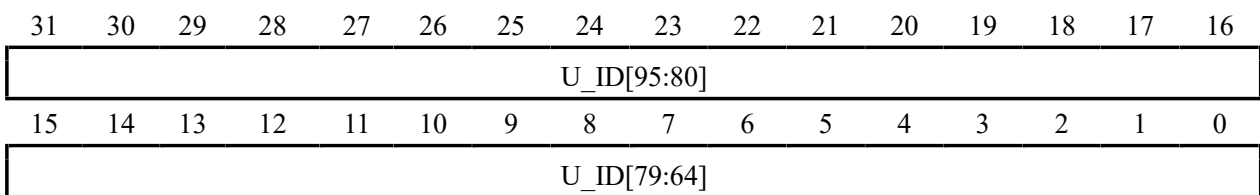
39.2.2 UID Register (ESIG_UNIID1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	U_ID[31:0]	RO	Bits 0-31 of the UID.	X

39.2.3 UID Register (ESIG_UNIID2)

Bit	Name	Access	Description	Reset value
[31:0]	U_ID[63:32]	RO	Bits 32-63 of the UID.	X

39.2.4 UID Register (ESIG_UNIID3)

Bit	Name	Access	Description	Reset value
[31:0]	U_ID[95:64]	RO	Bits 64-95 of the UID.	X

Chapter 40 Flexible Memory Controller (FMC)

This chapter's module description applies only to the CH32H417 microcontroller product.

The Flexible Memory Controller (FMC) is a controller for managing external memory, comprising: NAND memory controller, NOR/PSRAM memory controller, and synchronous DRAM (SDRAM/Mobile LPDDR SDRAM) controller.

40.1 Main Features

The FMC manages expanded connectivity to different types of memory, including: SDRAM, NAND Flash, and synchronous/asynchronous static memory. Its primary purposes are: meeting the access time requirements of external memory devices; and translating HB data communication transactions into appropriate external device protocols.

All external memories share address, data, and control signals, but each possesses an independent chip select signal. The FMC can access only one external device at a time.

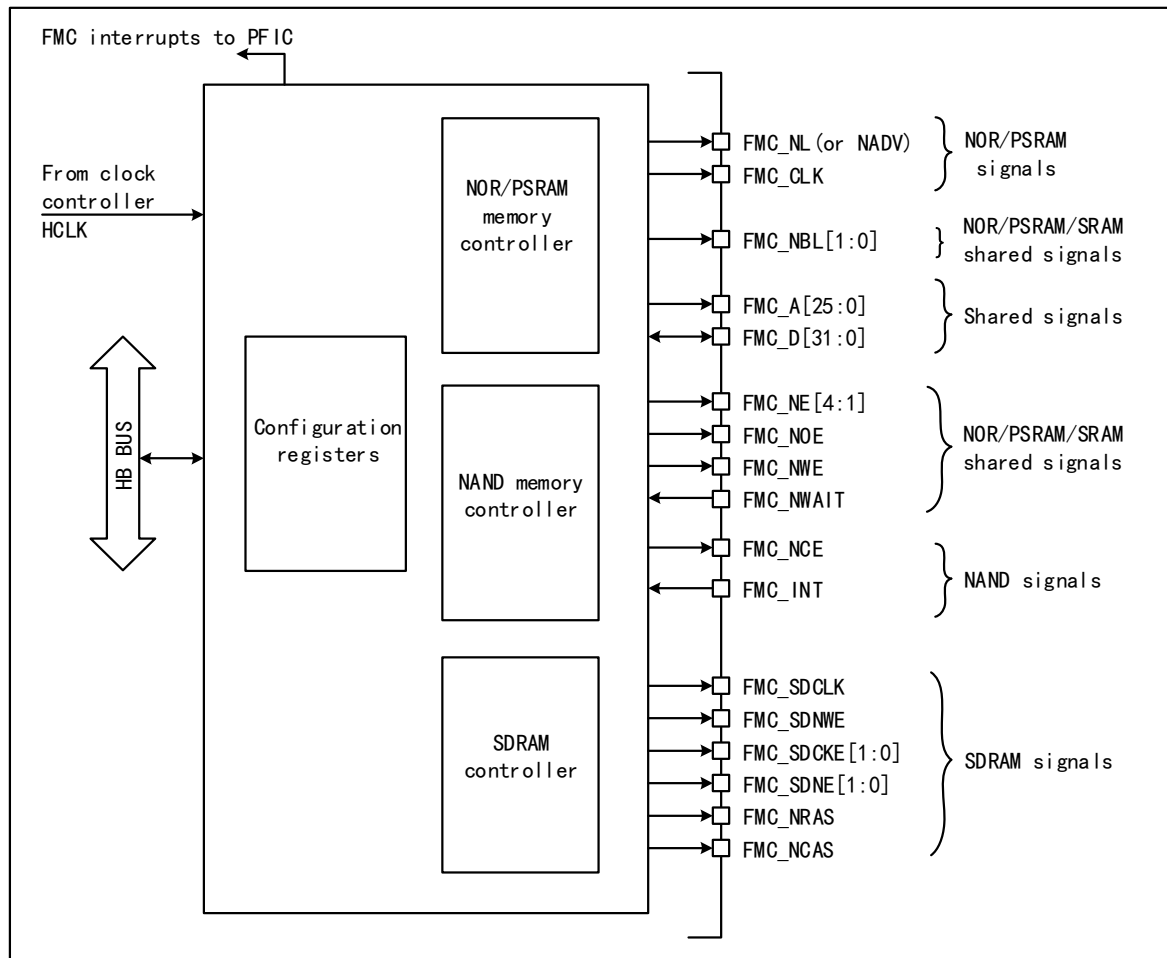
The main features of FMC controllers are as follows:

- Connect static memory mapped devices:
 - Static Random Access Memory (SRAM)
 - PSRAM (4 memory regions)
 - NOR Flash/OneNAND Flash
 - NAND Flash memory with hardware ECC capable of verifying up to 8KB of data
- Connect to synchronous DRAM (SDRAM/Mobile LPDDR SDRAM) memory
- External asynchronous wait control
- 16×32-bit depth write FIFO
- Each memory region configurable independently
- Independent chip select control for each memory region
- Support 8-bit, 16-bit, or 32-bit wide memory data buses
- Programmable continuous clock output for asynchronous and synchronous access
- SDRAM controller features a cacheable 6×32-bit depth read FIFO (6×14-bit address tag)
- Write enable and byte channel select outputs compatible with PSRAM, SRAM, and SDRAM devices
- Support burst mode for accelerated access to synchronous devices (e.g., NOR Flash, PSRAM, and SDRAM)

40.2 Overview

As shown in the figure below, the FMC comprises four main modules: the HB interface, SDRAM controller, NAND controller, and NOR Flash/PSRAM/SRAM controller.

Figure 40-1 Structural block diagram of FMC



40.3 HB Interface

The HB device interface permits the CPU and other HB bus masters to access external memory. HB transactions are translated into external device protocols. Specifically, when the selected external memory width is 16 bits or 8 bits, the 32-bit-wide transaction in the HB is split into multiple consecutive 16-bit or 8-bit accesses. Between consecutive accesses, the FMC chip select (FMC_NEx) does not toggle, except when operating in access mode D with extended mode enabled.

The FMC generates an HB error under the following conditions:

- Reading or writing to an FMC memory region that is not enabled.
- Violating the SDRAM address range (accessing reserved address ranges).
- Writing to a write-protected SDRAM memory region (WP bit 1 in the R32_FMC_SDCRx register).
- Reading or writing to NOR Flash memory regions when the FACCEN bit in the R32_FMC_BCRx register is reset.

The impact of this HB error depends on the HB master device attempting the read/write access.

- If it is the DMA controller, a DMA transfer error is generated and the corresponding DMA channel is automatically disabled.

The HB clock (HCLK) serves as the reference clock for the FMC.

40.3.1 Supported Memory and Transactions

Common transaction regulars

The HB transaction data width requested is 8, 16, 32, or 256 bits, whereas the external device accessed has a fixed data width. This difference can cause data width mismatch problems. To avoid this problem, the transaction regulars to follow are as follows:

- HB transaction data width equals memory data width: Normal operation.
- HB transaction data width exceeds memory width: The FMC splits the HB transaction into multiple smaller contiguous memory accesses to match the external data width. The FMC chip select (FMC_NEx) does not toggle between contiguous accesses.
- HB transaction data width is less than memory width: Transfer consistency may be affected, depending on the type of external device.
 - Accessing devices with byte selection capability (ROM, SRAM, SDRAM, PSRAM): The FMC permits read and write transactions, accessing the correct data through its byte select lines NBL[1:0]. The byte to be written is addressed via NBL[1:0]. All memory bytes are read (NBL[1:0] remains low during read transactions), with invalid bytes subsequently discarded.
 - Accessing devices without byte selection capability (NOR, NAND Flash): This occurs when byte access is required for 16-bit wide FLASH memory. Since these devices do not support byte-mode access (only allowing 16-bit words to be read from or written to Flash), read transactions are permitted but write transactions are not (the controller reads the entire 16-bit memory word but uses only the required byte).

NOR Flash/PSRAM and SDRAM rewinding support

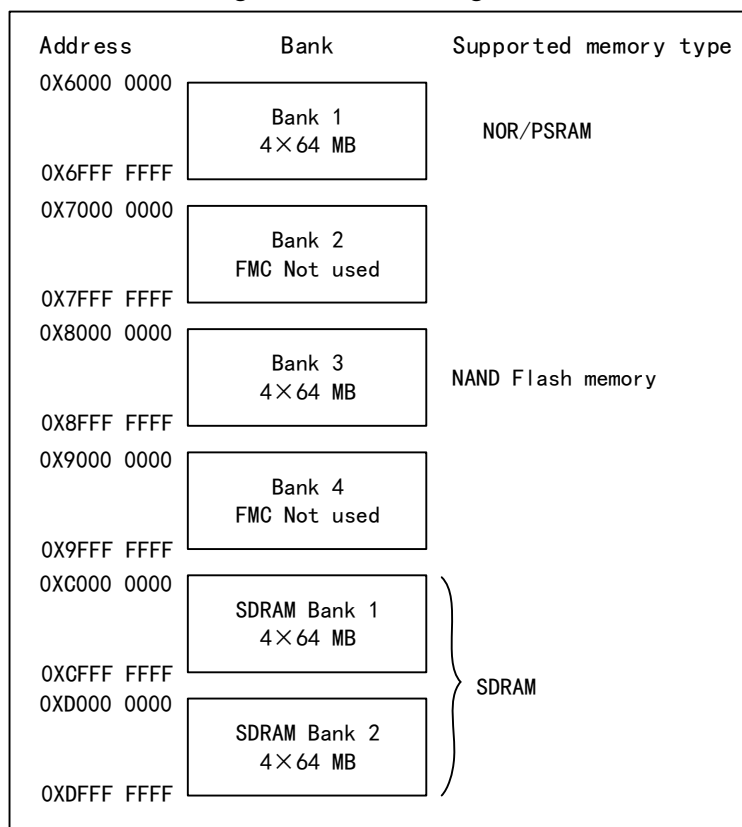
Since not all masters can issue rollback transactions, synchronous memory must be configured in a linear burst mode of undefined length. If a master generates HB rollback transactions:

- Read operations will be divided into two linear burst transactions.
- If the write FIFO is enabled, write operations will be divided into two linear burst transactions; if the write FIFO is disabled, write operations will be divided into multiple linear burst transactions.

40.4 External Device Address Mapping

From the FMC perspective, the external memory is divided into fixed-size storage areas of 256MB each, as shown in the following figure:

Figure 40-2 FMC storage area



Memory Area 1 can connect up to 4 NOR Flash or PSRAM devices. This memory area is divided into four NOR/PSRAM/SR sub-areas with four dedicated chip select signals: Memory Area 1 NOR/PSRAM1, Memory Area 1 NOR/PSRAM2, Memory Area 1 NOR/PSRAM3, and Memory Area 1 NOR/PSRAM4. Memory Area 3 is used to connect NAND Flash devices. The MPU memory characteristics of this space must be reconfigured into the device via software. Memory regions 4 and 5 are used to connect SDRAM devices (one device per memory region).

For each memory region, the memory type to be used can be configured by the user application through configuration registers.

The FMC memory area mapping can be modified via the BMP[1:0] bits in the R32_FMC_BCR1 register. The table below shows configurations for swapping NOR/PSRAM memory areas with SDRAM memory areas or remapping SDRAM Memory Area 2, thereby enabling access to the SDRAM memory area across two distinct address maps.

Table 40-1 FMC storage area mapping option

Start address-End address	BMP[1:0]=00 (Default mapping)	BMP[1:0]=01 exchange NOR/PSRAM to SDRAM storage area
0x60000000-0x6FFFFFFF	NOR/PSRAM storage area	SDRAM storage area 1
0x70000000-0x7FFFFFFF	FMC unused	
0x80000000-0x8FFFFFFF	NAND storage area	NAND storage area
0x90000000-0x9FFFFFFF	FMC unused	
0xC0000000-0xCFFFFFFF	SDRAM storage area 1	NOR/PSRAM storage area

0xD0000000-0xDFFFFFFF	SDRAM storage area 2	SDRAM storage area 2
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40.4.1 NOR/PSRAM Address Mapping

HADDR[27:26] bits are used to select one of the four storage areas shown in Table 40-1.

Table 40-2 NOR/PSRAM memory area selection

HADDR[27:26] ⁽¹⁾	Selected storage area
00	Storage area 1NOR/PSRAM1
01	Storage area 1NOR/PSRAM2
10	Storage area 1NOR/PSRAM3
11	Storage area 1NOR/PSRAM4

Note 1: HADDR is HB's internal address line, but also participates in addressing external memory.

The HADDR[25:0] bits contain the external memory address. Since HADDR is a byte address and memory is addressed in words, the address actually sent to memory varies depending on the memory data width, as shown in the table below.

Table 40-3 NOR/PSRAM memory area address

Memory width ⁽¹⁾	The data address sent to the memory	Maximum memory capacity (bits)
8 位	HADDR[25:0]	64MB×8=512Mb
16 位	HADDR[25:1]>>1	64MB/2×16=512Mb
32 位	HADDR[25:2]>>2	64MB/4×32=512Mb

Note 1: When the external memory width is 16 bits, the FMC uses the internal HADDR[25:1] address as the external memory address FMC_A[24:0]. When the memory width is 32 bits, the FMC will use the internal HADDR[25:2] address for external addressing. Regardless of the external memory width, FMC_A[0] must be connected to the external memory address A[0].

40.4.2 NAND Flash Address Mapping

The NAND memory area is divided into several memory areas, as shown in the following table.

Table 40-4 NAND memory mapping and timing register

Start address	End address	FMC storage area	Storage space	Timing register
0x88000000	0x8BFFFFFF	Storage area 3-NAND Flash	Characteristic region	R32_FMC_PATT(0x8C)
0x80000000	0x83FFFFFF		Universal area	R32_FMC_PMEM(0x88)

For NAND Flash memory, the storage space of general area and characteristic area is divided into three parts, all of which are located in the lower 256KB, as shown in the following table:

Table 40-5 NAND storage area selection

Partial name	HADDR[17:16]	Address range
Address area	1X	0x020000-0x03FFFF
Command area	01	0x010000-0x01FFFF

Data area	00	0x000000-0x0FFFF
-----------	----	------------------

Data Area: The first 64KB within the general/feature storage space; Command Area: The second 64KB within the general/feature storage space; Address Area: The next 128KB within the general/feature storage space.

Application software utilizes these three areas to access the NAND Flash memory:

- To send a command to NAND Flash, the software must write the command value to any memory cell within the command area.
- To specify a read or write address for NAND Flash, the software must write the address value to any memory cell within the address area. Since the address length can be 4 or 5 bytes (depending on the actual memory size), specifying a complete address requires multiple consecutive write operations to the address area.
- Read or write data: The software reads data from or writes data to any storage cell within the data region.
- Since NAND Flash memory automatically increments addresses, it is unnecessary to increment addresses within the data region to access consecutive memory locations.

Because NAND Flash memory will automatically increment the address, it can access continuous memory locations without incrementing the address of the data area.

40.4.3 SDRAM Address Mapping

HADDR[28] bit (internal HB address line 28) is used to select one of the two storage areas, as shown in Table 40-6.

Table 40-6 SDRAM storage area selection

HADDR[28]	Selected storage area	Control register	Timing register
0	SDRAM Bank1	R32_FMC_SDCR1	R32_FMC_SDTR1
1	SDRAM Bank2	R32_FMC_SDCR2	R32_FMC_SDTR2

The following table shows the SDRAM mapping for 13-bit row and 11-bit column configurations.

Table 40-7 SDRAM address mapping

Memory width ⁽¹⁾	Internal storage area	Line address	Column address ⁽¹⁾	Maximum memory capacity (MB)
8 位	HADDR[25:24]	HADDR[23:11]	HADDR[10:0]	64MB:4x8Kx2K
16 位	HADDR[26:25]	HADDR[24:12]	HADDR[11:1]	128MB:4x8Kx2Kx2
32 位	HADDR[27:26]	HADDR[25:13]	HADDR[12:2]	256MB:4x8Kx2Kx4

Note 1: When connecting a 16-bit memory, the FMC internally uses the HADDR[11:1] internal HB address lines for external addressing. When connecting a 32-bit memory, the FMC internally uses the HADDR[12:2] address lines for external addressing. Regardless of the external memory width, FMC_A[0] must be connected to the external memory address A[0].

Note 2: AutoPrecharge is not supported. FMC_A[10] must be connected to the external memory address A[10], but must always be held at a “low level”.

HADDR[27:0] bits are converted to external SDRAM addresses based on the SDRAM controller configuration. Data size: 8, 16, or 32 bits; column size: 8, 9, 10, or 11 bits; row size: 11, 12, or 13 bits; number of internal storage regions: 2 or 4 internal storage regions. The table below shows the SDRAM address mapping for different SDRAM controller configurations.

Table 40-8 SDRAM address mapping with 8-bit data bus width⁽¹⁾⁽²⁾

Row size configura tion	HADDR (HB internal address line)																															
	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
11-bit row size configura tion	Res.							Bank [1:0]		Row[10:0]										Column[7:0]												
	Res.						Bank [1:0]		Row[10:0]											Column[8:0]												
	Res.					Bank [1:0]		Row[10:0]											Column[9:0]													
	Res.				Bank [1:0]		Row[10:0]											Column[10:0]														
12-bit row size configura tion	Res.						Bank [1:0]		Row[11:0]											Column[7:0]												
	Res.					Bank [1:0]		Row[11:0]											Column[8:0]													
	Res.				Bank [1:0]		Row[11:0]											Column[9:0]														
	Res.			Bank [1:0]		Row[11:0]											Column[10:0]															
13-bit row size configura tion	Res.					Bank [1:0]		Row[12:0]											Column[7:0]													
	Res.				Bank [1:0]		Row[12:0]											Column[8:0]														
	Res.			Bank [1:0]		Row[12:0]											Column[9:0]															
	Res.		Bank [1:0]		Row[12:0]											Column[10:0]																

Note 1: BANK[1:0] corresponds to the storage area address BA[1:0]. When using only two internal storage areas, BA1 must always be set to 0.

Note 2: Accessing the reserved (Res.) address range will generate an HB error.

Table 40-9 SDRAM address mapping with 16-bit data bus width⁽¹⁾⁽²⁾

Row size configuration	HADDR (HB internal address line)																											
	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
11-bit row size configuration	Res.							Bank [1:0]		Row[10:0]										Column[7:0]								BM0 ⁽³⁾
	Res.							Bank [1:0]		Row[10:0]										Column[8:0]								BM0
	Res.							Bank [1:0]		Row[10:0]										Column[9:0]								BM0

	Res.	Bank [1:0]	Row[10:0]			Column[10:0]	BM0	
12-bit row size configu ration	Res.		Bank [1:0]	Row[11:0]			Column[7:0]	BM0
	Res.		Bank [1:0]	Row[11:0]			Column[8:0]	BM0
	Res.		Bank [1:0]	Row[11:0]			Column[9:0]	BM0
	Res.	Bank [1:0]	Row[11:0]			Column[10:0]		BM0
13-bit row size configu ration	Res.		Bank [1:0]	Row[12:0]			Column[7:0]	BM0
	Res.		Bank [1:0]	Row[12:0]			Column[8:0]	BM0
	Res.	Bank [1:0]	Row[12:0]			Column[9:0]		BM0
	Res	Bank [1:0]	Row[12:0]			Column[10:0]		BM0

Note 1: BANK[1:0] corresponds to the storage area address BA[1:0]. When using only two internal storage areas, BA1 must always be set to 0.

Note 2: Accessing reserved space (Res.) will generate an HB error.

Note 3: BM0 is the byte mask for 16-bit accesses.

Table 40-10 SDRAM address mapping with 21-bit data bus width⁽¹⁾⁽²⁾

Row size configu ration	HADDR (HB internal address line)																										
	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
11-bit row size configu ration	Res.				Bank [1:0]		Row[10:0]										Column[7:0]							BM[1: :0] ⁽³⁾			
	Res.			Bank [1:0]		Row[10:0]										Column[8:0]							BM[1: :0]				
	Res.		Bank [1:0]		Row[10:0]										Column[9:0]							BM[1: :0]					
	Res.	Bank [1:0]		Row[10:0]										Column[10:0]							BM[1: :0]						
12-bit row size configu ration	Res.			Bank [1:0]		Row[11:0]										Column[7:0]					BM[1: :0]						
	Res.		Bank [1:0]		Row[11:0]										Column[8:0]					BM[1: :0]							
	Res.		Bank [1:0]		Row[11:0]										Column[9:0]					BM[1: :0]							
	Res	Bank		Row[11:0]										Column[10:0]					BM[1: :0]								

		[1:0]				:0]
13-bit row size configu ration	Res.		Bank [1:0]	Row[12:0]		BM[1 :0]
	Res.		Bank [1:0]	Row[12:0]		BM[1 :0]
	Res		Bank [1:0]	Row[12:0]		BM[1 :0]
	Bank [1:0]			Row[12:0]		BM[1 :0]

Note 1: BANK[1:0] corresponds to the storage area address BA[1:0]. When using only two internal storage areas, BA1 must always be set to 0.

Note 2: Accessing reserved space (Res.) will generate an HB error.

Note 3: BM[1:0] serves as the byte mask for 32-bit accesses.

40.5 NOR Flash/PSRAM Controller

The FMC generates appropriate signal timing to drive the following memory types:

- Asynchronous SRAM and ROM: 8-bit, 16-bit, 32-bit;
- PSRAM (Cellular RAM): Asynchronous mode, burst mode with synchronous access, multiplexed or non-multiplexed;
- NOR Flash: Asynchronous mode, burst mode with synchronous access, multiplexed or non-multiplexed.

The FMC outputs unique chip select signals NE[4:1] for each memory region. All other signals (address, data, and control) are shared. The FMC supports multiple devices through programmable timing, including:

- Programmable continuous clock (FMC_CLK) output
- Programmable wait cycles (up to 15 clock cycles)
- Programmable bus turnaround cycles (up to 15 clock cycles)
- Independent read and write timing and protocols to support various memory types and timing requirements
- Programmable output enable and write enable delays (up to 15 clock cycles)

To generate the FMC_CLK continuous clock, memory area 1 must also be configured to support synchronous mode (refer to Section 40.5.6: NOR/PSRAM Control Registers). Since all synchronous memory uses the same clock, the HB data size must match the memory data width (MWID) when generating continuous output clocks and performing synchronous accesses. Otherwise, the FMC_CLK frequency will vary based on the occurrence of HB data transactions (refer to Section 40.5.5: Synchronous Transactions for information on the FMC_CLK division ratio formula).

Each memory region has a fixed size of 64MB. Each memory region is configured via dedicated registers (refer to Section 40.5.6: NOR/PSRAM Control Registers). Programmable memory parameters include access times (as shown in the table below) and support for wait management (used for burst mode access to NOR Flash and PSRAM).

When reading data using the NOR Flash/PSRAM controller, only DMA 256-bit reads are permitted.

Table 40-11 Programmable access parameters of NOR/PSRAM

Parameter	Function	Access mode	Max.	Min.	Unit
Address establishment	Duration of address establishment phase	Asynchronization	0	15	HB clock cycle (HCLK)

Address retention	Duration of the address holding phase	Asynchronous, multiplexed I/O	1	15	HB clock cycle (HCLK)
Data establishment	Duration of the data establishment phase	Asynchronous	1	256	HB clock cycle (HCLK)
Bus turnover	Duration of bus turnaround phase	Asynchronous and synchronous read/write	0	15	HB clock cycle (HCLK)
Clock frequency division ratio	The number of HB clock cycles (HCLK) required to construct one memory clock cycle (CLK)	Synchronous	2	16	HB clock cycle (HCLK)
Data delay	The number of clock cycles issued to the memory before the first data of the burst is issued.	Synchronous	2	17	Memory clock cycle (HCLK)

40.5.1 External Memory Interface Signal

Signals commonly used to connect NOR Flash, SRAM, and PSRAM are listed in Tables 40-12, 40-13, 40-14, and 40-15. Signals prefixed with “N” indicate low-active signals.

Table 40-12 Non-multiplexed I/O NOR Flash

FMC signal name	I/O	Function
CLK	O	Clock (for synchronous access)
A[25:0]	O	Address bus
D[31:0]	I/O	Bidirectional data bus
NE[x]	O	Chip selected, x=1~4
NOE	O	Output enable
NWE	O	Write enable
NL(=NADV)	O	Latch enable (For some NOR Flash devices, this signal is also called address valid (NADV).)
NWAIT	I	NOR Flash of FMC waits for input signal.

The maximum capacity is 512Mb(26 address lines).

Table 40-13 16-bit multiplexed I/O NOR Flash

FMC signal name	I/O	Function
CLK	O	Clock (for synchronous access)
A[25:16]	O	Address bus
AD[15:0]	I/O	16-bit multiplexed, bidirectional address/data bus (16-bit address A[15:0] multiplexed with the data bus)
NE[x]	O	Chip selected, x=1~4
NOE	O	Output enable
NWE	O	Write enable

NL (=NADV)	O	Latch enable (For some NOR Flash devices, this signal is also called address valid (NADV).)
NWAIT	I	NOR Flash of FMC waits for input signal.

The maximum capacity is 512Mb.

Table 40-14 Non-multiplexed I/O PSRAM/SRAM

FMC signal name	I/O	Function
CLK	O	Clock (for synchronous access)
A[25:0]	O	Address bus
D[31:0]	I/O	Bidirectional data bus
NE[x]	O	Chip selected, x=1~4 (In PSRAM applications, it is referred to as NCE (Cellular RAM, or CRAM))
NOE	O	Output enable
NWE	O	Write enable
NL (=NADV)	O	Address valid signal for PSRAM input only (Memory Signal Name: NADV)
NWAIT	I	PSRAM sends to FMC and waits for input signal
NBL[1:0]	O	Byte channel output. Bytes 0 to 3 control (High byte and low byte enable)

The maximum capacity is 512Mb.

Table 40-15 16-bit multiplexed I/O PSRAM

FMC signal name	I/O	Function
CLK	O	Clock (for synchronous access)
A[25:16]	O	Address bus
AD[15:0]	I/O	16-bit multiplexed, bidirectional address/data bus (16-bit address A[15:0] and D[15:0] multiplexed with the data bus)
NE[x]	O	Chip selected, x=1~4 (In PSRAM applications, it is referred to as NCE (Cellular RAM, or CRAM))
NOE	O	Output enable
NWE	O	Write enable
NL (=NADV)	O	Address valid signal for PSRAM input only (Memory Signal Name: NADV)
NWAIT	I	PSRAM sends to FMC and waits for input signal
NBL[1:0]	O	Byte channel output. Bytes 0 to 1 control (High byte and low byte enable)

The maximum capacity is 512Mb(26 address lines).

40.5.2 Supported Memory and Transactions

The following table shows examples of devices, access modes, and transactions supported when the memory data bus width for NOR Flash, PSRAM, and SRAM is 16 bits. Transactions not permitted (or unsupported) by the FMC are shown in gray in this example.

Table 40-16 NOR Flash/PSRAM: examples of supported memory and transactions

Device	Mode	R/W	HB Data size	Memory Data size	Whether to allow	Notes
NOR Flash (multiplexed I/O and non-multiplexed I/O)	Asynchronous	R	8	16	YES	-
	Asynchronous	W	8	16	NO	-
	Asynchronous	R	16	16	YES	-
	Asynchronous	W	16	16	YES	-
	Asynchronous	R	32	16	YES	Divided into 2 FMC visits
	Asynchronous	W	32	16	YES	Divided into 2 FMC visits
	Asynchronous page	R	-	16	NO	Do not support
	Synchronous	R	8	16	NO	-
	Synchronous	R	16	16	YES	-
	Synchronous	R	32	16	YES	-
PSRAM (multiplexed I/O and non- multiplexed I/O)	Asynchronous	R	8	16	YES	-
	Asynchronous	W	8	16	YES	Use byte channel NBL[1:0]
	Asynchronous	R	16	16	YES	-
	Asynchronous	W	16	16	YES	-
	Asynchronous	R	32	16	YES	Divided into 2 FMC visits
	Asynchronous	W	32	16	YES	Divided into 2 FMC visits
	Asynchronous page	R	-	16	NO	Do not support
	Synchronous	R	8	16	NO	-
	Synchronous	R	16	16	YES	-
	Synchronous	R	32	16	YES	-
	Synchronous	W	8	16	YES	Use byte channel NBL[1:0]
	Synchronous	W	16/32	16	YES	-
SRAM and ROM	Asynchronous	R	8/16	16	YES	-
	Asynchronous	W	8/16	16	YES	Use byte channel NBL[1:0]
	Asynchronous	R	32	16	YES	Divided into 2 FMC visits
	Asynchronous	W	32	16	YES	Divided into 2 FMC visits, Use byte channel NBL[1:0]

40.5.3 General Timing Regular

Signal synchronization

- All controller output signals change on the rising edge of the internal clock (HCLK);
- In synchronous mode (read/write), all output signals change on the rising edge of HCLK. Regardless of the CLKDIV value, all output signals change as follows:
 - NOEL/NWEL/NEL/NADV/L/NADVH/NBL address valid outputs change on the falling edge of the FMC_CLK clock.
 - NOEH/NWEH/NEH/NOEH/NBLH address valid outputs change on the rising edge of the FMC_CLK clock.

40.5.4 NOR Flash/PSRAM Controller Asynchronous Transaction

Asynchronous static memory (NOR Flash, PSRAM, SRAM)

- Signals are synchronized via the internal clock HCLK. This clock is not sent to the memory;
- FMC always samples data before disabling the chip select signal NE. This ensures compliance with memory data hold timing requirements (typically 0ns from chip select enable high to minimum data conversion time).
- If extended mode is enabled (EXTMOD bit set to 1 in the R32_FMC_BCRx register), up to four extended modes (A, B, C, D) are available. Modes A, B, C, and D can be mixed for read and write operations. For example, a read operation can be performed in mode A while a write operation occurs in mode B.
- If extended mode is disabled (EXTMOD bit cleared in the R32_FMC_BCRx register), the FMC can operate in either Mode 1 or Mode 2 as described below:
 - When selecting SRAM/PSRAM memory type, Mode 1 is the default mode (MTYP = 0x0 or 0x01 in the R32_FMC_BCRx register);
 - When selecting NOR memory type, Mode 2 is the default mode (MTYP = 0x10 in the R32_FMC_BCRx register).

Mode 1-SRAM/PSRAM (CRAM)

As shown in the figure below, the supported modes for read and write transactions, along with the required R32_FMC_BCRx and R32_FMC_BTRx/R32_FMC_BWTRx register configurations.

Figure 40-3 Mode 1 read access waveform

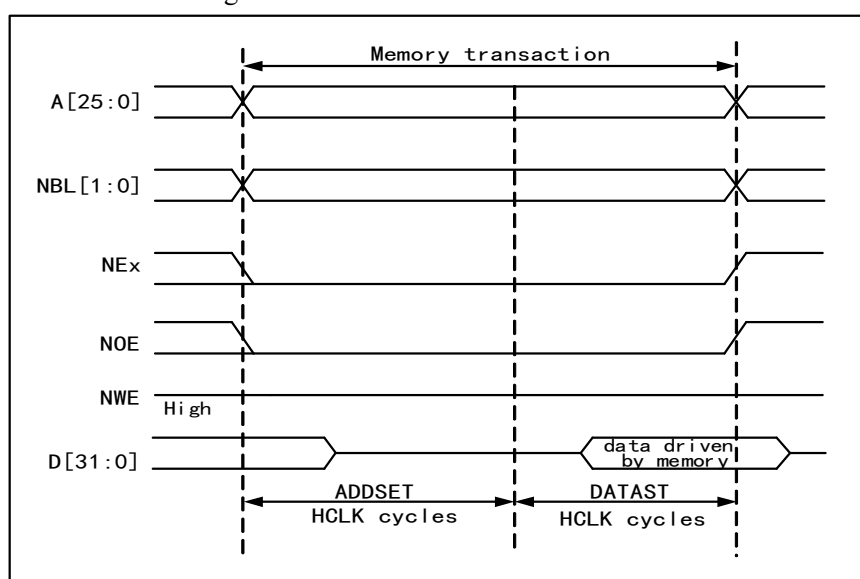
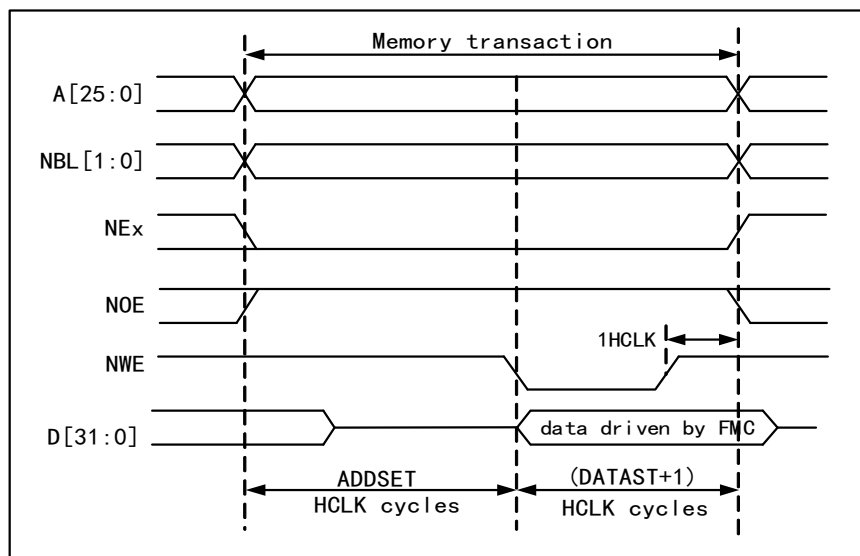


Figure 40-4 Mode 1 write access waveform



Writing one HCLK cycle at the end of a transaction helps ensure address and data hold time after the NEW rising edge. Due to this HCLK cycle, the DATAST value must be greater than zero ($\text{DATAST} > 0$).

Table 40-17 R32_FMC_BCRx bit field

Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00
[25:24]	BMP	0x0 or 0x1
[23:20]	Reserved	0x0
19	CBURSTRW	0x0 (has no effect on asynchronous mode)
[18:16]	CPSIZE	0x0 (has no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise, it remains at 0.
14	EXTMOD	0x0
13	WAITEN	0x0 (has no effect on asynchronous mode)
12	WREN	Make the settings as needed.
11	Reserved	0x0
10	WRAPMOD	0x0
9	WAITPOL	It only makes sense when bit 15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Irrelevant
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	Make the settings as needed, excluded 0x2 (NOR Flash)
1	MUXE	0x0
0	MBKEN	0x1

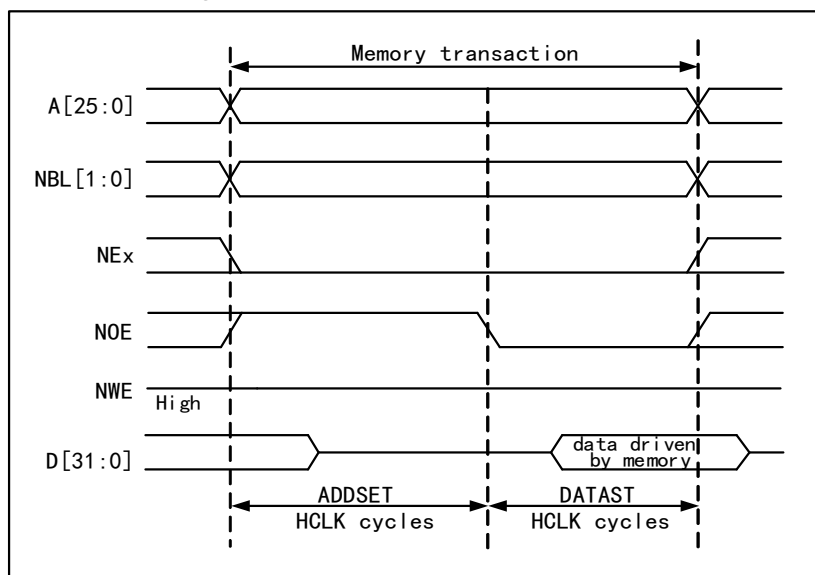
Table 40-18 FMC_BTRx bit field

Bit number	Bit name	The value to set
------------	----------	------------------

[31:30]	Reserved	0x00
[29:28]	ACCMOD	Irrelevant
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST+1 HCLK cycles write accesses, DATAST HCLK cycles for read accesses).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 0.

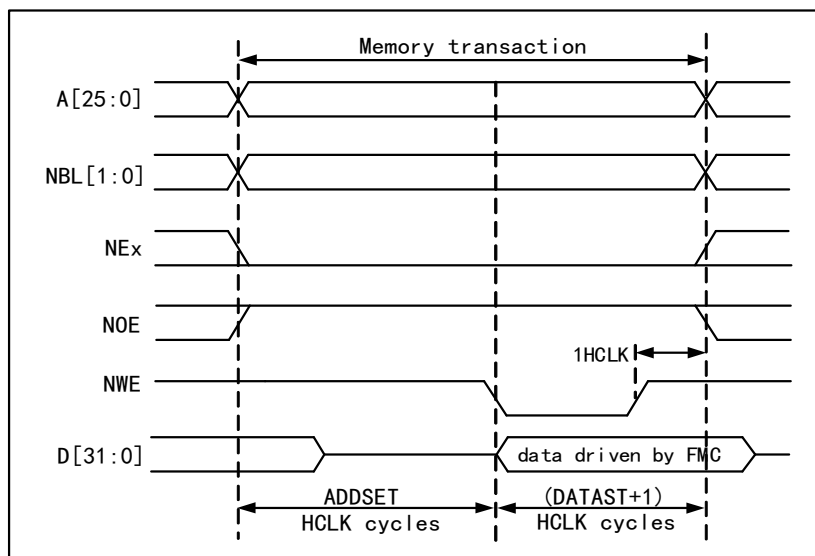
Mode A-SRAM/PSRAM (CRAM) OE switching

Figure 40-5 Mode A read access waveform



Note: NBL[1:0] is low during read access.

Figure 40-6 Mode A write access waveform



The difference from Mode 1 lies in the switching of NOE and independent reading and writing timing.

Table 40-19 R32_FMC_BCRx bit field

Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00
[25:24]	BMP	0x0 or 0x1
[23:20]	Reserved	0x0
19	CBURSTRW	0x0 (has no effect on asynchronous mode)
[18:16]	CPSIZE	0x0 (has no effect on asynchronous mode)
15	ASYNCAWAIT	Set to 1 if the memory supports this feature. Otherwise, it remains at 0.
14	EXTMOD	0x1
13	WAITEN	0x0 (has no effect on asynchronous mode)
12	WREN	Make the settings as needed.
11	Reserved	Irrelevant
10	WRAPMOD	0x0
9	WAITPOL	It only makes sense when bit 15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Irrelevant
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	Make the settings as needed, excluded 0x2 (NOR Flash)
1	MUXEN	0x0
0	MBKEN	0x1

Table 40-20 R32_FMC_BTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00

[29:28]	ACCMOD	Irrelevant
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST+1 HCLK cycles).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET+1 HCLK cycles). The minimum value for ADDSET is 0.

Table 40-21 R32_FMC_BWTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00
[29:28]	ACCMOD	0x00
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST HCLK cycles).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 0.

Mode 2/B-NOR Flash

Figure 40-7 Mode 2 and Mode B read access waveform

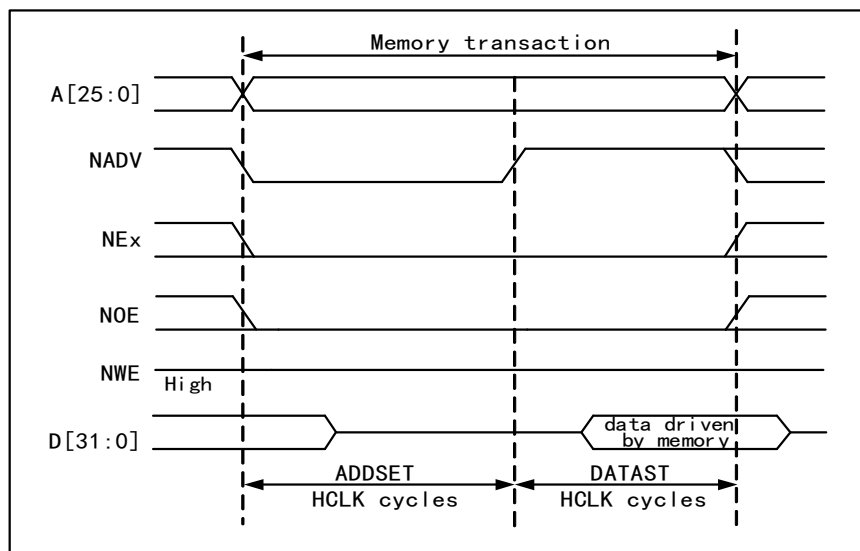


Figure 40-8 Mode 2 write access waveform

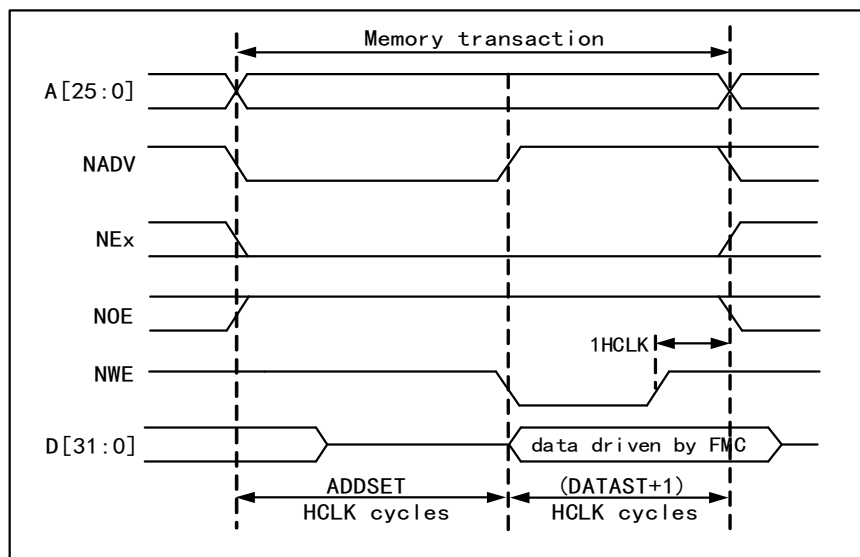
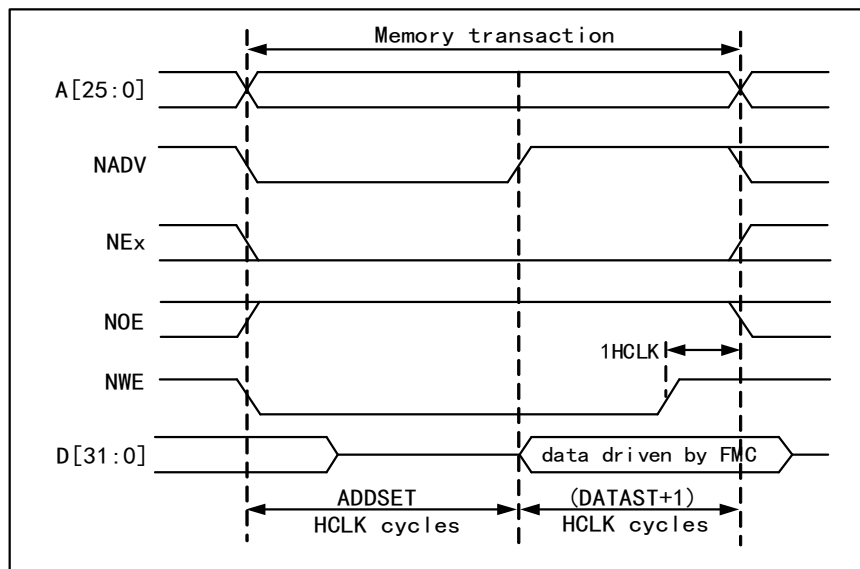


Figure 40-9 Mode B write access waveform



The difference from Mode 1 lies in the NWE switching and independent reading and writing timing when setting the extended mode (Mode B).

Table 40-22 R32_FMC_BCRx bit field

Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00
[25:24]	BMP	0x0 or 0x1
[23:20]	Reserved	0x0
19	CBURSTRW	0x0 (has no effect on asynchronous mode)
[18:16]	CPSIZE	0x0 (has no effect on asynchronous mode)
15	ASYNCAWAIT	Set to 1 if the memory supports this feature. Otherwise, it remains at 0.

14	EXTMOD	Mode B is 0x1, Mode 2 is 0x0
13	WAITEN	0x0 (has no effect on asynchronous mode)
12	WREN	Make the settings as needed.
11	Reserved	Irrelevant
10	WRAPMOD	0x0
9	WAITPOL	It only makes sense when bit 15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	0x2 (NOR Flash)
1	MUXEN	0x0
0	MBKEN	0x1

Table 40-23 R32_FMC_BTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00
[29:28]	ACCMOD	0x1 if set to extended mode.
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST HCLK cycles).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 0.

Table 40-24 R32_FMC_BWTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00
[29:28]	ACCMOD	0x1 if set to extended mode.
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST HCLK cycles).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 0.

Note: the R32_FMC_BWTRx register is valid only when the extended mode (mode B) is set, otherwise its contents are "irrelevant".

Mode C-NOR Flash-OE switching

Figure 40-10 Mode C write access waveform

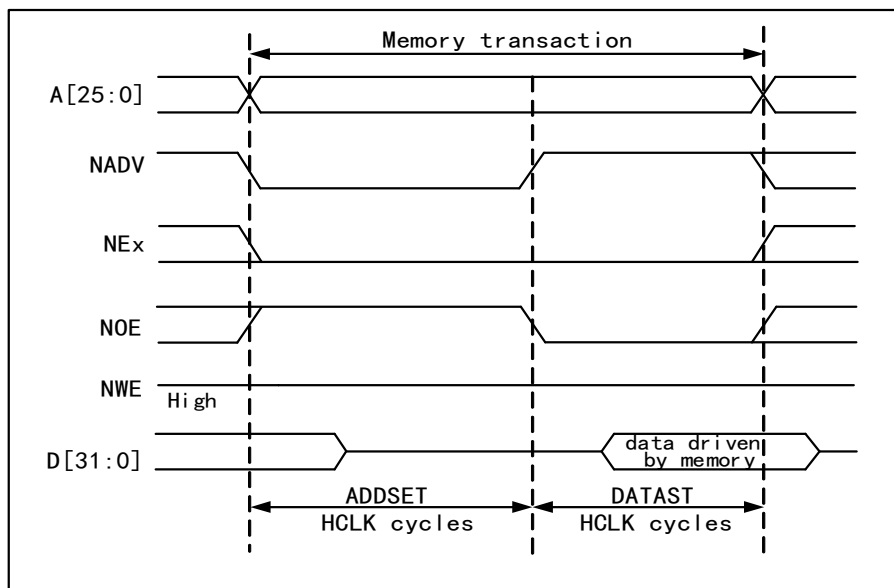
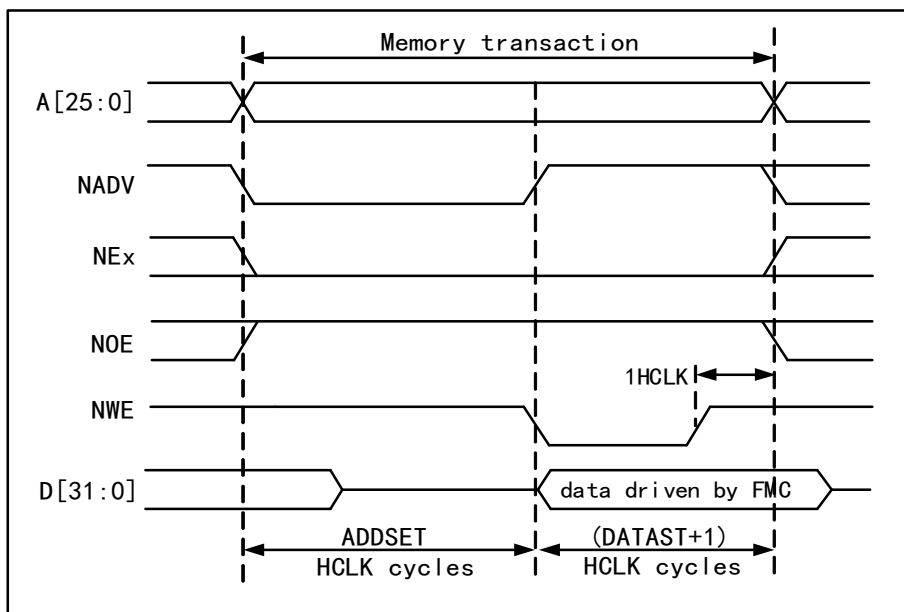


Figure 40-11 Mode C write access waveform



The difference from Mode 1 lies in the switching of NOE and independent reading and writing timing.

Table 40-25 R32_FMC_BCRx bit field

Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00
[25:24]	BMP	0x0 or 0x1
[23:20]	Reserved	0x0
19	CBURSTRW	0x0 (has no effect on asynchronous mode)

[18:16]	CPSIZE	0x0 (has no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise, it remains at 0.
14	EXTMOD	0x1
13	WAITEN	0x0 (has no effect on asynchronous mode)
12	WREN	Make the settings as needed.
11	Reserved	Irrelevant
10	WRAPMOD	0x0
9	WAITPOL	It only makes sense when bit 15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	0x2 (NOR Flash)
1	MUXEN	0x0
0	MBKEN	0x1

Table 40-26 R32_FMC_BTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00
[29:28]	ACCMOD	0x1 if set to extended mode.
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST+1 HCLK cycles).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET+1 HCLK cycles). The minimum value for ADDSET is 0.

Table 40-27 R32_FMC_BWTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00
[29:28]	ACCMOD	0x2
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST HCLK cycles).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 0.

Mode D-extended address asynchronous access

Figure 40-12 Mode D write access waveform

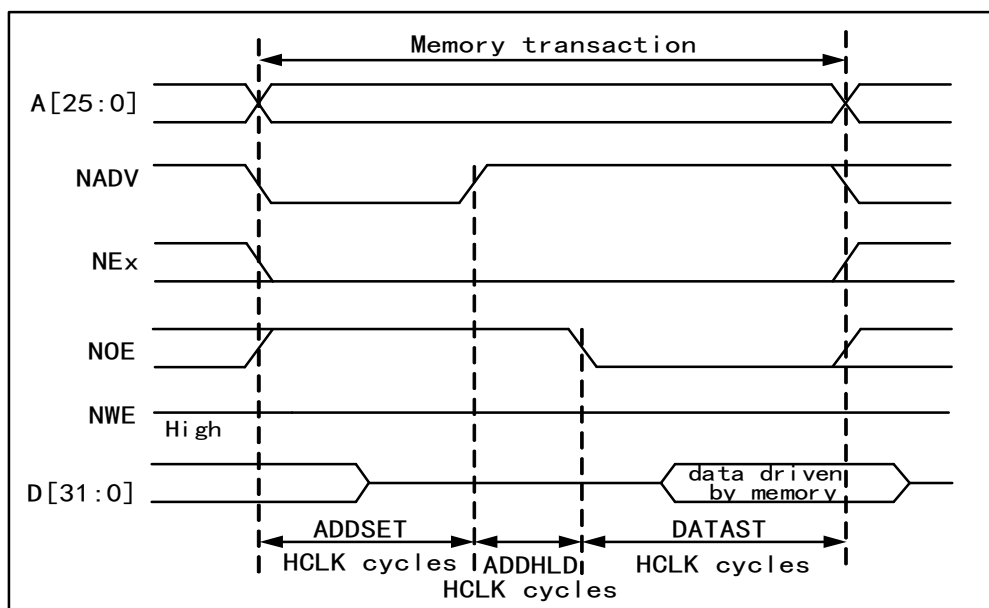
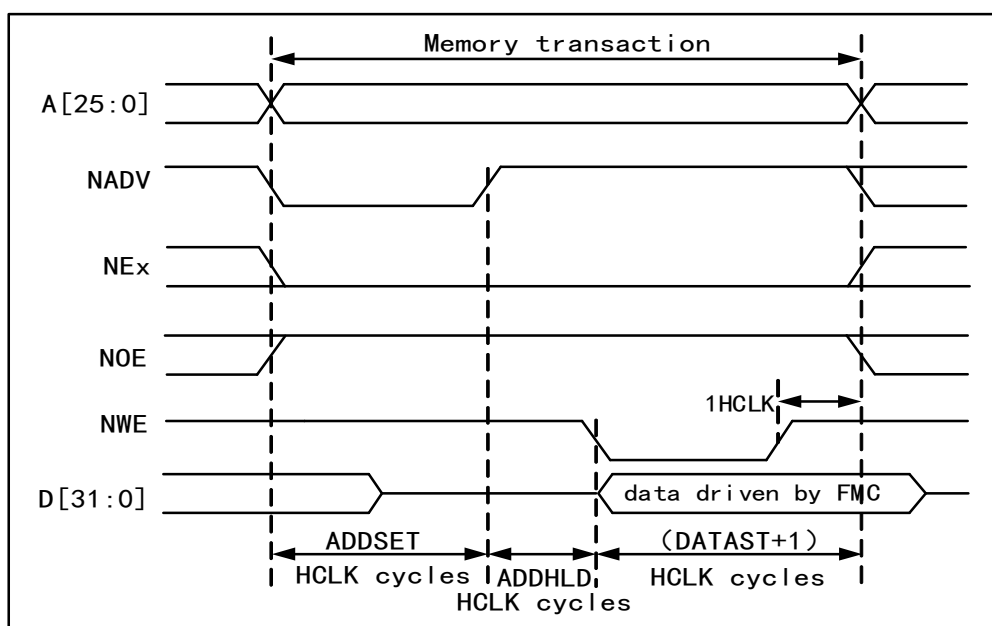


Figure 40-13 Mode D write access waveform



The difference from Mode 1 lies in the switching of NOE and independent reading and writing timing after the change of NADV.

Table 40-28 R32_FMC_BCRx bit field

Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00
[25:24]	BMP	0x0 or 0x1

[23:20]	Reserved	0x0
19	CBURSTRW	0x0 (has no effect on asynchronous mode)
[18:16]	CPSIZE	0x0 (has no effect on asynchronous mode)
15	ASYNCAWAIT	Set to 1 if the memory supports this feature. Otherwise, it remains at 0.
14	EXTMOD	0x1
13	WAITEN	0x0 (has no effect on asynchronous mode)
12	WREN	Make the settings as needed.
11	WAITCFG	Irrelevant
10	Reserved	0x0
9	WAITPOL	It only makes sense when bit 15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	Make the settings as needed.
1	MUXEN	0x0
0	MBKEN	0x1

Table 40-29 R32_FMC_BTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00
[29:28]	ACCMOD	0x3
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST+1 HCLK cycles).
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 1.

Table 40-30 R32_FMC_BWTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x00
[29:28]	ACCMOD	0x3
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	The duration of the second access phase (DATAST+1 HCLK cycles).
[7:4]	ADDHLD	Duration of the intermediate phase of the write access (ADDHLD HCLK cycles).

[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 0.
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Multiplex mode-multiplex asynchronous access NOR Flash

Figure 40-14 Multiplex read access waveforms

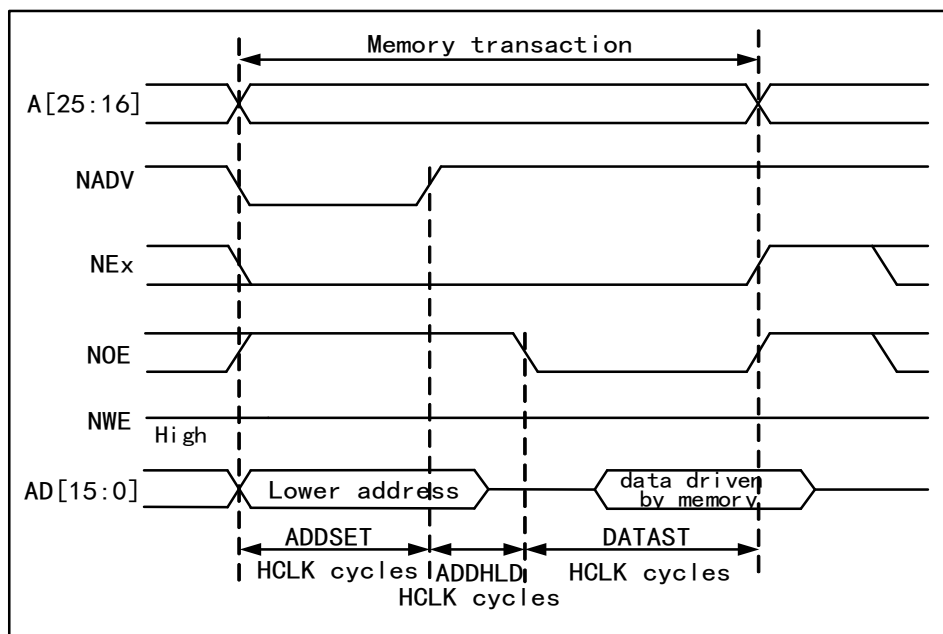
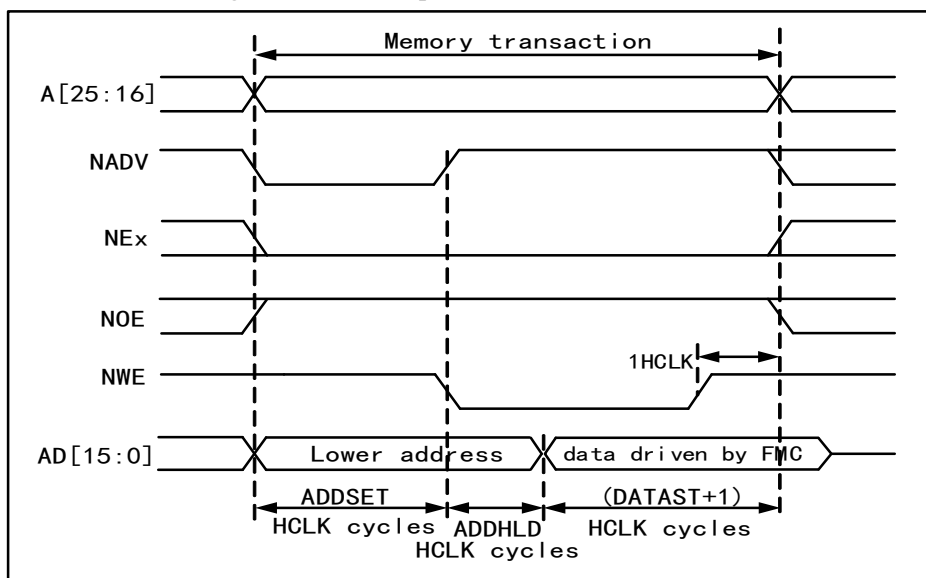


Figure 40-15 Multiplex write access waveforms



The difference from mode D is that low address bytes are driven on the data bus.

Table 40-31 R32_FMC_BCRx bit field

Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00

[25:24]	BMP	0x0 or 0x1
[23:20]	Reserved	0x0
19	CBURSTRW	0x0 (has no effect on asynchronous mode)
[18:16]	CPSIZE	0x0 (has no effect on asynchronous mode)
15	ASYNCWAIT	Set to 1 if the memory supports this feature. Otherwise, it remains at 0.
14	EXTMOD	0x0
13	WAITEN	0x0 (has no effect on asynchronous mode)
12	WREN	Make the settings as needed.
11	WAITCFG	Irrelevant
10	Reserved	0x0
9	WAITPOL	It only makes sense when bit 15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	0x2 (NOR Flash)
1	MUXEN	0x0
0	MBKEN	0x1

Table 40-32 R32_FMC_BTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	Irrelevant
[23:20]	CLKDIV	Irrelevant
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	Duration of the second access phase (read access is DATAST HCLK cy and write access is DATAST+1 HCLK cycles).
[7:4]	ADDHLD	Duration of access intermediate stage (ADDHLD HCLK cycles).
[3:0]	ADDSET	Duration of the first access phase (ADDSET HCLK cycles). The minimum value for ADDSET is 1.

WAIT management in asynchronous access

If the asynchronous memory sends a WAIT signal indicating that it is not ready to receive or provide data, the ASYNCWAIT bit in the R32_FMC_BCRx register must be set.

If the WAIT signal is in an active state (the level depends on the WAITPOL bit), the second access phase (data establishment phase) controlled by the DATAST bit will be extended until WAIT becomes invalid. Unlike the data establishment phase, the first access phase (address establishment and address holding phase) controlled by ADDSET and ADDHLD bits is insensitive to WAIT, so the first access phase will not be extended.

The data setup phase must be configured to detect WAIT 4 HCLK cycles before the end of the memory transaction. The following situations must be considered:

- 1) The WAIT signal sent by the memory is aligned with the NOE/NEW signal:

$$\text{DATAST} \geq (4 * \text{HCLK}) + \text{max_wait_assertion_time}$$

2) The WAIT signal sent by the memory is aligned with NEx (or the NOE/NWE signal does not flip):

If

$$\text{max_wait_assertion_time} > \text{address_phase} + \text{hold_phase}$$

Then:

$$\text{DATAST} \geq (4 * \text{HCLK}) + (\text{max_wait_assertion_time} - \text{address_phase} - \text{hold_phase})$$

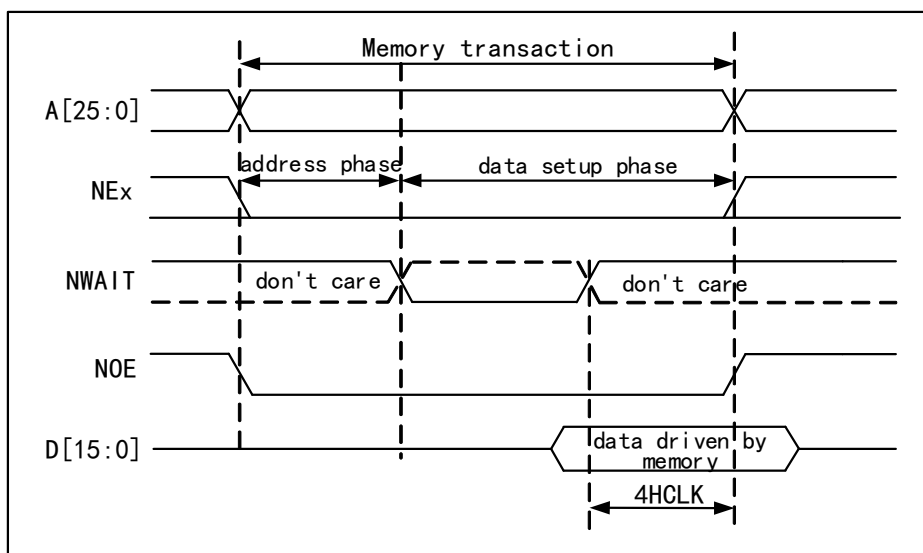
Otherwise

$$\text{DATAST} \geq 4 * \text{HCLK}$$

Where max_wait_assertion_time is the longest time it takes for the memory to enable the wait signal after NEx/NOE/NWE goes low.

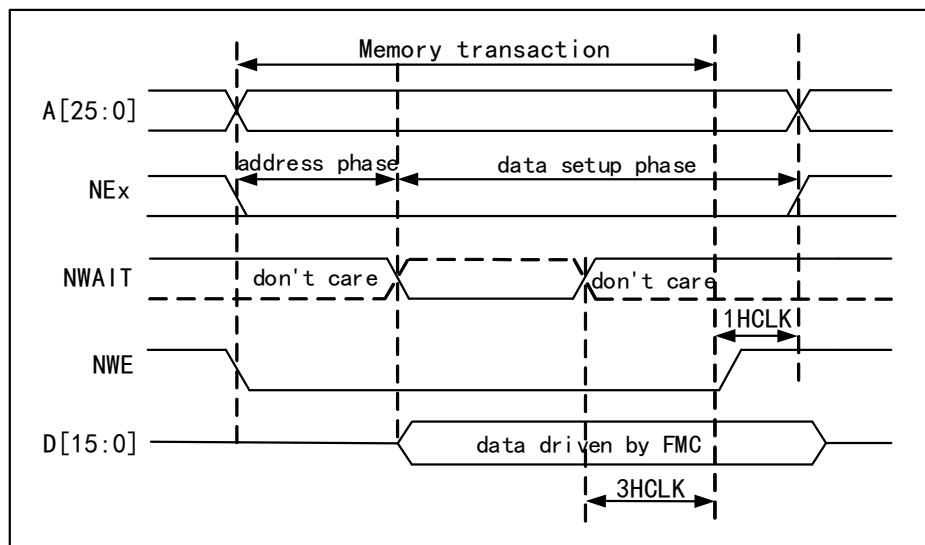
Figures 40-15 and 40-16 show the number of increased HCLK clock cycles in the memory access phase after the asynchronous memory releases WAIT (independent of the above situation).

Figure 40-16 Asynchronous waiting during reading access waveform



Note: The polarity of NWAIT depends on the setting of WAITPOL bit in R32_FMC_BCRx register.

Figure 40-17 Asynchronous waiting during writing access waveform



Note: The polarity of NWAIT depends on the setting of WAITPOL bit in R32_FMC_BCRx register.

40.5.5 Synchronous Transaction

The memory clock FMC_CLK is a divisor of HCLK. It depends on the value of CLKDIV and the MWID/HB data size, as shown in the following formula:

$$\text{FMC_CLK Frequency division ratio} = \max(\text{CLKDIV}+1, \text{MWID (HB data size)})$$

When MWID is 16-bit or 8-bit, the frequency division ratio of FMC_CLK is always defined by the setting value of CLKDIV.

When MWID is 32-bit, the frequency division ratio of FMC_CLK also depends on the size of HB data.

Example:

- LKDIV=1, MWID=32 bits, when HB data size=8 bits, FMC_CLK=HCLK/4.
- CLKDIV=1, MWID=16 bits, when HB data size =8 bits, FMC_CLK=HCLK/2.

NOR Flash specifies the minimum time from NADV enable to CLK high. In order to meet this limitation, FMC will not send the clock to the memory within the first internal clock cycle of synchronous access (before NADV is enabled). This ensures that the rising edge of the memory clock appears in the middle of the NADV low pulse.

Data delay and NOR delay

Data delay is the number of cycles to wait before sampling data. The value of DATLAT must match the delay value specified in the NORFLASH configuration register. When NADV is low in the data delay count, FMC will not count the clock period.

Some NOR Flash count the NADV low period in the data delay count, so the exact relationship between the NOR Flash delay and the FMC DATLAT parameter can be any of the following:

- NOR Flash delay =(DATLAT+2) CLK clock cycles.
- NOR Flash delay =(DATLAT+3) CLK clock cycles.

Recently, some memories will enable NWAIT in the delay phase. In this case, you can set DATLAT to the minimum value. Then, FMC will sample the data and wait long enough to evaluate whether the data is valid. In this way, FMC can detect the delay time in the memory, so as to process the real data.

Other memories will not enable NWAIT during the delay. In this case, the delay of FMC and memory must be set correctly, otherwise invalid data may be misused as valid data, or valid data may be lost in the initial stage of

memory access.

Single burst transmission

When the selected storage area is configured in synchronous burst mode, for example, if an HB single burst transaction is requested from a 16-bit memory, FMC will execute a burst transaction with a length of 1 (if the HB transmission is 16 bits) or a burst transaction with a length of 2 (if the HB transmission is 32 bits), and then disable the chip selection signal at the last data strobe.

Compared with asynchronous read operation, this is not the most efficient transmission method in terms of cycle. However, random asynchronous reading requires reprogramming the memory access mode first, so the total time will be longer.

Cross the border page of Cellular RAM1.5

Cellular RAM1.5 does not allow burst access to cross page boundaries. When the CPSIZE bit in the R32_FMC_BCR1 register is configured according to the memory page size to reach the memory page size, the FMC controller will automatically separate the burst access.

Waiting management

For synchronous NOR Flash, NWAIT will be evaluated after the configured delay period (equivalent to (DATLAT+2) CLK clock cycles).

If NWAIT is valid (low level when WAITPOL=0 and high level when WAITPOL=1), the waiting state will be inserted until NWAIT is invalid (high level when WAITPOL=0 and low level when WAITPOL=1).

When NWAIT is invalid, the data will be considered valid immediately (bit WAITCFG=1) or at the next clock edge (bit WAITCFG=0).

During the waiting period when the NWAIT signal is inserted, the controller will continue to send clock pulses to the memory to keep the chip select and output enable signals valid. But the data is not considered valid.

In burst mode, NOR Flash NWAIT signal has two timing configurations:

- Flash sends out NWAIT signal (the default value after reset) one data period before the waiting period.
- Flash sends out NWAIT signal during the waiting period.

FMC supports these 2 configurations of NOR Flash waiting period, and each chip selection is configured through the WAITCFG bit (x=0..3) pin of R32_FMC_BCRx register.

Figure 40-18 Waiting configuration waveform

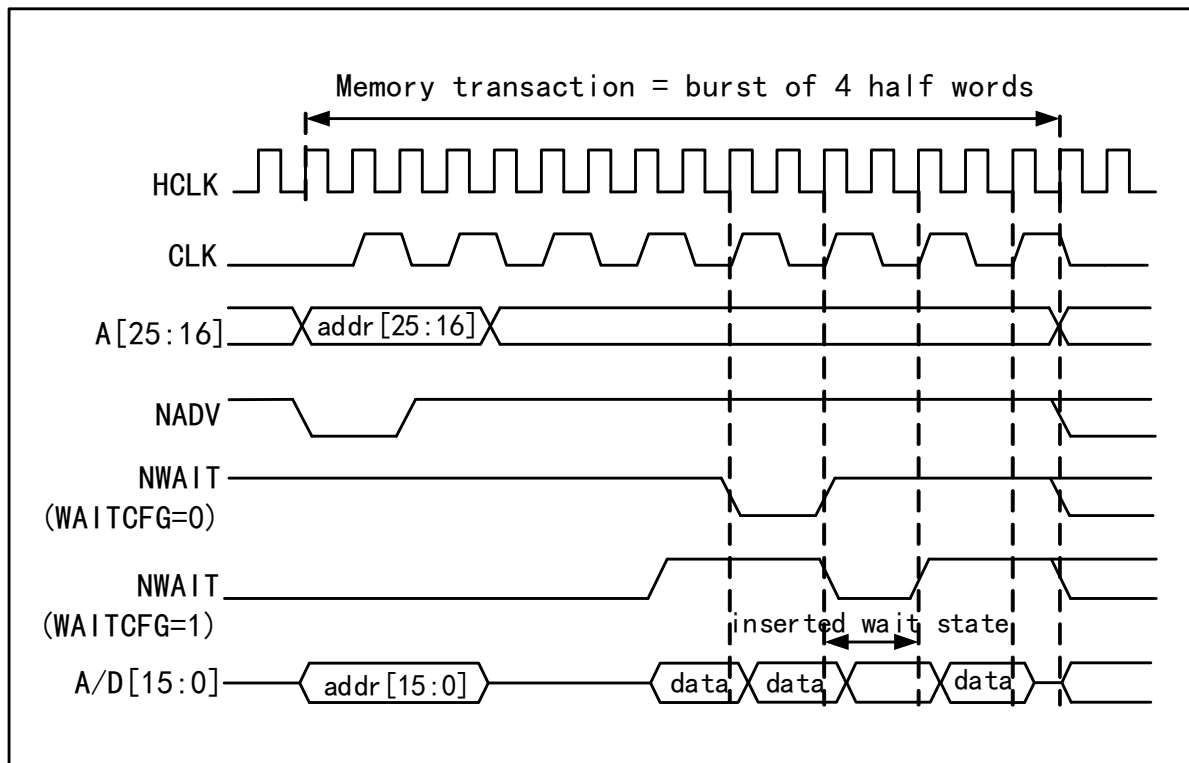
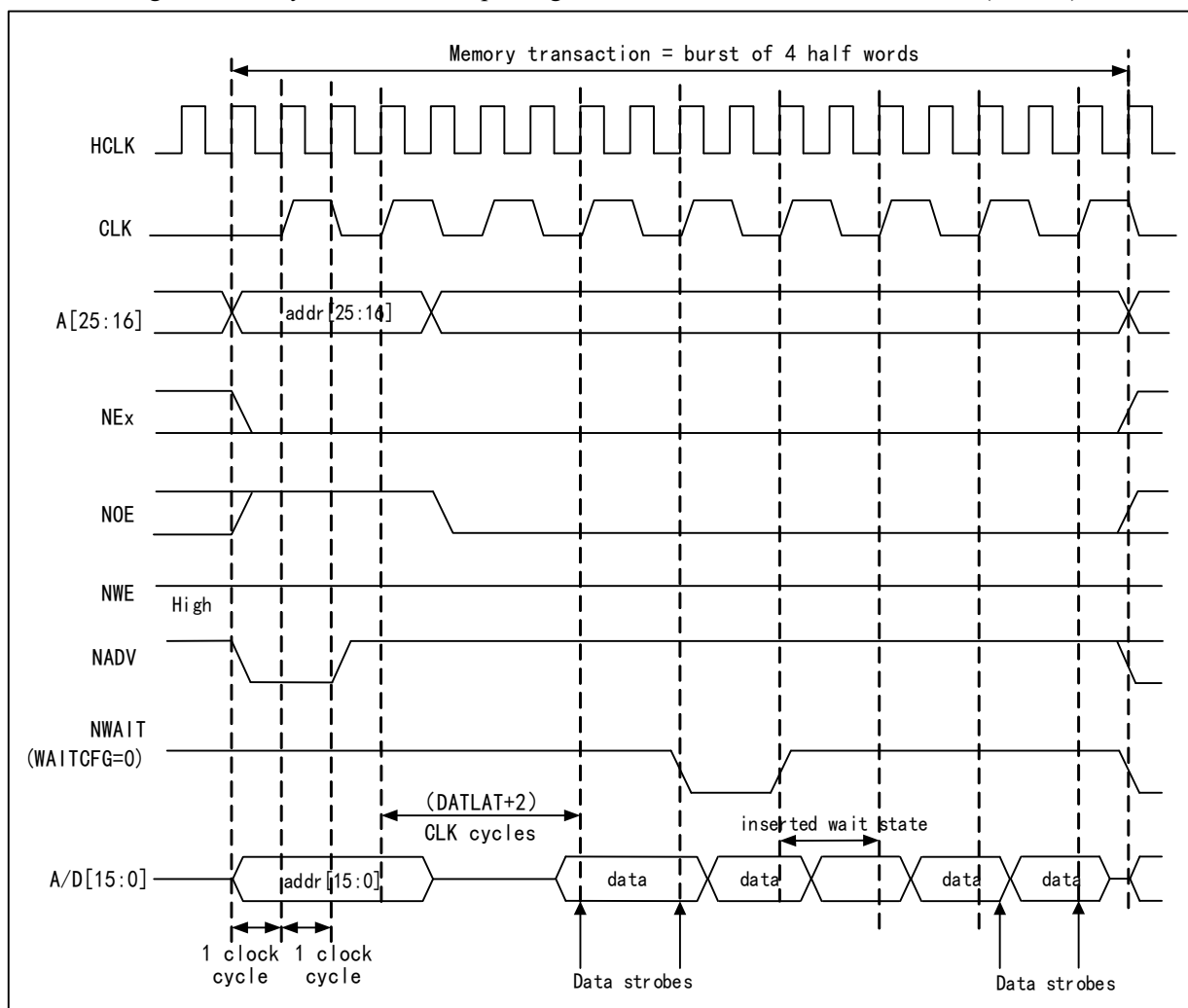


Figure 40-19 Synchronous multiplexing read mode waveform-NOR, PSRAM (CRAM)



Note: Byte channel outputs NBL are not shown. They remain high for NOR access and low for PSRAM(CRAM) access.

Table 40-33 R32_FMC_BCRx bit field

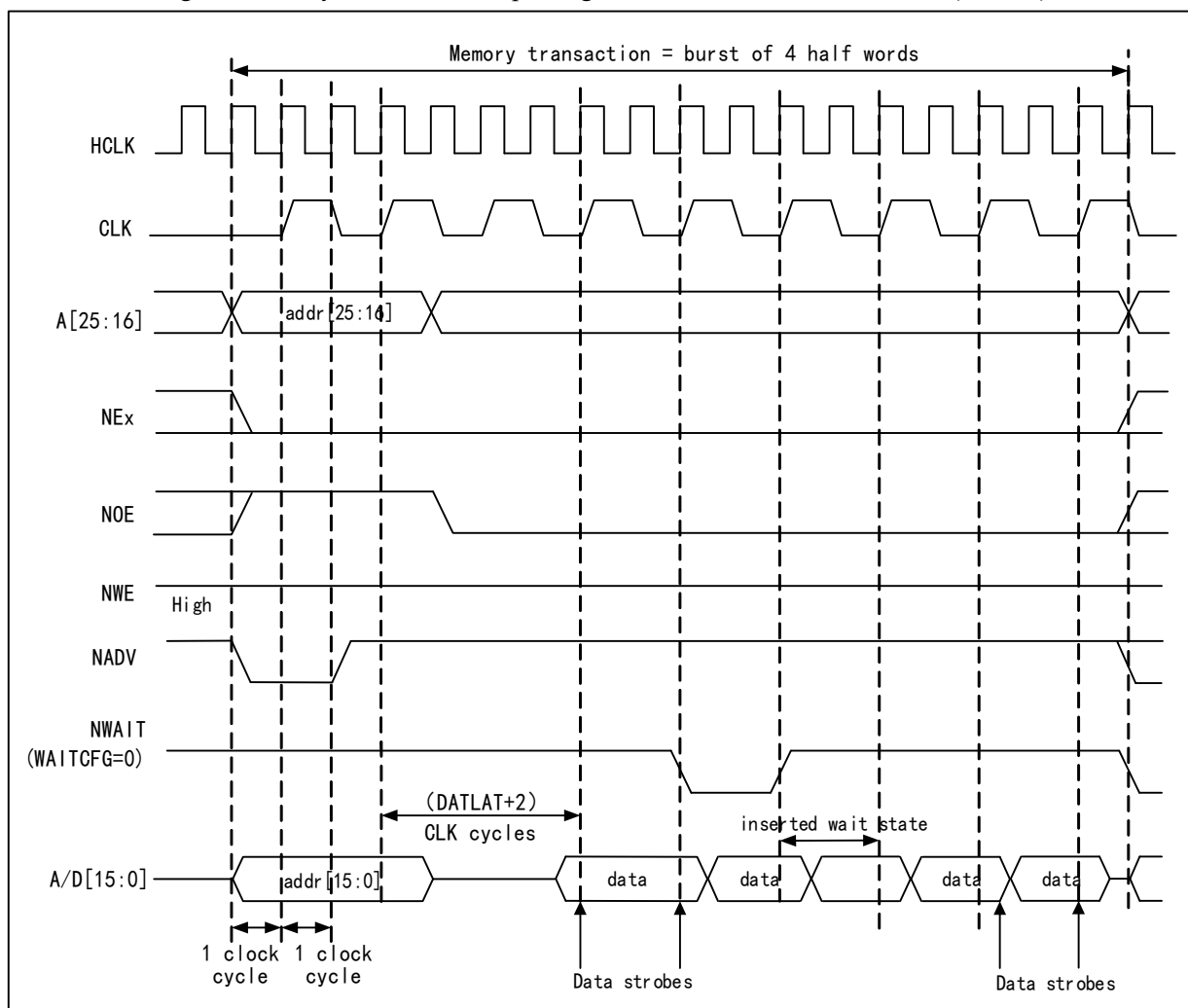
Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00
[25:24]	BMP	0x0 or 0x1
[23:20]	Reserved	0x0
19	CBURSTRW	There is no effect on synchronous reading.
[18:16]	CPSIZE	0x0 (has no effect on asynchronous mode)
15	ASYNCAWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	Set to 1 if the memory supports this feature, otherwise keep to 0.
12	WREN	There is no effect on synchronous reading.
11	WAITCFG	Whether it is set or not depends on the memory situation.
10	Reserved	0x0
9	WAITPOL	Whether it is set or not depends on the memory situation.

8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Set if memory supports it (NOR Flash).
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	0x1 or 0x2
1	MUXEN	Make the settings as needed.
0	MBKEN	0x1

Table 40-34 R32_FMC_BTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	Data delay
[23:20]	CLKDIV	0x0, CLK=HCLK (Not supported) 0x1, CLK=2*HCLK ...
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	Irrelevant
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Irrelevant

Figure 40-19 Synchronous multiplexing write mode waveform-PSRAM (CRAM)



Note 1: The memory must send out the NWAIT signal one cycle in advance, and WAITCFG must be programmed to 0 accordingly.

Note 2: Byte Channel (NBL) outputs are not shown, and they remain low when NEx is active.

Table 40-35 R32_FMC_BCRx bit field

Bit number	Bit name	The value to set
31	FMC_EN	0x1
[30:26]	Reserved	0x00
[25:24]	BMP	0x0 or 0x1
[23:20]	Reserved	0x0
19	CBURSTRW	0x1
[18:16]	CPSIZE	Set as needed (0x1 for CRAM1.5)
15	ASYNCAWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	Set to 1 if the memory supports this feature, otherwise keep to 0.
12	WREN	0x1
11	WAITCFG	0x0
10	Reserved	0x0

9	WAITPOL	Whether it is set or not depends on the memory situation.
8	BURSTEN	There is no effect on synchronous writing.
7	Reserved	0x1
6	FACCEN	According to the memory support.
[5:4]	MWID	Make the settings as needed.
[3:2]	MTYP	0x1
1	MUXEN	Make the settings as needed.
0	MBKEN	0x1

Table 40-36 R32_FMC_BTRx bit field

Bit number	Bit name	The value to set
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	Data delay
[23:20]	CLKDIV	0x0, CLK=HCLK (Not supported) 0x1, CLK=2*HCLK
[19:16]	BUSTURN	The time interval between NEx going high and NEx going low (BUSTURN HCLK)
[15:8]	DATAST	Irrelevant
[7:4]	ADDHLD	Irrelevant
[3:0]	ADDSET	Irrelevant

40.5.6 NOR/PSRAM Control Register

Table 40-37 NOR/PSRAM-related registers

Name	Access address	Description	Reset value
R32_FMC_BCR1	0x40025400	SRAM/NOR-Flash chip select control register 1	0x00003052
R32_FMC_BTR1	0x40025404	SRAM/NOR-Flash chip select timing register 1	0x0FFFFFFF
R32_FMC_BCR2	0x40025408	SRAM/NOR-Flash chip select control register 2	0x000030D2
R32_FMC_BTR2	0x4002540C	SRAM/NOR-Flash chip select timing register 2	0x0FFFFFFF
R32_FMC_BCR3	0x40025410	SRAM/NOR-Flash chip select control register 3	0x000030D2
R32_FMC_BTR3	0x40025414	SRAM/NOR-Flash chip select timing register 3	0x0FFFFFFF
R32_FMC_BCR4	0x40025418	SRAM/NOR-Flash chip select control register 4	0x000030D2
R32_FMC_BTR4	0x4002541C	SRAM/NOR-Flash chip select timing register 4	0x0FFFFFFF
R32_FMC_BWTR1	0x40025504	SRAM/NOR-Flash write timing register 1	0x0FFFFFFF
R32_FMC_BWTR2	0x4002550C	SRAM/NOR-Flash write timing register 2	0x0FFFFFFF
R32_FMC_BWTR3	0x40025514	SRAM/NOR-Flash write timing register 3	0x0FFFFFFF
R32_FMC_BWTR4	0x4002551C	SRAM/NOR-Flash write timing register 4	0x0FFFFFFF

40.5.6.1 SRAM/NOR-Flash Chip Select Control Register 1 (R32_FMC_BCR1)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FMC	Reserved					BMP		Reserved				CBU	CPSIZE[2:0]		

$\overline{\text{EN}}$											RST RW				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ASY NCW AIT	EX TM OD	WA ITE N	WR EN	WAIT CFG	Rese rved	WAIT POL	BURS TEN	Rese rved	FA C CE N	MWID[1:0]		MTYP[1:0]		MU XE N	MB KE N

Bit	Name	Access	Description	Reset value
31	FMC_EN	RW	NAND/NOR/PSRAM enable bit: 1: Enable; 0: Disable.	0
[30:26]	Reserved	RO	Reserved	0
[25:24]	BMP	RW	Address remapping to realize the address switching function of Table 40-1.	0
[23:20]	Reserved	RO	Reserved	0
19	CBURSTRW	RW	Synchronous write enable: 1: Write in synchronous mode; 0: Always write in asynchronous mode.	0
[18:16]	CPSIZE[2:0]	RW	CRAM page size: 000: No burst separation after crossing the page boundary; 001: 128-byte; 010: 256-byte; 011: 512-byte; 100: 1024-byte; Others: Reserved.	0
15	ASYNCAWAIT	RW	Waiting signal during asynchronous transmission: This bit enables/disables FMC to use the wait signal, even during asynchronous protocol. 1: Consider NWAIT signal when running asynchronous protocol; 0: NWAIT signal is not considered when running asynchronous protocol.	0
14	EXTMOD	RW	Extended mode enable: FMC can configure the write timing of non-multiplexed asynchronous access in R32_FMC_BWTR register. This configuration is enabled by the EXTMOD bit, so that the read and write operations adopt different timings. 1: Consider the value in R32_FMC_BWTR register; 0: The value in R32_FMC_BWTR register is not considered. <i>Note: If extended mode is disabled, FMC can operate</i>	0

			<i>in mode 1 or mode 2, as follows:</i> <i>–When the SRAM/PSRAM memory type is selected, Mode 1 is the default mode (MTYP=0x0 or 0x01);</i> <i>–When the NOR memory type is selected, mode 2 is the default mode (MTYP=0x10).</i>	
13	WAITEN	RW	Wait enable bit: 1: Enable the NWAIT signal (consider its level, and if enabled, insert the waiting period after the configured delay period); 0: Disable NWAIT signal (regardless of its level, do not insert the waiting period after the configured Flash delay period).	1
12	WREN	RW	Write enable bit: 1: FMC enables writing in the storage area; 0: FMC prohibits writing in the storage area, and HB error will be reported if writing is performed.	1
11	WAITCFG	RW	Waiting time sequence configuration: 1: NWAIT signal is valid during the waiting period (not applicable to PSRAM); 0: NWAIT signal is valid one data period before the waiting period.	0
10	Reserved	RO	Reserved	0
9	WAITPOL	RW	Waiting signal polarity bit: 1: NWAIT active high; 0: NWAIT active low.	0
8	BURSTEN	RW	Synchronous read enable bit: 1: Enable synchronous read mode. Read access in synchronous mode; 0: Disable synchronous reading mode. Read access in asynchronous mode.	0
7	Reserved	RO	Reserved	0
6	FACCEN	RW	Flash access enable: 1: Enable the corresponding NOR Flash access; 0: Disable the corresponding NOR Flash access.	1
[5:4]	MWID[1:0]	RW	Memory data bus width: 00: 8-bit; 01: 16-bit; 10: 32-bit; 11: Reserved.	01b
[3:2]	MTYP[1:0]	RW	Memory type: 00: SRAM (For storage areas 2 ~ 4, the default value after reset); 01: PSRAM (CRAM); 10: NOR Flash/OneNAND Flash (For storage area	00b

			1, the default value after reset); 11: Reserved.	
1	MUXEN	RW	Address/data multiplexing enable bits: When this bit is set to 1, the address and data values are multiplexed on the data bus, which is only valid for NOR and PSRAM memories. 1: Address/data are multiplexed on the data bus; 0: Address/data is not multiplexed.	1
0	MBKEN	RW	Storage area enable bit: Enable storage area. Enable storage area 1 after reset, and disable all other storage areas. Access to forbidden storage areas will cause errors on HB bus. 1: Enable the corresponding storage area; 0: Disable the corresponding storage area;	0

40.5.6.2 SRAM/NOR-Flash Chip Select Control Register x (R32_FMC_BCRx) (x=2/3/4)

Offset address: $0x8*(x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												CBURSTRW	CPSIZE[2:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ASYNCWAIT	EXTMOD	WAITEN	WREN	WAITCFG	Reserved	WAITPOL	BURSTEN	Reserved		MWID[1:0]		MTYP[1:0]		MUXEN	MBKEN

Bit	Name	Access	Description	Reset value
31	Reserved	RW	Reserved	0
[30:26]	Reserved	RO	Reserved	0
[25:24]	Reserved	RW	Reserved	0
[23:20]	Reserved	RO	Reserved	0
19	CBURSTRW	RW	Synchronous write enable: 1: Write in synchronous mode; 0: Always write in asynchronous mode.	0
[18:16]	CPSIZE[2:0]	RW	CRAM page size: These bits are used for CellularRAM1.5, which does not allow burst access to cross the address boundary between pages. If these bits are configured, the FMC controller will automatically separate the burst access when the memory page size is reached. 000: No burst separation after crossing the page boundary; 001: 128-byte; 010: 256-byte;	0

			011: 512-byte; 100: 1024-byte; Others: Reserved.	
15	ASYNCAWAIT	RW	Waiting signal during asynchronous transmission: This bit enables/disables FMC to use the wait signal, even during asynchronous protocol. 1: Consider NWAIT signal when running asynchronous protocol; 0: NWAIT signal is not considered when running asynchronous protocol.	0
14	EXTMOD	RW	Extended mode enable: FMC can configure the write timing of non-multiplexed asynchronous access in R32_FMC_BWTR register. This configuration is enabled by the EXTMOD bit, so that the read and write operations adopt different timings. 1: Consider the value in R32_FMC_BWTR register; 0: The value in R32_FMC_BWTR register is not considered. <i>Note: If extended mode is disabled, FMC can operate in mode 1 or mode 2, as follows:</i> –When the SRAM/PSRAM memory type is selected, Mode 1 is the default mode (MTYP=0x0 or 0x01); –When the NOR memory type is selected, mode 2 is the default mode (MTYP=0x10).	0
13	WAITEN	RW	Wait enable bit: 1: Enable the NWAIT signal (consider its level, and if enabled, insert the waiting period after the configured delay period); 0: Disable NWAIT signal (regardless of its level, do not insert the waiting period after the configured Flash delay period).	1
12	WREN	RW	Write enable bit: 1: FMC enables writing in the storage area; 0: FMC prohibits writing in the storage area, and HB error will be reported if writing is performed.	1
11	WAITCFG	RW	Waiting time sequence configuration: 1: NWAIT signal is valid during the waiting period (not applicable to PSRAM); 0: NWAIT signal is valid one data period before the waiting period.	0
10	Reserved	RO	Reserved	0
9	WAITPOL	RW	Waiting signal polarity bit: 1: NWAIT active high;	0

			0: NWAIT active low.	
8	BURSTEN	RW	Synchronous read enable bit: 1: Enable synchronous read mode. Read access in synchronous mode; 0: Disable synchronous reading mode. Read access in asynchronous mode.	0
7	Reserved	RO	Reserved	1
6	Reserved	RW	Reserved	1
[5:4]	MWID[1:0]	RW	Memory data bus width: 00: 8-bit; 01: 16-bit; 10: 32-bit; 11: Reserved.	01b
[3:2]	MTYP[1:0]	RW	Memory type: 00: SRAM (For storage areas 2 ~ 4, the default value after reset); 01: PSRAM (CRAM); 10: NOR Flash/OneNAND Flash (For storage area 1, the default value after reset); 11: Reserved.	0
1	MUXEN	RW	Address/data multiplexing enable bits: When this bit is set to 1, the address and data values are multiplexed on the data bus, which is only valid for NOR and PSRAM memories. 1: Address/data are multiplexed on the data bus; 0: Address/data is not multiplexed.	1
0	MBKEN	RW	Storage area enable bit: Enable storage area. Enable storage area 1 after reset, and disable all other storage areas. Access to forbidden storage areas will cause errors on HB bus. 1: Enable the corresponding storage area; 0: Disable the corresponding storage area;	0

40.5.6.3 SRAM/NOR-Flash Chip Select Timing Register x (R32_FMC_BTRx) (x=1/2/3/4)

Offset address: $0x04 + 0x8 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		ACCMOD [1:0]		DATLAT[3:0]			CLKDIV[3:0]			BUSTURN[3:0]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAST[7:0]								ADDHLD[3:0]			ADDSET[3:0]				

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved.	0

[29:28]	ACCMOD[1:0]	RW	Access mode: Refers to the synchronous interview mode, as shown in the time series diagram. These bits are valid only when the EXTMOD bit in the R32_FMC_BCRx register is set. 00: Access mode A; 01: Access mode B; 10: Access mode C; 11: Access mode D.	0
[27:24]	DATLAT[3:0]	RW	Data delay of synchronous memory: For synchronous access with read/write burst mode enabled (BURSTEN/CBURSTRW bit 1), this field defines the number of memory clock cycles to be sent to the memory before reading and writing the first data (+2): This timing parameter is expressed in FMC_CLK period instead of HCLK period. This value is independent of asynchronous access mode. 0000: Data delay of 2 CLK clock cycles during the first burst access; 1111: Data delay of 17 CLK clock cycles during the first burst access.	1111b
[23:20]	CLKDIV[3:0]	RW	Clock division ratio of FMC_CLK signal: 0000: Reserved; 0001: FMC_CLK cycle = 2*HCLK cycle; 0010: FMC_CLK cycle = 3*HCLK cycle; 1111: FMC_CLK cycle = 16*HCLK cycle. In asynchronous NOR Flash, SRAM or PSRAM access mode, this value is irrelevant. <i>Note: For information about the frequency division ratio formula of FMC_CLK, please refer to Section 40.5.5: Synchronous transactions.</i>	1111b
[19:16]	BUSTURN[3:0]	RW	Duration of bus turnaround phase: Writing these bits through software can add delay at the end of a write-read (and read-write) transaction. Bus turnaround delay can also be inserted between two consecutive read/write transactions to different memory areas. When two consecutive read transactions point to the same FMC storage area, only one HCLK clock cycle will be inserted regardless of the configured BUSTURN timing. When two consecutive write transactions point to the same FMC storage area, bus turnaround delay is not inserted.	1111b

			Bus turnaround delay can match the shortest time between consecutive transactions (t_{EHEL} from NEx high to NEx low) and the maximum time required for the memory to release the data bus after a read access (t_{EHQZ}): (BUSTRUN+1) HCLK cycle $\geq t_{EHELmin}$, (BUSTRUN+2) HCLK cycle $\geq t_{EHQZmax}$ (If EXTMOD=0) (BUSTRUN+2) HCLK cycle $\geq \max(t_{EHELmin}, t_{EHQZmax})$ (If EXTMOD=1). 0000: Duration of BUSTURN phase = Add 0 HCLK clock cycle; ... 1111: Duration of BUSTURN phase =15*HCLK clock cycle.	
[15:8]	DATAST[7:0]	RW	Duration of data phase: Writing these bits through software can define the duration of the data phase (please refer to Figure 40-3 ~ Figure 40-15), which is suitable for asynchronous access mode: 0000 0000: Reserved; 0000 0001: Duration of the DATAST phase =1*HCLK clock cycle; 0000 0010: Duration of the DATAST phase =2*HCLK clock cycle; ... 1111 1111: Duration of the DATAST phase =255*HCLK clock cycle. Information about the duration of each memory type and access mode data phase (please refer to Figure 40-3 ~ Figure 40-15). For example: In Mode 1, Write Access, and DATAST=1 conditions, the duration of the data phase = DATAST+1 = 2 HCLK clock cycles. <i>Note: In synchronous access mode, this value is irrelevant.</i>	1111 1111b
[7:4]	ADDHLD[3:0]	RW	Duration of address holding phase: Writing these bits by software can define the duration of the address holding phase (please refer to Figure 40-3 ~ Figure 40-15), which is suitable for mode D or multiplexing access: 0000: Reserved; 0001: Duration of ADDHLD phase=1*HCLK clock cycle;	1111b

			0010: Duration of ADDHLD phase = 2 * HCLK clock cycle; ... 1111: Duration of ADDHLD phase = 15 * HCLK clock cycle. For information about the duration of the address holding phase of each access mode, please refer to the corresponding pictures (please refer to Figure 40-3 ~ Figure 40-15). <i>Note: In synchronous access mode, this value is not used because the duration of the address holding phase is always 1 memory clock cycle.</i>	
[3:0]	ADDSET[3:0]	RW	Duration of address setting phase: Writing these bits via software defines the duration of the address setup phase (refer to Figures 40-3 through 40-15), applicable to SRAM, ROM, asynchronous NOR Flash, and PSRAM: 0000: Duration of ADDSET phase = 0 * HCLK clock cycle; ... 1111: Duration of ADDSET phase = 15 * HCLK clock cycle; For information about the duration of the address holding phase of each access mode, please refer to the corresponding pictures (please refer to Figure 40-3 ~ Figure 40-15). <i>Note: In synchronous access mode, this value is irrelevant.</i> <i>In multiplex mode or Mode D, the minimum value for ADDSET is 1.</i>	1111b

Note: PSRAM (CRAM) exhibits unpredictable data latency durations due to internal refresh cycles. Consequently, these memories emit NWAIT signals throughout the delay phase to extend latency as required. For PSRAM (CRAM), the DATLAT field must be set to 0. This instructs the FMC to immediately exit the delay phase, commence sampling NWAIT from the memory, and initiate read or write operations once the memory is ready.

This method also applies to the latest generation of synchronous Flash. Unlike earlier Flash types, these Flash devices generate an NWAIT signal (refer to the specific Flash datasheet for details).

40.5.6.4 SRAM/NOR-Flash Write Timing Register x (R32_FMC_BWTRx) (x=1/2/3/4)

Offset address: 0x104 + 0x8*(x-1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		ACCMOD [1:0]		Reserved								BUSTURN[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAST[7:0]								ADDHLD[3:0]				ADDSET[3:0]			

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved.	0
[29:28]	ACCMOD[1:0]	RW	Access mode: Refers to the synchronous interview mode, as shown in the time series diagram. These bits are valid only when the EXTMOD bit in the R32_FMC_BCRx register is set. 00: Access mode A; 01: Access mode B; 10: Access mode C; 11: Access mode D.	0
[27:20]	Reserved	RO	Reserved	0
[19:16]	BUSTURN[3:0]	RW	Duration of bus turnaround phase: Writing these bits via software can introduce a delay at the end of a write-read transaction, thereby matching the minimum time interval between two consecutive transactions (t_{EHEL} from ENx high to ENx low): (BUSTURN+1) HCLK cycle $\geq t_{EHELmin}$, 0000: Duration of BUSTURN phase = Add 0 HCLK clock cycle; ... 1111: Duration of BUSTURN phase =15*HCLK clock cycle.	1111b
[15:8]	DATAST[7:0]	RW	Duration of data phase: Writing these bits via software defines the duration of the data phase (refer to Figures 40-3 through 40-15), applicable to asynchronous SRAM, PSRAM, and NOR Flash access modes: 0000 0000: Duration of the DATAST phase =1*HCLK clock cycle; 0000 0001: Duration of the DATAST phase =2*HCLK clock cycle; 0000 0010: Duration of the DATAST phase =3*HCLK clock cycle; ... 1111 1111: Duration of the DATAST phase =256*HCLK clock cycle.	1111 1111b
[7:4]	ADDHLD[3:0]	RW	Duration of address holding phase: Writing these bits by software can define the duration of the address holding phase (please refer to Figure 40-3 ~ Figure 40-15), which is suitable for asynchronous multiplexing access:	1111b

			0000: Reserved; 0001: Duration of ADDHLD phase=1*HCLK clock cycle; 0010: Duration of ADDHLD phase =2*HCLK clock cycle; ... 1111: Duration of ADDHLD phase =15*HCLK clock cycle. <i>Note: In synchronous NOR Flash access mode, this value is not used because the duration of the address holding phase is always 1 Flash clock cycle.</i>	
[3:0]	ADDSET[3:0]	RW	Duration of address setting phase: Writing these bits via software defines the duration of the address setup phase (refer to Figures 40-3 through 40-15), applicable to SRAM, ROM, asynchronous NOR Flash, and PSRAM: 0000: Duration of ADDSET phase=0*HCLK clock cycle; ... 1111: Duration of ADDSET phase =15*HCLK clock cycle; <i>Note: In synchronous NOR Flash access mode, this value is not used because the duration of the address holding phase is always 1 Flash clock cycle.</i> <i>In multiplex mode or Mode D, the minimum value for ADDSET is 1.</i>	1111b

40.6 NAND Flash Controller

The FMC generates corresponding signal timing to drive the following device types: 8-bit and 16-bit NAND Flash. The NAND memory area is configured via dedicated registers (Section 40.6.7). Programmable memory parameters include access timing (as shown in the table below) and ECC configuration.

Table 40-38 Programmable NAND Flash access parameters

Parameter	Function	Access mode	Unit	Min.	Max.
Memory setup time	Number of clock cycles (HCLK) required for address establishment before the command is enabled.	Read/write	HB clock cycle (HCLK)	1	255
Memory waiting	Minimum duration of command enable (in HCLK clock cycle)	Read/write	HB clock cycle (HCLK)	2	255
Memory retention	The number of clock cycles (HCLK) that the address (and data if a write access is made) must be held after the command is disabled.	Read/write	HB clock cycle (HCLK)	1	254

Memory data bus high impedance state	The number of clock cycles (HCLK) during which the data bus remains in a high impedance state after a write access is started.	Write	HB clock cycle (HCLK)	1	255
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40.6.1 External Memory Interface Signal

8-bit NAND Flash

Table 40-39 8-bit NAND Flash

FMC signal name	I/O	Function
A[17]	O	NAND Flash address latch enable (ALE) signal
A[16]	O	NAND Flash command latch enable (CLE) signal
D[7:0]	I/O	8-bit multiplex bidirectional address/data bus
NCE	O	chip select (CS)
NOE (=NRE)	O	Output enable (memory signal name: read enable, NRE)
NWE	O	Write enable
NWAIT/INT	I	Input the NAND Flash ready/busy signal of FMC.

Since FMC can manage enough address cycles, there is no capacity limit in theory.

Note: The prefix "n" identifies a signal that is active low.

16-bit NAND Flash

Table 40-40 8-bit NAND Flash

FMC signal name	I/O	Function
A[17]	O	NAND Flash address latch enable (ALE) signal
A[16]	O	NAND Flash command latch enable (CLE) signal
D[15:0]	I/O	16-bit multiplex bidirectional address/data bus
NCE	O	chip select (CS)
NOE (=NRE)	O	Output enable (memory signal name: read enable, NRE)
NWE	O	Write enable
NWAIT/INT	I	Input the NAND Flash ready/busy signal of FMC.

Since FMC can manage enough address cycles, there is no capacity limit in theory.

Note: The prefix "n" identifies a signal that is active low.

40.6.2 Memory and Transaction Supported by NAND Flash

The following table describes the supported devices, access modes and transactions. Transactions not allowed (or not supported) by NAND Flash controller are displayed in gray.

Table 40-41 Supported memory and transactions

Device	Mode	R/W	HB Data size	Memory data size	Allow/disallow	Notes
8-bit NAND	Asynchronous	R	8	8	YES	-
	Asynchronous	W	8	8	YES	-

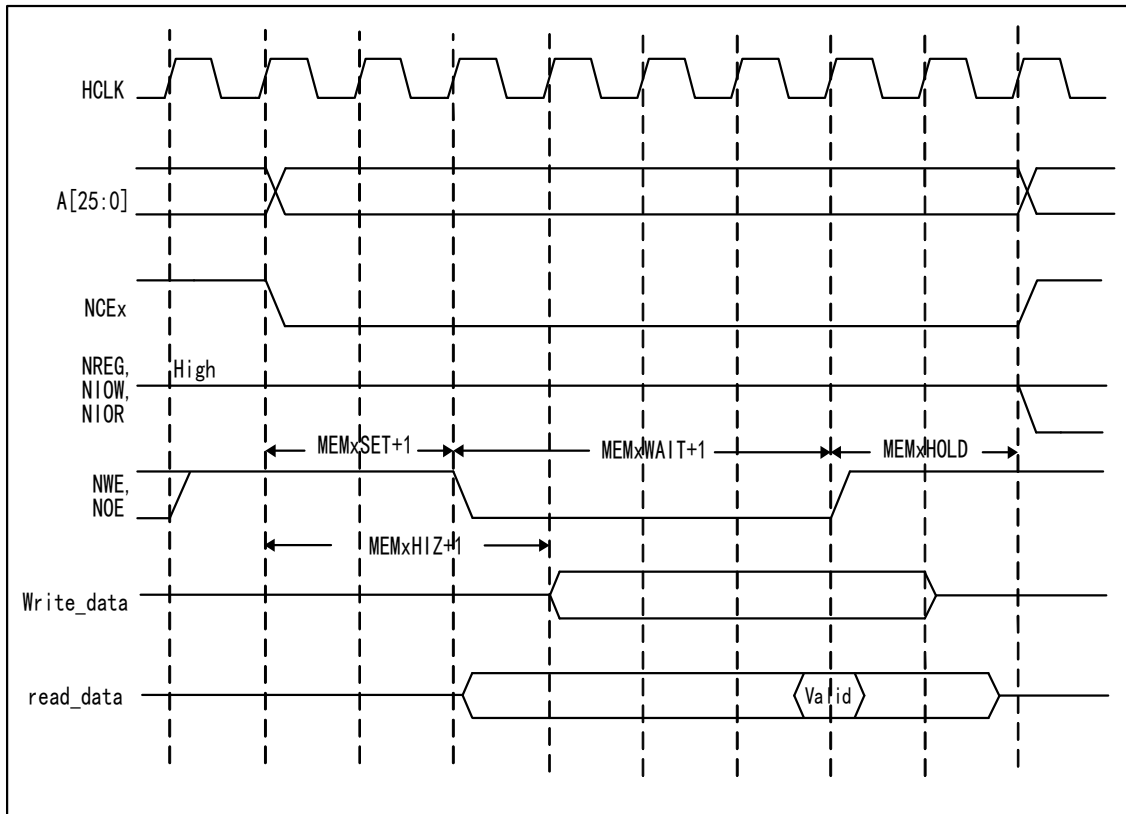
	nous					
	Asynchro nous	R	16	8	YES	Divided into 2 FMC visits
	Asynchro nous	W	16	8	YES	Divided into 2 FMC visits
	Asynchro nous	R	32	8	YES	Divided into 2 FMC visits
	Asynchro nous	W	32	8	YES	Divided into 2 FMC visits
16-bit NAND	Asynchro nous	R	8	16	YES	-
	Asynchro nous	W	8	16	No	-
	Asynchro nous	R	16	16	YES	-
	Asynchro nous	W	16	16	YES	-
	Asynchro nous	R	32	16	YES	Divided into 2 FMC visits
	Asynchronous	W	32	16	YES	Divided into 2 FMC visits

40.6.3 NAND Flash Timing Diagram

NAND Flash storage area is managed by the following set of registers:

- Control Register: FMC_PCR
- Interrupt Status Register: FMC_SR
- ECC Register: FMC_ECCR
- Timing Register for General Memory Space: R32_FMC_PMEM
- Timing Register for Feature Memory Space: R32_FMC_PATT

Figure 40-21 General memory access waveform of NAND Flash controller



Note: During write access, NOE remains high (disabled). During read access, NWE remains high (disabled).

Each timing configuration register contains 4 parameters: three define the number of HCLK cycles for the 3 access phases of the NAND Flash, while the fourth defines the timing for initiating data bus drive during write access. The diagram above illustrates the timing parameter definitions for general memory access; access timing for feature memory spaces follows a similar pattern.

40.6.4 NAND Flash Operation

The Command Latch Enable (CLE) signal and Address Latch Enable (ALE) signal for NAND Flash devices are driven by the address signals from the FMC controller. This means that to send a command or address to the NAND Flash, the CPU must perform a write operation to a specific address within its memory space.

A typical page read operation from a NAND Flash device requires the following steps:

- 1) Based on NAND Flash characteristics (PWID bit indicates NAND Flash data bus width, PTYP=1, set PWAITEN=0 or 1 as required; For timing configuration, refer to Section 40.4.2: NAND Flash Address Mapping), configure the FMC_PCR and FMC_PMEM registers (for certain devices, configure the R32_FMC_PATT register; refer to Section 40.6.5: NAND Flash Pre-Wait Function). This process configures and enables the corresponding storage regions.
- 2) The CPU performs a byte write operation to the general memory space, where the data byte equals one Flash command byte (e.g., 0x00 for Samsung NAND Flash devices). During the write enable period (a low-level pulse on NWE), the LE input of the NAND Flash is active. Consequently, the written byte is interpreted as a command for the NAND Flash. Once this command is latched by the memory device, it need not be rewritten during subsequent page read operations.
- 3) By writing the four bytes STARTAD[7:0], STARTAD[16:9], STARTAD[24:17], and STARTAD[25] (for 64Mb*8-bit NAND Flash) into the general memory space or feature space (three bytes for smaller capacity devices), the CPU can send the start address (STARTAD) for the read operation. During the write enable period

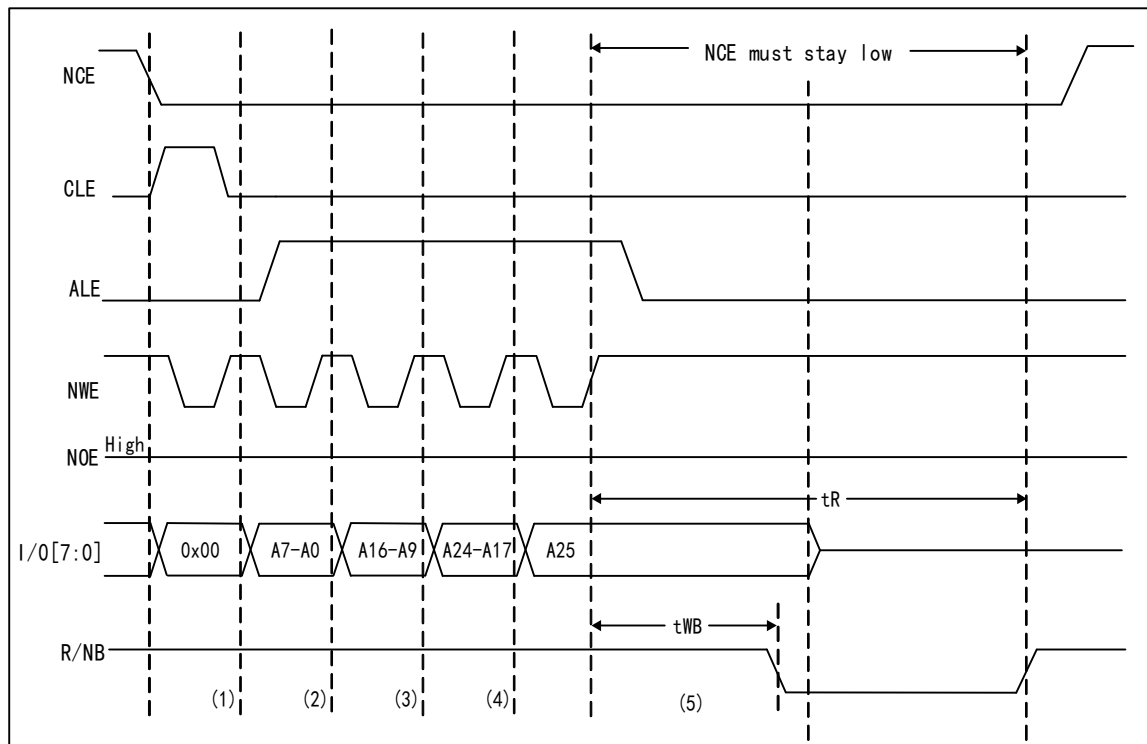
(a low-level pulse on NWE), the ALE input of the NAND Flash device is active, causing the written byte to be treated as the starting address for the read operation. Utilizing the feature memory space, an alternative timing configuration of the FMC can be employed to implement the pre-wait functionality required by certain NAND Flash devices (for details, refer to Section 40.6.5: NAND Flash Pre-Wait Functionality).

- 4) The controller waits for the NAND Flash to become ready (R/NB signal high) before initiating a new access to the same or another storage area. During this wait period, the controller maintains the NCE signal active (low).
- 5) The CPU can then execute byte-read operations from general memory space, thereby reading NAND Flash pages (data field + spare field) byte by byte.
- 6) The next NAND Flash page can be read without any CPU command or address write operation. This can be achieved in three distinct ways:
 - Perform the operation described in Step 5
 - Randomly access a new address by restarting the operation in Step 3
 - Send a new command to the NAND Flash device by restarting Step 2

40.6.5 NAND Flash Pre-waiting Function

Some NAND Flash devices need the controller to wait for the R/NB signal to go low after writing the last part of the address, as shown in the following figure.

Figure 40-22 Accessing non-CE-unrelated NAND flash



- 1) The CPU writes byte 0x00 to address 0x70010000.
- 2) The CPU writes bytes A7 through A0 to address 0x70020000.
- 3) The CPU writes bytes A16 through A9 to address 0x70020000.
- 4) The CPU writes bytes A24 to A17 at address 0x70020000.
- 5) The CPU writes byte A25 at address 0x78020000: The FMC performs a write access via the R32_FMC_PATT2 timing definition, where $ATTHOLD \geq 7$ (assuming $(7+1) * HCLK = 112ns > t_{WB}$ maximum value). This ensures NCE remains low until R/NB conversions low again, then goes high (only NAND Flash requiring NCE signal

control needs this behavior).

When this functionality is required, the t_{WB} timing can be ensured by programming the MEMHOLD value. However, all read or write accesses to NAND Flash by the CPU incur a hold delay of (MEMHOLD + 1) HCLK cycles (inserted between the rising edge of the NWE signal and the next access).

To overcome this timing constraint, use the feature memory space. Program its timing register with an ATTHOLD value that satisfies the t_{WB} timing and maintain MEMHOLD at its minimum value. Subsequently, the CPU must perform all NAND Flash read and write operations using the general memory space, except when writing the last address byte to the NAND Flash device. In this case, the CPU must execute a write operation to the feature memory space.

40.6.6 Error Correction Code (ECC) Calculation (NAND Flash)

The FMC NAND card controller incorporates two error correction code (ECC) calculation hardware modules, one for each storage area. These modules reduce the host CPU workload during software-based ECC processing.

Both ECC modules are identical and associated with storage areas 2 and 3, respectively. Consequently, hardware ECC calculations do not apply to memory connected to storage area 4.

During read or write operations to NAND Flash, the ECC algorithm employed in the FMC corrects single-bit errors and detects double-bit errors for every 256, 512, 1024, 2048, 4096, or 8192 bytes. This operation is based on the Hamming code algorithm and includes the calculation of row and column parity.

Whenever the NAND Flash storage area is active, the ECC module monitors the NAND Flash data bus and read/write signals (NCE and NWE).

ECC operates as follows:

- When accessing NAND Flash memory regions 2 or 3, it latches data appearing on the D[15:0] bus and uses it for ECC computation.
- When accessing any other address in the NAND Flash, the ECC logic enters idle state and performs no operation. Therefore, write operations defining NAND Flash commands or addresses are invalid during ECC computation.

After the host CPU completes the read/write operation for the required number of bytes to the NAND Flash, it must read the R32_FMC_ECCR register to retrieve the computed value. Following this read operation, these registers should be cleared by resetting the ECCEN bit to zero. To compute a new data block, the ECCEN bit in the FMC_PCR register must be set to 1.

To perform ECC calculation:

- 1) Enable the ECCEN bit in the R32_FMC_PCR register.
- 2) Write data to the NAND Flash page. During NAND page write, the ECC module calculates the ECC value.
- 3) Read the ECC value provided in the R32_FMC_ECCR register and store it in a variable.
- 4) Clear and then enable the ECCEN bit in the R32_FMC_PCR register. Read back the written data from the NAND page. During the NAND page read operation, the ECC module will compute the ECC value.
- 5) Read the new ECC value provided in the R32_FMC_ECCR register.
- 6) If the two read ECC values match, no correction is needed. Otherwise, an ECC error exists, and the software correction routine will return information indicating whether the error can be corrected.

40.6.7 NAND Flash Control Register

Table 40-42 NAND Flash-related registers

Name	Access address	Description	Reset value
R32_FMC_PCR	0x40025480	NAND Flash control register	0x00000018
R32_FMC_SR	0x40025484	FIFO status and interrupt register	0x00000040
R32_FMC_PMEM	0x40025488	General-purpose memory space timing register	0xFCFCFCFC
R32_FMC_PATT	0x4002548C	Characteristic memory space timing register	0xFCFCFCFC
R32_FMC_ECCR	0x40025494	ECC result register	0x00000000

40.6.7.1 NAND Flash Control Register (R32_FMC_PCR)

Offset address: 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												ECCPS[2:0]		TAR[3:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAR[2:0]			TCLR[3:0]				Reserved		ECCE N	PWID[1:0]		PTYP	PBKE N	PWAI TEN	Reserv ed

Bit	Name	Access	Description	Reset value
[31:20]	Reserved	RO	Reserved	0
[19:17]	ECCPS[2:0]	RW	ECC page size: 000: 256-byte; 001: 512-byte; 010: 1024-byte; 011: 2048-byte; 100: 4096-byte; 101: 8192-byte.	0
[16:13]	TAR[3:0]	RW	Delay from ALE to RE: Set the time from ALE low to RE low in HB clock cycles (HCLK): $t_{ar} = (TAR + SET + 2) * THCLK$, where THCLK is the HCLK clock period. 0000: 1 HCLK cycle (default); 1111: 16 HCLK cycles. <i>Note: Depending on the address space, SET is MEMSET or ATTSET.</i>	0
[12:9]	TCLR[3:0]	RW	CLE to RE Delay: Set the time from CLE low to RE low in HB clock cycles (HCLK): $t_{clr} = (TCLR + SET + 2) * THCLK$, where THCLK is the HCLK clock period 0000: 1 HCLK cycle (default); 1111: 16 HCLK cycles. <i>Note: Depending on the address space, SET is MEMSET or ATTSET.</i>	0

[8:7]	Reserved	RO	Reserved	0
6	ECCEN	RW	ECC logic enable bit: 1: Enable ECC logic; 0: Disable and reset ECC logic (reset to default value).	0
[5:4]	PWID[1:0]	RW	Data bus width: 00: 8 bits; 01: 16 bits; 10: Reserved; 11: Reserved.	01b
3	PTYP	RW	Memory type: Defines the type of device attached to the corresponding memory area: 1: NAND Flash (default value after reset); 0: Reserved, must retain reset value.	1
2	PBKEN	RW	NAND Flash Memory Area Enable Bit: Enables the specified memory area. Accessing a disabled memory area will cause an error on the HB bus. 1: Enables the corresponding memory area; 0: Disables the corresponding memory area (default value after reset).	0
1	PWAITEN	RW	Wait feature enable bit: 1: Enabled; 0: Disabled.	0
0	Reserved	RO	Reserved	0

40.6.7.2 FIFO Status and Interrupt Register (R32_FMC_SR)

Offset address: 0x84

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									FEMPT	IFEN	ILEN	IREN	IFS	ILS	IRS

Bit	Name	Access	Description	Reset value
[31:7]	Reserved	RO	Reserved	0
6	FEMPT	RO	FIFO empty: 1: FIFO is empty; 0: FIFO is not empty.	1
5	IFEN	RW	Interrupt falling edge detection enable bit: 1: Enable interrupt falling edge detection request; 0: Disable interrupt falling edge detection request.	0
4	ILEN	RW	Interrupt high-level detection enable bit:	0

			1: Enable interrupt high-level detection request; 0: Disable interrupt high-level detection request.	
3	IREN	RW	Rising edge detection enable bit: 1: Rising edge detection request enabled; 0: Rising edge detection request disabled.	0
2	IFS	RW0	Interrupt falling edge status: This flag is set to 1 by hardware, reset by software, and cleared by writing 0. 1: An interrupt falling edge has occurred; 0: No interrupt falling edge has occurred.	0
1	ILS	RW0	Interrupt high level status: This flag is set to 1 by hardware, reset by software, and cleared by writing 0. 1: An interrupt high level has occurred; 0: No interrupt high level has occurred.	0
0	IRS	RW0	Rising edge interrupt status: This flag is set to 1 by hardware, reset by software, and cleared by writing 0. 1: Rising edge interrupt occurred; 0: No rising edge interrupt occurred.	0

40.6.7.3 General-purpose Memory Space Timing Register (R32_FMC_PMEM)

Offset address: 0x88

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEMHIZ[7:0]								MEMHOLD[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MEMWAIT[7:0]								MEMSET[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	MEMHIZ[7:0]	RW	General-purpose memory data bus high-impedance state duration: 00000000: 1 HCLK cycle; 11111110: 255 HCLK cycles; 11111111: Reserved.	1111 1100b
[23:16]	MEMHOLD[7:0]	RW	General-purpose memory hold time: 00000000: Reserved; 00000001: 1 HCLK cycle; 11111110: 254 HCLK cycles; 11111111: Reserved.	1111 1100b
[15:8]	MEMWAIT[7:0]	RW	General-purpose memory wait time: 00000000: Reserved; 00000001: 2 HCLK cycles (+ wait cycle introduced when NWAIT is disabled);	1111 1100b

			11111110: 255 HCLK cycles (+ wait cycle introduced when NWAIT is disabled); 11111111: Reserved.	
[7:0]	MEMSET[7:0]	RW	General-purpose memory setup time: 00000000: 1 HCLK cycle; 11111110: 255 HCLK cycles; 11111111: Reserved.	1111 1100b

40.6.7.4 Characteristic Memory Space Timing Register (R32_FMC_PATT)

Offset address: 0x8C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ATTHIZ[7:0]								ATTHOLD[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ATTWAIT[7:0]								ATTSET[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	ATTHIZ[7:0]	RW	Characteristic memory data bus high-resistance state duration: 00000000: 0 HCLK cycles; 11111110: 255 HCLK cycles; 11111111: Reserved.	1111 1100b
[23:16]	ATTHOLD[7:0]	RW	Feature memory hold time: 00000000: Reserved; 00000001: 1 HCLK cycle; 11111110: 254 HCLK cycles; 11111111: Reserved.	1111 1100b
[15:8]	ATTWAIT[7:0]	RW	Feature memory wait time: 00000000: Reserved; 00000001: 2 HCLK cycles (+ wait cycle introduced when NWAIT is disabled); 11111110: 255 HCLK cycles (+ wait cycle introduced when NWAIT is disabled); 11111111: Reserved.	1111 1100b
[7:0]	ATTSET[7:0]	RW	Characteristic memory setup time: 00000000: 1 HCLK cycle; 11111110: 255 HCLK cycles; 11111111: Reserved.	1111 1100b

40.6.7.5 ECC Result Register (R32_FMC_ECCR)

Offset address: 0x94

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ECC															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ECC															

Bit	Name	Access	Description	Reset value																					
[31:0]	ECC	RO	ECC Result: This field contains the value calculated by the ECC computation logic, as shown in the table below.	0																					
			<table><tr><th>ECCPS[2:0]</th><th>Page size in bytes</th><th>ECC bit</th></tr><tr><td>000</td><td>256</td><td>ECC[21:0]</td></tr><tr><td>001</td><td>512</td><td>ECC[23:0]</td></tr><tr><td>010</td><td>1024</td><td>ECC[25:0]</td></tr><tr><td>011</td><td>2048</td><td>ECC[27:0]</td></tr><tr><td>100</td><td>4096</td><td>ECC[29:0]</td></tr><tr><td>101</td><td>8192</td><td>ECC[31:0]</td></tr></table>		ECCPS[2:0]	Page size in bytes	ECC bit	000	256	ECC[21:0]	001	512	ECC[23:0]	010	1024	ECC[25:0]	011	2048	ECC[27:0]	100	4096	ECC[29:0]	101	8192	ECC[31:0]
			ECCPS[2:0]		Page size in bytes	ECC bit																			
			000		256	ECC[21:0]																			
			001		512	ECC[23:0]																			
			010		1024	ECC[25:0]																			
			011		2048	ECC[27:0]																			
			100		4096	ECC[29:0]																			
101	8192	ECC[31:0]																							

40.7 SDRAM Controller

40.7.1 Main Features of SDRAM Controller

The main features of SDRAM controller are as follows:

- Support word, half-word, and byte access
- Power-down mode
- CAS delays 1, 2, 3
- Self-refresh mode
- Programmable timing parameters
- Software-configurable SDRAM power-up initialization
- 2 independently configurable SDRAM storage regions
- Multi-region ping-pong access
- SDRAM clock can be HCLK, HCLK/2, or HCLK/3
- Supports automatic refresh operation with programmable refresh rate
- 8-bit, 16-bit, 32-bit memory data bus width
- Automatic storage region boundary management
- Read FIFO cache with 6 lines × 32-bit depth (6 × 14-bit address tags)
- 13-bit address row, 11-bit address column, 4 internal storage regions: 4*16M*32bit (256MB), 4*16M*16bit (128MB), 4*16M*8bit (64MB)

40.7.2 SDRAM External Memory Interface Signal

At startup, the SDRAM I/O pin used to connect the FMCSDRAM controller with the external SDRAM device must be configured through the user application. Unused SDRAM controller I/O pins of application programs can be used for other purposes.

Table 40-43 SDRAM signal

SDRAM signal	I/O type	Description	Multiplexing function
SDCLK	O	SDRAM clock	-

SDCKE[1:0]	O	SDCKE0: SDRAM memory area 1 clock enable SDCKE1: SDRAM Memory Area 2 clock enable	-
SDNE[1:0]	O	SDNE0: SDRAM memory area 1 chip enable SDNE1: SDRAM Memory Area 2 chip enable	-
A[12:0]	O	Address	FMC_A[12:0]
D[31:0]	I/O	Bidirectional data bus	FMC_D[31:0]
BA[1:0]	O	Storage area address	FMC_A[15:14]
NRAS	O	Row address strobe	-
NCAS	O	Column address strobe	-
SDNWE	O	Write enable	-
DQM[3:0]	O	Write access output byte mask (Memory signal name: DQM[3:0])	-

40.7.3 SDRAM Controller Function Description

All SDRAM controller outputs (signal, address and data) can be configured to change on the falling edge of the memory clock (FMC_SDCLK).

SDRAM initialization

The initialization sequence is managed by software. If 2 memory regions are used, the target memory region bits CTB1 and CTB2 in the R32_FMC_SDCMR register must be set to 1, and the initialization sequences for memory region 1 and memory region 2 must be generated at the same time:

- 1) Program the memory device characteristics into the R32_FMC_SDCRx registers. The SDRAM clock frequency, RBURST, and RPIPE characteristics must be programmed into the R32_FMC_SDCR1 register.
- 2) Program the memory device timing into the R32_FMC_SDTRx registers. TRP and TRC timing must be programmed into the R32_FMC_SDTR1 register.
- 3) Set MODE[2:0] in the R32_FMC_SDCMR register to “001” and configure the target memory region bits (CTB1 and/or CTB2) in the R32_FMC_SDCMR register to begin clocking the memory (SDCKE driven high).
- 4) Wait for the specified delay cycle. Typical delay is 100μs.
- 5) Set MODE[2:0] in the R32_FMC_SDCMR register to “010” and configure the target memory area bits (CTB1 and/or CTB2) in the R32_FMC_SDCMR register to send the “Full Precharge” command.
- 6) Set MODE[2:0] in the R32_FMC_SDCMR register to “011” and configure the target storage area (CTB1 and/or CTB2) and the number of Non-Repeatable Automatic Refresh (NRFS) commands in the R32_FMC_SDCMR register.
- 7) Configure the MRD field according to the SDRAM device. Set MODE[2:0] in the R32_FMC_SDCMR register to “100” and configure the target storage area bits (CTB1 and/or CTB2) in the R32_FMC_SDCMR register to send the “Load Mode Register” command and program the SDRAM device. Specifically:
 - a) The CAS delay must be selected based on the value in R32_FMC_SDCR1/2;
 - b) The burst length (BL) must be set to 1 by configuring the MRD[2:0] bits in the R32_FMC_SDCMR register to 000.

If the mode registers for the two SDRAM storage regions differ, this step must be repeated twice—once for each storage region—with the corresponding target storage region bit set to 1.

- 8) Program the refresh rate in the R32_FMC_SDRTR register. The refresh rate corresponds to the delay between refresh cycles. Its value must be compatible with the SDRAM device.

At this stage, the SDRAM device is ready to receive commands. If a system reset occurs during an SDRAM access,

the data bus may still be driven by the SDRAM device. Therefore, the SDRAM device must be reinitialized after reset before the NOR Flash/PSRAM/SRAM or NAND Flash controller can issue new access commands.

Note: If two SDRAM devices are connected to the FMC, simultaneous access to both devices via the command mode register (loading the mode register command) will issue access commands based on the timing parameters (TMRD and TRAS timing) configured for SDRAM memory region 1 in the R32_FMC_SDTR1 register.

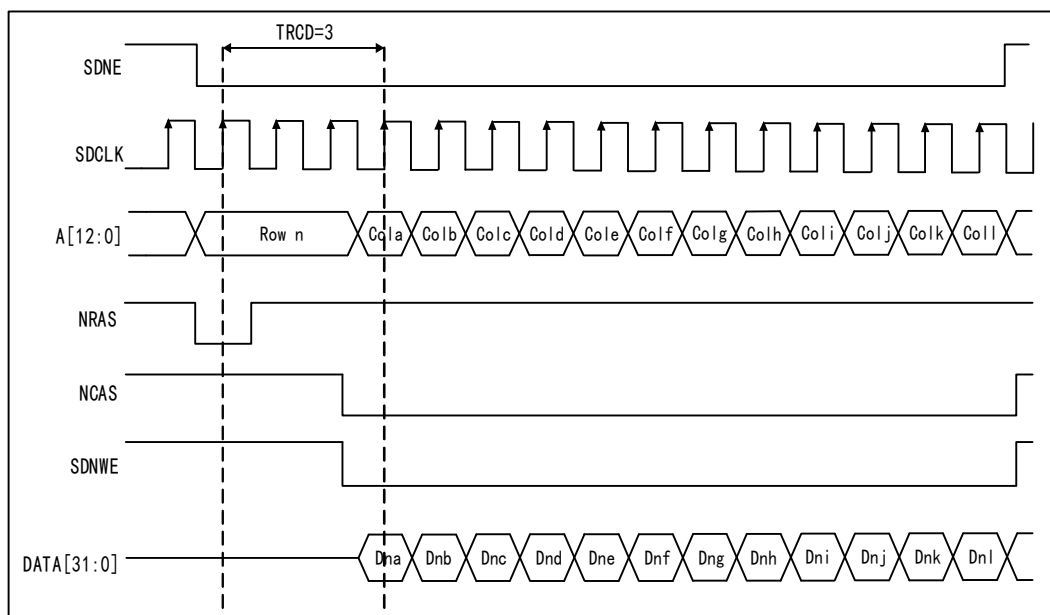
SDRAM controller write cycle

SDRAM controller can receive single and burst write requests and treat them as a single memory access. In both cases, the SDRAM controller will track the valid lines of each storage area, so as to be able to make continuous write access to different storage areas (multi-storage area ping-pong access). Before performing any write access, the WP bit in R32_FMC_SDCRx register must be cleared to prohibit write protection of SDRAM storage area.

The SDRAM controller always checks for the next access:

- 1) If the next access occurs on the same row or another valid row, the write operation proceeds directly.
- 2) If the next access points to an invalid row, the SDRAM controller generates a precharge command, activates the new row, and initializes the write command.

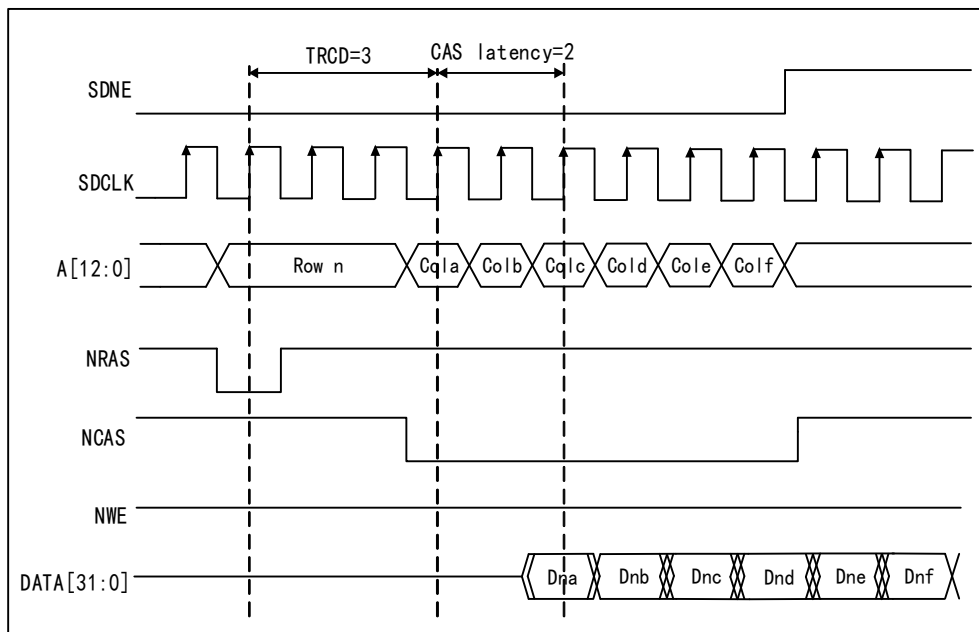
Figure 40-23 Burst write SDRAM access waveform



SDRAM controller read cycle

SDRAM controller can receive single and burst read requests and treat them as a single memory access. In both cases, the SDRAM controller will track the valid lines of each storage area, so as to be able to perform continuous read access to different storage areas (multi-storage area ping-pong access).

Figure 40-24 Burst read SDRAM access waveform



The FMC SDRAM controller features a cacheable read FIFO (6 lines $\times$ 32 bits) that pre-fetches read data during CAS latency cycles and RPIPE delays according to the following formula. The RBURST bit in the R32_FMC_SDCR1 register must be set to 1 to accept the next read access. Expected data count = CAS latency + 1 + (RPIPE latency) / 2, example:

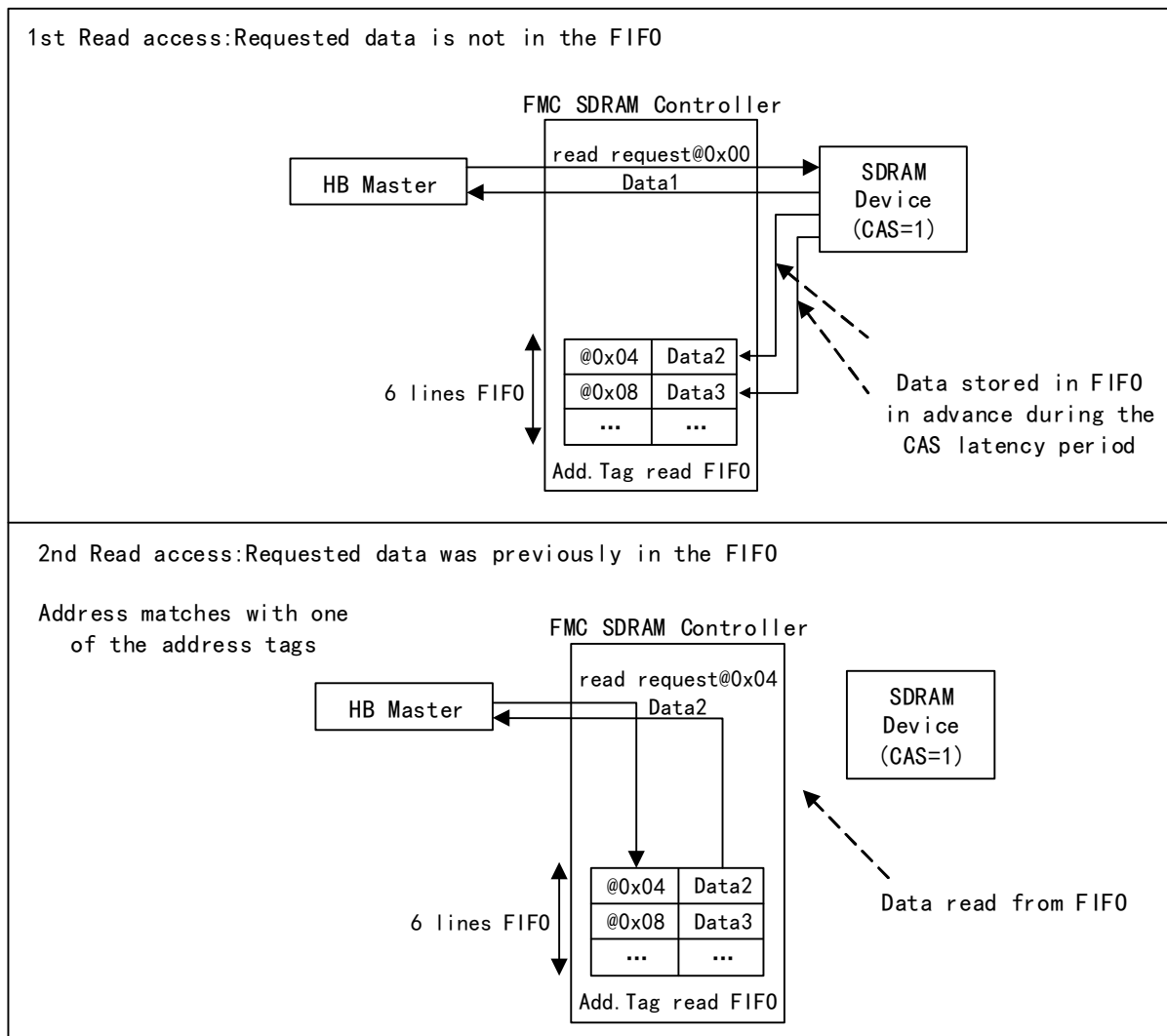
- CAS latency = 3, RPIPE latency = 0: Four data items (uncommitted) stored in the FIFO.
- CAS latency = 3, RPIPE latency = 2: Five data items (uncommitted) stored in the FIFO.

Each row in the read FIFO has a 14-bit address tag to identify its contents: 11 bits for the column address, 2 bits for selecting the internal memory region and valid row, and 1 bit for selecting the SDRAM device. During HB burst reads, if the end of the row is reached prematurely, the data read ahead (not yet committed) is not stored in the read FIFO. For single read accesses, the data is correctly stored in the read FIFO.

Whenever a read request occurs, the SDRAM controller checks:

- Whether the address matches one of the address tags. If a match is found, data is read directly from the FIFO, the corresponding address tag/row content is cleared, and the remaining data in the FIFO is compressed to avoid empty rows.
- Otherwise, a new read command is sent to memory and the FIFO is updated with the new data. If the FIFO is full, earlier data will be lost.

Figure 40-25 Read access logic diagram with RBURST set to 1 (CAS=1, RPIPE=0)



During a read access or precharge command, the read FIFO will refresh and prepare to fill with new data.

Upon receiving the first read request, if the current access has not yet reached a row boundary, the SDRAM controller will accept the next read access during the CAS latency cycle and RPIPE latency (if configured). This is achieved by incrementing the memory address. The following conditions must be met:

- The RBURST control bit in the R32_FMC_SDCR1 register must be set to 1
Address management depends on the next HB request:
- The next HB request is consecutive (HB burst)
In this case, the SDRAM controller will increment the address.
- The next HB request is not consecutive
 - If the new read request targets the same row as the previous request or another valid row, the new address is transmitted to the memory while the master device halts operations during the CAS latency period, awaiting new data from the memory.
 - If the new read request targets an invalid row, the SDRAM controller generates a precharge command, activates the new row, and initializes the read command.

If RURST is set to 0, the read FIFO is not used.

Row and storage area boundary management

When a read/write access crosses a row boundary, if the next read/write access is continuous and the current access has been executed to the row boundary, the SDRAM controller will perform the following operations:

- 1) Precharge the active row;
- 2) Activate the new row;
- 3) Initiate the read/write command.

For all column and data bus width configurations, automatic row activation at row boundaries is supported.

The SDRAM controller may insert additional clock cycles between the following commands as required:

- Inserted between the precharge and activate commands to match the TRP parameter (only when the next access targets another row within the same memory area).
- Inserted between the activate and read commands to match the TRCD parameter.

These parameters are defined in the FMC_SDRx register.

For information on row boundary crossing reads and burst write accesses, refer to the diagram below.

Figure 40-26 Read access of interbank boundary

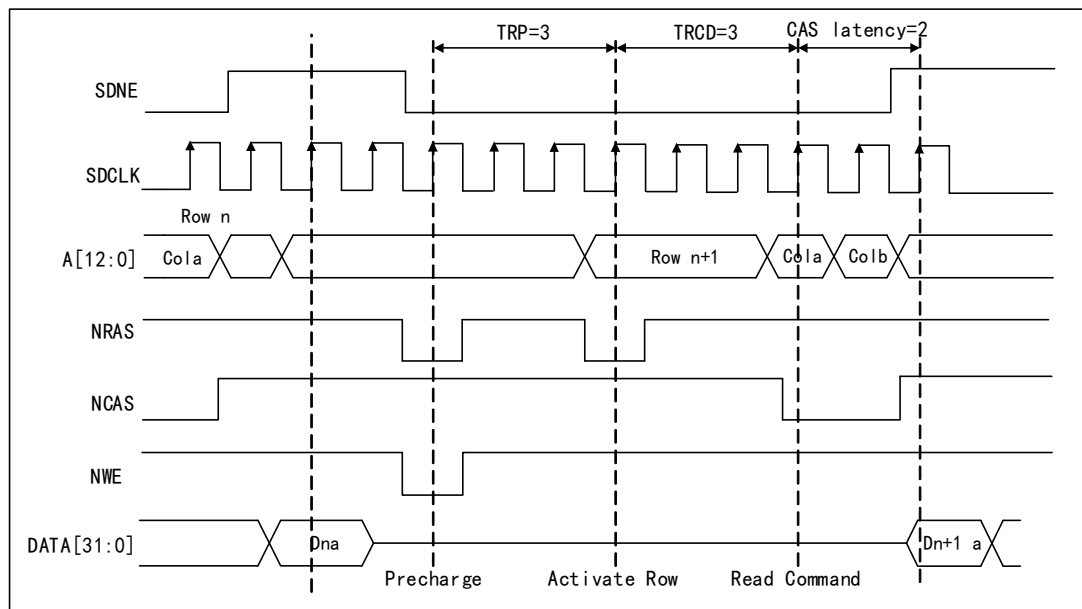
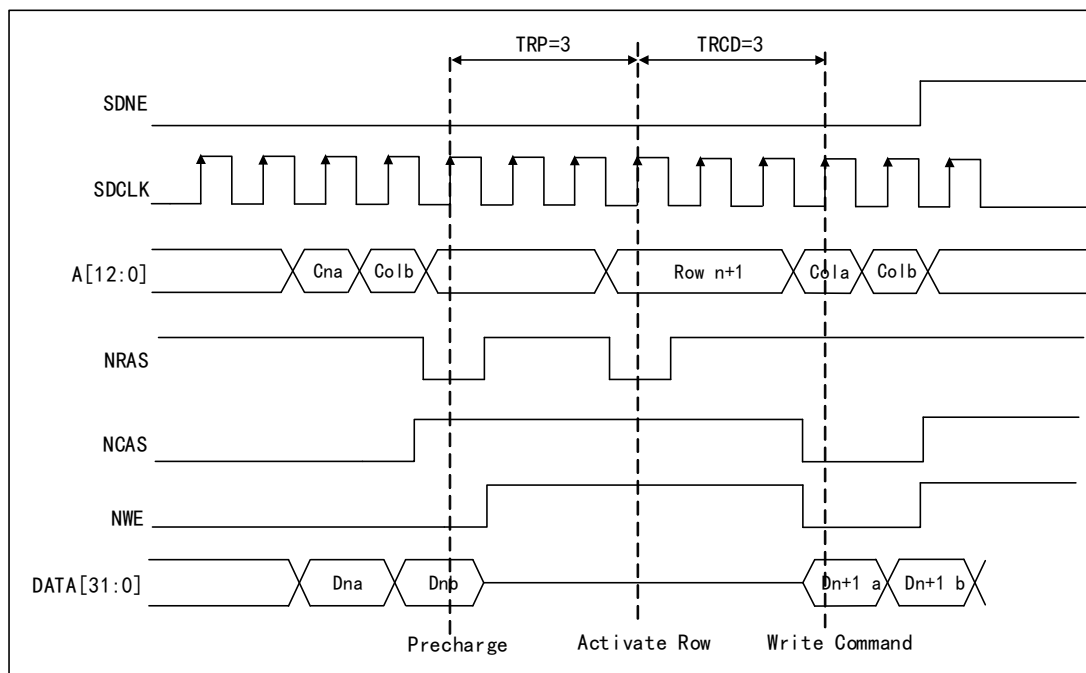


Figure 40-27 Write access of interbank boundary



If the next access is continuous and the current access crosses the storage area boundary, the SDRAM controller will activate the first row of the next storage area and issue a new read/write command. There are two possible situations:

- If the current memory area is not the last memory area, the active row in the new memory area must be precharged. For various row/column and data bus width configurations, the automatic activation of the next row at the boundary of the storage area is supported.
- If the current storage area is the last storage area, the automatic activation of the next row is only supported when addressing SDRAM devices with 13-bit rows, 11-bit columns, 4 internal storage areas and 32-bit data bus. Otherwise, SDRAM address range will be violated and HB error will be generated.
- For the SDRAM memory with 13-bit row address, 11-column address, 4 internal storage areas and 32-bit bus width, the SDRAM controller will continue the read/write operation through the second SDRAM device (assuming that the second SDRAM device is initialized):
 - The SDRAM controller will activate the first row (assuming that there is a valid row in the first internal storage area after precharging the valid row) and issue a new read/write command;
 - If the first row is activated, the SDRAM controller will only issue read/write commands.

SDRAM controller refresh cycle

The automatic refresh command is used to refresh the contents of SDRAM devices. SDRAM controller will send automatic refresh command periodically. It uses an internal COUNTER to load the count value in the R32_FMC_SDRTR register. This value defines the number of memory clock cycles (refresh rate) between refresh cycles. When the value of the counter reaches zero, an internal pulse will be generated.

If there is an ongoing memory access, the automatic refresh request will be delayed. However, when memory access and automatic refresh request occur at the same time, the automatic refresh request is given priority.

If the memory is accessed during the automatic refresh, the access request is cached and processed after the automatic refresh is completed.

If a new automatic refresh request appears when the last automatic refresh request has not been completed, the RE (refresh error) bit in the status register will be set to 1. If this bit is enabled (REIE=1), an interrupt will be generated.

If the SDRAM rows are not idle (not all rows are closed), the SDRAM controller will generate a PALL (pre-charge all) command before automatic refresh.

If the automatic refresh command is generated by the R32_FMC_SDCMR command mode register (mode bit = "011"), the PALL command (mode bit = "010") must be issued first.

40.7.4 Low-power Mode

2 low-power modes are available:

- Self-Refresh Mode

The SDRAM device performs automatic refresh cycles to retain data without requiring an external clock.

- Power-Down Mode

The SDRAM controller performs automatic refresh cycles.

Self-refresh mode

This mode is selected by setting the MODE position to '101' and configuring the target storage area bits (CTB1 and/or CTB2) in the R32_FMC_SDCMR register.

The SDRAM clock ceases operation after the TRAS delay, while the internal refresh timer only stops counting when one of the following conditions is met:

- A self-refresh command was issued to both devices.
- One of the devices was not activated (the SDRAM storage area was not initialized).

Before entering self-refresh mode, the SDRAM controller automatically issues a PALL command.

If the write data FIFO is not empty, all data will be sent to memory before self-refresh mode activates, and the BUSY status flag remains set to 1.

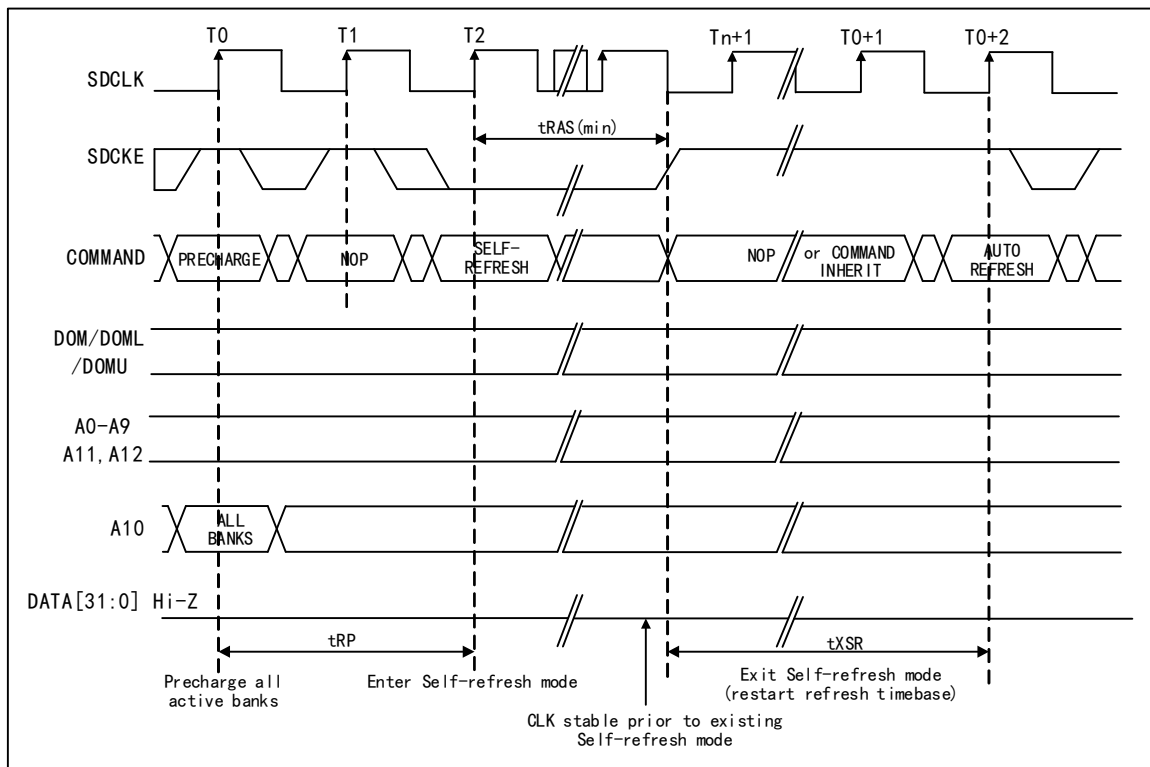
In self-refresh mode, all inputs to the SDRAM device are invalid except for SDCKE, which must remain low.

SDRAM devices must remain in self-refresh mode for a minimum duration equal to the TRAS time and must be capable of sustaining self-refresh mode for extended periods. To ensure this minimum duration, the BUSY status flag remains high throughout the TRAS delay period following self-refresh activation.

The SDRAM controller immediately generates a command sequence to exit self-refresh mode upon selection of an SDRAM device. After memory access completes, the selected device remains in normal mode.

To exit self-refresh mode, the MODE bit must be set to "000" (normal mode) and the target storage area bits (CTB1 and/or CTB2) within the R32_FMC_SDCMR register must be configured.

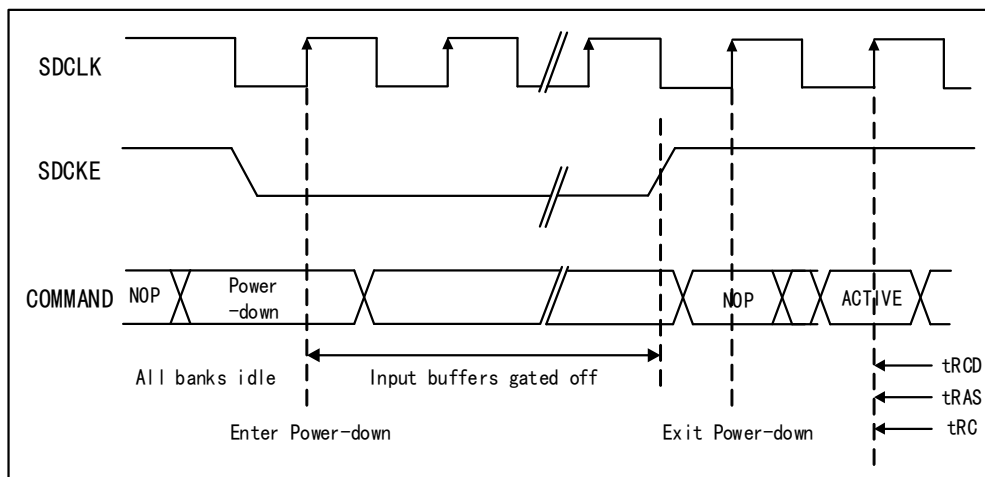
Figure 40-28 Self-refresh mode



Power-down mode

Select the MODE by setting the mode position to "110" and configuring the target memory area bits (CTB1 and/or CTB2) in the R32_FMC_SDCMR register.

Figure 40-29 Power-down mode



If the write data FIFO is not empty, all data will be sent to memory before power-down mode is activated.

The SDRAM controller will exit power-down mode immediately after an SDRAM device is selected. After memory access completes, the selected SDRAM device will remain in normal mode.

During power-down mode, all input/output buffers of the SDRAM device will be disabled, except for SDCKE, which remains low.

SDRAM devices can't stay in power-down mode for longer than refresh cycle, and they can't perform self-refresh cycle. Therefore, the SDRAM controller performs refresh by:

- 1) Exit power-down mode and drive SDCKE high.
- 2) Generate the PALL command. (provided an activation line exists in power-down mode)
- 3) Generate an auto-refresh command
- 4) Drive SDCKE low again to return to power-down mode.

To exit power-down mode, the MODE bit must be set to “000” (normal mode) and the target storage area bits (CTB1 and/or CTB2) must be configured in the R32_FMC_SDCMR register.

40.7.5 SDRAM Control Register

Table 40-44 SDRAM-related registers

Name	Access address	Description	Reset value
R32_FMC_SDCR1	0x40025540	SDRAM control register 1	0x000002D0
R32_FMC_SDCR2	0x40025544	SDRAM control register 2	0x000002D0
R32_FMC_SDTR1	0x40025548	SDRAM timing register 1	0x0FFFFFFF
R32_FMC_SDTR2	0x4002554C	SDRAM timing register 2	0x0FFFFFFF
R32_FMC_SDCMR	0x40025550	SDRAM command mode register	0x00000000
R32_FMC_SDRTR	0x40025554	SDRAM refresh timer register	0x00000000
R32_FMC_SDSR	0x40025558	SDRAM status register	0x00000000
R32_SDRAM_MISC	0x40025580	SDRAM custom register	0x00000000

40.7.5.1 SDRAM Control Register x (R32_FMC_SDCRx) (x=1/2)

Offset address: $0x140 + 4 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	RPIPE[1:0]		RB UR ST	SDCLK[1:0]		WP	CAS[1:0]		NB	MWID[1:0]		NR[1:0]		NC[1:0]	

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
[14:13]	RPIPE[1:0]	RW	Pipeline read: 00: 0 HCLK clock cycle delay; 01: 1 HCLK clock cycle delay; 10: 2 HCLK clock cycle delay; 11: Reserved. <i>Note: The corresponding bit in the R32_FMC_SDCR2 register is irrelevant.</i>	0
12	RBURST	RW	Burst read: 0: Do not manage single read requests as burst requests; 1: Always manage single read requests as burst requests.	0

			<i>Note: The corresponding bit in the R32_FMC_SDCR2 register is irrelevant.</i>	
[11:10]	SDCLK[1:0]	RW	SDRAM clock configuration: 00: SDCLK clock disabled; 01: SDCLK period = HCLK period; 10: SDCLK period = 2*HCLK period; 11: SDCLK period = 3*HCLK period; <i>Note: The corresponding bits in the R32_FMC_SDCR2 register are irrelevant.</i>	11b
9	WP	RW	Write protection: 1: Ignore write access; 0: Allow write access.	0
[8:7]	CAS[1:0]	RW	CAS Delay: 00: Reserved; 01: 1 cycle; 10: 2 cycles; 11: 3 cycles.	10b
6	NB	RW	Number of internal storage areas: 1: 4 internal storage areas; 0: 2 internal storage areas.	0
[5:4]	MWID[1:0]	RW	Memory data bus width: 00: 8-bit; 01: 16-bit; 10: 32-bit; 11: Reserved.	10b
[3:2]	NR[1:0]	RW	Address bits: 00: 11-bit; 01: 12-bit; 10: 13-bit; 11: Reserved.	0
[1:0]	NC[1:0]	RW	Column address bits: 00: 8-bit; 01: 9-bit; 10: 10-bit; 11: 11-bit.	0

Note: Before modifying the setting of RBURST or RPIPE or disabling the SDCLK clock, the user must first send the PALL command to ensure that the ongoing operation is completed first.

40.7.5.2 SDRAM Timing Register x (R32_FMC_SDTRx) (x=1/2)

Offset address: $0x148 + 4 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TRCD[3:0]				TRP[3:0]				TWR[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

TRC[3:0]	TRAS[3:0]	TXSR[3:0]	TMRD[3:0]
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Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:24]	TRCD[3:0]	RW	Row-to-column delay: 0000: 1 cycle; 0001: 2 cycles; 1111: 16 cycles.	1111b
[23:20]	TRP[3:0]	RW	Row precharge delay: TRP timing is configured only in the R32_FMC_SDTR1 register. If 2 SDRAM devices are used, TRP must be configured using the timing of the slower device. 0000: 1 cycle; 0001: 2 cycles; 1111: 16 cycles. <i>Note: The corresponding bit in the R32_FMC_SDTR2 register is irrelevant.</i>	1111b
[19:16]	TWR[3:0]	RW	Recovery delay: 0000: 1 cycle; 0001: 2 cycles; 1111: 16 cycles. <i>Note: TWR must be set to match the write recovery time (t_{WR}) defined in the SDRAM data manual to ensure: $TWR \geq TRAS - TRCD$, and $TWR \geq TRC - TRCD - TRP$ Example: $TRAS = 4$ cycles, $TRCD = 2$ cycles. Therefore, $TWR \geq 2$ cycles. TWR must be set to 0x1. If two SDRAM devices are used, R32_FMC_SDTR1 and R32_FMC_SDTR2 must be configured with the same TWR timing (corresponding to the slower SDRAM device).</i>	1111b
[15:12]	TRC[3:0]	RW	Cycle delay: Configure TRC timing only in the R32_FMC_SDTR1 register. If two SDRAM devices are used, configure TRC with the timing of the slower device. 0000: 1 cycle; 0001: 2 cycles;	1111b

			1111: 16 cycles. <i>Note: TRC must match the TRC and TRFC (auto-refresh cycle) timing defined in the SDRAM device data sheet.</i> <i>Note: The corresponding bit in the R32_FMC_SDTR2 register is a dummy bit.</i>	
[11:8]	TRAS[3:0]	RW	Self-refresh time: 0000: 1 cycle; 0001: 2 cycles; 1111: 16 cycles.	1111b
[7:4]	TXSR[3:0]	RW	Exit self-refresh delay: 0000: 1 cycle; 0001: 2 cycles; 1111: 16 cycles. <i>Note: If two SDRAM devices are used, the same TXSR timing must be configured for both R32_FMC_SDTR1 and R32_FMC_SDTR2 (corresponding to the slower SDRAM device).</i>	1111b
[3:0]	TMRD[3:0]	RW	Loading mode register to active: 0000: 1 cycle; 0001: 2 cycles; 1111: 16 cycles.	1111b

Note: If two SDRAM devices are connected, when both devices are accessed simultaneously via the command mode register (loading the mode register command), the access command will be issued using the timing parameters (TMRD and TRAS timing) configured for memory region 1 in the R32_FMC_SDTR1 register. Configure only the TRP and TRC timing in the R32_FMC_SDTR1 register. When using two SDRAM devices, configure the TRP and TRC timing based on the slowest device.

40.7.5.3 SDRAM Command Mode Register (R32_FMC_SDCMR)

Offset address: 0x150

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										MRD[12:7]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MRD[6:0]						NRFS[3:0]			CTB1		CTB2	MODE[2:0]			

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
[21:9]	MRD[12:0]	RW	Mode register definition: This 13-bit field defines the contents of the SDRAM mode register. The mode register can be programmed	0

			using the mode register load command.	
[8:5]	NRFS[3:0]	RW	Auto-refresh count: The number of consecutive auto-refresh commands issued when these bits define MODE="011". 0000: 1 auto-refresh cycle; 0001: 2 auto-refresh cycles; 1110: 15 auto-refresh cycles; 1111: 16 auto-refresh cycles.	0
4	CTB1	RW	Command target memory area 1: 1: Command sent to SDRAM memory area 1; 0: Command not sent to SDRAM memory area 1.	0
3	CTB2	RW	Command target memory area 2: 1: Command sent to SDRAM memory area 2; 0: Command not sent to SDRAM memory area 2.	0
[2:0]	MODE[2:0]	RW	Command mode: 000: Normal mode; 001: Clock configuration enable; 010: PALL ("Precharge All Layers") command; 011: Auto refresh command; 100: Load mode register; 101: Self refresh command; 110: Power down command; 111: Reserved. <i>Note: After a command is issued, at least one command target storage area bit (CTB1 or CTB2) must be set to 1; otherwise, the command will be ignored.</i> <i>Note: If 2 SDRAM storage areas are used, the self-refresh and PALL commands must be sent to both devices simultaneously by setting both CTB1 and CTB2 to 1; otherwise, the command will be ignored.</i> <i>Note: If only one SDRAM storage area is used and the command is issued by setting its associated CTB bit to 1, the CTB bit for the unused storage area must remain at 0.</i>	0

40.7.5.4 SDRAM Refresh Timer Register (R32_FMC_SDRTR)

Offset address: 0x154

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	REIE	COUNT[12:0]												CR E	

Bit	Name	Access	Description	Reset value
[31:15]	Reserved	RO	Reserved	0
14	REIE	RW	RES Interrupt Enable: 1: Generates interrupt when RE=1; 0: Disables interrupt.	0
[13:1]	COUNT[12:0]	RW	Refresh timer count: This 13-bit field defines the refresh rate of the SDRAM device, expressed in memory clock cycles. This field must be set to at least 41 SDRAM clock cycles. Refresh Rate = (COUNT + 1) × SDRAM Clock Frequency; COUNT = (SDRAM Refresh Period / Number of Lines) - 20.	0
0	CRE	RO	Clear refresh error flag: This bit clears the Refresh Error flag (RE) in the status register. 1: Clears the Refresh Error flag; 0: No operation.	0

Note: The programmed COUNT value cannot be equal to the sum of the following time series: $TWR + TRP + TRC + TRCD + 4$ memory clock cycles.

40.7.5.5 SDRAM Status Register (R32_FMC_SDSR)

Offset address: 0x158

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										BUS Y	MODES2[1:0]	MODES1[1:0]	R E		

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	BUSY	RO	Busy status: 1: SDRAM controller is not ready to accept new requests; 0: SDRAM controller is ready to accept new requests.	0
[4:3]	MODES2[1:0]	RO	Status modes for storage area 2: 00: Normal mode; 01: Self-refresh mode; 10: Power-down mode.	0
[2:1]	MODES1[1:0]	RO	Status modes for storage area 1: 00: Normal mode;	0

			01: Self-refresh mode; 10: Power-down mode.	
0	RE	RO	Refresh error flag: 1: Refresh error detected; 0: No refresh error detected. An interrupt is generated when REIE=1 and RE=1.	0

40.7.5.6 SDRAM Custom Register (R32_SDRAM_MISC)

Offset address: 0x180

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														En_Ba nk2	En_Ba nk1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Enhanc e_read _mode	Reserved					Phase_Sel					NRFS_CNT				

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0
17	En_Bank2	RW	Enable SDRAM2.	0
16	En_Bank1	RW	Enable SDRAM1.	0
15	Enhance_read_mode	RW	1: Enhanced/Prefetch Mode; 0: Normal Read Mode. The controller will prefetch data from the next 8, 16, or 32 addresses (depending on the SDRAM bit width, sufficient to obtain 256 bits of data). When addresses are contiguous, this enhances read performance. <i>Note: When this bit is enabled, the RBURST bit must be disabled.</i>	0
[14:8]	Reserved	RW	Reserved	0
[7:4]	Phase_Sel[3:0]	RW	Phase shift of the SDR_CLK output: Bit[7]: Output phase inversion; Bit[6:4]: Phase incrementally increases by approximately 0.4ns.	0
[3:0]	NRFS_CNT[3:0]	RW	Define the number of cycles per controller self-refresh: 0000: 1 cycle; 0001: 2 cycles; ... 1110: 15 cycles; 1111: 16 cycles.	0

Chapter 41 Encryption Module (ECDC)

The system has built-in block cipher algorithm module, which supports AES and SM4 block cipher algorithms and electronic codebook (ECB) and counter (CTR) modes. The module completes one encryption and decryption process with 128-bit data size as the basic unit, and provides a DMA encryption and decryption mode for data in memory and a single encryption and decryption mode for SFR registers.

41.1 Main Features

- ECB and CTR modes for SM4 algorithm with 128-bit keys
- ECB and CTR modes for AES algorithm with 128/192/256-bit keys
- Support direct encryption of individual 128-bit data via software writing to SFRs
- Support DMA-based (memory-to-memory) encryption/decryption of software-specified data blocks

41.2 AES/SM4 Algorithm

The Advanced Encryption Standard (AES) algorithm is a block cipher that employs a symmetric block cipher system, ranking among the most widely used symmetric-key encryption algorithms. The SM4 block cipher algorithm is typically a dedicated block cipher designed for wireless local area networks (WLANs) and trusted computing environments, though it can also be utilized for data encryption protection in other contexts.

During data encryption and decryption, keys must be loaded. For the AES algorithm, depending on the key length set to 128/192/256 bits, the user key must be expanded into an expansion key of $11 \times 128/13 \times 128/15 \times 128$ bits, respectively. In contrast, the SM4 algorithm expands a 128-bit user key into a 32×32 -bit expanded key. These expanded keys are stored in internal registers for convenient access during encryption and decryption operations.

41.3 ECB and CTR Mode

AES/SM4 supports two modes, electronic codebook (ECB) mode and counter (CTR) mode, in which the security performance of CTR mode is higher than that of ECB mode. As shown in figure 41-1 and 41-2 working block diagram.

Figure 41-1 ECB mode encryption and decryption process

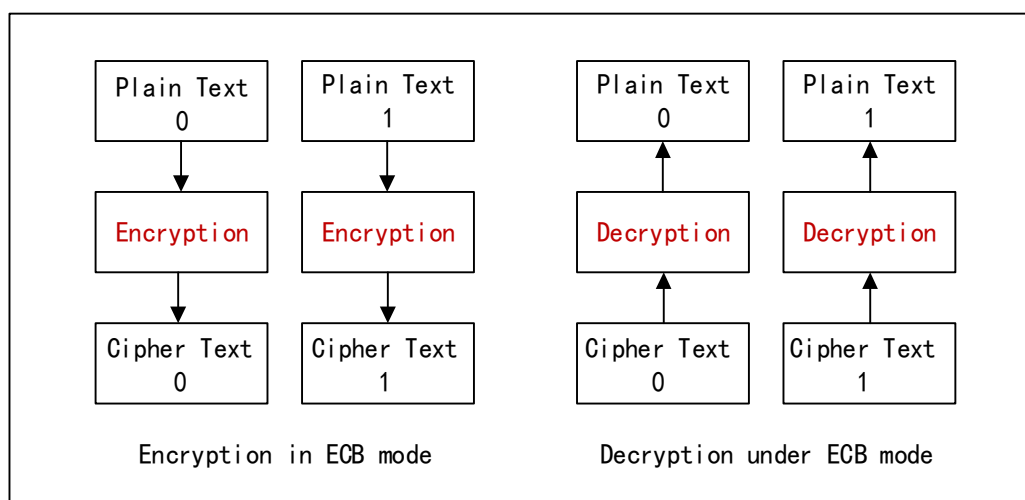
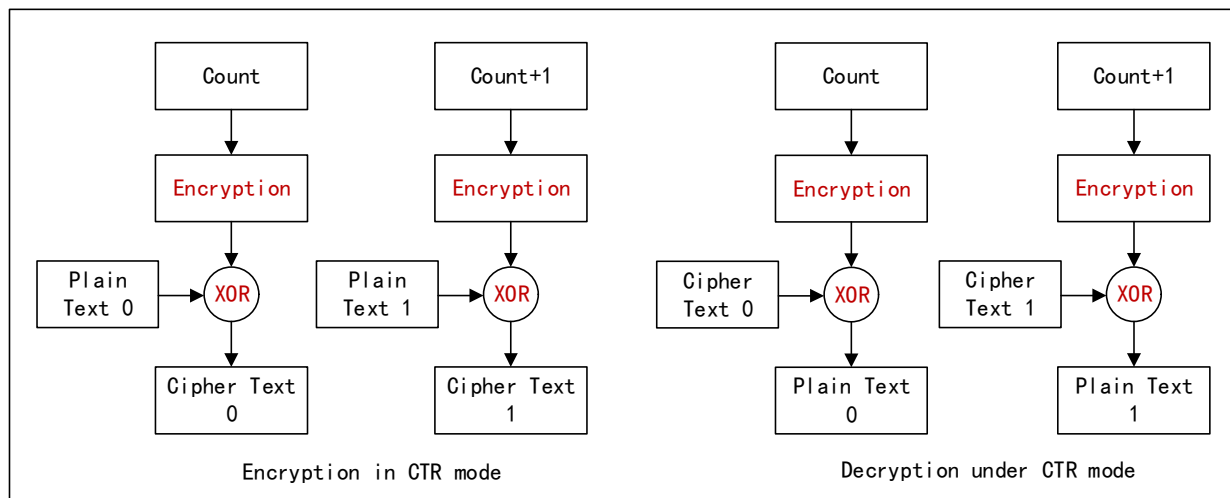


Figure 41-2 CTR mode encryption and decryption process



In ECB mode, plaintext and ciphertext are directly mapped one-to-one; the plaintext is encrypted verbatim to produce the ciphertext. In CTR mode, however, a 128-bit counter must be preloaded. This counter is encrypted, and the encrypted counter is then XORed with the plaintext to generate the ciphertext. It is important to note that in CTR decryption mode, only the counter is encrypted—it is not decrypted.

41.4 Module Application

41.4.1 Initialization of Encryption and Decryption Module

- Clock configuration: set RB_ECDC_CLKDIV_MASK bit in R32_ECEC_CTRL register to configure the clock for module operation. If you don't use this module, you can set it to 0 and turn it off to reduce power consumption.
- Data size end: set RB_ECDC_DAT_MOD bit of R32_ECEC_CTRL register, and you can choose the storage mode of the size end of the encrypted and decrypted data. According to the size of a packet of 128bits(16 bytes).
- Key Expansion: During encryption and decryption, an expansion key is used instead of the user key. After setting the user key length and content, the hardware must be instructed to perform key expansion. Once expanded, the key is stored in the module's internal registers. Only then can normal data encryption and decryption proceed. Configure the R32_ECEC_CTRL register: set RB_ECDC_WRSRAM_EN, RB_ECDC_ALGRM_MOD, RB_ECDC_CIPHER_MOD, and RB_ECDC_KLEN_MASK to enable SRAM data encryption/decryption, select the encryption/decryption algorithm, specify the block cipher mode, and define the key length. Then configure the user key register group R32_ECDC_KEY by populating it with the user key content. After setting RB_ECDC_KEYEX_EN in the R32_ECEC_CTRL register to 1 and then clearing it to 0, the hardware initiates the key expansion function.
- Initial counter value: When using CTR mode for encryption/decryption, an initial counter value is required. Set the 128-bit counter value in the CTR mode counter register group R32_ECDC_IV. Each successful completion of a data encryption/decryption cycle (128-bit data) increments the internal counter value by 1. This counter value is not needed in ECB mode.

41.4.2 Encryption and Decryption Configuration

The encryption and decryption data flow of the module includes two ways: memory to memory and one-time encryption and decryption of 128-bit data. Please refer to the configuration in Table 41-1 for details.

Table 41-1 Configuration of different data flow modes

Encrypted and decrypted data stream	RB_ECDC_WRSRAM_EN	RB_ECDC_NORMAL_EN	RB_ECDC_MODE_SEL	Others
SRAM data encryption	1	1	0	Need to configure register R32_ECDC_SRC_ADDR, R32_ECDC_DST_ADDR and R32_ECDC_SRAM_LEN
SRAM data decryption	1	1	1	
128bits data single encryption	-	1	0	Write R32_ECDC_SGSD register and fill in the original data (plaintext or ciphertext). Read R32_ECDC_SGRT register, and read the encrypted and decrypted result data.
128bits data single decryption	-	1	1	

Memory-to-memory data conversion begins when non-zero data is written to the R32_ECDC_SRAM_LEN length register. By checking the RB_ECDC_IF_WRSRAM bit in the R32_ECDC_INT_FG register, a value of 1 indicates the memory-to-memory data conversion is complete, clearing the RB_ECDC_IF_WRSRAM bit status. If the RB_ECDC_IE_WRSRAM function in the R32_ECDC_CTRL register is enabled, an interrupt service is triggered simultaneously with setting the RB_ECDC_IF_WRSRAM flag.

Single-pass encryption/decryption must be configured in big-endian mode (RB_ECDC_DAT_MOD=1) for execution. The single-pass encryption/decryption conversion of 128-bit data begins with a write operation to the upper 32 bits of the R32_ECDC_SGSD register. By checking the RB_ECDC_IF_SINGLE bit in the R32_ECDC_INT_FG register, a value of 1 indicates completion of the single 128-bit data conversion, clearing the RB_ECDC_IF_SINGLE bit status. If the RB_ECDC_IE_SINGLE function in the R32_ECDC_CTRL register is enabled, an interrupt service is triggered simultaneously with the setting of the RB_ECDC_IF_SINGLE flag.

41.5 Register Description

Table 41-2 ECDC channel-related registers

Name	Access address	Description	Reset value
R32_ECDC_CTRL	0x40016C00	ECDC control register	0x00000020
R32_ECDC_INT_FG	0x40016C04	ECDC interrupt flag register	0x00000000
R32_ECDC_KEY_255T224	0x40016C08	User key register 7	0xFFFFFFFF
R32_ECDC_KEY_223T192	0x40016C0C	User key register 6	0xFFFFFFFF

R32_ECDC_KEY_191T160	0x40016C10	User key register 5	0xFFFFFFFF
R32_ECDC_KEY_159T128	0x40016C14	User key register 4	0xFFFFFFFF
R32_ECDC_KEY_127T96	0x40016C18	User key register 3	0xFFFFFFFF
R32_ECDC_KEY_95T64	0x40016C1C	User key register 2	0xFFFFFFFF
R32_ECDC_KEY_63T32	0x40016C20	User key register 1	0xFFFFFFFF
R32_ECDC_KEY_31T0	0x40016C24	User key register 0	0xFFFFFFFF
R32_ECDC_IV_127T96	0x40016C28	CTR mode count value register 3	0xFFFFFFFF
R32_ECDC_IV_95T64	0x40016C2C	CTR mode count value register 2	0xFFFFFFFF
R32_ECDC_IV_63T32	0x40016C30	CTR mode count value register 1	0xFFFFFFFF
R32_ECDC_IV_31T0	0x40016C34	CTR mode count value register 0	0xFFFFFFFF
R32_ECDC_SGSD_127T96	0x40016C40	Single encryption/decryption of raw data 3	0x00000000
R32_ECDC_SGSD_95T64	0x40016C44	Single encryption/decryption of raw data 2	0x00000000
R32_ECDC_SGSD_63T32	0x40016C48	Single encryption/decryption of raw data 1	0x00000000
R32_ECDC_SGSD_31T0	0x40016C4C	Single encryption/decryption of raw data 0	0x00000000
R32_ECDC_SGRT_127T96	0x40016C50	Single encryption and decryption result 3	0x00000000
R32_ECDC_SGRT_95T64	0x40016C54	Single encryption and decryption result 2	0x00000000
R32_ECDC_SGRT_63T32	0x40016C58	Single encryption and decryption result 1	0x00000000
R32_ECDC_SGRT_31T0	0x40016C5C	Single encryption and decryption result 0	0x00000000
R32_ECDC_SRC_ADDR	0x40016C60	Encrypt and decrypt SRAM area source address	0x00000000
R32_ECDC_DST_ADDR	0x40016C64	Encrypt and decrypt SRAM destination address	0x00000000
R32_ECDC_SRAM_LEN	0x40016C68	Encrypt and decrypt SRAM length	0x00000000

41.5.1 ECDC Control Register (R32_ECDC_CTRL)

Offset address: 0x00

Bit	Name	Access	Description	Reset value
[31:26]	Reserved	RO	Reserved	0
25	RB_ECDC_AES/SM4_CLOCK_EN	RW	1: Enable encryption/decryption hardware clock; 0: Disable encryption/decryption hardware clock; <i>Note: Set to 1 when using encryption/decryption; set to 0 when not in use to conserve power.</i>	0
24	RB_ECDC_CLOCK_SELECT	RW	1: Encryption/decryption operates under the pllclk (maximum supported frequency: 400 MHz); 0: Operating frequency is the clock signal indicated by the [6:4] division ratio. <i>Note: The ECDC clock must be greater than or equal to the HCLK clock.</i>	0
[23:19]	Reserved	RO	Reserved	0
18	RB_ECDC_IE_WRSRAM	RW	Memory-to-memory encryption/decryption completes interrupt enable. 1: Enable corresponding interrupt; 0: Disable corresponding interrupt.	0

17	RB_ECDC_IE_SINGLE	RW	Single-cycle encryption/decryption completion interrupt enable. Completion of 128-bit register conversion. 1: Enable corresponding interrupt; 0: Disable corresponding interrupt.	0
[16:14]	Reserved	RO	Reserved	0
13	RB_ECDC_DAT_MODE	WO	Endianness Selection for Encryption/Decryption Data Size. 1: Big-endian mode; 0: Little-endian mode. <i>Note: Must be configured as big-endian mode for single 128-bit encryption/decryption operations.</i>	0
12	Reserved	RO	Reserved	0
[11:10]	RB_ECDC_KLEN_MASK	RW	Key length settings. 00: 128 bits; 01: 192 bits; 10: 256 bits; 11: Reserved.	0
9	RB_ECDC_CIPHER_MOD	RW	Block cipher mode selection. 1: CTR Mode; 0: ECB Mode.	0
8	RB_ECDC_ALGRM_MOD	RW	Encryption/Decryption algorithm mode selection. 1: AES; 0: SM4.	0
7	RB_ECDC_WRSRAM_EN	RW	Enable SRAM data encryption/decryption. Requires coordination with control bit RB_ECDC_KEYEX_EN. 1: Enabled; 0: Disabled.	0
[6:4]	RB_ECDC_CLKDIV_MASK	RW	Encryption/Decryption clock division factor. Calculation: $EDclk = PLLCLK / ED_CLK_PRE$. Minimum value 2. Writing 1 disables the ECDC module's operating clock; writing 0 enables it. <i>Note: The ECDC clock must be greater than or equal to the HCLK clock.</i>	010b
3	RB_ECDC_MODE_SEL	RW	Encryption/Decryption mode selection. 1: Decryption mode; 0: Encryption mode.	0
2	Reserved	RW	Reserved	0
1	RB_ECDC_NORMAL_EN	RW	Enable encryption/decryption operation in normal encryption/decryption mode. After the software detects completion of key expansion, set this bit to 1 before initiating encryption/decryption. 1: Encryption/decryption enabled;	0

			0: Disabled.	
0	RB_ECDC_KEYEX_EN	RW	Key expansion function enable control bit, activated by a high-level pulse. After configuring the key length, this bit must be set to 1 and the key expansion process must complete before encryption/decryption can proceed. <i>Note: Application code drivers must set this bit high and then low.</i>	0

41.5.2 ECDC Interrupt Flag Register (R32_ECDC_INT_FG)

Offset address: 0x04

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
18	RB_ECDC_IF_WRSRAM	RW1Z	Memory-to-Memory encryption/decryption completion interrupt flag: Write 1 to clear 0. 1: Encryption/decryption completion event; 0: No event.	0
17	RB_ECDC_IF_SINGLE	RW1Z	Single encryption/Decryption Completion Interrupt Flag: Write 1 to clear 0. 1: Single encryption/decryption completion event; 0: No event.	0
[16:0]	Reserved	RW	Reserved	0

41.5.3 User key Register Group (R32_ECDC_KEY)

Offset address: 0x08, 0x0C, 0x10, 0x14, 0x18 0x1C, 0x20, 0x24

Bit	Name	Access	Description	Reset value
[31:0]	R32_ECDC_KEY_255T224	RW	User key 224-256 bits.	X
[31:0]	R32_ECDC_KEY_223T192	RW	User key 192-223 bits.	X
[31:0]	R32_ECDC_KEY_191T160	RW	User key 160-191 bits.	X
[31:0]	R32_ECDC_KEY_159T128	RW	User key 128-159 bits.	X
[31:0]	R32_ECDC_KEY_127T96	RW	User key 96-127 bits.	X
[31:0]	R32_ECDC_KEY_95T64	RW	User key 64-95 bits.	X
[31:0]	R32_ECDC_KEY_63T32	RW	User key 32-63 bits.	X
[31:0]	R32_ECDC_KEY_31T0	RW	User key 0-31 bits.	X

41.5.4 CTR Mode Count Value Register group (R32_ECDC_IV)

Offset address: 0x28, 0x2C, 0x30, 0x34

Bit	Name	Access	Description	Reset value
[31:0]	R32_ECDC_IV_127T96	RW	Count value 96-127 bits.	X
[31:0]	R32_ECDC_IV_95T64	RW	Count value 64-95 bits.	X
[31:0]	R32_ECDC_IV_63T32	RW	Count value 32-63 bits.	X
[31:0]	R32_ECDC_IV_31T0	RW	Count value 0-31 bits.	X

41.5.5 Single Encryption/Decryption of Raw Data Register group (R32_ECDC_SGSD)

Offset address: 0x40, 0x44, 0x48, 0x4C

Bit	Name	Access	Description	Reset value
[31:0]	R32_ECDC_SGSD_127T96	RW	Raw data 96-127 bits.	0
[31:0]	R32_ECDC_SGSD_95T64	RW	Raw data 64-95 bits.	0
[31:0]	R32_ECDC_SGSD_63T32	RW	Raw data 32-63 bits.	0
[31:0]	R32_ECDC_SGSD_31T0	RW	Raw data 0-31 bits.	0

Note: Internal encryption and decryption is fixed at 128bit size. When R32_ECDC_SGSD_31T0 register is written, hardware automatically starts single encryption and decryption conversion. RB_ECDC_IF_SINGLE indicates whether conversion is completed.

41.5.6 Single Encryption/Decryption Result Register group (R32_ECDC_SGRT)

Offset address: 0x50, 0x54, 0x58 0x5C

Bit	Name	Access	Description	Reset value
[31:0]	R32_ECDC_SGRT_127T96	RW	Data 96-127 bits.	0
[31:0]	R32_ECDC_SGRT_95T64	RW	Data 64-95 bits.	0
[31:0]	R32_ECDC_SGRT_63T32	RW	Data 32-63 bits.	0
[31:0]	R32_ECDC_SGRT_31T0	RW	Data 0-31 bits.	0

Note: The encryption and decryption result data is stored in a fixed big-end mode, with a structure of 128bits.

41.5.7 Encryption/Decryption SRAM Area Source Address (R32_ECDC_SRC_ADDR)

Offset address: 0x60

Bit	Name	Access	Description	Reset value
[31:0]	ECDC_SRAM_SRC_ADDR	RW	The source start address of the encrypted SRAM data.	0

Note: This DMA address is in SRAM area.

41.5.8 Encryption/Decryption SRAM Area Destination Address (R32_ECDC_DST_ADDR)

Offset address: 0x64

Bit	Name	Access	Description	Reset value
[31:0]	ECDC_SRAM_DST_ADDR	RW	The starting address for encrypting/decrypting SRAM data, measured in 128-bit units. Writing a non-zero value initiates the encryption/decryption process for the SRAM data. (32-byte alignment)	0

41.5.9 Encryption/Decryption SRAM Length (R32_ECDC_SRAM_LEN)

Offset address: 0x68

Bit	Name	Access	Description	Reset value
[31:13]	Reserved	RO	Reserved	0
[12:0]	ECDC_SRAM_LEN	RW	Encrypt/decrypt SRAM data length, measured in 128-bit units. When writing a non-zero value, the encryption/decryption process for SRAM data will be initiated.	0

Chapter 42 Digital Filter for $\Sigma\Delta$ Modulator (DFSDM)

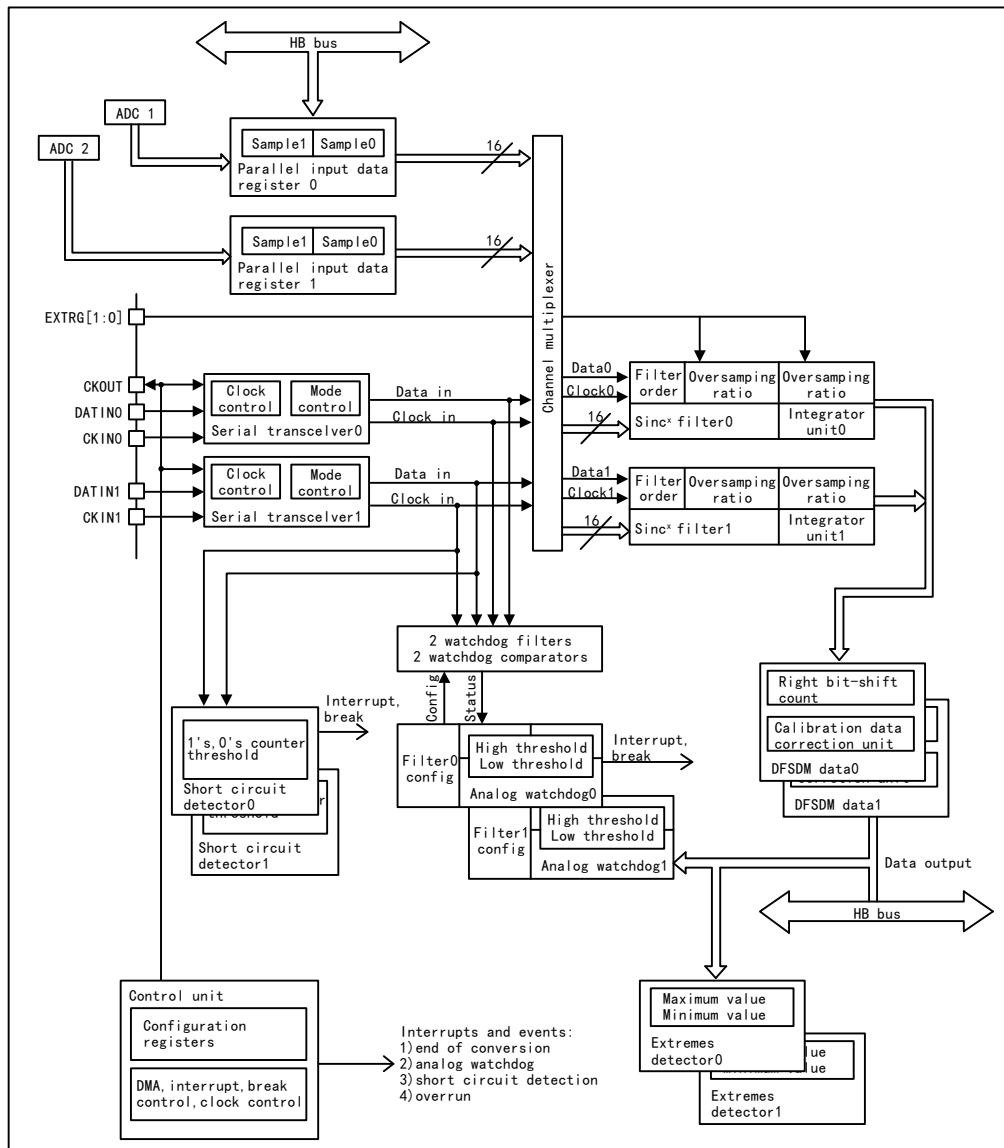
DFSDM is a high-performance module dedicated to connecting external $\Sigma\Delta$ modulator to MCU. It contains two external digital serial interfaces and 2 digital filters, and has flexible $\Sigma\Delta$ digital processing options, which can provide the final resolution of ADC up to 24 bits. DFSDM can also select parallel data stream input from internal ADC peripherals or device memory.

42.1 Main Features

- Provide 2 multiplexed digital serial input channels
- 2 internal digital parallel channels supporting optional inputs
- Support adjustable digital signal processing
- Support up to 24-bit output data resolution
- Support signed data formats
- Support automatic data offset correction (offset value stored by user in register)
- Include 2 conversion modes: single conversion mode and continuous conversion mode
- Conversion initiation supported via software trigger, internal timer, external event, or synchronization with the first DFSDM filter
- Supports analog watchdog
- Built-in short-circuit detector to identify saturated analog input values (lower and upper limits)
- Support open-circuit generation upon analog watchdog events and short-circuit monitoring events
- Built-in extreme value detector
- Support interrupts and DMA

42.2 Overview

Figure 42-1 Structural block diagram of DFSDM



42.3 Function Description

42.3.1 DFSDM Reset and Clock

DFSDM Switch Control

Global enablement of the DFSDM interface is achieved by setting the DFSDMEN bit in the R32_DFSDM_CH0CFGR1 register to 1. Once the DFSDM interface is globally enabled, all configured input channels y ($y=0/1$) and digital filters DFSDM_FLT x ($x=0/1$) will begin operation when their respective enable bits (channel enable bit CHEN in R32_DFSDM_CH y CFGR1 and DFSDM_FLT x enable bit DFEN in R32_DFSDM_FLT x CR1) are set to 1.

The digital filter DFSDM_FLT x ($x=0/1$) is enabled by setting DFEN in the R32_DFSDM_FLT x CR1 register to 1. After enabling DFSDM_FLT x (DFEN=1), the Sinc x digital filter unit and integrator unit will be reinitialized.

Clearing DFEN immediately halts all ongoing conversions and places DFSDM_FLT x into stop mode. All register

settings remain unchanged except for R32_DFSDM_FLTxAWSR and R32_DFSDM_FLTxISR, which are reset. Enable channel y (y=0/1) by setting CHEN in the R32_DFSDM_CHyCFGR1 register to 1. Once a channel is enabled, serial data is received from an external $\Sigma\Delta$ modulator or a parallel internal data source (CPU/DMA write operations to ADCs or memory).

Before stopping the system clock and entering stop mode, the DFSDM must be globally disabled (by setting DFSDMEN=0 in the R32_DFSDM_CH0CFGR1 register).

DFSDM Clock

The internal DFSDM clock f_{DFSDMCLK} drives channel transceivers, digital processing modules (including digital filters and integrators), and subsequent auxiliary modules (such as analog watchdogs, short-circuit detectors, extreme-value detectors, and control modules). This clock is generated by the RCC module, sourced from either the peripheral clock HCLK or the audio clock. The DFSDM clock automatically stops in stop mode (when all DFSDM_FLTx, x=0/1, and DFEN=0).

The DFSDM serial channel transceiver can accept an external serial clock to sample external serial data streams. If standard SPI encoding is used, the internal DFSDM clock must be at least four times faster than the external serial clock.

The DFSDM supports providing an external output clock signal to drive the clock input of an external $\Sigma\Delta$ modulator. This output clock signal is delivered via the CKOUT pin. The frequency of the output clock signal must fall within the range specified in the device manual and can be derived from either the DFSDM clock or the audio clock (refer to the CKOUTSRC bit in the R32_DFSDM_CH0CFGR1 register) through a divider. The programmable range of the divider is 2 to 256 (CKOUTDIV bit in the R32_DFSDM_CH0CFGR1 register). The audio clock is obtained by selecting PLL_CLK via the SYSPLL_SEL[2:0] field in the RCC configuration, which in turn configures PLLCLK.

42.3.2 Serial Channel Transceiver

This peripheral features 2 multiplexed serial data channels that can be selected for conversion via individual filters, analog watchdogs, or short-circuit detectors. These serial transceivers receive data streams from external $\Sigma\Delta$ modulators. Data streams can be transmitted in SPI format (refer to the SITP bit in the R32_DFSDM_CHyCFGR1 register).

The channel is enabled by setting the CHEN bit in the R32_DFSDM_CHyCFGR1 register to 1.

Channel Input Selection

Serial inputs (data and clock signals) for the DATINy and CKINy pins can be redirected from subsequent channel pins. The redirection functionality for serial input channels is configured via the CHINSEL bit in the R32_DFSDM_CHyCFGR1 register.

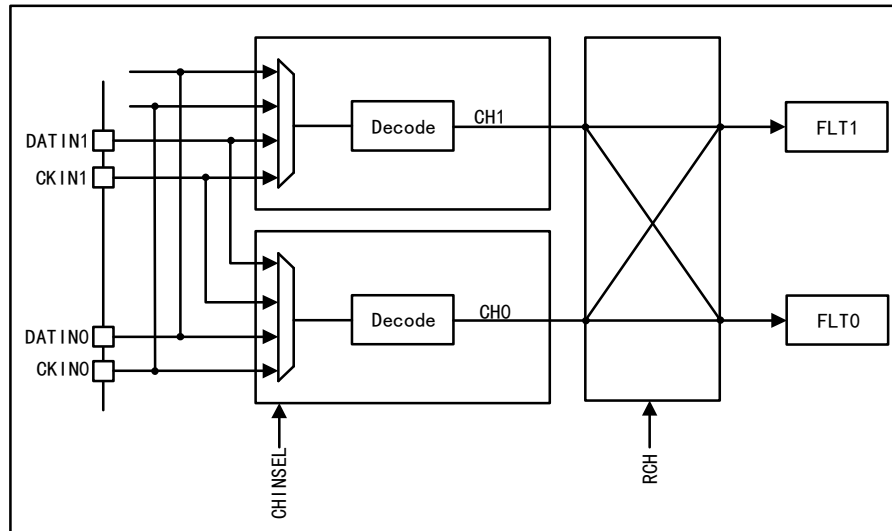
Channel redirection can be used to acquire audio data from PDM (Pulse Density Modulation) stereo microphones. A PDM stereo microphone contains one data signal and one clock signal, where the data signal provides information for the left and right audio channels (sampling occurs on the rising edge of the clock for the left channel and on the falling edge for the right channel).

Configuration of PDM microphone input serial channel;

- PDM microphone signals (data, clock) will be connected to the DFSDM input serial channel y (DATINy, CKOUT) pins
- Configure channel y as follows: CHINSEL=0 (input from given channel pins: DATINy and CKINy)
- Configure channel (y-1) (mod 2) as follows:
CHINSEL=1 (input from subsequent channel ((y-1)+1) pins: DATINy and CKINy)
- Channel y: SITP=0 (data gated on rising edge) – left audio channel for channel y

- Channel (y-1): SITP=1 (data gated on falling edge) – right audio channel for channel (y-1)
- 2 DFSDM filters will be assigned to channels y and (y-1) (for filtering left and right channels of PDM microphones)

Figure 42-2 Input channel pin redirection



Clock Signal Output

A clock signal can be provided at the CKOUT pin to drive the clock input of an external $\Sigma\Delta$ modulator. The CKOUT signal frequency is derived from either the DFSDM clock or the audio clock (refer to the CKOUTSRC bit in the R32_DFSDM_CH0CFGR1 register) via a prescaler (refer to the CKOUTDIV bit in the R32_DFSDM_CH0CFGR1 register). If the output clock is stopped, the CKOUT signal is placed in a low state (the output clock can be stopped by setting CKOUTDIV=0 in the R32_DFSDM_CHyCFGR1 register or by setting DFSDMEN=0 in the R32_DFSDM_CH0CFGR1 register). The output clock stop is executed at the following times:

- When CKOUTSRC=0, 4 system clock cycles after DFSDMEN cleared
- When CKOUTSRC=1, one system clock cycle and three audio clock cycles after DFSDMEN is cleared.
- Before changing CKOUTSRC, the software must wait for CKOUT to stop to avoid glitch signals on the CKOUT pin. The output clock signal frequency must be within the range of 0 to 20MHz.

Before changing CKOUTSRC, the software must wait for CKOUT to stop, so as to avoid generating glitch signal on CKOUT pin. The frequency of the output clock signal must be in the range of 0 ~ 20MHz.

SPI Data Input Format Operation

In SPI format, data streams are transmitted in serial format through data and clock signals. The data signal is always provided by the datum pin. The clock signal can be provided from the outside through CKINy pin or from the inside through the signal from CKOUT signal source.

If an external clock source is selected (SPICKSEL[1:0]=0), the data signal is sampled (DATINy pin) on the rising or falling edge of the clock according to the SITP bit setting in the R32_DFSDM_CHyCFGR1 register.

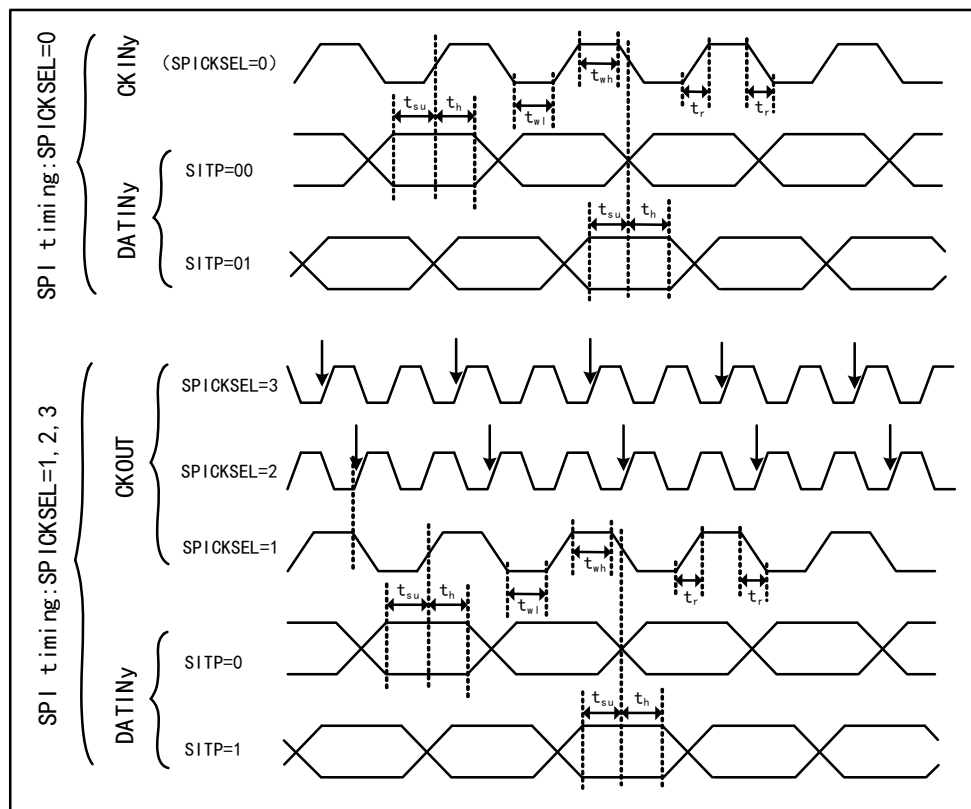
Internal clock source—refer to SPICKSEL[1:0] in the R32_DFSDM_CHyCFGR1 register:

- CKOUT signal:
 - For connecting an external $\Sigma\Delta$ modulator, which directly uses its clock input (from CKOUT) to generate the output serial communication clock.
 - Sampling point: Rising edge/falling edge, depending on SITP settings.

- CKOUT/2 signal (generated on the rising edge of CKOUT):
 - Used to connect an external $\Sigma\Delta$ modulator, which divides the clock input (from CKOUT) by 2 to generate the output serial communication clock. This output clock is active on each rising edge of the clock input.
 - Sampling point: Every second falling edge of CKOUT.
- CKOUT/2 signal (generated on the falling edge of CKOUT):
 - Used to connect an external $\Sigma\Delta$ modulator, which divides the clock input (from CKOUT) by 2 to generate the output serial communication clock. This output clock change is valid on each falling edge of the clock input.
 - Sampling point: Every second rising edge of CKOUT.

Note: The internal clock source can only be used when an external $\Sigma\Delta$ modulator employs the CKOUT signal as its clock input (to achieve synchronous clock and data operation). Using the internal clock source eliminates the need for a CKINy pin connection (the CKINy pin can be repurposed for other functions). When using SPI encoding, the clock source signal frequency must fall within the range of 0 to 20 MHz and be less than $f_{DFSDMCLK}/4$.

Figure 42-3 Channel transceiver timing diagram



Clock Loss Detection

This feature detects the presence or absence of clock signals on channel serial clock inputs, ensuring proper operation of conversion processes and error reporting. Clock absence detection is enabled or disabled for each input channel y via the CKABEN bit in the R32_DFSDM_CHyCFGR1 register. If clock absence detection is enabled for a given channel, it is continuously monitored. If an input clock error occurs, the clock loss flag is set to 1 (CKABF[y]=1) and an interrupt is triggered (when CKABIE=1). After the clock miss flag is cleared (via CLRCKABF in the R32_DFSDM_FLT0ICR register), the clock miss status bit CKABF[y] is reset. When the corresponding channel y is disabled, CKABF[y] is additionally set to 1 by hardware (if CHEN[y]=0, CKABF[y] remains set).

Upon occurrence of a clock-out event, the data conversion (and/or analog watchdog and short-circuit detector) will provide erroneous data. When clock-out is reported, the user should handle this event and discard the associated data.

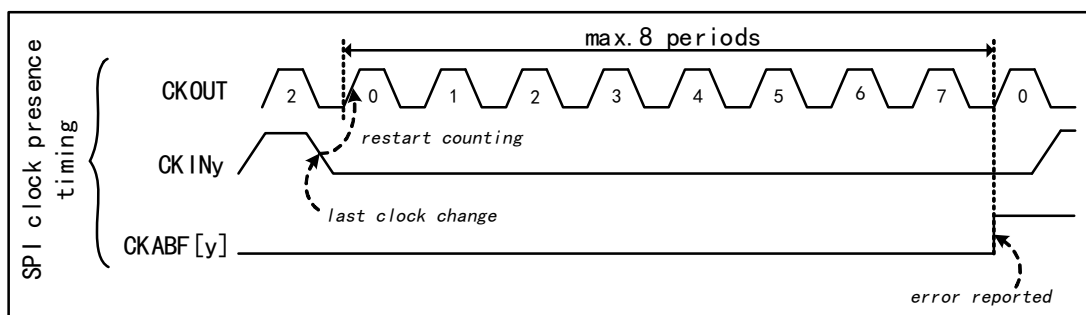
The clock-out function is only available when the system clock is used for the CKOUT signal (CKOUTSRC=0 in the R32_DFSDM_CH0CFGR1 register).

When the transceiver is out of synchronization, the clock loss flag will be set and cannot be cleared via the CLRCKABF[y] bit in the R32_DFSDM_FLT0ICR register. The software operation sequence associated with the clock loss detection function should be as follows:

- Enable the specified channel by setting CHEN=1
- Attempt to clear the clock absent flag (by setting CLRCKABF=1) until the clock absent flag is actually cleared (CKABF=0). At this point, the transceiver is synchronized (signal clock is valid) and capable of receiving data
- Enable the clock absent function CKABEN=1 and the associated interrupt CKABIE=1 to detect whether the SPI clock is lost

When using the SPI data format, clock loss detection is based on comparing the CKOUT signal generated from the external input clock with the output clock. The external input clock signal in the input channel must change once every 8 signal cycles of the CKOUT signal (controlled by the CKOUTDIV field in the R32_DFSDM_CH0CFGR1 register).

Figure 42-4 SPI clock missing timing diagram



If a clock recovery error occurs during input, the clock absent flag is set to 1 (CKABF[y]=1) and an interrupt is triggered (when CKABIE=1). After the clock absent flag is cleared (via CLRCKABF in the R32_DFSDM_FLT0ICR register), the clock absent flag is reset.

External Serial Clock Frequency Measurement

By measuring the serial clock input frequency of the channel, the data rate from the external $\Sigma\Delta$ modulator can be obtained in real time, which is critical for application purposes.

The external serial clock input frequency can be measured using a timer that counts the DFSDM clock (f_{DFSDMCLK}) over one conversion duration. Counting begins at the first input data clock following conversion triggering (either normal or injected) and ends at the last input data clock before conversion completion (when the conversion end flag is set to 1). Upon conversion completion (JEOCF=1 or REOCF=1), the time per conversion (the interval between the first and last serial samples) is updated in the CNVCNT[27:0] counter of register R32_DFSDM_FLTxCNVTIMR. The user can then calculate the data rate based on the digital filter settings (FORD, FOSR, IOSR, FAST). External serial frequency measurement ceases only when the filter is bypassed (FOSR=0, only the integrator is active, and CNVCNT[27:0] in the R32_DFSDM_FLTxCNVTIMR register equals 0).

For parallel data input (see Section 42.3.4: Parallel Data Input), the measured frequency represents the average input data rate during a single conversion.

Note: When a conversion is interrupted (e.g., by disabling/enabling the selected channel), the interruption duration is also counted in CNVCNT[27:0]. Therefore, to obtain accurate conversion timing results, it is recommended not to interrupt conversions. It is recommended to measure the clock in continuous fast mode.

Conversion time:

Injection conversion or regular conversion at FAST=0 (or first conversion at FAST=1):

For Sincx filters (x=1..5):

$$t = \text{CNVCNT} / f_{\text{DFSDMCLK}} = [\text{FOSR} * (2\text{IOSR} + \text{FORD} + 1) + 2] / f_{\text{CKIN}}$$

For FastSinc filters:

$$t = \text{CNVCNT} / f_{\text{DFSDMCLK}} = [\text{FOSR} * (\text{FOSR} * (2\text{IOSR} + 5) + 2)] / f_{\text{CKIN}}$$

Regular conversion when FAST=1 (except first conversion):

For Sincx and FastSinc filters:

$$t = \text{CNVCNT} / f_{\text{DFSDMCLK}} = [\text{FOSR} * 2\text{IOSR}] / f_{\text{CKIN}}$$

If FOSR=FOSR[9:0]+1=1 (filter bypassed, only integrator active), then:

$$t = 2^{\text{IOSR}} / f_{\text{CKIN}} \text{ (...but CNVCNT=0)}$$

Among them:

- 1) f_{CKIN} is the channel input clock frequency (at the CKINy pin of the given channel) or the input data rate (for parallel data input);
- 2) FOSR is the filter oversampling rate: $\text{FOSR} = \text{FOSR}[9:0] + 1$ (refer to the R32_DFSDM_FLTxFCR register);
- 3) 2^{IOSR} is the integrator oversampling rate (refer to the R32_DFSDM_FLTxFCR register);
- 4) FORD is the filter order: $\text{FORD} = \text{FORD}[2:0]$ (refer to the R32_DFSDM_FLTxFCR register).

Channel Offset Settings

Each channel has its own offset setting (in the register), which will eventually be subtracted from each conversion result (injection or normal) of a given channel. Offset correction is performed after the data is shifted to the right. The offset value is stored as a 24-bit signed value in the OFFSET[23:0] field of R32_DFSDM_CHyCFGR2 register.

Right Shift of Data

To align the result with a 24-bit value, each channel defines a series of right shifts that will be applied to each conversion result (injection or normal) of a given channel. The right shift number is stored in bits DTRBS[4:0] of R32_DFSDM_CHyCFGR2 register.

Shift right rounds the result to the nearest integer value. The sign of the shifted result is preserved, so as to obtain the result data in a valid 24-bit signed format.

42.3.3 Configuration Input Serial Interface

The following parameters must be configured for the input serial interface:

- Output clock prescaler. With the programmable prescaler, the output clock (range 2 to 256) can be generated from the DFSDM clock. This setting is determined by the CKOUTDIV[7:0] bits in the R32_DFSDM_CH0CFGR1 register.
- Type of serial interface and clock phase when input. Select SPI interface and sampling edge of input clock. This option is specified by the SITP bit in the R32_DFSDM_CHyCFGR1 register.
- Input clock source selection. You can select an external source from the CKINy pin or an internal source from the CKOUT pin. This selection is determined by the spikesel [1: 0] fields in the R32_DFSDM_CHyCFGR1 register.
- Final data right shift setting. Defines the right shift number of the final data to ensure that the result is aligned

with the 24-bit value. This configuration is defined by the DTRBS[4:0] bits in the R32_DFSDM_CHyCFGR2 register.

- Channel offset configuration. Set the analog offset (that is, the offset of the connected external $\sigma \delta$ modulator) for a specific serial channel. This value is set by bits OFFSET[23:0] in the R32_DFSDM_CHyCFGR2 register.
- Channel short detector and activation of clock missing function. In the R32_DFSDM_CHyCFGR1 register, the short-circuit detector of a given serial channel is enabled or disabled by the SCDEN bit, and the clock missing detection is enabled or disabled by the CKABEN bit.
- Threshold configuration of analog watchdog filter and short circuit detector. Used to set the parameters of channel analog watchdog filter and channel short circuit detector. These configurations are defined in the R32_DFSDM_CHyAWSCDR register.

42.3.4 Parallel Data Input

Each input channel provides a register for 16-bit parallel data input (except serial data input). Each 16-bit parallel input can only come from an internal data source:

- Internal ADC results (ADC1 and ADC2 only)
- Direct CPU/DMA write operation

Use the DATMPX[1:0] field of R32_DFSDM_CHyCFGR1 register to select whether to use serial or parallel data input for a given channel. DATMPX[1:0] also defines parallel data source: internal ADC or write operation directly by CPU/DMA.

Each channel contains a 32-bit data input register (R32_DFSDM_CHyDATINR), in which 16 bits of data can be written. The data is in a 16-bit signed format. These data can also be used as the input of a digital filter that accepts 16-bit parallel data.

If serial data input is selected (DATMPX[1:0]=0), the R32_DFSDM_CHyDATINR register is write-protected.

Input from internal ADC (ADC1 and ADC2 only)

For parallel ADC data inputs (DATMPX[1:0]=1), the ADC[y+1] result is assigned to channel y input (ADC1 populates the R32_DFSDM_CH0DATINR register, ADC2 populates the R32_DFSDM_CH1DATINR register). The conversion complete event from ADC[y+1] triggers an update to channel y's data (the parallel data from ADC[y+1] is used as the next sample for the digital filter). When the conversion complete event occurs, the data from ADC[y+1] is written to the R32_DFSDM_CHyDATINR register (the INDAT0[15:0] field).

The data packing mode setting (DATPACK[1:0] in the R32_DFSDM_CHyCFGR1 register) has no effect on the ADC data input.

Note: ADC Extension Specification: When the internal ADC is configured in interleaved mode (e.g., ADC1 interleaved with ADC2 - refer to ADC specification), each result from ADC1 or ADC2 is transmitted to the same 16-bit bus (i.e., ADC1's bus), which is connected to DFSDM Channel 0 (fixed connection). Therefore, DFSDM channel 0 will carry double the input data rate (even samples from ADC1, odd samples from ADC2). Channel 1, associated with ADC2, will remain idle.

Input from memory (written directly by CPU/DMA)

When CPU or DMA (DATMPX[1:0]=2) directly writes data into R32_DFSDM_CHyDATINR register, these data can be used as input to process digital data streams from memory or peripherals.

Data can be written into R32_DFSDM_CHyDATINR register by CPU or DMA:

1) CPU data writing:

Input data is directly written into R32_DFSDM_CHyDATINR register by CPU.

2) DMA data writing:

DMA should be configured as a memory-to-memory transfer mode to transfer the data in the buffer area of the memory to the R32_DFSDM_CHyDATINR register. The target memory address is the address of the R32_DFSDM_CHyDATINR register. Data is transferred from memory to DFSDM parallel input at DMA transfer speed.

This DMA is different from the DMA used to read the conversion result of DFSDM. These two DMA's can be used at the same time. The first DMA (configured as memory-to-memory transfer) is used for input data writing, and the second DMA (configured as peripheral-to-memory transfer) is used for data result reading.

The access to R32_DFSDM_CHyDATINR can be 16-bit or 32-bit wide, allowing one or two samples to be loaded in one write operation, respectively. The 32-bit input data register (R32_DFSDM_CHyDATINR) can be filled with one or two 16-bit data samples, depending on the data packing operation mode defined in the DATPACK[1:0] field of R32_DFSDM_CHyCFGR1 register:

1) Standard Mode (DATPACK[1:0]=00b):

Only one sample is stored in the INDAT0[15:0] field of the R32_DFSDM_CHyDATINR register, serving as input data for channel y. The upper 16 bits (INDAT1[15:0]) are ignored and write-protected. The digital filter must perform one input sample (via INDAT0[15:0]) to clear the data register after CPU/DMA fill. This mode combines with 16-bit CPU/DMA access to the R32_DFSDM_CHyDATINR register to load one sample per write operation.

2) Interleaved Mode (DATPACK[1:0]=01b):

The R32_DFSDM_CHyDATINR register functions as two sample buffers. The first sample is stored in INDAT0[15:0], and the second sample is stored in INDAT1[15:0]. The digital filter must perform two input samples through channel y to clear the R32_DFSDM_CHyDATINR register. This mode is combined with 32-bit CPU/DMA access to the R32_DFSDM_CHyDATINR register to load two samples per write operation.

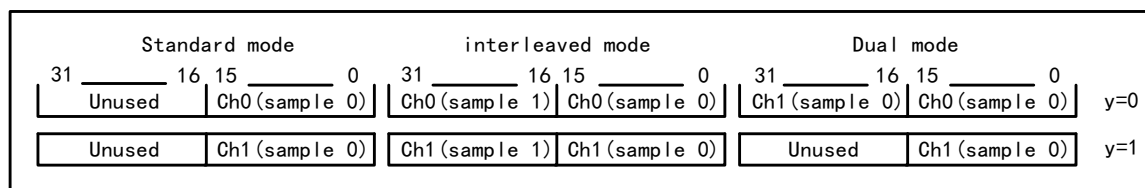
3) Dual-Channel Mode (DATPACK[1:0]=10b):

Write 2 samples into the R32_DFSDM_CHyDATINR register. Data in INDAT0[15:0] is used for channel y, while data in INDAT1[15:0] is used for channel y+1. Data in INDAT1[15:0] is automatically copied to INDAT0[15:0] of the next (y+1) channel data register (DFSDM_CH[y+1]DATINR). The digital filter must perform two samples (one from channel y and one from channel (y+1)) to clear the R32_DFSDM_CHyDATINR register.

Dual-channel mode configuration (DATPACK[1:0]=10b) is only possible for even channel numbers (y=0). If an odd channel (y=1) is set to dual-channel mode, both the INDAT0[15:0] and INDAT1[15:0] sections for that channel are write-protected. If an even channel is set to dual-channel mode, the next odd channel must be set to standard mode (DATPACK[1:0]=00b) to properly interface with the even channel.

Please refer to the figure below for the data mode of R32_DFSDM_CHyDATINR register and the allocation of channel data samples.

Figure 42-5 R32_DFSDM_CHyDATINR register operation mode and allocation



First enable the selected input channel (channel y) to initiate data acquisition (starting conversion for channel y). Then, a write operation to the R32_DFSDM_CHyDATINR register is required to load one or two samples. Failure to do so will result in the loss of written data during subsequent processing.

For example: In single conversion and interleaved mode, do not begin writing paired data samples to R32_DFSDM_CHyDATINR before initiating the single conversion (any data present in R32_DFSDM_CHyDATINR prior to conversion start will be discarded).

42.3.5 Channel Selection

There are 2 multiplex channels, which can be selected for injection channel group conversion and/or conventional channel conversion.

The injection channel group can select any or all of the 2 channels. JCHG[1:0] in the R32_DFSDM_FLTxJCHGR register selects the injection group channel, where JCHG[y]=1 indicates that channel y has been selected.

Injection conversion can work in scan mode (JSCAN=1) or single mode (JSCAN=0). In scan mode, each selected channel will be converted in turn. The lowest channel (channel 0, if selected) is converted first, and then the next higher channel is converted until all channels selected by JCHG[1:0] are converted. In single mode (JSCAN=0), only one selected channel will be converted, and the channel selection will be moved to the next channel. When JSCAN=0, writing to JCHG[1:0] will reset the channel selection to the lowest channel selected.

Injection conversion can be initiated by software or triggers. Such conversions are never interrupted by regular conversions.

Conventional channels only select one of the two channels. The RCH bit in the R32_DFSDM_FLTxCR1 register indicates the selected channel.

General conversions can only be initiated by software (not by triggers). When an injection conversion is requested, the continuous regular conversion sequence is temporarily interrupted.

Attempting a conversion on a disabled channel (CHEN=0 in the R32_DFSDM_CHyCFGR1 register) will cause the conversion to never complete, as no input data is provided (no clock signal). In this case, either enable the given channel (CHEN=1 in the R32_DFSDM_CHyCFGR1 register) or stop the conversion (by setting DFEN=0 in the R32_DFSDM_FLTxCR1 register).

42.3.6 Digital Filter Configuration

DFSDM includes a Sinc^x-type digital filter implementation. This Sinc^x filter performs filtering on the input digital data stream, resulting in a reduced output data rate (decimation) and increased output data resolution. The Sinc^x digital filter can be configured to achieve the desired output data rate and desired output data resolution. Configurable parameters include:

- Filter order/type: (Refer to FORD[2:0] bits in the R32_DFSDM_FLTxFCR register):
 - FastSinc
 - Sinc¹
 - Sinc²
 - Sinc³
 - Sinc⁴
 - Sinc⁵
- Filter oversampling/decimation rate: (Refer to FOSR[9:0] bits in the R32_DFSDM_FLTxFCR register):
 - FOSR=1-1024——Applicable to FastSinc filters and Sinc^x filters where x=FORD=1..3
 - FOSR=1-215——Applicable to Sinc^x filters where x=FORD=4
 - FOSR=1-73——Applicable to Sinc^x filters where x=FORD=5

The filter supports the following transfer functions (impulse response in the H domain):

- Sinc^x filter type:

$$H(z) = \left(\frac{1-z^{-FOSR}}{1-z^{-1}} \right)^x$$

- FastSinc filter type:

$$H(z) = \left(\frac{1-z^{-\text{FOSR}}}{1-z^{-1}} \right)^2 \times (1 + z^{-(2 \times \text{FOSR})})$$

Figure 42-6 Example: Sinc3 filter response

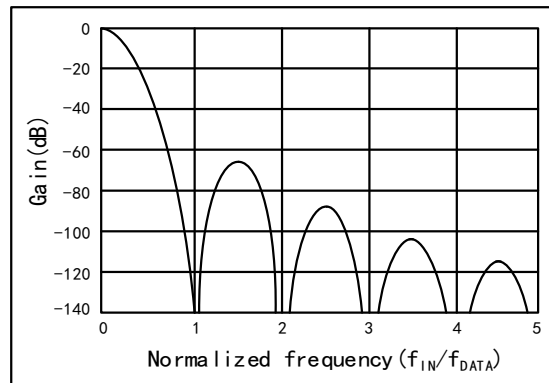


Table 42-1 Maximum output resolution of filter (peak data value from filter output), based on some FOSR values

FOSR	Sinc ¹	Sinc ²	FastSinc	Sinc ³	Sinc ⁴	Sinc ⁵
x	+/-x	+/-x ²	+/-2x ²	+/-x ³	+/-x ⁴	+/-x ⁵
4	+/-4	+/-16	+/-32	+/-64	+/-256	+/-1024
8	+/-8	+/-64	+/-128	+/-512	+/-4096	-
32	+/-32	+/-1024	+/-2048	+/-32768	+/-1048576	+/-33554432
64	+/-64	+/-4096	+/-8192	+/-262144	+/-16777216	+/-1073741824
128	+/-128	+/-16384	+/-32768	+/-2097152	+268435456	Under the condition of full-scale input, the result will overflow (> 32-bit signed integer value)
256	+/-256	+/-65536	+/-131072	+16777216		
1024	+/-1024	+/-1048576	+/-2097152	+1073741824		

42.3.7 Integrator Unit

The integrator will further improve the decimation rate and resolution of the data from the digital filter. For a given number of data samples from the filter, the integrator simply sums them.

The over-sampling rate parameter of the integrator defines how many data are summed as a data output of the integrator. IOSR can be set in the range of 1~256 (refer to the description of IOSR[3:0] bits in R32_DFSDM_FLTxFRCR register).

Table 42-2 Maximum output resolution of integrator (peak data value from integrator output), based on some IOSR values, FOSR=256 and Sinc3 filter type (maximum data).

IOSR	Sinc ¹	Sinc ²	FastSinc	Sinc ³	Sinc ⁴	Sinc ⁵
x	+/-FOSR.x	+/-FOSR ² .x	+/-2FOSR ² .x	+/-FOSR ³ .x	+/-FOSR ⁴ .x	+/-FOSR ⁵ .x
4	-	-	-	+/-67108864	-	-
32	-	-	-	+/-536870912	-	-
128	-	-	-	+/-2147483648	-	-

256	-	-	-	$\pm 2^{32}$	-	-
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42.3.8 Analog Watchdog

The analog watchdog is used to trigger an external signal (open circuit or interrupt) when it reaches or exceeds a given upper threshold or reaches or falls below a given lower threshold. Interrupt/event/open circuit generation can then be invoked.

Each analog watchdog monitors the serial data receiver output (after the watchdog filter is simulated on each channel) or the data output register (current injection or general conversion result) according to the AWFSEL bit setting (in the R32_DFSDM_FLTxCr1 register). Whether the input channels need to be monitored by analog watchdog X can be selected by AWDCH[1:0] in R32_DFSDM_FLTxCr2 register.

Analog watchdog conversions on input channels are independent of standard conversions. In standard conversion mode, the analog watchdog employs its own filter and signal processing on each input channel, unaffected by injection or regular master conversions. In continuous conversion mode, analog watchdog conversions are performed on selected input channels to maintain channel monitoring even when injection or regular master conversions are paused (RCIP=0, JCIP=0).

The upper threshold and lower threshold registers are compared against a given data value (set via the AWHT[23:0] bits of the R32_DFSDM_FLTxAWHTR register and the AWLT[23:0] bits of the R32_DFSDM_FLTxAWLTR register).

You can choose two situations to compare the threshold register with the data value:

- Option 1: In this case, the input data is taken from the final output data register (AWFSEL=0). This choice has the following characteristics:
 - High input data resolution (up to 24 bits)
 - Slow response time—not suitable for fast-response applications such as overcurrent detection
 - For comparison purposes, final data requires shift and offset correction
 - Final data becomes available only after executing a normal or injected main conversion
 - Operates with parallel input data sources (DATMPX[1:0] $\neq$ 0 in the R32_DFSDM_CHyCFGR1 register)
- Option 2: In this case, the input data is taken from the output of any serial data receiver (AWFSEL=1). This choice has the following characteristics:
 - Serial data input is processed through a dedicated analog watchdog Sincx channel filter. The filter's oversampling rate (1..32) and filter order (1..3) are configurable (refer to the AWFOSR[4:0] and AWFORD[1:0] bits in the R32_DFSDM_CHyAWSCDR register).
 - Lower resolution (up to 16 bits)
 - Fast response time—suitable for applications requiring rapid response, such as overcurrent/overvoltage detection
 - Data is provided in continuous mode, independent of regular or injected main conversion activity

During input channel monitoring (AWFSEL=1), data for threshold comparison is sourced from the channel selected via the AWDCH[1:0] field (R32_DFSDM_FLTxCr2 register). The filter result from each selected channel is compared against a threshold pair (AWHT[23:0]/AWLT[23:0]). At this point, only the upper 16 bits (AWHT[23:8]/AWLT[23:8]) define the 16-bit threshold compared against the analog watchdog filter output, as the data from the analog watchdog filter has a maximum resolution of 16 bits. In this case (AWFSEL=1), the AWHT[7:0]/AWLT[7:0] bits are not compared.

The parameters for the analog watchdog filter configuration of each input channel (filter order AWFORD[1:0] and filter oversampling rate AWFOSR[4:0]) are set in the R32_DFSDM_CHyAWSCDR register.

The analog watchdog filter data for channel y can be read via firmware from the WDATA[15:0] field of the

DFSDM_CHyWDATR register. If CHEN=1 in the R32_DFSDM_CHyCFGR1 register, this analog watchdog filter data is continuously converted at a data rate determined by the analog watchdog filter settings and the channel input clock frequency.

The analog watchdog filter operates similarly to a conventional fast continuous conversion (without integrator). The number of serial samples required for the analog watchdog filter to output a result (with channel input clock frequency fCKIN) is:

First conversion:

For Sinc^x filters (x=1..5): Number of samples = $\lceil F_{OSR} * (F_{ORD} + 1) + 2 \rceil$

For FastSinc filters: Number of samples = $\lceil F_{OSR} * 4 + 2 + 1 \rceil$

Subsequent conversion:

For Sinc^x and FastSinc filters: Number of samples = $\lceil F_{OSR} * F_{ORD} \rceil$

Among them:

F_{OSR}, or Filter Oversampling Ratio: F_{OSR} = AWFOSR[4:0] + 1 (refer to the R32_DFSDM_CHyAWSCDR register)

F_{ORD}, or Filter Order: F_{ORD} = AWFORD[1:0] (refer to the R32_DFSDM_CHyAWSCDR register)

For output data register monitoring (AWFSEL=0), comparison occurs after completing the final data right shift and offset correction (refer to the OFFSET[23:0] and DTRBS[4:0] fields in the R32_DFSDM_CHyCFGR2 register). The comparison occurs after each injection or regular conversion of the channel selected by the AWDCH[1:0] field (in the R32_DFSDM_FLTxCR2 register).

The status of simulated watchdog events is reflected in the R32_DFSDM_FLTxAWSR register, which locks the given event. The AWHTF[y]=1 flag indicates whether the AWHT[23:0] value has been exceeded on channel y. The AWLFTF[y]=1 flag indicates whether the AWLT[23:0] value has been exceeded on channel y. Events latched in the R32_DFSDM_FLTxAWSR register are cleared by writing a 1 to the corresponding clear bit CLRAWHTF[y] or CLRAWLTF[y] in the R32_DFSDM_FLTxAWCFR register.

42.3.9 Short Circuit Detector

The purpose of the short-circuit detector is to signal within an extremely fast response time when an analog signal reaches saturation (exceeds full-scale range) and remains there for a specified duration. This feature detects short-circuit or open-circuit faults (e.g., overcurrent or overvoltage). Interrupts/events/open-circuit generation can be invoked.

The short-circuit detector's input data is derived from the channel transceiver output.

Each input channel features an incrementing counter that tallies consecutive 0s or 1s on the serial data receiver output. If the received data stream changes (data signal conversions from 1 to 0 or from 0 to 1), the counter is reset. A short-circuit event is triggered when the counter reaches the short-circuit threshold register value (SCDT[7:0] bits in the R32_DFSDM_CHyAWSCDR register). Each input channel is equipped with a dedicated short-circuit detector. Continuous monitoring for any channel can be enabled by setting the SCDEN bit (in the R32_DFSDM_CHyCFGR1 register). Each channel has its own short-circuit detector configuration (threshold bits SCDT[7:0], status bits SCDF[1:0], status clear bits CLRSCDF[1:0]). When the corresponding channel y is disabled (CHEN[y]=0), the status flag SCDF[y] can also be cleared by hardware.

For each channel, short-circuit detector events can be assigned to the break output signals DFSDM_BK[1:0]. Two break outputs can be assigned to a single short-circuit detector event. The break signals are assigned to a given channel's short-circuit detector event via the BKSCD[1:0] field in the R32_DFSDM_CHyAWSCDR register.

If parallel input data channels are selected (DATMPX[1:0] ≠ 0 in the R32_DFSDM_CHyCFGR1 register), the short-circuit detector cannot be used.

A total of two break outputs are available (shared with the analog watchdog function).

42.3.10 Extreme Value Detector

The Extreme Value Detector captures the minimum and maximum values (peak-to-peak values) of the final output data word.

If the output data word exceeds the value stored in the maximum value register of the extremum detector (EXMAX[23:0] bits in the R32_DFSDM_FLTxEXMAX register), this register is updated with the current output data word value, and the channel storing its data is indicated in the EXMAXCH bit (of the R32_DFSDM_FLTxEXMAX register).

If the output data word is less than the value stored in the extremum detector minimum register (EXMIN[23:0] bits in the R32_DFSDM_FLTxEXMIN register), this register is updated with the current output data word value, and the channel storing its data is indicated by the EXMINCH bit (in the R32_DFSDM_FLTxEXMIN register).

The minimum and maximum value register values can be refreshed by software (by reading the R32_DFSDM_FLTxEXMAX or R32_DFSDM_FLTxEXMIN registers). After refresh, the minimum value data register (R32_DFSDM_FLTxEXMIN) is filled with 0x7FFFFFFF (maximum positive value), and the maximum value register (R32_DFSDM_FLTxEXMAX) is filled with 0x800000 (minimum negative value).

The extremum detector performs comparison operations only after completing right shift and offset data correction. For each extremum detector, the input channels involved in extremum calculation are selected by the EXCH[1:0] bits (in the R32_DFSDM_FLTxCR2 register).

42.3.11 Data Unit Module

The data unit module is the final module in the entire processing path: external $\Sigma\Delta$ modulator—serial transceiver—Sinc filter—integrator—data unit block.

The output data rate depends on the serial data stream rate and the filter and integrator settings. The maximum output data rate is:

$$\text{Data rate [samples/sec]} = \frac{f_{CKIN}}{F_{OSR} * (2^{IOSR} + F_{ORD} + 1) + 2}; \text{FAST}=0, \text{Sinc}^x \text{ filter}$$

$$\text{Data rate [samples/sec]} = \frac{f_{CKIN}}{F_{OSR} * (2^{IOSR} + 5) + 2}; \text{FAST}=0, \text{FastSinc filter}$$

Or

$$\text{Data rate [samples/sec]} = \frac{f_{CKIN}}{F_{OSR} * 2^{IOSR}}; \text{FAST}=1$$

Maximum output data rate for parallel data input:

$$\text{Data rate [samples/sec]} = \frac{f_{DATAIN_RATE}}{F_{OSR} * (2^{IOSR} + F_{ORD} + 1) + 2}; \text{FAST}=0, \text{Sinc}^x \text{ filter}$$

Or

$$\text{Data rate [samples/sec]} = \frac{f_{DATAIN_RATE}}{F_{OSR} * (2^{IOSR} + 5) + 2}; \text{FAST}=0, \text{FastSinc filter}$$

Or

$$\text{Data rate [samples/sec]} = \frac{f_{DATAIN_RATE}}{F_{OSR} * 2^{IOSR}}; \text{FAST}=1 \text{ or any bypass case } (F_{OSR}=1)$$

Where: f_{DATAIN_RATE} is the input data rate of ADC or CPU/DMA.

The right shift of the final data is performed in this module, because the final data width is 24 bits, and the data from the processing path can be up to 32 bits. For each selected input channel, this right shift can be configured in the range of 0-31 bits (refer to bits DTRBS[4:0] in R32_DFSDM_CHyCFGR2 register). Shift right rounds the result to the nearest integer value. The sign of the shifted result is preserved, so as to obtain the result data in a valid 24-

bit signed format.

Next, the results are offset corrected. Subtract the offset correction value (OFFSET[23:0] stored in register R32_DFSDM_CHyCFGR2) from the output data of a given channel. The data in the OFFSET[23:0] field is set by the software through the corresponding calibration program.

Since all operations in digital processing are performed on 32-bit signed registers, the following conditions must be met to ensure that the results will not overflow:

- 1) $FOSR^{FORD} \cdot 2^{IOSR} \leq 2^{31}$, suitable for Sincx filter ($x=1..5$);
- 2) $FOSR^2 \cdot 2^{IOSR} \leq 2^{31}$, suitable for FastSinc filter.

Note: When the filter and integrator are bypassed ($IOSR[3:0]=0$, $FOSR[9:0]=0$), the input data rate (f_{DATAIN_RATE}) must be limited to be able to read all the output data: $f_{DATAIN_RATE} \leq f_{HCLK}$; Among them, f_{HCLK} is the bus frequency of DFSDM peripheral connection.

42.3.12 Signed Data Format

Each channel of the DFSDM input serial channel can be connected to an external $\Sigma\Delta$ modulator. An external $\Sigma\Delta$ modulator is configured with two differential inputs (positive and negative), which can be used for either differential measurements or single-ended signal measurements.

It is always assumed that the $\Sigma\Delta$ modulator's output uses a signed format (i.e., the 0/1 data stream from the $\Sigma\Delta$ modulator represents the values -1 and +1).

Signed Data Format in Registers: Data stored in the final output data register, analog watchdog register, extreme value detector register, and offset correction register are all in signed format. The most significant bit (MSB) of the output data word indicates the sign of the value (using two's complement format).

42.3.13 Start Conversion

The method to start the injection conversion is as follows:

- Software: write 1 to JSWSTART in R32_DFSDM_FLTxCR1 register.
- Trigger: with JEXTEN activated, select the trigger signal through JEXTSEL[3:0] and determine the effective edge (refer to R32_DFSDM_FLTxCR1 register).
- If JSYNC is set to 1, DFSDM_FLT0 will start injection conversion synchronously: for DFSDM_FLTx($x>0$), injection conversion will start automatically in DFSDM_FLT0; The injection conversion is triggered by software (set to 1 by the JSWSTART bit in the R32_DFSDM_FLT0CR2 register). Each injection conversion in DFSDM_FLTx($x>0$) will be performed according to its local settings (such as JSCAN and JCHG).
- Only one injection conversion is allowed at a specific time. Therefore, if an injection conversion request has been issued but not yet completed, any subsequent startup request will be ignored.

The way to start a regular conversion is as follows:

- Software: write 1 to RSWSTART in R32_DFSDM_FLTxCR1 register.
- If RSYNC=1, use DFSDM_FLT0 to start synchronously: for DFSDM_FLTx($x>0$), normal conversion will start automatically in DFSDM_FLT0; The normal conversion is triggered by software (set to 1 by the RSWSTARTT bit in the R32_DFSDM_FLT0CR2 register). Every regular conversion in DFSDM_FLTx($x>0$) will be performed according to its local configuration settings (RCONT, RCH, etc.).

42.3.14 Continuous and Fast Continuous Mode

If RCONT in the R32_DFSDM_FLTxCR1 register is set to 1, normal conversion will be performed in continuous mode. RCONT=1 means that after writing 1 to RSWSTART, the channel selected by the RCH bit is repeatedly

converted.

Normal conversion performed in continuous mode can be stopped by writing 0 to RCONT. After the RCONT is cleared, the ongoing conversion will be stopped immediately.

In continuous mode, the data rate can be increased by setting bit 1 of the FAST position in the R32_DFSDM_FLTxCR1 register. At this time, if a continuous conversion is performed on a channel, the filter does not need to be refilled with new refresh data because the internal filter data is valid. This data was previously obtained by sampling continuous data. The speed increase is related to the selected filter order. After initiating continuous conversion with RSWSTART=1, the initial conversion in Fast mode (FAST=1) is performed fully (identical to FAST=0), followed by subsequent conversions completed at shorter intervals.

Conversion time in continuous mode:

When FAST=0 (or fast conversion when FAST=1):

For Sinc^x filter:

$$t = \text{CNVCNT} / f_{\text{DFSDMCLK}} = [F_{\text{OSR}} * (2^{\text{IOSR}} + F_{\text{ORD}} + 1) + 2] / f_{\text{CKIN}}$$

For FastSinc filter:

$$t = \text{CNVCNT} / f_{\text{DFSDMCLK}} = [F_{\text{OSR}} * (2^{\text{IOSR}} + 5) + 2] / f_{\text{CKIN}}$$

When FAST=1 (except for the first conversion):

For Sinc^x and FastSinc filter:

$$t = \text{CNVCNT} / f_{\text{DFSDMCLK}} = [F_{\text{OSR}} * 2^{\text{IOSR}}] / f_{\text{CKIN}}$$

When FOSR=FOSR[9:0]+1=1 (filter bypass, only integrator is effective):

$$t = 2^{\text{IOSR}} / f_{\text{CKIN}} \text{ (...but CNVCNT=0)}$$

Continuous mode is not applicable to injection conversion. Injection conversion can be initiated via timer triggering to simulate continuous mode with precise timing.

If a regular continuous conversion is in progress (RCONT=1), and if a write access is made to the R32_DFSDM_FLTxCR1 register requesting a regular continuous conversion (RCONT=1), the regular continuous conversion will restart from the next conversion cycle (similar to a new regular continuous conversion being applied to a new channel selection, even if no change is made to the R32_DFSDM_FLTxCR1 register).

42.3.15 Request Priority

Injected conversions take precedence over regular conversions. An ongoing regular conversion is immediately interrupted by an injected conversion request; this regular conversion will resume after the injected conversion completes.

If an injected conversion is pending or in progress, no other injected conversion can be initiated: requests to start an injected conversion (via JSWSTART or trigger) are ignored as long as bit JCIP (in the R32_DFSDM_FLTxISR register) is set to 1.

Similarly, if a regular conversion is pending or in progress, no other regular conversion can be initiated: requests to start a regular conversion (using RSWSTART) are ignored as long as bit RCIP (in the R32_DFSDM_FLTxISR register) is set to 1.

However, if an injection conversion is requested during a regular conversion, the regular conversion stops immediately and the injection conversion begins. The regular conversion restarts afterward, and this delayed restart is indicated by the RPEND bit.

Injection conversion takes precedence over regular conversion because it can temporarily interrupt a continuous sequence of regular conversions. Upon completing the injection conversion sequence, if RCONT remains set to 1, the continuous regular conversion will restart (the RPEND bit will issue a delayed start signal based on the result

of the first regular conversion).

Priority is also critical when initiating related operations through identical write operations to DFSDM, or if multiple operations remain pending at the conclusion of other operations. For example, assuming an injection conversion is in progress (JCIP=1), a single write operation to R32_DFSDM_FLTxCR1 writes 1 to RSWSTART to request a regular conversion. Upon completion of the injection sequence, the priority indicates execution of the subsequent regular conversion, with the corresponding delayed start indicated via the RPEND bit.

42.3.16 Power Consumption Optimization in Running Mode

To reduce power consumption, the DFSDM filter and integrator automatically enter idle state (RCIP=0, JCIP=0) when not used for conversion.

42.3.17 Trigger Signal and Trigger Source

Table 42-3 Connection between trigger signal and trigger source

Trigger name	Trigger source	Trigger name	Trigger source
DFSDM_JTRG0	TIM1_TRGO	DFSDM_JTRG8	TIM9_TRGO
DFSDM_JTRG1	TIM2_TRGO	DFSDM_JTRG9	TIM10_TRGO
DFSDM_JTRG2	TIM3_TRGO	DFSDM_JTRG10	TIM11_TRGO
DFSDM_JTRG3	TIM4_TRGO	DFSDM_JTRG11	TIM12_TRGO
DFSDM_JTRG4	TIM5_TRGO	DFSDM_JTRG12	EXTI11
DFSDM_JTRG5	TIM6_TRGO	DFSDM_JTRG13	EXTI15
DFSDM_JTRG6	TIM7_TRGO	DFSDM_JTRG14	LPTIM1
DFSDM_JTRG7	TIM8_TRGO	DFSDM_JTRG15	LPTIM2

42.3.18 DFSDM Open Circuit Connection

Table 42-4 DFSDM open circuit connection

Open circuit name	Open circuit target
DFSDM_BK[0]	TIM1 open circuit 1
DFSDM_BK[1]	TIM1 open circuit 2

42.4 Interrupt

In order to enhance the performance of central processing unit (CPU), a set of interrupts related to CPU events have been implemented:

- Injection conversion end interrupt:
 - Enabled via the JEOCIE bit in the R32_DFSDM_FLTxCR2 register
 - Indicated via the JEOCF bit in the R32_DFSDM_FLTxISR register
 - Cleared by reading the R32_DFSDM_FLTxJDATAR register (injection data)
 - Indicates which channel completed conversion and reports it in the JDATACH bit of the R32_DFSDM_FLTxJDATAR register
- Regular conversion end interrupt:
 - Enabled via the REOCIE bit in the R32_DFSDM_FLTxCR2 register
 - Indicated via the REOCF bit in the R32_DFSDM_FLTxISR register
 - Cleared by reading the R32_DFSDM_FLTxRDATAR register (regular data)

- Indicates which channel completed conversion and reports it in the RDATACH bit of the R32_DFSDM_FLTxDATAR register
- Injection conversion data overflow interrupt:
 - This interrupt occurs when injected conversion data is not read from the R32_DFSDM_FLTxDATAR register (via CPU or DMA) and is overwritten by newly injected conversion data.
 - Enabled via the JOVRIE bit in the R32_DFSDM_FLTxCR2 register
 - Indicated via the JOVRF bit in the R32_DFSDM_FLTISR register
 - Cleared by writing a 1 to the CLRJOVRF bit in the R32_DFSDM_FLTICR register
- Regular conversion data overflow interrupt:
 - This interrupt occurs when conventional conversion data is not read from the R32_DFSDM_FLTxDATAR register (via CPU or DMA) and is overwritten by a new conventional conversion.
 - Enabled via the ROVRIE bit in the R32_DFSDM_FLTxCR2 register
 - Indicated via the ROVRF bit in the R32_DFSDM_FLTISR register
 - Cleared by writing a 1 to the CLRROVRF bit in the R32_DFSDM_FLTICR register
- Analog watchdog interrupt:
 - This interrupt occurs when converted data (output data or data from the analog watchdog filter—depending on the setting of the AWFSEL bit in the R32_DFSDM_FLTxCR1 register) exceeds the upper threshold in the R32_DFSDM_FLTxAWHTR/R32_DFSDM_FLTxAWLTR register or falls below the corresponding lower threshold.
 - Enabled via the AWDIE bit in the R32_DFSDM_FLTxCR2 register (for the selected channel AWDCH[1:0])
 - Indicated by the AWDF bit in the R32_DFSDM_FLTISR register
 - Analog watchdog upper or lower threshold errors are individually indicated via the AWHTF[1:0] and AWLTF[1:0] fields in the R32_DFSDM_FLTAWSR register
 - Cleared by writing a 1 to the corresponding CLRAWHTF[1:0] or CLRAWLTF[1:0] bit in the R32_DFSDM_FLTAWCFR register
- Short circuit detector interrupt:
 - This interrupt occurs when stable data counts exceed the threshold in the R32_DFSDM_CHyAWSCDR register
 - Enabled via the SCDIE bit in the R32_DFSDM_FLTxCR2 register (for channels selected by the SCDEN bit in the R32_DFSDM_CHyCFGR1 register)
 - Indicated via the SCDF[1:0] bits in the R32_DFSDM_FLTISR register (also reports the channel where the short-circuit detector event occurred)
 - Cleared by writing a 1 to the corresponding CLRSCDF[1:0] bit in the R32_DFSDM_FLTICR register
- Channel clock missing interrupt:
 - This interrupt occurs when a clock loss is detected on the CKINy pin (refer to Section 42.3.2: Serial Channel Transceiver Clock Loss Detection)
 - Enabled via the CKABIE bit in the R32_DFSDM_FLTxCR2 register (for the channel selected by the CKABEN bit in the R32_DFSDM_CHyCFGR1 register)
 - Indicated via the CKABF[y] bit in the R32_DFSDM_FLTISR register
 - Cleared by writing a 1 to the CLRCKABF[y] bit in the R32_DFSDM_FLTICR register

Table 42-5 DFSDM interrupt request

Interrupt event	Event flag	Event/interrupt clearing method	Interrupt enable control bit
End of injection conversion	JEOCF	Read R32_DFSDM_FLTxJDATAR	JEOCIE
End of regular conversion	REOCF	Read R32_DFSDM_FLTxRDATAR	REOCIE
Injection data overflow	JOVRF	Write CLRJOVRF=1	JOVRIE
Regular data overflow	ROVRF	Write CLRROVRF=1	ROVRIE
Analog watchdog	AWDF, AWHTF[1:0], AWLTF[1:0]	Write CLRAWHTF[1:0]=1 Write CLRAWLTF[1:0]=1	AWDIE, (AWDCH[1:0])
Short circuit detector	SCDF[1:0]	Write CLRSCDF[1:0]=1	SCDIE, (SCDEN)
Channel clock missing	CKABF[1:0]	Write CLRCKABF[1:0]=1	CKABIE, (CKABEN)

42.5 DFSDM DMA Transfer

To reduce CPU intervention, DMA transfers can be used to transfer conversions to memory. DMA transfers for injected conversions are enabled by setting the JDMAEN bit in the R32_DFSDM_FLTxCR1 register. DMA transfers for regular conversions are enabled by setting the RDMAEN bit in the R32_DFSDM_FLTxCR1 register.

Note: With DMA transfers, interrupt flags (JEOCF or REOCF bits in the R32_DFSDM_FLTxISR register) are automatically cleared at the end of an injection or regular conversion because DMA is reading the R32_DFSDM_FLTxJDATAR or R32_DFSDM_FLTxRDATAR registers.

42.6 Register Description

Table 42-6 DFSDM channel-related registers

Name	Access address	Description	Reset value
R32_DFSDM_CH0CFGR1	0x40017000	DFSDM channel 0 configuration register 1	0x00000000
R32_DFSDM_CH0CFGR2	0x40017008	DFSDM channel 0 configuration register 2	0x00000000
R32_DFSDM_CH0AWSCDR	0x40017010	DFSDM channel 0 analog watchdog and short circuit detector register	0x00000000
R32_DFSDM_CH0WDATR	0x40017018	DFSDM channel 0 watchdog filter data register	0x00000000
R32_DFSDM_CH0DATINR	0x40017020	DFSDM channel 0 data input register	0x00000000
R32_DFSDM_CH1CFGR1	0x40017004	DFSDM channel 1 configuration register 1	0x00000000
R32_DFSDM_CH1CFGR2	0x4001700C	DFSDM channel 1 configuration register 2	0x00000000
R32_DFSDM_CH1WSCDR	0x4001714	DFSDM channel 1 analog watchdog and short circuit detector register	0x00000000

R32_DFSDM_CH1WDATR	0x400171C	DFSDM channel 1 watchdog filter data register	0x00000000
R32_DFSDM_CH1DATINR	0x40017024	DFSDM channel 1 data input register	0x00000000

Table 42-7 DFSDM filter-related registers

Name	Access address	Description	Reset value
R32_DFSDM_FLT0CR1	0x40017028	DFSDM filter 0 control register 1	0x00000000
R32_DFSDM_FLT0CR2	0x40017030	DFSDM filter 0 control register 2	0x00000000
R32_DFSDM_FLT0ISR	0x40017038	DFSDM filter 0 interrupt and status register	0x00000000
R32_DFSDM_FLT0ICR	0x40017040	DFSDM filter 0 interrupt flag clear register	0x00000000
R32_DFSDM_FLT0JCHGR	0x40017048	DFSDM filter 0 injection channel set selection register	0x00000001
R32_DFSDM_FLT0FCR3	0x40017050	DFSDM filter 0 control register 3	0x00000000
R32_DFSDM_FLT0JDATAR	0x40017058	DFSDM filter 0 injection set data register	0x00000000
R32_DFSDM_FLT0RDATAR	0x40017060	DFSDM filter 0 regular channel data register	0x00000000
R32_DFSDM_FLT0AWHTR	0x40017068	DFSDM filter 0 analog watchdog higher threshold register	0x00000000
R32_DFSDM_FLT0AWLTR	0x40017070	DFSDM filter 0 analog watchdog lower threshold register	0x00000000
R32_DFSDM_FLT0AWSR	0x40017078	DFSDM filter 0 analog watchdog status register	0x00000000
R32_DFSDM_FLT0AWCFR	0x40017080	DFSDM filter 0 analog watchdog clear flag register	0x00000000
R32_DFSDM_FLT0EXMAX	0x40017088	DFSDM filter 0 extreme value detector maximum register	0x80000000
R32_DFSDM_FLT0EXMIN	0x40017090	DFSDM filter 0 extremum value detector minimum register	0x00000000
R32_DFSDM_FLT0CNVTIMR	0x40017098	DFSDM filter 0 conversion timer register	0x00000000
R32_DFSDM_FLT1CR1	0x4001702C	DFSDM filter 1 control register 1	0x00000000
R32_DFSDM_FLT1CR2	0x40017034	DFSDM filter 1 control register 2	0x00000000
R32_DFSDM_FLT1ISR	0x4001703C	DFSDM filter 1 interrupt and status register	0x00000000
R32_DFSDM_FLT1ICR	0x40017044	DFSDM filter 1 interrupt flag clear register	0x00000000
R32_DFSDM_FLT1JCHGR	0x4001704C	DFSDM filter 1 injection channel set selection register	0x00000001
R32_DFSDM_FLT1FCR3	0x40017054	DFSDM filter 1 control register 3	0x00000000
R32_DFSDM_FLT1JDATAR	0x4001705C	DFSDM filter 1 injection set data	0x00000000

		register	
R32_DFSDM_FLT1RDATAR	0x40017064	DFSDM filter 1 regular channel data register	0x00000000
R32_DFSDM_FLT1AWHTR	0x4001706C	DFSDM filter 1 analog watchdog higher threshold register	0x00000000
R32_DFSDM_FLT1AWLTR	0x40017074	DFSDM filter 1 analog watchdog lower threshold register	0x00000000
R32_DFSDM_FLT1AWSR	0x4001707C	DFSDM filter 1 analog watchdog status register	0x00000000
R32_DFSDM_FLT1AWCFR	0x40017084	DFSDM filter 1 analog watchdog clear flag register	0x00000000
R32_DFSDM_FLT1EXMAX	0x4001708C	DFSDM filter 1 extreme value detector maximum register	0x80000000
R32_DFSDM_FLT1EXMIN	0x40017094	DFSDM filter 1 extremum value detector minimum register	0x00000000
R32_DFSDM_FLT1CNVTIMR	0x4001709C	DFSDM filter 1 conversion timer register	0x00000000

42.6.1 DFSDM Channel y Configuration Register 1 (R32_DFSDM_CHyCFGR1) (y=0/1)

Offset address: $0x00 + 0x04 * y$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DFS DME N	CKO UTSR C	Reserved						CKOUTDIV[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATPACK [1:0]		DATMPX [1:0]		Reserved			CHIN SEL	CH EN	CK AB EN	SC DE N	Reserv ed	SPICKSE L [1:0]		Reser ved	SI TP

Bit	Name	Access	Description	Reset value
31	DFSDMEN	RW	DFSDM interface global enable: 1: Enable DFSDM interface; 0: Disable DFSDM interface. If the DFSDM interface is enabled, it operates according to the enabled y-channel and enabled x-filter settings (the CHEN bit in R32_DFSDM_CHyCFGR1 and the DFEN bit in R32_DFSDM_FLTxCR1). Clearing data by setting DFSDMEN=0: - All registers R32_DFSDM_FLTxISR are reset (x=0, 1); - All registers R32_DFSDM_FLTxAWSR are reset (x=0, 1). <i>Note: DFSDMEN exists only in the</i>	0

			<i>R32_DFSDM_CH0CFGR1 register (channel y=0).</i>	
30	CKOUTSRC	RW	Serial clock source selection for output: 1: Output clock source is derived from the audio clock; 0: Output clock source is derived from the peripheral clock. This value can only be modified when DFSDMEN=0 (in the R32_DFSDM_CH0CFGR1 register). Note: CKOUTSRC exists only in the R32_DFSDM_CH0CFGR1 register (channel y=0).	0
[29:24]	Reserved	RO	Reserved	0
[23:16]	CKOUTDIV[7:0]	RW	Serial clock divider output: 0: Disables clock generation output (CKOUT signal set to low); 1–255: Defines the system clock division for serial clock output on the CKOUT signal, ranging from 2 to 256 (divider value = CKOUTDIV + 1). CKOUTDIV also defines the threshold for clock loss detection. This value can only be modified when DFSDMEN=0 (in the R32_DFSDM_CH0CFGR1 register). If DFSDMEN=0 (in the R32_DFSDM_CH0CFGR1 register), the CKOUT signal is set to low (the setting takes effect within one DFSDM clock cycle after DFSDMEN=0). Note: CKOUTDIV exists only in the R32_DFSDM_CH0CFGR1 register (channel y=0).	0
[15:14]	DATPACK[1:0]	RW	R32_DFSDM_CHyDATINR Register Data Encapsulation Mode: 00: Standard: Input data in the R32_DFSDM_CHyDATINR register is stored only in INDAT0[15:0]. To clear the R32_DFSDM_CHyDATINR register, the DFSDM filter must read one sample from channel y. 01: Interleaved: Input data in the R32_DFSDM_CHyDATINR register is stored as two samples: - The first sample resides in INDAT0[15:0] (assigned to channel y); - The second sample resides in INDAT1[15:0] (assigned to channel y). To clear the R32_DFSDM_CHyDATINR register, the digital filter must read two samples from channel y (the INDAT0[15:0] portion is read as the first sample,	0

			while the INDAT1[15:0] portion is read as the next sample). 10: Dual Channel: Input data in the R32_DFSDM_CHyDATINR register is stored as two samples: - The first sample resides in INDAT0[15:0] (assigned to channel y); - The second sample resides in INDAT1[15:0] (assigned to channel y+1). To clear the R32_DFSDM_CHyDATINR register, the first sample must be read from channel y by the digital filter, and the second sample must be read from channel y+1 by another digital filter. Dual-channel mode is supported only when the channel count is even (y=0). For odd channel counts (y=1), the R32_DFSDM_CHyDATINR register is write-protected. If an even channel is set to dual-channel mode, the next odd channel must enter standard mode (DATPACK[1:0]=0) to properly interface with the even channel. 11: Reserved. This value can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register).	
[13:12]	DATMPX[1:0]	RW	Channel y input data multiplexer: 00: Channel y input data is sourced from external serial input, representing a 1-bit value. The R32_DFSDM_CHyDATINR register is write-protected; 01: Channel y input data is sourced from the internal ADCy+1 output register update, representing a 16-bit value (if ADCy+1 exists). ADC output data is written to the INDAT0[15:0] portion of the R32_DFSDM_CHyDATINR register; 10: Channel y input data is sourced from the internal R32_DFSDM_CHyDATINR register (achieved via direct CPU/DMA write operations). One or two 16-bit data samples may be written, depending on the DATPACK[1:0] bitfield configuration; 11: Reserved. This value can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register). <i>Note: DATMPX[1:0]=1 applies only to ADC1 and ADC2.</i>	0
[11:9]	Reserved	RO	Reserved	0

8	CHINSEL	RW	Channel input selection: 1: Channel input is taken from the pin of the next channel (channel (y+1) modulo 2); 0: Channel input is taken from the pin of the same channel y. This value can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register).	0
7	CHEN	RW	Channel y Enable: 1: Enable channel y; 0: Disable channel y. If channel y is enabled, serial data reception will commence based on the specified channel settings.	0
6	CKABEN	RW	Channel y clock loss detector enable: 1: Enable the channel Y clock loss detector; 0: Disable the channel Y clock loss detector.	0
5	SCDEN	RW	Channel y short-circuit detector enable: 1: Input channel y will remain protected by the short-circuit detector; 0: Input channel y will not be protected by the short-circuit detector.	0
4	Reserved	RO	Reserved	0
[3:2]	SPICKSEL[1:0]	RW	Channel y SPI Clock Selection: 00: Clock sourced from external CKINy input—sampling point based on SITP; 01: Clock sourced from internal CKOUT output—sampling point based on SITP; 10: Clock sourced from internal CKOUT—sampling point occurring on every second falling edge of CKOUT. For connecting an external $\Sigma\Delta$ modulator, which divides the clock input (from CKOUT) by 2 to generate an output serial communication clock (this output clock change is valid on each clock input rising edge). 11: Clock sourced from internal CKOUT output—at the sampling point of every second rising edge of CKOUT. For connecting an external $\Sigma\Delta$ modulator, which divides the clock input (from CKOUT) by 2 to generate the output serial communication clock (this output clock change is active on each falling edge of the clock input). This value can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register).	0

1	Reserved	RO	Reserved	0
0	SITP	RW	Channel y serial interface type: 1: SPI data selected on falling edge; 0: SPI data selected on rising edge. This value can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register).	0

42.6.2 DFSDM Channel y Configuration Register 2 (R32_DFSDM_CHyCFGR2) (y=0/1)

Offset address: $0x08 + 0x04 * y$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OFFSET[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OFFSET[7:0]								DTRBS[4:0]				Reserved			

Bit	Name	Access	Description	Reset value
[31:8]	OFFSET[23:0]	RW	Channel y 24-bit calibration offset: An OFFSET is applied to each conversion result for Channel Y. This value is set by software.	0
[7:3]	DTRBS[4:0]	RW	Channel y data right shift: Valid range 00000-01000 defines the shift amount for integrator output data—specifies how many bits to shift right to obtain the final result. The shift is executed before offset correction. Data shifting rounds the result to the nearest integer value. The sign of the shifted result is preserved (to yield valid 24-bit signed format result data). This value can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register). Other: Invalid.	0
[2:0]	Reserved	RO	Reserved	0

42.6.3 DFSDM Channel y Analog Watchdog and Short Circuit Detector Register (R32_DFSDM_CHyAWSCDR) (y=0/1)

Offset address: $0x10 + 0x04 * y$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								AWFORD [1:0]		Reserved	AWFOSR[4:0]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		BKSCD[1:0]		Reserved				SCDT[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:22]	AWFORD[1:0]	RW	Channel y analog watchdog Sinc filter order: 00: FastSinc filter type; 01: Sinc1 filter type; 10: Sinc2 filter type; 11: Sinc3 filter type. This bit can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register).	0
21	Reserved	RO	Reserved	0
[20:16]	AWFOSR[4:0]	RW	Channel y analog watchdog filter oversampling rate: 0 to 31: Defines the length of the Sinc-type filter, with a valid range of 1 to 32 (AWFOSR+1). This value also represents the decimation factor for the analog data rate. This bit can only be modified when CHEN=0 (in the R32_DFSDM_CHyCFGR1 register). <i>Note: If AWFOSR=0, the filter has no effect (filter bypassed).</i>	0
[15:14]	Reserved	RO	Reserved	0
[13:12]	BKSCD[1:0]	RW	Channel y short circuit detector open circuit signal assignment: 00: Open circuit signal not assigned to channel short circuit detector; 01: Open circuit 0 signal assigned to channel 0 short circuit detector; 10: Open circuit 1 signal assigned to channel 1 short circuit detector; 11: Open circuit signals assigned to corresponding channel short circuit detectors.	0
[11:8]	Reserved	RO	Reserved	0
[7:0]	SCDT[7:0]	RW	Short-circuit detector threshold for channel y: These bits can be written by software to define the counter threshold for the short-circuit detector. If this value is reached, a short-circuit detector event occurs on the given channel.	0

42.6.4 DFSDM Channel y Watchdog Filter Data Register (R32_DFSDM_CHyWDATR) (y=0/1)

Offset address: $0x18 + 0x04 * y$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDATA[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:0]	WDATA[15:0]	RO	Input channel y watchdog data: Data converted by the analog watchdog filter on input channel y. This channel performs continuous data conversion (untriggered), with conversion resolution subject to certain limitations (OSR=1...32/sinc order=1...3).	0

42.6.5 DFSDM Channel y Data Input Register (R32_DFSDM_CHyDATINR) (y=0/1)

Offset address: $0x20 + 0x04 * y$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INDAT1[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INDAT0[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	INDAT1[15:0]	RW	Input data for channel y or channel y+1: When DATMPX[1:0]=1 or DATMPX[1:0]=2, the input parallel channel data will be processed through a digital filter. Data may be written by CPU/DMA (when DATMPX[1:0]=2) or directly by the internal ADC (when DATMPX[1:0]=1). When DATPACK[1:0]=0 (Standard Mode), INDAT0[15:0] is write-protected (not used for input sampling). When DATPACK[1:0]=1 (interleaved mode), the second channel y data sample is stored in INDAT1[15:0]. The first channel y data sample is stored in INDAT0[15:0]. The DFSDM_FLTx filter sequentially reads these 2 samples as the 2 channel y data samples. When DATPACK[1:0]=2 (dual mode), For even y-channels: Samples in INDAT1[15:0] are automatically copied to INDAT0[15:0] of channel (y+1). For odd y-channels: INDAT1[15:0] is write-protected. INDAT0[15:1] uses a 16-bit signed format.	0
[15:0]	INDAT0[15:0]	RW	Channel y input data: When DATMPX[1:0]=1 or DATMPX[1:0]=2, the input parallel channel data will be processed through a digital filter.	0

			Data may be written by CPU/DMA (when DATMPX[1:0]=2) or directly by the internal ADC (when DATMPX[1:0]=1). When DATPACK[1:0]=0 (Standard Mode), channel y data is sampled and stored in INDAT0[15:0]. When DATPACK[1:0]=1 (interleaved mode), the first channel y data sample is stored in INDAT0[15:0]; the second channel y data sample is stored in INDAT1[15:0]. The DFSDM_FLTx filter sequentially reads these two samples as the two channel y data samples. When DATPACK[1:0]=2 (dual mode), For even y-channels: The y-channel data sample is stored in INDAT0[15:0]. For odd y-channels: INDAT0[15:0] is write-protected. INDAT0[15:0] uses a 16-bit signed format.	
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42.6.6 DFSDM Filter x Control Register 1 (R32_DFSDM_FLTxCR1) (x=0/1)

Offset address: 0x28 + 0x04 * x

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	AWFSEL	FAST	Reserved				RCH	Reserved		RDMAEN	Reserved	RSYNC	RCON T	RSWSTART	Reserved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	JEXTEN [1:0]		Reserved	JEXTSEL[3:0]			Reserved			JDMAEN	JSCAN	JSYNC	Reserved	JSWSTART	DFEN

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
30	AWFSEL	RW	Analog watchdog fast mode selection: 1: Analog watchdog for channel transceiver values (after the watchdog filter); 0: Analog watchdog for data output values (after the digital filter). Comparison occurs after offset correction and shifting.	0
29	FAST	RW	Fast conversion mode selection for regular conversions: 1: Enable fast conversion mode; 0: Disable fast conversion mode. During regular conversions in continuous mode, enabling fast mode accelerates each conversion (except the first) compared to standard mode. This bit	0

			has no effect on non-consecutive conversions. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1).	
[28:25]	Reserved	RO	Reserved	0
24	RCH	RW	Regular channel selection: 1: Select channel 1 as the regular channel; 0: Select channel 0 as the regular channel. Write these bits when RCIP=1 for effect at the start of the next regular conversion. This is particularly useful in continuous mode (when RCONT=1). It also affects pending regular conversions (paused due to an ongoing injection conversion).	0
[23:22]	Reserved	RO	Reserved	0
21	RDMAEN	RW	Enable DMA channel to read regular conversion data: 1: DMA channel enabled to read regular data; 0: DMA channel disabled to read regular data. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1).	0
20	Reserved	RO	Reserved	0
19	RSYNC	RW	Use DFSDM_FLT0 to synchronously initiate regular conversion: 1: Initiate regular conversion in DFSDM_FLT0 while simultaneously initiating regular conversion in this DFSDM_FLTx; 0: Do not use DFSDM_FLT0 to synchronously initiate regular conversion. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1).	0
18	RCONT	RW	Continuous mode selection for regular conversion: 1: Repeats conversion of the regular channel after each conversion request; 0: Converts the regular channel only once per conversion request. During continuous regular conversion, writing a 0 to this bit immediately terminates continuous mode.	0
17	RSWSTART	RW1T	Software initiation of regular channel conversion: 1: Writing a 1 issues a request to initiate regular channel conversion and sets RCIP to 1. If RCIP is already set to 1, writing to RSWSTART has no effect. If RSYNC is 1, writing a 1 has no effect. 0: Writing a 0 has no effect. This bit is always read as 0.	0
[16:15]	Reserved	RO	Reserved	0
[14:13]	JEXTEN[1:0]	RW	Injection conversion trigger enable and trigger edge	0

			selection: 00: Disable trigger detection; 01: Each rising edge of the selected trigger generates a request to initiate injection conversion; 10: Each falling edge of the selected trigger generates a request to initiate injection conversion; 11: Both rising and falling edges of the selected trigger generate a request to initiate injection conversion. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1).																			
12	Reserved	RO	Reserved	0																		
[11:8]	JEXTSEL[3:0]	RW	Trigger signal selection for initiating injection conversion: 0x0 to 0xF: Trigger input selected according to the table below (internal or external trigger). This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1). <i>Note: Synchronous triggering has a maximum delay of one fDFSDMCLK clock cycle (with defined jitter). Asynchronous triggering has a delay of 2–3 fDFSDMCLK clock cycles (with jitter up to one cycle).</i> <table><tr><th></th><th>DFSDM_FL T0</th><th>DFSDM_FL T1</th></tr><tr><td>0x00</td><td>dfsdm_ jtrg0</td><td>dfsdm_ jtrg0</td></tr><tr><td>0x01</td><td>dfsdm_ jtrg1</td><td>dfsdm_ jtrg1</td></tr><tr><td>...</td><td></td><td></td></tr><tr><td>0xE</td><td>dfsdm_ jtrg14</td><td>dfsdm_ jtrg14</td></tr><tr><td>0xF</td><td>dfsdm_ jtrg15</td><td>dfsdm_ jtrg15</td></tr></table>		DFSDM_FL T0	DFSDM_FL T1	0x00	dfsdm_ jtrg0	dfsdm_ jtrg0	0x01	dfsdm_ jtrg1	dfsdm_ jtrg1	...			0xE	dfsdm_ jtrg14	dfsdm_ jtrg14	0xF	dfsdm_ jtrg15	dfsdm_ jtrg15	0
	DFSDM_FL T0	DFSDM_FL T1																				
0x00	dfsdm_ jtrg0	dfsdm_ jtrg0																				
0x01	dfsdm_ jtrg1	dfsdm_ jtrg1																				
...																						
0xE	dfsdm_ jtrg14	dfsdm_ jtrg14																				
0xF	dfsdm_ jtrg15	dfsdm_ jtrg15																				
[7:6]	Reserved	RO	Reserved	0																		
5	JDMAEN	RW	Enable the DMA channel to read data from the injection channel group: 1: DMA channel enabled to read injection data; 0: DMA channel disabled to read injection data. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1).	0																		
4	JSCAN	RW	Injection conversion scan conversion mode:	0																		

			1: Perform sequential conversion on the injection channel group starting from the lowest selected channel; 0: Performs a single channel conversion on the injected channel group, then selects the next channel within that group. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1). If JSCAN=0, writing to JCHG resets the channel selection to the minimum selected channel.	
3	JSYNC	RW	Using DFSDM_FLT0 JSWSTART to trigger synchronous injection conversion startup: 1: When triggering injection conversion startup via JSWSTART in DFSDM_FLT0, simultaneously initiate injection conversion in that DFSDM_FLTx; 0: Do not use DFSDM_FLT0 to synchronously initiate injection conversion. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1).	0
2	Reserved	RO	Reserved	0
1	JSWSTART	RW1T	Initiate injection channel group conversion: 1: Writing 1 issues a request to convert the injection channel group and simultaneously sets JCIP to 1. If JCIP=1 is already set, the write operation to JSWSTART has no effect. If JSYNC=1, writing 1 has no effect; 0: Writing 0 has no effect. This bit always reads as 0.	0
0	DFEN	RW	DFSDM_FLTx enable: 1: Enables DFSDM_FLTx. If enabled, DFSDM_FLTx begins operation according to its configuration; 0: Disable DFSDM_FLTx. All conversions assigned to DFSDM_FLTx stop immediately, and all DFSDM_FLTx functions cease. Data cleared by setting DFEN=0: - Register R32_DFSDM_FLTxISR is reset; - Register R32_DFSDM_FLTxAWSR is reset.	0

42.6.7 DFSDM Filter x Control Register 2 (R32_DFSDM_FLTxCR2) (x=0/1)

Offset address: 0x30 + 0x04 * x

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														AWDCH[1:0]	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						EXCH[1:0]	Reserved	CKABIE	SCDIE	AWDIE	ROVRIE	JOVRIE	REOCIE	JEOCIE	

Bit	Name	Access	Description	Reset value
[31:18]	Reserved	RO	Reserved	0
[17:16]	AWDCH[1:0]	RW	Analog watchdog channel selection: These bits select the input channels continuously monitored by the analog watchdog: 00: Disables the analog watchdog on channels 0 and 1; 01: Enables the analog watchdog on channel 0; 10: Enables the analog watchdog on channel 1; 11: Enables the analog watchdog on channels 0 and 1.	0
[15:10]	Reserved	RO	Reserved	0
[9:8]	EXCH[1:0]	RW	Extreme value detector channel selection: These bits select which input channel the extreme value detector will use. 00: The extreme value detector does not accept data from Channel 0 or Channel 1; 01: The extreme value detector accepts data from Channel 0; 10: The extreme value detector accepts data from Channel 1; 11: The extreme value detector accepts data from both Channel 0 and Channel 1.	0
7	Reserved	RO	Reserved	0
6	CKABIE	RW	Clock miss interrupt enable: 1: Enable detection of clock misses on the input channel; 0: Disable detection of clock misses on the input channel. Refer to the CKABF[1:0] description in R32_DFSDM_FLTISR. Note: CKABIE exists only in the R32_DFSDM_FLT0CR2 register (filter x=0).	0
5	SCDIE	RW	Short-circuit detector interrupt enable: 1: Enable short-circuit detector interrupt; 0: Disable short-circuit detector interrupt. Refer to the SCDF[1:0] description in R32_DFSDM_FLTISR. Note: SCDIE exists only in the R32_DFSDM_FLT0CR2 register (Filter x=0).	0

4	AWDIE	RW	Enable analog watchdog interrupt: 1: Enable analog watchdog interrupt; 0: Disable analog watchdog interrupt. Refer to the AWDF description in R32_DFSDM_FLTISR.	0
3	ROVRIE	RW	Enable regular data overflow interrupt: 1: Enable regular data overflow interrupt; 0: Disable regular data overflow interrupt. Refer to the ROVRF description in R32_DFSDM_FLTISR.	0
2	JOVRIE	RW	Enable data overflow interrupt: 1: Enable data overflow interrupt; 0: Disable data overflow interrupt. Refer to the JOVRF description in R32_DFSDM_FLTISR.	0
1	REOCIE	RW	Enable/Disable regular conversion end interrupt: 1: Enable regular conversion end interrupt; 0: Disable regular conversion end interrupt. Refer to the REOCF description in R32_DFSDM_FLTISR.	0
0	JEOCIE	RW	Injection conversion end interrupt enable: 1: Enable injection conversion end interrupt; 0: Disable injection conversion end interrupt. Refer to the JEOCF description in R32_DFSDM_FLTISR.	0

42.6.8 DFSDM Filter x Interrupt and Status Register (R32_DFSDM_FLTxISR) (x=0/1)

Offset address: $0x38 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						SCDF[1:0]		Reserved						CKABF[1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser ved	RCIP	JCIP	Reserved								AWD F	ROV RF	JOV RF	REOC F	JEOC F

Bit	Name	Access	Description	Reset value
[31:26]	Reserved	RO	Reserved	0
[25:24]	SCDF[1:0]	RO	Short circuit detector flags: 00: No short circuit detector events occurred on either Channel 0 or Channel 1; 01: The short circuit detector counter on Channel 0 reached the value programmed in the R32_DFSDM_CHyAWSCDR register; 10: The short circuit detector counter on Channel 1	0

			reached the value programmed in the R32_DFSDM_CHyAWSCDR register; 11: The short-circuit detector counters on both Channel 0 and Channel 1 have reached the value programmed in the R32_DFSDM_CHyAWSCDR register; This bit is set to 1 by hardware and can be cleared by software using the CLRSCDF bit in the R32_DFSDM_FLTxICR register. When CHEN[y]=0 (given channel is disabled), SCDF can also be cleared by hardware. <i>Note: SCDF[1:0] exists only in the R32_DFSDM_FLT0ISR register (filter x=0).</i>	
[23:18]	Reserved	RO	Reserved	0
[17:16]	CKABF[1:0]	RW	Clock absence indicator: 00: Clock signal present on both Channel 0 and Channel 1; 01: No clock signal present on Channel 0; 10: No clock signal present on Channel 1; 11: No clock signal present on either Channel 0 or Channel 1. When no clock signal is detected on a channel, this bit is set to 1 by hardware. When CHEN=0 (see R32_DFSDM_CHyCFGR1 register), it is held in the clock missing state by hardware. When the transceiver is not yet synchronized, it is held in the clock absent state by hardware. It can be cleared by software using the corresponding CLRCKABF bit in the R32_DFSDM_FLTxICR register. <i>Note: CKABF[1:0] exists only in the R32_DFSDM_FLT0ISR register (filter x=0).</i>	0
15	Reserved	RO	Reserved	0
14	RCIP	RW	Regular conversion in progress: 1: Regular channel conversion is in progress, or a regular conversion request is pending processing; 0: No request to convert the regular channel has been issued. When RCIP=1, ignore requests to initiate regular conversion.	0
13	JCIP	RW	Injection conversion is in progress: 1: The injection channel group conversion is in progress (1 written to JSWSTART) or an injection conversion request is pending processing (due to trigger detection);	0

			0: No request to inject the conversion into the channel group has been issued (neither software nor trigger has issued a request). When JCIP=1, ignore requests to initiate injection conversion.	
[12:5]	Reserved	RO	Reserved	0
4	AWDF	RW	Analog Watchdog: 1: The analog watchdog module detects a voltage exceeding the value programmed in the R32_DFSDM_FLTxAWLTR or R32_DFSDM_FLTxAWHTR register; 0: No analog watchdog event occurred. This bit is set to 1 by hardware and cleared by software as follows: Clear all source flag bits AWHTF[1:0] and AWLTF[1:0] in the R32_DFSDM_FLTxAWSR register (achieved by writing a 1 to the clear bit in the R32_DFSDM_FLTxAWCFR register).	0
3	ROVRF	RW	Regular conversion overflow flag: 1: A regular conversion overflow has occurred, meaning the regular conversion has completed but REOCF remains set to 1. RDATAR is unaffected by the overflow; 0: No regular conversion overflow has occurred. This bit is set by hardware and can be cleared by software using the CLRROVRF bit in the R32_DFSDM_FLTxICR register.	0
2	JOVRF	RW	Injection conversion overflow flag: 1: An injection conversion overflow has occurred, meaning the injection conversion has completed but JEOCF remains set to 1. JDATAR is unaffected by the overflow; 0: No injection conversion overflow has occurred. This bit is set to 1 by hardware and can be cleared by software using the CLRJOVRF bit in the R32_DFSDM_FLTxICR register.	0
1	REOCF	RW	Regular conversion end flag: 1: Regular conversion completed and data is ready for reading; 0: Regular conversion not completed. This bit is set to 1 by hardware. When software or DMA reads R32_DFSDM_FLTxRDATAR, this bit is cleared.	0
0	JEOCF	RW	Injection conversion completion flag:	0

			1: Injection conversion completed and data is ready for read; 0: Injection conversion incomplete. This bit is set to 1 by hardware. When software or DMA reads R32_DFSDM_FLTxDATAR, this bit is cleared.	
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Note: For each flag bit, the interrupt can be enabled by setting the corresponding bit in R32_DFSDM_FLTxCR2. If an interrupt is called, the flag must be cleared before exiting the interrupt service program. When DFEN=0, all bits of R32_DFSDM_FLTxISR are automatically reset.

42.6.9 DFSDM Filter x Interrupt Flag Clear Register (R32_DFSDM_FLTxICR) (x=0/1)

Offset address: $0x40 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						CLRSCDF [1:0]		Reserved						CLRCKABF [1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CLRR OVRF	CLRJ OVRF	Reserved	

Bit	Name	Access	Description	Reset value
[31:26]	Reserved	RO	Reserved	0
[25:24]	CLRSCDF[1:0]	RW	Reset short circuit detector flags: 00: Writing 0 has no effect; 01: Clear short-circuit detector flag on channel 0; 10: Clear short-circuit detector flag on channel 1; 11: Clear short-circuit detector flags on both channels 0 and 1. <i>Note: CLRSCDF[1:0] exists only in the R32_DFSDM_FLT0ICR register (filter x=0).</i>	0
[23:18]	Reserved	RO	Reserved	0
[17:16]	CLRCKABF[1:0]	RW	Clear clock absent flag: 00: Writing 0 has no effect; 01: Clears the clock absent flag on channel 0; 10: Clears the clock absent flag on channel 1; 11: Clears the clock absent flags on both channels 0 and 1. When the transceiver is not yet synchronized, the clock absent flag is set to 1 and cannot be cleared by CLRCKABF. <i>Note: CLRCKABF[1:0] exists only in the R32_DFSDM_FLT0ICR register (filter x=0).</i>	0
[15:4]	Reserved	RO	Reserved	0
3	CLRRORF	RW	Clear the conventional conversion overflow flag:	0

			1: Writing 1 clears the ROVRF bit in the R32_DFSDM_FLTxISR register; 0: Writing 0 has no effect.	
2	CLRJOVRF	RW	Clear the injection conversion overflow flag: 1: Writing 1 clears the JOVRF bit in the R32_DFSDM_FLTxISR register; 0: Writing 0 has no effect.	0
[1:0]	Reserved	RO	Reserved	0

Note: The bit of R32_DFSDM_FLTxICR is always read as 0.

42.6.10 DFSDM Filter x Injection Channel Group Selection Register (R32_DFSDM_FLTxJCHGR) (x=0/1)

Offset address: $0x48 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														JCHG[1:0]	

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved	0
[1:0]	JCHG[1:0]	RW	Injection channel group selection: 00: Neither Channel 0 nor Channel 1 belongs to the injection group; 01: Channel 0 belongs to the injection group; 10: Channel 1 belongs to the injection group; 11: Both Channel 0 and Channel 1 belong to the injection group. If JSCAN=1, each selected channel is converted sequentially. The smallest channel (Channel 0, if selected) is converted first, and the largest selected channel is converted last. If JSCAN=0, only one selected channel is converted, then the channel selection moves to the next channel. If JSCAN=0, writing to JCHG resets the channel selection to the smallest selected channel. At least one channel must always be selected for the injection group. If a write operation sets all JCHG bits to zero, the write operation is ignored.	01b

42.6.11 DFSDM Filter x Control Register 3 (R32_DFSDM_FLTxFCR3) (x=0/1)

Offset address: $0x50 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

FORD[2:0]			Reserved			FOSR[9:0]									
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												IOSR[3:0]			

Bit	Name	Access	Description	Reset value
[31:29]	FORD[2:0]	RO	Sinc filter order: 000: FastSinc filter type; 001: Sinc ¹ filter type; 010: Sinc ² filter type; 011: Sinc ³ filter type; 100: Sinc ⁴ filter type; 101: Sinc ⁵ filter type; 110, 111: Reserved. This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1).	0
[28:26]	Reserved	RO	Reserved	0
[25:16]	FOSR[9:0]	RW	Sinc filter oversampling rate: 0 to 1023: Defines the length of the Sinc-type filter, with a valid range of 1 to 1024 (FOSR = FOSR[9:0] + 1). This value also represents the decimation factor for the filter output data rate. This bit can only be modified when DFEN = 0 (R32_DFSDM_FLTxCR1). <i>Note: If FOSR=0, the filter has no effect (filter bypassed).</i>	0
[15:4]	Reserved	RO	Reserved	0
[3:0]	IOSR[3:0]	RW	The integrator oversampling rate is 2 ^{IOSR} . This defines how many sinc filter samples are combined to form the integrator output data. The integrator output data rate will be reduced by this value (the additional data extraction rate). This bit can only be modified when DFEN=0 (R32_DFSDM_FLTxCR1). <i>Note:</i> (1) If IOSR=0, the integrator has no effect (integrator bypassed). (2) In continuous non-fast mode, IOSR must be greater than 0.	0

42.6.12 DFSDM Filter x Injection Group Data Register 3 (R32_DFSDM_FLTxJDATAR) (x=0/1)

Offset address: 0x58 + 0x04 * x

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
JDATA[23:9]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JDATA[8:0]								Reserved							JDATA[0]

Bit	Name	Access	Description	Reset value
[31:8]	JDATA[23:0]	RO	Injection group conversion data: Data obtained after converting each channel within the injection group is stored in this field. Data is valid when JEOCF=1. Reading this register clears the corresponding JEOCF bit.	0
[7:1]	Reserved	RO	Reserved	0
0	JDATA[0]	RO	Recently converted injection channels: After converting each channel within the injection group, JDATA[0] is updated to indicate which channel has been converted. Consequently, the data retained in JDATA[23:0] corresponds to the channel specified by JDATA[0].	0

Note: DMA can be used to read data from this register. You can use halfword access to read only the most significant bit (MSB) of the converted data. Reading this register also clears JEOCF of R32_DFSDM_FLTISR. Therefore, if DMA is activated to read data from the register, the firmware cannot read the register.

42.6.13 DFSDM Filter x Regular Channel Data Register (R32_DFSDM_FLTxDATAR) (x=0/1)

Offset address: 0x60 + 0x04 * x

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDAT[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDAT[7:0]								Reserved		RPEND	Reserved			RDAT[0]	

Bit	Name	Access	Description	Reset value
[31:8]	RDAT[23:0]	RO	Regular conversion data: Upon completion of each regular conversion, the corresponding data is stored in this register. Data is valid when REOCF=1. Reading this register clears the corresponding REOCF bit.	0
[7:5]	Reserved	RO	Reserved	0
4	RPEND	RO	Regular channel pending data: Regular data in RDAT[23:0] is delayed due to an injection channel trigger occurring during conversion.	0
[3:1]	Reserved	RO	Reserved	0

0	RDATACH	RO	Recently converted regular channels: Each time a regular conversion completes, RDATACH is updated to indicate which channel was converted (since during regular conversion, the regular channel selection RCH in the R32_DFSDM_FLTxCr1 register can be updated). Therefore, the data retained in RDATA[23:0] corresponds to the channel indicated by RDATACH.	0
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Note: Half-word access can be used to read only the most significant bit (MSB) of the converted data. Reading this register will also clear REOCF of R32_DFSDM_FLTISR.

42.6.14 DFSDM Filter x Analog Watchdog Higher Threshold Register (R32_DFSDM_FLTxAWHTR) (x=0/1)

Offset address: $0x68 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AWHT[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AWHT[7:0]								Reserved					BKAHW[1:0]		

Bit	Name	Access	Description	Reset value
[31:8]	AWHT[23:0]	RW	Higher threshold of the analog watchdog: These bits are written by software to define the higher threshold of the analog watchdog.	0
[7:2]	Reserved	RO	Reserved	0
[1:0]	BKAHW[1:0]	RW	Assignment of circuit breaker signals for analog watchdog higher threshold events: 00: Circuit breaker signal is not assigned to the analog watchdog higher threshold event; 01: Circuit breaker 0 signal is assigned to the analog watchdog higher threshold event; 10: Circuit breaker 1 signal is assigned to the analog watchdog higher threshold event; 11: Both circuit breaker signals are assigned to the analog watchdog higher threshold event.	0

42.6.15 DFSDM Filter x Analog Watchdog Lower Threshold Register (R32_DFSDM_FLTxAWLTR) (x=0/1)

Offset address: $0x70 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AWLT[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

AWLT[7:0]	Reserved	BKAWL[1:0]
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Bit	Name	Access	Description	Reset value
[31:8]	AWLT[23:0]	RW	Lower threshold of the analog watchdog: These bits are written by software to define the lower threshold of the analog watchdog.	0
[7:2]	Reserved	RO	Reserved	0
[1:0]	BKAWL[1:0]	RW	Assignment of circuit breaker signals for analog watchdog lower threshold events: 00: Circuit breaker signal is not assigned to the analog watchdog lower threshold event; 01: Circuit breaker 0 signal is assigned to the analog watchdog lower threshold event; 10: Circuit breaker 1 signal is assigned to the analog watchdog lower threshold event; 11: Both circuit breaker signals are assigned to the analog watchdog lower threshold event.	0

42.6.16 DFSDM Filter x Analog Watchdog Status Register (R32_DFSDM_FLTxAWSR) (x=0/1)

Offset address: $0x78 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						AWHTF[1:0]		Reserved						AWLTF[1:0]	

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
[9:8]	AWHTF[1:0]	RO	Analog watchdog higher threshold indicator: 00: No higher threshold error on either Channel 0 or Channel 1; 01: Higher threshold error present on Channel 0; 10: Higher threshold error present on Channel 1; 11: Higher threshold errors present on both Channel 0 and Channel 1. This bit is set to 1 by hardware. It can be cleared by software using the corresponding CLRAWHTF bit in the R32_DFSDM_FLTxAWCFR register.	0
[7:2]	Reserved	RO	Reserved	0
[1:0]	AWLTF[1:0]	RO	Analog watchdog lower threshold indicator: 00: No lower threshold error on either Channel 0 or Channel 1; 01: Lower threshold error present on Channel 0;	0

			10: Lower threshold error present on Channel 1; 11: Lower threshold errors present on both Channel 0 and Channel 1. This bit is set to 1 by hardware. It can be cleared by software using the corresponding CLRAWHTF bit in the R32_DFSDM_FLTxAWCFR register.	
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42.6.17 DFSDM Filter x Analog Watchdog Clear Flag Register (R32_DFSDM_FLTxAWCFR)

(x=0/1)

Offset address: $0x80 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CLRAWHTF [1:0]		Reserved						CLRAWLTF [1:0]	

Bit	Name	Access	Description	Reset value
[31:10]	Reserved	RO	Reserved	0
[9:8]	CLRAWHTF[1:0]	RW1Z	Clear analog watchdog higher threshold flags: 00: Writing 0 has no effect; 01: Clears the higher threshold error flag for channel 0; 10: Clears the higher threshold error flag for channel 1; 11: Clears the higher threshold error flags for both channel 0 and channel 1.	0
[7:2]	Reserved	RO	Reserved	0
[1:0]	CLRAWLTF[1:0]	RW1Z	Clear analog watchdog lower threshold flags: 00: Writing 0 has no effect; 01: Clears the lower threshold error flag for channel 0; 10: Clears the lower threshold error flag for channel 1; 11: Clears the lower threshold error flags for both channel 0 and channel 1.	0

42.6.18 DFSDM Filter x Extreme Value Detector Maximum Register (R32_DFSDM_FLTxEVMAX) (x=0/1)

Offset address: $0x88 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVMAX[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVMAX[7:0]								Reserved						EX MA XC H	

Bit	Name	Access	Description	Reset value
[31:8]	EXMAX[23:0]	RZ	Maximum value detector: These bits are set to 1 by hardware to indicate the maximum value converted by DFSDM_FLTx. Reading this register resets the EXMAX[23:0] bits to the value (0x800000).	0
[7:1]	Reserved	RO	Reserved	0
0	EXMAXCH	RZ	Maximum value detector data channel: These bits indicate which channel's data is stored in EXMAX[23:0]. They can be cleared by reading this register.	0

42.6.19 DFSDM Filter x Extreme Value Detector Minimum Register (R32_DFSDM_FLTxEXMIN) (x=0/1)

Offset address: $0x90 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EXMIN[23:8]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXMIN[7:0]								Reserved						EXMINCH	

Bit	Name	Access	Description	Reset value
[31:8]	EXMIN[23:0]	RO	Minimum value detector: These bits are set to 1 by hardware to indicate the minimum value converted by DFSDM_FLTx. Reading this register resets the EXMIN[23:0] bits to the value (0x800000).	0
[7:1]	Reserved	RO	Reserved	0
0	EXMINCH	RO	Minimum value detector data channel: These bits indicate which channel's data is stored in EXMIN[23:0]. They can be cleared by reading this register.	0

42.6.20 DFSDM Filter x Conversion Timer Register (R32_DFSDM_FLTxCNVTIMR) (x=0/1)

Offset address: $0x98 + 0x04 * x$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				CNVCNT[27:16]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNVCNT[15:0]															

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:0]	CNVCNT[27:0]	RO	28-bit timer counts conversion time: $CNVCNT = (CNVCNT[27:0] + 1)$ $t = CNVCNT / f_{DFSDMCLK}$; The timer's input clock is derived from the DFSDM clock (system clock $f_{DFSDMCLK}$). The conversion time measurement begins at the start of each conversion and ends after every 36 conversions (i.e., the interval between the first and last serial samples). The conversion time measurement only stops when the filter is bypassed ($FOSR[9:0]=0$), and $CNVCNT[27:0]=0$. The timing is as follows: If $FAST=0$ (or the first conversion in $FAST=1$ continuous mode), then: $t = [FOSR * (2^{IOSR} + FORD + 1)] / f_{CKIN}$ (for Sincx filters); $t = [FOSR * (2^{IOSR} + 5)] / f_{CKIN}$ (for FastSinc filter); If $FAST=1$ in continuous mode (except for the first conversion), then: $t = [FOSR * 2^{IOSR}] / f_{CKIN}$; If $FOSR = FOSR[9:0] + 1 = 1$ (filter bypassed, only integrator active), then: $CNVCNT = 0$ (timing stop, conversion time: $t = 2 \cdot IOSR / f_{CKIN}$); where f_{CKIN} is the channel input clock frequency (at the $CKINy$ pin of the given channel); in the case of parallel data input (from an internal ADC or from a CPU/DMA write operation), it represents the input data rate. <i>Note: When a conversion is interrupted (e.g., by disabling/enabling the selected channel), the timer also includes this interruption time in its calculation.</i>	0

Chapter 43 LCD-TFT Display Controller (LTDC)

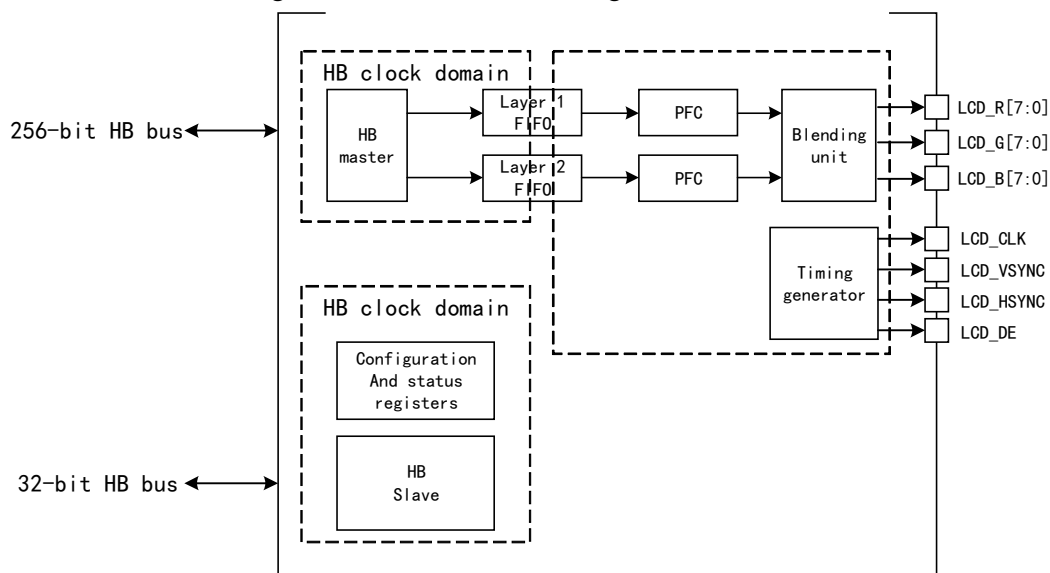
LCD-TFT (Thin Film Transistor-Liquid Crystal Display) Display Controller (LTDC) mainly provides parallel digital RGB, horizontal synchronization, vertical synchronization, pixel clock and data enable signals, which can be used as output signals to interface with different LCD and TFT panels.

43.1 Main Features

- Provide 2 display layers, each containing a proprietary 8×256-bit FIFO
- Support color lookup tables (CLUT) with up to 256 colors per layer
- Support programmable timing for different display panels
- Support programmable background color
- Support programmable polarity for HSYNC, VSYNC, and data enable signals
- Each display layer supports up to 8 input colors
- Flexible blending between 2 layers using alpha values (per-pixel or constant)
- Support chroma key (transparent color)
- Support programmable window position and size
- Support thin-film transistor (TFT) color displays
- Support up to 3 programmable interrupt timers

43.2 Overview

Figure 43-1 Structural block diagram of LTDC



- (1) PFC: pixel format converter, performing the pixel format conversion from the selected input pixel format of a layer to words.
 (2) ltcd_ker_ck: pixel clock.

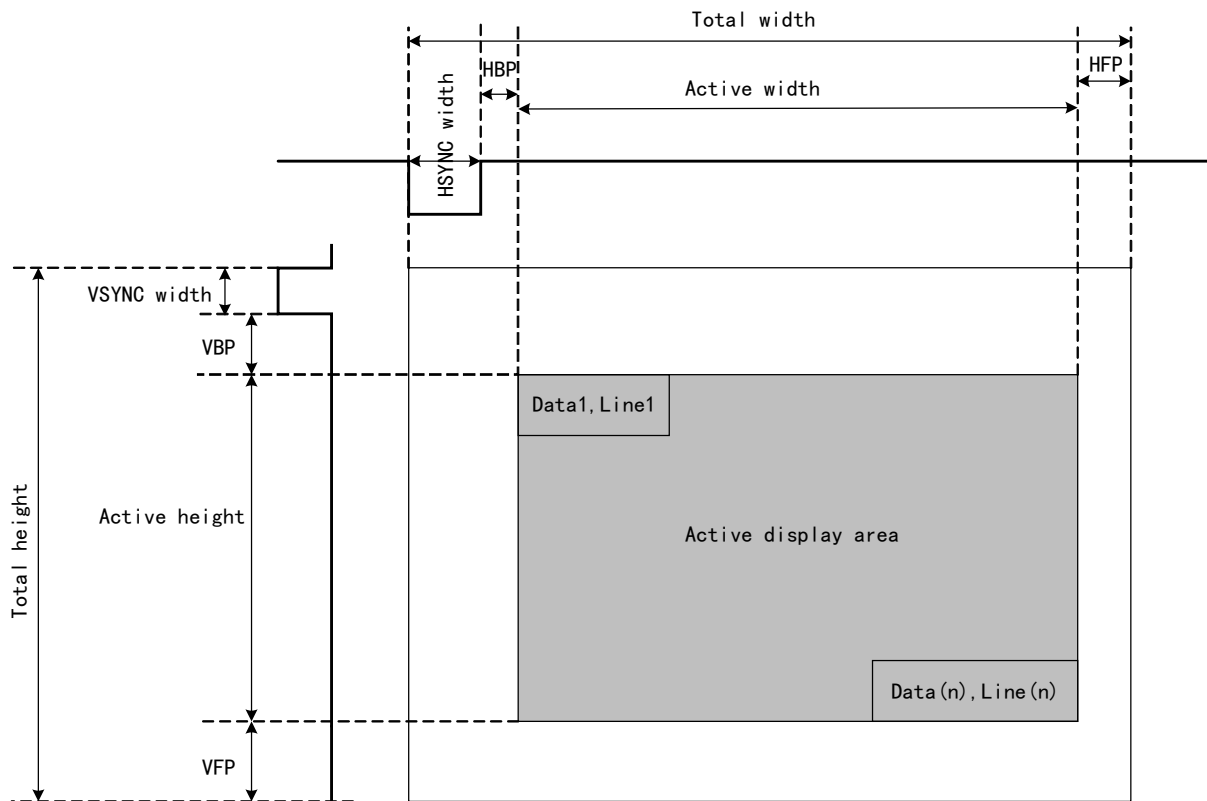
43.3 Function Description

43.3.1 LTDC Synchronization Timing

The LTDC interface is a synchronous data interface comprising pixel clock, pixel data, and horizontal and vertical

sync signals. Figure 43-2 illustrates the configurable timing parameters generated by the LTDC synchronous timing generator module.

Figure 43-2 LTDC synchronization timing



Note: (1) HSYNC and VSYNC denote horizontal and vertical sync widths, respectively; HBP and HFP denote horizontal back-edge period and horizontal front-edge period, respectively; VBP and VFP denote vertical back-edge period and vertical front-edge period, respectively.

(2) When LTDC is enabled, the generated timing uses the $X/Y=0/0$ position as the first horizontal sync pixel within the vertical sync region, followed by the back edge, active data display region, and front edge.

(3) When LTDC is disabled, the timing generator module resets to $X=\text{total width}-1$ and $Y=\text{total height}-1$, holding the previous pixel during the vertical sync phase and before FIFO refresh. Consequently, only blanking data is continuously output.

An example of synchronization timing configuration is as follows:

- 1) LCD-TFT timing (to be extracted from the panel datasheet);
- 2) Horizontal and vertical sync width: 0xA pixels, 0x2-1 lines;
- 3) Horizontal and vertical trailing edge: 0x14 pixels, 0x2 lines;
- 4) Effective width and effective height: 0x140 pixels, 0xF0 lines (320x240);
- 5) Horizontal leading edge: 0xA pixels;
- 6) Vertical leading edge: 0x4 lines;

The value programmed in the LTDC timing register will be:

- 1) R32_LTDC_SSCR register: Programmed to 0x00090001. (HSW[11:0] is 0xA-1 and VSH[10:0] is 0x2-1);
- 2) R32_LTDC_BPCR register: Programmed to 0x001D0003. (AHBP[11:0] is 0x1D (0xA + 0x14-1), AVBP[10:0] is 0x3 (0x2 + 0x2-1));

- 3) R32_LTDC_AWCR register: Programmed to 0x015D00F3; (AAW[11:0] is 0x15D (0xA + 0x14 + 0x140-1), AAH[10:0] is 0xF3 (0x2 + 0x2 + 0xF0-1));
- 4) R32_LTDC_TWCR register bit[27:16]: Programmed to 0x167. (TOTALW[11:0] is 0x167 (0xA + 0x14 + 0x140 + 0xA-1));
- 5) R32_LTDC_TWCR register bit[10:0]: Programmed to 0x0F7. (TOTALH[10:0] is 0xF7 (0x2 + 0x2 + 0xF0 + 0x4-1)).

43.3.2 Pixel Format

Programmable pixel format is used for data stored in the frame buffer of the layer. Up to eight input pixel format can be configured for each layer through the R32_LTDC_LxPFCR register.

The ARGB8888 format requires 8 bits of data per channel (Alpha, Red, Green and Blue). However, some channels in ARGB1555 and ARGB4444 formats are less than 8 bits. The pixel processing unit converts it into ARGB8888 by copying the high bits and filling them into the low bits. When processing RGB888 and RGB565 formats, the pixel processing unit assumes Alpha=255, and if the number of bits of the channel is less than 8, the copy high bits are also padded to the low bits.

43.3.3 Color Look-up Table (CLUT)

CLUT can be enabled for each layer at runtime through the CLUTEN bit in the R32_LTDC_LxCr register. CLUT is only suitable for index colors when using L8, AL44 and AL88 input pixel format.

First, CLUT must load R, G and B values to replace the original R, G and B values of the corresponding pixels (index colors). Each color (RGB value) has its own corresponding address in CLUT. The values of r, g and b and their respective addresses are programmed by R32_LTDC_LxCLUTWR register.

When using L8 and AL88 to input pixel format, CLUT must be loaded with 256 colors. The address of each color is configured in the CLUTADD bit of R32_LTDC_LxCLUTWR register.

When using AL44 to input pixel format, CLUT must only load 16 colors. The address of each color must be filled by repeating the 4-bit L channel into 8 bits, as follows:

- L0 (Index Color 0), at address 0x00;
- L1, at address 0x11;
- L2, at address 0x22;
-;
- L15, at address 0xFF.

43.3.4 Color Frame Buffer

44.3.4.1 Color Frame Buffer Address

The color frame buffer of each layer has a start address, which is configured by R32_LTDC_LxCFBAR register. When a layer is enabled, the data will be obtained from the color frame buffer.

44.3.4.2 Color Frame Buffer Length

Each layer is set with the total length (in bytes) and the number of lines of the color frame buffer, which can be configured by the R32_LTDC_LxCFBLR and R32_LTDC_LxCFBLNR registers respectively.

The length and number of rows are set to prevent data beyond the end of the frame buffer from being prefetched into the FIFO corresponding to the layer.

If it is set below the required bytes, a FIFO underflow interrupt will be generated (if previously enabled).

If it is set to more bytes than actually needed, useless data read from FIFO will be discarded and will not be displayed.

44.3.4.3 Color Frame Buffer Spacing

The color frame buffer of each layer has a configurable spacing, which is the distance (in bytes) from the beginning of one line to the beginning of the next line. It is configured through the R32_LTDC_LxCFBLR register.

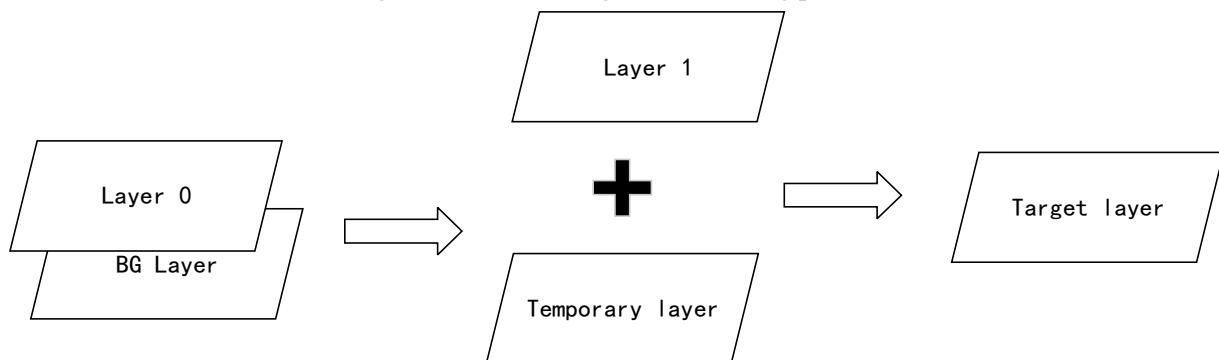
43.3.5 Layer Window and Blending Function

Each layer supports windowing functionality as well as two-layer blending. LTDC first performs windowing operations on each layer, then blends the two layers into a single frame.

Window functions define a display window, with each layer having independent window position configuration registers: R32_LTDC_LxWHPCR and R32_LTDC_LxWVPCR. Window configuration defines the display window within a layer. Pixels within the window retain their original values, while pixels outside the window are replaced with the pixel value defined by the R32_LTDC_LxDCCR register.

The blending order is fixed, proceeding from bottom to top. The blending unit first blends layer 0 with the background layer to produce a temporary layer, then blends layer 1 with the temporary layer to generate the final layer. The ARGB values of the background layer are defined in the R32_LTDC_BCCR register. If a layer is disabled, the blending function uses that layer's default color. To prevent the default color from appearing when a layer is disabled, set the blending coefficient for that layer in the R32_LTDC_LxBFCR register to its reset value. The blending process is illustrated in Figure 43-3.

Figure 43-3 Block diagram of blending process



General blending formula: $BC = BF1 * C + BF2 * CS$;

BC represents the blended color; BF1 is blending factor 1; C is the current layer's color; BF2 is blending factor 2; CS is the color of the next layer to be blended.

The blending factors for the current pixel have two possible values, determined by register configuration. One is the pixel's alpha multiplied by a constant Alpha, and the other is a constant Alpha.

Note: The constant Alpha value is programmed into the register and implemented by hardware as 255 equal divisions.

Example: To enable only the first layer, configure BF1 to the constant Alpha and BF2 to 1 minus the constant Alpha.

Constant Alpha: The constant Alpha programmed into the LxCACR register is 240 (0xF0).

Therefore, the constant Alpha value is $240/255 = 0.94$

C: The red channel of the current layer is 0x80.

Cs: The red channel of the background is 0x30.

The first layer blends with the background color.

*Final red channel result = Constant Alpha * C + (1 - Constant Alpha) * Cs*

*= $0.94 * 0x80 + (1 - 0.94) * 0x30 = 0x7B$.*

The green and blue channels follow the same principle.

43.3.6 Layer Window and Blending Function

Color keys (RGB) can be configured to represent transparent pixels. At runtime, the color key value can be configured and used to replace the pixel RGB value.

When the color key is enabled, the current pixel (the pixel after format conversion and before CLUT is mixed separately) will be compared with the color key. If the current pixel matches the programmed RGB value, all channels (ARGB) of the pixel are set to 0.

Color keys are configured through the LTDC_LxCKCR register. The programmed value depends on the pixel format, because it will be compared with the current pixel after the pixel format is converted to ARGB888.

For example, if medium yellow (50% red +50% green) is used as the transparent color key, then:

- In RGB565, the mid-yellow value is 0x8400. Set LTDC_LxCKCR to 0x848200.
- In ARGB8888, the mid-yellow value is 0x808000. Set LTDC_LxCKCR to 0x808000.
- In all CLUT-based color modes (L8, AL88, AL44), set one of the palette entries to mid-yellow 0x808000 and set LTDC_LxCKCR to 0x808000.

43.3.7 Reloaded shadow register

Some configuration registers perform shadow operations. When writing to the active register, or at the beginning of the vertical blanking period after the configuration phase of R32_LTDC_SRCR register, the shadow register value can be reloaded into the active register immediately. If the immediate overload configuration is selected, overloading should only be activated after all new registers have finished writing.

The shadow register should not be modified again before the overload is completed. Reading the shadow register will return the actual effective value. The newly written value can only be read after the overload occurs.

If enabled in the R32_LTDC_IER register, a register overload interrupt can be generated.

Shadow registers are Layer 1 and Layer 2 registers, except R32_LTDC_LxCLUTWR register.

43.3.8 Interrupt

The LTDC provides three maskable interrupts, which are logically ORed into a single interrupt vector.

Interrupt sources can be individually enabled or disabled via the R32_LTDC_IER register; corresponding interrupt flags can be read through the R32_LTDC_ISR register; and the corresponding interrupt flag bits can be cleared by setting the R32_LTDC_ICR register to 1.

Interrupts are generated when the following events occur:

- Line interrupt: Generated upon reaching a programmed line. The interrupt position is programmed in the R32_LTDC_LIPCR register;
- Register reload interrupt: Generated during shadow register reload execution within the vertical blanking interval;
- FIFO underflow interrupt: Generated when requesting pixels from an empty layer FIFO.

43.4 Register Description

Table 43-9 LTDC-related registers

Name	Access address	Description	Reset value
R32_LTDC_SSCR	0x40014800	LTDC synchronous size configuration register	0x00000000
R32_LTDC_BPCR	0x40014804	LTDC trailing-edge configuration register	0x00000000

R32_LTDC_AWCR	0x40014808	LTDC effective width configuration register	0x00000000
R32_LTDC_TWCR	0x4001480C	LTDC total width configuration register	0x00000000
R32_LTDC_GCR	0x40014810	LTDC global control register	0x00002220
R32_LTDC_SRCR	0x40014814	LTDC shadow reload configuration register	0x00000000
R32_LTDC_BCCR	0x40014818	LTDC background color configuration register	0x00000000
R32_LTDC_IER	0x4001481C	LTDC interrupt enable register	0x00000000
R32_LTDC_ISR	0x40014820	LTDC interrupt status register	0x00000000
R32_LTDC_ICR	0x40014824	LTDC interrupt clear register	0x00000000
R32_LTDC_LIPCR	0x40014828	LTDC line interrupt position configuration register	0x00000000
R32_LTDC_CPSR	0x4001482C	LTDC current position status register	0x00000000
R32_LTDC_CDSR	0x40014830	LTDC current display status register	0x0000000F
R32_LTDC_L1CR	0x40014834	LTDC layer 1 control register	0x00000000
R32_LTDC_L1WHPCR	0x40014838	LTDC layer 1 window horizontal position configuration register	0x00000000
R32_LTDC_L1WVPCR	0x4001483C	LTDC layer 1 window vertical position configuration register	0x00000000
R32_LTDC_L1CKCR	0x40014840	LTDC layer 1 color key configuration register	0x00000000
R32_LTDC_L1PFCR	0x40014844	LTDC layer 1 pixel format configuration register	0x00000000
R32_LTDC_L1CACR	0x40014848	LTDC layer 1 constant Alpha configuration register	0x00000000
R32_LTDC_L1DCCR	0x4001484C	LTDC layer 1 default color configuration Register	0x00000000
R32_LTDC_L1BFCR	0x40014850	LTDC layer 1 blending coefficient configuration register	0x00000600
R32_LTDC_L1CFBAR	0x40014854	LTDC layer 1 color frame buffer address register	0x00000000
R32_LTDC_L1CFBLR	0x40014858	LTDC layer 1 color frame buffer length register	0x00000000
R32_LTDC_L1CFBLNR	0x4001485C	LTDC layer 1 color frame buffer line number register	0x00000000
R32_LTDC_L1CLUTWR	0x40014860	LTDC layer 1 CLUT write register	0x00000000
R32_LTDC_L2CR	0x40014864	LTDC layer 2 control register	0x00000000
R32_LTDC_L2WHPCR	0x40014868	LTDC layer 2 window horizontal position configuration register	0x00000000
R32_LTDC_L2WVPCR	0x4001486C	LTDC layer 2 window vertical position configuration register	0x00000000
R32_LTDC_L2CKCR	0x40014870	LTDC layer 2 color key configuration register	0x00000000
R32_LTDC_L2PFCR	0x40014874	LTDC layer 2 pixel format configuration register	0x00000000
R32_LTDC_L2CACR	0x40014878	LTDC layer 2 constant Alpha configuration register	0x00000000
R32_LTDC_L2DCCR	0x4001487C	LTDC layer 2 default color configuration Register	0x00000000
R32_LTDC_L2BFCR	0x40014880	LTDC layer 2 blending coefficient configuration	0x00000600

		register	
R32_LTDC_L2CFBAR	0x40014884	LTDC layer 2 color frame buffer address register	0x00000000
R32_LTDC_L2CFBLR	0x40014888	LTDC layer 2 color frame buffer length register	0x00000000
R32_LTDC_L2CFBLNR	0x4001488C	LTDC layer 2 color frame buffer line number register	0x00000000
R32_LTDC_L2CLUTWR	0x40014890	LTDC level 2 CLUT write register	0x00000000

43.4.1 LTDC Synchronous Size Configuration Register (R32_LTDC_SSCR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				HSW[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				VSH[10:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	HSW[11:0]	RW	Horizontal synchronization width (in pixel clock period): This field defines the number of horizontal synchronization pixels minus 1.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	VSH[10:0]	RW	Vertical synchronization height (in horizontal scanning lines): This field defines the vertical synchronization height minus 1. It represents the number of horizontal synchronization lines.	0

43.4.2 LTDC Trailing-Edge Configuration Register (R32_LTDC_BPCR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				AHBP[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				AVBP[10:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	AHBP[11:0]	RW	Accumulation level trailing edge (in pixel clock cycles): This field defines the width of the accumulation level trailing edge (horizontal sync pixels plus trailing edge pixels minus one). The trailing edge is the interval between when the	0

			horizontal sync signal becomes inactive and the start of the next scan line's active display.	
[15:11]	Reserved	RO	Reserved	0
[10:0]	AVBP[10:0]	RW	Accumulated vertical Trailing Edge (Per Horizontal Scan Line): This field defines the accumulated vertical trailing edge width (vertical sync line plus vertical trailing edge line minus 1). The vertical trailing edge is the number of horizontal scan lines contained between the start of a frame and the start of the first valid scan line of the next frame.	0

43.4.3 LTDC Effective Width Configuration Register (R32_LTDC_AWCR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					AAW[11:0]										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					AAH[10:0]										

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	AAW[11:0]	RW	Cumulative effective width (in pixel clock cycles): This field defines the cumulative effective width (horizontal sync pixels plus horizontal trailing edge pixels minus one effective pixel). The effective width represents the number of pixels within the active display area of the panel scan line.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	AAH[10:0]	RW	Cumulative effective height (per horizontal scan line): This field defines the cumulative height (vertical sync line plus vertical trailing edge line plus effective height line minus 1). Effective height represents the number of valid lines within the panel.	0

43.4.4 LTDC Total Width Configuration Register (R32_LTDC_TWCR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					TOTALW[11:0]										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					TOTALH[10:0]										

Bit	Name	Access	Description	Reset value
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[31:28]	Reserved	RO	Reserved	0
[27:16]	TOTALW[11:0]	RW	Total width (in pixel clock cycles): This field defines the accumulated total width (horizontal sync pixel plus horizontal trailing edge pixel plus active width plus horizontal leading edge pixel minus 1).	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	TOTALH[10:0]	RW	Total height (per horizontal scan line): This field defines the cumulative height (vertical sync line plus vertical trailing edge line plus effective height plus vertical leading edge line minus 1).	0

43.4.5 LTDC Global Control Register (R32_LTDC_GCR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HSP OL	VSPOL	DEPOL	PCPOL	Reserved											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														LTDC EN	

Bit	Name	Access	Description	Reset value
31	HSPOL	RW	Horizontal sync polarity: 1: Horizontal sync active low; 0: Horizontal sync active high.	0
30	VSPOL	RW	Vertical sync polarity: 1: Vertical sync active high; 0: Vertical sync active low.	0
29	DEPOL	RW	Non-data enable polarity: 1: Non-data enable polarity active high; 0: Non-data enable polarity active low.	0
28	PCPOL	RW	Pixel clock polarity: 1: Pixel clock active high; 0: Pixel clock active-low.	0
[27:1]	Reserved	RO	Reserved	0x444
0	LTDCEN	RW	LCD-TFT controller enable bit: 1: Enable; 0: Disable.	0

43.4.6 LTDC Shadow Reload Configuration Register (R32_LTDC_SRCR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														VB R	IM R

Bit	Name	Access	Description	Reset value
[31:2]	Reserved	RO	Reserved	0
1	VBR	RW	Vertical blanking Reload: 1: Shadow register reloads during the vertical blanking period (starting at the first line after the active display area); 0: No effect. Note: This bit is set to 1 by software and cleared to 0 by hardware only after reloading.	0
0	IMR	RW	Immediate reload: 1: Shadow register immediate reload; 0: No effect. Note: This bit is set to 1 by software and is only cleared by hardware after reload.	0

43.4.7 LTDC Background Color Configuration Register (R32_LTDC_BCCR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								BCRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BCGREEN[7:0]								BCBLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	BCRED[7:0]	RW	Background red value.	0
[15:8]	BCGREEN[7:0]	RW	Background green value.	0
[7:0]	BCBLUE[7:0]	RW	Background blue value.	0

43.4.8 LTDC Interrupt Enable Register (R32_LTDC_IER)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												RRIE	Reser ved	FUIE	LIE

Bit	Name	Access	Description	Reset value
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[31:4]	Reserved	RO	Reserved	0
3	RRIE	RW	Register reload interrupt enable: 1: Enable; 0: Disable.	0
2	Reserved	RO	Reserved	0
1	FUIE	RW	FIFO underflow interrupt enable: 1: Enable; 0: Disable.	0
0	LIE	RW	Line interrupt enable: 1: Enable; 0: Disable.	0

43.4.9 LTDC Interrupt Status Register (R32_LTDC_ISR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												RRIF	Reserved	FUIF	LIF

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	RRIF	RO	Register overload interrupt flag: 1: Generates a register overload interrupt during vertical blanking reload (and on the first line after reaching the active area); 0: Does not generate a register overload interrupt.	0
2	Reserved	RO	Reserved	0
1	FUIF	RO	FIFO underflow interrupt flag: 1: A FIFO underflow interrupt occurs when one of the layer FIFOs is empty and pixel data is read from the FIFO. 0: No FIFO underflow interrupt has occurred.	0
0	LIF	RO	Line interrupt flag: 1: Generates a line interrupt when reaching the programmed line; 0: No line interrupt generated.	0

43.4.10 LTDC Interrupt Clear Register (R32_LTDC_ICR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CRRIF	Reserved	CFUIF	CLIF

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	CRRIF	RW1Z	Register reload interrupt clear flag, write 1 to clear 0, write 0 is invalid.	0
2	Reserved	RO	Reserved	0
1	CFUIF	RW1Z	FIFO underflow interrupt clear flag, write 1 to clear 0, write 0 is invalid.	0
0	CLIF	RW1Z	Line interrupt clear flag, write 1 to clear 0, write 0 is invalid.	0

43.4.11 LTDC Line Interrupt Position Configuration Register (R32_LTDC_LIPCR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						LIPOS[10:0]									

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
[10:0]	LIPOS[10:0]	RW	Line interrupt position	0

43.4.12 LTDC Current Position Status Register (R32_LTDC_CPSR)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CXPOS[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYPOS[15:0]															

Bit	Name	Access	Description	Reset value
[31:16]	CXPOS[15:0]	RW	Current X position.	0
[15:0]	CYPOS[15:0]	RW	Current Y position.	0

43.4.13 LTDC Current Display Status Register (R32_LTDC_CDSR)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												HSYN CS	VSYN CS	HDE S	VDES

Bit	Name	Access	Description	Reset value
[31:4]	Reserved	RO	Reserved	0
3	HSYNCS	RO	Horizontal synchronization display status: 1: Active high; 0: Active low.	1
2	VSYNCS	RO	Vertical synchronization display status: 1: Active high; 0: Active low.	1
1	HDES	RO	Horizontal data enabled display status: 1: Active high; 0: Active low.	1
0	VDES	RO	Vertical data enable display status: 1: Active high; 0: Active low.	1

43.4.14 LTDC Layer x Control Register (R32_LTDC_LxCR) (x=1/2)

Offset address: $0x34 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CL UT EN	Reserved	CO LK EN	LE N

Bit	Name	Access	Description	Reset value
[31:5]	Reserved	RO	Reserved	0
4	CLUTEN	RW	Color lookup table enable: 1: Enable; 0: Disable. <i>Note: CLUT is only relevant for L8, AL44, and AL88 pixel formats.</i>	0
[3:2]	Reserved	RO	Reserved	0
1	COLKEN	RW	Color key enable: 1: Enable; 0: Disable.	0
0	LEN	RW	Layer enable: 1: Enable; 0: Disable.	0

43.4.15 LTDC Layer x Window Horizontal Position Configuration Register (R32_LTDC_LxWHPCR) (x=1/2)

Offset address: $0x38 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WHSPPOS[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WHSTPOS[11:0]											

Bit	Name	Access	Description	Reset value
[31:28]	Reserved	RO	Reserved	0
[27:16]	WHSPPOS[11:0]	RW	Window horizontal end position: WHSPPOS[11:0] $\geq$ AHBP[11:0] + 1.	0
[15:12]	Reserved	RO	Reserved	0
[11:0]	WHSTPOS[11:0]	RW	Window horizontal start position: WHSTPOS[11:0] $\leq$ AAW[11:0].	0

For example, the R32_LTDC_BPCR register is configured as 0x000E0005 (AHBP[11:0] is 0xE), and the R32_LTDC_AWCR register is configured as 0x028E01E5 (AAW[11:0] is 0x28E). To configure the horizontal position of a 630x460 window (with a horizontal starting offset of 5 pixels within the valid data area).

- 1) First pixel of the layer window: WHSTPOS[11:0] should be programmed to 0x14 (0xE + 1 + 0x5);
- 2) Last pixel of the layer window: WHSPPOS[11:0] should be programmed to 0x28A.

43.4.16 LTDC Layer x Window Vertical Position Configuration Register (R32_LTDC_LxWVPCR) (x=1/2)

Offset address: $0x3C + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WVSPPOS[10:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WVSTPOS[10:0]											

Bit	Name	Access	Description	Reset value
[31:27]	Reserved	RO	Reserved	0
[26:16]	WVSPPOS[10:0]	RW	Window vertical end position: WVSPPOS[10:0] $\geq$ AVBP[10:0] + 1.	0
[15:11]	Reserved	RO	Reserved	0
[10:0]	WVSTPOS[10:0]	RW	Window vertical start position: WVSTPOS[10:0] $\leq$ AAH[10:0].	0

For example, the R32_LTDC_BPCR register is configured as 0x000E0005 (AVBP[10:0] is 0x5), and the R32_LTDC_AWCR register is configured as 0x028E01E5 (AAH[10:0] is 0x1E5). To configure the vertical position of a 630x460 window (with a vertical starting offset of 8 lines in the valid data area):

- 1) First line of the layer window: WVSTPOS[10:0] should be programmed to 0xE (0x5 + 1 + 0x8);
- 2) Last line of the layer window: WVSPPOS[10:0] should be programmed to 0x1DA.

43.4.17 LTDC Layer x Color Keyboard Configuration Register (R32_LTDC_LxCKCR) (x=1/2)Offset address: $0x40 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CKRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CKGREEN[7:0]								CKBLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	CKRED[7:0]	RW	Color keyboard red value.	0
[15:8]	CKGREEN[7:0]	RO	Color keyboard green value.	0
[7:0]	CKBLUE[7:0]	RW	Color keyboard blue value.	0

43.4.18 LTDC Layer x Pixel Format Configuration Register (R32_LTDC_LxPFCR) (x=1/2)Offset address: $0x44 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													PF[2:0]		

Bit	Name	Access	Description	Reset value
[31:3]	Reserved	RO	Reserved	0
[2:0]	PF[2:0]	RW	Pixel format: 000: ARGB8888; 001: RGB888; 010: RGB565; 011: ARGB1555; 100: ARGB4444; 101: L8 (8-bit Luminance); 110: AL44 (4-bit Alpha, 4-bit Luminance); 111: AL88 (8-bit Alpha, 8-bit Luminance).	0

43.4.19 LTDC Layer x Constant Alpha Configuration Register (R32_LTDC_LxCACR) (x=1/2)Offset address: $0x48 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CONSTA[7:0]							

Bit	Name	Access	Description	Reset value
[31:8]	Reserved	RO	Reserved	0
[7:0]	CONSTA[7:0]	RW	Constant Alpha: The constant Alpha used when configuring this domain as hybrid. The constant Alpha is divided into 255 equal parts by the hardware implementation. Example: If the programmed constant Alpha is 0xFF, then the constant Alpha value is $255/255 = 1$.	0xFF

43.4.20 LTDC Layer x Default Color Configuration Register (R32_LTDC_LxDCCR) (x=1/2)

Offset address: $0x4C + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DCALPHA[7:0]								DCRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCGREEN[7:0]								DCBLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	DCALPHA[7:0]	RW	Default Alpha value.	0
[23:16]	DCRED[7:0]	RW	Default color red.	0
[15:8]	DCGREEN[7:0]	RW	Default color green.	0
[7:0]	DCBLUE[7:0]	RW	Default color blue.	0

43.4.21 LTDC Layer x Default Color Configuration Register (R32_LTDC_LxBFCR) (x=1/2)

Offset address: $0x50 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				BF1[2:0]				Reserved				BF2[2:0]			

Bit	Name	Access	Description	Reset value
[31:11]	Reserved	RO	Reserved	0
[10:8]	BF1[2:0]	RW	Blending coefficient 1: 100: Constant Alpha; 110: Pixel Alpha * Constant Alpha; Others: Preserved.	110b
[7:3]	Reserved	RO	Reserved	0
[2:0]	BF2[2:0]	RW	Blending coefficient 2: 100: Constant Alpha; 110: Pixel Alpha * Constant Alpha; Others: Preserved.	0

43.4.22 LTDC Layer x Color Frame Buffer Address Register (R32_LTDC_LxCFBAR) (x=1/2)Offset address: $0x54 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CFBADD[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFBADD[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	CFBADD[31:0]	RW	Start address of color frame buffer.	0

43.4.23 LTDC Layer x Color Frame Buffer Length Register (R32_LTDC_LxCFBLR) (x=1/2)Offset address: $0x58 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			CFBP[12:0]												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CFBLL[12:0]												

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved	0
[28:16]	CFBP[12:0]	RW	Color frame buffer interval (in bytes): This field defines the increment from the start of one row of pixels to the start of the next row.	0
[15:13]	Reserved	RO	Reserved	0
[12:0]	CFBLL[12:0]	RW	Color frame buffer line length: Defines the length of a row of pixels (in bytes) + 31. The row length is calculated as: Effective Width × Bytes per Pixel + 31.	0

43.4.24 LTDC Layer x Color Frame Buffer Line Number Register (R32_LTDC_LxCFBLNR) (x=1/2)Offset address: $0x5C + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CFBLNBR[10:0]										

Bit	Name	Access	Description	Reset value
[31:12]	Reserved	RO	Reserved	0
[10:0]	CFBLNBR[10:0]	RW	Number of lines in frame buffer: This field defines the number of lines in the buffer,	0

			corresponding to the effective height.	
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Note: The line number and line length settings define the amount of data obtained from each frame for each layer. If configured below the required bytes, a FIFO underflow interrupt will be generated (if enabled). On the other hand, the starting address and spacing settings define the correct starting position of each line in the memory.

43.4.25 LTDC Layer x CLUT Write Register (R32_LTDC_LxCLUTWR) (x=1/2)

Offset address: $0x60 + 0x30 \times (x-1)$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CLUTADD[7:0]								RED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN[7:0]								BLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	CLUTADD[7:0]	WO	Configure the CLUT address of each RGB value.	0
[23:16]	RED[7:0]	WO	Red value.	0
[15:8]	GREEN[7:0]	WO	Green value.	0
[7:0]	BLUE[7:0]	WO	Blue value.	0

Note: (1) The CLUT write register should only be configured during the blanking period or when the layer is disabled.

(2) The CLUT can be enabled or disabled via the R32_LTDC_LxCR register.

(3) The CLUT is only relevant for the L8, AL44, and AL88 pixel formats.

Chapter 44 Graphics Processing Hardware accelerator (GPHA)

Image Processing Hardware Accelerator (GPHA) is a DMA dedicated to image processing, which supports indexed color mode or direct color mode and all classical color coding schemes. In addition, the GPHA module also has its own exclusive CLUT (Color Lookup Table).

44.1 Main Features

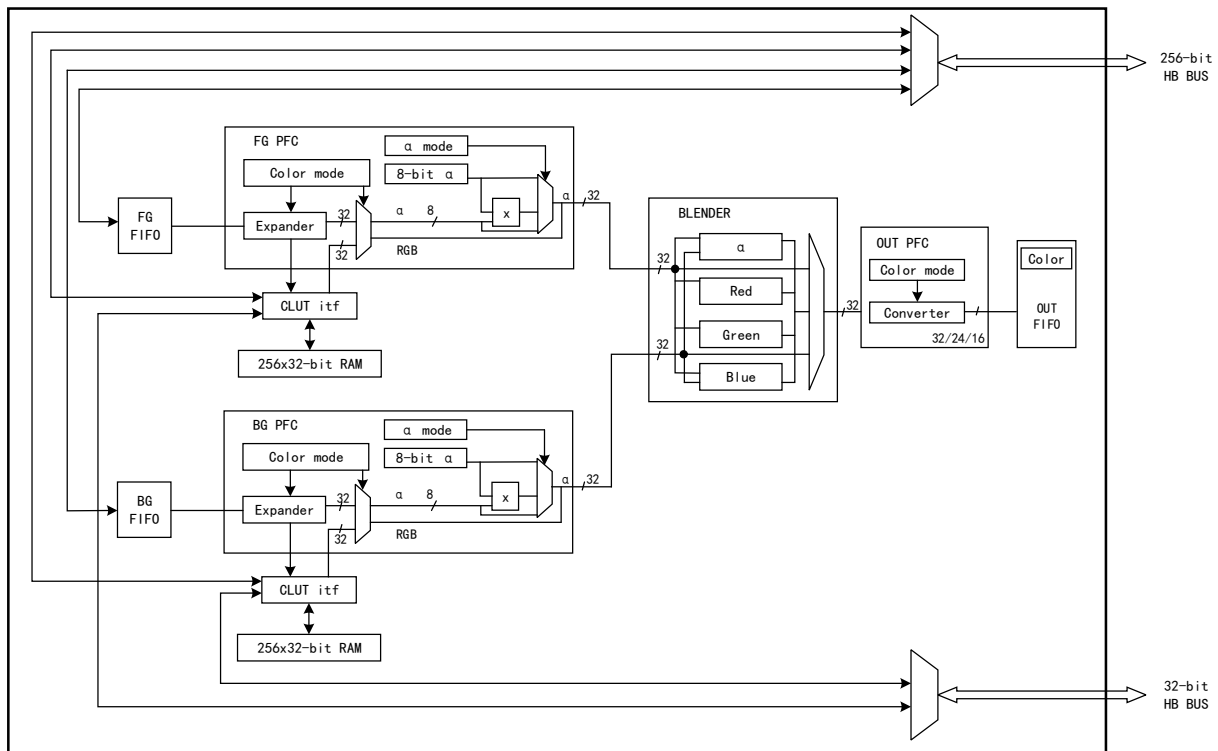
- Support single HB master bus architecture
- HB slave programming interface supports 8/16/32-bit access (except for 32-bit CLUT access)
- Support programmable source and destination region sizes and offsets
- Support programmable source and destination addresses across entire memory space
- Support adjustable alpha values (source value, fixed value, modulation value)
- Support hybrid operations with up to 2 sources
- Support programmable source and target color formats, accommodating up to 11 color formats with a maximum of 32 bits per pixel
- Support two internal memories for storing CLUTs in indirect color mode
- Support programmable CLUT size, with CLUTs automatically loaded via CPU or programmed manually
- Support internal timer control for HB bandwidth
- Support three operating modes: register-to-memory, memory-to-memory with pixel format conversion, and memory-to-memory with pixel format conversion and mixing
- Support filling specified portions or the entire area of the target image with a specific color
- Support copying portions or the entire content of the source image to corresponding portions or the entire target image
- Support copying portions or the entire content of the source image to portions or the entire target image with pixel format conversion
- Support blending two source image sections and/or entire images with differing pixel formats, then copying the result to sections or the entire target image with differing color formats
- Support canceling or suspending GPHA operations
- Support generating interrupts upon bus errors or access conflicts
- Support generating interrupts upon process completion

44.2 Function Description

44.2.1 Overview

The foreground layer (FG)FIFO and background layer (BG)FIFO of GPHA acquire the input data to be copied and processed. These FIFO acquire pixels according to the preset color format in the corresponding pixel format converter (PFC), perform color format conversion, wherein the foreground layer and background layer can be processed separately or mixed, and finally output the converted color data. As shown in Figure 44-1, it shows the structural block diagram of GPHA in detail.

Figure 44-1 GPHA structure block diagram



GPHA controller performs direct memory transfer. As a HB master, it can control HB bus matrix to start HB transactions. HB slave port is used to program GPHA controller.

GPHA supports the following 3 modes of operation:

- Register-to-memory;
- Memory-to-memory with pixel format conversion;
- Memory-to-memory with pixel format conversion and blending.

GPHA functions include the following operations:

- Support filling specified portions or the entire area of the target image with a specific color;
- Support copying portions or the entire content of the source image to corresponding portions or the entire target image;
- Support copying portions or the entire content of the source image to portions or the entire target image with pixel format conversion;
- Support blending portions and/or the entire content of two source images with different pixel formats, then copying the result to portions or the entire target image with different color formats.

44.2.2 GPHA Control

The GPHA controller can be configured via the R32_GPHA_CTLR register to perform the following operations:

- 1) Select the GPHA operating mode;
- 2) Enable/disable GPHA interrupts;
- 3) Start/suspend/abort an ongoing data transfer.

44.2.3 GPHA Foreground Layer FIFO and Background Layer FIFO

The coordinated operation of the GPHA Foreground Layer (FG) FIFO and Background Layer (BG) FIFO is critical to the efficient image processing of the GPHA. These FIFOs acquire input data for replication and processing,

fetching pixels according to the color format preset in the corresponding Pixel Format Converter (PFC). They can be programmed via the following registers:

- GPHA Foreground Layer Memory Address Register (R32_GPHA_FGMAR)
- GPHA Foreground Layer Offset Register (R32_GPHA_FGOR)
- GPHA Foreground Memory Address Register (R32_GPHA_BGMAR)
- GPHA Foreground Offset Register (R32_GPHA_BGBOR)
- GPHA Number of Lines Register (Number of Lines and Pixels per Line) (R32_GPHA_NLR)

GPHA works in register-to-memory mode and does not activate any FIFO. When GPHA works in memory-to-memory mode and supports pixel format conversion (no mixed operation), BGFIFO will not be activated.

44.2.4 GPHA Foreground Layer and Background Layer Pixel Format Converter (PFC)

The GPHA Foreground and Background Layer Pixel Format Converter (PFC) performs pixel format conversion to generate 32-bit values per pixel. Additionally, the PFC possesses the capability to modify the alpha channel.

This converter performs color format conversion in the first stage. The CM[3:0] bits of the R32_GPHA_FGPFCCR register and the CM[3:0] bits of the R32_GPHA_BGPFCCR register configure the raw color formats for foreground layer pixels and background layer pixels, respectively. The supported input color modes are listed in Table 44-1.

Table 44-1 Color modes supported during input

CM[3:0]	Color mode
0000	ARGB8888
0001	RGB888
0010	RGB565
0011	ARGB1555
0100	ARGB4444
0101	L8
0110	AL44
0111	AL88
1000	L4
1001	A8
1010	A4

The Alpha channel is an extra channel used to represent the transparency of each pixel in the image. It is often used with red, green and blue (RGB) channels to form RGBA model.

The following is a description of the encoding methods of color formats appearing in Table 44-1:

Alpha value field: transparent, with 0xFF value corresponding to opaque pixels and 0x00 corresponding to transparent pixels;

R field: Represents red;

G field: Represents green;

B field: Represents blue;

L field: Represents luminance; this field serves as the CLUT index for retrieving 3 or 4 RGB/ARGB components.

If the original format is in direct color mode, it is expanded to 8 bits per channel by copying the most significant bit (MSB) to the least significant bit (LSB). This ensures the conversion exhibits good linearity.

If the original format does not include an alpha channel, the alpha value is automatically set to 0xFF (opaque).

If the original format is in indirect color mode, a CLUT is required, and each pixel format converter is associated with a CLUT containing 256 32-bit entries.

For specific alpha modes A4 and A8, neither color information is stored nor indexed. The colors used to generate the image are fixed, with the foreground layer pixel color defined in the R32_GPHA_FGCOLR register and the background layer pixel color defined in the R32_GPHA_BGCOLR register.

The sequence of fields in the system memory is shown in Table 44-2.

Table 44-2 Data sequence in memory

Color mode	@+3	@+2	@+1	@+0
ARGB8888	A ₀ [7:0]	R ₀ [7:0]	G ₀ [7:0]	B ₀ [7:0]
RGB888	B ₁ [7:0]	R ₀ [7:0]	G ₀ [7:0]	B ₀ [7:0]
	G ₂ [7:0]	B ₂ [7:0]	R ₁ [7:0]	G ₁ [7:0]
	R ₃ [7:0]	G ₃ [7:0]	B ₃ [7:0]	R ₂ [7:0]
	R ₁ [4:0]G ₁ [5:3]	G ₁ [2:0]B ₁ [4:0]	R ₀ [4:0]G ₀ [5:3]	G ₀ [2:0]B ₀ [4:0]
ARGB1555	A ₁ [0]R ₁ [4:0]G ₁ [4:3]	G ₁ [2:0]B ₁ [4:0]	A ₀ [0]R ₀ [4:0]G ₀ [4:3]	G ₀ [2:0]B ₀ [4:0]
ARGB4444	A ₁ [3:0]R ₁ [3:0]	G ₁ [3:0]B ₁ [3:0]	A ₀ [3:0]R ₀ [3:0]	G ₀ [3:0]B ₀ [3:0]
L8	L ₃ [7:0]	L ₂ [7:0]	L ₁ [7:0]	L ₀ [7:0]
AL44	A ₃ [3:0]L ₃ [3:0]	A ₂ [3:0]L ₂ [3:0]	A ₁ [3:0]L ₁ [3:0]	A ₀ [3:0]L ₀ [3:0]
AL88	A ₁ [7:0]	L ₁ [7:0]	A ₀ [7:0]	L ₀ [7:0]
L4	L ₇ [3:0]L ₆ [3:0]	L ₅ [3:0]L ₄ [3:0]	L ₃ [3:0]L ₂ [3:0]	L ₁ [3:0]L ₀ [3:0]
A8	A ₃ [7:0]	A ₂ [7:0]	A ₁ [7:0]	A ₀ [7:0]
A4	A ₇ [3:0]A ₆ [3:0]	A ₅ [3:0]A ₄ [3:0]	A ₃ [3:0]A ₂ [3:0]	A ₁ [3:0]A ₀ [3:0]

The ARGB8888 mode supports 32-bit alignment for 24-bit RGB888. After generating the 32-bit value, the alpha channel can be adjusted based on the AM[1:0] bits of the R32_GPHA_FGPFCCR/R32_GPHA_BGPFCCR registers. Specific configurations are detailed in Table 44-3.

Table 44-3 Alpha mode configuration

AM[1:0]	Alpha mode
00	Do not make adjustments
01	Replace with the ALPHA[7:0] value of the R32_GPHA_FGPFCCR/R32_GPHA_BGPFCCR register
10	Replace with the product of the original alpha value and the quotient obtained by dividing the ALPHA[7:0] value from the R32_GPHA_FGPFCCR/R32_GPHA_BGPFCCR register by 255.
11	Reserved

Note: To enable support for the alternate format, set the AI bit in the R32_GPHA_FGPFCCR/R32_GPHA_BGPFCCR register to 1 to perform a bit inversion operation on the incoming alpha value. This operation also applies to the alpha values stored in the CLUT and the R32_GPHA_FGPFCCR/R32_GPHA_BGPFCCR register. Additionally, setting the RBS bit in the R32_GPHA_FGPFCCR/R32_GPHA_BGPFCCR register to 1 enables swapping the R and B fields. This operation

also applies to the RGB order used in the CLUT and the R32_GPHA_FGCOLR/R32_GPHA_BGCOLR registers.

44.2.5 GPHA Foreground Layer and Background Layer CLUT Interface

CLUT is a table for storing color mapping, which maps the index value of each pixel to a specific color value. CLUT interface can manage the access to CLUT memory and the automatic loading of CLUT. CLUT interface supports the following access methods:

- PFC reads the CLUT during pixel format conversion;
- The CPU accesses the CLUT via the HB from the device port to read or write data;
- When performing automatic CLUT loading, CLUT writing is executed through the HB master device port.

CLUT memory loading can be realized by automatic loading and manual loading:

Autoloading

- 1) Program the CLUT address into the R32_GPHA_FGCMAR register (foreground layer CLUT) or the R32_GPHA_BGCMAR register (background layer CLUT).
- 2) Program the CLUT size into the CS[7:0] bits of the R32_GPHA_FGPFCCR register (foreground layer CLUT) or the R32_GPHA_BGPFCCR register (background layer CLUT).
- 3) Set the START bit in the R32_GPHA_FGPFCCR register (foreground layer CLUT) or R32_GPHA_BGPFCCR register (background layer CLUT) to 1 to initiate the transfer. During auto-loading, the CPU must not access the CLUT. If a conflict occurs and the CAEIE bit in the R32_GPHA_CTLR register is set to 1, a CLUT access error interrupt is generated.

Manual loading

The application must manually program the local CLUT memory via the GPHA HB from the device port; this CLUT maps to the CLUT in the dedicated SRAM.

- 1) Program the CLUT pixel address into the FG_CLUT_INDEX bit of the R32_GPHA_FGCRWS register (foreground layer CLUT) or the BG_CLUT_INDEX bit of the R32_GPHA_FGCRWS register (background layer CLUT).
- 2) Program the CLUT pixel value into the R32_GPHA_FGCDAT register (foreground layer CLUT) or the R32_GPHA_BGCDAT register (background layer CLUT). The CPU automatically loads this pixel value into the corresponding CLUT memory.

The CLUT format supports 24-bit or 32-bit. The CCM bit in the R32_GPHA_FGPFCCR register (for the foreground layer CLUT) or the R32_GPHA_BGPFCCR register (for the background layer CLUT) configures the CLUT color mode. Refer to Table 44-4 for specific configuration details.

Table 44-4 Supported CLUT color modes

CCM	CLUT color modes
0	32-bit ARGB8888
1	24-bit RGB888

Table 44-5 shows the organization of CLUT data in the system memory.

Table 44-5 CLUT data order in memory

CLUT color modes	@+3	@+2	@+1	@+0
ARGB8888	A ₀ [7:0]	R ₀ [7:0]	G ₀ [7:0]	B ₀ [7:0]
RGB888	B ₁ [7:0]	R ₀ [7:0]	G ₀ [7:0]	B ₀ [7:0]

	G ₂ [7:0]	B ₂ [7:0]	R ₁ [7:0]	G ₁ [7:0]
	R ₃ [7:0]	G ₃ [7:0]	B ₃ [7:0]	R ₂ [7:0]

44.2.6 GPHA Mixer

The GPHA mixer mixes the source pixels in pairs to calculate the result pixels.

Mixing will be performed according to the following formula:

$$C_{OUT} = \frac{C_{FG}\alpha_{FG} + C_{BG}\alpha_{BG} - C_{BG}\alpha_{Mult}}{\alpha_{OUT}} \quad (\text{Quotient will be rounded down});$$

Where C=R or G or B,

$$\alpha_{Mult} = \frac{\alpha_{FG}\alpha_{BG}}{255}$$

$$\alpha_{OUT} = \alpha_{FG} + \alpha_{BG} - \alpha_{Mult}$$

The mixer does not need any configuration registers. Whether the mixer is used or not depends on the GPHA operation mode defined in MODE[1:0] bits of R32_GPHA_CTLR register.

44.2.7 GPHA Output PFC

The output PFC converts the pixel format from 32 bits to the specified output format, and the output format can be configured through bits CM[2:0] in R32_GPHA_OPFC CCR register.

Table 44-6 Supported color modes in output

CM[2:0]	Color modes
000	ARGB8888
001	RGB888
010	RGB565
011	ARGB1555
100	ARGB4444

Note: To enable support for the alternate format, set the AI bit of the R32_GPHA_OPFCCR register to 1 to invert the calculated alpha value. This operation also applies to the alpha value used in the R32_GPHA_OCOLR register. Additionally, setting the RBS bit in the R32_GPHA_OPFCCR register to 1 enables swapping the R and B fields. This operation also applies to the RGB order used in the R32_GPHA_OCOLR register.

44.2.8 GPHA Output FIFO

The output FIFO programs pixels according to the color format defined in the output PFC. The target area can be defined by the following registers:

- GPHA Output Memory Address Register (R32_GPHA_OMAR)
- GPHA Output Offset Register (R32_GPHA_OOR)
- GPHA Number of Lines Register (Number of Lines and Pixels per Line) (R32_GPHA_NLR)

When the GPHA operates in register-to-memory mode, the configured output rectangle will be filled with the color specified in the GPHA output color register (R32_GPHA_OCOLR), which contains a fixed 32-bit, 24-bit, or 16-bit value. The color format selection can be configured via the CM[2:0] bits of the R32_GPHA_OPFCCR register.

The sequence of fields in the system memory is shown in Table 44-7.

Table 44-7 Data Sequence in Memory

Color modes	@+3	@+2	@+1	@+0
-------------	-----	-----	-----	-----

ARGB8888	A ₀ [7:0]	R ₀ [7:0]	G ₀ [7:0]	B ₀ [7:0]
RGB888	B ₁ [7:0]	R ₀ [7:0]	G ₀ [7:0]	B ₀ [7:0]
	G ₂ [7:0]	B ₂ [7:0]	R ₁ [7:0]	G ₁ [7:0]
	R ₃ [7:0]	G ₃ [7:0]	B ₃ [7:0]	R ₂ [7:0]
RGB565	R ₁ [4:0]G ₁ [5:3]	G ₁ [2:0]B ₁ [4:0]	R ₀ [4:0]G ₀ [5:3]	G ₀ [2:0]B ₀ [4:0]
ARGB1555	A ₁ [0]R ₁ [4:0]G ₁ [4:3]	G ₁ [2:0]B ₁ [4:0]	A ₀ [0]R ₀ [4:0]G ₀ [4:3]	G ₀ [2:0]B ₀ [4:0]
ARGB4444	A ₁ [3:0]R ₁ [3:0]	G ₁ [3:0]B ₁ [3:0]	A ₀ [3:0]R ₀ [3:0]	G ₀ [3:0]B ₀ [3:0]

Note: Supports 32-bit aligned RGB888 via ARGB8888 mode.

44.2.9 GPHA HB Master Port Timer

An 8-bit timer is embedded in the port of HB master device, which aims to limit the bandwidth of crossbar matrix optionally. This timer is driven by HB clock, and is responsible for counting the dead zone between 2 consecutive accesses, so as to limit the bandwidth usage.

The enabling of the timer and the dead-time value can be configured through the EN bits and DT[7:0] bits of R32_GPHA_AMTCR register. Please refer to Chapter 44.2.15 for the setting of HB dead-time.

44.2.10 GPHA Transaction

GPHA transaction consists of a series of data transmission sequences with a preset number. The quantity and width of data can be programmed by software.

Each GPHA data transmission project needs at most 4 steps:

- 1) Load data from the memory location specified by the R32_GPHA_FGMAR register and perform pixel format conversion as defined in the R32_GPHA_FGPFCCR register.
- 2) Load data from the memory location specified by the R32_GPHA_BGMAR register and perform pixel format conversion as defined in the R32_GPHA_BGPFCCR register.
- 3) Process alpha values using PFC operations to obtain alpha channel information, then blend all retrieved pixels.
- 4) Perform pixel format conversion on the blended pixels according to the R32_GPHA_OPFCCR register, then program the converted data into memory cells addressed via the R32_GPHA_OMAR register.

44.2.11 GPHA Configuration

Source and destination data transmission can address peripherals and memory in the entire 4GB area (address range between 0x00000000 and 0xFFFFFFFF).

GPHA can work in the following three modes, and the working mode can be selected through bits MODE[1:0] of R32_GPHA_CTLR register:

- Register to Memory
- Memory to Memory with PFC Execution
- Memory to Memory with PFC and Hybrid Execution

44.2.11.1 Register to Memory

Register-to-memory mode is used to fill user-defined areas with predefined colors.

The color format is set in the R32_GPHA_OPFCCR register.

The GPHA does not acquire data from any source. It only writes the color defined in the R32_GPHA_OCOLR register to the region defined by R32_GPHA_OMAR addressing and the R32_GPHA_NLR and R32_GPHA_OOR registers.

44.2.11.2 Memory to Memory and Perform PFC

In this mode, GPHA performs pixel format transform on the source data and stores the result in the target storage unit.

For the size of the region to be transferred, the source region size is defined by the R32_GPHA_NLR and R32_GPHA_FGOR registers, while the destination region size is defined by the R32_GPHA_NLR and R32_GPHA_OOR registers.

Data is retrieved from the location defined by the R32_GPHA_FGMAR register and processed by the foreground layer PFC. The original pixel format is configured via the GPHA_FGPFCCR register.

If the original pixel format is in direct color mode, all color channels are expanded to 8 bits.

If the pixel format is in indirect color mode, the corresponding CLUT must be loaded into the CLUT memory.

CLUT loading can be performed automatically in the following sequence:

- 1) Set the CLUT address in the R32_GPHA_FGCMAR register.
- 2) Set the CLUT size in the CS[7:0] bits of the R32_GPHA_FGPFCCR register.
- 3) Set the CLUT format (24-bit or 32-bit) in the CCM bit of the R32_GPHA_FGPFCCR register.
- 4) Enable CLUT loading by setting the START bit in the R32_GPHA_FGPFCCR register to 1.

Upon completion of CLUT loading, the CTCIF flag in the R32_GPHA_ISR register will be set; if the CTCIE bit in the R32_GPHA_CTLR register is set to 1, an interrupt will also be generated. The automatic CLUT loading process cannot occur concurrently with traditional GPHA transfers.

The CLUT can also be filled by the CPU or by any other host device via the HB port. The CLUT cannot be accessed during GPHA transfers or when using CLUT (indirect color format).

44.2.11.3 Memory to Memory and Perform PFC and Mixing

In this mode, two source images will be captured in the foreground layer FIFO and the background layer FIFO (defined in R32_GPHA_FGMAR register and R32_GPHA_BGMAR register respectively).

2 pixel format converters must be configured as described in memory-to-memory mode. Since these 2 pixel format converters are independent and have their own CLUT memory, their configurations can be different.

After each pixel is converted into 32 bits by the corresponding PFC, it will be mixed according to the formula in 44.3.6. The output PFC will encode the obtained 32-bit pixel value according to the specified output format, and the encoded data will be written into the target storage unit addressed by GPHA_OMAR.

44.2.11.4 Configuration Error Detection

GPHA will check whether the configuration is correct before each transmission. When starting a new transfer/autoload, if a configuration error is detected, the hardware will set a configuration error interrupt flag. If the CEIE bit in the R32_GPHA_CTLR register is set, an interrupt will also be generated.

Detectable errors are configured as follows:

- Memory transfer (except register-to-memory mode): The MA bit in the R32_GPHA_FGMAR register does not match the CM bit in the R32_GPHA_FGPFCCR register;
- Memory transfer (except register-to-memory mode): The CM bit in the R32_GPHA_FGPFCCR register is invalid;
- Memory transfer (except register-to-memory mode): The PL bit in the R32_GPHA_NLR register is odd, while the CM bit in the R32_GPHA_FGPFCCR register is A4 or L4;
- Memory transfer (except register-to-memory mode): The LO bit in the R32_GPHA_FGOR register is odd, while the CM bit in the R32_GPHA_FGPFCCR register is A4 or L4;
- Memory Transfer (Hybrid Mode Only): The MA bit in the R32_GPHA_BGMAR register does not match the CM

bit in the R32_GPHA_BGPFCCR register;

- Memory Transfer: The CM bit in the R32_GPHA_BGPFCCR register is invalid (Hybrid Mode Only);
- Memory Transfer (Hybrid Mode Only): The PL bit in the R32_GPHA_NLR register is odd, while the CM bit in the R32_GPHA_BGPFCCR register is A4 or L4;
- Memory Transfer (Hybrid Mode Only): R32_GPHA_BGOR register LO bit is odd while R32_GPHA_BGPFCCR register CM bit is A4 or L4;
- Memory Transfer: The MA bit in the R32_GPHA_OMAR register does not match the CM bit in the R32_GPHA_OPFCCR register;
- Memory Transfer: The CM bit in the R32_GPHA_OPFCCR register is invalid;
- Memory Transfer: The NL bit in the R32_GPHA_NLR register is set to 0;
- Memory Transfer: PL bit in R32_GPHA_NLR register = 0;

44.2.12 GPHA Transmission Control (Start, Suspend, Stop and Complete)

After configuring the GPHA, setting the START bit to 1 in the R32_GPHA_CTLR register initiates data transfer. Upon completion, the START bit automatically resets and the TCIF flag is set in the R32_GPHA_ISR register. If the TCIE bit in the R32_GPHA_CTLR register is set to 1, an interrupt is also generated.

The user application can suspend the GPHA at any time by setting the SUSP bit in the R32_GPHA_CTLR register to 1. The transaction can then be aborted by setting the ABORT bit in the GPHA_CTLR register to 1, or resumed by clearing the SUSP bit in the R32_GPHA_CTLR register.

The user application may terminate a transaction in progress at any time by setting bit 1 of the ABORT field in the R32_GPHA_CTLR register. In this case, the TCIF flag will not be set.

The CLUT automatic transfer process may also be terminated or suspended by setting the START bit in the R32_GPHA_FGPFCCR register or the R32_GPHA_BGPFCCR register itself.

44.2.13 Watermark

The watermark can be programmed to generate an interrupt when the last pixel in a specified row is written to the target storage area.

The row number is defined in bits LW[15:0] of the R32_GPHA_LWR register.

Upon completion of the last pixel transfer in that row, the R32_GPHA_ISR will set the TWIF flag; an interrupt will also be generated if the TWIE bit in the R32_GPHA_CTLR register is set to 1.

44.2.14 Error Management

Triggerable error: CLUT access conflict (CPU attempts to access CLUT during CLUT loading or GPHA transfer execution), indicated by the CAEIF flag in the R32_GPHA_ISR register.

This flag correlates with the interrupt enable flags (TEIE and CAEIE) in the R32_GPHA_CTLR register, which control whether interrupts are generated when required.

44.2.15 HB Dead-time

To limit HB bandwidth usage, a dead-time can be programmed between 2 consecutive HB accesses.

The dead-time function is enabled by setting bit EN in the R32_GPHA_AMTCR register to 1.

The dead-time value is stored in bits DT[7:0] of the R32_GPHA_AMTCR register. This value represents the minimum number of cycles allowed between 2 consecutive transactions on the HB bus.

Updates to the dead-time value during GPHA operation take effect during the next HB transmission.

44.2.16 Interrupt

Table 44-8 Interrupt request

Interrupt event	Event flag	Enable control bit
Configuration error	CEIF	CEIE
CLUT transmission completed	CTCIF	CTCIE
CLUT access error	CAEIF	CAEIE
Transmission watermark	TWF	TWIE
Transmission completed	TCIF	TCIE

44.3 Register Description

Table 44-9 GPHA-related registers

Name	Access address	Description	Reset value
R32_GPHA_CTLR	0x40016800	GPHA control register	0x00000000
R32_GPHA_ISR	0x40016804	GPHA interrupt status register	0x00000000
R32_GPHA_IFCR	0x40016808	GPHA interrupt flag clear register	0x00000000
R32_GPHA_FGMAR	0x4001680C	GPHA foreground layer memory address register	0x00000000
R32_GPHA_FGOR	0x40016810	GPHA foreground layer offset register	0x00000000
R32_GPHA_BGMAR	0x40016814	GPHA background layer memory address register	0x00000000
R32_GPHA_BGOR	0x40016818	GPHA background layer offset register	0x00000000
R32_GPHA_FGPFCCR	0x4001681C	GPHA foreground layer PFC control register	0x00000000
R32_GPHA_FGCOLR	0x40016820	GPHA foreground layer color register	0x00000000
R32_GPHA_BGPFCCR	0x40016824	GPHA background layer PFC control register	0x00000000
R32_GPHA_BGCOLR	0x40016828	GPHA background layer color register	0x00000000
R32_GPHA_FGCMAR	0x4001682C	GPHA foreground layer CLUT memory address register	0x00000000
R32_GPHA_BGCMAR	0x40016830	GPHA background layer CLUT memory address register	0x00000000
R32_GPHA_OPFCCR	0x40016834	GPHA output PFC control register	0x00000000
R32_GPHA_OCOLR	0x40016838	GPHA output color register	0x00000000
R32_GPHA_OMAR	0x4001683C	GPHA output memory address register	0x00000000
R32_GPHA_OOR	0x40016840	GPHA output offset register	0x00000000

R32_GPHA_NLR	0x40016844	GPHA line number register	0x00000000
R32_GPHA_LWR	0x40016848	GPHA line watermark register	0x00000000
R32_GPHA_AMTCR	0x4001684C	GPHA HB master timer configuration register	0x00000000
R32_GPHA_FGCWRS	0x40016850	GPHA foreground layer CLUT read-write setting register	0x00000000
R32_GPHA_FGCDAT	0x40016854	GPHA foreground layer CLUT read-write data register	0x00000000
R32_GPHA_BGCWRS	0x40016858	GPHA background layer CLUT read-write setting register	0x00000000
R32_GPHA_BGCDAT	0x4001685C	GPHA background layer CLUT read-write data register	0x00000000

44.3.1 GPHA Control Register (R32_GPHA_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													MODE[2:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CEIE	CTCIE	CAEIE	TWIE	TCIE	Reserved							ABORT	SUSP	START

Bit	Name	Access	Description	Reset value
[31:19]	Reserved	RO	Reserved	0
[18:16]	MODE[2:0]	RW	GPHA operating modes: 001: Memory-to-Memory with PFC Execution (FG acquisition only when FG PFC is active); 010: Memory-to-Memory with Hybrid Execution (FG and BG acquisition during PFC and hybrid execution); 011: Register-to-Memory (No FG or BG, output stage activation only); Others: Reserved.	0
[15:14]	Reserved	RO	Reserved	0
13	CEIE	RW	Configure interrupt enable: 1: CE interrupt enabled; 0: CE interrupt disabled.	0
12	CTCIE	RW	CLUT transfer complete interrupt enable: 1: CTC interrupt enabled; 0: CTC interrupt disabled.	0
11	CAEIE	RW	CLUT access error interrupt enable: 1: CAE interrupt enabled; 0: CAE interrupt disabled.	0
10	TWIE	RW	Transmission watermark interrupt enable:	0

			1: TW interrupt enabled; 0: TW interrupt disabled.	
9	TCIE	RW	Transmission complete interrupt enable: 1: Enable TC interrupt; 0: Disable TC interrupt.	0
[8:3]	Reserved	RO	Reserved	0
2	ABORT	RW	Termination: 1: Request transmission termination; 0: Do not request transmission termination.	0
1	SUSP	RW	Suspended: 1: Transmission suspended; 0: Transmission not suspended.	0
0	START	RW	Start: This bit may be used to initiate GPHA based on parameters loaded into various configuration registers. This bit will be automatically reset under the following conditions: - At the end of a transmission; - When data transmission is aborted via the user application setting the ABORT bit in the GPHA_CTLR register to 1; - When a data transmission error occurs; - When data transmission fails to start due to configuration errors or another transmission operation already in progress (CLUT auto-load).	0

44.3.2 GPHA Interrupt Status Register (R32_GPHA_ISR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										CEIF	CTCIF	CAEIF	TWIF	TCIF	Reserved

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	CEIF	RO	Configuration error interrupt flag: When the START bit is set to 1 in the R32_GPHA_CTLR, R32_GPHA_FGPFCCR, or R32_GPHA_BGPFCCR register, and an erroneous configuration is programmed, this bit is set to 1.	0
4	CTCIF	RO	CLUT transfer complete interrupt flag: This bit is set to 1 when copying the CLUT from	0

			system memory to GPHA internal memory is complete.	
3	CAEIF	RO	CLUT access error interrupt flag: This bit is set to 1 if the CPU accesses the CLUT during automatic copying of the CLUT from system memory to the GPHA internal memory.	0
2	TWIF	RO	Watermark transmission interrupt flag: This flag is set to 1 when the last pixel of the watermark row completes transmission.	0
1	TCIF	RO	Transmission complete interrupt flag: This bit is set to 1 when the GPHA transmission operation completes (data transmission only).	0
0	Reserved	RO	Reserved	0

44.3.3 GPHA Interrupt Flag Clear Register (R32_GPHA_IFCR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										CCEIF	CCTCIF	CAECIF	CTWIF	CTCIF	Reserved

Bit	Name	Access	Description	Reset value
[31:6]	Reserved	RO	Reserved	0
5	CCEIF	RW1Z	Clear the configuration error interrupt flag, write 1 to clear 0, and write 0 is invalid.	0
4	CCTCIF	RW1Z	Clear the CLUT transmission completion interrupt flag, write 1 to clear 0, and write 0 is invalid.	0
3	CAECIF	RW1Z	Clear the interrupt flag of CLUT access error, write 1 to clear 0, and write 0 is invalid.	0
2	CTWIF	RW1Z	Clear the transmission watermark interrupt flag, write 1 to clear 0, and write 0 is invalid.	0
1	CTCIF	RW1Z	Clear the transmission completion interrupt flag, write 1 to clear 0, and write 0 is invalid.	0
0	Reserved	RO	Reserved	0

44.3.4 GPHA Foreground Layer Memory Address Register (R32_GPHA_FGMAR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MA[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MA[31:0]	RW	Storage address of foreground layer image: This register allows write operations only when data transmission is prohibited. Once the data transfer is started, the register will change to a read-only state. Address alignment needs to match the image format, for example, when the format of 32 bits per pixel is adopted, the address alignment must be 32 bits; When the format of 16 bits per pixel is adopted, the address alignment must be 16 bits; However, when the format of 4 bits per pixel is adopted, the address alignment must be 8 bits.	0

44.3.5 GPHA Foreground Layer Offset Register (R32_GPHA_FGOR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		LO[13:0]													

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	LO[13:0]	RW	Line offset of foreground layer image: This value is used to generate addresses. The row offset is appended to the end of each line to determine the starting address of the next line. This register permits write operations only when data transfer is disabled. Once data transfer begins, this field becomes read-only. If the image format is 4 bits per pixel, the row offset value must be an even number.	0

44.3.6 GPHA Foreground Layer Memory Address Register (R32_GPHA_BGMAR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MA[15:0]															

Bit	Name	Access	Description	Reset value
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[31:19]	MA[31:0]	RW	Storage address of background layer image: This register allows write operations only when data transmission is prohibited. Once the data transfer process is activated, the register will change to read-only state. Address alignment needs to match the image format, for example, when the format of 32 bits per pixel is adopted, the address alignment must be 32 bits; When the format of 16 bits per pixel is adopted, the address alignment must be 16 bits; However, when the format of 4 bits per pixel is adopted, the address alignment must be 8 bits.	0
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44.3.7 GPHA Background Layer Offset Register (R32_GPHA_BGOR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		LO[13:0]													

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0
[13:0]	LO[13:0]	RW	Line offset of background layer image: This value is used to generate the address. A line offset is added to the end of each line to determine the starting address of the next line. This register allows write operations only when data transmission is prohibited. Once the data transfer is started, the domain will become read-only. If the image format is 4 bits per pixel, the line offset value must be even.	0

44.3.8 GPHA Foreground Layer PFC Control Register (R32_GPHA_FGPFCCR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ALPHA[7:0]								Reserved	RB S	AI	Reserved	AM[1:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CS[7:0]								Reserved	ST AR T	CC M	CM[3:0]				

Bit	Name	Access	Description	Reset value
[31:24]	ALPHA[7:0]	RW	Alpha value: This field defines the alpha value, which may either replace the original alpha value or be multiplied by it, depending on the alpha value selected via the AM[1:0] bits. <i>Note: This field permits write operations only when data transfer is disabled. Once data transfer begins, this field becomes read-only.</i>	0
[23:22]	Reserved	RO	Reserved	0
21	RBS	RW	Red-Blue swap: 1: Swap mode (BGR or ABGR); 0: Standard mode (RGB or ARGB).	0
20	AI	RW	Invert Alpha: 1: Invert Alpha; 0: Normal Alpha. <i>Note: Once transmission begins, this bit becomes read-only.</i>	0
[19:18]	Reserved	RO	Reserved	0
[17:16]	AM[1:0]	RW	Alpha mode: 00: Do not modify the alpha channel value of the foreground layer image; 01: Replace the alpha channel value of the original foreground layer image with ALPHA[7:0]; 10: Replace the alpha channel value of the original foreground layer image with the product of ALPHA[7:0] and the original alpha channel value; Others: Preserve. <i>Note: This field permits write operations only when data transmission is disabled. Once data transmission begins, this field becomes read-only.</i>	0
[15:8]	CS[7:0]	RW	CLUT size: The number of CLUT entries used for the foreground layer image equals CS[7:0] + 1. <i>Note: Once CLUT transfer is initiated, this field becomes read-only.</i>	0
[7:6]	Reserved	RO	Reserved	0
5	START	RW	Initiate the automatic loading process for CLUT: 1: Enable; 0: Disable. This bit automatically resets under the following conditions: - At the end of transmission; - When transmission is aborted by setting the	0

			ABORT bit in the R32_GPHA_CTLR register via the user application; - When a transmission error occurs; - When transmission fails to start due to configuration errors or another transmission operation (data transmission or automatic background layer CLUT transmission) is already in progress.	
4	CCM	RW	CLUT color mode: 1: RGB888; 0: ARGB8888. <i>Note: This field permits write operations only when data transmission is disabled. Once transmission begins, this bit becomes read-only.</i>	0
[3:0]	CM[3:0]	RW	Color mode: 0000: ARGB8888; 0001: RGB888; 0010: RGB565; 0011: ARGB1555; 0100: ARGB4444; 0101: L8; 0110: AL44; 0111: AL88; 1000: L4; 1001: A8; 1010: A4; Other: Reserved. <i>Note: This field permits write operations only when data transmission is prohibited. Once transmission begins, this bit becomes read-only.</i>	0

44.3.9 GPHA Foreground Layer Color Register (R32_GPHA_FGCOLR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								RED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN[7:0]								BLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	RED[7:0]	RW	The red value of A4 or A8 mode of the foreground layer image.	0
[15:8]	GREEN[7:0]	RW	The green value of A4 or A8 mode of the foreground layer image.	0

[7:0]	BLUE[7:0]	RW	The blue value of A4 or A8 mode of the foreground layer image.	0
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Note: R32_GPHA_FGCOLR register only allows write operation when data transmission is prohibited. Once the transfer is started, this bit will become read-only.

44.3.10 GPHA Background Layer PFC Control Register (R32_GPHA_BGPFCCR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ALPHA[7:0]								Reserved		RB S	AI	Reserved		AM[1:0]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CS[7:0]								Reserved		ST AR T	CC M	CM[3:0]			

Bit	Name	Access	Description	Reset value
[31:24]	ALPHA[7:0]	RW	Alpha value: This field defines the alpha value, which may either replace the original alpha value or be multiplied by it, depending on the alpha value selected via the AM[1:0] bits. <i>Note: This field permits write operations only when data transfer is disabled. Once data transfer begins, this field becomes read-only.</i>	0
[23:22]	Reserved	RO	Reserved	0
21	RBS	RW	Red-Blue swap: 1: Swap mode (BGR or ABGR); 0: Normal mode (RGB or ARGB). <i>Note: Once transmission begins, this bit becomes read-only.</i>	0
20	AI	RW	Invert Alpha: 1: Invert Alpha; 0: Normal Alpha. <i>Note: Once transmission begins, this bit becomes read-only.</i>	0
[19:18]	Reserved	RO	Reserved	0
[17:16]	AM[1:0]	RW	Alpha mode: 00: Do not modify the alpha channel value of the background layer image; 01: Replace the alpha channel value of the original background layer image with ALPHA[7:0]; 10: Replace the alpha channel value of the original background layer image with the product of ALPHA[7:0] and the original alpha channel value;	0

			Others: Preserve. <i>Note: This field permits write operations only when data transmission is disabled. Once data transmission begins, this field becomes read-only.</i>	
[15:8]	CS[7:0]	RW	CLUT size: The number of CLUT entries used for the background layer image equals CS[7:0] + 1. <i>Note: Once CLUT transfer is initiated, this field becomes read-only.</i>	0
[7:6]	Reserved	RO	Reserved	0
5	START	RW	Initiate the automatic loading process for CLUT: 1: Enable; 0: Disable. This bit automatically resets under the following conditions: - At the end of a transfer; - When the transfer is aborted via the user application setting the ABORT bit in the R32_GPHA_CTLR register to 1; - When a transfer error occurs; - When the transfer fails to start due to configuration errors or another transfer operation (data transfer or automatic background layer CLUT transfer) is already in progress.	0
4	CCM	RW	CLUT color modes: 1: RGB888; 0: ARGB8888. <i>Note: This domain only allows write operations when data transmission is prohibited. Once the transfer is started, this bit will become read-only.</i>	0
[3:0]	CM[3:0]	RW	Color modes: 0000: ARGB8888; 0001: RGB888; 0010: RGB565; 0011: ARGB1555; 0100: ARGB4444; 0101: L8; 0110: AL44; 0111: AL88; 1000: L4; 1001: A8; 1010: A4; Other: Reserved. <i>Note: This field permits write operations only when</i>	0

			<i>data transmission is prohibited. Once transmission begins, this bit becomes read-only.</i>	
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44.3.11 GPHA Background Layer Color Register (R32_GPHA_BGCOLOR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								RED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN[7:0]								BLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	Reserved	RO	Reserved	0
[23:16]	RED[7:0]	RW	The red value of A4 or A8 mode of the background layer image.	0
[15:8]	GREEN[7:0]	RW	The green value of A4 or A8 mode of the background layer image.	0
[7:0]	BLUE[7:0]	RW	The blue value of A4 or A8 mode of the background layer image.	0

Note: R32_GPHA_BGCOLOR register only allows write operation when data transmission is prohibited. Once the transfer is started, this bit will become read-only.

44.3.12 GPHA Foreground Layer CLUT Memory Address Register (R32_GPHA_FGCMAR)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MA[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MA[31:0]	RW	CLUT storage address for foreground layer image: This register permits write operations only when data transfer is disabled. Once CLUT transfer is initiated, the register conversions to read-only status. If the foreground layer CLUT format is 32-bit, the address must be 32-bit aligned.	0

44.3.13 GPHA Background Layer CLUT Memory Address Register (R32_GPHA_BGCMAR)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

MA[15:0]

Bit	Name	Access	Description	Reset value
[31:19]	MA[31:0]	RW	CLUT storage address for background layer image: This register permits write operations only when data transfer is disabled. Once CLUT transfer is initiated, the register conversions to read-only status. If the foreground layer CLUT format is 32-bit, the address must be 32-bit aligned.	0

44.3.14 GPHA Output PFC Control Register (R32_GPHA_OPFCCR)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										RB S	AI	Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CM[2:0]			

Bit	Name	Access	Description	Reset value
[31:22]	Reserved	RO	Reserved	0
21	RBS	RW	Red-Blue swap: 1: Swap mode (BGR or ABGR); 0: Standard mode (RGB or ARGB). <i>Note: Once transmission begins, this bit becomes read-only.</i>	0
20	AI	RW	Invert Alpha: 1: Invert Alpha; 0: Normal Alpha. <i>Note: Once transmission begins, this bit becomes read-only.</i>	0
[19:3]	Reserved	RO	Reserved	0
[2:0]	CM[2:0]	RW	Color modes: 000: ARGB8888; 001: RGB888; 010: RGB565; 011: ARGB1555; 100: ARGB4444; Other: Reserved. <i>Note: This field permits write operations only when data transmission is prohibited. Once transmission begins, this bit becomes read-only.</i>	0

44.3.15 GPHA Output Color Register (R32_GPHA_OCOLR)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ALPHA[7:0]								RED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN[7:0]								BLUE[7:0]							

Bit	Name	Access	Description	Reset value
[31:24]	ALPHA[7:0]	RW	Alpha channel of output color.	0
[23:16]	RED[7:0]	RW	Red value of output image.	0
[15:8]	GREEN[7:0]	RW	Green value of output image.	0
[7:0]	BLUE[7:0]	RW	Blue value of output image.	0

44.3.16 GPHA Output Memory Address Register (R32_GPHA_OMAR)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MA[15:0]															

Bit	Name	Access	Description	Reset value
[31:19]	MA[31:0]	RW	Output FIFO storage address: This register permits write operations only when data transfer is disabled. Once data transfer is activated, the register conversions to read-only mode. Address alignment must match the image format. For example, with a 32-bit per pixel format, address alignment must be 32 bits; with a 16-bit per pixel format, address alignment must be 16 bits. Note: If the output pixel format is RGB888, the number of output pixels must be a multiple of 4.	0

44.3.17 GPHA Output Offset Register (R32_GPHA_OOR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		LO[13:0]													

Bit	Name	Access	Description	Reset value
[31:14]	Reserved	RO	Reserved	0

[13:0]	LO[13:0]	RW	Output line offset: This value is used to generate addresses. The row offset is appended to the end of each line to determine the starting address of the next row. This register permits write operations only when data transfer is disabled. Once data transfer begins, this field becomes read-only.	0
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44.3.18 GPHA Line Number Register (R32_GPHA_NLR)

Offset address: 0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		PL[13:0]													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NL[15:0]															

Bit	Name	Access	Description	Reset value
[31:30]	Reserved	RO	Reserved	0
[29:16]	PL[13:0]	RW	Pixels per line in the area to be transmitted: This register allows write operations only when data transmission is prohibited. Once the data transfer is started, the domain will become read-only. If the image format is 4 bits per pixel, the line offset value must be even. <i>Note: For chips where the fifth digit of the lot number is less than 3, the values of the PL[13:0] bits must be greater than 0 when MODE[2:0] in the R32_GPHA_CTLR register is set to 011b.</i>	0
[15:0]	NL[15:0]	RW	Number of lines in the area to be transmitted: This register allows write operations only when data transmission is prohibited. Once the data transfer is started, the domain will become read-only.	0

44.3.19 GPHA Line Watermark Register (R32_GPHA_LWR)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LW[15:0]															

Bit	Name	Access	Description	Reset value
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[31:16]	Reserved	RO	Reserved	0
[15:0]	LW[15:0]	RW	Line watermark: An interrupt is generated when the last pixel of the watermark line is transferred. Write operations are permitted only when data transfer is disabled. Once data transfer begins, this field becomes read-only.	0

44.3.20 GPHA HB Master Timer Configuration Register (R32_GPHA_AMTCR)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DT[7:0]								Reserved						EN	

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
[15:8]	DT[7:0]	RW	The dead-time value inserted between two consecutive accesses on the port of HB master device, expressed in the number of HB clock cycles. These bits represent the minimum number of cycles allowed between two consecutive HB accesses.	0
[7:1]	Reserved	RO	Reserved	0
0	EN	RW	Enable dead-time function: 1: Enable; 0: Disable.	0

44.3.21 GPHA Foreground Layer CLUT Read-write Setting Register (R32_GPHA_FGCWRS)

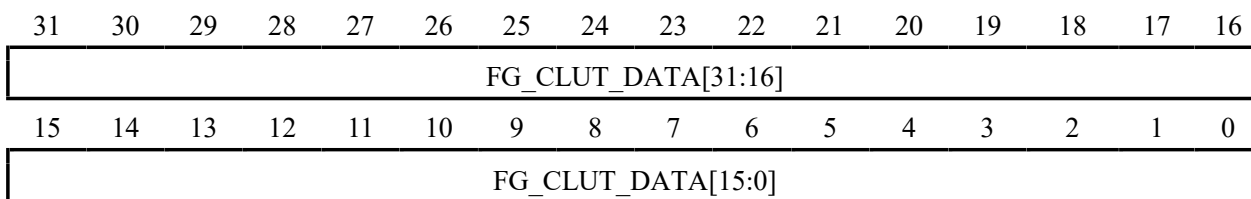
Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								FG_CLUT_EN	FG_CLUT_INDEX[7:0]						

Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
8	FG_CLUT_EN	RW	CPU lookup table read/write enable: 1: Enable; 0: Disable.	0
[7:0]	FG_CLUT_INDEX[7:0]	RW	Set the lookup table address.	0

44.3.22 GPHA Foreground Layer CLUT Read-write Data Register (R32_GPHA_FGCDAT)

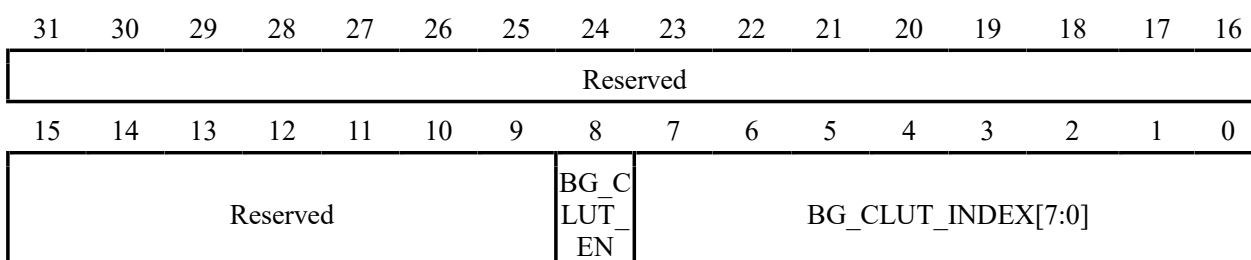
Offset address: 0x54



Bit	Name	Access	Description	Reset value
[31:0]	FG_CLUT_DATA[31:0]	RW	Read data or write look-up table SRAM data.	0

44.3.23 GPHA Background Layer CLUT Read-write Setting Register (R32_GPHA_BGCWRS)

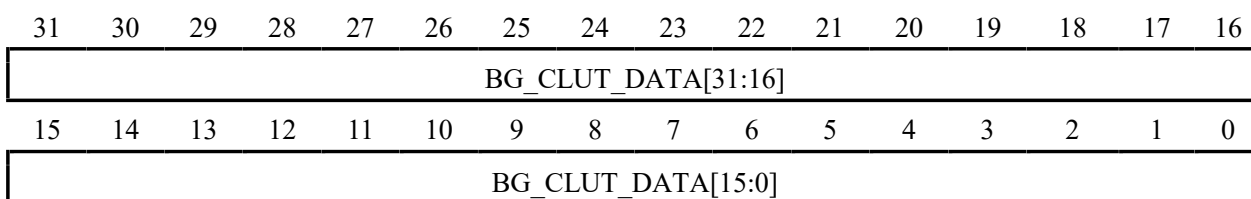
Offset address: 0x58



Bit	Name	Access	Description	Reset value
[31:9]	Reserved	RO	Reserved	0
8	BG_CLUT_EN	RW	CPU lookup table read/write enable: 1: Enable; 0: Disable.	0
[7:0]	BG_CLUT_INDEX[7:0]	RW	Set the lookup table address.	0

44.3.24 GPHA Background Layer CLUT Read-write Data Register (R32_GPHA_BGCDAT)

Offset address: 0x5C



Bit	Name	Access	Description	Reset value
[31:0]	BG_CLUT_DATA[31:0]	RW	Read data or write look-up table SRAM data.	0

Chapter 45 Debug Support (DBG)

45.1 Main Features

This register allows the MCU to be configured in the debug state. Includes:

- Counters supporting Independent Watchdog (IWDG)
- Counters supporting Window Watchdog (WWDG)
- Counter supporting timer
- Support I2CSMBus timeout control
- Support CAN communication

45.2 Register Description

45.2.1 RISC-V Debug MCU Configuration Register (R32_DBGMCU_CR)

Offset address: 0x7C0(CSR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	CAN3_STOP	CAN2_STOP	CAN1_STOP	I2C4_SMBUS_TIMEOUT	I2C3_SMBUS_TIMEOUT	LPTIM2_STOP	LPTIM1_STOP	TIM12_STOP	TIM11_STOP	TIM10_STOP	TIM9_STOP	TIM8_STOP	TIM7_STOP	TIM6_STOP	TIM5_STOP
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM4_STOP	TIM3_STOP	TIM2_STOP	TIM1_STOP	I2C2_SMBUS_TIMEOUT	I2C1_SMBUS_TIMEOUT	WWDG_STOP	IWDG_STOP	Reserved							

Bit	Name	Access	Description	Reset value
31	Reserved	RO	Reserved	0
30	CAN3_STOP	RW	CAN3 debug stop bit. When the core enters debug mode, CAN2 stops operation. 1: The receiving register of CAN3 does not continue to receive data; 0: CAN3 continues normal operation.	0
29	CAN2_STOP	RW	CAN2 debug stop bit. When the core enters debug mode, CAN2 stops operation. 1: The receiving register of CAN2 does not continue to receive data; 0: CAN2 continues normal operation.	0
28	CAN1_STOP	RW	CAN1 debug stop bit. When the core enters debug mode, CAN2 stops operation. 1: The receiving register of CAN1 does not continue to receive data; 0: CAN1 continues normal operation.	0

27	I2C4_SMBUS_TIMEOUT	RW	I2C4_SMBUS timeout mode debug stop bit. Disables SMBUS timeout mode when the core enters debug state. 1: Freezes SMBUS timeout control; 0: Operates identically to normal mode.	0
26	I2C3_SMBUS_TIMEOUT	RW	I2C3_SMBUS debug stop bit for timeout mode. Disables SMBUS timeout mode when the core enters debug state. 1: Freezes SMBUS timeout control; 0: Operates identically to normal mode.	0
25	LPTIM2_STOP	RW	Low-power timer 2 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Low-power timer 2 counter stops working; 0: Low-power timer 2 counter continues normal operation.	0
24	LPTIM1_STOP	RW	Low-power timer 1 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Low-power timer 1 counter stops working; 0: Low-power timer 1 counter continues normal operation	0
23	TIM12_STOP	RW	Timer 12 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 12 counter stops working; 0: Timer 12 counter continues normal operation.	0
22	TIM11_STOP	RW	Timer 11 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 11 counter stops working; 0: Timer 11 counter continues normal operation.	0
21	TIM10_STOP	RW	Timer 10 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 10 counter stops working; 0: Timer 10 counter continues normal operation.	0
20	TIM9_STOP	RW	Timer 9 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 9 counter stops working; 0: Timer 9 counter continues normal operation.	0
19	TIM8_STOP	RW	Timer 8 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 8 counter stops working; 0: Timer 8 counter continues normal operation.	0
18	TIM7_STOP	RW	Timer 7 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 7 counter stops working; 0: Timer 7 counter continues normal operation.	0
17	TIM6_STOP	RW	Timer 6 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 6 counter stops working; 0: Timer 6 counter continues normal operation.	0

16	TIM5_STOP	RW	Timer 5 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 5 counter stops working; 0: Timer 5 counter continues normal operation.	0
15	TIM4_STOP	RW	Timer 4 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 4 counter stops working; 0: Timer 4 counter continues normal operation.	0
14	TIM3_STOP	RW	Timer 3 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 3 counter stops working; 0: Timer 3 counter continues normal operation.	0
13	TIM2_STOP	RW	Timer 2 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 2 counter stops working; 0: Timer 2 counter continues normal operation.	0
12	TIM1_STOP	RW	Timer 1 debug stop bit. When the core enters debug mode, the counter stops functioning. 1: Timer 1 counter stops working; 0: Timer 1 counter continues normal operation.	0
11	I2C2_SMBUS_TIMEOUT	RW	SMBUS timeout mode debug stop bit. Stops SMBUS timeout mode when the core enters debug state. 1: Freezes the SMBUS timeout control. 0: Same as normal mode operation.	0
10	I2C1_SMBUS_TIMEOUT	RW	SMBUS timeout mode debug stop bit. Stops SMBUS timeout mode when the core enters debug state. 1: Freezes the SMBUS timeout control. 0: Same as normal mode operation.	0
9	WWDG_STOP	RW	WWDG debug stop bit. The debug WWDG stops working when the core enters the debug state. 1: WWDG counter stops working. 0: WWDG counter continues normal operation.	0
8	IWDG_STOP	RW	IWDG debug stop bit. The debug IWDG stops working when the core enters the debug state. 1: IWDG counter stops working. 0: IWDG counter continues normal operation.	0
[7:0]	Reserved	RW	Reserved	0

Chapter 46 Flash Memory and User Option Bytes (FLASH)

46.1 Flash Memory Organization

The flash memory organization inside the chip is as follows:

Table 46-1 Flash organizational structure 1 (user area 960K+BOOT area 56K)

Block	Name	Address range	Size (byte)
Main memory	Page 0	0x08000000 – 0x080000FF	256
	Page 1	0x08000100 – 0x080001FF	256
	Page 2	0x08000200 – 0x080002FF	256
	Page 3	0x08000300 – 0x080003FF	256
	Page 4	0x08000400 – 0x080004FF	256
	Page 5	0x08000500 – 0x080005FF	256
	Page 6	0x08000600 – 0x080006FF	256
	Page 7	0x08000700 – 0x080007FF	256
	...	...	...
	Page 3839	0x080EFF00 – 0x080EFFFF	256
Information block	Boot program code	0x1FFF0000 – 0x1FFFDFFF	56K
	User option bytes	0x1FFFF800 – 0x1FFFF8FF	256

Note: The above main memory area is used for the user's application storage, and the write protection is divided in 8K bytes (32 pages) units; except that the "Vendor configuration word" area is factory locked, the user is inaccessible. Other areas can be operated by users under certain conditions.

Table 46-2 Flash organizational structure 2 (user area 480K+BOOT area 28K)

Block	Name	Address range	Size (byte)
Main memory	Page 0	0x08000000 – 0x080000FF	256
	Page 1	0x08000100 – 0x080001FF	256
	Page 2	0x08000200 – 0x080002FF	256
	Page 3	0x08000300 – 0x080003FF	256
	Page 4	0x08000400 – 0x080004FF	256
	Page 5	0x08000500 – 0x080005FF	256
	Page 6	0x08000600 – 0x080006FF	256
	Page 7	0x08000700 – 0x080007FF	256
	...	...	...
	Page 1919	0x08077F00 – 0x08077FFF	256
Information block	Boot program code	0x1FFF0000 – 0x1FFF6FFF	28K
	User option bytes	0x1FFFF800 – 0x1FFFF8FF	256

Note: The above main memory area is used for the user's application storage, and the write protection is divided in 4K bytes (16 pages) units; except that the "Vendor configuration word" area is factory locked, the user is inaccessible. Other areas can be operated by users under certain conditions.

46.2 Flash Memory Programming and Safety

46.2.1 Two Program/Erase Methods

- **Standard Programming:** This is the default programming mode (compatibility mode). In this mode, the CPU executes programming in single 2-byte increments. When DBMODE=1, the chip performs erase operations in single 8K-byte increments; when DBMODE=0, the chip performs erase operations in single 4K-byte increments.
- **Fast Programming:** This method employs a page operation mode (recommended). After unlocking via a specific sequence, it executes a single 256-byte programming operation. When DBMODE=1, the chip performs a single 64K-byte erase operation; when DBMODE=0, the chip performs a single 32K-byte erase operation.

46.2.2 Security-preventing against Illegal Access (read, write and erase)

- Page write protection
- Read protection

When the chip is in read protection:

- 1) When DBMODE=1, pages 0-31 (8K bytes) of the chip's main memory are automatically write-protected and are not controlled by the R32_FLASH_WPR register. When read protection is disabled, all main memory pages are controlled by the R32_FLASH_WPR register.
- 2) When DBMODE=0, pages 0-15 (4K bytes) of the chip's main memory are automatically write-protected and not controlled by the R32_FLASH_WPR register. When read protection is disabled, all main memory pages are controlled by the R32_FLASH_WPR register.
- 3) The system boot code area, SWD or SDI mode, and RAM regions cannot be erased or programmed. The user-selectable word area can be erased or programmed. If an attempt is made to disable read protection (programming user words), the chip will automatically erase the entire user area.

46.3 FLASH Enhanced Read Mode

FLASH Enhanced Read Mode is suitable for user programs to run in FLASH, which can improve the access efficiency of FLASH. To enable this mode, set bit 1 of the EHMOD field in the R32_FLASH_ACTLR register. Then, read the ENHANCE_STATUS bit; if bit 1 of ENHANCE_STATUS is set, the enhanced read mode is active. To disable this mode, first clear bit 1 of EHMOD, then set RSENACT to 1.

Note: When using FLASH enhanced reading mode, you should pay attention to the following points:

- 1) Before performing any erase or program operations on the FLASH (including user programming such as read protection removal), you must first exit Enhanced Read Mode. Failure to do so will cause erase and program operations to fail.
- 2) Before entering Stop Mode, you must first exit Enhanced Read Mode. Failure to do so may cause Stop Mode to malfunction.
- 3) After a power reset or system reset completes, the chip automatically exits Enhanced Read Mode under hardware control.

46.4 Register Description

Table 46-3 FLASH-related registers

Name	Access address	Description	Reset value
R32_FLASH_ACTLR	0x40022000	Access control register	0x00000000
R32_FLASH_KEYR	0x40022004	FPEC key register	0xFFFFFFFF
R32_FLASH_OBKEYR	0x40022008	OBKEY register	0xFFFFFFFF
R32_FLASH_STATR	0x4002200C	Status register	0x00008000
R32_FLASH_CTLR	0x40022010	Control register	0x00008080
R32_FLASH_ADDR	0x40022014	Address register	0x00000000
R32_FLASH_OBR	0x4002201C	Option byte register	0xFFFFFFFFFE
R32_FLASH_WPR	0x40022020	Write protection register	0xFFFFFFFF
R32_FLASH_MODEKEYR	0x40022024	Extension key register	0xFFFFFFFF
R32_BOOT_MODEKEYR	0x40022028	BOOT key register	0xFFFFFFFF
R32_FLASH_CFGR0	0x4002202C	Configuration register 1	0xFFFFFFFF

46.4.1 Access Control Register (R32_FLASH_ACTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLASH_ST	FLASH_READY	Reserved		FLASH_RD_MD	Reserved		FLASH_LP	EHMOD	ENHANCE_STATUS	Reserved				SCK_CFG	

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
15	FLASH_ST	RO	FLASH low-power status indicator bit: 1: FLASH is in low-power mode; 0: FLASH is in normal operating mode.	0
14	FLASH_READY	RO	FLASH ready indicator bit: 1: FLASH is ready; 0: FLASH is not ready.	0
[13:12]	Reserved	RO	Reserved	0
11	FLASH_RD_MD	RW	FLASH read mode control bit: 1: Continuous read mode; 0: Single read mode. This bit is for standard unlocking.	0
[10:9]	Reserved	RO	Reserved	0
8	FLASH_LP	RW	FLASH low-power mode control bit: 1: FLASH low-power mode enabled; 0: FLASH low-power mode disabled.	0

7	EHMOD	RW	FLASH enhanced read mode control bit: This mode improves access efficiency when the program runs from FLASH. 1: Enables FLASH Enhanced Read Mode; 0: Disables FLASH Enhanced Read Mode. Requires coordination with the RSENACT bit. To exit, first clear the EHMODO bit to 0, then set RSENACT to 1. This bit performs standard unlocking.	0
6	ENHANCE_STATUS	RO	Enhanced status: 1: Command transmission completed; 0: Waiting.	0
[5:2]	Reserved	RO	Reserved	0
[1:0]	SCK_CFG	RW	FLASH wait state count: 00: HCLK; FLASH read operation -- Recommended HCLK $\leq$ 75MHz, FLASH erase/write operation -- Recommended HCLK $\leq$ 60MHz; 01: HCLK/2; FLASH read operation -- Recommended HCLK $\leq$ 150MHz, FLASH erase/write Operation -- Recommended HCLK $\leq$ 120MHz; 10: HCLK/4; FLASH read operation -- HCLK maximum is 150MHz, not recommended for this setting, FLASH erase/write operation -- Recommended HCLK $\leq$ 150MHz; 11: Reserved.	0

46.4.2 FPEC Key Register (R32_FLASH_KEYR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEYR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	KEYR[31:0]	WO	FPEC key, the unlock key used to enter FPEC includes: RDPRT key = 0x000000A5; KEY1 = 0x45670123;	X

			KEY2 = 0xCDEF89AB.	
--	--	--	--------------------	--

46.4.3 OBKEY Register (R32_FLASH_OBKEYR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OBKEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OBKEYR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	OBKEYR[31:0]	WO	Selection word key for entering a selection word key to disarm OPTWRE. KEY1 = 0x45670123; KEY2 = 0xCDEF89AB.	X

46.4.4 Status Register (R32_FLASH_STATR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BOOT_LOCK	BOOT_MODE	BOOT_STATUS	BOOT_AVA	Reserved						EOP	WRP_RTER	Reserved		WRBSY	BSY

Bit	Name	Access	Description	Reset value
[31:16]	Reserved	RO	Reserved	0
15	BOOT_LOCK	RW1	BOOT area lock: 1: Locked; 0: Unlocked.	1
14	BOOT_MODE	RW	When combined with BOOT_AVA, it controls switching between the user area and BOOT area: 1: After a software reset, it can switch to either the BOOT area or the user area; 0: After a software reset, it can switch to the user area.	0
13	BOOT_STATUS	RW0	Source of the currently executing program, write 0 to reset: 1: Indicates a program loaded from the BOOT area; 0: Indicates a program loaded from the user area.	0
12	BOOT_AVA	RO	Program initialization state: 1: Indicates booting from the BOOT sector; 0: Indicates booting from the user sector.	0
[11:6]	Reserved	RO	Reserved	0
5	EOP	RW1Z	Indicates the end of the operation, write 1 to clear. The hardware is set every time it is successfully	0

			erased or programmed.	
4	WRPRterr	RW1Z	Indicates write protection error, write 1 to clear. If you program a write-protected address, the hardware will set it.	0
[3:2]	Reserved	RO	Reserved	0
1	WRBSY	RO	This bit is used during fast page programming to indicate that programming data is being written. During page programming, when writing data, this bit is set to '1' and the hardware automatically clears '0'; If this bit is '0', it means that writing the next data is allowed.	0
0	BSY	RO	Indicates busy status: 1: Indicates that a flash operation is in progress; 0: The operation is finished.	0

46.4.5 Control Register (R32_FLASH_CTLR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									RSE NACT	PGST RT	Reserved	BER	Reserved	FTP G	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLO CK	Reserved	EOPIE	Reserved	ERRIE	OBWRE	Reserved	LOCK	STRT	OBER	OBPG	Reserved	PER	PG		

Bit	Name	Access	Description	Reset value
[31:23]	Reserved	RO	Reserved	0
22	RSENACT	WO	To exit Enhanced Read Mode, the hardware automatically clears the mode. This operation requires coordination with the EHMOD bit. The exit procedure involves first clearing the EHMOD bit to 0, then setting RSENACT to 1.	0
21	PGSTRT	WO	Start. Set to 1 to start a page programming, and the hardware will be cleared automatically. <i>This bit enables standard unlocking.</i>	0
[20:19]	Reserved	RO	Reserved	0
18	BER	RW	When DBMODE=1, perform a 64KB erase. When DBMODE=0, perform a 32KB erase. <i>This bit enables standard unlock + fast unlock.</i>	0
17	Reserved	RO	Reserved	0
16	FTPG	RW	Perform fast page programming operation. <i>This bit enables standard unlock + fast unlock.</i>	0
15	FLOCK	RW1	Fast Program Lock. Writes only '1'. When this bit is '1', quick program/erase mode is disabled. Hardware clears this bit to '0' upon detecting the correct unlock sequence. Software sets to '1' to re-lock.	1

			<i>This bit enables standard unlocking.</i>	
[14:13]	Reserved	RO	Reserved	0
12	EOPIE	RW	Interrupt control after operation completion (EOP bit set in R32_FLASH_STATR register): 1: Enable interrupt generation; 0: Disable interrupt generation. <i>This bit enables standard unlocking.</i>	0
11	Reserved	RO	Reserved	0
10	ERRIE	RW	Error status interrupt control (PGERR/WRPRTERR set in R32_FLASH_STATR register): 1: Allows interrupt generation; 0: Disables interrupt generation. <i>This bit enables standard unlocking.</i>	0
9	OBWRE	RW0	User option word lock, software reset: 1: Indicates the user-selected word is programmable. Requires writing the correct sequence to the R32_FLASH_OBKEYR register followed by hardware setting; 0: User-selected word relocks after software reset. <i>This bit enables standard unlocking.</i>	0
8	Reserved	RO	Reserved	0
7	LOCK	RW1	Lock. Writable only as '1'. When this bit is '1', it indicates that FPEC and R32_FLASH_CTLR are locked and cannot be written. After detecting the correct unlock sequence, the hardware clears this bit to '0'. Following an unsuccessful unlock operation, this bit remains unchanged until the next system reset. <i>This bit enables standard unlocking.</i>	1
6	STRT	RW1	Start. Set to 1 to start an erase operation, and the hardware automatically clears 0(BSY changes to '0'). <i>This bit enables standard unlocking.</i>	0
5	OBER	RW	User option word erasure is performed. <i>This bit enables standard unlocking.</i>	0
4	OBPG	RW	Perform user option word programming. <i>This bit enables standard unlocking.</i>	0
[3:2]	Reserved	RO	Reserved	0
1	PER	RW	When DBMODE=1, perform a sector (8KB) erase operation. When DBMODE=0, perform a sector (4KB)	0

			erase operation. <i>This bit enables standard unlocking.</i>	
0	PG	RW	Perform standard programming operations. <i>This bit enables standard unlocking.</i>	0

46.4.6 Address Register (R32_FLASH_ADDR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FAR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	FAR	WO	The flash address, which is the programming address for programming and the starting address for erasure. The register cannot be written when the BSY bit in the FLASH_STATR register is '1'.	0

46.4.7 Option Byte Register (R32_FLASH_OBR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							DATA1[7:0]							DATA0[7:6]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA0[5:0]						US ART DLE N	USB FSD LEN	Reserved					WDC _SW	RDP RT	OBE RR

Bit	Name		Access	Description	Reset value
[31:26]	Reserved		RO	Reserved	0
[25:18]	DATA1[7:0]		RO	Data byte 1	X
[17:10]	DATA0[7:0]		RO	Data byte 0	X
9	USER	USARTDLEN	RO	BOOT enables USART keyless download functionality: 1: Enabled; 0: Disabled.	1
8		USBFSLEN	RO	BOOT enables USBFS download functionality: 1: Enabled; 0: Disabled.	1

[7:3]		Reserved	RO	Reserved	11111b
2		IWDG_SW	RO	Independent watchdog (IWDG) hardware enable bit, active low.	1
1	RDPRT		RO	Read protection status. 1: Indicates that read protection is currently enabled for the flash memory.	1
0	OBERR		RO	Character selection error. 1: Indicates that the selected character and its complement do not match.	0

46.4.8 Write Protection Register (R32_FLASH_WPR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WRP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WRP[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	WRP[31:0]	RO	Flash memory write protection status. 1: Write protection disabled; 0: Write protection enabled. When DBMODE=1, each bit of the chip represents 8KB (32 pages) of storage for write protection status, with the first 8KB in write-protected state before read protection status. When DBMODE=0, each bit of the chip represents 4KB (16 pages) of storage for write protection status, with the first 4KB in write-protected state before read protection status.	X

46.4.9 Extension Key Register (R32_FLASH_MODEKEYR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODEKEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODEKEYR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	MODEKEYR[31:0]	WO	Enter the following sequence to unlock the fast program/erase mode: KEY1 = 0x45670123; KEY2 = 0xCDEF89AB.	X

46.4.10 BOOT Key Register (R32_BOOT_MODEKEYR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BOOTKEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BOOTKEYR[15:0]															

Bit	Name	Access	Description	Reset value
[31:0]	BOOTKEYR[31:0]	WO	Enter the following sequence to unlock the fast program/erase mode: KEY1 = 0x45670123; KEY2 = 0xCDEF89AB.	X

46.4.11 Configuration Register 1 (R32_FLASH_CFGR0)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DB MO DE	Reserved											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bit	Name	Access	Description	Reset value
[31:29]	Reserved	RO	Reserved	x
28	DBMODE	RO	FLASH mode status bit: 1: Double FLASH Mode; 0: Single FLASH Mode.	x
[27:0]	Reserved	RO	Reserved	x

46.5 Flash Operation Procedure**46.5.1 Read Operation**

Direct addressing is in the general address space, and the user can access the content of the flash memory module and get the corresponding data through any read operation of 8/16/32-bit data.

46.5.2 Flash Memory Unlock

After the system reset, the flash memory controller (FPEC) and FLASH_CTLR register will be locked and cannot be accessed. The flash memory controller module can be unlocked by writing the sequence to the R32_FLASH_KEYR register.

Unlock sequence:

1) Write KEY1 = 0x45670123 to the FLASH_KEYR register (must operate KEY1 first);

2) Write KEY2 = 0xCDEF89AB to the FLASH_KEYR register (must operate KEY2 secondly).

The above operations must be performed sequentially and continuously. Otherwise, it is an error operation, which will lock the FPEC module and FLASH_CTLR register and generate a bus error until the next system reset.

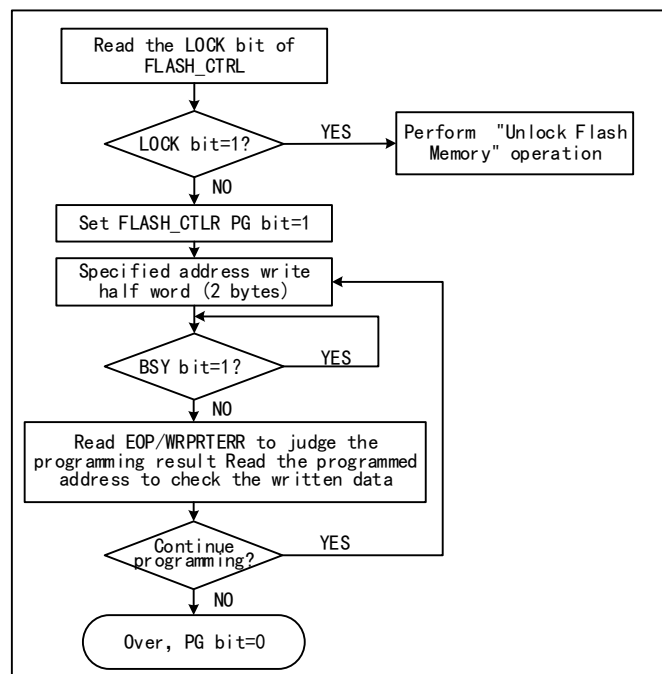
The flash memory controller (FPEC) and the R32_FLASH_CTLR register can be locked again by setting the "LOCK" bit in the R32_FLASH_CTLR register to 1.

46.5.3 Main Memory Standard Program

Standard programming writes 2 bytes per operation. When the PG bit in the R32_FLASH_CTLR register is set to '1', each write of a half-word (2 bytes) to a flash address initiates one programming cycle. Writing any non-half-word data causes FPEC to generate a bus error. During programming, the BSY bit is '1'; upon completion, BSY becomes '0' and EOP becomes '1'.

Note: When the BSY bit is '1', write operations to any register are prohibited.

Figure 46-1 FLASH program

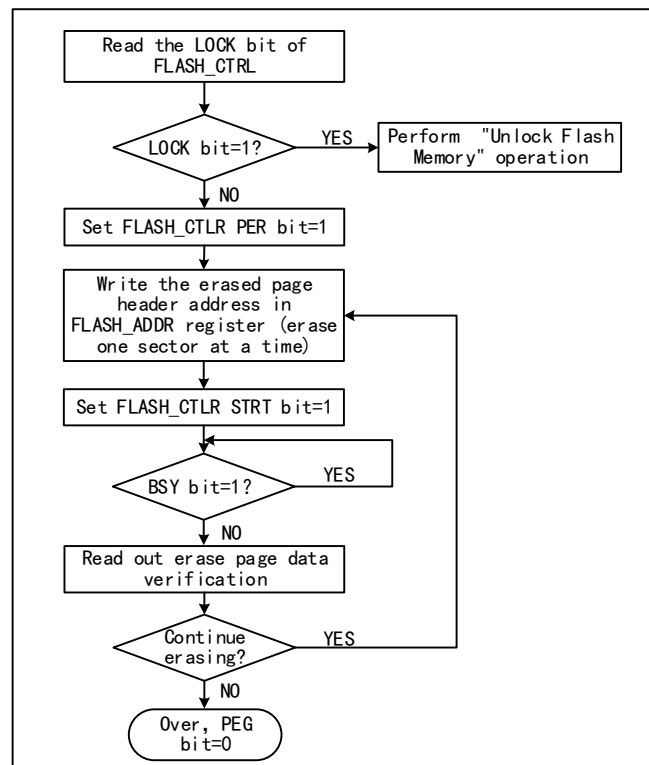


- 1) Check the LOCK bit in the R32_FLASH_CTLR register. If it is 1, you need to perform the "Release Flash Memory Lock" operation.
- 2) Set the PEG bit in the FLASH_CTLR register to '1' to enable the standard page erasure mode.
- 3) Write the half-word to be programmed to the specified flash memory address (even address).
- 4) Wait for the BSY bit to become '0' or for the EOP bit in the R32_FLASH_STATR register to become '1', indicating programming completion. Then clear the EOP bit to '0'.
- 5) Check the R32_FLASH_STATR register for errors, or read the programmed address data for verification.
- 6) To continue programming, repeat steps 3-5. To end programming, clear the PG bit.

46.5.4 Main Memory Standard Erase

Flash memory is erased by sector.

Figure 46-2 FLASH page erasure



- 1) Check the LOCK bit in the R32_FLASH_CTRL register. If it is 1, you need to perform the "Release Flash Memory Lock" operation.
- 2) Set the PEG bit in the FLASH_CTRL register to '1' to enable the sector erasure mode.
- 3) Write the page heading address of the page to be erased to the FLASH_ADDR register.
- 4) Set the STAT bit in the FLASH_CTRL register to '1' to start an erase action.
- 5) When the BYS bit changes to '0' or the EOP bit in the FLASH_STATR register to be '1', it indicates the end of erasure. Clear the EOP bit to 0.
- 6) Read the page of erasure page for verification.
- 7) To erase the sector continuously, you can repeat steps 3-5 to end erasing and clear the PEG bit to 0.

Note: After successful erasure, words read -0xe339e339, half words read -0xe339, even address bytes read -0xe39, and odd addresses read 0xe3.

46.5.5 Fast Programming Mode Unlock

Fast programming mode operation can be unlocked by writing a sequence to the R32_FLASH_MODEKEYR register. After unlocking, the block bit of the R32_FLASH_CTRL register will be cleared to 0, indicating that quick erasure and programming operations can be performed. Lock again by setting the "FLOCK" bit of the R32_FLASH_CTRL register to 1.

Unlock sequence:

- 1) Write KEY1=0x45670123 to the FLASH_MODEKEYR register.
- 2) Write KEY2=0xCDEF89AB to the FLASH_MODEKEYR register.

The above operations must be performed sequentially and continuously, otherwise the error operation will be locked and cannot be unlocked until the next system reset.

Note: Fast programming requires unlocking the "LOCK" and "FLOCK" layers.

46.5.6 Main Memory Fast Programming

The fast programming (256 bytes) is made according to the page.

- 1) Check the LOCK bit of the R32_FLASH_CTLR register. If it is '1', you need to perform the "unlock flash" operation.
- 2) Check the R32_FLASH_CTLR register FLOCK bit, if it is '1', you need to perform a "quick programming mode unlock" operation.
- 3) Check the BSY bit of the R32_FLASH_STATR register to confirm that there are no other programming operations in progress.
- 4) Set the FTPG bit of the R32_FLASH_CTLR register to '1' to enable fast page programming mode.
- 5) Write data to a FLASH address in a 32-bit manner, such as.
* (uint32_t*) 0x80000000=0x12345678.
- 6) Wait for WRBSY of R32_FLASH_STATR register to be '1', and write the next data.
- 7) Repeat steps 5-6 64 times
- 8) Set the FTPG bit in R32_FLASH_CTLR register to '1' to start fast page programming.
- 9) Waiting for the BSY bit to change to '0' or the EOP bit of the FLASH_STATR register to '1' means that a quick page programming is completed and the EOP bit is cleared to 0.
- 10) Query the R32_FLASH_STATR register to see if there are any errors, or read the programming address data check.
- 11) Continue fast page programming to repeat steps 5-10 and end the programming to clear the FTPG bit 0.

46.5.7 Main Memory Fast Erasure

Fast erase erases in blocks.

- 1) Check the R32_FLASH_CTLR register LOCK bit, if it is '1', you need to perform the "unlock flash memory" operation.
- 2) Check the R32_FLASH_CTLR register FLOCK bit, if it is '1', you need to perform a "fast programming mode unlock" operation.
- 3) Check the BSY bit of the R32_FLASH_STATR register to confirm that there are no other programming operations in progress.
- 4) Set the FTER bit of the R32_FLASH_CTLR register to '1', and turn on the fast page erasure mode function.
- 5) Write the first address of the quick erase page to the R32_FLASH_ADDR register.
- 6) Set the STAT bit of the R32_FLASH_CTLR register to '1' to start a quick page erase (256byte) action.
- 7) Wait for the BSY bit to change to '0' or the EOP bit of the R32_FLASH_STATR register to '1' means the erasure is over, and the EOP bit is cleared to 0.
- 8) Query the R32_FLASH_STATR register to see if there are any errors, or read the erase page address data check.
- 9) Continue to quickly erase the page and repeat steps 5-8 to end the erase to clear the FTER bit 0.

Note: After successful erasure, the byte reads -0xe339e339, the half-byte reads -0xe339, the even-address byte reads -0x39, and the odd-address byte reads 0xe3.

46.6 User Option Bytes

The User Option Bytes are solidified in FLASH and will be reloaded into the corresponding register after the system reset, and the user can erase and program at will. The user option word information block has a total of 8 bytes (4 bytes for write protection, 1 byte for read protection, 1 byte for configuration options, 2 bytes for user data storage), and each bit has the inverted code bit for checking during loading. The structure and meaning of the selected word

information are described below.

Table 46-4 32-bit option bytes format division

[31:24]	[23:16]	[15:8]	[7:0]
Inverse code of option bytes 1	Option bytes 1	Inverse code of option bytes 0	Option bytes 0

Table 46-5 User option bytes information structure

Address Bit	[31:24]	[23:16]	[15:8]	[7:0]
0x1FFFF800	nUSER	USER	nRDPR	RDPR
0x1FFFF804	nData1	Data1	nData0	Data0
0x1FFFF808	nWRPR1	WRPR1	nWRPR0	WRPR0
0x1FFFF80C	nWRPR3	WRPR3	nWRPR2	WRPR2

Name/Byte			Description	Reset value
RDPR			Read protection control. It configures whether the code in the flash memory can be read. 0xA5: If this byte is 0xA5 (nRDP must be 0x5A), it means that the current code is in a non-read protected state and can be read; Other values: Code read protection status, unreadable; pages 0-31 pages (4K) will be automatically write-protected and not controlled by WRPR0.	0xA5
USER	7	USARTDL EN	BOOT enable USART keyless download functionality ^[1] : 1: Enable; 0: Disable.	1
	6	USBFSDL EN	BOOT enable USBFS download functionality ^[1] : 1: Enable; 0: Disable.	1
	[5:1]	Reserved	Reserved	0x1F
	0	IWDGSW	Independent watchdog (IWDG) hardware enable bit: 1: IWDG function is enabled by software, but hardware is not allowed. 0: IWDG function is turned on by the hardware (determined by the LSI clock).	1
Data0–Data1			Stores 2 bytes of user data.	X
WRPR0 - WRPR3			Write protection control bit. Each bit is used to control the write protection status of 1 sector (When DBMODE=1: 8K bytes/sector; When DBMODE=0: 4K bytes/sector) in the main memory: 1: Disable write protection. 0: Enable write protection.	0xFFFFFFFF F

	4 bytes are used to protect main memory. WRPO: Sector 0-7 storage write protection control; WRP1: Sector 8-15 storage write protection control; WRP2: Sector 16-23 storage write protection control; WRP3: Bits 0-6 provide write protection for sectors 24-30; Bit 7 provides write protection for sectors 31-119.	
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Note [1]: Please refer to the instruction manual of CH32H417 evaluation board for the specific operation mode and precautions of this function.

46.6.1 User Option Bytes Unlock

The user option bytes operation can be unlocked by writing the sequence to the R32_FLASH_OBKEYR register. After unlocking, the OBWRE bit in the R32_FLASH_CTLR register will be set to 1, indicating that user option bytes can be erased and programmed. By setting the OBWRE bit in the R32_FLASH_CTLR register, it will be cleared to 0 by software to lock again.

Unlocking sequence

- 1) Write KEY1 = 0x45670123 to the R32_FLASH_OBKEYR register;
- 2) Write KEY2 = 0xCDEF89AB to the R32_FLASH_OBKEYR register.

Note: The user needs to unlock the 2 layers: "LOCK" and "OBWRE" for word selection.

46.6.2 User Option Bytes Programming

Only standard programming mode is supported, and half word (2 bytes) is written at a time. In the actual process, when programming the user-selected word, FPEC only uses the low byte in the half word, and automatically calculates the high byte (the high byte is the anti-code of the low byte), and then starts the programming operation, which will ensure that the byte in the user-selected word and its anti-code are always correct.

- 1) Check the LOCK bit in the R32_FLASH_CTLR register. If it is set to 1, perform the "Unlock Flash" operation.
- 2) Check the BSY bit in the R32_FLASH_STATR register to confirm no other programming operations are in progress.
- 3) Check the OBWRE bit in the R32_FLASH_CTLR register. If it is 0, perform the "User Option Byte Unlock" operation.
- 4) Set the OBPG bit in the R32_FLASH_CTLR register to '1'.
- 5) Write the half-word (2 bytes) to be programmed to the specified address.
- 6) Wait for the BSY bit to become '0' or for the EOP bit in the R32_FLASH_STATR register to become '1', indicating programming completion. Then clear the EOP bit.
- 7) Read the programming address data for verification.
- 8) To continue programming, repeat steps 5-7. To end programming, clear the OBPG bit.

Note: when "read protected" in the modified selection word is changed to "unprotected" state, an entire erase main storage area operation is automatically performed. If you modify a selection other than read Protection, the whole erase operation will not occur.

46.6.3 User Option Bytes Erasure

Erase the entire 128-byte user option bytes area directly.

- 1) Check the LOCK bit in the R32_FLASH_CTLR register. If it is set to 1, perform the "Unlock Flash" operation.
- 2) Check the BSY bit in the R32_FLASH_STATR register to confirm no programming operation is in progress.

- 3) Check the OBWRE bit in the R32_FLASH_CTLR register. If it is 0, perform the “User Option Byte Unlock” operation.
- 4) Set the OBER bit in the R32_FLASH_CTLR register to ‘1’. Subsequently, set the STRT bit in the R32_FLASH_CTLR register to ‘1’ to initiate user-selected word erase.
- 5) Wait for the BSY bit to become ‘0’ or for the EOP bit in the R32_FLASH_STATR register to become ‘1’ indicating erase completion, then clear the EOP bit to 0.
- 6) Read the erased address data for verification.
- 7) Clear the OBER bit to 0 to conclude the operation.

Note: After successful erasure, the character reads -0xe339e339, the half-character reads -0xe339, and the byte reads -0x39.

46.6.4 Read Protection Release

Whether the flash memory is read-protected or not is determined by the user option byte. Read the R32_FLASH_OBR register. When the RDPRT bit is '1', it indicates that the current flash memory is in read-protected state, and the flash operation is protected by a series of read-protected states. The process of unprotecting read is as follows:

- 1) Erase the entire user option byte area, read the protection field RDPR, and the read protection is still valid.
- 2) The user chooses words to program and writes the correct RDPR code 0xA5 to remove the read protection of the flash memory. (This step will first cause the system to automatically erase the entire piece of flash memory.).
- 3) A power-on reset is performed to reload the selected bytes (including the new RDPR code), and the read protection is removed.

46.6.5 Write Protection Release

Whether flash memory is write-protected is determined by the user-selectable bits. Reading the R32_FLASH_WPR register reveals that each bit represents a sector of flash memory space. A bit value of ‘1’ indicates an unprotected state, while ‘0’ indicates write protection. The process to remove write protection is as follows:

- 1) Erase the entire user option byte area.
- 2) Write the correct RDPR code 0xA5 to allow read access.
- 3) Perform a system reset, reload the selected bytes (including the new WRPR [3:0] bytes), and write protection is removed.

Chapter 47 Universal High-speed Interface (UHSIF)

The module description in this chapter only applies to CH32H417 microcontroller products.

The chip incorporates a universal high-speed interface (UHSIF) capable of operating at a maximum transmission clock frequency of 125MHz. It supports 8-bit, 16-bit, or 32-bit data widths, with a theoretical maximum data transfer rate of 500Mbytes/s.

For specific applications, please refer to and utilize the provided subroutine library.